FRM Part II 2026

Credit Risk Measurement and Management

Gold-source question bank

1247 questions, sorted by chapter and learning objective

Scope and LO wording: SchweserNotes 2026 (final authority). Every question solved and independently verified.

Contents

1 Reading 19. Fundamentals of Credit Risk 6
1.1 LO 19.a . . . 6
1.2 LO 19.b . . . 9
1.3 LO 19.c . . . 10
1.4 LO 19.d . . . 12
1.5 LO 19.e . . . 14

2 Reading 20. Governance 16
2.1 LO 20.a . . . 16
2.2 LO 20.b . . . 18
2.3 LO 20.c . . . 21
2.4 LO 20.d . . . 25
2.5 LO 20.e . . . 28

3 Reading 21. Credit Risk Management 32
3.1 LO 21.a . . . 32
3.2 LO 21.b . . . 35
3.3 LO 21.c . . . 37
3.4 LO 21.d . . . 39
3.5 LO 21.e . . . 41
3.6 LO 21.f . . . 42
3.7 LO 21.g . . . 44
3.8 LO 21.h . . . 46
3.9 LO 21.i . . . 48
3.10 LO 21.j . . . 49

4 Reading 22. Capital Structure in Banks 52
4.1 LO 22.a . . . 52
4.2 LO 22.b . . . 54
4.3 LO 22.c . . . 55
4.4 LO 22.d . . . 60
4.5 LO 22.e . . . 63
4.6 LO 22.f . . . 65
4.7 LO 22.g . . . 70
4.8 LO 22.h . . . 73
4.9 LO 22.i . . . 77

5 Reading 23. Introduction to Credit Risk Modeling and Assessment 79
5.1 LO 23.a . . . 79
5.2 LO 23.b . . . 83
5.3 LO 23.c . . . 86
5.4 LO 23.d . . . 91
5.5 LO 23.e . . . 98
5.6 LO 23.f . . . 101
5.7 LO 23.g . . . 106

6 Reading 24. Credit Scoring and Rating 118
6.1 LO 24.a . . . 118

6.2 LO 24.b . . . 120
6.3 LO 24.c . . . 124
6.4 LO 24.d . . . 127

7 Reading 25. Credit Scoring and Retail Credit Risk Management 131
7.1 LO 25.a . . . 131
7.2 LO 25.b . . . 133
7.3 LO 25.c . . . 135
7.4 LO 25.d . . . 138
7.5 LO 25.e . . . 140
7.6 LO 25.f . . . 143
7.7 LO 25.g . . . 145
7.8 LO 25.h . . . 147

8 Reading 26. Country Risk: Determinants, Measures, and Implications 151
8.1 LO 26.a . . . 151
8.2 LO 26.b . . . 153
8.3 LO 26.c . . . 154
8.4 LO 26.d . . . 156
8.5 LO 26.e . . . 158
8.6 LO 26.f . . . 160
8.7 LO 26.g . . . 163

9 Reading 27. Estimating Default Probabilities 165
9.1 LO 27.a . . . 165
9.2 LO 27.b . . . 167
9.3 LO 27.c . . . 168
9.4 LO 27.d . . . 170
9.5 LO 27.e . . . 174
9.6 LO 27.f . . . 178
9.7 LO 27.g . . . 180
9.8 LO 27.h . . . 182
9.9 LO 27.i . . . 184
9.10 LO 27.j . . . 186
9.11 LO 27.k . . . 188
9.12 LO 27.l . . . 190
9.13 LO 27.m . . . 202
9.14 LO 27.n . . . 211

10 Reading 28. Credit Value at Risk 214
10.1 LO 28.a . . . 214
10.2 LO 28.b . . . 216
10.3 LO 28.c . . . 221
10.4 LO 28.d . . . 226
10.5 LO 28.e . . . 232
10.6 LO 28.f . . . 236

11 Reading 29. Portfolio Credit Risk 240
11.1 LO 29.a . . . 240
11.2 LO 29.b . . . 248

11.3 LO 29.c . . . 250
11.4 LO 29.d . . . 256
11.5 LO 29.e . . . 260
11.6 LO 29.f . . . 262
11.7 LO 29.g . . . 264
11.8 LO 29.h . . . 265

12 Reading 30. Credit Risk 268
12.1 LO 30.a . . . 268
12.2 LO 30.b . . . 270
12.3 LO 30.c . . . 272
12.4 LO 30.d . . . 274
12.5 LO 30.e . . . 276
12.6 LO 30.f . . . 278
12.7 LO 30.g . . . 280

13 Reading 31. Credit Derivatives 284
13.1 LO 31.a . . . 284
13.2 LO 31.b . . . 289
13.3 LO 31.c . . . 291
13.4 LO 31.d . . . 293
13.5 LO 31.e . . . 295
13.6 LO 31.f . . . 298
13.7 LO 31.g . . . 300
13.8 LO 31.h . . . 302

14 Reading 32. Derivatives 304
14.1 LO 32.a . . . 304
14.2 LO 32.b . . . 306
14.3 LO 32.c . . . 311
14.4 LO 32.d . . . 314
14.5 LO 32.e . . . 318
14.6 LO 32.f . . . 320
14.7 LO 32.g . . . 321
14.8 LO 32.h . . . 324
14.9 LO 32.i . . . 326
14.10 LO 32.j . . . 331

15 Reading 33. Counterparty Risk and Beyond 333
15.1 LO 33.a . . . 333
15.2 LO 33.b . . . 337
15.3 LO 33.c . . . 341
15.4 LO 33.d . . . 343
15.5 LO 33.e . . . 350
15.6 LO 33.f . . . 352
15.7 LO 33.g . . . 357
15.8 LO 33.h . . . 359

16 Reading 34. Netting, Close-Out, and Related Aspects 363
16.1 LO 34.a . . . 363

16.2 LO 34.b . . . 365
16.3 LO 34.c . . . 371
16.4 LO 34.d . . . 382
16.5 LO 34.e . . . 385
16.6 LO 34.f . . . 389

17 Reading 35. Margin (Collateral) and Settlement 392
17.1 LO 35.a . . . 392
17.2 LO 35.b . . . 393
17.3 LO 35.c . . . 401
17.4 LO 35.d . . . 405
17.5 LO 35.e . . . 407
17.6 LO 35.f . . . 409
17.7 LO 35.g . . . 411
17.8 LO 35.h . . . 416
17.9 LO 35.i . . . 419
17.10 LO 35.j . . . 421
17.11 LO 35.k . . . 426

18 Reading 36. Central Clearing 428
18.1 LO 36.a . . . 428
18.2 LO 36.b . . . 429
18.3 LO 36.c . . . 431
18.4 LO 36.d . . . 434
18.5 LO 36.e . . . 435
18.6 LO 36.f . . . 438
18.7 LO 36.g . . . 441
18.8 LO 36.h . . . 444
18.9 LO 36.i . . . 447
18.10 LO 36.j . . . 449

19 Reading 37. Future Value and Exposure 454
19.1 LO 37.a . . . 454
19.2 LO 37.b . . . 470
19.3 LO 37.c . . . 471
19.4 LO 37.d . . . 473
19.5 LO 37.e . . . 481
19.6 LO 37.f . . . 482
19.7 LO 37.g . . . 487
19.8 LO 37.h . . . 489
19.9 LO 37.i . . . 497

20 Reading 38. CVA 499
20.1 LO 38.a . . . 499
20.2 LO 38.b . . . 501
20.3 LO 38.c . . . 503
20.4 LO 38.d . . . 508
20.5 LO 38.e . . . 511
20.6 LO 38.f . . . 514
20.7 LO 38.g . . . 516

20.8 LO 38.h . . . 521
20.9 LO 38.i . . . 523
20.10 LO 38.j . . . 529
20.11 LO 38.k . . . 531
20.12 LO 38.l . . . 532
20.13 LO 38.m . . . 541
20.14 LO 38.n . . . 542
20.15 LO 38.o . . . 544
20.16 LO 38.p . . . 546
20.17 LO 38.q . . . 548

21 Reading 39. The Evolution of Stress Testing Counterparty Exposures 551
21.1 LO 39.a . . . 551
21.2 LO 39.b . . . 553
21.3 LO 39.c . . . 555
21.4 LO 39.d . . . 558
21.5 LO 39.e . . . 561
21.6 LO 39.f . . . 564
21.7 LO 39.g . . . 565
21.8 LO 39.h . . . 569

22 Reading 40. Structured Credit Risk 572
22.1 LO 40.a . . . 572
22.2 LO 40.b . . . 574
22.3 LO 40.c . . . 578
22.4 LO 40.d . . . 579
22.5 LO 40.e . . . 583
22.6 LO 40.f . . . 587
22.7 LO 40.g . . . 592
22.8 LO 40.h . . . 594
22.9 LO 40.i . . . 595
22.10 LO 40.j . . . 601
22.11 LO 40.k . . . 603
22.12 LO 40.l . . . 604
22.13 LO 40.m . . . 606

23 Reading 41. An Introduction to Securitization 608
23.1 LO 41.a . . . 608
23.2 LO 41.b . . . 612
23.3 LO 41.c . . . 616
23.4 LO 41.d . . . 619
23.5 LO 41.e . . . 621
23.6 LO 41.f . . . 622
23.7 LO 41.g . . . 624
23.8 LO 41.h . . . 631

Reading 19. Fundamentals of Credit Risk

Sylvain Bouteille and Diane Coogan-Pushner, The Handbook of Credit Risk Management: Originating, Assessing, and Managing Credit Exposures, 2nd Edition (Hoboken, NJ: John Wiley & Sons, 2022). Chapter 1.

LO	Learning objective	Qs
19.a	define credit risk and explain how it arises using examples.	6
19.b	explain the differences between insolvency, default, and bankruptcy.	4
19.c	identify and describe transactions that generate credit risk.	3
19.d	describe the entities that are exposed to credit risk and explain circumstances under which exposure occurs.	3
19.e	discuss the motivations for managing or taking on credit risk.	3

LO 19.a

Define credit risk and explain how it arises using examples.

Q1

AnalystPrep Q.5772

REBUILT — Option A used 'insolvency' (a balance-sheet concept distinct from default per LO 19.b) as the credit risk; reworded A to the contractual-default definition and tightened the stem.

A venture capitalist invests in convertible bonds issued by a loss-making technology startup. The bonds pay a fixed coupon and may later be converted into equity. Which risk is the investor exposed to as a creditor, as distinct from the risk of holding the equity?

A. The startup fails to honor its contractual interest or principal payments, because of inability, unwillingness or delay.
B. Market conditions reduce the value of the equity received on conversion.
C. Technological obsolescence erodes the competitiveness of the startup's product.
D. Changes in tax law reduce the startup's after-tax profitability.

Answer: A

Step by step

1. Credit risk is the possibility of losing money because a counterparty is unable, unwilling or untimely in honoring a financial obligation.
2. As bondholder the investor holds a claim to coupons and principal, so the credit risk is the issuer failing to pay them.
3. Compare each option: only A is a failure to honor a contractual obligation.

Why A is correct: Credit risk is the loss arising from a counterparty's inability, unwillingness or nontimeliness in honoring its obligation; for a bond, that means missed coupon or principal payments.

Why the others are wrong

- **B:** Describes market/equity valuation risk on the conversion option, not failure to pay an obligation.
- **C:** Obsolescence is a business risk; it can lead to credit loss only indirectly, via the ability to pay, and is not the definition of credit risk.
- **D:** Tax law change is regulatory/tax risk; it too only affects credit risk indirectly.

Q2

AnalystPrep Q.5777

A bank issues fixed-coupon corporate bonds to fund an infrastructure project. For an investor buying these bonds, which statement best describes credit risk?

A. The market value of the bond falls because interest rates rise.
B. The investor loses money because the bank fails to meet its coupon and principal obligations.
C. Higher operating costs reduce the bank's profitability.
D. A downturn in the banking sector weakens economic conditions generally.

Answer: B

Step by step

1. Credit risk = loss if a counterparty fails to honor a financial obligation (inability, unwillingness or delay).
2. For a bond the obligations are the coupons and principal; B describes losing money when the issuer does not

pay them.
3. A is a price effect of rates, C and D are background drivers rather than the risk itself.

Why B is correct: Credit risk is the possibility of loss from the obligor's failure to honor its payment obligations, which for a bond are coupons and principal.

Why the others are wrong

- **A:** Interest rate (market) risk: price changes with rates even if the issuer pays in full.
- **C:** A cause that may raise the chance of non-payment, not the definition of credit risk itself.
- **D:** Systemic/sector conditions are a driver of default, not the risk of loss from non-payment.

Q3 Schweser Module Quiz 19.1 Q1

Which of the following sets of factors is most critical in helping creditors assess credit risk?

A. Amount of credit risk, probability of counterparty default, recovery amount and timing.
B. Foreign currency exposure, amount of credit risk, amount of illiquid counterparty assets.
C. Probability of counterparty default, counterparty management strength, recovery amount.
D. Recovery amount and timing, amount of uninsured assets, probability of counterparty insolvency.

Answer: A

Step by step

1. For each credit-risk transaction the three fundamental questions are: (1) how much can be lost (amount of credit risk/exposure), (2) what is the probability of default, (3) how much is recovered, when and at what cost.
2. Only A lists exactly these three.

Why A is correct: Creditors assess exposure amount, probability of counterparty default, and recovery amount and timing.

Why the others are wrong

- **B:** Currency exposure and illiquid assets are not among the three core factors; only 'amount of credit risk' is right.
- **C:** Replaces the exposure amount with management strength, a qualitative input rather than a core factor.
- **D:** Uninsured assets are not a core factor, and the probability that matters is of default, not insolvency (a balance-sheet condition).

Q4 AnalystPrep Q.5775

On a peer-to-peer lending platform, individual investors lend directly to borrowers without a bank intermediary. What is the investor's primary risk with respect to the borrower's commitment to repay?

A. Liquidity risk from the platform's matching mechanism.
B. Credit risk that the borrower fails to repay the loan.
C. Regulatory risk from future changes in peer-to-peer lending laws.
D. Market risk from fluctuations in platform loan interest rates.

Answer: B

Step by step

1. The investor has lent funds and is owed repayment.
2. The risk that the borrower does not honor that obligation is credit risk by definition.
3. The other risks exist but do not concern the borrower's commitment.

Why B is correct: Lending exposes the lender to loss if the borrower is unable, unwilling or late in repaying: credit risk.

Why the others are wrong

- **A:** Liquidity risk is about the ability to exit or match positions, not borrower repayment.
- **C:** Regulatory risk concerns law changes, not the borrower's repayment.
- **D:** Market (rate) risk concerns price/rate moves, not failure to pay.

Q5 AnalystPrep Q.5774

REBUILT — Stem was vague ('most critically') and options were weakly differentiated; reworded to ask which information bears directly on credit risk using the three assessment questions.

A financial services firm is deciding whether to extend a short-term credit line to a corporate client. Which consid-

eration bears most directly on the firm's credit risk?

A. The historical volatility of the client's stock price.

B. The likelihood that the client fails to repay, based on its repayment record and current ability to meet obligations.

C. The client's long-term strategic growth plans.

D. The general trend of the economy, regardless of the client's circumstances.

Answer: B

Step by step

1. Credit risk is loss if the counterparty fails to honor its obligation.
2. Assessing it means estimating the probability of default (repayment record, capacity to pay now), along with exposure and recovery.
3. B addresses default probability directly; A is market risk, C is long-dated and irrelevant to a short-term line, D is an indirect macro factor.

Why B is correct: Credit risk is driven by the borrower's probability of failing to pay, which depends on its repayment history and present ability to meet obligations.

Why the others are wrong

- **A:** Equity price volatility is a market-risk measure and says little about the ability to repay.
- **C:** Long-term growth plans are not relevant to repayment of a short-term line.
- **D:** Macro trends matter only through their effect on this client's ability to pay; ignoring the client makes it indirect.

Q6

BT answers P2.T6.24.1.1

REBUILT — Original options C and D both turned on whether the Jan 21 limit or the Jan 25 purchase raised the bank's risk, and the source explanation relied on an unstated 50% usage assumption; rebuilt so only the Jan 25 drawdown option is correct.

Jason applies for a credit card with Chazz Bank on January 15. Chazz Bank approves the card on January 21 with a $5,000 spending limit. On January 25 Jason uses the card to buy a washer and dryer worth $2,500. His first bill arrives January 31 and is due February 15. Using the reading's broad definition of credit risk (which includes credit exposure), which of the following statements is TRUE?

A. Jason created credit risk for himself when he applied for the card on January 15.

B. Jason became exposed to credit risk on January 25 because he now owes $2,500 to the bank.

C. Chazz Bank has no credit risk until Jason misses the February 15 due date.

D. Chazz Bank's credit exposure to Jason increased on January 25, when his purchase drew on the credit extended to him.

Answer: D

Key correction: Source key (C) said 'Chazz Bank increased its credit risk on January 21'; that option was defensible but option D ('did not increase on January 25') was also arguable and the source explanation assumed a 50% usage rate not given in the stem. The rebuilt item has a single clearly correct option, the January 25 drawdown.

Step by step

1. Credit risk is the risk that a creditor loses money if a counterparty fails to honor an obligation (inability, unwillingness or delay); broadly it includes the exposure to that possible loss.
2. The party that supplies credit and is owed money bears the credit risk. Here the creditor is Chazz Bank and the obligor is Jason.
3. On January 25 Jason's $2,500 purchase turns part of the $5,000 limit into an actual amount owed: the bank's money is now outstanding and at risk of non-payment.
4. Hence the bank's credit exposure rose on January 25. Jason, as the debtor, does not bear credit risk on the amount he owes.

Why D is correct: Credit exposure arises for the party owed money; the January 25 purchase converts available credit into an outstanding receivable for Chazz Bank, increasing its exposure.

Why the others are wrong

- **A:** An application creates no obligation; no credit has been extended on January 15, and a debtor does not bear credit risk on its own borrowing (confuses obligor with creditor).
- **B:** Reverses the roles: Jason is the obligor owing the money, so the bank, not Jason, is exposed to the risk of non-payment.
- **C:** Treats credit risk as arising only at default; credit risk exists from the moment credit is extended, because the risk is the possibility of loss, not its realization.

LO 19.b

Explain the differences between insolvency, default, and bankruptcy.

Q7

Schweser Module Quiz 19.1 Q2 · AnalystPrep Q.5781

Acquaria Corporation's year-end balance sheet shows $280 million in assets and $320 million in debt to creditors. Acquaria's management estimates that it will continue to be able to meet its upcoming payment obligations. The company is best characterized as being:

A. bankrupt.
B. insolvent.
C. in default.
D. nonperforming.

Answer: B

Step by step

1. Liabilities = $320 million exceed assets = $280 million, so equity = 280 − 320 = −$40 million (negative).
2. Insolvency is defined as liabilities exceeding assets, so Acquaria is insolvent.
3. Management expects to meet upcoming payments, so no contractual obligation has been missed: not in default.
4. No court protection has been sought: not bankrupt.

Why B is correct: Insolvency is a balance-sheet condition (liabilities greater than assets). It can exist before any payment is missed and before any court filing.

Why the others are wrong

- **A:** Bankruptcy is a legal procedure of seeking court protection; negative equity alone does not trigger it (confuses insolvency with bankruptcy).
- **C:** Default requires failing to meet a contractual obligation; the firm still expects to pay.
- **D:** Nonperforming describes a loan on which payments are overdue; no payment has been missed here.

Q8

AnalystPrep Q.6179

REBUILT — Original stem said the firm was 'seeking legal protection', which blurs the bankruptcy distinction; replaced with an explicit statement that no court filing has been made.

XYZ Corporation's liabilities exceed its assets, and it has failed to make several scheduled payments to its creditors. It is negotiating informally with creditors but has not filed for protection with any court. Which statement most accurately describes XYZ's status?

A. XYZ is insolvent but not in default or bankrupt.
B. XYZ is in default but not insolvent or bankrupt.
C. XYZ is both insolvent and in default, but not bankrupt.
D. XYZ is bankrupt but not insolvent or in default.

Answer: C

Step by step

1. Liabilities exceed assets: insolvent.
2. Scheduled payments have been missed: failure to meet contractual obligations, so in default.
3. Bankruptcy requires seeking legal protection through a court; XYZ has not filed, so it is not bankrupt.

Why C is correct: Insolvency (balance sheet), default (missed obligations) and bankruptcy (court procedure) are distinct; XYZ meets the first two only.

Why the others are wrong

- **A:** Ignores the missed payments, which constitute default.
- **B:** Ignores that liabilities exceed assets, which is insolvency.
- **D:** Bankruptcy needs a court filing, which has not occurred; it also contradicts the stated insolvency and default.

Q9

BT notes P2.T6.24.1.1

REBUILT — Balance sheet had OCR-damaged cash figure and unclear labels, and one option (default due to insolvency) was arguable given the firm was solvent; reconstructed the table (cash 52,000) and replaced that option.

Old Crow Corporation's balance sheet at 12/31/2022 (in $):

Assets	Amount	Liabilities	Amount
Cash	52,000	Accounts payable	27,000
Accounts receivable	35,000	Short-term debt	94,000
Inventory	20,000	Accrued expenses	20,000
Prepaid expenses	13,000	Rent payable	9,000
Total current assets	120,000	Total current liabilities	150,000
Long-term assets	80,000	Long-term liabilities	45,000
Total assets	200,000	Total liabilities	195,000

Short-term debt is the current portion of a loan from New Orleans Bank. On 1/1/2023 Old Crow was unable to repay this loan and has not sought court protection. Each of the following statements is true EXCEPT which one?

A. Old Crow defaulted on 1/1/2023, and the default could be due to illiquidity.

B. Old Crow went bankrupt on 12/31/2022 because its current assets exceeded its current liabilities.

C. Old Crow's default on 1/1/2023 does not by itself make it bankrupt.

D. Old Crow was solvent on 12/31/2022.

Answer: B

Key correction: Source did not provide a key; intended answer B.

Step by step

1. Solvency: total assets 200,000 > total liabilities 195,000 (equity = +5,000), so D is true.
2. Liquidity: current assets 120,000 < current liabilities 150,000, a 30,000 shortfall, so inability to repay a short-term loan is consistent with illiquidity; A is true.
3. Failing to repay the loan on 1/1/2023 is a default; bankruptcy needs a court filing, so C is true.
4. B is false twice over: bankruptcy is a legal procedure (none occurred) and current assets did not exceed current liabilities (120,000 < 150,000).

Why B is correct: Bankruptcy is a court procedure, not a balance-sheet condition, and the stated premise (current assets exceed current liabilities) is also false.

Why the others are wrong

- **A:** True: default occurred and the current-asset shortfall makes illiquidity a plausible cause.
- **C:** True: default and bankruptcy are distinct; bankruptcy requires seeking court protection.
- **D:** True: assets exceed liabilities by 5,000, so the firm was solvent despite being illiquid.

Q10

AnalystPrep Q.5780

REBUILT — Option B was loosely worded; reworded to the court-procedure definition (court steps in after default and filing) and tightened distractors.

Which option most accurately defines bankruptcy as it applies to a corporation?

A. A condition in which the corporation's liabilities exceed its assets.

B. A legal procedure, typically following default, in which the corporation seeks court protection and the court negotiates with management and creditors toward liquidation or reorganization.

C. A failure to meet a contractual obligation, such as a missed payment.

D. An informal, out-of-court agreement between the debtor and its creditors to restructure its debt.

Answer: B

Step by step

1. Bankruptcy is a legal process: the entity (typically already in default) files for court protection.
2. The court negotiates with management, creditors and other stakeholders; the outcome is liquidation (Chapter 7 in the US) or reorganization (Chapter 11).
3. Match: option B.

Why B is correct: Bankruptcy is distinguished from insolvency and default by the involvement of a court.

Why the others are wrong

- **A:** Describes insolvency (negative equity), a balance-sheet state, not a legal process.
- **C:** Describes default; default usually precedes but is not bankruptcy.
- **D:** An out-of-court restructuring involves no court; bankruptcy does.

LO 19.c

Identify and describe transactions that generate credit risk.

Q11 AnalystPrep Q.5771

REBUILT — Original stem did not say whether the financing was with or without recourse, making option C defensible; specified non-recourse factoring and tightened the options.

A finance company provides non-recourse accounts receivable financing: it advances cash to a manufacturer against the manufacturer's invoices and keeps a reserve percentage as a cushion. Which transaction type is this, and what is the finance company's principal credit exposure?

A. Receivables; the risk that the manufacturer's customers (the account debtors) fail to pay their invoices on time or at all

B. Lending to the manufacturer only; the risk that the manufacturer's own financial condition deteriorates

C. Derivatives; the risk that interest rates move against the fee the finance company charges

D. Prepayment; the risk that the manufacturer fails to deliver goods it was paid for

Answer: A

Step by step

1. Identify the transaction: the finance company buys/advances against invoices owed by buyers who have days or weeks to pay, which is a receivables transaction.
2. Because the financing is non-recourse, the finance company cannot claw back losses from the manufacturer.
3. So the cash it advanced is repaid only if the account debtors pay; their non-payment is the credit loss.
4. The reserve is merely a cushion against that loss, not a removal of the exposure.

Why A is correct: In non-recourse receivables financing the repayment source is the buyers' payments, so the credit risk sits with the account debtors.

Why the others are wrong

- **B:** Confuses the exposure with a plain loan to the manufacturer; with non-recourse terms the manufacturer's solvency is not the main source of loss.
- **C:** Fee-rate/interest changes are market risk, and the transaction is not a derivative.
- **D:** Prepayment risk is faced by the party that pays in advance for goods; here cash is advanced against goods already delivered.

Q12 BT BT Mock Q12 · BT answers P2.T6.24.1.3 · BT notes P2.T6.24.1.3

REBUILT — Original asked on which dates risk 'increased', which is ambiguous for the 13 Feb payment receipt (which extinguishes a receivable); reframed to ask which transactions generate credit risk.

A steel wholesaler (i) buys an insurance policy on 4 January, (ii) pays a supplier in full on 31 January for steel to be delivered on 2 February, (iii) delivers steel to two customers on 8 and 9 February, who pay on 13 February, and (iv) on 13 February deposits the proceeds at ABC Bank. Which of the following correctly identifies the transactions that generate credit risk for the wholesaler?

A. None of them; credit risk arises only from lending money

B. Only the insurance policy and the prepayment to the supplier

C. Only the sales to customers on credit and the bank deposit

D. All four: the insurance policy, the prepayment, the credit sales and the bank deposit

Answer: D

Step by step

1. Insurance policy (4 Jan): a contingent claim; the policyholder has no claim at inception, but if a covered loss occurs and the insurer fails to pay, a credit loss results.
2. Prepayment (31 Jan): the supplier might fail to deliver the steel it was paid for.
3. Credit sales (8-9 Feb): receivables; customers might not pay by the due date.
4. Bank deposit (13 Feb): claim on the bank; the wholesaler might not have timely access to funds.
5. Each is a standard credit-risk transaction type (contingent claim, prepayment, receivable, deposit), so all four generate credit risk.

Why D is correct: Per GARP, credit risk arises from lending, leases, receivables, prepayments, claims on assets held by others (deposits), contingent claims and derivatives, not only from lending.

Why the others are wrong

- **A:** Wrongly restricts credit risk to traditional lending; none of these is a loan but all are exposures.
- **B:** Omits receivables and deposits; treats only the January items as exposures although a credit sale and a deposit are also claims on counterparties.
- **C:** Omits the contingent claim on the insurer and the prepayment, forgetting that paying in advance or buying

insurance leaves you reliant on the counterparty.

Q13 Schweser Module Quiz 19.2 Q1

A bank has entered into a $25 million, 6-month repurchase agreement with an investment grade corporate client, collateralized by $26 million notional value, 10-year state bonds. The bank has:

A. no credit risk because the repurchase agreement matures before the bonds.
B. no credit risk because the client is rated investment grade; therefore, counterparty default is unlikely.
C. no credit risk because the notional value of the bonds exceeds the value of the repurchase agreement.
D. credit risk because if the client defaults, the bank may not be able to sell the bonds to cover the full amount of the repurchase agreement.

Answer: D

Step by step

1. A repurchase agreement is a transaction type that creates credit risk (asset-based lending/derivative-like exposure).
2. If the client defaults, the bank must liquidate the bonds to recover its $25 million.
3. The bonds' market value can fall (10-year bonds carry price risk) or they may be hard to sell, so proceeds may be below $25 million.
4. Collateral and a good rating reduce, but do not eliminate, the exposure.

Why D is correct: Collateral reduces but does not remove credit risk because its liquidation value is uncertain at the time of default.

Why the others are wrong

- **A:** Maturity mismatch is irrelevant to whether exposure exists; the bank is exposed through the 6-month life.
- **B:** Investment grade lowers default probability but does not make it zero.
- **C:** Notional (face) value is not the realizable market value; a 4% cushion can be eroded by price drops or illiquidity.

LO 19.d

Describe the entities that are exposed to credit risk and explain circumstances under which exposure occurs.

Q14 BT answers P2.T6.24.2.2

Ron is a risk manager for RLA, an insurance company. Below is a conversation between Ron and his colleague Warner about credit risk.
- Ron tells Warner, "RLA faces significant credit risk in our investment portfolio. We are taking on too much risk in emerging markets debt and need to find a way to hedge this risk."
- Warner responds, "I am more concerned about the credit risk we take on when originating new policies, in particular when we extend installment terms and grace periods to our corporate customers."
- Ron responds, "No, Warner, what we should worry about is our emerging market credit risk and the credit risk we are taking on from buying reinsurance."
- Warner responds, "Reinsurance? Reinsurance does not create credit risk. Reinsurance helps us mitigate our risk exposure."

In the above conversation, each of the statements is likely true (i.e., prima facie plausible) EXCEPT which of the statements is most likely incorrect?

A. Ron's statement about emerging market debt.
B. Warner's statement about originating new policies.
C. Ron's statement about reinsurance.
D. Warner's statement about reinsurance.

Answer: D

Step by step

1. Insurers have credit exposure through underwriting, investments and reinsurance.
2. Investments: holding emerging market debt exposes RLA to issuer default (Ron's first statement is plausible).
3. Underwriting: installment terms and grace periods mean premiums are received late or not at all, so RLA bears

the risk of non-payment (Warner's first statement is plausible).
4. Reinsurance: ceding claims transfers insurance risk but creates a reinsurance recoverable, a contingent claim on the reinsurer; if the reinsurer is slow or fails to pay, RLA suffers a credit loss (Ron's second statement is plausible).
5. Warner's claim that reinsurance creates no credit risk is therefore the incorrect statement.

Why D is correct: Reinsurance replaces insurance risk with credit risk on the reinsurer, because the reinsurance recoverable is only as good as the reinsurer's ability and willingness to pay.

Why the others are wrong

- **A:** True statement: bonds in the investment portfolio expose the insurer to issuer default. Rejecting it ignores investments as a source of insurer credit risk.
- **B:** True statement: granting installment terms and grace periods creates premium-collection (non-payment) risk in underwriting. Rejecting it ignores underwriting as a source of credit risk.
- **C:** True statement: the reinsurance recoverable is a contingent claim on the reinsurer and is a credit exposure; rejecting it repeats Warner's error of treating risk transfer as risk elimination.

Q15

BT notes P2.T6.24.1.1

REBUILT — Original left unclear whether the washer/dryer was charged to the card and the 'credit risk' definition drifted into EAD modelling not in the reading; clarified the facts and keyed the answer to who is exposed and when.

Jason applies for a credit card with Chazz Bank on January 15. Chazz Bank approves the card on January 21 with a $5,000 spending limit, and Jason may use it immediately. On January 25 Jason charges a $2,500 washer and dryer to the card; Chazz Bank pays the merchant. Jason's first bill arrives January 31, due February 15. Assume credit risk is defined broadly to include exposure. Which of the following statements is TRUE?

A. Jason creates credit risk for himself on January 15 by applying for the card.
B. Jason's credit risk increased on January 21 when the card was approved.
C. Chazz Bank became exposed to credit risk on January 21, when it committed to the $5,000 limit.
D. Chazz Bank's credit risk did not increase on January 25 because the card limit was already approved.

Answer: C

Step by step

1. Credit risk is borne by the party that may not be paid: the lender or provider of credit, not the borrower.
2. Jan 15: an application creates no commitment, so no one is exposed yet (rules out A).
3. Jan 21: approval creates a credit commitment of $5,000 that Jason can draw at will; the bank is now exposed to him (C true). Jason holds a right to borrow, not an exposure, so B is false.
4. Jan 25: the $2,500 charge is paid by the bank, converting undrawn commitment into an actual receivable, so the bank's exposure rises; D is false.

Why C is correct: Exposure arises for the party extending credit, and it starts when the commitment is made, not only when funds are drawn.

Why the others are wrong

- **A:** The borrower is the obligor, not the party exposed; an application also creates no commitment. Mistake: treating the borrower as bearing credit risk.
- **B:** Jason owes money and is the source of risk to the bank; approval gives him an option to borrow, not a claim that can go unpaid. Mistake: confusing borrower with creditor.
- **D:** The Jan 25 charge turns undrawn limit into a drawn receivable of $2,500, increasing actual exposure. Mistake: assuming exposure only counts at approval, ignoring drawdowns.

Q16

Schweser Module Quiz 19.2 Q2

Which of the following options would a corporation least likely select to mitigate its receivables credit risk?

A. Factoring.
B. Insurance.
C. Derivatives.
D. Documentary credit.

Answer: C

Step by step

1. Corporations typically mitigate receivables credit risk in three ways: insure the receivables, sell them (factor-

ing), or secure foreign sales with documentary credit.
2. Derivatives are not a typical tool for corporate receivables, so they are the least likely choice.

Why C is correct: Insurance, factoring and documentary credit are the standard receivables credit mitigants; derivatives are not.

Why the others are wrong

- **A:** Factoring (selling receivables to another company) is a standard mitigant, so it is likely to be used.
- **B:** Buying insurance on accounts receivable is a standard mitigant.
- **D:** Documentary credit is a bank guarantee commonly used to secure foreign transactions.

LO 19.e

Discuss the motivations for managing or taking on credit risk.

Q17

BT answers P2.T6.24.2.1

Tan Company, an international motorcycle distributor, is experiencing a decline in sales due to a global economic slowdown. The company seeks strategies to enhance sales volumes while minimizing additional credit risk exposure. Which of the following methods would allow Tan Company to achieve an increase in sales, taking on the least amount of credit risk?

A. Purchase insurance on their accounts receivable.
B. Set up a financing arm that helps customers secure credit from other sources.
C. Require collateral from customers.
D. Sell receivables at the end of every month.

Answer: B

Step by step

1. Goal: more sales (so credit must be made available to customers) while adding as little credit risk as possible.
2. Helping customers obtain financing from outside lenders lets Tan make the sale and be paid by the lender, so the customer credit risk sits with the lender rather than on Tan's balance sheet (unlike in-house vendor financing, where Tan keeps the risk).
3. Insurance, collateral and receivable sales either leave Tan extending the credit itself first (and retaining residual or interim exposure) or make terms tougher; none shifts the credit decision to a third party while supporting sales.

Why B is correct: Arranging third-party customer financing supports sales growth while the credit exposure is borne by the outside lender, so Tan's incremental credit risk is minimal.

Why the others are wrong

- **A:** Insurance leaves Tan extending the credit itself, with exposure to the insurer's ability to pay and to any uninsured portion; it is a mitigant of risk already taken, not a way of removing the customer exposure from the sale.
- **C:** Requiring collateral reduces loss given default but makes terms tougher, which tends to reduce rather than increase sales.
- **D:** Selling receivables only offloads exposure after Tan has extended the credit; month-end sales leave risk in the interim and do not boost sales.

Q18

BT Mock Q17

REBUILT — Original option A gave a true conclusion (C highest ROE after losses) with a false reason, and option B's claim was loosely worded; options rewritten so exactly one statement is correct.

Firm A, Firm B and Firm C are three bottled-water companies, each with a mix of debt and equity. Each has total assets of USD 1,000 million, borrows at 5%, pays tax at 25%, and generated EBIT of USD 100 million before credit losses. In 2022 each suffered credit losses of USD 60 million, which reduced EBIT.

Firm	Debt	Equity	Interest expense
A	700	300	35
B	500	500	25
C	300	700	15

Which of the following statements is correct?

A. With the credit losses, Firm A has the highest ROE because it uses the most leverage.
B. With the credit losses, Firm C has the lowest ROE because it has the most equity.

C. Without credit losses, Firm A would have had the highest ROE.
D. Without credit losses, Firm C would have had the highest ROE because it has the lowest interest expense.

Answer: C
Step by step
1. With losses, EBIT = 100 - 60 = 40. Net income = (EBIT - interest) x (1 - 0.25): A = (40-35)(0.75) = 3.75; B = (40-25)(0.75) = 11.25; C = (40-15)(0.75) = 18.75.
2. ROE with losses: A = 3.75/300 = 1.25%; B = 11.25/500 = 2.25%; C = 18.75/700 = 2.68%. Firm C is highest, Firm A lowest.
3. Without losses, EBIT = 100: A = (65)(0.75)/300 = 16.25%; B = (75)(0.75)/500 = 11.25%; C = (85)(0.75)/700 = 9.11%.
4. Without losses ROA is 10% > 5% cost of debt, so leverage raises ROE and Firm A (most debt) is highest. With losses ROA falls to 4% < 5%, so leverage hurts and the least-levered firm does best.

Why C is correct: Leverage amplifies ROE when asset returns exceed the borrowing cost (10% vs 5% before losses) and depresses it when they fall below (4% after losses); hence A is highest only in the no-loss case.

Why the others are wrong
- **A:** With losses A has the lowest ROE (1.25%); after losses ROA (4%) is below the 5% debt cost, so leverage hurts.
- **B:** C has the most equity but the highest ROE with losses (2.68%) because it bears the least interest.
- **D:** C has the lowest interest expense but also the lowest no-loss ROE (9.11%) since its equity cushion earns only 10% ROA while A and B gear up at 5% debt.

Q19

BT notes P2.T6.24.2.3

REBUILT — Original had OCR-damaged table, no tax treatment for a negative pre-tax income, and options not matching the question; rebuilt with clean consistent data and numeric options.

Firm A, Firm B and Firm C each have total assets of USD 1,000 million and pre-tax EBIT of USD 90 million before credit losses. Each borrows at 4% and pays tax at 25% on positive pre-tax income.

Firm	Debt	Equity	Credit losses
A	600	400	50
B	400	600	30
C	200	800	10

EBIT is reduced by credit losses; interest is paid on debt. Which firm has the highest ROE after credit losses, and what is it?

A. Firm A, 12.4%
B. Firm B, 5.5%
C. Firm C, 6.75%
D. Firm C, 9.0%

Answer: C
Step by step
1. Interest: A = 0.04(600) = 24; B = 0.04(400) = 16; C = 0.04(200) = 8.
2. EBIT after losses: A = 40; B = 60; C = 80. Pre-tax income: 16; 44; 72.
3. Net income at 75%: A = 12.0; B = 33.0; C = 54.0.
4. ROE = NI/equity: A = 12/400 = 3.00%; B = 33/600 = 5.50%; C = 54/800 = 6.75%.
5. Firm C has the highest ROE at 6.75%.

Why C is correct: Credit losses hit the heavily levered, high-loss firm hardest; Firm C has the smallest losses and interest burden, so its after-tax income relative to equity is highest.

Why the others are wrong
- **A:** 12.4% is Firm A's ROE ignoring credit losses ((90-24)(0.75)/400); it also is not the highest after losses (A is actually lowest at 3.0%).
- **B:** 5.5% is Firm B's correct ROE, but B is not the highest; C is higher.
- **D:** 9.0% is Firm C's pre-tax return on equity (72/800), forgetting to deduct 25% tax.

Reading 20. Governance

Sylvain Bouteille and Diane Coogan-Pushner, The Handbook of Credit Risk Management: Originating, Assessing, and Managing Credit Exposures, 2nd Edition (Hoboken, NJ: John Wiley & Sons, 2022). Chapter 2.

LO	Learning objective	Qs
20.a	define risk management responsibilities in an organization and explain the three lines of defense framework for effective risk management and control.	5
20.b	explain the processes that lead to risk taking including credit origination, credit risk assessment, and credit approval processes.	6
20.c	discuss the following key principles underlying best practice for the governance system of credit risk: Guidelines, Skills, Limits, and Oversight.	8
20.d	describe the most common parameters of a credit-sensitive transaction.	6
20.e	describe the roles of the credit committee in an organization.	7

LO 20.a

Define risk management responsibilities in an organization and explain the three lines of defense framework for effective risk management and control.

Q20

AnalystPrep Q.5791

REBUILT — Original was a generic scenario with giveaway distractors; rewrote around the GARP three-lines framework so exactly one option is defensible.

A bank launches a new product line. Under the three lines of defense framework, which assignment of risk management responsibilities is most appropriate?

A. Business owners own and manage the risks; enterprise risk management, compliance and legal monitor and oversee them and set policies; internal and external auditors provide independent assurance.

B. Enterprise risk management owns and manages all risks; business owners monitor the risk function; auditors set policies and procedures.

C. Business owners both manage the risks and provide the independent assurance over their own risk controls; the risk function only advises.

D. Internal and external auditors own and manage the risks and report to the business owners; compliance and legal provide independent assurance.

Answer: A

Step by step

1. Recall the GARP framework (IIA, 2013): first line = business owners who own and manage risk.
2. Second line = enterprise risk management, compliance and legal, who establish policies and procedures and oversee the first line.
3. Third line = internal audit, external auditors and audit committees giving independent assurance over the first two lines.
4. Only A matches all three assignments.

Why A is correct: Risk is owned by the business (first line), overseen by independent risk/compliance/legal functions that set policy (second line), and independently assured by auditors (third line).

Why the others are wrong

- **B:** Reverses the first and second lines and puts policy-setting in the third line; ERM oversees, it does not own, the risk.
- **C:** Removes independence: business owners cannot assure their own controls, and the second line oversees (not merely advises) the first.
- **D:** Confuses the third line with the first and misplaces assurance with compliance/legal, which belong to the second line.

Q21

AnalystPrep Q.5792

REBUILT — Original had giveaway distractors and an unspecific stem; reframed to test the CRO/second-line coordinating and oversight role against the first line.

A retailer is implementing a new inventory system with data security and supply chain risks. The CRO heads the second line of defense. Which action best fits the CRO's responsibility under the three lines of defense?

A. Establish policies and risk measurement standards and oversee the business and IT owners who manage the

risks, while retaining overall responsibility for an integrated view of risk.
B. Take over day-to-day management of the system's risks from the business owners so that responsibility is centralized.
C. Delegate all risk management for the system to the IT department and rely on its reports without challenge.
D. Perform independent audits of the system implementation and report the results directly to the audit committee.

Answer: A
Step by step
1. The CRO leads the second line: enterprise risk management, with compliance and legal.
2. Second-line duties: set policies and procedures and monitor and oversee risks taken by the first line.
3. Risk ownership stays with first-line business owners (here IT, supply chain, finance) who manage their risks.
4. Independent assurance via audits is third-line, so D is excluded. A is the only option consistent with the CRO's role.

Why A is correct: The second line sets policies and provides management oversight of the first line, which keeps ownership of risk.
Why the others are wrong
- **B:** Takes ownership away from the first line; the second line oversees rather than manages the risks.
- **C:** Passing everything to IT without challenge abandons the second-line oversight function.
- **D:** Describes the third line (audit); the CRO's second-line role is not independent assurance.

Q22 Mock 1 Q17

REBUILT — Premise of a 'fourth line' is not in the GARP framework, which places external auditors in the third line; rewritten to test which parties sit in the third line.

Under the three lines of defense framework described by GARP, which party belongs to the third line of defense?
A. Business line staff who originate credit transactions.
B. The credit risk management department.
C. External auditors.
D. The compliance function.

Answer: C
Key correction: Source key (B) treated external auditors as a fourth line; GARP puts external auditors in the third line, so the original premise is invalid.
Step by step
1. First line: business owners who own and manage risk (A).
2. Second line: enterprise risk management, compliance and legal (B, D).
3. Third line: internal audit, external auditors and special audit committees providing independent assurance.
4. So external auditors (C) are the third-line party.

Why C is correct: The third line is the independent assurance layer, and GARP includes external auditors in it.
Why the others are wrong
- **A:** Business line staff are the first line (owners of risk).
- **B:** Credit risk management is part of the second-line risk oversight function.
- **D:** Compliance is second-line; confusing independence from the business with independence from management.

Q23 Schweser Module Quiz 20.1 Q1

To ensure an adequate second line of defense, organizations should:
A. include the compliance and legal functions in risk oversight.
B. look primarily to business owners because they own and manage the risks.
C. ensure that the first line of defense has proper oversight of the second line.
D. ensure that the audit function provides an independent risk management role.

Answer: A
Step by step
1. The second line consists of enterprise risk management, compliance and legal.
2. These functions set policies and procedures and oversee the first line.

3. A matches this composition; B, C and D describe the first line, a reversed oversight direction, and the third line.

Why A is correct: The second line is built from ERM, compliance and legal functions that oversee risk-taking.

Why the others are wrong

- **B:** Describes the first line (business owners), not the second.
- **C:** Reverses the oversight direction: the second line oversees the first.
- **D:** Independent assurance is the third line (audit), not the second.

Q24 AP Mock 3 Q36

Which of the following is most likely a role and responsibility of the second line of defense for the overall risk management of a firm under risk governance?

A. Making decisions for managing risks.
B. Establishing risk management methods and measurement methods.
C. Overseeing the risk management activities of the third line of defense.
D. Reports independently to the board of directors.

Answer: B

Step by step

1. Second line = ERM, compliance, legal: establish policies and procedures and oversee the first line.
2. Setting risk management and measurement methods fits that role (B).
3. Decision-making on risks is first line (A); oversight flows from second to first, not to the third (C); independent reporting is the third line (D).

Why B is correct: The second line designs the risk framework (policies, methods, measurement) and oversees the first line.

Why the others are wrong

- **A:** Business owners make and manage risk decisions: first line.
- **C:** The third line is independent; the second line does not oversee audit, it oversees the first line.
- **D:** Independent reporting to the board characterizes the third line (audit).

LO 20.b

Explain the processes that lead to risk taking including credit origination, credit risk assessment, and credit approval processes.

Q25 AnalystPrep Q.5793

Credit origination is the first of the processes that lead to risk taking in a financial institution. Which of the following best describes the key activity in the credit origination process?

A. Evaluating the financial stability of a client and setting credit limits
B. Identifying potential credit transactions and understanding client needs
C. Approving or rejecting credit transactions based on risk assessments
D. Monitoring and reviewing existing credit transactions for compliance

Answer: B

Step by step

1. The processes that lead to risk taking are (1) credit origination, (2) credit risk assessment, and (3) credit approval.
2. Origination is the front-office activity of sourcing and structuring transactions: finding opportunities and understanding what the client needs.
3. Match each option to its process: A is assessment, C is approval, D is post-approval monitoring.
4. Only B describes sourcing and structuring, so B is origination.

Why B is correct: Origination is carried out by originators who identify and structure transactions for clients; analysis of creditworthiness and the yes/no decision come in the later assessment and approval processes.

Why the others are wrong

- **A:** Evaluating a client's financial strength and setting limits is credit risk assessment, the second process, not origination.

- **C:** Approving or rejecting transactions is credit approval, the third process, performed under delegated authority.
- **D:** Monitoring existing transactions takes place after approval and is not part of the origination process.

Q26

BT notes P2.T6.24.3.2

REBUILT — Approval matrix was OCR-jumbled; reconstructed it, stated that each figure is the maximum exposure that level can approve, and made the cumulative-approval point explicit.

Julie is a credit officer at Old Crow Bank and wants approval for a $135M loan to a counterparty rated R3. The bank's delegation of authority sets, for each rating, the maximum exposure (in US$ millions) that each level may approve. Delegation is cumulative.

Rating	Head of Trading	Head of Risk	Credit Committee
R1	135	160	185
R2	110	135	160
R3	85	110	135
R4	60	85	110
R5	35	60	85
R6	10	35	60

Who must approve this loan?

A. Head of Trading only
B. Head of Risk only
C. Credit Committee only
D. Head of Trading, Head of Risk, and Credit Committee

Answer: D

Step by step

1. Read the R3 row: Head of Trading 85, Head of Risk 110, Credit Committee 135.
2. $135M exceeds 85, so the Head of Trading cannot approve alone.
3. $135M exceeds 110, so the Head of Risk cannot approve alone.
4. $135M does not exceed 135, so the Credit Committee is the lowest level with sufficient authority.
5. Delegation of authority is cumulative: each intervening level must approve before the transaction reaches the credit committee, so all three approve.

Why D is correct: The exposure requires credit committee authority, and because delegation is cumulative the transaction cannot skip the Head of Trading and Head of Risk.

Why the others are wrong

- **A:** Treats the lowest level as sufficient, ignoring that 135 exceeds its R3 limit of 85.
- **B:** Ignores that 135 exceeds the Head of Risk R3 limit of 110, and omits the other levels.
- **C:** Correctly sees that only the credit committee has enough authority but forgets approval is cumulative; the deal cannot go straight to the committee.

Q27

Mock 2 Q33

REBUILT — Original relied on a three-question framework not found in the reading and its answer was only loosely supported; rebuilt to test the role of risk management in credit approval.

A bank's guidelines require input from risk management on transactions above certain size and rating thresholds. Which statement best describes the role of the risk management function in the credit approval process?

A. It holds a veto over every transaction because it is the second line of defense.
B. It has no approval authority; it advises decision makers, and its recommendation is required when the criteria in the guidelines are met.
C. It approves all transactions below the business unit head's limit so that originators are not slowed down.
D. It originates the transaction jointly with the business so that risk is built into the structure from the start.

Answer: B

Step by step

1. Business managers are accountable for their units' top and bottom lines, so approval authority sits with the business hierarchy and committees.
2. Risk management is not a profit center and has no approval authority; it plays an advisory role.
3. Guidelines state which transactions need a risk recommendation (for example, those above an exposure thresh-

old), and that recommendation must be obtained.
4. Vetoes by risk managers are rare, so A is wrong; B matches this description.

Why B is correct: Approval authority is delegated to the business hierarchy and committees, while risk management advises and provides a recommendation required by the guidelines for transactions meeting stated criteria.

Why the others are wrong

- **A:** Overstates the role: risk managers rarely hold a veto; they recommend and escalate disagreement.
- **C:** Gives risk management approval authority, which it does not have, and confuses it with a delegated approver.
- **D:** Confuses risk management with origination; originators source transactions, and joint origination would compromise independence.

Q28

AnalystPrep Q.5948

REBUILT — Stem tightened to ask which scenario shows a failure of credit risk assessment, and the options were sharpened so only one is clearly a breakdown of the process.

A bank's internal audit is reviewing weaknesses in its credit risk assessment and approval processes. Which of the following situations most clearly represents a failure of those processes?

A. A borrower with a good credit history is offered a loan at a slightly higher rate after the bank raises its base rate.

B. A large loan is approved for a new business without a thorough review of the business plan and market viability.

C. The bank tightens lending criteria, slightly reducing the number of approved mortgages.

D. A small personal loan to a long-standing customer is approved after income documentation and affordability checks are completed.

Answer: B

Step by step

1. Credit risk assessment should analyze the key risks of a counterparty before a decision is made.
2. A and C are routine pricing and policy changes and do not skip any assessment step.
3. D is a small loan where assessment was performed (documentation and affordability checks).
4. B approves a large exposure to an unproven counterparty without analyzing the plan or market, so assessment was effectively bypassed.

Why B is correct: A large exposure to a new business was approved without the analysis that credit risk assessment requires, so risk was taken without a basis for judging it.

Why the others are wrong

- **A:** A rate change that follows base rates is ordinary pricing, not a process failure.
- **C:** Tightening criteria is a conservative policy decision that lowers risk.
- **D:** Assessment was carried out and the exposure is small, so the relationship alone is not the basis for approval.

Q29

AnalystPrep Q.5949 · AnalystPrep Q.5950

REBUILT — Source scenarios had two defensible answers (collateral-based auto lending vs owner-credit-based lending; start-up projections); rebuilt around the origination incentive problem described in the reading.

At a bank, originators are paid mainly on loan volume and margin, and they prepare the case for each transaction themselves. Which outcome is the most direct risk that this arrangement creates in the credit origination process?

A. Originators tend to push for higher volume and returns and present deals favorably, increasing the number of higher-risk transactions.

B. Originators are unable to find enough clients because they spend too much time on documentation.

C. The credit committee is unable to meet because originators are overburdened.

D. Guidelines become too concise because originators prefer short documents.

Answer: A

Step by step

1. Incentive systems typically reward top-line growth and sometimes return on risk-adjusted capital.
2. As a result originators push for volume, margins and returns and show their transactions in the most favorable light.
3. This tends to increase the number of higher-risk transactions that reach assessment and approval.
4. Therefore adequate credit risk assessment and approval must exist to counter this bias; A is the direct risk.

Why A is correct: Volume- and margin-based rewards bias originators towards riskier deals and favorable pre-

sentation, which is why independent assessment and approval are needed.

Why the others are wrong

- **B:** Has no link to the incentive problem; the concern is too much, not too little, origination.
- **C:** Committee scheduling is not a consequence of originator incentives.
- **D:** Concise guidelines are a desirable feature, not a consequence of incentives.

Q30 AnalystPrep Q.5950

REBUILT — Source was a vague red-flag scenario; rebuilt as a numerical delegation-of-authority application including the risk assessment recommendation rule.

A bank's approval process has the following maximum exposure (US$ millions) that each level can approve for a transaction of tenor under five years. Delegation is cumulative, and the credit risk assessment unit must give a recommendation for every R3 transaction above $100M.

Rating	Head of Trading	Transaction Committee	Executive Risk Committee
R3	75	150	250

A three-year transaction with an R3 counterparty generates an exposure of $130M. Which approval path is required?

A. Head of Trading only

B. Transaction Committee only, with no risk assessment recommendation

C. Head of Trading and Transaction Committee, with a recommendation from the credit risk assessment unit

D. Head of Trading, Transaction Committee and Executive Risk Committee, with a recommendation from the credit risk assessment unit

Answer: C

Step by step

1. Exposure is $130M, which exceeds the Head of Trading limit of $75M.
2. It is below the Transaction Committee limit of $150M, so the Transaction Committee is the required top approver (no need to reach the $250M executive committee).
3. Delegation is cumulative, so the Head of Trading must also approve before the Transaction Committee.
4. $130M exceeds $100M for R3, so the credit risk assessment unit's recommendation is required.

Why C is correct: The top approver is the lowest level whose limit covers $130M, all lower levels approve too, and the exposure triggers a risk assessment recommendation.

Why the others are wrong

- **A:** 130 exceeds the Head of Trading limit of 75.
- **B:** Skips the cumulative lower level and the required recommendation (130 > 100).
- **D:** Escalates unnecessarily to the Executive Risk Committee although 130 is within the Transaction Committee limit of 150.

LO 20.c

Discuss the following key principles underlying best practice for the governance system of credit risk: Guidelines, Skills, Limits, and Oversight.

Q31 AnalystPrep Q.5804

REBUILT — Original had an OCR artifact in option A and giveaway distractors; rewrote options around the four GARP/Schweser attributes of good guidelines.

Which of the following best describes the characteristics that credit guidelines (credit policies) should have under best-practice governance?

A. Understandable, concise, precise (specific, with real-life scenarios) and easily accessible to users, for example through a short summary on the intranet

B. Written in formal legal language for enforceability, and comprehensive enough to cover every conceivable scenario in detail

C. Kept deliberately general and brief so that originators retain maximum flexibility, with precise interpretation left to each business unit

D. Restricted to senior credit officers to prevent misuse, with less experienced staff given only verbal instructions

Answer: A

Step by step

1. Best-practice guidelines must be understandable, concise, precise and accessible.
2. Understandable: plain, non-legalistic language. Concise: reasonable length. Precise: specific wording with real-life scenarios, without sacrificing precision for brevity. Accessible: easily available, e.g., a 1-2 page intranet summary.
3. Only option A contains all four attributes.

Why A is correct: Guidelines define how transactions must be conducted, so they must be usable by everyone who applies them: clear, short, specific and easy to find.

Why the others are wrong

- **B:** Legal jargon violates 'understandable' and an exhaustive document violates 'concise'.
- **C:** Brevity must not come at the cost of precision; vague guidelines cannot ensure consistent decisions or limit breaches.
- **D:** Guidelines must be accessible to all users; restricting them defeats their purpose.

Q32

AnalystPrep Q.5812

REBUILT — Original relied on generic wording; rebuilt around the specific GARP/Schweser requirements for independent oversight (reporting line, compensation, profit-center separation).

Which of the following arrangements best reflects effective oversight of credit risk?

A. Risk managers sit within the business profit center to stay close to originators, and their bonuses depend on the unit's profitability

B. Risk managers are located outside profit centers, their compensation is independent of business profitability, and they report to the CRO, who has access to the board's risk and audit committees

C. Risk managers report to the business unit head who sponsors the transactions, since that person best understands the risks

D. Risk managers are fully independent and avoid contact with business units and clients so that their judgment cannot be influenced

Answer: B

Step by step

1. Effective oversight needs independence, strong qualifications, closeness to the business and an open mind.
2. Independence: risk management is not in a profit center, pay is independent of business profitability, and it reports to the CRO (who reports to the CEO and has access to board risk/audit committees) rather than to business heads.
3. Risk managers must still understand the business, so isolation is not required. Only B satisfies both independence and closeness.

Why B is correct: Independence is achieved structurally (outside profit centers, pay not tied to profit, reporting to the CRO), while retaining business understanding.

Why the others are wrong

- **A:** Placing risk in a profit center and linking pay to profitability destroys independence.
- **C:** Reporting to business unit heads compromises independence; risk managers report to the CRO.
- **D:** Independence must be balanced with solid understanding of business units, profit drivers and transaction structures; isolation is not required.

Q33

AnalystPrep Q.5814 · AnalystPrep Q.5816

REBUILT — Two near-identical vague scenario questions merged and rebuilt around concrete GARP/Schweser rules on sponsorship, approval and review of guidelines.

A bank is designing the governance process for its credit guidelines. Which of the following is most consistent with best practice?

A. The CRO's office drafts and maintains the guidelines, which are sponsored by the CFO or CRO and approved by the board including its risk committee; they are reviewed periodically and after events that cause losses

B. The head of originations drafts and approves the guidelines, since business units best understand the products, and they are revised only when regulators require

C. The guidelines are drafted by the most junior risk staff to keep them simple, approved by the CRO alone, and reviewed every ten years

D. The guidelines are outsourced to an external consultant who drafts, approves and reviews them, leaving inter-

nal staff focused on operations

Answer: A

Step by step

1. The CRO typically owns the guidelines; the CRO's office drafts, seeks approval for and maintains them, and drafters must have sufficient seniority and expertise.
2. Best practice is for the CFO or CRO to sponsor them, with final approval by the board including a risk committee.
3. Guidelines should be reviewed periodically and after loss events or major changes (e.g., M&A, regulation), so only A fits.

Why A is correct: Guidelines protect the organization, so ownership sits with senior risk management, approval with the board, and review is both periodic and event-driven.

Why the others are wrong

- **B:** Originators have a conflict of interest, and review should not wait for regulators.
- **C:** Drafters need seniority and expertise, board approval is expected, and ten years is far too infrequent.
- **D:** Management remains ultimately responsible; outsourcing ownership and approval is not best practice.

Q34

AnalystPrep Q.5817

REBUILT — Original compared vague 'sophisticated' scenarios with AI giveaways; rebuilt around the GARP/Schweser rule on the advisory role and skills of risk management.

Which of the following statements about the role of risk management in the approval of credit transactions is most accurate?

A. Risk management holds final approval authority and a veto over all transactions because it is independent of the business

B. Risk management is an advisory function without approval authority; guidelines specify where its input is required, and dissenting views are typically expressed in written memos escalated to the credit committee

C. Risk management should avoid advising originators, since helping close deals would compromise its independence

D. Risk management becomes a profit center when it recommends transactions, so recommendations must be avoided

Answer: B

Step by step

1. Risk management is not a profit center and has no approval authority; it advises the business.
2. Guidelines describe when its input or recommendation is required, and risk managers typically have no veto.
3. If they strongly disagree, they write a memo that is escalated to the credit committee, whose members can hear their opinion. Only B matches.

Why B is correct: The skills principle delegates approval to experienced professionals; risk management supports them in an advisory role and escalates dissent rather than vetoing.

Why the others are wrong

- **A:** Risk management typically has no approval authority or veto right.
- **C:** Risk managers should build good relations with originators and help them succeed; advising does not impair independence.
- **D:** Risk management is never a profit center; recommending transactions does not change that.

Q35

Schweser Module Quiz 20.2 Q1

A firm's risk management unit is updating its guidelines relating to transaction approval and delegation. Risk managers are worried that even when transactions fall within available credit limits, foreign exchange volatility may cause breaches in these limits. Which of the following guideline parameters would be most appropriate to deal with this type of outcome?

A. The guidelines should include a carve-out for breaches that are due to foreign exchange rate volatility.

B. Breaches in the credit limit due to foreign exchange rate movements should be immediately escalated to the Chief Risk Officer (CRO).

C. The guidelines should treat each breach of the credit limit as an error by the originator, because currency movements should be anticipated.

D. Breaches of the credit limit due to foreign exchange rate movements should be immediately escalated to the risk committee of the board of directors.

Answer: A

Step by step

1. Guidelines should be written so that breaches are infrequent, but some breaches arise naturally rather than from intentional misconduct.
2. Breaches caused by FX movements are the standard example the guidelines should tolerate via a carve-out.
3. Treating them as originator error or escalating each one to the CRO or board would be punitive or inefficient.

Why A is correct: FX moves are outside the originator's control, so guidelines should carve out such unintentional breaches.

Why the others are wrong

- **B:** Escalating every FX-driven breach to the CRO is unnecessary; a carve-out handles it.
- **C:** Blaming the originator treats an unintentional, external breach as misconduct.
- **D:** Escalating to the board's risk committee is excessive for routine, tolerated breaches.

Q36

Schweser Module Quiz 20.2 Q2

Which of the following statements about risk managers' reporting structure is most accurate?

A. To be most effective, risk managers should report to the business unit heads.
B. To preserve their independence, risk managers should report directly to the CRO.
C. Risk managers should have dotted-line reporting to both the CRO and business unit heads.
D. To ensure effective risk governance, risk managers should report directly to the board of directors.

Answer: B

Step by step

1. Independence requires that risk management is not in a profit center and reports to the CRO rather than to business unit heads.
2. The CRO in turn reports to the CEO and has access to the board's risk and audit committees, giving risk influence at the top.
3. Hence B.

Why B is correct: Reporting to the CRO insulates risk managers from business profit pressures while keeping risk influential at the top of the firm.

Why the others are wrong

- **A:** Reporting to business heads compromises independence.
- **C:** Dotted-line reporting to business heads dilutes independence; not recommended.
- **D:** The board provides broad oversight; day-to-day risk managers report through the CRO, who has board access.

Q37

AnalystPrep Q.5809

REBUILT — Original rewarded an 'AI-driven' buzzword option not found in the reading; rebuilt around the GARP/Schweser definition and setting of limits.

Which of the following statements about credit limits within a firm's governance framework is most accurate?

A. Limits represent the maximum loss the firm will accept in absolute dollar terms; there is an overall limit with smaller limits for counterparties, sectors, industries or countries, set using exposure assessment and management judgment
B. Limits are set solely from model outputs, and management judgment is excluded to ensure objectivity
C. Limits apply only to individual counterparties, because an aggregate organization-wide limit is impossible to define
D. Limits are set by originators individually, since they know their clients best, and are not preapproved to avoid slowing deals

Answer: A

Step by step

1. Limits (credit lines) are the maximum loss acceptable in absolute dollar terms.
2. They include an aggregate limit and smaller limits for counterparties, sectors, industries or countries, and also limits for specific businesses.
3. They are set by assessing risk exposures combined with management's experience and intuition; originators generally like preapproved limits. Only A is consistent.

Why A is correct: Limits convert risk appetite into absolute dollar loss ceilings, cascaded from an overall limit to sub-limits, based on exposure analysis and management judgment.

Why the others are wrong
- **B:** Management experience and intuition are explicitly part of limit setting.
- **C:** Organizations have aggregate limits plus sub-limits; limits are not restricted to counterparties.
- **D:** Originators typically prefer preapproved limits, and limits are not set by originators alone.

Q38 AnalystPrep Q.5810

REBUILT — Original rewarded vague 'multi-stage process' language; rebuilt around the GARP/Schweser two-step delegation of authority (parameters then approval level).

A bank is designing its delegation of authority for credit approvals. Which of the following is most consistent with best practice?

A. Assign risk parameters (exposure amount, credit quality and tenor) to each transaction, then delegate approval authority based on those parameters, with higher approval levels for riskier transactions

B. Send every transaction, regardless of size, to the credit committee to ensure uniform treatment

C. Base approval level on the seniority of the originator, so that the most senior originators can approve any transaction

D. Allow lower approval levels for longer-tenor and lower-credit-quality transactions because they earn higher margins

Answer: A

Step by step
1. Delegation of authority has two steps: (1) assign risk parameters to each transaction, (2) delegate authority based on those parameters.
2. Parameters are the amount of exposure, credit quality and tenor.
3. Approval level rises with transaction risk: large, long-tenor or low-quality deals need senior management, while simple low-risk deals can be approved by one individual.

Why A is correct: Matching approval authority to measured transaction risk puts decisions with appropriately skilled people while keeping the CRO and committees free for high-risk deals.

Why the others are wrong
- **B:** Impractical for the CRO or committees to approve everything; authority is delegated down.
- **C:** Authority depends on transaction risk, not on the originator's seniority, and would create a conflict of interest.
- **D:** The opposite is true: longer tenor, larger size and lower credit quality require higher approval levels.

LO 20.d

Describe the most common parameters of a credit-sensitive transaction.

Q39 Schweser Module Quiz 20.2 Q3

A firm recently developed a model to help summarize the creditworthiness of each of its counterparties. Which of the following risk parameters of a credit-sensitive transaction does this most likely represent?

A. Tenor.

B. Credit quality.

C. Maximum loss.

D. Amount of exposure.

Answer: B

Step by step
1. The three common parameters of a credit-sensitive transaction are amount of exposure, credit quality and tenor.
2. Credit quality is the scale or model a firm develops to summarize the creditworthiness of each counterparty.
3. The model in the stem summarizes counterparty creditworthiness, so it is credit quality.

Why B is correct: Credit quality is, by definition, the firm's summary measure (scale or model) of counterparty creditworthiness.

Why the others are wrong
- **A:** Tenor is the length of time the firm is exposed until the last obligation is due, not a creditworthiness measure.

- **C:** Maximum loss is not a separate parameter; it is the idea behind amount of exposure, and a creditworthiness model does not estimate it.
- **D:** Amount of exposure is the estimated maximum amount that can be lost; the model measures counterparty quality, not size of loss.

Q40 AnalystPrep Q.5820 · AnalystPrep Q.5821

REBUILT — Two near-identical source items had options unrelated to the LO (dividends, market growth, etc.); merged and rebuilt to test the definition of amount of exposure with options drawn from the parameter set.

A bank is approving a large term loan to an industrial manufacturer. The credit officer must record, for the loan, the estimate of the maximum amount of money the bank could lose on the transaction. Which transaction parameter is the officer determining?

A. Tenor.
B. Credit quality.
C. Amount of exposure.
D. Counterparty diversification.

Answer: C

Step by step

1. The common parameters of a credit-sensitive transaction are amount of exposure, credit quality and tenor.
2. Amount of exposure is the estimate of the maximum amount a company can lose; how to compute it depends on the nature of the transaction.
3. The officer is estimating the maximum possible loss, so the parameter is amount of exposure.

Why C is correct: Amount of exposure is defined as the estimated maximum amount of money the firm can lose on the transaction.

Why the others are wrong

- **A:** Tenor is the period over which the credit exposure exists, not the size of the potential loss.
- **B:** Credit quality is a scale summarizing the counterparty's creditworthiness (likelihood of loss), not the size of the loss.
- **D:** Counterparty diversification is not one of the three standard transaction parameters.

Q41 AnalystPrep Q.5818

REBUILT — Source asked which method best assesses credit quality, with an arguable answer and irrelevant distractors; rewritten to test the definition of credit quality against the other two parameters.

Which of the following best describes the credit quality parameter of a credit-sensitive transaction?

A. The period from the closing of the agreement until the last principal repayment is due.
B. The estimated maximum amount of money the firm can lose on the transaction.
C. A scale developed by the firm that summarizes the creditworthiness of each counterparty.
D. The counterparty's share of its industry's total revenue.

Answer: C

Step by step

1. Credit quality is one of three common parameters (with amount of exposure and tenor).
2. Each company develops its own scale to summarize the creditworthiness of each counterparty.
3. Option C states this; A is tenor and B is amount of exposure.

Why C is correct: Credit quality is the firm's own scale summarizing counterparty creditworthiness, which is assessed to gauge the risk of loss.

Why the others are wrong

- **A:** This defines tenor, not credit quality.
- **B:** This defines amount of exposure, not credit quality.
- **D:** Market share is a single business characteristic, not the firm's creditworthiness scale, and is not a transaction parameter.

Q42 AnalystPrep Q.5819

REBUILT — Source relied on irrelevant-distractor options about loan tenor; rebuilt as a computational item testing the definition of tenor.

A bank closes a loan on January 1, 2026. Principal is repaid in five equal annual installments, with the final installment due on January 1, 2031; interest is paid quarterly. What is the tenor of this loan as a transaction parameter?

A. 5 years.
B. 3 years.
C. 1 year.
D. 0.25 years.

Answer: A

Step by step

1. Tenor is the period during which there is credit exposure, ending when the remaining financial obligation is paid.
2. For a loan it runs from closing of the loan agreement to the due date of the last principal repayment.
3. Jan 1, 2026 to Jan 1, 2031 = 5 years.

Why A is correct: Tenor for a loan is measured from closing to the last principal repayment, here 5 years.

Why the others are wrong

- **B:** 3 years is the weighted-average life of the principal repayments (0.2 x (1+2+3+4+5)); it is not the tenor.
- **C:** 1 year is the spacing between principal installments, not the full period of exposure.
- **D:** 0.25 years is the interest payment interval; interest dates do not determine tenor.

Q43 AnalystPrep Q.5951

REBUILT — Source tested loan duration with subjective borrower-quality scenarios (and secondary-mapped to another reading); rebuilt as a clean tenor comparison.

A bank compares four loans of equal size to counterparties of equal credit quality. Which loan has the longest tenor?

Loan	Closing date	Final principal repayment
W	Mar 1, 2025	Mar 1, 2030
X	Jan 1, 2024	Jan 1, 2031
Y	Jun 1, 2026	Jun 1, 2030
Z	Jan 1, 2025	Jan 1, 2029

A. W
B. X
C. Y
D. Z

Answer: B

Step by step

1. Tenor = time from closing to the last principal repayment.
2. W = 5.0 years; X = 7.0 years; Y = 4.0 years; Z = 4.0 years.
3. X has the longest tenor, 7 years.

Why B is correct: With amount and credit quality equal, tenor is the differentiating parameter; X is exposed for 7 years, the longest.

Why the others are wrong

- **A:** W spans 5 years; it is neither the earliest-closing nor the longest-lasting loan.
- **C:** Y has a later final repayment date than Z, but it spans only 4 years; comparing end dates instead of closing-to-final-repayment length gives this error.
- **D:** Z has the earliest maturity and spans only 4 years.

Q44 AnalystPrep Q.5952

REBUILT — Source was a vague risk-factor comparison relying on mitigating-factor judgement with no clear parameter; rebuilt as a mapping of the three transaction parameters.

A bank's credit file for a transaction records: (i) 'estimated maximum loss: $40 million'; (ii) 'internal counterparty grade: 3 on a 1-10 scale'; (iii) 'final principal repayment due 6 years after closing'. Which of the following correctly maps (i), (ii) and (iii) to the transaction parameters?

A. (i) amount of exposure; (ii) credit quality; (iii) tenor.

B. (i) credit quality; (ii) amount of exposure; (iii) tenor.
C. (i) amount of exposure; (ii) tenor; (iii) credit quality.
D. (i) tenor; (ii) credit quality; (iii) amount of exposure.

Answer: A
Step by step
1. (i) The estimated maximum loss is the amount of exposure.
2. (ii) A grade on an internal scale summarizing counterparty creditworthiness is credit quality.
3. (iii) The time until the last principal repayment is the tenor.

Why A is correct: Amount of exposure measures potential loss, credit quality ranks counterparty creditworthiness, and tenor measures the time to the last obligation.
Why the others are wrong
- **B:** Swaps exposure and credit quality: the dollar loss estimate is exposure, the grade is credit quality.
- **C:** Swaps credit quality and tenor: the grade is not a time period and the repayment horizon is not a creditworthiness scale.
- **D:** Misassigns all but one: the loss estimate is not a time period and the 6-year horizon is not a dollar amount.

LO 20.e

Describe the roles of the credit committee in an organization.

Q45

AnalystPrep Q.5822 · AnalystPrep Q.5823

REBUILT — Merged two near-identical items and removed the unsupported 'sets credit policies' wording and weak distractors; now tests the committee's core role of deciding on high-risk transactions.

A bank's credit committee is meeting to consider a large, high-risk loan to a corporate client. The originator has already prepared the approval package and the bank's relationship team has discussed indicative terms with the client. Which of the following is the primary role of the credit committee in this situation?

A. Preparing the credit analysis and approval package for the transaction
B. Negotiating the pricing and terms of the loan directly with the client
C. Making the approval decision (approve or decline) on the transaction
D. Providing independent assurance on the bank's credit processes as part of the third line of defense

Answer: C
Step by step
1. The credit committee consists of senior executives and exists to make decisions on important or high-risk transactions.
2. The package is prepared by the originator and circulated in advance; the committee reviews it and decides to approve or decline.
3. Preparing the package and negotiating terms are originator/front-office tasks; independent assurance is internal/external audit.

Why C is correct: The credit committee is the decision-making body for high-profile, high-risk transactions; originators prepare and market deals, committee members review and decide.
Why the others are wrong
- **A:** Package preparation is the originator's job; the committee reviews the package.
- **B:** Negotiating with clients is a front-office function and would compromise the committee's objectivity.
- **D:** Independent assurance is the role of audit (third line); the committee is a decision body within the approval process.

Q46

AP Mock 3 Q45

REBUILT — Original asked which task a committee 'most likely' undertakes using operational distractors and an unsupported policy-revision key; rebuilt around the Schweser/GARP membership rules (expertise, key functions represented, personal membership with limited delegation).

A bank's credit committee charter is being reviewed against best practice. Which of the following provisions is most consistent with best practice for committee membership?

A. Members should be senior decision-makers from the key functions (business, risk, tax and accounting, com-

pliance, legal) and may delegate attendance to junior staff freely

B. Members should be senior decision-makers from the key functions (business, risk, tax and accounting, compliance, legal), with only limited ability to delegate their authority

C. Membership should be limited to risk management and the business unit so that decisions are made quickly

D. Membership should be filled by the originators of the transactions presented so that the best product knowledge is available

Answer: B

Step by step

1. Best practice: the committee comprises senior executives with deep risk understanding.
2. It should include the key functions: business units, risk management, tax and accounting, compliance and legal.
3. Membership is personal; members are decision-makers with only limited ability to delegate.
4. Only B satisfies both the breadth and the limited-delegation requirement.

Why B is correct: A diverse, senior, decision-making membership ensures no aspect of a deal is overlooked, and limited delegation keeps accountability with the individuals tasked with deciding.

Why the others are wrong

- **A:** Right breadth but wrong on delegation: free delegation to junior staff undermines expertise and accountability.
- **C:** Omits compliance, legal, tax and accounting; speed is not a reason to remove key functions.
- **D:** Originators bring the proposal; having them decide their own deals creates a conflict of interest.

Q47

AnalystPrep Q.5825

REBUILT — Original key (strategic stance on high-risk segments) is not supported by the reading and distractors were unrelated; rebuilt to test the 'committee can say no / track record of approvals and declines' principle.

Over the past three years a bank's credit committee has approved every transaction submitted by originators. Which statement best assesses this record against best practice?

A. It indicates an efficient committee, because originators prepare packages carefully

B. It indicates a well-functioning committee, because unanimous approvals show that the guidelines are clear

C. It is a warning sign, because a well-functioning committee balances business needs with risk and has a track record of both approving and declining transactions

D. It is appropriate, because the committee's role is limited to ratifying decisions already made by business unit heads

Answer: C

Step by step

1. A well-functioning committee balances business needs against risk and shows a record of approving and declining transactions.
2. Originators must understand that the committee can say no; a committee that never says no is not efficiently managed.
3. A 100% approval record therefore signals possible rubber-stamping, so C.

Why C is correct: The committee's value lies in independent challenge; never declining indicates it is not exercising that role.

Why the others are wrong

- **A:** Treats approval rate as efficiency; the reading says a committee that never says no is not efficiently managed.
- **B:** Confuses clear guidelines with real challenge; unanimous approval of everything is not evidence of effectiveness.
- **D:** The committee is the decision-maker, not a ratifier of earlier decisions.

Q48

AnalystPrep Q.5828

REBUILT — Original key (recording minutes as the factor for a 'balanced review') was loosely worded; rebuilt to ask directly about meeting minutes and their purpose.

Which of the following best describes how a credit committee should handle the record of its meetings?

A. Brief notes of the final decision only, kept by the originator to avoid delay

B. Minutes recorded with detail and accuracy and distributed shortly after the meeting, since they are among the first documents management and auditors request if a transaction goes wrong

C. Minutes prepared only for declined transactions, because approvals are documented in the approval package

D. Minutes kept confidential to committee members and never distributed, to protect candid discussion

Answer: B

Step by step

1. Discussions must be recorded in detail and accurately in minutes.
2. Minutes are distributed quickly after the meeting.
3. If a transaction goes sour, management and auditors ask first for the credit committee minutes.
4. Only B states all of this.

Why B is correct: Detailed, timely minutes document the rationale and dissent and provide an audit trail for later review.

Why the others are wrong

- **A:** Final decision only omits the discussion and reasoning; the originator should not own the record.
- **C:** Approvals are the cases most likely to be scrutinized later; all discussions must be recorded.
- **D:** Minutes must be distributed; withholding them defeats their purpose as a record for management and auditors.

Q49

BT Mock Q22

REBUILT — Original had the approval documents arriving the same day, but also lacked a business unit member and had a vague 'vote' distractor; rebuilt with a business member added and clearer distractors so exactly one flaw remains.

Old Crow Bank held a credit committee meeting. The notes record: all members present: Jan (Chief Risk Officer), Eddie (Head of Compliance), Alyssa (Head of Legal), Don (Head of Tax), Ashton (CFO) and Sam (Head of the originating business unit). Jan opened by confirming that everyone had received the approval documents by email from credit officer Doug, who originated the loan. Everyone confirmed receiving the documents that morning, a few hours before the meeting. Jan gave a five-minute overview and then invited each member in turn to give an opinion; all members spoke and all supported approval. Minutes were circulated the next day. Which is the most valid criticism of this meeting?

A. The approval documents should have been distributed well in advance of the meeting.

B. Jan dominated the discussion and did not facilitate it objectively.

C. The committee lacked a representative of the business unit.

D. The minutes should have been distributed before the decision, not after.

Answer: A

Step by step

1. Approval packages should be distributed well in advance so members can read them and request more information.
2. Here the documents arrived only hours before the meeting, so independent review was not possible, which risks rubber-stamping a deal presented by the originator.
3. Jan spoke five minutes then invited every member's opinion; Sam attended; minutes after the meeting is the norm.

Why A is correct: Committee effectiveness depends on adequate time to review the package; same-day delivery defeats it.

Why the others are wrong

- **B:** A short overview followed by inviting every member's views is proper chairing.
- **C:** Sam, head of the originating business unit, is present.
- **D:** Minutes record the meeting and are distributed shortly after it, not before the decision.

Q50

BT answers P2.T6.24.3.3 · BT notes P2.T6.24.3.3

REBUILT — Original key (poor documentation) was not supported because notes were taken, and the documents arrived the same day as the meeting, making a second option arguable; rebuilt so documents are sent a week ahead and the only flaw is the absent minutes.

Old Crow Bank held a credit committee meeting yesterday. The record states:
- All members present: Jan (Chief Risk Officer), Eddie (Head of Compliance), Alyssa (Head of Legal), Don (Head of Tax), Ashton (CFO), Sam (Head of the originating business unit).
- Jan confirmed that everyone had received, a week earlier, the approval documents from credit officer Doug, who originated the loan.
- Jan gave a five-minute overview and invited opinions. All except Alyssa supported approval; Alyssa raised concern about the counterparty's trustworthiness.

- Jan respected Alyssa's view but disagreed and called a vote. The four other supporters voted with Jan and the loan was approved.
- No minutes were prepared; nobody has been asked to write up the meeting.

What is the most probable issue with the meeting?

A. The approval documents should have been distributed earlier.
B. Jan monopolized the conversation and did not facilitate the discussion.
C. The meeting was not recorded in minutes distributed shortly afterward.
D. The vote was not unanimous, so the loan should have been rejected.

Answer: C

Key correction: Source key (C) agreed with the rebuilt answer, but the original stem showed notes were taken and documents arrived the same day as the meeting, so the item was ambiguous; stem rebuilt.

Step by step

1. Documents were sent a week ahead: adequate time (rules out A).
2. Jan gave a brief overview, invited opinions, and Alyssa's dissent was heard: facilitation was fine (rules out B).
3. If there is no consensus a vote can be held; unanimity is not required (rules out D).
4. Meetings must be recorded in detailed minutes distributed quickly; none exist, which is the flaw: C.

Why C is correct: Detailed minutes are the audit trail of the decision and dissent and are the first thing management and auditors request if the loan fails.

Why the others are wrong

- **A:** A week is ample review time; this was not an issue here.
- **B:** The chair let all speak, including the dissenter, then called a vote as appropriate.
- **D:** Consensus is not required; when it is absent, a vote is the proper mechanism.

Q51 AnalystPrep Q.5826

REBUILT — Original key (assessing aggregate impact of strategic shift) is outside the reading's committee role; rebuilt to test the chair's role in soliciting opinions, facilitating objectively and calling a vote when no consensus.

At a credit committee meeting, the chair strongly favors a proposed transaction. Two members express reservations about the counterparty's governance, and the discussion shows no consensus. Which action by the chair is most consistent with best practice?

A. Close the discussion and record the transaction as approved because the chair's view carries the most weight
B. Ask the dissenting members to resolve their concerns privately with the originator and approve the deal in the meantime
C. Allow all members to present their views objectively and then call a vote
D. Defer the decision to the CEO to avoid disagreement within the committee

Answer: C

Step by step

1. The chair should solicit opinions and facilitate the discussion dispassionately.
2. Where there is no consensus, the chair directs the committee to a vote.
3. Hence allow all views and then vote: C.

Why C is correct: A respected chair gives every party a hearing, and the vote is the prescribed mechanism when consensus is lacking.

Why the others are wrong

- **A:** Imposes the chair's preference and ignores the dissent; the chair must be objective.
- **B:** Moves the concerns out of the committee record and approves before they are addressed.
- **D:** The committee is the decision body; escalating to avoid a vote abdicates its role.

Reading 21. Credit Risk Management

Hennie van Greuning and Sonja Brajovic Bratanovic, Analyzing Banking Risk, 4th Edition (World Bank Group, 2020). Chapter 7.

LO	Learning objective	Qs
21.a	describe key elements of an effective lending or financing policy.	7
21.b	explain the importance and challenges of setting exposure and concentration limits.	4
21.c	describe the scope and allocation processes of a bank's credit facility and explain bank-specific policies and actions to reduce credit risk.	3
21.d	discuss factors that should be considered during the credit asset classification process.	4
21.e	describe and explain loan loss provisions and loan loss reserves.	3
21.f	identify and explain the components of expected loss and differentiate between expected loss and unexpected loss.	3
21.g	explain the requirements for estimating expected loss under IFRS 9.	4
21.h	describe a workout procedure for loss assets and compare the following two approaches used to manage loss assets: retaining loss assets and writing off loss assets.	3
21.i	explain the components of credit risk analysis.	3
21.j	explain the components of credit risk management capacity, and identify key questions that the board of directors of a bank should ask.	3

LO 21.a

Describe key elements of an effective lending or financing policy.

Q52

AnalystPrep Q.5830

REBUILT — Options A and C were both defensible risk-reduction moves; reworded them so they are clearly poor policy and D is the single best answer.

A regional bank with a mix of commercial and consumer loans operates in several economic regions. It is updating its lending policy in response to changing economic conditions. Which modification would most effectively enhance its credit risk management?

A. Accepting a wider range of hard-to-value assets as collateral without changing advance rates.
B. Applying the same interest rate to every loan type to simplify the portfolio.
C. Withdrawing from every region in which defaults are above the bank's average.
D. Adapting loan terms and products to regional economic conditions and borrower profiles.

Answer: D

Step by step

1. A lending policy should be flexible and adapt to changing conditions, with terms (pricing, maturity, appraisal, collateral) set by loan type and customer.
2. Regional economic differences change borrower repayment capacity, so terms should reflect them.
3. Test each option against this: only D tailors terms to risk; A loosens protection, B ignores risk differences, C abandons markets instead of managing the risk.

Why D is correct: An effective lending policy sets terms by type of loan and borrower and adapts to local conditions, so risk is priced and controlled where it arises.

Why the others are wrong

- **A:** Looser collateral standards without matching advance rates (LTV) raise, not reduce, loss given default.
- **B:** Rates should cover cost of funds, supervision, overhead and probable losses, which differ by loan type; a uniform rate ignores this.
- **C:** An all-or-nothing exit is crude risk avoidance and sacrifices diversification and business; the policy should manage regional risk through terms and limits.

Q53

AnalystPrep Q.5832

A regional bank serves small businesses, large corporations and individual consumers in urban and rural settings. Which policy revision would best enhance its credit risk management while still meeting the needs of its varied clientele?

A. Introducing a uniform loan maturity policy across all customer segments to simplify portfolio management.
B. Customizing loan pricing models to reflect the risk and potential return of different customer segments and loan types.

C. Focusing predominantly on expanding unsecured consumer lending to drive growth.
D. Reducing the variety of loan products offered to streamline operations and reduce complexity.

Answer: B
Step by step
1. Policy should price loans by type so that rates cover cost of funds, supervision, overhead, probable losses and a reasonable profit.
2. Segments differ in risk and return, so pricing must differ.
3. B does this while still serving all segments; A, C, D ignore segment risk or reduce service.

Why B is correct: Risk-based pricing by segment and loan type compensates the bank for expected losses and costs while serving a diverse clientele.
Why the others are wrong
- **A:** Maturities should be set by loan type, with repayment source and collateral life in mind; a uniform maturity ignores this.
- **C:** Concentrating on unsecured consumer loans adds concentration and loss severity risk and abandons other clients.
- **D:** Cutting products reduces complexity but fails the stated need to serve varied clients and does nothing to risk-differentiate.

Q54 AnalystPrep Q.5833

A mid-sized bank's credit policy lacks a structured approach for handling high-risk client actions and accounts that have exceeded limits or become delinquent. Which policy enhancement would most effectively address this gap?
A. Implementing an automated system to immediately close accounts that become delinquent.
B. Developing a structured and independent oversight process for managing workouts and high-risk client actions.
C. Increasing interest rates uniformly for all accounts showing signs of becoming high-risk.
D. Transferring all high-risk accounts to a third-party agency for immediate debt recovery.

Answer: B
Step by step
1. Identify the gap: no structured process for prioritizing high-risk clients and handling collections, workouts and restructuring.
2. Sound policy provides prioritization of high-risk client actions, collections/workout of delinquent accounts, and independent oversight of the workout process.
3. Only B supplies that; the others are blanket mechanical responses.

Why B is correct: Policy should include prioritization of high-risk client actions and independent oversight of collections and workouts, giving case-by-case, objective handling.
Why the others are wrong
- **A:** Automatic closure ignores case-by-case workout or restructuring and may destroy recoverable value.
- **C:** A uniform rate rise ignores differing risk, may worsen repayment capacity, and is not a process.
- **D:** Blanket outsourcing removes bank control and oversight and is not tailored to each account.

Q55 BT answers P2.T6.24.4.1

Old Crow Regional Bank sees rising demand for healthcare loans. It has a strong small business lending team but few healthcare loans; its lending officers' experience is in consumer loans and they lack healthcare expertise, although healthcare revenues tend to be predictable. What approach should the bank take?
A. Decline healthcare loan applications because the lending officers lack experience.
B. Maintain current lending policies without adjustments for healthcare-specific risks.
C. Proceed with new loan applications cautiously while providing training to lending officers.
D. Channel deposits toward consumer loans to align with the staff's expertise.

Answer: C
Key correction: Source (BT) explanation for B and D mentions technology and commercial real estate sectors, apparently copy errors; the key C is correct.
Step by step

1. A lending policy sets types of credit based on officer expertise, deposit structure and credit demand.
2. Demand exists and the sector is relatively stable, so the bank should respond rather than refuse, but must build expertise and sector-specific risk standards.
3. C does both; A forgoes business, B ignores the new risk, D ignores demand.

Why C is correct: Credit types should match officers' expertise, but where demand is attractive the policy should be adapted and staff trained, with caution until expertise develops.

Why the others are wrong

- **A:** Refusing outright forgoes profitable business; the gap can be closed by training.
- **B:** Unchanged policy gives no sector-specific appraisal standards or limits for an unfamiliar sector.
- **D:** Ignores credit demand and the predictable healthcare revenues, and increases concentration in consumer lending.

Q56

Schweser Module Quiz 21.1 Q1

Within their credit risk management plan, banks are either required to include, or at least should strongly consider including, a:

A. classification of risks organized by collectability.
B. loan loss provision equal to at least 8% of their loan portfolio.
C. duration target for the bank's portfolio that is not higher than 10.
D. rigid set of policies designed to address every possible risk source.

Answer: A

Step by step

1. Lending policies fall into three categories: limiting credit risk, classifying assets by credit risk exposure and collectability, and creating provisions for potential losses.
2. A matches the second category.
3. B and C are specific numeric targets not mandated; D contradicts the need for flexibility.

Why A is correct: Regulators expect policies that classify assets by exposure and collectability; specific numeric targets are left to each bank, and policies must be flexible.

Why the others are wrong

- **B:** No fixed 8% provision exists; provision levels reflect each bank's risk (8% is a Basel capital ratio, not a provision rule).
- **C:** No required duration cap; this is an invented numeric target.
- **D:** Policies must be flexible to changing conditions, not rigid.

Q57

AnalystPrep Q.5857

A regional bank frequently accepts securities as collateral for loans and is revising its loan-to-value (LTV) policy. Which change would most effectively manage the risk from fluctuating market values of pledged securities?

A. Setting a higher LTV ratio for less marketable securities.
B. Regularly adjusting the LTV ratio based on the periodic pricing and marketability of the securities.
C. Applying a uniform LTV ratio across all securities.
D. Implementing more lenient margin requirements for new or innovative securities to encourage borrowing.

Answer: B

Step by step

1. Policy should relate margin requirements to the marketability of securities and set a timetable for periodic repricing of collateral.
2. Therefore LTV should be adjusted as prices and marketability change (B).
3. Less marketable or new securities need lower LTV (more margin), not higher; a uniform ratio ignores differing risk.

Why B is correct: Periodic repricing and marketability-based margins keep a buffer against falling collateral values.

Why the others are wrong

- **A:** Reversed: less marketable securities warrant a lower LTV, because they are harder to liquidate at value.
- **C:** Uniform LTV ignores differences in volatility and liquidity across securities.
- **D:** Leniency on untested securities increases collateral shortfall risk.

Q58 MK-AP Mock 3 Q47 · AnalystPrep Q.5860

A community bank has restructured several loans because of borrowers' financial difficulties. Management is considering the appropriate accounting treatment. According to sound financial accounting practice, how should the bank reflect these restructured loans in its financial statements?

A. Keep the recorded investment of restructured loans unchanged to maintain consistency in the loan portfolio.
B. Increase the recorded investment of restructured loans to cover potential future losses.
C. Measure the restructured loan by reducing its recorded investment to net realizable value, charging the reduction to the income statement.
D. Record the reduction in loan value as a deferred charge, to be recognized in future financial periods.

Answer: C

Step by step

1. Renegotiated debt is a loan restructured to reduce interest or principal payments because of the borrower's deteriorated financial position.
2. The concessions given reduce what the bank can expect to collect, so the loan is worth less than its carrying amount.
3. The bank measures the restructured loan by reducing its recorded investment to net realizable value, taking account of the cost of all concessions at the restructuring date.
4. The reduction is charged to the income statement of the period in which the loan is restructured.

Why C is correct: Concessions impair the loan's value, so the carrying amount is written down to net realizable value and the loss is recognized immediately in earnings.

Why the others are wrong

- **A:** Leaving the carrying amount unchanged ignores the concessions and overstates assets and income.
- **B:** Increasing the recorded investment is backwards: the concessions lower the expected recovery, and it would inflate assets.
- **D:** Deferring the reduction as a charge to future periods delays loss recognition (evergreening); the charge belongs to the restructuring period.

LO 21.b

Explain the importance and challenges of setting exposure and concentration limits.

Q59 GARP PE1 Q27 (mock)

A bank's chief credit officer is reviewing the bank's exposure to VH3 Limited, a failed corporate client. Reported loans to VH3 were USD 7.5 billion, within the 15% of capital single-client limit, but total claims in bankruptcy turned out to be USD 10 billion (20% of capital), breaching the regulatory single-counterparty limit. Which of the following exposures should have been captured in the calculation of risk relating to VH3 to comply with single-client limits?

A. Loans to companies in which VH3 has a 10% to 15% ownership position
B. Exposures to insurers who have issued guarantees for the obligations of VH3
C. A commodity swap position with VH3 where the bank is the fixed-price receiver
D. Claims on affiliates of the bank that have exposures to VH3

Answer: C

Step by step

1. Single-client limits cover all credit exposure to the client, including less direct forms (contingent liabilities, guarantees, acceptances, L/Cs, derivatives), not just funded loans.
2. The swap is a contract with VH3 itself: when commodity prices fall, the bank (fixed-price receiver) is in the money and VH3 owes it a payment, so the bank has counterparty credit exposure to VH3.
3. This exposure was omitted from the USD 7.5 billion of loans, which is why true exposure (USD 10 billion, 20% of capital) exceeded the limit.
4. A connected group requires control (voting rights of roughly 15-20% or more); a 10-15% stake does not clearly qualify, so A is not aggregated; B and D are exposures to other parties.

Why C is correct: Large-exposure limits are measured on total credit exposure to the single client, including derivative and other off-balance-sheet claims on that client; the swap is a direct claim on VH3.

Why the others are wrong

- **A:** A 10-15% ownership stake is below the typical 15-20% voting-right threshold that makes companies a connected group with VH3, so these loans are not aggregated.
- **B:** The guarantor is the insurer; that is an exposure to the insurer (a mitigant for VH3 risk), not an additional exposure to VH3. Confuses credit protection with exposure.
- **D:** Claims on the bank's own affiliates are intra-group exposures and are related-party issues for the bank, not exposures to VH3.

Q60

AnalystPrep Q.5842

A community bank lends primarily to local businesses and individual consumers. Because the local economy is interconnected and the bank has significant exposure to a few key sectors, what should be a primary focus of its risk management strategy to limit credit risk?

A. Concentrating loans in the most profitable local industry to maximize returns.
B. Regularly reassessing and adjusting credit limits for individual consumers based on their account history.
C. Monitoring the economic health of key sectors and adjusting lending practices to manage exposure.
D. Outsourcing risk assessment of loan applications to external experts for a more objective analysis.

Answer: C

Step by step

1. The problem is sector/geographic concentration: small, regionally oriented banks are especially exposed to overexposure to an economic sector.
2. The remedy is concentration limits that are reviewed and reported frequently, plus monitoring sector health and adjusting lending when a sector weakens.
3. Only C addresses sector concentration directly.

Why C is correct: Concentration risk to a sector is managed by setting sector limits and continuously monitoring sector conditions, adjusting lending accordingly.

Why the others are wrong

- **A:** Adds to concentration; chasing return in one industry increases exposure to a single sector shock.
- **B:** Consumer-level limit management is useful but addresses obligor-level risk, not the sector concentration described.
- **D:** Outsourcing assessments changes who analyses applications but does nothing to cap or monitor sector exposure.

Q61

AnalystPrep Q.5845

REBUILT — Option B (board approval) was defensible alongside C; reworded the stem to ask for the standard regulatory limit and replaced B with a clearly weaker control.

A mid-sized bank is reassessing its lending to related parties (its parent, major shareholders, subsidiaries, affiliates, directors and executive officers), who may influence credit decisions and obtain more favorable loan amounts, maturities, rates or collateral terms. Which policy is most consistent with how regulators address this risk?

A. Exclude all related parties from receiving any form of credit.
B. Allow the CEO to approve each related-party loan individually, with no aggregate cap.
C. Restrict total credit to related parties to a specified percentage of the bank's Tier 1 capital.
D. Charge related parties a higher interest rate as a risk premium and leave volumes uncapped.

Answer: C

Step by step

1. Related parties can exert control over credit decisions, so credit may be granted outside policy or on better-than-market terms.
2. Regulators typically cap total credit to related parties at a percentage of Tier 1 (or qualifying) capital; if none exists, the bank should set a comparable internal limit.
3. Only C is an aggregate limit tied to capital.

Why C is correct: The standard control is an aggregate cap on related-party credit as a percentage of Tier 1 capital, which bounds exposure irrespective of how favorable individual terms are.

Why the others are wrong

- **A:** Total prohibition is not what regulators require; related-party lending is permitted within limits.
- **B:** Individual approval by an insider-influenced executive does not limit aggregate exposure and leaves the bias

problem in place.
- **D:** Pricing does not cap exposure and does not address biased decision-making or non-price terms (maturity, collateral).

Q62 Schweser Module Quiz 21.1 Q2

A risk analyst is reviewing his bank's credit risk management policies. He notes that banks have limits on related-party financing decisions. His colleague agrees and correctly adds that:

A. the term "single client" only refers to a legal person.
B. all banks have a limit of 10% of capital for a single client.
C. loan collateral should be included in exposure sizing using mark-to-market principles.
D. acceptances and letters of credit should be factored into the risk exposure of a single client.

Answer: D

Step by step

1. Single client = an individual, a legal person, or a connected group, so A is wrong.
2. Countries typically impose limits in a 10%-25% of capital range, not a universal 10%, so B is wrong.
3. Collateral should not be considered when sizing exposures, so C is wrong.
4. Indirect credit (contingent liabilities, guarantees, acceptances, letters of credit) is included in exposure, so D is correct.

Why D is correct: Exposure to a single client includes less direct forms of credit such as acceptances and letters of credit.

Why the others are wrong

- **A:** Too narrow; a single client also includes individuals and connected groups that require aggregation.
- **B:** Limits vary by jurisdiction, commonly 10%-25% of capital; 10% is only the bottom of the range.
- **C:** Collateral is not netted against exposure in sizing, and valuation is uncertain; the claim reverses the rule.

LO 21.c

Describe the scope and allocation processes of a bank's credit facility and explain bank-specific policies and actions to reduce credit risk.

Q63 Schweser Module Quiz 21.1 Q3

A regional bank wants to decrease risk in its lending practices. It is considering several lending policy changes. Which of the following items should be included in this new risk-reduction-focused direction?

A. All loans should have a maximum maturity schedule of 10 years.
B. The Board should approve all individual delinquency workout plans.
C. Financial projections from business clients should match the maturity of their loan.
D. Credit officers should have the flexibility to extend any type of credit that a customer needs as long as they find it to be a reasonable risk for the bank.

Answer: C

Step by step

1. A lending policy reduces risk by specifying required financial information; for loans with maturity over one year, officers prepare projections whose horizon equals the loan maturity, so repayment from cash flow can be assessed.
2. Test A: maturities should be set per loan type (repayment source, purpose, collateral life), not one blanket cap.
3. Test B: the Board receives delinquency reports and policy, but does not approve each workout plan.
4. Test D: the policy must specify which credit instruments are offered; unrestricted officer discretion raises risk.
5. Only C is a risk-reducing policy element.

Why C is correct: Matching the horizon of financial projections to the loan maturity lets the bank confirm that the loan can be repaid from projected cash flow, which is a standard risk-reducing lending-policy element.

Why the others are wrong

- **A:** A policy should set clear maximum maturities for each loan type based on repayment source, loan purpose and collateral life; a single 10-year cap for all loans is arbitrary and not what the policy calls for.
- **B:** The Board should be informed of delinquencies and the process for handling them, but a solid policy means

it need not approve every individual workout plan.
- **D:** Lending policy should limit the credit instruments offered (based on officer expertise, deposit structure and demand) and set lending authority limits; open-ended officer discretion increases risk.

Q64

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A credit risk manager at a bank with capital of USD 400 million is planning the annual loan portfolio review. In addition to a random sample of loans, the bank's policy requires that certain loans always be included in the detailed review. All loans below are current, classified standard, and have unaltered terms. Which of the following loans must be included in the detailed review regardless of the random sample?

A. Two current facilities of USD 14 million and USD 11 million to the same corporate borrower.
B. A single USD 19 million loan to a manufacturer that has no other facility with the bank.
C. A single USD 18 million loan to a trading company that is 20 days past due.
D. A USD 8 million loan to a municipality that is the largest exposure in the bank's public sector book.

Answer: A

Step by step

1. 5% of capital = 0.05 x USD 400 million = USD 20 million.
2. Aggregate exposure to the corporate borrower = 14 + 11 = USD 25 million = 6.25% of capital, above the threshold.
3. Every other loan is below USD 20 million, is not more than 30 days past due, and has no related-party, rescheduling or classification trigger.

Why A is correct: A detailed portfolio review must include all loans to borrowers whose aggregate exposure exceeds 5% of the bank's capital. Aggregate means all facilities to the same borrower are added, so the two facilities together (USD 25 million, 6.25% of capital) trigger inclusion even though each is individually below USD 20 million.

Why the others are wrong

- **B:** USD 19 million is 4.75% of capital, below the 5% threshold, and the loan has none of the other triggers; it is reviewed only if the random sample selects it. Error: rounding up toward the threshold.
- **C:** USD 18 million is 4.5% of capital, and 20 days is short of the more-than-30-days-past-due trigger. Error: treating any delinquency as a trigger.
- **D:** Being the largest loan within a segment is not a trigger; the test is aggregate exposure relative to bank capital. Error: judging size relative to the segment rather than to capital.

Q65

GARP PE2 Q73

REBUILT — Original mixed duration-gap, VaR and LCR topics with a vague keyed option; rebuilt to test the lending-policy collateral provisions of LO 21.c with a single defensible answer.

Quant Banking Corporation (QBCo) plans to sell its UK government bonds and use the proceeds to extend BRL loans to Brazilian corporations. The loans will be held to maturity and secured by pledged high-quality foreign sovereign securities. The CRO is drafting the lending policy for this new loan category. Which of the following policy provisions would most effectively reduce QBCo's credit risk on these collateralized loans?

A. Set margin requirements (maximum loan-to-market-value of pledged securities) for each type of security based on its marketability, and assign responsibility and a timetable for periodic repricing of the collateral.
B. Value the pledged securities once at origination and rely on that value until maturity, because the loans are held to maturity.
C. Apply one uniform loan-to-value limit to all pledged securities regardless of type, to keep the policy simple.
D. Allow loan maturities to extend beyond the maturity of the pledged securities, because a longer tenor improves the borrower's ability to repay.

Answer: A

Step by step

1. Collateral protects the bank only if its current market value covers the loan, so the policy must control both the size of the loan relative to collateral and how that coverage is monitored.
2. Margin requirements should be established for all security types and related to the marketability of each security (more liquid securities can support a higher loan-to-value ratio).
3. The policy should include a provision for periodic repricing of the collateral, with assigned responsibility and

a timetable, so margin shortfalls are detected.
4. Only option A contains both elements; B, C and D each weaken collateral protection.

Why A is correct: Lending policy reduces credit risk on secured loans by setting security-specific margin requirements tied to marketability and by requiring periodic repricing of the collateral.

Why the others are wrong

- **B:** Holding a loan to maturity does not stabilise collateral value; stale valuation hides margin shortfalls when security prices move, and policy should require periodic repricing.
- **C:** A uniform limit ignores that margin requirements should differ by security type and marketability, so riskier or less liquid collateral would be over-lent against.
- **D:** Maturities should be limited by the source of repayment and the useful life of the collateral; a tenor beyond the collateral's life leaves the loan unsecured late in its term.

LO 21.d

Discuss factors that should be considered during the credit asset classification process.

Q66

AnalystPrep Q.5870

REBUILT — Original options overlapped (macro, collateral and peer-PD factors were all defensible); restated around the single emphasis stated in the reading and made distractors distinct.

When a bank assigns credit asset classifications to its loans, which factor receives the primary emphasis in judging the likelihood of repayment?

A. The borrower's ability and willingness to meet obligations out of prospective operating cash flow
B. The realizable value of the collateral pledged against the loan
C. The borrower's credit score relative to the average score of its industry peers
D. The length of time since the loan was originated

Answer: A

Step by step

1. Asset classification assigns assets to grades according to their likelihood of repayment.
2. The reading stresses that classification is about more than payments received: it looks at the borrower's cash flows and expected ability to repay.
3. Primary emphasis is therefore on the client's ability and willingness to repay from prospective operating cash flow.
4. Collateral is a secondary source of repayment, looked to only when the primary source has failed.

Why A is correct: Classification rests chiefly on whether the borrower's prospective operating cash flow can service the debt; collateral and other factors are secondary.

Why the others are wrong

- **B:** Collateral is a secondary source of repayment; relying on it first confuses the fallback with the primary source (full cash security only supports a Standard grade in the special case of cash or cash equivalents).
- **C:** A peer-relative credit score is a statistical input, not the primary emphasis; it ignores the borrower's own cash flows and circumstances.
- **D:** Loan age since origination is not a classification driver; delinquency age (days past due) matters, not time since origination.

Q67

AnalystPrep Q.5871 · BT answers P2.T6.24.4.2.

REBUILT — Two near-identical vignettes merged; 'missed two quarterly payments' was ambiguous between 90 and 180 days, so the delinquency was fixed at 100 days.

A regional bank lent to a ten-year-old manufacturer to buy new machinery, secured by the factory and equipment. Sector weakness has cut sales and cash flow, operating costs have risen and margins have fallen. The borrower has made no interest or principal payments for 100 days. Operating income is no longer sufficient to service the debt and the bank is looking to the collateral for repayment. How should the loan be classified?

A. Standard (pass)
B. Specially mentioned (watch)
C. Substandard
D. Doubtful

Answer: C

Step by step

1. The loan is nonperforming: no payments for 100 days, which exceeds the 90-day threshold.
2. Nonperforming loans delinquent over 90 days are typically substandard.
3. The primary source of repayment (operating cash flow) has failed and the bank looks to secondary sources (collateral), a hallmark of substandard.
4. Doubtful would need over 180 days delinquency (not sufficiently secured) or a more pronounced expectation of loss; 100 days is well short.

Why C is correct: A nonperforming loan more than 90 days delinquent whose primary repayment source has failed, with the bank turning to collateral, is substandard.

Why the others are wrong

- **A:** Standard requires debt service capacity to be beyond any doubt; a loan 100 days past due with failing cash flow does not meet that.
- **B:** Specially mentioned covers potential weakness that has not yet jeopardized repayment; here the weakness is well defined and payments have stopped.
- **D:** Doubtful applies at over 180 days delinquent (unless sufficiently secured) with a more pronounced expected loss; mistaking two missed quarterly payments for 180 days gives this answer.

Q68

Mock 1 Q28

REBUILT — Original said only 'over 180 days', which could also exceed one year (Loss); fixed the delinquency at 200 days and cleaned the he/she inconsistency.

A risk analyst reviews a bank's loan book and finds a nonperforming loan that has been delinquent for 200 days. The loan is not sufficiently secured. Under the standard five-category asset classification, how should it be classified?

A. Loss
B. Doubtful
C. Substandard
D. Specially mentioned

Answer: B

Step by step

1. Delinquency of 200 days is above the 180-day threshold but below one year.
2. Nonperforming loans over 180 days delinquent are classified doubtful unless sufficiently secured; this loan is not secured, so doubtful applies.
3. Loss applies at over one year delinquent (unless sufficiently secured), so 200 days is too early.

Why B is correct: Over 180 days but under one year past due and not sufficiently secured places the loan in doubtful: loss is expected but some recovery is still possible.

Why the others are wrong

- **A:** Loss requires delinquency over one year; 200 days is far short, and some chance of repayment remains.
- **C:** Substandard is the typical grade for over 90 days; it ignores the further deterioration beyond 180 days.
- **D:** Specially mentioned is for potential weaknesses on loans that are not yet well-defined problems; it is far too mild for a 200-day nonperforming loan.

Q69

Schweser Module Quiz 21.1 Q4

Three and a half months ago, XYZ Manufacturing lost its single largest customer, and the company stopped servicing all debt payments to ABC Bank. The bank has seized some collateral, but it is working with XYZ as the company plans to find new customers and build a better future. For now, the loans to XYZ Manufacturing should most likely be classified as:

A. loss.
B. doubtful.
C. substandard.
D. specially mentioned.

Answer: C

Step by step

1. 3.5 months is about 105 days of nonpayment, so the loans are nonperforming and over 90 days delinquent.
2. Over 90 days delinquent typically means substandard.

3. The bank has begun using secondary sources (seized collateral) because the primary source failed, which is also characteristic of substandard.
4. Doubtful needs over 180 days and loss over one year; neither is reached.

Why C is correct: Roughly 105 days of nonpayment with the bank turning to collateral fits substandard: past 90 days but well short of the 180-day and one-year thresholds.

Why the others are wrong

- **A:** Loss requires delinquency over one year (unless sufficiently secured); also XYZ still has a path to repayment.
- **B:** Doubtful requires over 180 days delinquency; 105 days is not enough.
- **D:** Specially mentioned is for potential weakness; payments have already stopped for more than 90 days, so the weakness is well defined.

LO 21.e

Describe and explain loan loss provisions and loan loss reserves.

Q70

AnalystPrep Q.5873

REBUILT — Original keyed loan loss reserves as the buffer for unprecedented default spikes, but reserves cover expected (not unexpected) losses and the options mixed in unrelated Basel buffers; rewritten to test the provisions-versus-reserves distinction in LO 21.e.

A bank's analyst is reviewing the balance sheet during an economic downturn. The analyst wants to confirm that she correctly understands how the bank's loan loss provisions differ from its loan loss reserves. Which statement is correct?

A. Provisions are the cumulative balance recorded on the balance sheet, while reserves are the amount set aside in a single period.

B. Reserves are intended to absorb unexpected losses in the extreme tail, while provisions cover expected losses.

C. Provisions are counted as Tier 2 capital, but reserves are not counted as regulatory capital.

D. Reserves are loan loss provisions accumulated over multiple years and recorded on the bank's financial statements.

Answer: D

Key correction: Source key (AnalystPrep) said C (loan loss reserves) to an unprecedented default spike; that framing is flawed because reserves are built from expected-loss provisions and extreme unexpected losses are absorbed by capital. The question was rebuilt around the definition.

Step by step

1. Provisions are resources set aside, from the asset classification process, to offset potential future losses.
2. Reserves are those provisions accumulated over multiple years and shown on the financial statements.
3. Both provisions and reserves are normally counted as Tier 2 capital.
4. Unexpected loss (tail event, e.g. 99th percentile) is by definition not captured by the expected-loss based provisioning, so reserves do not target it.
5. Only D matches this.

Why D is correct: Reserves are the accumulation of annual loan loss provisions over time and are reported on the balance sheet.

Why the others are wrong

- **A:** Reverses the terms: the cumulative balance is the reserve, the period charge is the provision.
- **B:** Confuses reserves with capital: reserves are built from expected-loss provisions; unexpected tail losses are not captured by the expected loss formula.
- **C:** Both provisions and reserves are normally counted as Tier 2 capital, so neither is excluded.

Q71

Schweser Module Quiz 21.2 Q1

Two senior credit managers are reviewing their bank's loan loss reserve procedures. Which of the following elements is most important when determining the appropriate level of reserves?

A. Current loan loss reserves.

B. The bank's experience with previous losses.

C. The current number of nonperforming loans in aggregate.

D. A list of all loans that are delinquent for more than 180 days.

Answer: B

Step by step

1. Factors in setting an adequate reserve include credit quality policies and procedures, previous loss experience, loan book growth, quality of lending management, collection and recovery policies, and changes in the economic and business environment.
2. Of the choices, only the bank's experience with previous losses is on this list.
3. The existing reserve balance is an output, not an input; the aggregate NPL count and a 180-day delinquency list are classification outputs and not the key reserve drivers.

Why B is correct: Prior loss experience is a core input for estimating how much loss the portfolio is likely to produce, and therefore how large the reserve should be.

Why the others are wrong

- **A:** Using the current reserve to set the reserve is circular; reserves should not be biased by what is already on the books.
- **C:** The aggregate count of nonperforming loans is a classification statistic, not a driver of the reserve policy; it ignores loss severity and history.
- **D:** A delinquency cutoff list is a classification output, not a driver of reserve adequacy; it ignores loss history, growth and the economic environment.

Q72

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A risk analyst at a regional bank is reviewing the adequacy of the bank's loan loss allowance. Over the past three years the bank's realized losses have been very low. During the same period the loan book grew 40% through a new lending segment, the bank hired many inexperienced lending officers, and collection and recovery procedures for the segment were relaxed. Which of the following is the most appropriate conclusion for the review?

A. The allowance can be held at its current level because low realized losses show that the existing provisioning policy is adequate.

B. The allowance should reflect current risk factors that differ from the historical experience, so the low past losses should not by themselves support a lower allowance.

C. The allowance should be set from the aggregate number of nonperforming loans on the books, because it is the only objective measure of expected losses.

D. The allowance should be reduced, because loan loss reserves are counted as Tier 2 capital and already provide a buffer against losses.

Answer: B

Step by step

1. Determinants of an adequate reserve include credit policies and procedures, prior loss experience, loan growth, quality of lending management, collection and recovery practices, and economic conditions.
2. Three of these factors (growth, inexperienced staff, relaxed recovery practices) have changed adversely and differ from the historical experience.
3. An allowance review should highlight current loss factors that differ from history, so low past losses alone do not justify holding or cutting the allowance.

Why B is correct: The adequacy assessment looks at current factors likely to cause losses that differ from historical loss experience, such as loan growth, management quality and changes in collection practices. Historical losses are one input, not the only one.

Why the others are wrong

- **A:** Ignores current factors that differ from history (growth, management quality, collection practices). Error: relying solely on past loss experience.
- **C:** Reserve setting weighs credit policies, prior losses, loan growth, management quality, recovery practices and economic conditions, not a single count of nonperforming loans. Error: using one classification statistic as the sole driver.
- **D:** Reserves count toward Tier 2 capital, but that does not reduce the need for adequate provisions; capital treatment is not a determinant of the allowance. Error: confusing capital treatment with reserve adequacy.

LO 21.f

Identify and explain the components of expected loss and differentiate between expected loss and unexpected loss.

Q73 Schweser Module Quiz 21.2 Q2

Which of the following components of expected loss reflects the type of product (e.g., secured versus unsecured, senior versus subordinated) used for the loan?

A. Unexpected loss.
B. Loss given default.
C. Exposure at default.
D. Probability of default.

Answer: B

Step by step

1. Expected loss is built from three components: EL = PD × EAD × LGD.
2. PD answers 'who' the bank lent to (borrower quality).
3. EAD answers 'how long' the loan has been outstanding (how much is drawn/owed at default).
4. LGD answers 'what product' was used, since the product determines collateral, seniority and hence the loss severity.
5. The product component is therefore LGD.

Why B is correct: The loan product drives collateral and seniority, which determine the fraction of exposure lost if default occurs, i.e., LGD.

Why the others are wrong

- **A:** Unexpected loss is not a component of expected loss; it is the tail loss (e.g., 99th percentile) not captured by EL.
- **C:** EAD reflects how long the loan has been outstanding / the amount exposed at default, not the product.
- **D:** PD reflects who the borrower is (credit quality), not the product.

Q74 BT answers P2.T6.24.5.1 · BT notes P2.T6.24.5.1

REBUILT — Source defined unexpected loss as realized loss minus EL ($25,000), which conflicts with the GARP definition (tail loss under a high-stress scenario); rebuilt to test EL calculation plus the correct UL definition.

Old Crow Bank approves a $450,000 home loan. Over the next 12 months the borrower has a 10% probability of default, and the recovery rate given default is 80%. Which of the following correctly gives the expected loss and describes unexpected loss?

A. EL is $36,000; UL is the loss under a high-stress (tail) scenario not captured by EL.
B. EL is $9,000; UL is the actual loss in excess of EL once the borrower defaults (e.g., an actual loss of $34,000 less $9,000 = $25,000).
C. EL is $9,000; UL is the loss under a high-stress (tail) scenario, e.g., at the 99th percentile, not captured by EL.
D. EL is $45,000; UL is the loss under a high-stress (tail) scenario not captured by EL.

Answer: C

Key correction: Source key (BT) said $25,000 (actual loss of $34,000 minus EL of $9,000); under the GARP definition, UL is the tail loss under a high-stress scenario, so that approach is not correct.

Step by step

1. LGD = 1 - recovery rate = 1 - 0.80 = 20%.
2. EL = PD × EAD × LGD = 0.10 × 450,000 × 0.20 = 9,000.
3. Unexpected loss is the loss incurred under a high-stress scenario, typically a point in the tail of the credit loss distribution (e.g., 99th percentile), i.e., loss not captured by EL.
4. So EL = $9,000 with UL defined as the tail loss.

Why C is correct: EL uses LGD (1 - recovery) with PD and EAD, giving $9,000, and UL is the tail-scenario loss beyond what EL captures, not a realized loss minus EL.

Why the others are wrong

- **A:** Used the recovery rate as LGD: 0.10 × 450,000 × 0.80 = 36,000.
- **B:** EL is right, but UL is mis-defined as realized loss minus EL; UL is a stress/tail-percentile loss measure, not a single outcome less the mean.
- **D:** Ignored LGD entirely: 0.10 × 450,000 = 45,000 (this is PD x EAD).

Q75 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A credit analyst at a bank is estimating the 12-month expected loss on a corporate revolving credit facility. The facility has a limit of USD 5.0 million, of which USD 2.5 million is currently drawn. The analyst estimates the borrower's probability of default at 2% and expects the drawn balance at the time of default to be USD 3.2 million. Expected recoveries on a defaulted balance are 55%. What is the expected loss on the facility?

A. USD 22,500
B. USD 35,200
C. USD 28,800
D. USD 45,000

Answer: C

Step by step

1. LGD = 1 - recovery rate = 1 - 0.55 = 45%.
2. EAD = expected drawn balance at default = USD 3.2 million.
3. $EL = PD \times EAD \times LGD = 0.02 \times 3{,}200{,}000 \times 0.45 = 28{,}800$.

Why C is correct: Expected loss is $PD \times EAD \times LGD$. EAD is the estimated exposure at the time of default (USD 3.2 million), given that a line may not be fully drawn, and LGD is the loss percentage, 45%, so EL = USD 28,800.

Why the others are wrong

- **A:** Uses the currently drawn USD 2.5 million as EAD: 0.02 x 2.5 million x 0.45. EAD is the expected exposure at default, not today's balance.
- **B:** Uses the 55% recovery rate as LGD: 0.02 x 3.2 million x 0.55. LGD is 1 - recovery rate = 45%.
- **D:** Uses the full USD 5.0 million limit as EAD: 0.02 x 5.0 million x 0.45. A credit line may not be fully drawn at default.

LO 21.g

Explain the requirements for estimating expected loss under IFRS 9.

Q76 AP Mock 2 Q70 · AnalystPrep Q.5875

A bank's audit committee is discussing IFRS 9 and its expected credit loss model. Which of the following scenarios exemplifies compliance with IFRS 9's expected loss estimation model?

A. Allocating a uniform loss provision percentage across the entire loan portfolio based on average historical loss data.
B. Estimating expected losses on a performing loan based on potential losses over its entire lifetime, regardless of its current performance status.
C. Assessing expected losses on underperforming assets by evaluating the 12-month loss outlook and adjusting provisions annually.
D. Applying a differentiated approach, where performing assets are assessed for 12-month expected losses and underperforming or non-performing assets for lifetime expected losses.

Answer: D

Step by step

1. IFRS 9 (effective 2018) replaced the IAS 39 incurred loss model with a three-stage expected loss model.
2. Stage 1 (performing, not in arrears): 12-month expected loss, interest on the gross amount.
3. Stage 2 (arrears or significant deterioration) and Stage 3 (nonperforming): lifetime expected loss.
4. Only option D differentiates by asset quality in this way.

Why D is correct: IFRS 9 ties the provisioning horizon to credit quality: 12-month expected loss for performing assets, lifetime expected loss once an asset is delinquent or nonperforming.

Why the others are wrong

- **A:** A flat historical-average provision ignores the stage-based, forward-looking design; it resembles the old backward-looking approach.
- **B:** Lifetime loss on a performing loan is wrong; performing (Stage 1) assets use only the 12-month horizon.
- **C:** Reverses the model: underperforming (Stage 2) and nonperforming (Stage 3) assets require lifetime, not 12-month, expected loss.

Q77 AnalystPrep Q.5876

A long-standing corporate client's loan has deteriorated significantly in credit quality and is now nonperforming. Which provisioning approach correctly aligns with IFRS 9 Stage 3 requirements?

A. Adopting a steady-state provisioning method based on the long-standing relationship with the client.
B. Continuation of provisioning based on initial PD and LGD at loan origination.
C. Provisions based on lifetime expected losses, with effective interest on the net (carrying) amount.
D. Relying on external credit ratings to determine the necessary provisions.

Answer: C

Step by step

1. The loan is nonperforming, so it sits in Stage 3.
2. Stage 3 requires lifetime expected credit loss recognition.
3. Interest is recognised on the net (carrying) amount, i.e. after the provision, unlike Stages 1 and 2 (gross amount).
4. Only C states both features.

Why C is correct: Stage 3 combines lifetime expected loss with effective interest on the net carrying amount, reflecting impairment.

Why the others are wrong

- **A:** Relationship history does not change the required methodology; IFRS 9 is driven by current credit quality, not tenure.
- **B:** Using origination PD/LGD ignores the required forward-looking update of expected loss as credit quality deteriorates.
- **D:** External ratings may be an input but are not the IFRS 9 requirement; the standard prescribes lifetime expected loss and net-amount interest.

Q78 Schweser Module Quiz 21.2 Q3

REBUILT — Stem was ambiguous (delinquent but 'not yet nonperforming') and option D 'Stage 4' is a giveaway; reworded and replaced D with a plausible conceptual distractor.

A bank's loan is delinquent but not yet nonperforming, because a short-term macroeconomic event is temporarily preventing repayment. Under IFRS 9, which treatment is required for this loan?

A. Stage 1: 12-month expected loss with effective interest on the gross amount.
B. Stage 2: lifetime expected loss with effective interest on the gross amount.
C. Stage 3: lifetime expected loss with effective interest on the net (carrying) amount.
D. Stage 2: 12-month expected loss with effective interest on the net (carrying) amount.

Answer: B

Step by step

1. The loan is in arrears, so it cannot be Stage 1 (performing, no delinquency).
2. It is not yet nonperforming, so it is not Stage 3.
3. Stage 2 applies to assets with any level of delinquency or significant credit deterioration.
4. Stage 2 uses lifetime expected loss with effective interest on the gross amount.

Why B is correct: Any delinquency, even temporary, moves an asset to Stage 2, which requires lifetime expected loss while interest stays on the gross amount.

Why the others are wrong

- **A:** Stage 1 is only for performing assets with no arrears; the delinquency rules it out.
- **C:** Stage 3 is reserved for nonperforming assets; the loan is not yet nonperforming, so interest stays on the gross amount.
- **D:** Mixes up the stage features: Stage 2 requires lifetime (not 12-month) expected loss and interest on the gross (not net) amount.

Q79 AnalystPrep Q.3191

REBUILT — Original options were arbitrary and EAD/LR units were unclear; clarified the inputs (EAD as a percentage of $50 million) and rebuilt distractors from realistic discounting/aggregation errors.

A bank holds a $50 million two-year loan portfolio. Under an IFRS 9 lifetime expected credit loss estimate, the marginal probability of default (PD), exposure at default (EAD, as a percentage of the $50 million) and loss rate

(LR) in each year are:

Marginal PD	EAD	LR	
Year 1	7.0%	75%	80%
Year 2	9.5%	35%	80%

The portfolio's effective interest rate is 7.9%, and default is assumed to occur in the middle of each year. What is the present value of the lifetime expected credit loss?

A. $3.0886 million
B. $3.2083 million
C. $3.4300 million
D. $3.1214 million

Answer: B

Step by step

1. Year 1 ECL = 50 x 7.0% x 75% x 80% = 2.1000 million; year 2 ECL = 50 x 9.5% x 35% x 80% = 1.3300 million.
2. Mid-year default: discount year 1 for 0.5 years and year 2 for 1.5 years.
3. PV1 = $2.1000/1.079^{0.5} = 2.0217$ million.
4. PV2 = $1.3300/1.079^{1.5} = 1.1866$ million.
5. Total = $2.0217 + 1.1866 = 3.2083$ million.

Why B is correct: Each year's expected loss (PD x EAD x LR) is discounted at the effective rate from the mid-year default date, and the discounted amounts are summed.

Why the others are wrong

- **A:** Discounted for full years (1 and 2) instead of mid-year (0.5 and 1.5): 2.1/1.079 + 1.33/1.079^2 = 3.0886.
- **C:** Summed the undiscounted expected losses (2.10 + 1.33) and skipped discounting.
- **D:** Used the wrong timing for year 2 (2.5 years instead of 1.5) while correctly using 0.5 for year 1: 2.0217 + 1.33/1.079^2.5 = 3.1214.

LO 21.h

Describe a workout procedure for loss assets and compare the following two approaches used to manage loss assets: retaining loss assets and writing off loss assets.

Q80

AP Mock 3 Q46 · AnalystPrep Q.5834

A regional bank is reassessing how it manages credit risk on high-risk borrowers and problem loans. It needs more effective handling of accounts that have exceeded credit limits, become delinquent, or entered default. Which policy revision would best enhance the bank's ability to manage these situations?

A. Outsourcing the management of all high-risk accounts to external consultants for immediate action.
B. Introducing a dynamic credit limit system that automatically adjusts based on the borrower's repayment behavior.
C. Establishing a specialized internal team dedicated to the workout of delinquent or defaulted accounts with structured and independent oversight.
D. Implementing a policy of immediate legal action against all borrowers whose accounts become delinquent.

Answer: C

Step by step

1. The problem is the handling of accounts already past limit, delinquent or in default, which is the domain of the workout function.
2. Sound practice is a dedicated workout unit with responsible staff, senior-management involvement and independent review of its recoveries and time to recovery.
3. Each borrower is treated on its own merits (more collateral or guarantees, advice/restructuring, third-party partner, or out-of-court liquidation), which a specialized team can do and blanket rules cannot.
4. Option C provides exactly this, so C is best.

Why C is correct: An expert, dedicated workout unit with independent oversight lets each problem borrower be handled case by case and lets the bank evaluate recovery success, which is what effective management of problem accounts requires.

Why the others are wrong

- **A:** Wholesale outsourcing gives up control and the bank's own accountability; it is not a workout procedure and does not build in-house oversight of recoveries.
- **B:** Dynamic limits are a preventive/monitoring tool for performing borrowers; they do not deal with accounts that are already delinquent or in default.
- **D:** Immediate legal action against all delinquent borrowers ignores that each loan should be considered on its own merits; restructuring or added collateral may recover more, and litigation (liquidation) is only one possible step.

Q81

BT answers P2.T6.24.5.2 · BT notes P2.T6.24.5.2

Converse Bank and Old Crow Bank each made a $1,000,000 loan to a technology startup, Janzee. Janzee defaulted and is unlikely to repay. Old Crow retains the loss asset on its balance sheet (carrying a matching loan loss reserve) in the hope of some later recovery, while Converse writes the loan off against its reserve. All else is equal. An analyst wants the two banks' financial statements to be comparable and believes the Janzee loan will never be recovered. What adjustment should he make?

A. Reduce Old Crow's asset values only.
B. Increase Old Crow's loan loss reserves only.
C. Reduce Old Crow's asset values and its loan loss reserves.
D. Reduce Old Crow's asset values and increase its loan loss reserves.

Answer: C

Step by step

1. Converse wrote the loan off: the loan and the reserve held against it were both removed from its balance sheet.
2. Old Crow retained it: the loan stays in assets and the reserve against it stays on the books, so both are larger than Converse's.
3. If the loan is truly unrecoverable, Old Crow should be put on the write-off basis: remove the $1,000,000 loan from assets and remove the corresponding reserve.
4. Both items fall by the same amount, so net assets are unchanged and the two banks become comparable.

Why C is correct: Retention inflates both gross assets and the loan loss reserve; writing the loss off removes both, so the analyst must reduce Old Crow's assets and reserves.

Why the others are wrong

- **A:** Reducing assets alone leaves the reserve for the dead loan on the books and would understate Old Crow's net assets.
- **B:** Increasing the reserve moves in the wrong direction; the reserve is already too large because it still includes the retained loss asset.
- **D:** Cutting assets is right but raising the reserve is wrong; write-off removes the reserve against the loan rather than adding to it.

Q82

Schweser Module Quiz 21.2 Q4

REBUILT — Reworded option B from 'write-down model' to 'write-off' to match the GARP/Schweser terminology; substance and answer unchanged.

A financial analyst for ABC Bank notices that loan loss reserves are small relative to the size of the bank's loan portfolio. He tells a coworker this is a red flag. The coworker replies that there may be a simple explanation that management could clear up. The coworker is suggesting that ABC Bank could be following the:

A. British model and retaining loss assets on the balance sheet.
B. U.S. model and promptly writing off loss assets against the reserve.
C. U.S. model and providing internal consulting to help the borrower improve its ability to pay.
D. U.S. model and allowing more time for collection efforts or collateral enhancement to work out.

Answer: B

Step by step

1. Two approaches exist: retain loss assets until all collection remedies are exhausted (British tradition) or promptly write them off against the reserve (U.S. tradition).
2. Retention keeps loss assets on the books, so the reserve looks unusually large.
3. Prompt write-off removes loss assets from the books, so the reserve looks small relative to the loan portfolio.
4. A small reserve therefore fits the U.S. write-off approach, so B.

Why B is correct: Immediate write-off of loss assets against the reserve removes them from the books, making the reserve appear small relative to the portfolio without implying under-provisioning.

Why the others are wrong

- **A:** Reversed: the British retention approach keeps loss assets on the books and makes the reserve look unusually large, not small.
- **C:** Consulting with the borrower is a workout step, not a balance-sheet treatment of loss assets, and it does not explain a small reserve.
- **D:** Allowing more time for collection is the rationale for retention, which would enlarge the reserve; it is not the U.S. write-off approach.

LO 21.i

Explain the components of credit risk analysis.

Q83

Schweser Module Quiz 21.2 Q5

REBUILT — Original option B (macroeconomic conditions) and D were weak/loosely related; reworded stem and options so only the segmentation/distribution item is a listed component.

A bank's portfolio-level credit risk analysis should include which of the following components?

A. Segmentation of the loan portfolio by currency, maturity, economic sector and borrower category.
B. A narrative opinion from the credit officer on each large borrower's management quality.
C. The bank's interest rate hedging policy for its funding book.
D. An appraisal of the collateral held against each individual loan, performed at origination.

Answer: A

Step by step

1. Portfolio-level (aggregate) credit risk analysis asks which products are lent, to whom, and for how long.
2. Required components: a summary of major loan types (number of customers, average maturity, average interest rate); distribution of the portfolio with segmentation (currency, short vs long maturity, economic sector, state-owned/private, corporate/retail); a list of loans with government or other guarantees; a review of loans by risk classification; an analysis of nonperforming loans (aging of past-due loans by customer type and economic activity).
3. Option A is the segmentation/distribution component, so A is correct.
4. Options B, C and D are individual-borrower, market-risk or origination-level items, not portfolio-level credit analysis.

Why A is correct: Portfolio analysis summarizes and segments the whole book (loan types, distribution, guarantees, risk classes, nonperforming loans) rather than assessing individual obligors.

Why the others are wrong

- **B:** Credit officer commentary on individual borrowers is obligor-level analysis; an aggregate review does not need it.
- **C:** Interest rate hedging policy concerns market/ALM risk, not a component of credit risk analysis of the loan book.
- **D:** Loan-by-loan collateral appraisal is part of the lending/appraisal process, not the aggregate portfolio analysis; confuses individual and portfolio levels.

Q84

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A risk analyst is reviewing a bank's loan portfolio and breaks the nonperforming loans down by the year in which the loans were originally granted. The nonperforming ratio for loans granted in one particular year is far higher than for the years before and after it. What is the primary purpose of this analysis?

A. To estimate the loss given default for each lending product by comparing secured and unsecured loans.
B. To determine how long each loan has been past due so that it can be assigned to the appropriate asset classification category.
C. To identify the root cause of each individual borrower's deterioration and decide whether the situation is reversible.
D. To measure loan losses per period of origination, in order to determine whether the bank's disciplines for granting credit were maintained.

Answer: D

Step by step

1. Portfolio-level credit risk analysis includes nonperforming loans with specific reference to the loss experience per vintage.
2. Vintage = the period in which the loans were granted.
3. A spike for one vintage signals that credit-granting standards slipped in that period.

Why D is correct: Analyzing nonperforming loans by vintage measures loan losses per period granted. A vintage with abnormally high losses indicates that credit-granting discipline was not maintained in that period.

Why the others are wrong

- **A:** LGD reflects the product lent and is estimated from past recoveries; grouping by origination year says nothing directly about product. Error: confusing vintage with product-based LGD.
- **B:** That describes aging of past-due loans (30, 90, 180, 360 days), not grouping by year of origination. Error: confusing vintage with delinquency age.
- **C:** That is the case-by-case assessment of nonperforming loans. Vintage analysis is a portfolio-level view by period granted. Error: confusing portfolio-level analysis with individual case review.

Q85

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

An analyst reviews a bank's loan portfolio profile over the last five periods. The share of loans with maturities under 90 days has fallen steadily, while the share with maturities above one year has risen. Customer mix and product mix have also shifted over the same period. Which of the following statements about this finding is correct?

A. It answers the question of how long the bank is lending for, may be driven by the shifts in customers and products, and interacts strongly with liquidity risk management.

B. It matters only for loans on the nonperforming list, because maturity is relevant to credit risk only after delinquency.

C. It reflects macroeconomic trends alone, because maturity is a contractual feature unaffected by changes in customers or products.

D. It shows to whom the bank is lending and should be reported only with the customer profile.

Answer: A

Step by step

1. Portfolio analysis asks what products were lent, to whom, and for how long.
2. The maturity profile answers how long; the customer profile answers to whom; the loan distribution profile answers what product.
3. Changes in maturity structure can be influenced by shifts in customers and products, and maturity interacts strongly with liquidity risk management.

Why A is correct: The loan maturity profile shows for how long the bank lends. Its changes may result from shifts in customers, products, risk factors or macro trends, and it interacts strongly with liquidity risk management.

Why the others are wrong

- **B:** Maturity structure is part of aggregate portfolio analysis for all loans, not just nonperforming ones. Error: restricting maturity analysis to delinquent loans.
- **C:** Changes in maturity structure can be influenced by shifts in customers and lending products as well as by risk factors and macro trends. Error: treating maturity as independent of portfolio mix.
- **D:** The customer profile answers to whom; the maturity profile answers for how long. Error: mixing up the three portfolio questions (who, what product, how long).

LO 21.j

Explain the components of credit risk management capacity, and identify key questions that the board of directors of a bank should ask.

Q86

BT answers P2.T6.24.5.3 · BT notes P2.T6.24.5.3

REBUILT — Original had no clearly unique answer (pricing and 'internal' models were not actually omitted/misplaced in a defensible way, and staffing is a capacity-review item); rebuilt as an omission question against the board's key credit-risk questions.

The board of directors of a regional bank meets quarterly. Minutes of the last meeting show the board discussed

only the following credit risk items:
- Whether the bank's loans and deposits are priced competitively.
- How diversified the bank's sources of interest income are.
- How accurate the bank's PD, LGD and EAD estimates have been.
- How frequently the loan portfolio is stress tested.
- Whether all concentration risks are adequately disclosed.

Which of the following standard board questions on credit risk management was NOT addressed at the meeting?

A. Is the bank's liquidity profile sustainable in stress scenarios?
B. Are the bank's loans and deposits priced competitively?
C. How accurate have the PD, LGD and EAD estimates been?
D. Are all concentration risks adequately disclosed?

Answer: A

Key correction: Source key (C) said the board should have discussed loan pricing and internal-model accuracy; that item is not defensible as the single answer, because the minutes' items map to the board's standard questions in a way that does not make C uniquely correct, so the item was rebuilt.

Step by step

1. The board's standard credit risk questions: pricing of loans and deposits; diversification of interest income; security of loan income; investment returns vs risk taken; liquidity under stress; which internal rating models are used; accuracy of PD, LGD and EAD; attitude to problematic loans; stress-test frequency; disclosure of concentration risk; availability of required information.
2. Map the minutes: pricing (covered), interest income diversification (covered), PD/LGD/EAD accuracy (covered), stress-test frequency (covered), concentration disclosure (covered).
3. The liquidity-under-stress question is not in the minutes, so A is the omission.

Why A is correct: Every other listed option appears in the minutes; the only standard board question on the list that was not discussed is whether the bank's liquidity position survives stressed scenarios.

Why the others are wrong

- **B:** Discussed (first bullet); picking it means misreading the minutes.
- **C:** Discussed (third bullet); picking it confuses the omission question with a topic that was covered.
- **D:** Discussed (fifth bullet); picking it means overlooking the concentration-risk disclosure item.

Q87

Schweser Module Quiz 21.2 Q6

REBUILT — Option B (information available to loan officers) was arguably also part of the information-flow review; replaced B and C with clearly out-of-scope items so only D is correct.

The credit risk management capacity review process should include:

A. staff diversity metrics.
B. the bank's interest rate hedging policy.
C. the bank's share price performance over the last year.
D. the timeliness of information flows from loan officers to senior management and the Board.

Answer: D

Step by step

1. A capacity review covers three areas: the lending process, staffing (number, age, experience, responsibilities, training) and information flows.
2. Information flows: is information reaching senior managers, the Board and the risk committee timely, accurate, complete and cost-effective?
3. D is an information-flow item; A, B and C are not part of the three review areas.

Why D is correct: Information flows, including whether data reaches the Board in a timely, accurate and cost-effective way, are one of the three components of a credit risk management capacity review.

Why the others are wrong

- **A:** Staffing review looks at number, age, experience, responsibilities and training, not diversity metrics; confuses HR reporting with capacity review.
- **B:** Hedging policy is a market-risk/ALM topic, not a lending process, staffing or information-flow component.
- **C:** Share price performance does not measure lending process, staffing or information flows.

Q88 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A consultant is carrying out the staffing part of a credit risk management capacity review at a bank whose credit department has recently expanded quickly. Which of the following is the most appropriate scope of the staffing assessment?

A. Confirm only the total headcount of the credit department, since the quality of lending skills cannot be assessed from staffing data.

B. Identify the staff involved in origination, appraisal, supervision and monitoring, including their number, levels, age, experience and responsibilities, and review the quality, frequency and adequacy of their training.

C. Compare the compensation of credit staff with peer banks, because pay is the main indicator of the staff's adherence to policy.

D. Assess the staff's length of service alone, since experience makes a review of training programs unnecessary.

Answer: B

Step by step

1. A capacity review covers three areas: lending process, staffing and information flows.
2. Staffing: identify staff in origination, appraisal, supervision and monitoring, and their number, levels, age, experience and specific responsibilities.
3. Also analyze skills and qualifications in relation to policies and procedures, and review ongoing training programs and their adequacy.

Why B is correct: The staffing component identifies the credit staff and their number, levels, age, experience and responsibilities, analyzes skills against policies and procedures, and reviews the adequacy of training; the quality and frequency of training is a good indicator of lending skills.

Why the others are wrong

- **A:** The staffing review covers number, levels, age, experience, responsibilities and training; training quality and frequency is a good indicator of lending skills. Error: reducing the review to headcount.
- **C:** Compensation benchmarking is not a component; the review evaluates whether staffing is adequate and diligent in following policies, analyzing skills and qualifications against procedures. Error: substituting pay for qualifications and training.
- **D:** Training programs are explicitly reviewed and their adequacy assessed regardless of experience. Error: treating tenure as a substitute for training.

Reading 22. Capital Structure in Banks

Gerhard Schroeck, Risk Management and Value Creation in Financial Institutions (New York, NY: John Wiley & Sons, 2002). Chapter 5, pages 170–186.

LO	Learning objective	Qs
22.a	evaluate a bank's economic capital relative to its level of credit risk.	3
22.b	identify and describe important factors used to calculate economic capital for credit risk: probability of default, exposure, and loss rate.	3
22.c	define and calculate expected loss (EL).	13
22.d	define and calculate unexpected loss (UL).	7
22.e	estimate the variance of default probability assuming a binomial distribution.	3
22.f	calculate UL for a credit asset portfolio and the UL contribution of each asset under various scenarios of portfolio composition, asset characteristics and size.	10
22.g	describe how economic capital is derived.	7
22.h	explain how the credit loss distribution is modeled.	7
22.i	describe challenges to quantifying credit risk.	3

LO 22.a

Evaluate a bank's economic capital relative to its level of credit risk.

Q89

GARP PE2 Q50 · Mock 4 Q51

A risk analyst at a credit ratings agency is evaluating the economic capital for credit risk of two competing regional banks, Bank ABC and Bank XYZ. The two banks have the same credit asset exposure, duration of credit exposure, credit rating, and expected loss. Assuming the average pairwise default correlation between credit assets of Bank ABC is lower than that of Bank XYZ, and the two banks assess their risk appetite at the same predetermined confidence level, which of the following statements would be correct?

A. If the confidence level for both banks is increased, the level of economic capital needed will increase for Bank ABC and for Bank XYZ.

B. If the confidence level for both banks is decreased, the level of economic capital needed will increase for Bank ABC but will decrease for Bank XYZ.

C. If the confidence level for both banks is increased, the level of economic capital needed will decrease for Bank ABC but will increase for Bank XYZ.

D. If the confidence level for both banks is kept unchanged, the level of economic capital needed for Bank ABC and for Bank XYZ will be equal.

Answer: A

Step by step

1. Economic capital is the distance between the loss at the chosen confidence level and expected loss: EC = CM × UL_P.
2. Both banks have the same expected loss, exposure, duration and rating, so the difference lies only in portfolio unexpected loss UL_P.
3. Higher average default correlation (Bank XYZ) means less diversification, so $UL_{P,XYZ} > UL_{P,ABC}$.
4. At the same confidence level the capital multiplier is the same, so $EC_{XYZ} > EC_{ABC}$; the banks do not need equal capital (D is false).
5. Raising the confidence level moves the loss quantile further into the tail, so CM rises and EC rises for both banks; lowering it reduces EC for both. Only A is correct.

Why A is correct: EC = CM x UL_P, and the capital multiplier increases monotonically with the confidence level, so a higher confidence level raises required capital for every bank, regardless of its correlation.

Why the others are wrong

- **B:** Lowering the confidence level lowers CM, so capital falls for both banks; it cannot rise for one and fall for the other (confuses the direction of the effect and treats the banks as reacting oppositely).
- **C:** Both banks face the same change in CM, so capital cannot fall for the lower-correlation bank when confidence rises; the candidate wrongly thinks diversification offsets a higher confidence level.
- **D:** Ignores correlation: with higher correlation Bank XYZ has larger portfolio UL, so it needs more capital than Bank ABC even at the same confidence level and equal EL.

Q90 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A risk manager at a regional bank is reviewing the bank's economic capital framework. Management has decided to target a higher credit rating, which requires the bank to remain solvent at a higher confidence level than currently used. The composition and riskiness of the bank's credit portfolio are unchanged. Which of the following statements about the effect on the bank's economic capital for credit risk is correct?

A. Economic capital is unchanged because the riskiness of the credit assets, not the confidence level, determines the capital requirement.

B. The unexpected loss of the portfolio increases, so economic capital increases in proportion to the change in unexpected loss.

C. The capital multiplier applied to unexpected loss increases, so economic capital increases even though unexpected loss is unchanged.

D. The capital multiplier applied to unexpected loss decreases, because a larger share of the loss distribution is then covered by expected loss.

Answer: C

Step by step

1. Economic capital depends on two parameters: the confidence level and the riskiness of the assets.
2. Economic capital = UL x capital multiplier (CM).
3. A higher confidence level (higher target rating) means a larger CM, covering more of the loss distribution.
4. With assets unchanged, UL is unchanged, so economic capital rises through the higher CM.

Why C is correct: A higher target rating requires a higher confidence level, which raises the CM applied to an unchanged UL, so economic capital increases.

Why the others are wrong

- **A:** Economic capital depends on both the confidence level and asset riskiness; ignoring the confidence level is the error.
- **B:** UL is the standard deviation of credit losses and depends on the assets (PD, LR, exposure, correlations), not on the confidence level; it does not change.
- **D:** A higher confidence level raises, not lowers, the CM; expected loss is not what the multiplier covers.

Q91 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

An analyst at a rating agency compares two banks, Bank M and Bank N, that both set economic capital at the same 99.95% confidence level. Bank M holds a diversified portfolio of highly rated corporate loans. Bank N holds a portfolio of the same total size but concentrated in lower-rated borrowers in a single industry. Which of the following statements is correct?

A. Bank M needs more economic capital because a diversified portfolio has a higher expected loss that must be covered by capital.

B. Bank N needs less economic capital because its higher expected loss is already priced into its loan interest rates, reducing unexpected loss.

C. Both banks need the same economic capital because the same confidence level implies the same capital multiplier and therefore the same capital.

D. Bank N needs more economic capital because its riskier assets produce a higher unexpected loss to which the same capital multiplier is applied.

Answer: D

Step by step

1. Economic capital = UL x CM.
2. Same confidence level: CM is the same for both banks (to a first approximation).
3. Bank N has riskier, more concentrated (higher correlated) assets, so its portfolio UL is higher.
4. Same CM x higher UL gives higher economic capital for Bank N.

Why D is correct: At a given confidence level, riskier and more concentrated assets give a higher UL, so Bank N needs greater economic capital.

Why the others are wrong

- **A:** Diversification lowers UL; expected loss is not what economic capital covers, and nothing indicates M has higher EL.

- **B:** Pricing in expected loss does not reduce unexpected loss; UL is volatility around EL and is what economic capital must cover.
- **C:** The same confidence level fixes the multiplier, but economic capital also depends on UL, which differs with asset riskiness.

LO 22.b

Identify and describe important factors used to calculate economic capital for credit risk: probability of default, exposure, and loss rate.

Q92 AnalystPrep Q.5955

A bank's risk team is discussing how scenarios affect the components of expected loss, EL = PD × EA × LR. Which scenario would most directly impact the loss rate (loss given default) of a corporate loan?

A. The borrower operates in a rapidly growing industry sector.
B. The loan is structured with a variable interest rate.
C. The borrower pledges substantial real estate and equipment as collateral for the loan.
D. The loan's maturity is extended.

Answer: C

Step by step

1. Identify the three EL components: PD (likelihood of default), EA (exposure at default), LR (fraction of exposure lost, LR = 1 − RR).
2. LR depends largely on collateral: secured loans lose less after the collateral is sold than unsecured loans.
3. Pledged real estate and equipment raise recoveries, so LR falls. Only option C acts on LR directly.

Why C is correct: Loss rate is largely a function of collateral; substantial collateral raises recovery and lowers LR.

Why the others are wrong

- **A:** Industry conditions relate to the borrower's creditworthiness and so mainly PD, not the fraction lost once default occurs.
- **B:** A floating rate mainly affects the interest cash flows and outstanding amount (exposure), not the share of exposure lost after default.
- **D:** A longer maturity gives more time to default (PD typically rises with horizon) and can change exposure; it does not directly change recoveries.

Q93 AnalystPrep Q.6184

An analyst is calculating economic capital for credit risk using borrower-specific information. Which set of factors correctly identifies the key inputs?

A. Credit conversion factor, expected exposure, and probability of recovery.
B. Liquidity coverage ratio, net stable funding ratio, and loss avoidance rate.
C. Probability of default, exposure amount, and loss rate.
D. Default adjustable rate, mark-to-market exposure, and salvage value rate.

Answer: C

Step by step

1. Economic capital for credit risk is driven by unexpected loss, which is built from expected loss: EL = PD × EA × LR.
2. The three factors are probability of default, exposure amount (exposure at default), and loss rate (loss given default).
3. Only option C lists these three.

Why C is correct: PD, EA and LR are the three factors that determine expected and unexpected credit loss and hence economic capital.

Why the others are wrong

- **A:** Mixes exposure-related terms and uses probability of recovery instead of PD; PD and loss rate are both missing.
- **B:** LCR and NSFR are liquidity regulations, not credit loss components; confuses liquidity with credit risk.

- **D:** Contains non-standard terms invented to resemble the real ones; no recognized PD, EA or LR measure.

Q94 Schweser Module Quiz 22.1 Q1

XYZ Bank is forecasting the expected loss on a loan to a mid-size corporate borrower. It estimates a 75% loss of the exposure if the borrower fails to perform its obligation. This risk measure is the:

A. probability of default.
B. loss rate.
C. unexpected loss.
D. exposure amount.

Answer: B

Step by step

1. The 75% is a percentage of exposure lost conditional on default.
2. The fraction lost given default is the loss rate (LR), also called loss given default; LR $= 1 -$ RR, so recovery is 25%.
3. PD would be the likelihood of default; EA a dollar amount; UL the standard deviation of losses.

Why B is correct: Loss rate is the percentage of exposure lost if default occurs.

Why the others are wrong

- **A:** PD is the likelihood that default occurs, not the severity once it occurs.
- **C:** Unexpected loss is the standard deviation of credit losses around EL, not a conditional loss percentage.
- **D:** Exposure amount is the dollar amount at risk at default, not a percentage loss.

LO 22.c

Define and calculate expected loss (EL).

Q95 AnalystPrep Q.3667

REBUILT — Option D ($40 million) had no realistic derivation; replaced with the PD+LR-sum error.

American International Bank has extended a loan to a corporate client. The credit analyst's note gives: exposure amount = USD 100 million, loss rate = 10%, probability of default = 20%. What is the expected loss on the loan?

A. USD 2 million
B. USD 20 million
C. USD 10 million
D. USD 30 million

Answer: A

Step by step

1. EL $=$ PD $\times$ EA $\times$ LR
2. $0.20 \times 100 \times 0.10 = 2.0$ million

Why A is correct: Expected loss is the product of the probability of default, the exposure amount and the loss rate.

Why the others are wrong

- **B:** Multiplied only PD by EA (0.20×100), ignoring the loss rate.
- **C:** Multiplied only LR by EA (0.10×100), ignoring the probability of default.
- **D:** Added PD and LR and applied the sum to the exposure ($(0.20 + 0.10) \times 100$) instead of multiplying the three components.

Q96 Mock 1 Q17 (AP)

REBUILT — Distractors 7,800 and 20,000 had no realistic derivation; replaced with single-bond and recovery-as-loss-rate errors.

A portfolio consists of two bonds, A and B. The probability of joint default of A and B is 1.45%, with 25% default correlation.

Bond	Value	Default probability	Recovery rate
A	GBP 1,000,000	4%	70%
B	GBP 800,000	6%	60%

Determine the expected credit loss for the portfolio.

A. GBP 12,000
B. GBP 19,200
C. GBP 31,200
D. GBP 56,800

Answer: C

Step by step

1. Loss rates: A = 30%, B = 40%
2. $EL_A = 1{,}000{,}000 \times 0.04 \times 0.30 = 12{,}000$
3. $EL_B = 800{,}000 \times 0.06 \times 0.40 = 19{,}200$
4. Sum = 31,200; joint default probability and correlation do not affect EL

Why C is correct: Expected loss is additive across exposures and independent of default correlation.

Why the others are wrong

- **A:** Only bond A.
- **B:** Only bond B.
- **D:** Used recovery rates as loss rates: 28,000 + 28,800.

Q97

AnalystPrep Q.3684

REBUILT — Distractor C ($4.5m) had no realistic derivation; replaced with the recovery-rate-for-loss-rate error.

An investor holds a $200 million bond portfolio: AA-rated bonds ($120 million) and BB-rated bonds ($80 million). One-year default probabilities are 4% (AA) and 6% (BB), and defaults are independent. Recovery rates are 65% for AA and 40% for BB. What is the one-year expected loss on the portfolio?

A. $1,680,000
B. $4,560,000
C. $5,040,000
D. $2,880,000

Answer: B

Step by step

1. Loss rate = 1− recovery rate: AA = 35%, BB = 60%
2. $EL_{AA} = 120{,}000{,}000 \times 0.04 \times 0.35 = 1{,}680{,}000$
3. $EL_{BB} = 80{,}000{,}000 \times 0.06 \times 0.60 = 2{,}880{,}000$
4. Portfolio EL is additive: 1,680,000 + 2,880,000 = 4,560,000

Why B is correct: EL is the sum of the individual ELs, each PD x EA x (1 - RR); independence does not affect expected loss.

Why the others are wrong

- **A:** Counted only the AA bonds.
- **C:** Used the recovery rates as loss rates: 120m × 0.04 × 0.65 + 80m × 0.06 × 0.40 = 5,040,000.
- **D:** Counted only the BB bonds.

Q98

AnalystPrep Q.1830

REBUILT — Distractors $500,000 and $2,000,000 had no realistic derivation; replaced with the no-recovery and recovery-as-loss-rate errors.

Albert Cook, FRM, owns a $50 million bond portfolio: AA-rated bonds ($30 million) and BB-rated bonds ($20 million). The 1-year default probabilities are 1% (AA) and 2.5% (BB). Recovery rates are 80% for AA bonds and 60% for BB bonds. What is the 1-year expected credit loss?

A. $800,000
B. $260,000
C. $540,000
D. $200,000

Answer: B

Step by step

1. Loss rates: AA = 1 − 0.80 = 20%; BB = 1 − 0.60 = 40%
2. $EL_{AA} = 30{,}000{,}000 \times 0.01 \times 0.20 = 60{,}000$

3. $EL_{BB} = 20{,}000{,}000 \times 0.025 \times 0.40 = 200{,}000$
4. Total = 260,000

Why B is correct: Portfolio EL is the sum of each bond's PD x EA x LR, with LR = 1 - recovery rate.

Why the others are wrong

- **A:** Ignored recoveries (LR = 100%): 300,000 + 500,000.
- **C:** Used recovery rates in place of loss rates: 240,000 + 300,000.
- **D:** Counted only the BB bonds.

Q99 AnalystPrep Q.5878

A bank trainer asks new risk analysts which of the following correctly represents the calculation of expected loss for a loan in the bank's portfolio.

A. The loss amount calculated based on the bank's past experiences with similar loan types.
B. The aggregate of all losses estimated across the entire loan portfolio for a financial year.
C. The product of the loan's probability of default, exposure at default, and loss given default.
D. The highest potential loss for the loan under the most adverse economic conditions.

Answer: C

Step by step

1. EL = PD × EA × LR (LR = LGD)
2. Compare each option with this definition; only C matches.

Why C is correct: Expected loss is the product of the three credit-risk components: PD, exposure and loss rate.

Why the others are wrong

- **A:** History only informs the inputs; it is not the EL calculation itself.
- **B:** A portfolio-wide aggregate, not the per-loan formula; it omits PD x EA x LR.
- **D:** Describes a worst-case loss, not the expected (average) loss; unexpected loss is the volatility around EL.

Q100 AnalystPrep Q.6187

Loan A has an exposure of $800,000, a probability of default of 1.2% and a loss given default of 40%. Loan B has an exposure of $1,200,000, a probability of default of 0.8% and a loss given default of 50%. What is the expected loss on the portfolio of these two loans?

A. $8,640
B. $4,800
C. $3,840
D. $4,320

Answer: A

Step by step

1. $EL_A = 800{,}000 \times 0.012 \times 0.40 = 3{,}840$
2. $EL_B = 1{,}200{,}000 \times 0.008 \times 0.50 = 4{,}800$
3. Portfolio EL = 3,840 + 4,800 = 8,640

Why A is correct: Portfolio expected loss is the sum of the individual expected losses.

Why the others are wrong

- **B:** Only Loan B's EL.
- **C:** Only Loan A's EL.
- **D:** Averaged the two ELs (8,640 / 2) instead of summing.

Q101 BT notes P2.T6.505.2

Which of the following is TRUE about expected credit loss?

A. Expected loss itself constitutes risk
B. Expected loss represents credit loss levels that do NOT fluctuate year to year
C. Expected loss is calculated from the bottom up (transaction by transaction) and reimbursed through adequate loan pricing
D. Expected loss is the risk that arises due to the variation in loss levels and therefore justifies the rationale for economic capital

Answer: C

Step by step

1. EL is an actuarial-type average loss, computed transaction by transaction.
2. It is a foreseeable cost of lending, recovered through risk-based loan pricing.

Why C is correct: EL is the predictable cost of doing business, priced into loans; only the variability around it (UL) is risk requiring capital.

Why the others are wrong

- **A:** EL is foreseeable, so it is a cost, not a risk; risk is the volatility around EL.
- **B:** Actual losses fluctuate with the cycle; EL is the long-run average, not a constant annual loss.
- **D:** This describes unexpected loss, which is what economic capital covers.

Q102 BT notes P2.T6.505.3

Expected loss (EL) has three components: probability of default (PD), exposure amount (EA), and loss rate (LR). Each of the following is true EXCEPT:

A. Exposure amount (EA) is the standard deviation of credit losses estimated at the end of the horizon, excluding outstanding interest payments

B. Probability of default (PD) is the probability that a borrower will default before the end of a predetermined period of time (the estimation horizon typically chosen is one year)

C. Loss rate (LR) represents the ratio of actual losses incurred at the time of default (including all costs associated with the collection and sale of collateral) to the exposure amount

D. Expected loss (EL) is equal to the product of: the probability of default up to time H (horizon); the exposure amount at time H; and the loss rate experienced at time H; i.e., EL(H) = PD(H)*EA(H)*LR(H)

Answer: A

Step by step

1. Check each statement against the definitions of the EL components.
2. B, C and D match the definitions and EL(H) = PD(H) x EA(H) x LR(H).
3. A is false: EA is a dollar exposure at default (including promised interest), not a standard deviation (volatility of losses is unexpected loss).

Why A is correct: EA is the amount owed at the time of default, including promised interest; a standard deviation of losses is the unexpected loss.

Why the others are wrong

- **B:** True; this is the definition of PD.
- **C:** True; LR is the fraction of exposure lost after recoveries and collection costs.
- **D:** True; this is the EL formula.

Q103 Schweser Module Quiz 22.1 Q3

REBUILT — Answer options were missing in the source (stored as open); rebuilt as a four-option MCQ.

If the recovery rate (RR) increases and the probability of default (PD) decreases, what is the effect on expected loss (EL), all else equal?

A. EL will increase because a higher RR raises the loss rate.

B. EL will be indeterminate because the two changes have offsetting effects.

C. EL will be unchanged because EL depends only on the exposure amount.

D. EL will decrease.

Answer: D

Step by step

1. $EL = PD \times EA \times LR$ with $LR = 1 - RR$
2. Higher RR lowers LR, so EL falls.
3. Lower PD also lowers EL; both effects work in the same direction.

Why D is correct: Both changes reduce a multiplicative factor of EL, so EL must fall.

Why the others are wrong

- **A:** Reverses the relation: LR = 1 - RR, so a higher RR lowers the loss rate.
- **B:** The effects are not offsetting; both reduce EL.
- **C:** EL also depends on PD and LR, not on EA alone.

Q104

Schweser Module Quiz 22.1 Q4

REBUILT — Distractors $68,000 and $272,000 had no realistic derivation; replaced with the recovery-as-loss-rate, full-limit and no-loss-rate errors.

Big Bank has a $20,000,000 credit facility with Upstart Corp., of which $18,000,000 is currently outstanding. Big Bank estimates a one-year probability of default of 2% and, because the collateral is unique with limited resale opportunities, an 80% loss rate. The expected loss for Big Bank is closest to:

A. $72,000
B. $288,000
C. $320,000
D. $360,000

Answer: B

Step by step

1. Exposure is the amount outstanding, $18,000,000 (not the facility limit).
2. EL = 18,000,000 × 0.02 × 0.80 = 288,000

Why B is correct: EL = PD x EA x LR, using the drawn amount as the exposure.

Why the others are wrong

- **A:** Used 20% (recovery) as the loss rate: 18m × 0.02 × 0.20.
- **C:** Used the full $20m limit: 20m × 0.02 × 0.80.
- **D:** Omitted the loss rate: 18m × 0.02.

Q105

AnalystPrep Q.3087

REBUILT — Original options included unexplained values (e.g. $567,500, $180,094.6); replaced with distractors from realistic errors and removed the irrelevant 95% credit VaR wording.

A risk analyst holds a portfolio of two bonds issued by Taylor Pharma and Simon Chemicals. Taylor Pharma's bond has a value of $7 million, a one-year default probability of 6%, and a recovery rate of 80%. Simon Chemicals' bond has a value of $9 million, a one-year default probability of 5%, and a recovery rate of 90%. The expected loss from default on the portfolio over one year is closest to:

A. $741,000
B. $84,000
C. $129,000
D. $870,000

Answer: C

Step by step

1. Loss rate = 1 - recovery rate: Taylor 20%, Simon 10%.
2. Taylor: 7,000,000 × 0.06 × 0.20 = 84,000.
3. Simon: 9,000,000 × 0.05 × 0.10 = 45,000.
4. Portfolio EL is the sum of individual ELs: 84,000 + 45,000 = 129,000.

Why C is correct: Expected loss is exposure times default probability times loss rate (1 - recovery), and portfolio EL is additive across assets regardless of correlation.

Why the others are wrong

- **A:** Used the recovery rate as the loss rate: 7m × 0.06 × 0.8 + 9m × 0.05 × 0.9 = 741,000.
- **B:** Included only Taylor Pharma's expected loss and omitted Simon Chemicals.
- **D:** Ignored recovery (loss rate of 100%): 7m × 0.06 + 9m × 0.05 = 870,000.

Q106

AnalystPrep Q.3685

REBUILT — Original distractors (18,424; 12,424) were not traceable to realistic errors; replaced them and dropped the irrelevant credit VaR sentence.

A portfolio consists of two bonds. The probability that the two bonds jointly default is 2%, and the default correlation is 25%. Bond 1 has a value of USD 500,000, a default probability of 5%, and a recovery rate of 50%. Bond 2 has a value of USD 300,000, a default probability of 3%, and a recovery rate of 30%. The expected credit loss of the portfolio over one year is:

A. USD 18,800
B. USD 12,500
C. USD 15,200
D. USD 34,000

Answer: A

Step by step

1. Bond 1: LGD = 50%; 500,000 × 0.05 × 0.50 = 12,500.
2. Bond 2: LGD = 70%; 300,000 × 0.03 × 0.70 = 6,300.
3. EL is additive, so portfolio EL = 12,500 + 6,300 = 18,800.
4. Joint default probability and correlation affect unexpected loss, not expected loss.

Why A is correct: Portfolio expected loss is the sum of the individual expected losses; correlation does not enter.

Why the others are wrong

- **B:** Counted only Bond 1 (12,500).
- **C:** Used the recovery rate of Bond 2 as its loss rate: 12,500 + 300,000 × 0.03 × 0.30 = 15,200.
- **D:** Ignored recovery (loss rate 100%): 500,000 × 0.05 + 300,000 × 0.03 = 25,000 + 9,000 = 34,000.

Q107

AnalystPrep Q.5540

REBUILT — Replaced the unexplained $240,000 option with the recovery-as-loss-rate error.

A credit portfolio contains several bonds, each with a notional value of $10 million. Bond A has a one-year default probability of 1% and a recovery rate of 40%. Bond B has a one-year default probability of 2% and a recovery rate of 60%. The default probabilities of all other bonds are negligible, and Bonds A and B are correlated. The portfolio's one-year expected loss is:

A. $160,000
B. $80,000
C. $60,000
D. $140,000

Answer: D

Step by step

1. Bond A: LGD = 60%; 10m × 0.01 × 0.60 = 60,000.
2. Bond B: LGD = 40%; 10m × 0.02 × 0.40 = 80,000.
3. Sum (correlation irrelevant for EL): 140,000.

Why D is correct: EL is the sum of exposure x PD x (1 - recovery) over the bonds; correlation only affects unexpected loss.

Why the others are wrong

- **A:** Used recovery as the loss rate: 40,000 + 120,000 = 160,000.
- **B:** Included only Bond B.
- **C:** Included only Bond A.

LO 22.d

Define and calculate unexpected loss (UL).

Q108

AP Mock 3 Q59

REBUILT — Original distractors were unexplained values; replaced with options that each reflect a specific realistic error.

American Banking Group approved a loan to Hazard Corp five years ago with a total commitment of USD 120 million. Today the outstanding balance is USD 80 million. The credit risk department reports:

Parameter	Value
Probability of default over the next year	1%
Loss rate if the customer defaults	25%
Standard deviation of the probability of default	4%
Standard deviation of the loss rate	15%

What is the unexpected loss for this loan today?

A. USD 2.163 million
B. USD 1.200 million
C. USD 0.800 million
D. USD 1.442 million

Answer: D

Step by step

1. Exposure today is the outstanding balance: EA = 80 million (not the 120 million commitment).
2. PD term: $0.01 \times 0.15^2 = 0.000225$.
3. Loss-rate term: $0.25^2 \times 0.04^2 = 0.0625 \times 0.0016 = 0.000100$.
4. Sum = 0.000325; square root = 0.018028.
5. UL = $80 \times 0.018028 = 1.442$ million.

Why D is correct: UL is the standard deviation of credit losses, UL = $\text{EA}\sqrt{PD\ \sigma_{LR}^2 + LR^2\ \sigma_{PD}^2}$, applied to the exposure outstanding today with both variance terms included.

Why the others are wrong

- **A:** Used the 120 million commitment as exposure (120 x 0.018028 = 2.163); only the drawn balance is at risk here.
- **B:** Dropped the $LR^2\sigma_{PD}^2$ term: $80\sqrt{0.000225} = 1.200$.
- **C:** Dropped the $PD\ \sigma_{LR}^2$ term: $80\sqrt{0.0001} = 0.800$.

Q109

AnalystPrep Q.3680

REBUILT — Original distractors included 1.442, which belongs to a different problem, plus unexplained values; replaced with options built from realistic errors.

Five years ago, American Banking Group approved a loan to Hazard Corp. with a total commitment of USD 120 million. Today, the outstanding balance of the loan is USD 80 million. The credit risk department provided the following:

- Standard deviation of the probability of default: 14%
- Standard deviation of the loss rate: 15%
- Probability of default over the next year: 2%
- Loss rate if the customer defaults: 25%

What is the unexpected loss for this loan today?

A. USD 3.274 million
B. USD 4.911 million
C. USD 2.800 million
D. USD 1.697 million

Answer: A

Step by step

1. Exposure is the outstanding balance, EA = 80 million.
2. PD term: $0.02 \times 0.15^2 = 0.00045$.
3. Loss-rate term: $0.25^2 \times 0.14^2 = 0.0625 \times 0.0196 = 0.001225$.
4. Sum = 0.001675; square root = 0.040927.
5. UL = $80 \times 0.040927 = 3.274$ million.

Why A is correct: UL combines the variability from the loss rate given default and from the default event itself, scaled by the amount actually exposed today.

Why the others are wrong

- **B:** Used the 120 million commitment (120 x 0.040927 = 4.911) instead of the 80 million outstanding.
- **C:** Dropped the $PD\ \sigma_{LR}^2$ term: $80\sqrt{0.001225} = 2.800$.
- **D:** Dropped the $LR^2\sigma_{PD}^2$ term: $80\sqrt{0.00045} = 1.697$.

Q110

AnalystPrep Q.3683

A bank has booked a loan with a total commitment of $100,000, of which 80% is currently outstanding. The default probability over the next year is 2% and the loss given default (LGD) is 40%. The standard deviation of LGD is 30% and the standard deviation of the probability of default is 14%. Drawdown on default (the fraction of the undrawn amount expected to be drawn) is 70%. Determine the expected and unexpected losses for the bank.

A. Expected loss = USD 640, unexpected loss = USD 5,621
B. Expected loss = USD 640, unexpected loss = USD 6,604
C. Expected loss = USD 752, unexpected loss = USD 6,604
D. Expected loss = USD 752, unexpected loss = USD 5,621

Answer: C

Step by step

1. Exposure at default: $80{,}000 + 0.70 \times 20{,}000 = 94{,}000$.
2. $EL = 94{,}000 \times 0.02 \times 0.40 = 752$.
3. Under the root: $0.02 \times 0.30^2 + 0.40^2 \times 0.14^2 = 0.0018 + 0.003136 = 0.004936$; square root $= 0.070257$.
4. $UL = 94{,}000 \times 0.070257 = 6{,}604$.

Why C is correct: The drawn amount plus the expected drawdown on default is at risk, so EA is 94,000; both EL and UL are computed on that amount.

Why the others are wrong

- **A:** Used only the current drawn balance of 80,000 for both EL (640) and UL (5,621), ignoring drawdown on default.
- **B:** EL used 80,000 (640) but UL used 94,000; the exposure measure is inconsistent.
- **D:** EL correctly used 94,000, but UL was computed on 80,000 (5,621); the exposure measure is inconsistent.

Q111

AnalystPrep Q.5877

REBUILT — Original keyed answer ('additional losses beyond expected') only loosely describes UL; rewrote as a precise definition question based on UL being the standard deviation of credit losses.

In the standard bottom-up credit risk framework, the stand-alone unexpected loss (UL) of a loan is best described as:

A. the mean of the credit loss distribution, equal to exposure times PD times loss rate
B. the standard deviation of actual credit losses around the expected loss
C. the variance of credit losses around the expected loss
D. the part of credit losses that can be priced into the loan's interest rate

Answer: B

Step by step

1. EL is the mean of the loss distribution; it is predictable, so it can be priced and reserved.
2. UL measures the unpredictable variability of actual losses around EL.
3. In statistical terms UL is the standard deviation (not the variance) of the loss distribution, and economic capital is held against it.

Why B is correct: UL is calculated as the standard deviation of the same distribution whose mean is EL, capturing the variation of actual losses around EL.

Why the others are wrong

- **A:** This describes expected loss (EA x PD x LR), the mean, not the dispersion.
- **C:** Variance is in squared dollars; UL is defined as the standard deviation, the square root of the variance.
- **D:** Expected losses can be priced into the loan rate; unexpected losses cannot be anticipated and require a cushion of economic capital.

Q112

BT notes P2.T6.506.2

REBUILT — Original options were unexplained values; replaced with options reflecting specific errors and stated that default is a binomial event.

A bank's credit exposure to a customer consists of the following:
- Exposure amount (EA): $50.0 million
- Probability of default (PD): 2.0%
- Loss rate (LR): 50.0%
- Standard deviation of loss rate: 40.0%

Assume default is a binomial (Bernoulli) event. The expected loss is $0.5 million. Which is nearest to the exposure's unexpected loss (UL)?

A. $2.83 million
B. $4.50 million
C. $6.33 million
D. $2.87 million

Answer: B

Step by step

1. Binomial PD variance: $\sigma_{PD}^2 = PD(1 - PD) = 0.02 \times 0.98 = 0.0196$.
2. PD term: $0.02 \times 0.40^2 = 0.0032$.

3. Loss-rate term: $0.50^2 \times 0.0196 = 0.0049$.
4. Sum = 0.0081; square root = 0.09.
5. UL = $50 \times 0.09 = 4.50$ million.

Why B is correct: UL uses $\sigma_{PD}^2 = PD(1 - PD)$ for the default event plus the loss-rate variance, giving 9% of exposure.

Why the others are wrong

- **A:** Dropped the default-variance term: $50\sqrt{0.0032} = 2.83$.
- **C:** Added the two standard-deviation terms ($50 \times (0.0566 + 0.07)$) instead of taking the root of the sum of variances.
- **D:** Used $\sigma_{PD}^2 = PD^2 = 0.0004$ instead of $PD(1 - PD)$: $50\sqrt{0.0032 + 0.0001} = 2.87$.

Q113 BT notes P2.T6.921.2

REBUILT — Original options were unexplained values; replaced with options reflecting specific errors and stated that default is a binomial event.

A credit portfolio contains a position with an adjusted exposure of $30.0 million and a default probability of 4.0%. The Portfolio Manager estimates a loss given default (LGD) of 40.0% with a standard deviation of LGD of 40.0%. Assume default is a binomial event. What is the position's unexpected loss (UL)?

A. $2.351 million
B. $2.400 million
C. $3.360 million
D. $4.752 million

Answer: C

Step by step

1. $\sigma_{PD}^2 = 0.04 \times 0.96 = 0.0384$.
2. PD term: $0.04 \times 0.40^2 = 0.0064$; LGD term: $0.40^2 \times 0.0384 = 0.006144$.
3. Sum = 0.012544; square root = 0.112.
4. UL = $30 \times 0.112 = 3.360$ million.

Why C is correct: UL is the root of the summed default and loss-rate variance contributions, 11.2% of exposure.

Why the others are wrong

- **A:** Dropped the $PD\ \sigma_{LR}^2$ term (0.0064), keeping only the LGD-variance term: 30 x sqrt(0.006144) = 2.351.
- **B:** Dropped the $LGD^2\sigma_{PD}^2$ term (0.006144), keeping only the PD term: 30 x sqrt(0.0064) = 2.400.
- **D:** Added the two standard-deviation terms (0.08 + 0.0784) instead of root-sum-of-squares: 30 x 0.1584 = 4.752.

Q114 Schweser Module Quiz 22.1 Q2

Which of the following statements about expected loss (EL) and unexpected loss (UL) is true?

A. EL always exceeds UL.
B. UL always exceeds EL.
C. EL and UL are parameterized by the exact same set of variables.
D. EL is directly related to exposure.

Answer: D

Step by step

1. EL = EA $\times$ $PD \times LR$ is linear in EA, so it rises proportionally with exposure.
2. UL also needs σ_{LR} and σ_{PD}, so the variable sets differ.
3. Neither 'always' statement holds: UL typically exceeds EL, but both are zero when PD is zero.

Why D is correct: EL is exposure multiplied by PD and LR, so it increases directly with exposure.

Why the others are wrong

- **A:** UL typically exceeds EL for risky loans, and 'always' fails anyway since both equal zero when PD = 0.
- **B:** UL typically exceeds EL, but 'always' is false; when PD = 0 both are zero.
- **C:** UL additionally depends on the standard deviations of PD and LR, which EL does not use.

LO 22.e

Estimate the variance of default probability assuming a binomial distribution.

Q115 AnalystPrep Q.6186

REBUILT — Rebuilt during solving to make the item self-contained with exactly one defensible answer; the verifier then made a minor notation fix.

A portfolio manager analyzes a subset of a loan portfolio made up of 500 loan exposures. Each loan has a one-year probability of default of 1.5%, and defaults are assumed independent so that each loan is a two-state (default / no default) outcome. Assuming a binomial distribution, what is the variance of the number of defaults in the subset?

A. 0.1125
B. 7.5
C. 2.72
D. 7.3875

Answer: D

Step by step

1. Each loan is a Bernoulli (binomial with one trial) variable with variance $\sigma_{PD}^2 = PD(1-PD) = 0.015 \times 0.985 = 0.014775$.
2. With 500 independent loans, variances add: 500×0.014775.
3. Var $= n \cdot PD \cdot (1-PD) = 500 \times 0.015 \times 0.985 = 7.3875$.

Why D is correct: In the two-state model the default indicator is Bernoulli with variance $PD(1-PD)$; for independent loans the variance of the default count is n times that, 7.3875.

Why the others are wrong

- **A:** Used $n \cdot PD^2 = 500 \times 0.015^2 = 0.1125$ instead of $PD(1-PD)$.
- **B:** This is the expected number of defaults $n \cdot PD = 7.5$ (forgot the $1-PD$ factor); it is a mean, not a variance.
- **C:** Took the standard deviation $\sqrt{7.3875} \approx 2.72$ instead of the variance.

Q116 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A credit analyst at a bank models the default of an individual corporate borrower as a two-state event over a one-year horizon. The borrower has a 2% probability of default. Assuming default follows a binomial (Bernoulli) distribution, what is the variance of the default probability?

A. 0.0196
B. 0.0004
C. 0.0200
D. 0.1400

Answer: A

Step by step

1. Default is a two-state event, so variance of PD = PD x (1 - PD).
2. = 0.02 x 0.98 = 0.0196.

Why A is correct: Variance = PD(1 - PD) = 0.02 x 0.98 = 0.0196.

Why the others are wrong

- **B:** 0.0004 is PD squared (0.02^2); it omits the (1 - PD) term.
- **C:** 0.0200 is PD itself; it ignores the factor (1 - PD).
- **D:** 0.1400 is the standard deviation (square root of 0.0196), not the variance.

Q117 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A credit officer compares two loans that each follow a two-state default model over one year. Loan X has a probability of default of 10% and Loan Y has a probability of default of 80%. Assuming binomial default, which of the following statements about the variance of default probability is correct?

A. Loan X has the higher variance, 0.09, because its probability of default is closer to zero and therefore more certain to be a non-default.
B. Loan Y has the higher variance, 0.16, compared with 0.09 for Loan X.
C. Loan Y has the higher variance, 0.64, because variance equals the squared probability of default.
D. The two loans have equal variance, because 10% and 80% are not symmetric but both are two-state outcomes.

Answer: B

Step by step

1. Variance = PD(1 - PD).
2. Loan X: 0.10 x 0.90 = 0.09.
3. Loan Y: 0.80 x 0.20 = 0.16.
4. 0.16 > 0.09, so Loan Y has the higher variance (variance peaks at PD = 50%).

Why B is correct: Variance of Loan Y = 0.16 exceeds Loan X = 0.09, because 80% is closer to the 50% maximum of PD(1 - PD) than 10% is.

Why the others are wrong

- **A:** 0.09 is correct for X but the conclusion is reversed; a low PD does not imply higher variance, and Y's variance is larger.
- **C:** 0.64 is PD squared, not PD(1 - PD).
- **D:** Variance depends on PD(1 - PD), which differs (0.09 vs 0.16); having two states does not equalize it.

LO 22.f

Calculate UL for a credit asset portfolio and the UL contribution of each asset under various scenarios of portfolio composition, asset characteristics and size.

Q118

AP Mock 3 Q50

An analyst has calculated expected loss for each bond in a corporate portfolio and must now estimate unexpected losses. The bonds have varying probabilities of default and loss given default, with associated volatilities. Why can the analyst not simply aggregate the unexpected losses of individual bonds to estimate the total unexpected loss for the portfolio?

A. Because the unexpected loss is calculated using the average of the expected losses, and the aggregation of averages does not give the unexpected loss.

B. Because the aggregation of individual unexpected losses assumes perfect correlation among the bonds, which is rarely the case in practice.

C. Because the bond portfolio is subject to market risk, which needs to be calculated separately from the credit risk of individual bonds.

D. Because the unexpected loss calculation requires the inclusion of recovery rates, which have not been accounted for individually for each bond.

Answer: B

Step by step

1. Expected loss is linear: $EL_P = \sum EL_i$, so EL can simply be added.
2. Unexpected loss is a standard deviation: $UL_P = \sqrt{\sum_i \sum_j UL_i UL_j \rho_{ij}}$.
3. Setting $\rho_{ij} = 1$ for all pairs gives $UL_P = \sum UL_i$, the maximum possible value.
4. Since default correlations are below 1 in practice, $UL_P < \sum UL_i$; summing overstates portfolio risk and ignores diversification.

Why B is correct: UL is a standard deviation, so it adds only when all pairwise correlations equal 1; with $\rho < 1$ the portfolio UL is lower than the sum of the individual ULs.

Why the others are wrong

- **A:** UL is not computed from an average of ELs; it is built from PD, LR and their volatilities. EL being additive is exactly why EL, unlike UL, can be summed.
- **C:** Market risk is a separate issue and is irrelevant to why credit ULs do not add; the problem is the correlation cross-terms.
- **D:** Recovery is already embedded in the loss rate used for each bond's UL; omission of recovery is not the reason aggregation fails.

Q119

AnalystPrep Q.6188

Prime Bank has a portfolio with two credit assets, Asset X and Asset Y. The unexpected loss (UL) of Asset X is $556,417 and Asset Y is $387,732, with a portfolio correlation of 0.4. Given these values, calculate the risk contribution (RC) of Asset X to the portfolio.

A. \$497,784
B. \$556,417
C. \$795,317
D. \$297,533

Answer: A
Step by step
1. $UL_P = \sqrt{556{,}417^2 + 387{,}732^2 + 2(0.4)(556{,}417)(387{,}732)} = \$795{,}317$
2. $RC_X = UL_X \times \dfrac{UL_X + \rho\, UL_Y}{UL_P}$
3. Numerator: $556{,}417 + 0.4 \times 387{,}732 = 711{,}510$
4. $RC_X = 556{,}417 \times 711{,}510/795{,}317 = \$497{,}784$
5. Check: $RC_Y = \$297{,}533$ and $RC_X + RC_Y = 795{,}317 = UL_P$.

Why A is correct: Risk contribution is the partial derivative of UL_P with respect to UL_X times UL_X, i.e. $UL_X(UL_X + \rho UL_Y)/UL_P$; it is below the stand-alone UL because of diversification.
Why the others are wrong
- **B:** This is Asset X's stand-alone UL; it ignores the diversification benefit (it would hold only for $\rho = 1$).
- **C:** This is the portfolio UL, not the contribution of one asset.
- **D:** This is the risk contribution of Asset Y (the other asset), obtained by swapping the roles of X and Y.

Q120 BT notes P2.T6.506.3

A bank has extended two loans to customers in the same industry. Both loans have an exposure amount (EA) of \$50.0 million, default probability (PD) of 2.0%, loss rate (LR) of 50.0%, and standard deviation of loss rate of 60.0%, such that each loan has an expected loss of \$500,000 and an unexpected loss of \$5.5 million. The credit portfolio consists of these two loans, and the default correlation between them is 28.0%. Which is nearest to the risk contribution of each asset to the portfolio's unexpected loss?

A. \$3.33 million
B. \$4.40 million
C. \$5.37 million
D. \$5.50 million

Answer: B
Step by step
1. $UL_P = \sqrt{5.5^2 + 5.5^2 + 2(0.28)(5.5)(5.5)} = \sqrt{30.25 \times 2.56} = \$8.80$ million
2. $RC_1 = UL_1(UL_1 + \rho UL_2)/UL_P = 5.5 \times (5.5 + 0.28 \times 5.5)/8.8 = 5.5 \times 7.04/8.8 = \$4.40$ million
3. By symmetry $RC_2 = \$4.40$ million; $RC_1 + RC_2 = 8.80 = UL_P$.

Why B is correct: Because the loans are identical, each contributes half of $UL_P = 8.8$, i.e. \$4.40 million, which is below the stand-alone \$5.5 million.
Why the others are wrong
- **A:** Not produced by the risk-contribution formula; it understates the contribution, and two such RCs (\$6.66 million) would not sum to the portfolio UL of \$8.8 million.
- **C:** Not produced by the formula; two such contributions would total \$10.74 million, which exceeds the correct portfolio UL of \$8.8 million.
- **D:** This is the stand-alone UL; it ignores diversification (correct only if $\rho = 1$), and the two RCs would sum to \$11 million rather than \$8.8 million.

Q121 BT notes P2.T6.920.2

Consider a credit portfolio that includes many loans. To derive economic capital (EC) for credit risk, we need to quantify four measures: expected losses (EL), unexpected losses (UL), unexpected loss contribution (ULC), and economic capital (EC). In regard to these four measures, each of the following definitions or descriptions is true EXCEPT which one, which is inaccurate?

A. Expected losses (EL) can be viewed as payments to an insurance pool, do not by themselves constitute risk, and are reimbursed through adequate loan pricing
B. Unexpected losses (UL) is the standard deviation of credit losses around the expected loss average
C. Unexpected loss contribution (ULC) is the first partial derivative of the portfolio's unexpected loss (portfolio

UL) with respect to the position's weight, w(i)

D. Economic capital (EC) for credit risk is the difference between the expected outcome and the unexpected, negative outcome at a certain confidence level

Answer: C

Step by step

1. Marginal contribution: $ULMC_i = \partial UL_P / \partial UL_i$, the derivative with respect to the loan's UL, not its weight.
2. Total contribution: $ULC_i = ULMC_i \times UL_i$; the ULCs sum to UL_P.
3. Check the others: EL is an expected cost reimbursed through pricing (A); UL is the standard deviation of losses around EL (B); EC is the distance between the expected and the unexpected outcome at a confidence level (D). All are true.
4. So C is the inaccurate statement.

Why C is correct: ULC is defined as $(\partial UL_P / \partial UL_i) \times UL_i$, so describing it as the derivative with respect to the weight is wrong.

Why the others are wrong

- **A:** True: EL is predictable, behaves like an insurance premium, and is covered through loan pricing and reserves.
- **B:** True: UL is the volatility (standard deviation) of credit losses around EL.
- **D:** True: EC is the gap between the expected loss and the loss at the chosen confidence level.

Q122

BT notes P2.T6.921.3

REBUILT — Math in the stem re-typeset in LaTeX after OCR damage; content and numbers unchanged.

Consider a credit portfolio that contains the following two positions:
- $300.0 million exposure with default probability (PD) of 2.0%, loss given default (LGD) of 20.0% and σ(LGD) of 10.0%
- $100.0 million exposure with default probability (PD) of 6.0%, loss given default (LGD) of 60.0% and σ(LGD) of 30.0%

The default correlation between the positions is 0.180. The individual exposures' unexpected losses are $16.0 million and $9.40 million, so the portfolio's unexpected loss is $\sqrt{16.0^2 + 9.40^2 + 2(0.180)(16.0)(9.40)} \approx$ $20.0 million. Which are NEAREST to the respective risk contributions (RC) of the exposures, also known as each exposure's unexpected loss contribution (ULC)?

A. $14.2 and $5.8 million

B. $11.3 and $8.7 million

C. $9.5 and $10.5 million

D. $8.1 and $11.9 million

Answer: A

Step by step

1. $UL_P \approx 20.0$ (given): $16.0^2 + 9.4^2 + 2(0.18)(16.0)(9.4) = 398.5$, square root = 19.96.
2. $RC_1 = 16.0 \times (16.0 + 0.18 \times 9.4)/20.0 = 16.0 \times 17.692/20.0 \approx$ $14.2 million
3. $RC_2 = 9.4 \times (9.4 + 0.18 \times 16.0)/20.0 = 9.4 \times 12.28/20.0 \approx$ $5.8 million
4. Check: $14.2 + 5.8 = 20.0 = UL_P$.

Why A is correct: Each RC is $UL_i(UL_i + \rho UL_j)/UL_P$; the larger-UL position takes the larger share and the RCs sum to the portfolio UL.

Why the others are wrong

- **B:** Not generated by the formula; it spreads risk too evenly between the positions (the RCs must reflect the 16.0 vs 9.4 stand-alone ULs).
- **C:** A near-equal split, which ignores the difference in stand-alone ULs ($16.0m vs $9.4m).
- **D:** Reverses the allocation, giving the smaller-UL exposure the larger contribution.

Q123

BT notes P2.T6.922.3

REBUILT — Garbled general-form equation restored in LaTeX; content unchanged.

Consider a large $20.0 million portfolio of 100 loans. In its general form, the portfolio's unexpected loss is $UL_P = \sqrt{\sum_{i=1}^{n} \sum_{j=1}^{n} UL_i \, UL_j \, \rho_{ij}}$. However, each loan in this portfolio has approximately the same characteristics and size (about $200,000). For modeling purposes, the pairwise correlation is set constant, $\rho_{ij} = 0.160$ for all $i \neq j$. These assumptions greatly simplify the calculation of the portfolio's unexpected loss and each loan's contribution

to portfolio risk. Which of the following statements is TRUE?

A. A practical problem with using the general form (i.e., specifying the correlation matrix) is that default correlations are very difficult to observe

B. Under the simplifying assumptions, each loan's risk contribution (unexpected loss contribution, ULC) is conveniently 16.0% of its individual unexpected loss, UL

C. If we attempted to estimate the portfolio's unexpected loss by specifying the pairwise correlation matrix of each ρ_{ij}, we would require 100! or 9.3E+157 correlation pairs

D. When estimating the portfolio's unexpected loss and its component contributions, banks prefer these analytical approaches over numerical procedures because the latter are cumbersome and prone to estimation errors

Answer: A

Step by step

1. Number of distinct pairs: $n(n-1)/2 = 100 \times 99/2 = 4{,}950$, not 100!, so C is false.
2. For a homogeneous portfolio $ULC_i \approx UL_i\sqrt{\rho} = \sqrt{0.16}\,UL_i = 40\%$ of UL (exactly $\sqrt{0.16 + 0.84/100} = 41.0\%$ for n = 100), not 16%, so B is false.
3. The analytical approach is cumbersome and error-prone, so banks use numerical procedures; D reverses this and is false.
4. Default correlations are very difficult, if not impossible, to observe, so A is true.

Why A is correct: The full correlation-matrix approach requires default correlations that are very hard to observe, which is why simplifying assumptions are used.

Why the others are wrong

- **B:** Confuses ρ with $\sqrt{\rho}$: ULC is about $\sqrt{0.16} = 40\%$ of UL, not 16%.
- **C:** Uses n! instead of n(n-1)/2 = 4,950 distinct pairs.
- **D:** Reversed: banks use numerical procedures precisely because the analytical approach is cumbersome and prone to estimation error.

Q124

GARP PE1 Q25 · Mock 3 Q55

REBUILT — Source gave sigma(PD)=19.6% (rounded), so results differed slightly by rounding; stem now states the variance of PD (0.0384) so the figures are exact.

An analyst in the credit risk department of a bank is estimating the unexpected loss contributions (ULC) of individual loans to overall credit portfolio risk. The portfolio consists of many loans that have approximately the same characteristics and size. Additional information about each loan is provided below:

Exposure amount: CNY 3,000,000

Constant pairwise default correlation: 0.25

Annual probability of default (PD): 4% (variance of PD = PD(1 - PD) = 0.0384)

Annual loss rate (LR): 15%

Standard deviation of LR: 9%

What is the correct estimate of the ULC of a single loan to the overall credit portfolio risk?

A. CNY 18,000

B. CNY 25,851

C. CNY 51,701

D. CNY 103,402

Answer: C

Step by step

1. $UL = EA\sqrt{PD\,\sigma_{LR}^2 + LR^2\,\sigma_{PD}^2} = 3{,}000{,}000\sqrt{0.04(0.0081) + 0.0225(0.0384)}$
2. $= 3{,}000{,}000\sqrt{0.000324 + 0.000864} = 3{,}000{,}000 \times 0.034467 =$ CNY 103,402
3. Large homogeneous portfolio: $ULC = UL\sqrt{\rho} = 103{,}402 \times \sqrt{0.25} = 103{,}402 \times 0.5$
4. ULC = CNY 51,701.

Why C is correct: In a large portfolio of similar loans, each loan contributes only $\sqrt{\rho}$ of its stand-alone UL, i.e. half of CNY 103,402.

Why the others are wrong

- **A:** This is the expected loss, $3{,}000{,}000 \times 0.04 \times 0.15$, not a UL measure.
- **B:** Multiplies UL by $\rho = 0.25$ instead of $\sqrt{\rho} = 0.5$.
- **D:** This is the stand-alone UL of one loan; it ignores the diversification effect in the portfolio.

Q125 GARP PE2 Q38 · Mock 4 Q39

A credit risk manager at a bank is estimating the unexpected loss (UL) of the bank's portfolio of loans and the UL contributions of the individual loans to the overall portfolio UL. The portfolio consists of a large number of loans and the manager assumes that the loans have approximately the same characteristics and size, with a constant pairwise default correlation of 0.32. Assuming the UL of each loan is USD 10,500, what is the approximate UL contribution of each loan to the portfolio UL?

A. USD 0
B. USD 1,075
C. USD 3,360
D. USD 5,940

Answer: D

Step by step

1. For a large homogeneous portfolio, $ULC \approx UL\sqrt{\rho}$.
2. $\sqrt{0.32} = 0.5657$
3. $ULC = 10{,}500 \times 0.5657 \approx$ USD 5,940

Why D is correct: With many similar loans, specific risk diversifies away and each loan retains $\sqrt{\rho}$ of its stand-alone UL.

Why the others are wrong

- **A:** Wrongly assumes all risk diversifies away; with a positive correlation of 0.32 a systematic component remains.
- **B:** Uses ρ^2 (10,500 × 0.1024).
- **C:** Uses ρ (10,500 × 0.32) instead of $\sqrt{\rho}$.

Q126 Schweser Module Quiz 22.1 Q5

REBUILT — The source exhibit with loan data was missing, so the question was unanswerable; I reconstructed a self-contained two-asset table and gave numeric options.

Bigger Bank has two loans outstanding with the features below. Assume the correlation between the two assets' losses is 0.2 and that $\text{UL}_i = \text{EA}_i\sqrt{\text{PD}_i\,\sigma^2_{LR,i} + \text{LR}_i^2\,\sigma^2_{PD,i}}$. What is the unexpected loss of the portfolio (ULP)?

Item	Asset 1	Asset 2
Exposure at default (EA)	$3,000,000	$2,000,000
Probability of default (PD)	2%	4%
Loss rate (LR)	50%	40%
Std. dev. of PD	14%	20%
Std. dev. of LR	25%	30%

A. $308,788
B. $435,266
C. $337,892
D. $323,667

Answer: C

Step by step

1. Asset 1: $\text{UL}_1 = 3{,}000{,}000\sqrt{0.02(0.25)^2 + 0.5^2(0.14)^2} = 3{,}000{,}000\sqrt{0.00125 + 0.0049} = 3{,}000{,}000(0.078422) = \$235{,}266$
2. Asset 2: $\text{UL}_2 = 2{,}000{,}000\sqrt{0.04(0.30)^2 + 0.4^2(0.20)^2} = 2{,}000{,}000\sqrt{0.0036 + 0.0064} = 2{,}000{,}000(0.10) = \$200{,}000$
3. Portfolio: $\text{UL}_P = \sqrt{\text{UL}_1^2 + \text{UL}_2^2 + 2\rho\,\text{UL}_1\text{UL}_2}$
4. $= \sqrt{235{,}266^2 + 200{,}000^2 + 2(0.2)(235{,}266)(200{,}000)} = \sqrt{1.1417 \times 10^{11}} \approx \$337{,}892$
5. Check: this is below the undiversified sum of $435,266 and above the zero-correlation value of $308,788, as expected for $0 < \rho < 1$.

Why C is correct: Portfolio UL is the standard deviation of combined losses, so the standalone ULs are combined with the correlation cross-term rather than simply added; with a correlation of 0.2 the result lies between the zero-correlation and perfect-correlation values.

Why the others are wrong

- **A:** Sets the correlation to zero ($\sqrt{\text{UL}_1^2 + \text{UL}_2^2}$), dropping the cross-term $2\rho\,\text{UL}_1\text{UL}_2$.
- **B:** Adds the two standalone ULs, which is correct only when the correlation is 1 and ignores diversification.
- **D:** Omits the factor 2 on the cross-term, using $\text{UL}_1^2 + \text{UL}_2^2 + \rho\,\text{UL}_1\text{UL}_2$, which understates ULP.

Q127 GARP PE1 Q36

A senior risk manager at a financial institution is presenting to a group of junior analysts on the capital requirements for the firm's banking book. The manager demonstrates the calculation of economic capital, at the firm's chosen confidence level of 99.9%, for a portfolio consisting of credit assets that have the same size and similar characteristics. The manager uses the following information:
Number of credit assets in the portfolio: 12
Capital multiplier at the 99.9% confidence level: 7.3
Unexpected loss of each credit asset: JPY 85 million
Default correlation between any pair of credit assets: 0.3
What would the manager be correct to estimate as the economic capital for credit risk for this portfolio?

A. JPY 2.234 billion
B. JPY 4.078 billion
C. JPY 4.457 billion
D. JPY 7.446 billion

Answer: C
Step by step
1. $UL_P = UL\sqrt{n + \rho(n^2 - n)} = 85\sqrt{12 + 0.3 \times 132} = 85\sqrt{51.6} = 85 \times 7.183 =$ JPY 610.6m.
2. EC = UL_P x CM = 610.6 x 7.3 = JPY 4,457m = JPY 4.457bn.

Why C is correct: Portfolio UL for identical assets is UL x sqrt(n + rho(n^2 - n)); EC is that UL times the capital multiplier.
Why the others are wrong
- **A:** Used n x rho x UL x CM = 12 x 0.3 x 85 x 7.3 = 2.234bn, treating correlation as a simple scalar on the sum.
- **B:** Used n x sqrt(rho) x UL x CM = 4.078bn, which drops the diagonal (own-variance) terms incorrectly.
- **D:** Simple sum of ULs (rho = 1): 12 x 85 x 7.3 = 7.446bn, ignoring diversification.

LO 22.g

Describe how economic capital is derived.

Q128 AP Mock 2 Q59

Two regional banks, PQR and LMN, have identical credit exposure, duration, credit ratings and expected loss rates. However, the average pairwise default correlation between credit assets is lower for Bank PQR than for Bank LMN. Both assess risk appetite at the same confidence level. Which of the following statements is correct?

A. If the confidence level for both banks is increased, the economic capital required for Bank PQR and Bank LMN will both increase.
B. If the confidence level for both banks is decreased, the economic capital required will increase for Bank PQR but decrease for Bank LMN.
C. If the confidence level for both banks is increased, the economic capital required will decrease for Bank PQR but increase for Bank LMN.
D. If the confidence level remains unchanged, the economic capital required for Bank PQR and Bank LMN will be the same.

Answer: A
Step by step
1. EL is the same for both banks (identical exposure, ratings, loss rates), and correlation does not change a mean.
2. Economic capital = tail loss at the confidence level minus EL = CM x UL. Higher confidence raises CM, so EC rises for both banks.
3. Lower default correlation lowers portfolio UL, so Bank PQR needs less capital than LMN at any confidence level.

4. Hence only A is correct.

Why A is correct: EC is a multiple (CM) of portfolio UL; a higher confidence level raises CM for every bank, while correlation only changes the size of UL.

Why the others are wrong

- **B:** Lowering the confidence level reduces CM and so reduces EC for both banks; no bank moves in the opposite direction.
- **C:** Raising confidence increases CM for both banks; PQR's EC does not fall (it merely stays below LMN's).
- **D:** Ignores correlation: lower pairwise correlation gives PQR a smaller portfolio UL, hence less EC at the same confidence level.

Q129 AnalystPrep Q.6191

A bank is estimating its economic capital for credit risk. The credit risk models are designed to estimate expected and unexpected losses over a one-year period. The bank wants to ensure that the capital reserves are set based on the credit risk present in its portfolio. How does the bank proceed with deriving economic capital based on the expected and unexpected losses as provided by its credit risk models?

A. By setting aside capital reserves equal to the sum of expected and unexpected losses
B. By setting aside capital reserves equal to the unexpected losses, assuming no diversification benefit
C. By estimating the excess capital reserves needed subject to a predetermined confidence level to cover unexpected losses
D. By allocating resources to manage expected losses and disregarding unexpected losses

Answer: C

Step by step

1. Banks cover expected loss with credit reserves/provisions and pricing.
2. Capital for unexpected loss is the excess over EL needed at a chosen confidence level (EC = loss at that quantile - EL).
3. So the bank estimates excess capital at a predetermined confidence level: C.

Why C is correct: Economic capital is the excess capital, at a chosen confidence level, needed to absorb unexpected loss beyond the expected loss already reserved for.

Why the others are wrong

- **A:** Adds EL to UL; EL is already provisioned, and UL alone is not the capital figure without a confidence level.
- **B:** Sets capital equal to UL with no confidence-level scaling (CM) and wrongly ties it to a no-diversification assumption.
- **D:** Ignores unexpected loss, which is exactly what economic capital is meant to buffer.

Q130 BT notes P2.T6.507.1

Which best describes the relationship between economic capital and unexpected loss?

A. Economic capital is a multiple of unexpected loss
B. Economic capital is unrelated to unexpected loss
C. Economic capital is equal to (synonymous with) unexpected loss
D. Economic capital is unexpected loss plus credit value at risk (CVaR)

Answer: A

Step by step

1. UL is the standard deviation of losses.
2. EC is the difference between the loss at the chosen confidence level and EL, expressed as a multiple of UL: EC = CM x UL.
3. So EC is a multiple of UL (A).

Why A is correct: Capital is set at a high confidence level that lies several standard deviations (UL) beyond EL, so EC = CM x UL.

Why the others are wrong

- **B:** EC is directly derived from UL via the capital multiplier.
- **C:** Treats CM as 1; one standard deviation is far short of a 99.9%+ tail quantile.
- **D:** EC is not UL plus another VaR-type figure; adding CVaR double counts the tail.

Q131 BT notes P2.T6.922.1

Consider a credit portfolio that contains three positions. The exposure (EAD) of each position is $10.0 million. Further, our model assumes the shape of the loss distribution (aka, the credit risk of each exposure) is identical for each exposure, although their means vary as follows:
- Exposure #1 has a default probability of 2.0% and unexpected loss (UL) of $597,000
- Exposure #2 has a default probability of 4.0% and unexpected loss (UL) of $840,000
- Exposure #3 has a default probability of 6.0% and unexpected loss (UL) of $1,023,500

The pairwise default correlation is 0.40 among each exposure pair, such that the portfolio's unexpected loss is $1,920,250. In regard to Exposure #1, its risk contribution is given by $597,000 x [$597,000 + ($840,000 x 0.40) + ($1,023,500 x 0.40)] / $1,920,250 = $417,348. Because the capital multiplier, CM, is set at 5.50 to reflect a specified confidence level, the economic capital for Exposure #1, EC(#1) = $417,348 x 5.50 = $2,295,415, or about $2.30 million. Which of the following is NEAREST, respectively, to the required economic capital for the second and third exposures, EC(#2) and EC(#3)?

A. $434,000 and $568,000
B. $651,000 and $852,000
C. $3.6 and $4.7 million
D. $5.9 and $7.7 million

Answer: C

Step by step

1. $RC_i = UL_i \times \sum_j UL_j \rho_{ij} / UL_P$.
2. RC(#2) = 840,000 x [840,000 + 0.4(597,000) + 0.4(1,023,500)] / 1,920,250 = 651,003.
3. RC(#3) = 1,023,500 x [1,023,500 + 0.4(597,000) + 0.4(840,000)] / 1,920,250 = 851,900.
4. EC = RC x CM: EC(#2) = 651,003 x 5.50 = 3,580,515 (about $3.6m); EC(#3) = 851,900 x 5.50 = 4,685,447 (about $4.7m).

Why C is correct: Economic capital for each exposure is its UL contribution scaled by the capital multiplier, EC_i = RC_i x CM.

Why the others are wrong

- **A:** Figures are below the risk contributions and not the result of any consistent step; capital must exceed RC because CM > 1.
- **B:** These are the risk contributions (~$651k, ~$852k); the answer forgot to multiply by CM = 5.50.
- **D:** Roughly 9 times the risk contributions; no consistent step produces this, since the correct EC is RC x 5.50 = $3.6m and $4.7m.

Q132 Mock 2 Q30

Using a confidence level of 99.9% over a 1-year time horizon, under the IRB framework a bank has an estimated VaR equal to USD 230 million. The expected loss on the portfolio is USD 130 million. A risk analyst is preparing a note for the risk committee. Which of the following is correct?

A. Economic capital is USD 100 million; and if the bank raised its confidence level from 99.9% to 99.97%, the required economic capital would increase.
B. Economic capital is USD 100 million; and if the bank raised its confidence level from 99.9% to 99.97%, the required economic capital would decrease because a higher confidence level implies a more conservative expected loss.
C. Economic capital is USD 230 million; and if the bank raised its confidence level from 99.9% to 99.97%, the required economic capital would increase.
D. Economic capital is USD 360 million; and if the bank raised its confidence level from 99.9% to 99.97%, the required economic capital would remain unchanged.

Answer: A

Step by step

1. Economic capital = loss at confidence level (VaR) - EL = 230 - 130 = USD 100m.
2. Raising confidence to 99.97% pushes the VaR quantile further into the tail while EL is unchanged, so EC increases.

Why A is correct: EC is the unexpected-loss cushion VaR - EL, and VaR grows with the confidence level while EL does not.

Why the others are wrong

- **B:** Direction wrong: EL does not change with confidence level, VaR rises, so EC rises.

- **C:** Fails to subtract EL (EC = VaR, 230m); EL is covered by reserves, not capital.
- **D:** Adds EL to VaR (360m) and wrongly claims EC is insensitive to confidence level.

Q133 Schweser Module Quiz 22.2 Q1

The type of capital used to buffer a bank from unexpected losses is known as:

A. economic capital.
B. regulatory capital.
C. unexpected capital.
D. risk-adjusted capital.

Answer: A

Step by step

1. Banks hold reserves for expected losses.
2. The excess capital held against unexpected loss is called economic capital.

Why A is correct: Economic capital is by definition the excess capital that buffers the bank against unexpected credit losses.

Why the others are wrong

- **B:** Regulatory capital is the minimum set by supervisors, not the internally estimated unexpected-loss buffer.
- **C:** Not a recognised capital term.
- **D:** Not the standard term; risk-adjusted capital is not the name for the UL buffer.

Q134 BT notes P2.T7.1.1

Consider the following statements about economic capital (EC):
I. Economic capital must equal or exceed regulatory capital
II. Economic capital should exceed equity capital
III. Economic capital should equal to the discounted stream of future excess return on capital
IV. Economic capital absorbs expected losses (EL) and should equal the firm's reserves
V. Economic capital absorbs unexpected losses (UL) up to a certain confidence level and over a specified time horizon
Which of the above statements is (are) TRUE?

A. I only
B. V only
C. I, II, and III
D. All are true

Answer: B

Step by step

1. I false: EC is an internal estimate and need not exceed regulatory capital.
2. II false: no requirement that EC exceed equity.
3. III false: describes a valuation/EVA concept, not the definition of EC.
4. IV false: EL is covered by reserves/provisions; EC covers UL.
5. V true: EC absorbs UL up to a confidence level over a time horizon.

Why B is correct: Economic capital covers unexpected loss (beyond EL) to a given confidence level and horizon; EL is handled by reserves.

Why the others are wrong

- **A:** Statement I is not a defining property; internal EC can fall below regulatory capital.
- **C:** Includes I, II and III, none of which is true.
- **D:** Statements I to IV are false (IV confuses EL reserves with EC).

LO 22.h

Explain how the credit loss distribution is modeled.

Q135 AnalystPrep Q.3679

The shape of the beta distribution is determined by its two shape parameters, α and β. If $\alpha = \beta$, the shape of the beta distribution is:

A. Symmetric
B. U-shaped
C. Inverted
D. Skewed to the left

Answer: A

Step by step

1. The beta distribution on [0, 1] is characterized by shape parameters α and β.
2. When $\alpha = \beta$, the density is a mirror image about 0.5, so the distribution is symmetric.
3. When $\alpha \neq \beta$ it is skewed (left skew if $\beta < \alpha$, right skew if $\beta > \alpha$).

Why A is correct: Equal shape parameters give a density that is symmetric around its mean; unequal parameters produce skewness.

Why the others are wrong

- **B:** A U-shape needs both parameters below 1, which is not implied by $\alpha = \beta$.
- **C:** 'Inverted' is not a standard description of a beta shape and is not implied by equal parameters.
- **D:** Skewness requires $\alpha \neq \beta$; equal parameters remove the asymmetry.

Q136 AnalystPrep Q.6192

Zenith Bank is reviewing its economic capital for credit risk and plans to model the credit loss distribution with a beta distribution. What is a key reason for using the beta distribution?

A. It is symmetrical and so aligns with the normal distribution of events
B. It is flexible and can be symmetric or skewed depending on its shape parameter values
C. It focuses only on extreme values and is therefore suited to credit risk analysis
D. It has a fixed mean, which simplifies the calculation of expected loss

Answer: B

Step by step

1. Credit losses are highly skewed (gains capped at promised payments, large losses rare), so a normal distribution is inappropriate.
2. The beta distribution has mass between 0 and 1, matching losses of 0% to 100%.
3. Its shape parameters let it be symmetric or skewed, and it is fully characterized by EL(P) and UL(P).

Why B is correct: The beta distribution is bounded between 0% and 100% and flexible enough to be symmetric or skewed, which suits skewed credit losses.

Why the others are wrong

- **A:** Credit losses are not normally distributed; the beta is only symmetric when the two shape parameters are equal, and symmetry is not the reason for using it.
- **C:** The beta describes the whole bounded distribution; in fact fitting its tail is the difficult part and needs Monte Carlo support.
- **D:** The mean is not fixed; it is set to the portfolio's EL(P) when fitting.

Q137 BT notes P2.T6.507.2

Schroeck notes that the beta distribution, defined between 0 and c (here c = 1 for loss rates of 0% to 100%), is extremely flexible in the shapes it can accommodate. Which of the following is TRUE about the beta distribution in credit risk modeling?

A. It is flexible because it has four parameters, one for each moment, allowing precise calibration of kurtosis
B. The most convenient and sufficiently realistic calibration sets the shape parameters α and β equal to each other
C. It is characterized by the portfolio's expected loss, EL(P), and unexpected loss, UL(P); the challenge is fitting the tail, which depends on the ratio EL(P)/UL(P)
D. Even when accurately fitted it gives no way to determine the capital multiplier, which therefore requires a different distribution

Answer: C

Step by step

1. The beta distribution is fully characterized by two parameters: its mean EL(P) and its standard deviation UL(P).
2. The difficulty is fitting the tail; the tail of the fitted beta depends on the ratio EL(P)/UL(P).
3. Hence C is true; A, B and D each misstate the parameters, the calibration or the capital multiplier.

Why C is correct: Two parameters (EL and UL) pin down the beta, but its tail fit depends on EL/UL, which is why tail fitting is the weak point.

Why the others are wrong

- **A:** The beta has two shape parameters, not four, and they are set by EL and UL rather than by each moment.
- **B:** Setting $\alpha = \beta$ forces symmetry, which contradicts the skewed shape of credit losses; the parameters are fitted to EL(P) and UL(P).
- **D:** Once the tail is fitted, the capital multiplier is read from the chosen quantile of the same distribution; no other distribution is needed.

Q138

BT notes P2.T6.922.1

Schroeck states that estimating economic capital requires choosing a loss distribution because only the tail matters, that credit risks are highly skewed, and that the beta distribution (between 0 and c = 1) is often recommended. Regarding the beta distribution, each of the following is true EXCEPT:

A. It is symmetric for all values of α and β and can be skewed only if a third parameter is added

B. It is a continuous approximation of a binomial distribution (the sum of independent two-point distributions)

C. For high-quality portfolios (EL(P) > UL(P)) it has too fat a tail and is likely to overestimate economic capital

D. For low-quality portfolios (EL(P) < UL(P)) it has too thin a tail and is likely to underestimate economic capital

Answer: A

Key correction: Source gave no answer key; solved independently and checked against the GARP text, including the EL/UL conditions in C and D, which are as stated in the reading.

Step by step

1. A: the beta is symmetric only when $\alpha = \beta$; with $\alpha \neq \beta$ it is skewed using just two parameters. FALSE.
2. B: the beta is a continuous approximation of a binomial distribution. True.
3. C: high quality (EL(P) > UL(P)): tail too fat, economic capital overestimated. True.
4. D: lower quality (EL(P) < UL(P)): tail too thin, economic capital underestimated. True. The EXCEPT answer is A.

Why A is correct: Skewness comes from unequal α and β; no third parameter is required, so A is the false statement.

Why the others are wrong

- **B:** A true statement, so not the exception.
- **C:** A true statement per the reading (EL > UL gives a too-fat tail), so not the exception.
- **D:** A true statement (EL < UL gives a too-thin tail), so not the exception.

Q139

AnalystPrep Q.6194

Global Trust Bank is setting aside economic capital based on its credit risk profile and uses Monte Carlo simulation in its modeling. In which situation are simulation techniques most useful?

A. Estimating expected losses over horizons longer than one year

B. Calculating economic capital under the assumption of perfect correlation among risk factors

C. Dealing with liquid assets where market prices directly reveal risk correlations

D. Modeling extreme credit losses, where a standard distribution such as the beta distribution may not fit the tail adequately

Answer: D

Step by step

1. Economic capital depends on the far tail of the credit loss distribution.
2. The beta distribution is a convenient analytical fit but is hard to fit exactly in the tail.
3. Combining it with Monte Carlo simulation is the practical way to fit the tail, so D.

Why D is correct: Monte Carlo simulation is combined with the beta distribution to fit the extreme tail where the analytical fit is weak.

Why the others are wrong

- **A:** Horizon length is not what motivates simulation here; the issue is tail shape.
- **B:** Perfect correlation is a degenerate assumption that needs no simulation.
- **C:** Credit assets are generally illiquid with no market prices revealing correlations; simulation is not tied to liquid assets.

Q140 AnalystPrep Q.3678 · AnalystPrep Q.6196

REBUILT — Merged two near-identical tail-fitting items (Q.3678 and Q.6196); Q.6196 distractors (extreme value theory) were arguably defensible, so Q.3678 wording and options were kept.

A key concern in using the beta distribution to measure credit risk is the difficulty of fitting it exactly to the tail of the portfolio's loss profile. The tail-fitting exercise is best accomplished by combining the beta distribution with:

A. A normal distribution
B. A uniform distribution
C. A Monte Carlo simulation
D. A VaR calculation

Answer: C

Step by step

1. The beta provides an analytical fit to the body of the loss distribution but fits the tail poorly.
2. The tail is best fitted by pairing the analytical beta solution with a numerical procedure: Monte Carlo simulation.
3. A normal is symmetric and thin-tailed, a uniform is unrealistic, and VaR is a measure, not a fitting tool.

Why C is correct: Monte Carlo simulation generates many loss scenarios and so captures the extreme tail outcomes that the analytical beta fit misses.

Why the others are wrong

- **A:** The normal is symmetric and understates skewed credit tails; it does not improve the tail fit.
- **B:** A uniform distribution assigns equal likelihood to all losses, which is unrealistic for credit losses.
- **D:** VaR is a tail risk measure computed from a distribution; it is not a technique for fitting one.

Q141 AnalystPrep Q.6199

REBUILT — Original asked a vague 'related challenge' with a dubious key about computational justification; rebuilt to test why beta is combined with Monte Carlo for the loss tail (22.h).

In modelling the credit loss distribution for economic capital, a beta distribution is commonly applied but is combined with Monte Carlo simulation. What is the reason for this?

A. The beta distribution cannot be defined on the 0% to 100% loss range.
B. The beta distribution can only be symmetric, so it cannot represent skewed credit losses.
C. Monte Carlo simulation replaces the need to estimate the portfolio's expected loss and unexpected loss.
D. A beta distribution characterised by the portfolio's EL and UL is hard to fit exactly to the tail of the loss distribution, so the tail is fitted by combining it with Monte Carlo simulation.

Answer: D

Step by step

1. Credit losses are highly skewed; the beta distribution (mass between 0 and 1) is flexible and fully characterised by ELp and ULp.
2. Economic capital depends on the far tail (e.g., 99.97% confidence), which is difficult to fit exactly with the beta distribution.
3. So the tail is best fitted by combining the analytical beta solution with Monte Carlo simulation.
4. Option D.

Why D is correct: Capital is driven by the tail, which a beta fitted to ELp and ULp captures imperfectly, so a numerical Monte Carlo procedure supplements it.

Why the others are wrong

- **A:** Wrong: beta is defined on [0,1], which maps naturally to 0-100% losses.
- **B:** Wrong: beta is very flexible; it is symmetric only when its shape parameters are equal and can be skewed otherwise.
- **C:** Wrong: ELp and ULp are still the inputs characterising the beta; simulation helps with the tail only.

LO 22.i

Describe challenges to quantifying credit risk.

Q142

AP Mock 3 Q60

REBUILT — Original option B tied the one-year horizon to illiquidity and distractors were loosely worded; rebuilt to test the one-year-horizon limitation directly with one clearly correct option.

A bank uses a bottom-up framework to quantify credit risk and derive economic capital, and its internal credit model uses a one-year estimation horizon. Which statement best describes the resulting limitation of the framework?

A. A one-year horizon prevents the model from reflecting default correlation among borrowers.

B. A one-year horizon is required because credit loss distributions cannot be defined over any other period.

C. Ideally the model would capture expected and unexpected changes in borrowers' credit quality over several years, but multi-period solutions are very complex, so a one-year horizon is used in practice.

D. A one-year horizon is used because credits are liquid and can be repriced daily, so longer horizons add nothing.

Answer: C

Step by step

1. The bottom-up framework is analytically able to include multi-period credit-quality changes (expected and unexpected) and their correlations.
2. Numerical solutions for the multi-period case become very complex and cumbersome.
3. Therefore almost all internal credit risk models use only a one-year estimation horizon, which is a recognised limitation.
4. Option C states exactly this trade-off.

Why C is correct: Credit exposures are multi-period in nature, but modelling multi-year changes in credit quality is cumbersome, so practitioners settle for a one-year horizon, which is a simplification.

Why the others are wrong

- **A:** Wrong: the approach does consider default correlation within credit risk; the horizon is unrelated to that. Confuses the horizon caveat with the correlation caveat.
- **B:** Wrong: distributions can be defined over other periods; the one-year choice is for practicality, not a mathematical requirement.
- **D:** Wrong: the framework assumes credits are illiquid, not liquid; the one-year horizon is a practical simplification, not a consequence of liquidity.

Q143

AnalystPrep Q.6197

REBUILT — Original keyed an 'overestimation of credit losses' that is not in the reading; rebuilt around the actual caveat (only risk contribution measured, not correlation with risk factors priced in liquid markets).

A bank's bottom-up approach to quantifying credit risk assumes that credits are illiquid assets. Which of the following is the resulting challenge?

A. The approach ignores default correlation among the borrowers in the credit portfolio.

B. The approach measures only each credit's risk contribution to the existing credit portfolio, not its correlation with risk factors as priced in liquid markets.

C. Because illiquid credits can be hedged at market prices, expected losses are eliminated and no economic capital is required.

D. The approach cannot compute expected loss because EL requires a quoted market price for each loan.

Answer: B

Key correction: Source key (B) said credit losses are 'overestimated'; the reading states no direction of bias: the limitation is that only internal risk contribution is captured, not correlation with liquid-market risk factors.

Step by step

1. Assuming credits are illiquid means each credit is valued by its contribution to the credit portfolio's loss (its internal 'beta').
2. It is therefore not linked to risk factors as priced in liquid markets.
3. As bank credit becomes more liquid and traded, a market-value approach (modelling changes in loan portfolio value) would be more suitable.
4. Option B states this.

Why B is correct: Illiquidity assumption means risk is measured only by internal portfolio contribution, omitting the correlation with market-priced risk factors that matters once loans trade.

Why the others are wrong

- **A:** Wrong: the approach does take default correlation within credit risk into account; the omitted correlation is with liquid-market risk factors.
- **C:** Wrong: illiquidity makes hedging harder, and expected loss is covered by reserves while unexpected loss needs economic capital.
- **D:** Wrong: EL = EA x PD x LR uses internal estimates, no market price is needed.

Q144 BT notes P2.T6.507.3

REBUILT — Source answer was recovered from stray text and option B was arguably defensible; rewrote B as a clearly false statement so exactly one option is the exception.

Each of the following is a recognised challenge in quantifying credit risk with the bottom-up approach to economic capital EXCEPT:

A. The approach assumes that credits are illiquid assets.

B. The approach assumes credits are liquid and fully captures their correlation with risk factors as priced in capital markets.

C. Internal credit risk models typically use only a one-year estimation horizon rather than a multi-period horizon.

D. Market and operational risk are separated from credit risk and are measured and managed in different departments of the bank.

Answer: B

Step by step

1. The reading lists three limitations: (1) credits are presumed illiquid, so only internal risk contribution is measured; (2) a one-year horizon is used in practice; (3) market and operational risk are separated from credit risk and managed in different departments.
2. A, C and D restate these.
3. B asserts the opposite of limitation (1), so it is not a challenge but a false statement.

Why B is correct: B contradicts the first caveat: the approach assumes illiquid credits and does not capture correlation with liquid-market risk factors.

Why the others are wrong

- **A:** This is limitation 1, so it is a valid challenge (not the exception).
- **C:** This is limitation 2, so it is a valid challenge (not the exception).
- **D:** This is limitation 3, so it is a valid challenge (not the exception).

Reading 23. Introduction to Credit Risk Modeling and Assessment

Michalis Doumpos, Christos Lemonakis, Dimitrios Niklis, and Constantin Zopounidis, Analytical Techniques in the Assessment of Credit Risk: An Overview of Methodologies and Applications (Springer, 2019). Chapter 1.

LO	Learning objective	Qs
23.a	explain the capital adequacy, asset quality, management, earnings, and liquidity (CAMEL) system used for evaluating the financial condition of a bank.	8
23.b	describe quantitative measurements and factors of credit risk, including probability of default, loss given default, exposure at default, expected loss, and time horizon.	7
23.c	estimate risk-weighted assets and capital adequacy ratio of a financial institution.	12
23.d	describe the judgmental approaches, empirical models, and financial models to predict default.	15
23.e	apply the Merton model to calculate default probability and the distance to default and describe the limitations of using the Merton model.	5
23.f	compare and contrast different approaches to credit risk modeling, such as those related to the Merton model, Credit Risk Plus (CreditRisk+), CreditMetrics, and the Moody's KMV model.	12
23.g	apply risk-adjusted return on capital (RAROC) to measure the performance of a loan.	22

LO 23.a

Explain the capital adequacy, asset quality, management, earnings, and liquidity (CAMEL) system used for evaluating the financial condition of a bank.

Q145

AnalystPrep Q.5932

REBUILT — Capital adequacy was not mentioned, so it was also a candidate; added a capital statement so only earnings remains weak.

A bank is preparing for a regulatory CAMEL assessment. It has diversified its loan portfolio, implemented robust risk management protocols, improved liquidity by holding more high-quality liquid assets, and reports a capital ratio comfortably above the regulatory minimum. However, it is still recovering from a series of underperforming investments that have hurt its profitability. Which CAMEL component should the risk management team focus on improving next?

A. Capital adequacy
B. Asset quality
C. Management
D. Earnings

Answer: D

Step by step

1. Map each fact to a CAMEL component: capital ratio above the minimum = capital adequacy (satisfactory).
2. Diversified loan portfolio = asset quality (addressed).
3. Robust risk management protocols = management (addressed).
4. More high-quality liquid assets = liquidity (addressed).
5. Underperforming investments hurting profitability = earnings (stability, net interest margin, ROA, future earnings potential) remains the weak area.

Why D is correct: Every component except earnings is described as satisfactory; poor investment results damage profitability, which is what the earnings component evaluates.

Why the others are wrong

- **A:** The capital ratio is stated to be comfortably above the minimum, so capital adequacy is not the weakness.
- **B:** Diversifying the loan portfolio addresses asset quality; the problem is investment performance, not delinquent loans.
- **C:** Robust risk management protocols indicate management is being strengthened; confusing weak investment results with management quality.

Q146

AnalystPrep Q.5933

Following an economic downturn, a regional bank is reviewed for stability and regulatory compliance. The review shows a robust net interest margin and fee income, indicating healthy profitability, but also a significant increase in loan defaults, especially in the commercial real estate portfolio. Which CAMEL component should be the primary focus of the bank's efforts?

A. Capital adequacy
B. Asset quality
C. Management
D. Liquidity

Answer: B
Step by step
1. Healthy net interest margin and fee income show earnings are fine.
2. Rising loan defaults in commercial real estate mean loans are showing delinquency/non-performance.
3. Whether loans are performing or delinquent is assessed under asset quality, so that is the primary focus.

Why B is correct: Asset quality examines whether loans are performing or showing signs of delinquency; the rise in defaults is a direct asset quality problem.
Why the others are wrong
- **A:** Capital adequacy is about capital relative to the regulatory minimum; defaults may eventually erode capital, but the stated problem is the loans themselves.
- **C:** No management or control deficiency is described.
- **D:** No funding or short-term liquidity problem is described; defaults are a loan-performance issue.

Q147
Schweser Module Quiz 23.1 Q1

Which subcomponent of the CAMEL analysis tool is focused on delinquent loans?
A. Liquidity
B. Earnings
C. Asset quality
D. Capital adequacy

Answer: C
Step by step
1. CAMEL's asset quality review examines whether the bank's loans are performing or showing signs of delinquency.
2. So delinquent loans fall under asset quality.

Why C is correct: Asset quality is the component that evaluates loan performance and delinquency.
Why the others are wrong
- **A:** Liquidity concerns short-term liquidity events and interest rate/liquidity risk on investments, not loan delinquency.
- **B:** Earnings looks at core earnings stability, net interest margin and ROA.
- **D:** Capital adequacy compares capital against the regulatory minimum.

Q148
AnalystPrep Q.5936

REBUILT — Original used unexplained numeric CAMEL ratings (scale not defined in the reading); replaced with explicit capital ratios and earnings data.

Bank Alpha and Bank Beta operate in the same regional market, and the regulatory minimum capital-to-risk-weighted-assets ratio is 10%. Alpha has a capital ratio of 14% and modest but positive earnings (ROA 0.4%). Beta has a capital ratio of 9.5% and strong earnings (ROA 1.6%). Which bank is better positioned to undertake significant lending expansion (which raises risk-weighted assets) while remaining in regulatory compliance?
A. Bank Alpha
B. Bank Beta
C. Both banks are equally positioned
D. Neither bank is positioned for lending expansion

Answer: A
Step by step
1. Lending expansion raises risk-weighted assets, so capital adequacy is the binding constraint.
2. Alpha: 14% vs 10% minimum = 4 percentage points of headroom.
3. Beta: 9.5% vs 10% minimum = already 0.5 points below the minimum.
4. Strong earnings may rebuild capital over time, but cannot cure an existing shortfall now. Alpha is better positioned.

Why A is correct: Alpha has a capital cushion above the regulatory minimum to absorb additional risk-weighted assets, while Beta is already non-compliant.

Why the others are wrong

- **B:** Overweights earnings; Beta's strong ROA does not offset a capital ratio already below the minimum.
- **C:** Ignores the 4.5-point difference in capital headroom.
- **D:** Alpha has headroom above the minimum, so it can expand.

Q149

AnalystPrep Q.5937

REBUILT — Original relied on undefined numeric ratings; replaced with explicit capital and liquidity data, and made the loss-absorption mechanism explicit.

Bank X and Bank Y are assessed under CAMEL. The regulatory minimum capital ratio is 10% and the minimum liquidity coverage ratio (LCR) is 100%. Bank X: capital ratio 16%, LCR 105%. Bank Y: capital ratio 10.5%, LCR 180%. A sudden downturn will cause substantial credit losses on loans. Which bank is better positioned to absorb the resulting losses without seeking external capital?

A. Bank X, because of its stronger capital adequacy
B. Bank Y, because of its stronger liquidity
C. Both banks are equally positioned
D. Neither bank can absorb the downturn losses

Answer: A

Step by step

1. Credit losses reduce capital, so capital adequacy determines loss-absorbing capacity.
2. Capital headroom: X = 16% - 10% = 6 points; Y = 10.5% - 10% = 0.5 points.
3. Y's high LCR protects against short-term funding stress but does not absorb credit losses.
4. X is therefore better positioned; X still meets its LCR minimum (105% >= 100%).

Why A is correct: Loan losses are absorbed by capital; X has far more headroom above the regulatory minimum while still meeting its liquidity requirement.

Why the others are wrong

- **B:** Liquid assets help meet short-term obligations but do not absorb credit losses; Y's capital would breach the minimum quickly.
- **C:** Ignores the large difference in capital headroom (6 vs 0.5 points).
- **D:** X has a substantial capital buffer and meets its LCR requirement.

Q150

AnalystPrep Q.5935

A community bank has steady growth and profitability. However, its latest audit shows total capital relative to risk-weighted assets has fallen below the regulatory minimum, raising concern about its ability to absorb losses and support growth, including several planned new branches. Which CAMEL component should the board focus on?

A. Capital adequacy
B. Asset quality
C. Management
D. Earnings

Answer: A

Step by step

1. The problem is capital relative to risk-weighted assets versus the regulatory minimum.
2. Comparing capital to the regulatory minimum is the capital adequacy component.
3. Remedies: raise capital or reduce risk-weighted assets.

Why A is correct: Capital adequacy tests whether the bank holds at least the regulatory minimum capital against its risk.

Why the others are wrong

- **B:** Asset quality concerns loan performance; no delinquency problem is described.
- **C:** Management concerns risk structure, strategy and controls; not the issue stated.
- **D:** Earnings are described as healthy; retained earnings could help, but the deficiency itself is capital.

Q151 BT answers P2.T6.24.6.1 · BT notes P2.T6.24.6.1

Maple Consultants made the following observations on Marble Bank using the CAMEL system:
- Marble Bank's capital-to-risk-weighted-assets ratio (CRAR) is 15%, well above the regulatory minimum of 10%.
- A significant portion of its loan portfolio is long-term, fixed-rate mortgages, which have shown an increase in delinquencies over the past year.
- The management team was recently restructured with several new appointments from within the banking sector, though their effectiveness in implementing strategy remains to be seen.
- Despite a challenging economy, net income has increased consistently over three years, indicating robust earnings capacity.
- Its liquidity coverage ratio (LCR) is below the required 100%, indicating potential challenges meeting short-term obligations without additional funding.

Each option below combines two CAMEL elements. All of the following accurately reflect Marble Bank's condition, EXCEPT:

A. Capital adequacy is strong; management changes are recent but promising.
B. Asset quality is becoming a concern due to rising delinquencies in mortgages; earnings have been consistently strong.
C. Liquidity is below regulatory requirements, and asset quality shows signs of deterioration.
D. Management has been effective in strategy implementation; liquidity coverage ratios meet the required standards.

Answer: D

Step by step

1. Check each claim against the facts.
2. A: CRAR 15% > 10% minimum (strong); new management is recent, effectiveness unproven - consistent.
3. B: rising mortgage delinquencies = asset quality concern; net income up three years = strong earnings - consistent.
4. C: LCR below 100% = liquidity shortfall; delinquencies = asset quality deterioration - consistent.
5. D: effectiveness of management is yet to be seen (not shown to be effective) and LCR is below 100% - both claims are inaccurate. D is the exception.

Why D is correct: Both halves of D contradict the facts: management effectiveness is unproven and the LCR is below the 100% requirement.

Why the others are wrong

- **A:** Accurate: CRAR of 15% exceeds the 10% minimum and the management change is recent with outcomes pending.
- **B:** Accurate: delinquencies point to asset quality concern and rising net income to strong earnings.
- **C:** Accurate: LCR below 100% is a liquidity weakness and delinquencies indicate deteriorating asset quality.

Q152 AnalystPrep Q.5934

The chief risk officer of a bank reviews its latest financial reports. They show an increased loan-to-deposit ratio, reliance on wholesale funding, and a decrease in holdings of government securities and cash reserves. Which CAMEL component should be prioritized for immediate improvement?

A. Capital adequacy
B. Asset quality
C. Management
D. Liquidity

Answer: D

Step by step

1. High loan-to-deposit ratio and wholesale funding reliance mean funding is less stable.
2. Lower government securities and cash mean fewer liquid assets to meet short-term obligations.
3. These indicate short-term liquidity risk, assessed under the liquidity component.

Why D is correct: The evidence (funding mix and falling liquid assets) bears directly on the bank's ability to meet short-term obligations.

Why the others are wrong

- **A:** No capital shortfall relative to the regulatory minimum is described.
- **B:** No loan delinquency or performance problem is described.
- **C:** Management quality is not the stated issue; the data are balance-sheet liquidity indicators.

LO 23.b

Describe quantitative measurements and factors of credit risk, including probability of default, loss given default, exposure at default, expected loss, and time horizon.

Q153

AnalystPrep Q.2231

Highlands Bank forms part of a Bank Holding Company (BHC). The bank is computing loss estimates on a number of its business lines. What are the components that the bank should take into account when estimating losses?

A. Probability of default (PD), time value of money (TM), and loss given default (LGD).
B. Probability of default (PD), loss given default (LGD), and exposure at default (EAD).
C. Probability of default (PD), time value of money (TM), and exposure at default (EAD).
D. Loss given default (LGD), exposure at default (EAD), and credit rating (CR).

Answer: B

Step by step

1. Expected loss over the horizon (usually one year) is EL = PD × EAD × LGD.
2. So the three required components are PD, EAD and LGD.
3. Only option B lists exactly these three.

Why B is correct: Credit loss estimates are built from how likely default is (PD), how much is exposed when it occurs (EAD) and what fraction is lost (LGD).

Why the others are wrong

- **A:** Replaces EAD with 'time value of money', which is not a component of the expected loss formula; without EAD the dollar loss cannot be computed.
- **C:** Replaces LGD with 'time value of money'; omits the loss severity needed to turn exposure into a loss.
- **D:** Omits PD and adds credit rating; a rating is an input to estimating PD, not a loss component itself.

Q154

AnalystPrep Q.5957

A bank's risk management team is evaluating the Exposure at Default (EAD) for various types of loans to enhance its credit risk models. Based on the nature of these loan products, which one would typically present the greatest challenge in estimating the EAD accurately?

A. A car loan with a fixed repayment schedule and a set interest rate over a five-year term.
B. A home equity line of credit (HELOC) with variable withdrawals and repayments.
C. A student loan with a grace period followed by fixed monthly payments after graduation.
D. A small business loan with a balloon payment at the end of a three-year term.

Answer: B

Step by step

1. EAD is the amount due to be repaid at the time of default; it is easy to specify when payment dates and amounts are fixed.
2. Car loan, student loan and balloon loan all have contractually scheduled balances, so the outstanding amount at any future date is predictable.
3. A HELOC is a revolving facility: the borrower draws and repays at will, so the balance at default is uncertain.
4. The HELOC is therefore the hardest EAD to estimate.

Why B is correct: EAD is hard to estimate when the outstanding balance fluctuates with borrower behavior, as with revolving credit lines.

Why the others are wrong

- **A:** Fixed schedule and rate make the balance at any date known; treating a long maturity as 'uncertainty' confuses term with balance variability.
- **C:** The grace period delays payments but the balance path is still determined by the contract.
- **D:** A balloon payment is large but its timing and size are fixed, so EAD is predictable.

Q155 AnalystPrep Q.5956

As part of its annual credit risk assessment, a bank is evaluating the Exposure at Default (EAD) for different loan products. Which loan type would most likely present a more complex challenge in accurately estimating the EAD?

A. A fixed-term personal loan with a set repayment schedule.
B. A standard home mortgage with a fixed interest rate and monthly payments.
C. A credit card account with a revolving credit facility and variable outstanding balances.
D. A corporate bond issuance with predetermined interest payments and a maturity date.

Answer: C

Step by step

1. EAD is straightforward when payment dates and amounts are fixed (a simple loan or a bond).
2. The personal loan, fixed-rate mortgage and bond all have scheduled payments, so the outstanding amount at any future date is known.
3. A credit card balance varies with spending and repayment, so the amount outstanding at default is uncertain.
4. The credit card is the hardest to estimate.

Why C is correct: Credit cards have outstanding amounts that vary over time, making the exposure at the (unknown) default date difficult to predict.

Why the others are wrong

- **A:** Fixed repayment schedule gives a predictable balance.
- **B:** Fixed-rate amortization gives a known balance path; the long term does not create balance uncertainty.
- **D:** Predetermined coupons and a fixed maturity make the principal outstanding known.

Q156 AnalystPrep Q.6198

REBUILT — Original keyed answer (one-year models give an 'unrealistic' view of downturn resilience) was vague and not stated in the reading; rebuilt to test the stated convention for the credit risk time horizon.

A bank's credit risk model produces an expected loss estimate over a set period. Which of the following statements about the time horizon in credit risk modeling is most accurate?

A. The horizon is usually one year, consistent with the standard basis for financial reporting, although longer horizons can be used, especially for long-dated exposures where uncertainty rises.
B. The horizon must always equal the remaining maturity of each loan, so a one-year horizon is never appropriate.
C. The horizon is fixed at 90 days, the period after which a payment is considered in default.
D. The horizon has no effect on the estimated PD because default probabilities are independent of the period considered.

Answer: A

Step by step

1. Credit risk models estimate expected loss over a specific period, usually set to one year to match financial reporting.
2. Longer horizons can be considered, but uncertainty increases with the horizon.
3. Option A states both points; B, C and D each contradict the convention.

Why A is correct: One year is the standard horizon (aligned with financial reporting), with longer horizons possible where appropriate.

Why the others are wrong

- **B:** Overstates: one year is the standard even for loans with longer maturities; maturity-matched horizons are an option, not a rule.
- **C:** Confuses the 90-days-past-due default definition with the modeling horizon.
- **D:** PD is a probability over a period, so a longer horizon gives a higher cumulative PD.

Q157 BT answers P2.T6.24.6.2 · BT notes P2.T6.24.6.2

REBUILT — Original asked which of five statements is 'incorrect' with answer 'All of them' (awkward and unclear); rebuilt as a single-best-correct-statement item on PD, EAD, LGD, EL and horizon.

A credit risk manager reviews statements from a junior analyst about the core components of credit risk. Which statement is correct?

A. PD is the likelihood that the obligor will not meet scheduled payments over the period, with default commonly defined as a payment delayed 90 days or more.

B. LGD is a dollar amount equal to the minimum guaranteed recovery on the loan, fixed at origination.
C. EAD depends mainly on the borrower's credit rating and is the same for a term loan and a revolving credit card.
D. Expected loss is estimated solely from the historical default rate and does not depend on exposure or recovery.

Answer: A
Step by step
1. Expected loss = PD × EAD × LGD, usually over one year.
2. PD is a probability that the obligor fails to pay as scheduled (default: payment delayed 90 days or more); A matches.
3. LGD is a percentage of EAD (1 - recovery rate), not a guaranteed dollar recovery; B is wrong.
4. EAD is mostly driven by loan characteristics, not the obligor, and varies for revolving facilities; C is wrong.
5. EL needs PD, EAD and LGD, not only default history; D is wrong.

Why A is correct: PD is defined as the probability of failing to meet scheduled payments, with default typically meaning 90 or more days delay.
Why the others are wrong
- **B:** Confuses LGD (loss percentage of EAD, the complement of recovery rate) with a fixed dollar recovery.
- **C:** EAD depends on loan characteristics rather than the obligor, and revolving exposures like credit cards fluctuate.
- **D:** Ignores EAD and LGD; EL is the product of all three components.

Q158
Schweser Module Quiz 23.1 Q2

A credit risk analyst is trying to determine the percentage of a loan that is expected to be lost if a specific borrower were to default. Which of the following metrics should he apply?
A. Loss given default.
B. Exposure at default.
C. Probability of default.
D. Recovery default rate.

Answer: A
Step by step
1. The question asks for a percentage loss conditional on default.
2. LGD is the percentage of the exposure expected to be lost if default occurs (LGD = 100% - recovery rate).
3. EAD is a dollar amount and PD is a likelihood, so neither fits.

Why A is correct: LGD is by definition the fraction of exposure lost given that default occurs.
Why the others are wrong
- **B:** EAD is the dollar exposure at the time of default, not a percentage loss.
- **C:** PD is the probability of default, not the loss severity.
- **D:** A recovery rate is the complement (100% - LGD); it measures what is recovered, not what is lost.

Q159
AnalystPrep Q.6200

REBUILT — Original ('most crucial combination of factors') was vague; rebuilt as a numeric expected-loss calculation using PD, EAD and LGD.
A bank has a $5,000,000 loan outstanding to a borrower with a one-year probability of default of 2% and a loss given default of 45%. Assuming the exposure at default equals the amount outstanding, what is the one-year expected loss?
A. $45,000
B. $55,000
C. $100,000
D. $2,250,000

Answer: A
Step by step
1. EL = PD × EAD × LGD
2. = 0.02 × 5,000,000 × 0.45
3. = 100,000 × 0.45 = 45,000

Why A is correct: Expected loss is the product of default probability, exposure and loss severity.

Why the others are wrong

- **B:** Used the recovery rate (1 - 45% = 55%) instead of LGD: 100,000 × 0.55.
- **C:** Forgot to multiply by LGD (PD x EAD only).
- **D:** Ignored PD: EAD x LGD = 5,000,000 x 0.45.

LO 23.c

Estimate risk-weighted assets and capital adequacy ratio of a financial institution.

Q160

BT notes P2.T6.24.6.3

Assess Company XYZ's capital adequacy over the last three years, as measured by the three key capital ratios.

Company XYZ - Excerpt from Annual Report Disclosure

($m)	2017	2016	2015
Common Equity Tier 1 Capital	146,424	142,367	137,100
Additional Tier 1 Capital	22,639	20,443	17,600
Tier 2 Capital	22,456	27,564	38,200
Total Regulatory Capital	191,519	190,374	192,900
Credit Risk RWAs	960,763	989,639	968,600
Market Risk RWAs	44,100	36,910	49,600
Operational Risk RWAs	293,825	256,300	224,300
Total RWAs	1,298,688	1,282,849	1,242,500

Based on the above, Company XYZ's capital adequacy over the last three years, as measured by the three key capital ratios, signals which of the following conditions?

A. Mixed
B. Declining
C. Improving
D. Not enough data to determine

Answer: A

Step by step

1. Each ratio = capital measure divided by Total RWAs.
2. CET1 ratio: 146,424/1,298,688 = 11.27% (2017); 142,367/1,282,849 = 11.10% (2016); 137,100/1,242,500 = 11.03% (2015) - rising.
3. Tier 1 ratio (CET1 + AT1): 13.02%; 12.69%; 12.45% - rising.
4. Total capital ratio (Tier 1 + Tier 2): 191,519/1,298,688 = 14.75%; 190,374/1,282,849 = 14.84%; 192,900/1,242,500 = 15.53% - falling.
5. Two ratios improve while the total capital ratio declines, so the picture is mixed.

Why A is correct: Higher-quality capital (CET1 and Tier 1) has grown faster than RWAs, but Tier 2 capital shrank so much that total capital relative to RWAs fell.

Why the others are wrong

- **B:** Looks only at the total capital ratio (15.53% to 14.75%) and ignores that the CET1 and Tier 1 ratios rose.
- **C:** Looks only at CET1 and Tier 1 ratios and ignores the fall in the total capital ratio caused by the decline in Tier 2 capital.
- **D:** All required data (capital by tier and total RWAs) is given; each ratio can be computed.

Q161

Schweser Module Quiz 23.1 Q3

The internal ratings-based (IRB) approach for calculating a bank's capital adequacy ratio (CAR) has several underlying elements. Which of the following statements relates to the IRB approach?

A. Idiosyncratic risk is explicitly factored into the model.
B. Estimates of probability of default are made by regulators.
C. Maturity adjustments should be higher for borrowers with high probabilities of default.
D. The maturity parameter reflects that capital requirements should increase with maturity.

Answer: D

Step by step

1. The IRB approach uses the asymptotic risk factor (ASFR) model, which assumes a fully diversified portfolio so only systematic risk remains - eliminates A.
2. Under IRB, PD (and, under advanced IRB, LGD) come from the bank's own historical data and internal ratings, not from the regulator - eliminates B.
3. The maturity adjustment is higher for low-PD borrowers (more downgrade potential), not high-PD ones - eliminates C.
4. Longer-maturity loans are riskier (downgrade more likely), so required capital rises with maturity - D is correct.

Why D is correct: The maturity parameter is built so that capital requirements increase with loan maturity, because longer-dated borrowers are more likely to be downgraded.

Why the others are wrong

- **A:** The ASFR model assumes idiosyncratic risk is diversified away; only systematic risk is modeled.
- **B:** Regulators prescribe risk weights in the standardized approach; under IRB the bank estimates PD from internal data.
- **C:** Reversed: low-PD borrowers have higher downgrade potential, so their maturity adjustment is larger; a high-PD borrower already has default risk reflected in its estimate.

Q162

AnalystPrep Q.3102

REBUILT — Original gave an unusual 'riskiness' framing and a CAR expressed as a multiple (4.61); rebuilt with explicit risk weights and a percentage CAR across three exposure classes.

Bank M has $60 million of Tier 1 capital and $20 million of Tier 2 capital. Its assets are $400 million of corporate loans (risk weight 100%), $300 million of residential mortgages (risk weight 50%) and $200 million of loans to highly rated banks (risk weight 20%). Using the standardized approach, what is Bank M's capital adequacy ratio (CAR)?

A. 15.69%
B. 13.56%
C. 10.17%
D. 8.89%

Answer: B

Step by step

1. RWA = 400 x 100% + 300 x 50% + 200 x 20% = 400 + 150 + 40 = $590 million.
2. Capital = Tier 1 + Tier 2 = 60 + 20 = $80 million.
3. CAR = 80 / 590 = 13.56%.

Why B is correct: CAR divides total regulatory capital (Tier 1 + Tier 2) by risk-weighted, not book, assets.

Why the others are wrong

- **A:** Applied the simple average of the risk weights (56.67%) to total assets: RWA = 510, 80/510 = 15.69%; ignores exposure sizes.
- **C:** Used Tier 1 capital only: 60/590 = 10.17%; CAR uses Tier 1 plus Tier 2.
- **D:** Divided by unweighted total assets: 80/900 = 8.89%; forgot to apply risk weights.

Q163

AnalystPrep Q.5958

Given the importance of maintaining financial stability, a regional bank is assessing its Capital Adequacy Ratio (CAR) to comply with Basel II requirements. The bank has calculated its Tier 1 and Tier 2 capital and assessed the risk weights of its assets. Which of the following correctly describes how the bank should calculate its CAR?

A. By dividing the sum of Tier 1 and Tier 2 capital by its total assets.
B. By dividing the sum of Tier 1 and Tier 2 capital by its Risk-Weighted Assets (RWA).
C. By adding the risk weights of all assets and dividing by the sum of Tier 1 and Tier 2 capital
D. By subtracting Tier 2 capital from Tier 1 capital and dividing by its total liabilities.

Answer: B

Step by step

1. CAR = (Tier 1 + Tier 2 capital) / RWA, compared with the regulatory minimum (8% under Basel II).
2. RWA is the average of assets weighted by risk, so assets are multiplied by their risk weights and summed.

Why B is correct: Capital is measured by Tier 1 plus Tier 2 and is scaled against risk-weighted assets so it matches

the risk of the asset base.

Why the others are wrong

- **A:** Uses unweighted total assets, ignoring asset risk.
- **C:** Inverts the ratio and sums risk weights (percentages) rather than weighted asset values.
- **D:** Tier 2 is added to Tier 1, not subtracted, and the denominator is RWA, not liabilities.

Q164

AnalystPrep Q.5961

REBUILT — Original asked a vague 'most likely to impact CAR' question where dividend policy is also defensible; rebuilt as a calculation of how higher risk weights change CAR.

A bank has total capital (Tier 1 + Tier 2) of $120 million and risk-weighted assets of $1,000 million, of which $200 million of exposures carry a 100% risk weight. The regulator raises the risk weight on that asset class to 150%, with no change in capital or exposures. What is the bank's CAR after the change?

A. 10.91%
B. 12.00%
C. 13.33%
D. 8.00%

Answer: A

Step by step

1. Initial CAR = 120/1,000 = 12.00%.
2. The $200m exposure moves from 200 x 100% = 200 to 200 x 150% = 300 of RWA, an increase of $100m.
3. New RWA = 1,000 + 100 = $1,100 million.
4. New CAR = 120 / 1,100 = 10.91%.

Why A is correct: CAR = capital / RWA; with capital fixed, a higher risk weight raises RWA and lowers the CAR.

Why the others are wrong

- **B:** Assumes a risk weight change has no effect; it raises RWA, so CAR must fall.
- **C:** Applied the change in the wrong direction (RWA of 900): 120/900 = 13.33%.
- **D:** Scaled all RWA by 1.5: 120/1,500 = 8.00%; only the $200m class is re-weighted.

Q165

BT notes P2.T7.24.10.1

REBUILT — Original relied on Basel II risk weights (50%, 0%, 35%) that were missing from the stem; rebuilt with these weights supplied and both RWA figures asked for.

An analyst at Bank DeNiro is computing risk-weighted assets under Basel I and under a Basel II standardized approach. The balance sheet shows the following.

Asset	Book value	Basel I weight	Basel II weight
Corporate loans (A rated)	$100 million	100%	50%
OECD government loans (AAA)	$10 million	0%	0%
Residential mortgages (uninsured)	$50 million	50%	35%

What are the bank's RWA under Basel I and under Basel II, respectively?

A. $125.0 million and $67.5 million
B. $67.5 million and $125.0 million
C. $160.0 million and $67.5 million
D. $135.0 million and $67.5 million

Answer: A

Step by step

1. Basel I RWA = 100 x 100% + 10 x 0% + 50 x 50% = 100 + 0 + 25 = $125.0 million.
2. Basel II RWA = 100 x 50% + 10 x 0% + 50 x 35% = 50 + 0 + 17.5 = $67.5 million.

Why A is correct: RWA is the sum of each exposure times its risk weight; the lower Basel II weights on A-rated corporates and mortgages reduce RWA.

Why the others are wrong

- **B:** Swapped the two regimes' results.
- **C:** Used unweighted book value ($160m) for Basel I.
- **D:** Applied a 100% weight to the OECD government loan under Basel I (100+10+25 = 135).

Q166

AnalystPrep Q.5938

REBUILT — Original compared two Tier 1 ratios with no threshold, so 'neither' could be defended; rebuilt as a CAR comparison against the Basel III 10.5% requirement.

Under Basel III the minimum CAR is 10.5%. Bank A has Tier 1 capital of $80 million, Tier 2 capital of $20 million and RWA of $1,000 million. Bank B has Tier 1 capital of $100 million, Tier 2 capital of $30 million and RWA of $1,100 million. Which statement is correct?

A. Only Bank A meets the minimum.
B. Only Bank B meets the minimum.
C. Both banks meet the minimum.
D. Neither bank meets the minimum.

Answer: B

Step by step

1. Bank A CAR = (80 + 20)/1,000 = 10.00% < 10.5%.
2. Bank B CAR = (100 + 30)/1,100 = 11.82% > 10.5%.
3. Only Bank B meets the requirement.

Why B is correct: The CAR uses Tier 1 plus Tier 2 over RWA, and only Bank B's 11.82% exceeds 10.5%.

Why the others are wrong

- **A:** Reversed the result; Bank A's 10.00% is below 10.5%.
- **C:** Would hold under the 8% Basel II minimum, but the question specifies 10.5%.
- **D:** Would result from using Tier 1 only (8.00% and 9.09%); CAR includes Tier 2.

Q167

AnalystPrep Q.5959

A large commercial bank is calculating its Capital Adequacy Ratio (CAR) in line with Basel II guidelines. The bank's Tier 1 capital stands at $200 million, Tier 2 capital at $100 million, and its Risk-Weighted Assets (RWA) are valued at $2 billion. What is the bank's CAR?

A. 10%
B. 15%
C. 5%
D. 20%

Answer: B

Step by step

1. Capital = 200 + 100 = $300 million.
2. RWA = $2,000 million.
3. CAR = 300 / 2,000 = 15%.

Why B is correct: CAR = (Tier 1 + Tier 2) / RWA.

Why the others are wrong

- **A:** Used Tier 1 only: 200/2,000 = 10%.
- **C:** Used Tier 2 only: 100/2,000 = 5%.
- **D:** Double counted Tier 1 (200 + 200 = 400): 400/2,000 = 20%.

Q168

AnalystPrep Q.5960

REBUILT — Original did not state the Basel II minimum in the stem; added the 8% figure so the compliance judgment is self-contained.

In an effort to enhance its risk management practices, a mid-sized investment bank is reviewing its Capital Adequacy Ratio (CAR) in accordance with Basel II guidelines. The bank's Tier 1 capital is $150 million, its Tier 2 capital is $50 million, and the total Risk-Weighted Assets (RWA) are $1.5 billion. What is the bank's CAR, and does it meet the 8% minimum requirement set by Basel II?

A. 10% and it meets the minimum requirement.
B. 13.33% and it does not meet the minimum requirement.
C. 13.33% and it meets the minimum requirement.
D. 10% and it does not meet the minimum requirement.

Answer: C

Step by step

1. Capital = 150 + 50 = $200 million.
2. CAR = 200 / 1,500 = 13.33%.
3. 13.33% > 8%, so the bank meets the Basel II minimum.

Why C is correct: CAR counts Tier 1 plus Tier 2 over RWA and 13.33% exceeds the 8% minimum.

Why the others are wrong

- **A:** 10% is Tier 1 only (150/1,500); the ratio is wrong even though the conclusion happens to be right.
- **B:** Correct ratio but wrong conclusion: 13.33% is above 8%.
- **D:** Uses Tier 1 only (10%) and also wrongly concludes it fails an 8% minimum.

Q169

AnalystPrep Q.5969

A large banking group is implementing the advanced internal ratings-based (A-IRB) approach under Basel II. Its risk team is reviewing the asymptotic single risk factor (ASRF) model that underlies the risk-weighted asset (RWA) calculation. Which statement accurately reflects the principles of the ASRF model?

A. It assumes idiosyncratic risks are predominant over systematic risk in the credit portfolio.

B. It requires banks to hold a completely undiversified credit portfolio for an accurate RWA calculation.

C. It assumes the portfolio is fully diversified, so idiosyncratic risks cancel out and only systematic risk remains.

D. It treats each credit exposure independently, without regard to dependence on the general state of the economy.

Answer: C

Step by step

1. The ASRF model underlies the IRB formula and allows capital to be computed loan by loan and then summed.
2. This additivity works because the portfolio is assumed to be so granular and diversified that borrower-specific (idiosyncratic) risk diversifies away.
3. Only the single systematic factor (state of the economy) drives portfolio losses, so option C is correct.

Why C is correct: Full diversification removes idiosyncratic risk, leaving one systematic factor, which is what makes loan-by-loan capital additive.

Why the others are wrong

- **A:** Reverses the assumption: idiosyncratic risk is assumed to vanish, not dominate.
- **B:** Confuses the model with its opposite; the model assumes a diversified portfolio, it does not require an undiversified one.
- **D:** Ignores the asset correlation parameter, which explicitly links each borrower to the general state of the economy.

Q170

AnalystPrep Q.5970

A bank using the advanced IRB approach under Basel II is reviewing the asset correlation parameter in the RWA formula. Which statement correctly describes how the asset correlation depends on borrower characteristics?

A. Asset correlation decreases with firm size, because larger firms have higher idiosyncratic risk.

B. Asset correlation is constant regardless of PD and firm size.

C. Asset correlation increases with PD, because higher-PD firms are more exposed to the systematic factor.

D. Asset correlation decreases as PD increases and increases with firm size, because larger firms are more closely tied to the general economy.

Answer: D

Step by step

1. Higher PD implies a greater idiosyncratic component of default, so asset correlation falls as PD rises.
2. Larger firms are more influenced by general economic conditions, while smaller firms default more for idiosyncratic reasons, so correlation rises with size.
3. Only D states both relationships correctly.

Why D is correct: Basel's correlation formula embeds two empirical observations: correlation falls with PD and rises with firm size.

Why the others are wrong

- **A:** Reverses the size effect; larger firms are more systematic, not more idiosyncratic.
- **B:** The correlation is a function of PD (and size), not a constant.
- **C:** Reverses the PD effect; higher PD means more idiosyncratic risk and lower correlation.

Q171 BT answers P2.T7.24.10.3 · BT notes P2.T7.24.10.3

REBUILT — Merged duplicates, removed the garbled maturity-adjustment footnote and replaced the weak distractors (book value, expected loss) with results of realistic formula errors; LO changed to 23.c (RWA estimation).

Bank Fortuna is estimating risk-weighted assets (RWA) for a retail loan portfolio under the Basel II IRB approach. Use RWA $= 12.5 \times$ EAD $\times$ LGD $\times$ (WCDR $-$ PD) $\times$ MA.

Parameter	Value
Exposure at default (EAD)	\$400,000,000
Loss given default (LGD)	35%
Worst-case default rate (WCDR)	10%
Probability of default (PD)	2%
Maturity adjustment (MA)	1 (none applies to retail)

What is the RWA?

A. \$140.0 million
B. \$11.2 million
C. \$175.0 million
D. \$35.0 million

Answer: A

Step by step

1. WCDR $-$ PD $= 10\% - 2\% = 8\%$.
2. Capital charge $K = 0.35 \times 0.08 \times 1 = 0.028$.
3. RWA $= 12.5 \times 400 \times 0.028 = \$140$ million.

Why A is correct: IRB capital covers only unexpected loss (WCDR minus PD), and RWA is the capital charge scaled by 12.5 (the inverse of the 8% minimum ratio).

Why the others are wrong

- **B:** Omits the 12.5 multiplier: $400 \times 0.35 \times 0.08 = 11.2$ million is the capital requirement, not RWA.
- **C:** Uses WCDR (10%) instead of WCDR - PD: $12.5 \times 400 \times 0.35 \times 0.10 = 175$ million, double counting expected loss.
- **D:** Uses PD (2%) instead of WCDR - PD: $12.5 \times 400 \times 0.35 \times 0.02 = 35$ million.

LO 23.d

Describe the judgmental approaches, empirical models, and financial models to predict default.

Q172 AnalystPrep Q.5974

A financial institution is considering the limitations of judgmental approaches in credit risk assessment and discussing potential challenges. Based on the nature of these approaches, which of the following statements accurately reflects a key limitation of using judgmental approaches in credit risk assessment?

A. They are overly reliant on real-time market data, making them less suitable for long-term risk assessment.
B. They rely heavily on advanced statistical models, making them difficult to implement without technical expertise.
C. They face challenges in assessing the quality of results and examining scenarios affecting creditworthiness.
D. They require constant updates of empirical data, making them unsuitable for dynamic market conditions.

Answer: C

Step by step

1. Judgmental approaches (5C analysis, expert systems) rely on qualitative expert input.
2. Their drawbacks (GARP): it is very difficult to validate the quality of results, to update the process to changing conditions, and to examine scenarios affecting a single borrower's creditworthiness or the whole portfolio's risk.
3. Only C matches this list.

Why C is correct: Because judgmental approaches rest on expert opinion rather than testable statistical relationships, their results cannot be validated easily and scenarios affecting creditworthiness are hard to examine.

Why the others are wrong

- **A:** Real-time market data is an input of empirical and financial models, not judgmental ones.

- **B:** Advanced statistical models characterize empirical models; judgmental approaches use expert judgment and need no statistical machinery.
- **D:** Constant empirical data updates describe the data needs of empirical models; judgmental approaches do not depend on empirical data.

Q173 AnalystPrep Q.5975

A bank is evaluating the limitations of empirical models in credit risk assessment. The credit risk team is discussing potential challenges associated with relying on empirical models for predicting defaults. Which of the following statements accurately reflects a key limitation of using empirical models in credit risk assessment?

A. They predominantly focus on qualitative factors, ignoring important quantitative data.
B. Empirical models require real-time market data updates, making them less suitable for long-term assessments.
C. These models face challenges due to their reliance on historical data, which may not fully describe future risks and are often considered backward-looking.
D. They are overly dependent on expert judgment, leading to subjective assessments.

Answer: C

Step by step

1. Empirical models fit patterns in historical loans and defaults.
2. Their key weakness is that the past may not describe future outcomes, particularly in volatile or uncertain periods, so they are backward-looking.
3. C states this; the others describe other approaches or reverse the facts.

Why C is correct: Empirical models are calibrated on historical data, so they may fail to capture future risks when conditions change.

Why the others are wrong

- **A:** Reversed: empirical models are quantitative; a qualitative focus is the judgmental approach.
- **B:** Reversed: empirical models are not tied to real-time market updates (their information is in fact not updated on a real-time basis); the key limitation is reliance on backward-looking historical data, not a real-time data requirement.
- **D:** Dependence on expert judgment is a feature of judgmental approaches, not data-driven empirical models.

Q174 AnalystPrep Q.5976

REBUILT — Option B (machine learning on real-time market trends) is partly true of empirical models, creating a second defensible answer; replaced it with a clear financial-model description.

A financial institution is assessing the strengths and weaknesses of empirical models for credit risk assessment. The team is exploring various aspects of these models in predicting borrower defaults. Which of the following statements correctly highlights a key feature of empirical models in credit risk assessment?

A. Empirical models primarily utilize expert judgment and qualitative assessments in predicting defaults.
B. They rest on normative economic theory and market prices of traded securities, so they apply only to corporate borrowers.
C. Empirical models use historical data, such as past loans and defaults, to establish patterns and predict future default probabilities.
D. They are based on theoretical financial frameworks and rely less on historical data analysis.

Answer: C

Step by step

1. Empirical models search historical data (loans accepted, rejected, in good standing, in default) for patterns linking borrower characteristics and risk factors to default.
2. The patterns are used to estimate future default probabilities.
3. A describes the judgmental approach; B and D describe financial (theory/market-based) models.

Why C is correct: Empirical models are data-driven: they learn the relationship between risk factors and default from past loan outcomes.

Why the others are wrong

- **A:** Expert judgment and qualitative assessment define the judgmental approach.
- **B:** Describes financial (market) models, which rely on theory and market data and cover corporate borrowers only.

- **D:** Being theory-based and relying little on historical data describes financial models, the opposite of empirical models.

Q175 AnalystPrep Q.5978

A regional bank, specializing in consumer and small business lending, has a long history of transactions and has accumulated a substantial amount of data over the years. This data includes detailed records of loan applications, approvals, rejections, repayments, and defaults. The bank is now looking to enhance its credit risk assessment process. Given the bank's extensive historical data and the nature of its lending activities, which type of model would be most appropriate for the bank to adopt for its credit risk assessment?

A. Judgmental approaches
B. Empirical models
C. Financial models based on economic theories
D. Real-time market analysis models

Answer: B

Step by step

1. The bank has extensive records of applications, approvals, rejections, repayments and defaults for consumer and small business loans.
2. This is exactly the input that empirical models use to find default patterns.
3. Financial models need market data on traded securities and apply only to corporate borrowers, so they do not fit small consumer and business loans.

Why B is correct: A large internal history of loan outcomes allows a data-driven empirical model to be estimated and validated.

Why the others are wrong

- **A:** Judgmental approaches are best when history is scarce; here they would waste the available data and be less consistent and transparent.
- **C:** Financial models require market prices and apply only to corporate borrowers, not retail and small business loans.
- **D:** Market-based analysis does not use the bank's own loan history and is unavailable for unlisted small borrowers.

Q176 AnalystPrep Q.5979

A small community bank, primarily serving a rural area, is looking to update its credit risk assessment process. The bank has a close relationship with its customers and a deep understanding of the local economy, but it lacks extensive historical data on loans and defaults. The credit portfolio includes a mix of agricultural loans, small business loans, and personal loans, often to customers with unique financial situations that are not well-documented statistically. In this scenario, which type of model would be most suitable for the bank to use for credit risk assessment?

A. Empirical models
B. Judgmental approaches
C. Financial models based on economic theories
D. Machine learning prediction models

Answer: B

Step by step

1. The bank has little historical data and borrowers with unique, poorly documented situations.
2. Empirical models (including machine learning) need extensive data; financial models need market data.
3. Judgmental approaches rely on analyst experience and local knowledge and work when history is lacking.

Why B is correct: Judgmental approaches add most value where historical data are scarce and loans are unique, using the analyst's local expertise.

Why the others are wrong

- **A:** Empirical models need substantial historical default data, which the bank lacks.
- **C:** Financial models rely on market data and apply only to corporate borrowers with traded securities; rural small loans have none.
- **D:** Machine learning is a type of empirical (data-driven) modeling and likewise needs substantial historical data, which the bank lacks.

Q177 AnalystPrep Q.5980

A multinational corporation is analyzing the credit risk of a select group of corporate bonds issued by companies with detailed financial disclosures. The financial analysts are particularly interested in evaluating how changes in a firm's asset value relative to its liabilities impact the probability of default. The bonds in question are issued by companies with transparent capital structures, allowing for the modeling of firm-specific financial dynamics. Which type of model would be most appropriate for analyzing the credit risk of these corporate bonds?

A. Reduced-form models
B. Empirical models
C. Structural models
D. Real-time market analysis models

Answer: C

Step by step

1. The focus is on how a firm's asset value relative to its liabilities drives default, using firm-specific financial dynamics.
2. Structural models treat default as endogenous, determined by the firm's internal structure (Merton model).
3. Reduced-form models treat default as an exogenous random event and use market data on bonds and credit derivatives.

Why C is correct: Structural models link default to the firm's asset value versus its debt obligations.

Why the others are wrong

- **A:** Reduced-form models treat default as an exogenous random (Poisson-type) event and ignore the asset-liability structure.
- **B:** Empirical models search historical data for statistical patterns and do not model asset value versus liabilities.
- **D:** Not a standard model category; market inputs alone do not model the asset-liability structure.

Q178 AnalystPrep Q.5981

A hedge fund is looking to assess the credit risk of a range of financial instruments, including derivatives such as credit default swaps (CDS) and collateralized debt obligations (CDOs). These instruments are complex and have varying levels of risk based on market conditions and the creditworthiness of underlying assets. Given the complexity and market-based nature of these instruments, which type of credit risk assessment model would be most suitable for the hedge fund to use in this situation?

A. Judgmental approaches
B. Data-driven empirical models
C. Financial models
D. Machine learning-based credit scoring models

Answer: C

Step by step

1. CDS and CDOs are traded, market-priced instruments on corporate credit.
2. Financial models are theory-based and use market data (bond and credit derivative prices), capturing market-driven risk.
3. Judgmental and historical-data models do not capture market sensitivity.

Why C is correct: Financial models are built on market data and financial theory, matching market-priced complex instruments.

Why the others are wrong

- **A:** Judgmental approaches are qualitative and not suited to complex market instruments.
- **B:** Empirical models are backward-looking and based on historical loan data, so less responsive to market conditions.
- **D:** Machine learning scoring is a form of empirical modeling, with the same historical-data limits.

Q179 AnalystPrep Q.5982

A global investment bank is evaluating the credit risk of a diversified portfolio of publicly traded companies. The bank aims to assess the default risk of these companies based on their market capitalization, stock price volatility, and economic conditions. The assessment requires a model that can link a company's credit risk to its market-driven financial metrics. In this scenario, which type of model would provide a significant advantage over other

model types for assessing the credit risk of publicly traded companies?

A. Judgmental approaches
B. Data-driven empirical models
C. Financial models
D. Real-time consumer behavior analysis models

Answer: C

Step by step

1. The task is to link credit risk to market capitalization, equity volatility and economic conditions of listed firms.
2. Financial (market) models use market data and apply to corporate borrowers; the Merton model, for example, uses equity value and volatility.
3. Judgmental and empirical approaches do not tie default directly to market values.

Why C is correct: Financial models rely on market data and theory, so they connect default risk to equity-market variables for public firms.

Why the others are wrong

- **A:** Judgmental approaches are expert-based and do not use market metrics systematically.
- **B:** Empirical models use historical default patterns and are backward-looking rather than anchored to market values.
- **D:** Consumer behavior models fit retail credit, not listed corporations.

Q180

AnalystPrep Q.5983

REBUILT — Original stem was loose enough that structural models were also defensible; rebuilt so default is explicitly an exogenous random event calibrated to bond and CDS market data.

An asset manager wants to estimate default risk for a portfolio of corporate bonds issued by diverse firms. The model should treat default as a random event driven by exogenous factors (such as a Poisson jump process) and be calibrated to market data on bonds and credit derivatives, rather than to each firm's internal capital structure. Which type of model fits this description?

A. Judgmental approaches
B. Empirical models
C. Reduced-form financial models
D. Structural financial models

Answer: C

Step by step

1. Default driven by a random exogenous event (Poisson jump) is the definition of a reduced-form model.
2. Reduced-form inputs are market data on bonds and credit derivatives.
3. Structural models instead assume default is endogenous to the firm's structure (Merton).

Why C is correct: Reduced-form models treat default as an exogenous random event calibrated from market prices of bonds and credit derivatives.

Why the others are wrong

- **A:** Judgmental approaches are qualitative expert assessments, not a model of random default events.
- **B:** Empirical models use historical borrower and loan data rather than a theory of default arrival or credit-derivative prices.
- **D:** Structural models make default endogenous to the firm's asset value and capital structure, contradicting the stem.

Q181

BT answers P2.T6.24.6.3 · BT notes P2.T6.24.6.3

REBUILT — Original asked which of three loosely worded statements was accurate (CreditMetrics is a portfolio VaR model, not a default-prediction approach); rebuilt as clear statements on the scope of each approach to predicting default.

Mancus Lending Group is selecting an approach to predict borrower default. Three board members make statements. Moe: "A judgmental approach relies on expert assessment and can be applied to both consumer and corporate loans." Larry: "Empirical models are data-driven but, because they rely on market data, they can only be applied to corporate borrowers." Curly: "Financial models such as the Merton model rely on financial market data and therefore can only be applied to corporate borrowers." Which statements are correct?

A. Moe only
B. Larry and Curly only

C. Moe and Curly only
D. Moe, Larry and Curly

Answer: C
Key correction: Source key (D, all three accurate) relied on a Larry statement describing CreditMetrics as a historical-data default estimation approach, which is not one of the three default-prediction approaches; the item was rebuilt.
Step by step
1. Judgmental approach: qualitative expert input (e.g., 5C analysis) on financial statements, sector, economy; works for consumer and corporate loans. Moe is correct.
2. Empirical models use historical data on accepted, rejected, performing and defaulted loans from internal or external databases; they apply to both consumer and corporate loans. Larry is wrong (it is financial models that depend on market data).
3. Financial (market) models are based on financial theory and rely on financial market data; GARP describes them as focused on corporate debt, so they are used for corporate borrowers. Curly is correct.
4. Moe and Curly only: C.

Why C is correct: Only financial models rely on market data and are focused on corporate borrowers; judgmental and empirical approaches cover both consumer and corporate loans.
Why the others are wrong
- **A:** Omits Curly; the Merton model relies on market data and is focused on corporate borrowers.
- **B:** Accepts Larry's claim, confusing empirical models with financial models, and wrongly rejects Moe.
- **D:** Accepts Larry's claim that empirical models are corporate-only, confusing data-driven with market-data-driven.

Q182

BT answers P2.T6.703.3

REBUILT — Source was based on De Laurentis, had two arguably false statements and an EXCEPT format; rebuilt on the structural vs reduced-form distinction in the reading.

Which of the following statements about structural and reduced-form financial models of default is correct?

A. Structural models assume default is triggered by a random exogenous event such as a Poisson jump.
B. Reduced-form models assume default depends on the firm's asset value relative to its debt.
C. Structural models assume default is an endogenous process linked to firm characteristics such as asset value and debt, as in the Merton model.
D. Both structural and reduced-form models are market models that can be applied to consumer as well as corporate borrowers.

Answer: C
Step by step
1. Structural: default is endogenous, driven by the firm's internal structure (asset value vs. debt); Merton is the example.
2. Reduced-form: default is a random exogenous event (e.g., Poisson jump); inputs are market data on bonds and credit derivatives.
3. Financial models rely on market data and are focused on corporate debt, so option D is wrong.
4. Only C states the structural definition correctly.

Why C is correct: Structural models explain default through the firm's own asset-debt structure (endogenous), whereas reduced-form models treat default as an exogenous random event.
Why the others are wrong
- **A:** Swaps the definitions: the exogenous Poisson jump describes reduced-form models.
- **B:** Swaps the definitions: asset value versus debt is the structural (Merton) mechanism.
- **D:** Financial models require market data and are limited to corporate borrowers; judgmental and empirical approaches cover consumers.

Q183

Schweser Module Quiz 23.2 Q1

REBUILT — Answer options were missing from the source; reconstructed options and asked for the pair of approaches described.

A financial analyst and a risk manager are discussing ways to derive the probability of default for a bank's loan portfolio. The analyst prefers models that rely on the perspectives of internal experts. The manager acknowledges the value of internal experts but prefers a model that is data driven and can assess loans for both consumer and corporate customers. Which types of approaches are the analyst and the manager, respectively, describing?

A. Judgmental approach; empirical model
B. Empirical model; judgmental approach
C. Judgmental approach; financial model
D. Financial model; empirical model

Answer: A
Step by step
1. Analyst: reliance on internal experts' perspectives = judgmental (expert system) approach.
2. Manager: data driven (historical loan data) and applicable to consumer and corporate customers = empirical model.
3. Financial models are market-data based and corporate-only, so they cannot be what the manager describes.

Why A is correct: Judgmental approaches are expert based; empirical models are historical-data based and cover both consumer and corporate loans.

Why the others are wrong
- **B:** Reverses the two approaches.
- **C:** Financial models rely on market data and apply only to corporates, so they do not fit the manager's consumer-and-corporate, data-driven preference.
- **D:** Financial models are theory/market-data based, not expert based, so they do not fit the analyst's preference for expert judgment.

Q184 AnalystPrep Q.5973

A financial institution is restructuring its credit risk assessment process and has decided to incorporate judgmental approaches for certain credit portfolios. Which of the following best describes a fundamental aspect of judgmental approaches in credit risk assessment?

A. They rely heavily on real-time market data to assess the creditworthiness of borrowers.
B. Judgmental approaches require extensive use of credit scoring models based on empirical data.
C. These approaches emphasize quantitative assessment using historical financial ratios and trends.
D. They involve qualitative analysis, focusing on aspects like the borrower's character and capacity.

Answer: D
Step by step
1. Judgmental (expert-system) approaches rely on credit analysts' qualitative judgment.
2. The standard framework is 5C analysis: character, capacity, capital, collateral, conditions.
3. D matches; A, B, C describe financial or empirical models.

Why D is correct: Judgmental approaches are built on expert qualitative assessment (e.g., 5C analysis) rather than on models or market data.

Why the others are wrong
- **A:** Real-time market data drive financial (market) models, not judgmental approaches.
- **B:** Scoring models based on empirical data are empirical models.
- **C:** Systematic quantitative analysis of historical data is the empirical approach; judgmental approaches are primarily qualitative.

Q185 AnalystPrep Q.1785

REBUILT — Source used 'reduced form' in the statistical-model sense and asked an open-ended best-practice question; rebuilt around the stated weakness of empirical models relying on historical data.

A bank has built an empirical default-prediction model using historical loan data from a long period of stable economic conditions. The risk manager is asked to identify the most important limitation to flag before the model is used. Which is the most appropriate?

A. Empirical models cannot be validated, because their structure and predictive power cannot be tested against actual data.
B. Because the model relies on historical data, it may not predict future outcomes well, especially in periods of high uncertainty and volatility.
C. Empirical models can be applied only to corporate borrowers because they require market data.
D. Empirical models cannot incorporate new risk factors such as corporate governance or real-time market information.

Answer: B

Step by step

1. Empirical models are estimated on past loan outcomes.
2. Their main weakness: historical patterns may not hold in the future, particularly in volatile or uncertain times; some inputs (financial statements) are updated only quarterly.
3. A, C, D contradict stated advantages: validation is possible, they cover consumer and corporate loans, and they can use ML and new factors.

Why B is correct: The main weakness of empirical models is dependence on historical data, which may fail to predict outcomes when conditions change.

Why the others are wrong

- **A:** Validation of structure and predictive power is an advantage of empirical models over judgmental ones.
- **C:** Market-data dependence limiting use to corporates is a feature of financial models; empirical models cover consumer and corporate loans.
- **D:** Empirical (ML) models can add new risk factors such as governance practices and real-time market data.

Q186

BT notes P2.T6.302.1

REBUILT — Original asked for 'three types' of model but listed four options (including a macro-factor/actuarial taxonomy not in the reading); rebuilt on the structural vs reduced-form distinction in the reading.

A credit analyst builds a model in which default is a random event that can occur at any time, driven by an exogenous Poisson jump process, and calibrates it to market data on bonds and credit derivatives. Which type of model is this?

A. Structural financial model
B. Empirical (statistical scoring) model
C. Judgmental model
D. Reduced-form financial model

Answer: D

Key correction: Source key (C, Factor) rested on a four-type taxonomy not used in the 2026 reading; the question was rebuilt on structural vs reduced-form.

Step by step

1. Structural models treat default as endogenous, tied to firm characteristics such as asset value and debt (e.g., Merton).
2. Reduced-form models treat default as a random exogenous event (e.g., Poisson jump process) and take bond and credit-derivative market data as main inputs.
3. The description matches reduced-form.

Why D is correct: Default driven by an exogenous random (Poisson) event and calibrated to bond/CDS prices defines reduced-form models.

Why the others are wrong

- **A:** Structural models make default endogenous, based on asset value versus debt, not an exogenous jump.
- **B:** Empirical models use historical borrower data and statistical scoring, not market-implied jump processes.
- **C:** Judgmental models rely on expert assessment of qualitative and quantitative factors.

LO 23.e

Apply the Merton model to calculate default probability and the distance to default and describe the limitations of using the Merton model.

Q187

AnalystPrep Q.4364

REBUILT — Original had two defensible statements (A and D) and a lognormal/normal subtlety; replaced with a clean statement about the Merton model's inputs and unknowns.

Which of the following statements about the inputs of the Merton model is CORRECT?

A. The current asset value A and asset volatility σ_A are unknowns that must be solved from the equity value and equity volatility equations.
B. The real-world probability of default is obtained by substituting the risk-free rate r for the expected asset return μ.
C. The risk-neutral probability of default is $N(d_2)$.

D. The face value of debt L is usually taken as the firm's long-term debt.

Answer: A

Step by step

1. The two equations (equity as a call option on assets, and the equity-volatility relation) contain two unknowns: A and σ_A.
2. Everything else is observed: equity market cap E, equity volatility, T, face value L (short-term debt), and r (T-bill yield).
3. So the unknowns are solved jointly, typically by an iterative numerical method.

Why A is correct: Asset value and asset volatility are not directly observable, so they are backed out from the observable equity value and equity volatility.

Why the others are wrong

- **B:** Reversed: the real-world PD substitutes μ for r, not the other way round.
- **C:** $N(d_2)$ is the probability shareholders repay (do not default); the risk-neutral PD is $N(-d_2)$.
- **D:** In practice L is taken as the short-term debt on the balance sheet, not long-term debt.

Q188

AnalystPrep Q.5984

REBUILT — Original 'core concept' stem had a vague option set; reframed to ask directly what N(-d2) measures.

In the Merton model, the borrower's equity is a call option on the firm's assets with strike equal to the face value of debt L. The quantity $N(-d_2)$ computed with the risk-free rate r represents:

A. the probability that the shareholders exercise the option and repay the debt

B. the risk-neutral probability that the shareholders do not exercise the option, i.e., that the firm defaults

C. the real-world probability of default, since asset growth is set to μ

D. the expected loss on the debt as a percentage of face value

Answer: B

Step by step

1. Equity payoff at T is $\max(A_T - L, 0)$, a call with strike L.
2. Shareholders exercise (repay) if $A_T > L$; this has probability $N(d_2)$.
3. Default is the complement: $N(-d_2) = 1 - N(d_2)$.
4. Using r as the asset growth rate makes this a risk-neutral measure.

Why B is correct: Default occurs when the option expires out of the money; with assets growing at r, its probability is the risk-neutral $N(-d_2)$.

Why the others are wrong

- **A:** This is $N(d_2)$, the complement of the PD.
- **C:** The real-world PD has the same form but requires substituting μ for r; with r it is risk-neutral.
- **D:** $N(-d_2)$ is only a probability; expected loss also needs LGD and EAD.

Q189

AnalystPrep Q.5986

REBUILT — Original distractors (operating costs, linear relationship to indices, variable-rate debt) were giveaways; replaced with plausible but wrong model assumptions.

Which of the following is an assumption of the Merton (Black-Scholes-Merton) model as applied to credit risk?

A. The firm's asset returns have constant volatility over the life of the loan.

B. The firm's assets can jump suddenly in value, triggering default at a random time.

C. The firm has several debt issues with different maturities, and default can occur on any of them at any time.

D. The risk-free rate is a stochastic process that is correlated with asset returns.

Answer: A

Step by step

1. Merton uses Black-Scholes mechanics: assets follow a diffusion with constant σ_A and constant r.
2. The firm has a single loan with face value L maturing at T; default can only be observed at T.
3. Jumps and random default times belong to reduced-form models; multiple maturities are not modeled.

Why A is correct: Constant asset volatility (together with a constant risk-free rate) is a core Black-Scholes-Merton assumption.

Why the others are wrong

- **B:** Jump/random-default-time behaviour describes reduced-form models; Merton assumes a continuous diffusion.
- **C:** The simple Merton setup has one loan maturing at T; multiple debt issues are a complication it does not handle.
- **D:** The risk-free rate is assumed constant, not stochastic.

Q190

AnalystPrep Q.5988

REBUILT — Original was a qualitative item with giveaway distractors; rebuilt as a calculation of the real-world PD under constant asset volatility.

A firm has asset value $A = 120$ and a single zero-coupon debt with face value $L = 100$ maturing in $T = 1$ year. Assume constant asset volatility $\sigma_A = 20\%$, expected asset return $\mu = 8\%$, and risk-free rate $r = 3\%$. Using the Merton model, the real-world probability of default is closest to:

A. 7.9%
B. 9.5%
C. 11.3%
D. 16.8%

Answer: C

Step by step

1. $d_2 = [\ln(A/L) + (\mu - \sigma^2/2)T]/(\sigma\sqrt{T})$
2. $\ln(1.2) = 0.1823$; $\mu - \sigma^2/2 = 0.08 - 0.02 = 0.06$
3. $d_2 = (0.1823 + 0.06)/0.20 = 1.2116$ (the distance to default in standard deviations)
4. PD $= N(-1.2116) = 11.3\%$

Why C is correct: Real-world PD substitutes μ for r in d_2 and equals $N(-d_2)$.

Why the others are wrong

- **A:** Used $d_1 = d_2 + \sigma\sqrt{T} = 1.4116$ instead of d_2: $N(-1.4116) = 7.9\%$.
- **B:** Forgot the $-\sigma^2/2$ term: $d = 1.3116$, $N(-d) = 9.5\%$.
- **D:** Used the risk-free rate (risk-neutral PD): $d_2 = 0.9616$, $N(-d_2) = 16.8\%$.

Q191

AnalystPrep Q.6009

REBUILT — Original answer option was truncated and distractors were weak; replaced with the Schweser/GARP limitation of the default point.

Which of the following is a recognised limitation of the Merton model when estimating a borrower's probability of default?

A. The default point is defined only as the firm's short-term debt, which can understate the obligations of a firm with large long-term debt.
B. It requires a rating transition matrix, which is hard to obtain for many borrowers.
C. It relies on observed bond prices and credit derivative spreads, which are unavailable for most firms.
D. It assumes default is driven by an exogenous Poisson jump that cannot be linked to firm fundamentals.

Answer: A

Step by step

1. Merton sets the default point L equal to short-term debt on the balance sheet.
2. Long-term liabilities are ignored (the Moody's KMV approach uses short-term debt plus half of long-term debt for this reason).
3. Other options describe CreditMetrics (transition matrix) or reduced-form models (market spreads, Poisson jumps).

Why A is correct: The model's default point ignores long-term debt, so it may misstate default risk for firms with substantial long-term obligations.

Why the others are wrong

- **B:** Transition matrices are the CreditMetrics input, not Merton's.
- **C:** Merton uses equity market data; bond and CDS prices are the inputs of reduced-form models.
- **D:** Exogenous Poisson-jump default is the reduced-form assumption; Merton is structural and ties default to asset value versus debt.

LO 23.f

Compare and contrast different approaches to credit risk modeling, such as those related to the Merton model, Credit Risk Plus (CreditRisk+), CreditMetrics, and the Moody's KMV model.

Q192

AP Mock 3 Q62

An analyst has already computed the distance to default for a firm and now uses the Moody's KMV (EDF) model to convert it into a probability of default. Which of the following best describes how KMV maps distance to default to a probability of default?

A. By applying the standard normal distribution to the distance to default.
B. By using a Poisson distribution based on the firm's past default history.
C. By deriving an empirical distribution of default frequencies from historical data.
D. By assigning a constant default probability regardless of the distance to default.

Answer: C

Step by step

1. The Merton model maps distance to default to PD with the standard normal CDF, $\text{PD} = N(-\text{DD})$.
2. The Moody's KMV EDF model replaces this step: firms with a given DD are grouped in a historical default database and the observed default frequency for that DD becomes the EDF.
3. So KMV uses an empirical distribution of default frequencies.

Why C is correct: The key difference from Merton is the mapping step: KMV uses historical default experience (an empirical distribution) instead of the normal distribution.

Why the others are wrong

- **A:** This is the Merton mapping, $N(-\text{DD})$, which KMV deliberately does not use.
- **B:** A Poisson distribution is the CreditRisk+ choice for default counts, not KMV's DD-to-PD mapping.
- **D:** A constant PD would make the distance to default irrelevant, which contradicts the purpose of computing it.

Q193

AnalystPrep Q.5991

An analyst assesses a corporation's credit risk with the Moody's KMV model. The company's short-term debt is much larger than its long-term debt. How does the KMV model define the default point used to compute distance to default?

A. The total amount of the company's equity.
B. All short-term debt plus half of long-term debt.
C. The volatility of the company's stock.
D. The current market value of the company's assets.

Answer: B

Step by step

1. In the Merton model the default point is the face value of (short-term) debt.
2. KMV defines the default point as short-term debt plus 50% of long-term debt, since long-term obligations are only partly due within the horizon.
3. Equity amount, stock volatility and asset value are model inputs, but none of them is the default point.

Why B is correct: The KMV default point (short-term debt + half of long-term debt) better approximates the borrower's actual obligations than Merton's debt face value.

Why the others are wrong

- **A:** Equity is the residual claim (a call on assets), used to infer asset value and volatility, not the default barrier.
- **C:** Equity volatility is an input for backing out asset volatility; it is not a debt amount.
- **D:** Asset value is the variable compared against the default point in the DD formula; it is not the default point itself.

Q194

AnalystPrep Q.5992

REBUILT — Original options were giveaways (dividend payout, current ratio, interest coverage); rewrote distractors as realistic confusions with other models' inputs.

A credit analyst applies the Moody's KMV model to a listed corporation that has just issued new equity and restructured its capital. The analyst needs to back out the unobservable asset value and asset volatility to compute the

distance to default. Which pair of inputs, observable in the market, is used for this purpose?

A. The book value of equity and the volatility of accounting earnings.
B. The borrower's credit rating and the rating transition matrix.
C. The market value of equity (market capitalization) and the volatility of equity.
D. The recovery rate and the credit spread for each rating grade.

Answer: C

Step by step

1. KMV/Merton treat equity as a call option on the firm's assets.
2. This gives two equations linking equity value E and equity volatility σ_E to the two unknowns, asset value A and asset volatility σ_A.
3. Therefore the market value of equity and equity volatility, both observed from the stock market, are the key inputs.

Why C is correct: Because equity is a call option on assets, observed equity value and equity volatility allow A and σ_A to be solved, and both change when equity is issued or the firm restructures.

Why the others are wrong

- **A:** Accounting (book) figures are not what the market-based KMV approach uses; it relies on market capitalization and market volatility.
- **B:** Ratings and transition matrices are CreditMetrics inputs.
- **D:** Recovery rates and grade-specific spreads are CreditMetrics revaluation inputs, not KMV inputs for asset value.

Q195 AnalystPrep Q.6003

A risk analyst is considering CreditMetrics for a client's long-term fixed-income portfolio with various maturities. Which of the following limitations of CreditMetrics should concern the analyst most in this context?

A. The model's inability to incorporate economic forecasting into its analysis.
B. The use of fixed interest rates when calculating recovery rates.
C. The complexity of the model requiring extensive computational resources.
D. The exclusion of equity instruments from the model's credit risk assessment.

Answer: B

Step by step

1. Two criticisms of CreditMetrics are the assumption that credit grades are homogeneous and the use of fixed interest rates.
2. The fixed-rate assumption is most damaging over long horizons because interest rates over long periods can be quite volatile.
3. Hence for a long-term portfolio, B is the key limitation.

Why B is correct: CreditMetrics uses fixed interest rates, which does not capture rate volatility and becomes more problematic the longer the horizon.

Why the others are wrong

- **A:** Not a stated limitation of the model.
- **C:** Complexity is not the recognised limitation; the method is described as promoting transparency and systematic risk management.
- **D:** CreditMetrics is designed for a wide range of products; equity exclusion is not a cited weakness.

Q196 AnalystPrep Q.6006

An analyst at an asset manager is considering CreditRisk+ for a diverse corporate loan portfolio. Which feature of CreditRisk+ is most distinctive?

A. It monitors fluctuations in the equity market prices of the borrowers.
B. It treats default frequency as a continuous random variable described by a Poisson distribution.
C. It evaluates the historical performance and dividend yields of the borrowers.
D. It assesses the impact of changing global economic conditions on the portfolio.

Answer: B

Step by step

1. CreditRisk+ (Credit Suisse, 1997) ignores the borrower's capital structure and equity prices.
2. It treats the default frequency as a continuous random variable, with defaults rare and independent.
3. The Poisson distribution suits this because its mean equals its variance; the model needs only default events and little data.

Why B is correct: CreditRisk+ models rare, independent defaults with a Poisson distribution (mean = variance), without using equity prices or ratings.

Why the others are wrong

- **A:** Equity-price monitoring belongs to KMV-style (market-based) models.
- **C:** Dividend yields and company performance are not CreditRisk+ inputs.
- **D:** CreditRisk+ is not built around macroeconomic conditions.

Q197

AnalystPrep Q.6008

An analyst compares CreditMetrics and CreditRisk+ for a bank holding debt securities and loans. Which statement correctly reflects a key difference between them?

A. CreditRisk+ assesses credit risk from market capitalization, unlike CreditMetrics.
B. CreditMetrics assesses losses only in bankruptcy, while CreditRisk+ considers a broader range of credit events.
C. CreditRisk+ distinguishes only two credit events, bankruptcy and bad credit condition, so losses arise only on default.
D. Both models equally emphasize macroeconomic factors in credit risk assessment.

Answer: C

Step by step

1. CreditMetrics values debt across rating migrations (upgrades, downgrades, default) in a mark-to-market framework.
2. CreditRisk+ recognises only two credit events, bankruptcy and bad credit condition, and losses occur only on default.
3. Hence C is correct, and B reverses the two models.

Why C is correct: CreditRisk+ is a default-only model (two credit events), whereas CreditMetrics captures losses from rating changes as well.

Why the others are wrong

- **A:** CreditRisk+ does not use capital structure or market capitalization; that is KMV.
- **B:** Reverses the models: CreditMetrics is the broader one.
- **D:** Neither model is defined by an emphasis on macroeconomic factors.

Q198

BT answers P2.T6.24.7.2 · BT notes P2.T6.24.7.2

REBUILT — Original stem was a long scenario in which several models plausibly fit (all described as using market data); replaced by three distinguishing features from the reading.

A risk manager wants a model for a portfolio of corporate borrowers that (i) uses market data so default probabilities update continuously, (ii) maps distance to default to PD using historical default frequencies rather than the normal distribution, and (iii) sets the default point at short-term debt plus half of long-term debt. Which model fits all three requirements?

A. CreditMetrics
B. Moody's KMV (EDF) model
C. CreditRisk+
D. Merton model

Answer: B

Key correction: Source key (D, Moody's-KMV) is retained as the same model; the original stem allowed the Merton model, which also uses equity data, as a competing answer.

Step by step

1. Merton: uses N(-DD) and a default point equal to debt face value, so fails (ii) and (iii).
2. CreditMetrics: PD from rating peer analysis and transition matrices (mark-to-market), no DD or default point; fails (ii) and (iii).
3. CreditRisk+: actuarial Poisson default-only model ignoring capital structure; fails all three.
4. Moody's KMV: equity-market-based DD, empirical default-frequency mapping, default point = short-term debt + 50% of long-term debt.

Why B is correct: Only KMV combines market-driven inputs, an empirical DD-to-EDF mapping and a default point that includes half of long-term debt.

Why the others are wrong

- **A:** CreditMetrics relies on ratings and transition matrices, with no distance to default or default point.
- **C:** CreditRisk+ is a Poisson, default-only model that ignores capital structure and market-based DD.
- **D:** Merton uses the standard normal distribution and the debt face value as default point.

Q199

GARP PE2 Q71 · Mock 4 Q72

A risk manager at a bank is conducting an internal training program for newly hired credit risk analysts. The manager gives an overview of credit risk models, compares the bank's internal credit rating models to the rating agencies' models, and explains the assumptions, structure, and performance measures of these models as applied to counterparty default prediction. Assuming a stable market environment, which of the following statements would the manager be correct to make?

A. The Credit Risk Plus (CreditRisk+) model relies on the use of a company's capital structure and credit ratings to predict default of the company.

B. Unlike the Merton model that produces default predictions based on asset and liability values, the agencies' rating models produce default predictions from a substantially larger set of business factors.

C. Unlike the agencies' rating models, the KMV model is only used to compute the risk of pure debt contracts and is not applicable to portfolios that include derivatives contracts.

D. A good rating model is expected to provide accurate borrower default predictions without the need to be back-tested.

Answer: B

Step by step

1. Merton uses only the market value of assets, asset volatility and the face value of debt (liabilities) to produce a default probability.
2. Rating-agency (judgmental) models also weigh country, industry, competitive and other business factors, so B is true.
3. CreditRisk+ does not use capital structure and does not use credit ratings, so A is false.
4. A rating model must be backtested to confirm its predictive accuracy, so D is false; the claim about KMV in C is unsupported.

Why B is correct: Merton is a narrow structural model (asset value vs. debt), whereas agency ratings blend many qualitative and quantitative factors.

Why the others are wrong

- **A:** CreditRisk+ ignores capital structure and models only default events, not ratings; the statement confuses it with KMV.
- **C:** KMV is not limited to pure debt; the claim is invented (derivatives exposure is handled by approaches such as CreditMetrics).
- **D:** Any rating model's performance must be validated by backtesting; no model is assumed accurate a priori.

Q200

Schweser Module Quiz 23.2 Q2

REBUILT — Option B (risk-free rate vs customized value) was arguably defensible; replaced it with a clearly false option so only C is correct, per GARP's two differences (empirical default distribution, default point).

The Merton model differs from the Moody's KMV Expected Default Frequency (EDF) approach in two key areas. Which of the following statements describes one of those differences?

A. The Merton model uses the market value of the borrower's equity, while Moody's KMV uses its book value.

B. The Merton model is based on an option-pricing framework, while Moody's KMV is based on a rating transition matrix.

C. Moody's KMV defines the default point as short-term debt plus half of long-term debt, while the Merton model uses short-term debt only.

D. The Merton model uses historical data to derive an empirical distribution of default frequencies, while Moody's KMV maps distance to default using the standard normal distribution.

Answer: C

Step by step

1. Merton: default point L = short-term debt; DD is mapped to PD with the standard normal distribution.

2. KMV EDF: default point = short-term debt + 1/2 long-term debt; DD is mapped to PD using an empirical distribution of historical default frequencies.
3. Compare options: only C states a true difference.

Why C is correct: The two KMV differences are the empirical DD-to-PD mapping and the broader default point including half of long-term debt.

Why the others are wrong

- **A:** Both models use the market value of equity (backed out of option pricing); book value is not used.
- **B:** KMV is also a structural option-based model; the rating transition matrix belongs to CreditMetrics.
- **D:** Reversed: KMV uses the historical empirical distribution and Merton the standard normal.

Q201

Schweser Module Quiz 23.2 Q3

REBUILT — Original stem cited 'volatility', which points to Merton/KMV; reworded to state the CreditMetrics features (peer set, rating transitions, mark-to-market) unambiguously.

A credit risk manager wants a method that estimates default risk by comparing a borrower with the default records of similar borrowers (a peer set), using the borrower's credit rating, a rating transition matrix and a mark-to-market valuation of the loan. Which model should the manager select?

A. Merton model
B. CreditRisk+ model
C. Moody's KMV model
D. CreditMetrics model

Answer: D

Step by step

1. Peer-set analysis with a rating transition matrix and mark-to-market revaluation is the defining feature of CreditMetrics.
2. Merton and KMV are option-pricing (structural) models using asset value and debt; CreditRisk+ models default events only, with no ratings.
3. Answer: CreditMetrics.

Why D is correct: CreditMetrics uses credit ratings, transition probabilities, recovery rates and bond-market spreads to revalue loans.

Why the others are wrong

- **A:** Merton uses option pricing on asset value and short-term debt, not peer comparison.
- **B:** CreditRisk+ considers only default events and does not use credit ratings.
- **C:** KMV is a structural model using asset value, asset volatility and a default point, not peer ratings.

Q202

AnalystPrep Q.6002

A credit risk analyst at a regional bank is evaluating the use of the CreditMetrics model to assess the bank's loan portfolio. The analyst is aware of certain criticisms and limitations of the model. Which of the following points would be a valid concern when using the CreditMetrics model for this purpose?

A. The inability of the model to assess credit risk in real-time market conditions.
B. The assumption of homogeneity in credit grades within the model.
C. The lack of focus on market-based instruments like derivatives in the model.
D. The model's reliance on highly complex algorithms that are difficult to interpret.

Answer: B

Step by step

1. CreditMetrics assumes all borrowers within a given credit grade are homogeneous (same transition and default behavior).
2. It also computes recovery rates using fixed interest rates, ignoring their volatility.
3. It is a mark-to-market model and covers derivatives such as swaps, so A and C are false; D is not a recognized criticism.

Why B is correct: Treating every obligor in a rating grade as identical ignores differences in individual credit quality.

Why the others are wrong

- **A:** It is a mark-to-market model that revalues loans using current spreads, so it does reflect market conditions.
- **C:** It explicitly accommodates market-based tools such as swaps and forwards.

- **D:** Transparency is cited as a strength; complexity is not a standard criticism.

Q203

AnalystPrep Q.6000

REBUILT — Original compared CreditMetrics to Black-Scholes (outside the LO) with a giveaway key; rebuilt as CreditMetrics vs CreditRisk+ comparison.

Which of the following statements correctly distinguishes the CreditMetrics model from the CreditRisk+ model?

A. CreditMetrics is a mark-to-market model using rating transitions, whereas CreditRisk+ considers only default events and does not use credit ratings.

B. CreditRisk+ uses a rating transition matrix to revalue loans, whereas CreditMetrics considers only bankruptcy.

C. CreditMetrics assumes defaults follow a Poisson distribution, whereas CreditRisk+ revalues loans using bond-market spreads.

D. CreditRisk+ requires extensive data on the borrower's capital structure, whereas CreditMetrics requires only default counts.

Answer: A

Step by step

1. CreditMetrics: inputs are rating, transition matrix, recovery rates, yield spreads; output is a distribution of loan value changes (mark-to-market), including upgrades and downgrades.
2. CreditRisk+ (Credit Suisse, 1997): considers only default events; Poisson distribution; small data requirement; ignores capital structure.
3. Only A matches.

Why A is correct: CreditMetrics values rating migrations as well as default, while CreditRisk+ is a default-only actuarial model.

Why the others are wrong

- **B:** Roles reversed: the transition matrix belongs to CreditMetrics and default-only focus to CreditRisk+.
- **C:** Roles reversed: the Poisson default distribution is a CreditRisk+ feature; spread-based revaluation is CreditMetrics.
- **D:** CreditRisk+ ignores capital structure and needs little data; CreditMetrics needs ratings, transitions and spreads.

LO 23.g

Apply risk-adjusted return on capital (RAROC) to measure the performance of a loan.

Q204

AnalystPrep Q.6010

REBUILT — Distractors were not tied to realistic mistakes; replaced them with results of specific errors and kept the correct answer.

A bank has issued a corporate loan of €1,000,000 at an interest rate of 5%. The loan has a duration of 4 years. The bank's risk team expects interest rates to rise by 1% (0.01). The loan generates a 0.2% commission and the spread between the loan's interest rate and the bank's cost of funds is 0.3%. There are no operating costs, expected losses or taxes. The bank's cost of capital is 12%. What is the loan's RAROC, and is the loan profitable?

A. RAROC = 12.50%, and the loan is profitable.

B. RAROC = 13.13%, and the loan is profitable.

C. RAROC = 7.88%, and the loan is not profitable.

D. RAROC = 11.90%, and the loan is not profitable.

Answer: B

Step by step

1. Capital at risk: $|\Delta L| = L \times D \times \Delta i/(1+i) = 1{,}000{,}000 \times 4 \times 0.01/1.05 = 38{,}095$.
2. Revenue $= (s+f) \times L = (0.3\% + 0.2\%) \times 1{,}000{,}000 = 5{,}000$ (no losses, costs or tax).
3. RAROC $= 5{,}000/38{,}095 = 13.12\%$.
4. 13.13% exceeds the 12% cost of capital, so the loan is profitable.

Why B is correct: A loan is profitable when RAROC (after-tax revenue divided by the duration-based change in loan value) exceeds the bank's cost of capital.

Why the others are wrong

- **A:** Omits the $1+i$ divisor: 5,000/40,000 = 12.50%.

- **C:** Counts only the 0.3% spread and leaves out the 0.2% commission: 3,000/38,095 = 7.88%.
- **D:** Multiplies by $1 + i$ instead of dividing: 5,000/42,000 = 11.90%, which wrongly falls below the 12% hurdle.

Q205

AnalystPrep Q.6011

REBUILT — Distractors were not tied to realistic mistakes; replaced them with results of specific errors and kept the correct answer.

A financial analyst is calculating the RAROC for a commercial loan of €750,000 with an interest rate of 7% and a remaining duration of 10 years. The spread between the loan rate and the bank's cost of funds is 7%. The bank expects interest rates to rise by 1% (0.01). The loan has operating costs of €5,000, expected losses of €10,000, and the bank earns a fee of 0.15% of the loan amount. The bank's cost of capital is 20% and its tax rate is 30%. What is the RAROC, and is the loan profitable?

A. RAROC = 55.10%, the loan is profitable.
B. RAROC = 36.05%, the loan is profitable.
C. RAROC = 38.57%, the loan is profitable.
D. RAROC = 43.57%, the loan is profitable.

Answer: C

Step by step

1. Capital at risk: $750{,}000 \times 10 \times 0.01/1.07 = 70{,}093$.
2. Spread income $= 0.07 \times 750{,}000 = 52{,}500$; fee $= 0.0015 \times 750{,}000 = 1{,}125$.
3. Revenue $= (52{,}500 + 1{,}125 - 10{,}000 - 5{,}000)(1 - 0.30) = 38{,}625 \times 0.70 = 27{,}037.5$.
4. RAROC $= 27{,}037.5/70{,}093 = 38.57\%$, above the 20% cost of capital, so profitable.

Why C is correct: Revenue is $(s + f - l - c)(1 - x)$ and the denominator is the duration-based loan value change $LD\Delta i/(1 + i)$; 38.57% exceeds 20%.

Why the others are wrong

- **A:** Forgets the 30% tax: 38,625/70,093 = 55.10%.
- **B:** Omits the $1 + i$ divisor in capital at risk (denominator 75,000): 27,037.5/75,000 = 36.05%.
- **D:** Forgets to deduct the 5,000 operating costs: (52,500 + 1,125 - 10,000)(0.7)/70,093 = 43.57%.

Q206

AnalystPrep Q.6013

REBUILT — Distractors were not tied to realistic mistakes; replaced them with results of specific errors and kept the correct answer.

A bank is evaluating a business loan of €2,000,000 with a fixed interest rate of 4% and a duration of 6 years. Analysts predict an interest rate increase of 0.75% (0.0075) over the coming year. Using the duration approximation, what is the change in the loan's value, ΔL?

A. -€90,000
B. -€14,423
C. -€86,538
D. -€93,600

Answer: C

Step by step

1. $\Delta L = -L \times D \times \Delta i/(1 + i)$.
2. $= -2{,}000{,}000 \times 6 \times 0.0075/1.04$.
3. $= -90{,}000/1.04 = -86{,}538$.

Why C is correct: The duration approximation divides L x D x Δi by (1 + i); a rate rise lowers the loan's value, giving a negative change.

Why the others are wrong

- **A:** Omits the $1 + i$ divisor: 2,000,000 x 6 x 0.0075 = -90,000.
- **B:** Leaves out the duration factor: 2,000,000 x 0.0075/1.04 = -14,423.
- **D:** Multiplies by $1 + i$ instead of dividing: 90,000 x 1.04 = -93,600.

Q207

AnalystPrep Q.6014

REBUILT — Distractors were not tied to realistic mistakes; replaced them with results of specific errors and kept the correct answer.

A bank is assessing a corporate loan of €1,500,000 using RAROC. The interest rate spread is 2.5% above the bank's cost of funds, the loan origination fee is 1%, expected loan losses are €20,000, operating costs are €15,000 and the

tax rate is 30%. What is the numerator of RAROC (loan revenues)?

A. €17,500
B. €5,250
C. €12,250
D. €22,750

Answer: C

Step by step

1. Spread income = 0.025 × 1,500,000 = 37,500; fee = 0.01 × 1,500,000 = 15,000.
2. Pre-tax: 37,500 + 15,000 − 20,000 − 15,000 = 17,500.
3. After tax: 17,500 × (1 − 0.30) = 12,250.

Why C is correct: Loan revenues = $(s + f - l - c)(1 - x)$: the pre-tax margin is reduced by the tax rate.

Why the others are wrong

- **A:** Pre-tax figure; the 30% tax is not applied.
- **B:** Computes the tax amount (17,500 x 0.30) instead of the after-tax figure (17,500 x 0.70).
- **D:** Omits the €15,000 operating costs: (37,500 + 15,000 - 20,000)(0.7) = 22,750.

Q208

AnalystPrep Q.6015

REBUILT — Qualitative comparison did not state capital at risk or the loss and cost inputs; converted to a numerical RAROC comparison with equal inputs.

A bank compares two loans, each of €2,000,000, with expected loan losses of 0.5% of principal, operating costs of 0.4% of principal, a tax rate of 25%, and capital at risk of €300,000 each. Loan A has an origination fee of 1.0% and an interest spread of 3.0%; Loan B has an origination fee of 0.5% and an interest spread of 4.0%. What are the RAROCs of Loans A and B?

A. Loan A 15.50%; Loan B 18.00%
B. Loan A 20.67%; Loan B 24.00%
C. Loan A 10.50%; Loan B 15.50%
D. Loan A 20.00%; Loan B 22.50%

Answer: A

Step by step

1. Loan A margin = 3.0% + 1.0% − 0.5% − 0.4% = 3.1%; after tax = 3.1% × 0.75 = 2.325%; revenue = 46,500.
2. Loan B margin = 4.0% + 0.5% − 0.5% − 0.4% = 3.6%; after tax = 2.70%; revenue = 54,000.
3. RAROC_A = 46,500/300,000 = 15.50%; RAROC_B = 54,000/300,000 = 18.00%.
4. Loan B's higher total of fee plus spread (4.5% vs 4.0%) gives it the higher RAROC.

Why A is correct: With equal losses, costs, tax and capital at risk, the loan with the larger fee-plus-spread income (B: 4.5% vs A: 4.0%) earns the higher RAROC.

Why the others are wrong

- **B:** Ignores tax: 62,000/300,000 and 72,000/300,000.
- **C:** Leaves out the origination fees: (spread - losses - costs)(1-x) gives 31,500 and 46,500.
- **D:** Ignores expected losses and operating costs: (spread + fee)(0.75) gives 60,000 and 67,500.

Q209

Schweser Module Quiz 23.2 Q4

REBUILT — Distractors replaced with results of specific mistakes; stem and answer unchanged.

ABC Bank has a loan with a value of $750,000. The commission is 0.15%, the spread between the loan's interest rate and the bank's cost of capital is 0.45%, operating costs are 0.3%, and the tax rate is 12%. The loan is expected to perform with no risk of default. The loan has a duration of 2.85 and an interest rate of 9.5%. Interest rates are expected to increase by 1.25%. What is the loan's RAROC?

A. 7.41%
B. 9.22%
C. 6.77%
D. 8.11%

Answer: D

Step by step

1. Revenue = 750,000 × (0.15% + 0.45% − 0.30%) × (1 − 0.12) = 750,000 × 0.003 × 0.88 = 1,980.

2. Capital at risk = 750,000 × 2.85 × 0.0125/1.095 = 24,400.68.
3. RAROC = 1,980/24,400.68 = 8.11%.

Why D is correct: RAROC is after-tax loan revenue divided by the duration-based loss in loan value; expected loss is zero for this performing loan.

Why the others are wrong

- **A:** Omits the $1 + i$ divisor: 1,980/26,718.75 = 7.41%.
- **B:** Forgets the tax: 2,250/24,400.68 = 9.22%.
- **C:** Multiplies by $1 + i$ instead of dividing: 1,980/29,257 = 6.77%.

Q210

AP Mock 2 Q38

REBUILT — Used a non-GARP RAROC variant (return on economic capital, hurdle rate, undefined adjusted RAROC, unused data); rebuilt with the GARP revenue formula and unexpected-loss capital at risk.

A bank is reviewing a small-business lending division with a €150 million loan portfolio (EAD = portfolio value, LGD = 60%). Spread income is 5.5% of the portfolio, fees are 0.5%, expected losses are PD x LGD x EAD with PD = 7%, operating costs are €1.25 million, and the tax rate is 25%. Capital at risk is the unexpected loss $\alpha \times \text{LGD} \times \text{EAD}$, where $\alpha = 5\sigma$ and the standard deviation of historical default rates is $\sigma = 1.5\%$. The bank's cost of capital is 12%. What should management do?

A. Keep the division; RAROC is 16.11%, above 12%.
B. Keep the division; RAROC is 21.48%, above 12%.
C. Close the division; RAROC is 9.67%, below 12%.
D. Close the division; RAROC is 7.78%, below 12%.

Answer: A

Key correction: Source key (B) said keep because RAROC 5.71% > 4.6% hurdle using a return-on-economic-capital variant; under the GARP/Schweser RAROC formula the data were rebuilt, so the stated answer no longer applies.

Step by step

1. Spread = 0.055 × 150 = 8.25m; fees = 0.005 × 150 = 0.75m; expected loss = 0.07 × 0.60 × 150 = 6.30m.
2. Pre-tax = 8.25 + 0.75 − 6.30 − 1.25 = 1.45m; after tax = 1.45 × 0.75 = 1.0875m.
3. $\alpha = 5 \times 1.5\% = 7.5\%$; capital at risk = 0.075 × 0.60 × 150 = 6.75m.
4. RAROC = 1.0875/6.75 = 16.11% > 12%: keep the division.

Why A is correct: RAROC (after-tax revenue net of expected loss and costs over unexpected-loss capital) is 16.11%, above the 12% cost of capital, so the division adds value.

Why the others are wrong

- **B:** Ignores tax: 1.45/6.75 = 21.48%; the decision is right but the RAROC is wrong.
- **C:** Leaves LGD out of capital at risk (0.075 x 150 = 11.25m): 1.0875/11.25 = 9.67%, wrongly below the hurdle.
- **D:** Omits the fee income: (8.25 - 6.30 - 1.25)(0.75)/6.75 = 7.78%, wrongly below the hurdle.

Q211

AnalystPrep Q.2734

REBUILT — Used a non-GARP RAROC variant (gross revenue, return on economic capital, transfer price); rebuilt with the GARP loan-revenue formula and unexpected-loss capital at risk.

A bank makes a €2,000,000 loan (EAD = loan value, LGD = 45%). The spread over the cost of funds is 0.9%, the origination fee is 0.4%, expected loan losses are 0.35% of the loan and operating costs are 0.25% of the loan. The tax rate is 25%. Capital at risk is the unexpected loss $\alpha \times \text{LGD} \times \text{EAD}$, with default rates assumed normal with standard deviation 2%, at the 99.5% confidence level ($\alpha = 2.6\sigma$). What is the loan's RAROC?

A. 29.91%
B. 10.10%
C. 22.44%
D. 33.65%

Answer: C

Key correction: Source key (A, 10%) used a return-on-economic-capital RAROC variant not in the GARP/Schweser formula; the data were rebuilt and the stated answer no longer applies.

Step by step

1. Pre-tax margin = 0.9% + 0.4% − 0.35% − 0.25% = 0.70%; revenue = 0.007 × 2,000,000 × 0.75 = 10,500.
2. $\alpha = 2.6 \times 2\% = 5.2\%$.
3. Capital at risk = 0.052 × 0.45 × 2,000,000 = 46,800.

4. RAROC = 10,500/46,800 = 22.44%.

Why C is correct: Revenue $(s + f - l - c)(1 - x)$ is divided by the unexpected loan loss $\alpha \times$ LGD $\times$ EAD.

Why the others are wrong

- **A:** Ignores tax: 14,000/46,800 = 29.91%.
- **B:** Omits LGD from capital at risk (0.052 x 2,000,000 = 104,000): 10,500/104,000 = 10.10%.
- **D:** Omits expected loan losses: (0.9% + 0.4% - 0.25%)(0.75)(2,000,000)/46,800 = 33.65%.

Q212

AnalystPrep Q.2964

REBUILT — Used a non-GARP RAROC variant (tax as expense, return on risk capital) with unrealistic numbers; rebuilt with GARP revenue formula and duration-based capital at risk.

A bank holds a €400 million loan with a duration of 4.2 years and an interest rate of 4.5%. Interest rates are expected to rise by 2% (0.02). Annual spread income is €11 million, fees are €2 million, expected losses are €3 million, operating costs are €2.5 million and the tax rate is 28%. What is the loan's RAROC?

A. 6.53%
B. 16.07%
C. 23.51%
D. 16.79%

Answer: D

Key correction: Source key (C, 0.6279) treated tax as a numerator deduction with a return-on-risk-capital add-back; under the GARP/Schweser formula the data were rebuilt, so the stated answer no longer applies.

Step by step

1. Revenue = $(11 + 2 - 3 - 2.5)(1 - 0.28) = 7.5 \times 0.72 = 5.40$m.
2. Capital at risk = $400 \times 4.2 \times 0.02/1.045 = 32.15$m.
3. RAROC = $5.40/32.15 = 16.79\%$.

Why D is correct: After-tax revenue is divided by the duration-based change in loan value, with modified duration dividing by (1 + i).

Why the others are wrong

- **A:** Multiplies pre-tax income by the tax rate (7.5 x 0.28 = 2.10) instead of $1 - x$: 2.10/32.15 = 6.53%.
- **B:** Omits the $1 + i$ divisor (denominator 33.6m): 5.40/33.6 = 16.07%.
- **C:** Leaves out expected losses: (11 + 2 - 2.5)(0.72)/32.15 = 23.51%.

Q213

AnalystPrep Q.6012

A senior analyst stresses that the denominator of the risk-adjusted return on capital (RAROC) ratio must be measured carefully. Which statement best describes the RAROC denominator?

A. The loan amount, which represents the revenue the loan can generate.
B. The capital at risk, i.e., the capital the bank needs to cover the risk of default or loss in value on the loan.
C. The bank's cost of capital, which is the benchmark for judging the loan's return.
D. The interest rate spread between the loan rate and the bank's cost of funds.

Answer: B

Step by step

1. RAROC = loan revenues / capital at risk.
2. Numerator: spread, fees, expected losses, operating costs and tax.
3. Denominator: capital at risk, either the change in loan value $\Delta L = -LD\Delta i/(1 + i)$ or the unexpected loss $\alpha \times$ LGD $\times$ EAD.
4. The bank's cost of capital is not in the ratio; it is the hurdle that RAROC is compared with.

Why B is correct: RAROC compares loan income with the capital that must be held against the loan's risk, so the denominator is the capital at risk.

Why the others are wrong

- **A:** The loan amount is not the denominator, and the spread and fees belong in the numerator; this confuses exposure with capital at risk.
- **C:** The cost of capital is the benchmark RAROC is compared against (loan is profitable if RAROC exceeds it), not a component of the ratio.
- **D:** The spread is one component of the numerator (revenue), not the denominator.

Q214

AnalystPrep Q.6016

REBUILT — Original left capital at risk and tax implicit; stated they are equal for both loans so exactly one option is defensible.

A bank compares two loans using RAROC. The loans have identical principal, spread, fees, tax rate and capital at risk. Loan X has expected loan losses of EUR 30,000 and operating costs of EUR 5,000. Loan Y has expected loan losses of EUR 20,000 and operating costs of EUR 10,000. Which statement is correct?

A. Loan X has the higher RAROC because its operating costs are lower.
B. Loan Y has the higher RAROC because its total of expected losses and operating costs is lower.
C. The loans have the same RAROC because the differences in losses and costs offset each other.
D. The ranking cannot be determined without the spread and fees.

Answer: B

Step by step

1. Revenue = $(s + f - l - c)(1 - x)$ over the same capital at risk.
2. Loan X: $l + c = 30{,}000 + 5{,}000 = 35{,}000$.
3. Loan Y: $l + c = 20{,}000 + 10{,}000 = 30{,}000$.
4. Spread, fees and tax are identical, so the lower total deduction (Y) gives higher revenue and higher RAROC.

Why B is correct: Expected losses and operating costs are both deducted from the numerator; Y deducts EUR 5,000 less in total, so its RAROC is higher.

Why the others are wrong

- **A:** Looks only at operating costs and ignores that X's EUR 10,000 higher expected loss outweighs its EUR 5,000 cost saving.
- **C:** Treats the two items as offsetting; they do not (35,000 vs 30,000).
- **D:** The common terms cancel in the comparison, so the ranking depends only on the differing items.

Q215

BT answers P2.T6.24.7.3 · BT notes P2.T6.24.7.3

REBUILT — Source had OCR-garbled inputs and an extra 'return on economic capital' item not in the GARP RAROC numerator; removed that item and treated the EUR 1m transaction cost as an operating cost, so the figures are recomputed.

A bank considers a 1-year loan of EUR 420 million with these terms:
- Loan rate 6% per annum; cost of funds 0.5% per annum
- Expected loss 3% of face value
- Operating costs 1% of face value, plus a one-off transaction cost of EUR 1 million
- Capital at risk (economic capital) 5% of the loan
- Tax rate 35%

The after-tax RAROC is closest to:

A. 16.4%
B. 22.9%
C. 25.2%
D. 0.8%

Answer: A

Step by step

1. Spread revenue = $(6\% - 0.5\%) \times 420 = 23.1$ m.
2. Deduct expected loss $3\% \times 420 = 12.6$ m, operating costs $1\% \times 420 = 4.2$ m and transaction cost 1.0 m: $23.1 - 12.6 - 4.2 - 1.0 = 5.3$ m pre-tax.
3. After tax: $5.3 \times (1 - 0.35) = 3.445$ m.
4. Capital at risk = $5\% \times 420 = 21$ m.
5. RAROC = $3.445/21 = 16.4\%$.

Why A is correct: RAROC is after-tax spread income net of expected loss and costs, divided by capital at risk (not by the loan amount).

Why the others are wrong

- **B:** Used the full 6% loan rate (25.2 m) instead of the spread over funding cost: $(25.2 - 12.6 - 4.2 - 1) \times 0.65/21 = 22.9\%$.
- **C:** Forgot to deduct tax: $5.3/21 = 25.2\%$.
- **D:** Divided by the loan amount rather than capital at risk: $3.445/420 = 0.8\%$.

Q216

BT answers P2.T7.24.11.3 · BT notes P2.T7.24.11.3

REBUILT — Source figures were internally inconsistent (projected 8.23%, actual -1.40%/-7.40% did not match the data) and the answer was a vague qualitative judgment; rebuilt as a numeric RAROC computation versus a hurdle rate.

A bank originates USD 100 million of loans. Per unit of loan value: expected revenue (spread plus fees) is 7%, expected loss is 3%, operating costs are 2.5% and the tax rate is 30%. Capital at risk is 40% of the loans. At year-end, operating costs turned out to be 3.5%, other items as expected. The bank's cost of capital is 2%. Which of the following correctly gives the projected RAROC, the actual RAROC, and the conclusion?

A. Projected 2.63%, actual 0.88%; the loan met the hurdle at origination but not in actuality.
B. Projected 3.75%, actual 1.25%; the loan met the hurdle at origination but not in actuality.
C. Projected 7.88%, actual 6.13%; the loan met the hurdle in both cases.
D. Projected 1.05%, actual 0.35%; the loan failed the hurdle in both cases.

Answer: A

Step by step

1. Projected after-tax income: $(7 - 3 - 2.5)\% \times (1 - 0.30) = 1.05\%$ of loans = USD 1.05 m.
2. Capital at risk $= 40\% \times 100 =$ USD 40 m, so projected RAROC $= 1.05/40 = 2.63\%$.
3. Actual: $(7 - 3 - 3.5)\% \times 0.70 = 0.35\%$, so $0.35/40 = 0.88\%$.
4. Compare with the 2% cost of capital: 2.63% > 2% (profitable); 0.88% < 2% (not profitable).

Why A is correct: RAROC divides after-tax net revenue by capital at risk; the higher operating costs cut it below the 2% hurdle.

Why the others are wrong

- **B:** Forgot the 30% tax: $1.5/40 = 3.75\%$, $0.5/40 = 1.25\%$.
- **C:** Omitted expected loss from the numerator: $4.5 \times 0.7/40 = 7.88\%$, $3.5 \times 0.7/40 = 6.13\%$.
- **D:** Divided the after-tax income by the loan amount (100) instead of capital at risk (40), giving 1.05% and 0.35%; then both fall below 2%.

Q217

BT answers P2.T7.24.12.1

REBUILT — Original used a Crouhy-style RAROC (adds a return on economic capital to the numerator); rebuilt to the GARP/Schweser convention, loan revenues = (s+f-l-c)(1-x) divided by capital at risk. Expected loss typo (7% -> 1%) also fixed.

Ant Commercial Bank is considering a loan that would be fully funded by deposits:
- Loan amount: $700 million
- Interest rate received on loan: 6.5%
- Interest rate paid on deposits (cost of funds): 0.5%
- Expected loss: 1.0% of face value of loan
- Annual operating costs: 2.0% of face value of loan
- Other transfer costs: 0.25% of face value of loan
- Capital at risk (economic capital): 20% of the loan amount
- Effective tax rate: 39%

Using the RAROC framework in which loan revenues = (spread + fees - expected losses - operating costs)(1 - tax rate), what is the after-tax RAROC for this loan?

A. 1.68%
B. 8.39%
C. 9.15%
D. 13.75%

Answer: B

Key correction: Source key (A, 10.83%) added a 4% pre-tax return on economic capital to the numerator and used a different cost structure; under the GARP/Schweser formula the correct answer is 8.39%.

Step by step

1. Spread $s = 6.5\% - 0.5\% = 6.0\%$; no fees, so $f = 0$.
2. Costs $c = 2.0\% + 0.25\% = 2.25\%$.
3. Pre-tax margin $= 6.0 - 1.0 - 2.25 = 2.75\%$; after tax $= 2.75\% \times 0.61 = 1.6775\%$.
4. Loan revenues $= 1.6775\% \times 700 =$ \$11.7425 million.
5. Capital at risk $= 20\% \times 700 =$ \$140 million.
6. RAROC $= 11.7425/140 = 8.39\%$.

Why B is correct: RAROC is after-tax risk-adjusted loan revenue divided by capital at risk, not by the loan amount.

Why the others are wrong

- **A:** Divided revenue by the $700 million loan instead of the $140 million capital at risk (11.7425/700).
- **C:** Left out the 0.25% transfer cost: (6.0-1.0-2.0)(0.61)(700)/140.
- **D:** Ignored the 39% tax: 2.75% x 700 / 140.

Q218

MK GARP PE1 Q73 · Mock 3 Q8

REBUILT — Original used a Crouhy-style RAROC (adds a return on economic capital to the numerator); rebuilt to the GARP/Schweser convention, loan revenues = (s+f-l-c)(1-x) divided by capital at risk.

A regional commercial bank is considering a 1-year loan fully funded by deposits:
- Loan amount: JPY 4.2 billion
- Interest rate paid on deposits: 0.4%
- Interest rate received on loan: 3.2%
- Expected loss: 2.0% of face value
- Annual operating costs: 0.5% of face value
- Capital at risk (economic capital): 10.0% of the loan
- Effective tax rate: 38%
- Other transfer costs: none

Using loan revenues = (s + f - l - c)(1 - x), what is the after-tax RAROC?

A. 0.19%
B. 1.86%
C. 3.00%
D. 14.26%

Answer: B

Key correction: Sources keyed B (2.73%), which adds a 1.4% return on economic capital to the numerator; under the GARP/Schweser formula the correct value is 1.86%.

Step by step

1. Spread $s = 3.2\% - 0.4\% = 2.8\%$.
2. Pre-tax margin $= 2.8 - 2.0 - 0.5 = 0.3\%$; after tax $= 0.3\% \times 0.62 = 0.186\%$.
3. Revenues $= 0.186\% \times 4{,}200 =$ JPY 7.812 million.
4. Capital at risk $= 10\% \times 4{,}200 =$ JPY 420 million.
5. RAROC $= 7.812/420 = 1.86\%$.

Why B is correct: Expected loss and operating costs absorb most of the 2.8% spread; the small after-tax remainder is measured against capital at risk.

Why the others are wrong

- **A:** Divided by the loan (4,200) instead of capital at risk (420).
- **C:** Ignored tax: 0.3% x 4,200 / 420.
- **D:** Omitted expected loss: (2.8-0.5)(0.62)(4,200)/420.

Q219

Mock 4 Q18

REBUILT — Original used a Crouhy-style RAROC (adds a return on economic capital to the numerator); rebuilt to the GARP/Schweser convention, loan revenues = (s+f-l-c)(1-x) divided by capital at risk. A 0.5% origination fee was added so that the fee term is tested.

A regional commercial bank is considering a loan fully funded by deposits:
- Loan amount: CNY 3.8 billion
- Interest rate paid on deposits: 0.6%
- Interest rate received on loan: 4.1%
- Origination fee: 0.5% of face value
- Expected loss: 3.0% of face value
- Annual operating costs: 0.3% of face value
- Capital at risk (economic capital): 15.0% of the loan
- Effective tax rate: 38%

What is the after-tax RAROC for this loan?

A. 0.43%
B. 0.83%
C. 2.89%
D. 4.67%

Answer: C

Step by step

1. Spread $s = 4.1\% - 0.6\% = 3.5\%$; fee $f = 0.5\%$.
2. Pre-tax margin $= 3.5 + 0.5 - 3.0 - 0.3 = 0.7\%$; after tax $= 0.7\% \times 0.62 = 0.434\%$.
3. Revenues $= 0.434\% \times 3{,}800 =$ CNY 16.492 million.
4. Capital at risk $= 15\% \times 3{,}800 = 570$ million.
5. RAROC $= 16.492/570 = 2.89\%$.

Why C is correct: Revenue includes the fee and is net of expected loss, costs and tax; it is divided by capital at risk.

Why the others are wrong

- **A:** Divided by the full loan (3,800) rather than capital at risk.
- **B:** Left out the 0.5% origination fee, giving a 0.2% margin: 0.2% x 0.62 x 3,800 / 570.
- **D:** Ignored tax: 0.7% x 3,800 / 570.

Q220

BT answers P2.T7.606.3 · BT notes P2.T7.606.3

REBUILT — Original used a Crouhy-style RAROC (adds a return on economic capital to the numerator); rebuilt to the GARP/Schweser convention, loan revenues = (s+f-l-c)(1-x) divided by capital at risk.

A bank has a $1.0 billion corporate loan portfolio earning 5.0% per annum, funded by $1.0 billion of retail deposits with a transfer-priced cost of 1.40%. Expected loss is 1.0% per annum ($10 million), direct operating cost is $6.0 million per annum, and the effective tax rate is 25.0%. Economic capital (capital at risk) of $80.0 million (8.0% of the loan amount) is required. Using loan revenues = (s + f - l - c)(1 - x), what is the RAROC?

A. 1.50%
B. 18.75%
C. 19.50%
D. 25.00%

Answer: B

Key correction: Sources keyed C (19.50%), which adds $0.8 million of interest earned on the economic capital invested in risk-free securities; under the GARP/Schweser formula that term is not included and the answer is 18.75%.

Step by step

1. Spread $s = 5.0\% - 1.4\% = 3.6\%$, i.e. $36 million.
2. Subtract expected loss $10m and operating cost $6m: pre-tax revenue $= 36 - 10 - 6 =$ $20 million.
3. After tax $= 20 \times 0.75 =$ $15 million.
4. RAROC $= 15/80 = 18.75\%$.

Why B is correct: After-tax revenue net of expected loss and costs, $15 million, is earned on $80 million of capital at risk.

Why the others are wrong

- **A:** Divided by the $1 billion loan instead of the $80 million capital at risk: 15/1,000.
- **C:** Added $0.8 million of risk-free return on equity capital to the numerator (20.8 x 0.75 / 80), which is not part of the GARP formula.
- **D:** Ignored the 25% tax: 20/80.

Q221

BT notes P2.T7.1.3

REBUILT — Original used a Crouhy-style RAROC; rebuilt to the GARP/Schweser duration-based capital-at-risk method. Added a tax rate and rate-change scenario so the data are self-contained.

A bank holds a $1,000 million loan with a duration of 3.0 years and an interest rate of 5.0%. The loan is funded by deposits costing 2.0%. Expected loss is 1.0% of the loan, operating costs are $6.0 million, there are no fees, and the tax rate is 25%. The bank defines capital at risk as the duration-approximated fall in loan value if interest rates rise by 2.0 percentage points. What is the loan's RAROC?

A. 17.50%
B. 18.38%
C. 24.50%
D. 31.50%

Answer: B

Step by step

1. Capital at risk $= -\Delta L = L \cdot D \cdot \Delta i/(1 + i) = 1{,}000 \times 3 \times 0.02/1.05 =$ $57.143 million.

2. Margin = (5.0 − 2.0) − 1.0 − 0.6 = 1.4% (operating costs = 6/1000 = 0.6%).
3. After tax = 1.4% × 0.75 = 1.05%; revenue = $10.5 million.
4. RAROC = 10.5/57.143 = 18.38%.

Why B is correct: Capital at risk is the duration-based loss in loan value, which includes the 1/(1+i) factor, and RAROC divides after-tax revenue by it.

Why the others are wrong

- **A:** Dropped the 1/(1+i) factor: capital at risk 1,000 x 3 x 0.02 = 60, so 10.5/60.
- **C:** Ignored tax: 14/57.143.
- **D:** Omitted expected loss: (3.0-0.6)(0.75)(10)/57.143 = 18/57.143.

Q222

BT notes P2.T7.1.4

REBUILT — Original used a Crouhy-style RAROC; rebuilt to the GARP/Schweser unexpected-loan-loss method (capital at risk = alpha x LGD x EAD). Source had no printed answer and used equity-return on capital; data were made self-contained.

A bank makes a $1,000 million loan at 6.0%, funded by deposits costing 4.8% (spread 1.2%). It also earns a 0.2% fee. Expected loss is 0.5% of the loan, operating costs are 0.3%, and the tax rate is 25%. Capital at risk is the unexpected loan loss $\alpha \times \text{LGD} \times \text{EAD}$, where LGD is 45%, EAD is $1,000 million, and $\alpha = 2.6\sigma$ with $\sigma = 2.5\%$ the standard deviation of historical default rates. What is the RAROC?

A. 6.92%
B. 15.38%
C. 20.51%
D. 40.00%

Answer: B

Step by step

1. $\alpha = 2.6 \times 2.5\% = 6.5\%$.
2. Unexpected loss = 0.065 × 0.45 × 1,000 = $29.25 million.
3. Margin = 1.2 + 0.2 − 0.5 − 0.3 = 0.6%; after tax = 0.6% × 0.75 = 0.45%; revenue = $4.5 million.
4. RAROC = 4.5/29.25 = 15.38%.

Why B is correct: The denominator is the unexpected loan loss, scaled by LGD and EAD with alpha = 2.6 sigma.

Why the others are wrong

- **A:** Omitted LGD from the denominator: 4.5/(0.065 x 1,000) = 6.92%.
- **C:** Ignored tax: 6/29.25.
- **D:** Used sigma (2.5%) instead of alpha (2.6 sigma): 4.5/(0.025 x 0.45 x 1,000).

Q223

BT answers P2.T7.24.13.1

REBUILT — Original used a Crouhy-style RAROC (adds a return on economic capital to the numerator); rebuilt to the GARP/Schweser convention, loan revenues = (s+f-l-c)(1-x) divided by capital at risk. Removed the CAPM beta adjustment (not in the 2026 LO) and kept the hurdle-rate decision.

A bank is considering a $200 million loan with these annual terms: interest earned on loan 10%, interest paid on deposits 4%, transfer cost 1% of loan, direct operating costs 0.8% of loan, expected loss 2.0%, effective tax rate 32%, and capital at risk of 50% of the loan. The bank's cost of capital (hurdle rate) is 6%. Using loan revenues = (s + f - l - c)(1 - x), which statement is correct?

A. RAROC is 8.43%, above the 6% hurdle rate, so the loan is profitable.
B. RAROC is 4.40%, below the 6% hurdle rate, so the loan is not profitable.
C. RAROC is 2.99%, below the 6% hurdle rate, so the loan is not profitable.
D. RAROC is 1.50%, below the 6% hurdle rate, so the loan is not profitable.

Answer: C

Key correction: Source key (B) used the CAPM-adjusted RAROC with a risk-free benchmark; adjusted RAROC is outside the GARP/Schweser LO, and the standard decision rule is RAROC versus cost of capital.

Step by step

1. Spread $s = 10\% - 4\% = 6\%$; costs $c = 1\% + 0.8\% = 1.8\%$.
2. Margin = 6 − 2 − 1.8 = 2.2%; after tax = 2.2% × 0.68 = 1.496%; revenue = $2.992 million.
3. Capital at risk = 50% × 200 = $100 million.
4. RAROC = 2.992/100 = 2.99% < 6%, so the loan is unprofitable.

Why C is correct: A loan is profitable only if RAROC exceeds the bank's cost of capital; here 2.99% is below 6%.

Why the others are wrong

- **A:** Used the full 10% loan rate instead of the spread (no deposit cost), which inflates the margin.
- **B:** Ignored tax: 2.2% x 200 / 100 = 4.40%.
- **D:** Divided by the $200 million loan instead of capital at risk: 2.992/200 = 1.5%.

Q224

Mock 1 Q35 · Schweser Module Quiz 59.1 Q2

REBUILT — Source RAROC used economic capital, a risk-free return on capital and a 60m business line (Mock 1) / OR-reading data (Schweser 59.1), which differ from the GARP Ch.5 loan RAROC convention; rebuilt as a duration-based loan RAROC with revenue (s + f - l - c)(1 - x).

A bank makes a $2,000,000 loan with the following characteristics. Spread between the loan rate and the bank's cost of funds: 0.80%. Fees attributable to the loan: 0.20%. Expected loan losses: 0.25%. Operating costs: 0.15%. Tax rate: 20%. The loan has a duration of 4.0 years and an interest rate of 8%, and interest rates are expected to rise by 1.0 percentage point over the one-year horizon. Using the change in loan value from the duration approximation as capital at risk, what is the loan's RAROC?

A. 12.00%
B. 16.20%
C. 12.96%
D. 18.36%

Answer: C

Key correction: Both source items used an economic-capital RAROC (revenues less costs and EL plus a risk-free return on capital, divided by capital); the keys (B = 17.27% and B = 10.03%) are not applicable to the GARP loan RAROC convention used in the rebuilt item.

Step by step

1. Revenue rate: $(s+f-l-c)(1-x) = (0.80\%+0.20\%-0.25\%-0.15\%)(1-0.20) = 0.60\% \times 0.80 = 0.48\%$.
2. Loan revenue: $0.0048 \times 2{,}000{,}000 = 9{,}600$.
3. Capital at risk: $|\Delta L| = L \times D \times \frac{\Delta i}{1+i} = 2{,}000{,}000 \times 4.0 \times \frac{0.01}{1.08} = 74{,}074$.
4. RAROC $= 9{,}600/74{,}074 = 12.96\%$.

Why C is correct: RAROC divides after-tax risk-adjusted loan revenue (spread plus fees less expected losses and operating costs, taxed) by the capital at risk, measured as the duration-implied fall in loan value, $LD\Delta i/(1+i)$.

Why the others are wrong

- **A:** 12.00% divides revenue by $LD\Delta i = 80{,}000$, forgetting to discount the rate change by $(1+i)$.
- **B:** 16.20% omits the tax adjustment: 0.60% x 2,000,000 = 12,000, divided by 74,074.
- **D:** 18.36% leaves out expected loan losses (uses 1.00% - 0.15% = 0.85%, taxed at 20%), overstating revenue.

Q225

Mock 1 Q35 · Schweser Module Quiz 59.1 Q2

REBUILT — Source used economic capital with a risk-free return on capital; rebuilt to the GARP Ch.5 convention where capital at risk is the unexpected loan loss α x LGD x EAD and revenue is (s + f - l - c)(1 - x).

A bank is evaluating a loan with exposure at default of $10,000,000 and loss given default of 40%. Spread over the bank's cost of funds: 0.60%. Fees: 0.25%. Expected loan losses: 0.20%. Operating costs: 0.15%. Tax rate: 25%. The bank measures capital at risk as the unexpected loan loss, using an unexpected default rate factor $\alpha = 5\%$ derived from historical default rates on similar loans. What is the loan's RAROC?

A. 7.50%
B. 25.00%
C. 26.25%
D. 18.75%

Answer: D

Step by step

1. Revenue rate: $(0.60\% + 0.25\% - 0.20\% - 0.15\%)(1 - 0.25) = 0.50\% \times 0.75 = 0.375\%$.
2. Loan revenue: $0.00375 \times 10{,}000{,}000 = 37{,}500$.
3. Capital at risk (unexpected loss): $\alpha \times \text{LGD} \times \text{EAD} = 0.05 \times 0.40 \times 10{,}000{,}000 = 200{,}000$.
4. RAROC $= 37{,}500/200{,}000 = 18.75\%$.

Why D is correct: With the historical-default approach, the RAROC denominator is the unexpected loan loss $\alpha \times$ LGD $\times$ EAD, and the numerator is after-tax revenue net of expected losses and operating costs.

Why the others are wrong

- **A:** 7.50% uses $\alpha \times \text{EAD} = 500{,}000$ as capital, leaving out LGD.

- **B:** 25.00% ignores tax: 0.50% x 10,000,000 = 50,000, divided by 200,000.
- **C:** 26.25% omits expected loan losses from the numerator: (0.85% - 0.15%) x 0.75 x 10,000,000 = 52,500, divided by 200,000.

Reading 24. Credit Scoring and Rating

Michalis Doumpos, Christos Lemonakis, Dimitrios Niklis, and Constantin Zopounidis, Analytical Techniques in the Assessment of Credit Risk: An Overview of Methodologies and Applications (Springer, 2019). Chapter 2.

LO	Learning objective	Qs
24.a	compare the credit scoring system to the credit rating system in assessing credit quality and describe the different types of each system.	5
24.b	differentiate between through-the-cycle credit rating system and point-in-time credit rating system.	8
24.c	describe the process for developing credit risk scoring and rating models.	5
24.d	describe rating agencies' assignment methodologies for issue and issuer ratings, and identify the main criticisms of the credit rating agencies' ratings.	7

LO 24.a

Compare the credit scoring system to the credit rating system in assessing credit quality and describe the different types of each system.

Q226

AP Mock 3 Q48

In comparing credit scoring and credit rating systems, an analyst must consider not just their output but also their purpose and audience. Which statement best distinguishes the typical application and audience of each?

A. Credit scoring systems are designed for public use and facilitate investors' assessments using ordinal grades, while credit rating systems provide numerical scores for internal use.

B. Credit scoring systems are crafted for internal decision-making by financial institutions using historical data and numerical scores, whereas credit rating agencies produce ordinal grades for public dissemination.

C. Credit scoring and credit rating systems both employ numerical values and ordinal grades interchangeably, serving both public and internal institutional purposes.

D. Both credit scoring and credit rating systems are exclusively used for public dissemination, offering guidance through a combination of automated models and expert judgment.

Answer: B

Step by step

1. Credit scoring: automated statistical models, numerical output, built and used internally by lenders (banks, corporate clients).
2. Credit rating: ordinal letter grades (AAA, AA, ...), built from models plus expert judgment, publicly available (e.g., S&P, Moody's, Fitch).
3. Only B pairs internal/numerical for scoring with public/ordinal for rating.

Why B is correct: Scores are numerical outputs of internal automated models; ratings are ordinal grades disseminated publicly.

Why the others are wrong

- **A:** Reverses the two systems: scoring is internal and numerical, ratings are public and ordinal.
- **C:** The outputs are not interchangeable; scoring gives numbers, rating gives ordinal grades, and audiences differ.
- **D:** Credit scores are used internally, not publicly disseminated, and scoring relies on automated models while judgment is characteristic of ratings.

Q227

AnalystPrep Q.5882

When considering credit scoring and credit rating systems, the methodologies and scope of each system are of particular interest to analysts. Which of the following correctly describes a characteristic that differentiates the scope of credit scoring systems from that of credit rating systems?

A. Credit scoring systems broadly evaluate the creditworthiness of entities across numerous markets, while credit rating systems are focused on individual consumers and small businesses.

B. Credit scoring systems commonly assess individual consumer and small business creditworthiness, whereas credit rating systems are often utilized to rate corporations, sovereigns, and structured finance products.

C. Both credit scoring and credit rating systems are primarily used for assessing the credit quality of large corporations and structured finance products, avoiding the consumer credit market.

D. Credit scoring systems only evaluate the creditworthiness of large corporate clients, while credit rating systems

have a more versatile application, including consumers, small businesses, and large enterprises.

Answer: B

Step by step

1. Scope of scoring: individual consumers and small businesses (high-volume lending).
2. Scope of rating: corporations, sovereigns and structured finance products (bonds, large corporate loans).
3. B matches this; A and D swap the populations; C wrongly says both avoid consumers.

Why B is correct: Scoring targets consumers and small businesses; ratings target large corporates, sovereigns and structured products.

Why the others are wrong

- **A:** Swaps the two scopes: ratings are not designed for consumers and small businesses.
- **C:** Scoring is mainly used for consumer and small-business credit, so it does not avoid the consumer market.
- **D:** Reverses the scopes: scoring is not limited to large corporates, and ratings are not typically applied to consumers.

Q228 AnalystPrep Q.5885

Credit rating systems incorporate varying elements into their evaluation of credit risk. When comparing credit scoring and credit rating systems, which feature is most typically associated with credit rating as opposed to credit scoring systems?

A. Utilizing automated models based solely on historical credit performance data to predict creditworthiness.

B. Evaluating broad categories such as management quality, industry risk, and competitive position to inform the rating.

C. Focusing on the assessment of individual consumers and small businesses using internal financial institutional data.

D. Regular updates based strictly on financial ratios and recent credit history

Answer: B

Step by step

1. Scoring is based on automated analytical models, using mostly quantitative historical data (A, D describe this).
2. Ratings combine analytical and judgmental assessment, including qualitative factors such as management quality, industry risk and competitive position.
3. Consumer/small-business focus with internal data (C) is scoring.
4. So B is the feature most associated with ratings.

Why B is correct: Ratings blend quantitative analysis with expert judgment on qualitative factors; scoring is automated and data-driven.

Why the others are wrong

- **A:** Automated, purely historical-data models describe credit scoring.
- **C:** Individual consumers and small businesses with internal institutional data describe scoring, not ratings.
- **D:** Updates driven strictly by ratios and recent credit history describe a mechanical scoring approach; ratings include qualitative judgment and are not purely ratio-driven.

Q229 BT answers P2.T6.24.8.1 · BT notes P2.T6.24.8.1

RiverBank, a commercial lender, is revamping its credit assessment procedures by introducing a Credit Scoring System (CSS) for personal and small business loans and a Credit Rating System (CRS) for larger corporate loans, bond issues, and municipal financing. This is in response to criticism of the previous method, which combined quantitative analytics with loan officer discretion and was deemed inconsistent and slow.

The bank aims to:

- Minimize subjective judgment in loan approvals via CSS.
- Use both systems for robust risk management, including scenario analyses and stress testing.
- Ensure consistency and transparency across customer evaluations using CRS.
- Enhance loan appraisal efficiency using CSS to curtail time and costs.

Which of the following BEST describes the distinct advantages of implementing the Credit Scoring System (CSS) and Credit Rating System (CRS) at RiverBank?

A. CSS and CRS will eliminate the need for loan officers, reducing personnel costs, while increasing market transparency with public credit evaluations.

B. The CSS will enable fast personal loan evaluations through automated numerical scoring, while the CRS will

facilitate standardized risk reporting for large-scale borrowers, attracting external stakeholders.

C. CSS will provide a publicly accessible rating for individual borrowers, increasing market competition, and CRS will replace existing risk management systems to streamline the process.

D. Both CSS and CRS aim to automate the lending process, removing qualitative assessments and reducing risk management's role in loan approvals.

Answer: B

Step by step

1. CSS is meant for personal and small-business loans: automated numerical scoring cuts subjectivity, time and cost.
2. CRS is meant for large corporate, bond and municipal borrowers: a common framework gives consistency, transparency and standardized reporting to external stakeholders.
3. Both systems still support scenario analysis and stress testing; neither removes human oversight entirely.
4. B alone states the distinct advantages correctly.

Why B is correct: Scores are fast, automated and internal for retail loans; ratings give a standardized, transparent framework for large borrowers and external users.

Why the others are wrong

- **A:** Systems reduce subjectivity but do not eliminate loan officers, and CSS is internal, not a public evaluation.
- **C:** Credit scores are internal, not publicly accessible, and CRS supplements rather than replaces risk management systems.
- **D:** Rating involves analytical plus judgmental assessment, so qualitative input is not removed, and risk management's role is not reduced.

Q230 Schweser Module Quiz 24.1 Q1

ABC Bank needs to conduct a risk assessment for a large manufacturing firm that is publicly traded. Which of the following statements regarding this assessment is correct? The bank should use a:

A. credit score to reduce subjectivity.

B. credit score to enhance transparency.

C. credit rating to reduce the cost of appraisal.

D. credit rating despite the limitations for stress testing.

Answer: C

Step by step

1. A large, publicly traded manufacturer is a corporate borrower, so a credit rating (not a score) is the appropriate tool.
2. That eliminates A and B.
3. Benefits of rating systems include reduced subjectivity, scenario analysis/stress testing capability, transparency and reduced time and cost of appraisal.
4. C cites a true benefit; D is false because ratings do enable stress testing.

Why C is correct: Large listed firms are rated, and a recognized benefit of rating systems is lower time and cost of appraisal.

Why the others are wrong

- **A:** A score is for individuals and small businesses, not a large public manufacturer.
- **B:** Scores are for smaller borrowers; the right tool here is a rating.
- **D:** Rating is right, but the claimed limitation is false: rating/scoring systems enable scenario analysis and stress testing.

LO 24.b

Differentiate between through-the-cycle credit rating system and point-in-time credit rating system.

Q231 AnalystPrep Q.5388

REBUILT — Original stem was ambiguous about whether a 2009-2024 benign-period migration matrix is TTC or PIT and had a 2023/2025 typo; rebuilt as a self-contained TTC vs PIT comparison in a severe recession.

Bank A uses a through-the-cycle (TTC) rating system and Bank B uses a point-in-time (PIT) rating system. Both banks lend to identical borrowers, and all borrowers were rated at the start of the year when economic conditions were benign. During the year the country enters a severe recession. Which of the following is the most likely observation about rating changes at year-end?

A. Bank A's ratings show substantially more downgrades than Bank B's, because TTC ratings react most quickly to the recession.

B. Bank B's ratings show substantially more downgrades than Bank A's, because PIT ratings reflect current conditions while TTC ratings are robust to short-term changes.

C. Both banks' ratings show the same downgrades, because the borrowers and their fundamentals are identical.

D. Bank A's ratings show more downgrades than Bank B's, because TTC ratings look ahead across a full cycle and anticipate the recession.

Answer: B

Step by step

1. TTC ratings have a long-term orientation covering at least one business cycle, so they are robust to short-term changes and updated less frequently.
2. PIT ratings focus on the current, short-term condition of the borrower (usually within one year), are more volatile and are updated promptly as new information arrives.
3. A severe recession is a short-term shock to current conditions, so Bank B's PIT ratings migrate down sharply while Bank A's TTC ratings change little.

Why B is correct: PIT ratings respond quickly to current conditions, while TTC ratings are designed to be stable through the cycle, so the recession shows up mainly in the PIT bank's downgrades.

Why the others are wrong

- **A:** Reverses the two philosophies: TTC ratings are the stable, slow-moving ones.
- **C:** Identical borrowers do not imply identical ratings; the rating philosophy determines how much current conditions feed through.
- **D:** Confuses the long horizon of TTC with forecasting a downturn; TTC looks through the cycle and deliberately does not move with the recession.

Q232 AnalystPrep Q.5886

Which of the following best characterizes a through-the-cycle (TTC) credit rating approach when compared to a point-in-time (PIT) approach?

A. TTC assessments react swiftly to short-term economic fluctuations and are frequently updated to reflect the current credit condition of the borrower.

B. PIT ratings take a long-term perspective, often encompassing a full business cycle, and aim to remain stable against economic ups and downs.

C. TTC ratings have a long-term view and tend to be less sensitive to short-term economic conditions than PIT ratings.

D. Both TTC and PIT ratings prioritize near-term economic indicators and are subject to frequent revisions.

Answer: C

Step by step

1. TTC: long-term orientation covering at least one business cycle, robust to short-term changes, less frequent updates.
2. PIT: current and short-term condition (about one year or less), more volatile, updated promptly.
3. Only C attributes the long-term, less sensitive profile to TTC.

Why C is correct: TTC looks across the whole cycle, so it is stable and less sensitive to short-term conditions than PIT.

Why the others are wrong

- **A:** Describes PIT, not TTC.
- **B:** Swaps the labels: the long-term, stable profile belongs to TTC.
- **D:** Wrong for TTC; only PIT emphasizes near-term indicators and frequent revision.

Q233 AnalystPrep Q.5887

For an investor with a short-term trading strategy, which of the following credit rating systems would likely be more informative?

A. A TTC rating system that provides a long-term view of credit risk based on a borrower's performance across business cycles.

B. A PIT rating system that reflects the borrower's current and near-term credit risk and adjusts quickly to market and economic changes.

C. A credit rating system that remains static over time, providing consistency rather than responsiveness.

D. A hybrid system that combines TTC and PIT elements and is updated semi-annually.

Answer: B

Step by step

1. Short horizon means current credit condition matters most.
2. PIT ratings capture current, short-term condition and are updated promptly.
3. So the PIT system (B) is more informative.

Why B is correct: PIT ratings are timely and reflect current conditions, which is what a short-horizon investor needs.

Why the others are wrong

- **A:** TTC smooths through the cycle and is slow to reflect current deterioration, so it is less useful over a short horizon.
- **C:** A static system by definition ignores new information.
- **D:** A semi-annual update is not timely enough and is less responsive than PIT.

Q234 AnalystPrep Q.5888

Which of the following describes a scenario where a financial institution might prefer a TTC credit rating system over a PIT system?

A. It wants to reflect real-time changes in credit risk in its trading book due to market dynamics.

B. It seeks a stable credit outlook for its loan portfolio even when there are short-term fluctuations in the economy.

C. It needs to adjust its credit risk models frequently to respond to immediate macroeconomic news affecting specific borrowers.

D. It needs daily credit risk assessment for speculative trading strategies requiring quick decisions.

Answer: B

Step by step

1. TTC ratings are robust to short-term changes and updated less often, and suit long-term loans.
2. Choice B wants stability despite short-term swings, matching TTC.
3. A, C and D all want rapid responsiveness, which is the PIT feature.

Why B is correct: TTC delivers stable, cycle-neutral ratings suited to long-term loan books.

Why the others are wrong

- **A:** Real-time reflection of conditions is a PIT feature.
- **C:** Frequent adjustment to immediate news is PIT behavior.
- **D:** Daily, short-horizon assessment for speculative trading is PIT use.

Q235 AnalystPrep Q.5889

Which of the following statements is true regarding the frequency of updates in credit rating systems?

A. TTC rating systems are updated more frequently than PIT systems to give a timely reflection of economic conditions.

B. PIT rating systems are typically updated less frequently, coinciding with significant shifts in the business cycle.

C. PIT rating systems are updated more frequently than TTC systems to incorporate current conditions and short-term trends.

D. Both TTC and PIT systems are updated on a fixed annual schedule regardless of economic or borrower changes.

Answer: C

Step by step

1. TTC ratings are robust to short-term changes and subject to less frequent updates.
2. PIT ratings are updated in a timely manner to reflect new information on the borrower.
3. Hence PIT is updated more often than TTC.

Why C is correct: PIT ratings track current information and so are updated more often than the slow-moving TTC ratings.

Why the others are wrong

- **A:** Reverses the frequencies.
- **B:** Describes TTC update behavior but attributes it to PIT.
- **D:** Neither system is defined by a fixed annual schedule; PIT updates as information arrives.

Q236

AnalystPrep Q.5890

Which of the following might be a consequence of relying on a TTC credit rating system for external reporting compared to a PIT system?

A. Stakeholders might perceive an overstatement of credit risk during downturns because of the system's sensitivity to short-term fluctuations.

B. External reports would likely show less volatility in credit risk metrics over time, giving a more consistent long-term perspective.

C. Financial reports might display rapid changes in credit risk assessments that mirror the market's perception during economic instability.

D. Reported credit risk may change frequently, requiring frequent updates to external communication and possibly confusing stakeholders.

Answer: B

Step by step

1. TTC ratings are robust to short-term changes and updated less often, so they are less volatile.
2. PIT ratings are more volatile and updated promptly.
3. Less volatile reported metrics (B) follow from TTC.

Why B is correct: Because TTC ratings are stable through the cycle, reported credit risk metrics are smoother and more consistent.

Why the others are wrong

- **A:** Sensitivity to short-term fluctuations is a PIT trait.
- **C:** Rapid, market-mirroring changes are characteristic of PIT.
- **D:** Frequent changes in reported risk are a PIT consequence.

Q237

BT notes P2.T6.24.8.2 · BT answers P2.T6.24.8.2

MatrixBank uses a through-the-cycle (TTC) rating system to evaluate borrower creditworthiness. Due to recent economic volatility, it is considering supplementing it with a point-in-time (PIT) system, and the board is divided. The CRO will compare the two systems using two borrowers: Firm A, a long-standing client with cyclically stable revenues, and Firm B, a tech startup with rapid growth but facing current market uncertainty. Which of the following BEST reflects the distinctive features and appropriate application of TTC and PIT systems?

A. TTC is more suitable for Firm B, giving a stable outlook that mitigates the startup's uncertain growth; PIT best serves Firm A by frequently updating its rating in line with stable market conditions.

B. PIT is preferred for both firms because real-time risk assessment is critical for all borrowers, regardless of revenue stability.

C. TTC should be applied to Firm A, given its stable long-term revenue and a consistent rating less affected by market fluctuations; PIT is ideal for Firm B, reflecting the immediate environment affecting the startup's rapidly changing situation.

D. MatrixBank should use TTC exclusively for all clients because its long-term stability suits a conservative strategy, despite the recent volatility.

Answer: C

Step by step

1. TTC: long-term, cycle-spanning, stable, less frequently updated; fits a borrower with stable long-run fundamentals (Firm A).
2. PIT: current, short-term, volatile, promptly updated; fits a borrower whose condition is changing quickly (Firm

B).
3. C matches each system to the borrower it suits.

Why C is correct: TTC suits the stable long-term profile of Firm A while PIT captures the fast-changing, uncertain position of Firm B.

Why the others are wrong

- **A:** Swaps the roles: it uses stable TTC for the rapidly changing firm and frequent PIT updates for the stable firm.
- **B:** Ignores that TTC is appropriate for borrowers with stable long-term fundamentals; PIT is not best for every borrower.
- **D:** Ignores the value of PIT responsiveness, which is exactly what the board wants to add given recent volatility.

Q238 Schweser Module Quiz 24.1 Q2

A lender is considering using a through-the-cycle approach when assessing the creditworthiness of a potential borrower. Which of the following statements regarding this approach is correct? This method:

A. will capture default risk in the best way possible.
B. should be used if the loan is short term in nature.
C. should be used if the loan is long term in nature.
D. is a robust option because it relies on internal data only.

Answer: C

Step by step

1. TTC covers an entire business cycle and is long term in nature.
2. Therefore it is best suited to long-term loans; PIT suits short-term loans.
3. Only C is consistent with this.

Why C is correct: A TTC horizon spans a full business cycle, so it matches the long maturity of long-term loans.

Why the others are wrong

- **A:** Real-time default risk is captured better by PIT, not TTC.
- **B:** Short-term loans are better suited to PIT.
- **D:** TTC is used mainly by rating agencies and is not defined by reliance on internal data; internal bank systems tend to be PIT.

LO 24.c

Describe the process for developing credit risk scoring and rating models.

Q239 AnalystPrep Q.5892

REBUILT — Original had giveaway distractors (stock market trends, one-time process); rewrote the stem and options so each wrong option reflects a realistic misconception about the implementation stage.

A bank has completed the fitting and out-of-sample validation of a new corporate credit rating model and has mapped its scores to rating classes. It is now moving to the final stage of the development process, implementation. Which of the following statements best describes how the system's effectiveness is ascertained once it is in use?

A. Because the model passed validation, no further testing is needed; the system only needs to be re-examined if the bank changes its business strategy.
B. The system is continuously monitored, reviewed and recalibrated as needed, with the focus on identifying any degradation in the quality of the risk estimates it produces.
C. Effectiveness is judged by benchmarking the system against rating agency ratings and adjusting it until it reproduces those external ratings exactly.
D. Effectiveness is judged by re-running the fitting step on the original training sample each period to confirm that the estimated coefficients have not changed.

Answer: B

Step by step

1. The development process has five steps: data collection and preprocessing, model fitting, model validation, definition and validation of the risk rating, and implementation, monitoring and review.
2. In implementation the system produces real-time risk estimates for new loan applications.

3. The system must be monitored and reviewed continuously and recalibrated as required; the risk management focus is detecting degradation in the quality of the risk estimates.
4. Therefore B is correct.

Why B is correct: Implementation is not a one-time event: ongoing monitoring, review and recalibration are part of the final stage, aimed at detecting deterioration in the quality of risk estimates as conditions change.

Why the others are wrong

- **A:** Wrong: passing validation does not end the process; monitoring is inherent to the implementation stage, and degradation can trigger re-evaluation. Confuses validation with ongoing review.
- **C:** Wrong: benchmarking is an alternative validation technique, and its goal is to identify and understand deviations from external ratings, not to replicate them exactly.
- **D:** Wrong: re-fitting on the same training sample says nothing about performance on new applications; monitoring assesses live results, not stability of in-sample coefficients.

Q240

AnalystPrep Q.5894

REBUILT — Original was a vague 'key checkpoint' item with weak distractors; rebuilt as a scenario testing out-of-sample plus out-of-time validation against GARP's description.

An analyst fits a default prediction model using borrower data through 2019. To assess the model's expected predictive performance and its robustness over time, which validation approach is most appropriate?

A. Measure the model's fit on the same 2019-and-earlier borrowers used to estimate the coefficients, because these cases are known to contain the correct default outcomes.

B. Apply the model to a holdout of borrowers not used in fitting but drawn from the same period (out-of-sample test only).

C. Apply the model to different borrowers in later periods (e.g., 2020, 2021), ideally within a walk-forward scheme, so that out-of-sample and out-of-time tests are combined.

D. Compare the model's ratings with rating agency ratings and recalibrate until the two agree exactly for every borrower.

Answer: C

Step by step

1. Validation checks the fitted model on data not used in fitting, using the same risk attributes but different cases.
2. Out-of-sample tests apply the model to borrowers not in the build sample; out-of-time tests apply it to a different period than the one used for fitting.
3. Because models are built at one point and used in future periods, both dimensions matter; ideally the tests are combined, and walk-forward testing (rolling the window forward) makes results more robust.
4. Option C does this.

Why C is correct: Validation must use different borrowers and different time periods than fitting, so combining out-of-sample and out-of-time (walk-forward) tests best measures real-world predictive performance and robustness.

Why the others are wrong

- **A:** Wrong: in-sample fit measures how well the model describes the training data, not how well it predicts; it ignores overfitting.
- **B:** Wrong: an out-of-sample test alone ignores the time dimension; the model's robustness over time is only verified by out-of-time tests, which should ideally be combined with it.
- **D:** Wrong: benchmarking against external ratings is a qualitative alternative to backtesting; its aim is to identify and explain discrepancies, not to force exact agreement.

Q241

Schweser Module Quiz 24.1 Q3

A risk analyst is considering the best method to generate a credit score that updates in near real time while considering the payment and purchase history of a customer. Which of the following methods for developing credit risk scoring and rating models should he use?

A. Behavioral scoring.

B. Social lending scoring.

C. Account-level profit scoring.

D. Customer-level profit scoring.

Answer: A

Step by step

1. The requirement has two parts: scores must be refreshed frequently and must be driven by observed payment and purchase behavior of existing customers.
2. Behavioral scoring (consumer lending) monitors existing customers' payment and purchase behavior dynamically and is updated frequently (usually monthly), unlike static application scores.
3. Profit scoring instead estimates the profitability of an account or customer, and social (P2P) lending scoring addresses online lending platforms; neither is defined by dynamic behavioral updating.
4. Choose behavioral scoring.

Why A is correct: Behavioral scoring dynamically tracks existing customers' payment and purchase behavior and provides frequently updated default-risk estimates.

Why the others are wrong

- **B:** Social lending scoring adapts models to peer-to-peer platforms (big data, no collateral); it is not the method defined by dynamic payment and purchase monitoring of existing customers.
- **C:** Account-level profit scoring measures profit on a single financial product, ignoring the customer's wider behavior; its focus is profitability, not updated default risk.
- **D:** Customer-level profit scoring aggregates profit across all of a customer's accounts; it estimates profitability rather than refreshing behavior-based risk estimates.

Q242 Schweser Module Quiz 24.2 Q1

Moody's is considering a revision to its credit rating model. When it checked the model derived from model training data against out-of-sample data, it found a few issues to address. Which of the following steps in the credit rating development process was Moody's likely on when it noticed this issue?

A. Model fitting.
B. Preprocessing.
C. Model validation.
D. Validation of the risk rating.

Answer: C

Step by step

1. The development process has five steps: (1) data collection and preprocessing, (2) model fitting, (3) model validation, (4) definition and validation of the risk rating, (5) implementation.
2. Fitting uses the training data to estimate the model; validation then tests the fitted model on a separate out-of-sample (holdout) data set with the same risk attributes.
3. Testing the fitted model against out-of-sample data is therefore step 3, model validation.

Why C is correct: Applying a model derived from training data to a separate out-of-sample data set is the definition of the model validation step.

Why the others are wrong

- **A:** Model fitting builds the model on the training data; the issue here was found by testing on different data, after fitting.
- **B:** Preprocessing is data collection and cleaning that precedes any modeling, so no model exists to check yet.
- **D:** Validation of the risk rating (step 4) checks that the rating classes map to empirical PDs and are well dispersed; it comes after the model itself has been validated.

Q243 BT answers P2.T6.702.2

REBUILT — Original tested De Laurentis rating-system features (monotonic, specificity) that are not in Schweser or GARP Ch.24; rebuilt on the Basel specification requirements for scoring/rating systems covered in both.

Which of the following statements about the Basel Committee's specification requirements for a credit scoring or rating system is correct?

A. Credit grades should be designed so as to manage regulatory capital requirements.
B. Borrowers in the same credit grade must be treated identically regardless of transaction-level differences.
C. Borrower ratings should be updated on a reasonable basis, in practice no less frequently than an annual review.
D. Borrower ratings need be reviewed only when a default or downgrade event occurs.

Answer: C

Step by step

1. Basel's five requirements: meaningful differentiation of risk, continuous evaluation of borrowers, operational oversight, correct selection of risk assessment attributes, collecting a substantial database.
2. Continuous evaluation means ratings are updated on a reasonable basis, in practice at least an annual rerating review (statement C).
3. Meaningful differentiation requires grades based on risk differences, not on managing regulatory capital, and allows borrowers in the same grade to be treated differently on transaction-level differences (A and B are false).
4. Event-driven-only review contradicts continuous evaluation (D false).

Why C is correct: The continuous-evaluation requirement obliges institutions to update borrower ratings regularly, at least annually.

Why the others are wrong

- **A:** Reverses the meaningful-differentiation requirement: grades must reflect risk differences, not be an attempt to manage regulatory capital.
- **B:** Reverses the rule: borrowers in the same grade may be treated differently because of transaction-level differences.
- **D:** Ratings must be reviewed periodically (at least annually), not only after a default or downgrade trigger.

LO 24.d

Describe rating agencies' assignment methodologies for issue and issuer ratings, and identify the main criticisms of the credit rating agencies' ratings.

Q244

AnalystPrep Q.1778

REBUILT — Original rule of thumb was vague and mixed debt with cash-flow margins; rebuilt as a concrete comparison of profitability and solvency ratios per GARP.

A rating analyst compares four otherwise similar firms using financial ratios. Which firm's ratios would, other things equal, normally support the strongest credit rating?

A. Higher operating profit margins and a lower debt-to-equity ratio

B. Higher operating profit margins and a higher debt-to-equity ratio

C. Lower operating profit margins and a lower debt-to-equity ratio

D. Lower operating profit margins and a higher debt-to-equity ratio

Answer: A

Step by step

1. Profitability ratios (e.g., profit margins) measure the ability to generate profit from assets and capital: higher is better.
2. Solvency ratios (debt to assets, debt to equity) measure debt burden: lower is better.
3. The firm that is best on both dimensions is the one with higher margins and lower debt-to-equity: A.

Why A is correct: Stronger profitability and a lighter debt burden both indicate a safer financial structure and so a better rating.

Why the others are wrong

- **B:** Strong margins but the heavier debt burden raises default risk; treats leverage as a plus.
- **C:** Low debt helps, but weak margins mean weaker ability to generate profit and service debt.
- **D:** Worst on both dimensions: weak profitability and high leverage give the weakest rating.

Q245

AnalystPrep Q.5896

REBUILT — Original wording was circular (answer restated the question); rebuilt to test issuer versus issue-specific ratings.

Which statement correctly distinguishes an issuer credit rating from an issue-specific rating?

A. An issuer rating reflects the terms, collateral and use of a specific loan, whereas an issue-specific rating is a general view of the entity.

B. An issuer rating is a forward-looking opinion on an obligor's overall creditworthiness and can help the entity access capital markets or funding, whereas an issue-specific rating also considers the terms, collateral and use of a specific obligation.

C. An issue-specific rating ignores collateral and loan terms and always equals the issuer rating.

D. An issuer rating is a backward-looking summary of historical default statistics used only for regulatory capital.

Answer: B

Step by step

1. Issuer credit rating: forward-looking opinion on an obligor's creditworthiness; a good rating improves access to capital markets/funding and lowers financing costs.
2. Issue-specific rating: opinion on the obligor's ability to meet a specific obligation, considering duration, collateral, use of the loan and other risk features of that debt.
3. Only B states both correctly.

Why B is correct: Issuer ratings concern the entity; issue ratings concern a particular obligation and incorporate its specific terms.

Why the others are wrong

- **A:** Swaps the definitions of the two rating types.
- **C:** Issue ratings do consider collateral and terms, so they can differ from the issuer rating.
- **D:** Issuer ratings are forward-looking, and their uses are broader than regulatory capital.

Q246

AnalystPrep Q.5897

REBUILT — Original had a giveaway distractor ('Subprime'); rebuilt to test the investment/speculative boundary.

On the Moody's / S&P scale, which rating is the lowest category still classified as investment grade?

A. Baa / BBB

B. Ba / BB

C. A / A

D. Caa / CCC

Answer: A

Step by step

1. Investment grades: Aaa/AAA, Aa/AA, A/A, Baa/BBB.
2. Speculative grades begin at Ba/BB and run down through B, Caa/CCC, Ca/CC, C and D.
3. The lowest investment-grade category is Baa/BBB.

Why A is correct: The investment/speculative boundary lies between Baa/BBB and Ba/BB.

Why the others are wrong

- **B:** Ba/BB is the highest speculative category, one step below the boundary.
- **C:** A/A is investment grade but two categories above the lowest.
- **D:** Caa/CCC is poor quality with high default likelihood, deep in speculative territory.

Q247

AnalystPrep Q.5898

Credit rating agencies face various criticisms regarding their methodologies and practices. Which of the following criticisms points out a potential concern regarding the transparency of the rating process?

A. Ratings may not be as proactive as desired and are often downgraded after a financial downturn has become apparent.

B. CRAs are compensated by the entities they rate, leading to potential biases in the ratings provided.

C. Methodologies and factors influencing the final rating are not always made clear to investors or the entities being rated.

D. Ratings can lead to herding behavior among investors, exacerbating financial market volatility.

Answer: C

Step by step

1. Transparency criticism: the way CRAs prepare ratings is generally not publicly disclosed.
2. Match each option to its criticism; only C is about disclosure of methodology.

Why C is correct: Lack of transparency means the rating process and drivers are not disclosed.

Why the others are wrong

- **A:** Describes poor predictive ability/timeliness, not transparency.
- **B:** Describes conflict of interest (issuer-pays), a related but distinct criticism.
- **D:** Describes herding/market volatility, not a transparency issue.

Q248 AnalystPrep Q.5900

In relation to credit rating agencies, the aspect of "conflict of interest" has been a persistent concern. What does this criticism entail?

A. CRAs face inherent conflicts because they are publicly traded entities, with obligations to maximize shareholder value over rating accuracy.

B. CRAs have a conflict of interest due to the fact that they are typically compensated by the entities they rate, which may lead to biased ratings.

C. CRAs are paid by the investors who use the ratings, which gives them an incentive to publish overly pessimistic ratings.

D. CRAs are conflicted as they have to balance the interests of rated entities with those of investors using the ratings for investment decisions.

Answer: B

Step by step

1. The conflict-of-interest criticism: CRA revenues depend on fees paid by the entities being rated.
2. This creates an incentive for favorable ratings; B states this directly.

Why B is correct: Issuer-pays fees give CRAs an incentive to bias ratings toward the rated entity.

Why the others are wrong

- **A:** Ownership structure is not the conflict of interest GARP identifies; the conflict arises from fees paid by rated entities.
- **C:** Reverses the fee flow: CRA revenues depend on fees paid by the entities being rated, not by investors, and the bias incentive is toward favorable ratings.
- **D:** Too vague; it does not identify the revenue-model source of the conflict.

Q249 BT answers P2.T6.24.8.3 · BT notes P2.T6.24.8.3

REBUILT — Original had three obviously absurd distractors (CRAs 'praised'); rebuilt as a 'which is NOT a criticism' item using the five GARP criticisms.

Credit rating agency (CRA) ratings have attracted several well-known criticisms. Which of the following is NOT one of the main criticisms of CRA ratings?

A. Rating processes are not publicly disclosed, and fees paid by rated entities create conflicts of interest.

B. Ratings were too optimistic during economic expansions and too pessimistic during recessions.

C. Ratings are overly conservative, restricting debt availability and discouraging lending.

D. Ratings have been weaker predictors of corporate default than accounting- and market-based models.

Answer: C

Step by step

1. GARP's main criticisms: lack of transparency and accountability together with conflict of interest (fees paid by rated entities), poor predictive ability, pro-cyclicality, and promoting debt explosion.
2. A = transparency/conflict of interest; B = pro-cyclicality; D = poor predictive ability.
3. C is the opposite of the debt-explosion criticism (ratings made debt more accessible, feeding credit bubbles), so it is not a criticism.

Why C is correct: The criticism is that ratings and scoring made debt too accessible, not that they restrict it.

Why the others are wrong

- **A:** This is a genuine criticism (transparency and conflict of interest).
- **B:** This is the pro-cyclicality criticism.
- **D:** This is the poor-predictive-ability criticism (e.g., Lehman, Enron, 2007-08 crisis).

Q250 AnalystPrep Q.1777

Nowadays, rating agencies aim to provide an accurate credit analysis of every party. Their main purpose is to surmount the problem of information asymmetry. What does the term "information asymmetry" refer to?

A. The availability of complete and transparent information to market participants for accurate evaluation of the other party's default risk.

B. The unavailability of complete and transparent information to market participants for accurate evaluation of a counterparty's default risk.

C. The unavailability of complete and transparent information to governments regarding business entities.

D. The unavailability of complete information for use by internal and external auditors so as to accurately evaluate

an entity's chances of winding up.

Answer: B

Step by step

1. Information asymmetry: one party (the borrower) knows more about its credit quality than the other party.
2. Market participants therefore cannot fully evaluate counterparty default risk; ratings bridge this gap.
3. B states exactly this.

Why B is correct: Lack of complete, transparent information on a counterparty's default risk is the asymmetry that ratings address.

Why the others are wrong

- **A:** States the opposite: complete information would mean no asymmetry.
- **C:** Restricts the problem to governments; the issue is faced by market participants.
- **D:** Restricts it to auditors and winding-up chances, which is not the credit-evaluation problem.

Reading 25. Credit Scoring and Retail Credit Risk Management

Michel Crouhy, Dan Galai, and Robert Mark, The Essentials of Risk Management, 2nd Edition (New York, NY: McGraw-Hill, 2014). Chapter 9.

LO	Learning objective	Qs
25.a	analyze the credit risks and other risks generated by retail banking.	5
25.b	explain the differences between retail credit risk and corporate credit risk.	4
25.c	discuss the "dark side" of retail credit risk and the measures that attempt to address the problem.	5
25.d	define and describe credit risk scoring model types, key variables, and applications.	3
25.e	discuss the key variables in a mortgage credit assessment and describe the use of cutoff scores, default rates, and loss rates in a credit scoring model.	7
25.f	discuss the measurement and monitoring of a scorecard performance including the use of cumulative accuracy profile (CAP) and the accuracy ratio (AR) techniques.	3
25.g	describe the customer relationship cycle and discuss the trade-off between creditworthiness and profitability.	4
25.h	discuss the benefits of risk-based pricing of financial services.	6

LO 25.a

Analyze the credit risks and other risks generated by retail banking.

Q251

AnalystPrep Q.2009

Retail banking serves small businesses and consumers who finance property and small projects. Although most retail facilities are secured, some are not. Which of the following retail loans is normally unsecured?

A. Home mortgage loans
B. Home equity loans
C. Credit card revolving loans
D. Small business loans

Answer: C

Step by step

1. Classify each product by collateral.
2. Home mortgages and home equity loans are secured by residential property.
3. Small business loans are secured by business assets or the owners' personal guarantees.
4. Credit card revolving loans have no collateral, so they are the unsecured product.

Why C is correct: Credit card revolving loans are the retail product with no collateral; the other three are backed by property, business assets or guarantees.

Why the others are wrong

- **A:** Mortgages are secured by the financed residential property; confusing high balance with unsecured status.
- **B:** Home equity loans and HELOCs are secured by the borrower's equity in the home.
- **D:** Small business loans are secured by the firm's assets or by the owners' personal guarantees; treating them as unsecured because the borrower is a small firm is the error.

Q252

BT answers P2.T6.25.5.1 (rebuilt)

REBUILT — Original scenario asked for the 'least accurate' statement with several arguable options (concentration vs systemic risk, borrower behavior not in the reading); replaced with a single-answer classification of retail banking risks per LO 25.a.

A retail bank notes four events. Which of the following correctly matches the event to the type of risk it represents?
I. Mortgage origination volume falls sharply when interest rates rise.
II. A bank's leased-auto portfolio loses value because the assumed residual values at lease end were too high.
III. Regulators threaten to remove the bank's franchise after it is seen to act unfairly toward customers.
IV. Customer fraud is detected through newly introduced fraud-detection processes.

A. I. Business risk; II. Asset valuation risk; III. Reputation risk; IV. Operational risk
B. I. Asset valuation risk; II. Business risk; III. Operational risk; IV. Reputation risk
C. I. Interest-rate risk; II. Business risk; III. Reputation risk; IV. Credit risk
D. I. Business risk; II. Operational risk; III. Asset valuation risk; IV. Reputation risk

Answer: A

Step by step

1. I: changes in mortgage volume as rates move are volume risk, a component of business risk.
2. II: wrong residual-value assumptions on leased autos are a valuation (market-type) risk, i.e., asset valuation risk.
3. III: standing with regulators and customers is reputation risk.
4. IV: customer fraud and fraud-detection processes are operational risk, managed within the business line.
5. Only option A matches all four.

Why A is correct: Volume changes are business risk, residual-value errors are asset valuation risk, regulatory/customer standing is reputation risk, and fraud is operational risk.

Why the others are wrong

- **B:** Swaps I with II and III with IV: mortgage volume is not a valuation issue and fraud is not reputation risk.
- **C:** Treats volume change as interest-rate risk and customer fraud as credit risk; fraud is operational, and credit risk is borrower default.
- **D:** Calls residual-value error operational risk and regulatory standing asset valuation risk; residual values are valuation risk and regulator relations are reputation risk.

Q253

BT answers P2.T6.603.3 · BT notes P2.T6.603.3

The principal risk for a retail credit business is credit risk, but retail banking faces other risks as well. In addition to reputation risk, four non-credit risks are interest-rate risk, asset valuation risk, operational risk and business risk. They are described below without identification:

I. This risk arises on both the asset and liability sides of the balance sheet. It is transferred from the retail business lines to the bank's treasury and managed in the asset/liability and liquidity (ALM) unit.

II. These risks are managed as an inherent part of the business. An example is fraud detection.

III. Senior managers should focus on these risks, which include business volume, strategic risks and merger and acquisition (M&A) decisions.

IV. This risk is a special form of market risk affecting the retail line's profitability. It includes mortgage prepayment risk and is often managed centrally by the treasury unit.

Which of the following correctly matches each risk to its description?

A. I. Asset valuation risk, II. Business risk, III. Interest-rate risk, IV. Operational risk

B. I. Asset valuation risk, II. Interest-rate risk, III. Operational risk, IV. Business risk

C. I. Interest-rate risk, II. Asset valuation risk, III. Operational risk, IV. Business risk

D. I. Interest-rate risk, II. Operational risk, III. Business risk, IV. Asset valuation risk

Answer: D

Step by step

1. I: on both sides of the balance sheet, transferred to treasury/ALM = interest-rate risk.
2. II: managed within the business, fraud detection = operational risk.
3. III: volume, strategic and M&A decisions = business risk.
4. IV: special form of market risk including prepayment risk = asset valuation risk.
5. Only D has I=interest-rate, II=operational, III=business, IV=asset valuation.

Why D is correct: Each description maps uniquely: ALM transfer is interest-rate risk, fraud detection is operational, volume/strategy/M&A is business risk, and prepayment as a market-risk subclass is asset valuation risk.

Why the others are wrong

- **A:** Calls the ALM-transferred risk asset valuation risk and puts fraud detection under business risk; every pairing is wrong.
- **B:** Mislabels I as asset valuation and II as interest-rate; fraud detection is operational and strategic/M&A is business, not operational.
- **C:** Gets I right but swaps the rest: fraud is operational (not asset valuation), strategic/M&A decisions are business risk (not operational), and prepayment is asset valuation (not business).

Q254

Schweser Module Quiz 25.1 Q1

Which of the following statements is most accurate regarding risks incurred by retail lenders?

A. Reputation risk is more of a concern for the borrower than the lender.

B. Business risk relates to the day-to-day operational risks of the business.

C. Credit risk relates to the potential for a lender to default on its obligation.

D. Refinancing a mortgage when rates decrease is an example of asset valuation risk.

Answer: D

Step by step

1. Test each statement against the definitions.
2. Falling rates lead borrowers to prepay and refinance mortgages, reducing the value of the lender's mortgage portfolio: prepayment risk, a form of asset valuation risk.
3. A, B and C each misstate a definition (see distractors).

Why D is correct: Mortgage prepayment when rates fall is the textbook example of asset valuation risk, a form of market risk.

Why the others are wrong

- **A:** Reputation risk concerns the lender's standing with customers and regulators, not the borrower.
- **B:** Day-to-day risks of running the business are operational risk; business risk is strategic and volume risk.
- **C:** Credit risk is the risk that the borrower defaults, not the lender.

Q255

AnalystPrep Q.2017

What characterized the retail lending market in the run-up to the 2007/2008 financial crisis?

A. Enforcement of strict underwriting standards, elevated interest rates to discourage speculative borrowing, and controlled competition among a few dominant lenders.

B. Somewhat relaxed underwriting criteria combined with cyclically adjusted interest rates and a competitive landscape dominated by traditional banking institutions.

C. Loosely applied underwriting standards, historically low interest rates fueled by expansive monetary policies, and fierce competition among a broad array of lending institutions.

D. Conservatively modeled underwriting processes that inadequately captured borrower risk, variable interest rates influenced by short-term economic indicators, and a fragmented competitive environment with aggressive non-bank entrants.

Answer: C

Step by step

1. GARP: low interest rates, poorly structured broker incentives and an increasingly competitive environment led to lowered underwriting standards.
2. Subprime loans often had limited or no income verification and high LTVs.
3. Only C matches loose standards, low rates and fierce competition.

Why C is correct: Low rates and intense competition pushed lenders to loosen underwriting, producing high-LTV loans to weaker borrowers.

Why the others are wrong

- **A:** Describes the opposite: standards were lax and rates low.
- **B:** Understates the laxity and wrongly implies cyclically adjusted rates and traditional-bank dominance.
- **D:** Underwriting was loose, not conservative; rates were low rather than tightly variable.

LO 25.b

Explain the differences between retail credit risk and corporate credit risk.

Q256

AnalystPrep Q.2010

Why is the credit exposure of retail banking different from that of corporate banking?

A. Credit exposures of retail banking arrive in large pieces but individual exposures by a single consumer are not expensive enough to threaten the bank.

B. Credit exposures of retail banking arrive in bite-sized pieces; hence, an individual consumer's exposure is not big enough to threaten the bank.

C. Corporate exposures arrive in bite-sized pieces which must be repaid before a new facility can be offered; hence, individual corporate exposures rarely threaten the bank.

D. Credit exposures of corporates arrive in large pieces, each of which must be repaid before a new facility can be offered; hence, individual corporate exposures rarely threaten the bank.

Answer: B

Step by step

1. The defining feature of retail credit exposure is that it arrives in small (bite-sized) pieces across a very large number of customers.
2. Therefore the default of any single retail customer is never expensive enough to threaten the bank.
3. Corporate and commercial portfolios contain large exposures to single names and concentrations in sectors or regions, so one default can matter materially.
4. Only B states the retail feature correctly and consistently.

Why B is correct: Retail exposures are small and numerous, so no single default is large enough to threaten the bank; corporate exposures are large and concentrated.

Why the others are wrong

- **A:** Self-contradictory: retail exposures do not arrive in large pieces; they are small, which is why a single default is not threatening.
- **C:** Reverses the roles: corporate exposures are large, not bite-sized, and the repay-before-new-facility idea is irrelevant to the distinction.
- **D:** Correctly says corporate exposures are large but then draws the wrong conclusion; large single-name exposures can threaten the bank.

Q257

BT answers P2.T6.603.1 · BT notes P2.T6.603.1 · BT Mock Q37

Retail exposures include credit cards, installment loans (e.g., personal finance, educational loans, auto loans, leasing), revolving credits (e.g., overdrafts, home equity lines of credit) and residential mortgages. Each of the following tends to be a stronger feature of retail credit risk rather than corporate credit risk EXCEPT which is more typical of corporate credit risk?

A. Portfolio tends to behave like a well-diversified portfolio in normal markets.

B. Default by a single customer is never expensive enough to threaten a bank.

C. A rise in defaults is often signaled in advance by a change in customer behavior.

D. Portfolio risk is dominated by the risk that credit losses will rise to unexpected levels due to concentrations of exposures.

Answer: D

Step by step

1. Classify each statement as retail or corporate.
2. A: granular retail books behave like diversified portfolios in normal markets, so expected loss is predictable and can be priced - retail.
3. B: retail exposures are small, so a single default cannot threaten the bank - retail.
4. C: changes in customer behaviour give early warning of rising defaults, allowing pre-emptive action - retail.
5. D: risk dominated by unexpected losses from concentrated exposures is the corporate/commercial feature - this is the EXCEPT.

Why D is correct: Corporate portfolios contain large single-name and sector concentrations, so their risk is dominated by unexpected-loss surprises; the other three are retail tendencies.

Why the others are wrong

- **A:** A retail feature: diversification in normal markets makes expected loss estimable; picking it misses that the question asks for the corporate trait.
- **B:** A retail feature: small exposures mean no single default threatens the bank.
- **C:** A retail feature: behavioural changes signal rising defaults early; corporate problems usually surface too late.

Q258

BT answers P2.T6.25.5.2

REBUILT — Original relied on a missing portfolio table and an ambiguous option C ('reduces systemic exposure'); rewrote as a self-contained question with four clean statements, exactly one false.

Silver Lining Bank has a retail loan portfolio and a corporate loan portfolio of equal total size. Which of the following statements comparing retail and corporate credit risk is LEAST accurate?

A. A default by a single retail customer is unlikely to threaten the bank, whereas a default by a large corporate borrower can have a significant impact.

B. Expected losses on the retail portfolio can be estimated reliably and priced into customer charges as a cost of doing business, whereas corporate portfolio risk is dominated by the risk of unexpected losses.

C. Changes in customer behavior often signal a rise in retail defaults early enough for preemptive action, whereas

corporate problems often become known too late to correct.

D. Because the two portfolios are equal in size, they have comparable risk profiles, so a loss rate above expectations would hurt each portfolio equally.

Answer: D

Step by step

1. Test each statement against the retail-vs-corporate differences.
2. A: true - retail exposures are small; corporate exposures are large and single defaults matter.
3. B: true - retail diversification makes expected loss predictable and priceable; corporate risk is dominated by unexpected losses.
4. C: true - retail behaviour changes give early warning; commercial problems often surface too late.
5. D: false - risk depends on granularity, concentration and predictability, not on total size; losses above expectations are far more damaging for a concentrated corporate book.

Why D is correct: Portfolio size does not determine risk: retail portfolios are granular and diversified, corporate portfolios are concentrated, so they do not have comparable risk profiles.

Why the others are wrong

- **A:** Accurate statement of the exposure-size difference; chosen only by someone who forgets the question asks for the LEAST accurate.
- **B:** Accurate: expected-loss predictability is a retail feature; unexpected loss dominates corporate risk.
- **C:** Accurate: early behavioural signals allow preemptive action in retail, rarely in commercial lending.

Q259 Mock 1 Q41

Which of the following is most accurate about early warning signals of retail and commercial exposures?

A. Neither retail nor commercial exposures provide early warning signals of default that allow banks time to take action to minimize PD and LGD.

B. Both retail and commercial exposures provide early warning signals of default that allow banks time to take action to minimize PD and LGD.

C. Commercial exposures but not retail exposures provide early warning signals of default that allow banks time to take action to minimize PD and LGD.

D. Retail exposures but not commercial exposures provide early warning signals of default that allow banks time to take action to minimize PD and LGD.

Answer: D

Key correction: Source key (Mock 1) said C; correct is D because changes in retail customer behaviour signal rising defaults in time for preemptive action, whereas commercial problems often become known too late.

Step by step

1. Retail: changes in customer behaviour signal a potential rise in defaults, giving the bank time for preemptive action (e.g., marketing to lower-risk customers, raising rates on higher-risk customers).
2. Commercial: portfolios typically do not offer such signals; problems may not be known until it is too late to correct them.
3. So retail but not commercial exposures provide usable early warnings - D.

Why D is correct: Retail portfolios show behavioural early warnings that allow action to limit PD and LGD; commercial problems typically surface too late.

Why the others are wrong

- **A:** Wrong because retail does give early behavioural signals.
- **B:** Wrong because commercial exposures typically do not give timely warnings.
- **C:** Reverses the two: it attributes early warning to commercial lending and denies it for retail, the opposite of the Schweser/GARP treatment.

LO 25.c

Discuss the “dark side” of retail credit risk and the measures that attempt to address the problem.

Q260

AnalystPrep Q.2012

REBUILT — Option D (property as sole collateral) was arguably defensible as a systematic risk; replaced it with a clearly non-problematic behavior, and moved the LO to 25.c (dark side).

Retail lending entails offering facilities to a large group of borrowers in relatively small amounts, thereby naturally bringing about diversification. However, certain lending behaviors can erode the benefits of diversification and result in market-wide financial turmoil, as the 2007-2009 financial crisis showed. Which of the following behaviors is MOST likely to result in less than perfect protection from systematic risks?

A. International lending
B. Relaxation of lending requirements, such as advancing money without checking incomes
C. Multi-currency lending
D. Charging borrowers a risk-based interest rate

Answer: B

Step by step

1. Diversification fails when many credits become exposed to the same hidden systematic factor.
2. Advancing money without checking incomes (relaxed standards) across the industry introduces a hidden systematic risk into whole credit portfolios.
3. In economic trouble this causes sudden jumps in default rates and falls in collateral values, as in the 2007-2009 crisis.
4. Geographic or currency spread (A, C) and risk-based pricing (D) do not create this common vulnerability.

Why B is correct: Industry-wide loosening of underwriting standards creates a common hidden systematic risk that makes defaults rise together, undermining diversification.

Why the others are wrong

- **A:** International lending spreads exposures across economies and, if anything, adds diversification.
- **C:** Multi-currency lending also spreads exposures; it is not the behavior GARP identifies as eroding diversification.
- **D:** Risk-based pricing aligns price with risk and does not create a common systematic exposure.

Q261

AnalystPrep Q.5431

REBUILT — Easy-credit and higher-rate options were giveaways and 'self-regulation' was vague; rewrote distractors as realistic but wrong responses.

A bank wants to manage the "dark side" of retail credit risk, the danger that losses suddenly rise to unexpected levels because of a systematic risk factor. Which of the following is the most appropriate measure?

A. Relaxing underwriting standards to grow market share before competitors do.
B. Using stress tests to gauge how damaging each plausible worst-case scenario would be for large conventional portfolios.
C. Relying on diversification across a large number of small loans to remove the systematic risk.
D. Setting capital equal to the expected loss rate observed in the last benign period.

Answer: B

Step by step

1. The dark side arises from an unforeseen systematic factor that hits many retail credits at once (defaults jump, collateral values fall).
2. Such risk cannot be given a reliable single risk number, so the bank must examine severe-but-plausible scenarios.
3. GARP: for large portfolios vulnerable to sharp changes in multiple risk factors, banks must use stress tests to gauge how devastating each worst-case scenario might be.
4. Hence B.

Why B is correct: Stress testing reveals the loss impact of correlated, adverse scenarios that historical data and diversification cannot capture.

Why the others are wrong

- **A:** Looser standards worsen the problem (more high-risk borrowers) rather than address it.
- **C:** Diversification gives only imperfect protection against a systematic factor; in 2007-09 house prices fell nearly everywhere.
- **D:** Benign-period expected loss ignores unexpected systematic losses, so capital would be inadequate.

Q262 BT answers P2.T6.603.2 · BT notes P2.T6.603.2

REBUILT — Option D was an implausible, inconsistent distractor (75% vs 15% across versions); replaced distractors so exactly one measure is correct.

The dark side of retail risk management has four prime causes: innovative products lack historical loss data; sensitivity to the macroeconomic environment; consumer defaults are the product of a social and legal system that changes; and operational issues that can have a systematic effect. Which of the following measures attempts to address the problem?

A. Banks should assign a precise risk number to these wild-card risks and price them into the loans.
B. Banks should exclude every new type of retail product from their portfolios.
C. Banks should use stress tests to gauge how devastating each plausible worst-case scenario might be.
D. Banks should expand quickly into new products such as subprime lending to accumulate loss data.

Answer: C

Step by step

1. GARP says it is almost by definition difficult to put a risk number to wild-card risks (rules out A).
2. Banks should limit, not eliminate, exposure to new risks: a little exposure generates information, a lot makes the bank a hostage to fortune (rules out B and D).
3. For large portfolios vulnerable to sharp changes in several risk factors, banks use stress tests of plausible worst cases.
4. Answer C.

Why C is correct: Because the risks cannot be reliably quantified or avoided entirely, banks limit exposure and use stress tests to measure worst-case impact.

Why the others are wrong

- **A:** Wild-card risks cannot be given a reliable risk number; this misreads the problem.
- **B:** Over-corrects: banks should only limit the number of portfolios vulnerable to new risks, not avoid them all.
- **D:** Rapid, large exposure to unproven products makes the bank a hostage to fortune; exposure should be limited.

Q263 Schweser Module Quiz 25.1 Q2

The dark side of retail credit risk is perpetuated by all of the following factors EXCEPT:

A. capital set aside to protect a bank in the event of default.
B. process flaws resulting in high-risk applicants receiving credit.
C. new products that do not have sufficient historical loss data.
D. a social acceptance of bankruptcy and borrowers ”walking away” from their obligations.

Answer: A

Step by step

1. List the causes: new products lacking loss data, macro sensitivity, evolving social/legal system encouraging default, operational flaws in a semi-automated process.
2. B = operational flaw; C = lack of loss data; D = social/legal acceptability of default.
3. A (capital) is a response to retail risk, not a cause.

Why A is correct: Capital absorbs losses; it does not generate or perpetuate the dark side.

Why the others are wrong

- **B:** A listed cause: operational flaw granting credit to higher-risk individuals.
- **C:** A listed cause: no historical loss data for innovative products.
- **D:** A listed cause: social and legal environment that encourages defaults.

Q264 BT answers P2.T6.25.5.3

REBUILT — Distractors B-D were giveaways contradicting the scenario; rewrote them as plausible misreadings while keeping the systematic-risk theme.

A retail lender holds small consumer loans spread across many regions and income groups. During a severe recession defaults rise simultaneously across all segments, and losses far exceed the level the credit model projected using historical data. Which statement best describes the structural risk exposed?

A. A common macroeconomic risk factor made borrower defaults move together, so diversification across many small loans gave only imperfect protection.
B. The losses were mainly idiosyncratic, so the portfolio should be split into even smaller loans to remove them.
C. The loss reflects expected loss, which should simply have been built into loan pricing.
D. The model was sound; the excess losses show only that borrowers’ credit scores were too high at origination.

Answer: A

Step by step

1. Simultaneous defaults across regions and segments point to a systematic factor, not borrower-specific events.
2. GARP: in 2007-09 house prices fell at about the same time across a large economy, so diversification offered less than perfect protection.
3. This is the dark side: losses rise unexpectedly because of an unforeseen systematic factor.
4. Choose A.

Why A is correct: A systematic risk factor cannot be diversified away, so losses exceeded the estimate based on benign history.

Why the others are wrong

- **B:** Splitting loans further does not remove systematic risk; losses were correlated, not idiosyncratic.
- **C:** Expected loss is predictable and priced; the issue here is unexpected losses beyond the estimated threshold.
- **D:** Not supported; the failure was in ignoring correlation under stress, not score levels at origination.

LO 25.d

Define and describe credit risk scoring model types, key variables, and applications.

Q265

Schweser Module Quiz 25.1 Q3

Which of the following statements is correct regarding credit risk scoring models?

A. A pooled model will result in scores ranging from 300 to 850.

B. A custom model is cheaper to implement than credit bureau scores.

C. Multiple requests for new credit will reduce an applicant's credit score.

D. An example of a characteristic in a scoring model is the applicant's current gross salary of $50,000.

Answer: C

Step by step

1. Credit files record an inquiry each time a lender accesses the file for a new-credit request, and lenders see these inquiries.
2. Many recent inquiries signal credit-seeking behaviour and lower the score, so C is correct.
3. Check the others: 300-850 is the range of generic credit bureau (FICO) scores; pooled models are vendor-built for an industry, costing more than bureau scores; custom models are in-house and the costliest; a characteristic is the question (e.g., gross salary with current employer) and $50,000 is the attribute.

Why C is correct: Each access of the credit file for new credit leaves an inquiry, and multiple inquiries are treated as a negative factor, lowering the score.

Why the others are wrong

- **A:** Confuses pooled models with credit bureau scores; the 300-850 range describes bureau (FICO-type) scores, while a pooled model is built by outside vendors using data from many lenders with similar portfolios.
- **B:** Reverses the cost ranking: bureau scores are cheapest and fastest; pooled models cost more; custom models, built in-house on the lender's own data, cost the most.
- **D:** Confuses characteristic with attribute: the characteristic is the question ('gross salary with current employer') and the $50,000 response is the attribute.

Q266

BT answers P2.T6.25.6.1

REBUILT — Original relied on a lost assessment table for two applicants; rebuilt as a self-contained application scorecard with explicit characteristics, attributes, points and cutoff.

A bank uses the following application scorecard for consumer loans. Each applicant's score is the sum of the points for the attribute matching each characteristic. Applications scoring at or above the cutoff of 70 are approved; those below are rejected.

Characteristic	Attribute	Points
Residential status	Own	20
Residential status	Rent	10
Years with current employer	Less than 2	5
Years with current employer	2 to 5	15
Years with current employer	More than 5	25
Prior delinquencies	None	30
Prior delinquencies	1 or 2	10
Credit inquiries, last 6 months	0 or 1	15
Credit inquiries, last 6 months	2 or 3	5
Credit inquiries, last 6 months	4 or more	-10

Applicant X: renter, 8 years with current employer, no prior delinquencies, 5 credit inquiries in the last 6 months. Applicant Y: homeowner, 3 years with current employer, no prior delinquencies, 2 credit inquiries in the last 6 months.
Which of the following is the correct outcome?

A. Both X and Y are approved.
B. X is rejected and Y is approved.
C. X is approved and Y is rejected.
D. Both X and Y are rejected.

Answer: B

Step by step

1. Applicant X: Rent 10 + more than 5 years 25 + no delinquencies 30 + 4 or more inquiries (-10) = 55.
2. Applicant Y: Own 20 + 2 to 5 years 15 + no delinquencies 30 + 2 or 3 inquiries 5 = 70.
3. Cutoff is 70 with 'at or above' approved: X (55) is below the cutoff and is rejected; Y (70) meets it and is approved.
4. Hence X is rejected and Y is approved.

Why B is correct: Summing the attribute points for each characteristic gives X = 55 and Y = 70; only Y reaches the cutoff of 70, and the heavy inquiry penalty is what pulls X down despite long job tenure.

Why the others are wrong

- **A:** Treats X's -10 for 4 or more inquiries as +10 (score 75), so X wrongly clears the cutoff.
- **C:** Reverses the outcome by weighting only employment tenure (X has more than 5 years) and ignoring that X scores 55 while Y scores 70.
- **D:** Applies a strict 'above 70' test, so Y at exactly 70 is rejected; the rule states 'at or above' the cutoff is approved.

Q267

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A risk manager at a regional auto finance company is selecting a scoring approach for a new subprime auto loan product. The company has accumulated several years of its own application and repayment data on similar loans, wants a model that is specific to the company and screens for the applicant profile of this product, and is willing to bear the highest development cost to get it. Which of the following approaches is the most appropriate?

A. A credit bureau score, because it is developed in-house from the lender's own population of credit applications.
B. A pooled model, because it is built from the company's own application pool and is therefore company specific.
C. A custom model, because it is developed from the lender's own application data and screens for a specific applicant profile for the lender's specific product.
D. A pooled model, because it costs less than both a credit bureau score and a custom model while using data from several lenders.

Answer: C

Step by step

1. Identify the need: a company-specific model built on the company's own application data for one product, with cost not the binding constraint.
2. Credit bureau (FICO-type) scores are generic scores supplied by the credit bureaus; they are low cost, quickly installed and give a broad overview of overall creditworthiness. They are not built from the lender's own application pool.
3. Pooled models are built by outside vendors using data from a range of lenders with similar portfolios; they cost more than generic scores but less than custom models, and they can be tailored to an industry but are not

company specific.
4. Custom models are usually developed in-house from the lender's own unique population of credit applications and are tailored to screen for a specific applicant profile for a specific lender's product; they are the most costly of the three.

Why C is correct: Only a custom model is built from the lender's own application pool and is specific to the lender and its product. The company has that data and accepts the higher cost, so a custom model fits the stated need.

Why the others are wrong

- **A:** Credit bureau scores are generic scores produced and supplied by the credit bureaus (e.g., using the Fair Isaac methodology), not developed in-house from the lender's own applications. Error: attributing the custom model's data source to the bureau score.
- **B:** Pooled models are built by outside vendors from data collected from several lenders; they can be tailored to an industry but are not company specific. Error: confusing pooled models with custom models.
- **D:** Pooled models cost more than generic credit bureau scores (though less than custom models), and being multi-lender they are not company specific in any case. Error: misranking the cost of pooled models against bureau scores.

LO 25.e

Discuss the key variables in a mortgage credit assessment and describe the use of cutoff scores, default rates, and loss rates in a credit scoring model.

Q268

AnalystPrep Q.2011

The Consumer Financial Protection Act (CFPA) requires lenders to assess a borrower's ability to repay a mortgage, and a loan labeled a qualified mortgage (QM) is presumed to satisfy this requirement. Which of the following characterizes a qualified mortgage?

A. The debt-to-income (DTI) ratio must be greater than 43%.
B. It must not have excess upfront points or fees.
C. It must have a minimum term of 30 years.
D. It must have interest-only features.

Answer: B

Step by step

1. Recall the QM features: no excess upfront points and fees; no toxic features (negative amortization, terms longer than 30 years, interest-only periods); a cap on the share of income going to debt (DTI below about 43%); no balloon payments.
2. Compare each option: only B matches a QM feature.

Why B is correct: A qualified mortgage is a plain-vanilla loan, so it carries no excess upfront points or fees, which lets the lender presume the borrower can repay.

Why the others are wrong

- **A:** Reverses the DTI rule: a QM caps DTI (about 43% or less); it does not require it to be above 43%.
- **C:** A 30-year term is a maximum, not a minimum; terms longer than 30 years are excluded.
- **D:** Interest-only periods are 'toxic' features that disqualify a loan from QM status.

Q269

AnalystPrep Q.5469

REBUILT — Original asked which model tweak is 'most effective' with several defensible answers (raising the cutoff just trades false goods for false bads); rewritten to test how cutoffs are set from loss-rate and profitability experience.

A bank has set the cutoff score for its mortgage scorecard. Over time it accumulates data on the loss rate and profitability of each score band. According to the standard approach to cutoff scores, which of the following is the most appropriate way for the bank to adjust the cutoff?

A. Estimate loss rates and profitability from a long history that spans a full economic cycle, and move the cutoff to the level that best balances risk and profitability.
B. Raise the cutoff to the highest feasible level so that no defaults occur, since a higher score means lower risk.
C. Base the cutoff on the last twelve months of experience only, because recent data best reflect current conditions.
D. Lower the cutoff whenever losses are low, because a lower cutoff always increases profit without affecting the loss rate.

Answer: A

Step by step

1. A higher score means lower risk, so the cutoff is the threshold that decides who is lent to and on what terms.
2. Too low a cutoff raises default risk; too high a cutoff eliminates profitable low-risk borrowers.
3. Historical loss rates and profitability per product line are estimated from experience; longer horizons that capture a full economic cycle are preferred.
4. The bank then adjusts the cutoff to maximize the balance between risk and profitability: option A.

Why A is correct: The cutoff is a risk-profitability trade-off, best calibrated on loss and profit experience that covers a whole economic cycle.

Why the others are wrong

- **B:** A very high cutoff rejects many good, profitable borrowers (false bads); the goal is balance, not zero defaults.
- **C:** A short window may capture only a benign or stressed phase; longer horizons covering a full cycle are preferred.
- **D:** Lowering the cutoff admits riskier applicants and raises the loss rate; low past losses do not make the trade-off disappear.

Q270

AnalystPrep Q.5470

REBUILT — Original was a vague 'most relevant factor' question where the keyed answer was not clearly better than the others; replaced with a numerical loss-rate/LGD/PD calculation from the reading.

A bank groups its mortgage portfolio into score bands. For one band, the historical annual loss rate is 2.4% and the loss given default (LGD) is 60%. What is the implied probability of default (PD) for the band?

A. 1.44%
B. 0.96%
C. 2.4%
D. 4.0%

Answer: D

Step by step

1. Loss rate = PD × LGD, so PD = loss rate/LGD.
2. PD = 2.4%/0.60 = 4.0%.

Why D is correct: The implied PD is a by-product of the historical loss rate and the LGD: PD = loss rate divided by LGD.

Why the others are wrong

- **A:** Multiplied the loss rate by LGD (2.4% × 0.60 = 1.44%) instead of dividing.
- **B:** Multiplied the loss rate by (1 - LGD) = 0.40, confusing LGD with recovery (2.4% × 0.40 = 0.96%).
- **C:** Treated the loss rate as the PD, ignoring that only part of a defaulted exposure is lost.

Q271

BT answers P2.T6.25.6.2

REBUILT — Applicant table was missing; reconstructed it from the source explanation (cutoff 700, applicant values) and made the cutoff part of the stem.

A bank applies a mortgage credit assessment with a FICO cutoff score of 700. Two applicants have the following profiles:

Variable	Applicant X	Applicant Y
FICO score	730	680
Debt-to-income (DTI) ratio	32%	48%
Loan-to-value (LTV) ratio	75%	90%
Documentation	Full doc	Stated income

Which of the following best describes the relative credit risk of the two applicants?

A. Applicant X is likely lower risk and Applicant Y is likely higher risk.
B. Applicant Y is likely lower risk and Applicant X is likely higher risk.
C. Both are likely high risk.
D. Neither is likely to have higher risk.

Answer: A

Step by step

1. FICO: higher score means lower default risk. X (730) is above the 700 cutoff; Y (680) is below it.

2. DTI: lower is better. X 32% vs Y 48%, so Y is more stretched.
3. LTV: lower is better (more borrower equity). X 75% vs Y 90%.
4. Documentation: full doc (X) verifies income and assets; stated income (Y) does not verify income.
5. Every variable favors X, so X is lower risk and Y higher risk.

Why A is correct: Applicant X is better on every key variable (FICO, DTI, LTV, documentation), and Y falls below the cutoff.

Why the others are wrong

- **B:** Reverses the direction of the variables, e.g. treating a lower FICO or higher LTV as safer.
- **C:** X clears the FICO cutoff and has moderate DTI and LTV, so it is not high risk.
- **D:** The variables clearly differ in X's favor; ignoring the cutoff and ratios would suggest no difference.

Q272 BT answers P2.T6.604.1 · BT notes P2.T6.604.1

In applying for a mortgage loan, a borrower who is seeking approval prefers a lower number for each of the following variables in her application, ceteris paribus, EXCEPT for which does she not necessarily prefer a lower number?

A. DTI ratio
B. FICO score
C. LTV ratio
D. Number of recent inquiries

Answer: B

Step by step

1. DTI (monthly debt payments / gross monthly income): lower means more capacity, so lower is better.
2. LTV (loan / appraised value): lower means more equity cushion, so lower is better.
3. Number of recent inquiries: more inquiries signal credit-seeking and risk, so fewer is better.
4. FICO score: a higher score means lower default risk, so the borrower prefers a higher number.

Why B is correct: FICO is the only variable where a higher number indicates lower default risk.

Why the others are wrong

- **A:** A lower DTI means a smaller debt burden relative to income, which the borrower wants.
- **C:** A lower LTV means a larger down payment and less lender exposure, which the borrower wants.
- **D:** Fewer recent credit inquiries signal less credit stress, so lower is preferred.

Q273 BT answers P2.T6.604.2 · BT notes P2.T6.604.2

Suppose the population of credit applicants, which is ordinarily unknowable, divides into either "good" or "bad" accounts to use Crouhy's terminology. Good accounts have a mean credit score of 730.0 with a standard deviation of 25.0. Bad accounts have a mean credit score of 600.0 with a standard deviation of 40.0. Both distributions are approximated by a normal distribution. A bank is evaluating whether to set its cutoff score at 680.0; i.e., applicants with a score of 680.0 or greater will be approved. There are two types of underwriting error: a "false bad" is the rejection of a good account; a "false good" is the acceptance of a bad account. Which of the following is a TRUE statement?

A. This cutoff of 680.0 ensures an error rate of less than 1.0%
B. The bank can reduce the fraction of false bads by increasing the cutoff score
C. The bank can reduce the fraction of false goods by increasing the cutoff score
D. The bank can simultaneously reduce the fraction of errors (i.e., both false bads and false goods) by increasing the cutoff score

Answer: C

Step by step

1. False bad rate $= P(\text{good} < 680) = \Phi((680 - 730)/25) = \Phi(-2.0) = 2.28\%$.
2. False good rate $= P(\text{bad} \geq 680) = 1 - \Phi((680 - 600)/40) = 1 - \Phi(2.0) = 2.28\%$. Each error rate already exceeds 1%, so A is false.
3. Raising the cutoff moves the line right: fewer bad accounts lie above it (fewer false goods) but more good accounts lie below it (more false bads).
4. So only C is true; B and D claim a benefit on false bads that moving right cannot give.

Why C is correct: A higher cutoff cuts off more of the bad distribution but also more of the good distribution, so false goods fall while false bads rise.

Why the others are wrong

- **A:** Each error rate at 680 is about 2.28% (2 standard deviations in each tail), which exceeds 1%.
- **B:** Reversed direction: raising the cutoff increases false bads; lowering it reduces them.
- **D:** The two errors move in opposite directions as the cutoff changes (overlapping distributions), so one cannot reduce both.

Q274 Schweser Module Quiz 25.2 Q1

In assessing the key variables associated with a potential mortgage loan, a bank will charge a higher interest rate if the borrower has a relatively:

A. high FICO score.
B. high loan-to-value ratio.
C. low debt-to-assets ratio.
D. low debt-to-income ratio.

Answer: B

Step by step

1. Higher risk to the lender warrants a higher rate.
2. A high LTV means little borrower equity, so more risk (and larger loss on default): higher rate.
3. A high FICO, low debt-to-assets and low DTI all signal lower risk and so lead to lower rates.

Why B is correct: A high LTV means thin borrower equity and larger potential loss for the lender, so it is compensated with a higher rate.

Why the others are wrong

- **A:** A high FICO score signals low default risk, which supports a lower rate.
- **C:** Low leverage relative to assets signals lower risk, a lower rate.
- **D:** Low DTI means debt payments are small relative to income, lower risk and a lower rate.

LO 25.f

Discuss the measurement and monitoring of a scorecard performance including the use of cumulative accuracy profile (CAP) and the accuracy ratio (AR) techniques.

Q275 BT answers P2.T6.25.6.3

REBUILT — Original CAP exhibit and areas were illegible and the key's arithmetic was inconsistent; rebuilt with explicit areas and an AR computation.

A retail bank compares two credit scorecards using the cumulative accuracy profile (CAP). In the CAP chart, the area between the perfect-model line and the random-model (45-degree) line is 0.40. The area between Provider A's CAP curve and the random-model line is 0.34, and the area between Provider B's CAP curve and the random-model line is 0.18. The accuracy ratio (AR) is the area between the model's curve and the random line divided by the area between the perfect line and the random line. What are the accuracy ratios of Provider A and Provider B?

A. A = 0.34; B = 0.18
B. A = 0.68; B = 0.36
C. A = 0.85; B = 0.45
D. A = 0.40; B = 0.40

Answer: C

Key correction: Source key (B) was based on an illegible exhibit and inconsistent arithmetic (0.75/0.4 shown as 0.375); the original item could not be verified, so it was rebuilt.

Step by step

1. Accuracy ratio: $\text{AR} = \dfrac{\text{area between model and random line}}{\text{area between perfect and random line}}$.
2. Provider A: $0.34/0.40 = 0.85$.
3. Provider B: $0.18/0.40 = 0.45$.
4. Provider A is much closer to the perfect model (AR near 1), so it discriminates better.

Why C is correct: The AR scales each model's improvement over random by the maximum possible improvement (the perfect model's 0.40), so a value near 1 means near-perfect discrimination.

Why the others are wrong

- **A:** Reports the raw areas and forgets to divide by the perfect-model area of 0.40.

- **B:** Divides by 0.50 (the area under the random line / half the unit square) instead of 0.40.
- **D:** Uses the perfect-model area as if it were each model's result; nothing in the data supports equal ratios.

Q276 BT answers P2.T6.604.3 · BT notes P2.T6.604.3

About the cumulative accuracy profile (CAP), each of the following statements is true EXCEPT, which is not?

A. A perfect credit scoring model generates an accuracy ratio (AR) of 1.0, which is the upper bound on the AR

B. A purely random model that cannot differentiate between good and bad customers is likely to generate an accuracy ratio (AR) of 0.40 to 0.60; i.e., 50% +/- 10%

C. The CAP curve, which plots the actual rating model as a cumulative percentage of defaults, is monotonically increasing (aka, nondecreasing or weakly increasing)

D. The CAP curve plots the fraction of defaulted customers (y-axis) against the fraction of entire customer population sorted by score from highest risk (left) to lowest risk (right)

Answer: B

Step by step

1. The random model is the 45-degree line: it captures defaults in proportion to the population share and cannot discriminate.
2. The AR measures the area between the model curve and the random line relative to the perfect-model area, so for the random model the area is zero and AR $\approx$ 0, not 0.40-0.60.
3. A: the perfect model has AR = 1, the maximum. C: cumulative default share can only stay flat or rise. D: the population is sorted from riskiest to safest on the x-axis, defaults on the y-axis.
4. So B is the false statement.

Why B is correct: A random model has no discriminatory power, so its CAP lies on the 45-degree line and its AR is about 0, not 50%.

Why the others are wrong

- **A:** True: AR = 1 for the perfect model and AR cannot exceed this.
- **C:** True: a cumulative curve never decreases.
- **D:** True: this is the standard CAP construction (riskiest accounts first).

Q277 BT answers P2.T7.20.4.3

REBUILT — Original relied on a missing distribution exhibit and on ROC/AUC material outside the 2026 scorecard-performance LO; rebuilt as a self-contained CAP benchmark-line question.

A bank's scorecard predicts that 10% of accounts will default over the next 12 months. Accounts are ranked from riskiest to safest by score. Actual results show that the riskiest 10% of accounts contain 70% of all defaults that occurred. Which statement about the cumulative accuracy profile (CAP) at the riskiest 10% of the population is correct?

A. The perfect model line shows 10% of defaults and the random model line shows 100%, so the scorecard is between them.

B. The perfect model line shows 100% of defaults and the random model line shows 10%; the scorecard's 70% lies between them, so it is better than random but not perfect.

C. The perfect model line shows 100% of defaults and the random model line shows 50%; the scorecard is better than random.

D. The perfect model line shows 70% of defaults, which equals the scorecard, so the accuracy ratio is 1.

Answer: B

Step by step

1. Perfect model: if 10% of accounts will default, all defaults come from the riskiest 10%, so the line reaches 100% at x = 10%.
2. Random model: defaults are spread in proportion to the population, so the riskiest 10% hold 10% of defaults.
3. Observed scorecard: 70%, which is above 10% and below 100%.
4. The curve lies between random and perfect, so $0 < AR < 1$.

Why B is correct: The perfect line captures all defaults in the riskiest slice equal to the default rate, the random line captures defaults pro rata, and an observed curve between them signals partial discriminatory power.

Why the others are wrong

- **A:** Swaps the two benchmarks: perfect is the higher line (100%), random the lower (10%).

- **C:** Assumes the random model captures 50% of defaults; random captures defaults in proportion to population share, i.e. 10%.
- **D:** Confuses the observed 70% with the perfect line; the perfect line is at 100%, so AR is below 1.

LO 25.g

Describe the customer relationship cycle and discuss the trade-off between creditworthiness and profitability.

Q278 AnalystPrep Q.5471

REBUILT — Original option C described a genuine account-management activity and could be defended; reworded options so only the screening-stage initiative is correct.

A bank is refining its customer relationship cycle, which runs from marketing initiatives to screening applications, managing accounts and cross-selling. The team wants to improve the screening stage in particular. Which initiative best fits the screening stage of the cycle?

A. Increasing the frequency of mailed offers to all prospects regardless of their likely response.
B. Using application scorecards to make the accept/reject decision, set the initial credit line, and assign risk-based pricing tiers.
C. Revising credit lines and pricing for existing accounts based on observed payment behavior and renewing or collecting on them.
D. Offering every existing customer the full range of the bank's other products.

Answer: B

Step by step

1. The cycle has four stages: marketing, screening applications, managing accounts, cross-selling.
2. Screening = decide accept/reject on the basis of scorecards, grant the initial credit line, and set pricing appropriate to the applicant's risk.
3. Option B lists exactly these screening activities.
4. A is marketing, C is account management, D is cross-selling.

Why B is correct: Screening applicants consists of using scorecards to accept or reject applications, setting the initial credit line and setting risk-appropriate pricing.

Why the others are wrong

- **A:** Marketing-stage activity, and untargeted mailing contradicts the targeting/response-scoring logic of that stage.
- **C:** This is account management (modifying lines/pricing, renewals, collections), a later stage; picking it confuses screening with managing accounts.
- **D:** Cross-selling to existing customers closes the loop; it is also indiscriminate rather than based on detailed customer knowledge.

Q279 AnalystPrep Q.5472

REBUILT — Original overlapped heavily with the screening item; rebuilt to test the closing-the-loop role of cross-selling and the stage ordering of the cycle.

RiverBank describes its customer relationship cycle to new risk analysts. Which statement about the cycle is most accurate?

A. Cross-selling is the final stage; it uses detailed knowledge of existing customers to induce them to buy additional products, closing the loop on the cycle.
B. Cross-selling precedes account management, because customers must hold several products before credit lines can be authorized.
C. Screening applications only decides accept or reject; pricing is set later, during account management.
D. Marketing initiatives target only new prospects; existing customers are handled solely in account management.

Answer: A

Step by step

1. The cycle runs: marketing initiatives, screening applications, managing accounts, cross-selling.
2. Cross-selling uses detailed knowledge of existing customers to offer them additional products; it closes the loop.

3. B reverses the order of the last two stages.
4. C is wrong: screening includes setting the initial credit line and risk-appropriate pricing.
5. D is wrong: marketing targets both new and existing customers.

Why A is correct: Cross-selling is the last stage and closes the loop by leveraging what the bank has learned about existing customers.

Why the others are wrong

- **B:** Reverses the stage order; account management comes before cross-selling.
- **C:** Ignores that screening also grants the initial credit line and sets pricing for the client's risk level (managing accounts later may modify pricing).
- **D:** Marketing covers both new and existing customers (targeting, tailoring product/message), so this restricts it wrongly.

Q280

BT answers P2.T6.25.7.1

REBUILT — Source exhibit (segment credit scores/profitability) was lost and the options were inconsistent; rebuilt as a self-contained expected-profit comparison illustrating the creditworthiness vs profitability trade-off.

A card issuer compares two segments using per-account annual figures (EAD = exposure at default, LGD = loss given default).

Segment	Typical credit score	Annual revenue	Servicing cost	PD	EAD	LGD
X (transactors)	780	$120	$30	0.5%	$5,000	80%
Y (revolvers)	650	$600	$40	6.0%	$5,000	80%

Expected profit per account = revenue - servicing cost - expected loss. Which option gives the correct expected profits and the correct implication?

A. X = $70, Y = $320; the lower-score segment Y is more profitable despite higher credit risk.
B. X = $90, Y = $560; Y is far more profitable.
C. X = $65, Y = $260; Y is more profitable.
D. X = $20, Y = $240; X is more profitable because it has the lower expected loss.

Answer: A

Step by step

1. Expected loss = PD × EAD × LGD: X = $0.005 \times 5000 \times 0.8 = \$20$; Y = $0.06 \times 5000 \times 0.8 = \$240$.
2. Profit X = $120 - 30 - 20 = \$70$.
3. Profit Y = $600 - 40 - 240 = \$320$.
4. Y has the weaker credit score yet is more profitable because it revolves balances and pays interest; X is safer but earns little.
5. Lenders therefore need profit scoring as well as credit scoring.

Why A is correct: Profit must be measured net of expected loss; the riskier revolver segment generates much higher net profit, illustrating that creditworthiness and profitability are not the same thing.

Why the others are wrong

- **B:** Ignores expected loss (revenue less servicing cost only).
- **C:** Computed expected loss as PD x EAD, omitting the 80% LGD: losses of $25 and $300 give $65 and $260.
- **D:** Reports expected loss alone and treats low risk as high profitability, forgetting revenue and costs.

Q281

AnalystPrep Q.5473

REBUILT — Original was a soft 'best practice' item with giveaway distractors set in an investment firm; rebuilt as a clear scorecard-to-purpose matching question from the LO's content.

A lender uses several scorecards alongside credit bureau scores to evaluate creditworthiness and profitability across the customer relationship cycle. Which pairing of scorecard and use is correct?

A. Application score: supports the decision whether to extend credit to a new applicant.
B. Behavior score: assigns the probability that a customer will respond to a marketing offer.
C. Response score: assesses existing customers' credit usage and past delinquencies.
D. Attrition score: evaluates existing customers' potential profitability.

Answer: A

Step by step

1. Application scores support the accept/reject decision for new applicants (screening stage).

2. Behavior scores assess existing customers' credit usage and delinquencies (account management).
3. Response scores give the probability a customer responds to an offer (marketing).
4. Revenue scores evaluate existing customers' potential profitability; attrition scores estimate the likelihood that existing customers will close accounts, not renew a credit, or reduce their balances.

Why A is correct: Application scores are the scorecards used at screening to decide whether to extend credit to a new applicant.

Why the others are wrong

- **B:** Swaps behavior and response scores: the description given is a response score.
- **C:** Swaps response and behavior scores: the description given is a behavior score.
- **D:** The profitability measure for existing customers is the revenue score; attrition scores estimate the likelihood that customers close accounts, don't renew, or reduce balances.

LO 25.h

Discuss the benefits of risk-based pricing of financial services.

Q282

AnalystPrep Q.5941

A regional bank charges one interest rate to all applicants regardless of risk. Which situation correctly illustrates adverse selection in this lending process?

A. A borrower deliberately conceals a poor financial history to obtain a loan.
B. Low-risk borrowers are deterred by the high rate, leaving mostly high-risk borrowers applying.
C. A borrower cannot repay because of an unexpected job loss.
D. A borrower overestimates their ability to meet repayments and defaults.

Answer: B

Step by step

1. Adverse selection arises when a single price relative to risk attracts high-risk customers and pushes away low-risk ones.
2. B describes exactly this skewing of the applicant pool.
3. A is concealment of information, C is an unforeseen shock, D is borrower misjudgment.

Why B is correct: A uniform price that is high relative to low-risk borrowers' risk drives them away and leaves a riskier pool, which is adverse selection (and motivates risk-based pricing).

Why the others are wrong

- **A:** Misrepresentation of information by the borrower (fraud/information asymmetry), not a skewed applicant pool caused by pricing.
- **C:** An unexpected shock to the borrower's income, not a selection effect.
- **D:** Borrower misjudgment of capacity, not a selection effect of pricing.

Q283

BT answers P2.T6.25.7.2

REBUILT — Original segment table was missing and the OCR numbers were garbled (a 70% rate); rebuilt with a self-contained two-segment example and a quantified incremental-income question.

A retail bank has 10,000 loans of $10,000 each and charges a flat 10% rate to every customer. LGD is 50% for all loans. Competitors price prime borrowers at 8%, so under the flat rate half of the prime customers leave; subprime customers stay at any price considered. The bank is considering risk-based pricing (8% for prime, 14% for subprime), after which all customers are retained.

Segment	Loans	PD
Prime	6,000	1%
Subprime	4,000	6%

What is the incremental annual risk-adjusted income (interest income less expected loss) from adopting risk-based pricing?

A. $3.25 million
B. $3.40 million
C. $0.40 million
D. $0 (revenue is only redistributed between segments)

Answer: A

Step by step

1. Expected loss rate = PD x LGD: prime $1\% \times 50\% = 0.5\%$; subprime $6\% \times 50\% = 3\%$.
2. Flat pricing: only 3,000 prime remain. Prime $3{,}000 \times 10{,}000 \times (10\% - 0.5\%) = 2.85$m; subprime $4{,}000 \times 10{,}000 \times (10\% - 3\%) = 2.80$m; total \$5.65m.
3. Risk-based pricing: prime $6{,}000 \times 10{,}000 \times (8\% - 0.5\%) = 4.50$m; subprime $4{,}000 \times 10{,}000 \times (14\% - 3\%) = 4.40$m; total \$8.90m.
4. Incremental income $= 8.90 - 5.65 = 3.25$ million dollars.

Why A is correct: Risk-based pricing retains low-risk customers who would be lost to competitors under a single price (reducing adverse selection) and charges high-risk customers a rate that covers their higher expected loss.

Why the others are wrong

- **B:** Compares interest income only (\$10.4m vs \$7.0m) and ignores expected loss.
- **C:** Compares the two schemes with no prime attrition under the flat rate (\$8.9m vs \$8.5m), ignoring adverse selection.
- **D:** Assumes total income is merely redistributed; it ignores that the correct price retains prime customers and compensates for subprime risk.

Q284

BT answers P2.T6.25.7.3

REBUILT — Original relied on a missing calculation table; rebuilt with explicit segment data so risk-adjusted profit by score band can be computed.

A bank evaluating the purchase of a loan book analyses profitability by score band, as in risk-based pricing. Funding cost is 3%, operating expense is 1% of the loan and the cost of equity on allocated capital is 12%. Average loan size is \$20,000.

Segment	Loans	Rate	PD	LGD	Capital held
Prime	3,000	7%	1%	40%	8% of loan
Subprime	2,000	12%	8%	50%	30% of loan

Which statement best describes the result?

A. Subprime adds more value because its 9% spread over funding cost exceeds prime's 4% spread.

B. Subprime adds more value because after expected losses it earns about 4.0% versus about 2.6% for prime.

C. Prime adds more value: after losses, operating costs and capital costs it earns about 1.64% of the loan versus about 0.40% for subprime.

D. Neither segment is worth analysing separately because both earn a positive margin.

Answer: C

Step by step

1. Prime margin $= 7\% - 3\% - (1\% \times 40\%) - 1\% - (8\% \times 12\%) = 7 - 3 - 0.40 - 1 - 0.96 = 1.64\%$.
2. Subprime margin $= 12\% - 3\% - (8\% \times 50\%) - 1\% - (30\% \times 12\%) = 12 - 3 - 4 - 1 - 3.6 = 0.40\%$.
3. Dollar profit: prime $3{,}000 \times 20{,}000 \times 1.64\% \approx 0.98$ million dollars; subprime $2{,}000 \times 20{,}000 \times 0.40\% = 0.16$ million dollars.
4. Prime is clearly more valuable after risk and capital costs, so prices must be set band by band on a risk-adjusted basis.

Why C is correct: Higher nominal rates on riskier customers can be more than consumed by expected losses and the capital charge; risk-based pricing evaluates profitability at each score band so that price reflects PD, LGD, EAD, capital and costs.

Why the others are wrong

- **A:** Looks only at the nominal spread over funding cost, ignoring expected loss, operating cost and capital.
- **B:** Deducts expected loss but forgets the capital cost (subprime 4.0% vs prime 2.6%), which reverses the ranking.
- **D:** Positive margins do not make the segments equal; the segment-level difference in risk-adjusted profit is exactly the information risk-based pricing uses.

Q285

BT answers P2.T6.605.1 · BT notes P2.T6.605.1

REBUILT — Original asked about 'trends in retail banking' that sits across scoring LOs and had an arguable option; rebuilt as a clear question on the problem that risk-based pricing addresses (single price and adverse selection).

A lender charges the same price for a retail product to all customers regardless of risk. Which outcome is most likely?

A. Low-risk customers are drawn in by the price while high-risk customers are driven away.

B. High-risk customers are attracted by a price that is low relative to their risk, while low-risk customers are

pushed away by a price that is high relative to their risk.

C. The portfolio's average probability of default falls because the single price screens out risky applicants.

D. Revenue is maximised because the lender avoids differentiating among score bands.

Answer: B

Step by step

1. A single price is cheap for high-risk customers relative to their PD/LGD, and expensive for low-risk customers.
2. High-risk customers therefore take up the product and low-risk customers go to competitors with lower prices.
3. This is adverse selection, which risk-based pricing (different prices for different risk) is designed to reduce.

Why B is correct: A one-price-fits-all policy mis-prices risk, producing adverse selection; risk-based pricing aligns price with risk.

Why the others are wrong

- **A:** Reverses the effect; it is the high-risk customers who find the flat price attractive.
- **C:** Adverse selection raises, not lowers, the average PD of the portfolio.
- **D:** Ignores the fact that the flat price loses low-risk business and under-charges high-risk business; nothing about it maximises risk-adjusted revenue.

Q286

BT answers P2.T6.605.3 · BT notes P2.T6.605.3

REBUILT — Source answer B referred to machine learning, which is not in the reading; rebuilt option B as a clearly false statement (risk-based pricing still requires PD/LGD/EAD estimation).

Which of the following is NOT a benefit of risk-based pricing of retail financial services?

A. It lets managers map pricing strategies to metrics such as profit/loss, revenue, market share and risk-adjusted return at each score band.

B. It removes the need to estimate PD, LGD and EAD at the account level because prices are set by score band alone.

C. It lets managers evaluate trade-offs among profitability, market share and risk in order to increase shareholder value.

D. For the same product, it lets the bank charge higher prices to higher-risk customers and lower prices to lower-risk customers.

Answer: B

Step by step

1. Risk-based pricing charges different prices for the same product according to customer risk (D).
2. Prices are mapped to profit, revenue, market share and risk-adjusted return by score band (A) and used to weigh profitability, market share and risk to raise shareholder value (C).
3. Pricing depends on key factors including take-up probability, PD, LGD, EAD, capital allocated, cost of equity and operating expenses, so these must still be estimated; B is not a benefit.

Why B is correct: Risk-based pricing is built on account-level economics (PD, LGD, EAD, capital, costs); it does not remove the need for them.

Why the others are wrong

- **A:** True benefit: pricing strategies can be mapped to profit, revenue, market share and risk-adjusted return by score band.
- **C:** True benefit: management can evaluate trade-offs among profitability, market share and risk to increase shareholder value.
- **D:** True benefit: differentiated prices for the same product by risk are the essence of risk-based pricing.

Q287

Schweser Module Quiz 25.2 Q2

By implementing risk-based pricing on its mortgage products, a bank will likely charge a:

A. higher interest rate to a customer with a higher FICO score.

B. lower interest rate to a customer with a lower credit bureau score.

C. higher interest rate to a customer with a higher probability of default.

D. lower interest rate to a customer positioned on a lower relative score band.

Answer: C

Step by step

1. Risk-based pricing ties the rate to risk: higher PD means a higher rate.
2. Higher scores (FICO/credit bureau) indicate lower risk and so lower rates; lower score bands indicate higher risk and so higher rates.

Why C is correct: The riskier the customer (higher PD, lower score), the higher the price charged.

Why the others are wrong

- **A:** A higher FICO score signals lower risk, so a lower rate is charged.
- **B:** A lower credit bureau score signals higher risk, so a higher rate is charged.
- **D:** A lower relative score band means higher risk, so a higher rate.

Reading 26. Country Risk: Determinants, Measures, and Implications

Aswath Damodaran, Country Risk: Determinants, Measures, and Implications – The 2022 Edition (2022).

LO	Learning objective	Qs
26.a	identify and explain the different sources of country risk.	4
26.b	evaluate the methods for measuring country risk and discuss the limitations of using those methods.	3
26.c	compare and contrast foreign currency defaults and local currency defaults.	3
26.d	explain the consequences of a country's default.	5
26.e	discuss measures of sovereign default risk and describe components of a sovereign rating.	3
26.f	describe the shortcomings of the sovereign rating systems of rating agencies.	5
26.g	compare the use of credit ratings, market-based credit default spreads, and CDS spreads in predicting default.	3

LO 26.a

Identify and explain the different sources of country risk.

Q288 BT answers P2.T6.24.9.1 · BT notes P2.T6.24.9.1

Stan is evaluating an investment in a small island country named 1bit. The nation is governed by a robust, unelected leader who has held power for the past 15 years. A publication rated 1bit as a 44 after surveying 400 economists. Given the information provided, what potential concern(s) should an investor be wary of and what publication most likely rated 1bit?

A. Continuous risk, Euromoney
B. Discontinuous risk, Euromoney
C. Continuous risk, The Economist
D. Discontinuous risk, The Economist

Answer: B

Step by step

1. Step 1: A single unelected leader in power for 15 years indicates an authoritarian regime.
2. Step 2: Risk in dictatorships is discontinuous: policies change rarely, but changes are severe and hard to hedge (wars, expropriation). Risk in democracies is continuous but generally low.
3. Step 3: The publication that surveys 400 economists and scores countries from 0 to 100 is Euromoney. The Economist instead assesses currency, sovereign debt and banking risk.
4. Step 4: Discontinuous risk and Euromoney gives B.

Why B is correct: Authoritarian rule implies infrequent but severe policy change (discontinuous risk), and the 400-economist 0-100 survey is Euromoney's country risk score.

Why the others are wrong

- **A:** Right publication, but labels authoritarian risk as continuous; continuous (frequent, mild) risk is the democratic pattern.
- **C:** Both parts wrong: continuous risk is the democracy pattern, and The Economist does not use a 400-economist survey.
- **D:** Correct risk type but wrong publication; The Economist scores currency, sovereign debt and banking risk, with lower scores meaning lower risk.

Q289 Mock 1 Q53

Which of the following statements is incorrect regarding the factors that influence a country's sovereign default risk?

A. There may be more pressure on the leaders of dictatorships if the country defaults.
B. Countries with more diversified economies are more likely to have stable tax receipts.
C. Rating agencies and other market participants look at implicit guarantees by other entities.
D. Governments make significant commitments to their citizens, including health care and pension payments.

Answer: A

Step by step

1. Step 1: The question asks for the incorrect statement.
2. Step 2: Political risk factor: default exposes leaders to pressure, but dictators face less political backlash than elected leaders, so there may be LESS pressure on dictatorships. Statement A reverses this.
3. Step 3: B, C and D match the listed determinants: diversification stabilizes tax receipts; implicit backing (e.g., EU members) affects assessed default risk; pension and health commitments raise default risk.

Why A is correct: Dictators face less political backlash from default, which is why autocracies may be more likely to default; statement A says the opposite.

Why the others are wrong

- **B:** True: diversified economies (e.g., India versus Jamaica or Peru) have more stable tax receipts.
- **C:** True: implicit, not explicit, backing from stronger economies lowered assessed default risk for Spain, Greece and Portugal.
- **D:** True: greater pension and health care commitments increase default risk.

Q290

Mock 2 Q42

An emerging markets analyst is preparing a briefing on political risk and its impact on investment returns. The analyst considers the effects of governmental regimes, corruption and policy continuity on company profitability and investor returns. Which of the following statements is most likely incorrect?

A. It is not clear whether authoritarian or democratic governmental regimes produce higher economic growth.

B. Some investors prefer the relative stability of investing in companies from countries where a single leader controls decision-making.

C. Countries with democratic governmental policies exhibit discontinuous risk because policies can change with each election cycle.

D. Countries with high government corruption impose an implicit tax on investors because corruption directly reduces company profits and returns.

Answer: C

Key correction: Source key (Mock 2) said D; correct is C because corruption does act as an implicit tax that directly reduces company profits (and indirectly investor returns), so D is true, while democratic risk is continuous, not discontinuous, so C is false.

Step by step

1. Step 1: Identify the false statement.
2. Step 2: A is true: studies are mixed on whether authoritarian or democratic systems grow faster.
3. Step 3: B is true: some investors like autocracies because policies are locked in and predictable.
4. Step 4: D is true: corruption is like an implicit tax that directly reduces company profits and returns and indirectly reduces investor returns.
5. Step 5: C is false: democratic risk is continuous (frequent, generally low); discontinuous risk (rare, severe change) characterizes dictatorships.

Why C is correct: Democracies carry continuous, generally low risk; discontinuous risk belongs to authoritarian regimes, so C is the incorrect statement.

Why the others are wrong

- **A:** True statement: evidence on growth under the two regimes is mixed.
- **B:** True statement: policy stability under single-leader rule appeals to some investors.
- **D:** True statement: corruption is an implicit tax that directly cuts company profits and returns; rejecting it as 'indirect only' is the error behind the source key.

Q291

Schweser Module Quiz 26.1 Q1

A portfolio manager for a hospital foundation is considering the inclusion of sovereign bonds in the fixed income portion of the foundation's portfolio. Much to the surprise of his colleagues, the manager plans to purchase the bonds of a country that has long been under authoritarian rule. He cites "lower political risk" when asked about his investment decision. Which of the following statements is most likely what the manager meant by his assertion of lower risk?

A. Authoritarian regimes are more likely to control corruption in government agencies.

B. Government policies that may affect debt repayment are often more stable under an authoritarian regime.

C. Relative to a democracy, risks are greater on a day-to-day basis, but the effects are less detrimental overall.

D. In most authoritarian countries, property rights are protected and property disputes are settled quickly.

Answer: B

Step by step

1. Step 1: Some investors prefer autocracies because policies are locked in and predictable, whereas elections in a democracy can change policy significantly.
2. Step 2: So the 'lower risk' claim refers to policy stability, which matches B.
3. Step 3: A and D are not supported: corruption and weak legal systems are risks of authoritarian states.
4. Step 4: C reverses the pattern: dictatorship risk is less frequent but more severe (discontinuous).

Why B is correct: Authoritarian policy is locked in and predictable, so policy changes affecting debt repayment are less frequent than under democratic election cycles.

Why the others are wrong

- **A:** Corruption and lack of legal systems are risks of authoritarian states that can outweigh the benefits.
- **C:** Reversed: day-to-day (continuous) risk is greater in democracies; dictatorships have rarer but more severe discontinuous risk.
- **D:** Weak legal systems and unreliable property protection are typical authoritarian risks, not benefits.

LO 26.b

Evaluate the methods for measuring country risk and discuss the limitations of using those methods.

Q292 Schweser Module Quiz 26.1 Q2

In an attempt to understand country risk, an analyst at Global Funds examines multiple sources of information to determine the truest measure of risk. She considers sovereign risk ratings, default risk spreads, and composite measures of risk. Which of the following sources relies on surveys of several hundred economists to measure sovereign risk?

A. Political Risk Services.
B. The Economist.
C. The World Bank.
D. Euromoney.

Answer: D

Step by step

1. Match each provider to its method: Euromoney surveys 400 economists who rate country risk factors on a 0 to 100 scale (higher means lower risk).
2. PRS scores 22 political, economic and financial variables for 100+ countries.
3. The Economist assesses currency, sovereign debt and banking risk.
4. The World Bank compiles measures from several sources for 215 countries across six governance areas.
5. Only Euromoney relies on an economist survey, so the answer is D.

Why D is correct: Euromoney's country risk score comes from a survey of about 400 economists, unlike the variable-based or compiled approaches of the other providers.

Why the others are wrong

- **A:** PRS builds a composite from 22 variables across political, economic and financial dimensions, not an economist survey.
- **B:** The Economist scores currency, sovereign debt and banking risk using its own assessment, not a survey of economists.
- **C:** The World Bank compiles indicators from several sources (corruption, government effectiveness, etc.), not a survey of economists.

Q293 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A risk analyst at an asset manager is comparing two emerging-market countries, X and Y, using three country risk services. The scores are shown below. Euromoney ranks countries from 0 to 100 with higher scores indicating lower risk. The Economist uses a scale on which lower scores indicate lower risk. The World Bank scores are scaled around zero, with positive numbers indicating less risk.

Provider	Country X	Country Y
Euromoney	62	48
The Economist	55	40
World Bank	+0.7	-0.4

Which of the following correctly identifies the providers whose scores indicate that Country X is less risky than Country Y?

A. Euromoney only.
B. Euromoney, The Economist, and the World Bank.
C. The Economist and the World Bank only.
D. Euromoney and the World Bank only.

Answer: D

Step by step

1. Euromoney: higher score means lower risk, so 62 > 48 means X is less risky.
2. The Economist: lower score means lower risk, so 55 > 40 means X is riskier.
3. World Bank: positive means less risk, so +0.7 vs -0.4 means X is less risky.
4. Only Euromoney and the World Bank rank X as less risky.

Why D is correct: The providers use different, non-standardized scales, so each score must be read in its own direction; only Euromoney and the World Bank point to X being less risky.

Why the others are wrong

- **A:** Wrong: reverses the World Bank sign convention (treats positive as more risk), so omits the World Bank.
- **B:** Wrong: assumes higher scores mean lower risk for every provider, ignoring that The Economist's scale runs the other way.
- **C:** Wrong: swaps the Euromoney and Economist conventions (treats higher Economist scores as lower risk and higher Euromoney scores as higher risk).

Q294 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A strategist at a pension fund reviews The Economist's country risk scores, on which lower scores indicate lower risk. Country M has a score of 70 and Country N has a score of 35. Which of the following conclusions is most appropriate given the limitations of country risk services?

A. Country M is riskier than Country N, but it cannot be concluded that M is twice as risky as N.
B. Country M is twice as risky as Country N because 70 is twice 35.
C. Country M is less risky than Country N because it has the higher score.
D. Country M and Country N cannot be ranked because the scores are not standardized across providers.

Answer: A

Step by step

1. The Economist: lower score = lower risk, so M (70) is riskier than N (35).
2. Scores are better used as rankings than as true scores.
3. A ratio of 70/35 = 2 has no meaning as a measure of relative risk, so 'twice as risky' cannot be concluded.

Why A is correct: Country risk scores are ordinal: they support a ranking within one provider but not a proportional statement about risk.

Why the others are wrong

- **B:** Wrong: treats the score as a cardinal measure and reads the ratio 70/35 = 2 as relative risk.
- **C:** Wrong: applies the convention of Euromoney/PRS (higher = lower risk) to The Economist.
- **D:** Wrong: lack of standardization hinders comparisons across providers, not rankings within a single provider's scale.

LO 26.c

Compare and contrast foreign currency defaults and local currency defaults.

Q295 BT answers P2.T6.24.9.2 · BT notes P2.T6.24.9.2

Jim is a wealthy citizen of 1bit and is not allowed to hold assets outside of his home country. 1bit is a small autocratic country that is currently burdened with high national debt at 9.5x GDP denominated in IBT (1bit's home currency). As an investor, what is Jim's primary risk, and how can he hedge this risk?

A. Inflation, by purchasing gold and other hard assets
B. Inflation, by purchasing 1bit debt
C. Deflation, by purchasing gold and other hard assets
D. Deflation, by purchasing 1bit debt

Answer: A

Step by step

1. Debt is denominated in 1bit's own currency, so the government can in principle repay by printing more of it rather than defaulting.
2. With debt at 9.5x GDP, default is costly and printing money is the likely route; this debases the currency and pushes inflation up, possibly sharply.
3. Jim cannot hold foreign assets, so he cannot escape the currency; the hedge is real assets (gold, real estate, hard assets) whose value rises with inflation.
4. 1bit debt is a fixed nominal claim in the currency being debased, so it would lose real value rather than hedge.

Why A is correct: High local currency debt tends to be inflated away by printing money, so the primary risk is inflation, hedged with hard (real) assets.

Why the others are wrong

- **B:** Right risk, wrong hedge: 1bit debt pays fixed nominal amounts in the currency being debased, so it adds to the inflation exposure.
- **C:** Deflation is unlikely because it would raise the real value of the debt, which the government is trying to avoid; also wrong risk even though the hedge asset is right.
- **D:** Wrong risk (deflation) and wrong hedge; deflation would increase the real debt burden, making printing more likely, not less.

Q296 Schweser Module Quiz 26.2 Q1

Which of the following statements regarding foreign currency defaults is true?

A. African countries are responsible for the greatest number of defaults in the last 50 years.
B. Before the 20th century, no country had ever defaulted on funds borrowed in a foreign currency.
C. Latin America accounted for more than 60% of foreign currency defaults in the 1990s.
D. Countries are more likely to default on funds borrowed from foreign banks than on sovereign bond issues.

Answer: D

Step by step

1. Evaluate each statement against the historical evidence on foreign currency defaults.
2. A: Latin America, not Africa, accounts for the largest share of defaults (in dollar terms) over the last 50 years. False.
3. B: Defaults occurred in seven periods from 1824-1834 to 1998-2003, so 19th-century foreign defaults existed. False.
4. C: Latin America accounted for at least 60% of defaults in each of the last five decades except the 1990s. False.
5. D: Countries have been more likely to default on bank debt than on sovereign bonds. True.

Why D is correct: Studies show sovereigns default more often on foreign bank loans than on bonds they have issued.

Why the others are wrong

- **A:** Confuses Africa with Latin America, which is the epicenter of sovereign defaults.
- **B:** Ignores that sovereign defaults go back to 1824-1834 and earlier; defaults are not a 20th-century phenomenon.
- **C:** Reverses the exception: the 1990s is the only decade in which Latin America was below 60%.

Q297 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A sovereign credit analyst is assessing a government with a floating exchange rate that has issued debt in its own currency. The government has also borrowed heavily in foreign currency, and these borrowings have been used to fund local currency assets. Which of the following statements best explains why this government might default on its local currency debt rather than print money to repay it?

A. Printing money is not possible because the country is required to hold gold reserves to back its currency.

B. Printing money would raise inflation and devalue the local currency, reducing the value of local currency assets while leaving the foreign currency liabilities unchanged.

C. Printing money would be inexpensive because devaluation reduces the real value of foreign currency liabilities.

D. Printing money is prohibited because foreign currency lenders have priority over holders of local currency debt.

Answer: B

Step by step

1. Compare costs of default with costs of printing money (inflation, devaluation).
2. Foreign currency debt funding local currency assets raises the cost of printing: devaluation shrinks asset values while foreign liabilities are unchanged.
3. So the more foreign debt funds local assets, the less attractive printing becomes, and default on local debt becomes a more likely choice.

Why B is correct: The greater the extent to which foreign currency debt funds local currency assets, the more negative the consequences of printing money, which helps explain local currency defaults.

Why the others are wrong

- **A:** Wrong: the gold standard ended in 1971 and the scenario describes a floating-rate country, so it is not the explanation.
- **C:** Wrong: devaluation increases, not decreases, the local-currency burden of foreign currency liabilities.
- **D:** Wrong: the reading gives no such legal prohibition; the constraint is the economic cost of printing.

LO 26.d

Explain the consequences of a country's default.

Q298 AP Mock 3 Q49

REBUILT — Original distractors were giveaways and the stem was vague; rewrote options to include the 8% / 15-year findings and a plausible 'short-lived' distractor.

A study of sovereign defaults examines the effect of a default on trade between the defaulting country and its trading partners. Which of the following conclusions is most consistent with the research on the consequences of default?

A. Bilateral trade rises as renegotiated terms improve the country's competitiveness, and the effect persists for up to 15 years.

B. Bilateral trade is essentially unaffected because trade agreements are insulated from sovereign credit events.

C. Trade retaliation reduces bilateral trade (an average drop of about 8%), the effect can last up to 15 years, and export-oriented industries are hit hardest.

D. Trade retaliation reduces bilateral trade, but the effect is short-lived and disappears within about one year of the default.

Answer: C

Step by step

1. Recall the research findings on sovereign default: trade retaliation follows default.
2. One study finds an average 8% drop in bilateral trade after default, with effects lasting up to 15 years.
3. Export-oriented industries are the most affected.
4. Only C matches all three findings.

Why C is correct: Default invites trade retaliation by trading partners, producing a persistent fall in bilateral trade (about 8%, up to 15 years) concentrated in exporters.

Why the others are wrong

- **A:** Reverses the direction of the effect; default reduces, not raises, bilateral trade.
- **B:** Ignores trade retaliation; the research finds a material drop in bilateral trade.
- **D:** Gets the direction right but confuses the trade effect with the GDP effect, which is short-lived (within the first

year); trade effects last up to 15 years.

Q299 AnalystPrep Q.5913

Historical research on the consequences of sovereign default (via the after-effects of sharp currency devaluations that often accompany default) most likely indicates which of the following about political change?

A. A negligible increase in the likelihood of a change in leadership, as defaults are typically unforeseen.
B. A reduced likelihood of political turnover because of increased national solidarity during crises.
C. A significant rise in the probability of a change in the top leader and an even larger rise for the finance minister or head of the central bank.
D. No consistent trend in political turnover, so other factors play the decisive role.

Answer: C

Step by step

1. Default is often accompanied by sharp currency devaluations.
2. Studies of devaluations (1971-2003) find a 45% increase in the probability of a change in the president or prime minister.
3. They also find a 64% increase in the probability of a change in the finance minister or central bank head.
4. This matches C.

Why C is correct: Default and the devaluations that accompany it increase the likelihood of political change, with a 45% rise for the top leader and 64% for financial leadership.

Why the others are wrong

- **A:** Understates the effect; the increases are large (45% and 64%), not negligible.
- **B:** Wrong direction; default raises, not lowers, turnover.
- **D:** The research does find a consistent, significant increase in turnover.

Q300 AnalystPrep Q.5927

How is a sovereign default most likely to influence the banking sector of the defaulting country?

A. The banking sector is fortified by international support and regulatory reform following the default.
B. The banking sector is largely insulated from the default because of regulatory protections.
C. Demand for commercial loans increases as banks finance recovery from the default.
D. The probability of a banking crisis increases, as default makes the banking system more fragile.

Answer: D

Step by step

1. A study of 149 countries (1975-2000) finds the probability of a banking crisis is 14% following a sovereign default.
2. This is 11 percentage points higher than for non-defaulting countries.
3. Hence default makes banking systems more fragile (D).

Why D is correct: Banks hold sovereign debt and depend on confidence, so default raises the probability of a banking crisis (14%, 11 points above non-defaulters).

Why the others are wrong

- **A:** Reverses the effect; default weakens rather than strengthens banks.
- **B:** Banks are not insulated; evidence shows a significantly higher crisis probability.
- **C:** Default is associated with financial fragility and weaker credit conditions, not a rise in loan demand; not a finding of the research.

Q301 BT answers P2.T6.24.9.3 · BT notes P2.T6.24.9.2

REBUILT — Original stem referenced an undefined entity and a 74% banking-crisis figure not in the reading, and mixed in a political-risk taxonomy; rebuilt with self-contained options that test the consequences of default.

An investor, Jim, anticipates that the government of a country will default on its sovereign debt. Based on research on the consequences of default, which position is most consistent with that view if Jim can invest only in that country's markets?

A. Short the equity of domestic banks, because default makes a banking crisis more likely.
B. Hold domestic cash and long equities, because default is usually followed by faster GDP growth.

C. Long domestic equities, because trade retaliation raises export demand.
D. Long the domestic currency, because defaults are typically followed by currency appreciation.

Answer: A
Step by step
1. Research finds a banking crisis is more likely after default (14% probability, 11 points above non-defaulters).
2. Default is also followed by GDP decline of 0.5%-2.0%, sharp currency devaluations, and capital market turmoil.
3. Therefore a short position in domestic bank equity benefits from the expected fragility.
4. Options B, C, D each assume the opposite of the research findings.

Why A is correct: Default makes banking systems more fragile, so shorting bank equities is consistent with an anticipated default.
Why the others are wrong
- **B:** GDP growth falls 0.5%-2.0% after default and capital markets are in turmoil, so a long equity stance is inconsistent.
- **C:** Trade retaliation reduces, not increases, exports (bilateral trade falls about 8%).
- **D:** Sharp currency devaluations often follow default, so a long currency position loses.

Q302 Schweser Module Quiz 26.2 Q2

Following a sovereign default:
A. borrowing costs rise 1.0% to 2.5%, and the country is more likely to suffer a banking crisis.
B. countries are more likely to suffer banking crises and GDP growth generally falls 2.0% to 3.5%, but the drop is usually short-lived.
C. ratings fall one to two notches, and GDP growth falls between 0.5% to 2.0%.
D. trade retaliations are likely to occur but are short-lived, lasting only six months to one year.

Answer: C
Step by step
1. Ratings of past defaulters are one to two notches lower: C's first clause is correct.
2. GDP growth falls 0.5%-2.0% (short-lived): C's second clause is correct.
3. A: borrowing costs rise only 0.5%-1.0%.
4. B: GDP decline is 0.5%-2.0%, not 2.0%-3.5%.
5. D: trade retaliation can last up to 15 years.

Why C is correct: C states both the rating (one to two notches) and GDP (0.5%-2.0%) findings correctly.
Why the others are wrong
- **A:** Overstates the borrowing-cost increase (actual 0.5%-1.0%).
- **B:** Overstates the GDP decline (actual 0.5%-2.0%).
- **D:** Trade retaliation effects can last up to 15 years, not six months to a year.

LO 26.e

Discuss measures of sovereign default risk and describe components of a sovereign rating.

Q303 AP Mock 2 Q65

The differentiation between local and foreign currency ratings is fundamental to the assessment of country risk by credit rating agencies. What aspect of a country's economic and financial system primarily drives this distinction?
A. The country's reliance on foreign aid and grants, which impacts the balance of payments and external debt levels.
B. A country's fiscal deficit as a percentage of GDP, which signals fiscal health and borrowing needs in different currencies.
C. Monetary policy independence, where countries with greater control over their currency have a higher potential for divergence between local and foreign currency ratings.
D. The volume of international trade conducted by the country, dictating the need for separate assessments for local and foreign currency transactions.

Answer: C

Step by step

1. Rating agencies rate a sovereign's local currency debt and foreign currency debt separately.
2. A sovereign can print its own currency to repay local currency debt, but cannot print foreign currency.
3. The more monetary policy independence a country has (floating exchange rate, deep domestic bond market), the more it can rely on printing local currency, so the local currency rating can sit well above the foreign currency rating.
4. Countries that give up monetary independence (dollarization, joining a monetary union, hard peg) see the two ratings converge.

Why C is correct: The gap between local and foreign currency ratings is primarily a function of monetary policy independence: the ability to print the local currency lowers local currency default risk but does not help with foreign currency obligations.

Why the others are wrong

- **A:** Foreign aid affects the balance of payments and external position, but it is not the main driver of the local-versus-foreign rating gap; it affects both ratings.
- **B:** The fiscal deficit signals fiscal health and bears on both ratings alike; it does not explain why the two ratings differ.
- **D:** Trade volume influences foreign exchange earnings, but the size of trade does not create a separate rating; the ability to print the currency does.

Q304

BT answers P2.T6.24.10.1 · BT notes P2.T6.24.10.1

REBUILT — Duplicate BT answers/notes records merged; original table had unverifiable figures (300% pension, OCR-garbled labels) and a central bank independence row and PRS score not in the Schweser/GARP factor list, so the table was rebuilt around the listed default-risk factors with one clearly best country.

A sovereign analyst compares four countries to judge which is likely to have the highest sovereign credit rating. The data (revenues in USD millions):

Metric	Aldoria	Bretonia	Calvera	Dunmark
Government debt (% of GDP)	120%	118%	121%	119%
Pension and health care commitments (% of GDP)	60%	20%	18%	22%
Share of tax receipts from largest sector	22%	45%	15%	40%
Political system	Democracy	Democracy	Democracy	Autocracy
Government revenue 2021	18.45	9.80	10.50	4.35
Government revenue 2022	19.65	15.60	11.20	6.23
Government revenue 2023	21.00	11.20	12.00	4.90

Which country most likely has the highest sovereign credit rating?

A. Aldoria
B. Bretonia
C. Calvera
D. Dunmark

Answer: C

Step by step

1. Indebtedness is nearly identical (118%-121% of GDP), so it does not separate the countries; high debt alone does not imply a low rating (e.g., the US, Japan, France are highly rated).
2. Pension and health care commitments: Aldoria 60% of GDP is far above the others (18%-22%), so it has the highest default risk on this factor; Calvera is lowest at 18%.
3. Stability of tax receipts: a diversified base is more stable. Largest-sector share is lowest for Calvera (15%) and highest for Bretonia (45%) and Dunmark (40%). Revenue volatility (std. dev./mean): Aldoria about 5.3%, Calvera about 5.5% (population standard deviation), Dunmark about 15.3%, Bretonia about 20.3%.
4. Political risk: Dunmark is an autocracy, where leaders face less pressure to avoid default, which raises default risk.
5. Calvera combines the lowest pension burden, the most diversified and stable revenue, and a democracy, so it most likely has the highest rating: C.

Why C is correct: Sovereign default risk depends on debt level, pension/social commitments, size and stability of tax receipts, political risk and outside backing; with debt similar, Calvera is best on every other factor.

Why the others are wrong

- **A:** Aldoria has stable and growing revenue, but its pension and health care commitments of 60% of GDP are about three times the others; ignoring social commitments leads to this choice.

- **B:** Bretonia has low commitments but depends on one sector (45% of receipts) and has the most volatile revenue (it jumped to 15.6 then fell to 11.2); choosing it focuses on low commitments and the slightly lowest debt ratio only.
- **D:** Dunmark has low commitments but concentrated receipts (40%), volatile revenue, and an autocratic system with weaker political pressure against default; it is not the strongest on any factor.

Q305 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A fixed income analyst compares how rating agencies assess sovereign credit risk. Which of the following statements regarding the objective of sovereign ratings is correct?

A. Both agencies focus on default risk faced by official creditors such as the IMF and the World Bank.
B. Moody's ratings measure only the probability of default, while S&P ratings measure both the probability and the severity of default.
C. S&P ratings measure only the probability of default, while Moody's ratings measure both the probability and the severity of default.
D. Both agencies define default only as a missed payment of principal or interest on the due date.

Answer: C

Step by step

1. S&P ratings capture the probability of default.
2. Moody's ratings capture probability and severity (expected recovery) of default.
3. Agencies focus on banks and private bondholders, not official creditors, and define default to include restructuring.

Why C is correct: S&P measures probability of default; Moody's also incorporates expected recovery, i.e., the severity of default.

Why the others are wrong

- **A:** Wrong: ratings focus on banks and private bondholders, not official creditors like the IMF and World Bank.
- **B:** Wrong: reverses the two agencies.
- **D:** Wrong: default also includes rescheduling or restructuring of debt, not just outright nonpayment.

LO 26.f

Describe the shortcomings of the sovereign rating systems of rating agencies.

Q306 AP Mock 3 Q75 · AnalystPrep Q.5909

A global investment firm's CRO raises concerns about relying on sovereign ratings from major agencies. Which statement most accurately captures a significant shortcoming of sovereign rating systems?

A. The ratings tend to be excessively volatile, leading to frequent and unnecessary market disruptions.
B. Sovereign ratings typically lag behind market indicators and fail to provide timely warnings of impending crises.
C. Rating agencies consistently overestimate the creditworthiness of developed economies while underestimating emerging markets.
D. The methodologies used are too transparent, allowing governments to manipulate economic data to achieve better ratings.

Answer: B

Step by step

1. Recall the listed shortcomings: upward bias, herd behavior, rating changes not timely (too little, too late), overreaction/vicious cycle, and ratings failures.
2. The timing criticism is that agencies take too long to change ratings, so changes arrive too late to protect investors in a crisis.
3. Option B states exactly this; the other options contradict or are not among the recognized shortcomings.

Why B is correct: Investors need timely default-risk assessments, but agencies are criticized for changing ratings too slowly, after market indicators have already moved.

Why the others are wrong

- **A:** The criticism is that ratings are slow, not too volatile; the overreaction criticism applies only to crisis periods and is not general volatility.
- **C:** Home bias is limited to US-based agencies over-rating the US; there is no systematic over-rating of all developed and under-rating of all emerging markets (the general bias is upward).
- **D:** Governments hiding bad news is an information problem, not excessive transparency of methodology.

Q307

AnalystPrep Q.5910

REBUILT — Option D also described herd-type cascading changes (two defensible answers); rewrote D and B so only C describes herd behavior.

Herd behavior among credit rating agencies reduces the value of their sovereign risk evaluations. Which of the following best describes how herd behavior manifests?

A. Agencies rely heavily on public financial data and international economic indicators, so their assessments lack depth.

B. Agencies review their methodologies on a fixed annual calendar, so their rating changes are spread evenly through the year.

C. When one agency raises or lowers a sovereign's rating, the other agencies tend to follow suit.

D. Agencies assign higher ratings to sovereigns that generate more fee revenue for them.

Answer: C

Step by step

1. Herd behavior concerns the independence of the agencies' views.
2. The documented symptom is that after one agency upgrades or downgrades a sovereign, the others tend to follow.
3. So the agencies' opinions are not independent, which reduces the benefit of having three separate ratings. This is option C.

Why C is correct: Following another agency's rating action means the assessments are not independent, so multiple ratings add little information.

Why the others are wrong

- **A:** Reliance on common public data is a cause cited for herd behavior (limited resources), not the behavior itself.
- **B:** Calendar-driven timing is not herd behavior and is not a documented criticism.
- **D:** Fee-driven higher ratings describe revenue bias (upward bias), not herd behavior.

Q308

AnalystPrep Q.5911

REBUILT — Original keyed a corporate-template methodology criticism that is not among the shortcomings in the LO; replaced with a vicious-cycle scenario testing the same LO.

During a sovereign debt crisis, a rating agency downgrades the country by several notches, more than fundamentals appear to justify. The downgrade lifts the country's borrowing costs and deepens the crisis. Which criticism of rating agencies does this best illustrate?

A. Herd behavior

B. Rating changes that are not timely

C. Ratings failures

D. Overreaction leading to a vicious cycle

Answer: D

Step by step

1. The agency lowers ratings by more than seems warranted once the crisis has begun.
2. The downgrade raises funding costs and worsens the crisis, creating a feedback effect.
3. This is the overreaction / vicious cycle criticism.

Why D is correct: Excessive downgrades during a crisis feed back into the crisis itself, making it worse.

Why the others are wrong

- **A:** Herd behavior is other agencies copying one agency's action; the scenario describes a feedback effect on the country.
- **B:** The timing criticism is that agencies act too slowly; here the agency acts aggressively.
- **C:** Ratings failures refers to multiple rating changes in a short period admitting an initially wrong assessment; here it is one overly large downgrade with a feedback effect.

Q309 AnalystPrep Q.5912 · AnalystPrep Q.5916

REBUILT — Source keys cited the issuer-pays conflict, which GARP/Schweser say is much less of a problem for sovereigns; rewrote options so the sub-sovereign revenue bias is the answer.

Sovereign ratings are often criticized as biased upward. Which of the following is a recognized contributor to that upward bias?

A. The issuer-pays conflict of interest, which is the main driver for sovereign ratings because sovereigns pay large rating fees.

B. Agencies earn significant revenue rating sub-sovereign issuers, which can discourage harsh sovereign ratings because a sovereign downgrade is followed by sub-sovereign downgrades.

C. Rating analysts frequently leave to work for the sovereigns they rate, which creates a conflict of interest.

D. Agencies charge investors subscription fees for sovereign ratings and want to keep subscribers satisfied.

Answer: B

Key correction: Source keys (AnalystPrep Q.5912 said B, Q.5916 said C) named the issuer-pays model; per GARP/Schweser that conflict is much less of a problem for sovereigns, and the upward bias is attributed to revenue bias via sub-sovereign ratings and governments' selective disclosure.

Step by step

1. Sovereign ratings are generally provided free to users, so revenue comes from issuers or related businesses.
2. The issuer-pays conflict is much less of a problem for sovereigns: revenue from them is small while the reputational damage of over-rating is large, so A is wrong.
3. Agencies earn significant revenue from sub-sovereigns (states, cities); a sovereign downgrade is followed by sub-sovereign downgrades, so sub-sovereigns resist it, contributing to upward bias.
4. Analysts moving to the sovereigns they rate is uncommon and not a problem, so C is wrong.

Why B is correct: Revenue bias from rating sub-sovereigns (and governments' selective disclosure) pushes ratings upward; the plain issuer-pays conflict is a weak argument for sovereigns.

Why the others are wrong

- **A:** The issuer-pays conflict is the usual explanation for corporate ratings; for sovereigns revenue is small relative to reputational loss.
- **C:** This is uncommon and is not considered a conflict-of-interest problem.
- **D:** Sovereign ratings are provided gratis to most users; there are no investor subscription fees driving the bias.

Q310 BT answers P2.T6.24.10.2 · BT notes P2.T6.24.10.1

REBUILT — Mention of passive ETFs created a plausible incentive-problem reading (two defensible answers); replaced with an unambiguous statement of no other business lines.

John, a US-based analyst, is rating the country of 1bit. 1bit's debt is all issued by its federal government (no sub-sovereign debt). John's data come solely from 1bit's own finance department, and his agency has no index, portfolio-evaluation, or other commercial businesses linked to 1bit. Which criticism of rating agencies is MOST justified here?

A. Home bias effect

B. Information problems

C. Revenue bias

D. Incentive problems

Answer: B

Step by step

1. Home bias: US-based agencies over-rating the US; 1bit is foreign, so it does not apply.
2. Revenue bias: arises from fees on sub-sovereign issuers; 1bit has none.
3. Incentive problems: arise from agencies' other businesses or analysts seeking jobs; none here.
4. Information problems: the data come only from the government, which can hold back bad news, so ratings may be too favorable. Answer B.

Why B is correct: Relying on government-supplied data allows selective disclosure of good news only, which is the information problem.

Why the others are wrong

- **A:** Home bias is over-rating of the agency's own country; the rated country is foreign.
- **C:** Revenue bias needs sub-sovereign issuer revenue, and 1bit has no sub-sovereign debt.
- **D:** No other agency businesses or job-seeking analysts are involved.

LO 26.g

Compare the use of credit ratings, market-based credit default spreads, and CDS spreads in predicting default.

Q311

AnalystPrep Q.6202

REBUILT — Original asked which method is 'most robust' with no defined criterion and a vague stem; rebuilt to test the specific rating-versus-CDS comparison in predicting sovereign default.

An analyst compares sovereign credit ratings with sovereign CDS spreads as indicators of default risk. Which of the following statements is most accurate?

A. Ratings adjust more quickly to new information than CDS spreads, so ratings are the better predictor of default.

B. CDS spreads adjust more quickly to market information than ratings, and changes in CDS spreads have been found to lead changes in ratings.

C. A CDS spread isolates pure sovereign credit risk, so it can be read directly as a default probability estimate.

D. CDS spreads are preferable to ratings because the sovereign CDS market is deep and highly liquid.

Answer: B

Step by step

1. Ratings are updated infrequently by agencies, so they lag new information about a sovereign.
2. CDS prices are set continuously in the market, so they adjust quickly to new information on default risk.
3. Empirically, changes in CDS spreads lead changes in sovereign bond yields and sovereign ratings, and CDS spreads are more timely and dynamic than ratings.
4. Limitations of CDS: prices are affected by counterparty and liquidity risk, and the narrow CDS market can suffer liquidity problems. This rules out C and D.

Why B is correct: CDS spreads are market-based and update continuously, whereas ratings are revised infrequently, so CDS changes lead rating changes and predict sovereign risk better.

Why the others are wrong

- **A:** Reverses the timeliness relationship: ratings are the slower, infrequently updated measure.
- **C:** CDS prices are also affected by counterparty and liquidity risk, so a spread is not a pure default-risk measure and cannot be read directly as a default probability.
- **D:** The CDS market is narrow and prone to liquidity problems, which can distort prices; liquidity is a limitation, not an advantage.

Q312

AnalystPrep Q.5917

REBUILT — Original had two defensible answers (thin trading and sensitivity to sentiment/non-credit factors are both limitations); rebuilt so only the credit/market/liquidity-mixture limitation is correct.

An analyst plans to use sovereign credit default swap (CDS) spreads as a stand-alone gauge of a country's default risk. Which of the following is a key limitation of this approach?

A. Sovereign CDS spreads are not observable because CDS contracts are privately negotiated.

B. CDS spreads are affected by counterparty and liquidity risk as well as default risk, so using them to measure only default risk could overstate that risk.

C. CDS spreads adjust more slowly to new information than sovereign credit ratings do.

D. CDS spreads cannot be compared across countries because there is no risk-free security to serve as a benchmark.

Answer: B

Key correction: Source key (AnalystPrep) said thin trading (illiquidity); that is also a recognized limitation but option D in the source (sensitivity to global sentiment, i.e. non-credit factors) was equally defensible, so the item was rebuilt with a single correct option.

Step by step

1. Recall the limitations of CDS spreads as predictors of country default risk: (1) counterparty risk and liquidity risk embedded in CDS prices can move spreads for reasons unrelated to default, and (2) the narrow CDS market can make an individual CDS susceptible to liquidity problems.
2. Option B states this: the spread is a mixture of default, counterparty and liquidity effects, so reading it as pure default risk could overstate that risk.
3. Check the others against the facts: CDS prices are quoted and observable (A), CDS spreads adjust faster than ratings (C), and the missing risk-free benchmark problem belongs to local-currency bond default spreads, not

CDS (D).

Why B is correct: A CDS spread is a market price that also reflects counterparty and liquidity risk, so treating it as a pure measure of default risk can overstate default risk.

Why the others are wrong

- **A:** CDS spreads are observable market quotes; confusing OTC trading with unobservable prices.
- **C:** Reverses the evidence: CDS spreads react more quickly than ratings (e.g., Greece 2009-2010), which is an advantage, not a limitation.
- **D:** Mixes up local-currency bond default spreads (no risk-free local benchmark) with CDS, which are quoted directly as a spread on the notional.

Q313 AnalystPrep Q.6201

REBUILT — Original asked for the 'most effective' measure with debatable options (reserves also defensible); rebuilt to test the documented comparison of CDS spreads versus ratings and default spreads.

Which of the following statements best describes the evidence on predicting sovereign default using credit ratings, bond default spreads and CDS spreads?

A. Sovereign ratings change before bond default spreads widen, so ratings give the earliest warning of default.

B. CDS spreads adjust more quickly to new information than ratings do, and are considered better predictors of sovereign risk than ratings.

C. CDS spreads have been proven to be better at assessing default risk than market-based default spreads.

D. Default spreads are uncorrelated with ratings, so the two measures contain unrelated information.

Answer: B

Step by step

1. Default spreads are leading indicators: they widen before downgrades and narrow before upgrades, so ratings lag spreads (rules out A).
2. CDS spreads adjust faster than ratings (e.g., Greece 2009-2010, when CDS spreads changed far more often than ratings) and are better predictors of sovereign risk than ratings (B).
3. CDS have not been shown to be more beneficial than market-based default spreads (rules out C).
4. Default spreads are positively correlated with ratings and with default (rules out D).

Why B is correct: CDS spreads are market prices that update continuously, whereas ratings are revised infrequently, so CDS spreads signal changes in sovereign risk earlier.

Why the others are wrong

- **A:** Reverses the timing: spreads move before ratings, and rating changes mostly follow market information.
- **C:** Overstates the evidence: CDS contain significant default information but have not been proven superior to market-based default spreads.
- **D:** Default spreads are positively correlated with ratings and with default; ratings also affect pricing when they change.

Reading 27. Estimating Default Probabilities

John C. Hull, Risk Management and Financial Institutions, 6th Edition (John Wiley & Sons, 2023). Chapter 17.

LO	Learning objective	Qs
27.a	compare agencies' ratings to internal credit rating systems.	3
27.b	describe Altman's Z-score methodology and apply it to predict a firm's default and to classify a sample of firms by credit quality.	3
27.c	describe the relationship between borrower rating and probability of default.	3
27.d	describe a rating migration matrix and calculate the probability of default, cumulative probability of default, and marginal probability of default.	7
27.e	define the hazard rate and use it to define probability functions for default time as well as to calculate conditional and unconditional default probabilities.	11
27.f	describe recovery rates and their dependencies on default rates.	3
27.g	define a credit default swap (CDS) and explain its mechanics including the obligations of both the default protection buyer and the default protection seller.	3
27.h	describe CDS spreads and explain how CDS spreads can be used to estimate hazard rates.	6
27.i	define and explain CDS-bond basis.	3
27.j	compare default probabilities calculated from historical data with those calculated from credit yield spreads.	5
27.k	describe the difference between real-world and risk-neutral default probabilities and determine which one to use in the analysis of credit risk.	4
27.l	calculate the value of a firm's debt and equity, the volatility of firm value, and the volatility of firm equity using the Merton model.	28
27.m	calculate distance to default and default probability using the Merton model.	19
27.n	assess the quality of the default probabilities produced by the Merton model, the Moody's KMV model, and the Kamakura model.	3

LO 27.a

Compare agencies' ratings to internal credit rating systems.

Q314

BT answers P2.T6.24.13.1

REBUILT — Original had a giveaway distractor (internal ratings 'accessible to the public') and an ambiguous 'advantage' framing; rewrote as a single-correct-statement comparison of agency vs internal ratings.

A bank lends to many small and mid-sized firms that have no publicly traded debt. Which of the following statements correctly compares agency credit ratings with the bank's internal credit rating system?

A. Agency ratings are designed to be stable and 'rate through the cycle', whereas an internal system for small and mid-sized firms can be built on profitability, liquidity and leverage ratios and a cash flow statement derived from the borrower's financial statements.

B. Agency ratings change frequently so that they reflect each piece of new market information, whereas internal ratings are updated only at annual reviews.

C. Agencies routinely rate small and mid-sized private firms, so a bank has little need for an internal system for these borrowers.

D. Internal rating systems rely mainly on accounting profits, because profits rather than cash are what repay a loan, whereas agency ratings focus on cash flow.

Answer: A

Step by step

1. Agency ratings are published mainly for firms with publicly traded debt; many small and mid-sized firms have none, so banks build internal systems.
2. Agencies aim for rating stability (avoid reversals, rate through the economic cycle), so ratings change only when a long-term change in creditworthiness is believed to have occurred.
3. Internal systems typically use profitability (e.g., ROE or ROA), liquidity (quick/current ratio) and leverage (debt-to-equity) ratios, and convert financial statements into a cash flow statement to judge repayment ability.
4. Only statement A matches both points.

Why A is correct: Agency ratings favour long-term stability and cover mainly issuers of public debt, while internal ratings fill the gap for smaller firms using financial ratios and cash flow analysis.

Why the others are wrong

- **B:** Reverses the facts: agencies deliberately avoid frequent changes, so ratings change infrequently.

- **C:** Agencies generally do not rate small and mid-sized firms without public debt, which is exactly why internal systems exist.
- **D:** Banks recognise that cash, not profits, repays a loan and so convert statements into cash flow statements; the statement also wrongly attributes a cash flow focus to agencies.

Q315 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A portfolio manager at a bond fund is told that her fund may hold only investment-grade bonds. She observes that a rating agency did not change an issuer's rating even though a sharp economic downturn raised the issuer's probability of default over the next six months, with little effect on its default probability over the next three to five years. Which of the following best explains the agency's behavior?

A. Agencies measure only the probability of default over the next six months, which the downturn left unchanged.
B. Agencies base ratings on the issuer's equity price, which does not respond to short-term economic conditions.
C. Agencies update ratings only at fixed annual reviews and ignore information released between reviews.
D. Agencies try to rate through the cycle, and stable ratings spare rule-bound bond investors from heavy trading and transaction costs caused by frequent rating changes.

Answer: D

Step by step

1. Agency ratings are designed to be stable and to take a long-term view of creditworthiness ('rate through the cycle').
2. A short-lived rise in near-term default risk with little change over three to five years does not justify a rating change.
3. Bond traders are major users of ratings and many are bound by rating-based rules; frequent changes would force costly trading.

Why D is correct: Option D states the through-the-cycle objective and the practical reason for it given in the reading.

Why the others are wrong

- **A:** Ratings reflect a long-term view; saying they use only a six-month horizon reverses the actual practice, and the stem says the six-month risk did increase.
- **B:** Agency ratings are not equity-price based; it is Moody's KMV and Kamakura estimates that use equity prices and respond quickly. Confuses agency ratings with market-based estimates.
- **C:** Ratings can change when information is released; the stability is a deliberate objective, not a fixed review calendar.

Q316 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A regional bank is building an internal credit rating system for privately held manufacturing borrowers that are too small to be rated by any rating agency. Which of the following approaches is most consistent with how such systems are typically built?

A. Score each borrower on profitability, liquidity and solvency measures such as return on equity, the quick ratio and debt-to-equity, and adjust the borrower's financial statements to derive a cash flow statement.
B. Adopt the rating of the largest rated firm in the same industry, since agencies rate whole industries rather than individual issuers.
C. Use the equity-market-based default probability from a vendor model, because small private firms have market-traded shares.
D. Rely on accounting net income alone, because a loan is repaid out of reported profits rather than out of cash.

Answer: A

Step by step

1. Agencies typically do not rate small and mid-sized firms, so lenders build their own systems.
2. Internal systems use profitability (e.g., ROE), liquidity (e.g., quick ratio) and solvency (e.g., debt-to-equity) measures.
3. Borrowings are repaid with cash, so company-prepared statements are adjusted to produce a cash flow statement.

Why A is correct: Option A matches the reading: ratio-based scoring plus a cash flow statement to judge repayment

ability.

Why the others are wrong

- **B:** Agency ratings are issuer-specific and are not transferable from an industry peer; assumes agencies rate industries.
- **C:** Private firms have no traded equity price, so market-based vendor estimates are unavailable; this misapplies the KMV/Kamakura approach.
- **D:** Loans are repaid with cash, not accounting profit, which is why a cash flow statement is built; this reverses the logic.

LO 27.b

Describe Altman's Z-score methodology and apply it to predict a firm's default and to classify a sample of firms by credit quality.

Q317

BT answers P2.T6.24.13.2 · GARP book Example 9.1 (Hull)

REBUILT — Original compared Z-score with a vaguely described internal rating (option A wording ambiguous, 'just above at-risk threshold' undefined); rebuilt as a self-contained Z-score calculation and classification.

A publicly traded manufacturer reports: working capital 170,000; retained earnings 300,000; EBIT 60,000; sales 2,200,000; market value of equity 380,000; book value of total liabilities 240,000; total assets 670,000. Using Altman's original Z-score for public manufacturers, $Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 0.999X_5$, where X_1 = working capital/total assets, X_2 = retained earnings/total assets, X_3 = EBIT/total assets, X_4 = market value of equity/book value of total liabilities, X_5 = sales/total assets, what is the Z-score and the resulting credit classification?

A. Z of about 5.46; default unlikely (Z above 3.0)

B. Z of about 2.18; reasonable probability of default (Z between 1.8 and 2.7)

C. Z of about 1.23; high likelihood of default (Z below 1.8)

D. Z of about 4.51; default unlikely (Z above 3.0)

Answer: A

Step by step

1. Ratios: $X_1 = 170/670 = 0.254$; $X_2 = 300/670 = 0.448$; $X_3 = 60/670 = 0.0896$; $X_4 = 380/240 = 1.583$; $X_5 = 2{,}200/670 = 3.284$.
2. Weighted terms: $1.2(0.254) = 0.305$; $1.4(0.448) = 0.627$; $3.3(0.0896) = 0.296$; $0.6(1.583) = 0.950$; $0.999(3.284) = 3.281$.
3. Sum: $Z = 0.305 + 0.627 + 0.296 + 0.950 + 3.281 = 5.46$.
4. Zones: above 3.0 unlikely to default; 2.7 to 3.0 on alert; 1.8 to 2.7 good chance of default; below 1.8 very high probability of financial distress. 5.46 > 3.0, so default is unlikely.

Why A is correct: Z is a weighted sum of five accounting ratios; the high sales turnover and market-value cushion over liabilities push Z well above the 3.0 safe cutoff.

Why the others are wrong

- **B:** 2.18 omits the sales/assets term (0.999 X5); the correct Z includes it, and sales turnover is the largest contributor here.
- **C:** 1.23 keeps only the first three terms (X1 to X3) and drops both X4 and X5, understating Z.
- **D:** 4.51 leaves out the 0.6 X4 term (market equity/liabilities contribution 0.95); the classification is the same but the score is wrong.

Q318

Schweser Module Quiz 27.1 Q1

REBUILT — Financial data table and Z-score equation were missing from the OCR; restored the inputs (recovered from the source explanation) and the equation into the stem.

Nielsen Corp. reports: working capital 525,000; retained earnings 1,120,000; EBIT 480,000; market value of equity 4,215,000; book value of total liabilities 1,850,000; sales 1,760,000; total assets 5,100,000. Altman's Z-score is $Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 0.999X_5$ with X_1 = working capital/total assets, X_2 = retained earnings/total assets, X_3 = EBIT/total assets, X_4 = market value of equity/book value of total liabilities, X_5 = sales/total assets. Guidelines: Z > 3.0 no default likely; 2.7 to 3.0 potential default (on alert); 1.8 to 2.7 reasonable probability of default; Z < 1.8 high likelihood of default. What is the most appropriate classification of Nielsen?

A. No default is likely.

B. Potential default.
C. Reasonable probability of default.
D. High likelihood of default.

Answer: C
Step by step
1. $X_1 = 525/5{,}100 = 0.1029$; $X_2 = 1{,}120/5{,}100 = 0.2196$; $X_3 = 480/5{,}100 = 0.0941$; $X_4 = 4{,}215/1{,}850 = 2.2784$; $X_5 = 1{,}760/5{,}100 = 0.3451$.
2. Terms: 1.2(0.1029)=0.1235; 1.4(0.2196)=0.3075; 3.3(0.0941)=0.3106; 0.6(2.2784)=1.3670; 0.999(0.3451)=0.3448.
3. $Z = 2.4534$.
4. 2.4534 lies between 1.8 and 2.7, so Nielsen has a reasonable probability of default.

Why C is correct: The Z-score of about 2.45 falls in the 1.8 to 2.7 band, which Altman's guidelines label a good (reasonable) chance of default.

Why the others are wrong
- **A:** Z of about 4.36 (above 3.0) results from dividing every ratio by total liabilities instead of total assets, overstating the score.
- **B:** The 'potential default' band is 2.7 to 3.0; 2.45 is below it. This comes from misremembering the band boundaries.
- **D:** A score below 1.8 arises if X4 is inverted (liabilities/equity, Z about 1.35) or computed as market equity/total assets (Z about 1.58).

Q319

BT answers P2.T6.704.3

REBUILT — Original relied on a missing De Laurentis exhibit with inconsistent figures; rebuilt with explicit balance-sheet data and a stress on sales, EBIT and market equity.

A public manufacturer has total assets 1,000; working capital 200; retained earnings 300; EBIT 100; market value of equity 800; book value of total liabilities 500; sales 1,500. Using Altman's Z-score ($Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 0.999X_5$, with X4 = market equity/book liabilities), a stress test cuts sales by 40%, EBIT by 50% and market value of equity by 25%; no other accounts change. Guidelines: Z > 3.0 default unlikely; 2.7 to 3.0 on alert; 1.8 to 2.7 reasonable probability of default; Z < 1.8 high likelihood of default. What is the stressed Z-score and classification?

A. Z of about 3.28; default still unlikely
B. Z of about 2.68; reasonable probability of default
C. Z of about 2.44; reasonable probability of default
D. Z of about 1.66; high likelihood of default

Answer: C
Step by step
1. Base: $X_1 = 0.20, X_2 = 0.30, X_3 = 0.10, X_4 = 800/500 = 1.60, X_5 = 1.50$; $Z = 0.24 + 0.42 + 0.33 + 0.96 + 1.4985 = 3.45$ (default unlikely).
2. Stressed: EBIT 50, equity 600, sales 900 give $X_3 = 0.05, X_4 = 600/500 = 1.20, X_5 = 0.90$.
3. Terms: 1.2(0.2)=0.24; 1.4(0.3)=0.42; 3.3(0.05)=0.165; 0.6(1.2)=0.72; 0.999(0.9)=0.8991.
4. $Z = 2.4441 \approx 2.44$, inside 1.8 to 2.7: reasonable probability of default.

Why C is correct: Each shock lowers one weighted term (-0.165, -0.24, -0.599), cutting Z by about 1.0 from 3.45 to 2.44, moving the firm from the safe zone into the 1.8 to 2.7 band.

Why the others are wrong
- **A:** 3.28 shocks only EBIT (Z falls by 0.165) and ignores the sales and equity declines.
- **B:** 2.68 applies the EBIT and sales shocks but leaves market equity unchanged, so X4 stays 1.60 instead of 1.20.
- **D:** 1.66 treats the declines as the remaining fraction (keeping only 25% of equity, 40% of sales), shocking far too much.

LO 27.c

Describe the relationship between borrower rating and probability of default.

Q320 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A credit analyst uses the cumulative default rate table below (percent, by years since the rating was assigned) from a rating agency.

BBB: Year 1 0.16; Year 2 0.43; Year 3 0.75; Year 4 1.14

B: Year 1 3.34; Year 2 7.80; Year 3 11.75; Year 4 14.89

CCC/C: Year 1 28.30; Year 2 38.33; Year 3 43.42; Year 4 46.36

Which of the following statements about the probability of default in each individual year (the marginal probability) is correct?

A. For B, the probability of default in each year falls steadily from year 1 through year 4.

B. For BBB, the probability of default in each year rises from year 1 through year 4.

C. For CCC/C, the probability of default in year 2 is 38.33%, which is higher than in year 1.

D. For BBB, the probability of default in year 3 is 0.75%.

Answer: B

Step by step

1. Marginal PD in a year = cumulative rate at end of that year minus cumulative rate at end of the prior year.
2. BBB: 0.16, 0.27, 0.32, 0.39 (rising).
3. B: 3.34, 4.46, 3.95, 3.14 (rises in year 2, then falls).
4. CCC/C: 28.30, 10.03, 5.09, 2.94 (falling).
5. BBB year 3: 0.75 - 0.43 = 0.32%.

Why B is correct: Marginal PDs for BBB are 0.16%, 0.27%, 0.32% and 0.39%, which increase every year, consistent with investment-grade issuers deteriorating with time.

Why the others are wrong

- **A:** B-rated marginal PD rises from 3.34% to 4.46% in year 2 before declining, so it does not fall steadily; ignores the year-1 to year-2 increase.
- **C:** 38.33% is the cumulative rate; the year 2 marginal PD is 38.33 - 28.30 = 10.03%, lower than year 1. Confuses cumulative with marginal.
- **D:** 0.75% is the cumulative rate through year 3; the year 3 marginal is 0.32%. Same cumulative/marginal confusion.

Q321 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A bank holds 1,000 equally sized loans to borrowers that were all rated BB at the start of year 1. Using the cumulative default rates for BB of 0.63% (year 1), 1.93% (year 2) and 3.46% (year 3), and ignoring any rating changes, approximately how many of these borrowers are expected to default during year 3 only?

A. About 35

B. About 19

C. About 15

D. About 6

Answer: C

Step by step

1. Marginal PD in year 3 = 3.46% - 1.93% = 1.53%.
2. Expected defaults = 1,000 x 1.53% = 15.3, i.e. about 15.

Why C is correct: The year 3 marginal default probability is the difference in cumulative rates, 1.53%, applied to 1,000 loans.

Why the others are wrong

- **A:** 3.46% x 1,000 = about 35 is the cumulative default count through year 3, not the year-3-only count.
- **B:** 1.93% x 1,000 is the cumulative count through year 2, not the year 3 default count.
- **D:** About 6 is the year-1 count (0.63% x 1,000); uses the wrong year.

Q322 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A risk manager observes that for an A-rated issuer the probability of default during each of years 1 to 4 is 0.05%, 0.08%, 0.09% and 0.11%, whereas for an issuer initially rated CCC/C the corresponding probabilities are 28.30%, 10.03%, 5.09% and 2.94%. Which of the following best explains the CCC/C pattern?

A. A CCC/C issuer is initially considered creditworthy, so the longer time passes the greater the chance its financial health declines.

B. Rating agencies lower the rating of CCC/C issuers each year, which mechanically reduces the default probability of those that remain.

C. Cumulative default rates for CCC/C issuers decline over time, so fewer issuers default in later years.

D. A CCC/C issuer faces a critical period in the first year or two; if it survives, its financial health is likely to have improved.

Answer: D

Step by step

1. For poorly rated issuers the next year or two are critical.
2. Survivors of that period are likely to be healthier, so the marginal PD declines with time.
3. The investment-grade pattern (rising) arises because creditworthy issuers can only deteriorate with time.

Why D is correct: Survival through the critical early period signals improved financial health, so conditional on survival the annual default probability falls.

Why the others are wrong

- **A:** This explains the rising pattern of the A-rated issuer, not the falling CCC/C pattern.
- **B:** The pattern is not caused by rating downgrades; it reflects the survivors' improving health.
- **C:** Cumulative default rates never decline; only the marginal (annual) rates decline. Confuses cumulative and marginal.

LO 27.d

Describe a rating migration matrix and calculate the probability of default, cumulative probability of default, and marginal probability of default.

Q323 AnalystPrep Q.1856

REBUILT — Stem did not say the 5% is an annual rate applied to surviving loans; clarified it and replaced the vague integer options with precise values and realistic error distractors.

A pool of 100 loans is followed with no replacement of defaulted loans. Each year, 5% of the loans that are still performing at the start of that year default (a constant annual conditional default rate of 5%). What is the expected number of defaults during year 5?

A. 5.00

B. 4.07

C. 3.87

D. 4.29

Answer: B

Step by step

1. Survivors at the start of year t are $100 \times 0.95^{t-1}$.
2. Start of year 5 follows four years of survival: $100 \times 0.95^4 = 81.45$ loans.
3. Expected defaults in year 5: $81.45 \times 0.05 = 4.07$.

Why B is correct: Because defaulted loans are not replaced, the 5% conditional rate applies to a shrinking pool: $100 \times 0.95^4 \times 0.05 = 4.07$.

Why the others are wrong

- **A:** Applies 5% to the original 100 loans, ignoring that earlier defaults have left the pool.
- **C:** Uses $100 \times 0.95^5 \times 0.05 = 3.87$, i.e. five years of survival before year 5 (an off-by-one in the exponent).
- **D:** Gives the year-4 figure $100 \times 0.95^3 \times 0.05 = 4.29$, using three years of survival instead of four.

Q324 AnalystPrep Q.3051

What is the probability that a bond rated Caa-C defaults during the third year (unconditional, as seen today)? Average cumulative default rates (%), 1970-2012, from Moody's.

Term (years)	1	2	3	4	5	7	10	15	20
Aaa	0.000	0.013	0.013	0.037	0.106	0.247	0.503	0.935	1.104
Aa	0.022	0.069	0.139	0.256	0.383	0.621	0.922	1.756	3.135
A	0.063	0.203	0.414	0.625	0.870	1.441	2.480	4.255	6.841
Baa	0.177	0.495	0.894	1.369	1.877	2.927	4.740	8.628	12.483
Ba	1.112	3.083	5.424	7.934	10.189	14.117	19.708	29.172	36.321
B	4.051	9.608	15.216	20.134	24.613	32.747	41.947	52.217	58.084
Caa-C	16.448	27.867	36.908	44.128	50.366	58.302	69.483	79.178	81.248

A. 36.908%
B. 9.041%
C. 52.124%
D. 45.949%

Answer: B

Step by step

1. Table entries are cumulative default probabilities.
2. Cumulative PD to end of year 3 = 36.908%; to end of year 2 = 27.867%.
3. Marginal (unconditional) year-3 PD = 36.908% − 27.867% = 9.041%.

Why B is correct: The probability of defaulting in a specific year, seen from time zero, is the difference between consecutive cumulative default rates.

Why the others are wrong

- **A:** Reads the 3-year cumulative PD as if it were the year-3 marginal PD.
- **C:** Adds the B-rated 3-year cumulative rate (15.216%) to the Caa-C 3-year cumulative rate (36.908%), using the wrong row.
- **D:** Adds the marginal PD (9.041%) to the cumulative PD (36.908%) instead of subtracting the prior cumulative PD.

Q325 AnalystPrep Q.3052

What is the probability that a Ba-rated bond survives (does not default) until the end of year 2? Average cumulative default rates (%), 1970-2012, from Moody's.

Term (years)	1	2	3	4	5	7	10	15	20
Aaa	0.000	0.013	0.013	0.037	0.106	0.247	0.503	0.935	1.104
Aa	0.022	0.069	0.139	0.256	0.383	0.621	0.922	1.756	3.135
A	0.063	0.203	0.414	0.625	0.870	1.441	2.480	4.255	6.841
Baa	0.177	0.495	0.894	1.369	1.877	2.927	4.740	8.628	12.483
Ba	1.112	3.083	5.424	7.934	10.189	14.117	19.708	29.172	36.321
B	4.051	9.608	15.216	20.134	24.613	32.747	41.947	52.217	58.084
Caa-C	16.448	27.867	36.908	44.128	50.366	58.302	69.483	79.178	81.248

A. 94.576%
B. 98.888%
C. 96.917%
D. 95.805%

Answer: C

Step by step

1. Ba cumulative PD to end of year 2 = 3.083%.
2. Survival = 100% − 3.083% = 96.917%.

Why C is correct: Survival to time t is one minus the cumulative default probability to t.

Why the others are wrong

- **A:** Uses the 3-year cumulative PD (5.424%): 100 - 5.424 = 94.576%, the wrong horizon.
- **B:** Uses the 1-year cumulative PD (1.112%): 100 - 1.112 = 98.888%, the wrong horizon.
- **D:** Subtracts the sum of the year-1 and year-2 cumulative PDs (1.112 + 3.083 = 4.195%), double counting year-1 defaults.

Q326 AnalystPrep Q.3053

A risk analyst holds a bond rated Caa-C. Using the table, the probability that the bond defaults during the fourth year, conditional on no earlier default, is closest to:

Average cumulative default rates (%), 1970-2012, from Moody's.

Term (years)	1	2	3	4	5	7	10	15	20
Aaa	0.000	0.013	0.013	0.037	0.106	0.247	0.503	0.935	1.104
Aa	0.022	0.069	0.139	0.256	0.383	0.621	0.922	1.756	3.135
A	0.063	0.203	0.414	0.625	0.870	1.441	2.480	4.255	6.841
Baa	0.177	0.495	0.894	1.369	1.877	2.927	4.740	8.628	12.483
Ba	1.112	3.083	5.424	7.934	10.189	14.117	19.708	29.172	36.321
B	4.051	9.608	15.216	20.134	24.613	32.747	41.947	52.217	58.084
Caa-C	16.448	27.867	36.908	44.128	50.366	58.302	69.483	79.178	81.248

A. 69.94%
B. 19.56%
C. 63.092%
D. 11.44%

Answer: D

Step by step

1. Unconditional year-4 PD = 44.128% − 36.908% = 7.220%.
2. Survival to end of year 3 = 100% − 36.908% = 63.092%.
3. Conditional PD = 7.220/63.092 = 11.44%.

Why D is correct: The conditional PD divides the year-4 marginal PD by the probability of having survived the first three years.

Why the others are wrong

- **A:** Divides the 4-year cumulative PD (44.128%) by the 3-year survival probability: 69.94%.
- **B:** Divides the marginal PD (7.220%) by the 3-year cumulative PD (36.908%) instead of the 3-year survival probability (63.092%): 19.56%.
- **C:** Reports the survival probability to end of year 3, not a default probability.

Q327 AnalystPrep Q.5433

REBUILT — Original counts were ambiguous about timing and the year-4 default count came out too small to distinguish marginal from conditional PD; rebuilt with time-0 to year-4 counts and conditional/cumulative distractors.

A risk manager tracks a cohort of 230 corporate issuers, none of which is replaced after default. The number of issuers not yet in default is:

Time	End of year 0	End of year 1	End of year 2	End of year 3	End of year 4
Non-defaulted issuers	230	215	190	185	170

What is the estimated 1-year marginal (unconditional) probability of default in year 4?

A. 8.11%
B. 6.52%
C. 10.87%
D. 26.09%

Answer: B

Step by step

1. Year-4 defaults = 185 − 170 = 15.
2. Marginal PD is measured against the original cohort: 15/230 = 6.52%.

Why B is correct: The marginal (unconditional) PD is the share of the original cohort that defaults in the given year, as seen at time zero.

Why the others are wrong

- **A:** Divides by the 185 survivors, giving the conditional PD (15/185 = 8.11%), not the marginal PD.
- **C:** Uses year-2 defaults (215 − 190 = 25) over 230 = 10.87%: wrong year.
- **D:** Uses cumulative defaults through year 4 (230 − 170 = 60) over 230 = 26.09%, i.e. cumulative rather than marginal PD.

Q328

GARP PE2 Q78 · Mock 4 Q79

REBUILT — Source option A confused a joint path probability (1.8%) with the conditional year-2 PD (2%), making the item ambiguous and the explanation inconsistent; replaced with a clean 2-year cumulative PD calculation from the same matrix.

A 1-year credit migration matrix is given (rows = current rating, columns = rating at end of year; D = default, which is absorbing):

Current rating	A	B	C	D
A	90%	8%	0%	2%
B	5%	76%	10%	9%
C	1%	14%	69%	16%

Assuming the matrix is constant over time, what is the probability that a bond currently rated B defaults within 2 years?

A. 15.94%
B. 18.00%
C. 17.54%
D. 17.44%

Answer: C

Step by step

1. Default in year 1: 9%.
2. Survive year 1 then default in year 2, by end-of-year-1 rating: via A $0.05 \times 0.02 = 0.10\%$; via B $0.76 \times 0.09 = 6.84\%$; via C $0.10 \times 0.16 = 1.60\%$.
3. Total $= 9\% + 0.10\% + 6.84\% + 1.60\% = 17.54\%$.

Why C is correct: Two-year cumulative PD = year-1 PD plus the sum over surviving ratings of (migration probability) x (that rating's default probability).

Why the others are wrong

- **A:** Omits the path through C (1.60%): 9 + 0.10 + 6.84 = 15.94%.
- **B:** Simply doubles the 1-year PD (2 x 9%), ignoring that defaulted bonds cannot default again and that year-2 PD depends on the migration.
- **D:** Omits the path through A (0.10%): 9 + 6.84 + 1.60 = 17.44%.

Q329

Schweser Module Quiz 27.1 Q2

REBUILT — Exhibit was missing from the source; reconstructed the B-rated cumulative default row (year 3 = 11.75%, year 4 = 14.89% as in the source explanation) and replaced weak distractors.

The following is an excerpt from a rating migration matrix for a B-rated bond (cumulative default probabilities):

Year	1	2	3	4
Cumulative PD	4.02%	8.09%	11.75%	14.89%

What is the probability that the bond will default during the fourth year conditional on no earlier default?

A. 3.14%
B. 3.56%
C. 3.69%
D. 21.09%

Answer: B

Step by step

1. Unconditional year-4 PD = $14.89\% - 11.75\% = 3.14\%$.
2. Survival through year 3 = $100\% - 11.75\% = 88.25\%$.
3. Conditional PD = $3.14/88.25 = 3.56\%$.

Why B is correct: Conditional on surviving three years, the year-4 default probability is the marginal PD divided by the 3-year survival probability.

Why the others are wrong

- **A:** Stops at the unconditional (marginal) PD without conditioning on survival.
- **C:** Divides by survival to end of year 4 ($100 - 14.89 = 85.11\%$) instead of year 3: 3.14/85.11 = 3.69%.
- **D:** Divides the marginal PD by the 4-year cumulative PD (14.89%), which is not a survival probability.

LO 27.e

Define the hazard rate and use it to define probability functions for default time as well as to calculate conditional and unconditional default probabilities.

Q330

BT answers P2.T6.307.3 · BT notes P2.T6.307.3

REBUILT — OCR-garbled formulas and option B and a missing derivative exhibit; restated the distributions cleanly and kept B as the single false statement.

A constant hazard rate λ gives the cumulative default-time distribution $F(t) = P[t^* \le t] = 1 - e^{-\lambda t}$ and the survival distribution $1 - F(t) = P[t^* > t] = e^{-\lambda t}$. Each of the following statements is true EXCEPT, which is false?

A. As t grows large, $1 - F(t)$ converges to zero, so even a highly rated company (low hazard rate) will eventually default.

B. Given survival to time t, the conditional probability of default over a very short interval $(t, t + dt)$ is $F(t)/e^{-\lambda t}$.

C. The marginal default probability (density), the derivative of the cumulative default-time distribution, is $F'(t) = \lambda e^{-\lambda t}$.

D. The derivative of the survival function is $-F'(t) = -\lambda e^{-\lambda t} < 0$, i.e. survival probability falls over time.

Answer: B

Step by step

1. Differentiate $F(t) = 1 - e^{-\lambda t}$ with the chain rule: $F'(t) = \lambda e^{-\lambda t}$, so C is true.
2. Survival $= 1 - F(t)$ has derivative $-F'(t) = -\lambda e^{-\lambda t} < 0$, so D is true.
3. As $t \to \infty$, $e^{-\lambda t} \to 0$ for any $\lambda > 0$, so A is true.
4. Conditional default probability over dt given survival is $F'(t)\,dt/[1 - F(t)] = \lambda e^{-\lambda t} dt / e^{-\lambda t} = \lambda\,dt$. This is the definition of the hazard rate.
5. B states $F(t)/e^{-\lambda t}$, the ratio of the cumulative default probability to the survival probability; it is not $\lambda\,dt$. B is false.

Why B is correct: The hazard rate is the conditional default probability per unit time given survival, $\lambda\,dt$; it equals density divided by survival, not cumulative probability divided by survival.

Why the others are wrong

- **A:** True: survival decays exponentially to zero; any positive hazard rate implies eventual default. Not the exception.
- **C:** True: the density is the derivative of F by the chain rule. Not the exception.
- **D:** True: the survival function is decreasing, so its derivative is negative. Not the exception.

Q331

AP Mock 3 Q38

REBUILT — Two distractors (9.14%, 8.27%) did not correspond to realistic errors; replaced with the unconditional PD, the density and the 2-year cumulative PD.

A corporation's hazard rate is 0.09 per year. Given that the company survived the first year, what is the probability that it will default before the end of the second year, assuming a constant hazard rate?

A. 7.87%

B. 8.61%

C. 8.23%

D. 16.47%

Answer: B

Step by step

1. Constant hazard rate means defaults are memoryless: the conditional one-year PD is the same every year.
2. Conditional PD $= 1 - e^{-0.09} = 1 - 0.91393 = 8.61\%$.
3. Check: $[e^{-0.09} - e^{-0.18}]/e^{-0.09} = (0.91393 - 0.83527)/0.91393 = 8.61\%$.

Why B is correct: Conditioning on survival to year 1 divides the unconditional year-2 default probability by the year-1 survival probability, which for a constant hazard gives $1 - e^{-\lambda}$.

Why the others are wrong

- **A:** Unconditional (marginal) year-2 PD $e^{-0.09} - e^{-0.18}$; failed to condition on survival to year 1.
- **C:** Used the density $\lambda e^{-\lambda t}$ at $t = 1$ (0.09×0.9139), which is not a probability over a year.
- **D:** Cumulative two-year PD $1 - e^{-0.18}$; ignored that default in year 1 is excluded by the condition.

Q332 AnalystPrep Q.6066

A risk manager is reviewing a company with a 7-year CDS. The company's constant hazard rate is estimated at 2.1% per annum. What is the probability that the company will default in the fifth year?

A. 0.0191
B. 0.9003
C. 0.0997
D. 0.9194

Answer: A

Step by step

1. Survival to end of year 4: $e^{-0.021\times4} = 0.9194$.
2. Survival to end of year 5: $e^{-0.021\times5} = 0.9003$.
3. Unconditional default in year 5 $= 0.9194 - 0.9003 = 0.0191$.

Why A is correct: The unconditional probability of default during a year is the drop in the survival probability over that year.

Why the others are wrong

- **B:** Survival probability to the end of year 5, not the probability of default in year 5.
- **C:** Cumulative default probability to year 5, $1 - 0.9003$; includes defaults in years 1-4.
- **D:** Survival probability to the end of year 4.

Q333 AnalystPrep Q.6070

A risk manager is estimating expected CDS payments. If the market implies a constant hazard rate of 2% per annum, what is the probability of default by the end of the second year?

A. 0.0392
B. 0.0198
C. 0.9608
D. 0.9802

Answer: A

Step by step

1. Cumulative default probability $Q(t) = 1 - e^{-\lambda t}$.
2. $Q(2) = 1 - e^{-0.02\times2} = 1 - 0.9608 = 0.0392$.

Why A is correct: With a constant hazard rate, the probability of default by time t is one minus the exponential survival probability.

Why the others are wrong

- **B:** One-year default probability $1 - e^{-0.02}$; used $t = 1$ instead of 2.
- **C:** Two-year survival probability, not default probability.
- **D:** One-year survival probability.

Q334 BT answers P2.T6.24.13.3

REBUILT — Source had OCR-damaged data and options C and D (0.32% vs 0.321%) were both defensible; rebuilt with new cumulative PDs and clearly distinct options.

Cumulative default probabilities for a bond are 3% by the end of year 1, 10% by the end of year 2, and 20% by the end of year 3. Given that the bond has survived to the end of year 2, what is the probability that it defaults during year 3?

A. 20.00%
B. 10.00%
C. 11.11%
D. 10.31%

Answer: C

Step by step

1. Unconditional (marginal) year-3 PD $= 20\% - 10\% = 10\%$.
2. Survival to end of year 2 $= 1 - 10\% = 90\%$.
3. Conditional PD $= 10\%/90\% = 11.11\%$.

Why C is correct: A conditional default probability divides the marginal default probability by the survival probability at the start of the period.

Why the others are wrong

- **A:** Cumulative three-year PD; includes defaults in years 1 and 2.
- **B:** Unconditional year-3 PD; does not condition on survival to year 2.
- **D:** Divided by year-1 survival (97%) instead of year-2 survival (90%): 10%/97%.

Q335

BT answers P2.T6.703.2

REBUILT — OCR-damaged table was reconstructed and the 'not enough information' option was a giveaway; converted to a numeric question on conditional vs unconditional PD in year 3.

A rating class shows the following pattern of defaults over five years:

Year	0	1	2	3	4	5
Names at t	1000	990	976	958	947	922
Defaults, cumulated	0	10	24	42	53	78

What are the conditional (forward) and unconditional (marginal) default probabilities in year 3?

A. Conditional 1.844%; unconditional 1.800%
B. Conditional 1.800%; unconditional 1.844%
C. Conditional 1.879%; unconditional 1.800%
D. Conditional 4.200%; unconditional 1.800%

Answer: A

Step by step

1. Defaults during year 3 = 42 − 24 = 18.
2. Unconditional PD = 18/1000 = 1.800% (relative to the original cohort).
3. Names surviving to the start of year 3 = 1000 − 24 = 976.
4. Conditional PD = 18/976 = 1.844%. Conditional exceeds unconditional because the denominator is smaller.

Why A is correct: Unconditional PD uses the original number of names as denominator; conditional PD uses only names surviving to the start of the year.

Why the others are wrong

- **B:** Labels swapped; conditional PD is always at least as large as unconditional PD after year 1.
- **C:** Divided by year-3 end survivors (958) instead of start-of-year survivors (976).
- **D:** Used cumulative defaults (42/1000) as the conditional figure.

Q336

BT notes P2.T6.506.1

REBUILT — Row for loan (d) lost its exposure and LGD in OCR; supplied exposure of $200 million and LGD of 50% so exactly one loan has the highest expected loss.

A bank holds four short-term loans. The one-year probability of default is a hazard rate (default intensity), so the default probability over the remaining term T is $1 - e^{-\lambda T}$.

Loan	Remaining term (months)	Exposure at default ($ millions)	One-year PD (hazard rate)	LGD
a	3	100	4.00%	90%
b	6	120	3.00%	60%
c	9	150	2.00%	60%
d	12	200	1.00%	50%

Which loan has the highest expected loss in dollar terms over its remaining term?

A. Loan (a)
B. Loan (b)
C. Loan (c)
D. Loan (d)

Answer: C

Step by step

1. Loan a: PD = $1 - e^{-0.04\times0.25} = 0.995\%$; EL = 100 × 0.995% × 90% = \$0.896m.
2. Loan b: PD = $1 - e^{-0.03\times0.5} = 1.489\%$; EL = 120 × 1.489% × 60% = \$1.072m.
3. Loan c: PD = $1 - e^{-0.02\times0.75} = 1.489\%$; EL = 150 × 1.489% × 60% = \$1.340m.
4. Loan d: PD = $1 - e^{-0.01} = 0.995\%$; EL = 200 × 0.995% × 50% = \$0.995m.

5. Largest is loan c at $1.34 million.

Why C is correct: Expected loss is EAD times PD over the remaining term times LGD; the PD must be scaled to each loan's remaining term via the hazard rate.

Why the others are wrong

- **A:** Ignored the remaining term and used the annual PD (EL $= 100 \times 4\% \times 90\% = 3.6$), which makes the short loan with the highest annual PD look worst.
- **B:** Second-highest EL ($1.07m); same PD as c but a smaller exposure.
- **D:** Largest exposure and a low PD, but only $0.995m after the 50% LGD; picking it relies on exposure size alone.

Q337

Mock 1 Q56

REBUILT — Original table mixed forecast exposures with spreads and did not say which CDS maturity the 375 bp referred to; stem now states the 3-year CDS spread and recovery explicitly.

MRS Brokers has a three-year derivative exposure to a counterparty (expected exposure AUD 18 million each year). Market CDS quotes for the counterparty by maturity, with the recovery rate assumed for each maturity, are:

Item	1-year	2-year	3-year
CDS spread (bps)	150	225	375
Recovery rate (%)	90	85	75

Using the approximation that the average hazard rate equals the CDS spread divided by (1 - recovery rate), and assuming a constant hazard rate over the first three years, what is the cumulative probability of default over three years?

A. 13.93%
B. 36.24%
C. 15.00%
D. 25.92%

Answer: B

Step by step

1. Use the 3-year point: spread $s = 3.75\%$, recovery $R = 75\%$.
2. Average hazard rate $\bar{\lambda} \approx s/(1-R) = 0.0375/0.25 = 0.15$ (15% per year).
3. Cumulative PD over $T = 3$: $1 - e^{-\bar{\lambda}T} = 1 - e^{-0.45}$.
4. $e^{-0.45} = 0.63763$, so cumulative PD = 36.24%. The exposure figure is irrelevant to PD.

Why B is correct: The 3-year CDS spread and recovery give the 3-year average hazard rate of 15%; the cumulative default probability is $1 - e^{-\bar{\lambda}T}$ with $T = 3$.

Why the others are wrong

- **A:** 13.93% = $1 - e^{-0.15}$: one-year probability, forgot to multiply the hazard rate by 3 years.
- **C:** 15.00% is the hazard rate itself, not a probability over three years.
- **D:** 25.92% = $1 - e^{-0.30}$: used a two-year horizon (0.15 x 2 = 0.30). Note the 2-year spread and recovery also give a 15% hazard rate, so only the horizon is wrong.

Q338

BT notes P2.T6.301.2

REBUILT — OCR-garbled formula and option D (printed P[t*=t]) repaired; wording made self-contained and exactly one false statement.

Default time is modeled as a random variable t^*. The conditional default probability from time t over interval τ is $P[t < t^* < t + \tau \mid t^* > t] = P[t^* > t \text{ and } t^* < t + \tau]/P[t^* > t]$.

Which of the following statements about this formula is INCORRECT?

A. t is the time from which default is being viewed, which may be today or a future date.
B. τ is the length of the interval over which the default probability is measured.
C. t^* is the random variable giving the time at which default occurs.
D. $P[t^* > t]$ is the unconditional probability of default before time t, i.e. over (0, t).

Answer: D

Step by step

1. t^* is the (random) default time; t is the starting date; τ is the interval length: A, B, C are correct.
2. $t^* > t$ means default has NOT happened by t, so $P[t^* > t]$ is the survival probability to t, $1 - F(t)$.
3. The unconditional probability of default over (0, t) is $P[t^* \le t] = F(t)$, the complement.
4. The formula divides the joint probability (survive to t and default in the interval) by the survival probability, so

D is false.

Why D is correct: The denominator $P[t^* > t]$ is the probability of surviving to t, not of defaulting before t.

Why the others are wrong

- **A:** True: t is the starting point and may be now or a future date.
- **B:** True: τ is the length of the window in which default is measured.
- **C:** True: t^* is the random default time.

Q339 BT answers P2.T6.307.2 · BT notes P2.T6.307.2

Suppose the hazard rate (default intensity, λ) is constant and equal to 0.090 per year. Each of the following is true EXCEPT which is false?

A. The unconditional one-year default probability is approximately 8.6%.

B. The two-year cumulative default probability is approximately 16.5%.

C. The probability of the joint event of survival through the first year and default in the second year is approximately 7.9%.

D. The conditional one-year default probability, given survival through the first year, is approximately 17.3%.

Answer: D

Step by step

1. One-year PD $= 1 - e^{-0.09} = 8.61\%$ (A true).
2. Two-year cumulative PD $= 1 - e^{-0.18} = 16.47\%$ (B true).
3. Survive year 1 and default in year 2: $e^{-0.09} - e^{-0.18} = 0.9139 - 0.8353 = 7.87\%$ (C true).
4. Conditional PD in year 2 given survival: $7.87\%/91.39\% = 8.61\%$, not 17.3%, so D is false.

Why D is correct: With a constant hazard rate the process is memoryless, so the conditional one-year default probability equals the unconditional 8.61%.

Why the others are wrong

- **A:** True: $1 - e^{-0.09} = 8.61\%$.
- **B:** True: $1 - e^{-0.18} = 16.47\%$.
- **C:** True: 16.47% - 8.61% = 7.87%; this is the joint probability, not yet divided by the survival probability.

Q340 BT answers P2.T6.307.1 · BT notes P2.T6.307.1

REBUILT — Source BT notes had no answer and two distractors (8.42%, 17.00%) corresponded to no realistic error; replaced with plausible mistakes.

A municipal bond analyst observes about 6.0 US municipal defaults per year and assumes defaults follow a Poisson process with a constant average of 6.0 per year. What is the probability that the next municipal default occurs within one month?

A. 7.99%

B. 39.35%

C. 50.00%

D. 60.65%

Answer: B

Step by step

1. Intensity $\lambda = 6$ per year, so over $t = 1/12$ year the expected count is $\lambda t = 0.5$.
2. Time to the next default is exponential: $P[t^* \le t] = 1 - e^{-\lambda t}$.
3. $1 - e^{-0.5} = 1 - 0.60653 = 39.35\%$.

Why B is correct: Time to the first event in a Poisson process is exponential, so $P = 1 - e^{-\lambda t}$ with $\lambda t = 0.5$.

Why the others are wrong

- **A:** 7.99% = $1 - e^{-1/12}$: used 1/12 as the intensity, forgetting to multiply by 6.
- **C:** 50.00% = 6/12: treated the expected number of defaults in the month as a probability.
- **D:** 60.65% = $e^{-0.5}$: the probability of NO default in the month (survival), not default.

LO 27.f

Describe recovery rates and their dependencies on default rates.

Q341 BT answers P2.T6.24.14.2

REBUILT — Option D ('inversely proportional') was also defensible as a description of a negative relationship and the stem was padded; reworded D into a clearly wrong liquidity claim and shortened the stem.

A credit risk analyst is refining loss estimates for a corporate bond portfolio during an economic downturn in which default rates rise sharply. Which of the following best describes how average recovery rates behave in this environment?

A. Recovery rates rise, because the larger number of defaults gives lenders more experience to improve their workout strategies.

B. Recovery rates fall, because defaults rise with a weak economy that depresses the prices of defaulted bonds and collateral.

C. Recovery rates are unaffected, because they are determined only by the issuer's original credit rating.

D. Recovery rates are unaffected, because defaulted securities are highly liquid in a downturn and are always resold at face value.

Answer: B

Step by step

1. Recovery rate = price of the defaulted bond about a month after default, as a fraction of face value.
2. A weak economy produces more defaults, and a larger supply of defaulted bonds and falling collateral values push post-default prices down.
3. So default rates and recovery rates are negatively correlated: high default rate comes with low recovery rate.
4. Therefore, in a downturn, recovery rates fall (the 'double hit' for lenders: more defaults and bigger losses per default).

Why B is correct: Recovery rates are negatively correlated with default rates because the same weak economy that raises defaults also depresses the value of defaulted bonds and collateral.

Why the others are wrong

- **A:** Reverses the sign of the relationship; more defaults do not improve recoveries, and the market effect dominates any learning effect.
- **C:** Ignores the macro dependence; seniority and rating matter, but average recovery rates also move with the default cycle.
- **D:** Defaulted securities are illiquid and trade at depressed prices in a downturn, and recoveries are not tied to face value.

Q342 AnalystPrep Q.1802

REBUILT — Original asked for a dollar amount with a giveaway option ($2.5m exceeds total assets); recast as the recovery rate on the subordinated debt to test the seniority dependence of recovery.

A firm has $2.0m of senior debt and $0.5m of subordinated debt. It is liquidated and its assets realize $2.25m, which is distributed strictly by seniority. What is the recovery rate (as a percentage of face value) on the subordinated debt?

A. 12.5%

B. 25%

C. 50%

D. 100%

Answer: C

Step by step

1. Senior debt is paid first: $\min(2.25, 2.0) = 2.0$ m, fully repaid.
2. Residual for subordinated holders: $2.25 - 2.0 = 0.25$ m.
3. Recovery rate = $0.25/0.5 = 50\%$ of the subordinated face value.
4. This illustrates that senior claims recover more than junior claims (senior 100% here versus 50%).

Why C is correct: Subordinated holders receive only what is left after the senior claim is satisfied in full, $0.25m of a $0.5m claim, or 50%.

Why the others are wrong

- **A:** Divided the $0.25m residual by the senior debt ($2.0m) instead of the subordinated claim ($0.5m).
- **B:** Read the $0.25m residual as a percentage figure rather than dividing by the $0.5m claim.
- **D:** Ignored seniority and assumed the subordinated debt is repaid in full, although assets cover only part of its claim after the senior debt.

Q343 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A credit risk manager estimates the annual expected loss rate on a corporate loan book as the default probability multiplied by (1 minus the recovery rate). In normal conditions the default probability is 2% and the recovery rate is 40%. The manager now builds a recession stress scenario in which the default probability is 5%. Given the historical relationship between default rates and recovery rates, which expected loss rate is most appropriate if the recovery rate in the stress scenario is 30%?

A. 3.5%
B. 1.5%
C. 1.2%
D. 3.0%

Answer: A

Step by step

1. Recovery rates are negatively correlated with default rates, so recovery should be lowered (not held at 40%) in the stress.
2. Expected loss = 5% x (1 - 0.30) = 3.5%.

Why A is correct: Using the lower recovery rate consistent with higher defaults gives 5% x 70% = 3.5%, which captures the compounding effect of high defaults and low recoveries.

Why the others are wrong

- **B:** 5% x 30% = 1.5% multiplies by the recovery rate instead of the loss given default.
- **C:** 2% x (1 - 0.40) = 1.2% is the unstressed loss rate.
- **D:** 5% x (1 - 0.40) = 3.0% holds recovery at its normal level, ignoring the negative default-recovery correlation and understating loss.

LO 27.g

Define a credit default swap (CDS) and explain its mechanics including the obligations of both the default protection buyer and the default protection seller.

Q344 BT notes P2.T6.24.10.1 · BT answers P2.T6.24.10.3

REBUILT — Original options B and D were essentially identical and the first source gave no answer; made the stem explicit that no credit event occurs and rewrote options so exactly one is correct.

Chicago Firefighters Pension (CFP) owns Belize government bonds with a par value of $10,000,000 and has bought 10-year credit default swap protection on them from Old Crow Bank at a CDS spread of 300 basis points per year (3% of par). In year 8, the market value of the bonds falls to $8,000,000 because of a decrease in market demand for the bonds. Belize continues to make all of its scheduled payments and no credit event is declared. What is the cash flow between CFP and Old Crow Bank for year 8?

A. CFP receives a net payment of $1.7M ($2M payout less $300,000 premium).
B. CFP receives a payment of $2M from Old Crow Bank.
C. CFP pays $300,000 to Old Crow Bank and receives no payout.
D. No payments are made by either party because the swap is marked to market and terminated when the bond price falls.

Answer: C

Step by step

1. A CDS pays the protection buyer only if a credit event occurs (failure to pay, restructuring, bankruptcy).
2. The fall to $8M is caused by lower demand, not by a default or restructuring, so no credit event has occurred and the protection seller owes nothing.
3. The buyer keeps paying the fixed premium until a credit event or maturity: $0.03 \times 10{,}000{,}000 = 300{,}000$ dollars per year.
4. Year 8 cash flow: CFP pays $300,000 and receives $0.

Why C is correct: A CDS is triggered by a credit event, not by a market-price decline; the buyer continues paying the premium for the life of the contract unless a credit event occurs.

Why the others are wrong

- **A:** Assumes the $2M price drop (par minus market value) is paid out as if a credit event had occurred, then nets the premium: 2.0 - 0.3 = 1.7.
- **B:** Same payout error as A, and additionally ignores that the premium is still due.
- **D:** Treats the CDS like a margined, mark-to-market contract that ends on a price decline; a CDS continues until a credit event or maturity and the premium remains payable.

Q345

BT notes P2.T6.24.22.2

REBUILT — Original asked for a PV of expected payoff without a discount rate and had a garbled default-probability table; rebuilt as a self-contained cash-settlement and final-accrual question testing CDS mechanics.

A protection buyer holds a 5-year CDS on a corporate reference entity with a notional principal of $100 million. The buyer pays a CDS spread of 90 basis points per year, in quarterly installments paid in arrears on the standard payment dates. The reference entity suffers a credit event exactly one month after the most recent payment date. The contract is cash settled, and an ISDA-organized auction sets the mid-market value of the cheapest-to-deliver bond at $35 per $100 of face value. What is the net amount received by the protection buyer at settlement, after paying the required final accrual?

A. $64,925,000
B. $65,075,000
C. $65,000,000
D. $34,925,000

Answer: A

Step by step

1. Cash-settlement payoff = notional $\times(1 - \text{CTD price}) = 100{,}000{,}000 \times (1 - 0.35) = 65{,}000{,}000$ dollars, paid by the seller.
2. Annual premium = $0.0090 \times 100{,}000{,}000 = 900{,}000$; quarterly = $225,000.
3. Premiums are in arrears, so the buyer owes accrual for the 1 month since the last payment date: $225{,}000 \times 1/3 = 75{,}000$ dollars.
4. Payments then cease. Net to buyer = $65,000,000 - $75,000 = $64,925,000.

Why A is correct: On a credit event the seller pays notional times (1 minus the auction price of the cheapest-to-deliver bond), and the buyer pays a final accrued premium for the part-period before premiums stop.

Why the others are wrong

- **B:** Adds the accrued premium to the payoff instead of subtracting it; the accrual is owed by the buyer.
- **C:** Ignores the final accrual payment that is due because premiums are paid in arrears.
- **D:** Uses the bond's post-default value (35%) as the payoff instead of the loss (65%), then subtracts the accrual.

Q346

Schweser Module Quiz 27.2 Q1

REBUILT — Clarified that the protection seller pays the full notional on the defaulted name (recovery ignored), which the original left implicit; numbers and answer unchanged.

The five-year CDX NA IG index (125 companies) is quoted as bid 141 bps and ask 143 bps. An investor sells $1 million of protection on each company. At the beginning of the third year, before the annual protection payment is made, one of the companies defaults. Assume no other defaults, and that the seller pays the full $1 million notional on the defaulted name (ignore the value of the bond received in exchange). The investor's cash flow for the third year is closest to:

A. $748,400 inflow.
B. $748,400 outflow.
C. $773,200 inflow.
D. $773,200 outflow.

Answer: A

Step by step

1. The investor is a protection seller, so receives the bid spread: $0.0141 \times 1{,}000{,}000 \times 125 = 1{,}762{,}500$ dollars.
2. With one default, the premium falls by $1{,}762{,}500/125 = 14{,}100$ dollars.
3. The seller pays $1,000,000 on the defaulted name.
4. Net = $1{,}762{,}500 - 14{,}100 - 1{,}000{,}000 = 748{,}400$ dollars, an inflow.

Why A is correct: The seller collects premium on the 124 surviving names and pays the notional on the defaulted one; premium exceeds the payout, so the net is an inflow.

Why the others are wrong
- **B:** Correct magnitude but wrong sign; the premium received on 124 names exceeds the single $1M payout.
- **C:** Uses the ask spread (143 bps) instead of the bid; a protection seller receives the bid. 0.0143 x 1M x 125 - 14,300 - 1M = 773,200.
- **D:** Uses the ask spread and also gets the sign wrong.

LO 27.h

Describe CDS spreads and explain how CDS spreads can be used to estimate hazard rates.

Q347

GARP PE2 Q62 · Mock 4 Q63

A bond issued by Company GRN trades at a spread of 250 bps over the risk-free rate with a recovery rate of 75%. A risk analyst applies a constant hazard rate model and assumes that under a stress scenario the spread widens to 480 bps and the recovery rate falls to 40%. Under the stress scenario, what is the probability that GRN does not default over the next 3 years?

A. 69.8%
B. 78.7%
C. 86.6%
D. 74.1%

Answer: B

Step by step
1. Hazard rate $\lambda \approx s/(1-R)$, using stressed inputs.
2. $\lambda = 0.0480/(1-0.40) = 0.08$.
3. Three-year survival $= e^{-0.08\times 3} = e^{-0.24} = 78.66\% \approx 78.7\%$.

Why B is correct: The spread compensates for expected loss per year, $\lambda(1-R)$, so $\lambda = s/(1-R)$; survival is $e^{-\lambda t}$.

Why the others are wrong
- **A:** Divided the spread by R instead of 1-R (0.048/0.40 = 12%): $e^{-0.36} = 69.8\%$.
- **C:** Ignored recovery ($\lambda = 4.8\%$): $e^{-0.144} = 86.6\%$.
- **D:** Used unstressed inputs ($\lambda = 0.025/0.25 = 10\%$): $e^{-0.30} = 74.1\%$.

Q348

AP Mock 1 Q23

REBUILT — Original options were not tied to identifiable errors and the hazard-rate convention was unstated; stated the constant-hazard/credit-triangle convention and rebuilt distractors.

A 2-year corporate bond yields 7% and the risk-free rate is 3%. The recovery rate is 40%. Assume the average hazard rate is constant and estimated by the credit triangle $\lambda \approx s/(1-R)$, with continuous-time survival $Q(t) = e^{-\lambda t}$. What is the unconditional (marginal) probability that default occurs during the second year?

A. 0.0645
B. 0.0603
C. 0.1248
D. 0.0667

Answer: B

Step by step
1. Spread $s = 7\% - 3\% = 4\%$.
2. $\lambda = 0.04/(1-0.40) = 0.0667$.
3. $Q(1) = e^{-0.0667} = 0.9355$; $Q(2) = e^{-0.1333} = 0.8752$.
4. Default in year 2 (unconditional) $= Q(1) - Q(2) = 0.9355 - 0.8752 = 0.0603$.

Why B is correct: The unconditional probability of default in year 2 is the survival probability at the start of the year minus that at the end, both measured from time zero.

Why the others are wrong
- **A:** Conditional probability of default in year 2 given survival to year 1, $1 - e^{-\lambda} = 0.0645$; the question asks for the unconditional value.
- **C:** Two-year cumulative default probability $1 - Q(2) = 0.1248$; this includes year 1.
- **D:** The hazard rate itself (0.0667), not a probability over the year.

Q349

BT answers P2.T6.24.14.1

REBUILT — Stem asked for an 'unconditional' PD though the hazard rate is a conditional intensity, and contained irrelevant Merton/bond data; reworded to ask for the average hazard rate from the 5-year CDS spread.

A firm's 5-year CDS spread is 240 basis points and the expected recovery rate is 40%. Using the approximation that the CDS spread equals the average hazard rate times the loss given default, what is the average hazard rate (default intensity) over the five years?

A. 2.4%
B. 4.0%
C. 6.0%
D. 3.9%

Answer: B

Key correction: Source key (B) is retained; the source wording 'unconditional PD' was loose because 4% is the hazard rate (conditional intensity).

Step by step

1. $s \approx \lambda(1 - R)$, so $\lambda = s/(1 - R)$.
2. $\lambda = 0.0240/(1 - 0.40) = 0.0240/0.60$.
3. $\lambda = 4.0\%$ per year.

Why B is correct: The CDS spread compensates the seller for expected loss per year, which is hazard rate times loss given default (1 - R).

Why the others are wrong

- **A:** Uses the spread as the hazard rate, ignoring recovery.
- **C:** Divides by the recovery rate (0.0240/0.40) instead of the loss rate (1 - R).
- **D:** Converts the hazard rate into a one-year default probability $1 - e^{-0.04} = 3.9\%$ (and is not the intensity asked for).

Q350

Mock 2 Q6

REBUILT — Distractor 6.4% corresponded to no realistic error; replaced with spread times loss given default.

The credit spread on a one-year B-rated corporate bond relative to the maturity-matched T-bill is 2.2%. The recovery rate in default is 75%. What is the hazard rate (default intensity) of the bond?

A. 2.2%
B. 2.9%
C. 0.55%
D. 8.8%

Answer: D

Step by step

1. $\lambda \approx s/(1 - R)$.
2. $\lambda = 0.022/(1 - 0.75) = 0.022/0.25$.
3. $\lambda = 8.8\%$.

Why D is correct: With high recovery, a given spread must be explained by a high default probability, so the hazard rate is the spread divided by the loss given default.

Why the others are wrong

- **A:** Spread alone, which assumes zero recovery (R = 0).
- **B:** Divides by the recovery rate (0.022/0.75) instead of the loss rate.
- **C:** Multiplies the spread by the loss rate (0.022 x 0.25) instead of dividing.

Q351

BT answers P2.T6.308.1 (Malz CDS hazard-rate equation, exhibit missing)

REBUILT — Original depended on a missing equation exhibit from Malz; rebuilt as a self-contained forward hazard rate calculation from a CDS term structure (GARP Example 9.3 method).

The 3-year and 5-year CDS spreads for a company are 90 and 120 basis points respectively, and the expected recovery rate is 40%. Using the approximation average hazard rate = spread/(1 - R), what is the estimated average hazard rate between year 3 and year 5?

A. 1.75%
B. 2.75%

C. 1.65%
D. 0.50%

Answer: B
Step by step
1. 3-year average hazard rate: $0.0090/0.60 = 1.50\%$.
2. 5-year average hazard rate: $0.0120/0.60 = 2.00\%$.
3. Cumulative hazard: $5 \times 0.0200 = 0.1000$ and $3 \times 0.0150 = 0.0450$.
4. Forward rate $= (0.1000 - 0.0450)/2 = 2.75\%$.

Why B is correct: Average hazard rates times maturity give cumulative hazard; the difference over the two extra years, divided by 2, is the year 3-5 average hazard rate.
Why the others are wrong
- **A:** Simple average of the 3-year and 5-year hazard rates (1.50% and 2.00%), ignoring the maturity weights.
- **C:** Same bootstrap but using spreads without dividing by (1 - R): $(5 \times 1.20\% - 3 \times 0.90\%)/2 = 1.65\%$.
- **D:** Unweighted difference of the two average hazard rates (2.00% - 1.50%).

Q352 Mock 1 Q19

A risk manager values a credit default swap by computing default probabilities. Holding the market CDS spread constant, the risk-neutral probability of default (hazard rate) will be:
A. unrelated to the recovery rate.
B. proportional to the notional principal.
C. positively related to the recovery rate.
D. negatively related to the recovery rate.

Answer: C
Step by step
1. Credit triangle: $\lambda \approx s/(1 - R)$.
2. With s fixed, a higher R lowers $1 - R$, so λ rises.
3. E.g. $s = 2\%$: $R = 40\% \Rightarrow 3.33\%$; $R = 60\% \Rightarrow 5.0\%$.

Why C is correct: A fixed spread must compensate for the same expected loss, so a smaller loss given default requires a higher probability of default.
Why the others are wrong
- **A:** The recovery rate appears in the denominator $1 - R$, so it matters.
- **B:** Notional scales the payoff and premium equally; it cancels out of the hazard rate.
- **D:** Intuitive but reversed: higher recovery lowers loss per default, so more defaults are implied by the same spread.

LO 27.i

Define and explain CDS-bond basis.

Q353 Mock 1 Q44

QuantTech has an outstanding zero-coupon bond with one year remaining to maturity. The bond has a face value of USD 1,000 and is currently trading at 93% of face value (USD 930). Assume the bond is considered to have a zero recovery rate. If the corresponding credit default swap (CDS) spread is less than the bond yield spread (a negative CDS-bond basis), which position should a trader take to exploit the arbitrage opportunity?
A. Sell the QuantTech corporate bond and buy CDS protection to earn more than the risk-free rate.
B. Buy the QuantTech corporate bond and buy CDS protection to earn more than the risk-free rate.
C. Sell the QuantTech corporate bond and sell CDS protection to borrow at less than the risk-free rate.
D. Buy the QuantTech corporate bond and sell CDS protection to borrow at less than the risk-free rate.

Answer: B
Step by step
1. Bond yield = 1,000 / 930 - 1 = 7.53%; the bond yield spread is this yield less the risk-free rate.

2. CDS-bond basis = CDS spread - bond yield spread. Here CDS spread < bond yield spread, so the basis is negative: credit risk is cheaper to hedge via the CDS than the bond pays for bearing it.
3. Buying the bond earns the (wide) bond spread; buying CDS protection removes the default risk at the lower CDS cost.
4. The hedged position is approximately risk-free yet earns the bond yield minus the CDS spread, i.e. more than the risk-free rate.

Why B is correct: When the CDS spread is below the bond yield spread, buying the bond and buying protection locks in a near risk-free return above the risk-free rate (the bond spread exceeds the cost of protection).

Why the others are wrong

- **A:** Shorting the bond and also buying protection leaves the trader with offsetting-in-name but same-direction credit positions (both gain on default) and pays the wide bond spread; it does not capture the arbitrage.
- **C:** Reverses both legs, which would be the trade for a CDS spread greater than the bond spread (borrow below risk-free), the opposite sign of the basis.
- **D:** Buying the bond and selling protection doubles up long credit exposure rather than hedging it; it earns both spreads but is not an arbitrage and carries double default risk.

Q354

Schweser Module Quiz 27.2 Q2

In the context of arbitrage trades, if the CDS spread is significantly greater than the bond yield spread (a positive CDS-bond basis), what is the most appropriate action by the investor?

A. Buy the corporate bond and buy CDS protection.
B. Buy the corporate bond and sell CDS protection.
C. Sell the corporate bond and buy CDS protection.
D. Sell the corporate bond and sell CDS protection.

Answer: D

Step by step

1. CDS-bond basis = CDS spread - bond yield spread > 0, so protection is expensive relative to the bond's spread.
2. Short the corporate bond: the investor effectively borrows at the bond's (lower) yield spread over the risk-free rate, which is the cost of the borrowing.
3. Sell CDS protection: the investor receives the higher CDS spread and takes on issuer default risk, which offsets the credit exposure of the short bond.
4. Net: the position is roughly credit-neutral and the investor effectively borrows at less than the risk-free rate (the CDS spread received exceeds the bond spread paid).

Why D is correct: With a CDS spread above the bond spread, shorting the bond and selling protection collects more in CDS premium than is paid on the bond, giving funding below the risk-free rate.

Why the others are wrong

- **A:** Buys the bond and buys protection, the trade for a CDS spread below the bond spread (negative basis); here protection is too expensive.
- **B:** Long bond plus sold protection doubles the credit exposure instead of hedging it, so it is not an arbitrage.
- **C:** Short bond plus bought protection doubles the short credit position and pays the expensive CDS spread; it hedges nothing.

Q355

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A credit analyst notes that the CDS-bond basis for a corporate issuer, defined as the CDS spread minus the bond yield spread, has turned negative. Which of the following is a factor that tends to push the basis in a negative direction?

A. The CDS contract's restructuring clause may lead to a payoff even though the bond has not defaulted.
B. The protection seller in the CDS may itself default, creating counterparty risk for the protection buyer.
C. The CDS contains a cheapest-to-deliver bond option for the protection buyer.
D. The bond is trading at a price below par.

Answer: B

Step by step

1. Basis = CDS spread - bond yield spread.
2. Counterparty risk lowers what the buyer will pay for protection, so the CDS spread falls relative to the bond

spread: negative basis.
3. The cheapest-to-deliver option, the restructuring clause and bond prices below par all push the basis in a positive direction.

Why B is correct: Counterparty default risk in the CDS reduces the CDS spread relative to the bond yield spread, pushing the basis negative.

Why the others are wrong

- **A:** The restructuring clause can trigger payoffs without default, making protection more valuable and the CDS spread higher: positive direction.
- **C:** The cheapest-to-deliver option benefits the buyer, so the spread is higher: positive direction.
- **D:** Bond prices below par tend to give a positive basis; prices above par tend to give a negative basis.

LO 27.j

Compare default probabilities calculated from historical data with those calculated from credit yield spreads.

Q356

AnalystPrep Q.1805

REBUILT — Stated continuous compounding explicitly and replaced unmotivated distractors with error-based ones.

A firm's only debt is a zero-coupon bond with face value \$100 million and five years remaining to maturity. The bond currently has a market value of \$88 million. The risk-free rate implied by the zero-coupon bond price is 1% (continuously compounded). What is the credit spread on the bond, using continuous compounding?

A. 1.56%
B. 1.73%
C. 2.56%
D. 11.78%

Answer: A

Step by step

1. Continuously compounded yield: $y = -\frac{1}{T}\ln\left(\frac{D}{F}\right) = -\frac{1}{5}\ln(0.88)$.
2. $\ln(0.88) = -0.12783$, so $y = 0.12783/5 = 2.557\%$.
3. Credit spread $= y - r_f = 2.557\% - 1.00\% = 1.56\%$.

Why A is correct: The credit spread is the bond's yield minus the risk-free rate; the yield on a zero is $-\ln(D/F)/T$.

Why the others are wrong

- **B:** Uses simple (non-continuous) yield $(100/88 - 1)/5 = 2.73\%$ and subtracts 1%.
- **C:** Reports the bond yield 2.56% and forgets to subtract the 1% risk-free rate.
- **D:** Omits the $1/T$ factor: $-\ln(0.88) - 1\% = 12.78\% - 1\%$.

Q357

AnalystPrep Q.3060

REBUILT — Replaced unmotivated distractors with error-based ones.

A firm's only debt is a zero-coupon bond with face value \$200 million and six years remaining to maturity. The bond's current value is \$110 million. The risk-free rate implied by the zero-coupon bond price is 5% (continuously compounded). What is the credit spread on the bond?

A. 9.96%
B. 5.48%
C. 4.96%
D. 8.64%

Answer: C

Step by step

1. Yield: $y = -\frac{1}{6}\ln\left(\frac{110}{200}\right) = -\frac{1}{6}\ln(0.55)$.
2. $\ln(0.55) = -0.59784$, so $y = 9.964\%$.
3. Spread $= 9.964\% - 5\% = 4.96\%$.

Why C is correct: Credit spread = continuously compounded bond yield $-\ln(D/F)/T$ minus the risk-free rate.

Why the others are wrong

- **A:** Reports the yield of 9.96% without subtracting the 5% risk-free rate.

- **B:** Uses annual compounding: $(200/110)^{1/6}-1 = 10.48\%$, then subtracts 5%; the question specifies continuous compounding.
- **D:** Uses simple interest $(200/110 - 1)/6 = 13.64\%$ and subtracts 5%.

Q358

AnalystPrep Q.3196

REBUILT — Clarified that the bond was issued two years ago so 3 years remain; replaced unmotivated distractors with error-based ones.

A five-year zero-coupon bond with face value $100 million was issued two years ago and now has a market value of $80 million. The bond trades at a credit spread of 2% (continuous compounding). What is the risk-free rate implied by these data?

A. 9.44%
B. 7.44%
C. 5.44%
D. 2.46%

Answer: C

Step by step

1. Remaining maturity $T - t = 5 - 2 = 3$ years.
2. Bond yield: $y = -\frac{1}{3}\ln(0.80) = 0.22314/3 = 7.438\%$.
3. Spread $= y - r_f$, so $r_f = 7.438\% - 2\% = 5.44\%$.

Why C is correct: Because spread $= y - r_f$, the risk-free rate is the yield less the spread, using the remaining 3 years.

Why the others are wrong

- **A:** Adds the spread to the yield (7.44% + 2%) instead of subtracting it.
- **B:** Reports the bond yield, forgetting to subtract the 2% spread.
- **D:** Uses the original 5-year maturity: $-\ln(0.8)/5 = 4.46\%$, minus 2% = 2.46%.

Q359

BT answers P2.T6.403.3

REBUILT — Replaced unmotivated distractors with error-based ones.

Debt with five years to maturity ($T = 5.0$) has a face value of $1,000 and a current price of $834.00. If the risk-free rate is 2.0% per annum, which is nearest to the credit spread (both rates continuously compounded)?

A. 3.63%
B. 1.63%
C. 1.70%
D. 1.98%

Answer: B

Step by step

1. Yield: $y = -\frac{1}{5}\ln(834/1000) = 0.18150/5 = 3.630\%$.
2. Spread $= 3.630\% - 2.0\% = 1.63\%$.

Why B is correct: Spread = continuously compounded yield $-\ln(D/F)/T$ minus the risk-free rate.

Why the others are wrong

- **A:** Reports the yield and forgets to subtract the 2% risk-free rate.
- **C:** Uses annual compounding: $(1000/834)^{1/5} - 1 = 3.70\%$, minus 2%; the question specifies continuous compounding.
- **D:** Uses simple interest $(1000/834 - 1)/5 = 3.98\%$, minus 2%.

Q360

BT answers P2.T6.24.14.2

REBUILT — Original mixed a 5-year spread with a 7-year cumulative PD, bundled an unrelated CDS-bond basis point, and used vague options; rebuilt as a consistent 7-year historical vs spread-implied hazard rate comparison with exactly one correct option.

A risk analyst compares two estimates of the average 7-year default intensity for a BB-rated issuer. Historical data show a 7-year cumulative default probability of 4.0%. The issuer's 7-year average credit spread is 300 basis points, and the assumed recovery rate is 40%. Which statement correctly gives the two average hazard rates and the proper interpretation of their comparison?

A. Historical hazard rate about 0.58% per year versus spread-implied about 5.00% per year; the spread-implied

rate is much higher, so investors are compensated for more than the expected cost of default (systematic risk, liquidity)

B. Historical hazard rate about 0.58% per year versus spread-implied about 5.00% per year; the spread-implied rate is much higher, which shows investors are under-compensated for default risk

C. Historical hazard rate about 0.58% per year versus spread-implied about 3.00% per year; the spread-implied rate is higher, so investors are compensated for more than the expected cost of default

D. Historical hazard rate about 0.58% per year versus spread-implied about 7.50% per year; the spread-implied rate is higher, so investors are compensated for more than the expected cost of default

Answer: A

Step by step

1. Historical hazard rate from cumulative PD: $\lambda_{hist} = -\frac{1}{7}\ln(1-0.04) = 0.04082/7 = 0.00583$, i.e. about 0.58% per year.
2. Spread-implied hazard rate: $\lambda_{spread} \approx \frac{s}{1-R} = \frac{0.0300}{1-0.40} = 0.0500$, i.e. 5.00% per year.
3. Compare: 5.00% is roughly 8.6 times 0.58%, so the spread-implied rate is far above the historical rate.
4. Interpretation: spreads embed more than expected default loss because defaults are correlated with the economy (systematic, non-diversifiable risk), unsystematic risk cannot be fully diversified, and corporate bonds carry a liquidity premium. Investors earn excess return for bearing credit risk.

Why A is correct: Hazard rates from spreads (s/(1-R)) are materially higher than those from historical default data, and the gap reflects a risk premium (systematic risk and liquidity) rather than under-compensation.

Why the others are wrong

- **B:** Numbers are right but the conclusion is reversed: a spread-implied rate above the historical rate means investors are over-compensated relative to actuarial default cost, not under-compensated.
- **C:** Used s directly as the hazard rate (3.00%), forgetting to divide by $1 - R$ = 0.60.
- **D:** Divided the spread by the recovery rate (0.03/0.40 = 7.50%) instead of the loss given default $1 - R$.

LO 27.k

Describe the difference between real-world and risk-neutral default probabilities and determine which one to use in the analysis of credit risk.

Q361

AnalystPrep Q.5989

REBUILT — Source key/explanation was loose (claimed no risk-preference assumption is needed); rebuilt so the options test the precise meaning of risk-neutral (asset growth at the risk-free rate, higher PD than real-world).

An analyst estimates a firm's default probability with the Merton model and compares a risk-neutral calculation with a real-world calculation, holding the face value of debt, asset volatility and maturity constant. Which statement correctly describes the risk-neutral calculation?

A. The firm's assets are assumed to grow at the risk-free rate, so the default probability is higher than in the real-world calculation.

B. The firm's assets are assumed to grow at the risk-free rate plus a market risk premium, so the default probability is higher than in the real-world calculation.

C. The firm's assets are assumed to grow at the risk-free rate, so the default probability is lower than in the real-world calculation.

D. The firm's assets are assumed to grow at the risk-free rate plus a market risk premium, so the default probability is lower than in the real-world calculation.

Answer: A

Key correction: Source key (AnalystPrep) said B of its own option set (no assumptions about risk preferences needed); that wording is imprecise. The rebuilt item tests the precise meaning: risk-neutral valuation sets expected asset growth equal to the risk-free rate.

Step by step

1. Risk-neutral world: expected growth rate of assets = r_f (no premium for bearing risk).
2. Real world: expected growth rate = r_f + market risk premium, so expected asset values are higher.
3. With the same face value of debt, higher asset values make a shortfall below the debt less likely, so the real-world PD is lower.

4. Hence the risk-neutral PD is higher than the real-world PD: choice A.

Why A is correct: Risk-neutral valuation drops the risk premium from asset growth, so assets drift up more slowly and are more likely to end below the debt's face value; risk-neutral PD exceeds real-world PD.

Why the others are wrong

- **B:** Attaches the real-world drift (risk-free rate plus premium) to the risk-neutral measure; that higher drift would lower, not raise, the PD.
- **C:** Gets the drift right but reverses the ranking of the two PDs; lower asset growth raises PD.
- **D:** Drift and ranking are both wrong: it describes the real-world calculation but labels it risk-neutral.

Q362

Schweser Module Quiz 27.3 Q1

REBUILT — Answer options A-D were lost in OCR; reconstructed as four combinations of (type of default probability for valuation, effect of a rising distance to default).

A junior analyst is drafting a research report and addresses two questions: (1) Which type of default probabilities should be used in valuation? (2) What is the effect on the likelihood of a company defaulting when its distance to default rises? Which conclusion should the report support?

A. Real-world default probabilities for valuation; default becomes less likely as distance to default rises.
B. Risk-neutral default probabilities for valuation; default becomes less likely as distance to default rises.
C. Risk-neutral default probabilities for valuation; default becomes more likely as distance to default rises.
D. Real-world default probabilities for valuation; default becomes more likely as distance to default rises.

Answer: B

Step by step

1. Valuation requires risk-neutral estimates (e.g., default rates backed out of credit spreads); no extra premium for bearing risk is needed.
2. Real-world estimates (historical data) are for scenario analysis, not valuation.
3. Distance to default is the number of standard deviations the firm's asset value sits above the default point; $PD = N(-DD)$ falls as DD rises.
4. So: risk-neutral for valuation, and default is less likely as DD rises: B.

Why B is correct: Valuation uses risk-neutral probabilities, and a larger distance to default puts assets further from the default boundary, so the default probability falls.

Why the others are wrong

- **A:** Uses real-world probabilities for valuation; they belong in scenario analysis.
- **C:** Right measure for valuation but misreads DD: a higher DD lowers, not raises, default likelihood.
- **D:** Wrong measure for valuation and wrong direction for DD.

Q363

BT answers P2.T6.24.14.2 · BT notes P2.T6.24.15.2

REBUILT — Original asked for one measure to 'assess credit risk' with a questionable key (risk-neutral) and a truncated option set; rebuilt with two explicit uses (CDS valuation vs scenario loss) and a numeric expected-loss check so exactly one option is right.

ACME Financial holds a $10 million exposure to Zigzag Corp. Zigzag's historical (real-world) default rate is 2% per year, while the risk-neutral default probability implied by its CDS spreads is 3% per year. Assume a 40% recovery rate. ACME wants to (i) value a CDS on Zigzag and (ii) estimate the expected one-year credit loss on the exposure in a scenario analysis of its portfolio. Which approach is appropriate?

A. Use the 3% risk-neutral probability for both; scenario expected loss is $180,000.
B. Use the 2% real-world probability for CDS valuation and the 3% risk-neutral probability for the scenario analysis; scenario expected loss is $180,000.
C. Use the 3% risk-neutral probability for CDS valuation and the 2% real-world probability for the scenario analysis; scenario expected loss is $120,000.
D. Use the 2% real-world probability for both; scenario expected loss is $120,000.

Answer: C

Key correction: Source key (BT notes) said B (risk-neutral as the measure for assessing the portfolio exposure); under the reading's rule risk-neutral is for valuation and real-world for scenario analysis, so the original keyed answer was questionable.

Step by step

1. Valuation (CDS pricing) uses risk-neutral probabilities: 3%.
2. Scenario analysis uses real-world probabilities: 2%.

3. Expected loss $= 10{,}000{,}000 \times 2\% \times (1 - 40\%) = \$120{,}000$.
4. Choice C matches both the measures and the loss.

Why C is correct: Risk-neutral probabilities embed the risk premium and are consistent with market prices, so they are used for valuation; real-world probabilities measure the actual likelihood of loss and are used in scenario analysis.

Why the others are wrong

- **A:** Applies risk-neutral to scenario analysis too; the 3% overstates actual loss likelihood ($10m \times 3\% \times 60\% = \$180{,}000$).
- **B:** Reverses the two uses: real-world for pricing and risk-neutral for scenarios, giving the inflated $180,000.
- **D:** Uses real-world probabilities for valuation too; a CDS priced off the historical 2% would be inconsistent with market spreads.

Q364 BT answers P2.T6.327.2

While there are important exceptions, which is the best general advice in regard to the selection of real or risk-neutral probabilities in credit exposure models?

A. Exposure management (e.g., scenario generation) should focus on risk-neutral, market-implied parameters, but pricing should focus on real, historical parameters, especially when counterparty risk is actively managed

B. Exposure management (e.g., scenario generation) should focus on real, historical parameters, but pricing should focus on risk-neutral, market-implied parameters, especially when counterparty risk is actively managed

C. Both exposure management and pricing should focus on risk-neutral, market-implied parameters

D. Both exposure management and pricing should focus on real, historical parameters

Answer: B

Step by step

1. Valuation/pricing: risk-neutral (market-implied) estimates, e.g., from credit spreads, since cash flows are discounted at the risk-free rate.
2. Scenario analysis / exposure management: real-world (historical) estimates.
3. Only B assigns each measure to its proper use.

Why B is correct: Pricing must be consistent with market prices (risk-neutral), whereas scenario and exposure analysis seeks realistic outcomes (real-world).

Why the others are wrong

- **A:** Swaps the two uses of the measures.
- **C:** Risk-neutral parameters overstate default likelihood and are not appropriate for scenario generation.
- **D:** Real-world parameters are inconsistent with market prices, so they are not suitable for pricing.

LO 27.l

Calculate the value of a firm's debt and equity, the volatility of firm value, and the volatility of firm equity using the Merton model.

Q365 AnalystPrep Q.2878

REBUILT — Stem did not say the put is written on the firm's assets and had no question mark; clarified it and replaced unmotivated distractors with error-based ones.

A firm has only one liability, a zero-coupon bond with face value $F = \$206.74$ million maturing in one year. A European put option on the firm's assets with exercise price F and the same maturity is currently priced at $19.37 million. The current price of a zero-coupon risk-free bond paying $1 in one year is $0.86. Using the Merton model, what is the market value of the firm's debt?

A. $197.17 million

B. $158.43 million

C. $187.37 million

D. $177.80 million

Answer: B

Step by step

1. In the Merton model, risky debt = risk-free debt minus a put on the firm's assets struck at F.
2. Risk-free value of the debt: $0.86 \times 206.74 = 177.80$ million.
3. Subtract the put: $177.80 - 19.37 = 158.43$ million.

Why B is correct: Bondholders own a risk-free bond but have written a put to shareholders (who can hand over the assets instead of paying F), so $D = P(0,T)F -$ put.

Why the others are wrong

- **A:** Adds the put to the risk-free bond value (177.80 + 19.37); the put is a liability of the bondholders, not an asset.
- **C:** Subtracts the put from the undiscounted face value (206.74 - 19.37), forgetting to discount F at the risk-free bond price.
- **D:** Ignores the put entirely and reports the risk-free value of the debt, overstating the value because default risk is not deducted.

Q366

AnalystPrep Q.4356

REBUILT — Original asked a garbled loss-given-default calculation mixing real-world expected return with the 5% rate; rebuilt as a risk-neutral Merton market-value-of-debt question with N(d1), N(d2) supplied.

A firm has asset value $V_0 = \$100$ million. Its only debt is a 3-year zero-coupon bond with face value \$80 million. The risk-free rate is 5% (continuously compounded) and asset volatility is 30%. Using the Merton model, with $N(d_1) = 0.8359$ and $N(d_2) = 0.6767$, what is the market value of the firm's debt?

A. \$68.9 million
B. \$63.0 million
C. \$37.0 million
D. \$46.6 million

Answer: B

Step by step

1. $80e^{-0.05\times3} = 80 \times 0.86071 = 68.86$ million.
2. Equity as call: $E_0 = 100(0.8359) - 68.86(0.6767) = 83.59 - 46.60 = 36.99$ million.
3. Debt $= V_0 - E_0 = 100 - 36.99 = 63.0$ million.

Why B is correct: Equity is a call on the assets, and with no other claims debt value is $V_0 - E_0$, which is below the risk-free value of 68.9 million.

Why the others are wrong

- **A:** Reports the risk-free present value of the debt, ignoring default risk (the upper bound for debt value).
- **C:** Reports the equity (call) value instead of subtracting it from firm value.
- **D:** Computes only $Fe^{-rT}N(d_2)$ and omits the $V_0N(-d_1)$ term of debt value.

Q367

Mock 3 Q4 · GARP PE1 Q77

REBUILT — Merged two identical sources; options had a 48.996 vs 48.997 rounding clash and unmotivated distractors, so options were rebuilt from realistic errors.

A fixed-income analyst applies the Merton model to a non-dividend-paying company whose assets have a current market value of CAD 85 million. The company's only debt is a zero-coupon bond with face value CAD 55 million maturing in 2 years. Additional information: $N(d_1) = 0.9495$; $N(d_2) = 0.8983$; annual risk-free rate 5% (continuously compounded); annual asset volatility 26%. What is the Merton-model market value of the company's debt?

A. CAD 48.997 million
B. CAD 49.766 million
C. CAD 53.699 million
D. CAD 36.003 million

Answer: A

Key correction: GARP PE1 Q77's printed key said B, but its own worked solution (and the arithmetic here) give A.

Step by step

1. $55e^{-0.05\times2} = 55 \times 0.904837 = 49.766$ million.
2. Equity as a call: $E_0 = 85(0.9495) - 49.766(0.8983) = 80.7075 - 44.704 = 36.003$ million.
3. Debt $= V_0 - E_0 = 85 - 36.003 = 48.997$ million.

Why A is correct: Equity is a call on the assets with strike equal to the debt face value; debt is the residual $V_0 - E_0$,

which lies below the risk-free value of 49.766.

Why the others are wrong

- **B:** Reports the risk-free PV of the debt ($55e^{-0.10}$); this is the upper bound and ignores default risk.
- **C:** Fails to discount the strike: $E = 80.7075 - 55(0.8983) = 31.30$, so debt $= 85 - 31.30 = 53.70$.
- **D:** Reports the equity value (the call price) rather than debt = V0 - E0.

Q368

AP Mock 1 Q3

Since it was founded ten years ago, Bright Technologies has paid no dividends and has been financed with 100% equity. Management recently issued a zero-coupon bond with a principal amount of USD 100 million maturing in exactly three years. If the value of the firm at maturity is USD 80 million, determine the values of the components of the firm's capital structure at the bond's maturity date.

A. Value of equity = USD 0; value of debt = USD 80 million.
B. Value of equity = USD 20 million; value of debt = USD 80 million.
C. Value of equity = USD 180 million; value of debt = USD 100 million.
D. Value of equity = USD 20 million; value of debt = USD 0.

Answer: A

Step by step

1. Equity payoff $E_T = \max(V_T - D, 0) = \max(80 - 100, 0) = 0$.
2. Debt payoff $= \min(V_T, D) = \min(80, 100) = 80$ million.
3. Check: $E_T + D_T = 0 + 80 = V_T$.

Why A is correct: Shareholders hold a call on the assets struck at the debt face value; with $V_T < D$ the call expires worthless and bondholders take all assets.

Why the others are wrong

- **B:** Gives equity 20 (treating the 100 - 80 shortfall as equity); components then sum to 100, not the firm value of 80.
- **C:** Adds the face value to the firm value to get equity (80 + 100) and pays debt in full, ignoring that assets are insufficient.
- **D:** Reverses the claims: gives the bondholders nothing and equity 20, ignoring that debt has priority and equity is limited to max(V - D, 0).

Q369

AP Mock 2 Q41

A firm has a current value of USD 200 million. Its only outstanding debt is a three-year zero-coupon bond with a face value of USD 180 million. The annual risk-free interest rate is 5% and the volatility of the firm's value is 10%. Using the Merton model, what is the value of equity?

A. USD 42 million.
B. USD 90 million.
C. USD 46 million.
D. USD 20 million.

Answer: C

Step by step

1. Equity is a call on firm value with strike $D = 180$ and $T = 3$.
2. $d_1 = [\ln(200/180) + (0.05 + 0.5 \times 0.10^2) \times 3]/(0.10\sqrt{3}) = 1.561$
3. $d_2 = d_1 - 0.10\sqrt{3} = 1.388$; $N(d_1) = 0.9408$, $N(d_2) = 0.9174$
4. $E_0 = 200\,N(d_1) - 180e^{-0.05\times3}N(d_2) = 46.02$ million.

Why C is correct: Equity is a European call on firm assets struck at the debt face value, so Black-Scholes-Merton gives about USD 46 million.

Why the others are wrong

- **A:** Below the call's lower bound $V_0 - De^{-rT} = 200 - 154.9 = 45.1$, so it cannot be a valid equity value; reflects a mis-specified d1/d2 or N() lookup.
- **B:** Not a model output; roughly half of firm value with no basis in the call-option valuation.
- **D:** Undiscounted intrinsic value 200 - 180; ignores discounting of the debt and the option's time value.

Q370 AP Mock 3 Q63 · AnalystPrep Q.2677

For a firm financed partly by debt and partly by equity, in the Merton model the value of the debt:

A. increases if the volatility of the firm increases.
B. increases if the face amount of debt falls.
C. decreases if its time to maturity increases.
D. increases if the interest rate increases.

Answer: C

Step by step

1. Debt value = firm value minus equity: $D_0 = V_0 - E_0$, where E_0 is a call on V struck at the face value.
2. The call value rises with maturity (no dividends), so debt value falls when maturity lengthens.
3. Checking the others: higher volatility raises the call, lowering debt; higher rate raises the call, lowering debt; a lower face value reduces the promised payment and so the debt claim.

Why C is correct: Since $D_0 = V_0 - E_0$ and the equity call increases with maturity, debt value decreases as maturity increases.

Why the others are wrong

- **A:** Higher volatility raises the equity call value and so lowers debt value (equity holders gain from risk).
- **B:** A smaller face amount is a smaller promised payment, so the debt claim is worth less (the call strike falls and the call rises); sign reversed.
- **D:** A higher rate raises the call value and lowers the present value of the promised payment, so debt value falls.

Q371 AnalystPrep Q.1796

A firm has a principal amount due on its zero-coupon bond of \$60m and the firm has no other creditors. This amount is due at time T. Suppose the value of the firm at time T is \$50m. What is the value of equity at time T?

A. \$0 million
B. -\$10 million
C. \$10 million
D. \$110 million

Answer: A

Step by step

1. $E_T = \max(V_T - D, 0)$
2. $= \max(50 - 60, 0) = 0$

Why A is correct: Equity holders have limited liability, so equity cannot be negative; with firm value below the debt, all value goes to debt holders.

Why the others are wrong

- **B:** Ignores limited liability: reports $V_T - D = -10$ without the floor at zero.
- **C:** Takes the absolute shortfall $|50 - 60|$.
- **D:** Adds the firm value and the debt $50 + 60$.

Q372 AnalystPrep Q.1797

REBUILT — Stem called the equity payoff a 'call option payoff' in a muddled way; reworded to ask for the equity value at maturity.

A firm has \$120 million of debt (zero-coupon, face value) due at time T and no other creditors. In the Merton model, equity holders receive something only if firm value exceeds the face value of debt. What is the value of equity at maturity if the firm is worth \$160 million at time T?

A. \$40 million
B. \$0 million
C. -\$40 million
D. \$160 million

Answer: A

Step by step

1. $E_T = \max(V_T - D, 0)$
2. $= \max(160 - 120, 0) = 40$ million.

Why A is correct: Equity is a call on firm value struck at the debt face value; it is worth $V_T - D$ when V_T exceeds D.

Why the others are wrong
- **B:** Applies the default case ($V_T < D$) although the firm is solvent.
- **C:** Reverses the sign: $D - V_T$.
- **D:** Gives all firm value to equity, ignoring the debt holders' claim.

Q373 AnalystPrep Q.2667

Given that F represents the face value of the debt and V_T is the value of the firm at maturity, which formula correctly describes the payment to the debt holders at maturity in the Merton model?

A. $\max(V_T - F, 0)$
B. $V_T - \max(F - V_T, 0)$
C. $F - \max(V_T - F, 0)$
D. $F - \max(F - V_T, 0)$

Answer: D
Step by step
1. If $V_T \geq F$: debt holders receive F; $\max(F - V_T, 0) = 0$, so the formula gives F.
2. If $V_T < F$: they receive V_T; the formula gives $F - (F - V_T) = V_T$.
3. So payoff = $(\min(V_T, F)$ = F - \max(F - V_T, 0) \): risk-free debt minus a short put.

Why D is correct: Debt holders hold a risk-free claim F and are short a put on firm assets, which equals $\min(V_T, F)$.
Why the others are wrong
- **A:** This is the equity payoff (a call).
- **B:** Gives $2V_T - F$ when V_T < F and V_T when V_T > F; wrong in both states.
- **C:** Subtracts the call instead of the put; gives $2F - V_T$ when the firm is solvent.

Q374 AnalystPrep Q.2670

Use the simple Merton model. I. The principal amount due on the firm's debt is $25 million and the value of the firm at maturity is $35 million. Find the value of the firm's equity. II. The principal due is $25 million and the value of the firm at maturity is $20 million. Find the value of the firm's debt. (Answer as equity in I, debt in II.)

A. $10 million and 0
B. 0 and $10 million
C. $10 million and $20 million
D. $20 million and $10 million

Answer: C
Step by step
1. I: $E = \max(35 - 25, 0) = 10$.
2. II: debt $= \min(V, F) = 20$ (equivalently $25 - \max(25 - 20, 0)$).

Why C is correct: Equity gets the excess over face value; when firm value is below face value, debt holders get all of the firm value.
Why the others are wrong
- **A:** Gives debt 0 in II, forgetting that debt holders claim all remaining assets.
- **B:** Equity set to 0 although V exceeds the debt, and debt reported as 10; both parts are wrong.
- **D:** Swaps the two answers (20 for equity, 10 for debt).

Q375 AnalystPrep Q.3058

A risk analyst uses the Merton model and estimates the value of the firm to be $70 million when its debt matures. The face value of the firm's debt is $95 million. The payoffs to debt holders and equity holders at maturity are respectively:

A. $70 million to debt holders and $0 to equity holders
B. $0 to debt holders and $70 million to equity holders
C. $95 million to debt holders and $0 to equity holders
D. $70 million to debt holders and $25 million to equity holders

Answer: A

Step by step

1. Debt payoff $= \min(V_T, F) = \min(70, 95) = 70$.
2. Equity payoff $= \max(V_T - F, 0) = 0$.

Why A is correct: Firm value is below the face value, so debt holders take all assets and equity is worthless.

Why the others are wrong

- **B:** Gives the assets to equity holders, reversing the priority of claims.
- **C:** Debt cannot be paid in full as $V_T < F$.
- **D:** Pays equity the shortfall in absolute value; the payoffs 70 + 25 would also exceed firm value.

Q376

AnalystPrep Q.3061

REBUILT — Exact answer is 2.30%, source option 2.29% was a rounding artefact; options rebuilt so each wrong one reflects a realistic error.

A $2 million (face value) zero-coupon corporate bond has a remaining maturity of 7 years. Using the Merton model, the current value of the bond is $1.2 million. If the risk-free rate is 5% (continuously compounded), the credit spread is closest to:

A. 2.30%
B. 7.30%
C. 4.52%
D. 5.71%

Answer: A

Step by step

1. Yield $y = -\frac{1}{T}\ln(D_0/F) = -\frac{1}{7}\ln(1.2/2) = 7.30\%$.
2. Spread $= y - r = 7.30\% - 5\% = 2.30\%$.

Why A is correct: Credit spread is the continuously compounded yield implied by the Merton debt value less the risk-free rate.

Why the others are wrong

- **B:** 7.30% is the bond yield; the risk-free rate was not subtracted.
- **C:** Uses a simple annual rate (2/1.2 - 1)/7 = 9.52% minus 5%.
- **D:** Uses the loss fraction (2 - 1.2)/2/7 = 5.71% as spread, ignoring compounding and the risk-free rate.

Q377

AnalystPrep Q.3194

REBUILT — Source computed an expected-loss figure mislabelled 'LGD' with an internally inconsistent PD (11.025% vs implied 4.7%) and OCR-garbled formula; rebuilt as a Merton debt-value question on the same firm data.

A firm has a value of $16 million. It has issued a 7-year zero-coupon bond with face value $18 million. The risk-free rate is 7% (continuously compounded) and the firm's asset volatility is 17%. Using the Merton model, what is the current value of the firm's debt?

A. $5.65 million
B. $10.35 million
C. $11.03 million
D. $11.75 million

Answer: B

Step by step

1. $d_1 = [\ln(16/18) + (0.07 + 0.5 \times 0.17^2) \times 7]/(0.17\sqrt{7}) = 1.0524$; $d_2 = d_1 - 0.4498 = 0.6027$.
2. $N(d_1) = 0.8537$, $N(d_2) = 0.7266$.
3. $E_0 = 16 \times 0.8537 - 18e^{-0.49} \times 0.7266 = 5.65$.
4. $D_0 = V_0 - E_0 = 16 - 5.65 = 10.35$ million.

Why B is correct: Debt equals firm value minus the Merton equity call.

Why the others are wrong

- **A:** This is the equity value (5.65), not the debt value.
- **C:** Risk-free present value of the face value, $18e^{-0.49} = 11.03$; ignores default risk.
- **D:** Subtracts a miscomputed equity (using N(d1) on both terms, 4.25) from V.

Q378

AnalystPrep Q.4355

REBUILT — Distractors were implausible (0.5%, 0.025%); replaced with options from realistic mistakes.

A firm has a zero-coupon bond maturing in 5 years. The face value of the debt is $100 million and its current value is $88 million. Assuming a continuously compounded risk-free rate of 1.5%, what is the credit spread on the bond?

A. 1.057%
B. 2.557%
C. 0.900%
D. 4.057%

Answer: A

Step by step

1. Yield on the risky debt: $y = -\frac{1}{T}\ln\frac{D}{F} = -\frac{1}{5}\ln(0.88) = 0.025567$ (2.557%).
2. Credit spread = yield minus risk-free rate = 2.557% - 1.500% = 1.057%.

Why A is correct: The credit spread is the continuously compounded yield implied by the Merton debt value, $-\frac{1}{T}\ln(D/F)$, less the risk-free rate.

Why the others are wrong

- **B:** Forgot to subtract the risk-free rate; this is the full yield, not the spread.
- **C:** Used a simple-return approximation $0.12/5 = 2.4\%$ minus 1.5% = 0.90%, instead of the continuously compounded yield.
- **D:** Added the risk-free rate to the yield (2.557% + 1.5%) instead of subtracting it.

Q379

AnalystPrep Q.4357

REBUILT — Distractors replaced with options from realistic mistakes.

A corporate bond with face value $1,000 and 10 years to maturity is valued at $780 using the Merton model. The continuously compounded risk-free rate is 1%. What is the credit spread on the bond?

A. 2.485%
B. 1.485%
C. 1.200%
D. 3.485%

Answer: B

Step by step

1. Yield: $y = -\frac{1}{10}\ln(780/1000) = 0.024846$ (2.485%).
2. Credit spread = 2.485% - 1.000% = 1.485%.

Why B is correct: Spread = continuously compounded debt yield minus the risk-free rate.

Why the others are wrong

- **A:** Forgot to subtract the risk-free rate.
- **C:** Used a simple-return approximation: $0.22/10 = 2.2\%$ minus 1% = 1.2%, instead of the log (continuous) yield.
- **D:** Added the risk-free rate to the yield instead of subtracting it.

Q380

AnalystPrep Q.5442

REBUILT — Distractors replaced with options from realistic mistakes.

RedRose Group has zero-coupon bonds with face value $45,000,000 and current market value $40,000,000, maturing in 3 years. Comparable risk-free bonds yield 2.5% per annum, continuously compounded. The average credit spread on RedRose's bonds is closest to:

A. 1.43%
B. 3.93%
C. 1.67%
D. 6.43%

Answer: A

Step by step

1. Yield: $y = -\frac{1}{3}\ln(40/45) = 0.039261$ (3.926%).

2. Spread = 3.926% - 2.500% = 1.426%, about 1.43%.

Why A is correct: The spread is the Merton-implied debt yield, $-\frac{1}{T}\ln(D/F)$, less the risk-free yield.

Why the others are wrong

- **B:** Forgot to subtract the risk-free rate (reported the yield).
- **C:** Used simple return on market value: $(5/40)/3 = 4.17\%$ minus 2.5% = 1.67%, instead of the log yield.
- **D:** Added the risk-free rate to the yield instead of subtracting it.

Q381

AnalystPrep Q.5429

REBUILT — Options replaced so each wrong answer reflects a realistic slip.

In a Merton model for a non-dividend-paying firm with debt maturing in 10 years, an analyst computes $d_1 = 2.321$ and $d_2 = 2.143$. What is the implied volatility of firm (asset) value, assuming it is constant?

A. 5.63%
B. 17.8%
C. 1.78%
D. 56.3%

Answer: A

Step by step

1. $d_2 = d_1 - \sigma_V\sqrt{T}$, so $\sigma_V\sqrt{T} = 2.321 - 2.143 = 0.178$.
2. $\sigma_V = 0.178/\sqrt{10} = 0.178/3.1623 = 0.0563$, i.e. 5.63%.

Why A is correct: The gap between d1 and d2 equals asset volatility times the square root of time to maturity.

Why the others are wrong

- **B:** Took the difference d1 - d2 as the volatility, forgetting to divide by the square root of T.
- **C:** Divided by T = 10 instead of by the square root of T.
- **D:** Multiplied by $\sqrt{10}$ instead of dividing: 0.178 x 3.162 = 56.3%.

Q382

AnalystPrep Q.5553

REBUILT — Original question was vague and its key arguable; rewritten to test the Merton equity/debt structure unambiguously.

Which of the following statements about the Merton model is correct?

A. Equity is a call option on the firm's assets with strike price equal to the face value of the debt, and debt value equals asset value minus equity value.
B. Equity is a put option on the firm's assets with strike price equal to the face value of the debt.
C. Debt value equals asset value plus equity value.
D. Equity value is unaffected by the volatility of firm assets because the debt is fixed.

Answer: A

Step by step

1. At T, equity pays $\max(V_T - F, 0)$: a call on assets with strike F (the debt face value).
2. Assets = debt + equity, so debt value today = V0 - E0 (equivalently the PV of riskless debt minus a put on assets).
3. Higher asset volatility raises the call value, so option D is false.

Why A is correct: Shareholders hold a call on the firm's assets struck at the debt face value; creditors own the remainder of assets.

Why the others are wrong

- **B:** Confuses equity with the put; the put is the creditors' short position (default option), not the equity payoff.
- **C:** Sign error: assets are the sum of debt and equity, so debt = assets minus equity.
- **D:** Equity is a call, and call value rises with volatility of the underlying assets.

Q383

BT answers P2.T6.403.1

REBUILT — Options had a giveaway distractor (41.30); replaced with realistic mistakes.

A firm has asset value $V = 110$ million with asset volatility 30% per annum. Its only debt is a zero-coupon bond with face value $F = 80$ million maturing in 5 years. The risk-free rate is 4% (continuously compounded). The BSM value of a put on the firm's assets with strike F is $6.95 million. What is the current value of the firm's debt?

A. $6.95 million
B. $65.50 million

C. $58.55 million
D. $72.45 million

Answer: C
Step by step
1. PV of riskless debt = $80e^{-0.04\times5} = 80 \times 0.81873 = 65.50$.
2. Risky debt = riskless debt minus the put (creditors are short the default put): 65.50 - 6.95 = 58.55.

Why C is correct: Risky debt equals riskless debt less a put on firm assets struck at the face value.
Why the others are wrong
- **A:** Reported the put value itself rather than the debt value.
- **B:** Reported riskless debt, forgetting to subtract the put.
- **D:** Added the put instead of subtracting it.

Q384

BT answers P2.T6.403.2

REBUILT — Stem was ambiguous ('implied ceteris paribus ... EXCEPT') and the T->0 result requires V0 > F; reworded as a plain 'which is FALSE' item and stated V0 > F.

Under the Merton model the firm's only debt is a single long-term zero-coupon bond, and the current value of the firm's assets exceeds the bond's face value. Holding all other parameters constant, which of the following statements is FALSE?

A. A higher asset volatility increases the value of the firm's equity.
B. A higher asset volatility decreases the value of the firm's debt.
C. As the debt's time to maturity $T \to 0$, the value of the firm's equity decreases.
D. As the debt's time to maturity $T \to 0$, the value of the firm's debt decreases.

Answer: D
Step by step
1. Equity is a European call on firm assets with strike equal to the debt face value; debt $= V_0 - E_0$ (riskless bond minus a put on assets).
2. Higher σ_V raises option value, so equity rises and, with V_0 fixed, debt falls: A and B are true.
3. Numerical check ($V_0 = 120, F = 100, r = 5\%, \sigma_V = 30\%$): the call is 51.98 at $T = 5$ and 28.88 at $T = 1$, so equity falls as T shrinks: C is true.
4. Debt $= V_0 - E_0$ is 120 - 51.98 = 68.02 at $T = 5$ and 120 - 28.88 = 91.12 at $T = 1$: debt rises toward par as $T \to 0$, so D is false.

Why D is correct: Shorter maturity erodes the time value of the equity call, so the residual debt claim $V_0 - E_0$ gains value and is pulled up toward par, not down.
Why the others are wrong
- **A:** True statement: the equity call has positive vega, so higher asset volatility raises equity value.
- **B:** True statement: debt = assets minus equity, so when equity rises with volatility, debt falls (the embedded short put is worth more).
- **C:** True statement: with less time left the call loses time value, so equity falls toward V0 - F.
- **D:** This is the false statement and so the answer: debt value increases as T shrinks because the embedded short put loses time value.

Q385

BT answers P2.T6.404.1

REBUILT — OCR-damaged V ($700m) restored to $100m and giveaway distractors replaced with values from realistic mistakes.

A firm has two classes of debt (senior and subordinated), both five-year zero-coupon bonds. Assume the Merton framework with:
- Firm value V = $100.0 million, asset volatility 20.0% per annum
- Face value of senior debt F = $60.0 million
- Face value of subordinated debt U = $40.0 million
- Riskless rate = 3.0%
- Call value c(V, F, T) = c(100, 60, 5) = $49.33
- Call value c(V, F+U, T) = c(100, 100, 5) = $24.33

Which is NEAREST to the value of the subordinated debt?

A. $10.67
B. $25.00

C. $50.67
D. $75.67

Answer: B
Step by step
1. Equity is a call on assets struck at total debt: $E = c(V, F+U, T) = 24.33$.
2. Senior debt $= V - c(V, F, T) = 100 - 49.33 = 50.67$.
3. Subordinated debt $= V - E -$ senior $= 100 - 24.33 - 50.67 = 25.00$, i.e. $c(V, F, T) - c(V, F+U, T) = 49.33 - 24.33$.

Why B is correct: Subordinated holders own the slice of asset value between F and F+U, which is the difference of the two call values.
Why the others are wrong
- **A:** $10.67 = F - c(V,F,T) = 60 - 49.33$; it subtracts the call from the senior face value, which has no valuation meaning.
- **C:** $50.67 is the value of the senior debt (V - c(V,F,T)), not the subordinated debt.
- **D:** $75.67 = V - equity, the value of ALL debt (senior plus subordinated).

Q386

BT answers P2.T6.404.3

REBUILT — Heavy OCR corruption reconstructed; the post-change call values needed a Black-Scholes calculation and were not given, so they are now supplied.
A high-value firm has two classes of five-year zero-coupon debt. Under the Merton framework:
- Firm value V = $200.0 million, asset volatility 20.0% per annum
- Face value of senior debt F = $100.0 million
- Face value of subordinated debt U = $50.0 million
- Riskless rate initially 10.0%: c(200, 100, 5) = $139.40 and c(200, 150, 5) = $109.95

If the riskless rate falls to 2.0% (V, volatility and maturities unchanged), the call values become c(200, 100, 5) = $110.42 and c(200, 150, 5) = $72.02. Which of the following is the impact of the fall in the riskless rate?

A. The value of equity increases.
B. The value of the senior debt decreases.
C. The value of the subordinated debt increases.
D. None of the above.

Answer: C
Step by step
1. Equity $= c(V, F+U, T)$: from 109.95 to 72.02, a decrease, so A is wrong.
2. Senior debt $= V - c(V, F, T)$: from 200 - 139.40 = 60.60 to 200 - 110.42 = 89.58, an increase, so B is wrong.
3. Subordinated debt $= c(V, F, T) - c(V, F+U, T)$: from 139.40 - 109.95 = 29.45 to 110.42 - 72.02 = 38.40, an increase, so C is correct.
4. Check: 72.02 + 89.58 + 38.40 = 200.00 = V.

Why C is correct: A lower riskless rate lowers the value of the equity call, so with V fixed both debt classes gain value.
Why the others are wrong
- **A:** Reverses the sign: equity (a call) falls as r falls, 109.95 to 72.02.
- **B:** Reverses the sign: senior debt rises from 60.60 to 89.58; the error is thinking lower rates lower debt value while forgetting V is fixed and equity falls.
- **D:** Wrong because C is true: subordinated debt rises from 29.45 to 38.40.

Q387

BT notes P2.T6.24.15.3

REBUILT — Source answer 'about 20%' was loosely rounded (exact 19.74%) and its option C (19%) was nearly as close; options rebuilt around realistic errors.
RegalTech has total asset value $V_0 = \$240$ million and equity value $E_0 = \$150$ million with equity volatility $\sigma_E = 30\%$. In the Merton model, $N(d_1) = 0.95$ and $N(d_2) = 0.9968$. Using Ito's lemma, the asset volatility σ_V is closest to:

A. 19.7%
B. 30.0%
C. 18.8%
D. 45.6%

Answer: A

Step by step

1. Ito's lemma on $E = f(V)$ gives $\sigma_E E_0 = N(d_1)\,\sigma_V V_0$.
2. $\sigma_V = \dfrac{\sigma_E E_0}{N(d_1)V_0} = \dfrac{0.30 \times 150}{0.95 \times 240} = \dfrac{45}{228} = 19.74\%$.

Why A is correct: Equity's delta with respect to assets is $N(d_1)$, so equity volatility equals asset volatility scaled by $N(d_1)V_0/E_0$.

Why the others are wrong

- **B:** 30% is simply the equity volatility, ignoring the leverage and delta adjustment.
- **C:** 18.8% uses $N(d_2) = 0.9968$ in place of $N(d_1)$: 45 / (0.9968 x 240).
- **D:** 45.6% inverts the relationship: $0.30 \times 0.95 \times 240/150$.

Q388

BT notes P2.T6.303.1

REBUILT — Required external N(.) look-ups; supplied d1, d2 and normal CDF values and replaced distractors with realistic mistake values.

A firm's assets are worth \$120.0 million with asset volatility of 44.0% per annum and an expected asset return of 15.0%. Its only debt is a one-year zero-coupon bond with face value \$100.0 million. The risk-free rate is 3.0% per annum (continuously compounded). Use the Merton model with:

$d_1 = 0.7025,\ d_2 = 0.2625$

$N(d_1) = 0.7588,\ N(d_2) = 0.6036,\ N(-d_1) = 0.2412,\ N(-d_2) = 0.3964$

Which is NEAREST to the value of the firm's debt?

A. \$80.34
B. \$87.51
C. \$89.30
D. \$97.04

Answer: B

Step by step

1. Riskless PV of the debt $= 100e^{-0.03} = 97.04$.
2. Equity (call) $= 120(0.7588) - 97.04(0.6036) = 91.06 - 58.57 = 32.49$.
3. Debt $= V_0 - E_0 = 120 - 32.49 = 87.51$.
4. Check via put: $97.04 - [97.04(0.3964) - 120(0.2412)] = 97.04 - 9.53 = 87.51$. The 15% expected return is not used.

Why B is correct: Risky debt equals firm value minus equity (or the riskless bond minus a put on assets); pricing is risk-neutral, so the expected return is irrelevant.

Why the others are wrong

- **A:** \$80.34 uses the 15% expected return in d1, d2 and the discounting instead of the risk-free rate.
- **C:** \$89.30 fails to discount the face value in the equity call (uses 100 instead of 97.04), overstating the call's strike and understating equity.
- **D:** \$97.04 is the riskless value of the bond, ignoring the default put.

Q389

GARP PE1 Q22 · Mock 3 Q58

REBUILT — Merged two near-identical items (d2 printed 3.038905 vs 3.038805); options rebuilt so each wrong value comes from a specific scaling mistake.

A portfolio manager applies the Merton model to a non-dividend-paying firm whose shares are held in the fund's portfolio. Preliminary analysis gives:

Value of equity: USD 45 million

Value of the firm's only debt, maturing in 5 years: USD 60 million

d1 = 3.217790

d2 = 3.038905

Assuming a constant volatility of firm value, what is the estimate of that volatility?

A. 3.6%
B. 8.0%
C. 17.9%
D. 40.0%

Answer: B

Step by step

1. In the Merton model $d_2 = d_1 - \sigma_V\sqrt{T}$, so $\sigma_V\sqrt{T} = 3.217790 - 3.038905 = 0.178885$.
2. $\sigma_V = 0.178885/\sqrt{5} = 0.178885/2.2361 = 0.0800 = 8.0\%$.
3. The equity and debt values are not needed.

Why B is correct: The gap between d_1 and d_2 is exactly $\sigma_V\sqrt{T}$, so volatility follows once T is known.

Why the others are wrong

- **A:** 3.6% divides by $T = 5$ instead of $\sqrt{T}$.
- **C:** 17.9% is $d_1 - d_2$ taken as the volatility, forgetting to divide by $\sqrt{T}$.
- **D:** 40.0% multiplies by $\sqrt{T}$ instead of dividing.

Q390

Mock 1 Q7

REBUILT — Given N(d1)=0.9931 and N(d2)=0.9778 were inconsistent with the inputs (true values 0.9101 and 0.8145), so values and options were rebuilt.

A credit analyst applies the Merton model to a firm whose only debt is a five-year zero-coupon bond.

Parameter	Value
Current value of firm assets (V0)	USD 100 million
Principal due in five years (F)	USD 100 million
Risk-free rate (continuously compounded)	10% per annum
Asset volatility	20% per annum
N(d1)	0.9101
N(d2)	0.8145

Which of the following gives the current value of the firm's equity and the risk-neutral probability of default?

A. Equity of USD 41.61 million; default probability of 18.55%.
B. Equity of USD 41.61 million; default probability of 8.99%.
C. Equity of USD 58.39 million; default probability of 18.55%.
D. Equity of USD 58.39 million; default probability of 8.99%.

Answer: A

Step by step

1. $Fe^{-rT} = 100e^{-0.5} = 60.6531$.
2. Equity $= V_0N(d_1) - Fe^{-rT}N(d_2) = 100(0.9101) - 60.6531(0.8145) = 91.01 - 49.40 = 41.61$.
3. Risk-neutral default probability $= N(-d_2) = 1 - 0.8145 = 18.55\%$.

Why A is correct: Equity is a call on assets; default occurs when the call is not exercised, which has risk-neutral probability $N(-d_2)$.

Why the others are wrong

- **B:** Correct equity, but uses $1 - N(d_1) = 8.99\%$; $N(d_1)$ is the delta, not the exercise probability.
- **C:** Reports the value of the debt (100 - 41.61 = 58.39) as equity; probability is correct.
- **D:** Both errors: debt value reported as equity and 1 - N(d1) as the default probability.

Q391

AnalystPrep Q.2877

REBUILT — Replaced unexplainable options ($76M/$19M, $51M/$10M) with distractors reflecting realistic errors.

Lenny and Mo Inc. has issued debt requiring a payment of $360 million to debt holders at maturity and has no other creditors. The firm's total value at maturity is $400 million in scenario 1 and $350 million in scenario 2. What do equity holders receive at maturity in each scenario?

A. $40M in scenario 1 and $10M in scenario 2
B. $360M in scenario 1 and $350M in scenario 2
C. $40M in scenario 1 and $0M in scenario 2
D. $400M in scenario 1 and $350M in scenario 2

Answer: C

Step by step

1. Equity payoff is a call on firm value struck at the debt: $E_T = \max(V_T - D, 0)$.
2. Scenario 1: $400 - 360 = 40$.
3. Scenario 2: $\max(350 - 360, 0) = 0$; debt holders receive all 350 and the firm defaults.

Why C is correct: Equity holders have limited liability and are the residual claimants, so their payoff cannot be negative.

Why the others are wrong

- **A:** Takes the absolute difference (360 - 350 = 10) in scenario 2, ignoring limited liability.
- **B:** Shows the debt holders' claim (360, or all of 350) as equity's payoff.
- **D:** Reports total firm value, ignoring the debt claim ahead of equity.

Q392

AnalystPrep Q.1798

REBUILT — Original stem was garbled ('debt is a decreasing function of equity') and its key (lower principal) is not a clean Merton comparative static; rewritten as a clear comparative-statics question.

In the Merton model the firm's only debt is a zero-coupon bond and the firm's asset value exceeds the bond's face value. Holding other inputs constant, which of the following changes would DECREASE the value of the firm's debt?

A. An increase in the volatility of the firm's assets
B. A decrease in the risk-free interest rate
C. A shorter time to maturity of the debt
D. An increase in the current value of the firm's assets

Answer: A

Step by step

1. Debt $= V_0 - E_0$, where equity is a call on assets; equivalently riskless bond minus a put on assets.
2. Higher σ_V raises the call, so equity rises and debt falls (check with $V_0 = 120, F = 100, r = 5\%, T = 5$: debt 68.02 at $\sigma = 30\%$ vs 61.19 at 40%).
3. Lower r lowers the call, so debt rises (72.87 at $r = 3\%$ vs 68.02 at 5%).
4. Shorter maturity: the call loses time value and debt rises (91.12 at $T = 1$ vs 68.02 at $T = 5$).
5. Higher V_0 (130 vs 120): debt rises from 68.02 to 69.51 because the put is further out of the money.

Why A is correct: Debt holders are short a put on the assets, so anything that raises option value (higher volatility) transfers value from debt to equity.

Why the others are wrong

- **B:** Debt rises when rates fall: the equity call loses value, leaving more of the fixed asset value for debt.
- **C:** Debt rises as maturity shortens because the short put loses time value and debt is pulled toward par.
- **D:** More assets make the debt safer, so debt value rises (put delta falls in magnitude).

LO 27.m

Calculate distance to default and default probability using the Merton model.

Q393

BT answers P2.T6.405.1

REBUILT — Distractors (2.7, 3.3, 8.5) did not correspond to realistic errors; replaced them with results of common formula mistakes and stated that the expected return is used.

A firm has asset value of $400.0 million, expected asset return of 14.0% per annum and asset volatility of 36.0% per annum. Its only debt is a one-year zero-coupon bond with face value of $300.0 million. The risk-free rate is 4.0%. Using the expected return on assets (real-world drift), which is nearest to the firm's distance to default?

A. 0.73
B. 0.80
C. 1.01
D. 1.37

Answer: C

Step by step

1. $\ln(400/300) = 0.2877$.
2. Drift term: $(0.14 - 0.36^2/2) \times 1 = 0.14 - 0.0648 = 0.0752$.
3. Numerator $= 0.2877 + 0.0752 = 0.3629$; denominator $= 0.36\sqrt{1} = 0.36$.
4. $DD = 0.3629/0.36 = 1.008 \approx 1.01$.

Why C is correct: Distance to default is d2 with the asset drift: the number of asset-volatility standard deviations by which expected log asset value exceeds the log of debt at T.

Why the others are wrong

- **A:** 0.73 uses the risk-free rate (4%) instead of the 14% expected return, which gives the risk-neutral d2.
- **B:** 0.80 is $\ln(V/D)/\sigma$ only, ignoring the drift term entirely.

- **D:** 1.37 adds $\sigma^2/2$ instead of subtracting it in the drift term.

Q394

AP Mock 2 Q54

REBUILT — Stem did not specify the approximation formula, so market value of debt vs. face value and drift were ambiguous; stated the simplified formula explicitly.

A credit manager monitors counterparties using the Merton model. Use the simplified one-year distance to default $DD \approx \ln(V_0/F)/\sigma_V$, where F is the face value of debt (drift ignored).

Item	Bank A	Bank B	Bank C	Bank D
Market value of assets	USD 12m	USD 8m	USD 45m	USD 32m
Face value of debt	USD 3m	USD 2m	USD 14m	USD 9m
Market value of debt	USD 2.5m	USD 1.5m	USD 14m	USD 8m
Annual volatility of assets	8%	7%	9%	10%

Each counterparty's only liability is a one-year zero-coupon bond. Which counterparty is most likely to default?

A. Bank A
B. Bank B
C. Bank C
D. Bank D

Answer: D

Step by step

1. Use face value F (the amount owed at T), not market value of debt.
2. A: $\ln(12/3)/0.08 = 1.3863/0.08 = 17.33$.
3. B: $\ln(8/2)/0.07 = 1.3863/0.07 = 19.80$.
4. C: $\ln(45/14)/0.09 = 1.1676/0.09 = 12.97$.
5. D: $\ln(32/9)/0.10 = 1.2685/0.10 = 12.69$.
6. Lowest DD is Bank D, so it is the most likely to default.

Why D is correct: A lower distance to default means fewer standard deviations of cushion before assets fall below the debt, so higher default probability; Bank D has the smallest.

Why the others are wrong

- **A:** DD of 17.33 is large: assets are 4x debt and volatility is only 8%.
- **B:** Bank B has the highest DD (19.80), so it is the least likely to default; low volatility (7%) and 4x coverage dominate.
- **C:** Bank C's DD of 12.97 is close to, but above, D's 12.69; choosing it comes from comparing coverage ratios (3.2x vs 3.6x) without dividing by volatility, or from mis-rounding.

Q395

AP Mock 3 Q64

REBUILT — Stem did not specify the approximation formula; stated it explicitly.

A credit manager assesses default risk for four banks using the simplified one-year distance to default $DD \approx \ln(V_0/F)/\sigma_V$ (drift ignored).

Item	Bank A	Bank B	Bank C	Bank D
Annual volatility of asset value	9%	11%	10%	7%
Face value of debt	USD 80m	USD 120m	USD 90m	USD 60m
Market value of assets	USD 150m	USD 230m	USD 150m	USD 100m

Each bank's only liability is a one-year zero-coupon bond. Which bank is most likely to default?

A. Bank A
B. Bank B
C. Bank C
D. Bank D

Answer: C

Step by step

1. A: $\ln(150/80)/0.09 = 0.6286/0.09 = 6.98$.
2. B: $\ln(230/120)/0.11 = 0.6506/0.11 = 5.91$.
3. C: $\ln(150/90)/0.10 = 0.5108/0.10 = 5.11$.
4. D: $\ln(100/60)/0.07 = 0.5108/0.07 = 7.30$.
5. Lowest DD is Bank C.

Why C is correct: The bank with the smallest distance to default is closest to its default point; Bank C combines thin coverage (1.67x) with relatively high volatility.

Why the others are wrong

- **A:** DD of 6.98 is well above C's 5.11.
- **B:** Bank B has the highest volatility (11%), which tempts one to choose it, but its coverage (1.92x) gives DD 5.91, still above C's.
- **D:** Bank D has the same coverage as C (1.67x) but lower volatility (7%), so its DD is higher (7.30); choosing it ignores the volatility in the denominator.

Q396

AP Mock 2 Q76

REBUILT — Original relied on a Moody's KMV EDF-to-rating schedule (27.n) and an unspecified default point; rebuilt as a Merton real-world default probability with total debt as the barrier.

ICB Corp has assets of GBP 25 billion with 45% annual asset volatility. The expected return on assets is 8% and the risk-free rate is 3%. ICB has short-term debt of GBP 6 billion and long-term debt of GBP 4 billion; treat all GBP 10 billion as a single zero-coupon liability due in one year. Using the Merton model with the expected asset return, which is nearest to ICB's one-year probability of default?

A. 0.09%
B. 0.65%
C. 2.34%
D. 3.02%

Answer: C

Step by step

1. $\ln(25/10) = 0.9163$.
2. Drift: $0.08 - 0.45^2/2 = 0.08 - 0.10125 = -0.02125$.
3. $DD = (0.9163 - 0.02125)/0.45 = 1.989$.
4. $PD = N(-1.989) \approx 2.34\%$.

Why C is correct: Default probability is $N(-d_2)$, the chance that assets finish below total debt at T when assets drift at the expected return.

Why the others are wrong

- **A:** 0.09% uses debt of only GBP 6bn (short-term only), giving DD 3.12.
- **B:** 0.65% uses a KMV-style default point of 6 + 0.5(4) = GBP 8bn, not the full debt barrier (DD 2.48).
- **D:** 3.02% uses the 3% risk-free rate instead of the 8% expected return, giving the risk-neutral DD 1.88.

Q397

AP Mock 3 Q61

REBUILT — Stem did not say which formula/drift to use; specified the Merton distance to default with expected return.

An analyst computes the Merton distance to default $DD = \dfrac{\ln(A/L) + (\mu - \sigma^2/2)T}{\sigma\sqrt{T}}$ for three companies:

Company	Assets (A)	Face value of debt (L)	Asset volatility	Expected return (mu)	T
X	800,000	500,000	25%	7%	3 years
Y	1,000,000	600,000	20%	5%	4 years
Z	900,000	400,000	30%	6%	2 years

Based on distance to default, rank the companies from most likely to least likely to default.

A. X, Y, Z
B. Z, Y, X
C. Y, Z, X
D. Z, X, Y

Answer: A

Step by step

1. X: $[\ln(1.6) + (0.07 - 0.03125) \cdot 3]/(0.25\sqrt{3}) = (0.4700 + 0.1163)/0.4330 = 1.354$.
2. Y: $[\ln(1.6667) + (0.05 - 0.02) \cdot 4]/(0.20 \cdot 2) = (0.5108 + 0.1200)/0.4000 = 1.577$.
3. Z: $[\ln(2.25) + (0.06 - 0.045) \cdot 2]/(0.30\sqrt{2}) = (0.8109 + 0.0300)/0.4243 = 1.982$.
4. Lowest DD defaults first: X, Y, Z.

Why A is correct: The smaller the distance to default, the higher the default probability, so ordering by ascending

DD gives the ranking.

Why the others are wrong

- **B:** This is the reverse order (least to most likely), from ranking by highest DD first.
- **C:** Places Y first, but Y's DD (1.577) exceeds X's (1.354), so X is riskier.
- **D:** Places Z first, though Z has the largest DD (1.982) thanks to its high asset-to-debt ratio of 2.25.

Q398

AnalystPrep Q.2876

REBUILT — Options rebuilt so each wrong choice is a realistic error; correct value restated as 0.152 (source rounded 0.1515).

Logos International has asset value of $280 million and debt with face value of $478 million due in 7 years. The expected return on assets is 18% and asset volatility is 21%. Using the Merton model, which is nearest to the probability of default at maturity?

A. 0.848
B. 0.096
C. 0.152
D. 0.057

Answer: C

Step by step

1. $\ln(280/478) = -0.5348$.
2. Drift term: $(0.18 - 0.21^2/2) \times 7 = 0.1579 \times 7 = 1.1053$.
3. $d_2 = (-0.5348 + 1.1053)/(0.21\sqrt{7}) = 0.5705/0.5556 = 1.027$.
4. $PD = N(-1.027) = 1 - 0.8479 = 0.152$.

Why C is correct: Default occurs when assets finish below the face value of debt, with probability $N(-d_2)$.

Why the others are wrong

- **A:** 0.848 is $N(d_2)$, the probability of repayment, not default.
- **B:** 0.096 omits the $-\sigma^2/2$ term in the drift (d = 1.305).
- **D:** 0.057 adds $\sigma^2/2$ instead of subtracting it (d = 1.583).

Q399

AnalystPrep Q.3054

REBUILT — Source answer 2.74% came from rounding d to 1.92; exact value is 2.72%, and distractors replaced by realistic-error values.

The assets of Company X are worth $1,300,000. In three months it must repay $1,000,000 of debt. Asset volatility is 30% per annum and the expected return on assets is 15% per annum. Using the Merton model, which is nearest to the probability of default in three months?

A. 2.72%
B. 2.28%
C. 11.04%
D. 4.01%

Answer: A

Step by step

1. $T = 0.25$; $\ln(1.3) = 0.2624$.
2. Drift: $(0.15 - 0.045) \times 0.25 = 0.02625$.
3. $d_2 = (0.2624 + 0.02625)/(0.30 \times 0.5) = 0.2887/0.15 = 1.924$.
4. $PD = N(-1.924) \approx 2.72\%$.

Why A is correct: PD is $N(-d_2)$ with drift and volatility scaled to the 3-month horizon.

Why the others are wrong

- **B:** 2.28% omits the $-\sigma^2/2$ term (d = 1.999).
- **C:** 11.04% fails to scale drift and volatility to 0.25 years (annual formula, d = 1.225).
- **D:** 4.01% drops the drift term entirely (d = 1.749).

Q400

AnalystPrep Q.3056

REBUILT — Stem lacked the formula; stated the simplified DD explicitly.

A credit manager ranks four counterparties using the simplified one-year distance to default $DD \approx \ln(V_0/F)/\sigma_V$. Each firm's only liability is a one-year zero-coupon bond, and none pays dividends.

Item	Company P	Company Q	Company R	Company S
Market value of assets ($m)	100	180	200	250
Face value of debt ($m)	70	120	100	170
Annual asset volatility	10%	8%	6%	12%

Rank the counterparties from least likely to most likely to default.

A. S, Q, P, R
B. R, P, S, Q
C. S, P, R, Q
D. R, Q, P, S

Answer: D

Step by step

1. P: $\ln(100/70)/0.10 = 0.3567/0.10 = 3.57$.
2. Q: $\ln(180/120)/0.08 = 0.4055/0.08 = 5.07$.
3. R: $\ln(200/100)/0.06 = 0.6931/0.06 = 11.55$.
4. S: $\ln(250/170)/0.12 = 0.3857/0.12 = 3.21$.
5. Highest DD (least likely to default) first: R, Q, P, S.

Why D is correct: A larger distance to default means a bigger buffer in standard deviations, so descending DD gives least-to-most likely to default.

Why the others are wrong

- **A:** This puts S first, but S has the lowest DD (3.21) and is the most likely to default; it is essentially the reverse order with Q and P swapped.
- **B:** Places Q last, but Q has the second-highest DD (5.07), above both P (3.57) and S (3.21); a descending-DD order cannot put Q after P and S.
- **C:** Starts with S, which has the lowest DD (3.21) and is the most likely to default, and places Q last although Q has the second-highest DD (5.07); the order is largely inverted.

Q401

GARP PE1 Q6 · Mock 3 Q74

REBUILT — Stem lacked the formula; stated the simplified DD explicitly. Duplicate of two sources merged.

A credit manager uses a simplified Merton model to rank three non-dividend-paying counterparties with the one-year approximation $DD \approx \ln(V_0/F)/\sigma_V$. Each company's only liability is a one-year zero-coupon bond.

Company	P	Q	R
Market value of assets (EUR million)	100	150	250
Face value of debt (EUR million)	60	100	160
Annual asset volatility	10.0%	7.0%	8.0%

Rank the counterparties from most likely to least likely to default.

A. P; R; Q
B. Q; P; R
C. Q; R; P
D. R; Q; P

Answer: A

Step by step

1. P: $\ln(100/60)/0.10 = 0.5108/0.10 = 5.11$.
2. Q: $\ln(150/100)/0.07 = 0.4055/0.07 = 5.79$.
3. R: $\ln(250/160)/0.08 = 0.4463/0.08 = 5.58$.
4. Ascending DD (most likely to default first): P (5.11), R (5.58), Q (5.79).

Why A is correct: The lowest distance to default signals the highest default probability; ordering ascending gives P, R, Q.

Why the others are wrong

- **B:** Q first ranks Q as riskiest, but Q has the highest DD (5.79) because of its low 7% volatility.
- **C:** Also puts Q first; the lowest-volatility firm has the largest DD, not the smallest.
- **D:** This is the reverse of the correct ranking (least to most likely).

Q402

AnalystPrep Q.3193

REBUILT — Removed irrelevant LGD/EAD/bond-value wording and stated the real-world (expected-return) drift explicitly; replaced weak distractors with mistake-based ones.

A firm has issued a five-year zero-coupon bond. The firm's asset value is 1.1 times the face value of the bond, the expected return on the firm's assets is 15% per year, asset volatility is 19% per year, and the risk-free rate is 7.5%. Using the Merton model, the real-world probability of default over the five years is closest to:

A. 1.38%
B. 3.78%
C. 18.55%
D. 96.22%

Answer: B

Step by step

1. Only the ratio $V_0/D = 1.1$ is needed; $\ln 1.1 = 0.09531$.
2. Drift term: $(\mu - \sigma_V^2/2)T = (0.15 - 0.0361/2) \times 5 = 0.66975$.
3. $\sigma_V\sqrt{T} = 0.19 \times 2.23607 = 0.42485$.
4. $d_2 = (0.09531 + 0.66975)/0.42485 = 1.7772$ (distance to default).
5. PD $= N(-1.7772) = 3.78\%$.

Why B is correct: Default occurs if $V_T < D$, whose real-world probability is $N(-d_2)$ with the expected asset return as drift.

Why the others are wrong

- **A:** Added the variance term instead of subtracting it: $d_2 = 2.20$, PD 1.38%.
- **C:** Used the 7.5% risk-free rate as drift (risk-neutral PD, $d_2 = 0.895$), not the expected return.
- **D:** Reported $N(d_2)$, the probability of no default, instead of $N(-d_2)$.

Q403

AnalystPrep Q.5438

REBUILT — Fixed truncated stem, clarified expected-return drift, replaced the arbitrary 82.05% distractor with the risk-neutral PD and used exact rounding (24.28%).

Use Merton's model to estimate the real-world probability of default for a non-dividend-paying company whose only debt is a single zero-coupon long-term bond, using the following information.

Asset value	USD 17,701,000
Face value of debt	USD 16,710,000
Expected return on assets	12%
Asset volatility	22%
Risk-free rate	5%
Time to maturity of debt	1 year

The probability of default is closest to:

A. 17.95%
B. 24.28%
C. 35.23%
D. 75.72%

Answer: B

Step by step

1. $\ln(17{,}701/16{,}710) = 0.05760$.
2. $(\mu - \sigma^2/2)T = 0.12 - 0.0242 = 0.0958$; $\sigma\sqrt{T} = 0.22$.
3. $d_2 = (0.0576 + 0.0958)/0.22 = 0.6973$.
4. PD $= N(-0.6973) = 24.28\%$.

Why B is correct: The real-world PD uses the expected asset return as drift; PD is the lower-tail probability N(-d2).

Why the others are wrong

- **A:** Added the variance term instead of subtracting it ($d_2 = 0.917$).
- **C:** Used the 5% risk-free rate as drift (risk-neutral PD, $d_2 = 0.379$).
- **D:** Gave $N(d_2)$, the probability of survival, rather than $N(-d_2)$.

Q404

GARP PE2 Q6 · Mock 4 Q7

REBUILT — Original combined PD with a questionable "limitation" component and PD options that did not follow from realistic errors (risk-neutral PD is 12.60%, not 12.49%); asked for the PD only with mistake-based options.

A credit analyst uses the Merton model to estimate the real-world probability of default of a non-dividend-paying company whose debt is a single zero-coupon bond.

Value of the company's assets	CAD 400 million
Face value of debt	CAD 300 million
Expected return on assets	15%
Asset volatility	25%
Risk-free rate	3%
Remaining maturity of debt	1 year

The probability of default is closest to:

A. 3.03%
B. 5.20%
C. 12.60%
D. 94.80%

Answer: B

Step by step

1. $\ln(400/300) = 0.28768$.
2. $(\mu - \sigma^2/2)T = 0.15 - 0.03125 = 0.11875$; $\sigma\sqrt{T} = 0.25$.
3. $d_2 = (0.28768 + 0.11875)/0.25 = 1.6257$.
4. PD $= N(-1.6257) = 5.20\%$.

Why B is correct: Real-world PD uses the expected return (15%) as drift in d2 and takes the lower tail N(-d2).

Why the others are wrong

- **A:** Added the variance term ($d_2 = 1.876$) instead of subtracting it.
- **C:** Used the 3% risk-free rate as drift (risk-neutral PD, $d_2 = 1.146$).
- **D:** Reported $N(d_2)$, the survival probability.

Q405

AP Mock 2 Q42

REBUILT — Restated the garbled formula in standard form and replaced unexplained distractors with mistake-based ones.

Tenacity Limited has a five-year zero-coupon bond with face value USD 300 million; the rest of its capital is equity. The market value of the firm's assets is USD 380 million, the expected return on the assets is 20% per year, and asset volatility is 30% per year. Assume asset value is lognormal with constant volatility. Using $\text{PD} = N\left(\dfrac{\ln(D/V_0) - \mu T + 0.5\sigma^2 T}{\sigma\sqrt{T}}\right)$, the estimated probability of default is closest to:

A. 49.3%
B. 6.6%
C. 3.3%
D. 1.5%

Answer: B

Step by step

1. $\ln(300/380) = -0.23639$; $-\mu T = -0.20 \times 5 = -1.0$; $+0.5\sigma^2 T = 0.5 \times 0.09 \times 5 = 0.225$.
2. Numerator $= -0.23639 - 1.0 + 0.225 = -1.01139$.
3. $\sigma\sqrt{T} = 0.30 \times 2.23607 = 0.67082$.
4. Argument $= -1.5077$; PD $= N(-1.5077) = 6.6\%$.

Why B is correct: The formula is $N(-d_2)$ with the expected return as drift; the 5-year horizon scales both drift and variance terms.

Why the others are wrong

- **A:** Dropped the drift term entirely ($d_2 = 0.017$), so the 20% expected return had no effect.
- **C:** Omitted the $0.5\sigma^2 T$ term ($d_2 = 1.843$), understating PD.
- **D:** Added instead of subtracted the variance term ($d_2 = 2.179$), understating PD.

Q406

AP Mock 2 Q56

REBUILT — Replaced unexplained distractors with mistake-based options; clarified that real-world PD is requested.

A firm has a current asset value of USD 110 million. Its only debt is a three-year zero-coupon bond with face value USD 130 million. The risk-free rate is 5%, the expected return on the firm's assets is 20%, and asset volatility is 30%. Using the Merton model, the real-world probability of default over the three years is closest to:

A. 13.7%
B. 28.3%
C. 61.5%
D. 71.7%

Answer: B

Step by step

1. $\ln(110/130) = -0.16705$.
2. $(\mu - \sigma^2/2)T = (0.20 - 0.045) \times 3 = 0.465$.
3. $\sigma\sqrt{T} = 0.30 \times 1.73205 = 0.51962$.
4. $d_2 = (-0.16705 + 0.465)/0.51962 = 0.5734$; PD $= N(-0.5734) = 28.3\%$.

Why B is correct: Real-world PD uses the 20% expected return as drift; the 5% risk-free rate is not used.

Why the others are wrong

- **A:** Added the variance term instead of subtracting it ($d_2 = 1.093$).
- **C:** Used the 5% risk-free rate as drift ($d_2 = -0.293$, risk-neutral PD).
- **D:** Reported $N(d_2)$, the survival probability.

Q407

AnalystPrep Q.2674

REBUILT — Options were tightly clustered with no identifiable errors; replaced with mistake-based options and specified real-world PD (the 7% rate is a distractor).

Given the following information about a firm, calculate its real-world probability of default over the life of the debt using the Merton model.

Value of the firm	USD 40 million
Outstanding debt (zero-coupon, face value)	USD 25 million
Maturity of debt	5 years
Risk-free rate	7%
Volatility of the firm	30%
Expected return of the firm	10%

A. 3.7%
B. 13.3%
C. 18.8%
D. 86.7%

Answer: B

Step by step

1. $\ln(40/25) = 0.47000$.
2. $(\mu - \sigma^2/2)T = (0.10 - 0.045) \times 5 = 0.275$.
3. $\sigma\sqrt{T} = 0.30 \times 2.23607 = 0.67082$.
4. $d_2 = (0.47 + 0.275)/0.67082 = 1.1106$; PD $= N(-1.1106) = 13.3\%$.

Why B is correct: Real-world PD uses the expected return of 10% as drift; the risk-free rate is a distractor.

Why the others are wrong

- **A:** Added the variance term ($d_2 = 1.781$).
- **C:** Used the 7% risk-free rate as drift (risk-neutral PD, $d_2 = 0.887$).
- **D:** Reported $N(d_2)$, the survival probability.

Q408

AnalystPrep Q.5985

In the Merton model, what does the distance to default measure?

A. The difference between a firm's total assets and total liabilities.
B. The number of standard deviations by which the firm's asset value must change for default to occur at the debt's maturity.
C. The gap between the market value of equity and the face value of debt.

D. The time remaining until the firm's debt reaches maturity.

Answer: B

Step by step

1. In the Merton model default occurs if $V_T < D$.
2. Distance to default $= d_2 = \dfrac{\ln V_0 - \ln D + (\mu - \sigma_V^2/2)T}{\sigma_V\sqrt{T}}$, which is in units of standard deviations.
3. PD $= N(-d_2)$: a larger distance means a lower PD.

Why B is correct: DD standardises the cushion between the asset value and the default point by asset volatility, so it is a number of standard deviations.

Why the others are wrong

- **A:** An absolute asset-minus-liability gap is not scaled by volatility, so it is not a distance in standard deviations.
- **C:** Equity minus debt is an absolute dollar gap, unrelated to the volatility-scaled distance.
- **D:** Time to maturity T is an input to d2, not what d2 measures.

Q409

BT answers P2.T6.24.16.1 · BT notes P2.T6.24.15.3

REBUILT — Source key A paired DD 2.04 with PD 1.1% (inconsistent; 1.1% is N(-2.29)); rebuilt options so DD and PD are consistent and exactly one is correct.

FiberTech has asset value of USD 500 million and total debt of USD 300 million due in one year. Asset volatility is 25% and the risk-free rate is 3% per annum. Using the Merton model with the risk-free rate as the asset drift (risk-neutral setting, as in the GARP/Schweser example), which pair gives the distance to default and the probability of default?

A. Distance to default 2.04; probability of default 2.08%
B. Distance to default 2.29; probability of default 1.11%
C. Distance to default 2.04; probability of default 97.92%
D. Distance to default 1.92; probability of default 2.72%

Answer: A

Key correction: Source key (A) paired DD 2.04 with PD 1.1%, which is internally inconsistent: N(-2.04)=2.08%, while 1.1% corresponds to N(-2.29) using d1. Corrected option A to DD 2.04 / PD 2.08%.

Step by step

1. $\ln(500/300) = 0.51083$.
2. $(r - \sigma^2/2)T = 0.03 - 0.03125 = -0.00125$; $\sigma\sqrt{T} = 0.25$.
3. $d_2 = (0.51083 - 0.00125)/0.25 = 2.038 \approx 2.04$ = distance to default.
4. PD $= N(-2.038) = 2.08\%$.

Why A is correct: Distance to default is d_2 and PD is $N(-d_2)$; both come from the same d_2.

Why the others are wrong

- **B:** Used $d_1 = d_2 + \sigma\sqrt{T} = 2.29$ as the distance to default; $N(-d_1) = 1.11\%$.
- **C:** Correct DD but reported $N(d_2) = 97.92\%$, the survival probability.
- **D:** Subtracted the full 3% rate (instead of the net drift $r - \sigma^2/2 = -0.125\%$) from $\ln(V_0/D)$: $d = (0.51083 - 0.03)/0.25 = 1.92$, PD 2.72%.

Q410

BT notes P2.T6.303.2

REBUILT — Source used the 10% expected asset return as the drift in d2; the Merton model (Schweser/GARP) defines DD = d2 and the default probability N(-d2) with the risk-free rate, so the question now asks for the risk-neutral PD and options were rebuilt.

A firm's current asset value is $62.0 million with asset volatility of 38.0% per annum. Debt consists of a single zero-coupon bond with face value $40.0 million maturing in one year. The risk-free rate is 2.0% per annum (the expected return on the firm's assets is 10.0%). Using the Merton model, what is the firm's (risk-neutral) probability of default?

A. 8.14%
B. 11.00%
C. 15.48%
D. 12.44%

Answer: C

Key correction: Source key (D, 11.00%) used the 10% asset drift in d2; the Merton default probability N(-d2) uses the risk-free

rate, so the correct value is 15.48%.

Step by step

1. Merton: $d_2 = \dfrac{\ln(V_0/D) + (r - \sigma_V^2/2)T}{\sigma_V\sqrt{T}}$, PD = $N(-d_2)$.
2. $\ln(62/40) = 0.4383$.
3. $r - \sigma_V^2/2 = 0.02 - 0.38^2/2 = 0.02 - 0.0722 = -0.0522$.
4. $d_2 = (0.4383 - 0.0522)/0.38 = 1.016$ (the distance to default).
5. PD = $N(-1.016) = 15.48\%$.

Why C is correct: Distance to default is d_2 computed with the risk-free rate; the probability that assets end below the debt face value is $N(-d_2)$, here 15.48%.

Why the others are wrong

- **A:** 8.14% is $N(-d_1)$ with $d_1 = d_2 + \sigma\sqrt{T} = 1.396$; the wrong d term was used.
- **B:** 11.00% uses the 10% expected asset return as the drift instead of the 2% risk-free rate (d = 1.226), which gives a real-world-style number, not the Merton risk-neutral PD.
- **D:** 12.44% drops the drift term $(r - \sigma^2/2)T$ altogether: $\ln(62/40)/0.38 = 1.153$, $N(-1.153) = 12.44\%$.

Q411

Mock 2 Q72

REBUILT — Source computed DD as ln(expected future value / debt)/sigma, ignoring the Merton d2 formula and mixing in an expected value; rebuilt with a risk-free rate so DD = d2 is unambiguous.

A credit analyst has the following data for Firm ABC:

Parameter	Value
Current market value of assets (V0)	3,750
Face value of zero-coupon debt due in one year (D)	2,500
Annualized volatility of firm's assets	20%
Risk-free rate	4% per annum

According to the Merton model, what is the one-year distance to default (DD = d2)?

A. 2.03 standard deviations
B. 2.13 standard deviations
C. 2.23 standard deviations
D. 2.33 standard deviations

Answer: B

Key correction: Source key (C, 2.93) used a 4,500 expected asset value in ln(E[V]/D)/sigma; the Merton DD is d2 using the current value V0 and the risk-free rate.

Step by step

1. DD = $d_2 = \dfrac{\ln(V_0/D) + (r - \sigma^2/2)T}{\sigma\sqrt{T}}$.
2. $\ln(3750/2500) = \ln 1.5 = 0.4055$.
3. $r - \sigma^2/2 = 0.04 - 0.02 = 0.02$.
4. $d_2 = (0.4055 + 0.02)/0.20 = 2.127 \approx 2.13$.

Why B is correct: DD is the number of standard deviations the asset value must fall to reach the debt level, equal to d_2; here 2.13.

Why the others are wrong

- **A:** 2.03 omits the drift term and uses $\ln(V_0/D)/\sigma$ only.
- **C:** 2.23 adds rT but forgets to subtract $\sigma^2/2$ (uses r as the drift correction instead of $r - \sigma^2/2$).
- **D:** 2.33 is $d_1 = d_2 + \sigma\sqrt{T}$, using $+\sigma^2/2$ instead of $-\sigma^2/2$.

LO 27.n

Assess the quality of the default probabilities produced by the Merton model, the Moody's KMV model, and the Kamakura model.

Q412 BT answers P2.T6.24.16.2 · BT notes P2.T6.24.16.2

REBUILT — Original key (B) is not supported by the syllabus text and the stem was a long scenario; rebuilt as a single-concept question on ranking quality and the risk-neutral vs real-world distinction with exactly one correct option.

A credit analyst is preparing a board report comparing the Merton model, the Moody's KMV model and the Kamakura model as sources of default probabilities. Which statement best assesses the quality of these models' default probabilities?

A. All three produce reliable and reasonably accurate rankings of default probabilities; the Merton output is in theory a risk-neutral probability, and KMV and Kamakura restate it as a real-world probability.

B. The Merton model produces real-world default probabilities directly, while KMV and Kamakura convert them into risk-neutral probabilities.

C. Only the KMV and Kamakura models rank default probabilities reliably; the Merton model's rankings are poor because it ignores market data.

D. None of the three models ranks default probabilities well enough to be useful, so ratings from agencies should be used instead.

Answer: A

Key correction: Source key (BT answers) said B (KMV is superior because it uses more market data than Merton and Kamakura); the reading does not rank KMV above the others, and states all three rank default probabilities reliably.

Step by step

1. Empirical evidence shows Merton, Moody's KMV and Kamakura all give reliable, reasonably accurate rankings of default probabilities.
2. Merton's N(-d2) comes from an option pricing model, so it is in theory a risk-neutral default probability.
3. Because the ranking is good, a monotonic transformation can map the Merton output into a real-world probability; KMV and Kamakura sell this transformed output.
4. Only A states all of this correctly.

Why A is correct: The models share the Merton ranking, which is empirically good; KMV and Kamakura add a transformation from risk-neutral to real-world probabilities.

Why the others are wrong

- **B:** Reverses the direction: Merton's output is risk-neutral and KMV/Kamakura restate it as real-world.
- **C:** Wrongly says Merton rankings are poor; the KMV and Kamakura outputs are built from the Merton ranking, so its ranking quality is what they rely on.
- **D:** Overstates the weakness; empirical evidence supports the models' rankings.

Q413 BT answers P2.T6.24.16.3 · BT notes P2.T6.24.16.3

REBUILT — Original options were inconsistent with the stated estimates (Merton lowest, but options called Merton's higher) and the item needed unavailable industry data; rebuilt using the notes' numbers (2.5%, 1.8%, 2.1%) with a clear conceptual answer.

A quantitative analyst reviews five-year default probability estimates for a technology issuer from three models using the same financial statements: Merton 2.5%, Moody's KMV 1.8%, Kamakura 2.1%. Which interpretation is most consistent with how these models are assessed?

A. The Merton figure is in theory a risk-neutral probability, whereas the KMV and Kamakura figures are restated as real-world probabilities, so a somewhat higher Merton figure does not by itself signal that any model is wrong.

B. KMV's estimate is the lowest, so it is the most accurate because it uses market data.

C. Because the three estimates differ, all three models are unreliable and should be discarded.

D. Kamakura's estimate lies between the other two, so it must be the correct probability.

Answer: A

Step by step

1. Merton's N(-d2) is derived from option pricing, so it is a risk-neutral default probability.
2. KMV and Kamakura transform the Merton output into real-world default probabilities.
3. Risk-neutral and real-world probabilities are different quantities, so the 2.5% versus 1.8% and 2.1% gap is expected and not evidence of error.
4. The models are still judged by how well they rank default probabilities, which empirical evidence supports for all three.

Why A is correct: The Merton output is risk-neutral while the others are real-world, so levels differ by construction while rankings remain reliable.

Why the others are wrong

- **B:** Assumes a lower estimate means greater accuracy; accuracy is assessed on ranking quality, not on the lowest number.
- **C:** Treats different levels as inconsistency; the models measure different probability types and all rank well.
- **D:** Picking the middle value has no basis; being between two estimates says nothing about correctness.

Q414 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A risk manager ranks a bank's counterparties by default probability and also needs the likely frequency of actual defaults. She is considering the Merton model, Moody's KMV and Kamakura. Which of the following is the most accurate assessment of these models' default probabilities?

A. Merton output is risk-neutral and therefore cannot be used to rank counterparties by default risk.

B. KMV and Kamakura produce more reliable rankings than Merton because they do not use the Merton output.

C. All three rank default probabilities reliably, but only KMV and Kamakura restate the risk-neutral Merton output as a real-world probability.

D. Merton output is a real-world probability, while KMV and Kamakura convert it to a risk-neutral probability for pricing.

Answer: C

Step by step

1. Empirical evidence supports all three as giving reliable, reasonably accurate rankings.
2. The Merton output is risk-neutral because it derives from an option pricing model.
3. KMV and Kamakura take the Merton output and restate it as a real-world probability.

Why C is correct: Rankings are reliable for all three models, but only KMV and Kamakura express real-world probabilities, which is what is needed for default frequency.

Why the others are wrong

- **A:** Being risk-neutral affects the level, not the reliability of the ranking.
- **B:** KMV and Kamakura build on the Merton output, and empirical evidence supports all three for ranking.
- **D:** Reverses the roles: Merton is risk-neutral, and KMV and Kamakura convert to real-world.

Reading 28. Credit Value at Risk

John C. Hull, Risk Management and Financial Institutions, 6th Edition (John Wiley & Sons, 2023). Chapter 19.

LO	Learning objective	Qs
28.a	compare market risk value at risk (VaR) with credit VaR in terms of definition, time horizon, and tools for measuring them.	4
28.b	define and calculate credit VaR.	11
28.c	describe the use of rating transition matrices for calculating credit VaR.	9
28.d	describe the application of Vasicek's model to estimate capital requirements under the Basel II internal- ratings-based (IRB) approach.	13
28.e	explain the Vasicek's model, Credit Risk Plus (CreditRisk+) model, and the CreditMetrics ways of estimating the probability distribution of losses arising from defaults as well as modeling the default correlation.	10
28.f	define credit spread risk and assess its impact on calculating credit VaR.	6

LO 28.a

Compare market risk value at risk (VaR) with credit VaR in terms of definition, time horizon, and tools for measuring them.

Q415

AP Mock 2 Q4 · Schweser Module Quiz 28.1 Q1

A risk manager is explaining to the board why a bank's market risk VaR and its credit VaR are measured over different time horizons. Which statement correctly describes the typical horizons?

A. Both are typically measured over one day, because both kinds of risk events occur at high frequency.

B. Market risk VaR typically uses a longer horizon than credit VaR, because market prices move less often than credit events occur.

C. Market risk VaR typically uses a one-day horizon, while credit VaR typically uses a longer horizon, often one year.

D. Credit VaR typically uses a shorter horizon than market risk VaR, because defaults and downgrades occur more often than price changes.

Answer: C

Step by step

1. Market risk VaR covers prices that change continuously and positions that can be liquidated quickly, so the usual horizon is one day.
2. Credit risk VaR covers defaults, downgrades and spread changes on instruments often not held for trading; the usual horizon is one year.
3. Therefore the credit VaR horizon (about 1 year) is longer than the market VaR horizon (1 day): option C.

Why C is correct: Market VaR is usually computed over one day; credit VaR, for instruments not held for trading, is usually computed over a longer horizon of about one year.

Why the others are wrong

- **A:** Applies the market-risk convention to credit VaR; a one-day credit VaR is not the norm.
- **B:** Reverses the relationship; the longer horizon belongs to credit VaR.
- **D:** Reverses the relationship and wrongly claims credit events are more frequent than price changes.

Q416

AnalystPrep Q.6020

REBUILT — Removed unsupported claims (e.g., 'infrequency') from the correct option and tightened distractors so the key rests on the statement that credit VaR needs a more elaborate model incorporating credit correlation.

A risk manager is asked which tools are appropriate for measuring credit VaR as opposed to market risk VaR. Which of the following is the most appropriate approach for credit VaR?

A. A simple historical simulation of daily changes in portfolio value, as is the main tool for market risk VaR.

B. A more elaborate model that captures credit correlation and, where relevant, rating transitions and credit spread changes.

C. Assuming defaults of different companies are independent so that each exposure can be modeled separately.

D. Scaling a one-day market risk VaR by the square root of 250 and treating the result as the credit VaR.

Answer: B

Step by step

1. Market risk VaR is mainly computed by historical simulation over a one-day horizon.
2. Credit VaR depends on defaults, downgrades and spread changes over a longer horizon, and defaults are correlated (they cluster in downturns).
3. Hence a more elaborate model that incorporates credit correlation (and rating transitions) is normally needed: option B.

Why B is correct: Credit correlation is a key feature of credit losses, so credit VaR normally requires a more elaborate model than the historical simulation used for market VaR.

Why the others are wrong

- **A:** Applies the market-risk tool; a short history of daily changes rarely contains the defaults and correlated credit events that drive credit VaR.
- **C:** Ignores credit correlation; defaults are not independent and correlation rises in stressed markets, so independence understates tail loss.
- **D:** Square-root-of-time scaling of a one-day market VaR does not capture default, migration or correlation effects.

Q417

BT answers P2.T6.24.17.1

REBUILT — Shortened the stem, which gave away the answers, and kept the same four contrasts.

Warren Bank computes market risk VaR using a one-day horizon and historical simulation. For credit VaR it considers an elaborate model that captures defaults, downgrades, spread changes and credit correlation. It generates scenarios with real-world default probabilities and values the portfolio at the horizon with risk-neutral estimates. Which statement accurately contrasts market risk VaR with credit VaR?

A. Both use a one-day horizon, aligning the measurement of trading and non-trading instruments.

B. Market risk VaR is mainly calculated by historical simulation, whereas credit VaR usually needs a more elaborate model that captures credit correlation and several types of credit event.

C. Market risk VaR typically uses a longer horizon than credit VaR because of the need to capture long economic cycles.

D. Credit VaR uses real-world default probabilities to generate scenarios and risk-neutral probabilities to set regulatory capital.

Answer: B

Step by step

1. Horizon: market VaR about one day; credit VaR often one year. This eliminates A and C.
2. Tools: historical simulation is the main market-VaR tool; credit VaR needs a more elaborate model for correlation and defaults/downgrades/spread changes, so B is correct.
3. Probabilities: real-world default probabilities generate the scenarios, and risk-neutral estimates value the portfolio at the VaR horizon date, not set regulatory capital. This eliminates D.

Why B is correct: Credit VaR differs from market VaR in horizon (about one year vs one day) and in tools (an elaborate correlation-aware model vs historical simulation).

Why the others are wrong

- **A:** Ignores that credit VaR normally uses a one-year horizon for instruments not held for trading.
- **C:** Reverses the horizon relationship; credit VaR has the longer horizon.
- **D:** Misplaces risk-neutral probabilities: they value the portfolio on the horizon date; they do not determine regulatory capital.

Q418

AnalystPrep Q.6022

REBUILT — Original correct option was loosely worded (maximum loss, market-driven losses); rewrote options so the correct one matches the standard definition (loss not exceeded at a confidence level over a period).

Which of the following is the correct definition of credit risk VaR?

A. The average credit loss expected over the long run, regardless of the confidence level chosen.

B. The credit loss over a given time period that will not be exceeded with a given level of confidence.

C. The smallest credit loss across all outcomes, measured over the time period and ignoring the confidence level.

D. The credit loss over a given time period that is expected to be exceeded with the given level of confidence.

Answer: B

Step by step

1. Credit VaR is defined like market VaR: a loss quantile.
2. It is the credit loss over a certain period that will not be exceeded with a certain confidence level (e.g., 99.9% over one year).
3. Only B states this.

Why B is correct: Credit VaR is the loss level for a stated horizon that is not exceeded with the specified confidence, and may include losses from defaults, downgrades and/or spread changes.

Why the others are wrong

- **A:** Describes expected loss, a mean, not a quantile tied to a confidence level.
- **C:** Describes a minimum, not a high quantile, and drops the confidence level entirely.
- **D:** Reverses the confidence statement: at 99.9% confidence the loss is exceeded only 0.1% of the time, not 99.9%.

LO 28.b

Define and calculate credit VaR.

Q419

BT notes P2.T6.305.1

REBUILT — Removed the off-topic Merton commentary and stated explicitly that credit VaR here is the 99.9th percentile loss in excess of expected loss; replaced the options with realistic-error distractors.

A firm's only debt is a one-year zero-coupon bond with a face value of $800 million. A Merton-type model gives these results for the bond one year from now: expected loss (EL) = $8.40 million, probability of default = 10.6%, and the 0.001 quantile of the future bond value = $503.00 million. Credit VaR at the 99.9% confidence level is defined as the 99.9th percentile loss in excess of the expected loss (the unexpected loss). Which is closest to the credit VaR?

A. $79.25 million
B. $288.60 million
C. $297.00 million
D. $305.40 million

Answer: B

Step by step

1. Worst-case (99.9th percentile) value of the bond = $503.00 million, so the 99.9th percentile loss relative to face = 800 - 503 = $297.00 million.
2. Expected loss = $8.40 million, so the expected future value of the debt = 800 - 8.40 = $791.60 million.
3. Credit VaR (unexpected loss) = percentile loss - EL = 297.00 - 8.40 = $288.60 million.
4. Equivalently, expected future value minus worst-case value = 791.60 - 503.00 = $288.60 million.

Why B is correct: Under the stated definition, credit VaR is the loss at the 99.9% percentile measured from the mean of the loss distribution, i.e. percentile loss minus expected loss.

Why the others are wrong

- **A:** $79.25 million is the expected loss given default (EL/PD = 8.40/10.6%), a conditional average loss, not a percentile-based measure.
- **C:** $297.00 million is the 99.9th percentile loss measured from face value; it forgets to subtract the expected loss.
- **D:** $305.40 million adds the expected loss (297.00 + 8.40) instead of subtracting it.

Q420

AnalystPrep Q.6023

REBUILT — Original distractors were giveaways (profit, guaranteed loss); replaced them with plausible confusions such as market VaR, expected loss and total exposure.

In a risk management workshop, an instructor is asked to define credit VaR. Which definition is most accurate?

A. The average credit loss expected over the horizon, which does not depend on any confidence level.
B. The credit loss over a certain time period that will not be exceeded with a certain confidence level.
C. The loss from adverse market price movements that will not be exceeded over a one-day horizon with a certain confidence level.
D. The total exposure that would be lost if every counterparty in the portfolio defaulted over the horizon.

Answer: B

Step by step

1. Credit VaR is defined like market VaR: a loss quantile.
2. It is the credit-risk loss over a given period that will not be exceeded at a stated confidence level; some models count only defaults, others also downgrades and credit spread changes.
3. Compare each option with this definition: only B contains both a time period and a confidence level for credit losses.

Why B is correct: Credit VaR is a percentile of the credit loss distribution over a horizon (often one year), exactly as VaR is a percentile of the market loss distribution.

Why the others are wrong

- **A:** Describes expected loss (a mean) and ignores the confidence level that defines a VaR.
- **C:** Describes market risk VaR (market price risk, typically a one-day horizon), not credit VaR.
- **D:** Describes the maximum possible loss (100% confidence, all names default), not a loss at a chosen confidence level.

Q421

BT answers P2.T6.24.17.2.

REBUILT — Original used a non-standard, OCR-garbled formula (EL plus z times spread volatility); rebuilt as a clean credit VaR = percentile loss minus expected loss calculation with all inputs given.

A bank holds a $100 million corporate loan portfolio. The one-year probability of default for the portfolio is 1% and loss given default is 50%. The bank's model estimates that, with 99% confidence, the one-year default rate will not exceed 4.0%. Credit VaR at the 99% level is defined as the 99th percentile credit loss in excess of the expected loss. What is the credit VaR?

A. $500,000
B. $1,500,000
C. $2,000,000
D. $3,000,000

Answer: B

Step by step

1. Expected loss = 100,000,000 x 1% x 50% = $500,000.
2. 99th percentile loss = 100,000,000 x 4.0% x 50% = $2,000,000.
3. Credit VaR = 2,000,000 - 500,000 = $1,500,000.

Why B is correct: Credit VaR here is the loss at the 99% worst-case default rate (after applying LGD) measured above the expected loss.

Why the others are wrong

- **A:** $500,000 is only the expected loss.
- **C:** $2,000,000 is the 99th percentile loss; it forgets to subtract the expected loss.
- **D:** $3,000,000 omits LGD in both terms: 100m x (4% - 1%) = $3.0m, treating the loss as a full write-off.

Q422

Mock 4 Q38 · GARP PE2 Q37

REBUILT — Merged two identical sources; stated the credit VaR convention (percentile loss minus expected loss) in the stem so only one ranking is defensible.

A credit analyst at a bank estimates credit VaR (CVaR) for three loans. Credit VaR is defined as the 95th percentile of the unrecovered credit loss minus the expected loss.

Loan	Maturity (years)	Exposure (SGD)	Loss given default	S&P rating
S	2	55,000,000	0.8	BBB
T	3	36,000,000	0.9	BB-
U	4	50,000,000	0.7	A

The annual probability of default (PD) by rating class and maturity is:

Loan maturity (years)	2	3	4
PD (investment grade)	1.5%	2.5%	3.5%
PD (non-investment grade)	5.0%	12.0%	18.0%

Assuming the 95th percentile of unrecovered credit loss is the same for all three loans, which comparison of the 95% CVaR is correct?

A. CVaR of Loan S > CVaR of Loan U > CVaR of Loan T.
B. CVaR of Loan T > CVaR of Loan U > CVaR of Loan S.

C. CVaR of Loan T > CVaR of Loan S > CVaR of Loan U.
D. CVaR of Loan U > CVaR of Loan S > CVaR of Loan T.

Answer: A
Step by step
1. PDs: S is BBB (investment grade), 2 years: 1.5%; T is BB- (non-investment grade), 3 years: 12.0%; U is A (investment grade), 4 years: 3.5%.
2. EL(S) = 55,000,000 x 0.8 x 0.015 = 660,000.
3. EL(T) = 36,000,000 x 0.9 x 0.12 = 3,888,000.
4. EL(U) = 50,000,000 x 0.7 x 0.035 = 1,225,000.
5. With equal 95th percentile losses, CVaR = percentile loss - EL, so the loan with the smallest EL has the largest CVaR: EL ranks T > U > S, hence CVaR ranks S > U > T.

Why A is correct: CVaR is the common percentile loss less EL, so it moves inversely to expected loss.
Why the others are wrong
- **B:** Ranks CVaR in the same order as EL (T > U > S), forgetting that EL is subtracted.
- **C:** Puts S below T and U below S, which contradicts the EL figures: EL(U) = 1.225m exceeds EL(S) = 0.66m, so U's CVaR is lower than S's.
- **D:** Ranks U above S, which would require EL(U) < EL(S); in fact EL(U) = 1.225m > EL(S) = 0.66m (a larger exposure and maturity do not by themselves raise CVaR).

Q423 Schweser Module Quiz 28.1 Q2

During a period of slowing economic growth, an analyst will likely identify which of the following trends regarding credit correlation and financial institution risks?
A. A decrease in credit correlation and defaults.
B. An increase in credit correlation and defaults.
C. An increase in credit correlation and a decrease in defaults.
D. A decrease in credit correlation and an increase in defaults.

Answer: B
Step by step
1. Default rates are linked to the state of the economy, which is a main source of credit correlation.
2. In a slowdown most companies are adversely affected, so defaults rise and tend to occur together, raising credit correlation.
3. Higher correlation raises the risk of a financial institution's credit portfolio.

Why B is correct: A weak economy simultaneously raises default probabilities and the tendency of defaults to cluster, so both defaults and credit correlation increase.
Why the others are wrong
- **A:** Describes a strong economy, where companies are favorably affected.
- **C:** Defaults cannot fall in a downturn while correlation rises; both effects stem from the same weak economy.
- **D:** Defaults do rise, but correlation rises too, because common economic factors drive all firms.

Q424 AP Mock 3 Q3

REBUILT — Original left the credit VaR convention unstated (both USD 950,000 and USD 1 million defensible); stated that credit VaR is the 99th percentile loss minus expected loss.

A USD 1 million portfolio is split into 10 equal credit positions, all obligations of the same obligor. The obligor has a one-year default probability of 5% and the recovery rate is zero. Credit VaR at 99% confidence is defined as the 99th percentile loss minus the expected loss. What is the credit VaR?
A. USD 50,000
B. USD 950,000
C. USD 1 million
D. USD 0

Answer: B
Step by step
1. All ten positions are claims on one obligor, so they default together (perfect correlation): loss = 0 with proba-

bility 95%, loss = USD 1,000,000 with probability 5%.
2. The 99th percentile: P(loss = 0) = 95% < 99%, so the 99th percentile loss is USD 1,000,000.
3. Expected loss = 5% x 1,000,000 = USD 50,000.
4. Credit VaR = 1,000,000 - 50,000 = USD 950,000.

Why B is correct: Concentration in one obligor makes the 99th percentile equal to the full exposure; subtracting EL gives the unexpected loss.

Why the others are wrong

- **A:** USD 50,000 is only the expected loss.
- **C:** USD 1 million is the 99th percentile loss; it forgets to subtract the expected loss.
- **D:** Treats the ten positions as diversified so a loss is unlikely at 99%; but they are claims on one obligor and default together with 5% probability, which exceeds 1%.

Q425

AnalystPrep Q.5444

REBUILT — Source applied LGD to a value shortfall (unconventional, no sound basis); rebuilt as a single-bond credit VaR with a clearly defined 99th percentile loss and expected loss.

A portfolio consists of one 1-year zero-coupon bond of PYY Plc with a face value of $6,500,000. The bond's 1-year probability of default is 7% and its recovery rate is 88%. Credit VaR at the 99% confidence level is defined as the 99th percentile loss minus the expected loss. What is the bond's 1-year 99% credit VaR?

A. $54,600
B. $780,000
C. $725,400
D. $834,600

Answer: C

Step by step

1. LGD = 1 - 88% = 12%; loss if default = 6,500,000 x 12% = $780,000.
2. The loss is 0 with probability 93% and $780,000 with probability 7%. Because 7% > 1%, the 99th percentile loss is $780,000.
3. Expected loss = 7% x 780,000 = $54,600.
4. Credit VaR = 780,000 - 54,600 = $725,400.

Why C is correct: With a default probability above 1%, the 99th percentile loss is the full loss given default; subtracting EL gives the credit VaR.

Why the others are wrong

- **A:** $54,600 is only the expected loss.
- **B:** $780,000 is the 99th percentile loss; it forgets to subtract the expected loss.
- **D:** $834,600 adds the expected loss to the percentile loss instead of subtracting it.

Q426

BT answers P2.T6.24.11.3

REBUILT — Source claimed the 95th percentile of Binomial(50, 0.015) is 7 defaults; it is 2 (cumulative 96.1%). Recomputed with the correct percentile and stated zero recovery.

A portfolio has 50 credits, each with exposure $1,000,000 and zero recovery (total $50,000,000). Defaults are independent, so the number of defaults is binomial with n = 50 and a default probability of 0.015 per credit. Unexpected loss (UL) at the 95% confidence level is the 95th percentile loss minus the expected loss. What is UL?

A. $750,000
B. $1,250,000
C. $2,000,000
D. $2,250,000

Answer: B

Key correction: Source key (A: $6,250,000, from 7 defaults at the 95th percentile) is wrong: for Binomial(50, 0.015), P(X<=2) = 0.9608 >= 0.95 while P(X<=1) = 0.8273, so the 95th percentile is 2 defaults; correct UL is $1,250,000.

Step by step

1. Expected loss = 50 x 0.015 x $1,000,000 = $750,000.
2. Cumulative binomial probabilities: P(X<=0) = 0.4697, P(X<=1) = 0.8273, P(X<=2) = 0.9608.
3. The smallest k with P(X<=k) >= 95% is k = 2, so the 95th percentile loss = 2 x $1,000,000 = $2,000,000.
4. UL = 2,000,000 - 750,000 = $1,250,000.

Why B is correct: The 95th percentile of the default count is 2, giving a $2.0 million percentile loss; deducting the $0.75 million expected loss leaves the unexpected loss.

Why the others are wrong

- **A:** $750,000 is only the expected loss.
- **C:** $2,000,000 is the 95th percentile loss; it forgets to subtract the expected loss.
- **D:** Uses 3 defaults (the 99th percentile, cumulative 99.3%) instead of 2: 3,000,000 - 750,000 = $2,250,000.

Q427

BT notes P2.T6.305.2 · BT notes P2.T6.305.3

REBUILT — Source items used Malz's convention (credit VaR = unexpected loss, net of expected loss), which conflicts with the GARP/Hull definition; rebuilt to test the GARP definition of credit VaR.

Which of the following best defines credit VaR as used in the Basel/Vasicek framework?

A. The credit loss that will not be exceeded over a given time horizon with a specified confidence level.
B. The expected credit loss over a one-day horizon, estimated by historical simulation.
C. The worst-case loss minus the expected loss over the horizon, so that expected losses are excluded.
D. The change in portfolio value from one-day moves in credit spreads at a specified confidence level.

Answer: A

Key correction: Sources (Malz-style) keyed credit VaR as par minus expected loss minus the 0.001 quantile, i.e. unexpected loss; GARP/Hull define credit VaR as the credit loss not exceeded at the confidence level (e.g. WCDR x EAD x LGD), so the item was rebuilt on the GARP definition.

Step by step

1. Credit VaR parallels market VaR: a loss quantile over a horizon at a confidence level.
2. GARP: it is the credit loss that will not be exceeded over some time horizon with a specified confidence level.
3. Credit VaR typically uses a one-year horizon (market VaR often one day) and models more elaborate than historical simulation.
4. Choice A matches this definition.

Why A is correct: Credit VaR is a loss quantile: the credit loss not exceeded over the horizon at the chosen confidence level.

Why the others are wrong

- **B:** Confuses credit VaR with an expected loss and with market VaR's one-day horizon and historical simulation; credit VaR is a high-percentile loss over typically one year.
- **C:** Describes unexpected loss (Malz convention); in the GARP/Hull framework the credit VaR is the loss quantile itself, not net of expected loss.
- **D:** Describes a market-VaR-type measure of spread moves over one day, not the credit loss quantile over a long horizon.

Q428

Mock 2 Q56

REBUILT — Source subtracted expected loss (Malz convention), conflicting with GARP where credit VaR = WCDR x EAD x LGD; restated the stem with the standard LGD/WCDR definition so one answer is correct.

Neo Bank holds a homogeneous loan portfolio with total exposure (EAD) of USD 350 million. The expected annual default rate is 2.5% and the recovery rate on defaulted loans is 60%. At the 99.9% confidence level, the worst-case annual default rate (WCDR) is estimated at 10.25%. Using the standard definition of credit VaR as the credit loss that will not be exceeded at the confidence level, what is the bank's one-year 99.9% credit VaR?

A. USD 3.50 million.
B. USD 10.85 million.
C. USD 14.35 million.
D. USD 21.53 million.

Answer: C

Key correction: Source key (B) said USD 10.85m by subtracting expected loss (unexpected-loss convention); under the GARP/Hull definition credit VaR = WCDR × EAD × LGD = 14.35m.

Step by step

1. LGD = 1 − recovery = 1 − 0.60 = 0.40.
2. Credit VaR = WCDR × EAD × LGD = 0.1025 × 350 × 0.40.
3. = USD 14.35 million.
4. Expected loss (0.025 × 350 × 0.40 = 3.50m) is not deducted.

Why C is correct: Credit VaR is the loss quantile: the 99.9% worst-case default rate times exposure times loss given default.

Why the others are wrong

- **A:** This is the expected loss (2.5% default rate), not the 99.9% worst-case loss.
- **B:** Subtracts expected loss (14.35 - 3.50); that is unexpected loss, not the credit VaR as defined by GARP.
- **D:** Used the recovery rate (60%) instead of LGD (40%): 0.1025 x 350 x 0.60.

Q429

BT notes P2.T6.305.3

REBUILT — Original EXCEPT item relied on Malz's unexpected-loss convention and unverified OCR dollar figures; replaced by a two-loan credit VaR calculation using WCDR x EAD x LGD.

A bank has two loans. Loan 1: EAD USD 200 million, LGD 45%, one-year 99.9% WCDR 8%, PD 2%. Loan 2: EAD USD 100 million, LGD 60%, one-year 99.9% WCDR 12%, PD 3%. Using the Vasicek/GARP approach of summing individual loan loss percentiles, what is the one-year 99.9% credit VaR?

A. USD 10.80 million.
B. USD 13.60 million.
C. USD 14.40 million.
D. USD 28.00 million.

Answer: C

Step by step

1. Loan 1: $0.08 \times 200 \times 0.45$ = USD 7.20m.
2. Loan 2: $0.12 \times 100 \times 0.60$ = USD 7.20m.
3. Credit VaR = 7.20 + 7.20 = USD 14.40 million.

Why C is correct: Credit VaR is the sum over loans of WCDR x EAD x LGD; PD is not used.

Why the others are wrong

- **A:** Deducted expected loss (200x0.02x0.45 + 100x0.03x0.60 = 3.6m) giving 10.8m; GARP credit VaR is not net of EL.
- **B:** Used (1 - LGD) instead of LGD: 0.08x200x0.55 + 0.12x100x0.40.
- **D:** Omitted LGD: 0.08x200 + 0.12x100 = 28m is the defaulted exposure, not the loss.

LO 28.c

Describe the use of rating transition matrices for calculating credit VaR.

Q430

AnalystPrep Q.6004

An investment manager is using the Credit Metrics model to assess the credit risk of a corporate bond portfolio. The portfolio includes bonds with varying credit ratings and maturities. In the context of the Credit Metrics model, which aspect is most critical for the investment manager to analyze in order to understand the potential shifts in the portfolio's credit risk over the upcoming year?

A. The historical dividend yield trends of the corporations issuing the bonds.
B. The probability of changes in credit ratings of the bonds over time.
C. The influence of foreign exchange rates on the bond issuers' financial stability.
D. The impact of changing commodity prices on the bond issuers' market value.

Answer: B

Step by step

1. CreditMetrics values the portfolio at the one-year horizon after migrating each bond's rating.
2. Migration is driven by the rating transition matrix: probabilities of upgrade, unchanged, downgrade and default.
3. So the key item for shifts in portfolio credit risk is the probability of rating changes.

Why B is correct: CreditMetrics revalues each bond under every possible year-end rating, weighted by transition probabilities, to build the loss distribution.

Why the others are wrong

- **A:** Dividend yield history is an equity measure and is not an input to rating migration.

- **C:** FX effects on issuers are not what CreditMetrics models; it works from rating migrations.
- **D:** Commodity price effects on issuer market value belong to structural/market models, not the transition-matrix approach.

Q431 AnalystPrep Q.6025

While analyzing a bond portfolio, a risk manager considers the inclusion of credit rating transition matrices in the calculation of Credit VaR. For what purpose are credit rating transition matrices primarily used in this context?

A. To monitor the real-time market value changes of the bond portfolio for intraday trading strategies.

B. To predict the exact future credit ratings of the bonds at the end of the one-year time horizon based on historical patterns.

C. To estimate the likelihood of credit rating changes over time, thereby factoring in the potential for defaults or downgrades.

D. To calculate the variance-covariance matrix for interest rate changes impacting the bond portfolio's market risks.

Answer: C

Step by step

1. A transition matrix gives, for each starting rating, the probability of ending in each rating (including default) over a horizon, usually one year.
2. These probabilities weight the possible downgrade/default outcomes in the credit VaR calculation.
3. Option C describes exactly this; B claims exact prediction, which a probability matrix cannot give.

Why C is correct: Transition matrices are probability tables of rating migration and default; they feed credit VaR with the likelihood of each outcome.

Why the others are wrong

- **A:** Intraday market value monitoring is a market-risk task, not the role of a one-year migration matrix.
- **B:** The matrix gives probabilities, not the exact future rating of a bond.
- **D:** Interest-rate covariance matrices come from market data, not rating transitions.

Q432 AnalystPrep Q.6029

REBUILT — Original options A, B and C were all defensible descriptions; rewrote the options so only the full conditional-migration definition is correct.

A bank uses a ratings transition matrix for its corporate bond portfolio to calculate credit VaR. What do the entries in a one-year ratings transition matrix represent?

A. For a bond starting in a given rating, the probability of default only, with no information about other ratings.

B. For a company starting in a given rating, the probability of ending the year in each rating category, including the same rating and default, estimated from historical data.

C. Forecasts of each bond's future rating based on the current economic outlook, updated each quarter.

D. The proportion of the portfolio's bonds currently held in each rating category.

Answer: B

Step by step

1. Rows of the matrix are starting ratings; columns are ending ratings (including default).
2. Each entry is a conditional probability of migrating from the row rating to the column rating over one year.
3. Each row sums to 100%, and the largest entry is usually on the diagonal (rating unchanged).
4. Estimates come from historical rating data, so B is the correct definition.

Why B is correct: A transition matrix is a set of conditional historical migration probabilities, row by row, covering unchanged, upgrade, downgrade and default.

Why the others are wrong

- **A:** Describes only the default column; the matrix also covers upgrades, downgrades and no change.
- **C:** The matrix is based on historical data, not a forward-looking economic forecast.
- **D:** This is the current portfolio composition, not conditional migration probabilities.

Q433 AnalystPrep Q.6031

REBUILT — Options A, C and D were vague or not credit VaR methods; rewrote them so Monte Carlo sampling of the matrix is the one clearly correct method.

To calculate credit VaR for BBB-rated bonds, a bank uses a one-year ratings transition matrix and a CreditMetrics-type approach. How is the matrix typically used?

A. Each trial samples year-end ratings from the BBB row, each bond is revalued at its sampled rating (or default value), and credit VaR is read from the resulting loss distribution.

B. The BBB default probability is multiplied by the face value of the bonds and by the recovery rate.

C. The probabilities of the downgrade scenarios are averaged to give an expected loss, which is reported as credit VaR.

D. The BBB row is applied only to rating upgrades, since downgrades are handled by spreads.

Answer: A

Step by step

1. Monte Carlo: in each trial, sample the year-end rating of each bond from its row of the matrix (with correlation between companies).
2. Revalue each bond at the sampled rating, or at its recovery value if defaulted.
3. Sum the portfolio gain/loss for each trial and repeat many times.
4. Credit VaR is the loss at the chosen percentile of the simulated distribution.

Why A is correct: The matrix supplies sampling probabilities for each trial; the simulated loss distribution, not a single expected loss, gives credit VaR.

Why the others are wrong

- **B:** Gives a rough default-loss figure that multiplies by recovery instead of 1 minus recovery, and ignores migrations and the tail of the distribution.
- **C:** An expected loss is a mean, not a percentile of the loss distribution, and ignoring unchanged and upgrade outcomes distorts it.
- **D:** The matrix covers all migrations including downgrades and default; excluding them removes the main loss drivers.

Q434 AnalystPrep Q.6032

During a financial analysis training session, an instructor discusses how to extend a one-year credit rating transition matrix to estimate risk over different timeframes. Which method would appropriately project the transition probabilities for a two-year period?

A. By computing the geometric mean of the one-year transition probabilities.

B. By applying exponential smoothing to one-year transition data and historical two-year transition trends.

C. By squaring the one-year transition matrix to estimate the transition matrix for a two-year period.

D. By updating the one-year transition matrix with recent default and recovery rates before applying the matrix to the second year.

Answer: C

Step by step

1. Assuming rating changes are independent across periods, two successive one-year migrations are compounded.
2. Two-year matrix = T x T = T^2.
3. So squaring the one-year matrix gives the two-year transition matrix.

Why C is correct: Matrix multiplication compounds one-year migrations, so the n-year matrix is the nth power of the one-year matrix.

Why the others are wrong

- **A:** A geometric mean averages values rather than compounding migration paths.
- **B:** Exponential smoothing is a forecasting technique, not a way to extend a transition matrix.
- **D:** Updating with default and recovery rates does not extend the horizon; it also confuses recovery (loss given default) with migration probabilities.

Q435 AnalystPrep Q.6034

A risk manager needs to calculate the Credit VaR for a bond portfolio, which includes only 'A' rated bonds, using a rating transition matrix. If the manager wants to adjust this one-year transition matrix to estimate Credit VaR over a three-month period, what approach should be taken?

A. The original one-year transition matrix should be divided by four to approximate the transitions for a three-month period.
B. The transition probabilities in the matrix should be recalculated based on empirical data specific to three-month intervals.
C. The fourth root of the one-year transition matrix should be taken to estimate the transition matrix for a three-month period.
D. A new matrix should be created by assuming that the one-year probabilities remain constant over any shorter period.

Answer: C
Step by step
1. Three months is 1/4 of a year, so four quarterly matrices multiplied together must equal the one-year matrix: Q^4 = T.
2. Therefore Q = T^(1/4), the fourth root of the one-year matrix.
3. Dividing probabilities by four does not preserve this relation.

Why C is correct: Because the n-year matrix is the nth power, a sub-year matrix is the corresponding root of the one-year matrix.
Why the others are wrong
- **A:** Dividing every entry by four breaks the row sums (the diagonal would be far below 100%) and ignores compounding.
- **B:** Recalculating from three-month data is possible, but the reading's method derives the shorter-period matrix from the one-year matrix by taking a root.
- **D:** Keeping one-year probabilities for three months overstates migration and default risk; shorter periods have lower migration probabilities.

Q436

BT answers P2.T6.24.17.3

REBUILT — Original stem had probabilities summing above 100% and a vague 'strategy' question; rebuilt as a clear question on adjusting a one-year matrix to other horizons.

A risk team holds A-rated bonds and uses a one-year ratings transition matrix to compute credit VaR. They need transition matrices for a five-year horizon and for a one-month horizon. Which approach is correct, assuming rating changes are independent across periods?

A. Five-year matrix: multiply the one-year probabilities by 5; one-month matrix: divide them by 12.
B. Five-year matrix: take the fifth root of the one-year matrix; one-month matrix: raise it to the twelfth power.
C. Five-year matrix: raise the one-year matrix to the fifth power; one-month matrix: take the twelfth root of the one-year matrix.
D. Use the one-year matrix unchanged for both horizons because migration probabilities do not depend on the horizon.

Answer: C
Step by step
1. Longer horizon: compound successive one-year migrations, so the five-year matrix is T^5.
2. Shorter horizon: twelve one-month matrices multiply to the one-year matrix, so the one-month matrix is T^(1/12).
3. Longer horizons reduce the chance of keeping the rating and raise default probability; shorter horizons do the reverse.

Why C is correct: Powers extend the horizon and roots shorten it, under the independence (Markov) assumption.
Why the others are wrong
- **A:** Scaling probabilities linearly ignores compounding and can break the row sums (a 5x scale would exceed 100%).
- **B:** Roots and powers are reversed: the root shortens the horizon, the power lengthens it.
- **D:** Migration probabilities depend strongly on the horizon; for example default probability rises with time.

Q437 BT notes P2.T6.921.1

REBUILT — Replaced the weak distractor $2,250,000 with $2,652,000 (recovery rate used as loss given default) so each wrong option reflects a realistic error.

The following simplified credit rating transition matrix (aka, migration matrix) displays one-year conditional probabilities for only two credits (A and B). For example, the A-rated credit has an 80.0% probability of remaining A-rated at the end of the year and a 20.0% probability of being downgraded to B-rated but is not expected to default within one year. From year to year, migrations are independent; i.e., the matrix satisfies the Markov property.

Transition Matrix (T)

A	B	D	
A	80.0%	20.0%	0.0%
B	10.0%	70.0%	20.0%
D	0.0%	0.0%	100.0%

A bank has extended a three-year $15.0 million loan to a B-rated corporate borrower. The bank assumes the exposure at default (EAD) is the principal amount of $15.0 million and estimates a 40.0% recovery rate. If the relevant default probability is the three-year cumulative default probability, then what is the expected loss (EL)?

A. $1,800,000
B. $2,652,000
C. $3,978,000
D. $4,392,000

Answer: C

Step by step

1. Cube the transition matrix T (rating changes independent across years). The B row of T^3 is [17.1%, 38.7%, 44.2%], so the three-year cumulative PD = 44.20%.
2. Path check: B>D 20.0% + B>B>D 14.0% + B>B>B>D 9.8% + B>A>B>D 0.4% = 44.2%.
3. LGD = 1 - 40% = 60%.
4. EL = 15,000,000 x 44.20% x 60% = $3,978,000.

Why C is correct: Three-year cumulative PD comes from the cube of the one-year matrix, and EL = EAD x PD x (1 - RR).

Why the others are wrong

- **A:** 15m x 20% x 60% = $1.8m uses the one-year PD rather than the three-year cumulative PD.
- **B:** 15m x 44.2% x 40% = $2.652m multiplies by the recovery rate instead of the loss given default (1 - RR).
- **D:** 15m x 48.8% x 60% = $4.392m uses 1-(1-20%)^3, which assumes a constant 20% annual PD and ignores that a B bond may migrate to A (which cannot default directly within a year).

Q438 Schweser Module Quiz 28.1 Q3

Historical data shows that over a one-year period, there is a 91.93% chance that a company rated BBB will keep its rating. What is the percentage chance that this rating will remain unchanged over a four-year period?

A. 36.77%.
B. 67.72%.
C. 71.42%.
D. 95.97%.

Answer: C

Step by step

1. Assuming independent rating changes across years, the four-year probability of keeping the rating is the one-year probability raised to the 4th power (as in the reading).
2. 0.9193^4 = 0.7142.
3. So the probability is 71.42%.

Why C is correct: Compounding the same-rating probability over four independent years gives 0.9193^4.

Why the others are wrong

- **A:** 36.77% is far too low and is not the result of compounding the one-year probability.
- **B:** 67.72% = 1 - 4 x 8.07% subtracts the one-year migration probability linearly instead of compounding.
- **D:** 95.97% is higher than the one-year probability, which is impossible since the chance of keeping a rating falls as the horizon lengthens; it comes from shrinking the migration probability instead of compounding.

LO 28.d

Describe the application of Vasicek's model to estimate capital requirements under the Basel II internal- ratings-based (IRB) approach.

Q439

AnalystPrep Q.3238 · AnalystPrep Q.3239

REBUILT — Merged two near-identical Kimpala items; replaced giveaway options (e.g. 0.8784, 0.1216) with distractors from realistic formula errors and used unrounded z-values (answer about 9.1%).

Kimpala Leasing Bank holds \$690 million of retail loans. The bank's actuary estimates a one-year probability of default PD = 1% and a loss given default LGD = 40%. Using Vasicek's model as in the Basel II IRB approach with credit correlation $\rho = 0.1216$, what are the worst-case default rate (WCDR) at the 99.9% confidence level and the expected loss (EL)?

A. WCDR about 9.1%; EL \$2.76 million
B. WCDR about 9.1%; EL \$25.2 million
C. WCDR about 8.1%; EL \$2.76 million
D. WCDR about 9.1%; EL \$6.90 million

Answer: A

Step by step

1. $N^{-1}(0.01) = -2.326$ and $N^{-1}(0.999) = 3.090$.
2. $\text{WCDR} = N\left[\dfrac{N^{-1}(\text{PD}) + \sqrt{\rho}\,N^{-1}(X)}{\sqrt{1-\rho}}\right]$, with $\sqrt{0.1216} = 0.3487$ and $\sqrt{1-0.1216} = 0.9372$.
3. Argument $= (-2.326 + 0.3487 \times 3.090)/0.9372 = -1.332$.
4. $\text{WCDR} = N(-1.332) \approx 0.091$, i.e. about 9.1% (rounding the z-values to -2.33 and 3.09 gives 9.07%).
5. $\text{EL} = \text{EAD} \times \text{LGD} \times \text{PD} = 690 \times 0.40 \times 0.01 = \$2.76$ million.

Why A is correct: Vasicek's formula converts the average PD into the default rate that is exceeded only 0.1% of the time, using the correlation to scale the systematic factor; expected loss uses the unstressed PD.

Why the others are wrong

- **B:** Computes $690 \times 0.40 \times 0.0914 \approx \$25.2$ million, which is the 99.9th percentile loss, not the expected loss (used WCDR instead of PD).
- **C:** Subtracts PD from WCDR (0.0914 - 0.01), i.e. the unexpected default rate, rather than reporting the WCDR itself.
- **D:** Forgets LGD in EL: $690 \times 0.01 = \$6.9$ million is expected default exposure, not expected loss.

Q440

AnalystPrep Q.6024

REBUILT — Reworded stem for precision and replaced the 'multifactor Gaussian copula' option with CreditMetrics.

A junior risk analyst is asked which model is used to calculate the high-percentile default rates that determine regulatory capital under the Basel II internal-ratings-based approach. Which model should the analyst identify?

A. CreditMetrics with Monte Carlo simulation
B. Vasicek's Gaussian copula model
C. CreditRisk+
D. Historical simulation

Answer: B

Step by step

1. Basel II IRB capital uses the worst-case default rate at 99.9%.
2. That WCDR comes from Vasicek's one-factor Gaussian copula model.
3. CreditMetrics and CreditRisk+ are internal credit VaR methods, and historical simulation is not the regulatory method.

Why B is correct: Vasicek's model gives WCDR(T,X) from PD, correlation and confidence level, which is the IRB regulatory capital building block.

Why the others are wrong

- **A:** CreditMetrics models rating migration by simulation and is an internal credit VaR approach, not the Basel IRB formula.
- **C:** CreditRisk+ uses an insurance-style Poisson/negative binomial default model, not the regulatory formula.
- **D:** Historical simulation is a market risk technique and is not prescribed for IRB capital.

Q441 AnalystPrep Q.6038

A bank applies Vasicek's model to a commercial loan portfolio to estimate Basel II IRB capital. How does the credit correlation parameter ρ affect the capital requirement?

A. A higher ρ lowers capital, since defaults are more diversified across the portfolio.
B. A higher ρ lowers capital, since it implies better management of default risk.
C. A lower ρ raises capital, since default patterns are less predictable.
D. A higher ρ raises capital, since defaults are more likely to occur simultaneously.

Answer: D

Step by step

1. WCDR $= N[(N^{-1}(\text{PD}) + \sqrt{\rho}\,N^{-1}(X))/\sqrt{1-\rho}]$ with $N^{-1}(X) > 0$ at X = 99.9%.
2. A larger ρ raises the numerator's positive term and shrinks the denominator, so WCDR rises.
3. A higher WCDR means more capital; the intuition is that defaults cluster.

Why D is correct: Higher correlation means less diversification and a fatter tail of joint defaults, so the 99.9% default rate and capital increase.

Why the others are wrong

- **A:** Reverses the relationship; higher correlation means less, not more, diversification.
- **B:** Correlation is not a measure of management quality; higher ρ raises WCDR.
- **C:** Reverses the direction: lower ρ reduces WCDR and capital.

Q442 AnalystPrep Q.6040

A risk manager explains Vasicek's model with PD = 2%, $\rho = 12\%$ and confidence level X = 99.9%. What role does X play in determining capital under the model?

A. It sets the expected loss the capital allocation must cover.
B. It determines the worst-case default rate the bank must be able to withstand.
C. It is the minimum percentage of exposure that must be held as capital.
D. It dictates the degree of diversification the loan portfolio must achieve.

Answer: B

Step by step

1. X enters the formula through $N^{-1}(X)$, which scales the systematic-factor shock.
2. It selects the percentile of the default rate distribution, i.e. WCDR(T,X).
3. Expected loss uses PD, not X, and diversification is not set by X.

Why B is correct: The confidence level picks the percentile of the default rate distribution that defines the worst-case default rate.

Why the others are wrong

- **A:** Expected loss depends on PD, LGD and EAD only; X concerns the tail beyond expected loss.
- **C:** X is a probability level, not a capital ratio; capital also depends on LGD and EAD.
- **D:** Diversification is reflected in the correlation parameter, not in X.

Q443 AnalystPrep Q.6041

REBUILT — Clarified that the auditor is independently recomputing WCDR so that only the model inputs are the defensible answer.

A regulatory auditor wants to independently recompute the worst-case default rate (WCDR) that a bank derived from Vasicek's model for its corporate loan portfolio. Which inputs does the auditor need?

A. The bank's rating-grade structure and the mapping of ratings to PDs only.
B. The probability of default, the credit correlation, the confidence level, and how the first two were estimated.
C. Macroeconomic forecasts and stress scenarios used to adjust PDs and correlations.
D. Back-testing reports comparing the model with actual defaults over recent years.

Answer: B

Step by step

1. WCDR $= N[(N^{-1}(\text{PD}) + \sqrt{\rho}\,N^{-1}(X))/\sqrt{1-\rho}]$.
2. The only inputs are PD, the correlation and X (plus T).
3. Hence recomputing needs those three and the basis for PD and correlation.

Why B is correct: The formula is fully determined by PD, correlation and confidence level, so these plus their estimation methods are what is needed.

Why the others are wrong

- **A:** Rating-to-PD mapping informs PD but omits the correlation and confidence level.
- **C:** Stress scenarios may inform inputs but are not required to recompute WCDR.
- **D:** Back-testing assesses model performance, not the calculation of WCDR.

Q444

AnalystPrep Q.6543

REBUILT — Original asked which parameter 'directly determines' sensitivity to rho; PD also interacts with rho, so the key (confidence level) was not defensible. Rewrote as a clear conceptual question on the role of rho with computed values.

Vasicek's model gives $\text{WCDR}(T,X) = N\left[\frac{N^{-1}(\text{PD})+\sqrt{\rho}\,N^{-1}(X)}{\sqrt{1-\rho}}\right]$. A bank has a large portfolio with one-year PD of 2.0% and sets X = 99.9%. Evaluating the formula gives WCDR = 2.00% when $\rho = 0$ and WCDR = 22.63% when $\rho = 0.20$. Which statement about the role of the credit correlation ρ is correct?

A. At $\rho = 0$ the worst-case default rate equals the PD, and it rises as ρ increases because defaults become more clustered in the tail.

B. The worst-case default rate falls as ρ increases because higher correlation means greater diversification.

C. At $\rho = 0$ the worst-case default rate is already far above the PD because X is so high.

D. Once X is fixed at 99.9%, the worst-case default rate no longer depends on ρ.

Answer: A

Step by step

1. Set $\rho = 0$: $\text{WCDR} = N[N^{-1}(0.02)] = 2.00\%$, the PD itself.
2. Set $\rho = 0.20$: $N^{-1}(0.02) = -2.0537$, $\sqrt{0.2} \times 3.0902 = 1.3820$, numerator $= -0.6717$, divide by $\sqrt{0.8} = 0.8944$ gives -0.7510; $N(-0.7510) = 22.63\%$.
3. So WCDR increases with ρ (the $\sqrt{\rho}N^{-1}(X)$ term is positive for X > 50%), which reflects defaults clustering in a bad state of the economy.

Why A is correct: With $\rho = 0$ defaults are independent and a large portfolio's default rate converges to PD; correlation adds the positive $\sqrt{\rho}N^{-1}(X)$ term that raises the tail default rate.

Why the others are wrong

- **B:** Reverses the effect: higher correlation reduces diversification, so tail defaults increase (2.00% to 22.63%).
- **C:** With no correlation the $N^{-1}(X)$ term vanishes, so X has no effect and WCDR = PD = 2.00%.
- **D:** The ρ terms enter both the numerator and denominator, so WCDR changes with ρ at any X (2.00% vs 22.63% at the same X).

Q445

BT answers P2.T6.24.18.1

REBUILT — Original 'all correct EXCEPT' with Roman-numeral combinations was ambiguous (statement V is actually correct per GARP, only IV is false, and key set C included V). Rebuilt as a single-statement 'incorrect' item.

A bank uses Vasicek's Gaussian copula model for its loan portfolio. Which of the following statements about the model is INCORRECT?

A. The Xth percentile loss for a large portfolio of small loans is approximated by $\sum_i \text{WCDR}_i(T,X) \times \text{EAD}_i \times \text{LGD}_i$.

B. The credit correlation ρ can be estimated from the average correlation between equity returns of the companies in the portfolio (or of similar publicly traded companies if they are private).

C. The model is used to set the interest rate charged on new loans by assessing the portfolio's expected profit.

D. For regulatory capital the sum is evaluated with T = one year and X = 99.9%.

Answer: C

Key correction: Source key (C = III, IV, V) was inconsistent: statements III and V are correct, only IV is incorrect; rebuilt so the unique false statement is the answer.

Step by step

1. A: Gordy's result extends the single-loan formula to a large portfolio of small loans as the sum of WCDR x EAD x LGD - correct.
2. B: structural models imply ρ is about the correlation of asset returns, approximated by equity-return correlation - correct.
3. D: the Basel IRB capital is the summation with T = 1 year, X = 99.9% (times a maturity adjustment) - correct.
4. C: Vasicek's model estimates a high-percentile credit loss (capital); it is not a loan pricing model - incorrect.

Why C is correct: The model's application is estimating a worst-case loss percentile and hence capital, not pricing.

Why the others are wrong

- **A:** Correct statement of the portfolio loss approximation, so not the exception.
- **B:** Correct; equity-return correlation is the practical proxy for rho (the source wrongly treated this as false in its V).
- **D:** Correct; the regulatory parameters are T = 1 year and X = 99.9%.

Q446

BT answers P2.T6.311.2 · BT notes P2.T6.311.2

REBUILT — Source options had a correct 'unexpected loss' answer ($6.845m) that conflicts with GARP, where credit VaR is WCDR x EAD x LGD with no EL deduction; restated the question and options accordingly.

A bank holds a highly granular $100.0 million portfolio of retail exposures with an average one-year default probability of 2.0%, an average recovery rate of 50.0%, and a copula correlation of 0.130. Using Vasicek's model, what is the 99.9% one-year credit VaR (that is, WCDR × EAD × LGD)?

A. $6.845 million
B. $7.845 million
C. $8.845 million
D. $15.689 million

Answer: B

Key correction: Source key (B) of $6.845m subtracted expected loss ('Malz definition'); under the GARP/Hull definition used in this reading, credit VaR = $7.845m.

Step by step

1. $N^{-1}(0.02) = -2.0537$ and $N^{-1}(0.999) = 3.0902$.
2. Numerator $= -2.0537 + \sqrt{0.13} \times 3.0902 = -2.0537 + 1.1143 = -0.9394$; with $\sqrt{0.87} = 0.9327$ the ratio is -1.0073.
3. WCDR $= N(-1.0073) = 15.689\%$.
4. LGD = 1 - 0.50 = 50%, so VaR $= 100 \times 0.15689 \times 0.50 = \$7.845$ million.

Why B is correct: Credit VaR in this framework is the worst-case default rate times exposure times loss given default.

Why the others are wrong

- **A:** Subtracts expected loss (2% x 50% x $100m = $1.0m); the GARP/Hull credit VaR is the full percentile loss, not the unexpected loss.
- **C:** Adds the $1.0m expected loss instead of ignoring it.
- **D:** Forgets to multiply by LGD (100 x 15.689%).

Q447

Schweser Module Quiz 28.2 Q1

REBUILT — Replaced the unrelated ROE figures (8% and 6%) and 'ROE correlation' with equity-return correlation, which is the GARP proxy for rho.

Two publicly traded companies, A and B, are in a loan portfolio. The average correlation between their equity returns is 0.24. The credit correlation ρ most likely used in Vasicek's Gaussian copula model is closest to:

A. 0.12
B. 0.24
C. 0.48
D. 0.76

Answer: B

Step by step

1. Structural models imply the credit correlation equals the correlation between the companies' asset returns.
2. As an approximation this equals the correlation between equity returns, which is given as 0.24.
3. So $\rho \approx 0.24$.

Why B is correct: Vasicek's rho is estimated directly from the average correlation of equity returns of the companies (or similar public firms).

Why the others are wrong

- **A:** Halves the correlation (0.24 / 2); there is no such scaling.
- **C:** Doubles the correlation (0.24 x 2).
- **D:** Uses 1 - 0.24 (the complement).

Q448 AnalystPrep Q.6048

REBUILT — Source options were unitless and option B (46.61) did not match the exact result 46.63; added units, corrected the number and replaced a weak distractor.

A bank holds a portfolio of mortgage-backed securities worth $300 million. The average one-year probability of default is 2.5%, the average recovery rate is 40%, and the copula correlation is 0.20. Using Vasicek's model, what is the one-year credit VaR at the 99.9% confidence level?

A. $22.30 million
B. $46.63 million
C. $31.09 million
D. $42.13 million

Answer: B

Step by step

1. $N^{-1}(0.025) = -1.9600$, $N^{-1}(0.999) = 3.0902$, $\sqrt{0.2} = 0.4472$, $\sqrt{0.8} = 0.8944$.
2. Ratio $= (-1.9600 + 1.3820)/0.8944 = -0.6462$, so WCDR $= N(-0.6462) = 25.91\%$.
3. LGD = 1 - 0.40 = 0.60.
4. VaR $= 300 \times 0.2591 \times 0.60 =$ $46.63 million.

Why B is correct: Credit VaR = WCDR x EAD x LGD with the recovery rate converted to LGD.

Why the others are wrong

- **A:** No consistent calculation; it understates the tail loss (not WCDR x EAD x LGD).
- **C:** Uses LGD = 40% (the recovery rate) instead of 60%: 300 x 25.91% x 0.40 = 31.09.
- **D:** Subtracts expected loss (2.5% x 60% x 300 = 4.5) from the VaR.

Q449 AP Mock 3 Q65

REBUILT — Rebuilt options (source had 43.2 and others without units); retained the same inputs.

A bank has USD 120 million of commercial loans in a granular portfolio. The average one-year probability of default is 2%, the average recovery rate is 40%, and the copula correlation is 0.25. Using Vasicek's model with X = 99.9% and T = one year, the regulatory capital (the Xth percentile loss, WCDR × EAD × LGD) is closest to:

A. USD 13.37 million
B. USD 18.61 million
C. USD 20.05 million
D. USD 33.42 million

Answer: C

Key correction: Source key (C, 18.61) subtracted expected loss; under GARP/Hull the capital is the full percentile loss WCDR x EAD x LGD = 20.05.

Step by step

1. $N^{-1}(0.02) = -2.0537$; $\sqrt{0.25} \times 3.0902 = 1.5451$; numerator $= -0.5086$; $\sqrt{0.75} = 0.8660$ gives -0.5873.
2. WCDR $= N(-0.5873) = 27.85\%$.
3. LGD = 1 - 0.40 = 0.60.
4. Capital $= 120 \times 0.2785 \times 0.60 =$ USD 20.05 million.

Why C is correct: In this reading regulatory capital equals the 99.9% one-year worst-case loss, WCDR x EAD x LGD, with no deduction for expected loss.

Why the others are wrong

- **A:** Uses LGD = 40% (the recovery rate): 120 x 27.85% x 0.40 = 13.37.
- **B:** Deducts expected loss (PD x LGD x EAD = 1.44): 120 x (27.85% - 2%) x 0.60 = 18.61.
- **D:** Forgets LGD: 120 x 27.85% = 33.42.

Q450 BT answers P2.T7.517.3 · BT notes P2.T7.517.3

REBUILT — Original relied on a garbled PD-dependent correlation formula and Basel 12.5 RWA scaling not in this reading; fixed rho = 0.19 and asked for capital via WCDR x EAD x LGD x maturity adjustment.

A bank holds $100.0 million of loans to A-rated corporations in a granular portfolio. The one-year PD is 0.90%, LGD is 75%, the copula correlation is 0.19, and the loans' average maturity gives a maturity adjustment factor of 1.270. Under the IRB approach, using Vasicek's model with T = 1 year and X = 99.9%, the regulatory capital for these loans is closest to:

A. $9.66 million

B. $11.42 million
C. $12.27 million
D. $16.37 million

Answer: C
Step by step
1. $N^{-1}(0.009) = -2.3656$; $\sqrt{0.19} \times 3.0902 = 1.3470$; numerator $= -1.0186$; $\sqrt{0.81} = 0.9$ gives -1.1318.
2. WCDR $= N(-1.1318) = 12.886\%$.
3. Capital = WCDR × EAD × LGD × MA $= 0.12886 \times 100 \times 0.75 \times 1.270 =$ $12.27 million.

Why C is correct: Regulatory capital is the 99.9% one-year worst-case loss multiplied by the maturity adjustment factor.
Why the others are wrong
- **A:** Omits the maturity adjustment (9.66 = 100 x 0.75 x 12.886%).
- **B:** Deducts expected loss (0.90%) before applying the maturity adjustment.
- **D:** Forgets LGD: 100 x 12.886% x 1.270 = 16.37.

Q451

GARP PE1 Q19 · Mock 3 Q61

REBUILT — Original gave pre-computed worst-case default rates and deducted expected loss, which conflicts with the reading's Vasicek capital (WCDR x EAD x LGD); rebuilt to require the Vasicek WCDR calculation from PD and correlation.

A bank estimates credit capital for a large, granular portfolio of equally sized exposures using Vasicek's Gaussian copula model, with capital defined as the one-year, 99.9% percentile of the loss distribution (ignore any maturity adjustment). The portfolio data are:

Item	Value
Total exposure at default	EUR 200 million
1-year probability of default (PD)	2.0%
Credit correlation ρ	0.12
Loss given default (LGD)	60%

Useful values: $N^{-1}(0.02) = -2.054$, $N^{-1}(0.999) = 3.090$, $N^{-1}(0.95) = 1.645$.
What is the capital requirement?

A. EUR 6.82 million
B. EUR 15.27 million
C. EUR 17.67 million
D. EUR 29.46 million

Answer: C
Key correction: Original sources (answer C, EUR 9.62m) deducted expected loss from the 99.9% loss; under the reading, the Xth percentile of loss is WCDR x EAD x LGD with no deduction. The rebuilt item avoids the issue.
Step by step
1. $\text{WCDR} = N\left(\dfrac{N^{-1}(\text{PD}) + \sqrt{\rho}\,N^{-1}(X)}{\sqrt{1-\rho}}\right)$
2. Numerator: $-2.054 + \sqrt{0.12} \times 3.090 = -2.054 + 0.3464 \times 3.090 \approx -0.984$
3. Denominator: $\sqrt{1 - 0.12} = 0.9381$, so argument ≈ -1.049
4. WCDR $= N(-1.049) \approx 14.7\%$ (14.73% with unrounded inputs)
5. Loss percentile = 0.1473 × 200 × 0.60 = EUR 17.67 million

Why C is correct: Vasicek converts the 2% average PD into a 99.9% worst-case default rate of 14.73% given correlation 0.12; multiplying by EAD and LGD gives the 99.9% loss, which is the capital.
Why the others are wrong
- **A:** Uses the 95% percentile ($N^{-1}(0.95) = 1.645$, WCDR 5.68%) instead of 99.9%, giving 120 x 5.68% = 6.82.
- **B:** Deducts the expected loss (200 x 0.6 x 2% = 2.4) from the 99.9% loss; capital here is defined as the full percentile of loss, WCDR x EAD x LGD.
- **D:** Omits LGD (treats the whole exposure as lost): 200 x 14.73% = 29.46.

LO 28.e

Explain the Vasicek's model, Credit Risk Plus (CreditRisk+) model, and the CreditMetrics ways of estimating the probability distribution of losses arising from defaults as well as modeling the default correlation.

Q452

AP Mock 2 Q9

REBUILT — Option B bundled an unclear 'time to default' clause; reworded so the correct option states the one-factor copula and single correlation parameter cleanly.

A portfolio manager is exploring models for the probability distribution of losses from defaults and for credit correlation in a debt portfolio. How does Vasicek's model conceptualize default losses and default correlation?

A. Losses arise only from defaults, and default events are correlated through a multiple-factor Gaussian copula.

B. Losses arise only from defaults, and default correlation is captured by a one-factor Gaussian copula with a single credit (asset) correlation parameter.

C. Defaults follow a Poisson distribution, and losses are quantified without explicitly modeling default correlation.

D. Defaults are treated as independent events, so the loss distribution is estimated separately for each borrower.

Answer: B

Step by step

1. Vasicek's model considers only defaults (no downgrades).
2. It uses a one-factor Gaussian copula: one common systematic factor plus an idiosyncratic factor per borrower, so correlation is summarised by a single parameter ρ.
3. The worst case default rate is $\text{WCDR}(T,X) = N\left[\frac{N^{-1}(PD)+\sqrt{\rho}\,N^{-1}(X)}{\sqrt{1-\rho}}\right]$, with ρ approximated by the correlation of asset (or equity) returns.
4. Only B matches both features: default-only losses and a one-factor copula.

Why B is correct: Vasicek's model ties all borrowers to one systematic factor through the Gaussian copula, so default correlation is governed by the single parameter ρ.

Why the others are wrong

- **A:** Right on defaults-only, but the copula is one-factor, not multiple-factor; this mixes up Vasicek with richer factor models.
- **C:** Poisson default counts describe Credit Risk Plus, not Vasicek.
- **D:** Independence is the opposite of Vasicek's premise; the model exists to capture credit correlation.

Q453

AnalystPrep Q.5997

REBUILT — Original distractors (stock volatility, dividends, interest rates) were giveaways; replaced with inputs of the other two LO 28.e models.

A bank uses CreditMetrics to assess how the credit quality of its bond portfolio could change over the next year, including downgrades as well as defaults. Which input is most central to this model?

A. The volatility of the gamma-distributed expected number of defaults.

B. The worst case default rate at the 99.9% level, computed from each bond's PD and a single correlation parameter.

C. A rating transition matrix (internal or from rating agencies), with the credit spread term structure for each rating category.

D. The Poisson mean number of defaults per year for each rating bucket.

Answer: C

Step by step

1. CreditMetrics accounts for downgrades as well as defaults, so it needs a rating transition matrix giving one-year migration probabilities.
2. Monte Carlo trials sample each counterparty's year-end rating; non-default outcomes are revalued using the credit spread term structure for that rating.
3. Gamma-distributed default counts belong to Credit Risk Plus; the WCDR formula belongs to Vasicek.

Why C is correct: The transition matrix (with rating-specific spread curves) is what lets CreditMetrics capture rating migrations, not just defaults.

Why the others are wrong

- **A:** Gamma volatility of the default rate is a Credit Risk Plus input.
- **B:** WCDR from PD and rho is Vasicek's approach, which looks only at defaults.

- **D:** A Poisson default mean is a Credit Risk Plus input and ignores downgrades.

Q454 AnalystPrep Q.6547 · AnalystPrep Q.5999

Which feature distinguishes CreditMetrics from both Vasicek's model and Credit Risk Plus?

A. It ignores credit rating migrations and focuses solely on default events.
B. It incorporates mark-to-market revaluations due to credit downgrades.
C. It assumes fixed recovery rates for all obligors in the portfolio.
D. It uses a Poisson distribution to model default frequency.

Answer: B

Step by step

1. Vasicek's model and Credit Risk Plus estimate loss distributions from defaults only; downgrades are not considered.
2. CreditMetrics uses a rating transition matrix and credit spreads, so a downgrade lowers the revalued position and creates a loss even without default.
3. Therefore the distinguishing feature is revaluation for downgrades (B).

Why B is correct: Only CreditMetrics models rating migrations and the resulting value changes, in addition to defaults.

Why the others are wrong

- **A:** Reversed: ignoring migrations describes Vasicek and Credit Risk Plus.
- **C:** CreditMetrics does not assume a fixed recovery rate (recovery is sampled in simulation), and fixed recovery is not what separates it from the others.
- **D:** Poisson default frequency is Credit Risk Plus.

Q455 AnalystPrep Q.6007

In the Credit Risk Plus model, which distribution is used for the number of defaults when there is uncertainty about the default rate?

A. Poisson distribution
B. Gamma distribution
C. Negative binomial distribution
D. Normal distribution

Answer: C

Step by step

1. With a known small default probability and many loans, the number of defaults is Poisson.
2. If the expected number of defaults is itself uncertain and assumed gamma-distributed (mean μ, standard deviation σ), the Poisson becomes a negative binomial distribution.
3. As σ falls the negative binomial converges to the Poisson; as σ rises the chance of many defaults rises.

Why C is correct: Mixing a Poisson count with a gamma-distributed expected default number yields a negative binomial for the number of defaults.

Why the others are wrong

- **A:** Poisson applies only when the default rate is known/fixed.
- **B:** Gamma describes the uncertain expected number of defaults, not the default count itself.
- **D:** A normal distribution cannot capture the positively skewed, fat-tailed default count.

Q456 AnalystPrep Q.6043

REBUILT — Original option D (Monte Carlo simulation) is a technique, not a model, and the stem was loose; reworded around the insurance-style analytics and replaced D.

A head of risk wants a loan-portfolio default loss model that relies on analytic approximations well established in the insurance industry, with default counts Poisson-distributed and no copula linking borrowers. Which model fits?

A. Vasicek's model
B. CreditMetrics
C. Credit Risk Plus

D. Historical simulation

Answer: C

Step by step

1. Credit Risk Plus (Credit Suisse Financial Products) uses analytic approximations from the insurance industry.
2. Defaults are Poisson (negative binomial if the rate is uncertain); no Gaussian copula is used to link borrowers.
3. Vasicek and CreditMetrics both rely on a Gaussian copula.

Why C is correct: Credit Risk Plus is the actuarial-style, Poisson-based model without an explicit copula.

Why the others are wrong

- **A:** Vasicek uses a one-factor Gaussian copula with correlation parameter ρ.
- **B:** CreditMetrics uses a Gaussian copula with equity-return correlations and simulation.
- **D:** Historical simulation is the market VaR tool, not a credit default-loss model with Poisson counts.

Q457

AnalystPrep Q.6044

Which of the following statements is correct about the modeling of default correlation in CreditMetrics and Vasicek's model?

A. Vasicek's model calculates default correlation using multiple Gaussian copulas to reflect varying economic conditions across sectors.

B. CreditMetrics employs a single-factor model that simplifies the correlation of defaults to a common economic influence affecting all entities equally.

C. Vasicek's model uses a single systematic factor that represents overall economic conditions to model default correlations, affecting all entities in the portfolio.

D. CreditMetrics uses sector-based risk adjustments in addition to a Gaussian copula for modeling the default correlations specifically tied to individual industries.

Answer: C

Step by step

1. Vasicek: one Gaussian copula, one systematic factor plus idiosyncratic noise, one correlation parameter.
2. CreditMetrics: Gaussian copula on rating transitions with the correlation between two companies set to their equity-return correlation (pairwise, not one common factor).
3. Only C describes these correctly.

Why C is correct: Vasicek's default correlation comes from a single systematic factor affecting every borrower.

Why the others are wrong

- **A:** Vasicek uses one copula and one factor, not multiple copulas by sector.
- **B:** Attributes Vasicek's single-factor structure to CreditMetrics, which uses pairwise equity-return correlations.
- **D:** CreditMetrics has no sector-based adjustment layer in this treatment; it uses the copula with equity correlations.

Q458

AnalystPrep Q.6206

REBUILT — Option D (transition matrices built from internal default rates) was arguable since internal ratings may be used; replaced with a clearly false CreditMetrics statement and tightened the stem.

Which of the following statements about credit VaR models is accurate?

A. CreditMetrics uses a Gaussian copula to calculate the exposure at default of each counterparty from correlated equity returns.

B. Credit Risk Plus estimates default probability with a Gaussian copula that combines individual and systematic risk factors.

C. Vasicek's model applies a Gaussian copula to estimate the worst case default rate (WCDR) of a loan portfolio.

D. CreditMetrics assumes that rating changes of different counterparties are independent.

Answer: C

Step by step

1. Vasicek: Gaussian copula gives WCDR(T,X); portfolio loss percentile = sum of WCDR x EAD x LGD.
2. CreditMetrics: the copula correlates rating transitions (via equity returns), not EAD.
3. Credit Risk Plus: Poisson/negative binomial default counts, no Gaussian copula.
4. Hence C.

Why C is correct: WCDR from a Gaussian copula is precisely Vasicek's contribution.
Why the others are wrong
- **A:** The copula drives joint rating transitions; EAD is an exposure input, not an output.
- **B:** Credit Risk Plus uses Poisson/negative binomial, not a Gaussian copula.
- **D:** CreditMetrics explicitly models correlated transitions through the Gaussian copula.

Q459

AnalystPrep Q.6544 · AnalystPrep Q.6548

In CreditMetrics, how is the correlation between the rating transitions of two companies typically modeled?

A. By assuming independence between all rating transitions
B. Using a Gaussian copula with the correlation set equal to the correlation between the companies' equity returns
C. By directly linking default probabilities to macroeconomic factors
D. Through sector-specific Poisson distributions

Answer: B
Step by step
1. A joint distribution of rating changes is built with a Gaussian copula.
2. Two standard normal variables are sampled with correlation equal to the equity-return correlation of the two companies (e.g., 0.2).
3. Each variable is mapped to a new rating through the transition matrix thresholds.

Why B is correct: Equity-return correlation proxies credit correlation and drives the Gaussian copula for rating transitions.
Why the others are wrong
- **A:** Independence is not assumed; ratings momentum and common economic conditions make transitions dependent.
- **C:** Macro factors are not modeled directly in CreditMetrics.
- **D:** Poisson distributions belong to Credit Risk Plus.

Q460

BT answers P2.T6.24.18.2 · BT notes P2.T6.24.18.2

REBUILT — Original asked which team member 'best aligns' with a subjective need (Vasicek is the actual Basel basis, so the keyed answer was debatable) and Bob's statement mixed correct and incorrect claims; rebuilt as a single-correct-statement item on the three models.

A bank's risk team is comparing the Vasicek model, CreditRisk+ and CreditMetrics for estimating the one-year distribution of credit losses. Which of the following statements is most accurate?

A. Vasicek's model and CreditRisk+ both capture losses from rating downgrades as well as from defaults.
B. In CreditRisk+, greater uncertainty (volatility) in the default rate raises the probability of a large number of defaults and gives a positively skewed loss distribution.
C. In CreditRisk+, as the volatility of the default rate decreases, the distribution of the number of defaults moves further away from the Poisson distribution.
D. CreditMetrics models rating changes of different counterparties as independent, so no copula is needed.

Answer: B
Key correction: Source key (BT) said C (Clara); the original item was subjective since Alice's Vasicek statement is also valid, so it was replaced by an unambiguous statement question.
Step by step
1. Vasicek and CreditRisk+ consider defaults only; only CreditMetrics adds downgrades via a ratings transition matrix, so A is false.
2. CreditRisk+ allows the default rate to be uncertain (negative binomial). As volatility rises, defaults become more clustered (higher effective default correlation), large default counts become more likely and the loss distribution is positively skewed. B is true.
3. As the default-rate volatility $\sigma \to 0$, the negative binomial converges to the Poisson distribution, the opposite of C.
4. CreditMetrics links rating changes across counterparties with a Gaussian copula (correlation set from equity returns), so D is false.

Why B is correct: Default-rate uncertainty creates correlation among defaults, fattening the right tail of the loss distribution and making it positively skewed.
Why the others are wrong

- **A:** Only CreditMetrics handles downgrades; this confuses the three models' scope.
- **C:** Reverses the relationship: lower volatility moves the negative binomial toward (not away from) the Poisson.
- **D:** CreditMetrics explicitly uses a Gaussian copula to correlate rating transitions; assuming independence ignores credit correlation.

Q461 Schweser Module Quiz 28.2 Q2

If an analyst wants a credit risk model that accounts for both defaults and downgrades, she will most likely use which of the following models?

A. CreditMetrics.
B. Credit Risk Plus.
C. Vasicek's model.
D. Monte Carlo simulation.

Answer: A

Step by step

1. List which models consider only defaults: Vasicek (worst-case default rate via Gaussian copula) and CreditRisk+ (default counts).
2. CreditMetrics uses a ratings transition matrix, so it values losses from both downgrades and defaults.
3. Monte Carlo simulation is a technique used within models (e.g., CreditMetrics), not a model of its own.

Why A is correct: CreditMetrics combines a ratings transition matrix with a Gaussian copula, so it captures losses from downgrades as well as defaults.

Why the others are wrong

- **B:** CreditRisk+ models only the number/frequency of defaults, with no rating migration.
- **C:** Vasicek's model gives the worst-case default rate only; downgrades are ignored.
- **D:** Monte Carlo is a simulation technique applied within models, not a credit risk model that defines the loss types.

LO 28.f

Define credit spread risk and assess its impact on calculating credit VaR.

Q462 AnalystPrep Q.6545

A risk manager calculates a one-day 99% VaR for a portfolio of bonds in the trading book whose values are sensitive to credit spreads. She uses historical simulation based on the last 500 days of daily credit spread changes for each issuer. Which of the following is a key limitation of this approach?

A. It assumes credit spread changes follow a normal distribution, underestimating extreme movements in bond prices.
B. Because the issuers are alive today, they did not default in the sample period, so the calculation implicitly assumes no probability of default and may understate tail risk.
C. It fails to capture correlations between the spread changes of different issuers, overstating portfolio losses.
D. It requires historical recovery rate assumptions from past defaults, which skew the VaR.

Answer: B

Step by step

1. Historical simulation replays each of the 500 days of observed spread changes on today's portfolio.
2. Issuers that exist today did not default during the sample, so no scenario contains a default.
3. The method therefore implicitly assumes zero default probability, understating tail credit losses: option B.

Why B is correct: Scenarios are built only from past spread changes of surviving companies, so default losses never appear and the tail is understated.

Why the others are wrong

- **A:** Historical simulation is non-parametric and makes no normality assumption; this describes a parametric method.
- **C:** Applying the same-day spread changes to all issuers preserves their historical correlation, so correlation is captured, not ignored.
- **D:** No recovery-rate assumptions are needed because no default scenarios exist; the actual problem is that de-

faults are absent.

Q463 AnalystPrep Q.6546

REBUILT — Re-worded option B to refer to probabilities of loss from small rating changes (matching the GARP comparison) and re-mapped the item to the constant level of risk discussion (28.f).

A risk manager compares two strategies for a portfolio of BBB-rated bonds. Under a buy-and-hold strategy the bonds are held for one year without rebalancing. Under a constant level of risk strategy the portfolio is rebalanced monthly: any bond no longer rated BBB is sold and replaced with a BBB-rated bond. Relative to buy-and-hold, which statement best describes the effect of the constant level of risk strategy?

A. It increases the probability of large losses from defaults and reduces the probability of losses from small rating changes.

B. It reduces the probability of large losses from defaults and increases the probability of losses from small rating changes.

C. It increases both the probability of large losses from defaults and the probability of losses from small rating changes.

D. It has no effect on the probability of losses compared with buy-and-hold.

Answer: B

Step by step

1. Buy-and-hold keeps a downgraded bond until year-end, so it can slide to default: the one-year default probability for a BBB bond is 0.0017.
2. Constant level of risk sells downgraded bonds each month, so default requires a BBB bond to default within a month (0.0001 per month): $P(\text{one default}) = 12 \times 0.0001 \times 0.9999^{11} = 0.0012 < 0.0017$.
3. Monthly rebalancing creates twelve chances for a small downgrade: $P(\text{one downgrade to BB}) = 12 \times 0.0035 \times 0.9965^{11} = 0.0440 > 0.0378$ under buy-and-hold.
4. So large default/downgrade losses are less likely, small-rating-change losses more likely, and VaR is generally smaller.

Why B is correct: Rebalancing removes deteriorating bonds before they can reach default, but it realises small downgrade losses repeatedly over the year.

Why the others are wrong

- **A:** Reverses the effect: it is buy-and-hold, not constant level of risk, that leaves bonds exposed to large downgrades and default.
- **C:** Correct on small rating changes but wrong on large losses; rebalancing lowers the default probability (0.0012 vs 0.0017).
- **D:** The two strategies give different outcome probabilities (e.g. 0.0012 vs 0.0017 for default), so there is an effect.

Q464 AnalystPrep Q.6026

In a meeting with senior management, a credit risk officer is explaining the challenges of adjusting Value at Risk for credit spread risk. What significant risk should the officer highlight when discussing the limitations of using historical simulation for Credit VaR calculations?

A. The risk of overestimating credit spread changes due to the assumption of continuously increasing spreads over time.

B. The challenge of capturing jump-to-default risk, which arises from the possibility that a company could default abruptly without prior indication.

C. The issue of underestimating market liquidity risk due to a narrow focus on credit spreads instead of the broader market impacts on portfolio value.

D. The limitations in assessing interest rate risks as historical simulation for Credit VaR is strictly confined to credit spread changes and neglects rate fluctuations.

Answer: B

Step by step

1. Historical simulation of credit VaR uses past daily credit spread changes of the companies to which the bank is exposed.
2. Problem 1: credit spreads are not updated daily for all companies, so the history is of poor quality.
3. Problem 2: a company that is operating today did not default in the sample period, so the history implicitly assumes no default in the future (no jump-to-default).

4. Option B describes problem 2, so it is the limitation to highlight.

Why B is correct: Because surviving firms have no default in their spread history, a historical simulation of spread changes omits jump-to-default losses; this is one of the two stated weaknesses of the approach.

Why the others are wrong

- **A:** Nothing in the method assumes continuously rising spreads; scenarios are drawn from observed (up and down) changes.
- **C:** Liquidity risk is not the stated limitation; the issue is missing default events and stale spread data.
- **D:** Interest-rate risk is not the stated limitation; the method's weakness is the missing default/stale-spread issue, not an exclusion of rates.

Q465

BT answers P2.T6.24.18.3 · BT notes P2.T6.24.18.2

REBUILT — Original statements were loosely worded (VaR vs realized spread widening, 'long-run yield offset') and tested no specific rule; rebuilt around the Schweser/GARP rules for historical simulation and CreditMetrics-style credit spread VaR.

Three analysts discuss how credit spread risk is handled in credit VaR.

- Damian: "A historical simulation of credit spread changes is hampered because spreads are not updated daily for all companies, so the data may be of poor quality."
- Tanyth: "Because a company that is operating today has not defaulted in the past, a historical simulation implicitly assumes no default; a CreditMetrics-type approach using a rating transition matrix brings in downgrades and defaults."
- James: "To introduce credit correlation, the spread changes for different rating categories should be assumed uncorrelated so that diversification is preserved."

Which analyst's statement is incorrect?

A. Damian
B. Tanyth
C. James
D. None of the above

Answer: C

Step by step

1. Damian: spreads are not updated daily, so historical data quality is weak. Correct.
2. Tanyth: surviving firms have no past default, so history assumes no default; the transition-matrix (CreditMetrics) approach, sampled by Monte Carlo, adds rating changes and default. Correct.
3. James: credit correlation is introduced either with a Gaussian copula for rating changes or by assuming rating-category spread changes are perfectly or very highly correlated, so spreads of all ratings move in unison. Assuming zero correlation is wrong.
4. Only James is incorrect.

Why C is correct: Credit correlation in spread-based credit VaR comes from a Gaussian copula or from very high correlation between spread changes across rating categories, not from independence.

Why the others are wrong

- **A:** Damian's statement is a correct stated weakness of historical simulation (stale, non-daily spreads).
- **B:** Tanyth's statement is correct: no-default history is the second weakness and the transition matrix addresses it.
- **D:** Wrong because James's statement is incorrect; choosing it means treating the independence assumption as acceptable.

Q466

Schweser Module Quiz 28.2 Q3

When accounting for credit spreads and potential bond losses for a bond currently rated A, an analyst will likely assign the:

A. biggest spreads to situations where the bond rating increases.
B. lowest probability to situations where the bond rating increases.
C. lowest spreads to situations where the bond maintains its rating.
D. highest probability to situations where the bond maintains its rating.

Answer: D

Step by step

1. In a one-period rating transition matrix the diagonal (no change) entry is by far the largest, so the highest

probability goes to maintaining the rating.
2. Spreads rise as ratings fall: upgrade gives the lowest spreads, no change intermediate, downgrade the biggest.
3. The lowest probability is for default.

Why D is correct: Ratings are persistent, so the no-change outcome has the highest probability; spreads rank with credit quality.

Why the others are wrong

- **A:** Reversed: the biggest spreads go with downgrades, not upgrades.
- **B:** The lowest probability belongs to default, not to an upgrade.
- **C:** The lowest spreads go with upgrades; a retained rating carries intermediate spreads.

Q467 AnalystPrep Q.6210

REBUILT — Original asked a generic structured-credit hedging question not covered by the LO; rebuilt to test the buy-and-hold versus constant level of risk comparison for credit VaR.

A risk manager computes one-year credit VaR for a portfolio of BBB-rated bonds under two strategies. Under strategy 1 the bonds are held to the horizon whatever their rating becomes. Under strategy 2 any bond that is no longer BBB at each period end is sold and replaced with a BBB-rated bond. Which statement is most accurate?

A. Strategy 1 (buy-and-hold) generally produces larger losses from big downgrades and defaults, so its credit VaR tends to be larger than under strategy 2.

B. Strategy 2 (constant level of risk) generally produces larger losses from big downgrades and defaults, so its credit VaR tends to be larger than under strategy 1.

C. The two strategies give the same credit VaR because the rating transition matrix is identical.

D. Strategy 1 has the smaller credit VaR because ratings change little over one year.

Answer: A

Step by step

1. Strategy 1 is buy-and-hold: the original bond stays in the portfolio through downgrades, so a multi-notch downgrade or default is fully borne.
2. Strategy 2 is constant level of risk: deteriorated bonds are replaced by BBB bonds, so risk returns to its initial level and large cumulative downgrades or defaults are less likely to persist.
3. Therefore buy-and-hold has larger losses from big downgrades and defaults and a larger credit VaR; constant level of risk has a smaller credit VaR.

Why A is correct: Rebalancing to the original rating resets risk each period, so cumulative downgrade and default losses are smaller than if the original bonds are held.

Why the others are wrong

- **B:** Reverses the strategies; rebalancing reduces rather than increases exposure to deteriorated credits.
- **C:** The matrix is the same per period, but the strategies compound it differently (rebalanced versus held), so VaR differs.
- **D:** Small rating changes do give lower probabilities of loss, but the tail (large downgrades and defaults), which drives VaR, is worse under buy-and-hold.

Reading 29. Portfolio Credit Risk

Allan Malz, Financial Risk Management: Models, History, and Institutions (Hoboken, NJ: John Wiley & Sons, 2011). Chapter 8, Sections 8.1, 8.2, and 8.3.

LO	Learning objective	Qs
29.a	define and calculate default correlation for credit portfolios.	19
29.b	identify drawbacks in using the correlation-based credit portfolio framework.	4
29.c	assess the impact of correlation on a credit portfolio and its credit VaR.	15
29.d	describe the use of a single factor model to measure portfolio credit risk, including the impact of correlation.	9
29.e	define beta and calculate the asset return correlation of any pair of firms using the single factor model.	3
29.f	estimate the probability of a joint default of any pair of credits and the default correlation between any pair of credits using the single factor model.	3
29.g	describe how credit VaR can be calculated using a simulation of joint defaults.	3
29.h	assess the effect of granularity on credit VaR.	3

LO 29.a

Define and calculate default correlation for credit portfolios.

Q468

BT notes P2.T5.501.3 · BT answers P2.T5.501.3

REBUILT — Two distractors (6.20%, 8.72%) were not products of any realistic error; replaced with 3.33% (omitted the (1-PD) term) and 9.00% (perfect-correlation limit).

Commercial bank C has lent to companies A and B. Both A and B have a 9.0% probability of default, and their pairwise default correlation is 0.280. Which is nearest to the joint probability of default of A and B?

A. 0.81%
B. 3.10%
C. 3.33%
D. 9.00%

Answer: B

Step by step

1. Bernoulli default variables: $\sigma_A = \sigma_B = \sqrt{0.09 \times 0.91} = 0.2862$.
2. Covariance $= \rho\sigma_A\sigma_B = 0.280 \times 0.09 \times 0.91 = 0.022932$.
3. Joint PD $= \pi_A\pi_B + \text{Cov} = 0.0081 + 0.022932 = 0.031032 \approx 3.10\%$.

Why B is correct: Joint default probability equals the independence value $\pi_1\pi_2$ plus the covariance $\rho\,\sigma_1\sigma_2$, where each Bernoulli standard deviation is $\sqrt{\pi(1-\pi)}$.

Why the others are wrong

- **A:** 0.81% is $\pi_A\pi_B$, the joint PD under independence; it ignores the positive correlation.
- **C:** 3.33% uses $\rho\pi$ (0.28 x 0.09 = 0.0252) instead of $\rho\pi(1-\pi)$ as the covariance, i.e. it omits the (1-PD) factors.
- **D:** 9.00% is the joint PD when correlation is 1 (the maximum, equal to the marginal PD); correlation here is only 0.28.

Q469

Schweser Module Quiz 8.3 Q1

REBUILT — Wording 'expected loss under worst-case' was ambiguous and the source formula was missing; restated as the expected loss from the joint-default event and replaced unrealistic distractors.

A creditor makes a $4 million loan to Company X and a $4 million loan to Company Y. Each company has a 7% default probability, and the default correlation between them is 0.6. Assuming loss given default of 100%, the expected loss from the scenario in which both loans default (a loss of $8 million) is closest to:

A. $39,200
B. $351,680
C. $312,480
D. $560,000

Answer: B

Step by step

1. Joint PD $= \rho\,\pi(1-\pi) + \pi^2 = 0.6 \times 0.07 \times 0.93 + 0.07^2$.
2. $= 0.03906 + 0.0049 = 0.04396$.
3. Loss if both default = \$4m + \$4m = \$8m (LGD 100%).
4. Expected loss $= 0.04396 \times \$8{,}000{,}000 = \$351{,}680$.

Why B is correct: The worst case requires both loans to default, whose probability under correlation is the independence term plus the covariance term; multiply by the \$8 million exposure.

Why the others are wrong

- **A:** \$39,200 = 0.0049 × \$8m: uses the independent joint PD, ignoring the 0.6 correlation.
- **C:** \$312,480 = 0.03906 × \$8m: includes only the covariance term and omits $\pi_1\pi_2$.
- **D:** \$560,000 = 0.07 × \$8m: uses a single firm's PD, which equals the joint PD only if correlation were 1.

Q470

AnalystPrep Q.1831

REBUILT — Options were garbled by OCR; restated cleanly with each option a standard expression from the reading.

If two firms have independent default events, which of the following expressions is non-zero?

A. $\rho_{12}\sqrt{\pi_1(1-\pi_1)}\sqrt{\pi_2(1-\pi_2)}$

B. $\dfrac{\pi_{12} - \pi_1\pi_2}{\sqrt{\pi_1(1-\pi_1)}\sqrt{\pi_2(1-\pi_2)}}$

C. π_{12}/π_1

D. $\pi_{12} - \pi_1\pi_2$

Answer: C

Step by step

1. Independence means $\pi_{12} = \pi_1\pi_2$.
2. Then covariance $\pi_{12} - \pi_1\pi_2 = 0$ (option D), so default correlation $\rho_{12} = 0$ (option B).
3. Option A is ρ_{12} times the standard deviations, i.e. the covariance, which equals 0.
4. Option C $= \pi_1\pi_2/\pi_1 = \pi_2 > 0$: the conditional default probability equals the unconditional one.

Why C is correct: Independence sets covariance and correlation to zero, but the conditional probability π_{12}/π_1 reduces to π_2, which is positive.

Why the others are wrong

- **A:** Equals the covariance, which is 0 under independence (rho = 0).
- **B:** This is the default correlation itself; its numerator is 0 under independence.
- **D:** This is the covariance of the default indicators, exactly 0 under independence.

Q471

AnalystPrep Q.1837

REBUILT — Replaced the 0.03% distractor with 0.50% (omitted the (1-PD) terms) so each wrong option reflects a realistic error.

The probability of default for firm 1 is 5% and for firm 2 is 1%. Their default correlation is 0.2. What is the joint probability of default?

A. 0.50%

B. 0.05%

C. 0.06%

D. 0.48%

Answer: D

Step by step

1. $\sqrt{0.05 \times 0.95} = 0.21794$; $\sqrt{0.01 \times 0.99} = 0.09950$.
2. Covariance $= 0.2 \times 0.21794 \times 0.09950 = 0.004337$.
3. $\pi_1\pi_2 = 0.0005$.
4. Joint PD $= 0.0005 + 0.004337 = 0.004837 \approx 0.48\%$.

Why D is correct: Joint default probability is the independent product plus the correlation-scaled product of Bernoulli standard deviations.

Why the others are wrong

- **A:** 0.50% uses $\rho\sqrt{\pi_1\pi_2} + \pi_1\pi_2$ (0.2 x 0.02236 + 0.0005 = 0.497%), omitting the (1-PD) factors.
- **B:** 0.05% is the independence value pi1 x pi2; it ignores the correlation.
- **C:** 0.06% uses $\rho\pi_1\pi_2 + \pi_1\pi_2$ (0.2 x 0.0005 + 0.0005), scaling the PD product instead of the standard deviations.

Q472 AnalystPrep Q.2651

REBUILT — Replaced 5.78% and 7.04% (no realistic derivation) with errors that arise from the formula.

A bank has issued loans to two companies, A and B, with default probabilities of 5% and 10%, respectively. What is the joint probability of default if their default correlation is 0.6?

A. 3.92%
B. 4.74%
C. 4.42%
D. 0.80%

Answer: C

Step by step

1. $\sqrt{0.05 \times 0.95 \times 0.10 \times 0.90} = \sqrt{0.004275} = 0.06538$.
2. Covariance $= 0.6 \times 0.06538 = 0.03923$.
3. $\pi_A \pi_B = 0.005$.
4. Joint PD $= 0.005 + 0.03923 = 0.04423 \approx 4.42\%$.

Why C is correct: Add the independent joint probability to the correlation times the product of Bernoulli standard deviations.

Why the others are wrong

- **A:** 3.92% is only the covariance term; the $\pi_A \pi_B = 0.5\%$ is omitted.
- **B:** 4.74% uses $\sqrt{\pi_A \pi_B} = 0.0707$ (0.6 x 0.0707 = 4.24%) instead of the Bernoulli standard deviations, then adds $\pi_A \pi_B = 0.5\%$.
- **D:** 0.80% uses $\rho \pi_A \pi_B + \pi_A \pi_B$ (0.6 x 0.005 + 0.005), scaling the PD product instead of the standard deviations.

Q473 AnalystPrep Q.2884

REBUILT — Replaced 0.5% and 0.45% (no realistic derivation) with 0.09% (independence) and 2.50% (perfect-dependence bound).

Given $\pi_1 = 0.025$, $\pi_2 = 0.035$ and $\rho_{12} = 0.05$, determine the joint default probability of firms 1 and 2.

A. 0.23%
B. 0.09%
C. 0.14%
D. 2.50%

Answer: A

Step by step

1. $\sqrt{0.025 \times 0.975} = 0.15612$; $\sqrt{0.035 \times 0.965} = 0.18378$.
2. Covariance $= 0.05 \times 0.15612 \times 0.18378 = 0.001435$.
3. $\pi_1 \pi_2 = 0.000875$.
4. Joint PD $= 0.000875 + 0.001435 = 0.00231 \approx 0.23\%$.

Why A is correct: Joint PD equals the independence value 0.0875% plus the covariance 0.1435%.

Why the others are wrong

- **B:** 0.09% is pi1 x pi2 (0.0875%), the independence case; it ignores the correlation.
- **C:** 0.14% is only the covariance term (0.1435%); pi1 x pi2 is omitted.
- **D:** 2.50% is min(pi1, pi2), the joint PD under perfect dependence; correlation here is only 0.05.

Q474 BT notes P2.T5.22.11.2

REBUILT — Replaced distractors (zero/0.30/0.15) with realistic formula errors, kept the same data.

Consider two loans to Company X and Company Y with PD(X) = 10.0% and PD(Y) = 4.0%. The joint default probability is PD(X and Y) = 3.930%. Which is nearest to the implied default correlation?

A. 0.56
B. 0.60
C. 0.67
D. Zero

Answer: B

Step by step

1. Covariance $= 0.0393 - 0.10 \times 0.04 = 0.0353$.

2. $\sigma_X = \sqrt{0.10 \times 0.90} = 0.3$; $\sigma_Y = \sqrt{0.04 \times 0.96} = 0.19596$; product = 0.058788.
3. $\rho = 0.0353/0.058788 = 0.6005 \approx 0.60$.

Why B is correct: Default correlation is the covariance $\pi_{12} - \pi_1\pi_2$ divided by the product of the Bernoulli standard deviations.

Why the others are wrong

- **A:** 0.56 divides by $\sqrt{\pi_X \pi_Y} = 0.0632$, omitting the (1-PD) factors.
- **C:** 0.67 forgets to subtract pi_X x pi_Y: 0.0393/0.058788.
- **D:** Zero would require the joint PD to equal 0.4%; at 3.93% the defaults are strongly positively correlated.

Q475

AnalystPrep Q.1832

REBUILT — Stem ambiguity ('portfolio default probability of 0.2%', 'maximum') clarified as joint default probability of 0.2%; distractors replaced by formula errors.

Two credits rated BBB- and BBB+ have default probabilities of 0.3% and 0.4%, respectively. If the joint probability that both default is 0.2%, what is their default correlation?

A. 57.94%
B. 57.59%
C. 66.67%
D. 57.74%

Answer: B

Step by step

1. Covariance = $0.002 - 0.003 \times 0.004 = 0.001988$.
2. $\sigma_1\sigma_2 = \sqrt{0.003 \times 0.997 \times 0.004 \times 0.996} = \sqrt{1.19164 \times 10^{-5}} = 0.003452$.
3. $\rho = 0.001988/0.003452 = 0.5759$, i.e. 57.59%.

Why B is correct: Invert the joint-PD formula: correlation is (joint PD minus product of PDs) over the product of Bernoulli standard deviations.

Why the others are wrong

- **A:** 57.94% omits the subtraction of pi1 x pi2 (0.002/0.003452).
- **C:** 66.67% is the conditional probability 0.002/0.003, not a correlation.
- **D:** 57.74% omits both the subtraction and the (1-PD) factors: $0.002/\sqrt{0.003 \times 0.004}$.

Q476

AnalystPrep Q.1833

REBUILT — Stem said ratings differ yet 'equal' PDs, and options C/D were garbled; restated for two credits with identical PD and cleaned options.

Two credits have the same default probability π. If ρ_{12} is the default correlation between them, which expression gives their joint default probability?

A. $\pi_{12} = \rho_{12}\,\pi(1-\pi) + \pi^2$
B. $\pi_{12} = \rho_{12}$
C. $\pi_{12} = \pi^2$
D. $\pi_{12} = \pi^3$

Answer: A

Step by step

1. General: $\pi_{12} = \rho_{12}\sqrt{\pi_1(1-\pi_1)}\sqrt{\pi_2(1-\pi_2)} + \pi_1\pi_2$.
2. Set $\pi_1 = \pi_2 = \pi$: the root product becomes $\sqrt{\pi^2(1-\pi)^2} = \pi(1-\pi)$.
3. So $\pi_{12} = \rho_{12}\pi(1-\pi) + \pi^2$.

Why A is correct: With equal PDs the Bernoulli standard deviations are identical, so their product is the variance pi(1-pi).

Why the others are wrong

- **B:** Confuses joint probability with correlation; they are different quantities.
- **C:** Correct only if rho = 0 (independence).
- **D:** A three-way independent default probability; not relevant to two credits.

Q477 AnalystPrep Q.2693

REBUILT — Original distractors were arbitrary numbers with no traceable error; replaced with options that each reflect a realistic mistake.

Two credits have default probabilities $\pi_1 = 0.004$ and $\pi_2 = 0.010$. What is the joint probability of default if the default correlation is 10%?

A. 0.0668%
B. 0.0628%
C. 0.0672%
D. 0.0044%

Answer: A

Step by step

1. $\pi_{12} = \rho_{12}\sqrt{\pi_1(1-\pi_1)}\sqrt{\pi_2(1-\pi_2)} + \pi_1\pi_2$.
2. $\sqrt{0.004 \times 0.996} = 0.063119$; $\sqrt{0.010 \times 0.990} = 0.099499$; product = 0.006280.
3. Correlation term: 0.10 x 0.006280 = 0.000628.
4. Independence term: 0.004 x 0.010 = 0.00004.
5. $\pi_{12} = 0.000628 + 0.00004 = 0.000668 = 0.0668\%$.

Why A is correct: Joint default probability equals the independence value plus the correlation-driven excess, rho times the product of the Bernoulli standard deviations.

Why the others are wrong

- **B:** Omitted the independence term $\pi_1\pi_2$ (0.0628% = 0.000628 only).
- **C:** Ignored the $(1-\pi)$ terms inside the square roots: 0.10 x $\sqrt{0.004 \times 0.010}$ + 0.00004 = 0.0672%.
- **D:** Used the product of the variances instead of standard deviations: 0.10 x 0.003984 x 0.0099 + 0.00004 = 0.0044%.

Q478 AnalystPrep Q.3069

REBUILT — Original distractors were arbitrary numbers with no traceable error; replaced with options that each reflect a realistic mistake.

McLeod Bank holds two credits, rated CCC and BBB, with default probabilities over horizon t of 0.008 and 0.004. The joint probability that both default over t is 0.00035. What is the default correlation?

A. 5.65%
B. 6.22%
C. 5.62%
D. 0.318%

Answer: A

Step by step

1. Covariance = $0.00035 - 0.008 \times 0.004 = 0.000318$.
2. $\sigma_1 = \sqrt{0.008 \times 0.992} = 0.089084$; $\sigma_2 = \sqrt{0.004 \times 0.996} = 0.063119$; product = 0.005623.
3. $\rho_{12} = 0.000318/0.005623 = 0.0565 = 5.65\%$.

Why A is correct: Default correlation is the Bernoulli covariance divided by the product of the two standard deviations.

Why the others are wrong

- **B:** Did not subtract $\pi_1\pi_2$: 0.00035/0.005623 = 6.22%.
- **C:** Used $\sqrt{\pi_1\pi_2}$ as the denominator instead of the Bernoulli standard deviations: 0.000318/0.005657 = 5.62%.
- **D:** Reported the covariance (0.000318) as a percentage without dividing by the standard deviations.

Q479 AnalystPrep Q.3070

REBUILT — Original distractors were arbitrary numbers with no traceable error; replaced with options that each reflect a realistic mistake.

Consider a BBB+ credit with default probability 0.0065 and a BBB- credit with default probability 0.0125. If defaults are uncorrelated, the joint default probability is 0.00008. If the default correlation is 9%, the joint default probability is closest to:

A. 0.0008
B. 0.0009
C. 0.00008
D. 0.00009

Answer: B

Step by step

1. $\pi_{12} = \rho\sqrt{\pi_1(1-\pi_1)}\sqrt{\pi_2(1-\pi_2)} + \pi_1\pi_2$.
2. $\sqrt{0.0065 \times 0.9935} = 0.080357$; $\sqrt{0.0125 \times 0.9875} = 0.111099$; product = 0.008928.
3. Correlation term: 0.09 x 0.008928 = 0.000804; independence term: 0.0065 x 0.0125 = 0.0000813.
4. $\pi_{12} = 0.000804 + 0.0000813 = 0.000885 \approx 0.0009$.

Why B is correct: A 9% default correlation raises the joint default probability about eleven-fold relative to independence because the base probabilities are tiny.

Why the others are wrong

- **A:** Omitted the independence term (0.000804 only).
- **C:** Assumed independence, ignoring the correlation.
- **D:** Used variances rather than standard deviations in the correlation term: 0.09 x 0.006459 x 0.012344 + 0.0000813 = 0.00009.

Q480

AnalystPrep Q.4365

REBUILT — Original distractors were arbitrary numbers with no traceable error; replaced with options that each reflect a realistic mistake.

An investment firm holds two credits: one rated AAA with default probability 0.002 and one rated BBB with default probability 0.004 over horizon t. The joint default probability is 0.00018. What is the default correlation?

A. 0.0638
B. 0.0610
C. 0.000172
D. 21.63

Answer: B

Step by step

1. Covariance = $0.00018 - 0.002 \times 0.004 = 0.000172$.
2. $\sigma_1 = \sqrt{0.002 \times 0.998} = 0.044677$; $\sigma_2 = \sqrt{0.004 \times 0.996} = 0.063119$; product = 0.002820.
3. $\rho = 0.000172/0.002820 = 0.0610$.

Why B is correct: Correlation = Bernoulli covariance scaled by both standard deviations.

Why the others are wrong

- **A:** Forgot to subtract $\pi_1\pi_2$: 0.00018/0.002820 = 0.0638.
- **C:** Stopped at the covariance without dividing by the standard deviations.
- **D:** Divided by the product of the variances instead of the standard deviations: 0.000172/0.00000795 = 21.63.

Q481

AnalystPrep Q.5437

REBUILT — Original distractors were arbitrary numbers with no traceable error; replaced with options that each reflect a realistic mistake.

Two credits rated A- and BBB+ have one-year default probabilities of 2.5% and 3.3%. The joint default probability is 0.8%. What is the implied default correlation?

A. 28.68%
B. 25.73%
C. 24.98%
D. 0.72%

Answer: B

Step by step

1. Covariance = $0.008 - 0.025 \times 0.033 = 0.007175$.
2. $\sigma_1 = \sqrt{0.025 \times 0.975} = 0.156125$; $\sigma_2 = \sqrt{0.033 \times 0.967} = 0.178638$; product = 0.027890.
3. $\rho = 0.007175/0.027890 = 0.2573 = 25.73\%$.

Why B is correct: Default correlation equals the excess of joint default over the independent value, scaled by the Bernoulli standard deviations.

Why the others are wrong

- **A:** Did not subtract $\pi_1\pi_2$: 0.008/0.027890 = 28.68%.
- **C:** Used $\sqrt{\pi_1\pi_2}$ in the denominator (ignoring the $1-\pi$ factors): 0.007175/0.028723 = 24.98%.
- **D:** Reported the covariance (0.7175%) instead of the correlation.

Q482 BT answers P2.T6.24.11.1

REBUILT — Original distractors were arbitrary numbers with no traceable error; replaced with options that each reflect a realistic mistake.

An investment firm has exposure to a BBB credit with default probability 0.003 and a CCC credit with default probability 0.005 over horizon t. The joint default probability is 0.0002. The default correlation is closest to:

A. 0.0518
B. 0.0480
C. 0.000185
D. 12.43

Answer: B

Step by step

1. Covariance = $0.0002 - 0.003 \times 0.005 = 0.000185$.
2. $\sigma_1 = \sqrt{0.003 \times 0.997} = 0.054690$; $\sigma_2 = \sqrt{0.005 \times 0.995} = 0.070534$; product = 0.003857.
3. $\rho = 0.000185/0.003857 = 0.04796 \approx 0.0480$.

Why B is correct: Correlation is the Bernoulli covariance divided by the product of standard deviations.

Why the others are wrong

- **A:** Did not subtract $\pi_1\pi_2$: 0.0002/0.003857 = 0.0518.
- **C:** Reported the covariance without scaling.
- **D:** Divided by the product of the variances instead of the standard deviations: 0.000185/0.0000149 = 12.43.

Q483 BT answers P2.T6.309.1 · BT notes P2.T6.309.1

REBUILT — Original distractors were arbitrary numbers with no traceable error; replaced with options that each reflect a realistic mistake.

Consider two speculative credits, rated BB and BB-, with default probabilities of 2.00% and 3.00%. If their joint default probability is 0.40%, which is nearest to the implied default correlation?

A. 0.0034
B. 0.1675
C. 0.1424
D. 5.96

Answer: C

Step by step

1. Covariance = $0.0040 - 0.02 \times 0.03 = 0.0034$.
2. $\sigma_1 = \sqrt{0.02 \times 0.98} = 0.140000$; $\sigma_2 = \sqrt{0.03 \times 0.97} = 0.170587$; product = 0.023882.
3. $\rho = 0.0034/0.023882 = 0.1424$.

Why C is correct: Default correlation scales the excess joint default probability by the two Bernoulli standard deviations.

Why the others are wrong

- **A:** Reported the covariance, not divided by the standard deviations.
- **B:** Did not subtract $\pi_1\pi_2$: 0.0040/0.023882 = 0.1675.
- **D:** Divided by the product of variances (0.0196 x 0.0291 = 0.000570): 0.0034/0.000570 = 5.96.

Q484 Mock 1 Q2

REBUILT — Removed dependence on the unreproduced Bank Q exhibit and the pre-computed standard deviation; made the stem self-contained.

Two prime counterparties each have a one-year marginal default probability of 5.0%. Default of each is modelled as a Bernoulli (binomial) event, and the default correlation between the two is 0. Which of the following is closest to the joint probability of default of the two counterparties?

A. 0.25%.
B. 2.38%.
C. 5.00%.
D. 9.75%.

Answer: A

Step by step

1. Joint default probability: $\pi_{12} = \rho\,\sigma_1\sigma_2 + \pi_1\pi_2$.
2. Each $\sigma = \sqrt{0.05 \times 0.95} = 0.2179$, but with $\rho = 0$ the covariance term is 0.

3. $\pi_{12} = 0 + 0.05 \times 0.05 = 0.0025 = 0.25\%$.

Why A is correct: With zero default correlation defaults are independent, so the joint probability is just the product of the marginal probabilities.

Why the others are wrong

- **B:** 2.38% is the covariance term 0.5×0.0475 from a correlation of 0.5, not 0.
- **C:** 5.00% treats the two defaults as perfectly dependent (joint = marginal).
- **D:** 9.75% is the probability that at least one defaults, $0.05 + 0.05 - 0.0025$, not both.

Q485

Mock 1 Q3

REBUILT — Made stem self-contained (removed Bank Q reference) and replaced an unexplained 3.61% distractor with a realistic error (omitting one standard deviation).

Two prime counterparties each have a one-year marginal default probability of 5.0%. Default of each is modelled as a Bernoulli event, and the default correlation between the two is 0.5. Which of the following is closest to the joint probability of default of the two counterparties?

A. 2.13%.
B. 2.38%.
C. 2.63%.
D. 11.15%.

Answer: C

Step by step

1. $\sigma^2 = 0.05 \times 0.95 = 0.0475$, so $\sigma = 0.2179$.
2. Covariance $= \rho\sigma_1\sigma_2 = 0.5 \times 0.0475 = 0.02375$.
3. $\pi_{12} = 0.02375 + \pi_1\pi_2 = 0.02375 + 0.0025 = 0.02625 \approx 2.63\%$.

Why C is correct: Joint default probability equals the covariance ($\rho\sigma_1\sigma_2$) plus the independent-case product $\pi_1\pi_2$.

Why the others are wrong

- **A:** 2.13% subtracts $\pi_1\pi_2$ instead of adding it (0.02375 - 0.0025).
- **B:** 2.38% omits the $\pi_1\pi_2$ term, giving only the covariance.
- **D:** 11.15% multiplies ρ by a single standard deviation (0.5×0.2179) rather than by $\sigma_1\sigma_2$, then adds 0.25%.

Q486

Schweser Module Quiz 29.1 Q1

REBUILT — Option formulas were lost as images; rewrote them in text with realistic incorrect variants.

Which of the following best defines the default correlation ρ_{12} for a two-firm credit portfolio, where π_1, π_2 are the firms' default probabilities and π_{12} is the joint default probability?

A. $\rho_{12} = \dfrac{\pi_{12} - \pi_1\pi_2}{\sqrt{\pi_1(1-\pi_1)\,\pi_2(1-\pi_2)}}$

B. $\rho_{12} = \dfrac{\pi_{12} - \pi_1\pi_2}{\pi_1(1-\pi_1)\,\pi_2(1-\pi_2)}$

C. $\rho_{12} = \dfrac{\pi_{12}}{\sqrt{\pi_1(1-\pi_1)\,\pi_2(1-\pi_2)}}$

D. $\rho_{12} = \dfrac{\pi_{12} - \pi_1\pi_2}{\pi_1\pi_2}$

Answer: A

Step by step

1. Each default indicator is Bernoulli: $E[x_i] = \pi_i$, $\mathrm{Var}(x_i) = \pi_i(1-\pi_i)$, $E[x_1x_2] = \pi_{12}$.
2. Covariance $= E[x_1x_2] - E[x_1]E[x_2] = \pi_{12} - \pi_1\pi_2$.
3. Correlation = covariance divided by the product of standard deviations, i.e. the square root of the product of variances, giving option A.

Why A is correct: Default correlation is the covariance of the two Bernoulli default indicators scaled by the product of their standard deviations.

Why the others are wrong

- **B:** Divides by the product of variances, forgetting the square root (standard deviations).
- **C:** Omits the $\pi_1\pi_2$ term, using the raw joint probability instead of the covariance.
- **D:** Scales covariance by the product of default probabilities, which is not the standard deviations of the Bernoulli variables.

LO 29.b

Identify drawbacks in using the correlation-based credit portfolio framework.

Q487

BT answers P2.T6.309.2 · BT notes P2.T6.309.2

REBUILT — Option D stated a pairwise-correlation count (n(n-1)/2) that differs from the GARP text's N(N-1); reworded D to the non-negativity point only so exactly one option is false. LO changed to 29.b (drawbacks of the correlation framework).

According to Malz, each of the following is an important implication of default correlation in models of portfolio credit risk EXCEPT:

A. Default correlation is hard to measure or estimate from historical default data.
B. Default correlation has a tremendous impact on the credit portfolio's expected loss.
C. Default correlations are small in magnitude, so an "optically" small correlation can still have a large impact.
D. With many credits, all pairwise correlations are often set equal to a single parameter, which must be non-negative.

Answer: B

Step by step

1. Default is a rare event, so history is thin and estimates vary widely by period and industry: A is true.
2. Expected loss is the sum of exposure x default probability x LGD, which does not depend on correlation; correlation affects the volatility and extreme quantiles of loss (hence credit VaR): B is false.
3. Default probabilities are small, so default correlations are small in magnitude, yet they have a large impact: C is true.
4. A single common non-negative parameter is used to avoid a non-positive-definite correlation matrix: D is true.

Why B is correct: Correlation changes the shape and tail of the loss distribution but not its mean (expected loss).

Why the others are wrong

- **A:** True statement (data are scarce), so not the exception.
- **C:** True statement from the text, so not the exception.
- **D:** True statement (single non-negative parameter), so not the exception.

Q488

AnalystPrep Q.1835

REBUILT — Source key (1) used N(N-1)/2, contradicting the GARP/Schweser convention of N(N-1) pairwise correlations; options had no correct value, so options were rebuilt around 2.

Apple issues 8 five-year subordinated unsecured bonds with very close maturity dates. Shell issues 2 five-year senior bonds with very close maturity dates. A large investment bank buys all of them and builds a credit portfolio model using the default-correlation framework, in which the number of pairwise correlations to be specified is $N(N-1)$ for N distinct credit entities. How many pairwise default correlations are required?

A. 110
B. 90
C. 2
D. 1

Answer: C

Key correction: Source key (D, 1) used N(N-1)/2; the GARP/Schweser framework counts N(N-1) pairwise correlations, so the correct count is 2.

Step by step

1. Default correlation is defined between distinct obligors; bonds of the same issuer default together, so the 10 bonds represent only $N = 2$ entities (Apple, Shell).
2. The framework requires $N(N-1)$ pairwise correlations (GARP: 4 entities need 12).
3. $N(N-1) = 2 \times 1 = 2$.
4. The different seniority of the bonds affects recovery rates (one per bond), not the number of default correlations.

Why C is correct: Correlations are needed between entities, not between bonds; with two issuers the count is $2 \times 1 = 2$.

Why the others are wrong

- **A:** 110 = 11 × 10: counts 11 positions/entities, an off-by-one miscount.
- **B:** 90 = 10 × 9: treats each of the 10 bonds as a separate credit, ignoring that bonds of one issuer share the same

default event.
- **D:** 1 uses $N(N-1)/2$, the number of unordered pairs; the framework as presented requires $N(N-1)$.

Q489 BT answers P2.T6.24.11.2

REBUILT — Original options did not isolate a recognised drawback (keyed option A describes idiosyncratic default, not a cited drawback; D's explanation was truncated); rewritten around the three drawbacks in the reading.

A bank estimates default correlations among its corporate obligors from historical data and feeds them into a default-correlation-based credit portfolio model. Which of the following statements describes a recognised drawback of this framework?

A. Because firm defaults are rare events, estimated default correlations vary greatly with the data horizon and industry sampled.

B. Because default correlations are large in magnitude, the joint default probability of two firms is much larger than their individual default probabilities.

C. The framework cannot be applied to a portfolio of plain-vanilla bonds because it requires option-like payoffs.

D. The number of event outcomes grows only linearly with the number of credits, so the model understates portfolio diversification.

Answer: A

Key correction: Source key (A, sudden default despite low correlation) is not a drawback cited in the reading; the correct choice is the limited-data drawback.

Step by step

1. The recognised drawbacks are: (1) the large number of inputs (2^n event outcomes, n + 1 + n(n-1)/2 conditions), (2) positions such as guarantees, revolving credit lines, CDS basis trades and convertibles that fit poorly, and (3) limited data because defaults are rare.
2. Option A states drawback (3): rare defaults make estimated correlations unstable across horizons and industries.
3. B is false: estimated default correlations are small (about 0.05) and the joint default probability is even smaller.
4. C is false: the problem is with option-like instruments, not plain bonds.
5. D is false: outcomes grow as 2^n, i.e. exponentially.

Why A is correct: Defaults are rare, so there is little data to estimate default correlations, which therefore vary widely by time horizon and industry.

Why the others are wrong

- **B:** Reverses the fact: default correlations are small and joint default probabilities are smaller still.
- **C:** Reverses the fact: option-like contingent positions (guarantees, revolvers) are what fit poorly, not plain bonds.
- **D:** Mistakes exponential growth (2^n outcomes) for linear growth.

Q490 Schweser Module Quiz 29.1 Q2

Suppose a portfolio manager is using a default correlation framework for measuring credit portfolio risk. How many unique event outcomes are there for a credit portfolio with eight different firms?

A. 10

B. 56

C. 256

D. 517

Answer: C

Step by step

1. Each firm either defaults or does not (Bernoulli), so there are 2 outcomes per firm.
2. For $n = 8$ firms: $2^8 = 256$ unique event outcomes.

Why C is correct: Outcomes are all default/no-default combinations across firms, 2^n.

Why the others are wrong

- **A:** 10 is $n + 2$ or a guess; outcomes grow exponentially, not linearly.
- **B:** 56 is the number of conditions $(n + 1) + n(n-1)/2$ for $n = 10$ firms, not the number of outcomes for 8.
- **D:** 517 has no basis; it is not 2^n for any whole n ($2^9 = 512$).

LO 29.c

Assess the impact of correlation on a credit portfolio and its credit VaR.

Q491

AnalystPrep Q.1842

REBUILT — Source stated the 95th percentile of defaults as 4, but for Binomial(100, 0.03) it is 6 (P(X<=5)=0.919, P(X<=6)=0.969); stem and options were recomputed with the correct percentile.

A $1 million portfolio is divided into 100 equally weighted credits, each with default probability $\pi = 0.03$ and zero recovery. Default correlation is zero, so the number of defaults is binomial with $n = 100$ and $\pi = 0.03$. The 95th percentile of the number of defaults is 6. What is the credit VaR at the 95% confidence level?

A. $0.03 million
B. $0.06 million
C. $0.01 million
D. $0.09 million

Answer: A

Key correction: Source key (A, $0.01 million) relied on a 95th percentile of 4 defaults, which is incorrect for Binomial(100, 0.03); the correct percentile is 6, giving $0.03 million.

Step by step

1. Value per credit = $1,000,000 / 100 = $10,000.
2. Expected loss = 0.03 x $1,000,000 = $30,000.
3. 95th percentile loss = 6 defaults x $10,000 = $60,000.
4. Credit VaR = $60,000 − $30,000 = $30,000 = $0.03 million.

Why A is correct: Credit VaR is the loss quantile minus expected loss; with zero correlation the loss quantile comes from the binomial distribution of defaults.

Why the others are wrong

- **B:** $0.06 million is the loss quantile itself; the expected loss of $30,000 was not subtracted.
- **C:** $0.01 million results from using 4 defaults (4 x $10,000 - $30,000), a wrong percentile; Binomial(100, 0.03) needs 6 defaults to reach 95%.
- **D:** $0.09 million adds the expected loss to the quantile loss ($60,000 + $30,000) instead of subtracting it.

Q492

AnalystPrep Q.1857

REBUILT — Replaced out-of-scope 'tranche thinness' option with 'Expected loss' and tightened the stem.

In credit portfolio analysis, which of the following captures the systematic, non-diversifiable component of portfolio credit risk, i.e., the tendency for defaults to cluster?

A. Default correlation
B. Credit VaR
C. Expected loss
D. Granularity

Answer: A

Step by step

1. Defaults cluster when credits respond to common factors; this co-movement is measured by default correlation.
2. Higher default correlation means less diversification benefit; at correlation 1 the portfolio behaves like a single credit.
3. Credit VaR, expected loss and granularity do not themselves measure co-movement of defaults.

Why A is correct: Default correlation measures how defaults move together due to common (systematic) drivers, and this cannot be diversified away by adding more credits.

Why the others are wrong

- **B:** Credit VaR is a loss measure that depends on correlation; it is an outcome, not the systematic-risk concept itself.
- **C:** Expected loss (value x pi x LGD) is unaffected by correlation, so it cannot capture clustering of defaults.
- **D:** Granularity concerns how finely the portfolio is divided into positions (idiosyncratic concentration), not common-factor risk.

Q493 Schweser Module Quiz 29.1 Q3

REBUILT — Replaced implausible option B ($1,000) with the common mistake of not subtracting expected loss.

A portfolio has a notional value of $1,000,000 with 20 credit positions. Each credit has a default probability of 2% and a recovery rate of zero. All positions are obligations of the same obligor, so default correlation equals 1. What is the credit VaR at the 99% confidence level?

A. $0
B. $1,000,000
C. $20,000
D. $980,000

Answer: D

Step by step

1. Correlation 1: one-credit behavior, 2% chance of a $1,000,000 loss, 98% chance of zero.
2. Since 2% > 1% (= 1 - 99%), the 99% loss quantile is $1,000,000.
3. Expected loss = 0.02 x $1,000,000 = $20,000.
4. Credit VaR = $1,000,000 − $20,000 = $980,000.

Why D is correct: With default correlation of 1 the portfolio acts as one credit; when pi exceeds 1 minus the confidence level, the loss quantile is the full portfolio value, so credit VaR = full loss minus expected loss.

Why the others are wrong

- **A:** $0 applies the zero-loss-quantile case (pi below 1 - confidence) or ignores the single-credit logic; here pi = 2% exceeds the 1% tail.
- **B:** $1,000,000 is the loss quantile; expected loss was not subtracted.
- **C:** $20,000 is only the expected loss, not the credit VaR.

Q494 AnalystPrep Q.3073

Portfolio Y has a notional value of $1,000,000 with 30 credit positions. Each credit has a default probability of 3% and zero recovery. All positions are obligations of the same obligor, so default correlation equals 1. Determine the credit VaR at the 99% confidence level.

A. $970,000
B. $980,000
C. $30,000
D. $200,000

Answer: A

Step by step

1. Correlation 1: single-credit behavior; 3% chance of $1,000,000 loss, 97% chance of zero.
2. 3% > 1%, so the 99% loss quantile = $1,000,000.
3. Expected loss = 0.03 x $1,000,000 = $30,000.
4. Credit VaR = $1,000,000 − $30,000 = $970,000.

Why A is correct: With default correlation of 1 the portfolio acts as one credit; when pi exceeds 1 minus the confidence level, the loss quantile is the full portfolio value, so credit VaR = full loss minus expected loss.

Why the others are wrong

- **B:** $980,000 subtracts an expected loss of $20,000 (a 2% default probability) instead of the correct $30,000.
- **C:** $30,000 is just the expected loss.
- **D:** $200,000 has no basis in the data; it ignores the single-credit logic of correlation 1.

Q495 AnalystPrep Q.3072

A risk analyst at a bank holds a portfolio with a notional value of $4 million and 30 credit positions. Each credit has a default probability of 4% and a recovery rate of zero. The default correlation is 1. What is the credit VaR at the 99% confidence level?

A. $3.6 million
B. $0.16 million
C. $4 million
D. $3.84 million

Answer: D

Step by step

1. Correlation 1: one-credit behavior; 4% chance of $4 million loss, 96% chance of zero.
2. 4% > 1%, so the 99% loss quantile = $4,000,000.
3. Expected loss = 0.04 x $4,000,000 = $160,000.
4. Credit VaR = $4,000,000 − $160,000 = $3,840,000.

Why D is correct: With default correlation of 1 the portfolio acts as one credit; when pi exceeds 1 minus the confidence level, the loss quantile is the full portfolio value, so credit VaR = full loss minus expected loss.

Why the others are wrong

- **A:** $3.6 million subtracts 10% of the portfolio instead of the 4% expected loss.
- **B:** $0.16 million is only the expected loss.
- **C:** $4 million is the loss quantile; expected loss was not subtracted.

Q496 AnalystPrep Q.4366

A firm owns a portfolio of credits that exhibit a default correlation of 1. In this case:

A. The number of defaults is binomially distributed because there is no correlation among the credits
B. The number of defaults is Poisson distributed with a mean of 1 per unit time
C. The portfolio behaves as if it consisted of just one credit
D. Significant credit diversification may be achieved

Answer: C

Step by step

1. Correlation 1 means all credits default together or not at all.
2. So the portfolio has only two outcomes, total loss or none, like a single credit.
3. No diversification benefit exists.

Why C is correct: Perfectly correlated defaults remove any diversification, so the portfolio is equivalent to one credit.

Why the others are wrong

- **A:** The binomial description applies to default correlation of 0 (independent defaults), not 1.
- **B:** Nothing in a correlation of 1 implies a Poisson distribution; the outcome is two-point.
- **D:** Diversification is greatest at correlation 0 and absent at correlation 1.

Q497 AnalystPrep Q.4367

REBUILT — Replaced implausible distractors ($100,000, $1,000,000) with realistic mistakes.

A portfolio worth $100,000,000 consists of n credits, each with default probability 2% and zero recovery. The default correlation is 1. What is the credit VaR at the 95% confidence level?

A. $98,000,000
B. -$2,000,000
C. $2,000,000
D. $0

Answer: B

Step by step

1. Correlation 1: single-credit behavior; 2% chance of $100,000,000 loss, 98% chance of zero.
2. 2% < 5% (= 1 - 95%), so the 95% loss quantile is $0.
3. Expected loss = 0.02 x $100,000,000 = $2,000,000.
4. Credit VaR = $0 − $2,000,000 = −$2,000,000.

Why B is correct: Because pi is below the 5% tail, the loss quantile is zero and credit VaR equals minus the expected loss.

Why the others are wrong

- **A:** $98,000,000 applies the pi > tail rule ($100M - $2M), but pi = 2% is below the 5% tail.
- **C:** $2,000,000 drops the sign; VaR is 0 minus EL, which is negative.
- **D:** $0 reports the loss quantile and forgets that credit VaR is measured net of expected loss.

Q498 AnalystPrep Q.4368

A $100,000,000 portfolio consists of 50 credits, so each has a future value of $2,000,000 if it does not default. Default correlation is 0, $\pi = 0.02$, recovery is zero, and the number of defaults is binomial with $n = 50$ and $\pi = 0.02$. The 95th percentile of the number of defaults is 3. Determine the credit VaR at the 95% confidence level.

A. $10,000,000
B. $6,000,000
C. $4,000,000
D. $2,000,000

Answer: C

Step by step

1. Expected loss = 0.02 x $100,000,000 = $2,000,000.
2. 95th percentile loss = 3 x $2,000,000 = $6,000,000 (check: P(X<=2) = 0.922 < 0.95 <= P(X<=3) = 0.982).
3. Credit VaR = $6,000,000 − $2,000,000 = $4,000,000.

Why C is correct: Credit VaR is the loss quantile minus expected loss; here the quantile is 3 defaults.

Why the others are wrong

- **A:** $10,000,000 would require 6 defaults (6 x $2M - $2M), a quantile beyond the 95th percentile; it is also the loss from 5 defaults with no expected loss subtracted.
- **B:** $6,000,000 is the quantile loss; expected loss was not subtracted.
- **D:** $2,000,000 is just the expected loss, or equivalently the result of using only 2 defaults ($4M - $2M), a quantile one too low.

Q499 AnalystPrep Q.4369

A portfolio has a notional value of $10,000,000 with 10 credit positions. Each position has a default probability of 4% and a recovery rate of zero. All positions are obligations of the same obligor. Determine the credit VaR at the 99% confidence level.

A. $0
B. $9,600,000
C. $10,000,000
D. $5,000,000

Answer: B

Step by step

1. Same obligor implies default correlation 1, so the portfolio acts as one credit: 4% chance of $10,000,000 loss, 96% chance of zero.
2. 4% > 1%, so the 99% loss quantile = $10,000,000.
3. Expected loss = 0.04 x $10,000,000 = $400,000.
4. Credit VaR = $10,000,000 − $400,000 = $9,600,000.

Why B is correct: With default correlation of 1 the portfolio acts as one credit; when pi exceeds 1 minus the confidence level, the loss quantile is the full portfolio value, so credit VaR = full loss minus expected loss.

Why the others are wrong

- **A:** $0 uses the zero-quantile case, which applies only when pi is below 1 - confidence (1%).
- **C:** $10,000,000 is the loss quantile; expected loss was not subtracted.
- **D:** $5,000,000 assumes half the portfolio defaults, which is not an outcome when correlation is 1.

Q500 AnalystPrep Q.5538

REBUILT — Default probability (3%) equalled the 3% tail at 97% confidence, making the quantile ambiguous and the printed key arguably wrong; changed default probability to 4% so the tail is unambiguously inside the default probability.

A portfolio has a total value of $800,000 and consists of 8 equally valued loan positions. Each loan has a default probability of 4% and a recovery rate of zero. The loans are issued to different divisions of the same company, so the default correlation among them is 1. What is the credit VaR at the 97% confidence level?

A. $800,000
B. $768,000
C. -$32,000
D. $32,000

Answer: B

Key correction: Source key (C, $776,000) came from 3% default probability exactly equal to the 3% tail, an ambiguous boundary; the item was rebuilt with 4% to remove the ambiguity.

Step by step

1. Correlation of 1 means the portfolio behaves like a single credit: total loss of $800,000 with probability 4%, zero loss with probability 96%.
2. Expected loss $= 800{,}000 \times 0.04 = 32{,}000$.
3. The 97% loss quantile leaves a 3% tail. Since the default probability (4%) exceeds 3%, the 97th percentile of loss is the full $800,000.
4. Credit VaR $= 800{,}000 - 32{,}000 = 768{,}000$.

Why B is correct: With correlation 1 the loss is all-or-nothing; because $\pi > 1 -$ confidence, the quantile is the total loss, and credit VaR is that quantile less expected loss.

Why the others are wrong

- **A:** Reports the loss quantile and forgets to subtract expected loss; credit VaR is quantile minus EL.
- **C:** Uses a zero loss quantile (which applies only if the default probability is below the tail) and subtracts EL: $0 - 32{,}000$.
- **D:** Reports the expected loss itself rather than the quantile of loss less EL.

Q501 AnalystPrep Q.5539

A portfolio has a total value of $1,500,000 and consists of 15 equally valued corporate bonds. Each bond has a default probability of 2% and a recovery rate of zero. All bonds are issued by the same corporation, so the default correlation is 1. What is the credit VaR at the 95% confidence level?

A. $1,500,000
B. $0
C. $30,000
D. -$30,000

Answer: D

Step by step

1. Correlation of 1: the portfolio acts as one credit, with a 2% chance of a $1,500,000 loss and a 98% chance of zero loss.
2. Expected loss $= 1{,}500{,}000 \times 0.02 = 30{,}000$.
3. The tail at 95% is 5%. Since $2\% < 5\%$, the 95th percentile of loss is 0.
4. Credit VaR $= 0 - 30{,}000 = -30{,}000$.

Why D is correct: Because the default probability is smaller than the 5% tail, the loss quantile is zero, so credit VaR is the loss quantile minus EL, which is negative.

Why the others are wrong

- **A:** Takes the total-loss outcome as the quantile, which applies only if the default probability exceeded 5%, and ignores EL.
- **B:** Uses the zero loss quantile but forgets to subtract expected loss.
- **C:** Gets the magnitude right but the wrong sign; the quantile (0) is below EL, so VaR is negative.

Q502 AnalystPrep Q.5541

REBUILT — Original compared two unspecified portfolios and option A was self-contradictory; rebuilt with controlled equal inputs and clean options.

Portfolio A and Portfolio B each hold 100 equally sized loans with the same default probability and the same loss given default. Defaults in Portfolio A are uncorrelated (default correlation 0). Defaults in Portfolio B are highly correlated. Which statement is correct?

A. Both have the same expected loss, but Portfolio B has a more volatile loss distribution with a fatter right tail.
B. Portfolio B has a higher expected loss because correlated loans default more often.
C. Portfolio A has the higher volatility of loss because it has more independent sources of default.
D. Both have the same expected loss and the same loss volatility, since default correlation affects neither.

Answer: A

Step by step

1. Expected loss $= \sum$ exposure $\times \pi \times$ LGD depends only on the individual default probabilities, not on correlation,

so EL is equal.
2. Higher correlation makes defaults cluster: many or few defaults together, increasing loss volatility and extreme quantiles.
3. Hence B has the same EL but a more volatile, fatter-tailed loss distribution.

Why A is correct: Default correlation changes the shape (volatility and extreme quantiles) of the loss distribution, not the expected loss.

Why the others are wrong

- **B:** Confuses clustering of defaults with a higher default frequency; each loan's π is unchanged so EL is unchanged.
- **C:** Reverses the effect; independence diversifies and reduces loss volatility.
- **D:** Correct about EL but wrong that volatility is unaffected; correlation raises the variance of portfolio loss.

Q503

AnalystPrep Q.5543

REBUILT — Purely qualitative original with a loosely reasoned key; rebuilt as a numeric comparison of credit VaR under correlation 0 versus 1.

A $1,000,000 portfolio holds 50 equal credits, each with default probability 2% and zero recovery. With default correlation 0, the number of defaults is binomial and its 99th percentile is 4. With default correlation 1, the portfolio behaves as a single credit. By how much does the 99% credit VaR increase when correlation moves from 0 to 1?

A. $900,000
B. $920,000
C. $940,000
D. $980,000

Answer: B

Step by step

1. Position size $= 1{,}000{,}000/50 = 20{,}000$; EL $= 1{,}000{,}000 \times 0.02 = 20{,}000$ in both cases.
2. Correlation 0: 99th percentile loss $= 4 \times 20{,}000 = 80{,}000$; VaR $= 80{,}000 - 20{,}000 = 60{,}000$.
3. Correlation 1: tail 1% < 2%, so the 99th percentile loss is $1,000,000; VaR $= 1{,}000{,}000 - 20{,}000 = 980{,}000$.
4. Increase $= 980{,}000 - 60{,}000 = 920{,}000$.

Why B is correct: EL is unchanged by correlation, but the 99% loss quantile jumps from $80,000 to $1,000,000, so VaR rises by $920,000.

Why the others are wrong

- **A:** Subtracts the correlation-0 loss quantile ($80,000) rather than its VaR ($60,000) from $980,000.
- **C:** Uses $1,000,000 as the correlation-1 VaR (no EL subtraction) minus $60,000.
- **D:** Reports the correlation-1 credit VaR itself and forgets to subtract the correlation-0 VaR.

Q504

Mock 1 Q8

REBUILT — Loss table was inconsistent with the stated binomial (95th percentile is 3 defaults, not 4); table corrected and the question moved to the 99% level so the exposure/EL/WCL mechanics are unchanged.

A credit analyst is estimating the credit VaR of a homogeneous loan portfolio. The portfolio's characteristics and the loss distribution (number of defaults) from the bank's model are:

Portfolio parameter	Value
Total value of the credit portfolio	USD 600 million
Number of credits (equal size)	40
Probability of default per credit	3.0%
Loss given default	80%
Default correlation	0
Percentile of number of defaults	Number of defaults
95.0th	3
97.5th	4
99.0th	4
99.9th	6

What is the credit VaR at a 99% confidence level?

A. USD 21.6 million
B. USD 42.0 million
C. USD 33.6 million
D. USD 48.0 million

Answer: C

Step by step

1. Exposure per credit = 600/40 = 15 million.
2. Expected loss = 600 × 0.03 × 0.80 = 14.4 million.
3. 99th percentile: 4 defaults, so loss = 4 × 15 × 0.80 = 48.0 million.
4. Credit VaR = 48.0 − 14.4 = 33.6 million.

Why C is correct: Credit VaR is the loss at the chosen percentile of the loss distribution less expected loss, with LGD applied to both.

Why the others are wrong

- **A:** Used the 95th percentile (3 defaults): 36.0 − 14.4 = 21.6.
- **B:** Ignored LGD: 4 × 15 − 600 × 0.03 = 60 − 18 = 42.
- **D:** Reported the 99th percentile loss without subtracting expected loss.

Q505

Mock 2 Q41

Bank XYZ has a credit portfolio of 1,000 positions with a total value of USD 1 million. Each position has a default probability of 0.5%, recovery rate zero, and default correlation zero. The 99th percentile of the number of defaults is 11. At a 99% confidence level, the credit VaR is approximately:

A. USD 1,000
B. USD 5,000
C. USD 6,000
D. USD 11,000

Answer: C

Step by step

1. Position size = 1,000,000/1,000 = 1,000.
2. Expected loss = 1,000,000 × 0.005 = 5,000.
3. 99th percentile loss = 11 × 1,000 = 11,000.
4. Credit VaR = 11,000 − 5,000 = 6,000.

Why C is correct: Credit VaR equals the loss quantile (11 defaults) less expected loss (5 defaults' worth).

Why the others are wrong

- **A:** Reports a single position's value, confusing exposure size with VaR.
- **B:** Reports the expected loss instead of the quantile minus EL.
- **D:** Reports the 99th percentile loss and forgets to subtract expected loss.

LO 29.d

Describe the use of a single factor model to measure portfolio credit risk, including the impact of correlation.

Q506

AnalystPrep Q.3071

REBUILT — Source distractors were not tied to specific errors and the quantile convention was unstated; fixed N^-1(0.01) = -2.33 and rebuilt distractors from realistic formula errors.

A credit position has a correlation to the market factor of $\beta = 0.8$. In a single-factor model, what realized market value m is used to compute the probability of reaching a default threshold at the 99% confidence level (loss level $x = 0.01$, default probability $\pi = 0.01$)? Use $N^{-1}(0.01) = -2.33$.

A. -0.777
B. -0.932
C. -1.165
D. -1.398

Answer: C

Step by step

1. Loss-level relation: $m = [k - \sqrt{1-\beta^2}\,N^{-1}(x)]/\beta$ with $k = N^{-1}(\pi) = -2.33$ and $N^{-1}(0.01) = -2.33$.
2. $\sqrt{1-\beta^2} = \sqrt{1-0.64} = 0.6$.
3. Numerator: $-2.33 - 0.6(-2.33) = -2.33 + 1.398 = -0.932$.

4. $m = -0.932/0.8 = -1.165$.

Why C is correct: Setting the conditional default probability equal to the stated loss level and solving for the market factor gives the market realization at or below which losses reach that level.

Why the others are wrong

- **A:** Swapped the roles of beta and $\sqrt{1-\beta^2}$: $(-2.33 + 0.8 \cdot 2.33)/0.6 = -0.777$.
- **B:** Computed the numerator correctly but forgot to divide by beta.
- **D:** Multiplied -2.33 by $\sqrt{1-\beta^2} = 0.6$ (giving -1.398) and stopped, ignoring the k term and the division by beta.

Q507

Schweser Module Quiz 29.2 Q2

REBUILT — Original distractors were not tied to specific errors and the quantile convention was unstated; restated with N^-1(0.01) = -2.33 and rebuilt distractors.

A credit position has a correlation to the market factor of $\beta = 0.5$. In a single-factor model, what realized market value m is used to compute the probability of reaching a default threshold at the 99% confidence level ($\pi = 0.01$, loss level $x = 0.01$)? Use $N^{-1}(0.01) = -2.33$.

A. -0.624
B. -0.312
C. -1.249
D. -1.345

Answer: A

Step by step

1. $m = [k - \sqrt{1-\beta^2}\,N^{-1}(x)]/\beta$, with $k = -2.33$ and $N^{-1}(x) = -2.33$.
2. $\sqrt{1-0.25} = 0.8660$.
3. Numerator: $-2.33 - 0.8660(-2.33) = -2.33 + 2.018 = -0.312$.
4. $m = -0.312/0.5 = -0.624$.

Why A is correct: The market factor must be low enough that the conditional default probability equals the 1% loss level; solving the inverted conditional-PD formula gives -0.624.

Why the others are wrong

- **B:** Forgot to divide the numerator by beta.
- **C:** Divided by $\beta^2 = 0.25$ instead of β ($-0.312/0.25 = -1.249$).
- **D:** Swapped beta and $\sqrt{1-\beta^2}$: $(-2.33 + 0.5 \cdot 2.33)/0.866 = -1.345$.

Q508

BT answers P2.T6.310.2 · BT notes P2.T6.310.2

In the single-factor model, a firm's conditional cumulative default probability is $p(m) = N\left[\dfrac{k - \beta m}{\sqrt{1-\beta^2}}\right]$. A firm has $\beta = 0.60$ and $k = -1.645$, so its unconditional default probability is 5.0%. In an economic downturn the market factor shifts to $m = -1.41$. What is the conditional default probability?

A. 7.75%
B. 10.59%
C. 15.90%
D. 21.21%

Answer: C

Step by step

1. $\sqrt{1-\beta^2} = \sqrt{1-0.36} = 0.80$.
2. Distance to default: $k - \beta m = -1.645 - 0.60(-1.41) = -1.645 + 0.846 = -0.799$.
3. Standardize: $-0.799/0.80 = -0.9988$.
4. $p(m) = N(-0.9988) = 15.90\%$, up from 5.0% unconditional.

Why C is correct: The downturn shifts the default trigger to $k - \beta m = -0.799$ while the conditional standard deviation shrinks to 0.80, so the default probability rises from 5.0% to about 15.9%.

Why the others are wrong

- **A:** Used $\beta^2 m$ instead of βm in the numerator: $N[(-1.645 + 0.36 \cdot 1.41)/0.8]$.
- **B:** Divided by $1 - \beta^2 = 0.64$ instead of its square root: $N(-0.799/0.64)$.
- **D:** Omitted the $\sqrt{1-\beta^2}$ scaling: $N(-0.799) = 21.2\%$.

Q509 BT answers P2.T6.311.1 · BT notes P2.T6.311.1

REBUILT — Source numbers were OCR-damaged and the BT answers options did not contain the correct value; reconstructed with N^-1(0.01) = -2.326 and rebuilt options from realistic errors.

In a single-factor credit VaR model, a portfolio loss level x corresponds to a market factor value $m = [k - \sqrt{1-\beta^2}\,N^{-1}(x)]/\beta$. The default probability is 1.0%, so $k = N^{-1}(0.01) = -2.326$, and $\beta = 0.80$. Which is nearest to the probability that the portfolio loss is 0.01 or worse, i.e., the probability that the market factor is at or below the value m associated with a loss of 0.01?

A. 17.6%
B. 21.9%
C. 1.0%
D. 12.2%

Answer: D

Key correction: BT answers option C (72.24%) was OCR-damaged and is not the correct value; BT notes gave 12.24%. Correct is about 12.2%.

Step by step

1. $\sqrt{1-\beta^2} = 0.60$; $N^{-1}(0.01) = -2.326$.
2. $m = [-2.326 - 0.60(-2.326)]/0.80 = -0.930/0.80 = -1.163$.
3. Probability of a loss of 0.01 or worse $= N(m) = N(-1.163) \approx 12.2\%$.

Why D is correct: The unconditional probability of a given loss level equals the probability that the market factor is at or below the value that produces that loss, here $N(-1.163)$.

Why the others are wrong

- **A:** Forgot to divide by beta: $N(-0.930) = 17.6\%$.
- **B:** Swapped beta and $\sqrt{1-\beta^2}$: $m = -0.775$, $N = 21.9\%$.
- **C:** Confused the loss level (1%) with the probability of the market outcome; the required probability is N(m), not x.

Q510 BT notes P2.T6.304.1

REBUILT — Distractors included impossible betas above 1 (giveaways); rebuilt options from realistic errors with one correct pair.

In a single-factor model, $a = \beta m + \sqrt{1-\beta^2}\,\varepsilon$ with $m, \varepsilon \sim N(0,1)$ independent, and the firm defaults if $a < k$. A firm's systematic risk accounts for 70% of its total return variance (idiosyncratic risk 30%), and its unconditional default probability is 2.0%. What are the implied β and k?

A. $\beta = 0.55, k = -2.05$
B. $\beta = 0.70, k = -2.05$
C. $\beta = 0.84, k = -2.05$
D. $\beta = 0.84, k = +2.05$

Answer: C

Step by step

1. Systematic variance share $= \beta^2 = 0.70$, so $\beta = \sqrt{0.70} = 0.837 \approx 0.84$.
2. Unconditional default probability: $k = N^{-1}(0.02) = -2.054$.

Why C is correct: The weight on the market factor is the square root of its variance share, and the default threshold is the inverse normal of the unconditional PD (negative for PD below 50%).

Why the others are wrong

- **A:** Used the idiosyncratic share: $\sqrt{0.30} = 0.55$.
- **B:** Treated 70% as beta itself instead of β^2.
- **D:** Correct beta but wrong sign on the threshold; a 2% PD requires a negative threshold.

Q511 BT notes P2.T6.304.2

REBUILT — Distractors were not tied to specific errors; rebuilt options from realistic mistakes.

In a single-factor credit model, $a = \beta m + \sqrt{1-\beta^2}\,\varepsilon$, a firm has $\beta = 0.40$ and an unconditional default probability of 3.0%. In a modest downturn the market factor is $m = -1.0$. What is the conditional default probability?

A. 3.0%
B. 5.3%
C. 6.9%

D. 0.6%

Answer: B

Step by step

1. $k = N^{-1}(0.03) = -1.881$.
2. Distance to default: $k - \beta m = -1.881 + 0.40 = -1.481$.
3. $\sqrt{1 - 0.16} = 0.9165$; standardized: $-1.481/0.9165 = -1.616$.
4. $N(-1.616) \approx 5.3\%$.

Why B is correct: The downturn moves the distance to default closer to zero and the conditional standard deviation shrinks to 0.9165, lifting the PD from 3.0% to about 5.3%.

Why the others are wrong

- **A:** Reported the unconditional PD, ignoring the market shift.
- **C:** Left out the $\sqrt{1 - \beta^2}$ scaling: $N(-1.481) = 6.9\%$.
- **D:** Sign error on the shock: used $k + \beta m$, treating the downturn as a boom, giving $N(-2.49) \approx 0.6\%$.

Q512

BT notes P2.T6.304.3

REBUILT — Option C relied on an equity-beta statement not supported by the reading; replaced with a statement drawn from the model.

Each of the following is an assumption or implication of the single-factor credit model $a_i = \beta_i m + \sqrt{1 - \beta_i^2}\,\varepsilon_i$ EXCEPT:

A. m and ε_i are independent standard normal variables.

B. Because $\text{cov}[m, \varepsilon_i] = 0$, the asset return a_i is also standard normal.

C. The correlation between the asset returns of firms i and j is $\beta_i \beta_j$.

D. Conditioning is done by fixing the idiosyncratic shock ε (the single factor), which increases the standard deviation of the default distribution.

Answer: D

Step by step

1. A: the model assumes $m, \varepsilon \sim N(0, 1)$ with zero covariance: true.
2. B: variance of a_i is $\beta^2 + (1 - \beta^2) = 1$, mean 0: true.
3. C: $\text{Corr}(a_i, a_j) = \beta_i \beta_j$ as the shocks are independent: true.
4. D: the single factor is the market factor m; conditioning on m shifts the mean to βm and reduces the standard deviation from 1 to $\sqrt{1 - \beta^2}$. False.

Why D is correct: The model conditions on the market factor m, not on epsilon, and conditioning reduces rather than increases the standard deviation.

Why the others are wrong

- **A:** True: this is the model's basic assumption.
- **B:** True: the variances sum to one.
- **C:** True: asset correlation between two firms is $\beta_i \beta_j$.

Q513

GARP PE2 Q69 · Mock 4 Q70

REBUILT — Original did not state the LGD, so 'loss rate of 0% or 100%' only holds for LGD = 100%; stated LGD = 100% explicitly and tidied wording.

An analyst is assessing the credit VaR of a homogeneous portfolio using the single-factor model, in which each firm's asset return is $a_i = \beta_i m + \sqrt{1 - \beta_i^2}\,\varepsilon_i$. The portfolio has 1,000 firms, each 0.1% of the portfolio, there is no idiosyncratic risk across the portfolio (it is fully granular, so idiosyncratic shocks diversify away), and loss given default is 100% for every firm. Which of the following observations about the model and its parameters is correct?

A. When $\beta = 1$, the loss rate of the portfolio is either 0% or 100%.

B. For a given probability of default, the distance to default of a firm is low when its correlation to the market factor is low.

C. When the correlation to the market factor is zero, the portfolio loss rate is typically higher than the firm's probability of default.

D. The asset return correlation to the market factor is measured by the square root of β.

Answer: A

Step by step

1. Asset return: $a_i = \beta_i m + \sqrt{1-\beta_i^2}\,\varepsilon_i$; default occurs when $a_i \le k_i$, where $k_i = N^{-1}(PD)$.
2. With $\beta = 1$ the idiosyncratic term vanishes: $a_i = m$ for every firm, so all firms default if $m \le k$ and none otherwise.
3. Hence the portfolio loss rate is 100% (all default, LGD = 100%) or 0%: option A.
4. With $\beta = 0$ the firms are independent; the loss rate converges to PD (law of large numbers), not above it.
5. $k_i = N^{-1}(PD)$ depends only on PD, not on correlation; β itself is the correlation with the factor.

Why A is correct: At $\beta = 1$ all firms share exactly the same driver, so defaults are perfectly correlated and the portfolio is all-or-nothing.

Why the others are wrong

- **B:** Distance to default k = N^{-1}(PD) is set by PD alone; correlation changes the shape of the loss distribution, not k.
- **C:** With zero correlation and a granular portfolio, diversification makes the loss rate equal PD, not higher; confuses zero correlation with high risk.
- **D:** The correlation of a firm's return to the market factor is β itself; $\beta_i\beta_j$ is the correlation between two firms. The square root of β is not used.

Q514 Schweser Module Quiz 29.2 Q1

REBUILT — Option expressions were blank in OCR; reconstructed from the single-factor model and rewrote the stem and distractors to be unambiguous.

In the single-factor model $a_i = \beta_i m + \sqrt{1-\beta_i^2}\,\varepsilon_i$, where m and ε_i are independent standard normal variables, a specific value m of the market factor is realized. What are the mean and standard deviation of the conditional distribution of firm i's asset return?

A. Mean 0 and standard deviation 1, the same as the unconditional distribution.

B. Mean $\beta_i m$ and standard deviation $\sqrt{1-\beta_i^2}$.

C. Mean m and standard deviation β_i.

D. Mean $\beta_i m$ and standard deviation $1-\beta_i^2$.

Answer: B

Step by step

1. Fix m: the only random term left is ε_i, so $a_i \mid m = \beta_i m + \sqrt{1-\beta_i^2}\,\varepsilon_i$.
2. Mean: $E[a_i \mid m] = \beta_i m$ since $E[\varepsilon_i] = 0$.
3. Standard deviation: $\sqrt{1-\beta_i^2} \times 1 = \sqrt{1-\beta_i^2}$, below the unconditional value of 1 for $\beta_i > 0$.
4. The conditional distribution is therefore normal, with a shifted mean and reduced dispersion.

Why B is correct: Conditioning removes the systematic variance β_i^2, leaving only the idiosyncratic variance $1-\beta_i^2$ around the shifted mean $\beta_i m$.

Why the others are wrong

- **A:** This is the unconditional standard normal distribution; it ignores the information from the realized m.
- **C:** Mixes up the roles: the mean is scaled by β_i, not equal to m, and the dispersion comes from the idiosyncratic loading $\sqrt{1-\beta_i^2}$, not β_i.
- **D:** Correct mean but reports the conditional variance $1-\beta_i^2$ (0.36 at $\beta = 0.8$) instead of the standard deviation $\sqrt{1-\beta_i^2}$ (0.6).

LO 29.e

Define beta and calculate the asset return correlation of any pair of firms using the single factor model.

Q515

BT answers P2.T6.24.12.1

REBUILT — Original asked for a covariance using a market-factor variance of 0.05, inconsistent with the standard-normal single-factor model; rebuilt to ask for the asset return correlation of the pair (LO 29.e).

Under the single-factor credit model, each firm's asset return is $a_i = \beta_i m + \sqrt{1-\beta_i^2}\,\varepsilon_i$, where m and ε_i are independent standard normal variables, so beta serves as a correlation (unlike the unbounded CAPM beta). Firm X has $\beta_X = 0.8$ and Firm Y has $\beta_Y = 0.9$. What is the correlation between the asset returns of the two firms?

A. 0.72
B. 0.85
C. 1.70
D. 0.036

Answer: A

Step by step

1. Both returns are standard normal, so covariance equals correlation: $\text{Cov}(a_X, a_Y) = \beta_X\beta_Y \text{Var}(m)$ with $\text{Var}(m) = 1$.
2. Idiosyncratic shocks are independent of each other and of m, so only the common factor contributes.
3. $\rho_{XY} = 0.8 \times 0.9 = 0.72$.

Why A is correct: The pairwise asset correlation is the product of the two betas, $\beta_X\beta_Y$, because the market factor is the only shared source of risk.

Why the others are wrong

- **B:** Took the geometric mean $\sqrt{0.72} = 0.85$; the correlation is the product, not its square root.
- **C:** Added the betas (0.8 + 0.9); correlation cannot exceed 1.
- **D:** Multiplied the betas by an extra factor of 0.05 ($0.72 \times 0.05 = 0.036$), which wrongly treats a variance as scaling a correlation of standard normals.

Q516

BT answers P2.T6.310.1 · BT notes P2.T6.310.1

REBUILT — The two candidate duplicates test the same calculation but the betas were OCR-damaged in both (0.35/0.40 vs 0.85/0.87); merged into one item with consistent betas 0.85, 0.87, 0.56 and rebuilt distractors so each reflects a realistic error.

In a single-factor model, each firm's standardized asset return is $a_i = \beta_i m + \sqrt{1-\beta_i^2}\,\varepsilon_i$, where $m, \varepsilon_i \sim N(0,1)$, $\text{cov}[m, \varepsilon_i] = 0$ and $\text{cov}[\varepsilon_i, \varepsilon_j] = 0$ for $i \neq j$. A portfolio contains three credits with betas $\beta_1 = 0.85$, $\beta_2 = 0.87$ and $\beta_3 = 0.56$. What is the implied asset return correlation between credits 1 and 2?

A. 0.860
B. 0.260
C. 0.740
D. 0.476

Answer: C

Step by step

1. Each asset return has unit variance: $\text{var}(a_i) = \beta_i^2 + (1-\beta_i^2) = 1$, so correlation equals covariance.
2. Only the common factor links the two firms: $\text{cov}(a_1, a_2) = \beta_1\beta_2 \text{var}(m) = \beta_1\beta_2$.
3. $\rho_{1,2} = \beta_1\beta_2 = 0.85 \times 0.87 = 0.7395 \approx 0.740$.
4. Credit 3 ($\beta_3 = 0.56$) is irrelevant to the pairwise correlation of credits 1 and 2.

Why C is correct: The only shared source of randomness is the market factor, so the pairwise asset correlation is the product of the two betas, $\beta_1\beta_2 = 0.7395$.

Why the others are wrong

- **A:** 0.860 is the simple average of the betas (or roughly $\sqrt{\beta_1\beta_2}$); the correlation is the product, not the average or geometric mean.
- **B:** 0.260 is $\sqrt{1-\beta_1^2}\sqrt{1-\beta_2^2}$, the product of the idiosyncratic loadings, which are uncorrelated and contribute nothing to correlation.
- **D:** 0.476 is $\beta_1\beta_3 = 0.85 \times 0.56$, the correlation of credits 1 and 3 - the wrong pair was used.

Q517 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A credit analyst at an asset manager uses the single-factor model, in which firm i has asset return $a_i = \beta_i m + \sqrt{1-\beta_i^2}\,\varepsilon_i$, with m and each ε_i independent standard normal variables. Two firms in the same industry are assumed to have the same beta to the market factor, and the analyst estimates that the correlation between their asset returns is 0.36. Which of the following is the beta of each firm?

A. 0.13
B. 0.18
C. 0.36
D. 0.60

Answer: D

Step by step

1. Asset return correlation between firms i and j is $\beta_i\beta_j$.
2. With equal betas: $\beta^2 = 0.36$.
3. $\beta = \sqrt{0.36} = 0.60$.

Why D is correct: Because the idiosyncratic shocks are uncorrelated, the only common component is the market factor, so $\rho_{ij} = \beta_i\beta_j = \beta^2$. Beta is therefore the square root of 0.36, or 0.60.

Why the others are wrong

- **A:** 0.13 is 0.36^2; it squares the correlation instead of taking its square root.
- **B:** 0.18 is half of 0.36; it divides the correlation by 2, but with equal betas the correlation is β^2, so the correct step is a square root.
- **C:** 0.36 treats the correlation as equal to beta; this ignores that the correlation is β^2 when betas are equal.

LO 29.f

Estimate the probability of a joint default of any pair of credits and the default correlation between any pair of credits using the single factor model.

Q518 BT answers P2.T6.310.3 · BT notes P2.T6.310.3

REBUILT — Original referred to missing exhibit panels and had an illegible table entry; rebuilt with a self-contained table of joint PDs (computed from the bivariate normal) and the default-correlation formula stated.

Two credits each have a 1-year default probability of $\pi = 2.0\%$. Under the single-factor model, their joint default probability is $\pi_{12} = \Phi_2(k, k; \rho_a)$, the bivariate standard normal CDF at the default threshold $k = \Phi^{-1}(2\%)$, where ρ_a is the asset correlation. The default correlation is $\rho = (\pi_{12} - \pi^2)/(\pi(1-\pi))$. The table lists joint default probabilities implied by various asset correlations (solved numerically):

Asset correlation	Joint default probability
0.00	0.040%
0.05	0.053%
0.10	0.069%
0.15	0.088%
0.20	0.110%
0.25	0.136%

Given a default correlation of 4.90%, what is the implied asset correlation?

A. 0.10
B. 0.15
C. 0.20
D. 0.25

Answer: D

Key correction: BT answers listed option D among a corrupted set (0.70, 0.75, 0.20, 0.25); BT notes also gave D. D (0.25) is correct; the source's joint PD of 0.7360% was an OCR error for 0.1360%.

Step by step

1. Rearrange the default correlation formula: $\pi_{12} = \rho\,\pi(1-\pi) + \pi^2$.

2. $\rho\,\pi(1-\pi) = 0.0490 \times 0.02 \times 0.98 = 0.0000960$ (0.0960%).
3. $\pi^2 = 0.02^2 = 0.000400$ (0.0400%).
4. $\pi_{12} = 0.0960\% + 0.0400\% = 0.1360\%$.
5. Look up 0.136% in the table: it corresponds to an asset correlation of 0.25.

Why D is correct: The default correlation converts to a joint default probability of 0.136% via the covariance identity, and the bivariate normal table maps that joint PD to an asset correlation of 0.25. Asset correlation exceeds default correlation because default is a rare, discrete event.

Why the others are wrong

- **A:** 0.10 maps to a joint PD of 0.069% (default correlation about 1.5%); it comes from understating the joint PD, e.g. mis-scaling $\rho\,\pi(1-\pi)$.
- **B:** 0.15 maps to 0.088%; it is the nearest row to the joint PD of about 0.096% obtained by omitting the $\pi^2 = 0.040\%$ term (computing only $\rho\,\pi(1-\pi)$).
- **C:** 0.20 corresponds to 0.110%, i.e. a default correlation of about 3.6%; it is a near-miss from a rough read of the table rather than from $\pi_{12} = 0.136\%$.

Q519

GARP PE1 Q41 · Mock 3 Q39

The fixed-income trading desk of an investment bank uses several methodologies to estimate joint default probabilities of counterparties. The manager in charge of the desk asks an analyst to calculate the joint default probability of two firms (credits) over a 1-year period using the single-factor model. The portfolio consists of two firms (credits) that have the same beta to the market factor. The probability of default of each firm is 2.85%. If the analyst applies a default correlation between this pair of firms of 0.45, what is the joint default probability of the firms using the single-factor model?

A. 1.25%
B. 1.33%
C. 2.85%
D. 2.77%

Answer: B

Step by step

1. Joint default probability: $\pi_{12} = \rho\,\pi(1-\pi) + \pi^2$.
2. $\pi(1-\pi) = 0.0285 \times 0.9715 = 0.027688$.
3. $\rho\,\pi(1-\pi) = 0.45 \times 0.027688 = 0.012459$.
4. $\pi^2 = 0.0285^2 = 0.000812$.
5. $\pi_{12} = 0.012459 + 0.000812 = 0.013272 \approx 1.33\%$.

Why B is correct: Default correlation is defined as $(\pi_{12} - \pi^2)/(\pi(1-\pi))$ for identical credits, so π_{12} is the correlation-scaled variance plus the independent joint probability π^2.

Why the others are wrong

- **A:** 1.25% is $\rho\,\pi(1-\pi)$ alone; it omits the $\pi^2 = 0.081\%$ independence term.
- **C:** 2.85% is the single-name PD; it is the joint PD only under perfect default correlation (rho = 1), and it exceeds what rho = 0.45 implies.
- **D:** 2.77% is $\pi(1-\pi)$, the Bernoulli variance, used as if it were the joint probability without multiplying by ρ.

Q520

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A risk analyst at a bank estimates that two counterparties each have a 1-year default probability of 10%, and that the probability that both default within the year is 2%. Which of the following is the default correlation between the two counterparties?

A. 0.111
B. 0.100
C. 0.200
D. 0.010

Answer: A

Step by step

1. Default correlation $\rho = (\pi_{12} - \pi_1\pi_2)/\sqrt{\pi_1(1-\pi_1)\pi_2(1-\pi_2)}$.
2. Numerator: $0.02 - 0.10 \times 0.10 = 0.01$.

3. Denominator: $0.10 \times 0.90 = 0.09$.
4. $\rho = 0.01/0.09 = 0.111$.

Why A is correct: The default correlation is the excess of the joint default probability over the independence value (0.01), scaled by the Bernoulli standard deviations (variance 0.09 for each credit), giving 0.111.

Why the others are wrong

- **B:** 0.100 divides by π (0.10) instead of $\pi(1-\pi)$, omitting the $1-\pi$ factor in the variance.
- **C:** 0.200 divides the joint default probability by the single-name default probability and ignores the independence term $\pi_1\pi_2$.
- **D:** 0.010 is only the numerator (excess joint default probability); it was never scaled by the variance.

LO 29.g

Describe how credit VaR can be calculated using a simulation of joint defaults.

Q521

AnalystPrep Q.2887

REBUILT — Original options were vague and omitted the copula/correlation step; rewrote as an ordering question on the four-step copula simulation of credit VaR so exactly one sequence is correct.

A risk manager computes the credit VaR of a loan portfolio by simulating joint defaults with a copula, so that default correlation drives how often borrowers default together. Which sequence correctly lists the steps?

A. Simulate default times; define the copula function; compute portfolio distribution statistics; obtain scenario market values and profit-and-loss.

B. Define the copula function (including the correlation between credits); simulate default times; obtain market values and profit-and-loss for each scenario using the simulated default times; compute portfolio distribution statistics (e.g., the loss percentile) from the summed terminal values.

C. Obtain market values and profit-and-loss for each scenario; define the copula function; simulate default times; compute portfolio distribution statistics.

D. Define the copula function; obtain market values and profit-and-loss for each scenario; simulate default times; compute portfolio distribution statistics.

Answer: B

Step by step

1. Step 1: define the copula function, which fixes the marginal default behaviour and the correlation structure between credits.
2. Step 2: simulate default times from the copula, so correlation determines how often defaults cluster.
3. Step 3: for each scenario use the simulated default times to obtain terminal market values and profit-and-loss for each credit.
4. Step 4: add the terminal values across credits and compute distribution statistics; credit VaR is read from the chosen percentile of the simulated loss distribution.
5. This matches option B.

Why B is correct: Losses can only be valued once default times are simulated, and default times can only be simulated once the copula (with its correlation) is specified; VaR comes last from the aggregated distribution.

Why the others are wrong

- **A:** Simulates default times before the copula exists (no correlation structure), and computes statistics before the P&L data they need.
- **C:** Values the portfolio before any defaults have been simulated, so there are no scenario losses to value.
- **D:** Swaps steps 2 and 3: scenario P&L cannot be obtained until default times have been simulated.

Q522

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A risk manager is estimating the credit VaR of a two-loan portfolio by simulating joint defaults with a normal copula. Which of the following best describes how the simulation produces the portfolio loss distribution?

A. Independent default times are simulated for each loan, and the copula correlation is applied afterward by scaling the sum of the individual losses.

B. Correlated normal draws are generated for the credits and mapped, using each credit's own default probability, to simulated default times; terminal values for each scenario are then summed and the loss percentile is read

from the simulated distribution.

C. Credit VaR is computed separately for each loan from its own simulated loss distribution, and the portfolio credit VaR is the sum of the two.

D. The copula is used to simulate only the market factor, and the portfolio loss distribution is then obtained analytically without generating scenarios.

Answer: B

Step by step

1. Step 1: define the copula (normal copula with the specified correlation).
2. Step 2: simulate correlated draws and map them to each credit's default time.
3. Step 3: obtain market values and P&L per scenario using the simulated default times.
4. Step 4: add terminal values across loans and read the percentile of the loss distribution.

Why B is correct: The correlation is built into the simulated draws themselves, so defaults cluster in the same scenarios; portfolio statistics come from summing the scenario terminal values.

Why the others are wrong

- **A:** Correlation must enter the draws that determine default times; applying it afterward to independent simulations does not reproduce joint defaults.
- **C:** Summing stand-alone VaRs ignores the joint behaviour captured by the copula (default correlation) and generally overstates diversified VaR.
- **D:** The method simulates default times for each credit and aggregates scenario values; it is not an analytic calculation without scenarios.

Q523 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A risk manager simulates 1,000 scenarios of a portfolio of two $1,000,000 loans using a normal copula. The simulated 1-year portfolio losses are: $0 in 930 scenarios, $400,000 in 40 scenarios, $600,000 in 22 scenarios and $1,000,000 in 8 scenarios. Which of the following is the 99% credit VaR (loss level) implied by the simulation?

A. $1,000,000

B. $400,000

C. $600,000

D. $37,200

Answer: C

Step by step

1. Total scenarios 8+22+40+930 = 1,000; the worst 1% is 10 scenarios.
2. Losses of $1,000,000 occur in only 8 scenarios (0.8%), fewer than 1%, so they lie beyond the 99th percentile.
3. Adding the 22 scenarios at $600,000 gives 30 (3%) with loss at least $600,000, which exceeds 1%; the 99th percentile loss is $600,000.

Why C is correct: The 99% VaR is the loss level exceeded in no more than 1% of simulated scenarios; that level is $600,000.

Why the others are wrong

- **A:** $1,000,000 is the worst-case loss (both loans default), which is not the 99th percentile because it occurs in only 0.8% of scenarios.
- **B:** $400,000 is the 95% level (70 scenarios, 7%, have loss at least $400,000 and only 3% exceed it); it is too low for 99%.
- **D:** $37,200 is the expected loss, $(8 \times 1000 + 22 \times 600 + 40 \times 400)/1000$ in $ thousands; VaR is a percentile, not the mean.

LO 29.h

Assess the effect of granularity on credit VaR.

Q524 BT answers P2.T6.24.12.3 · BT notes P2.T6.24.12.3

Bastrop Bank specializes in making commercial loans to large real estate developments. Management is considering reducing its exposure to any one development and making smaller average loans to more small businesses. How would this likely affect the granularity of the portfolio and its credit VaR?

A. Granularity would decrease as the bank focuses on fewer loans with higher exposure.
B. Granularity would increase as the bank holds a larger number of smaller loans, which tends to lower credit VaR.
C. Granularity would not be affected by the loan size or number of borrowers.
D. The impact cannot be determined without knowing the current credit VaR.

Answer: B

Step by step

1. Granularity means each credit is a smaller share of the total portfolio, achieved by increasing the number of credits.
2. Replacing a few large developments with many smaller loans reduces the weight of each credit, so granularity rises.
3. A more granular portfolio has lower credit VaR (loss quantile less expected loss), because the loss converges toward the expected loss as idiosyncratic defaults diversify away.
4. The direction of the effect does not depend on the current credit VaR level.

Why B is correct: Spreading the same total exposure across many smaller loans makes the portfolio more granular, and greater granularity reduces credit VaR.

Why the others are wrong

- **A:** Reverses the relationship: fewer, larger loans mean lower granularity, but the bank is moving the other way.
- **C:** Ignores that granularity is defined by the weight of each credit, i.e. by loan size and number of credits.
- **D:** The direction of the effect is a general result and needs no starting VaR value.

Q525 BT answers P2.T6.309.3 · BT notes P2.T6.309.3

REBUILT — Option D (higher default correlation DECREASES credit VaR) was also false, giving two false statements; D now says INCREASE so only C is false.

A portfolio has a total value of $1.0 billion divided into n = 20 equal positions of $50 million. Each position has a default probability of 1.0% and a recovery rate of zero, and default correlation is zero. Credit VaR is defined at the 95% confidence level as the 95th percentile of credit loss less the expected loss. Varying granularity (n) or default correlation, one at a time and holding all else constant, each of the following is true EXCEPT:

A. The initial assumptions imply a 95% credit VaR of $40 million.
B. Increasing granularity from n = 20 to n = 100 positions decreases the 95% credit VaR.
C. As n approaches infinity with independent small positions, the 95% credit VaR tends toward $990 million.
D. An increase in default correlation increases the 95% credit VaR.

Answer: C

Key correction: Source key (C) was correct, but the printed option D (correlation increase DECREASES VaR) was also false; D was reworded. Source also quoted P(0 defaults) as 87.8%; correct is 0.99^20 = 81.8%.

Step by step

1. Expected loss = 1% x $1,000 million = $10 million for any n.
2. n = 20: defaults ~ Binomial(20, 1%). P(0) = 0.99^20 = 81.8%, P(<=1) = 98.3% >= 95%, so the 95th percentile is 1 default = $50 million loss. Credit VaR = 50 - 10 = $40 million, so A is true.
3. n = 100: each position is $10 million; P(<=2) = 92.1% < 95% and P(<=3) = 98.2%, so 3 defaults = $30 million loss. Credit VaR = 30 - 10 = $20 million < $40 million, so B is true.
4. As n grows with independent defaults, the loss converges to the expected loss ($10 million), so credit VaR tends toward 0, not $990 million. C is false.
5. Higher default correlation fattens the tail of the loss distribution (correlation 1 means the portfolio behaves like one credit), raising extreme quantiles, so D is true.

Why C is correct: Diversification in a granular, uncorrelated portfolio pushes the 95th percentile loss toward the expected loss, so credit VaR goes to zero rather than $990 million.

Why the others are wrong

- **A:** True: the 95th percentile is one default ($50m) and subtracting EL of $10m gives $40m; calling it false usually comes from forgetting to subtract EL or using P(0)=87.8%.
- **B:** True: at n=100 the quantile is 3 defaults x $10m = $30m, VaR $20m, lower than $40m; granularity reduces credit VaR.

- **D:** True: higher default correlation raises tail losses and credit VaR; the opposite direction confuses correlation with diversification.

Q526 AnalystPrep Q.5529

Global Finance Bank is assessing a collateralized loan obligation (CLO) whose underlying pool consists of a large number of small-to-medium-sized corporate loans. How does the higher loan granularity of this pool affect its risk profile compared with a pool of lower granularity?

A. It increases the systemic risk due to a higher number of loans.
B. It reduces the diversification benefit, thus increasing the probability of tail events.
C. It enhances diversification, thereby reducing the probability of tail events and concentration risk.
D. It leads to higher operational costs due to the complexity of managing a larger number of loans.

Answer: C

Step by step

1. Higher granularity means each loan is a smaller share of the pool.
2. A single default then has a small effect on total loss, so idiosyncratic defaults diversify away.
3. Loss converges toward the expected loss, so extreme losses become less likely and credit VaR falls; name concentration is reduced.

Why C is correct: More, smaller exposures diversify idiosyncratic default risk, lowering the probability of extreme losses and credit VaR.

Why the others are wrong

- **A:** The number of loans does not raise systemic risk; granularity removes idiosyncratic risk, so the loss effect is the opposite.
- **B:** Reverses the effect: granularity increases diversification and lowers tail probability.
- **D:** Operating cost may rise, but it is not the credit-risk effect asked about.

Reading 30. Credit Risk

John C. Hull, Options, Futures, and Other Derivatives, 11th Edition (Pearson, 2022). Chapter 24.

LO	Learning objective	Qs
30.a	assess the credit risks of derivatives.	4
30.b	define credit valuation adjustment (CVA) and debt valuation adjustment (DVA).	4
30.c	calculate the probability of default using credit spreads.	5
30.d	describe, compare, and contrast various credit risk mitigants and their role in credit analysis.	3
30.e	describe the significance of estimating default correlation for credit portfolios and distinguish between reduced form and structural default correlation models.	4
30.f	describe the Gaussian copula model for time to default and calculate the probability of default using the one-factor Gaussian copula model.	5
30.g	describe how to estimate credit VaR using the Gaussian copula and the CreditMetrics approach.	4

LO 30.a

Assess the credit risks of derivatives.

Q527

AnalystPrep Q.6212

REBUILT — Source said collateral reflected the value 9 days earlier although the margin period is 10 days; rewrote so the collateral is based on the value one full margin period (10 days) before default.

Bank A and Bank B have a bilateral swap under an ISDA Master Agreement with a zero-threshold, two-way collateral agreement. The margin period of risk (cure period) is 10 days: Bank B stops posting collateral 10 days before default, so Bank A holds collateral equal to the swap value 10 days before the default. That value was \$40 million. Bank B defaults at time T, just before the next margin call, when the swap value to Bank A is \$50 million. What is Bank A's credit exposure to Bank B at default?

A. \$0
B. \$10 million
C. \$40 million
D. \$50 million

Answer: B

Step by step

1. Collateral held by A = value 10 days before default = \$40 million.
2. Current value of the swap to A at default = \$50 million (positive, so B owes A).
3. Exposure = max(value at default - collateral held, 0) = $50 - 40 = \$10$ million.
4. A is an unsecured creditor for \$10 million.

Why B is correct: Collateral lags value by the margin period of risk, so only the 10-day change in value is uncollateralized.

Why the others are wrong

- **A:** Assumes collateral fully covers the exposure; ignores that collateral is based on the stale value from 10 days earlier.
- **C:** Reports the collateral held (\$40m) instead of the uncovered amount.
- **D:** Ignores the collateral entirely, treating A as unsecured for the full value.

Q528

AnalystPrep Q.6213

REBUILT — Source asked which outcome was 'possible' with loosely worded options; rewrote as a numeric exposure question using the Hull cure-period convention with an excess-collateral twist.

A bank and a nonfinancial counterparty have a two-way, zero-threshold collateral agreement: each side must post collateral equal to $\max(V, 0)$, where V is the value of the transactions to the other side. The cure period (margin period of risk) is 10 days, so the counterparty stops posting or returning collateral 10 days before default. In a simulation trial, the value of the outstanding transactions to the bank at time T is \$80 million, and 10 days earlier it was \$90 million. The counterparty defaults at T. What is the bank's exposure?

A. \$0
B. \$10 million
C. \$80 million

D. $90 million

Answer: A

Step by step

1. Collateral held by the bank = value 10 days earlier = $90 million.
2. Value owed to the bank at default = $80 million.
3. Collateral held ($90m) exceeds the amount owed ($80m), so the bank is fully covered.
4. Exposure = max(80 - 90, 0) = $0.

Why A is correct: Because the collateral held (based on the earlier $90m value) exceeds the current $80m value, nothing is uncollateralized and exposure is zero.

Why the others are wrong

- **B:** Takes the absolute difference between 90 and 80; the excess is held by the bank, not a loss to it.
- **C:** Ignores the collateral and treats the bank as unsecured for the full $80m.
- **D:** Reports the collateral held, which is not an exposure measure.

Q529 Schweser Module Quiz 30.2 Q1

In a bilaterally cleared derivatives transaction between two companies (Company A and Company B), Company B defaults. The value of the transactions for Company A is a positive $50,000, and the collateral posted by Company B is $30,000. In this situation, Company A is a(n):

A. secured creditor in the amount of $20,000.
B. secured creditor in the amount of $80,000.
C. unsecured creditor in the amount of $20,000.
D. unsecured creditor in the amount of $80,000.

Answer: C

Step by step

1. Value owed to A = $50,000; collateral held from B = $30,000.
2. Uncollateralized amount = 50,000 - 30,000 = $20,000.
3. The uncovered part is an unsecured claim against B's estate.

Why C is correct: When the nondefaulting party's positive value exceeds the defaulter's collateral, it is an unsecured creditor for the difference.

Why the others are wrong

- **A:** Right amount but wrong status; the $20,000 shortfall has no collateral behind it.
- **B:** Adds the collateral to the value (80,000) and mislabels it as secured.
- **D:** Adds the collateral to the value instead of subtracting it, and the status is right but the amount is wrong.

Q530 BT answers P2.T6.24.20.1 · BT notes P2.T6.24.20.1

REBUILT — Source was a vague 'all of the above' question with a giveaway answer and not testing the Hull ISDA loss conditions; rewrote to test when the nondefaulting party incurs a loss.

Under an ISDA Master Agreement, Company B defaults and Company A is the nondefaulting party. Collateral is posted by both sides. In which of the following situations will Company A most likely incur a loss, becoming an unsecured creditor?

A. Transactions are worth $20 (thousand) to A, and B has posted $35 (thousand) of collateral to A.
B. Transactions are worth $40 (thousand) to A, and B has posted $40 (thousand) of collateral to A.
C. Transactions are worth $30 (thousand) to B, and A has posted $50 (thousand) of collateral to B.
D. Transactions are worth $30 (thousand) to B, and A has posted $30 (thousand) of collateral to B.

Answer: C

Step by step

1. A loss arises in two cases: (1) value to the nondefaulting party is positive and exceeds the defaulter's collateral; (2) value to the defaulter is positive and is less than the collateral posted by the nondefaulting party.
2. A: value to A is 20 and A holds 35, so collateral covers the claim; no loss.
3. B: 40 vs 40 held; fully covered; no loss.
4. C: value to B is 30 but A posted 50; the excess of 20 must be recovered from B's estate, so A is an unsecured creditor for 20.

5. D: A posted 30 against 30 owed; nothing excess; no loss.

Why C is correct: A is exposed either when its claim exceeds collateral held, or when it has over-posted collateral to the defaulter, and C is the over-posting case.

Why the others are wrong

- **A:** Collateral held exceeds the claim, so A is protected (A would return the excess).
- **B:** Claim and collateral are equal, so no uncovered amount.
- **D:** Collateral posted exactly equals the amount A owes B, so there is no excess to recover.

LO 30.b

Define credit valuation adjustment (CVA) and debt valuation adjustment (DVA).

Q531

AnalystPrep Q.6060

An FRM candidate must explain how credit valuation adjustment (CVA) and debit (debt) valuation adjustment (DVA) affect the value of a bank's derivatives portfolio with a counterparty. Which statement is correct?

A. CVA increases the portfolio value by accounting for the potential gain from the counterparty's default, while DVA decreases it by considering the bank's own default risk.

B. CVA decreases the portfolio value by adjusting for the risk of counterparty default, and DVA increases it by considering the potential gain from the bank's own default.

C. CVA increases the portfolio value due to the risk of loss from the counterparty's potential default, and DVA has no influence since it only affects the counterparty's portfolio.

D. Both CVA and DVA decrease the portfolio value since they both adjust for the losses expected from defaults by the counterparty and the bank, respectively.

Answer: B

Step by step

1. CVA = PV of the expected cost to the bank if the counterparty defaults, so it is a loss and is subtracted from the no-default value.
2. DVA = PV of the expected cost to the counterparty if the bank defaults. That is a benefit to the bank (it will not have to make some payments), so it is added.
3. Value with defaults: $f = f_{nd} - \text{CVA} + \text{DVA}$.
4. Option B states exactly these two signs.

Why B is correct: CVA is a cost to the bank and lowers portfolio value; DVA is a cost to the counterparty, hence a benefit to the bank, and raises portfolio value.

Why the others are wrong

- **A:** Reverses both signs: treats CVA as a gain and DVA as a loss.
- **C:** Treats CVA as a value increase and ignores that DVA reduces the counterparty's claim, which raises the bank's value; DVA does affect the bank's portfolio.
- **D:** Forgets that DVA is a benefit to the bank; treats both adjustments as losses to the bank.

Q532

AnalystPrep Q.6217

REBUILT — Original treated the $100m no-default portfolio value as the exposure for both CVA and DVA; rebuilt with explicit, separate exposures and fixed options so exactly one is correct.

Bank Z has derivatives with a counterparty. The present value of the bank's expected exposure to the counterparty is $100 million, and the present value of the counterparty's expected exposure to the bank is $60 million (treat both as constant over the life of the transactions and independent of default). The expected recovery rate is 40%. The total risk-neutral probability of default over the life of the transactions is 5% for the counterparty and 3% for the bank. Which statement best describes the effect of CVA and DVA on the value of the derivatives portfolio to the bank?

A. CVA of $3.00 million reduces value and DVA of $1.08 million increases value, for a net decrease of $1.92 million.

B. CVA of $5.00 million reduces value and DVA of $1.80 million increases value, for a net decrease of $3.20 million.

C. CVA of $3.00 million reduces value and DVA of $1.08 million also reduces value, for a total decrease of $4.08 million.

D. CVA of $3.00 million increases value and DVA of $1.08 million reduces value, for a net increase of $1.92 million.

Answer: A

Step by step

1. CVA $= (1 - R) \times \text{PD}_{cp} \times \text{EE}_{bank} = 0.60 \times 0.05 \times 100 = 3.00$ million.
2. DVA $= (1 - R) \times \text{PD}_{bank} \times \text{EE}_{cp} = 0.60 \times 0.03 \times 60 = 1.08$ million.
3. Value with defaults $= f_{nd} - \text{CVA} + \text{DVA}$, so the net effect is $-3.00 + 1.08 = -1.92$ million (a decrease).

Why A is correct: CVA (loss from counterparty default) is subtracted and DVA (benefit from own default) is added; both use loss given default (1-R) times default probability times the relevant exposure.

Why the others are wrong

- **B:** Omits the recovery rate (uses 5% x 100 and 3% x 60), so both adjustments are overstated.
- **C:** Subtracts DVA as well; DVA is a benefit to the bank and is added.
- **D:** Reverses both signs (CVA added, DVA subtracted).

Q533

BT notes P2.T5.507.2

REBUILT — Original asked about reported income, which is outside the reading; reframed to the effect on DVA and on the value of the bank's derivatives portfolio, holding the counterparty's credit unchanged.

The credit spread of a bank suddenly widens, reflecting a higher probability that the bank itself will default. Holding everything else (including the counterparty's credit quality and the no-default value of the transactions) constant, what is the effect on the bank's DVA and on the value of its derivatives portfolio with that counterparty?

A. DVA decreases and portfolio value decreases.
B. DVA increases and portfolio value decreases.
C. DVA decreases and portfolio value increases.
D. DVA increases and portfolio value increases.

Answer: D

Step by step

1. DVA is the PV of the expected cost to the counterparty if the bank defaults; it depends on the bank's default probability (implied by its credit spread).
2. A wider bank spread raises the bank's default probability, so DVA increases.
3. Value with defaults $= f_{nd} - \text{CVA} + \text{DVA}$; CVA is unchanged, so a larger DVA raises the portfolio value to the bank.

Why D is correct: A higher own default probability raises DVA, which is a benefit to the bank and so increases the value of its derivatives portfolio.

Why the others are wrong

- **A:** Assumes higher own credit risk lowers DVA and value; confuses DVA with a cost to the bank.
- **B:** Gets DVA right but treats it as a deduction like CVA; DVA is added to value.
- **C:** Reverses the DVA direction: DVA rises with the bank's default probability, it does not fall.

Q534

Schweser Module Quiz 30.2 Q2

A bank is assessing the impact of a new transaction on CVA and DVA. If the new transaction is negatively correlated to existing transactions, the impact will likely be a(n):

A. increase in both CVA and DVA.
B. decrease in both CVA and DVA.
C. increase in CVA and decrease in DVA.
D. decrease in CVA and increase in DVA.

Answer: B

Step by step

1. CVA and DVA depend on the exposure of the whole netted portfolio.
2. A new transaction positively correlated with existing ones tends to add to exposure and raise both CVA and DVA.
3. A negatively correlated transaction (e.g., partly unwinding existing positions) offsets exposure, so both CVA and DVA likely decrease.

Why B is correct: Negative correlation with existing transactions reduces netted exposure, which lowers both CVA and DVA.

Why the others are wrong

- **A:** Applies the positive-correlation result.
- **C:** Assumes CVA and DVA move in opposite directions; the new trade's effect on exposure moves both together.

- **D:** Same error reversed; both adjustments are driven by the same exposure effect and move in the same direction.

LO 30.c

Calculate the probability of default using credit spreads.

Q535 AnalystPrep Q.6047

REBUILT — Option C ('0.') was a giveaway; replaced with the realistic error of ignoring the recovery rate, and options written consistently as percentages.

A risk analyst observes that a 1-year bond yields 4.00% above the risk-free rate, and the recovery rate is 40%. Using the approximation that the average hazard rate equals the credit spread divided by loss given default, the average hazard rate is closest to:

A. 6.67%
B. 2.40%
C. 4.00%
D. 10.00%

Answer: A

Step by step

1. Average hazard rate $\bar{\lambda} \approx s/(1-R)$.
2. Loss given default $= 1 - 0.40 = 0.60$.
3. $\bar{\lambda} = 0.04/0.60 = 0.0667$, i.e. 6.67% per year.

Why A is correct: The spread compensates for the expected annual loss, which equals hazard rate times LGD, so the hazard rate is spread divided by $(1-R)$.

Why the others are wrong

- **B:** Multiplied the spread by LGD (0.04×0.60) instead of dividing.
- **C:** Ignored recovery and treated the whole spread as the default probability (implicitly R = 0).
- **D:** Divided by the recovery rate ($0.04/0.40$) instead of by $1-R$.

Q536 BT answers P2.T6.24.20.2 · BT notes P2.T6.24.20.2

As a Risk Analyst at Investment Bank Plc, you are evaluating the credit risks associated with a derivative contract linked to a specific counterparty. You have been given the following financial metrics to work with: a 5-year credit spread for the counterparty is 200 basis points (bps), the prevailing risk-free rate is set at 5%, and the recovery rate in case of default is assumed to be 40% compared to 50% when it was last assessed by a different analyst. Your task involves calculating the Probability of Default (PD) using the specified financial metrics and then assessing the implications of the change in the recovery rate on both the Credit Valuation Adjustment (CVA) and the Debt Valuation Adjustment (DVA) for a derivative contract. Select the statement that correctly describes these impacts.

A. PD = 15.35%; The CVA will increase, reflecting higher credit risk of the counterparty, reducing the derivative's valuation, while DVA adjusts for the firm's own risk, potentially increasing the value if its credit worsens.
B. PD = 15.35%; The CVA will decrease as the counterparty's credit risk is manageable, increasing the derivative's valuation, while DVA remains unaffected as it only considers market conditions.
C. PD = 5.82%; The CVA will decrease as the counterparty's credit risk is manageable, increasing the derivative's valuation, while DVA remains unaffected as it only considers market conditions.
D. PD = 5.82%; The CVA will increase, reflecting higher credit risk of the counterparty, reducing the derivative's valuation, while DVA adjusts for the firm's own risk, potentially increasing the value if its credit worsens.

Answer: A

Step by step

1. Average hazard rate $\bar{\lambda} = s/(1-R) = 0.02/0.60 = 3.333\%$ per year (the 5% risk-free rate is not needed).
2. $\text{PD}_{5y} = 1 - e^{-\bar{\lambda}T} = 1 - e^{-0.03333\times 5} = 1 - e^{-0.1667} = 15.35\%$.
3. Lowering recovery from 50% to 40% raises loss given default, so CVA (the cost of counterparty default) increases and reduces the derivative's value.
4. DVA reflects the dealer's own default risk: if its own credit worsens, DVA rises and increases the derivative's value (a gain to the firm), so DVA does not stay unaffected.

Why A is correct: The hazard rate is spread over LGD, giving 15.35%; CVA is positive in the counterparty's credit

risk and reduces value, while DVA depends on the firm's own credit risk and increases value when it worsens.

Why the others are wrong

- **B:** Right PD, but wrongly claims CVA falls and DVA is unaffected; DVA depends on the firm's own credit spread, not just market conditions.
- **C:** PD of 5.82% comes from multiplying the spread by $(1 - R)$ $(1 - e^{-0.02\times0.6\times5})$ instead of dividing; CVA and DVA statements are also wrong.
- **D:** CVA/DVA statement is correct, but PD of 5.82% results from using $s \times (1 - R)$ rather than $s/(1 - R)$ for the hazard rate.

Q537

Schweser Module Quiz 30.1 Q1

If the unconditional default probability of a Ba-rated bond during Year 3 is 1.914% and the probability of survival through Year 2 is 97.413%, the probability of a default during Year 3, conditional on no earlier default, is closest to:

A. 0.673%.
B. 1.914%.
C. 1.965%.
D. 2.587%.

Answer: C

Step by step

1. Conditional PD = unconditional PD in Year 3 divided by survival probability through Year 2.
2. $0.01914/0.97413 = 0.019648 \approx 1.965\%$.

Why C is correct: Conditioning on survival to the start of Year 3 rescales the Year 3 unconditional default probability by dividing by the survival probability.

Why the others are wrong

- **A:** $2.587\% - 1.914\% = 0.673\%$: subtracted the unconditional PD from the cumulative default probability.
- **B:** Reported the unconditional probability without conditioning on survival.
- **D:** $1 - 0.97413 = 2.587\%$: gave the cumulative default probability through Year 2.

Q538

Schweser Module Quiz 30.1 Q2

REBUILT — Option B (1.55%) did not correspond to any identifiable error; replaced with 1.46% (hazard rate wrongly divided by the 2-year maturity).

A 2-year corporate bond yields 190 basis points above the risk-free rate. With a recovery rate of 35%, the average hazard rate for Years 1 and 2 is closest to:

A. 0.67%.
B. 1.46%.
C. 2.92%.
D. 5.43%.

Answer: C

Step by step

1. $\bar{\lambda} \approx s/(1 - R)$.
2. $s = 0.0190$, $1 - R = 0.65$.
3. $\bar{\lambda} = 0.0190/0.65 = 0.02923 \approx 2.92\%$ per year.

Why C is correct: The spread is the annual expected loss rate (hazard rate times LGD), so dividing by LGD of 65% gives the hazard rate.

Why the others are wrong

- **A:** Multiplied the spread by the recovery rate ($1.90\% \times 0.35 = 0.67\%$).
- **B:** Divided the correct 2.92% by the 2-year maturity; the hazard rate is already a per-year average.
- **D:** Divided by the recovery rate (1.90%/0.35) instead of by LGD.

Q539

BT notes P2.T6.301.3

REBUILT — OCR-garbled formula restored; options made consistent so exactly one is within tolerance, with distractors from realistic formula errors.

A single-period model gives the minimum coupon spread z an investor requires as compensation for default risk by setting the expected payoff equal to the risk-free payoff: $(1-\text{PD})(1+r+z)+\text{PD}\times R = 1+r$. The risk-free rate is 2.0%, the default probability is 3.0%, and the loss given default is 60.0% (so R = 40%). The minimum spread z is closest to:

A. 124 basis points
B. 180 basis points
C. 186 basis points
D. 192 basis points

Answer: D
Step by step
1. Recovery $R = 1 - 0.60 = 0.40$.
2. Solve for z: $z = \dfrac{(1+r) - \text{PD}\,R}{1-\text{PD}} - (1+r) = \dfrac{\text{PD}\,(1+r-R)}{1-\text{PD}}$.
3. $z = 0.03 \times (1.02 - 0.40)/0.97 = 0.0186/0.97 = 0.019175$.
4. So $z \approx 1.92\% \approx 192$ bp.

Why D is correct: The spread must offset the 3% chance of losing $(1 + r - R)$ per unit on a bond that otherwise pays $1 + r + z$ with probability 97%.
Why the others are wrong
- **A:** Used recovery instead of LGD in the shortcut: $\text{PD} \times R/(1 - \text{PD}) = 1.24\%$.
- **B:** Expected loss $\text{PD} \times \text{LGD} = 1.80\%$: ignores that the spread is earned only in the 97% survival state.
- **C:** $\text{PD} \times \text{LGD}/(1 - \text{PD}) = 1.856\%$: adjusts for survival but ignores that the lost interest r must also be compensated.

LO 30.d

Describe, compare, and contrast various credit risk mitigants and their role in credit analysis.

Q540

BT answers P2.T6.24.20.3 · BT notes P2.T6.24.20.3

REBUILT — Original used a confusing 'true EXCEPT' combination format with a garbled netting premise and an ambiguous statement iii; rebuilt as a single-statement 'which is INCORRECT' item using GARP's netting example and downgrade-trigger facts.

A bank has three uncollateralized derivative transactions with the same counterparty, valued from the bank's perspective at +$10 million, +$30 million and -$25 million. Which of the following statements about credit risk mitigants is INCORRECT?

A. If a netting agreement applies, the bank's exposure falls from $40 million (treating the transactions independently) to $15 million.
B. Under a collateral agreement, if the counterparty defaults, the bank may keep the collateral that the counterparty has posted.
C. Downgrade triggers of the type AIG had eliminate liquidity risk, because no collateral ever has to be posted when the triggers are in place.
D. The value of downgrade triggers is reduced if the counterparty has similar triggers in place with several dealers.

Answer: C
Key correction: Source key said A for an 'EXCEPT' combination question (statements i and ii); the intended false idea was the downgrade-trigger statement. Rebuilt so the false statement is isolated as C.
Step by step
1. Without netting, exposure counts only positive values: $10 + 30 + 0 = 40$ million.
2. With netting, transactions are one position: $10 + 30 - 25 = 15$ million, exposure $\max(15, 0) = 15$ million. A is correct.
3. Collateral agreements let the non-defaulting party keep collateral posted by the defaulter, reducing its loss. B is correct.
4. AIG's triggers meant it posted no collateral while rated above AA; once downgraded in September 2008 it faced immediate large collateral calls it could not meet. Triggers shift when collateral is demanded; they do not eliminate liquidity risk. C is incorrect.
5. Triggers lose value when the counterparty has them with multiple dealers (all call collateral at once) or when downgrades are large jumps. D is correct.

Why C is correct: Downgrade triggers only defer collateral requirements until a rating fall, at which point sudden collateral calls (as with AIG in 2008) can create severe liquidity strain, so claiming they eliminate liquidity risk is false.
Why the others are wrong

- **A:** True statement: gross positive exposures are 10+30=40 and the netted position is 10+30-25=15; picking it means misreading netting.
- **B:** True statement: the non-defaulting party can retain collateral posted by the defaulter; picking it confuses this with returning collateral to the defaulter.
- **D:** True statement: multiple dealers with triggers all demand collateral at once, so each trigger is worth less; picking it reflects misreading the reduction in value.

Q541 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A credit risk analyst at a bank reviews four uncollateralized derivative transactions with a single nonfinancial counterparty, all covered by one master netting agreement. The values of the transactions from the bank's perspective are +$18 million, -$7 million, +$12 million and -$20 million. By how much does the netting agreement reduce the bank's exposure to the counterparty, compared with treating the four transactions as independent?

A. $3 million
B. $57 million
C. $30 million
D. $27 million

Answer: D

Step by step

1. Without netting, only positive-value transactions are exposures: $18 + 12 = 30$ million (negative transactions count as zero).
2. With netting, the transactions are treated as one: $18 - 7 + 12 - 20 = 3$ million.
3. Reduction $= 30 - 3 = 27$ million.

Why D is correct: Exposure falls from $30 million (independent transactions, negatives count as zero) to $3 million (net), a reduction of $27 million.

Why the others are wrong

- **A:** $3 million is the exposure remaining after netting, not the reduction; it answers a different question.
- **B:** $57 million adds the absolute values of all four transactions, wrongly treating negative-value transactions as exposures of the bank.
- **C:** $30 million is the exposure without netting; it is the starting point, not the reduction.

Q542 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A risk manager at a derivatives dealer is assessing the credit risk mitigants available against a nonfinancial counterparty. The master agreement contains a downgrade trigger that lets the dealer require collateral or close out all transactions at market value if the counterparty's rating falls below BBB. The counterparty is currently rated A and has similar triggers with several other dealers. Which of the following statements is correct?

A. The trigger offers limited protection if the counterparty's rating drops directly from A to default, and the value of the trigger is reduced because other dealers hold similar triggers.
B. The trigger becomes more valuable as more dealers hold similar triggers, because the counterparty then has greater incentive to protect its rating.
C. If the counterparty defaults, the dealer may keep posted collateral only after lengthy legal proceedings establish its claim.
D. The trigger eliminates the dealer's credit exposure, because collateral or close-out is available as soon as the rating falls below BBB.

Answer: A

Step by step

1. Downgrade triggers do not protect against a large jump in rating, e.g., from A to default.
2. They work well only if little use is made of them; if the counterparty has triggers with many dealers, each dealer gets little protection (the AIG experience).
3. On default, the nondefaulting party is entitled to keep collateral posted by the other side without expensive, time-consuming legal proceedings.

Why A is correct: Triggers cannot help against a sudden jump to default, and widespread use by multiple dealers dilutes their value, as AIG showed in 2008.

Why the others are wrong

- **B:** Reverses the relationship: if many dealers hold triggers, a downgrade produces simultaneous collateral calls the counterparty cannot meet, so each trigger is worth less.
- **C:** Derivatives receive favorable treatment in default: the nondefaulting party may keep the collateral without lengthy legal proceedings.
- **D:** Triggers reduce but do not eliminate exposure; they offer no protection against a jump from A to default, and exposure can arise before the trigger level is reached.

LO 30.e

Describe the significance of estimating default correlation for credit portfolios and distinguish between reduced form and structural default correlation models.

Q543

AP Mock 1 Q26

REBUILT — Original option C was loosely worded (reduced-form correlation 'statistically'); rewrote options to match the Schweser/GARP definitions (hazard rates correlated with macro variables vs. correlated asset processes) so exactly one is correct.

A risk manager is explaining how reduced-form and structural models generate default correlation between two firms. Which statement is most accurate?

A. Reduced-form models assume the hazard rates of the firms follow stochastic processes that are correlated with macroeconomic variables, whereas structural models, similar to Merton's model, induce correlation by assuming the firms' asset values follow correlated stochastic processes.

B. Reduced-form models induce default correlation through correlated asset value processes, whereas structural models assume the firms' hazard rates are correlated with macroeconomic variables.

C. Structural models ignore systematic factors, whereas reduced-form models capture common market and sector influences through correlated hazard rates.

D. Reduced-form models can be calibrated to achieve any desired level of default correlation, whereas structural models can only produce low default correlations.

Answer: A

Step by step

1. Reduced-form models: hazard rates of different companies follow stochastic processes correlated with macroeconomic variables; a high hazard rate for A tends to coincide with a high hazard rate for B.
2. Structural (Merton-type) models: default occurs when asset value falls below a threshold; correlation is introduced by assuming correlated stochastic processes for the firms' assets.
3. Option A states exactly these two mechanisms; B swaps them, C wrongly denies systematic factors in structural models, and D reverses the correlation-range property.

Why A is correct: Each model class generates default correlation through a different channel: correlated hazard rates (reduced-form) versus correlated asset processes (structural).

Why the others are wrong

- **B:** Swaps the mechanisms: asset-value correlation belongs to structural models, macro-correlated hazard rates to reduced-form models.
- **C:** Structural models do include systematic factors through correlated asset processes; saying they ignore them is false.
- **D:** Reversed: it is structural models that allow correlation to be set as high as needed, while reduced-form models achieve only a low range of default correlations.

Q544

BT answers P2.T6.24.21.1 · BT notes P2.T6.24.20.3

REBUILT — Merged duplicate scenario (one source truncated); original options gave a model name without a clear reason and A and D were defensible; rewrote options to carry explicit reasoning tied to the high-correlation requirement.

A regional bank's loan portfolio is exposed to a downturn in the local mining industry. Two borrowers, Alpha Inc. (machinery supplier to miners) and Delta Inc. (logistics provider to miners), are likely to be hit by the same shock at nearly the same time. The risk manager wants a default correlation model that can represent a high probability of near-simultaneous default of these two firms. Which model is most suitable, and why?

A. A reduced-form model, because hazard rates correlated with macroeconomic variables can be calibrated to give as high a default correlation as needed over short horizons.

B. A structural model, because correlation can be introduced through correlated stochastic processes for the firms' assets and can be set as high as needed, at the cost of longer computation time.
C. A reduced-form model, because it is mathematically simple and the range of default correlations it can produce is wide.
D. A structural model, because it best reflects how default correlation trends move with the economic cycle and is quick to compute.

Answer: B
Step by step
1. The requirement is a high default correlation for two firms sharing an industry shock.
2. Reduced-form models link hazard rates to macro variables; it is unlikely two firms default in the same short period, so the achievable default correlation range is low.
3. Structural (Merton-type) models create correlation via correlated asset processes, and the correlation can be set as high as needed; the drawback is significant computation time.
4. Hence B.

Why B is correct: Only structural models let the asset correlation, and hence default correlation, be set as high as needed, which fits two firms exposed to the same industry shock.
Why the others are wrong
- **A:** Wrong because reduced-form models achieve only a low range of default correlations over short periods, even though hazard rates are correlated with macro variables.
- **C:** Right about mathematical simplicity but wrong that the correlation range is wide; it is low.
- **D:** Structural models are computationally slow, and the economic-cycle trend effect is a strength of reduced-form models.

Q545 Schweser Module Quiz 30.3 Q1

Which of the following statements is most accurate regarding reduced-form versus structural models used to estimate default correlation?
A. Structural models tend to take a long time to process.
B. Reduced-form models are more computationally intensive.
C. Reduced-form models allow for very high default correlations.
D. Economic cycle impacts on default correlation trends are best reflected in structural models.

Answer: A
Step by step
1. Structural models require modelling correlated asset processes, so computation time is significant.
2. Reduced-form models are mathematically straightforward, so they are less computationally intensive.
3. Reduced-form models give only a low range of default correlations; structural models can be set as high as needed.
4. Economic cycle effects on default correlation are captured by reduced-form models (hazard rates tied to macro variables).

Why A is correct: Structural models are computationally heavy, which is their main drawback relative to reduced-form models.
Why the others are wrong
- **B:** Reversed: reduced-form models are the simpler, faster ones.
- **C:** Reversed: reduced-form models achieve only low default correlations; structural models allow very high ones.
- **D:** Reversed: economic-cycle effects are a strength of reduced-form models, whose hazard rates are correlated with macro variables.

Q546 AnalystPrep Q.6052

REBUILT — Original distractors (bonus allocation, no capital impact) were giveaways; replaced with plausible misconceptions about default correlation.
An investment bank wants to quantify the benefit of diversification in its credit portfolio. Why is estimating default correlation among the obligors important?
A. Because default correlation drives the portfolio's loss distribution: positive correlation makes defaults cluster, so credit risk cannot be fully diversified away and tail losses are larger than under independence.
B. Because with positive default correlation, defaults of different obligors offset each other, so portfolio credit risk can be diversified away by adding more obligors.

C. Because default correlation matters only for pricing single-name exposures; it has no bearing on the loss distribution of a portfolio.
D. Because default correlation changes the expected loss of each obligor, while the variability of portfolio losses is unaffected.

Answer: A
Step by step
1. Default correlation is the likelihood that companies default around the same time (common geographic/industry shocks, economic conditions, contagion).
2. Higher correlation means defaults cluster, widening the loss distribution tail and limiting diversification.
3. Because of default correlation, credit risk cannot be fully diversified away, so it must be estimated to measure portfolio risk.

Why A is correct: Default correlation determines how clustered defaults are and therefore the shape of the portfolio loss distribution and the real diversification benefit.
Why the others are wrong
- **B:** Positive correlation means defaults occur together rather than offset, so diversification is limited.
- **C:** Correlation is irrelevant for a single name but is central to the portfolio loss distribution.
- **D:** Expected loss is the sum of individual expected losses and does not depend on correlation; it is the spread (variance, tail) of losses that does.

LO 30.f

Describe the Gaussian copula model for time to default and calculate the probability of default using the one-factor Gaussian copula model.

Q547 AnalystPrep Q.6054

REBUILT — Original used non-GARP notation (b_i) and loose percentile wording; restated with GARP's a_i and Z_i and explicit percentiles, answer unchanged.

Under the one-factor Gaussian copula model, $x_i = a_i F + \sqrt{1 - a_i^2}\, Z_i$, where F and Z_i are independent standard normal variables. A company has $a_i = 0.7$. In a given scenario the common factor F is at the 80th percentile of the standard normal distribution and the idiosyncratic factor Z_i is at the 60th percentile. The company's probability of default by time T is 10%. What are the company's creditworthiness value x_i and the default threshold for x_i?
A. $x_i = 0.77$; default threshold -1.28
B. $x_i = 0.38$; default threshold -1.65
C. $x_i = 1.77$; default threshold 1.28
D. $x_i = -0.38$; default threshold 1.60

Answer: A
Step by step
1. Percentiles: $N^{-1}(0.80) = 0.8416$, $N^{-1}(0.60) = 0.2533$.
2. $\sqrt{1 - 0.7^2} = \sqrt{0.51} = 0.7141$.
3. $x_i = 0.7(0.8416) + 0.7141(0.2533) = 0.5891 + 0.1809 = 0.770$.
4. Default occurs if $x_i < N^{-1}(Q_i(T)) = N^{-1}(0.10) = -1.2816$.
5. Since 0.77 > -1.28, no default in this scenario.

Why A is correct: The creditworthiness value is the weighted sum of the factor draws with weights a_i and $\sqrt{1 - a_i^2}$, and default is triggered when it falls below N^{-1} of the cumulative default probability.
Why the others are wrong
- **B:** Wrong on both parts: x_i does not follow from the formula, and -1.65 is $N^{-1}(5\%)$, not $N^{-1}(10\%)$, so it is the threshold for the wrong default probability.
- **C:** $x_i = 1.77$ is far above the formula value, and $+1.28$ is $N^{-1}(90\%)$, the wrong tail; the default threshold must be negative for a 10% default probability.
- **D:** Both parts wrong: a negative x_i does not result from two draws above the median, and $+1.60$ confuses the default threshold with an upper-tail value.

Q548 AnalystPrep Q.6214

REBUILT — Original asked a vague qualitative question with a dubious key and no computation; rebuilt as a one-factor conditional default probability calculation.

A risk analyst uses the one-factor Gaussian copula model with a common copula correlation $\rho = 0.25$ (so $a_i = \sqrt{\rho} = 0.5$) for all companies. Each company has a 10% cumulative probability of default over 5 years. Conditional on the common factor F being at its 1st percentile (F = -2.326), what is the probability of default over 5 years for a given company? ($N^{-1}(0.10) = -1.2816$)

A. 44.6%
B. 43.7%
C. 20.9%
D. 0.2%

Answer: A

Step by step

1. Conditional default probability: $Q(T \mid F) = N\left(\dfrac{N^{-1}[Q(T)] - aF}{\sqrt{1-a^2}}\right)$.
2. $a = \sqrt{0.25} = 0.5$; $\sqrt{1-0.25} = 0.8660$.
3. Numerator: $-1.2816 - 0.5(-2.326) = -1.2816 + 1.1632 = -0.1184$.
4. Argument: $-0.1184/0.8660 = -0.1367$.
5. $N(-0.1367) = 0.4456 \approx 44.6\%$.

Why A is correct: A very bad common factor (1st percentile) raises the conditional default probability far above the unconditional 10%; the formula uses $a = \sqrt{\rho}$ and divides by $\sqrt{1-\rho}$.

Why the others are wrong

- **B:** Divided by $1 - \rho = 0.75$ instead of its square root 0.866, giving $N(-0.158) = 43.7\%$.
- **C:** Used $\rho = 0.25$ instead of $a = \sqrt{\rho} = 0.5$ as the factor loading, understating the effect of F: $N(-0.808) = 20.9\%$.
- **D:** Got the sign of the factor term wrong (added aF instead of subtracting), as if F were at the 99th percentile, giving 0.2%.

Q549 BT answers P2.T6.24.21.2 · BT notes P2.T6.24.21.2

REBUILT — Original conceptual item had three giveaway distractors; rebuilt as a numeric mapping of a simulated normal sample to default year using the GARP Example 12.4 thresholds.

A risk analyst simulates defaults over 5 years for 10 companies using a Gaussian copula with pairwise copula correlation 0.2. Each company's cumulative default probability at 1, 2, 3, 4 and 5 years is 1%, 3%, 6%, 10% and 15%. For one company the simulated standard normal sample is $x_i = -1.60$. When does this company default?

A. In year 2
B. In year 3
C. In year 4
D. No default within 5 years

Answer: B

Step by step

1. Thresholds are N^{-1} of cumulative default probabilities: $N^{-1}(0.01) = -2.33$, $N^{-1}(0.03) = -1.88$, $N^{-1}(0.06) = -1.55$, $N^{-1}(0.10) = -1.28$, $N^{-1}(0.15) = -1.04$.
2. Compare the sample $x_i = -1.60$: it is below -1.55 and above -1.88.
3. So default falls in year 3 (between the 2-year and 3-year thresholds).

Why B is correct: The sample is mapped to a default time by comparing it with the inverse-normal thresholds of the cumulative default probabilities; -1.60 lies between the year-2 and year-3 thresholds.

Why the others are wrong

- **A:** Year 2 requires a sample between -2.33 and -1.88; -1.60 is higher than -1.88, so the default has not yet occurred by year 2.
- **C:** Year 4 requires a sample between -1.55 and -1.28; -1.60 is below -1.55, so default occurs earlier.
- **D:** No default requires a sample above $N^{-1}(15\%) = -1.04$; -1.60 is well below that.

Q550 BT answers P2.T6.24.21.2 · BT notes P2.T6.24.21.2

You are a risk analyst tasked with simulating default scenarios for a portfolio of 10 companies over the next 5 years using the Gaussian copula model. The cumulative probabilities of default for each company over the next 1, 2, 3, 4, and 5 years are set at 1%, 3%, 6%, 10%, and 15%, respectively, with a default correlation of 0.2 between each pair of companies. Using the Gaussian copula model, how would you determine if a default occurs within the first year for any company?

A. Simulate a value from a multivariate normal distribution for each company. If the value is less than the inverse cumulative normal distribution value at 1% (-2.33), default occurs within the first year.

B. Calculate the cumulative probability for each company independently without using a copula. If the probability is less than 1%, assume default occurs within the first year.

C. Use a single-factor model to compute default probability without transforming the values into a normal distribution. Default is assumed if the factor exceeds the threshold set for the first year.

D. Sum the individual default probabilities; if the total exceeds 10%, conclude that a default occurs within the portfolio's first year.

Answer: A

Step by step

1. Transform each company's time to default into a standard normal variable: $x_i = N^{-1}[Q_i(t)]$.
2. Sample correlated normals (pairwise correlation 0.2).
3. First-year threshold: $N^{-1}(0.01) = -2.33$; a sample below -2.33 means default within year 1.

Why A is correct: The Gaussian copula samples correlated normals and compares them with N^{-1} of the cumulative default probability for each horizon.

Why the others are wrong

- **B:** Treats companies independently, ignoring the copula correlation of 0.2 that drives joint default behaviour.
- **C:** Skips the transformation to normal variables, which is the essence of the Gaussian copula; the factor must be compared with an N^{-1} threshold.
- **D:** Adding probabilities across companies is meaningless for individual default timing and ignores correlation and the 1% threshold.

Q551 Schweser Module Quiz 30.3 Q2

The Gaussian copula model transforms which of the following factors into normal variables?

A. Recovery rates.

B. Times to default.

C. Credit spread risk.

D. Default probabilities.

Answer: B

Step by step

1. Times to default are not normally distributed.
2. The model maps each via $x_i = N^{-1}[Q_i(t_i)]$ into a standard normal variable (mean 0, SD 1).
3. The x_i are assumed multivariate normal; the copula correlation is the correlation of the x_i.

Why B is correct: The Gaussian copula transforms each company's time to default (percentile-to-percentile) into a standard normal variable so a correlation structure can be applied.

Why the others are wrong

- **A:** Recovery rates are not transformed; they enter only in loss calculations such as credit VaR.
- **C:** Credit spread risk is not a variable in the copula transform.
- **D:** Default probabilities are inputs to the transformation (via the cumulative distribution), not the variables being transformed.

LO 30.g

Describe how to estimate credit VaR using the Gaussian copula and the CreditMetrics approach.

Q552

AnalystPrep Q.6057

REBUILT — Original asked vague 'next steps' with a truncated option C and loosely defended simulation steps; rewritten as a clear comparison of the Gaussian copula (Vasicek) and CreditMetrics approaches with one defensible answer.

A risk analyst compares two ways to estimate the 1-year, 99.9% credit VaR of a bank's loan book: (i) the one-factor Gaussian copula approach applied to a large portfolio of similar loans, and (ii) the CreditMetrics approach. Which statement is most accurate?

A. Approach (i) converts the probability of default and the copula correlation into a 99.9% worst-case default rate, which is multiplied by $L(1 - RR)$; approach (ii) simulates rating changes, so its losses include downgrades as well as defaults.

B. Approach (ii) counts losses only from defaults, whereas approach (i) also captures losses from rating downgrades.

C. Approach (i) multiplies the unconditional default probability by $L(1 - RR)$, so the copula correlation does not affect the credit VaR.

D. Approach (ii) samples the rating changes of different counterparties independently, which is why it is computationally quick.

Answer: A

Step by step

1. One-factor Gaussian copula (Vasicek): for a large homogeneous portfolio, the worst-case default rate at confidence X is $V(T, X) = N\left[\frac{N^{-1}(Q(T)) + \sqrt{\rho}\, N^{-1}(X)}{\sqrt{1-\rho}}\right]$.
2. Credit VaR $= L \times (1 - RR) \times V(T, X)$, so both the PD $Q(T)$ and the copula correlation ρ drive the result.
3. CreditMetrics: Monte Carlo simulation of rating changes for all counterparties, using a Gaussian copula so that rating changes are correlated; revaluing after each trial gives a loss distribution that includes downgrades (upgrades count as negative losses).
4. Only statement A describes both approaches correctly.

Why A is correct: The copula formula translates PD and correlation into a stressed default rate, while CreditMetrics values rating migrations, so it captures downgrade losses at the cost of heavy computation.

Why the others are wrong

- **B:** Reverses the approaches: CreditMetrics is the one that captures downgrades, while the copula formula captures defaults only.
- **C:** Ignores the correlation; $V(T, X)$ rises with ρ and exceeds $Q(T)$ at high confidence levels, so using the unconditional PD understates the VaR.
- **D:** CreditMetrics correlates rating changes through a Gaussian copula (correlation set equal to equity return correlation) and is computationally intensive, not quick.

Q553

BT answers P2.T6.24.21.3 · BT notes P2.T6.24.21.3

REBUILT — Merged two identical conceptual items whose BBB thresholds were inconsistent as printed and whose answer was a giveaway; rebuilt as a CreditMetrics threshold-mapping exercise using the GARP thresholds with specific simulated draws.

In a CreditMetrics simulation, a AAA company (A) and a BBB company (B) have an equity return correlation of 0.2, which is used as the Gaussian copula correlation between their standard normal samples x_A and x_B. The 1-year thresholds implied by the transition matrix are:

Company	Sample range	Rating after 1 year
A (AAA)	$x_A < 1.2719$	AAA
A (AAA)	$1.2719 \le x_A < 2.4089$	AA
A (AAA)	$2.4089 \le x_A < 2.8070$	A
B (BBB)	$x_B < -3.7190$	AAA
B (BBB)	$-3.7190 \le x_B < -3.0618$	AA
B (BBB)	$-3.0618 \le x_B < -1.7866$	A

On one simulation trial the correlated samples are $x_A = 2.50$ and $x_B = -3.40$. What are the simulated ratings of the two companies?

A. A: AA; B: AA

B. A: A; B: AAA

C. A: A; B: AA

D. A: A; B: A

Answer: C

Step by step

1. Thresholds are inverse normals of cumulative transition probabilities, e.g. AAA row: $N^{-1}(0.8983) = 1.2719$, $N^{-1}(0.8983 + 0.0937) = 2.4089$, $N^{-1}(0.8983 + 0.0937 + 0.0055) = 2.8070$.
2. Company A: $2.4089 \leq 2.50 < 2.8070$, so it is downgraded from AAA to A.
3. Company B: $-3.7190 \leq -3.40 < -3.0618$, so it is upgraded from BBB to AA.
4. The correlation of 0.2 only affects how the pair (x_A, x_B) was drawn; with the samples given, each rating follows from its own thresholds. Repeating over many trials and revaluing the positions gives the loss distribution and credit VaR.

Why C is correct: Each sampled normal is mapped to a rating by the band in which it falls: 2.50 lies in the A band for the AAA company and -3.40 lies in the AA band for the BBB company.

Why the others are wrong

- **A:** Places 2.50 in the 1.2719-2.4089 band (AA); 2.50 exceeds 2.4089, so the AAA company falls to A.
- **B:** Assigns AAA to company B; that needs $x_B < -3.7190$, and -3.40 is above that cutoff.
- **D:** Gets A right but puts -3.40 in the A band; that band starts at -3.0618 and -3.40 is lower, so it is in the AA band.

Q554

GARP PE2 Q76

REBUILT — The stated 99.9% worst-case default rate of 11.92% is inconsistent with the Vasicek formula (it gives 10.73%); data corrected and options rebuilt around realistic errors.

A bank has a retail loan exposure of GBP 85 million. The 1-year average probability of default is 2.50% and the average recovery rate is 72%. Using the one-factor Gaussian copula approach with a copula correlation of 0.06, the 99.9% worst-case default rate over one year is approximately 10.73% (10.7337% unrounded). What is the 99.9% credit VaR of the exposure?

A. GBP 1.960 million
B. GBP 2.555 million
C. GBP 0.595 million
D. GBP 6.569 million

Answer: B

Key correction: Source key (GARP PE2) said D (GBP 2.837m) using a worst-case default rate of 11.92%; that rate does not follow from the inputs. Recomputed rate = 10.73%, giving GBP 2.555m.

Step by step

1. Check the worst-case default rate: $V = N\left[\frac{N^{-1}(0.025)+\sqrt{0.06}\,N^{-1}(0.999)}{\sqrt{0.94}}\right] = N\left[\frac{-1.9600+0.2449\times3.0902}{0.9695}\right] = N(-1.2409) \approx 10.7337\%$.
2. Loss given default $= 1 - RR = 1 - 0.72 = 28\%$.
3. Credit VaR $= L \times (1 - RR) \times V = 85 \times 0.28 \times 0.107337 =$ GBP 2.555 million.

Why B is correct: Credit VaR under the Gaussian copula is exposure times loss given default times the stressed (worst-case) default rate, which already embeds the correlation.

Why the others are wrong

- **A:** Subtracts the expected default rate: $85 \times 0.28 \times (0.107337 - 0.025) = 1.960$. The formula gives the loss not exceeded at 99.9% and does not net out expected loss.
- **C:** Computes expected loss, $85 \times 0.025 \times 0.28 = 0.595$, using the average PD instead of the worst-case default rate.
- **D:** Uses the recovery rate instead of LGD: $85 \times 0.1073 \times 0.72 = 6.569$.

Q555

Schweser Module Quiz 30.3 Q3

Assuming a loan portfolio of L, a recovery rate of RR, and the percentage of losses on a portfolio less than $V(T, X)$, which of the following formulas is used to estimate credit VaR?

A. $L \times RR \times V(T, X)$
B. $L \times (1 - RR)/V(T, X)$
C. $V(T, X)/[L \times (1 - RR)]$
D. $L \times (1 - RR) \times V(T, X)$

Answer: D

Step by step

1. $V(T, X)$ is the worst-case fraction of the portfolio defaulting with confidence X over T years.
2. Loss on a defaulted loan is the unrecovered part, $1 - RR$.
3. Credit VaR $= L \times (1 - RR) \times V(T, X)$.

Why D is correct: Credit VaR scales the portfolio size by the worst-case default rate and by the loss given default $1 - RR$.

Why the others are wrong

- **A:** Uses the recovery rate instead of loss given default, so it measures amounts recovered rather than lost.
- **B:** Divides by $V(T, X)$; the default rate must multiply exposure, not divide it.
- **C:** Inverts the formula and drops the exposure as a multiplier; it is a ratio, not a monetary loss.

Reading 31. Credit Derivatives

John C. Hull, Options, Futures, and Other Derivatives, 11th Edition (Pearson, 2022). Chapter 25.

LO	Learning objective	Qs
31.a	describe a credit derivative, credit default swap (CDS), total return swap, and collateralized debt obligation (CDO).	10
31.b	explain how to account for credit risk exposure in valuing a CDS.	6
31.c	identify the default probabilities used to value a CDS.	3
31.d	evaluate the use of credit indices and fixed coupons in pricing CDS transactions.	5
31.e	define CDS forwards and CDS options.	6
31.f	describe the process of valuing a synthetic CDO using the spread payments approach and the Gaussian copula model of time to default approach.	5
31.g	define the two measures of implied correlation: compound (tranche) correlation and base correlation.	4
31.h	discuss alternative approaches used to estimate default correlation.	3

LO 31.a

Describe a credit derivative, credit default swap (CDS), total return swap, and collateralized debt obligation (CDO).

Q556

AnalystPrep Q.1552

Other reasons for the financial crisis included residential mortgages, giving loans at lower interest rates, and also the collapse of the subprime mortgage market. All these led to heavy selling and buying of CDOs and Credit Default Swaps (CDSs).

What is the purpose of the insurance contract underlying a Credit Default Swap?

A. To protect against the default of the underlying asset.
B. To speculate on the market movement in CDOs.
C. To reduce the individual risk of any asset by selling a large number of CDSs.
D. To trade a large number of securities at once to hedge against risk.

Answer: A

Step by step

1. A CDS is an OTC contract in which the protection buyer pays a periodic spread to the protection seller.
2. If the reference entity defaults, the seller compensates the buyer for the loss (face value less recovery value).
3. So the contract works like insurance against default of the reference asset, transferring credit risk without selling the asset.

Why A is correct: A CDS is insurance against default of a reference entity or asset: the buyer pays a spread and the seller pays out on a credit event.

Why the others are wrong

- **B:** Speculation on CDOs is a possible use of derivatives but is not the purpose of the insurance contract; a CDS is tied to default of a reference entity, not CDO market moves.
- **C:** Selling many CDSs increases, not reduces, an institution's credit exposure; protection is obtained by buying CDS.
- **D:** Describes a bulk securities trade or portfolio hedge, not the single-purpose default protection a CDS provides.

Q557

AnalystPrep Q.6061

REBUILT — Stem said 'three' derivatives but four options were listed, and 'set payment' was imprecise; reworded to 'floating rate plus a spread' and removed the count.

A risk manager wants a credit derivative in which one party pays the total return on a reference asset (coupons plus any gain or loss in value) and receives a floating rate plus a spread, so that the other party gets the economic exposure of owning the asset without owning it. Which instrument fits this description?

A. Credit default swap (CDS)
B. Total return swap (TRS)
C. Physically settled CDS
D. Collateralized debt obligation (CDO)

Answer: B

Step by step

1. The description has total return (coupons plus price change) exchanged for a floating rate plus a spread.
2. That is the definition of a total return swap: the receiver gets the asset's economic performance and pays a financing-type leg.
3. A CDS pays only on a credit event; physical settlement is only a settlement method; a CDO is a tranched securitization of a portfolio, not a swap of returns.

Why B is correct: A TRS exchanges the total return on an asset for floating plus a spread, giving the receiver synthetic exposure (like a leveraged purchase of the asset).

Why the others are wrong

- **A:** A CDS pays only if a credit event occurs, in return for a periodic spread; it does not pass through coupons and price changes.
- **C:** Physical settlement is a way of settling a CDS after default (deliver the bond for par); it does not exchange total returns.
- **D:** A CDO is a structure that tranches the credit losses of a portfolio of bonds or CDS; it is not a swap of total return for floating rate.

Q558

AnalystPrep Q.6064

REBUILT — Source answer was debatable (TRS receiver and tranche investor also gain from improving credit); rebuilt as a clear question on which position gains when the entity's credit spread narrows.

A portfolio manager expects the credit quality of a reference entity to improve, so its CDS spread should narrow. All else equal, which position should gain from this view?

A. Buying protection on the entity through a single-name CDS
B. Selling protection on the entity through a single-name CDS
C. Buying protection on a first-to-default basket CDS that includes the entity
D. Being the total return payer on a swap referencing a bond of the entity

Answer: B

Step by step

1. Narrowing spreads mean the market-implied default risk falls.
2. The protection seller is receiving a spread that is now above the market spread and faces a lower chance of paying out, so the position's value rises (it is a long credit position, like owning the bond).
3. The protection buyer, the first-to-default buyer and the TRS payer (who has passed the bond's price risk to the receiver) are short credit and do not benefit from improvement.

Why B is correct: Selling protection is a long credit position (like owning the bond), so it gains when credit quality improves.

Why the others are wrong

- **A:** A protection buyer is short credit and gains when spreads widen, not narrow.
- **C:** A first-to-default protection buyer is also short credit and benefits from deterioration in any name.
- **D:** The TRS payer passes the bond's total return to the receiver, so it is effectively short the bond's exposure and does not gain when credit improves and the bond rises.

Q559

AnalystPrep Q.6065

One investor seeks a credit derivative that arranges credit risk into tiers of priority, giving distinct risk-return profiles for each tier. Which instrument is the most appropriate?

A. Single-name Credit Default Swap (CDS).
B. Total Return Swap (TRS).
C. Synthetic Collateralized Debt Obligation (CDO).
D. Add-up Basket CDS.

Answer: C

Step by step

1. The key feature is tranching: losses are allocated by seniority, so equity, mezzanine and senior tranches have different risk and return.
2. A (synthetic) CDO creates such tranches, with the originator selling CDS protection on a portfolio and passing

the losses to tranche investors by seniority.
3. CDS, TRS and add-up basket CDS have no priority structure.

Why C is correct: Only a CDO structure divides portfolio credit losses into tranches of different seniority, each with its own risk-return profile.

Why the others are wrong

- **A:** A single-name CDS covers one entity with a single payoff and no tranches.
- **B:** A TRS swaps the total return on an asset for floating plus a spread; there is no priority structure.
- **D:** An add-up basket CDS pays on every default in the basket but has no tranche priority; the buyer is paid per default, not by seniority.

Q560

BT answers P2.T6.24.14.3

REBUILT — Source stem was internally inconsistent (missing coupon, contradictory loss description); replaced by a self-contained CDS payoff question with a single false statement.

A bank holds $10 million par of Konkcore bonds and buys 5-year CDS protection on $10 million notional at a spread of 200 bps per year, paid quarterly. Konkcore later defaults, and the post-default auction values the bonds at 40% of par (recovery rate 40%). Which of the following statements is NOT correct?

A. Under cash settlement the protection seller pays the bank $6.0 million.
B. Under physical settlement the bank delivers the bonds and receives $10 million.
C. Whichever settlement is used, the bond-plus-CDS position leaves the bank with $10 million of par value (ignoring premiums paid and accrued premium).
D. The CDS payoff to the bank is $4.0 million, because the payoff equals the notional times the recovery rate.

Answer: D

Step by step

1. Cash settlement payoff = notional x (1 - recovery) = 10 x (1 - 0.40) = $6.0 million.
2. Physical settlement: deliver bonds with $10 million face and receive $10 million par.
3. Cash case: bonds worth 10 x 0.40 = $4.0 million plus $6.0 million CDS payoff = $10 million, same as physical.
4. So the payoff is notional x (1 - R), not notional x R; $4.0 million is the value of the defaulted bonds, not the CDS payout. Statement D is wrong.

Why D is correct: A CDS pays the loss given default, notional x (1 - recovery) = $6.0 million, so the statement that it pays the recovery value is incorrect.

Why the others are wrong

- **A:** Correct statement (10 x 0.6 = 6.0), so it is not the answer.
- **B:** Correct statement: physical settlement exchanges the defaulted bonds for par.
- **C:** Correct statement: 4.0 of bond value plus 6.0 of CDS payoff (or par under physical settlement) equals 10.
- **D:** This is the false statement: it confuses payoff with recovery value (uses R instead of 1 - R), giving $4.0 million rather than $6.0 million.

Q561

BT answers P2.T6.24.22.1 · BT notes P2.T6.24.22.1

REBUILT — Source scenario was garbled (TRS legs inconsistent, answer rationale weak); rebuilt as a clear question on TRS payer cash flows when the bond falls in value.

Sesame Street Bank owns a $200 million bond portfolio and enters a 5-year total return swap as the total return payer with a hedge fund. During the swap the bank pays the coupons earned on the bonds and receives SOFR plus 25 bps on $200 million. At maturity the portfolio's value has fallen by 15% and there has been no default. Which statement about the end-of-swap payment is correct?

A. The bank pays the hedge fund $30 million, since the bank bears the price decline.
B. The bank receives $30 million from the hedge fund, since the receiver pays the decline in value.
C. No payment is made, because the coupons paid by the bank offset the price decline.
D. The bank receives $170 million, the portfolio's value at maturity.

Answer: B

Key correction: Source key (B) said renegotiate the TRS to receive a variable rate; the scenario's TRS terms were inconsistent and the rationale not defensible from the reading, so the item was rebuilt around TRS mechanics.

Step by step

1. The decline in value = 15% x $200 million = $30 million.
2. In a TRS the payer pays the total return (coupons plus gain in value) and the receiver pays the floating leg plus

the loss in value.
3. Value fell, so the receiver (the hedge fund) pays the $30 million decline to the payer (the bank).
4. The bank has therefore passed the price risk to the hedge fund; the SOFR + 25 bps leg is paid separately during the life of the swap.

Why B is correct: The TRS receiver bears the bond's price changes, so when value falls the receiver pays the payer the 15% x $200 million = $30 million decline.

Why the others are wrong

- **A:** Reverses the direction: the payer pays gains and receives losses; the bank does not keep the price risk.
- **C:** Treats coupons as offsetting the price change; coupons are exchanged separately and the price change is settled in addition.
- **D:** Confuses the change in value ($30 million) with the portfolio's ending value ($200m x 0.85 = $170 million).

Q562

BT answers P2.T6.607.3

Financial Bank has a long position in bonds issued by Konkcore Corporation. To hedge the credit risk of this bond, Financial Bank enters a long position in an over-the-counter (OTC) credit default swap (CDS) with Counterparty Finservices, LLC. Which of the following is TRUE about the long position in the CDS?

A. If the price of Konkcore's bond decreases due to a rise in interest rates, the value of the CDS will increase

B. If the yield on Konkcore's bond increases due to a rating downgrade, the value of the CDS will increase

C. If the default correlation between Konkcore and Counterparty increases, the value of the CDS will increase

D. If Konkcore defaults on the bonds and settlement occurs in cash, the payoff on the CDS from Counterparty to Financial Bank is approximately the mid-market value of defaulted bonds as determined by auction process several days after the credit event

Answer: B

Step by step

1. Long CDS = protection buyer, short credit risk of Konkcore.
2. A downgrade widens Konkcore's credit spread (yield rises), so the market-implied default risk rises and the protection becomes more valuable: B is true.
3. A: a rate-driven bond price fall is not a credit event and does not change the CDS spread, so no gain.
4. C: higher correlation between reference and counterparty raises the chance the seller defaults exactly when payout is due, lowering the CDS value to the buyer.
5. D: the cash payoff is notional x (1 - recovery), i.e. par minus the auction value, not the defaulted bonds' value.

Why B is correct: A downgrade widens credit spreads, so the protection buyer's CDS gains value.

Why the others are wrong

- **A:** Rate-driven bond price declines are not credit-spread moves; the CDS hedges credit risk, not interest rate risk.
- **C:** Higher correlation between reference entity and protection seller increases counterparty risk, which lowers the CDS value to the buyer.
- **D:** The payoff is par minus the auction-determined value (the loss), not the value of the defaulted bonds.

Q563

BT answers P2.T6.608.1

REBUILT — OCR-damaged LIBOR (7.0%) restored to 1.0% from the source working, and the unexplained distractors were replaced with ones that each result from a specific mistake.

Mainway Bancorp holds a $500.0 million loan paying a fixed 5.0% per annum, semiannually. Mainway hedges the loan as the total return (TRS) payer on a swap with notional equal to the loan's par value: it pays the loan's interest plus the change in the loan's marked-to-market value, and receives LIBOR + 300 basis points. The swap settles semiannually. On the next settlement date LIBOR is 1.0% per annum, and the loan's mark-to-market value has fallen by 4.0%. Ignoring the loan itself and its coupon, what is Mainway's net cash flow on this settlement date from the TRS only?

A. Outflow of $22.5 million

B. Outflow of $2.5 million

C. Inflow of $7.5 million

D. Inflow of $17.5 million

Answer: D

Step by step

1. TRS payer pays the loan's coupon: $500 \times 5.0\%/2 = \$12.5$ million.
2. TRS payer also pays the change in value; the value fell 4.0%, so the payer receives $500 \times 4.0\% = \$20.0$ million.
3. Floating leg received: $500 \times (1.0\% + 3.0\%)/2 = \$10.0$ million.
4. Net $= -12.5 + 20.0 + 10.0 = +\$17.5$ million, an inflow.

Why D is correct: The TRS payer passes on the asset's total return, so a price decline is a payment to the payer; combined with the floating-rate receipt this more than offsets the coupon paid.

Why the others are wrong

- **A:** Treats the 4.0% price fall as an outflow (sign error): $-12.5 - 20.0 + 10.0 = -22.5$.
- **B:** Ignores the mark-to-market change entirely: $10.0 - 12.5 = -2.5$; the TRS also transfers price changes.
- **C:** Omits the floating-rate receipt: $20.0 - 12.5 = +7.5$.

Q564

Mock 2 Q40

REBUILT — Option C (TRS return payer) was also a defensible risk-reducing choice; changed it to the TRS return receiver so only D reduces credit risk.

An investor wishes to reduce the credit risk of his portfolio. The best approach for this investor is to:

A. buy a collateralized debt obligation (CDO) tranche.
B. sell protection in an index credit default swap (CDS).
C. enter into a total return swap (TRS) as the total return receiver.
D. buy protection in a credit default swap (CDS) on the portfolio constituents.

Answer: D

Key correction: Source key (Mock 2) said D, but its explanation conceded that a TRS return payer also reduces risk; option C was changed so D is uniquely correct.

Step by step

1. Reducing credit risk requires a position that gains when credit quality deteriorates, i.e. one that transfers default risk to someone else.
2. A CDS protection buyer pays a spread and receives a payoff on default of the reference entity, offsetting losses on matching holdings.
3. Buying a CDO tranche, selling index protection and receiving the total return on a TRS each add credit exposure.
4. Therefore buying CDS protection on the portfolio constituents is the answer.

Why D is correct: Buying CDS protection transfers the default loss on the reference entities to the protection seller without selling the underlying assets.

Why the others are wrong

- **A:** Buying a CDO tranche makes the investor a credit-risk taker (long the tranche's losses), increasing exposure.
- **B:** Selling protection means paying out on defaults of index names, adding credit risk rather than hedging it.
- **C:** The TRS receiver gets the asset's total return and bears its credit and market losses, replicating a leveraged long position in the asset.

Q565

Schweser Module Quiz 31.1 Q1

REBUILT — Replaced the unexplained 7.75% option with 6.00% (result of using the 5-year swap rate), so every distractor is a specific error.

Modus Corp's 7-year, 5% coupon bond is rated AA. The annual CDS spread on a 7-year bond is 3%. The swap spread is flat at 25 bps, while the swap fixed rate is 3% for 5 years and 4% for 7 years. To prevent arbitrage, Modus Corp's bond should most likely yield:

A. 7.00%.
B. 7.25%.
C. 6.00%.
D. 8.00%.

Answer: A

Step by step

1. CDS-bond basis = CDS spread - bond yield spread, and should be about zero for the same maturity to prevent arbitrage.
2. So the bond yield spread = CDS spread = 3.00%.
3. Bond yield spread = bond yield - swap fixed rate of the same maturity; the bond is 7-year, so the swap rate is 4.00%.

4. Bond yield = 4.00% + 3.00% = 7.00%.

Why A is correct: With a zero CDS-bond basis, the bond's yield must equal the same-maturity swap rate plus the CDS spread.

Why the others are wrong

- **B:** Adds the 25 bp swap spread, which is already embedded in the swap rate; it is irrelevant here.
- **C:** Uses the 5-year swap rate (3%) instead of the 7-year rate (4%): 3% + 3% = 6.00%.
- **D:** Adds the CDS spread to the 5% coupon (5% + 3%); the coupon is not the benchmark for the yield spread.

LO 31.b

Explain how to account for credit risk exposure in valuing a CDS.

Q566

Schweser Module Quiz 31.1 Q3

CDS spreads are calculated such that the PV of:

A. accrual payments is zero.
B. expected payoff is less than the PV of expected payments during the life of the CDS.
C. expected payoff is equal to the PV of expected payments during the life of the CDS.
D. expected payoff is greater than the PV of expected payments during the life of the CDS.

Answer: C

Step by step

1. A new CDS has zero value at initiation, so neither side pays to enter.
2. The buyer's side: PV of expected regular spread payments plus PV of the expected accrual payment made on default.
3. The seller's side: PV of expected payoff, (1 - recovery rate) times notional, on default.
4. The spread s is the value that sets PV of expected payments (including accrual) equal to PV of expected payoff.

Why C is correct: Because the swap has zero initial value, the spread is solved so that the PV of expected payoff equals the PV of expected payments (regular plus accrual) over the life of the CDS.

Why the others are wrong

- **A:** The accrual payment is the pro rata premium owed on default between payment dates and is generally positive; it is part of the payments leg, not set to zero.
- **B:** Payoff PV below payments PV would mean the buyer overpays, so the swap would start with positive value to the seller; the spread would then be too high.
- **D:** Payoff PV above payments PV would mean the buyer underpays, giving the swap positive value to the buyer at inception; the spread would be too low.

Q567

AnalystPrep Q.6067

When valuing a CDS, the risk manager assumes that a default, if it occurs within a year, happens at the middle of that year rather than at the end. The risk-free rate is 3% per annum with continuous compounding. Holding the default probability and the payoff amount constant, how does the present value of the expected payoff under the mid-year assumption compare with the end-of-year assumption?

A. It is higher, because the payoff is discounted over a shorter period.
B. It is lower, because the payoff is discounted over a longer period.
C. It is unchanged, because the assumption does not affect the discount factor.
D. It cannot be determined without knowing the spread payment rate.

Answer: A

Step by step

1. Mid-year default: discount factor $e^{-0.03\times0.5} = 0.9851$ for year 1 (convention: defaults at points midway between payment dates).
2. End-of-year default: $e^{-0.03\times1} = 0.9704$.
3. Same probability and payoff multiplied by a larger discount factor gives a larger PV, so the mid-year PV is higher.

Why A is correct: Discounting runs for a shorter time when default occurs mid-year, so the discount factor is larger and the PV of the expected payoff is higher.

Why the others are wrong

- **B:** Reverses the direction: mid-year is earlier than year-end, so the discounting period is shorter, not longer.
- **C:** The discount factor depends on the default time; e^{-rt} changes with t.
- **D:** The spread payment rate is irrelevant to the PV of the expected payoff, which depends only on default probability, loss given default and discounting.

Q568

AnalystPrep Q.6068

In valuing a CDS, the recovery rate is 40%. How is the recovery rate used to compute the expected payoff on a default event?

A. The probability of default in the period is multiplied by (1 - recovery rate) and by the notional.
B. The recovery rate is added to the probability of default to reflect the higher potential return.
C. The recovery rate is used as a direct multiplier of the probability of default and the notional.
D. The recovery rate is applied to the expected spread payments to estimate the actual cash flows.

Answer: A

Step by step

1. The protection seller pays notional minus the recovered value: $L(1-R)$.
2. With $R = 40\%$, loss given default is 60% of notional.
3. Expected payoff in a period = PD for that period $\times(1-R)\times$ notional; this is then discounted.

Why A is correct: The payoff on default is the loss given default, notional times (1 - R), weighted by the probability of default in that period.

Why the others are wrong

- **B:** Adding a rate to a probability is meaningless and ignores loss given default.
- **C:** Multiplies by R (40%) rather than (1 - R) (60%), so it measures the amount recovered instead of the amount paid out.
- **D:** Recovery affects the payoff leg, not the buyer's spread payments, which depend on survival.

Q569

AnalystPrep Q.6074

A credit analyst calculates the present value of expected payments and payoff for a CDS. What default timing convention is used within each payment period?

A. Defaults occur at the start of each period.
B. Defaults occur at random times, averaged over the life of the contract.
C. Defaults occur at the end of each period.
D. Defaults occur at the midpoint of each period.

Answer: D

Step by step

1. The standard valuation assumes defaults happen only at points midway between payment dates.
2. Hence payoff in year 2 is discounted from t = 1.5, and an accrual payment of s/2 is due on default in the year.

Why D is correct: The market convention is that defaults occur midway between payment dates, which is simple and usually gives good results.

Why the others are wrong

- **A:** Start-of-period default is not the convention and would overstate the PV of the payoff.
- **B:** A continuous-default average is not the convention; the model uses discrete mid-period default points.
- **C:** End-of-period default would understate the PV of the payoff and ignore the accrual payment.

Q570

AnalystPrep Q.6075

REBUILT — Original options were non-numeric and loosely worded ('notional minus recovery'); rebuilt with numbers so exactly one option is right.

A 5-year CDS has a constant hazard rate of 1.5% per annum for the reference entity. The risk-free rate is 3% per annum, continuously compounded, and the recovery rate is 40%. Defaults are assumed to occur midway through each year. Notional is $10 million. The present value of the expected payoff from a default in Year 3 is closest to:

A. $80,400
B. $79,200
C. $53,600

D. $244,900

Answer: A

Step by step

1. $PS_2 = e^{-0.015\times2} = 0.970446$, $PS_3 = e^{-0.015\times3} = 0.955997$.
2. Year-3 default probability = $0.970446 - 0.955997 = 0.014448$.
3. Expected payoff = $0.014448 \times (1 - 0.40) \times \$10\text{m} = \$86{,}688$.
4. Default at t = 2.5: $DF = e^{-0.03\times2.5} = 0.92774$.
5. PV = $86{,}688 \times 0.92774 = \$80{,}425 \approx \$80{,}400$.

Why A is correct: Year-3 PD times loss given default times notional, discounted from the mid-year point t = 2.5.

Why the others are wrong

- **B:** Discounted at t = 3 (DF 0.91393) instead of the mid-year t = 2.5.
- **C:** Used the recovery rate (40%) instead of loss given default (60%).
- **D:** Used the cumulative default probability to Year 3 (4.40%) instead of the Year-3 marginal probability.

Q571

BT answers P2.T6.24.22.2

REBUILT — Stem's $700M notional conflicted with the $100M used in the solution, the table was OCR-garbled, and bond-yield data was irrelevant; rebuilt with a consistent $100M notional and the mid-year discount factor.

A 5-year CDS has notional $100 million. The recovery rate is 35%. The hazard rate is a constant 6.5% (continuous), giving cumulative survival probabilities of 93.71%, 87.81% and 82.28% at the end of Years 1, 2 and 3. The risk-free rate is 5% per annum, continuously compounded. Defaults are assumed to occur midway through each year. What is the expected payoff and its present value for Year 3?

A. Payoff $3,594,500 and PV $3,172,000

B. Payoff $3,594,500 and PV $3,094,000

C. Payoff $5,530,000 and PV $4,880,000

D. Payoff $1,936,000 and PV $1,708,000

Answer: A

Key correction: Source key (A) is retained, but the original arithmetic used an end-of-year discount factor and a $100M notional despite the $700M in the stem.

Step by step

1. Year-3 PD = 87.81% − 82.28% = 5.53%.
2. Expected payoff = $5.53\% \times (1 - 35\\%) \times \$100\text{m} = \$3{,}594{,}500$.
3. Default at t = 2.5: $DF = e^{-0.05\times2.5} = 0.8825$.
4. PV = $3{,}594{,}500 \times 0.8825 = \$3{,}172{,}000$.

Why A is correct: Payoff is PD times loss given default (65%) times notional, and the PV discounts from t = 2.5 under the mid-year convention.

Why the others are wrong

- **B:** Same payoff but discounted at t = 3 ($e^{-0.15} = 0.8607$) instead of 2.5.
- **C:** Omitted loss given default: 5.53% x $100m with PV at 0.8825.
- **D:** Multiplied by the recovery rate (35%) instead of 1 - RR (65%).

LO 31.c

Identify the default probabilities used to value a CDS.

Q572

Schweser Module Quiz 31.1 Q2

Xeta Corp's hazard rate is estimated to be 2% over the next five years. The probability of default in Year 2 for Xeta is closest to:

A. 1.94%.

B. 1.98%.

C. 2.00%.

D. 2.23%.

Answer: A

Step by step

1. Hazard rate $\lambda = 2\%$; survival probability $\text{PS}_t = e^{-\lambda t}$.
2. Year 1 survival: $e^{-0.02\times 1} = 0.980199$.
3. Year 2 survival: $e^{-0.02\times 2} = 0.960789$.
4. Probability of default in Year 2 = $\text{PS}_1 - \text{PS}_2 = 0.980199 - 0.960789 = 0.019410 \approx 1.94\%$.

Why A is correct: The hazard rate gives unconditional default probability in a year as the fall in cumulative survival probability from the start to the end of that year, $\text{PD}_t = \text{PS}_{t-1} - \text{PS}_t$.

Why the others are wrong

- **B:** 1.98% is $1 - e^{-0.02}$, the Year 1 default probability (or the probability of default in any one year conditional on survival to its start); it ignores that Year 2 default requires surviving Year 1.
- **C:** 2.00% simply reads the hazard rate as the annual default probability; the hazard rate is an instantaneous intensity, and unconditional yearly PD declines over time.
- **D:** 2.23% overstates the answer and matches no valid use of the formula; Year 2 default probability cannot exceed the Year 1 probability of 1.98% when the hazard rate is constant, because survival decays.

Q573 AnalystPrep Q.6073

REBUILT — Original asked a vague role-of-spread question with a defensible distractor (fixed coupon); rebuilt to test the 31.c point directly: risk-neutral, market-implied default probabilities.

An analyst is valuing a CDS on a corporate issuer. Which default probabilities should be used, and how are they typically obtained?

A. Risk-neutral default probabilities, implied from market CDS spreads or bond prices.
B. Real-world default probabilities, estimated from historical default rates for the issuer's rating class.
C. Real-world default probabilities, estimated from the issuer's historical accounting ratios.
D. Risk-neutral default probabilities, set equal to the historical average annual default rate.

Answer: A

Step by step

1. A CDS is priced from market prices, so its default probabilities must be consistent with market prices (risk-neutral).
2. They are backed out from quoted CDS spreads or bond prices, given an assumed recovery rate.
3. Historical (real-world) probabilities are generally lower than risk-neutral ones and would misprice the CDS relative to the market.

Why A is correct: CDS valuation uses risk-neutral default probabilities implied by market spreads or bond prices, not real-world (historical) probabilities.

Why the others are wrong

- **B:** Historical rating-class default rates are real-world probabilities, which are not the ones used to value a CDS.
- **C:** Also real-world; accounting ratios do not give the market-implied default probability.
- **D:** Risk-neutral probabilities are market-implied, not set from historical averages; calling a historical average 'risk-neutral' is contradictory.

Q574 Schweser Module Quiz 31.2 Q1

Banko, Inc. entered into a $5 million notional, 5-year CDS as a protection buyer two years ago at a spread of 1.85%. The current 3-year CDS spread for the same reference entity is 2.30%, based on a PV of expected payoff of 0.0312 per $1 notional. The value of the CDS to Banko is closest to:

A. -$45,000.
B. -$30,900.
C. +$30,500.
D. +$53,000.

Answer: C

Step by step

1. At the current spread, PV of payments = PV of payoff: $0.023\,s = 0.0312$, so $s = 1.3565$.
2. PV of payments at the old 1.85% spread $= 0.0185 \times 1.3565 = 0.02510$.
3. Value to buyer $= 0.0312 - 0.0251 = 0.00610$ per $1.
4. $0.00610 \times \$5\text{m} \approx +\$30{,}500$.

Why C is correct: Spreads widened, so protection bought at 1.85% is worth more than a new contract at 2.30%; the buyer's gain is the PV of the 45 bp difference on the risky annuity.

Why the others are wrong

- **A:** Wrong sign: the buyer gains when spreads widen. The figure equals the 0.45% spread difference times the 2 elapsed years times notional, which ignores the risky annuity.
- **B:** Close in size but the wrong sign: this treats the buyer as losing when spreads widen, which is the protection seller's position.
- **D:** Overstates the gain; the correct value applies the 45 bp spread difference to the risky annuity of 1.3565 (about $30,500 on $5 million), not a larger factor.

LO 31.d

Evaluate the use of credit indices and fixed coupons in pricing CDS transactions.

Q575

AnalystPrep Q.6076

REBUILT — Original asked only for a loosely worded bid/ask interpretation; rebuilt as a numeric annual-cost question so the bid/ask direction is tested unambiguously.

A 5-year iTraxx Europe index (125 equally weighted investment-grade entities) is quoted by a market maker at bid 75 bp, ask 76 bp. A fund wants to buy protection on the entire index with a notional of EUR 1 million on each entity. What is the approximate annual premium the fund pays?

A. EUR 950,000
B. EUR 937,500
C. EUR 7,600
D. EUR 943,750

Answer: A

Step by step

1. The protection buyer deals at the dealer's offer (ask) of 76 bp; the bid (75 bp) is what the dealer pays to buy protection from a seller.
2. Cost per entity = $0.0076 \times 1{,}000{,}000 = 7{,}600$.
3. Protection is bought on all 125 entities: $7{,}600 \times 125 = 950{,}000$ per year.

Why A is correct: A buyer of index protection pays the ask spread on the notional of every constituent, so total premium is ask x notional per entity x 125.

Why the others are wrong

- **B:** Used the 75 bp bid ($0.0075 \times 1\text{m} \times 125 = 937{,}500$); the bid applies when selling protection.
- **C:** Computed the cost for a single entity and forgot to multiply by 125.
- **D:** Averaged bid and ask (75.5 bp); trades execute at the ask for a buyer.

Q576

AnalystPrep Q.6071

REBUILT — Original distractors were giveaways and the stem blurred default probabilities with spreads; rebuilt around what the index spread actually represents.

The 5-year CDX NA IG index (125 investment-grade names) is quoted at a spread of 60 bp. Which statement best describes how a trader should interpret this quote?

A. It is approximately the average of the CDS spreads of the 125 constituents, so it gauges the general level of credit risk in the market.
B. It is the CDS spread of the single constituent with the largest notional weight.
C. It is the sum of the 125 constituent CDS spreads.
D. It is the exact default probability of each constituent over five years.

Answer: A

Step by step

1. The index covers 125 equally weighted entities with equal protection on each.
2. The index spread is roughly (slightly below) the average of the constituents' CDS spreads, so 60 bp indicates an average credit cost across the portfolio.
3. It is a spread (an annual premium in bp), not a probability, and it is not tied to any one entity.

Why A is correct: Because every entity carries equal protection, the index spread behaves like an average of the constituent spreads and so serves as a market-wide credit gauge.

Why the others are wrong

- **B:** Constituents are equally weighted, so no entity has a largest weight; the index is not a single-name spread.
- **C:** Summing would give a total across names far above the quote; the quote is per unit of notional on each company, not a sum.
- **D:** Confuses a spread with a default probability; a spread embeds hazard rate and recovery and is not a probability for each name.

Q577

AnalystPrep Q.6077

REBUILT — Original asked to pick a formula from loosely stated options; rebuilt as a numeric up-front payment calculation using P = 100 - 100 x D x (s - c).

A standardized CDS index contract has a fixed coupon of 150 bp. The quoted spread is 200 bp and the duration of the spread payments is 4.0 years. For a notional of USD 10 million, what up-front payment is made?

A. The protection buyer pays USD 200,000
B. The protection seller pays USD 200,000
C. The protection buyer pays USD 50,000
D. The protection buyer pays USD 800,000

Answer: A

Step by step

1. Price $P = 100 - 100 \times D \times (s - c) = 100 - 100 \times 4.0 \times (0.0200 - 0.0150) = 98.0$.
2. Buyer pays $100 - P = 2.0$ per 100 of notional.
3. $2.0\% \times 10{,}000{,}000 = 200{,}000$, paid by the protection buyer because the true spread (200 bp) exceeds the coupon (150 bp).

Why A is correct: When the spread exceeds the fixed coupon, the coupon undercompensates the seller, so the buyer pays 100 - P up front equal to D x (s - c) of notional.

Why the others are wrong

- **B:** Reverses the direction; the seller pays only when the spread is below the coupon (P > 100).
- **C:** Omitted the duration: $0.005 \times 10\text{m} = 50{,}000$.
- **D:** Used the full spread instead of the spread minus coupon: $4.0 \times 0.02 \times 10\text{m} = 800{,}000$.

Q578

BT answers P2.T6.24.22.3 · BT notes P2.T6.24.22.3

REBUILT — Original was a vague judgmental 'which analysis to prioritize' item with no defensible unique answer; rebuilt as a numeric up-front settlement using the given spread, coupon and duration.

Alex is evaluating a 5-year iTraxx Europe index trade. The index is quoted at 45 bp, the standardized fixed coupon is 50 bp, and the duration of the contract is 4.6 years. Alex buys protection with a total notional of EUR 50 million. What is the up-front settlement?

A. Alex receives EUR 115,000
B. Alex pays EUR 115,000
C. Alex receives EUR 25,000
D. Alex pays EUR 1,150,000

Answer: A

Step by step

1. $P = 100 - 100 \times 4.6 \times (0.0045 - 0.0050) = 100 + 0.23 = 100.23$.
2. Because $P > 100$, the protection seller pays the buyer $P - 100 = 0.23$ per 100.
3. $0.0023 \times 50{,}000{,}000 = 115{,}000$ received by Alex, who is paying a coupon (50 bp) above the fair spread (45 bp).

Why A is correct: The coupon exceeds the spread, so the buyer overpays on the running leg and is compensated up front by D x (c - s) of notional.

Why the others are wrong

- **B:** Sign error; a spread below the coupon means the buyer receives, not pays.
- **C:** Left out duration: $0.0005 \times 50\text{m} = 25{,}000$.
- **D:** Multiplied duration by the coupon (50 bp) rather than the coupon-spread difference: $4.6 \times 0.005 \times 50\text{m} = 1{,}150{,}000$.

Q579 Schweser Module Quiz 31.2 Q2

If one of the entities in the CDX NA IG index defaults, the CDS index would most likely:

A. make a payoff and be discontinued.
B. continue with a replacement entity, with no change to the notional principal.
C. continue with 99 entities, and the notional principal of each entity would be reduced by 1%.
D. continue with 124 entities, and the notional principal of each entity would remain the same.

Answer: D

Step by step

1. CDX NA IG holds 125 equally weighted entities, each with the same protection notional.
2. On a default, the protection seller makes the usual (1 - RR) payoff on that entity's notional.
3. The contract continues on the remaining 124 entities at unchanged per-entity notional, and the annual premium falls by that entity's share (1/125).

Why D is correct: Only the defaulted name is settled and removed; the other entities and their notionals are untouched, so the index continues with 124 names.

Why the others are wrong

- **A:** The whole index is not terminated; only the defaulted entity is settled.
- **B:** Defaulted names are not replaced mid-life; replacements occur only when the index is updated (twice a year, for new contracts).
- **C:** One default leaves 124 of the 125 names, not 99, and the surviving entities keep their original notional rather than being reduced by 1%.

LO 31.e

Define CDS forwards and CDS options.

Q580 AnalystPrep Q.6078

A CDS forward is an agreement to enter into a CDS at a future date at an agreed-upon fixed spread. Which market participant is likely to find a CDS forward most appealing?

A. An investor expecting an immediate improvement in credit quality
B. A creditor looking to hedge against potential future credit deterioration
C. A speculative trader seeking instantaneous credit exposure
D. A protection seller aiming to mitigate risk in a widening credit spread environment

Answer: B

Step by step

1. A CDS forward is an obligation to buy or sell protection on a specified CDS (fixed spread and term) starting at a future date.
2. The forward locks in today's spread for future protection, so the protection buyer is shielded from spreads or credit quality worsening before the start date.
3. A creditor concerned about future deterioration therefore benefits: it fixes the cost of protection now.

Why B is correct: A forward fixes the spread on protection that starts later, so it suits a party that wants to hedge future credit deterioration.

Why the others are wrong

- **A:** Immediate exposure or an immediate view is handled by a standard CDS today, not by a contract that starts later.
- **C:** Instantaneous exposure requires a spot CDS; a forward starts only at the future date.
- **D:** A protection seller loses when spreads widen; a forward to sell protection at a fixed spread adds to that loss rather than mitigating it.

Q581 AnalystPrep Q.6079

REBUILT — Stem defined the call option in its own text (giveaway) and distractors were implausible; rewrote the stem and distractors.

An investor holds a CDS call option on a reference entity with a strike spread of 250 bps, exercisable in one year. What does the option give the holder?

A. The right to sell protection at 250 bps, which is valuable if credit quality improves
B. The right to buy protection at 250 bps, which is valuable if credit spreads widen
C. The obligation to buy protection at 250 bps in one year regardless of market spreads
D. The right to buy protection at 250 bps, which is valuable if credit spreads narrow

Answer: B
Step by step
1. A CDS call option is an option to buy protection (enter a CDS as protection buyer) at the strike spread.
2. It is exercised if the market CDS spread at expiry exceeds the 250 bps strike, since protection is then obtained below its market cost.
3. So the holder has the right, not the obligation, to buy protection, valuable when spreads widen.

Why B is correct: A call on a CDS is a right to buy protection at the strike, which pays off when spreads rise above the strike.
Why the others are wrong
- **A:** This describes the put option (right to sell protection), confusing call with put.
- **C:** An option carries no obligation; an obligation to buy protection at a future date is a CDS forward.
- **D:** The right is correct but the payoff direction is wrong: buying protection at 250 bps is attractive only when market spreads exceed 250 bps.

Q582 AnalystPrep Q.6080

A financial institution wants an instrument that gives it the choice, but not the obligation, to enter a CDS as protection buyer in the future depending on how market conditions evolve. Which instrument meets this requirement?
A. CDS forward
B. CDS call option
C. CDS put option
D. Physical settlement CDS

Answer: B
Step by step
1. A CDS forward is an obligation to enter the CDS, so it gives no choice.
2. A CDS option gives the right but not the obligation; the right to enter as protection buyer is the call.
3. Physical settlement is a way to settle after default, not optionality.

Why B is correct: Only options give the choice, and the call is the option to enter as protection buyer.
Why the others are wrong
- **A:** A forward commits both parties to enter the CDS, removing the choice.
- **C:** A put gives the right to sell protection, the opposite side of the trade.
- **D:** Physical settlement describes delivery of the bond after a credit event, not a right to enter a CDS.

Q583 AnalystPrep Q.6081

A corporation has credit exposure to a bond issuer with temporary liquidity problems. It does not need protection now, but wants the right to sell protection at an attractive spread if the issuer's credit improves. Which credit derivative fits?
A. CDS forward
B. Physical settlement CDS
C. CDS call option
D. CDS put option

Answer: D
Step by step
1. The corporation wants the right (not obligation) to sell protection: an option to enter a CDS as protection seller is a put.
2. It would be exercised if the market spread falls below the strike, since it then sells protection above the market spread.
3. A forward is an obligation; a call is the right to buy protection.

Why D is correct: A CDS put is the right to sell protection at a strike spread, valuable if credit improves and spreads

tighten.

Why the others are wrong

- **A:** A forward is an obligation to sell protection at the fixed spread, with no right to walk away if credit worsens.
- **B:** Physical settlement is a post-default delivery method, not an option.
- **C:** A call gives the right to buy protection, valuable when credit deteriorates, the opposite view.

Q584 BT answers P2.T6.24.23.1 · BT notes P2.T6.24.23.1

REBUILT — Two variants, one with no key and vague judgemental options; merged into one self-contained question with a single defensible answer.

A bank has sold, via a CDS forward, 5-year protection on Tech Co at 300 bps starting in one year. It also owns a CDS call option giving it the right to buy 5-year protection on Tech Co starting in one year at 250 bps (premium paid upfront). The bank expects Tech Co's CDS spreads and spread volatility to increase materially. Which action best positions the bank to benefit from this view?

A. Increase the CDS forward position (sell more forward protection)
B. Buy additional CDS call options
C. Write (short) CDS put options
D. Close out the call option and keep the forward

Answer: B

Step by step

1. Rising spreads hurt the forward seller of protection: it must provide protection at 300 bps while the market spread rises.
2. A long CDS call (right to buy protection at 250 bps) gains when the spread exceeds the strike, and as an option it also gains value from higher spread volatility.
3. Buying more calls therefore gains from both expected spread widening and rising volatility, with loss limited to the premium.
4. Shorting puts is short volatility, and more forward selling adds to losses if spreads widen.

Why B is correct: Long CDS calls benefit from higher spreads and higher volatility, with downside limited to the premium.

Why the others are wrong

- **A:** Selling more protection forward increases the loss when spreads widen.
- **C:** A short put position is short volatility and loses when volatility rises; it also does not gain from widening spreads.
- **D:** Closing the call removes the only position that profits from widening spreads and leaves the bank with the losing forward.

Q585 Schweser Module Quiz 31.2 Q3

Which of the following statements about CDS forwards and CDS options is most accurate?

A. CDS forwards entered at market rates require a premium payment, while CDS options do not.
B. CDS options require a premium payment, while CDS forwards entered at market rates do not.
C. Both CDS forwards entered at market rates and CDS options require a premium payment.
D. Neither CDS forwards entered at market rates nor CDS options require a premium payment.

Answer: B

Step by step

1. A forward is struck at the market forward spread, so it has zero value at inception and no upfront payment.
2. An option gives the holder a right, so the holder pays an upfront premium to the writer.
3. Hence options require a premium; market-rate forwards do not.

Why B is correct: Options carry a right and must be paid for; a forward at the market forward spread has zero initial value.

Why the others are wrong

- **A:** Reverses the two instruments.
- **C:** A forward at the market rate has zero value at inception, so no premium is paid.
- **D:** Options have positive value to the holder, so the writer requires a premium.

LO 31.f

Describe the process of valuing a synthetic CDO using the spread payments approach and the Gaussian copula model of time to default approach.

Q586

AnalystPrep Q.6082

REBUILT — Giveaway distractors (A, B, D were absurd) and a stem that leaked the answer; rewrote stem and options so each wrong option is a realistic misconception.

A risk manager values a tranche of a synthetic CDO using the spread payments approach. Which description of the method is most accurate?

A. Find the tranche spread that sets the PV of expected spread payments (on the declining tranche principal) equal to the PV of expected default-related payoffs.

B. Take the average CDS spread of the reference entities and apply it unchanged to every tranche.

C. Set the tranche spread equal to the observed bid-ask midpoint of the standard index tranche, with no model of defaults.

D. Set the tranche spread so that the PV of expected losses on the senior tranche equals that on the equity tranche.

Answer: A

Step by step

1. Spread payments are received only on the outstanding tranche principal, which falls as losses are absorbed.
2. PV of inflows $= s \times (A + B)$, where A is the PV of payments on the expected remaining principal and B is the final accrual on losses occurring mid-period.
3. PV of outflows (expected tranche losses, discounted from mid-period) $= C$.
4. Set $s(A + B) = C$ so the breakeven spread is $s = C/(A + B)$.

Why A is correct: The tranche spread is the breakeven that equates the PV of expected premium inflows with the PV of expected credit-loss payoffs, as for a single-name CDS.

Why the others are wrong

- **B:** Ignores seniority: the average pool spread is the aggregate across tranches, but each tranche earns a spread consistent with its own risk.
- **C:** Confuses a market quote with the valuation method; the approach models expected tranche losses and does not read off a spread.
- **D:** Tranches have different loss exposures, so equating expected losses across tranches is not the pricing condition; each tranche is priced on its own inflows = outflows.

Q587

AnalystPrep Q.6083

REBUILT — Distractors were weak; rewrote options to be realistic misconceptions about the Gaussian copula model.

In the one-factor Gaussian copula model of time to default used to value a synthetic CDO, the model's key role is to:

A. Assume defaults are independent so that tranche losses follow a simple binomial distribution unconditionally.

B. Link the marginal default-time distributions of reference entities through a correlation parameter, so the distribution of the number of defaults, and hence tranche losses, can be derived.

C. Assign a different hazard rate to each entity so that the model is heterogeneous.

D. Predict which reference entity will default first and when.

Answer: B

Step by step

1. Each entity has the same marginal default probability $Q(t)$ (homogeneous pool, constant hazard rate).
2. A copula with constant pairwise correlation ρ joins these marginals into a joint distribution of default times.
3. Conditional on the common factor F, defaults are independent, giving binomial probabilities of k defaults.
4. Integrating over F gives expected tranche principal and losses.

Why B is correct: Correlation through the copula turns identical marginal default probabilities into a joint loss distribution, which is what differentiates equity, mezzanine and senior tranche values.

Why the others are wrong

- **A:** Independence only holds conditional on the common factor; unconditionally defaults are correlated through F.
- **C:** The model is homogeneous: the same $Q(t)$ is assumed for all entities.
- **D:** The model gives probabilities of the number of defaults, not the identity or order of defaults.

Q588 AnalystPrep Q.6084

REBUILT — Option wording was vague; rewrote so the role of conditional probabilities is tested precisely.

In the one-factor Gaussian copula model, the conditional default probability $Q(t \mid F)$ is used because, given the common factor F:

A. The default time of each entity becomes exactly predictable.

B. Defaults of different entities become independent, so the probability of exactly k defaults follows a binomial distribution; results are then integrated over F.

C. Hazard rates are replaced by historical default frequencies.

D. Correlation is eliminated from the valuation, so the tranche price no longer depends on ρ.

Answer: B

Step by step

1. Unconditional PD: $Q(t) = 1 - e^{-\lambda t}$ with constant hazard rate λ.
2. Conditional PD $Q(t \mid F)$ depends on F, the systematic (market-wide) factor and on ρ.
3. Given F, defaults are independent, so $P(k \text{ defaults})$ is binomial in n and $Q(t \mid F)$.
4. Weight the conditional expected tranche principal/losses by the standard normal density of F and integrate (Gaussian quadrature).

Why B is correct: Conditioning on the common factor makes defaults independent, allowing a binomial calculation; correlation still enters via $Q(t \mid F)$ and the integration over F.

Why the others are wrong

- **A:** The model gives probabilities, never exact default times.
- **C:** Hazard rates implied from market spreads are still used for $Q(t)$; no switch to historical frequencies.
- **D:** Correlation is embedded in $Q(t \mid F)$ via ρ, so the price still depends on it.

Q589 BT answers P2.T6.24.23.2 · BT notes P2.T6.24.23.2

REBUILT — Merged two versions (700 vs 100 entities, inconsistent) and replaced the vague action-style options with precise statements of the valuation steps.

Omega Capital is valuing a synthetic CDO referencing 100 corporate entities, with equity (first 5% of losses), mezzanine and senior tranches, using the one-factor Gaussian copula model of time to default. Which step is the core of computing each tranche's expected loss and hence its breakeven spread?

A. Compute conditional default probabilities and expected tranche principals given the common factor, then integrate these over the standard normal distribution of the factor.

B. Compute the unconditional default probability $Q(t)$ only and apply it directly to the tranche attachment points.

C. Set the tranche spread equal to the average CDS spread of the 100 entities.

D. Assume each entity defaults independently so no integration is needed.

Answer: A

Step by step

1. Unconditional PD from the hazard rate: $Q(t) = 1 - e^{-\lambda t}$.
2. For a given factor value F, compute $Q(t \mid F)$ and the binomial probability of k defaults among $n = 100$.
3. Convert to expected tranche principal (and loss) conditional on F using attachment/detachment points.
4. Integrate over $F \sim N(0,1)$ (Gaussian quadrature) to get unconditional A, B, C, then $s = C/(A + B)$.

Why A is correct: The tranche values depend on the whole loss distribution, which is obtained by integrating conditional (binomial) results over the common factor.

Why the others are wrong

- **B:** $Q(t)$ alone gives the marginal default probability but not the distribution of the number of defaults, which depends on correlation.
- **C:** The average spread is the aggregate premium across tranches; tranche spreads differ with seniority.
- **D:** Independence ignores correlation, which drives the equity/senior tranche valuation difference.

Q590 Schweser Module Quiz 31.3 Q1

REBUILT — Option C ('binary CDSs') was ambiguous and arguably also true under GARP/Schweser; reworded options so only one is correct.

At inception, the tranche spreads of a synthetic CDO are set so that:

A. each tranche earns the same spread rate.

B. each tranche's spread rate reflects its seniority (risk), and the total spread payments across tranches equal the CDS premiums received on the reference pool.

C. senior tranches earn the highest spread rate because they have the largest notional.
D. the equity tranche earns the lowest spread rate because it is the smallest.

Answer: B
Step by step
1. Example: $100m pool, 2% average spread = $2m per year.
2. Equity ($5m) earns 10% = $0.5m; mezzanine ($20m) earns 5.625% = $1.125m; senior ($75m) earns 0.5% = $0.375m.
3. Sum = $0.5m + $1.125m + $0.375m = $2m, equal to the pool premium.
4. Rates fall with seniority even though notional rises.

Why B is correct: Spread rates are priced to risk, so they decline with seniority, while the dollar total is the pool's CDS premium.
Why the others are wrong
- **A:** Equal rates would ignore the different loss exposure; equity bears first losses and earns 10% vs 0.5% for senior.
- **C:** Senior has the largest notional but the lowest rate (0.5%) because it is last to absorb losses.
- **D:** Equity earns the highest rate since it bears first losses, regardless of its size.

LO 31.g

Define the two measures of implied correlation: compound (tranche) correlation and base correlation.

Q591

AnalystPrep Q.6085

REBUILT — Original keyed 'constant correlation per tranche for simplification', which misdescribes compound correlation; rewrote stem and options around the GARP definition.

In pricing a synthetic CDO with the one-factor Gaussian copula model and a 40% recovery rate, the copula correlation is the only unknown input. The compound (tranche) correlation for a given tranche is best described as the correlation that:

A. makes the model-calculated spread for that tranche equal to the spread quoted in the market, found by an iterative search.
B. makes the model price of the equity tranche $[0, a]$ consistent with the market, using the cumulative expected loss up to the tranche's detachment point.
C. is estimated directly from historical co-movements in the credit spreads of the reference entities.
D. is set equal across all tranches so that the model spreads of all tranches reproduce the market spreads simultaneously.

Answer: A
Step by step
1. With $R = 40\%$, the copula correlation is the sole free parameter, analogous to implied volatility in Black-Scholes-Merton.
2. For tranche q, search iteratively for the correlation at which the model spread equals the market-quoted spread for that tranche alone.
3. This value is the compound (tranche) correlation; it is solved tranche by tranche.

Why A is correct: Compound correlation is an implied correlation: it is backed out so that each tranche's model spread equals its market spread.
Why the others are wrong
- **B:** Describes base correlation (the 0-to-detachment tranche priced consistently with the market), not compound correlation.
- **C:** Describes a historical estimate; implied correlation is backed out of market quotes, not history.
- **D:** A single common correlation would only match all tranches if the model were consistent with the market; in practice compound correlations differ by tranche (the smile).

Q592

AnalystPrep Q.6086

REBUILT — Original correct option ('varying correlation assumptions for each tranche') also describes compound correlation, so it did not isolate base correlation; rewrote around the $[0, a_q]$ tranche definition.

A trader implies the base correlation for the tranche with detachment point a_q $(q > 1)$ in a synthetic CDO. Which statement best describes base correlation?

A. It is the correlation that makes the model spread of the $\{a_{q-1}, a_q\}$ tranche equal its quoted spread.

B. It is the correlation that makes the $\{0, a_q\}$ tranche price consistently with the market, with the tranche's expected loss built from the compound correlations of the tranches up to a_q.

C. It is the correlation of the most senior tranche, applied to every junior tranche beneath it.

D. It is the average of the compound correlations of all tranches in the capital structure.

Answer: B

Step by step

1. Compute the compound correlation of each tranche (iterative search).
2. Use these to get the PV of expected loss of each tranche as a % of tranche principal, C_q.
3. Convert to the PV of expected loss of the $\{0, a_q\}$ tranche as a % of total portfolio principal: $\sum C_p(a_p - a_{p-1})$.
4. Find (iteratively) the correlation giving that value for the $\{0, a_q\}$ tranche: this is the base correlation.

Why B is correct: Base correlation is defined for the $\{0, a_q\}$ tranche, so it is tied to the detachment point and built up from the expected losses implied by the compound correlations.

Why the others are wrong

- **A:** This is the compound (tranche) correlation, which targets a single tranche $\{a_{q-1}, a_q\}$.
- **C:** No such rule exists; the base correlation is implied separately for each detachment point.
- **D:** Averaging compound correlations is not the procedure; base correlation is solved from expected loss of the $\{0, a_q\}$ tranche.

Q593

BT answers P2.T6.24.23.3 · BT notes P2.T6.24.23.2

REBUILT — Merged duplicates; source statement ii misdescribed base correlation (attachment point, 'subordinate tranches'), iii belongs to alternative models, iv was vague, and the key was unreliable; rewrote the statements around the GARP definitions.

Global Credit Solutions is reviewing implied correlations for its synthetic CDO tranches. Which of the following statements are correct?

i. The compound correlation of a tranche is the correlation at which the model spread equals the market-quoted spread for that tranche, found by an iterative search.

ii. The base correlation for detachment point a_q is the correlation that prices the $\{0, a_q\}$ tranche consistently with the market, using expected losses obtained from the compound correlations of the tranches.

iii. If market prices were consistent with the one-factor Gaussian copula model, the implied correlations would be the same for all tranches.

iv. Compound correlations exhibit a skew rising with seniority, while base correlations exhibit a smile.

A. i only

B. i and ii only

C. i, ii and iii only

D. ii, iii and iv only

Answer: C

Key correction: Source key (BT answers) said B (i and ii only); the original statement ii was inaccurate and the item has been rewritten, so the corrected answer is C.

Step by step

1. i is correct: definition of compound correlation.
2. ii is correct: base correlation is defined for the $\{0, a_q\}$ tranche using the compound-correlation-based expected losses.
3. iii is correct: a pronounced smile or skew is inconsistent with a single-correlation Gaussian copula, a model limitation.
4. iv is incorrect: compound correlations show a smile; base correlations show a skew that rises with detachment point (swapped in the statement).
5. Correct: i, ii and iii only.

Why C is correct: Statements i-iii restate the GARP/Schweser definitions and the consistency implication; iv swaps smile and skew.

Why the others are wrong
- **A:** Omits ii and iii, which are both true.
- **B:** Omits iii, which correctly states that a consistent model would give equal implied correlations.
- **D:** Excludes i, a correct definition, and includes iv, which swaps smile and skew.

Q594 Schweser Module Quiz 31.3 Q2

Base correlation and compound correlation are both:

A. tranche correlations.
B. implied correlations.
C. not relevant for synthetic CDOs.
D. exhibiting a skew with correlations rising with tranche seniority.

Answer: B

Step by step
1. Both are backed out of market quotes for CDO tranches, like implied volatility.
2. So both are implied correlations.
3. Check others: only compound correlation is strictly the tranche correlation; both are relevant to synthetic CDOs; compound shows a smile, base shows a skew.

Why B is correct: Both measures are obtained by an iterative search that makes the model consistent with market prices, so both are implied correlations.

Why the others are wrong
- **A:** Only compound correlation is a per-tranche correlation; base correlation applies to the $\{0, a_q\}$ tranche.
- **C:** Both are used to value and quote synthetic CDO tranches.
- **D:** Only base correlations show a skew rising with seniority; compound correlations show a smile.

LO 31.h

Discuss alternative approaches used to estimate default correlation.

Q595 AnalystPrep Q.6087

REBUILT — Original stem was wordy and option B was only subtly wrong; tightened the stem and options so exactly one answer is defensible.

The standard market (one-factor Gaussian copula) model for valuing a CDO is homogeneous. How does a heterogeneous model differ, and what is the main implication for implementation?

A. It assumes identical default probabilities and copula correlations for all entities, which simplifies implementation.
B. It allows different time-to-default distributions for different entities, so the binomial formula for the probability of k defaults can no longer be used and a numerical procedure is needed.
C. It allows different time-to-default distributions for different entities, and the binomial formula for the probability of k defaults still applies, so valuation is no harder.
D. It models only default probabilities and ignores correlations, which reduces the computational burden.

Answer: B

Key correction: Source key (C) matched the idea but its wording was loose; the rebuilt item makes the binomial/numerical-procedure point explicit.

Step by step
1. The standard model is homogeneous: the same time-to-default distribution for every company and the same pairwise copula correlation.
2. A heterogeneous model relaxes this and lets the distributions (and correlations) differ by reference entity.
3. With identical entities the number of defaults follows a binomial distribution; with different entities it does not.
4. Therefore the probability of k defaults must be computed with a numerical procedure, making the model more complex.

Why B is correct: Heterogeneity destroys the symmetry that makes the binomial formula valid, so valuation requires numerical procedures.

Why the others are wrong

- **A:** Describes the homogeneous model, not the heterogeneous one.
- **C:** Correct on the relaxed assumption but wrong on implementation: the binomial formula no longer applies.
- **D:** Heterogeneous models still include correlation; they do not drop it.

Q596

AnalystPrep Q.6089

REBUILT — Original options B and D differed only in a trailing clause and option A cited historical data; rewrote options so each wrong one reflects a distinct realistic error.

In the implied copula approach of Hull and White, what is the simplest version of the model assumed, and what is implied from the market prices of CDO tranches?

A. Each company has its own hazard rate, and their distributions are implied from historical tranche prices.

B. Each company has its own hazard rate, and the individual hazard rates are implied from tranche prices.

C. A single average hazard rate applies to all companies over the life of the CDO, and its probability distribution is implied from current tranche prices.

D. A fixed average hazard rate applies to all companies, and it is estimated from historical default data rather than tranche prices.

Answer: C

Key correction: Source key (D) was correct for the original wording; after rewording the correct option is C.

Step by step

1. The implied copula is analogous to implied volatility: it is backed out of market quotes, not historical data.
2. The simplest version assumes one average hazard rate common to all companies in the portfolio over the CDO's life.
3. That average hazard rate has a probability distribution, which is implied from the pricing of tranches.
4. Option C states exactly this.

Why C is correct: The model implies a distribution for a single common average hazard rate from current tranche prices.

Why the others are wrong

- **A:** Assumes company-specific hazard rates and uses historical data; both contradict the approach.
- **B:** Assumes individual hazard rates; the simplest implied copula uses one average rate.
- **D:** Uses historical default data and a fixed rate; the approach implies a distribution from market tranche prices.

Q597

Schweser Module Quiz 31.3 Q3

The structural model of credit risk is most likely a(n):

A. dynamic model.

B. heterogeneous model.

C. implied copula model.

D. random recovery rate model.

Answer: A

Step by step

1. Dynamic models model the evolution of portfolio loss through time; they include structural, reduced-form and top-down types.
2. Structural models simulate correlated asset-value processes and declare default when an asset hits a barrier, so they are dynamic.
3. The other options are different alternatives to the Gaussian copula.

Why A is correct: Structural models are one of the dynamic model types that track loss evolution over time.

Why the others are wrong

- **B:** Heterogeneous models allow different time-to-default distributions per entity within a static copula framework.
- **C:** The implied copula is derived from market tranche prices, not from asset-value dynamics.
- **D:** Random recovery models reflect the negative link between default rates and recovery rates; unrelated to barrier-based default.

Reading 32. Derivatives

Jon Gregory, The xVA Challenge: Counterparty Credit Risk, Funding, Collateral, and Capital, 4th Edition (West Sussex, UK: John Wiley & Sons, 2020). Chapter 2.

LO	Learning objective	Qs
32.a	define derivatives and explain how derivative transactions create counterparty credit risk.	4
32.b	compare and contrast exchange-traded derivatives and over-the-counter (OTC) derivatives, and discuss the features of their markets.	10
32.c	describe the process of clearing a derivative transaction.	8
32.d	identify the participants and describe the use of collateralization in the derivatives market.	8
32.e	define the International Swaps and Derivatives Association (ISDA) Master Agreement, the risk-mitigating features it provides, and the default events it covers.	5
32.f	describe the features and use of credit derivatives and discuss potential risks they may create.	3
32.g	describe central clearing of OTC derivatives and discuss the roles, mandate, advantages, and disadvantages of the central counterparty (CCP).	6
32.h	explain the margin requirements for both centrally-cleared and non-centrally-cleared derivatives.	4
32.i	define special purpose vehicles (SPVs), derivatives product companies (DPCs), monolines, and credit derivatives product companies (CDPCs) and describe the limitations of using them as risk mitigating methods.	9
32.j	describe the approaches used and the challenges faced in modeling derivatives risk.	3

LO 32.a

Define derivatives and explain how derivative transactions create counterparty credit risk.

Q598

AnalystPrep Q.6153

In the wake of the Global Financial Crisis (GFC), a risk manager is evaluating the complex landscape of derivatives, mindful of Warren Buffett's characterization of these instruments as "financial weapons of mass destruction." The manager is focused on dissecting the subtle, yet potentially severe, risks that derivatives could introduce to a firm's financial stability. Which of the following statements most precisely articulates one of the significant risks tied to derivatives, as underscored by Warren Buffett?

A. Derivatives can inadvertently lead to interest rate mismatches, potentially exposing firms to significant interest rate volatility risks.

B. Derivatives, with their inherent counterparty risk, necessitate a substantial initial investment, which can tie up significant capital resources.

C. Derivatives can induce liquidity crises due to (rating) downgrade triggers, necessitating firms to meet unexpected cash demands during critical periods.

D. Derivatives might expose firms to foreign exchange risk, especially when the underlying assets are denominated in different currencies.

Answer: C

Step by step

1. Buffett's criticisms (GARP Ch.2): counterparty risk, overstated 'mark-to-myth' valuations, rating downgrade triggers, daisy-chain risk.
2. Downgrade triggers force extra collateral exactly when the firm is weakest, creating an unexpected cash demand and liquidity problems that can cause further downgrades (a death spiral).
3. Only C describes this; A, B and D are not among his points.

Why C is correct: Rating-downgrade triggers require greater collateralisation at the worst time, producing an unexpected cash demand, liquidity stress and possibly further downgrades.

Why the others are wrong

- **A:** Interest rate mismatch is a generic market risk, not one of Buffett's points; confuses derivatives risk with ordinary balance-sheet ALM risk.
- **B:** Wrong: derivatives need little or no upfront payment relative to notional (that is leverage); counterparty risk does not require a large initial investment.
- **D:** FX risk can arise from some derivatives, but it is a generic market risk and not among the risks Buffett highlighted.

Q599 AnalystPrep Q.6115

A junior trader is being trained on different financial instruments and their characteristics. The trainer presents a series of definitions and asks the trainee to identify which one correctly defines a derivative instrument. Which definition should the trainee choose?

A. A financial instrument whose value is based solely on the market movements of a single underlying stock or bond.

B. A financial instrument whose value is contingent upon, or derived from, a specified underlying asset or group of assets, like stocks, bonds, or market indices.

C. A financial tool that directly represents ownership in an underlying asset, such as shares in a company or government-issued debt securities.

D. An investment vehicle that guarantees a fixed return based on the performance of an underlying market index.

Answer: B

Step by step

1. A derivative's value depends on (is derived from) one or more underlying assets, rates or indices; it is a contract to buy/sell the underlying or make a payment in the future.
2. Check each option: A restricts to a single stock or bond; C describes ownership (a security); D implies a guaranteed fixed return.
3. Only B states value is contingent on a specified asset or group of assets.

Why B is correct: A derivative is a contract whose value is contingent on, or derived from, one or more underlying assets, rates or indices.

Why the others are wrong

- **A:** Too narrow: underlyings include groups of assets, indices, rates and commodities, not just one stock or bond.
- **C:** Describes a cash security (equity or bond) giving direct ownership; a derivative does not represent ownership of the underlying.
- **D:** Derivatives carry no guaranteed return; payoffs vary with the underlying and counterparty can default.

Q600 AnalystPrep Q.6156

In a masterclass on the complexities of financial derivatives, the instructor delves into the nuanced risks highlighted by Warren Buffett, particularly focusing on the concept of 'daisy-chain' risk. What does the term 'daisy-chain' risk primarily signify in the context of derivatives?

A. The risk of a single derivative position leading to disproportionate losses, exceeding the initial investment.

B. The risk arising from the accumulation of large receivables from interconnected counterparties, potentially leading to systemic risk.

C. The risk of derivatives being used to hedge against unfavorable price movements, resulting in a reduced exposure to market volatility.

D. The risk of regulatory changes leading to sudden adjustments in derivative valuations, impacting the overall financial market.

Answer: B

Step by step

1. Buffett's daisy-chain point (GARP Ch.2): derivatives defeat prudent counterparty diversification because large receivables from interconnected counterparties build up over time.
2. Hence a default by one party can cascade through the network, a systemic risk.
3. Only B matches this.

Why B is correct: Daisy-chain risk is the accumulation of large receivables from interconnected counterparties, so one failure can propagate through the system.

Why the others are wrong

- **A:** Describes leverage-driven loss magnification, a different Buffett/leverage theme, not daisy-chain risk.
- **C:** Describes the purpose of hedging (a benefit), not a risk.
- **D:** Regulatory/valuation-change risk is not what daisy-chain refers to; the term concerns interconnected counterparty exposures.

Q601 AnalystPrep Q.6157

In a lecture on the structural aspects of financial derivatives, a professor is emphasizing the role of leverage in derivatives and its potential impact on market dynamics. Why is leverage considered a key feature of derivatives, and what potential impact does it have on market dynamics?

A. Leverage in derivatives allows institutions to execute trades without any upfront payment, effectively reducing the cost of trading and limiting market disturbances.

B. Leverage in derivatives permits institutions to take on positions significantly larger than their initial investment, potentially amplifying both returns and risks.

C. Leverage in derivatives ensures a fixed return on investment, safeguarding institutions against the volatility of the financial markets.

D. Leverage in derivatives mandates a substantial initial investment, thereby minimizing the risk of default and stabilizing the financial market.

Answer: B

Step by step

1. Most derivatives require only a small (relative to notional) or no upfront payment, so exposure far exceeds the cash invested: leverage.
2. Leverage amplifies both gains and losses and can contribute to systemic risk.
3. B states this; A, C and D contradict it.

Why B is correct: Leverage lets institutions hold positions much larger than the upfront cash outlay, amplifying both returns and risks.

Why the others are wrong

- **A:** Wrong on consequences: low upfront cost makes trading cheap but does not limit market disturbances; it increases risk-taking (and 'no payment' ignores margin).
- **C:** Confuses leverage with a guarantee; leverage creates variable, amplified outcomes, not a fixed return.
- **D:** Reverses the concept: leverage means a small, not substantial, initial investment, and it raises rather than lowers risk.

LO 32.b

Compare and contrast exchange-traded derivatives and over-the-counter (OTC) derivatives, and discuss the features of their markets.

Q602 AP Mock 3 Q58

A portfolio manager is designing a bespoke hedge for a client with foreign currency exposure tied to unique cash flow timelines and amounts that do not align with standard contract sizes or maturities. The manager opts for an OTC derivative. What is the primary benefit in this scenario?

A. Access to a wide range of standardized contract sizes and maturities.

B. Availability of highly liquid secondary markets for the derivative.

C. Ability to customize the contract to match the unique hedging needs of the client.

D. Reduction of legal and operational risks associated with the derivative contract.

Answer: C

Step by step

1. The exposure has non-standard dates and amounts, so a standardized exchange contract would leave a mismatch (basis risk).
2. OTC contracts are privately negotiated between two counterparties, so size, maturity and other terms can be tailored.
3. Hence the primary benefit is customization.

Why C is correct: OTC derivatives are flexible and negotiable, allowing a precise match to the hedged exposure.

Why the others are wrong

- **A:** Standardized sizes and maturities are an exchange feature, and they are exactly what does not fit this client.
- **B:** OTC contracts are usually not actively traded in a secondary market, so liquidity is a weakness, not a benefit.
- **D:** OTC contracts need bilaterally negotiated documentation and carry bilateral counterparty risk, so legal and operational risk is higher, not lower.

Q603

AnalystPrep Q.1867

REBUILT — Original stem was a vague 'other main reason' prompt whose option C was a true statement but a disadvantage; rewrote as a clear advantage-of-OTC question with one defensible answer.

Over-the-counter (OTC) derivative markets have grown substantially, partly because of new products such as credit derivatives. Which of the following is the other main advantage of OTC derivatives that explains this growth?

A. Trading is centralized in one place, which raises liquidity and transparency.
B. Contract terms can be tailored accurately to the specific needs of the client.
C. A clearing house stands between buyer and seller, which removes counterparty risk.
D. Standardized terms make the contracts easy to trade in an active secondary market.

Answer: B

Step by step

1. OTC derivatives are privately negotiated, so in theory any transaction terms can be accommodated.
2. This flexibility lets users hedge exposures (dates, notional, structure) that exchange contracts cannot match, and supports new products such as credit derivatives.
3. Therefore B.

Why B is correct: The main advantage of OTC derivatives is their inherent flexibility; they need not be standardized.

Why the others are wrong

- **A:** Describes an exchange; OTC trading is decentralized and bilateral, with less transparency.
- **C:** Describes central clearing of exchange-traded contracts; OTC contracts are traditionally bilateral, so each party bears the other's counterparty risk.
- **D:** Standardization and active secondary markets are exchange features; OTC contracts are typically customized and not actively traded.

Q604

AnalystPrep Q.2296

REBUILT — Original had a trivial public/private option set; rebuilt to test the OTC negotiation and reporting features more rigorously.

Arrenberg Bank is a large dealer bank that executes many over-the-counter trades. Which statement best describes how its OTC trades are typically arranged and traded?

A. Terms are set by an exchange and the trade is guaranteed by a clearing house.
B. Terms are negotiated privately and directly between the two parties, with no exchange as intermediary.
C. Terms are posted publicly and any participant may accept them at the quoted price.
D. Terms are negotiated privately, but the trade must be reported publicly to an exchange before execution.

Answer: B

Step by step

1. OTC derivatives are traditionally privately negotiated and traded directly between two parties (dealer and end user, or two dealers).
2. No exchange or other intermediary is involved, and legal documentation is bilaterally negotiated.
3. Historically there was no trade reporting, so trades occurred in private.

Why B is correct: OTC trades are bilateral and private, which gives flexibility but lower transparency.

Why the others are wrong

- **A:** Describes exchange-traded derivatives, where the exchange sets standardized terms.
- **C:** Public posted prices accepted by anyone describe an exchange, not bilateral negotiation.
- **D:** OTC markets did not historically include trade reporting, and reporting is not a precondition to execution.

Q605

AnalystPrep Q.2300

Because of the nature of their lending business, banks widely hold a derivative whose terms (maturity, payment frequency, reference rate) can be matched precisely to a floating-rate loan. Which of the following derivatives is this?

A. Futures contracts
B. Options contracts
C. Forward contracts
D. Interest-rate swaps

Answer: D

Step by step

1. Banks borrow short (deposits) and lend, so they are exposed to interest rate risk, for example on floating-rate loans.
2. A swap precisely matched to the loan converts floating to fixed (or the reverse), hedging that risk.
3. Banks therefore commonly hold interest rate swaps.

Why D is correct: Interest rate swaps are the bespoke OTC instrument banks use to hedge the interest rate risk inherent in lending.

Why the others are wrong

- **A:** Futures are standardized exchange contracts and rarely match a loan's terms exactly, so they are not the characteristic bank hedge for loans.
- **B:** Options can be used, but they do not replicate the recurring floating-to-fixed exchange of payments that hedges a loan book.
- **C:** Forwards are single-date contracts and do not match the recurring floating-rate payments of a loan; swaps are the instrument used to hedge lending.

Q606

AnalystPrep Q.6116

A hedge fund wants to speculate on a widely traded commodity. It needs high liquidity for quick position adjustments, values transparent pricing for fair entry and exit, and prefers standard maturity dates matching its quarterly trading cycle. If it chooses an exchange-traded derivative, what is the primary implication for its trading strategy?

A. The fund will benefit from customized contracts tailored to its specific trading strategies.

B. The fund will have to manage the additional basis risk due to the lack of contract customization.

C. The fund will take advantage of high liquidity and transparent pricing for efficient trading.

D. The fund will face higher counterparty risk due to the lack of bilateral agreements.

Answer: C

Step by step

1. The fund's stated needs are liquidity, price transparency and standard maturities.
2. Exchange-traded derivatives are standardized, very liquid and have transparent, accessible prices.
3. These match the fund's needs, so the main implication is efficient trading (C).

Why C is correct: Exchange-traded contracts offer liquidity and price transparency, which the speculating fund values.

Why the others are wrong

- **A:** Exchange contracts are standardized, not customized; customization is an OTC feature.
- **B:** Basis risk from standardization matters to hedgers, but this fund speculates and wants standard maturities, so it is not the primary implication.
- **D:** Exchange trades are centrally cleared by a CCP that becomes the counterparty, which reduces, not raises, counterparty risk.

Q607

AnalystPrep Q.6120

REBUILT — Option B (hold to maturity) was a defensible alternative to the keyed answer; reworded so only the offsetting-trade answer fits the typical close-out of a standardized, liquid contract.

An investor holds a long position in an exchange-traded gold futures contract and wants to exit before expiry. Given the contract's standardization and the market's liquidity, how is the position typically closed?

A. The investor must locate a specific counterparty willing to take over the contract.

B. The investor must hold the contract to expiry and deliver or take delivery of the gold.

C. The investor executes an offsetting trade (sells an identical contract) in the liquid market.

D. The exchange itself takes over the position at the investor's request.

Answer: C

Step by step

1. Because contracts are identical (standardized), a long position is offset by selling an identical contract.
2. Liquid exchange markets make this quick and at a transparent price, so no negotiation with the original counterparty is needed.
3. Therefore the position is closed by an offsetting trade.

Why C is correct: Standardization and liquidity let exchange positions be closed by an offsetting trade at market prices.

Why the others are wrong

- **A:** Finding a specific counterparty is the difficulty of unwinding a custom OTC contract, not an exchange contract.
- **B:** Holding to expiry is not required and is not the typical way to exit; most positions are offset beforehand.
- **D:** The exchange/CCP stands as counterparty but does not take over a position at the holder's request.

Q608

AnalystPrep Q.6123

Which of the following is correct regarding OTC derivatives?

A. Unwinding these derivatives is typically simpler than standardized products due to their tailored nature.

B. The original counterparty may demand a premium for unwinding the derivative, capitalizing on their exclusive position.

C. Custom OTC derivatives can be easily transferred to a third party at market value without the original counterparty's consent.

D. The market for custom OTC derivatives is highly liquid, facilitating easy unwinding at fair market value.

Answer: B

Step by step

1. A custom OTC contract generally has no active secondary market.
2. Unwinding to avoid future counterparty risk must be done with the original counterparty, who may quote unfavorable terms because of its privileged position.
3. Novation to a third party typically needs the original counterparty's permission.

Why B is correct: Because custom contracts can only practically be unwound with the original counterparty, it can quote unfavorable terms.

Why the others are wrong

- **A:** Unwinding is harder, not easier, for tailored contracts.
- **C:** Novation typically requires the original counterparty's consent.
- **D:** Many OTC markets are illiquid; custom contracts are not actively traded.

Q609

BT answers P2.T6.24.24.2 · BT notes P2.T6.24.24.2

REBUILT — Source key D is not defensible (OTC swaps are now routinely collateralised and exchange-traded futures also carry credit exposure to the clearing house); rewrote the stem and options around the standardised-versus-customised distinction so exactly one option is correct.

Robot Inc., a large manufacturer, wants to hedge its exposure to commodity price fluctuations on a specific set of future purchase dates. It is comparing an exchange-traded commodity futures contract with an OTC commodity swap arranged bilaterally with a major bank. Which statement describes a key difference between the two instruments?

A. The futures contract has standardised terms such as maturity and size, which may leave Robot with basis risk, whereas the OTC swap can be tailored to Robot's exact dates and quantities.

B. The futures contract can be customised to Robot's exact dates and quantities, whereas the OTC swap must use standardised maturities.

C. The futures contract is traded privately between two parties and is therefore less liquid than the OTC swap.

D. The OTC swap is easy to terminate with any third party at a transparent market price, whereas the futures contract can only be terminated with the original counterparty.

Answer: A

Key correction: Source key (BT) said D; D is not a reliable distinguishing feature because OTC derivatives are now commonly collateralised and the sources' options were not clearly distinguishable on a single correct feature. Correct (rebuilt) key is A.

Step by step

1. Exchange-traded derivatives (futures, options) have standardised terms: maturity, size and strike are fixed by the exchange.
2. OTC derivatives are privately negotiated bilaterally, so any terms (maturity, size, strike) can be accommodated.
3. For a hedge needed on specific dates, standardised futures maturities create a mismatch (basis risk); a customised OTC swap can match the exposure exactly.
4. Hence A states the key difference; B, C and D each reverse a feature of the two markets.

Why A is correct: Standardisation is the defining feature of exchange-traded contracts and flexibility the main advantage of OTC contracts, so a bespoke hedge is better matched by the swap while futures may leave basis risk.

Why the others are wrong

- **B:** Reverses the features: it is the OTC swap, not the exchange-traded future, that can be customised.
- **C:** Confuses the venues: futures trade on an exchange with very good liquidity; it is OTC trades that are private, bilateral and less liquid.
- **D:** Reverses the features again: exchange contracts are easily closed out by an offsetting trade at transparent prices, whereas an OTC trade is typically unwound only with the original counterparty (or novated with its permission).

Q610

AnalystPrep Q.6119

REBUILT — Original asked about 'roll over vs close' with a vague best answer (C) and an assertion about liquidity points not supported by the reading; reworked to test the standardised-maturity feature and the resulting basis/maturity mismatch.

A corporate treasurer needs to hedge an interest rate exposure that ends on 15 May. The only exchange-traded U.S. Treasury bond futures available have standard quarterly delivery months (March, June, September, December). Which statement best describes how standardised maturities affect this hedge relative to a customised OTC derivative?

A. The futures hedge cannot be put in place because exchange-traded contracts cannot be used for hedging.

B. The futures hedge introduces basis risk because the contract maturity does not match the exposure date, a mismatch a customised OTC derivative could avoid.

C. The futures hedge eliminates all risk because standardised maturities guarantee a perfect offset on any date.

D. The futures hedge has higher counterparty risk than an OTC derivative because standardised maturities require a bilateral agreement.

Answer: B

Step by step

1. Exchange-traded futures allow only standard maturities (here quarterly months).
2. The exposure ends on 15 May, which falls between the March and June delivery months, so the nearest futures maturity does not coincide with the exposure date.
3. This maturity mismatch means the futures price and the hedged exposure need not move together: basis risk.
4. A customised OTC derivative can be set to mature exactly on 15 May, avoiding this basis risk (at the cost of less liquidity and bilateral counterparty risk).

Why B is correct: Standardised maturities restrict the exchange-traded hedger to dates that may not match the exposure, producing basis risk that a customised OTC contract can avoid.

Why the others are wrong

- **A:** Wrong: exchange-traded futures are widely used for hedging; they are simply less tailored.
- **C:** Ignores the mismatch: standard maturities do not guarantee an offset on any date, and basis risk remains.
- **D:** Confuses features: standardised exchange contracts are not bilateral agreements; counterparty risk is a feature of bilateral OTC trades, not a result of standard maturities.

Q611

AnalystPrep Q.1866

In today's economy, many derivatives are exchange-traded. Which statement correctly describes a benefit of trading over an exchange?

A. An exchange encourages market efficiency and boosts liquidity through centralised trading in a single place.

B. An exchange encourages investors to make more profits by taking advantage of arbitrage opportunities.

C. An exchange is less congested compared to over-the-counter markets.

D. An exchange encourages the clearing party to take a commission on each trade.

Answer: A

Step by step

1. An exchange centralises trading of standardised contracts in a single place.
2. This centralisation promotes market efficiency (transparent, accessible prices) and enhances liquidity, making it easy to enter and exit positions.
3. Check B-D: none is a benefit identified for exchanges; A matches the stated benefit.

Why A is correct: Centralised trading of standardised contracts gives transparent prices and ease of entering and exiting positions, which is market efficiency and liquidity.

Why the others are wrong

- **B:** An exchange does not exist to create arbitrage profits for investors; efficient pricing, if anything, reduces arbitrage.
- **C:** Congestion relative to OTC markets is not an exchange benefit; centralised trading concentrates, not reduces, activity.
- **D:** Commissions to the clearing party are not a benefit to the market participant; the clearing function exists to manage counterparty risk.

LO 32.c

Describe the process of clearing a derivative transaction.

Q612

AnalystPrep Q.6122

Regarding the involvement of clearing houses in the trading of derivatives, which of the following statements accurately reflects their usage in futures markets and over-the-counter (OTC) markets?

A. Never used in futures markets and sometimes used in OTC markets
B. Used in OTC markets, but not in futures markets
C. Always used in futures markets and sometimes used in OTC markets
D. Always used in both futures markets and OTC markets

Answer: C

Step by step

1. All derivatives exchanges have an associated CCP that performs clearing, so exchange-traded futures are always centrally cleared.
2. OTC derivatives have historically been cleared bilaterally; central clearing is increasingly applied, but it requires standardization, so not all OTC products can be centrally cleared.
3. Hence: always for futures, sometimes for OTC.

Why C is correct: Exchanges always have a CCP, whereas only sufficiently standardized OTC products are centrally cleared and a significant portion remains bilateral.

Why the others are wrong

- **A:** Reverses the facts for futures: every exchange has a clearing function.
- **B:** Reverses the usage: central clearing is universal for futures, not absent.
- **D:** Overstates OTC usage; non-standardized OTC trades cannot be centrally cleared and remain bilaterally cleared.

Q613

AnalystPrep Q.6124

What is the primary distinction between bilateral and central clearing in the context of clearing OTC derivative transactions?

A. Bilateral clearing involves a third party taking responsibility for managing counterparty risk, while central clearing is conducted between the two parties entering a trade.
B. Bilateral clearing facilitates the computation of payment obligations without the necessity for netting, while central clearing mandates netting.
C. Bilateral clearing requires the counterparties to handle risk management processes themselves, while central clearing involves a central counterparty (CCP) taking over these functions.
D. Bilateral clearing results in reduced operational costs for the counterparties compared to central clearing due to the absence of a third-party involvement.

Answer: C

Step by step

1. In bilateral clearing the two parties to the trade manage the clearing processes and risks themselves.
2. In central clearing the CCP becomes the counterparty to each side and performs risk management (e.g., collateralization).
3. Only C states this distinction correctly.

Why C is correct: The distinction is who performs risk management during clearing: the two counterparties (bilateral) versus the CCP (central).

Why the others are wrong

- **A:** Swaps the roles: the third party exists in central, not bilateral, clearing.
- **B:** Netting is not what separates the two; bilateral netting agreements exist and central clearing is not defined by mandatory netting.
- **D:** The cost claim is unsupported by the reading; the defining difference is risk-management responsibility.

Q614

AnalystPrep Q.6126

REBUILT — Original answer was correct only by elimination (bilateral netting also exists); reworded so the CCP-specific function is unambiguous.

Which of the following is a function performed by the central counterparty (CCP) in the central clearing of an OTC derivative, in contrast to bilateral clearing?

A. It takes over the management of counterparty risk and associated components such as collateralization.

B. It customizes the contract terms to meet the hedging needs of each client.

C. It executes the trade on an exchange to provide price discovery.

D. It negotiates the contract terms between buyer and seller.

Answer: A

Step by step

1. Clearing is the period between execution and settlement in which counterparty risk is managed.
2. In bilateral clearing the two parties manage this themselves; in central clearing a third party (the CCP) takes responsibility for counterparty risk and components such as collateralization.
3. Customization, negotiation and execution all occur before clearing and are not CCP functions.

Why A is correct: The CCP replaces the bilateral relationship during clearing and performs the risk management and collateral functions.

Why the others are wrong

- **B:** Customization is done by the counterparties/dealer when the trade is structured, before clearing.
- **C:** Execution and price discovery are exchange (trading) functions, not clearing.
- **D:** Negotiation of terms is bilateral and precedes clearing.

Q615

AnalystPrep Q.6127

REBUILT — Option D was truncated in the source and option C mixed correct and unsupported claims; rewrote options so exactly one is clearly correct.

A risk manager compares the clearing of exchange-traded and OTC derivatives. Which statement is accurate?

A. Customized OTC derivatives are routinely centrally cleared because their negotiable terms make the CCP process simpler.

B. Exchange-traded derivatives are predominantly bilaterally cleared because their standardized terms allow direct agreement between counterparties.

C. Exchange-traded derivatives are centrally cleared, while OTC derivatives have historically been bilaterally cleared and only sufficiently standardized ones can be centrally cleared.

D. Exchange-traded derivatives require individually tailored clearing arrangements for each pair of counterparties.

Answer: C

Step by step

1. All derivatives exchanges have an associated CCP, so exchange-traded contracts are centrally cleared.
2. OTC derivatives have historically been bilaterally cleared; central clearing is growing but requires standardization, so not all can be centrally cleared.
3. Only C is consistent with this.

Why C is correct: Central clearing is a natural extension of exchange trading and needs standardization, so customized OTC products stay bilateral.

Why the others are wrong

- **A:** Reverses the logic: customization prevents central clearing, which requires standardization.
- **B:** Exchange-traded contracts are centrally cleared, not bilateral.
- **D:** Exchange clearing is uniform through the CCP; there are no tailored bilateral arrangements.

Q616 AnalystPrep Q.6129

A derivatives specialist compares the settlement and clearing periods of exchange-traded and OTC derivatives. Which of the following statements is correct?

A. OTC derivatives typically settle immediately after trade execution, increasing the need for efficient clearing processes.

B. The clearing process for OTC derivatives usually spans a longer time horizon, often years or even decades, due to their long maturities.

C. Exchange-traded derivatives have increasingly longer settlement periods, leading to a demand for more complex clearing procedures.

D. OTC derivatives tend to have shorter maturities than exchange-traded products, thus their clearing has become less significant.

Answer: B

Step by step

1. Clearing is the period between execution and settlement.
2. Exchange-traded products settle in days or at most months; OTC derivatives are generally long-dated, so clearing lasts years and often decades.
3. This long clearing period is why OTC clearing is growing in importance.

Why B is correct: Clearing lasts until settlement, and OTC derivatives are long-dated, so their clearing horizon is years to decades.

Why the others are wrong

- **A:** OTC derivatives are not settled for a long time; they do not settle immediately.
- **C:** Exchange-traded settlement is short (days to months), not lengthening.
- **D:** OTC maturities are longer, and clearing is becoming more important, not less.

Q617 AnalystPrep Q.6130

Which of the following is a crucial activity conducted during the clearing of a derivative transaction?

A. Adjustment of the derivative contract's terms to reflect market conditions at the time of settlement.

B. Complete transfer of the underlying securities or cash between the parties involved in the transaction.

C. Calculation and often netting of payment obligations between the parties to streamline future settlements.

D. Initiation of a new derivative transaction as a hedge against the one that has just been executed.

Answer: C

Step by step

1. Clearing sits between execution and settlement; payment obligations are computed (and often netted) and the transaction is managed (margining, cash-flow payments) before settlement.
2. Netting of obligations is a typical clearing activity.
3. Transfer of cash/securities is settlement; contract terms are not adjusted; hedging is a trading decision.

Why C is correct: Clearing is the process by which payment obligations between firms are computed and often netted before settlement.

Why the others are wrong

- **A:** Contract terms are fixed at execution and are not adjusted during clearing.
- **B:** Transfer of cash or securities is settlement, the stage after clearing.
- **D:** Hedging is a trading decision unrelated to the clearing process.

Q618 BT answers P2.T6.24.24.3 · BT notes P2.T6.24.24.2

REBUILT — Original swap was ambiguous (standard interest rate swaps can be centrally cleared) and option D is not a valid process; specified a customized swap and tightened options.

Lay Corporation enters a customized, non-standardized 5-year OTC interest rate swap with a bank (notional $100 million). The swap's terms are tailored to Lay's cash flows, so it is not eligible for a CCP. Which statement best describes how this transaction is cleared?

A. It will be centrally cleared, with a clearinghouse becoming the counterparty to both Lay and the bank.

B. It will be bilaterally cleared: Lay and the bank themselves handle the risk management processes, such as collateral arrangements, and bear the counterparty risk.

C. Clearing does not apply to OTC derivatives because they are privately negotiated.

D. A regulator will collect margin from both parties for the life of the swap and manage the counterparty risk.

Answer: B

Step by step

1. Clearing is the period between execution and settlement, and it applies to every derivative, OTC included.
2. Central clearing requires standardization; a customized swap cannot go to a CCP, so it is bilaterally cleared.
3. In bilateral clearing the two parties take responsibility for the processes and risks (collateral, counterparty risk monitoring).

Why B is correct: A tailored swap cannot be centrally cleared, so the two counterparties manage the clearing process and risk themselves.

Why the others are wrong

- **A:** Central clearing needs standardization; this swap is customized.
- **C:** Clearing applies to all derivatives; for OTC it is simply bilateral, which is why long clearing periods matter.
- **D:** Regulators do not collect and manage margin; in bilateral clearing the parties do, in central clearing the CCP does.

Q619 Schweser Module Quiz 32.1 Q1

The process under which margin transactions are facilitated and computed is best referred to as:

A. clearing.
B. settlement.
C. execution.
D. collateralization.

Answer: A

Step by step

1. Clearing is the period between execution and settlement in which the transaction is managed (margining, cash-flow payments) and payment obligations are computed.
2. Margin computation therefore falls under clearing.

Why A is correct: Clearing is the stage in which margining is carried out and payment obligations are computed.

Why the others are wrong

- **B:** Settlement is the payment and transfer of money when the contract is closed out.
- **C:** Execution is entering into the contract, before clearing.
- **D:** Collateralization is posting assets as security, a risk-mitigation tool, not the process that computes margin.

LO 32.d

Identify the participants and describe the use of collateralization in the derivatives market.

Q620 AnalystPrep Q.6131

Which participant in the derivatives market is most likely to operate across various asset classes and significantly influence market liquidity?

A. A regional bank focused on making markets in certain products relevant to its operation.
B. A government entity using derivatives to hedge against risks associated with its financial strategies.
C. A large global bank, often known as a dealer, active across all major asset classes.
D. A small financial institution using derivatives primarily to manage risks inherent to its business model.

Answer: C

Step by step

1. Classify participants by portfolio size: large players, medium-sized players, end users.
2. Large players are global dealer banks trading FX, interest rate, equity, commodity and credit derivatives.
3. They are members of most exchanges and CCPs and provide the bulk of market liquidity.
4. Only choice C describes this group.

Why C is correct: Large global dealer banks trade across all asset classes and supply most OTC market liquidity.

Why the others are wrong

- **A:** A regional bank is a medium-sized player: it covers several asset classes but not all (typically no credit or exotic products) and is not a main liquidity provider.
- **B:** A government is an end user with a specific hedging need, not a liquidity provider.
- **D:** A small institution using derivatives to manage its own risks is an end user with a narrow, single-purpose use.

Q621 AnalystPrep Q.6133

Which type of OTC derivatives end user is especially known for receiving but not posting collateral in transactions with banks, subject to possible rating triggers?

A. High-frequency trading firms specializing in algorithmic derivatives trading.
B. Corporate clients engaged in large-scale hedging of operational risks.
C. Sovereigns, supranational entities, or multilateral development banks with high credit ratings.
D. Investment funds using derivatives for portfolio diversification and risk mitigation.

Answer: C

Step by step

1. Identify end users with exceptionally high credit quality: sovereigns, supranationals, multilateral development banks.
2. These have historically dictated terms under which the bank posts collateral but they do not.
3. Rating triggers may require them to post if they are downgraded.
4. Such arrangements are now seen as costly for banks because of funding and capital requirements.

Why C is correct: Exceptional credit quality lets sovereigns, supranationals and MDBs receive collateral without posting, subject to downgrade triggers.

Why the others are wrong

- **A:** Trading firms are not high-credit-quality end users and have no such special arrangement.
- **B:** Corporates are typically unable or unwilling to post collateral, but they do not get one-way terms from banks; they simply remain uncollateralised.
- **D:** Investment funds are ordinary end users without the exceptional credit quality that earns one-way collateral terms.

Q622 AnalystPrep Q.6135 · AnalystPrep Q.6139

A group of OTC derivative users, such as corporations or other end users, either cannot commit to collateralization or are not required to do so by their counterparties, so they do not post collateral in their bilateral trades. Which category of derivatives does this practice best describe?

A. Bilateral OTC collateralized derivatives.
B. Exchange-traded derivatives.
C. Bilateral OTC uncollateralized derivatives.
D. OTC centrally cleared derivatives.

Answer: C

Step by step

1. Four categories: exchange-traded, OTC centrally cleared, OTC collateralized, OTC uncollateralized.
2. Exchange-traded and centrally cleared trades always involve daily margin.
3. Bilateral collateralized trades have both sides posting collateral.
4. Bilateral trades where a party does not post collateral are OTC uncollateralized, the riskiest category.

Why C is correct: Bilateral OTC trades without collateral posting are, by definition, OTC uncollateralized derivatives.

Why the others are wrong

- **A:** In collateralized bilateral trades the parties do post cash or securities.
- **B:** Exchange-traded derivatives are settled daily with margin, so collateral is always posted.
- **D:** Central clearing requires daily variation margin, so posting is mandatory.

Q623 AnalystPrep Q.6136

What collateral requirement is typically associated with exchange-traded derivatives?

A. No collateral is posted because these instruments are considered self-hedging.
B. Collateral is posted only at contract maturity to cover the final settlement amount.
C. Margin must be posted against losses on a daily basis by all exchange members.
D. Collateral in the form of physical assets is required at the initiation of the contract.

Answer: C

Step by step

1. Exchanges have central clearing functions.
2. Margin must be posted against losses daily (variation margin), settled in cash.
3. This applies to all members and underpins the clearing guarantee of the exchange.

Why C is correct: Exchange-traded derivatives are effectively settled daily through margin posted by all members.

Why the others are wrong

- **A:** No such 'self-hedging' exemption exists; counterparty risk is removed by margining.
- **B:** Margin is exchanged daily, not only at maturity.
- **D:** Margin is mainly in cash and is posted against daily losses; physical assets are not the norm.

Q624 AnalystPrep Q.6137 · AnalystPrep Q.6134

What is the collateral requirement specifically associated with OTC centrally cleared derivatives?

A. Parties post securities as collateral on an annual basis.
B. Collateral in the form of variation margin is posted in cash on a daily basis by participants.
C. Collateral is not typically required because the trades are bespoke.
D. A single upfront collateral payment is made at initiation and adjusted at maturity.

Answer: B

Step by step

1. OTC centrally cleared trades are complex, illiquid trades not suited to exchanges.
2. They are nonetheless centrally cleared by a CCP.
3. OTC CCPs usually require daily collateralization in cash, called variation margin (as for exchange-traded contracts).

Why B is correct: A CCP collateralizes cleared OTC trades daily with cash variation margin.

Why the others are wrong

- **A:** Annual posting would leave exposures unmargined for far too long.
- **C:** Bespoke trades are the bilateral uncollateralized case; cleared trades are always collateralized.
- **D:** One upfront payment adjusted only at maturity ignores daily mark-to-market margining.

Q625 AnalystPrep Q.2694

REBUILT — Original relied on a ranking table not in the readings; reworded to ask which counterparty regularly posts collateral.

Which of the following OTC counterparties is most likely to post collateral regularly to its trading counterparties?

A. Private equity funds
B. Sovereigns
C. Corporates
D. Dealer banks

Answer: D

Key correction: Source key (D) is correct; stem reworded only.

Step by step

1. Large players (dealer banks) collateralize positions as long as the counterparty makes the same commitment.
2. Highly rated sovereigns often receive collateral without posting.
3. Corporates and many funds cannot or will not commit to margining, or post illiquid collateral infrequently.
4. Dealer banks therefore post the most collateral.

Why D is correct: Dealers are active collateral posters, whereas end users typically avoid or limit posting.

Why the others are wrong

- **A:** Private equity funds are end-user type counterparties that are often unable or unwilling to commit to posting

collateral.
- **B:** Sovereigns have historically received but not posted collateral, subject to rating triggers.
- **C:** Corporates are typically unable or unwilling to post, or post illiquid collateral infrequently.

Q626

BT answers P2.T6.24.25.1

REBUILT — Keyed answer C was not supportable (OTC swap collateral is bilateral and not necessarily initial plus variation margin); rewrote options so exactly one is correct.

Harder Manufacturing, an end user with a single commodity hedging need, can hedge with (1) an exchange-traded commodity futures contract or (2) a bilateral OTC commodity swap with a major bank. Which statement about collateralization is most accurate?

A. The futures contract requires daily margin posted against losses via the exchange's central clearing, whereas the OTC swap's collateral terms are negotiated bilaterally and may be limited or absent for an end user.

B. Neither the futures contract nor the swap needs collateral because Harder is hedging rather than speculating.

C. The futures contract requires collateral only at initiation, and the swap requires daily collateral.

D. Both the futures contract and the swap must be margined daily in cash on identical terms.

Answer: A

Key correction: Source key (C) said both require collateral with standardized terms for futures; collateral on a bilateral OTC swap is not guaranteed for an end user, so the item was rebuilt and the correct option is A.

Step by step

1. Exchange-traded futures are centrally cleared: margin is posted against losses daily.
2. A bilateral OTC swap is not centrally cleared; collateral depends on the agreement between the parties.
3. End users often cannot or will not post collateral, or post illiquid collateral less frequently, so the swap may be uncollateralized.
4. Only A states both facts.

Why A is correct: Futures are margined daily by the clearing function; bilateral OTC collateral is negotiated and often absent for end users.

Why the others are wrong

- **B:** Purpose (hedging) does not remove margin requirements on exchange-traded contracts.
- **C:** Reverses the pattern: futures are margined daily, not only at initiation, and bilateral swaps are not automatically margined daily.
- **D:** Identical daily cash terms apply only to cleared products; a bilateral swap with a bank has negotiated terms.

Q627

AnalystPrep Q.6132

REBUILT — Original tested G-SIB regulation, outside the LO; rebuilt to test the large-player/dealer share of OTC notional.

Large global dealer banks are the 'large players' in the derivatives market. Approximately what proportion of total OTC derivatives notional do they control?

A. 9%

B. 80%

C. 60%

D. 40%

Answer: B

Step by step

1. Large players are global banks acting as OTC dealers.
2. They are members of most exchanges and CCPs and provide most of the market liquidity.
3. The OTC market is highly concentrated: the largest dealers hold around four-fifths (80%) of notional.

Why B is correct: The OTC market is concentrated in a small number of dealers holding about 80% of notional.

Why the others are wrong

- **A:** 9% is the exchange-traded share of the total derivatives market.
- **C:** 60% is the centrally cleared share in the market breakdown.
- **D:** 40% is the bilateral share in the market breakdown.

LO 32.e

Define the International Swaps and Derivatives Association (ISDA) Master Agreement, the risk-mitigating features it provides, and the default events it covers.

Q628

AnalystPrep Q.6145

Which of the following statements accurately represents a default provision explicitly stated in the ISDA Master Agreement?

A. It initiates an automatic restructuring of the terms for distressed derivatives.
B. It enumerates specific events, such as bankruptcy, that constitute a default.
C. It provides a grace period for all defaults before any enforcement action.
D. It ensures equal profit distribution among counterparties in case of default.

Answer: B

Step by step

1. The ISDA Master Agreement defines events of default: failure to pay or deliver, breach of agreement, credit support default, misrepresentation, default under a specified transaction, cross-default, bankruptcy and merger without assumption.
2. A default terminates the agreement and starts the close-out process; nothing in it restructures trades or shares profits.
3. Only option B describes an enumerated list of default events.

Why B is correct: The Agreement lists specific events of default (e.g., bankruptcy, failure to pay) so that default is defined contractually, not left to legal uncertainty.

Why the others are wrong

- **A:** Default leads to termination and close-out, not automatic restructuring of the trades.
- **C:** Grace periods apply only to some events (e.g., certain payment failures), not to all defaults; 'all' makes it false.
- **D:** The Agreement calculates a single net close-out amount; it does not distribute profits equally.

Q629

AnalystPrep Q.6147

Which of the following is a default event as explicitly defined in the ISDA Master Agreement?

A. A significant fluctuation in the market value of the underlying asset.
B. The initiation of a merger or acquisition process by either party.
C. A failure to meet payment or delivery obligations by a counterparty.
D. The introduction of new regulatory requirements affecting the derivatives.

Answer: C

Step by step

1. Recall the ISDA events of default: failure to pay or deliver, breach of agreement, credit support default, misrepresentation, specified-transaction default, cross-default, bankruptcy, merger without assumption.
2. Only C matches an item on the list (and it is among the most common).

Why C is correct: Failure to pay or deliver is the first and most common listed ISDA event of default.

Why the others are wrong

- **A:** Market moves change exposure and are handled by collateral/margin, not by a default event.
- **B:** A merger alone is not a default; only a merger without assumption of obligations is.
- **D:** New regulation is not a default event; it is not a failure of the counterparty.

Q630

AnalystPrep Q.6143

Which of the following correctly describes a risk-mitigating feature explicitly outlined in the ISDA Master Agreement?

A. A requirement for the daily mark-to-market of positions.
B. Consolidation of all referenced transactions into a single net obligation.
C. Prescribed standardized interest rates for all derivative contracts.
D. The use of a specific third-party valuation service for all transactions.

Answer: B

Step by step

1. The ISDA Master Agreement's risk mitigants are: collateral terms (via the CSA), events of default and termination, netting of obligations, and the close-out process.
2. Netting combines all transactions under the agreement into one legal contract and one net amount on default, which matches B.
3. The other options are not features of the Agreement.

Why B is correct: Multiple transactions under one Master Agreement form a single legal contract, so on default they are netted into one obligation, reducing exposure.

Why the others are wrong

- **A:** Valuation frequency is a collateral (CSA) matter negotiated bilaterally, not a requirement of the Master Agreement itself.
- **C:** Commercial terms such as rates are set in each trade confirmation, not standardized by the Agreement.
- **D:** The Agreement does not mandate a particular valuation provider.

Q631 AnalystPrep Q.6144

Which of the following correctly reflects a provision explicitly included in the ISDA Master Agreement?

A. It mandates a uniform haircut percentage for all types of collateral.
B. It specifies mechanics around the close-out process in the event of default.
C. It automatically adjusts the margin requirements based on market volatility.
D. It requires all derivative transactions to be settled in a single, predetermined currency.

Answer: B

Step by step

1. Listed features: collateral terms, events of default/termination, netting, close-out process.
2. B corresponds to the close-out process, which defines how positions are valued and netted after a default.
3. Haircuts, margin levels and currencies are bilaterally negotiated terms, not fixed by the Agreement.

Why B is correct: The Agreement defines the close-out process, removing legal uncertainty about how transactions are valued and settled upon default.

Why the others are wrong

- **A:** Haircuts are negotiated in the credit support annex, and differ by collateral type; none is mandated uniformly.
- **C:** Margin terms are set contractually in the CSA; they are not automatically volatility-adjusted by the Agreement.
- **D:** Settlement currency is specified per transaction in the confirmation, not forced to one currency.

Q632 AnalystPrep Q.6148

Which of the following scenarios is considered a default event as outlined in the ISDA Master Agreement?

A. A transient delay in the settlement process due to technical issues.
B. The counterparty's credit rating downgrade below a specified threshold.
C. The breach of a covenant related to a specified transaction within the agreement.
D. A change in market interest rates impacting the valuation of the derivatives.

Answer: C

Step by step

1. ISDA events of default include breach of agreement and default under a specified transaction.
2. C is a breach of an agreement covenant, so it is a listed default event.
3. A, B and D are not on the list.

Why C is correct: Breach of agreement (including default under a specified transaction) is an enumerated ISDA event of default.

Why the others are wrong

- **A:** A brief operational delay is not a default event; failure to pay is an event of default only if the failure persists beyond the allowed grace period.
- **B:** A rating downgrade is not a listed event of default (it may trigger collateral terms in the CSA).
- **D:** Interest rate changes are market risk that alter exposure, not a default event.

LO 32.f

Describe the features and use of credit derivatives and discuss potential risks they may create.

Q633 AnalystPrep Q.6152

In response to evolving market dynamics and regulatory pressures post-GFC, a financial advisory firm is considering integrating credit derivatives into its risk management portfolio. The firm acknowledges the utility of instruments like Credit Default Swaps (CDS) in mitigating credit risk but remains cautious due to the complexities and potential risks associated with these derivatives. Which statement accurately captures a key feature and an inherent risk of credit derivatives, particularly CDS?

A. Credit derivatives, such as CDS, function exclusively to enhance investment returns, with negligible influence on the firm's credit risk profile.

B. Credit derivatives, while instrumental in transferring credit risk, inherently carry counterparty risk within the contractual framework.

C. Credit derivatives, by design, distribute credit risk so evenly across parties that the possibility of default becomes negligible.

D. Credit derivatives are generally immune to market fluctuations, ensuring stable valuation and risk assessment.

Answer: B

Step by step

1. A CDS transfers the credit risk of a reference entity from the protection buyer to the protection seller.
2. The buyer's exposure is replaced by exposure to the protection seller, who may itself default or fail to pay when the reference entity defaults.
3. So the derivative both transfers credit risk and creates counterparty risk (and, if sellers are concentrated, wrong-way and systemic risk).
4. Only option B states both the feature and the inherent risk.

Why B is correct: Credit derivatives transfer credit risk, but the protection buyer then bears the risk that the protection seller fails to perform, so counterparty risk is embedded in the contract.

Why the others are wrong

- **A:** Credit derivatives are mainly used to hedge or transfer credit risk, so they materially change the credit risk profile; claiming they only enhance returns confuses a secondary speculative use with their main purpose.
- **C:** Risk is transferred, not eliminated; default risk remains with the protection seller and the counterparty, so it does not become negligible.
- **D:** CDS values move with credit spreads and market conditions (mark-to-market, collateral calls), so they are not immune to market fluctuations.

Q634 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A risk manager at an asset manager holds a large bond position in Bank Z and buys credit default swap (CDS) protection on Bank Z from Dealer Y. Bank Z and Dealer Y operate in the same markets and their credit quality is highly correlated. The asset manager is considering whether the CDS is an effective hedge. Which of the following is the most accurate assessment?

A. The CDS fully removes credit risk from the position because the credit risk of Bank Z is transferred to Dealer Y.

B. The CDS transfers the credit risk of Bank Z but introduces counterparty risk to Dealer Y, which is most likely to be severe exactly when Bank Z defaults and protection is needed.

C. The CDS is an effective hedge because the correlation between Bank Z and Dealer Y reduces the probability that both fail together.

D. The CDS creates counterparty risk only if Dealer Y is not a bank, because bank counterparties are default remote.

Answer: B

Step by step

1. A CDS transfers the reference entity's credit risk to the protection seller but embeds counterparty risk in the contract itself.
2. The seller's ability to pay is most in doubt when the reference entity defaults; high correlation between Bank Z and Dealer Y makes this wrong-way risk acute.

3. Hence the hedge is only as good as Dealer Y's credit quality in the state where Bank Z defaults.

Why B is correct: CDSs can hedge credit risk but carry counterparty risk within the contract; with highly correlated names the protection seller is likely to be weak when payout is due.

Why the others are wrong

- **A:** Ignores the counterparty risk embedded in the CDS contract; the risk is transferred, not eliminated.
- **C:** Reverses the effect of correlation: positive correlation raises, not lowers, the joint failure probability.
- **D:** Dealers were not default remote (the pre-2008 belief proved wrong); counterparty risk applies to any protection seller.

Q635 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A regulator is reviewing the credit derivatives market and notes that it grew quickly in the decade before the global financial crisis (GFC) but has stalled since. A member of the review team is asked to summarize the features and risks of credit derivatives. Which of the following statements is correct?

A. Because the credit default swap is a complex instrument, its use has been confined to a few specialist dealers and it has had little effect on the trading of credit risk.

B. Credit derivatives can only be used to hedge credit risk and cannot be used to take on credit exposure, so they cannot contribute to market instability.

C. Credit derivatives are an efficient means of transferring credit risk, but if not used correctly they can be counterproductive, and the CDS contract itself contains counterparty risk.

D. Central clearing removes the risk of CDSs because CDSs are more liquid and less risky than other cleared products.

Answer: C

Step by step

1. The CDS is a simple core instrument that transformed credit risk trading and grew swiftly before the GFC.
2. It transfers credit risk efficiently but embeds counterparty risk; misuse (speculation, leverage) can be toxic.
3. CDSs are more illiquid and riskier than other cleared products, so a CCP's ability to handle them is a concern.

Why C is correct: GARP states credit derivatives are efficient at transferring credit risk but can be counterproductive and highly toxic if misused, and CDS contracts embed counterparty risk.

Why the others are wrong

- **A:** The CDS is described as simple and as having transformed credit risk trading; its use was widespread.
- **B:** Derivatives can also create synthetic credit exposure and leverage, and have catalysed market disturbances.
- **D:** CDSs are much more illiquid and risky than other cleared products, so CCP capacity to handle them was untested.

LO 32.g

Describe central clearing of OTC derivatives and discuss the roles, mandate, advantages, and disadvantages of the central counterparty (CCP).

Q636 AnalystPrep Q.6164

Which of the following statements best describes an advantage of central counterparties (CCPs) in the context of OTC derivatives?

A. CCPs enhance market liquidity by acting as market makers and setting prices for OTC derivatives.

B. CCPs contribute to financial stability by providing a structured, regulated environment for managing counterparty risk.

C. CCPs eliminate market risk in OTC derivatives by standardizing the underlying assets.

D. CCPs simplify the legal landscape of OTC derivatives by unifying international regulatory standards across jurisdictions.

Answer: B

Step by step

1. A CCP interposes itself as buyer to every seller and seller to every buyer, becoming the new counterparty to each clearing member.

2. It manages the resulting risk through multilateral netting, initial and variation margin, a default fund and orderly close-out.
3. These mechanisms reduce counterparty risk and mutualize losses, which supports systemic stability: option B.
4. Options A, C and D each attribute to the CCP a function it does not perform.

Why B is correct: A CCP replaces bilateral credit exposures with a margined, netted, loss-mutualizing structure under common rules, which lowers counterparty and systemic risk.

Why the others are wrong

- **A:** A CCP is not a market maker and does not set prices; this confuses clearing with trading/dealing.
- **C:** Standardization is needed for clearing, but market risk stays with the position holders; the CCP addresses counterparty (credit) risk only.
- **D:** CCPs often operate cross-border and face conflicting legal frameworks; they do not unify regulation (the reading lists this as a shortfall).

Q637

AnalystPrep Q.6165

REBUILT — Original keyed 'liquidity in stress', which is not a documented CCP disadvantage in the reading, while the cross-border legal option is a stated shortfall (two defensible answers); rebuilt around the documented bondholder-priority disadvantage.

Which of the following is a recognized disadvantage of central clearing of OTC derivatives through a CCP?

A. Because the CCP replaces the original counterparties, each clearing member's bilateral exposure to its original counterparty is left unchanged.

B. Prioritizing the CCP's derivatives counterparties can come at the expense of other creditors, such as bondholders.

C. Multilateral netting of offsetting positions increases the net exposure that must be collateralized.

D. Loss mutualization concentrates all default losses on the defaulting member's original counterparty.

Answer: B

Key correction: Original AnalystPrep key (B, liquidity in stress) was not supported by the reading and option C (cross-border legal complexity) was also defensible; the rebuilt item has a single supported answer.

Step by step

1. List the shortfalls of CCPs: (i) the CCP must be of very strong credit quality because its failure could cause a systemic shock; (ii) reducing risk for derivatives counterparties may come at the expense of other market participants such as bondholders; (iii) cross-border activity exposes the CCP to conflicting legal frameworks.
2. Option B restates shortfall (ii).
3. Options A, C and D contradict the mechanics of central clearing (novation to the CCP, netting, loss mutualization).

Why B is correct: Central clearing gives derivatives counterparties priority protection through collateral, which shifts risk onto non-cleared creditors such as bondholders.

Why the others are wrong

- **A:** Novation replaces the original counterparty with the CCP, so the bilateral exposure to the original counterparty is removed, not left unchanged.
- **C:** Multilateral netting reduces net exposures and therefore the collateral required.
- **D:** Loss mutualization spreads losses across all clearing members rather than concentrating them on one counterparty.

Q638

AnalystPrep Q.6175

Which entity is best described as follows? It is a third party that takes responsibility for managing counterparty risk and collateralization in derivatives markets, emerged as a natural extension of exchange trading for OTC derivatives, requires standardization of the contracts it clears, and offers netting benefits and fungibility. Its role in managing major defaults during the Global Financial Crisis drove its adoption.

A. Central counterparty (CCP)

B. Credit derivatives product company (CDPC)

C. Derivatives product company (DPC)

D. Special purpose vehicle (SPV)

Answer: A

Step by step

1. Identify the key features: third-party risk manager, collateralization, standardized contracts, netting, fungibility.
2. These are the defining features of a CCP, which stands between buyer and seller and uses margining and loss mutualization.
3. CDPCs, DPCs and SPVs are separate vehicles used to isolate or credit-enhance risk, not clearing houses.

Why A is correct: Central clearing extends the exchange model to OTC trades and needs standardized contracts, margin and netting; that is a CCP.

Why the others are wrong

- **B:** A CDPC is a credit-protection seller (monoline-like); it does not clear trades or net across members.
- **C:** A DPC is a AAA-rated, bankruptcy-remote dealer subsidiary; it is not a clearing house with margining and mutualization.
- **D:** An SPV isolates assets/risks from a parent; it does not interpose between all buyers and sellers.

Q639

BT answers P2.T6.24.26.1 · BT notes P2.T6.24.26.1

Global Hedge Fund holds a large portfolio of OTC derivatives, including credit default swaps. Regulators now mandate central clearing of standardized OTC contracts through central counterparties (CCPs). Which advantage of central clearing is most important to the fund given its large OTC exposures?

A. Reduced operational complexity from the standardization of contracts facilitated by CCPs.

B. Increased market transparency and pricing insights from the centralization of trade data.

C. Mitigation of counterparty credit risk and lower systemic risk exposure through CCP risk-management mechanisms.

D. Greater flexibility to structure and customize contracts, since CCPs accommodate non-standardized instruments.

Answer: C

Step by step

1. The CCP becomes the counterparty to each trade, so the fund's bilateral default risk is replaced by exposure to a margined, loss-mutualizing CCP.
2. For a large OTC/CDS holder, counterparty credit risk is the dominant concern (the GFC showed its importance), so mitigation of this risk is the key benefit.
3. Standardization and transparency are secondary side effects; customization is reduced, not increased.

Why C is correct: The primary purpose of central clearing is to reduce counterparty and systemic risk via netting, margining and loss mutualization.

Why the others are wrong

- **A:** Standardization is a requirement for clearing, not the principal benefit; it is secondary to credit-risk reduction.
- **B:** Transparency may be a by-product but is not the core purpose for a fund with large credit exposure.
- **D:** Central clearing reduces flexibility because contracts must be standardized; customization is a feature of bilateral markets.

Q640

Schweser Module Quiz 32.2 Q2

Which of the following is not an advantage of the central counterparty (CCP) in the centralized clearing process?

A. Loss mutualization.

B. Elimination of counterparty risk.

C. Improved operational efficiency.

D. Risk reduction through multilateral netting.

Answer: B

Step by step

1. Advantages of a CCP: multilateral netting reduces exposures; losses are mutualized across clearing members; the process is more efficient and orderly (e.g., auctioned close-out).
2. Counterparty risk is transferred to and concentrated in the CCP, not eliminated; a CCP failure could cause a systemic shock.
3. So B is not an advantage as stated.

Why B is correct: Central clearing significantly reduces counterparty risk relative to bilateral trading but cannot fully eliminate it.

Why the others are wrong

- **A:** Loss mutualization is a genuine CCP benefit (default losses spread over clearing members).
- **C:** Standardized, centralized processing and orderly close-out improve efficiency, so this is an advantage.
- **D:** Multilateral netting of offsetting positions is a core CCP advantage.

Q641 Schweser Module Quiz 32.2 Q1

Which of the following is not an improvement that centrally cleared markets offer relative to bilateral markets? Centrally cleared markets:

A. remain market neutral by netting trades.
B. offer more flexibility in contract selection.
C. formalize the default close-out process by mutualizing losses.
D. improve counterparty risk by replacing the original counterparty with a series of new counterparties.

Answer: B

Step by step

1. Check each statement against the CCP benefits: market neutrality via multilateral netting (A), formal close-out and loss mutualization (C), counterparty replaced by the CCP/other clearing members (D).
2. Bilateral markets allow any customized contract and negotiated collateral; central clearing requires standardized contracts and fixed, non-negotiable collateral rules.
3. Hence flexibility falls under central clearing, and B is not an improvement.

Why B is correct: Central clearing trades flexibility for standardization; contract and collateral choice are narrower than in bilateral markets.

Why the others are wrong

- **A:** A true benefit: the CCP nets buy-side against sell-side positions and stays market neutral.
- **C:** A true benefit: default close-out is formalized and losses are mutualized across members.
- **D:** A true benefit: the original counterparty is replaced by the CCP and, on default, by other clearing members.

LO 32.h

Explain the margin requirements for both centrally-cleared and non-centrally-cleared derivatives.

Q642 AnalystPrep Q.6167

Which of the following statements accurately reflects the distinction between margin requirements for centrally-cleared and non-centrally-cleared derivatives?

A. Both centrally-cleared and non-centrally-cleared derivatives require exactly the same initial and variation margin, as mandated by global regulatory standards.
B. Margin requirements for centrally-cleared derivatives are determined by market makers, while those for non-centrally-cleared derivatives are set by the individual trading parties.
C. Margin requirements for centrally-cleared derivatives are standardized and enforced by the CCP, whereas non-centrally-cleared derivatives involve bilateral agreements with potentially varied margin arrangements.
D. Non-centrally-cleared derivatives are subject to less stringent risk assessments than centrally-cleared derivatives, leading to more lenient margin requirements.

Answer: C

Step by step

1. Cleared trades: the CCP sets and enforces initial margin, variation margin and default fund contributions for all clearing members under its own rules.
2. Non-cleared trades: margin is agreed bilaterally between the two parties; regulators now require margining that loosely mimics central clearing, but terms are still set bilaterally and can differ by counterparty.
3. Compare each option with these facts: only C states both halves correctly.

Why C is correct: The CCP imposes one standardized margin regime on its members, while bilateral margining is agreed between the two counterparties (within regulatory minimums) and so varies.

Why the others are wrong

- **A:** Overstates regulatory convergence: bilateral rules loosely mimic, but do not equal, CCP margining, and amounts and terms differ.

- **B:** Market makers do not set margin; the CCP does for cleared trades, so the roles are wrongly attributed.
- **D:** Regulators have pushed non-cleared margining to mimic central clearing, so non-cleared trades are not subject to more lenient margin requirements.

Q643 BT answers P2.T6.24.26.2 · BT notes P2.T6.24.26.1

REBUILT — Original keyed A, which treats OTC margin as purely negotiated and ignores that post-crisis rules now mandate margin on bilateral trades, and option C was a different-LO CDPC claim; rewrote the options so exactly one statement matches the reading.

Crestview Bank trades both centrally cleared and over-the-counter (OTC) derivatives. Following the regulatory reforms after the 2007-2009 financial crisis, which of the following statements about margin on its trades is most accurate?

A. Clearing members of a CCP must post initial margin and variation margin and contribute to a default fund, and regulators have been introducing similar margin requirements for non-centrally-cleared bilateral trades.

B. Variation margin is mandatory only for centrally cleared derivatives, and remains entirely optional for non-centrally-cleared derivatives.

C. Initial margin is not required on centrally cleared trades because the CCP is the counterparty to every trade, but it is required on non-centrally-cleared trades.

D. Because bilateral OTC trades historically had little or no margin, regulators have chosen to keep non-centrally-cleared derivatives outside margin rules.

Answer: A

Key correction: Source key (BT answers) said A for the original wording, where A claimed OTC margins are simply negotiated bilaterally; that is outdated because regulators now mandate margin on non-cleared trades, so the option set was rewritten and the new A matches the reading.

Step by step

1. Cleared derivatives: initial margin is required on all trades, variation margin covers market-value changes, and members also contribute to a default fund to cover a clearing member default.
2. Non-cleared derivatives: historically no or only limited margin, but in recent years regulators globally (G20 2011, BCBS-IOSCO) have moved to require margin similar to that under central clearing.
3. A states both points; B, C and D each contradict one of them.

Why A is correct: Central clearing uses initial margin, variation margin and a default fund, and the post-crisis reforms extended similar margin requirements to bilateral trades.

Why the others are wrong

- **B:** Variation margin is required for cleared trades, but it is also now mandated for non-cleared trades, so it is not optional.
- **C:** Reverses the facts: initial margin is required on all cleared trades (it protects the CCP against a member default), and is also now required bilaterally.
- **D:** Ignores the post-crisis mandate: regulators moved to impose margin on bilateral trades precisely to reduce systemic risk and the incentive to avoid clearing.

Q644 BT answers P2.T7.20.6.2

In regard to changes to the regulation of OTC derivatives after the 2007-2009 financial crisis, which of the following BEST summarizes the rules for standardized transactions and for uncleared trades (the derivatives not covered by the rules for standardized transactions)?

A. The new regulations applied neither to standardized transactions nor to uncleared trades.

B. Standardized transactions had no new requirements, but uncleared trades between financial institutions must clear through a CCP.

C. Standardized transactions between financial institutions must clear through a CCP, but uncleared trades are exempt from margin requirements.

D. Standardized transactions between financial institutions must clear through a CCP, and uncleared trades between financial institutions require both initial and variation margin.

Answer: D

Step by step

1. 2009 G20 agreement: sufficiently standardized OTC derivatives must be centrally cleared (the clearing mandate).

2. November 2011 G20 (Cannes): added a mandate for margin on non-centrally-cleared derivatives, via BCBS-IOSCO standards, to give a strong incentive to clear.
3. So standardized trades go to a CCP and uncleared trades face margin (initial and variation); only D says both.

Why D is correct: Reform has two prongs: a clearing mandate for standardized trades and mandatory margin for those that remain bilateral.

Why the others are wrong

- **A:** Wrong: both categories were newly regulated.
- **B:** Reverses the logic: standardized trades are the ones that must be cleared, and uncleared trades are by definition not cleared (they face margin instead).
- **C:** Correct on clearing but wrong on margin: the 2011 G20 mandate specifically added margin for non-centrally-cleared derivatives.

Q645

BT answers P2.T7.20.6.3

REBUILT — Original mixed Hull-only items (SIMM, rehypothecation, ISDA close-out) outside the GARP/Schweser text for this LO; rebuilt as an 'except' question on CCP margin and default resources as described in the reading.

Each of the following statements about the margin and loss-absorbing resources of a central counterparty (CCP) is correct EXCEPT which one?

A. Variation margin is collected in cash so that it closely tracks market movements.

B. Initial margin is held to cover the worst-case liquidation or close-out costs above the variation margin.

C. A default fund is maintained to mutualise losses in the event of a severe clearing member default.

D. Like monolines and CDPCs, a CCP posts only limited variation margin and typically only in extreme situations such as a rating downgrade.

Answer: D

Step by step

1. A CCP's resources are cash variation margin, initial margin, a default fund, and a documented plan for the extreme case where these are exhausted.
2. A, B and C each restate one of these roles correctly.
3. D is false: it is monolines and CDPCs that posted only limited collateral, often only on a downgrade, or none at inception; CCPs require both initial and variation margin in all situations.

Why D is correct: CCPs, unlike monolines and CDPCs, impose margining (initial and variation) on members in all situations and run a matched book.

Why the others are wrong

- **A:** True statement, so not the exception: variation margin tracks market moves.
- **B:** True statement, so not the exception: initial margin covers close-out costs beyond variation margin.
- **C:** True statement, so not the exception: the default fund shares losses among members in a severe default.

LO 32.i

Define special purpose vehicles (SPVs), derivatives product companies (DPCs), monolines, and credit derivatives product companies (CDPCs) and describe the limitations of using them as risk mitigating methods.

Q646

AnalystPrep Q.6171

Within the OTC derivatives market, firms have used various structures to mitigate counterparty risk and manage credit exposure, including special purpose vehicles (SPVs), derivatives product companies (DPCs), monolines, and credit derivatives product companies (CDPCs). Which of the following statements is correct?

A. DPCs are legal entities created primarily for asset management and to finance large projects without exposing the entire firm to financial risk.

B. Monolines and CDPCs are primarily used to change bankruptcy rules so that clients receive their full investment if a counterparty becomes insolvent.

C. SPVs are designed to isolate a firm from financial risk, typically by transferring assets to the SPV for management or by using it to finance a large project without risking the entire firm.

D. CDPCs evolved to allow counterparties with less than triple-A credit quality to trade long-dated derivatives.

Answer: C

Step by step

1. An SPV (or SPE) is a legal entity, such as a company or limited partnership, created to isolate a firm from financial risk; assets are transferred to it for management or it finances a large project.
2. Match each option to the entity it really describes: A describes an SPV, B describes SPV bankruptcy-priority features, D describes the original purpose of DPCs.
3. Only C attributes the isolation/project-financing role to the correct entity (the SPV).

Why C is correct: The asset-transfer, project-financing and risk-isolation role is the defining feature of an SPV; each of the other structures has a different purpose.

Why the others are wrong

- **A:** Attributes the SPV's role to a DPC. A DPC is a separately capitalised, bankruptcy-remote, typically triple-A dealer subsidiary used as a high-quality derivatives counterparty.
- **B:** Changing bankruptcy priorities (e.g. flip provisions) is an SPV feature. Monolines and CDPCs are leveraged guarantors/protection sellers that do not post collateral in normal times.
- **D:** Facilitating long-dated trading for sub-AAA counterparties was the original purpose of DPCs. CDPCs extended the DPC idea with a monoline-like business model of selling credit protection.

Q647

AnalystPrep Q.6172

REBUILT — Original option B (DPCs facilitate long-dated trading for counterparties below triple-A) is also a GARP-supported description of DPCs, giving two defensible answers; option B was replaced with a false statement.

Derivatives product companies (DPCs) were developed to mitigate counterparty risk in OTC derivatives markets. Which of the following statements about DPCs is correct?

A. DPCs primarily change bankruptcy rules so that clients receive their full investment ahead of other claims if a counterparty becomes insolvent.

B. DPCs hold no capital of their own; their triple-A rating is always taken directly from the rating of the sponsoring parent.

C. DPCs are legal entities created to manage assets and finance large projects, isolating the entire firm from financial risk.

D. DPCs are typically triple-A-rated, separately capitalised, bankruptcy-remote subsidiaries of a major dealer, which give counterparties a degree of protection against failure of the DPC parent.

Answer: D

Step by step

1. A DPC is set up by one or more banks as a bankruptcy-remote subsidiary of a major dealer.
2. Unlike an SPV, it is separately capitalised to obtain a triple-A rating (via capital, margin and activity restrictions), so its counterparties are partly protected from failure of the parent.
3. Check each option: A and C describe SPV features; B contradicts the separate capitalisation. Only D matches.

Why D is correct: Separate capitalisation plus bankruptcy remoteness from the parent is what lets a DPC obtain a triple-A rating and shield counterparties from parent failure.

Why the others are wrong

- **A:** Altering bankruptcy priorities for investors (flip provisions) is an SPV mechanism, not the purpose of a DPC.
- **B:** Wrong because DPCs are separately capitalised and maintain the rating through capital, margining and activity restrictions; 'always the parent's rating' ignores this.
- **C:** Asset management and project financing with risk isolation describes an SPV.

Q648

AnalystPrep Q.6173

Monoline insurers were established as financial guarantee companies with strong credit ratings and provided 'credit wraps' to enhance the credit quality of obligations. They later entered single-name credit default swaps (CDS) and structured finance, and ran into major difficulties during the global financial crisis. Which of the following statements about monolines is most accurate?

A. Monolines were legal entities created to manage assets and finance large projects, isolating the entire firm from financial risk.

B. The primary function of monolines was to act as structured entities designed to mitigate counterparty risk in OTC derivatives markets.

C. Monolines were financial guarantee companies that provided 'credit wraps', and later also sold protection in the CDS and structured finance markets.

D. Monolines evolved to allow counterparties with less than triple-A credit quality to trade long-dated derivatives.

Answer: C

Step by step

1. Monolines are single-line financial guarantors with strong (typically triple-A) ratings that sold credit wraps.
2. They expanded into single-name CDS and structured finance to seek diversification and better returns.
3. Options A, B and D describe an SPV, a DPC/SPV-type structure and a DPC respectively.

Why C is correct: Providing credit wraps (financial guarantees) and then CDS/structured-finance protection is the defining business of monolines.

Why the others are wrong

- **A:** This is the SPV description (asset transfer and project financing).
- **B:** Monolines are guarantors who assume credit risk for a premium, not bankruptcy-remote structures built to reduce counterparty risk like SPVs/DPCs.
- **D:** Facilitating long-dated trading for sub-AAA counterparties was the original motivation for DPCs.

Q649 AnalystPrep Q.6174

A legal entity, typically a company or limited partnership, is created to isolate a firm from financial risk. A firm may transfer assets to it for management or use it to finance a large project. Jurisdictions may require that it is not owned by the firm on whose behalf it is set up. It aims to change bankruptcy rules so that, if a counterparty is insolvent, a client can receive its full investment before other claims are paid, which necessarily imposes a less favourable position on other parties. It is commonly used in structured notes to guarantee counterparty risk on the note's principal to a very high (typically triple-A) level, and rating agencies examine its mechanics and legal specifics before granting a rating. Which entity is described?

A. Credit derivative product company (CDPC)
B. Monoline insurance company
C. Derivatives product company (DPC)
D. Special purpose vehicle (SPV)

Answer: D

Step by step

1. Key clues: asset transfer, project financing, not owned by the sponsor, rearranged bankruptcy priorities (flip provisions), use in structured notes.
2. These are the defining features of an SPV (SPE).
3. A DPC is a separately capitalised dealer subsidiary; monolines and CDPCs are leveraged protection sellers.

Why D is correct: Altering bankruptcy priority and isolating assets in a separate, sponsor-independent entity used in structured notes is characteristic of an SPV.

Why the others are wrong

- **A:** CDPCs extend DPCs with a monoline-like business of selling credit protection; they do not rely on bankruptcy-priority changes.
- **B:** Monolines provide credit wraps (financial guarantees); they are not asset-isolating vehicles.
- **C:** A DPC is a separately capitalised, bankruptcy-remote subsidiary of a dealer used as a triple-A derivatives counterparty, not a priority-shifting structure for structured notes.

Q650 AnalystPrep Q.6176

This legal structure, used in OTC derivatives markets, aims to isolate firms from financial risk. Firms transfer assets to it for management or use it to finance large projects so that neither the firm nor a counterparty is put at risk. Its design is to change bankruptcy rules so that, if a counterparty is insolvent, a client can retrieve its full investment before other claims are paid. It is commonly used in structured notes to guarantee counterparty risk on the principal of the note, often at a triple-A level, and rating agencies assess its creditworthiness by examining its mechanics and legal specifics. What is this entity?

A. Derivatives product company (DPC)
B. Monoline insurance company
C. Credit derivative product company (CDPC)
D. Special purpose vehicle (SPV)

Answer: D

Step by step

1. Isolating assets, financing projects, changing bankruptcy priority and use in structured notes identify an SPV.
2. The rating depends on legal mechanics, which is also why SPVs carry legal (consolidation) risk.
3. DPCs, monolines and CDPCs do not rearrange bankruptcy priorities.

Why D is correct: Bankruptcy-priority engineering and asset isolation are the defining features of an SPV.

Why the others are wrong

- **A:** A DPC is a separately capitalised, bankruptcy-remote dealer subsidiary rated on capital, margin and activity restrictions; it does not shift bankruptcy priorities for note investors.
- **B:** A monoline sells financial guarantees (credit wraps); it is a leveraged insurer, not an asset-isolating vehicle.
- **C:** A CDPC resembles a monoline selling credit-derivative protection and does not use bankruptcy-rule changes.

Q651

AnalystPrep Q.6177

Which of the following statements correctly identifies a difference between a monoline insurance company and a special purpose vehicle (SPV)?

A. Monolines provide financial guarantees and were heavily involved in single-name CDS and structured finance, whereas SPVs typically manage assets or finance large projects to isolate firms from financial risk.

B. SPVs are primarily used to provide 'credit wraps' in the credit derivatives market, as monolines do.

C. Both monolines and SPVs are legally obliged to post collateral against declines in the market value of their contracts because of their credit ratings.

D. Both monolines and SPVs are designed to change bankruptcy rules so that clients can recover their full investment if a counterparty becomes insolvent.

Answer: A

Step by step

1. Monolines: financial guarantors that sold credit wraps and entered CDS/structured finance.
2. SPVs: legal entities used to isolate a firm from financial risk (asset management, project finance) and to shift bankruptcy priorities.
3. Test B, C, D against these: each wrongly merges the two entities. Only A separates them correctly.

Why A is correct: Monolines are guarantee/protection sellers while SPVs are risk-isolating legal structures, so A is the only accurate contrast.

Why the others are wrong

- **B:** Credit wraps are the business of monolines, not SPVs.
- **C:** Monolines and CDPCs typically did not have to post collateral in normal times thanks to their ratings, and nothing obliges SPVs to post collateral on this basis either.
- **D:** Bankruptcy-priority changes are an SPV feature; monolines are insurers and do not alter bankruptcy rules.

Q652

BT answers P2.T6.319.3 · BT notes P2.T6.319.3

REBUILT — Original option C asserted that triple-A ratings were achieved because monolines did not post collateral, reversing the causality in GARP (no posting was required because of the rating); reworded to the GARP-consistent claim.

Which of the following statements about triple-A-rated monoline insurers is correct?

A. The triple-A rating effectively eliminates counterparty risk to a monoline.

B. The triple-A rating implies that the monoline must post collateral against declines in the market value of its contracts.

C. In normal times monolines did not have to post collateral because of their triple-A ratings, but rating-downgrade clauses could force collateral posting just when their losses were rising, so counterparties depended on the rating.

D. CDPCs are safer than monolines because they exhibit right-way risk and do not create systemic risk.

Answer: C

Step by step

1. Monolines and CDPCs typically did not post collateral in normal times, relying on their excellent ratings.
2. In the GFC, MTM losses on protection sold caused rating concerns; downgrade clauses then triggered collateral calls they could not meet, so ratings fell rapidly and several failed.
3. Hence counterparties were relying on a rating that could vanish; A, B, D contradict this.

Why C is correct: The lack of collateral meant protection relied entirely on the rating, and downgrade-triggered posting created a wrong-way spiral in the crisis.

Why the others are wrong

- **A:** A triple-A rating lowers but does not eliminate counterparty risk; it collapsed for monolines in the GFC.
- **B:** Monolines typically did not post collateral in normal times; posting was triggered only by downgrades.
- **D:** CDPCs resemble monolines, are highly leveraged, and were exposed to wrong-way risk rather than right-way risk; many failed.

Q653

BT answers P2.T6.323.2 · BT notes P2.T6.323.2

REBUILT — Original relied on a 'default-remote entities' passage (moral hazard, SIFIs, CCP preference) not supported by the GARP chapter or Schweser LO 32.i; rebuilt as an EXCEPT question on the limitations of SPVs, DPCs, monolines and CDPCs.

Each of the following is a limitation of using special purpose vehicles (SPVs), derivatives product companies (DPCs), monolines or credit derivative product companies (CDPCs) as counterparty risk mitigants, EXCEPT:

A. An SPV transforms counterparty risk into legal risk, since a bankruptcy court may consolidate the SPV's assets with those of the sponsor.

B. A DPC's protection proved weaker than expected because its fate was linked to its parent: the Bear Stearns DPCs were wound down by JPMorgan after the parent's rescue, two Lehman DPCs filed for Chapter 11, and DPCs also carry market and operational risk.

C. Monolines and CDPCs did not post collateral in normal times, but downgrade clauses forced posting just as their insured losses rose.

D. DPCs reliably achieved an orderly, pre-packaged wind-down when their parents failed in the global financial crisis, removing counterparty risk for their clients.

Answer: D

Step by step

1. Test each statement against the documented limitations.
2. A is true: SPV risk is legal risk (consolidation; US courts have a history of consolidation rulings).
3. B is true: the Bear Stearns DPCs were wound down by JPMorgan after their parent's rescue and two Lehman DPCs filed Chapter 11, linking DPC fate to the parent; DPCs also face market and operational risk.
4. C is true: monolines/CDPCs lost ratings and were forced to post collateral when losses mounted.
5. D is false: the orderly unwinding did not hold in the GFC, so D is the exception.

Why D is correct: In the GFC, DPC wind-downs were not orderly, so the claim that they reliably removed counterparty risk is the one false statement.

Why the others are wrong

- **A:** True statement (a genuine limitation), so not the EXCEPT answer; chosen by those who think SPVs leave no residual risk.
- **B:** True statement; chosen by those who assume a DPC's rating implies safety despite the Lehman/Bear Stearns experience.
- **C:** True statement; chosen by those who think structures that post no collateral are risk-free.

Q654

Schweser Module Quiz 32.2 Q3

Which of the following statements is an enhancement offered by the central counterparty (CCP) structure relative to the special purpose vehicle (SPV), the derivative product company (DPC), and the monoline insurance models?

The CCP structure:

A. enables financial institutions to remove assets from their balance sheets.

B. enables counterparty risk to be outsourced, but in a nondiversified format.

C. spreads losses over a group of counterparties to minimize potential systemic risk.

D. enables a counterparty transaction originator to fail and not affect the other member firms.

Answer: C

Step by step

1. List what each earlier mechanism does: the SPV isolates assets/risk off the originator's balance sheet; the DPC is a separately capitalised triple-A subsidiary that protects against the parent's failure; the monoline insures/wraps risk but concentrates it in a few nondiversified insurers.
2. Identify what is new in a CCP: margining, a default fund and loss mutualization, so that a defaulting member's

losses are shared across all clearing members rather than falling on one entity.
3. Compare each option: A describes the SPV, B the monoline, D the DPC (isolation from the parent's failure); only C describes the CCP's distinguishing feature, loss mutualization.
4. Answer: C.

Why C is correct: A CCP uses collateralization and a default fund to mutualize losses across clearing members, diversifying counterparty risk and reducing the chance that one failure becomes systemic; SPVs, DPCs and monolines concentrate risk in a single entity.

Why the others are wrong

- **A:** Describes the SPV, which moves assets off the originator's balance sheet; it is not a CCP enhancement.
- **B:** Describes the monoline model, where risk is outsourced to a few insurers in a nondiversified (concentrated) form; the CCP does the opposite by mutualizing.
- **D:** Describes the DPC's protection from the parent's failure; in a CCP a member's default does affect the other members through the default fund and loss mutualization.

LO 32.j

Describe the approaches used and the challenges faced in modeling derivatives risk.

Q655

BT answers P2.T6.24.26.3 · BT notes P2.T6.24.26.3

REBUILT — Original asked for the 'best' dependency-modelling toolkit with B and D both defensible and no grounding in the reading; rebuilt around the reading's point that historical correlations are an unreliable guide to dependence in stress.

A bank models the joint behaviour of interest rates and credit spreads for its derivatives portfolio using a single correlation coefficient estimated from five years of calm-market data. A risk committee member questions the approach. Which of the following is the most valid criticism of relying on this historical correlation to model derivatives risk?

A. Historical correlations are unreliable only because VaR ignores dependence between risk factors altogether.

B. Correlations and dependence between risk factors can change substantially, especially in volatile or stressed periods, so a historical estimate may be a poor predictor of future outcomes.

C. Correlation is irrelevant to credit exposure measures such as potential future exposure, which depend only on the counterparty's default probability.

D. A single correlation number is sufficient because dependence between risk factors is stable over time and the same in normal and stressed markets.

Answer: B

Key correction: Source key (D) endorsed a combination of copulas, scenario analysis and stress testing; the item was a vague 'best approach' choice and B (t-copula) was also defensible, so it was replaced.

Step by step

1. Identify the modelling issue: dependence between risk factors (rates, credit spreads, collateral) is an input to derivatives risk models such as VaR, ES and PFE.
2. The reading states that historical correlations are often a poor predictor of future outcomes.
3. Correlations also change significantly in periods of high volatility and stress, and the dependence structure itself can change.
4. A coefficient estimated in calm markets therefore understates joint moves in stress; the valid criticism is statement B.

Why B is correct: Historical correlation is backward-looking and unstable: it shifts in volatile or stressed markets, so a calm-period estimate can badly misstate future dependence.

Why the others are wrong

- **A:** False premise: VaR and ES models do incorporate correlation and dependence; the weakness is the quality and stability of the inputs, not their absence.
- **C:** Confuses exposure with default probability alone; PFE depends on the joint behaviour of market risk factors (and wrong-way dependence with default), so dependence matters.
- **D:** Assumes stable dependence, which is the exact opposite of the reading; correlations typically rise in stress.

Q656 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A risk analyst at a bank uses historical simulation to compute the 1-day 95% VaR of a trading book. Of 100 daily outcomes, the five worst losses (USD million) are 9.0, 7.5, 6.0, 5.5, and 5.0. The analyst takes the 95% VaR as the fifth-worst loss and defines expected shortfall (ES) as the average of all losses equal to or greater than VaR. Which of the following correctly gives the VaR and the ES, and the correct interpretation?

A. VaR is USD 5.0 million and ES is USD 5.0 million; VaR already measures the worst possible loss.
B. VaR is USD 5.0 million and ES is USD 7.0 million; ES shows the average loss in the tail that VaR does not reveal.
C. VaR is USD 5.0 million and ES is USD 8.25 million; ES is the average of the two worst outcomes.
D. VaR is USD 5.0 million and ES is USD 6.6 million; ES shows the average loss in the tail that VaR does not reveal.

Answer: D

Step by step

1. VaR = fifth-worst loss = 5.0.
2. ES = (9.0 + 7.5 + 6.0 + 5.5 + 5.0)/5 = 33.0/5 = 6.6.
3. VaR says nothing about losses beyond the confidence level; ES gives their average.

Why D is correct: ES averages the losses at or beyond VaR: 6.6 million versus VaR of 5.0 million.

Why the others are wrong

- **A:** Treats VaR as the maximum loss; VaR is only the loss threshold at the confidence level and ignores larger tail losses.
- **B:** Averages only the four losses strictly beyond VaR (28.0/4 = 7.0), omitting the observation equal to VaR.
- **C:** Averages only the two worst outcomes (16.5/2 = 8.25), an arbitrary tail.

Q657 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A bank that has long applied backtested market-risk VaR models is extending its framework to measure counterparty risk using potential future exposure (PFE). A senior risk officer asks the model validation team how PFE modeling differs from VaR. Which of the following is the most accurate response?

A. PFE is the credit risk analogue of VaR but looks years ahead rather than a few days, which adds complexity and makes the models prone to misinterpretation and misuse.
B. PFE is easier to backtest than VaR because it is measured over a single day.
C. PFE measures the loss from a fall in value of the portfolio, whereas VaR measures only gains on the portfolio.
D. PFE avoids correlation and dependency problems because it is derived from the counterparty's default probability alone.

Answer: A

Step by step

1. PFE is the counterparty risk equivalent of VaR, measuring credit exposure from gains.
2. Its horizon is many years versus a short VaR horizon (a few days), so historical simulation and backtesting are harder.
3. Dependency between risk factors, including wrong-way risk, must also be modeled, adding complexity.

Why A is correct: GARP notes counterparty risk modeling is complex and prone to misinterpretation and misuse, partly because it looks years ahead rather than days.

Why the others are wrong

- **B:** PFE horizons are years, not one day; backtesting is easier for VaR with its short horizon.
- **C:** Reverses the concepts: PFE captures exposure due to a gain (credit exposure) not a loss on the portfolio.
- **D:** PFE depends on exposure dynamics and dependency between risk factors; correlation difficulties are even greater.

Reading 33. Counterparty Risk and Beyond

Jon Gregory, The xVA Challenge: Counterparty Credit Risk, Funding, Collateral, and Capital, 4th Edition (West Sussex, UK: John Wiley & Sons, 2020). Chapter 3.

LO	Learning objective	Qs
33.a	describe counterparty risk and differentiate it from lending risk.	8
33.b	describe transactions that carry counterparty risk and explain how counterparty risk can arise in each transaction.	9
33.c	identify and describe institutions that take on significant counterparty risk.	3
33.d	describe credit exposure, credit migration, recovery, mark-to-market, replacement cost, default probability, loss given default, and the recovery rate.	14
33.e	describe credit value adjustment (CVA) and compare the use of CVA and credit limits in evaluating and mitigating counterparty risk.	4
33.f	identify and describe the different ways institutions can quantify, manage, and mitigate counterparty risk.	12
33.g	identify and explain the costs of an OTC derivative.	3
33.h	explain the components of the X-Value Adjustment (xVA) term.	7

LO 33.a

Describe counterparty risk and differentiate it from lending risk.

Q658

BT notes P2.T6.302.3

REBUILT — Removed the 'According to Malz' attribution (source not in the reading) and reworded the stem so the question rests on the reading's counterparty-risk concept.

Which of the following statements about counterparty risk is TRUE?

A. It arises only when the contract has a positive net present value (NPV) to the party facing the risk.
B. It arises only when the contract has a negative net present value (NPV) to the party facing the risk.
C. Because it is a credit risk, it has no relationship to market risk.
D. Because it is a market risk, it has no relationship to default risk.

Answer: A

Step by step

1. Counterparty risk is the risk that the counterparty defaults before the contract ends, so that remaining payments are not made.
2. A party loses only if the counterparty owes it money, i.e. the contract is an asset (positive value) to it.
3. If the contract has negative value to the party, it owes the defaulter; the obligation is still due, so there is no credit loss.
4. The contract value is driven by market variables (rates, FX, etc.), so counterparty risk is a credit risk informed by market risk. Only A is correct.

Why A is correct: Loss on a counterparty default occurs only where the surviving party has a gain (positive value) on the contract, and that value is driven by market variables.

Why the others are wrong

- **B:** Reverses the sign: a negative-NPV contract is a liability, so a default by the counterparty does not create a loss.
- **C:** Counterparty risk is a credit risk, but its size depends on market variables, so it is strongly linked to market risk.
- **D:** The risk is a credit (default) risk; market variables only determine the exposure. Treating it as purely market risk ignores that a default is required.

Q659

AnalystPrep Q.1860

Which of the following statements is true about counterparty risk?

A. Counterparty risk is unilateral because only one counterparty bears a risk.
B. Counterparty risk is bilateral because cash flows can be negative or positive and both counterparties have a risk to the other counterparty.
C. The value of the contract in the future is certain.
D. Counterparty risk is not uncertain but depends upon the value of the underlying security which is fixed.

Answer: B

Step by step

1. Derivative cash flows can be positive or negative over the life of the contract.
2. Whichever party is 'winning' at the default date faces the risk that the 'losing' party defaults, so each party has risk to the other (bilateral).
3. Future contract value is uncertain, and the underlying is not fixed. Only B is correct.

Why B is correct: Because the contract value can be positive or negative, either party can end up as the creditor, making counterparty risk bilateral.

Why the others are wrong

- **A:** Describes lending risk (only the lender is exposed), not counterparty risk.
- **C:** The future value is typically highly uncertain; certainty would make the exposure resemble a loan.
- **D:** Exposure is uncertain precisely because the underlying market variables move; the underlying is not fixed.

Q660 BT answers P2.T6.900.1 · BT notes P2.T6.900.1 · BT Mock Q47

Julia is analyzing the credit risk of a contract her firm is considering entering. Which feature of her work most likely indicates that this is counterparty credit risk rather than traditional lending risk?

A. Interest rate fluctuations have little impact on the amount owed.
B. The exposure at default is 100% of the notional over most of the life of the contract.
C. Her firm is the only party assuming credit risk: if her firm defaults, the other party faces no losses.
D. She is employing Monte Carlo simulation because the future value of the contract is highly uncertain.

Answer: D

Step by step

1. Lending risk: amount at risk is known with reasonable certainty, and risk is unilateral.
2. Counterparty risk: future contract value is uncertain (positive or negative) and risk is bilateral.
3. A, B and C each describe a lending-type feature (stable amount, fixed full-principal exposure, one-way risk).
4. Needing simulation because future value is highly uncertain is the counterparty-risk feature: D.

Why D is correct: Highly uncertain, market-driven future exposure (needing simulation to model) is the hallmark of counterparty risk versus lending risk.

Why the others are wrong

- **A:** Low sensitivity to market variables means a predictable amount owed, which is lending-like.
- **B:** Exposure fixed at about the full notional is lending-like; derivative exposure is typically a small and variable fraction of notional.
- **C:** Unilateral risk is lending risk; counterparty risk is bilateral.

Q661 BT answers P2.T6.316.1 · BT notes P2.T6.316.1

Consider the following assertions about the differences between lending risk and counterparty risk:
I. Lending risk is a type of credit risk, but counterparty risk is a type of market risk.
II. Lending exposure is somewhat predictable, but counterparty credit exposure is highly uncertain.
III. Lending risk is unilateral, but counterparty risk is bilateral.
IV. Lending risk arises from an obligor's inability to meet a contractual obligation, but counterparty risk arises due to market variables.
V. Lending risk cannot be easily quantified, but counterparty risk is amenable to straightforward conventional quantification.
Which of the above are true differences between lending and counterparty risk?

A. None are true
B. Only II and III are true
C. I, II, and V are true
D. All are true

Answer: B

Step by step

1. I false: both are credit risk; counterparty risk is simply more sensitive to market variables.
2. II true: lending amount is roughly known; derivative exposure at default is uncertain.
3. III true: only the lender is at risk; in a derivative each party can be the creditor.
4. IV false: in both cases the loss is caused by the obligor being unable or unwilling to pay.

5. V false: counterparty risk is harder to quantify (uncertain, bilateral exposure). Only II and III: B.

Why B is correct: The two genuine differences are uncertain future exposure and bilateral (rather than unilateral) risk.

Why the others are wrong

- **A:** Ignores that II and III are genuine differences.
- **C:** Wrongly accepts I (market-risk classification) and V (counterparty risk is easier to quantify) and omits III.
- **D:** Accepts I, IV and V, which are false.

Q662

BT notes P2.T6.302.2

REBUILT — Removed the 'Malz asserts' attribution and framed as a standard EXCEPT question on counterparty risk concepts.

Each of the following statements about counterparty risk is true EXCEPT:

A. Counterparty risk arises when a party stands to gain on a contract and another party must fulfil a contractual obligation.

B. Counterparty risk is a special form of credit risk, typically relevant where credit risk is two-way rather than one-way.

C. Exchange-traded derivatives carry less counterparty risk than over-the-counter (OTC) derivatives.

D. Counterparty risk can be essentially eliminated if the trading partner is a prime broker.

Answer: D

Step by step

1. A true: a loss needs a gain (asset) on the contract plus a counterparty that owes performance.
2. B true: lending risk is unilateral; counterparty risk is typically bilateral.
3. C true: exchange-traded contracts are centrally cleared and margined daily, so they carry less counterparty risk.
4. D false: the identity of the counterparty does not remove the risk; collateral and clearing only reduce it. D is the exception.

Why D is correct: Counterparty risk can be reduced by collateral, clearing and netting, but it cannot be eliminated simply by choosing a prime broker as counterparty.

Why the others are wrong

- **A:** True statement, so it is not the exception.
- **B:** True statement (bilateral nature distinguishes it from lending risk).
- **C:** True statement: central clearing and daily margining reduce exchange-traded counterparty risk.

Q663

Schweser Module Quiz 33.1 Q3

Which of the following statements regarding counterparty credit risk is most accurate?

A. Counterparty risk is unilateral.

B. Over-the-counter (OTC) derivatives contain less counterparty risk than exchange-traded derivatives because the counterparty is known.

C. The precise future value of the contract is uncertain, but the counterparties are aware of whether the future value will be positive or negative.

D. Counterparty risk is typically associated with counterparty default prior to the settlement rather than default during the settlement process.

Answer: D

Step by step

1. A false: risk is bilateral.
2. B false: exchange-traded derivatives are centrally cleared and margined, so they carry less counterparty risk than OTC.
3. C false: neither the amount nor the sign of the future value is known.
4. D true: counterparty risk usually means pre-settlement risk; settlement risk is a separate concept.

Why D is correct: By convention counterparty risk refers to default before final settlement (pre-settlement risk).

Why the others are wrong

- **A:** That is lending risk; counterparty risk is bilateral.
- **B:** Reversed: central clearing and daily margining make exchange-traded risk lower; knowing the counterparty does not reduce risk.
- **C:** The sign of future value is also unknown, which is why either party may be exposed.

Q664

AnalystPrep Q.6142

REBUILT — Original asked about accounting and collateral practices, which are not covered by the counterparty-versus-lending-risk LO and are not in the reading; rebuilt around the credit-risk differences between the loan and the swap.

A firm borrows from a bank at a floating rate and hedges the interest rate risk by entering into a pay-fixed, receive-floating swap with the same bank. Which statement best describes the difference in the bank's credit risk on the two transactions?

A. Both exposures are about equal to the notional amount throughout, and both are unilateral.

B. The loan exposure is approximately known and unilateral, while the swap exposure is uncertain and bilateral, since either party may owe the other depending on rates.

C. The loan exposure is uncertain and bilateral, while the swap exposure is known and unilateral.

D. Both exposures are uncertain because both depend on the level of interest rates.

Answer: B

Step by step

1. Loan: bank's amount at risk is roughly the principal, known with reasonable certainty; only the bank can lose (unilateral).
2. Swap: only net payments are exchanged and its value depends on rates; it can be positive or negative to the bank over time, so exposure is uncertain.
3. Because the swap can be an asset or a liability to the bank, the firm is also exposed to the bank: risk is bilateral.
4. This matches B.

Why B is correct: Lending risk is known-amount and unilateral; derivative counterparty risk is uncertain-amount and bilateral.

Why the others are wrong

- **A:** Swap exposure is a variable fraction of notional, not the full notional, and is bilateral.
- **C:** Reverses the characterization of the two transactions.
- **D:** A floating-rate loan's principal is still approximately known; interest rate moves create only moderate uncertainty over the amount owed.

Q665

AnalystPrep Q.2266

REBUILT — Original referred to an undefined 'LaR framework' and asked an ambiguous 'paramount risk' question with no testable LO 33.a content; rebuilt to test the counterparty-risk vs lending-risk distinction.

A bank enters a 5-year interest rate swap with a corporate client and, separately, extends a 5-year fixed-rate term loan to another client. Which statement best differentiates the counterparty risk in the swap from the lending risk in the loan?

A. In the swap the amount at risk is uncertain and either party may end up as the one exposed to the other's default, whereas in the loan the principal at risk is reasonably known and only the lender is exposed.

B. In the swap only the party receiving the fixed payments is exposed to default, whereas in the loan both the lender and the borrower are exposed to the other's default.

C. In the swap the exposure equals the full notional amount at inception, whereas in the loan the exposure is limited to the interest payments due.

D. In the swap the exposure is certain and one-way, whereas in the loan the exposure is uncertain because the future value of the principal depends on market variables.

Answer: A

Key correction: Source key (A, credit risk) was for a stem that did not define 'LaR' and did not test LO 33.a directly; the original item was replaced.

Step by step

1. Lending risk: the principal at risk is known with reasonable certainty (fixed-rate loan) and the risk is unilateral - only the lender bears it.
2. Counterparty risk on a swap: the future value of the contract is uncertain in amount, because it depends on market variables.
3. It is also bilateral: whichever party has a positive mark-to-market value ('winning') is exposed to default by the party that is 'losing', and this can switch over time.
4. Option A states exactly these differences.

Why A is correct: Counterparty risk differs from lending risk because the exposure is uncertain in size and bilateral (either party can be the one at risk), while lending risk involves a known principal and one-way exposure.

Why the others are wrong

- **B:** Confuses direction: exposure depends on the sign of the swap's value, not on which leg is received, and a borrower is not exposed to the lender's default on a loan already funded.
- **C:** A swap involves no exchange of principal, so the exposure is not the notional; the loan's exposure is the principal, not just interest. This reverses the actual relationship.
- **D:** Reverses the two risk types: it is the swap exposure that is uncertain and two-way, while loan principal is known and one-way.

LO 33.b

Describe transactions that carry counterparty risk and explain how counterparty risk can arise in each transaction.

Q666

GARP PE1 Q45

REBUILT — Original options (9.45, 19.63) did not correspond to any sensible calculation; replaced with distractors from realistic mistakes (intrinsic value, discounted intrinsic value, Black-Scholes premium).

A derivative trading firm sells a European-style call option on stock JKJ with a time to expiration of 9 months, a strike price of EUR 45, an underlying asset price of EUR 67, and implied annual volatility of 27%. The option premium is paid in full upfront. The annual risk-free interest rate is 2.5%. What is the trading firm's counterparty credit exposure from this transaction?

A. EUR 0
B. EUR 22.00
C. EUR 22.84
D. EUR 23.02

Answer: A

Step by step

1. The firm is the option writer and has already received the premium in full at inception.
2. The buyer has no further payment obligation, so the firm can never be owed money by the buyer.
3. Credit exposure is $\max(V, 0)$, where V is the contract value to the firm. For a written option $V \leq 0$ always, so exposure $= 0$.
4. Inputs (S, K, volatility, rate) are irrelevant to the seller's exposure. For reference, the Black-Scholes premium is about EUR 23.02 and intrinsic value is $67 - 45 = 22$.

Why A is correct: With the premium paid upfront, the option writer has no claim on the buyer, so a buyer default cannot cause a loss; counterparty risk is borne only by the buyer.

Why the others are wrong

- **B:** Intrinsic value $S - K = 22$; treats the option's payoff as an amount owed to the seller, but it is a liability of the seller, not an asset.
- **C:** Discounted-strike value $67 - 45e^{-0.025\times0.75} \approx 22.84$ (forward-style value); again confuses the buyer's gain with the seller's receivable.
- **D:** The Black-Scholes premium (about 23.02) already received; the option's value is a liability to the seller, not exposure.

Q667

AnalystPrep Q.5435

REBUILT — Original distractors (5.75, 37.5) were unexplained giveaways; replaced with intrinsic value, Black-Scholes value and the stock price.

A European-style call option on Electric Inc. shares has a time to expiration of 3 months, a strike price of $25, an underlying asset price of $47, and implied annual volatility of 23%. The option was sold for an upfront premium, and the annual risk-free interest rate is 1.5%. What is the counterparty credit exposure of the company that sold the option from this transaction?

A. $0
B. $22.00
C. $22.09
D. $47.00

Answer: A

Step by step

1. The seller received the premium upfront, so the buyer owes nothing further.
2. Exposure $= \max(V, 0)$ with V the value to the seller. A short option position has $V \leq 0$, so exposure $= 0$.
3. Black-Scholes premium is about \$22.09 and intrinsic value is $47 - 25 = 22$; neither is an amount the buyer could fail to pay the seller.

Why A is correct: The seller's only claim (the premium) has already been settled, so default by the buyer causes no credit loss to the seller. The seller still bears market risk on the short position.

Why the others are wrong

- **B:** Intrinsic value $47 - 25$; this is what the seller owes the buyer, not what the buyer owes the seller.
- **C:** The option's model value; again a liability of the seller, mistaken for a receivable.
- **D:** The stock price; confuses the underlying with credit exposure.

Q668

AnalystPrep Q.1861

REBUILT — Option wording was garbled (options not clearly distinguishing settlement risk); rewrote options so only the timing-difference definition is correct.

Which of the following best defines settlement risk?

A. The risk that a counterparty defaults at some point between trade inception and final settlement date, causing loss of the remaining contractual payments.

B. The risk that arises at final settlement because the two parties perform their obligations at different times, so one may pay while the other fails to deliver.

C. The risk that a borrower fails to repay the principal amount of a loan.

D. The risk that the market value of collateral falls between margin calls.

Answer: B

Step by step

1. Settlement risk arises at settlement, when there is a difference in timing between each party's performance of its contractual obligations.
2. During that gap one party may have paid while the other defaults, giving a potentially very large loss (up to the full amount paid).
3. Presettlement risk (A) is the default risk during the life of the contract, which is what 'counterparty risk' normally means unless otherwise stated.

Why B is correct: Settlement risk is defined by the timing difference in performance at final settlement, which creates a large but short-lived exposure.

Why the others are wrong

- **A:** This is presettlement risk, the usual meaning of counterparty risk.
- **C:** This describes lending (credit) risk, not settlement risk.
- **D:** This is collateral market risk (gap risk in margining), not settlement risk.

Q669

AnalystPrep Q.1862

REBUILT — Option B was garbled and stem verbose; rewrote cleanly with the same GARP/Schweser relationship.

Counterparty risk is often divided into settlement risk and presettlement risk. Which of the following statements about the two is correct?

A. Settlement risk generates much smaller exposures than presettlement risk, but default at settlement is much more likely.

B. Settlement risk generates potentially much larger exposures, but default before the final settlement date is substantially more likely than default at settlement.

C. Settlement risk generates much larger exposures and is also much more likely to occur than presettlement default.

D. The two risks are similar in both size of exposure and likelihood, so they should be limited jointly.

Answer: B

Step by step

1. Settlement risk: exposure may be close to 100% of the notional paid, but it exists only for a very short time (days or hours).
2. Presettlement risk: exposure is smaller (replacement cost) but lasts for the whole life of the contract, so the

likelihood of default is substantially higher.
3. Hence large-but-unlikely (settlement) versus smaller-but-more-likely (presettlement).

Why B is correct: Settlement exposures are large but brief; presettlement exposures are smaller but persist over the contract's life, making default before settlement substantially more likely.

Why the others are wrong

- **A:** Reverses the exposure sizes and likelihood.
- **C:** Gets the size right but wrongly says settlement default is more likely; the exposure window is very short.
- **D:** Ignores the clear differences in exposure size and duration.

Q670 BT answers P2.T6.316.2 · BT notes P2.T6.316.2

Which of the following transactions is likely to carry the LEAST amount of counterparty risk?

A. An over the counter (OTC) derivatives contract where the exposure is large but the default probability is small
B. An over the counter (OTC) derivatives contract where the exposure is small but the default probability is large
C. An exchange-traded derivative contract where the exchange clears the trade via an attached central counterparty
D. A repurchase agreement (repo) with zero haircut applied to illiquid collateral

Answer: C

Step by step

1. Exchange-traded derivatives are cleared through the exchange's central counterparty, which becomes the counterparty and uses margining, so counterparty risk is minimal.
2. Both OTC contracts (A, B) are bilateral and uncollateralised as described, so they carry counterparty risk.
3. The repo with zero haircut and illiquid collateral has no cushion against borrower default or collateral decline, so risk is high.

Why C is correct: The central counterparty is the effective counterparty and is backed by margin, so exchange-traded contracts carry the least counterparty risk of the choices.

Why the others are wrong

- **A:** Bilateral OTC with large exposure; risk combines exposure and default probability, and neither is mitigated.
- **B:** Bilateral OTC; small exposure helps, but a large default probability still leaves material risk and no CCP.
- **D:** A zero haircut on illiquid collateral removes the main mitigant, so this is among the riskiest.

Q671 BT answers P2.T6.406.2

REBUILT — Stem referred to an unavailable repo exhibit; removed the reference (the question is answerable without it).

In a classic repurchase agreement (repo), Bank A is the repo seller (borrows cash and delivers securities) and Counterparty B is the repo buyer (lends cash and receives securities). Each of the following statements is true EXCEPT:

A. A higher haircut (overcollateralization) reduces the counterparty risk faced by the lender, Counterparty B.
B. The securities sold by Bank A become collateral legally owned by Counterparty B, who, if Bank A defaults, can liquidate them to recover the cash lent.
C. Because Bank A borrows cash, it bears no counterparty risk; only Counterparty B is exposed.
D. Counterparty B is exposed to credit risk (default of Bank A), market risk (collateral price volatility), liquidity risk (illiquid collateral), operational risk and legal risk.

Answer: C

Step by step

1. Check A: a haircut gives B a cushion against borrower default and collateral value decline: true.
2. Check B: title to the securities passes to the buyer, who can liquidate on default: true.
3. Check D: the lender faces credit, market, liquidity, operational and legal risk: true.
4. Check C: with a haircut, Bank A has delivered securities worth more than the cash received, so if B defaults and does not return the securities, A loses the excess. Counterparty risk exists for both sides, so C is false.

Why C is correct: Overcollateralization means the repo seller is itself exposed to the buyer failing to return the securities, so risk is two-sided.

Why the others are wrong

- **A:** True statement, so not the answer.
- **B:** True statement, so not the answer.

- **D:** True statement, so not the answer.

Q672 BT notes P2.T6.407.3 · BT answers P2.T6.407.3

Income Partners, LLC, the protection buyer, purchases a credit default swap (CDS) from Institutional Traders, LLC, which sells protection by writing the CDS. The CDS is marked-to-market, and the reference entity of the CDS is Acme Corporation. Each of the following is true EXCEPT, which is not?

A. Income Partners has more exposure than its counterparty, Institutional Traders, due to potential future exposure at high confidence levels

B. In regard to the exposure faced by Income Partners due to the CDS: the credit spread of Acme Corporation does NOT have an impact; however, the credit spread of Institutional Traders DOES impact this credit exposure

C. In the event of a bankruptcy by Institutional Traders, Income Partners will likely be pari passu with the firm's senior, unsecured bondholders

D. Exposure faced by Income Partners, with respect to Institutional Traders, will be measured gross of recovery; but the estimated recovery rate will impact the quantification of counterparty risk via the credit value adjustment (CVA)

Answer: B

Step by step

1. The CDS mark-to-market, and hence Income Partners' exposure, is driven by the reference entity's (Acme's) credit spread: widening gives the protection buyer a positive MtM.
2. Institutional Traders' own spread reflects its default probability, which enters the CVA but does not drive the contract's MtM or exposure. So B reverses the roles and is false.
3. A is true: the buyer can gain a lot if Acme's spread blows out (large PFE), whereas the seller's positive MtM is bounded; there is also wrong-way risk for the buyer.
4. C is true: derivative claims in bankruptcy rank pari passu with senior unsecured creditors. D is true: exposure is measured before recovery, with recovery entering the loss (CVA).

Why B is correct: Exposure on a CDS depends on the reference entity's spread, while the counterparty's spread affects its default probability and therefore CVA, not the exposure.

Why the others are wrong

- **A:** True: the protection buyer has the larger potential future exposure.
- **C:** True: derivative claimants rank with senior unsecured creditors.
- **D:** True: exposure is gross of recovery; recovery enters the loss through CVA.

Q673 Schweser Module Quiz 33.1 Q1

When considering counterparty credit risk, which of the following financial products has the largest outstanding notional amount in the marketplace?

A. Credit default swaps.

B. Foreign exchange forwards.

C. Interest rate swaps.

D. Repos and reverse repos.

Answer: C

Step by step

1. Counterparty risk arises mainly in OTC derivatives and securities financing transactions (repos, reverse repos).
2. OTC derivatives are significantly larger than securities financing, and interest rate swaps make up the bulk of the OTC derivatives market.

Why C is correct: Interest rate swaps account for the bulk of OTC derivative notional.

Why the others are wrong

- **A:** CDS notional is far smaller than interest rate swap notional.
- **B:** FX forwards carry large counterparty risk per trade but total notional is much smaller than swaps.
- **D:** Securities financing is much smaller than the OTC derivatives market.

Q674 AnalystPrep Q.1864

REBUILT — Original had the investor paying euros and receiving dollars, so a euro rise left the investor out of the money (no default loss) and the keyed $100,000 was a mark-to-market loss, not a counterparty-default loss; restated direction (investor receives euros), cleaned OCR currency symbols and rebuilt options.

An investor enters a forward foreign exchange contract to pay $2.6 million and receive €2 million at a specified future date (agreed rate 1.30 USD/EUR). Settlement risk would be the loss of the full $2.6 million if the investor paid the dollars but the euros were never delivered. Before the settlement date the EUR/USD rate rises from 1.30 to 1.35, and the counterparty then defaults before settlement. Ignoring collateral and recovery, what is the investor's pre-settlement loss from the default?

A. $130,000
B. $0
C. $100,000
D. $2,600,000

Answer: C

Key correction: Source key (C) agreed numerically, but its stated logic (contract out of the money for the investor) is inconsistent with a loss on default; direction was restated so that C is correct.

Step by step

1. Pre-settlement loss on default = replacement cost = current value of the contract to the non-defaulting party, if positive.
2. The investor receives €2m, now worth $2{,}000{,}000 \times 1.35 = \$2{,}700{,}000$.
3. The investor owes $2,600,000 under the contract ($2{,}000{,}000 \times 1.30$).
4. Contract value to the investor = $2{,}700{,}000 - 2{,}600{,}000 = \$100{,}000$ (in the money), equivalently $2{,}000{,}000 \times (1.35 - 1.30)$.
5. The counterparty's default removes this gain; the investor must replace the contract at the market rate, losing $100,000.

Why C is correct: Pre-settlement (replacement cost) loss is the positive mark-to-market value of the contract to the survivor, here the euro appreciation on the euro notional, not the full notional, which is the settlement-risk loss.

Why the others are wrong

- **A:** Applies the 0.05 rate move to the dollar notional (2.6m × 0.05) instead of the euro notional.
- **B:** Assumes the contract is out of the money for the investor; but the investor receives the appreciating currency, so it has positive value and a default causes a loss.
- **D:** Gives the settlement-risk exposure (the full dollar payment lost if paid but nothing received), not the pre-settlement replacement cost.

LO 33.c

Identify and describe institutions that take on significant counterparty risk.

Q675 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A risk analyst at a bank is classifying the firm's OTC derivatives counterparties by activity profile. One counterparty is a national government agency that enters into a small number of cross-currency swaps with only two dealers to hedge its foreign-currency debt. If it posts collateral, the collateral is usually illiquid. Which of the following best describes this counterparty?

A. A large derivatives player, because sovereign entities typically trade across commodities, equity, foreign exchange, interest rate, and credit derivatives.
B. A small derivatives player, because it trades with few counterparties, specializes in one asset class, and posts illiquid collateral if any.
C. A medium derivatives player, because it has a large number of clients but is less active than dealers in some asset classes.
D. A small derivatives player, because small players are always required to post liquid collateral to dealers to offset their lower trading volume.

Answer: B

Step by step

1. Sovereigns, large corporations, and smaller financial institutions with specific needs are small derivatives players.
2. The few trades, two dealers, single asset class, and illiquid collateral all match the small-player profile.

Why B is correct: Small derivatives players trade with few counterparties, have few OTC trades, specialize in one asset class, and post collateral (if at all) that is often illiquid.

Why the others are wrong

- **A:** Describes large dealer banks; a sovereign with specific hedging needs does not cover a very wide range of assets.
- **C:** Medium players are smaller banks or financial institutions with many clients and high volume, not a government agency with two dealers.
- **D:** Right category but wrong collateral claim; small players' collateral, if posted, is often illiquid.

Q676

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A senior risk manager is comparing the characteristics of three types of institutions that take on counterparty risk through OTC derivatives trading. Which of the following statements is correct?

A. Small players hold the largest number of OTC derivatives trades because they spread their activity across many asset classes.

B. Medium players are definitely required to post collateral, but they trade in a narrower range of asset classes than small players.

C. Large players trade with each other and many clients, cover a very wide range of asset classes, and post collateral against their positions.

D. Large players are typically sovereigns and corporations whose derivatives needs are specific and tied to a single asset class.

Answer: C

Step by step

1. Large players are the big dealer banks: many clients, very wide asset coverage, collateral posted.
2. Medium players are likely, not certain, to post collateral and cover a wide range, though less active than large players.
3. Small players have few trades and usually one asset class.

Why C is correct: Large derivatives players are large dealer banks that trade with each other and many clients across all major asset classes and post collateral.

Why the others are wrong

- **A:** Small players have few OTC trades and specialize in one asset class.
- **B:** Medium players are likely but not definitely posting collateral, and they cover a wide (not narrower) range of assets.
- **D:** Describes small players, not large dealers.

Q677

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A consultant presents a firm that provides central clearing services, trade compression, and collateral management software to derivatives market participants. A junior analyst states that the firm should be grouped with large derivatives players because it is critical to counterparty risk in the OTC market. Which of the following is the most accurate assessment?

A. The analyst is correct, because any firm involved in clearing must be a large dealer that takes on significant trading counterparty risk.

B. The analyst is incorrect, because the firm is a medium derivatives player that has a large number of clients and a high volume of OTC trades.

C. The analyst is correct, because trade compression and collateral management only exist inside large dealer banks.

D. The analyst is incorrect, because the firm is a third-party provider of products and services that reduce counterparty risk and improve efficiency, not a derivatives player taking on risk through trading.

Answer: D

Step by step

1. Derivatives players (large, medium, small) take on counterparty risk through trading.
2. Clearing, software, compression, and collateral management are services offered by third parties to reduce counterparty risk and improve efficiency.

Why D is correct: Third parties offering clearing, software, trade compression, and collateral management are distinct from derivatives players that take on counterparty risk by trading.

Why the others are wrong

- **A:** Offering clearing services does not make a firm a dealer; it is a third-party service.
- **B:** Medium players are smaller banks or financial institutions that trade a high volume of OTC derivatives; this firm provides services.
- **C:** These services are provided by external third parties, not only by dealers.

LO 33.d

Describe credit exposure, credit migration, recovery, mark-to-market, replacement cost, default probability, loss given default, and the recovery rate.

Q678

BT notes P2.T6.411.1 · BT answers P2.T6.411.1

Capital Bank enters a vanilla interest rate swap as the fixed-rate receiver with three annual payments. Therefore, Capital Bank is also the floating-rate payer, where the floating rate is the one-year spot rate. The spot rate curve is upward-sloping: the one-year rate is 2.0%, the two-year rate is 3.0%, and the three-year rate is 4.0%. Consequently, a fixed rate of 3.95% ensures an initial value of (approximately) zero for the swap. All rates are expressed with annual compounding to match the annual settlement cash flows:

Item	Value		
Notional (mm)	$100		
Fixed rate	3.95%		
Year	1.0	2.0	3.0
Spot rates	2.00%	3.00%	4.00%

If we can assume that the forward rates represent expected future spot rates (i.e., we can assume "realized forwards" rather than "unchanged term structure"), then which is nearest to the expected mark-to-market (expected value) of the swap to Capital Bank in one year, immediately after the first swap settlement?

A. -1.95 million
B. Zero
C. +0.73 million
D. +2.44 million

Answer: A

Key correction: BT answers version listed the options in a different order (keyed B = -1.95 million, OCR-printed as -7.95 million); same answer, re-lettered to A here.

Step by step

1. Implied forwards: $F_{1,2} = 1.03^2/1.02 - 1 = 4.0098\%$ and $F_{2,3} = 1.04^3/1.03^2 - 1 = 6.0292\%$. Under realized forwards these are the expected floating rates for the year-2 and year-3 settlements.
2. Expected net cash flow to the fixed receiver at year 2: $(3.95\% - 4.0098\%) \times 100\text{m} = -\$0.0598\text{m}$.
3. Expected net cash flow at year 3: $(3.95\% - 6.0292\%) \times 100\text{m} = -\$2.0792\text{m}$.
4. Discount from the year-1 viewpoint on the realized curve: $-0.0598/1.040098 = -\$0.0575\text{m}$ and $-2.0792/(1.040098 \times 1.060292) = -\$1.8854\text{m}$ (the year-1 two-year spot is $\sqrt{1.040098 \times 1.060292} - 1 = 5.0147\%$).
5. Expected MtM = $-0.0575 - 1.8854 = -\$1.943\text{m} \approx -1.95$ million (using the exact par rate 3.9475% gives -$1.9475m).

Why A is correct: With an upward-sloping curve, the receiver-fixed swap has positive expected net cash flows early and negative ones later; once the first (positive) settlement is paid, only the negative expected net flows remain, so the expected mark-to-market turns negative.

Why the others are wrong

- **B:** Assumes the swap keeps its zero inception value; the value drifts because the remaining expected floating rates (4.01%, 6.03%) exceed the 3.95% fixed rate after the first settlement drops off.

- **C:** A positive value results from a sign or perspective error (e.g., valuing it for the fixed payer, or mis-discounting); the receiver of 3.95% fixed against higher expected floating rates is out of the money.
- **D:** A positive value of this size comes from valuing it for the fixed payer or from keeping the low early floating rates (an unchanged-term-structure style assumption) instead of realized forwards; under the stated assumption the receiver's value is negative.

Q679

AP Mock 2 Q63

Prime Bank enters a GBP 200 million, five-year annual-pay interest rate swap, receiving 8% fixed against 12-month LIBOR. The swap rate declines by 150 basis points over the first year.
The current exposure at the end of year one is closest to:

A. GBP 5.10 million.
B. GBP 10.825 million.
C. GBP 10.28 million.
D. GBP 5.1387 million.

Answer: C

Step by step

1. After one year, 4 annual payments remain and the market swap rate is $8\% - 1.5\% = 6.5\%$.
2. Prime Bank receives 8% while a replacement swap would pay only 6.5%, an annual advantage of $200\text{m} \times 1.5\% =$ GBP 3m.
3. Annuity factor at 6.5% for 4 years: $(1 - 1.065^{-4})/0.065 = 3.4258$.
4. MtM = $3\text{m} \times 3.4258 =$ GBP 10.277m; current exposure = $\max(\text{MtM}, 0) =$ GBP 10.28m.

Why C is correct: Current exposure is the positive mark-to-market (replacement cost) of the swap: the present value of the 1.5% rate advantage on the notional over the 4 remaining years, discounted at the new 6.5% swap rate.

Why the others are wrong

- **A:** Roughly half the correct value; results from capturing only about two years of the GBP 3m advantage instead of all four remaining annual payments.
- **B:** Overstates the value, e.g. by mis-timing or under-discounting the remaining payments; the correct PV at 6.5% over 4 years is GBP 10.28m.
- **D:** Also about half the correct value; it does not capture the PV of all four remaining GBP 3m annual advantages.

Q680

AnalystPrep Q.1874

REBUILT — Stem said recovery is made by, and reduces the loss of, the issuer; corrected to the creditor per GARP's definition (percentage of the outstanding claim recovered when a counterparty defaults); options and key unchanged.

The recovery rate is the percentage of the outstanding claim that the creditor recovers when a counterparty defaults. The higher the recovery rate, the smaller the creditor's loss. A related variable is loss given default, defined in percentage terms as:
Loss Given Default = 1 - Recovery rate
Which of the following statements correctly describes the relationship between the two?

A. The higher the recovery rate from the counterparty, the lower the loss given default amount.
B. The higher the recovery rate from the counterparty, the higher will be the loss given default amount.
C. The lower the recovery rate from the counterparty, the lower will be the loss given default amount.
D. The recovery rate from the counterparty will always be equal to the loss given default amount.

Answer: A

Step by step

1. GARP: LGD in percentage terms is 100% minus the recovery rate, $\text{LGD} = 1 - R$.
2. Example: $R = 40\% \Rightarrow \text{LGD} = 60\%$; $R = 70\% \Rightarrow \text{LGD} = 30\%$.
3. So a higher recovery rate means a lower LGD: they move inversely, one-for-one.

Why A is correct: Because LGD is defined as 1 minus the recovery rate, every increase in recovery reduces LGD by the same amount.

Why the others are wrong

- **B:** Reverses the sign of the relationship; LGD falls, not rises, as recovery rises.
- **C:** Also reverses the relationship; a lower recovery rate raises LGD.
- **D:** They are equal only in the special case R = 50%; in general they sum to 100% rather than being equal.

Q681 AnalystPrep Q.1929

REBUILT — Stem did not say whose credit event occurred or from whose perspective the value was measured; clarified that the counterparty has defaulted and the value is negative to the institution, and tightened options so A is the single defensible answer.

In financial contracts, credit exposure depends on whether the contract's value is positive or negative. Suppose an institution's counterparty has defaulted and, at that time, the contract's value to the institution is negative. What does this scenario imply?

A. The institution owes the amount and is still legally obliged to settle it.
B. The institution is owed money by the counterparty, who is legally obliged to settle the amount.
C. The institution is released from its obligation and does not need to settle the amount it owes.
D. The institution is in financial distress and the contract should be terminated.

Answer: A

Step by step

1. Negative value to the institution means the PV of what it must pay exceeds the PV of what it expects to receive; it owes the counterparty.
2. GARP: with negative value, an institution is still obliged to honour its contractual payments and cannot gain from the default by being released from its liability.
3. So the institution has no credit exposure (no loss from the default) but must still settle what it owes.

Why A is correct: Credit exposure is asymmetric: a positive value is a claim on the defaulted counterparty, while a negative value remains a liability the survivor must pay to the defaulted party.

Why the others are wrong

- **B:** Describes positive value (a claim on the counterparty), the opposite of the scenario.
- **C:** Wrongly assumes the default frees the institution from its liability; except in special non-standard cases it cannot gain this way.
- **D:** A negative MtM on one contract says nothing about the institution's own distress, and termination does not cancel the amount owed.

Q682 AnalystPrep Q.1967

An important part of CVA calculation is the loss given default (LGD). What does LGD represent?

A. The amount of exposure expected to be recovered in the event of counterparty default.
B. The amount of exposure expected to be lost in the event of counterparty default.
C. The probability of counterparty default given that it defaulted on a previous contract.
D. The probability of making a huge loss following a counterparty default event.

Answer: B

Step by step

1. GARP: LGD in percentage terms is 100% minus the recovery rate.
2. Recovery is the share of the claim recovered; LGD is the share of the exposure lost if the counterparty defaults.
3. LGD is a loss severity, not a probability, and it is an input to CVA.

Why B is correct: LGD measures how much of the exposure is lost on default (1 minus recovery), which is why it multiplies exposure and default probability in CVA.

Why the others are wrong

- **A:** Describes the recovery rate, the complement of LGD.
- **C:** LGD is a loss severity, not a conditional default probability.
- **D:** LGD is not a probability of any loss size; it is the expected fraction of exposure lost given default.

Q683 BT answers P2.T6.317.2 · BT notes P2.T6.317.2

Why is credit migration crucial to measures of credit exposure (which is the best answer)?

A. Because it impacts the current mark-to-market (MtM)
B. Because it impacts the recovery
C. Because it impacts the term structure of probability
D. None of the above: credit migration has no impact on credit exposure

Answer: C

Step by step

1. Credit migration = discrete changes in credit quality, such as rating changes.
2. GARP: credit migrations are crucial since they influence the term structure of default probability (e.g., the default probability between years 4 and 5, not just over the next year).
3. Mean reversion means good credits tend to have rising default probability over time and poor credits falling, which shapes that term structure.

Why C is correct: Counterparty risk must be assessed over the whole life of the trades, and migration determines how the default probability evolves over that horizon, i.e., its term structure.

Why the others are wrong

- **A:** The current MtM depends on market variables and contract terms; migration matters for future default probability, not today's valuation.
- **B:** Recovery/LGD is a separate, highly uncertain input; migration is about changes in credit quality and default probability.
- **D:** Migration does matter: it changes the default probability term structure and can cause issues even before a default.

Q684

BT answers P2.T6.317.3 · BT notes P2.T6.317.2

A bank has previously entered into an interest rate swap with a counterparty. The swap has a remaining term of three years. The imminent (next) settlement due, which will net both parties' obligated exchange payments (i.e., the next "net coupon"), is a $300,000 payment to be made from the bank to the counterparty. The present value (PV) of all payments the bank expects to receive over the next three years is $5.6 million. The present value of all payments the bank expects to make, including the current $300,000 obligation, is $4.3 million. These present values (PVs) value the swap as two bonds; however, the payments will be netted. Each of the following is true EXCEPT, which is false?

A. Current MtM is (negative) -$300,000
B. The mark-to-market (MtM) value = $5.6 - $4.3 = positive (+) $1.3 million
C. The current exposure is $1.3 million
D. Replacement cost is unlikely to equal exactly the MtM but is often approximated by the MtM

Answer: A

Key correction: BT notes version gave no answer; the BT answers key (A) is correct.

Step by step

1. GARP: MtM is the PV of all payments an institution expects to receive less those it is obliged to make.
2. MtM = $5.6 - 4.3 = +\$1.3$m, so B is true.
3. Current exposure = max(MtM, 0) = $1.3m, so C is true.
4. Replacement cost includes costs such as bid-offer spreads, so it is not exactly MtM but is approximated by it: D is true.
5. The -$300,000 is only the next net cash flow (already inside the $4.3m PV), not the MtM, so A is false.

Why A is correct: MtM aggregates the PV of all remaining netted flows, not just the next coupon; the swap is worth +$1.3m to the bank even though its next net payment is outgoing.

Why the others are wrong

- **B:** True: MtM is PV of receipts minus PV of payments, 5.6 - 4.3 = +1.3m.
- **C:** True: with positive MtM and no collateral, current exposure equals the MtM of $1.3m.
- **D:** True: replacement cost differs from MtM because of costs such as bid-offer spreads, but MtM is the usual approximation.

Q685

BT answers P2.T6.318.1 · BT notes P2.T6.318.1

A bank holds an over-the-counter (OTC) derivatives position with a counterparty. The current mark-to-market (MtM) value of the bank's position is +$4.0 million, but the position is highly illiquid. There is no collateral or margin involved, and the net counterparty risk (net CVA adjustment) is zero. If the counterparty defaults, the bank expects a recovery rate of 35%. Which of the following is the best estimate of the position's replacement cost?

A. Replacement cost is about $1.4 million
B. Replacement cost is about $2.6 million
C. Replacement cost is less than $4.0 million
D. Replacement cost is more than $4.0 million

Answer: D

Step by step

1. MtM is the present value of expected inflows less expected payments: +$4.0 million, and is generally a close measure of replacement cost.
2. Replacement cost is the cost of entering an equivalent transaction with another counterparty, so it also includes transaction costs and bid-offer spreads.
3. For a highly illiquid position the bid-offer spread is large, so the cost to replace exceeds MtM.
4. The recovery rate affects the loss on default, not the cost of replacing the contract. Hence replacement cost is more than $4.0 million.

Why D is correct: Replacement cost approximates MtM but adds bid-offer and transaction costs, which are significant for illiquid contracts, so it is above the $4.0 million MtM.

Why the others are wrong

- **A:** Applies the 35% recovery rate to MtM ($4.0 \times 35\% = 1.4$); recovery is not part of replacement cost.
- **B:** Haircuts MtM by the loss given default ($4.0 \times 65\% = 2.6$); confuses the expected recovery with the cost of replacement.
- **C:** Assumes replacement can be done below MtM; transaction costs and bid-offer spreads push replacement cost up, not down.

Q686

BT answers P2.T6.320.3

REBUILT — Original joint-default claim was not supported by the reading and lacked assumptions; rebuilt to test the asymmetry of credit exposure (claim vs. obligation) in a hedged pair of swaps.

Financial institution XYZ has an interest rate swap with Counterparty A and hedges it with an identical offsetting swap with Counterparty B. At a future date the mark-to-market of the swap with A is +V (in XYZ's favor, V > 0) and therefore the swap with B is -V. Assume zero recovery, no collateral and no netting across counterparties. Which of the following statements is correct?

A. If either counterparty defaults, XYZ's loss is zero because the two swaps hedge each other

B. If only Counterparty B defaults, XYZ gains V because it is released from its obligation

C. If only Counterparty A defaults, XYZ loses V

D. If only Counterparty B defaults, XYZ loses V

Answer: C

Step by step

1. Credit exposure is the loss conditional on a counterparty's default: a positive value is a claim on the defaulter, a negative value is an obligation.
2. With A, XYZ's MtM is +V, so XYZ has a claim of V. If A defaults with zero recovery, XYZ loses V.
3. With B, XYZ's MtM is -V, so XYZ owes V. If B defaults, XYZ has no claim to lose and remains obliged to honor its payments, so the loss is zero (and no gain).
4. The offsetting swap hedges market risk but not counterparty default risk, since each swap is with a different counterparty.

Why C is correct: Exposure is max(MtM, 0) per counterparty: only the counterparty that owes XYZ can cause a loss, and a defaulter's creditor cannot gain by being released from its liability.

Why the others are wrong

- **A:** Confuses a market hedge with credit protection; the hedge is with a different counterparty, so A's default still leaves XYZ with a loss of V.
- **B:** Assumes default releases XYZ from its liability; the institution is still obliged to pay, so it cannot gain from the counterparty's default.
- **D:** Applies the loss to the wrong side; XYZ owes B (-V), so there is no claim on B and the loss is zero.

Q687

BT answers P2.T6.407.2

REBUILT — Original option C was ambiguous (credit exposure vs. current MtM); reworded so the false statement is clearly about potential exposure over the contract's life.

Which of the following statements about credit exposure, current exposure and potential exposure is FALSE?

A. Credit exposure is the loss conditional on the counterparty defaulting

B. If the mark-to-market (MtM) of a contract is negative today, current exposure is zero

C. If the MtM of a contract is negative today, the contract's credit exposure over its remaining life is necessarily zero

D. Credit exposure encompasses both current exposure and potential future exposure

Answer: C

Step by step

1. Credit exposure is the loss if the counterparty defaults, so it is conditional on default (A true).
2. Current exposure is the claim today: max(MtM, 0). With negative MtM it is zero (B true).
3. Credit exposure covers current claims and potential future claims (D true).
4. A contract that is out of the money today can become positive before default, so potential future exposure may be positive; C is false.

Why C is correct: Credit exposure includes potential future claims, so negative MtM today only sets current exposure to zero, not future exposure.

Why the others are wrong

- **A:** True statement: exposure is relevant only if the counterparty defaults.
- **B:** True statement: current exposure is max(MtM, 0), which is zero for negative MtM.
- **D:** True statement: credit exposure is the general term covering current and potential future exposure.

Q688

BT answers P2.T6.901.2 · BT notes P2.T6.901.2

REBUILT — Original asked for an 'EAD' approximation (replacement cost + PFE) not supported by the reading; rebuilt as an expected-loss calculation using replacement cost, default probability and loss given default.

A market maker has a fixed-for-floating interest rate swap with a notional of $80.0 million. The replacement cost (current exposure) is 3.0% of notional, or $2.4 million. The counterparty's one-year default probability is 1.0% and the expected recovery rate is 40% of the claim. Ignoring any further change in exposure, which of the following is the closest expected one-year credit loss on the current exposure?

A. $14,400
B. $24,000
C. $9,600
D. $480,000

Answer: A

Step by step

1. Exposure is the replacement cost, not the notional: $2,400,000.
2. Loss given default = 1 − recovery rate = 1 − 40% = 60%.
3. Expected loss = exposure × $PD \times LGD$ = 2,400,000 × 1% × 60% = 14,400.

Why A is correct: Expected loss multiplies the exposure at risk (replacement cost) by the probability of default and by LGD = 1 - recovery rate.

Why the others are wrong

- **B:** Omits LGD (2.4m × 1% = 24,000), implicitly assuming zero recovery.
- **C:** Uses the recovery rate instead of LGD (2.4m × 1% × 40% = 9,600).
- **D:** Applies PD and LGD to the notional (80m × 1% × 60% = 480,000); notional is not the amount at risk, replacement cost is.

Q689

BT notes P2.T6.406.3

REBUILT — Source had no answer and the 'greatest exposure' wording was indeterminate; rebuilt to ask which position has essentially no credit exposure after the premium is settled.

Assume any upfront premium has already been settled and no other payments are outstanding. Which of the following positions exposes the bank to essentially no credit exposure to its counterparty, because the bank has only obligations and no claims on it?

A. Long call option
B. Long put option
C. Long forward contract
D. Written (short) put option

Answer: D

Step by step

1. Credit exposure arises only where the bank can have a positive-value claim on the counterparty.

2. Long options (call or put): the holder has a claim equal to the option's value if it is in the money, so exposure is positive.
3. A forward can move to positive MtM, so a long forward has potential claims.
4. A written put: the premium has been received, and the bank's only future cash flow is a payment if the option is exercised, so it never has a claim on the counterparty.

Why D is correct: An option writer who has been paid the premium has only obligations; the buyer holds the claim, so only the buyer bears counterparty credit exposure.

Why the others are wrong

- **A:** Option buyers hold a claim when the call is in the money, so they bear credit exposure.
- **B:** A long put also has a positive claim when it is in the money, hence credit exposure.
- **C:** A forward has no upfront premium and its MtM can become positive, creating a claim, so exposure is possible.

Q690 Schweser Module Quiz 33.1 Q2

Liz Parker is a junior quantitative analyst who is preparing a report dealing with credit migration. An excerpt of her report contains the following statements:
I. Future default probability will likely increase over time, especially for periods far into the future.
II. When computing the default probability of a counterparty under a risk-neutral measure, we need to first determine the actual default probability.
Which of Parker's statements is (are) correct?

A. I only.
B. II only.
C. Both I and II.
D. Neither I nor II.

Answer: D

Step by step

1. Statement I: the default probability for a distant period tends to decrease, because default may already have occurred earlier (survival is unlikely). So I is false.
2. Statement II: the risk-neutral (market-implied) probability is calculated from market prices such as credit spreads; it does not require the actual (real-world) probability, which is estimated from historical data. So II is false.
3. Neither statement is correct.

Why D is correct: Far-future default probabilities tend to fall, and real-world and risk-neutral probabilities are two separate estimates.

Why the others are wrong

- **A:** Statement I reverses the tendency; default probability for far periods tends to decrease.
- **B:** Statement II conflates real-world and risk-neutral measures; the risk-neutral probability is implied from market prices.
- **C:** Both statements are incorrect.

Q691 BT answers P2.T6.317.1 · BT notes P2.T6.317.1

Each of the following statements is true about credit exposure EXCEPT, which is false?

A. Credit exposure defines the loss in the event of a counterparty default
B. Credit exposure does not assume counterparty default, i.e., is not conditional on default
C. Credit exposure includes the risk from current (actual) claims
D. Credit exposure includes the risk from possible future (potential) claims

Answer: B

Step by step

1. Exposure is the loss if a counterparty defaults (A true).
2. It is relevant only if the counterparty defaults, so quantification is conditional on default; B is false.
3. It covers current and potential future claims (C, D true).

Why B is correct: Credit exposure is a conditional loss measure: it is relevant only when the counterparty defaults.

Why the others are wrong

- **A:** True statement: this is the definition of credit exposure.
- **C:** True statement: current claims are part of exposure.

- **D:** True statement: potential future claims are part of exposure.

LO 33.e

Describe credit value adjustment (CVA) and compare the use of CVA and credit limits in evaluating and mitigating counterparty risk.

Q692 AP Mock 2 Q10 · AnalystPrep Q.1965

REBUILT — Merged two definitional CVA items; replaced the non-standard LVA and the nonsensical 'PFE minus EE' distractors with real xVA terms and a market-risk distractor.

A bank's derivatives desk adjusts the value of an OTC derivative to reflect the possibility that the counterparty defaults before meeting its obligations. This adjustment is the difference between the value of the contract assuming no counterparty default and its true (risky) value. Which of the following best describes this credit value adjustment (CVA)?

A. The market value of counterparty market risk.
B. The market value of counterparty credit risk.
C. The cost of posting initial margin over the life of the transaction.
D. The cost of holding regulatory capital against the transaction over its life.

Answer: B

Step by step

1. The risk-free value of a derivative assumes the counterparty never defaults.
2. The risky value deducts the expected loss from counterparty default.
3. The deduction (risk-free value minus risky value) is the CVA.
4. Because it is the price the market would charge for bearing the counterparty's default risk, CVA is the market value of counterparty credit risk.

Why B is correct: CVA is the portion of a derivative's valuation that accounts for counterparty default, i.e. the market price of counterparty credit risk.

Why the others are wrong

- **A:** Counterparty risk is credit (default) risk, not market risk; this confuses the market exposure that drives the loss with the source of the loss.
- **C:** This describes the margin value adjustment (MVA), a different xVA term.
- **D:** This describes the capital value adjustment (KVA), a different xVA term.

Q693 BT answers P2.T6.320.1 · BT notes P2.T6.320.1

Traditional counterparty credit risk management depends on credit limits, but a more sophisticated approach employs a credit value adjustment (CVA). According to Gregory, which is the key advantage of CVA over a credit line?

A. CVA can include potential future exposure (PFE), but credit limits cannot incorporate PFE
B. CVA incorporates a portfolio perspective, but credit limits are not portfolio credit risk management
C. CVA enables transaction decisions based on profitability at the trade level, whereas limits only control exposure at the portfolio level
D. CVA includes default probability, recovery, and correlation between counterparties, but credit limits cannot

Answer: C

Step by step

1. Credit limits are set on exposure (typically PFE) and act at the portfolio level; a trade is accepted or refused depending on whether it breaches the limit.
2. So with a limit the implicit price of counterparty risk is zero (accepted) or infinite (rejected).
3. CVA is the actual price of counterparty risk, defined at the trade and counterparty level.
4. Hence a trade can be judged on whether its revenue exceeds the CVA, i.e. on profitability. This is the key advantage.

Why C is correct: CVA prices counterparty risk into each trade, so decisions rest on profitability, whereas limits only cap exposure and ignore the return earned.

Why the others are wrong

- **A:** False: credit limits are normally expressed in terms of PFE, so they do incorporate it.
- **B:** False: credit limits act precisely at the portfolio level to avoid concentrations; it is CVA (a counterparty-level calculation) that does not capture concentration.
- **D:** Overstated: limits are set with regard to counterparty credit quality. The distinguishing advantage is pricing/profitability, not that limits ignore default probability altogether.

Q694 BT answers P2.T6.902.1 · BT notes P2.T6.902.1

Patricia is a risk analyst working on a counterparty risk management system for her international financial services firm. The Risk Committee has requested that the system assess counterparty risk at three levels: trade level, counterparty level, and portfolio level. In this context, Patricia is evaluating the pros and cons of credit limits and the credit value adjustment (CVA). Each of the following is true EXCEPT, which is false?

A. CVA and credit limits are complements in this three-level system
B. Credit limits act at the portfolio level by limiting exposures to avoid concentrations
C. CVA focuses on evaluating counterparty risk at the trade level and counterparty level
D. CVA encourages maximizing the number of trading counterparties in order to maximize the benefits of netting, but credit limits encourage minimizing this number to encourage smaller exposures

Answer: D

Step by step

1. CVA covers the trade level (trade features) and counterparty level (netting, collateral); credit limits act at the portfolio level. So C and B are true.
2. Because they cover different levels, they are complementary and used together. A is true.
3. Netting benefits are largest when trades are concentrated with few counterparties, so CVA encourages minimizing the number of counterparties.
4. Limits cap exposure per name, which encourages more counterparties, smaller exposures and diversification.
5. D reverses both statements, so D is the false one.

Why D is correct: CVA favors fewer counterparties (more netting) while credit limits favor more counterparties (diversification); D states the reverse.

Why the others are wrong

- **A:** True statement, so not the exception; CVA and limits are used together.
- **B:** True statement; limits operate at the portfolio level to avoid concentrations.
- **C:** True statement; CVA captures trade features and counterparty-level mitigants such as netting and collateral.

Q695 BT notes P2.T6.901.3

Herman, Ferry, and Quitzon LLC is a market maker with a position in an over-the-counter (OTC) derivatives contract. For this position, they have estimated the following values:
- The risk-free value of the position is +$32.0 million
- The credit valuation adjustment (CVA) is $5.0 million
- The debt valuation adjustment (DVA) is $8.0 million

Based only on this information, which of the following is the best estimate of the risky value of the position from Herman's perspective?

A. Risky value = $27.0 million
B. Risky value = $29.0 million
C. Risky value = $35.0 million
D. Without estimates for the counterparty's default probability (PD) and loss given default (LGD), we cannot estimate a risky value

Answer: C

Step by step

1. Risky value = risk-free value − CVA + DVA.
2. CVA is the cost of the counterparty's default risk; DVA is the benefit from the firm's own default risk (counterparty risk from the firm's own point of view).
3. $32.0 - 5.0 + 8.0 = 35.0$ million.

Why C is correct: CVA reduces and DVA increases the value, so the net adjustment is $-5 + 8 = +3$ and the risky value exceeds the risk-free value.

Why the others are wrong

- **A:** Subtracted only CVA, $32 - 5 = 27$, ignoring DVA.
- **B:** Subtracted the net adjustment the wrong way: $32 - (8 - 5) = 29$ treats DVA minus CVA as a cost. DVA is a benefit, so it is added, not subtracted.
- **D:** PD and LGD are inputs used to compute CVA and DVA, but both adjustments are already given, so the risky value follows directly.

LO 33.f

Identify and describe the different ways institutions can quantify, manage, and mitigate counterparty risk.

Q696

AP Mock 1 Q22 · AnalystPrep Q.5436

REBUILT — Tightened wording so that collateral is clearly the only strategy that can bring net exposure close to zero in a concentrated market.

A trading firm specialises in derivatives on scarce commodities. Together with a few other dominant firms it holds most of the notional contracts in this niche market. Management wants to reduce its net counterparty exposure to close to zero. Which strategy is most effective?

A. Expanding and diversifying the counterparty base.
B. Implementing cross-product netting agreements with each counterparty.
C. Requiring counterparties to post ample high-quality collateral to support the net exposure.
D. Buying credit default swaps on its counterparties.

Answer: C

Step by step

1. Goal: drive net exposure towards zero, not merely reduce it.
2. Diversification limits exposure to any single name, but the market has only a few dominant players, so there is little to diversify into.
3. Netting only reduces gross exposure to a net figure; the net amount remains unsecured.
4. CDS protection sellers are the same dominant firms (wrong-way risk), and hedging adds market risk and cost.
5. Sufficient collateral posted to support the net exposure theoretically reduces it to zero.

Why C is correct: Collateralization (margining) sized to the net exposure theoretically reduces exposure to zero, whereas the other tools only reduce or shift it.

Why the others are wrong

- **A:** Diversification is not feasible in a market with few dominant firms, and it only spreads exposure rather than removing it.
- **B:** Netting reduces exposure to the net amount, but the remaining net exposure is still unsecured.
- **D:** CDS shifts risk to protection sellers who are the same concentrated group (wrong-way risk) and creates market risk; it is not near-zero.

Q697

AnalystPrep Q.2303

REBUILT — Added the assumptions (no haircut, zero recovery on the uncovered part) needed for a single answer; fixed source typo.

Bodrum Bank has a $20m exposure to Aurora Bank, which has posted $15m of collateral that Bodrum holds. Assume the collateral can be liquidated at full value with no haircut, and that nothing is recovered on the uncollateralised part of the exposure. If Aurora defaults, how much does Bodrum lose?

A. $10m
B. $20m
C. $15m
D. $5m

Answer: D

Step by step

1. Exposure = $20m; collateral held = $15m.
2. Loss = exposure - collateral = 20 - 15 = $5m.

Why D is correct: Collateral offsets exposure dollar for dollar, so only the uncollateralised portion is at risk.

Why the others are wrong

- **A:** $10m has no basis; it is not 20 - 15 or any combination of the figures.
- **B:** Ignores the collateral entirely (gross exposure).
- **C:** Reports the collateral value as the loss instead of the exposure net of collateral.

Q698

BT answers P2.T6.24.24.1

REBUILT — Contradictory 'additional collateral to secure potential future exposure' and arguable central-clearing option replaced; options now have one clear answer.

G Corporation has a 3-year OTC interest rate swap with notional $50 million with Sunset Financial Bank. The swap currently has positive mark-to-market value to G, so Sunset owes G. Concerns about Sunset's creditworthiness have grown. Which action would best mitigate G's counterparty credit risk on this swap?

A. Terminate the swap immediately and leave the interest rate risk unhedged.

B. Require Sunset to post collateral under a collateral (margin) agreement covering the current mark-to-market exposure.

C. Negotiate a walkaway feature in the swap contract.

D. Increase the frequency of mark-to-market valuations without changing the contract terms.

Answer: B

Step by step

1. G is owed money (positive MtM), so G bears the exposure to Sunset.
2. Collateralization requires the counterparty to post collateral to support the net exposure, which offsets the loss if Sunset defaults.
3. Termination removes the hedge and trades counterparty risk for interest rate risk; walkaway favours the party with negative MtM (Sunset), not G; more frequent valuation only monitors.

Why B is correct: Collateral held against the positive MtM directly offsets the loss on Sunset's default.

Why the others are wrong

- **A:** Closing out leaves G unhedged against interest rates; it converts rather than mitigates risk and may realise a claim against a weak counterparty.
- **C:** A walkaway clause lets the party with negative MtM cancel; here that is Sunset, so it worsens G's position.
- **D:** Monitoring more frequently measures exposure but does not reduce it.

Q699

BT answers P2.T6.316.3 · BT notes P2.T6.316.3

REBUILT — Reworded options to match the reading (collateral creates operational/liquidity risk, hedging creates market risk, CCPs create systemic risk) so exactly one statement is false.

According to Gregory, each of the following statements about mitigation of counterparty risk is true EXCEPT:

A. Netting applies to every product category and transaction type, and its main downside is that it can increase exposure to a given counterparty.

B. Collateralization can significantly reduce counterparty risk but creates operational and liquidity risks.

C. Credit derivatives can hedge counterparty risk, but at a cost and with additional market risk.

D. Central counterparties simplify the problem of bilateral risks but can create systemic risk, since a CCP failure could be a significant systemic disturbance.

Answer: A

Step by step

1. Netting offsets positive and negative MtM values, so exposure can only fall or stay the same, never rise.
2. Its side effect is legal risk if a netting agreement is not enforceable in a jurisdiction.
3. B, C, D describe documented side effects: collateral gives operational/liquidity risk, hedging gives market risk, CCPs give systemic risk.

Why A is correct: Netting never increases exposure; the risk it creates is legal risk, so statement A is the false one.

Why the others are wrong

- **B:** True: collateral management creates operational and liquidity risk.
- **C:** True: hedging via CDS costs money and creates MtM (market) risk.
- **D:** True: CCPs centralise risk and their failure is systemic.

Q700 BT answers P2.T6.406.1 · BT notes P2.T6.406.1

Which of the following is TRUE about counterparty risk?

A. Counterparty risk excludes pre-settlement and settlement risks.

B. In contrast to counterparty risk, settlement risk is characterized by small exposures for long periods of time.

C. Counterparty risk mitigation does not eliminate risk but instead converts counterparty risk into other types of financial risk.

D. In the credit derivatives market, notional outstanding is one of the more accurate measures of counterparty risk in the system.

Answer: C

Step by step

1. Mitigants (netting, collateral, hedging, CCPs) leave residual risk and create new risks: legal, operational, liquidity, market, systemic.
2. So C is true; check the others against the definitions.

Why C is correct: No mitigant is perfect; each converts counterparty risk into other forms of financial risk.

Why the others are wrong

- **A:** Counterparty risk covers pre-settlement and settlement risk; it does not exclude them.
- **B:** Reversed: settlement risk is large exposure for a very short period, not small exposure for long.
- **D:** Notional amounts overstate actual risk; they are not an accurate measure of counterparty risk.

Q701 BT answers P2.T6.406.3 · BT notes P2.T6.406.3

According to Gregory, risk mitigation is imperfect: it cannot perfectly eliminate counterparty risk, and it creates new risks. Each of the following associations of mitigation method and the risk it creates is true EXCEPT:

A. Collateral leads to operational and liquidity risks through the need to run a complex collateral management function.

B. Central counterparties create systemic risk since the failure of a CCP could be a significant systemic disturbance.

C. Hedging may lead to additional market risk through the mark-to-market volatility of the hedging instruments.

D. Netting creates additional exposure in the form of basis risk because the mark-to-market of a position can become positive for both counterparties.

Answer: D

Step by step

1. Netting offsets values across a netting set, so it cannot increase exposure.
2. The risk netting creates is legal risk (unenforceable agreements), not basis risk.
3. A, B, C match the documented side effects.

Why D is correct: Netting's side effect is legal risk; it does not add exposure or basis risk.

Why the others are wrong

- **A:** True: collateral management creates operational and liquidity risk.
- **B:** True: CCPs concentrate risk, creating systemic risk.
- **C:** True: CDS hedges have MtM volatility, i.e. market risk.

Q702 BT notes P2.T6.900.2

Acme Financial Incorporated has an outstanding derivatives contract with its counterparty. Currently, neither posts collateral, but they mutually agree to a modification such that the revised contract now includes a bilateral collateral arrangement. Which risk has Acme reduced?

A. Acme reduced counterparty risk

B. Acme reduced operational risk

C. Acme reduced market/liquidity risk

D. All of the above (reduction in counterparty, operational, and market/liquidity risk)

Answer: A

Step by step

1. Collateral posted against the net exposure reduces the loss given counterparty default, so counterparty risk

falls (theoretically to zero if collateral is sufficient).
2. Running a collateral agreement requires margin calls, valuation, and collateral movements, which adds operational work and risk.
3. Collateral must be funded (liquidity risk) and its value can move with price and FX volatility (market risk); it can also carry legal risk (e.g., rehypothecation, segregation, enforceability).
4. Hence only counterparty risk is reduced; other risks increase.

Why A is correct: Collateralization transforms counterparty risk into operational, liquidity, market and legal risks rather than reducing them.

Why the others are wrong

- **B:** Operational risk rises because margin calls, valuations and collateral transfers must be managed; it does not fall.
- **C:** Market/liquidity risk rises because posted collateral must be funded and its value can move with price and FX volatility; it does not fall.
- **D:** Only counterparty risk is reduced; the other two risks are increased by the collateral arrangement.

Q703

GARP PE1 Q28 · Mock 3 Q52

A derivative trading firm only trades derivatives on rare commodities. The company and a handful of other firms, all of whom have large notional outstanding contracts with the company, dominate the market for such derivatives. The company's management would like to mitigate its overall counterparty exposure, with the goal of reducing it to almost zero. Which of the following methods, if implemented, could best achieve this goal?

A. Ensuring that sufficient collateral is posted by counterparties
B. Diversifying among counterparties
C. Cross-product netting on a single counterparty basis
D. Purchasing credit derivatives, such as credit default swaps

Answer: A

Step by step

1. Market structure: only a handful of firms trade these contracts, all of them already counterparties.
2. Diversification (B) is impractical: there are no additional counterparties to spread exposure across.
3. Netting (C) offsets positive and negative values per counterparty but leaves a net exposure, usually well above zero.
4. CDS (D) would have to be bought from the same few dealers, leaving residual risk (wrong-way risk) and market risk.
5. Sufficient collateral reduces net exposure to theoretically zero with each existing counterparty, so A.

Why A is correct: Sufficient collateral posted against net exposure can theoretically reduce counterparty exposure to zero regardless of market concentration.

Why the others are wrong

- **B:** Needs a larger pool of counterparties, which a concentrated market does not offer.
- **C:** Netting only reduces gross to net exposure per counterparty; it does not take exposure near zero.
- **D:** Protection sellers are the same few firms and CDS creates new market and wrong-way risk; it cannot bring exposure near zero.

Q704

Schweser Module Quiz 33.2 Q1

Ondine Financial, Inc., (Ondine) uses a variety of techniques to manage counterparty risk. It has entered into an interest rate swap with Scarbo, Inc. (Scarbo). Currently, Ondine's position in the swap has a -$1 million mark-to-market value. Based on the information provided, which of the following credit risk mitigation techniques would be most advantageous to Ondine if Scarbo defaults?

A. Close-out.
B. Collateralization.
C. Netting.
D. Walkaway.

Answer: D

Step by step

1. Ondine's MtM is -$1 million, so it owes value to Scarbo; Scarbo's default would not hurt Ondine through lost receivables.

2. A walkaway feature lets the non-defaulting party cancel the transaction on default, so Ondine need not pay the $1 million.
3. Close-out with netting would immediately realize the loss, so Ondine would pay the net $1 million.
4. Collateral would only matter for the party with positive exposure; Ondine has none.

Why D is correct: A walkaway feature is advantageous to the non-defaulting party when its MtM is negative, because it can cancel the obligation.

Why the others are wrong

- **A:** Close-out crystallizes the -$1 million and Ondine pays it.
- **B:** Collateral protects the party with positive exposure; Ondine is out of the money, so it gains nothing.
- **C:** Netting offsets against other trades; with only this swap the net amount owed is still $1 million.

Q705

Schweser Module Quiz 33.2 Q2

REBUILT — Hedging with CDS can also create systemic risk through feedback effects (per GARP), making two options defensible; replaced the hedging option with close-out and reworded the stem.

Which of the following methods of mitigating counterparty risk is most likely to generate systemic risk by concentrating counterparty risk in a limited number of entities?

A. Bilateral netting agreements.
B. Collateralization.
C. Close-out provisions.
D. Central counterparties.

Answer: D

Step by step

1. Netting, collateral and close-out all operate bilaterally between the two trading parties.
2. Central counterparties become the counterparty to every trade and so centralize counterparty risk in a few institutions.
3. If a CCP fails, the shock is spread across the whole financial system, so CCPs can create systemic risk (as well as operational and liquidity risk).

Why D is correct: Central counterparties concentrate risk in a few entities, so the failure of one could create a system-wide shock.

Why the others are wrong

- **A:** Netting reduces exposure bilaterally; its main added risk is legal (enforceability), not systemic.
- **B:** Collateral adds operational, liquidity and legal risk but remains bilateral and dispersed.
- **C:** Close-out is a bilateral procedure on default; it does not concentrate risk in any institution.

Q706

AP Mock 1 Q7

REBUILT — Original gave only contract values with no signs, so the netting benefit could not be shown and collateral was also defensible; rebuilt with signed MtM values and a netting calculation.

A hedge fund has the following mark-to-market positions with a single counterparty, AB&B, all covered by a legally enforceable cross-product netting agreement (positive = AB&B owes the fund):

Contract	MtM (USD millions)
A	+44.0
B	+88.0
C	-35.2
D	-3.3
E	+20.0

Assuming AB&B defaults with zero recovery, what is the fund's counterparty exposure to AB&B after netting?

A. USD 113.5 million
B. USD 152.0 million
C. USD 190.5 million
D. USD 38.5 million

Answer: A

Step by step

1. Sum of positive MtM: 44.0 + 88.0 + 20.0 = 152.0.

2. Sum of negative MtM: -35.2 - 3.3 = -38.5.
3. Net exposure = 152.0 - 38.5 = 113.5 (a net claim, so the fund loses this amount at zero recovery).
4. Netting reduced exposure from 152.0 gross to 113.5.

Why A is correct: Netting aggregates positive and negative values with the defaulting counterparty, so exposure is the net sum when positive.

Why the others are wrong

- **B:** Gross positive exposure; ignores netting of the negative contracts.
- **C:** Sum of absolute values of all contracts (44+88+35.2+3.3+20), wrongly adding negatives.
- **D:** Only the sum of the negative contracts (the fund's own liabilities), not the net.

Q707

AnalystPrep Q.1998

REBUILT — Original was vague and its key (gradual hedging) was not a clear best answer; rebuilt around the combination of netting, close-out and collateral to limit default losses.

A bank has many derivative contracts with each of several counterparties, with positive and negative values. Which of the following best prepares the bank so that the default of any one counterparty inflicts minimum financial damage?

A. Continuously monitoring each counterparty's credit rating.
B. Cross-product netting with close-out, supported by collateral sufficient to cover the net exposure.
C. Starting litigation as soon as a counterparty shows signs of distress.
D. Increasing the number of counterparties while leaving all exposures uncollateralized.

Answer: B

Step by step

1. Netting with close-out lets the bank offset what it owes against what it is owed, leaving a single net claim.
2. Collateral covering that net exposure theoretically reduces the remaining loss to zero.
3. Monitoring and litigation are reactive and do not reduce the exposure itself; diversification alone only limits concentration.

Why B is correct: Netting plus close-out cut exposure to the net amount, and collateral then covers that net amount, directly limiting loss on default.

Why the others are wrong

- **A:** Monitoring informs but does not reduce exposure at default.
- **C:** Litigation is slow, costly and after the fact; it does not reduce the loss.
- **D:** Diversification limits concentration, but each uncollateralized, un-netted exposure can still be fully lost.

LO 33.g

Identify and explain the costs of an OTC derivative.

Q708

AP Mock 1 Q19

Global Finance Bank (GFB) is training its derivatives team on the economic costs of OTC derivatives over their lifetime, and how the mark-to-market position affects those costs. The case study concerns an interest rate swap currently in a positive MtM position - the transaction is in the money to GFB. The swap is partially collateralised. Which of the following accurately describes the economic costs GFB would incur in this scenario?

A. Counterparty risk costs due to GFB's potential for default, and a funding benefit from the uncollateralized component of the MtM position.
B. Counterparty risk and funding costs arising from the uncollateralized component of the MtM position, and funding costs for the capital needed for the transaction and initial margin.
C. Legal and compliance costs for contract documentation, and market risk costs due to potential unfavorable movements in interest rates.
D. Brokerage and transaction fees for executing the swap, and premium costs for buying the derivative.

Answer: B

Step by step

1. A positive MtM means the swap is in the money to GFB: the counterparty owes GFB, so GFB holds the credit

exposure.
2. Only the uncollateralized part of that MtM is exposed, so counterparty risk (CVA-type) cost and a funding cost for carrying that unsecured receivable arise from the uncollateralized component.
3. For the collateralized part, the counterparty chooses the collateral type to post, so no further exposure cost arises there.
4. In any MtM scenario (positive or negative) there are also funding costs for the capital needed for the transaction and for initial margin.
5. Combining these gives option B.

Why B is correct: With positive MtM the uncollateralized portion is an unsecured claim on the counterparty, creating counterparty risk and funding costs, and capital and initial margin add funding costs in every case.

Why the others are wrong

- **A:** Reverses the direction: counterparty risk from one's own default and a funding benefit from uncollateralized MtM arise when MtM is negative (out of the money), not positive.
- **C:** Describes documentation and generic market risk; these are not the lifetime economic costs of an OTC derivative that depend on MtM and collateral in LO 33.g.
- **D:** Describes brokerage fees and an option premium; a swap has no upfront premium and these are not the economic costs identified for OTC derivatives.

Q709

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A derivatives risk manager at a bank reviews an interest rate swap whose mark-to-market (MtM) is currently negative (out of the money) for the bank. The counterparty does not post collateral. Which of the following best describes the economic costs and benefits the bank faces over the life of the swap?

A. Counterparty risk from the bank's own default and a funding benefit on the uncollateralized MtM, plus a funding cost for the capital and any initial margin.

B. Counterparty risk from the counterparty's default and a funding cost on the uncollateralized MtM, plus the funding cost of the capital held.

C. No counterparty risk and no funding impact, because a negative MtM means the bank cannot lose on the counterparty's default.

D. Counterparty risk from the bank's own default and a funding cost on the uncollateralized MtM, but no cost for capital or initial margin.

Answer: A

Step by step

1. Negative MtM: the bank owes the counterparty, so risk arises from the bank's own default.
2. The counterparty is uncollateralized, so the bank effectively gets a funding benefit.
3. Regardless of sign, funding the capital held and any initial margin posted is a cost.

Why A is correct: For an out-of-the-money position there is counterparty risk from one's own default and a potential funding benefit to the extent the counterparty is uncollateralized; in all cases there is a cost of funding capital and initial margin.

Why the others are wrong

- **B:** Describes the in-the-money case (risk from the counterparty's default, funding cost).
- **C:** Own-default counterparty risk and funding effects still exist, and capital/margin costs apply.
- **D:** Gets the funding effect backwards and omits the overall capital and initial margin costs that apply whatever the MtM sign.

Q710

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A bank enters a centrally cleared swap that requires it to post initial margin, and the swap's MtM is subsequently close to zero. A trainee argues that the bank has no economic costs on the position because the MtM is neither positive nor negative. Which of the following best evaluates the trainee's claim?

A. The claim is correct, because economic costs of a derivative arise only when MtM is positive.

B. The claim is incorrect, because initial margin represents over-collateralization that carries a funding cost, and funding the capital held against the transaction is a cost regardless of MtM sign.

C. The claim is incorrect, because posting initial margin creates a funding benefit that offsets the capital cost to zero.

D. The claim is correct, because initial margin is returned immediately and so carries no funding or capital implications.

Answer: B

Step by step

1. Initial margin is over-collateralization, funded at a cost.
2. Capital held against the transaction also has a funding cost whether MtM is positive or negative.

Why B is correct: Whether the portfolio has positive or negative value, there are costs from funding the capital held against the transaction and the initial margin posted.

Why the others are wrong

- **A:** Costs arise in both positive and negative MtM states, and margin/capital costs persist regardless.
- **C:** Initial margin is a funding cost, not a benefit; a funding benefit arises only when OTM and the counterparty is uncollateralized.
- **D:** Initial margin is held for the life of the trade and is a funding cost.

LO 33.h

Explain the components of the X-Value Adjustment (xVA) term.

Q711

AnalystPrep Q.4868

REBUILT — Original asked for an 'average funding cost' recovery (not a GARP/Schweser xVA definition) with a trivial 'marking to market' distractor; rebuilt to test the components of FVA.

A bank's valuation of a derivative includes a funding value adjustment (FVA). Which of the following best describes FVA?

A. It is the sum of a funding cost adjustment (FCA) and a funding benefit adjustment (FBA), so it can be either a cost or a benefit.

B. It is the cost of posting initial margin over the lifetime of the transaction.

C. It is always a cost, because funding a derivative can never generate a benefit.

D. It is the cost of holding regulatory capital over the lifetime of the transaction.

Answer: A

Step by step

1. FVA defines the cost and benefit arising from funding a transaction.
2. It is divided into two terms: funding cost adjustment (FCA) and funding benefit adjustment (FBA), so FVA = FCA + FBA.
3. A funded positive-MtM position creates a cost; a negative-MtM position against an uncollateralised counterparty can create a funding benefit.
4. Therefore FVA can be a cost or a benefit (answer A).

Why A is correct: FVA captures both the funding costs and the funding benefits of a transaction (FCA plus FBA), so its sign depends on the position.

Why the others are wrong

- **B:** This is the definition of MVA (margin value adjustment), not FVA.
- **C:** Ignores the funding benefit adjustment (FBA); xVA terms in general can be benefits, and FVA explicitly has a benefit leg.
- **D:** This is the definition of KVA (capital value adjustment).

Q712

AnalystPrep Q.5515

Global Finance Bank's CRO notes an adjustment to derivative valuation that reflects the cost differential arising from the bank's own financing requirements for its derivative positions, as opposed to assuming idealised, risk-free financing. Which xVA component is being described?

A. Funding Value Adjustment (FVA)

B. Credit Value Adjustment (CVA)

C. Capital Value Adjustment (KVA)

D. Margin Value Adjustment (MVA)

Answer: A

Step by step

1. The adjustment is driven by the bank's actual financing cost versus a risk-free funding assumption.
2. The xVA term defined as the cost and benefit resulting from funding a transaction is FVA.
3. CVA concerns counterparty default, KVA the cost of capital, MVA the cost of posting initial margin; none is about funding cost differentials.
4. Answer: A.

Why A is correct: FVA is by definition the cost (and benefit) of funding a transaction, relative to a funding-neutral base valuation.

Why the others are wrong

- **B:** CVA values the counterparty's default risk, not financing costs.
- **C:** KVA is the cost of holding regulatory capital over the life of the trade.
- **D:** MVA is the cost of posting initial margin; it is a margin item, not the general funding cost differential.

Q713

AnalystPrep Q.5516

REBUILT — Stem said 'collateral', which GARP assigns to ColVA when about embedded options; reworded to initial margin (MVA) and tightened the option set.

New regulations require Global Finance Bank to post initial margin on its OTC derivatives, which over-collateralises the positions. The Chief Trader wants a valuation adjustment that captures the cost of posting this initial margin over the lifetime of the transactions. Which xVA component is being referred to?

A. Margin Value Adjustment (MVA)
B. Credit Value Adjustment (CVA)
C. Funding Value Adjustment (FVA)
D. Capital Value Adjustment (KVA)

Answer: A

Step by step

1. The cost described is that of posting initial margin over the transaction's life.
2. The xVA term defined as the cost of posting initial margin over the lifetime of the transaction is MVA.
3. CVA is default risk, FVA general funding, KVA capital; none is specific to initial margin.
4. Answer: A.

Why A is correct: MVA is defined as the cost of posting initial margin over the lifetime of a transaction.

Why the others are wrong

- **B:** CVA prices the counterparty's default risk, not margin costs.
- **C:** FVA covers funding costs and benefits in general; the initial margin cost is separately captured by MVA.
- **D:** KVA is the cost of holding regulatory capital, not of posting margin.

Q714

AnalystPrep Q.5517

REBUILT — Verbose stem tightened and the non-syllabus 'Liquidity Value Adjustment' distractor replaced with ColVA.

Under Basel III, increased counterparty credit risk in stressed markets raises the regulatory capital Global Finance Bank must hold against its derivative portfolio. Which xVA component should the bank include to reflect the cost of holding this capital over the lifetime of the transactions?

A. Margin Value Adjustment (MVA)
B. Credit Value Adjustment (CVA)
C. Capital Value Adjustment (KVA)
D. Collateral Value Adjustment (ColVA)

Answer: C

Step by step

1. The cost in question is the cost of holding (regulatory) capital over the transaction's life.
2. This is the definition of KVA.
3. MVA is initial margin cost, CVA counterparty default risk, ColVA optionality and non-standard terms in collateral agreements.
4. Answer: C.

Why C is correct: KVA defines the cost of holding capital, typically regulatory capital, over the lifetime of a transaction.

Why the others are wrong

- **A:** MVA is the cost of posting initial margin, not of holding capital.
- **B:** CVA prices the counterparty default risk itself; the cost of the capital held against it is a separate adjustment (KVA).
- **D:** ColVA reflects costs/benefits of embedded options and non-standard terms in collateral agreements, not capital.

Q715

BT answers P2.T6.902.2 · BT notes P2.T6.902.2 · BT Mock Q57

REBUILT — Removed dependence on the missing figure by describing the CVA desk set-up in the stem; merged three identical copies.

A bank's CVA desk (market maker) enters an uncollateralised trade with a client and hedges it with a collateralised trade in the market. The client trade has a positive mark-to-market (MTM) value to the bank, so the collateralised hedge has a negative MTM. Market moves now make the client trade's positive MTM even larger. Which of the following implications is most likely TRUE?

A. Funding cost (FVA) is positive.
B. From the trader's (market maker's) perspective, CVA decreases.
C. Because the position is hedged, there is no associated regulatory cost of capital to the bank.
D. The trader (market maker) writes a contingent credit default swap (CCDS) to transfer its counterparty risk.

Answer: A

Step by step

1. The hedge has a more negative MTM, so the bank must post more collateral on it.
2. The client trade is uncollateralised, so the bank receives no collateral against its larger receivable; the extra collateral posted must be funded, giving a funding cost (positive FVA).
3. Larger positive exposure to the client raises, not lowers, counterparty risk, so CVA increases.
4. The collateralised hedge does not offset the counterparty risk (and its capital) on the uncollateralised client leg, and to transfer that risk the desk would buy, not write, a CCDS.

Why A is correct: The collateral posted on the growing negative hedge MTM has no offsetting collateral received on the uncollateralised client leg, so the bank incurs a funding cost.

Why the others are wrong

- **B:** Exposure to the client has increased, so the expected loss from client default (CVA) increases, not decreases.
- **C:** The hedge removes market risk but not the counterparty credit risk (and the associated regulatory capital) on the uncollateralised leg.
- **D:** To transfer counterparty risk the desk would buy protection via a CCDS; writing one would add risk.

Q716

BT answers P2.T6.900.3 · BT notes P2.T6.900.3

REBUILT — Original correct option listed 'liquidity' (LVA), which is not in the GARP/Schweser xVA list, and option D had unrecoverable market-share figures; rebuilt as a clean test of the xVA component definitions.

Which of the following statements about xVA terms is TRUE?

A. ColVA is the cost of posting initial margin over the lifetime of a transaction.
B. DVA (OVA) represents counterparty risk from the point of view of the counterparty's default.
C. KVA is the cost of holding capital, typically regulatory capital, over the lifetime of a transaction.
D. FVA captures only funding costs and never any funding benefits.

Answer: C

Step by step

1. Recall the definitions: CVA/DVA = counterparty risk (counterparty default and own default), FVA = funding cost and benefit, ColVA = embedded collateral optionality and non-standard terms, KVA = cost of holding capital, MVA = cost of posting initial margin.
2. A mixes up ColVA and MVA; B reverses whose default DVA reflects; D ignores FBA.
3. Only C matches the definition of KVA.

Why C is correct: KVA is defined as the cost of holding (typically regulatory) capital over the transaction's lifetime.

Why the others are wrong

- **A:** That is MVA; ColVA covers costs/benefits from collateral optionality (e.g. choice of currency or type of collateral) and non-standard collateral terms.
- **B:** DVA reflects counterparty risk from the party's own default point of view, not the counterparty's default (that is CVA).

- **D:** FVA = FCA + FBA; it includes funding benefits.

Q717 BT answers P2.T6.902.3 · BT notes P2.T6.902.3

Peter, a risk analyst, has determined that the basic valuation (no counterparty risk, funding, or capital effects) of his firm's uncollateralized derivative position is $50.0 million. He must now adjust for the xVA terms, each stated as a positive amount and described as follows:
- CVA = $5.0 million (cost)
- DVA = $2.0 million (benefit)
- FVA = $1.0 million unsecured funding cost (not a funding benefit)
- ColVA = 0 (not applicable because the contract is uncollateralized)
- KVA = $4.0 million (cost)
- MVA = $3.0 million (cost)

What is the correct final valuation?

A. $35.0 million
B. $39.0 million
C. $47.0 million
D. $55.0 million

Answer: B

Step by step

1. Costs reduce value and benefits increase it: CVA, FVA (here a cost), KVA and MVA are costs; DVA is a benefit (own default risk); ColVA = 0.
2. Total costs = $5.0 + 1.0 + 4.0 + 3.0 = 13.0$ million.
3. Total benefit = DVA = 2.0 million.
4. Final value = $50.0 - 13.0 + 2.0 = 39.0$ million.

Why B is correct: xVA terms adjust the basic valuation: counterparty risk (CVA), funding cost (FVA), capital (KVA) and margin (MVA) are deducted, while DVA, the party's own credit risk, is added as a benefit.

Why the others are wrong

- **A:** $50 - 5 - 2 - 1 - 4 - 3 = 35$: treats DVA as a cost and subtracts it instead of adding it.
- **C:** Applies only CVA and DVA ($50 - 5 + 2 = 47$) and ignores the FVA, KVA and MVA costs.
- **D:** Adds CVA instead of deducting it ($50 + 5 = 55$) and ignores DVA, FVA, KVA and MVA; costs reduce value.

Reading 34. Netting, Close-Out, and Related Aspects

Jon Gregory, The xVA Challenge: Counterparty Credit Risk, Funding, Collateral, and Capital, 4th Edition (West Sussex, UK: John Wiley & Sons, 2020). Chapter 6.

LO	Learning objective	Qs
34.a	explain the purpose of an International Swaps and Derivatives Association (ISDA) master agreement.	5
34.b	summarize netting and close-out procedures (including multilateral netting), explain their advantages and disadvantages, and describe how they fit into the framework of the ISDA master agreement.	14
34.c	describe the effectiveness of netting in reducing credit exposure under various scenarios.	24
34.d	describe the mechanics of termination provisions and trade compressions and explain their advantages and disadvantages.	4
34.e	provide examples of trade compression of derivative positions, calculate net notional exposure amount, and identify the party holding the net contract position in a trade compression.	8
34.f	identify and describe termination events and discuss their potential effects on parties to a transaction.	4

LO 34.a

Explain the purpose of an International Swaps and Derivatives Association (ISDA) master agreement.

Q718

AnalystPrep Q.1876

REBUILT — Options A and B were both defensible (B is a subset of A) and C/D were giveaways; reworded so exactly one option fully describes the ISDA purpose and each distractor reflects a realistic misconception.

Which of the following best describes the primary purpose of an ISDA Master Agreement in over-the-counter (OTC) derivatives transactions?

A. It provides a standardized legal framework between two counterparties that sets general terms on netting, collateral, events of default, and termination, reducing legal uncertainty and credit risk.

B. It sets the price, notional, and maturity of each individual derivative trade between the counterparties.

C. It is a regulatory rule issued by authorities that caps the size of OTC derivative positions each party may hold.

D. It is a separate contract for each transaction that is renegotiated from scratch whenever a new trade is added.

Answer: A

Step by step

1. The ISDA Master Agreement standardizes OTC documentation between two counterparties.
2. It forms a single legal framework (indefinite term) covering many transactions.
3. Its terms address collateral, netting, close-out, events of default and termination events.
4. The result is less legal uncertainty and lower credit risk; this matches A.

Why A is correct: The Master Agreement is a standardized bilateral framework for all trades between two parties, so netting, collateral and default/termination terms are agreed once and legal uncertainty falls.

Why the others are wrong

- **B:** Confuses the Master Agreement with a trade confirmation; individual trade economics (price, notional, maturity) are in confirmations.
- **C:** Treats ISDA as a regulator; it is a private industry-standard contract, not a position-limit rule.
- **D:** Reverses the purpose: the Master Agreement is one framework covering multiple trades, so new trades do not require renegotiating it.

Q719

AnalystPrep Q.6149

A financial institution is running workshops on the ISDA Master Agreement for its trading and risk teams. Which of the following statements best describes the ISDA Master Agreement?

A. It is a regulatory guideline that dictates the trading limits for derivatives.

B. It is a standardized contract that provides the terms for individual derivative trades.

C. It is a bilateral agreement framework governing OTC derivatives transactions between parties.

D. It is a mandatory legal document imposed by international trading commissions.

Answer: C

Step by step

1. Identify the nature of the document: a private, standardized, bilateral contract.
2. It governs all OTC derivative transactions between the two parties under one framework.
3. It is voluntary and is not a regulation or limit-setting rule.
4. Only C matches.

Why C is correct: The ISDA Master Agreement is a bilateral framework that sets standard terms (netting, collateral, default and termination) for multiple OTC trades between two parties.

Why the others are wrong

- **A:** Mistakes a private contract for regulation; it sets no trading limits.
- **B:** Too narrow: it is a framework for many trades, not terms for individual trades (those are in confirmations).
- **D:** It is voluntary and privately negotiated, not imposed by any commission.

Q720 AnalystPrep Q.6150

A financial institution is reviewing its derivatives contracts and wants its legal team to understand the essence of the ISDA Master Agreement. Which of the following statements best captures it?

A. It primarily serves as a database for recording all global derivatives transactions.

B. It is an individual contract tailored for single transactional use in derivatives trading.

C. It is a framework agreement that sets the standard terms for multiple OTC derivatives transactions between parties.

D. It is an international regulatory mandate that all derivatives traders must adhere to.

Answer: C

Step by step

1. The ISDA Master Agreement is a single legal contract between two counterparties.
2. It standardizes terms (netting, collateral, termination) for all OTC trades between them.
3. Hence it is a framework for multiple transactions: C.

Why C is correct: One master contract with standard terms covers every OTC trade between the same two parties, reducing legal uncertainty and credit risk.

Why the others are wrong

- **A:** Confuses it with a trade repository; ISDA Master is a contract, not a record database.
- **B:** It is not single-use; its purpose is to cover many trades under one agreement.
- **D:** It is voluntary and private, not a regulatory mandate.

Q721 BT answers P2.T6.323.1 · BT notes P2.T6.323.1

Which of the following is TRUE about an ISDA Master Agreement?

A. Both counterparties must be domiciled in the United States, as the Master Agreement is not enforceable in other jurisdictions

B. All transactions done under a Master Agreement between two counterparties are treated as a single legal whole with a single net value

C. Each additional trade between counterparties generally requires an update to the Master Agreement

D. A Master Agreement does mitigate market risk via the mark-to-market (MtM) provision but mitigates neither counterparty credit risk nor legal risk

Answer: B

Step by step

1. The Master Agreement forms one legal contract covering all trades between the two parties.
2. Under close-out netting, all trades are valued and combined into a single net amount.
3. A is false (ISDA has legal opinions in many jurisdictions); C is false (new trades just get confirmations under the existing agreement); D is false (it mitigates both legal and credit risk).
4. Answer B.

Why B is correct: A single legal agreement means all trades form one netting set, so on default only one net value is owed.

Why the others are wrong

- **A:** False: the agreement is internationally used, with legal opinions supporting netting in many jurisdictions.

- **C:** False: the framework is indefinite-term; each new trade needs only a confirmation.
- **D:** False: it is designed to reduce legal uncertainty and credit risk (via netting and collateral); it is not a market-risk tool.

Q722 BT answers P2.T6.903.1 · BT notes P2.T6.903.1

Becky's firm, Zapstreet International, is considering entering a set of significant OTC derivative trades with a single institution (the Counterparty). The Counterparty has proposed that the trades be governed by the bilateral framework of an International Swaps and Derivatives Association (ISDA) Master Agreement. Which of the following statements is TRUE?

A. Although the ISDA Master Agreement will mitigate legal risk, it cannot be expected to mitigate counterparty risk

B. If an individual transaction produces a trade confirmation, then the trade confirmation will nullify the Master Agreement

C. A key drawback of the ISDA Master Agreement is that its terms are "hard-wired" and not adjustable even if Zapstreet and the Counterparty both agree

D. The Master Agreement will include contractual terms regarding the posting of collateral, transactions referenced into single netting set(s), close-out process mechanics, and definitions of events of default

Answer: D

Step by step

1. Recall the contents of the Master Agreement: collateral terms, netting, close-out, events of default and termination events.
2. A is false: netting and collateral reduce counterparty risk.
3. B is false: confirmations supplement, not nullify, the Master Agreement.
4. C is false: parties customize via the Schedule.
5. Answer D.

Why D is correct: The Master Agreement standardizes collateral, netting, close-out and default definitions in one framework.

Why the others are wrong

- **A:** It mitigates credit risk too, via close-out netting and collateral provisions.
- **B:** Confirmations are part of the single agreement and sit under the Master; they do not cancel it.
- **C:** Terms can be tailored by mutual agreement through the Schedule and additional provisions.

LO 34.b

Summarize netting and close-out procedures (including multilateral netting), explain their advantages and disadvantages, and describe how they fit into the framework of the ISDA master agreement.

Q723 BT answers P2.T6.408.2 · BT notes P2.T6.408.1

Each of the following statements about a "netting set" is true EXCEPT:

A. A netting set is a set of trades that can be legally netted together in the event of a default

B. Within a netting set, expected exposure (EE) and credit value adjustment (CVA) are additive

C. Across netting sets, exposure is additive, whereas within a netting set the MtM values of trades can be added

D. A netting set may consist of a single trade, and a counterparty may have more than one netting set

Answer: B

Step by step

1. Netting set = trades that can be legally netted on default (A true).
2. Within a set, MtM values offset, so the set has one net exposure $\max(\sum \text{MtM}, 0)$; exposure is not the sum of trade-level positive exposures.
3. Because of this offsetting, trade-level EE (and hence CVA) are not additive within a netting set; additivity applies across netting sets.
4. So B is the false statement.

Why B is correct: Netting makes exposure non-additive within a netting set (only net MtM matters), while separate

netting sets do not offset and their exposures add.

Why the others are wrong

- **A:** True: this is the definition of a netting set.
- **C:** True: no offset across sets (additive), while MtM values are summed within a set.
- **D:** True: a single trade can form its own netting set, and several sets can exist per counterparty.

Q724 AnalystPrep Q.1877

What is the main purpose of close-out netting in derivatives trading?

A. To net cash flows falling due on the same day

B. To terminate all agreements between an insolvent counterparty and a solvent one and offset the transaction values into a single net amount

C. To net cash flows falling due after a specified number of days

D. To terminate all agreements between two parties by compensating only the non-defaulting party while penalising the defaulter

Answer: B

Step by step

1. Close-out netting applies on a counterparty's default/insolvency.
2. All transactions are terminated and valued, and the values are offset into one sum owed by one party.
3. This matches B.

Why B is correct: Close-out netting combines the right to terminate all contracts on default with the right to offset their values into a single claim.

Why the others are wrong

- **A:** Describes payment netting (same-day cash flows), not close-out netting.
- **C:** Also a cash-flow timing concept; close-out is triggered by default, not by a number of days.
- **D:** Close-out is a calculation of a net claim, not a punitive or one-sided compensation mechanism.

Q725 AnalystPrep Q.1878

An institution must pay a 205m floating swap payment to a counterparty on day Y and is due to receive a 200m fixed payment from the same counterparty on the same day. Under payment netting, the institution makes a single net payment of:

A. 5m

B. 405m

C. 200m

D. 205m

Answer: A

Step by step

1. Payment netting combines same-day cash flows between the parties.
2. Net = 205 - 200 = 5m payable by the institution.

Why A is correct: Payment netting reduces settlement risk by exchanging only the difference between offsetting payments.

Why the others are wrong

- **B:** Adds the two payments (205 + 200) instead of offsetting them.
- **C:** Pays only the fixed amount received, ignoring the floating leg.
- **D:** Pays the gross floating amount, i.e. no netting applied.

Q726 AnalystPrep Q.1882

Without netting, if a counterparty defaults the loss is the sum of the positive mark-to-market values of dealings with that counterparty. This means that:

A. derivatives with negative values must be settled, whereas those with positive values are only a claim in the bankruptcy process

B. derivatives with both positive and negative values must be claimed through the bankruptcy process

C. derivatives with negative values must be hedged afresh

D. derivatives with positive values must be liquidated immediately

Answer: A

Step by step

1. Without netting, the surviving party still owes the full amount on negative-MtM trades.
2. On positive-MtM trades it can only file a claim and recover a fraction.
3. Hence exposure is the sum of positives (additive).

Why A is correct: Absent netting, the defaulter's estate collects what it is owed while the survivor merely becomes an unsecured creditor on its gains.

Why the others are wrong

- **B:** Only positive-value trades are claims; the survivor must still pay on negative ones.
- **C:** Re-hedging is a market-risk action, not what the lack of netting implies for settlement.
- **D:** Nothing forces immediate liquidation of positive-value trades.

Q727 AnalystPrep Q.5508

Global Finance Bank has many OTC derivative transactions with various counterparties under ISDA Master Agreements. Which statement best summarises the role of netting under these agreements?

A. It mainly streamlines tax reporting and simplifies the legal aspects of cross-border trades

B. It combines cash flows from different contracts into a single net amount, reducing settlement risk and improving operational efficiency

C. It manages only market risk, aligning the risk profiles of the bank and its counterparties

D. It lets the bank pursue proprietary strategies without disclosure, preserving confidentiality

Answer: B

Step by step

1. Netting (set-off) combines cash flows from different contracts into one net amount.
2. This reduces settlement risk and improves operational efficiency; close-out netting does the same for contract values on default.

Why B is correct: Netting replaces many gross payments or claims with one net amount, cutting settlement and credit risk and operational burden.

Why the others are wrong

- **A:** Tax and cross-border legal simplification are not the purpose of netting.
- **C:** Netting mitigates credit/settlement exposure, not only market risk.
- **D:** Confidentiality is not a benefit; multilateral netting in fact requires trade disclosure.

Q728 AnalystPrep Q.5509

One of Global Finance Bank's derivatives counterparties defaults. Which statement best explains the role of close-out netting under the ISDA Master Agreement?

A. It reallocates the defaulted trades to other counterparties to maintain market stability

B. It accelerates all future payments due from the defaulting counterparty to give immediate liquidity

C. It allows the bank to terminate all outstanding transactions with the defaulter and net them into a single claim or obligation

D. It automatically transfers all of the bank's positions with the defaulter to a central clearinghouse

Answer: C

Step by step

1. On default the non-defaulting party terminates all transactions under the agreement.
2. Values are offset into one net sum: a claim if positive, an obligation if negative.

Why C is correct: Close-out netting terminates all contracts and offsets their values into one net amount, unlike acceleration which only makes future payments immediately due.

Why the others are wrong

- **A:** Close-out is a bilateral termination process; it does not reallocate trades.
- **B:** Describes an acceleration clause, which the reading distinguishes from close-out.
- **D:** Transfer to a clearinghouse is clearing/multilateral netting, not an automatic result of close-out.

Q729 BT answers P2.T6.24.25.2

REBUILT — Option D (governing law and jurisdiction) is arguably also true of an ISDA Master Agreement, giving two defensible answers; replaced D with a clearly false statement.

Globex Corporation, a multinational company, has entered into several OTC interest rate swaps with various banks to hedge its interest rate exposure. To ensure legal certainty and manage credit risk, Globex requires each counterparty bank to sign an ISDA Master Agreement before any derivative transaction is executed.

Which of the following statements accurately describes a key risk-mitigating feature of the ISDA Master Agreement in this scenario?

A. The ISDA Master Agreement eliminates the need for Globex to post collateral or margin on its OTC derivative transactions with the banks.

B. If a counterparty bank defaults, the ISDA Master Agreement allows Globex to close out and net all outstanding transactions with that bank into a single payment obligation, reducing its credit exposure.

C. The ISDA Master Agreement standardizes the pricing and valuation models used for the swaps, so that every transaction with every bank is valued consistently.

D. The ISDA Master Agreement removes the non-defaulting party's right to terminate transactions early, so that all swaps continue to maturity after a default.

Answer: B

Step by step

1. The ISDA Master Agreement forms a single legal contract covering all transactions with a counterparty, with netting and close-out provisions.
2. On a counterparty default, the non-defaulting party may terminate all transactions and offset positive and negative values into one net amount.
3. Globex's claim is therefore limited to the net amount rather than the sum of the positive-value trades, which reduces credit exposure.
4. Collateral terms (CSA), pricing and early-termination rights are not eliminated or standardized by the master agreement, so A, C and D fail.

Why B is correct: Close-out netting under the master agreement converts many trades into one net claim at default, so exposure is the net, not the gross, amount.

Why the others are wrong

- **A:** The master agreement provides a framework for collateral (via the credit support annex); it does not remove the need for it. Confuses having a collateral framework with not needing one.
- **C:** ISDA standardizes legal terms, not pricing or valuation models; each party values trades with its own models. Confuses legal with economic standardization.
- **D:** The agreement does the opposite: it gives the non-defaulting party the right to terminate (close out) all transactions on default.

Q730 BT answers P2.T6.324.2 · BT notes P2.T6.324.2

About netting and close-out, each of the following is true EXCEPT, which is false?

A. Netting is the right to offset amounts due at termination of individual contracts between the same counterparties when determining the final obligation.

B. Netting can increase the exposure to a given counterparty.

C. A disadvantage of multilateral netting is that it tends to mutualize and homogenize counterparty risk, creating a reduced incentive for institutions to scrutinize counterparty credit quality.

D. Close-out is the right of a counterparty to terminate contracts unilaterally under certain specified conditions.

Answer: B

Step by step

1. Check A: netting is the right to offset (net) amounts due at termination into a single sum, so A is true.
2. Check B: netting either reduces exposure or leaves it unchanged (when all positions are same-direction), but it can never increase it; B is false.
3. Check C: multilateral netting mutualizes counterparty risk and weakens each member's incentive to monitor others' credit quality, so C is true.
4. Check D: close-out is the right of one side to terminate contracts unilaterally under specified conditions, so D is true.

Why B is correct: Exposure with netting is max(net value, 0), which is never greater than the sum of positive values without netting.

Why the others are wrong

- **A:** True; it is the standard definition of the netting right, so it cannot be the exception.
- **C:** True; mutualization is the recognized drawback of multilateral netting, so choosing it misreads the EXCEPT format.
- **D:** True; this is the definition of close-out, so it is not the false statement.

Q731 BT answers P2.T6.903.3 · BT notes P2.T6.903.3

REBUILT — Original answer option cited a non-existent 'Product Coverage Method' (giveaway) and an arguable 'best approach' claim; rebuilt around the payment netting vs close-out netting distinction.

In regard to netting, each of the following statements is true EXCEPT, which is false?

A. Payment netting reduces pre-settlement risk by netting the mark-to-market values of contracts.
B. Payment netting reduces settlement risk by netting cash flows or obligations.
C. Close-out netting reduces pre-settlement risk by netting, in general, mark-to-market values.
D. Set-off rights involve the potential to offset or net obligations between two parties across different products.

Answer: A

Step by step

1. Payment netting combines cash flows due on the same date into one payment; it addresses settlement (and operational) risk, so B is true.
2. Close-out netting applies on default and nets contract mark-to-market values into one amount; it addresses pre-settlement risk, so C is true.
3. Set-off allows a party to apply amounts owed to it against amounts it owes, potentially across different agreements and products, so D is true.
4. A assigns the close-out netting role (pre-settlement, MtM values) to payment netting, so A is false.

Why A is correct: Payment netting nets cash flows and reduces settlement risk; only close-out netting nets MtM values to reduce pre-settlement risk.

Why the others are wrong

- **B:** True statement of payment netting's purpose; picking it ignores the EXCEPT format.
- **C:** True statement of close-out netting; it is the correct pairing of mechanism and risk.
- **D:** True; set-off can extend to offsetting obligations across other agreements and product types.

Q732 Schweser Module Quiz 34.1 Q1

Riggs Resources, LLC (Riggs) is a commodity trading firm with numerous trades outstanding with several counterparties. It is concerned with presettlement risk, the risk that a counterparty defaults before settlement. To reduce presettlement risk, it would be most beneficial for Riggs to:

A. have payment netting.
B. have close-out netting.
C. analyze potential losses as the sum of exposures.
D. have netting but not set-off.

Answer: B

Step by step

1. Presettlement risk is the loss from a counterparty defaulting before the trades settle.
2. Close-out netting terminates all contracts with the defaulted counterparty and nets their values into one amount, directly limiting that loss.
3. Payment netting only nets same-day cash flows, so it addresses settlement and operational risk instead.
4. Summing exposures ignores offsets, so it does not reduce risk.

Why B is correct: Close-out netting reduces the claim on a defaulted counterparty to the net MtM value, which is what limits presettlement loss.

Why the others are wrong

- **A:** Payment netting reduces settlement and operational risk, not presettlement risk; confuses cash flow netting with MtM netting.
- **C:** Summing exposures is the no-netting approach and overstates risk; netting makes exposures non-additive.
- **D:** Netting without set-off still does not single out close-out netting, the mechanism that nets mark-to-market values on default; set-off is a related right to offset obligations, not something to give up to reduce presettlement risk.

Q733 Schweser Module Quiz 34.1 Q2

Entity XYZ is netting its trades with Entity ABC. Which of the following techniques best describes this type of netting arrangement?

A. Multilateral netting.
B. Bilateral netting.
C. Close-out netting.
D. Additive exposure netting.

Answer: B

Step by step

1. Only two entities (XYZ and ABC) are involved in the netting arrangement.
2. Netting between two parties is bilateral netting.
3. Multilateral netting requires three or more parties, usually through a central entity such as a clearinghouse.

Why B is correct: A netting arrangement limited to two counterparties is, by definition, bilateral netting.

Why the others are wrong

- **A:** Multilateral netting involves multiple counterparties, typically via a clearinghouse; only two entities are present here.
- **C:** Close-out netting refers to netting contract values when the counterparty defaults; it is not defined by the number of parties.
- **D:** This is not a netting technique; without netting, exposures are additive.

Q734 Schweser Module Quiz 34.1 Q3

Assume the following current MtM values for five different transactions with the same counterparty, Entity ABC: +5, -4, +2, +3, and -6. What are the total exposures with and without netting, respectively?

A. 0, 10.
B. 20, 10.
C. 10, 0.
D. 10, 20.

Answer: A

Step by step

1. With netting: sum of MtMs = 5 - 4 + 2 + 3 - 6 = 0, and exposure = max(0, 0) = 0.
2. Without netting: only positive values count, so 5 + 2 + 3 = 10.
3. Answer, in the order requested (with, without): 0 and 10.

Why A is correct: Netting lets negative-value trades offset positive ones, so exposure falls from the sum of positives (10) to max(net, 0) = 0.

Why the others are wrong

- **B:** 20 sums the absolute values of all trades (5+4+2+3+6) and treats that as the netted figure; netting offsets rather than adds negatives.
- **C:** Reverses the order: 10 is the exposure without netting and 0 is with netting.
- **D:** Reverses the order for 10 and uses 20 (sum of absolute values) for without netting; negative-value trades are not exposures.

Q735 BT answers P2.T6.904.1 · BT notes P2.T6.904.1

REBUILT — Option B's figure was OCR-uncertain and could coincide with the true empirical value; fixed it to a clearly wrong 10% reduction and sharpened options C and D so exactly one statement is true.

According to Gregory, close-out netting is widely used in the over-the-counter (OTC) derivatives market. Which of the following statements about the impact of netting is TRUE?

A. Netted positions are inherently more volatile than their underlying gross positions, which can create systemic risk
B. Close-out netting reduces exposure in the OTC derivatives market by only about 10%
C. Close-out netting is a risk mitigant but has not had a material effect on the growth of the OTC derivatives market
D. When a counterparty is in distress, netting increases market participants' concern, which increases systemic risk

Answer: A

Step by step

1. Gregory: netting reduces overall credit exposure by close to 90% (not 10%), so B is false.
2. Netting is the single biggest counterparty risk mitigant and was critical to growth of the OTC market, so C is false.
3. With netting, institutions worry less about a distressed counterparty when net MtM is negative, which reduces systemic risk, so D is false.
4. Netted positions are smaller but inherently more volatile than gross positions (a net can swing from zero to large), which can create systemic risk: A is true.

Why A is correct: Netting shrinks exposure but a net position moves more erratically than the underlying gross positions, which is a recognised drawback.

Why the others are wrong

- **B:** Understates the effect: netting currently reduces exposure by close to 90%, not 10%.
- **C:** Gets the effect backwards: close-out netting is described as the biggest risk mitigant and critical to market growth.
- **D:** Reverses the direction: netting reduces concern about a distressed counterparty (little worry when current MtM is negative), reducing rather than increasing systemic risk.

Q736 AnalystPrep Q.1892

Some OTC derivatives contain walkaway features, which allow an institution to terminate transactions if the counterparty defaults. Such clauses do not reduce credit exposure, but they allow the institution to:

A. cease payments and avoid settling the amounts it owes the defaulting counterparty
B. terminate the contract and gain from the replacement cost estimated before the transaction
C. cease payments but pay only the net amount to the counterparty
D. cease payments and institute legal proceedings against the counterparty

Answer: A

Step by step

1. A walkaway clause lets the survivor walk away from its net liabilities to a defaulted counterparty.
2. It still claims on positive MtM exposure, so it gains where it would otherwise have paid.
3. It does not reduce exposure; it adds potential gains. Hence A.

Why A is correct: Walkaway clauses let the solvent party avoid paying net liabilities to the defaulter while still claiming on positive exposure.

Why the others are wrong

- **B:** No gain arises from a pre-transaction replacement cost estimate.
- **C:** Paying a net amount is ordinary close-out netting, not a walkaway.
- **D:** Legal proceedings are not a feature or benefit of the clause.

LO 34.c

Describe the effectiveness of netting in reducing credit exposure under various scenarios.

Q737 AP Mock 1 Q18

REBUILT — Original was a vague conceptual item with a loosely worded correct option; rebuilt as a numerical netting-vs-no-netting exposure calculation.

Global Finance Bank has three OTC derivative trades with one counterparty, all covered by an ISDA Master Agreement with enforceable close-out netting. The trades have mark-to-market values of +40, -25 and -5 (in millions). If the counterparty defaults, what is the bank's credit exposure with and without netting?

A. With netting 10; without netting 40
B. With netting 40; without netting 10
C. With netting 15; without netting 70
D. With netting 0; without netting 40

Answer: A

Step by step

1. With enforceable netting the trades form one claim: $\max(40 - 25 - 5, 0) = 10$.
2. Without netting, exposure is the sum of the positive MtM values only: $40 + 0 + 0 = 40$.
3. So netting reduces exposure from 40 to 10.

Why A is correct: Close-out netting collapses all trades into one net amount, so negative-MtM trades offset positive ones; without it only positives count and exposure is additive.

Why the others are wrong

- **B:** Reverses the two cases; netting can never increase exposure.
- **C:** 15 ignores the third trade (40-25); 70 sums absolute values, wrongly counting negative trades as exposure.
- **D:** Floors the net at zero incorrectly (the net is +10, not 0); the no-netting figure is right but the netted one is wrong.

Q738

AnalystPrep Q.1879

REBUILT — Original asked which multi-trade situation is 'highly unlikely' with a weak, questionable key; rebuilt as a numerical hedge/unwind netting example from the reading.

An entity has two equal and opposite trades with the same counterparty (an original trade and a reversing unwind) with mark-to-market values of +10 and -10. The counterparty defaults. Which statement about the entity's exposure is correct?

A. Without netting exposure is 10; with netting it is 0
B. Without netting exposure is 0; with netting it is 0
C. Without netting exposure is 10; with netting it is 10
D. Without netting exposure is 0; with netting it is 10

Answer: A

Step by step

1. Without netting the entity must settle the -10 trade but can only claim on the +10 trade in bankruptcy, so exposure is the sum of positive MtMs = 10.
2. With netting the values offset: $\max(10 - 10, 0) = 0$.

Why A is correct: Netting lets equal and opposite trades cancel, whereas without it exposure is the additive sum of positive MtM values.

Why the others are wrong

- **B:** Assumes the offset happens even without a netting agreement.
- **C:** Assumes netting gives no benefit; but it removes the exposure on offsetting trades.
- **D:** Reverses the effect; netting can never increase exposure.

Q739

AP Mock 1 Q35

A Spanish bank has the following derivative positions with a local company:

Position	Value (USD)
Long currency swaps	+33 million
Short credit default swaps	-22 million
Long interest rate swaps	+8 million
Short swaptions	-17 million

In the event that the counterparty defaults, what would be the loss to the financial institution if netting is used compared with the loss without netting?

A. With netting USD 2m; without netting USD 2m.
B. With netting USD 2m; without netting USD 41m.
C. With netting USD 41m; without netting USD 2m.
D. With netting USD 29m; without netting USD 41m.

Answer: B

Step by step

1. Without netting: sum of positive MtMs = $33 + 8 = 41$ m (negative trades are still payable and do not reduce the loss).
2. With netting: net value = $33 - 22 + 8 - 17 = 2$ m.
3. Exposure with netting = $\max(2, 0) = 2$ m.
4. Answer: 2m with netting vs 41m without (benefit of 39m).

Why B is correct: Under an enforceable netting agreement the loss is the positive part of the net close-out value; without netting the bank loses on every positive trade while still owing on the negative ones, so exposures add.

Why the others are wrong

- **A:** Treats both cases as the net value; ignores that without netting the negative trades cannot offset the positives.
- **C:** Reverses the two cases.
- **D:** 29 is not the netted value (33 - 22 + 8 - 17 = 2m); the net figure and the gross figure are both misstated.

Q740

AP Mock 2 Q44

ATC has nine contracts with a counterparty, all transacted in Sydney. Their current market values are:
120, 85, -65, -50, 80, -90, 75, 10, -140
ATC does not currently have a legally binding netting agreement. By how much would ATC's current credit exposure improve if it had a legally enforceable netting agreement?

A. 0.
B. 25.
C. 345.
D. 330.

Answer: C

Step by step

1. Without netting: $120 + 85 + 80 + 75 + 10 = 370$.
2. Net sum: $370 - (65 + 50 + 90 + 140) = 370 - 345 = 25$.
3. With netting: $\max(25, 0) = 25$.
4. Improvement = $370 - 25 = 345$.

Why C is correct: Netting replaces the sum of positive values with the positive part of the net sum, so the improvement is gross positive exposure minus net exposure.

Why the others are wrong

- **A:** Assumes netting gives no improvement.
- **B:** Reports the exposure remaining after netting (25), not the improvement.
- **D:** Arithmetic slip in the subtraction; not the difference between gross (370) and net (25) exposure.

Q741

AnalystPrep Q.1865

REBUILT — Source option C reported -$4.0m as netted 'exposure' (exposure cannot be negative); options rebuilt so netted exposure is zero.

A bank has four derivatives trades with a counterparty. From the bank's perspective, the mark-to-market (MtM) values are: -$20.0 million, -$15.0 million, +$14.0 million and +$17.0 million. The bank and the counterparty have an enforceable netting agreement. What is the bank's exposure (i) without netting and (ii) with netting?

A. $31.0 m without netting; zero with netting
B. $31.0 m without netting; $4.0 m with netting
C. $35.0 m without netting; zero with netting
D. $17.0 m without netting; zero with netting

Answer: A

Key correction: Source key (AnalystPrep) said C (-$4.0m with netting); exposure is floored at zero, so the correct netted exposure is 0.

Step by step

1. Without netting: $14 + 17 = 31$ m.
2. Net value: $-20 - 15 + 14 + 17 = -4$ m.
3. With netting: $\max(-4, 0) = 0$ (the bank owes a net 4m, so it has no credit exposure).

Why A is correct: Exposure is the positive part of what the counterparty owes: gross it is the sum of positive MtMs; netted it is the positive part of the net sum, which is zero when the net value is negative.

Why the others are wrong

- **B:** Uses the absolute net value 4 as exposure; a negative net MtM means no exposure.
- **C:** Sums the negative MtMs (35), which are the bank's liabilities, not its exposure.
- **D:** Counts only the single largest positive trade instead of summing all positive trades.

Q742 BT answers P2.T6.319.2 · BT notes P2.T6.319.2

A bank has positions in four derivatives trades with a counterparty. From the bank's perspective, the mark-to-market (MtM) values are: -$10.0 million, -$6.0 million, +$4.0 million and +$9.0 million. The bank and the counterparty have a netting agreement. What is the bank's exposure (i) without netting and (ii) with netting?

A. zero without netting; -$3.0 MM with netting
B. $13.0 MM without netting; zero with netting
C. $16.0 MM without netting; $3.0 MM with netting
D. $9.0 MM without netting; $3.0 MM with netting

Answer: B

Step by step

1. Without netting: $4 + 9 = 13$ MM.
2. Net value: $-10 - 6 + 4 + 9 = -3$ MM.
3. With netting: $\max(0, -3) = 0$.

Why B is correct: Without netting only positive MtMs count and they add; with netting the trades are summed first and exposure is the positive part of the total, here zero.

Why the others are wrong

- **A:** Reverses the logic: gross exposure is not zero, and exposure cannot be negative.
- **C:** 16 is the sum of the negative MtMs (10+6), and 3 is the absolute net value.
- **D:** Counts only the largest positive trade (9) and uses the absolute value of the net (3).

Q743 AnalystPrep Q.1880

Assume the following current mark-to-market values exist for Entity K in seven different transactions: +7, +5, +8, -7, -4, -2, -6. What is the total exposure, with and without netting, respectively?

A. 19, 1
B. 1, 19
C. 20, 19
D. 1, 20

Answer: D

Step by step

1. With netting: $7 + 5 + 8 - 7 - 4 - 2 - 6 = 1$, so exposure $\max(1, 0) = 1$.
2. Without netting: $7 + 5 + 8 = 20$.
3. Answer: 1 with, 20 without.

Why D is correct: With netting the values are summed (exposures are non-additive); without netting only positive values count.

Why the others are wrong

- **A:** 19 is the total of the negative MtMs, not an exposure; the order is also wrong.
- **B:** Puts the netted figure first, and 19 (sum of negatives) is not gross exposure.
- **C:** Quotes 20 as the netted figure and 19 (sum of negatives) as the gross; both are wrong.

Q744 AnalystPrep Q.1883

Consider five different transactions with counterparty B with MtM values of +4 million, +7 million, +2 million, -5 million and -3 million. What is the total exposure with and without netting?

A. +$13 million with netting and +$5 million without netting
B. +$13 million without netting and +$5 million with netting
C. +$7 million without netting and +$2 million with netting
D. +$7 million without netting and -$3 million with netting

Answer: B

Step by step

1. Without netting: $4 + 7 + 2 = 13$ m.
2. Net: $13 - 5 - 3 = 5$ m, $\max(5, 0) = 5$ m with netting.

Why B is correct: Gross exposure sums positive MtMs; netted exposure is the positive part of the total.

Why the others are wrong

- **A:** Swaps the two cases.
- **C:** Uses the single largest (7) and smallest (2) positive trades instead of summing and netting.
- **D:** Uses the largest trade and the smallest negative trade; exposure also cannot be negative.

Q745

AnalystPrep Q.1885

REBUILT — Original option D was malformed and the options were giveaways; rewrote options with a realistic overstatement distractor.

Without netting, the current size and liquidity of the OTC derivatives market would be unlikely to exist. Which statement best explains why?

A. Netting means overall credit exposure in the market grows at a lower rate than the notional growth of the market itself.

B. Netting means overall credit exposure in the market grows at a higher rate than the notional growth of the market.

C. Netting means overall credit exposure grows at the same rate as the notional growth of the market.

D. Netting eliminates all counterparty credit exposure, so market size is no longer limited by credit risk.

Answer: A

Step by step

1. Netting combines trade values with a counterparty into one net amount at close-out.
2. Net exposure is at most gross exposure, so exposure grows more slowly than notional.
3. Hence market size and liquidity could expand without credit exposure becoming prohibitive.

Why A is correct: Close-out netting makes exposures non-additive, so credit exposure scales more slowly than notional and counterparty risk stays manageable as volumes grow; netting is also recognised in capital rules.

Why the others are wrong

- **B:** Netting can only reduce or leave unchanged exposure, never raise its growth rate.
- **C:** Would imply no benefit from netting; exposure is reduced relative to notional.
- **D:** Overstates the benefit: netting reduces but does not eliminate exposure (a positive net amount can remain).

Q746

AnalystPrep Q.1941

From the following table, calculate the netting benefit, defined as expected exposure (EE) without netting minus EE with netting (EE = average exposure over the five equally likely scenarios).

Scenario	Trade 1 Future Value	Trade 2 Future Value	Total Exposure No Netting	Total Exposure Netting
Scenario 1	30	10	40	40
Scenario 2	20	5	25	25
Scenario 3	5	-5	5	0
Scenario 4	-25	-25	0	0
Scenario 5	-15	-15	0	0

A. 5

B. 1

C. 1.33

D. 2.5

Answer: B

Step by step

1. EE without netting = $(40 + 25 + 5 + 0 + 0)/5 = 14$.
2. EE with netting = $(40 + 25 + 0 + 0 + 0)/5 = 13$.
3. Benefit = $14 - 13 = 1$.

Why B is correct: Netting only helps in scenarios where trades have opposite signs (Scenario 3); the benefit is that scenario's reduction of 5 averaged over five scenarios.

Why the others are wrong

- **A:** Total reduction in Scenario 3 (5), not averaged over the five scenarios.
- **C:** Not supported by any correct step; an incorrect averaging of the reduction of 5.
- **D:** Divides the reduction of 5 by 2 (number of trades) instead of 5 (number of scenarios).

Q747 AnalystPrep Q.1942

REBUILT — Original interpretation options were ambiguous (positive vs negative values); rewrote to test the offsetting mechanism with one defensible answer.

From the table, find the netting benefit (EE without netting minus EE with netting, averaging over the five equally likely scenarios) and its correct interpretation.

Scenario	Trade 1 Future Value	Trade 2 Future Value	Total Exposure No Netting	Total Exposure Netting
Scenario 1	-30	35	35	5
Scenario 2	-25	15	15	0
Scenario 3	-15	5	5	0
Scenario 4	-25	-25	0	0
Scenario 5	-15	-15	0	0

A. 10; the benefit arises in scenarios where the trades have opposite signs, so the negative trade offsets the positive one

B. 5; the benefit arises in scenarios where the trades have opposite signs, so the negative trade offsets the positive one

C. 10; the benefit arises in scenarios where both trades are negative

D. 11; netting removes all exposure in every scenario

Answer: A

Step by step

1. EE without netting = $(35 + 15 + 5 + 0 + 0)/5 = 11$.
2. EE with netting = $(5 + 0 + 0 + 0 + 0)/5 = 1$.
3. Benefit = $11 - 1 = 10$, arising in Scenarios 1-3 where Trade 1 is negative and Trade 2 is positive.

Why A is correct: Netting offsets a positive trade value against a negative one; where both trades are negative there is no exposure to reduce and no benefit.

Why the others are wrong

- **B:** 5 is the total netted exposure (not averaged) and is not the benefit.
- **C:** Right number, wrong reason: in Scenarios 4-5 both trades are negative, exposure is already zero.
- **D:** 11 is only EE without netting; ignores the 1 of exposure remaining after netting (Scenario 1 still nets to 5).

Q748 AnalystPrep Q.3078

ABT Holdings is a large commercial bank that has three uncollateralized transactions with a counterparty worth +$10 million, +$30 million, and −$25 million. What will be the bank's exposure on the transactions if they are regarded as independent transactions and if there is a netting agreement in place?

A. $15 million if independent and $15 million if netting is in place

B. $40 million if independent and $15 million if netting is in place

C. $40 million if independent and $40 million if netting is in place

D. $65 million if independent and $40 million if netting is in place

Answer: B

Step by step

1. Independent (no netting): each trade's exposure is $\max(V_i, 0)$: $10 + 30 + 0 = 40$.
2. With netting: exposure is $\max(\sum V_i, 0) = \max(10 + 30 - 25, 0) = 15$.
3. Exposure without netting is $40 million; with netting it is $15 million.

Why B is correct: Without netting, the negative trade cannot offset the positive ones, so exposure is the sum of positive MtMs. With a legally enforceable netting agreement, values are combined into one net claim, so the −25 offsets part of the +40.

Why the others are wrong

- **A:** Treats the independent case as if netting applied (10 + 30 − 25 = 15), ignoring that without netting the negative trade is floored at zero.
- **C:** Assumes netting gives no benefit, i.e. also counts only the positive trades when netting is in place.
- **D:** Sums absolute values (10 + 30 + 25 = 65) for the independent case, counting the negative trade as exposure, and uses 40 for netting.

Q749 AnalystPrep Q.5434

REBUILT — Original distractors (1.5, 2.0, 1.0) were not tied to any realistic mistake and no recovery assumption was stated; replaced with error-based options and stated zero recovery.

ABC Bank has four open derivative positions with a hedge fund, all covered by a legally enforceable netting agreement. Assume no collateral and zero recovery. Current exposures (from ABC's perspective) are:

Position	Exposure ($)
Long futures contracts	−1,500,000
Long currency derivatives	−11,000,000
Long credit default swaps	3,500,000
Long swaptions	12,500,000

What is the loss to ABC Bank using netting if the hedge fund defaults?

A. $3.5 million
B. $5.0 million
C. $12.5 million
D. $16.0 million

Answer: A

Step by step

1. Net value: $-1.5 - 11.0 + 3.5 + 12.5 = 3.5$ million.
2. Net exposure is positive, so the loss on default is $\max(3.5, 0) = 3.5$ million.

Why A is correct: With close-out netting only the net amount is claimed: the positive trades (16.0) are offset by the negative trades (−12.5), leaving a claim of 3.5 million.

Why the others are wrong

- **B:** Forgets the −1.5 million futures position (3.5 + 12.5 − 11.0 = 5.0).
- **C:** Takes the largest single positive exposure (swaptions) and ignores both the CDS and all negative positions.
- **D:** Adds only the positive exposures (3.5 + 12.5 = 16.0), which is the loss with no netting.

Q750 BT Mock Q62 · BT answers P2.T6.903.2 · BT notes P2.T6.903.2

REBUILT — BT answers copy had OCR-damaged numbers (72 for 12) and the notes copy defined the benefit in reverse order; merged three versions, restored 12, fixed the definition and replaced the distractor 3.

Acme Financial has five derivative contracts with a counterparty, all in a single netting set. The current mark-to-market values are +12, −9, +20, −6 and +8 (in millions). Define the netting benefit as the current exposure as if netting did not apply minus the current exposure with netting. What is the netting benefit?

A. 0
B. 15
C. 25
D. 40

Answer: B

Step by step

1. Exposure without netting: $12 + 0 + 20 + 0 + 8 = 40$.
2. Exposure with netting: $\max(12 - 9 + 20 - 6 + 8, 0) = 25$.
3. Benefit $= 40 - 25 = 15$, which equals the sum of the negative values $9 + 6$ because the net stays positive.

Why B is correct: Netting lets the negative MtMs (−15 in total) offset positive ones; since the net is still positive, the full 15 is saved.

Why the others are wrong

- **A:** Ignores that negative trades offset positives under netting.
- **C:** This is the exposure with netting, not the reduction.
- **D:** This is the exposure without netting, not the reduction.

Q751 BT answers P2.T6.319.1 · BT notes P2.T6.319.1

REBUILT — BT answers copy listed $7.0 million twice as options; used one clean option set and replaced the unexplained $1.0 million option.

A bank has multiple derivative positions with the same counterparty XYZ under a legally binding bilateral netting agreement. The sum of the positions with positive value to the bank is +$7.0 million; the sum of those with negative value is −$5.0 million. Compared with the bank's exposure without netting, what is the reduction in exposure due

to netting?

A. $2.0 million
B. $5.0 million
C. $7.0 million
D. $12.0 million

Answer: B

Step by step

1. Without netting: exposure = 7.0 million (only positives count).
2. With netting: max(7.0 − 5.0, 0) = 2.0 million.
3. Reduction = 7.0 − 2.0 = 5.0 million.

Why B is correct: Netting removes the offsetting negative value (5.0) from the exposure while the net remains positive, so the gain equals 5.0.

Why the others are wrong

- **A:** Reports the exposure remaining after netting rather than the reduction.
- **C:** Reports the exposure without netting rather than the reduction.
- **D:** Adds the absolute values of the positive and negative sums (7 + 5), which is not an exposure measure.

Q752

BT answers P2.T6.324.1 · BT notes P2.T6.323.3

REBUILT — Stem carried OCR damage and confusing extra remarks about correlation; rewrote as a clean scenario table with the benefit defined as the reduction in expected exposure.

The table shows four equally likely scenarios and the mark-to-market (MtM) values of two trades with the same counterparty, covered by a netting agreement.

Scenario	Trade 1	Trade 2
Scenario 1	−20	+25
Scenario 2	−25	+15
Scenario 3	−15	−5
Scenario 4	−25	−15

The expected exposure without netting is (25 + 15 + 0 + 0)/4 = 10. What is the netting benefit to the expected exposure (expected exposure without netting minus expected exposure with netting)?

A. 0
B. 1.25
C. 8.75
D. 10.0

Answer: C

Step by step

1. Exposure with netting per scenario $\max(V_1 + V_2, 0)$: S1 $\max(5, 0) = 5$; S2 $\max(-10, 0) = 0$; S3 0; S4 0.
2. Expected exposure with netting = $(5 + 0 + 0 + 0)/4 = 1.25$.
3. Benefit = $10 - 1.25 = 8.75$.

Why C is correct: Netting only leaves exposure in Scenario 1, where the +25 exceeds the −20 by 5; in all other scenarios the net is negative and exposure is zero.

Why the others are wrong

- **A:** Assumes netting gives no reduction.
- **B:** This is the expected exposure with netting, not the reduction.
- **D:** This is the expected exposure without netting; it would be the benefit only if netting eliminated all exposure.

Q753

BT answers P2.T6.324.3 · BT notes P2.T6.324.2

REBUILT — Original omitted that only the sign (not size) of the initial MtM differs between options, making the MtM comparison ambiguous; added equal-size/equal-volatility assumption and reasoned from the netting-benefit formula.

Acme Bank has two trades with the same counterparty under a netting agreement and simulates scenarios for each trade's mark-to-market (MtM). Both trades have the same MtM volatility, and the initial MtM of each trade has the same absolute size in every case below (only its sign differs). The netting benefit is the expected exposure without netting (average of the sum of the positive MtMs) minus the expected exposure with netting (average of the positive part of the summed MtM). Under which set of assumptions will the netting benefit be GREATEST?

A. Negative initial MtM and positive correlation between the trades
B. Positive initial MtM and positive correlation between the trades
C. Negative initial MtM and zero correlation between the trades
D. Positive initial MtM and negative correlation between the trades

Answer: D
Step by step
1. Netting benefit $= E[X^+] + E[Y^+] - E[(X+Y)^+]$, where X, Y are the two trade MtMs and $x^+ = \max(x, 0)$.
2. Use the identity $x^+ = x + (-x)^+$: the benefit is unchanged if both MtMs flip sign, so the sign of the initial MtM does not matter (only its size). Options A and B are therefore equal.
3. Correlation drives the benefit: the lower the correlation, the more often one trade is in-the-money when the other is out-of-the-money, so the sum is pulled further below the sum of positives.
4. Illustration (normal MtMs, mean $\pm 1\sigma$, correlations +0.5 / 0 / -0.5, benefit in units of σ): +0.5 gives 0.060, 0 gives 0.116, -0.5 gives 0.158.
5. The greatest benefit comes with negative correlation (D).

Why D is correct: Netting only helps when positive and negative MtMs offset; negative correlation maximises offsetting, and the sign of the initial MtM is irrelevant because the benefit is symmetric under flipping all signs.

Why the others are wrong
- **A:** Positive correlation means both trades tend to be in or out of the money together, so there is little offsetting (benefit about 0.06); a negative initial MtM does not change this.
- **B:** Same benefit as A (the sign of initial MtM is irrelevant) and positive correlation gives the smallest netting benefit.
- **C:** Zero correlation gives a moderate benefit (about 0.12); the error is thinking a negative initial MtM adds benefit, when only the correlation differs from D.

Q754

BT answers P2.T6.408.1 · BT notes P2.T6.408.1

Consider five different transactions with a particular counterparty with current mark-to-market (MtM) values of +5, -3, +4, +6, -2. What is the total exposure, respectively, with netting and without netting?
A. +3 (with netting) and +10 (without netting)
B. +9 (with netting) and +10 (without netting)
C. +10 (with netting) and +15 (without netting)
D. +15 (with netting) and +20 (without netting)

Answer: C
Step by step
1. With netting: exposure $= \max(5 - 3 + 4 + 6 - 2,\ 0) = \max(10, 0) = 10$.
2. Without netting: exposure $= \sum \max(\text{MtM}_i, 0) = 5 + 0 + 4 + 6 + 0 = 15$.
3. Netting benefit = 15 - 10 = 5 (the sum of the negative MtMs in absolute value).

Why C is correct: Netted exposure is the positive part of the net MtM (10); gross exposure ignores negative trades and sums only positives (15).

Why the others are wrong
- **A:** Neither number is correct: 3 is not the net sum (10), and 10 is the net figure wrongly used as the no-netting exposure.
- **B:** Neither number comes from the correct sums: 9 is not the net value, and 10 (the net) is shown as the gross exposure.
- **D:** Shows the gross exposure (15) as the netted figure and 20 (sum of absolute values 5+3+4+6+2) as gross, treating negative MtMs as exposure.

Q755

BT notes P2.T6.320.3

REBUILT — Original left unclear which swap had +V and made 'either counterparty defaults' ambiguous; specified the swap with A has MtM +V and rewrote options so exactly one is correct.

Financial institution XYZ has an interest rate swap with Counterparty A and hedges it with an offsetting swap with a different Counterparty B. At a future time T the swap with A has mark-to-market +$V to XYZ, so the swap with B has mark-to-market -$V. There is no netting agreement between A and B (they are different counterparties). Which statement is TRUE if there is a default at time T?

A. If only B defaults, XYZ loses $V
B. If only A defaults, XYZ loses nothing
C. If both A and B default, XYZ loses $V
D. If both A and B default, XYZ loses $2V

Answer: C
Step by step
1. Exposure to A = $\max(+V, 0) = V$; exposure to B = $\max(-V, 0) = 0$.
2. Only A defaults: loss = V. Only B defaults: loss = 0 (XYZ owes B and has no claim on it).
3. Both default: loss = $V + 0 = V$; the negative leg creates no claim on B and no offset against A across different counterparties.
4. Therefore C is correct.

Why C is correct: Netting works only within a single counterparty; across A and B, the loss is the sum of each counterparty's exposure max(MtM,0), giving V + 0 = V.

Why the others are wrong
- **A:** Confuses which swap is in-the-money: B owes XYZ nothing at -V, so B's default costs nothing.
- **B:** Assumes the hedge offsets the loss; but the swap with A is a real receivable and A's default loses V (the hedge is with a different counterparty, so there is no netting).
- **D:** Doubles the exposure by counting the absolute value of both legs (V + V); the out-of-the-money leg produces no loss.

Q756

GARP PE1 Q43 · Mock 3 Q37

REBUILT — Options A and D were identical in the source; replaced D with a distinct plausible wrong option (28/44).

A financial institution has four open derivative positions with an investment company, all under one enforceable netting agreement. Current market values:

Position	Exposure (USD)
Long swaptions	32 million
Long credit default swaps	12 million
Long currency derivatives	-16 million
Long futures contracts	-8 million

If the investment company defaults, what is the institution's loss if netting is used compared with the loss if netting is not used?

A. Loss of USD 20 million if netting is used; loss of USD 24 million if netting is not used
B. Loss of USD 20 million if netting is used; loss of USD 44 million if netting is not used
C. Loss of USD 24 million if netting is used; loss of USD 32 million if netting is not used
D. Loss of USD 28 million if netting is used; loss of USD 44 million if netting is not used

Answer: B
Step by step
1. With netting: $\max(32 + 12 - 16 - 8,\ 0) = 20$ million.
2. Without netting: $\sum \max(\text{MtM}_i, 0) = 32 + 12 + 0 + 0 = 44$ million.
3. Netting benefit = 44 - 20 = 24 million = |-16| + |-8|.

Why B is correct: Netting lets the negative marks (-24) offset the positive marks (+44), leaving 20; without it, only positive-MtM positions count, giving 44.

Why the others are wrong
- **A:** Uses the netting benefit (24) as the no-netting loss instead of the gross exposure of positives (44).
- **C:** Uses the sum of negatives (24) as the netted loss and the largest single position (32) as the gross loss.
- **D:** Nets only the currency derivative (44 - 16 = 28) instead of all negative positions; the gross figure 44 is right but the netted figure is wrong.

Q757

Mock 1 Q45

REBUILT — Source table gave inconsistent (non-mirror) bilateral data between counterparties; rebuilt as one bank with three trades per counterparty under netting agreements.

A bank has three trades with each of four counterparties, each counterparty covered by a legally enforceable close-out netting agreement. Current MtM to the bank (USD millions):

Counterparty	Trade 1	Trade 2	Trade 3
A	+12	-4	+3
B	+9	+7	-12
C	+6	-1	+2
D	+5	-10	+2

Rank the counterparties by the bank's credit exposure with netting, from highest to lowest.

A. B, A, C, D
B. A, B, C, D
C. A, C, B, D
D. B, D, A, C

Answer: C

Step by step

1. Exposure with netting = $\max(\sum \text{MtM}, 0)$ per counterparty.
2. A: 12-4+3 = 11. B: 9+7-12 = 4. C: 6-1+2 = 7. D: 5-10+2 = -3, floored at 0.
3. Ranking highest to lowest: A (11), C (7), B (4), D (0).

Why C is correct: Netted exposure is computed counterparty by counterparty as the floored sum of MtMs, giving A 11 > C 7 > B 4 > D 0.

Why the others are wrong

- **A:** Ranks by gross exposure (sum of positive MtMs: B 16, A 15, C 8, D 7), ignoring netting.
- **B:** Ranks by largest single positive trade (12, 9, 6, 5), ignoring other trades and netting.
- **D:** Ranks by netting benefit (gross minus net: B 12, D 7, A 4, C 1) instead of the remaining exposure.

Q758

Mock 2 Q36

Barkon, Ltd. holds long positions in an interest rate swap portfolio with seven transactions, all with the same counterparty, with the following mark-to-market (MtM) values:
+8, -3, +2, +6, -4, -1, +5
What is Barkon's total exposure to the counterparty without netting and with netting, respectively?

A. 8 and 13.
B. 8 and 21.
C. 13 and 8.
D. 21 and 13.

Answer: D

Step by step

1. Without netting, exposure is additive over the positive MtM values only: $8 + 2 + 6 + 5 = 21$.
2. With a legally enforceable netting agreement, the MtMs are summed first: $8 - 3 + 2 + 6 - 4 - 1 + 5 = 13$.
3. Net exposure = $\max(13, 0) = 13$, which is below the gross figure of 21 (netting can reduce exposure but never increase it).
4. Without netting 21, with netting 13.

Why D is correct: Without netting only positive MtMs count and they add up; with netting the positive and negative MtMs offset, leaving the (positive) net sum.

Why the others are wrong

- **A:** Uses the single largest MtM (8) as the non-netted exposure and 13 for the netted figure; the numbers are in the wrong order and 8 is not the gross exposure (the sum of positive MtMs is 21).
- **B:** Uses 8 (the largest position) for the non-netted case and 21 (the gross figure) for the netted case, so both numbers are misassigned.
- **C:** Reverses the order: 13 is the netted figure and 21 the gross figure. Netting can never produce a larger exposure than no netting.

Q759

Mock 2 Q57

REBUILT — Original asked for an undefined 'netting benefit' in currency units; rebuilt using the GARP/BIS definition (1 minus netted exposure over gross exposure) and made the option set consistent.

Kirkland, Co. is drafting a netting agreement with its counterparty Shanty, Inc. and wants an exhibit showing the benefit of netting for two trades under four scenarios. The MtM values of the two trades are:

Scenario	Trade 1 MtM	Trade 2 MtM
Scenario 1	18	4
Scenario 2	12	-3
Scenario 3	8	-5
Scenario 4	-4	7

Gross exposure in a scenario is the sum of the positive MtMs; netted exposure is the larger of the net sum and zero. Defining the netting benefit as 1 minus (average netted exposure / average gross exposure) across the four scenarios, the netting benefit is closest to:

A. 24.5%
B. 32.4%
C. 36.2%
D. 75.5%

Answer: A

Step by step

1. Gross exposure per scenario (sum of positive MtMs): 22, 12, 8, 7; average $= 49/4 = 12.25$.
2. Netted exposure per scenario max(sum, 0): 22, 9, 3, 3; average $= 37/4 = 9.25$.
3. Netting benefit $= 1 - 9.25/12.25 = 1 - 0.755 = 24.5\%$.
4. The max operator is applied scenario by scenario before averaging.

Why A is correct: Netting benefit is the proportional reduction in expected exposure, 1 - net/gross, computed after applying the exposure floor in each scenario.

Why the others are wrong

- **B:** Divides the average absolute reduction (3.00) by the average netted exposure (9.25) instead of the average gross exposure (12.25).
- **C:** Averages the four scenario-level ratios (0%, 25%, 62.5%, 57.1%) instead of dividing average netted by average gross exposure.
- **D:** This is net/gross (9.25/12.25), the fraction of exposure that remains; the benefit is 1 minus this.

Q760 Schweser Module Quiz 34.1 Q4

Which of the following trading instruments would have the most beneficial effect on netting?

A. Options with up-front premiums.
B. Equity options.
C. FX options.
D. Futures.

Answer: D

Step by step

1. An instrument helps netting only if its MtM can be negative at some point in its life, so that it can offset positive MtMs of other trades.
2. Options with up-front premiums (equity options, FX options, swaptions, caps, floors) have MtM that is never negative for the holder.
3. Futures can move to a negative MtM, so they can offset positive exposure under a netting agreement.
4. Futures give the most beneficial effect.

Why D is correct: Netting benefit comes from negative MtM values offsetting positive ones; only futures among the choices can have negative MtM.

Why the others are wrong

- **A:** Premium-paid options have only non-negative MtM, so they have nothing to offset other positive exposures.
- **B:** Equity options with up-front premiums only have positive MtM, so they add exposure without offsetting it.
- **C:** FX options with up-front premiums likewise cannot have negative MtM; they are among the least helpful for netting.

LO 34.d

Describe the mechanics of termination provisions and trade compressions and explain their advantages and disadvantages.

Q761 AnalystPrep Q.1888

Which of the following best describes the consequence of exercising a break clause in a contract?

A. The defaulting party will have authority to terminate the transaction at its current replacement value.
B. The exercising party will have authority to terminate the transaction at its pre-defined replacement value.
C. The exercising party will have authority to trade the new transaction at a pre-defined replacement value.
D. The exercising party will have the authority to terminate the transaction at its current replacement value.

Answer: D

Step by step

1. A break clause (liquidity put or early termination option) lets a party terminate a transaction at specified future dates.
2. Termination is settled at the transaction's replacement value, i.e., its market value at the time of termination, not a value fixed in advance.
3. The right belongs to the party exercising the clause (often either party, since clauses are usually bilateral), not specifically to a defaulting party.
4. Only D combines the exercising party with the current replacement value.

Why D is correct: A break clause gives the exercising party the option to close the trade at its prevailing replacement (market) value, so it can exit without cost when the counterparty weakens.

Why the others are wrong

- **A:** Gives the right to the defaulting party; the clause is for the party worried about the counterparty's credit quality, not a right reserved to the defaulter.
- **B:** Uses a pre-defined value; the settlement is at the replacement value at the time of termination, so a value fixed at inception is wrong.
- **C:** Describes entering a new trade at a fixed price; a break clause terminates the existing trade and does not create a new one.

Q762 AnalystPrep Q.1889

A mandatory break clause implies that:

A. the transaction will certainly terminate at the date of the break clause regardless of the financial situations of both parties.
B. the transaction will not terminate at the date of the break clause regardless of the situations of both parties.
C. both parties have the option to terminate the trade transaction depending on the creditworthiness of each party.
D. the transaction will certainly terminate at the date of the break clause but give the defaulting party an option to accept it or reject it.

Answer: A

Step by step

1. Break clauses fall into three types: mandatory, optional and trigger-based.
2. Mandatory: the transaction terminates on the break-clause date with no discretion and no trigger.
3. Optional: one or both parties may choose to terminate at the pre-specified date.
4. Trigger-based: an event such as a ratings downgrade must occur before the clause can be exercised.
5. The description of the mandatory type matches A.

Why A is correct: A mandatory break clause terminates the trade automatically on the specified date, irrespective of either party's credit condition.

Why the others are wrong

- **B:** States the opposite; the trade does terminate on the date.
- **C:** Describes an optional break clause, where a party chooses whether to terminate.
- **D:** Adds an accept/reject right for the defaulting party; a mandatory clause gives no such choice.

Q763 BT answers P2.T6.408.3 · BT notes P2.T6.408.3

REBUILT — Original options contained garbled wording and claims outside the Schweser LO (rating-transition probabilities, multilateral netting mutualisation); rebuilt as a clean EXCEPT question on break clauses, walkaway, reset and trade compression.

Which of the following statements about termination features and trade compression is NOT correct?

A. A break clause is most useful if exercised early, before the counterparty's credit quality deteriorates substan-

tially, but firms often avoid early exercise to preserve relationships with counterparties.

B. A walkaway clause allows a party to avoid its net liabilities to a defaulted counterparty while still claiming if its own MtM exposure is positive, and it is criticised for creating moral hazard.

C. Trade compression reduces the gross notional and the number of trades, but it can do so only by changing the institution's overall risk profile.

D. A reset agreement resets a trade that is heavily in the money back to at the money, for example by exchanging the MtM value on each reset date.

Answer: C

Step by step

1. A: this is the banker's paradox: a break clause should be exercised early, but entities avoid early exercise to keep good relationships. True.
2. B: walkaway clauses let a party escape net liabilities to a defaulter while claiming on positive MtM; they are criticised for moral hazard and higher costs for the defaulter. True.
3. C: trade compression nets redundant trades across counterparties to reduce gross notional and trade count while leaving the overall risk profile unchanged. The statement says the opposite, so it is false.
4. D: a reset agreement resets heavily in-the-money trades to at the money, e.g., a resettable cross-currency swap exchanges MtM and resets the FX rate at each reset date. True.

Why C is correct: The defining feature of trade compression is that it cuts gross notional and the number of trades without changing the institution's overall risk profile, so C is the incorrect statement.

Why the others are wrong

- **A:** A correct statement of the banker's paradox, so it is not the exception.
- **B:** A correct description of a walkaway clause and its criticism, so it is not the exception.
- **D:** A correct description of a reset agreement, so it is not the exception.

Q764

BT answers P2.T6.904.3 · BT notes P2.T6.904.3

REBUILT — Original relied on unspecified modelling-complexity and monoline remarks and had two near-identical rating-linked options (ATE vs break clause, which Schweser treats as the same); rebuilt with clearer requirements and distinct options.

Doubledown Financial is entering a long-dated cross-currency swap under an ISDA Master Agreement. Its Board wants (1) to limit its credit exposure if the swap becomes deeply in the money, (2) the swap to stay in place for its full life as a hedge, and (3) no reliance on credit-rating triggers, because ratings lag credit deterioration and downgrades can cause sudden cliff-edge effects. Which of the following features is BEST suited to the Board's requirements?

A. A walkaway clause

B. A reset agreement

C. A trigger-based break clause linked to a rating downgrade

D. A mandatory break clause on a fixed future date

Answer: B

Step by step

1. Requirement 1: need a feature that reduces exposure as MtM becomes large. A reset agreement resets a heavily in-the-money trade to at the money by exchanging the MtM (and, for a cross-currency swap, resetting the FX rate to spot).
2. Requirement 2: after a reset the trade continues, so the hedge stays in place.
3. Requirement 3: resets are tied to payment dates or a market-value breach, not to credit ratings, so there is no lagging-indicator or cliff-edge dependence.
4. A fails requirement 1, C fails requirement 3, D fails requirement 2, so B is best.

Why B is correct: A reset agreement removes accumulated MtM exposure while keeping the trade alive and uses dates or market-value triggers rather than ratings.

Why the others are wrong

- **A:** A walkaway clause only lets a party escape net liabilities when the counterparty defaults; it does not reduce credit exposure on an in-the-money swap.
- **C:** A rating-triggered break clause depends on ratings, which lag credit deterioration and can cause cliff-edge effects, and it terminates rather than preserves the hedge.
- **D:** A mandatory break clause terminates the swap on the fixed date regardless of conditions, ending the hedge and not managing exposure before that date.

LO 34.e

Provide examples of trade compression of derivative positions, calculate net notional exposure amount, and identify the party holding the net contract position in a trade compression.

Q765

AnalystPrep Q.5439

An analyst has the following information on a bank's derivatives trades. All trades are on the same reference credit and have the same maturity.

Counterparty	Reference Credit	Notional ($)	Bank's Position
4G Inc.	Blue Seal Int.	300,000	Long protection
Lights Advisors	Blue Seal Int.	800,000	Short protection
Air London	Blue Seal Int.	250,000	Short protection
Texas Electric Plc.	Blue Seal Int.	900,000	Long protection

What is the bank's net contract notional value after trade compression?

A. $1,200,000
B. $950,000
C. $150,000
D. $900,000

Answer: C

Step by step

1. Trades on the same reference credit and maturity can be compressed into one net contract.
2. Long protection: 300,000 + 900,000 = 1,200,000.
3. Short protection: 800,000 + 250,000 = 1,050,000.
4. Net = 1,200,000 - 1,050,000 = 150,000 long protection, replacing four trades with one.

Why C is correct: Compression nets all long against all short notionals on the same credit and maturity, so only the 150,000 difference remains, as a long protection position.

Why the others are wrong

- **A:** Gross long-protection notional (1,200,000); the short trades were ignored.
- **B:** 1,200,000 - 250,000: offset only the Air London short and ignored Lights Advisors.
- **D:** The largest single trade (Texas Electric, 900,000) rather than the net of all trades.

Q766

AnalystPrep Q.5511

REBUILT — Original left the identity of the net-contract holder to an unstated convention; the stem now states the convention (net contract booked with the counterparty with the largest existing position in the net direction) so exactly one option is correct.

Global Finance Bank (GFB) is compressing credit default swaps on the same reference entity and maturity (notionals in millions):

Reference Entity	Notional	Position	Counterparty
XYZ Index	300	Long	Counterparty A
XYZ Index	100	Short	Counterparty B
XYZ Index	150	Short	Counterparty C
XYZ Index	50	Long	Counterparty D

Assume the net contract is booked with the counterparty that already holds the largest position in the net direction. After compression, what is GFB's net notional exposure and which counterparty holds the net contract position?

A. 100 short, held by Counterparty C.
B. 100 long, held by Counterparty A.
C. 0, with no counterparty holding a net position.
D. 100 long, held by Counterparty D.

Answer: B

Key correction: Source key (A) matched the amount and holder under the largest-position convention, which the original did not state; it is now explicit.

Step by step

1. Long: 300 + 50 = 350. Short: 100 + 150 = 250.
2. Net = 350 - 250 = 100 long.

3. The net direction is long; the long counterparties are A (300) and D (50).
4. By the stated convention the largest long, Counterparty A, holds the single 100 long contract.

Why B is correct: Compression nets the long and short notionals to 100 long and replaces the four trades with one contract placed with the counterparty already holding the largest position in that direction.

Why the others are wrong

- **A:** Short is the smaller side (250 vs 350); the direction of the net was reversed.
- **C:** Assumes offsetting positions cancel completely; the notionals (350 vs 250) do not match.
- **D:** Right amount and direction, but Counterparty D holds only 50 long; A holds the largest long position.

Q767

AnalystPrep Q.5512

REBUILT — Original mixed EUR/USD conversion with a flawed 'largest position' holder rule and ambiguous options; rebuilt as a same-currency compression with one trade of a different maturity that cannot be compressed, with the holder convention stated.

A bank holds the following CDS positions on the same index (notionals in millions). Only trades with the same reference entity and maturity can be compressed together. The net contract is booked with the counterparty that already holds the largest position in the net direction.

Reference Entity	Maturity	Notional	Position	Counterparty
DEF Index	20 Dec 2030	500	Long	Counterparty I
DEF Index	20 Dec 2030	400	Short	Counterparty J
DEF Index	20 Dec 2030	300	Short	Counterparty K
DEF Index	20 Jun 2031	200	Long	Counterparty L

What is the net notional of the compressible (December 2030) positions and which counterparty holds the net contract?

A. 200 long, held by Counterparty I.
B. 0, because all long and short positions offset.
C. 200 short, held by Counterparty J.
D. 200 short, held by Counterparty K.

Answer: C

Step by step

1. Only the three December 2030 trades are compressible; the June 2031 trade (L, 200 long) has a different maturity and is left untouched.
2. Long: 500. Short: 400 + 300 = 700.
3. Net = 500 - 700 = -200, i.e. 200 short.
4. Largest short position is Counterparty J (400), so J holds the net 200 short contract.

Why C is correct: Netting the same-maturity trades leaves 200 short, placed with the counterparty holding the largest short (J, 400).

Why the others are wrong

- **A:** Subtracted in the wrong order (500 - 700) and labelled the result long; the net direction is short.
- **B:** Included the June 2031 long of 200 in the netting (500 + 200 - 700 = 0), although a different maturity cannot be compressed.
- **D:** Correct size and direction, but K's short (300) is smaller than J's (400), so J holds the net contract.

Q768

BT answers P2.T6.904.2 · BT notes P2.T6.904.2 · BT Mock Q67

REBUILT — Original ring legs were missing in one version, answer options had internally inconsistent totals and the Mock key (C) was wrong; legs restated, the rule (keep each party's net, no reversals, maximum reduction) made explicit, and options rebuilt with consistent totals.

Three counterparties A, B and C hold fungible positions in a ring. Notional sizes and directions: A → 50 → B, B → 75 → C, C → 60 → A. Counting each position from both parties' points of view (twice), the current total notional is 370. Trade compression must leave every party's net position unchanged, may not reverse the direction of any position, and should minimise total notional. Which is the efficient solution?

A. B → 75 → C and C → 60 → A (new total notional = 270).
B. B → 25 → C and C → 10 → A (new total notional = 70).
C. A → 25 → B, B → 50 → C and C → 35 → A (new total notional = 220).
D. B → 10 → A and B → 15 → C (new total notional = 50).

Answer: B

Key correction: Source key (BT Mock) said option C of its original list (B -> 35 -> C -> 30 -> A), which does not preserve net positions; BT answers/notes key (subtract the smallest leg, 50, giving B -> 25 -> C -> 10 -> A, total 70) is correct.

Step by step

1. Net positions (inflow - outflow): A = 60 - 50 = +10; B = 50 - 75 = -25; C = 75 - 60 = +15.
2. Reduce each ring leg by the smallest leg (50): A→B = 0, B→C = 75 - 50 = 25, C→A = 60 - 50 = 10.
3. Check nets: A = +10, B = -25, C = 25 - 10 = +15, unchanged.
4. New total (counted twice) = 2 × (25 + 10) = 70, down from 2 × 185 = 370.

Why B is correct: Subtracting the smallest leg from every leg of the ring removes one trade entirely, keeps all net positions and directions, and gives the largest reduction without reversals.

Why the others are wrong

- **A:** Removed only the A→B leg without reducing the others: A's net becomes +60 instead of +10, so net positions change (total 2 × 135 = 270).
- **C:** Subtracted 25 from each leg: nets are preserved and no reversal, but it is not the maximum compression (2 × 110 = 220).
- **D:** Subtracted 60 from every leg, which reverses A→B into B→A; the stem forbids reversals (total 2 × 25 = 50).

Q769

BT answers P2.T6.24.27.1

REBUILT — Original table was OCR-damaged (one notional inferred) and the options were loosely worded; contract data restated, with a statement that fixed rates and dates are identical so the swaps compress fully.

Alpha Financial and Beta Investments have five 5-year interest rate swaps with each other, all with identical fixed rates, payment dates and maturity, so they are fully fungible. Notionals in USD millions:

Contract	Party long	Notional	Counterparty (short)
1	Alpha Financial	100	Beta Investments
2	Beta Investments	50	Alpha Financial
3	Alpha Financial	100	Beta Investments
4	Beta Investments	100	Alpha Financial
5	Alpha Financial	50	Beta Investments

After trade compression, what is the net notional amount and which party holds the net contract position?

A. USD 100m; Alpha Financial holds the net long position.

B. USD 100m; Beta Investments holds the net long position.

C. USD 50m; Alpha Financial holds the net long position.

D. Zero; the positions fully offset and no net contract remains.

Answer: A

Step by step

1. Alpha long: 100 + 100 + 50 = 250.
2. Beta long: 50 + 100 = 150.
3. Net = 250 - 150 = 100, in Alpha's favour.
4. All five contracts compress into one swap of 100 with Alpha long.

Why A is correct: Because the swaps are fungible and between the same two parties, all long and short notionals net to a single contract of 100 held long by the side with the larger gross long position.

Why the others are wrong

- **B:** Right amount, wrong direction: Beta's long total (150) is smaller than Alpha's (250).
- **C:** Netted only contracts 1 and 2 (100 - 50) and ignored contracts 3 to 5.
- **D:** Assumed each party having both directions means full cancellation; gross longs are 250 vs 150, not equal.

Q770

BT answers P2.T6.24.27.3 · BT notes P2.T6.24.27.3

REBUILT — Original relied on an unrecoverable exhibit and a flawed premise (a net position shrinking under compression); rebuilt with explicit directions from the GARP ring example (compress the 2-3-4 ring by its smallest leg) asking for the new gross notional and Party 4's unchanged net.

Five counterparties hold fungible contracts. 'X → Y: n' means X owes Y a notional of n. Positions: 1 → 2: 95; 1 → 5: 20; 2 → 3: 60; 3 → 4: 70; 4 → 2: 85; 4 → 5: 105; 5 → 2: 95; 5 → 4: 75. Compression reduces the ring 2 → 3 → 4 → 2 by its smallest leg and leaves every party's net position unchanged. After compression, what is the gross notional (each position counted once) and Party 4's net position?

A. Gross 545; Party 4 is a net payer of 45.

B. Gross 425; Party 4 is a net payer of 45.

C. Gross 425; Party 4 is a net receiver of 85.
D. Gross 425; Party 4 is a net payer of 105.

Answer: B
Key correction: Source key (A: 'net notional 65, Party 4 long') rested on reducing only the 3-4 position from 70 to 10, which would change Party 4's net; compression must preserve nets, so that premise is invalid.
Step by step
1. Ring legs: 2→3 = 60, 3→4 = 70, 4→2 = 85. Smallest = 60.
2. After: 2→3 = 0, 3→4 = 10, 4→2 = 25. Gross before = 95+20+60+70+85+105+95+75 = 605.
3. Gross after = 605 - 3 × 60 = 425.
4. Party 4 before: receives 70 + 75 = 145, pays 85 + 105 = 190, net payer 45.
5. After: receives 10 + 75 = 85, pays 25 + 105 = 130, net payer 45, unchanged.

Why B is correct: Removing the smallest ring leg (60) from all three legs cuts gross notional by 180 and leaves each party's net unchanged, so Party 4 remains a net payer of 45.
Why the others are wrong
- **A:** Subtracted 60 only once from the gross (605 - 60) instead of from each of the three ring legs.
- **C:** Looked only at Party 4's post-compression receivables (10 + 75 = 85) and ignored its payables.
- **D:** Reduced only the 3→4 leg by 60 (receipts 85) but left 4→2 at 85 (pays 190): net payer 105, which would change Party 4's net.

Q771

GARP PE2 Q18

REBUILT — Original asked for an ambiguous 'compressed notional value' (key counted only the NUMU net, ignoring the untouched WKL position); now asks explicitly for the bank's total notional after compression.

An analyst applies trade compression to single-name CDS contracts, all with the same maturity and with the bank's counterparties shown:

Counterparty	Reference credit	Notional (EUR m)	Bank's position	Coupon (bps)
Digital Corporation	NUMU Inc.	9	Long protection	180
Digital Corporation	WKL Company	6	Long protection	150
WKL Company	NUMU Inc.	7	Short protection	150

What is the bank's total notional across all contracts after compression?
A. EUR 1 million.
B. EUR 2 million.
C. EUR 8 million.
D. EUR 22 million.

Answer: C
Key correction: Source key (B, EUR 2m) is the compressed NUMU position only; the question asks for the total, which also includes the uncompressible WKL 6.
Step by step
1. Only trades on the same reference credit (and maturity) are netted; counterparty and coupon do not matter (coupons are blended).
2. NUMU: long 9 - short 7 = 2 long protection.
3. WKL Company reference: only one trade (long 6), nothing to offset, stays at 6.
4. Total after compression = 2 + 6 = EUR 8m (down from 22m).

Why C is correct: Compression nets long and short only within the same reference credit, leaving 2 on NUMU and the unoffset 6 on WKL.
Why the others are wrong
- **A:** Netted the WKL long (6) against the NUMU short (7) although they reference different credits.
- **B:** Counted only the compressed NUMU position and omitted the WKL 6.
- **D:** Gross notional before compression (9 + 6 + 7); no compression applied.

Q772

Mock 4 Q19

REBUILT — Source key (reduction of 90 by removing the smallest leg) is not the maximum compression because the remaining legs can still be netted; stem rewritten to ask for the minimum remaining gross notional and the net seller.

Three dealers hold credit default swap positions referencing the same entity, all with identical maturities and

coupons:

Protection buyer	Protection seller	Notional
Dealer A	Dealer B	EUR 50 million
Dealer B	Dealer C	EUR 30 million
Dealer C	Dealer A	EUR 40 million

The dealers submit these positions to a multilateral compression cycle that reduces gross notional as far as possible while leaving each dealer's net protection position unchanged. After compression, what is the total remaining gross notional, and which dealer is the net protection seller?

A. EUR 20 million; Dealer B is the net protection seller
B. EUR 20 million; Dealer A is the net protection seller
C. EUR 30 million; Dealer B is the net protection seller
D. EUR 10 million; Dealer B is the net protection seller

Answer: A

Key correction: Source key (C, reduction of EUR 90 million) said removing the smallest leg is the answer; correct is EUR 20 million remaining (reduction of EUR 100 million) because the leftover legs (20 and 10) can be netted further.

Step by step

1. Net protection bought minus sold: A = 50 - 40 = +10; B = 30 - 50 = -20; C = 40 - 30 = +10 (EUR million). Net positions sum to zero.
2. Gross notional before compression = 50 + 30 + 40 = 120.
3. Removing the smallest leg (30) from every leg leaves A-B 20, B-C 0, C-A 10, gross 30, but this is not minimal.
4. Minimum gross notional = total net protection sold = 20: Dealer B sells 10 to A and 10 to C, replacing all three trades.
5. Remaining gross notional 20; Dealer B is the net protection seller. Reduction = 120 - 20 = 100.

Why A is correct: Compression preserves each dealer's net position, so the minimum gross notional equals the sum of the net protection sold (20), all held by Dealer B against the net buyers A and C.

Why the others are wrong

- **B:** Right amount but wrong direction: A (+10) and C (+10) are net buyers; B (-20) is the net seller.
- **C:** Stops after removing the smallest leg (30) from each trade, leaving 20 + 10 = 30; the residual legs can be netted further.
- **D:** Counts only Dealer A's (or C's) net position of 10 and ignores the other net buyer, so it understates the notional needed to preserve net positions.

LO 34.f

Identify and describe termination events and discuss their potential effects on parties to a transaction.

Q773

AnalystPrep Q.1887

REBUILT — Original stem was a run-on and option A (changing terms) overlapped with reset agreements; reworded the stem and made each distractor map to a different feature.

A bank holds a long-dated derivative with a counterparty. Current exposure is small, but it could become large and unmanageable if the counterparty's credit quality declines over the life of the trade. The bank wants the contractual right to end the trade at specified future dates at its replacement value, so that it can exit if the counterparty weakens. Which feature best provides this?

A. A reset agreement that moves the trade back to at-the-money
B. A break clause (additional termination event)
C. A walkaway clause
D. Payment netting of the trade's cash flows

Answer: B

Step by step

1. The need is to terminate a long-dated trade at specified future dates at replacement value when the counterparty's credit quality declines.
2. This is the definition of a break clause (liquidity put or early termination option), often linked to an additional termination event.

3. Reset agreements readjust trade parameters; walkaway clauses apply only after default; payment netting reduces settlement risk, not future exposure growth.

Why B is correct: A break clause gives a party the option to terminate a (long-dated) transaction at specified future dates at replacement value, so exposure cannot keep growing as credit quality worsens.

Why the others are wrong

- **A:** Reset agreements reset a heavily in-the-money trade to at-the-money (e.g. reset MtM and FX rate); the trade stays in place, so the counterparty is not exited.
- **C:** A walkaway clause lets the non-defaulter avoid net liabilities only after default; it does not let the bank exit before deterioration.
- **D:** Payment netting combines same-day cash flows and reduces settlement risk; it does not terminate the trade.

Q774

AnalystPrep Q.1890

REBUILT — Original options were garbled and two did not describe a danger; rewrote the stem and options around the banker's paradox.

Which of the following best describes a potential danger (practical weakness) of additional termination events and other break clauses?

A. They can only be exercised after the counterparty has already defaulted, so they give no protection

B. Parties tend to avoid exercising them early to preserve the relationship, so they may not be used until the counterparty's credit quality has already declined substantially

C. They are always mandatory, so the trade is terminated whether or not the counterparty has deteriorated

D. They reset the trade to at-the-money, which transfers collateral to the weaker counterparty

Answer: B

Step by step

1. The value of a break clause lies in exercising it before credit quality falls substantially.
2. In practice, entities avoid early exercise to preserve good relationships with counterparties (banker's paradox).
3. Hence exposure to the weakened counterparty may remain, or grow, when the clause is finally used - the clause is least effective when most needed.

Why B is correct: Because early exercise damages relationships, break clauses tend to be exercised too late, after substantial deterioration, limiting their protective value.

Why the others are wrong

- **A:** Break clauses (ATEs) operate before default, when credit quality declines; they are not default-only.
- **C:** Break clauses can be mandatory, optional or trigger-based; they are not always mandatory.
- **D:** Resetting to at-the-money is a reset agreement, a different feature, and has no collateral-transfer-to-weaker-party effect.

Q775

BT answers P2.T6.323.3 · BT notes P2.T6.323.3

According to Gregory, each of the following is true about termination and walkaway features in credit contracts, EXCEPT which is false?

A. Termination events give an institution the right (option) to terminate a trade prior to their counterparty's creditworthiness deteriorating to the point of bankruptcy

B. A break clause (a.k.a. liquidity put or early termination option) is an agreement to terminate (break) a transaction at pre-specified dates in the future at market rates

C. Break clauses are often linked to additional termination events (ATEs), which are not in the standardized ISDA documentation and therefore are a result of negotiations

D. Walkaway features are parts of the standardized ISDA documentation and "should be utilized in almost all transactions" as they mitigate counterparty risk

Answer: D

Step by step

1. A: termination events let an institution exit before the counterparty is bankrupt - true.
2. B: break clauses (liquidity puts/ETOs) terminate at pre-specified future dates at replacement (market) value - true.
3. C: ATEs are negotiated and not part of standard ISDA documentation - true.
4. D: walkaway features are not part of standardized ISDA documentation, do not reduce credit exposure, and should be avoided - false, so D is the exception.

Why D is correct: Walkaway clauses were common before the 1992 ISDA Master Agreement, are not part of standard documentation, impose extra costs on the defaulter, and should be avoided.

Why the others are wrong

- **A:** True statement; the question asks for the false one.
- **B:** True statement; break clauses terminate at specified dates at replacement value.
- **C:** True statement; ATEs arise from negotiation rather than the standard form.

Q776 Schweser Module Quiz 34.2 Q1

Leverage, Inc., an investment bank, has numerous credit default swaps with XYZ Corp. Leverage has established a break clause with XYZ Corp. to reduce risk. The break clause is trigger-based and may be exercised once the trigger is satisfied. The CEO of Leverage is concerned about the banker's paradox. Which of the following statements best describes the CEO's concern?

A. To be effective, the break clause option should not be used too early.

B. The weak firm often recovers after the use of the break clause.

C. The break clause option is used too late, and the weak firm gets weaker.

D. The break clause option is used too early, and is unlikely to avoid systemic risk issues.

Answer: C

Step by step

1. Banker's paradox: for a break clause to be truly useful it should be exercised early, before a substantial decline in the counterparty's credit quality.
2. Entities typically avoid early exercise to preserve good relationships with counterparties.
3. So the clause tends to be exercised too late, after the weak firm has weakened further - choice C.

Why C is correct: The paradox is that the clause is most valuable if used early, yet relationship concerns delay use until credit quality has substantially declined.

Why the others are wrong

- **A:** Reverses the logic: effectiveness requires early exercise, not avoiding it.
- **B:** Recovery is not part of the paradox; the concern is delayed use and further deterioration.
- **D:** The paradox concerns use too late, not too early; systemic risk is not the issue.

Reading 35. Margin (Collateral) and Settlement

Jon Gregory, The xVA Challenge: Counterparty Credit Risk, Funding, Collateral, and Capital, 4th Edition (West Sussex, UK: John Wiley & Sons, 2020). Chapter 7.

LO	Learning objective	Qs
35.a	describe the rationale for collateral management.	3
35.b	describe the terms of a collateral agreement and features of a credit support annex (CSA) within the ISDA Master Agreement including threshold, initial margin, minimum transfer amount and rounding, haircuts, credit quality, and credit support amount.	17
35.c	calculate the credit support amount (margin) under various scenarios.	7
35.d	describe the role of a valuation agent.	3
35.e	describe the mechanics of collateral and the types of collateral that are typically used.	3
35.f	explain the process for the reconciliation of collateral disputes.	5
35.g	explain the features of a collateralization agreement.	11
35.h	differentiate between a two-way and one-way CSA agreement and describe how collateral parameters can be linked to credit quality.	5
35.i	explain aspects of collateral including funding, rehypothecation, and segregation.	5
35.j	explain how market risk, operational risk, and liquidity risk (including funding liquidity risk) can arise through collateralization.	9
35.k	describe the various regulatory capital requirements.	3

LO 35.a

Describe the rationale for collateral management.

Q777 AP Mock 1 Q31

In an internal training session on counterparty credit risk for OTC derivatives traders, the head of risk management asks about trades that have no collateral agreement in place. What is the collateral requirement typically associated with OTC uncollateralized derivatives?

A. Participants are required to post collateral based on credit ratings at predetermined intervals.
B. There is no routine collateral requirement; participants do not typically post collateral in these transactions.
C. Collateral is pooled among all participants to cover the obligations within the entire market segment.
D. A one-time collateral is posted at the start of the contract and released upon successful settlement.

Answer: B

Step by step

1. Uncollateralised (non-margined) means no CSA or margining arrangement covers the trades.
2. With no agreement, neither party posts or returns collateral as MtM changes; exposure is unsecured.
3. Compare with the options: A, C and D each describe some form of collateral or margin, so only B describes the absence of collateral.

Why B is correct: Uncollateralised trading means no margin is exchanged, so the positive-MtM party bears the full counterparty exposure with no routine collateral flows.

Why the others are wrong

- **A:** Describes a rating-linked (rating-triggered) CSA, which is a collateralised arrangement, not an uncollateralised one.
- **C:** Describes mutualised loss-sharing like a CCP default fund; bilateral OTC trades have no market-wide collateral pool.
- **D:** Describes an upfront initial margin / independent amount, which is itself collateral and not 'no collateral'.

Q778 BT notes P2.T6.905.1

Two counterparties (A and B) have a set of OTC derivative transactions between them. Prior to initiating the underlying transactions, they also entered into a bilateral collateral agreement. Each of the following is true EXCEPT, which is false?

A. If they followed best practices, collateral posting should be continuous for both of them.
B. As a counterparty risk mitigant, collateral posting is generally superior to a reset feature.
C. Variation margin is a type of collateral that aims to track the change in valuation (the change in MtM) of the

position over time.

D. As a bilateral agreement, in the case of a positive MtM the party will call for collateral, and in the case of a negative MtM the party will be required to post collateral itself.

Answer: A

Step by step

1. Test each statement against the collateral framework.
2. A: posting is not continuous in practice; to control operational cost, margin is posted periodically or in blocks under pre-agreed rules. So A is false.
3. B: a reset is only a weaker, periodic form of margining, so collateral is generally the stronger mitigant. True.
4. C: variation margin tracks MtM changes. True.
5. D: bilateral margining means the party with positive MtM calls and the party with negative MtM posts. True.

Why A is correct: Margin is not posted continuously because that would be operationally costly; calls occur at agreed frequencies and in blocks, so statement A is the false one.

Why the others are wrong

- **B:** True: a reset is essentially a periodic, infrequent settlement similar to margin, i.e. a weaker form of margining. Choosing it means confusing which mitigant is stronger.
- **C:** True: variation margin is the collateral that follows MtM changes. Choosing it confuses it with initial margin, which is independent of exposure.
- **D:** True: this is how bilateral (two-way) collateral works. Choosing it confuses bilateral with one-way arrangements.

Q779 Schweser Module Quiz 35.1 Q1

Which of the following features is least likely a benefit of collateralization?

A. Reduces capital requirements.

B. Allows for more competitive pricing of counterparty risk.

C. Reduces market, operational, and liquidity risk.

D. Reduces credit exposure.

Answer: C

Step by step

1. The motivations for collateral management are: reduce credit exposure to enable more trading, be able to trade with a counterparty, reduce capital requirements, and allow more competitive pricing of counterparty risk.
2. A, B and D match these motivations.
3. C is not a benefit: collateralization creates market risk (collateral value changes), operational risk (legal and processing issues) and liquidity risk (funding and selling collateral).

Why C is correct: Collateral reduces counterparty credit risk but introduces market, operational and liquidity risks, so claiming it reduces them is the false benefit.

Why the others are wrong

- **A:** A genuine benefit: collateral lowers exposure and hence regulatory capital. Choosing it overlooks the capital motivation.
- **B:** A genuine benefit: secured exposure allows tighter pricing of counterparty risk. Choosing it overlooks the pricing motivation.
- **D:** The primary benefit: collateral directly offsets positive MtM exposure.

LO 35.b

Describe the terms of a collateral agreement and features of a credit support annex (CSA) within the ISDA Master Agreement including threshold, initial margin, minimum transfer amount and rounding, haircuts, credit quality, and credit support amount.

Q780 BT notes P2.T6.24.28.3

REBUILT — Original had an OCR-garbled table, inferred amounts and a mismatch between listed and used asset order; rebuilt with clean data and an unambiguous posting order, and re-mapped to 35.b (haircuts).

Eagle Heights Bank must post collateral with a market-value credit of USD 30 million against an interest rate swap. A haircut of x% means only (1 - x%) of an asset's market value is credited. Haircuts: cash 0%, high-quality government bonds (1-5 years) 2%, high-quality corporate bonds (1-5 years) 4%. The bank posts its assets in this order: USD 10 million of cash, then all USD 12 million of its 2-year government bonds, then 4-year corporate bonds (USD 15 million available) for the remainder, posting only as much as is needed. What is the total market value of collateral posted?

A. USD 30.00 million
B. USD 30.58 million
C. USD 30.24 million
D. USD 31.25 million

Answer: B

Step by step

1. Cash credit = 10.00 (0% haircut); government bonds credit = $12 \times 0.98 = 11.76$; cumulative credit = 21.76.
2. Remaining requirement = $30 - 21.76 = 8.24$.
3. Corporate bonds needed (market value) = $8.24/0.96 = 8.5833$, below the 15 available.
4. Total posted = $10 + 12 + 8.5833 = 30.58$ million.

Why B is correct: Each asset's credit is market value times (1 - haircut), so the last tranche must be grossed up by dividing the remaining requirement by 0.96.

Why the others are wrong

- **A:** Ignores haircuts altogether (market value equals the requirement).
- **C:** Credits the government bonds net of the haircut but posts the last 8.24 at face value without dividing by 0.96 ($10 + 12 + 8.24$).
- **D:** Divides the whole USD 30 million requirement by the corporate-bond valuation percentage (30/0.96), ignoring the cash and government bond already posted.

Q781 AnalystPrep Q.1900

REBUILT — Original options A and C were both defensible and the stem was vague; rebuilt as a clear 'which is NOT set in a CSA' item based on the CSA parameter list.

Which of the following is NOT a term that is typically established in a credit support annex (CSA) appended to an ISDA Master Agreement?

A. The eligible collateral and the haircuts applied to non-cash collateral
B. The threshold, independent amount (initial margin) and minimum transfer amount
C. The method and timing of valuations and of margin transfers, and the procedure for resolving disputes
D. The events of default and termination events that permit close-out netting of the trades

Answer: D

Step by step

1. A CSA governs collateral: eligibility, haircuts, substitutions, valuation and transfer timing, dispute handling, rehypothecation, and credit support parameters (threshold, initial margin, MTA, rounding).
2. Options A, B and C are all items on that list.
3. Events of default and termination events, and the close-out netting they trigger, belong to the ISDA Master Agreement itself, not the CSA.

Why D is correct: The CSA is the collateral annex; the definition of default/termination events and close-out netting sits in the ISDA Master Agreement.

Why the others are wrong

- **A:** Typically set in the CSA (eligible collateral and haircuts); it is a true feature, so not the exception.
- **B:** These are the core CSA credit-support parameters, so they are set in the CSA.
- **C:** Valuation/transfer mechanics and dispute resolution are explicit CSA terms.

Q782 AnalystPrep Q.1902

REBUILT — Stem wording was unclear ('help investors check exposure risk'); reworded and added a numeric illustration while keeping the same concept.

A CSA specifies a threshold of $10 million for a counterparty and no initial margin. Which statement about the threshold is correct?

A. It is the minimum size of a single collateral transfer, below which no transfer is made.

B. It is the level of mark-to-market exposure below which no margin can be called; if exposure is $25 million, only $15 million can be called.

C. It is an extra amount of collateral that must be posted regardless of the portfolio value, so a $25 million exposure requires $35 million.

D. It is the level of mark-to-market exposure above which posted collateral must be liquidated.

Answer: B

Step by step

1. The threshold is the amount below which margin is not required, leaving that part of the exposure uncollateralised.
2. Above the threshold only the incremental amount can be called: credit support amount = $\max(25-10,0) = 15$ million dollars.
3. With zero initial margin nothing else is added.

Why B is correct: A threshold permits uncollateralised exposure up to a limit, reducing operational and liquidity burden at the cost of weaker risk reduction.

Why the others are wrong

- **A:** Describes the minimum transfer amount (MTA), not the threshold.
- **C:** Describes initial margin (independent amount), which works in the opposite direction to a threshold; also adds rather than subtracts.
- **D:** Threshold has nothing to do with liquidation; it governs when margin is called.

Q783 AnalystPrep Q.1903

Which of the following statements correctly describes the relationship between thresholds and initial margin (independent amounts) in a CSA?

A. They move in the same direction: both increase the collateral called.

B. They work in opposite directions: a threshold allows undercollateralisation, while initial margin creates overcollateralisation.

C. They are always equal for a given counterparty.

D. A threshold is always less than or equal to the initial margin.

Answer: B

Step by step

1. Credit support amount = $\max(V-K,0)+IM$, where K is the threshold and IM the initial margin.
2. A higher K reduces collateral called; a higher IM increases it.
3. Thus they act as opposites (IM behaves like a negative threshold), and in practice they are not seen together: usually one of them is zero.

Why B is correct: Threshold leaves exposure uncollateralised below a level; initial margin adds collateral independent of the portfolio value.

Why the others are wrong

- **A:** Confuses the signs: the threshold subtracts from the amount called, IM adds to it.
- **C:** No requirement links their sizes; usually one is zero.
- **D:** No such ordering exists; when IM is present the threshold is usually zero.

Q784 AnalystPrep Q.1910

REBUILT — Original options mixed percent and decimal notation ('(1 - A)%'); rebuilt with a concrete haircut of 4%.

A security posted as collateral under a CSA has a haircut of 4%. Which statement is correct?

A. For each $1 of the security posted, the recipient gives credit of $0.04.

B. For each $1 of the security posted, the recipient gives credit of about $1.04.

C. For each $1 of the security posted, the recipient gives credit of $0.96.

D. For each $1 of the security posted, the recipient gives credit of about $1.0417 (= 1/0.96).

Answer: C

Step by step

1. A haircut of x% means only (1 - x) of value is credited (the valuation percentage).
2. Valuation percentage = $1 - 0.04 = 0.96$.
3. Credit per \$1 posted = \$0.96; to meet a \$1 call the poster must deliver $1/0.96 = 1.0417$ dollars of the security.

Why C is correct: The haircut discounts the collateral's value, so credit is (1 - haircut) times market value.

Why the others are wrong

- **A:** Treated the haircut itself as the credit given instead of the amount deducted.
- **B:** Added the haircut (1 + 0.04), which would give more credit than the market value.
- **D:** Gives the amount of security needed per \$1 of required credit (1/0.96), not the credit received per \$1 posted.

Q785

AnalystPrep Q.5163

What is the purpose of a Credit Support Annex (CSA) in a derivatives trade?

A. To protect the trading party from any potential losses due to market volatility.
B. To provide a mechanism for netting and settlement of trades.
C. To create an obligation for trading parties to mutually agree on posting collateral.
D. To ensure that all derivative contracts are honored by both parties involved.

Answer: C

Step by step

1. Historically OTC contracts had no obligation to post collateral.
2. A CSA appended to an ISDA Master Agreement lets the parties agree contractual margin posting.
3. Therefore C describes its purpose; netting is handled by the master agreement and a CSA only mitigates, not eliminates, losses.

Why C is correct: A CSA converts collateral posting into a contractual agreement that mitigates counterparty credit risk.

Why the others are wrong

- **A:** Overstated: collateral mitigates credit exposure but does not protect against all market-volatility losses.
- **B:** Netting and settlement come from the ISDA Master Agreement, not the CSA.
- **D:** A CSA does not guarantee performance; it only provides collateral support.

Q786

AnalystPrep Q.6146

REBUILT — Option D (automatic adjustment with credit rating) is defensible since CSAs can link parameters to ratings; replaced it with a clearly wrong option and referred to the CSA.

A bank reviews the collateral terms of the credit support annex (CSA) appended to its ISDA Master Agreements. Which statement accurately reflects a feature of a CSA?

A. It mandates a fixed collateral amount for the duration of all derivative contracts.
B. It specifies the collateral terms, including events that trigger changes in collateral requirements.
C. It requires collateral to be posted exclusively in the form of government securities.
D. It requires the same threshold to apply to both parties regardless of their credit quality.

Answer: B

Step by step

1. A CSA must define all collateralization parameters and scenarios, including collateral changes triggered by events (e.g. rating downgrades).
2. Collateral is recalculated regularly with exposure, so it is not fixed (A), and eligible collateral is negotiated (cash and various securities), so C is wrong.
3. Thresholds can be asymmetric and linked to credit quality, so D is wrong.

Why B is correct: The CSA is a flexible, negotiated agreement that sets collateral terms and event-triggered changes.

Why the others are wrong

- **A:** Collateral is recalculated with portfolio value; it is not fixed.
- **C:** Cash is the most common collateral; eligible assets are negotiated.
- **D:** Thresholds can differ by party and be tied to credit rating.

Q787 BT answers P2.T6.325.1 · BT notes P2.T6.325.1

Counterparty FundCorp receives a collateral call of $100,000. Each of the following is true, except which is FALSE?

A. FundCorp can post $100,000 in cash

B. FundCorp can post about $108,700 in collateral with an 8.0% haircut

C. If FundCorp has already posted an Independent Amount (IA) of $60,000, then FundCorp can apply the IA and meet the margin call with only $40,000

D. Collateral with higher price volatility, ceteris paribus, will be assigned a higher haircut

Answer: C

Step by step

1. Cash has no haircut, so $100,000 meets the call (A true).
2. With an 8% haircut the valuation percentage is 0.92: $100{,}000/0.92 \approx 108{,}696 \approx 108{,}700$ (B true).
3. The independent amount (initial margin) is overcollateralisation held separately; it is not netted against the variation-margin call, so FundCorp still must deliver the full $100,000 (C false).
4. More volatile collateral gets a larger haircut (D true).

Why C is correct: Initial margin/independent amount is not netted against variation margin, so it cannot reduce the call.

Why the others are wrong

- **A:** True: cash carries a 0% haircut.
- **B:** True: $100{,}000/(1-0.08) \approx 108{,}696$.
- **D:** True: haircuts are driven by volatility and liquidity.

Q788 BT answers P2.T6.409.1 · BT notes P2.T6.409.1

Each of the following is true about a credit support annex (CSA) EXCEPT, which is not?

A. A CSA is appended within an ISDA Master Agreement and permits the counterparties to mitigate their credit risk by agreeing to various collateral posting

B. A CSA is critically defined by key parameters, including threshold, independent amount, and minimum transfer amount

C. In some trading relationships, CSAs are not used because one or both parties cannot commit to collateral posting

D. If a counterparty employs a credit value adjustment (CVA), then a CSA gives no benefit and is not necessary (a CSA would be redundant with CVA)

Answer: D

Step by step

1. A, B and C restate how a CSA is structured and why some parties (e.g. corporates) have none.
2. CVA prices the expected loss from counterparty risk; collateral under a CSA reduces the exposure and so reduces that CVA charge.
3. So a CSA is not redundant with CVA; D is false.

Why D is correct: CVA prices counterparty risk and a CSA reduces it; they complement each other.

Why the others are wrong

- **A:** True: CSA is an annex to the ISDA Master Agreement that provides for collateral posting.
- **B:** True: threshold, initial margin and MTA are key CSA parameters.
- **C:** True: non-posting entities (e.g. corporates) often have no CSA.

Q789 Schweser Module Quiz 35.1 Q2

Which of the following statements is least accurate regarding a credit support annex (CSA) and/or an ISDA Master Agreement?

A. ISDA documentation has helped standardize collateral management.

B. CSAs must define all collateralization parameters in order to work as intended.

C. Compared to the ISDA Master Agreement, CSAs were first to establish collateral standards.

D. CSAs are incorporated into an ISDA Master Agreement.

Answer: C

Step by step

1. A CSA is an annex that is incorporated into an ISDA Master Agreement (statement D is true).
2. Collateral management moved from no legal standards to highly standardized practice through ISDA documentation introduced in 1994 (statement A is true).
3. The CSA governs eligibility, haircuts, transfer timing, valuation, reuse, disputes and so on, and must define all parameters and scenarios to work as intended (statement B is true).
4. Because the CSA sits inside the ISDA framework, it cannot have established collateral standards before the ISDA Master Agreement; statement C is false and is therefore the least accurate.

Why C is correct: The CSA is an annex to the ISDA Master Agreement, so the ISDA framework came first and the CSA did not precede it as a standard-setter.

Why the others are wrong

- **A:** True: ISDA documentation (from 1994) is what standardized collateral management; choosing it confuses a true statement with the false one.
- **B:** True: a CSA must define every collateralization parameter and account for relevant scenarios; picking it misreads the 'least accurate' wording.
- **D:** True: a CSA is incorporated into the ISDA Master Agreement; picking it overlooks that the CSA is an annex, not a separate stand-alone agreement.

Q790

BT answers P2.T6.906.1

REBUILT — Removed the '(false!)' giveaway from option D and reworded it as a plausible but incorrect claim.

Collateral agreements are increasingly shaped by funding (FVA) and capital (KVA) costs in addition to pure counterparty risk (CVA), and by the shift toward central clearing and bilateral margin requirements. What is the general impact of these trends on collateral thresholds?

A. Thresholds are increasingly set to zero
B. Thresholds are increasingly set to infinity
C. There is no impact on thresholds
D. Thresholds are typically set equal to the initial margin

Answer: A

Step by step

1. A threshold is the exposure level below which no margin can be called; it leaves that part of the exposure uncollateralised.
2. Greater recognition of funding and capital costs, and regulatory bilateral margin rules, make uncollateralised exposure costly, so parties want full collateralisation.
3. Thresholds are therefore increasingly zero and can be non-zero only for parties exempt from the bilateral margin requirements.
4. Two-way CSAs with zero thresholds are standard between financial counterparties.

Why A is correct: Funding, capital and regulatory pressures remove the benefit of leaving exposure uncollateralised, so thresholds are increasingly zero.

Why the others are wrong

- **B:** An infinite threshold means a party never posts margin (one-way CSA); that is the opposite of the trend toward more margining.
- **C:** Wrong: the trends directly affect thresholds, which are increasingly zero.
- **D:** Thresholds and initial margin work in opposite directions (IM is like a negative threshold), so they are not seen together; when IM is present the threshold is zero, not equal to IM.

Q791

BT answers P2.T6.330.3

REBUILT — Original options 'operational costs' and 'liquidity risk' were both defensible; rebuilt so one option states the operational and liquidity-cost trade-off and the rest are clearly wrong.

Which of the following best describes the primary reason two counterparties agree to a non-zero (higher) threshold in a CSA?

A. To reduce the operational workload and liquidity cost of margining, accepting some uncollateralised exposure
B. To eliminate counterparty credit exposure entirely
C. To increase the amount of collateral held against the current exposure
D. To remove wrong-way risk from the collateral posted

Answer: A

Step by step

1. The threshold is the amount of exposure below which no margin can be called; only the excess above it is collateralised.
2. This leaves exposure up to the threshold uncollateralised, which weakens risk reduction.
3. In return, fewer and smaller margin calls reduce the operational burden and the liquidity costs of posting margin.
4. Parties accept the trade-off when they can tolerate uncollateralised risk up to a level (e.g., within a credit limit).

Why A is correct: A higher threshold means fewer margin calls, trading some risk mitigation for lower operational and liquidity costs.

Why the others are wrong

- **B:** A higher threshold leaves more exposure uncollateralised; it does not eliminate exposure.
- **C:** A higher threshold reduces collateral held, since margin is called only on exposure above the threshold; initial margin is what increases collateral.
- **D:** Wrong-way risk depends on the link between collateral and counterparty credit quality and is not addressed by the threshold.

Q792

AnalystPrep Q.1898

REBUILT — Original stem was ambiguous about whether $80 million was the credited value, and options were giveaway-prone; clarified stem and rebuilt distractors from realistic haircut errors.

Party Y has a negative mark-to-market position and must deliver collateral worth $80 million to its counterparty X, after any haircut (that is, X must be credited with $80 million). Y posts a security that carries a 2% haircut. What is the market value of the security that Y must post?

A. $78.40 million
B. $80.00 million
C. $81.60 million
D. $81.63 million

Answer: D

Step by step

1. A haircut of 2% means only $1 - 0.02 = 98\%$ of the market value is credited.
2. Required market value $= 80/0.98$.
3. $80/0.98 = 81.63$ million.
4. Check: $81.63 \times 0.98 = 80.00$.

Why D is correct: Because only 98% of the posted market value counts, the posting must be grossed up by dividing by $(1 - h)$, not by multiplying.

Why the others are wrong

- **A:** $80 \times 0.98 = 78.40$: multiplied by (1 - haircut), which understates what must be posted and would leave X under-credited.
- **B:** Ignores the haircut entirely, as if the security were cash.
- **C:** $80 \times 1.02 = 81.60$: added the haircut percentage to the amount rather than dividing by (1 - haircut); close but not exact.

Q793

AnalystPrep Q.3076

Zhao Lee is a trader at the largest commercial bank in mainland China. He has a large position in a US sovereign bond that has a haircut of 3% and is used for a collateral call of $500,000. What amount of bond is needed for $500,000 to be credited?

A. $515,464
B. $485,000
C. $484,536
D. $500,000

Answer: A

Step by step

1. Credit given per $1 of bond $= 1 - 0.03 = 0.97$.
2. Bond needed $= 500{,}000/0.97$.

3. = 515,464 (rounded to the nearest dollar).
4. Check: 515,464 × 0.97 ≈ 500,000.

Why A is correct: The haircut reduces the credit received for each dollar of bond, so the bond posted must exceed the call amount by dividing by (1 - haircut).

Why the others are wrong

- **B:** 500,000 × 0.97 = 485,000: multiplied instead of dividing, so the bond posted would be credited at less than $500,000.
- **C:** 500,000 − 15,464 = 484,536: subtracted the haircut shortfall (the excess of the correct answer over the call) from the call, moving in the wrong direction.
- **D:** Ignores the haircut, treating the bond as cash.

Q794

BT answers P2.T6.24.28.3

REBUILT — OCR-damaged asset list/table reconstructed (Treasury maturity set to 4 years to avoid bucket-boundary ambiguity, gold used for the final slice) and distractors rebuilt.

Eagle Heights Bank must post collateral with a post-haircut value of $45 million against an interest rate swap. Haircuts by asset class:

Asset class	Haircut
Cash in same currency	0%
High-quality government bond (<1 yr)	0.5%
High-quality government bond (1-5 yr)	2%
High-quality government bond (>5 yr)	4%
High-quality corporate bond (<1 yr)	1%
High-quality corporate bond (1-5 yr)	4%
High-quality corporate bond (>5 yr)	8%
Equities in a major index	15%
Gold	15%

The bank posts its assets in the following order, using as much of each as needed:

1	$10 million cash (USD)
2	$15 million US Treasury bonds, remaining maturity 4 years
3	$20 million high-quality corporate bonds, remaining maturity 4 years
4	$5 million gold bullion
5	$8 million S&P 500 stocks

What total market value of collateral must the bank post?

A. $45.00 million
B. $43.90 million
C. $46.29 million
D. $52.94 million

Answer: C

Step by step

1. Cash: 10 × (1 − 0) = 10.00.
2. Treasuries (1-5 yr, 2%): 15 × 0.98 = 14.70; cumulative 24.70.
3. Corporates (1-5 yr, 4%): 20 × 0.96 = 19.20; cumulative 43.90.
4. Shortfall = 45 − 43.90 = 1.10, taken from gold (15%): market value = 1.10/0.85 = 1.294 (within the $5 million available).
5. Total market value posted = 10 + 15 + 20 + 1.294 = 46.29 million.

Why C is correct: The first three assets are used in full and give 43.90 of credit; the remaining 1.10 must be grossed up for gold's 15% haircut.

Why the others are wrong

- **A:** Ignores haircuts, equating market value with credited value.
- **B:** This is the post-haircut credit of the first three assets only; it falls 1.10 short of the 45 requirement.
- **D:** 45/0.85 = 52.94: applied the 15% haircut to the entire requirement instead of only the last slice.

Q795 Schweser Module Quiz 35.2 Q3

Assume a sovereign bond has a haircut of 5% and is used for a collateral call of $100,000. What amount is credited if $100,000 of the bond is submitted, and what amount of the bond is needed for $100,000 to be credited, respectively?

A. $100,000; $106,263
B. $95,000; $100,000
C. $95,000; $105,263
D. $105,263; $95,000

Answer: C

Step by step

1. Valuation percentage = 1 − 5% = 95%.
2. Credit for $100,000 of bond = 100,000 × 0.95 = 95,000.
3. Bond needed for a $100,000 credit = 100,000/0.95 ≈ 105,263.

Why C is correct: A haircut of x% gives credit of only (1 - x) per unit posted, so the credit is below market value and the amount to post is grossed up by dividing by the valuation percentage.

Why the others are wrong

- **A:** Ignores the haircut on the credit ($100,000) and uses a wrong gross-up ($106,263 is not 100,000/0.95).
- **B:** Correct credit but fails to gross up for the haircut, assuming $100,000 of bond satisfies the call.
- **D:** Reverses the two calculations: divides for the credit and multiplies for the amount needed.

Q796 BT answers P2.T6.24.28.1 · BT notes P2.T6.24.28.1

REBUILT — Original NOT-true item hinged on cross-asset-class netting, which the GARP text does not state; rebuilt around the stated gross (no-netting) exchange of initial margin and re-mapped to 35.k where these rules sit.

Which of the following statements about the regulatory margin requirements for non-centrally cleared derivatives is NOT true?

A. Initial margin amounts posted by the two parties may be netted against each other, so that only the net amount is exchanged.
B. The minimum transfer amount for variation margin must not exceed 500,000 euros.
C. Variation margin must be exchanged regularly (for example daily) with a zero threshold.
D. Initial margin may be calculated using validated internal models or regulatory tables, based on an extreme but plausible move over a 10-day horizon at a 99% confidence level.

Answer: A

Key correction: Sources keyed C (the original cross-asset-class netting option); that option is replaced here. Schweser quotes a $650,000 transfer-amount cap whereas GARP states 500,000 euros (US-dollar equivalent is the same); GARP is used.

Step by step

1. Variation margin: exchanged regularly (e.g. daily), full margin with zero threshold, minimum transfer amount not above 500,000 euros, so B and C are true.
2. Initial margin: exchanged by both parties, calculated from validated internal models or regulatory tables using an extreme but plausible move at 99% confidence over 10 days, so D is true.
3. Initial margin is exchanged by both parties with no netting of amounts: each posts its own, so A is false.

Why A is correct: Initial margin is exchanged by both parties with no netting of amounts, so the two parties' initial margin amounts cannot be offset against each other.

Why the others are wrong

- **B:** True: the cap is 500,000 euros; picking this means misremembering the cap.
- **C:** True: variation margin is exchanged regularly and in full, with no threshold; confusing it with the thresholds common in ordinary CSAs would make it look false.
- **D:** True: this is the stated calibration (99%, 10-day, extreme but plausible), allowed via models or regulatory tables.

LO 35.c

Calculate the credit support amount (margin) under various scenarios.

Q797

AnalystPrep Q.5428

REBUILT — Original distractors had no logical origin; kept the data and answer, rebuilt options from realistic errors and clarified that transfers occur only when the credit support amount exceeds the MTA.

A hedge fund and a commercial bank have a one-way credit support annex under which only the hedge fund posts collateral, based on the mark-to-market value of the transaction. The terms are:

Item	Value ($)
Net exposure (mark-to-market)	87,000,000
Collateral already posted	50,000,000
Threshold amount	65,000,000
Minimum transfer amount	12,000,000
Rounding amount	10,000

The net exposure rises to $110,000,000 and the posted collateral is unchanged. How much additional collateral must the hedge fund post?

A. $60,000,000
B. $0
C. $45,000,000
D. $33,000,000

Answer: B

Step by step

1. Required collateral = $\max(110 - 65, 0) = 45$ million.
2. Collateral held = 50 million, so credit support amount = $45 - 50 = -5$ million (the fund is over-collateralised).
3. A negative amount means a return, and in any case the absolute value of 5 million is below the MTA of 12 million, so no transfer occurs.
4. Additional collateral to post = 0.

Why B is correct: Exposure above the threshold (45 million) is still covered by the 50 million already posted, so nothing further is called; the 5 million surplus is below the MTA so it is not even returned.

Why the others are wrong

- **A:** $110 - 50 = 60$: ignored the threshold, treating the full exposure as needing collateral.
- **C:** $110 - 65 = 45$: computed required collateral but forgot to deduct the 50 million already posted.
- **D:** $45 - 12 = 33$: subtracted the MTA from the required collateral, misusing the MTA as an additional threshold and ignoring existing collateral.

Q798

BT answers P2.T6.24.28.2

REBUILT — Source was internally inconsistent (MTA 750,000 vs 150,000 in options; collateral held by the wrong party); made the data consistent and rebuilt options.

Maplewood Financial and Downers Grove Capital have a two-way collateral agreement with a threshold of $7,000,000 (each party), a minimum transfer amount (MTA) of $750,000, and rounding to the nearest $25,000. No collateral has been exchanged to date. The latest valuation shows a mark-to-market value of +$7,020,000 for Maplewood Financial (and -$7,020,000 for Downers Grove Capital). What collateral movement results?

A. Downers Grove posts $25,000 to Maplewood
B. Downers Grove posts $750,000 to Maplewood
C. No collateral is transferred
D. Downers Grove posts $7,020,000 to Maplewood

Answer: C

Step by step

1. Credit support amount = $\max(7{,}020{,}000 - 7{,}000{,}000,\ 0) - 0 = 20{,}000$.
2. MTA = 750,000; $20{,}000 < 750{,}000$, so the call cannot be made.
3. Collateral transferred = 0.

Why C is correct: The MTA is a floor on any transfer; the amount above the threshold (20,000) is far below it, so no collateral moves.

Why the others are wrong

- **A:** Rounded 20,000 to the nearest 25,000 and called it, ignoring the MTA test (rounding is applied only once a transfer is due).
- **B:** Called the MTA amount itself; the MTA is a minimum trigger, not the amount transferred.

- **D:** Ignored the threshold and called the full mark-to-market value.

Q799

BT notes P2.T6.24.28.2

REBUILT — Source key (A) was wrong and the stem was inconsistent about who held the 45,000; restated that Maplewood holds 45,000 previously posted by Downers Grove and rebuilt options.

Maplewood Financial and Downers Grove Capital have a two-way collateral agreement with a threshold of $1,000,000 (each party), a minimum transfer amount (MTA) of $150,000, and rounding to the nearest $25,000. Maplewood currently holds $45,000 of collateral previously posted by Downers Grove. The latest valuation shows a mark-to-market value of +$1,105,000 for Maplewood (and -$1,105,000 for Downers Grove). What collateral movement results?

A. Downers Grove posts a further $150,000
B. Downers Grove posts a further $100,000
C. No collateral is transferred
D. Downers Grove posts a further $50,000

Answer: C

Key correction: Source key (BT notes) said A (a $150,000 transfer); correct is C because the credit support amount is only $60,000, below the $150,000 MTA.

Step by step

1. Required collateral = max(1,105,000 − 1,000,000, 0) = 105,000.
2. Credit support amount = 105,000 − 45,000 held = 60,000.
3. 60,000 < MTA 150,000, so no transfer is made.
4. Collateral transferred = 0.

Why C is correct: The MTA is tested against the credit support amount (required less held), which is 60,000 and so below the 150,000 MTA.

Why the others are wrong

- **A:** Treated the MTA as the amount to be posted; the MTA is only a trigger level.
- **B:** Rounded the full required 105,000 to 100,000, ignoring both the 45,000 already held and the MTA.
- **D:** Rounded the correct 60,000 down to 50,000 but ignored the MTA test.

Q800

BT answers P2.T6.410.3 · BT notes P2.T6.410.3

A financial institution (FI) enters a derivatives trade with a counterparty (CP) under a two-way CSA with identical parameters for both parties: daily valuation, a mutual threshold of $100,000, and a mutual minimum transfer amount (MTA) of $40,000. At inception T(0) the exposure is zero. FI then gains on the trade:
- T(1): FI's exposure to CP increases to $115,000
- T(2): the trade gains a further $35,000, so FI's exposure to CP is $150,000
The collateral posted by CP attracts a haircut of 20.0%. Which of the following accurately describes the first collateral call?

A. On T(1), the counterparty posts collateral with market value of $18,750
B. On T(1), the counterparty posts collateral with market value of $143,750
C. On T(2), the counterparty posts collateral with market value of $50,000
D. On T(2), the counterparty posts collateral with market value of $62,500

Answer: D

Step by step

1. T(1): credit support amount = 115,000 − 100,000 = 15,000 < MTA 40,000, so no call.
2. T(2): credit support amount = 150,000 − 100,000 = 50,000 > 40,000, so the first call is at T(2).
3. Collateral value to be credited = 50,000; with a 20% haircut the market value posted = 50,000/(1 − 0.20) = 62,500.

Why D is correct: The first time exposure less threshold exceeds the MTA is T(2); the haircut requires grossing up the 50,000 call by dividing by 0.80.

Why the others are wrong

- **A:** 15,000/0.8 = 18,750: called at T(1) ignoring that 15,000 is below the MTA.
- **B:** 115,000/0.8 = 143,750: ignored the threshold and the MTA at T(1).
- **C:** Correct timing and call amount but forgot the haircut, so the market value posted would be credited at only

40,000.

Q801

BT answers P2.T6.414.3 · BT notes P2.T6.414.3

REBUILT — Sources left the rounding convention for returns unstated and had weak distractors; stated that calls round up and returns round down and rebuilt options from realistic errors.

Consider a two-way CSA with identical parameters for both parties: threshold $1,000,000; independent amount zero; minimum transfer amount (MTA) $100,000; rounding to a multiple of $25,000 (collateral calls are rounded up, returns are rounded down); initial collateral held is zero. From the perspective of Party A, the portfolio mark-to-market is +$1,430,000 on day 1 and +$1,030,000 on day 2. Which is the correct sequence of collateral movements for Party A?

A. Day 1: $450,000 called (received) by Party A; day 2: $400,000 returned by Party A
B. Day 1: $450,000 called by Party A; day 2: no return, since the mark-to-market is still above the threshold
C. Day 1: $430,000 called by Party A; day 2: $420,000 returned by Party A
D. Day 1: $450,000 called by Party A; day 2: $450,000 returned by Party A

Answer: A

Step by step

1. Day 1: $1{,}430{,}000 - 1{,}000{,}000 = 430{,}000 >$ MTA 100,000; rounded up to 450,000, which is called.
2. Day 2: required collateral $= 1{,}030{,}000 - 1{,}000{,}000 = 30{,}000$; held $= 450{,}000$.
3. Credit support amount $= 30{,}000 - 450{,}000 = -420{,}000$; its absolute value exceeds the MTA of 100,000, so a return is due.
4. Rounded down to a multiple of 25,000: 400,000 returned (leaving 50,000 held).

Why A is correct: Collateral is called on the excess over the threshold, rounded up, and returned when collateral held exceeds the requirement, rounded down.

Why the others are wrong

- **B:** Wrongly assumes collateral is returned only when the mark-to-market falls below the threshold; surplus collateral over the 30,000 requirement is returned even while above it.
- **C:** Ignores rounding: 430,000 and 420,000 are the unrounded amounts.
- **D:** Returns all collateral held, although 30,000 above the threshold still needs to be collateralised.

Q802

BT answers P2.T6.906.3 · BT notes P2.T6.906.3

REBUILT — Source key B had the wrong direction (ToughFax is in-the-money, so it calls rather than GreenBam returns); corrected the answer and replaced the implausible $300,000 option with a realistic distractor.

ToughFax International has a two-way credit support annex (CSA) with its counterparty, GreenBam Financial, in a derivatives transaction. Both parties post collateral under mutual parameters: threshold of $1,000,000; minimum transfer amount (MTA) of $100,000; and rounding of $25,000. Before today's re-margin determination, ToughFax held no collateral, but GreenBam held $50,000 of collateral previously posted by ToughFax. Today the mark-to-market (MTM) value is +$1,094,000 from ToughFax's perspective (and -$1,094,000 from GreenBam's). What is the credit support amount?

A. Zero (neither call nor return)
B. GreenBam returns $150,000
C. ToughFax calls $150,000
D. ToughFax calls $94,000

Answer: C

Key correction: Source key (BT answers/notes) said B; correct is C because ToughFax has the positive exposure, so collateral flows from GreenBam to ToughFax (a call by ToughFax, which also nets off the $50,000 ToughFax had posted), not a return by GreenBam.

Step by step

1. Two-party formula (ToughFax's view): $\text{CSA} = \max(V - K_c, 0) - \max(-V - K_p, 0) - C$.
2. $\max(1{,}094{,}000 - 1{,}000{,}000, 0) = 94{,}000$; $\max(-1{,}094{,}000 - 1{,}000{,}000, 0) = 0$.
3. Credit support balance held by ToughFax is $C = -50{,}000$ (it has posted $50,000 to GreenBam).
4. $\text{CSA} = 94{,}000 - 0 - (-50{,}000) = 144{,}000$ (positive, so ToughFax may call).
5. $144{,}000 >$ MTA $= 100{,}000$, so the transfer is made.
6. Rounding to the $25,000 multiple: $144{,}000 \rightarrow 150{,}000$. ToughFax calls $150,000.

Why C is correct: The threshold is netted against the positive exposure and the collateral ToughFax already posted

is netted off the requirement, giving $144,000, which exceeds the MTA and rounds to $150,000 payable to ToughFax.

Why the others are wrong

- **A:** Compares only the $94,000 excess over the threshold with the $100,000 MTA and concludes no transfer; this ignores the $50,000 ToughFax already posted, which GreenBam must return, so the net amount is $144,000 and exceeds the MTA.
- **B:** Right size but wrong direction: ToughFax is in the money, so GreenBam must deliver collateral to ToughFax (the $50,000 return plus a $94,000 delivery); the amount flows to ToughFax, not as a $150,000 return of ToughFax's own collateral.
- **D:** Uses just the $94,000 excess over the threshold, ignoring the $50,000 of collateral to be netted and ignoring the MTA and rounding rules.

Q803 GARP PE1 Q16 · Mock 3 Q64

An investment bank has a one-way credit support annex (CSA) on a bilateral transaction with a hedge fund counterparty. Under the CSA, the mark-to-market value of the transaction forms the basis of the hedge fund's collateral requirements, given below:

Item	Value (CNY)
Mark-to-market value of net exposure	25,000,000
Mark-to-market value of collateral posted	10,800,000
Threshold amount	14,000,000
Minimum transfer amount	2,500,000
Rounding amount	10,000

Assuming the net exposure increases to CNY 27,000,000 and the mark-to-market value of collateral posted has not changed, how much additional collateral will the hedge fund have to post?

A. CNY 0
B. CNY 1,990,000
C. CNY 2,000,000
D. CNY 2,500,000

Answer: A

Step by step

1. Required collateral = exposure minus threshold: $27{,}000{,}000 - 14{,}000{,}000 = 13{,}000{,}000$.
2. Collateral already held: 10,800,000.
3. Credit support amount: $\max(27{,}000{,}000 - 14{,}000{,}000, 0) - 10{,}800{,}000 = 2{,}200{,}000$.
4. Compare with the MTA: $2{,}200{,}000 < 2{,}500{,}000$, so no call can be made.
5. Rounding applies only to a transfer that is actually triggered, so the additional collateral is CNY 0.

Why A is correct: The shortfall of CNY 2.2 million is below the minimum transfer amount of CNY 2.5 million, so no margin call can be made.

Why the others are wrong

- **B:** Equals the CNY 2,000,000 exposure increase less the CNY 10,000 rounding amount, which is not a valid calculation; it ignores that the CNY 2.2m shortfall is below the MTA, so no transfer occurs.
- **C:** Uses the CNY 2,000,000 increase in exposure (27m - 25m) as the call, ignoring the threshold, the existing collateral and the MTA.
- **D:** Assumes a call equal to the MTA is triggered; the MTA is a minimum size for a transfer, not an amount to post, and the shortfall is below it.

LO 35.d

Describe the role of a valuation agent.

Q804 BT answers P2.T6.409.2 · BT notes P2.T6.409.2

Each of the following is a typical calculation performed by a valuation agent EXCEPT which one?

A. To calculate credit exposure including the impact of netting
B. To calculate the threshold
C. To calculate the uncollateralized exposure
D. To calculate the delivery or return amount (the amount of collateral to be posted by either counterparty)

Answer: B

Step by step

1. The valuation agent calls for delivery or return of collateral and handles the margin calculations: credit exposure (net of netting), market value of posted collateral after haircuts, total uncollateralized exposure, and the credit support (delivery/return) amount.
2. A, C and D are on this list.
3. The threshold is a contractual parameter fixed in the CSA at inception, not something recalculated by the valuation agent each period.
4. The answer is B.

Why B is correct: The threshold is negotiated and written into the CSA; the valuation agent applies it but does not calculate it.

Why the others are wrong

- **A:** Is a core valuation agent task (current exposure with netting), so it is not the exception.
- **C:** Is a core valuation agent task; picking it confuses it with the fixed threshold parameter.
- **D:** The delivery/return (credit support) amount is the main output of the valuation agent, so it is not the exception.

Q805 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

Two dealer banks of broadly similar size have just signed a bilateral credit support annex. A collateral manager at one of them notes that the agreement names both banks as valuation agent. Which of the following is the most likely consequence of this arrangement, and the usual remedy?

A. Each bank calculates exposure but neither may call for collateral, so the central counterparty must be introduced as the remedy.
B. Only the bank with the larger positive exposure may call for collateral, so disputes are eliminated and no remedy is needed.
C. Each bank calls for collateral when it has positive exposure, which can lead to disputes and delays; appointing a third-party valuation agent is a common remedy.
D. Collateral calls are made only on the net of the two banks' calculations, so disputes cannot arise and no third party is required.

Answer: C

Step by step

1. When both counterparties are valuation agents, each calls for collateral when it has positive exposure and each performs its own calculations.
2. Differences between two sets of calculations create potential for disputes and delays in collateral movements.
3. A third-party valuation agent can process substitutions, resolve disputes and produce daily valuation reports.

Why C is correct: Dual valuation agents each call when in the money, which creates dispute and delay risk; a third-party agent is the standard remedy.

Why the others are wrong

- **A:** A valuation agent does call for collateral; a CCP is not the remedy for a bilateral agreement.
- **B:** Nothing restricts calls to the larger exposure; each agent calls when its own exposure is positive, and disputes are not eliminated.
- **D:** Netting of two calls is not how the dual-agent arrangement works; two independent calculations are the source of disputes.

Q806 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A collateral analyst acting as valuation agent for a bilateral OTC portfolio reports the following: net exposure after netting is USD 18 million; the market value of collateral previously posted is USD 8 million and carries a 10% haircut; the threshold is USD 5 million; no initial margin applies; and the minimum transfer amount is USD 0.5 million. Assuming the valuation agent applies the haircut to the posted collateral, what collateral amount should it call for?

A. USD 4.2 million
B. USD 5.0 million
C. USD 10.8 million
D. USD 5.8 million

Answer: D

Step by step

1. Adjust posted collateral for the haircut: 8 x (1 - 0.10) = 7.2.
2. Uncollateralized exposure = 18 - 7.2 = 10.8.
3. Exposure to be collateralized = 10.8 - threshold 5 = 5.8, which exceeds the 0.5 MTA, so call 5.8.

Why D is correct: Valuation agent values posted collateral net of its haircut, then removes the threshold: 18 - 7.2 - 5 = 5.8.

Why the others are wrong

- **A:** 18 - 8(1.10) - 5 = 4.2: grosses up collateral instead of applying the haircut.
- **B:** 18 - 8 - 5 = 5.0: ignores the haircut on posted collateral.
- **C:** 10.8 is the uncollateralized exposure; it fails to deduct the threshold.

LO 35.e

Describe the mechanics of collateral and the types of collateral that are typically used.

Q807 AnalystPrep Q.5506

REBUILT — Original distractors were giveaways (words like 'exclusive' and return-seeking rationales); rewrote options so each reflects a realistic misconception while keeping the same LO and answer.

Global Finance Bank is reviewing the collateral it will accept and post against its OTC derivatives portfolio, with the priority of keeping collateral liquid and free of price uncertainty so that margin calls are easy to meet. Which collateral approach is most consistent with this priority?

A. Rely mainly on cash, supplemented by high-quality government securities.
B. Rely mainly on mortgage-backed securities, since their yield offsets the credit risk of the derivatives.
C. Rely mainly on letters of credit, since a guarantee of payment removes the need for liquid assets.
D. Rely mainly on equities, since their expected appreciation improves the return on the collateral pool.

Answer: A

Step by step

1. The stated priority is liquidity and absence of price uncertainty.
2. Cash is the most common collateral type and has no price variability (haircut typically zero in the eligible currency).
3. Government securities are liquid and low-risk, so they complement cash.
4. MBS, letters of credit and equities are all permitted collateral types but carry price uncertainty, liquidity or rehypothecation problems.
5. Therefore A.

Why A is correct: Cash, supplemented by government securities, is the most liquid, least price-uncertain collateral, which is what the priority requires.

Why the others are wrong

- **B:** Confuses yield with collateral quality; MBS prices are volatile and can be illiquid in stress, so they fail the priority.
- **C:** Treats a guarantee as equivalent to cash; a letter of credit is not a liquid asset that can be delivered to meet calls and depends on the issuer.

- **D:** Chases return; equities have high price variability, require large haircuts, and collateral should protect, not earn, returns.

Q808

AnalystPrep Q.5162

REBUILT — Option D was garbled and options C/D were ambiguous about 'control'; reworded options so exactly one is correct.

Which statement correctly distinguishes a security interest arrangement from a title transfer arrangement for posting margin?

A. Under a security interest, legal ownership of the margin passes to the receiver; under title transfer it stays with the giver.

B. Under title transfer, legal ownership of the margin passes to the receiver (possibly with usage restrictions); under a security interest the giver retains ownership and the receiver holds a claim exercisable on default.

C. Under a security interest, the receiver may freely use the margin assets; under title transfer it may not.

D. Under title transfer, the giver keeps the margin assets and is protected if the receiver defaults; under a security interest it is an unsecured creditor.

Answer: B

Step by step

1. Security interest (pledge, typical of New York law): the giver continues to own the assets; the receiver has a claim on them on default events.
2. Title transfer (typical of English law): ownership passes outright, subject to possible usage restrictions.
3. Hence the receiver can generally use title-transferred assets freely, and the giver ranks as an unsecured creditor for their return.
4. Only B states this correctly.

Why B is correct: Title transfer moves legal ownership to the receiver, whereas a security interest leaves ownership with the giver and gives the receiver only a claim on default.

Why the others are wrong

- **A:** Reverses the two arrangements.
- **C:** Reverses who has freedom of use; the receiver of title-transferred assets can generally use them freely.
- **D:** Reverses the giver's protection: under title transfer the giver is an unsecured creditor of the receiver; a security interest is preferable for the giver.

Q809

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A risk manager at a derivatives dealer is reviewing the types of collateral accepted under its collateral agreements. Which of the following statements is most accurate?

A. Cash is the most common collateral but can become scarce in stress, while noncash collateral can introduce rehypothecation issues and price uncertainty.

B. Government agency securities are free of price uncertainty, which is why they are always preferred over cash and no issues arise from their reuse.

C. Letters of credit and equities are the most common collateral because they are the easiest to value and to rehypothecate.

D. Noncash collateral eliminates the need for daily marking to market because its value cannot change between collateral calls.

Answer: A

Step by step

1. Cash is most common but supply can be scarce in stress.
2. Noncash collateral (agency securities, etc.) may be preferred for liquidity but gives rise to rehypothecation problems and price uncertainty.

Why A is correct: Statement A matches the described strengths and weaknesses of cash and noncash collateral.

Why the others are wrong

- **B:** Agency securities carry price uncertainty and rehypothecation issues; cash is the most common type.
- **C:** Letters of credit and equity are less common, and are not the easiest to value.
- **D:** Noncash collateral is valued by marking to market, typically daily, because its value does change.

LO 35.f

Explain the process for the reconciliation of collateral disputes.

Q810 AnalystPrep Q.1907

Collateral management still relies heavily on manual processes, which often leads to valuation disputes over previously posted collateral. Which of the following correctly describes how such a dispute should be addressed?

A. If the disputed amount is higher than an acceptable level stated in the collateral agreement, the counterparties may "split the difference."

B. If the disputed amount is under an acceptable level stated in the collateral agreement, the counterparties may "split the difference"; otherwise, a case must be filed against the defaulting party.

C. If the disputed amount is under an acceptable level stated in the collateral agreement, the counterparties may "split the difference"; otherwise, the cause of the difference must be found.

D. If the disputed amount is under an acceptable level stated in the collateral agreement, the parties should ignore it; otherwise, the cause of the difference must be found.

Answer: C

Step by step

1. Small valuation differences (below the tolerance in the agreement) are settled pragmatically by splitting the difference.
2. Larger differences cannot be split: the undisputed amount is transferred, the disputed exposure stays uncollateralized, and the reason for the dispute must be identified.
3. Only C pairs 'split the difference' with a small amount and 'find the cause' for a larger one.

Why C is correct: Splitting the difference is a shortcut reserved for small differences; larger disputes require identifying the root cause (and, if unresolved, obtaining market-maker quotes).

Why the others are wrong

- **A:** Reverses the logic: splitting is for amounts below, not above, the tolerance.
- **B:** Right on small amounts, but a lawsuit against a 'defaulting party' is not part of the reconciliation process; a disputing party is not in default.
- **D:** Ignoring a small difference is not the practice; it is split, and the difference is still settled between the parties.

Q811 AnalystPrep Q.5504

Global Finance Bank (GFB) has a collateral dispute with a major OTC derivatives counterparty over the valuation of complex credit derivatives covered by their collateral agreement. As risk manager, which of the following is the standard first sequence of steps in reconciling the dispute?

A. Immediately request quotes from several market makers for the MtM value of the disputed trades and temporarily exclude the problematic assets from collateral calls.

B. Split the difference on the disputed amount regardless of its size to ensure a quick resolution and preserve the relationship.

C. Notify the counterparty of the intent to dispute by the end of the day following the collateral call, transfer all undisputed amounts, and identify the reason for the dispute.

D. Permanently exclude all complex credit derivatives from the collateral agreement to prevent future disputes.

Answer: C

Step by step

1. Step 1: the disputing party notifies the counterparty by the end of the day following the collateral call.
2. Step 2: all undisputed amounts are transferred and the reason for the dispute is identified.
3. Step 3 (only if unresolved): quotes are requested from several (usually four) market makers.
4. Option C is steps 1 and 2.

Why C is correct: The standard process is notify by end of next day, transfer undisputed amounts and diagnose the cause; market-maker quotes come only for unresolved disputes.

Why the others are wrong

- **A:** Market-maker quotes are the last step for unresolved disputes, not the first; excluding assets from calls is not part of the dispute process.
- **B:** Splitting is acceptable only when the difference is small; larger differences leave the exposure uncollateralized until resolved.

- **D:** A permanent exclusion is a trade-population design choice made when negotiating the agreement, not a step in reconciling a live dispute.

Q812 AnalystPrep Q.5505

During reconciliation of its OTC derivatives portfolio, GFB finds valuation discrepancies causing a dispute with a key counterparty. The dispute stems from different market data and market closing times used for valuation. Which action best aligns with standard practice for resolving such a dispute?

A. Insist that the counterparty accept GFB's valuation because its market data and processes are accurate.

B. Immediately split the difference, regardless of the reason for the discrepancy.

C. Review both parties' valuation methodologies, market data sources and the market closing times used for the disputed trades to identify the cause.

D. Automatically accept the counterparty's valuation to preserve the relationship and adjust GFB's processes accordingly.

Answer: C

Step by step

1. Market data and market close time are recognized sources of disputes.
2. Standard practice: transfer undisputed amounts and identify the reason for the dispute before anything else.
3. Therefore compare methodologies, data sources and closing times (C).

Why C is correct: Identifying the root cause of the discrepancy is the required step when the difference is not trivially small; the disagreement here is a data/timing difference that can be diagnosed.

Why the others are wrong

- **A:** Unilateral insistence resolves nothing and ignores that either side's data or timing may be the cause.
- **B:** Splitting applies only to small differences, not irrespective of size and cause.
- **D:** Blind acceptance is not a process step and does not identify why the valuations differ.

Q813 BT answers P2.T6.409.3 · BT notes P2.T6.409.3

Each of the following is a step in the process for the reconciliation of a collateral dispute EXCEPT:

A. The disputing party notifies its counterparty (or the third-party valuation agent) that it disputes the exposure or collateral calculation.

B. Both parties post collateral for the full amount of any disputed exposure, even if this means mutual overcollateralization, so that neither counterparty is partially exposed during the dispute.

C. The disputing party identifies the reason for the dispute and transfers the undisputed amount, and the parties attempt to resolve the dispute within a resolution time.

D. If the parties fail to resolve the dispute within the resolution time, they obtain MtM quotations from several market makers for the disputed components.

Answer: B

Step by step

1. Process: (1) notify of dispute by end of day after the call; (2) transfer undisputed amounts and identify the reason; (3) if unresolved, obtain quotes from several market makers.
2. A is step 1, C is step 2, D is step 3.
3. B is not a step: only the undisputed amount moves, so the disputed part stays uncollateralized until resolved.

Why B is correct: Only undisputed amounts are transferred; the disputed exposure remains partly uncollateralized until the cause is traced and corrected.

Why the others are wrong

- **A:** This is step 1 of the process, so it is not the exception.
- **C:** This is step 2 (transfer undisputed amount, find the reason), so it is not the exception.
- **D:** This is step 3 for unresolved disputes, so it is not the exception.

Q814 BT answers P2.T6.905.3 · BT notes P2.T6.905.3

REBUILT — Original asked for the 'most proactive solution with respect to the valuation agent' with vague options; reworded to ask which practice most reduces the chance of disputes and made distractors realistic.

A large bank trades OTC derivatives with a smaller counterparty under a collateral agreement. Given the decen-

tralized, non-transparent OTC market, significant disagreements about collateral requirements can occur. Which of the following is the most effective way to reduce the chance of collateral disputes?

A. Name both parties as valuation agents so each can call for collateral on its own valuation.

B. Reconcile portfolios and valuations regularly, for example daily, and consider practice reconciliations before trading.

C. Wait until a dispute arises and then split the difference regardless of its size.

D. Name the smaller counterparty as sole valuation agent, since it has the most to lose from a miscalculation.

Answer: B

Step by step

1. Reconciling trades minimizes the chance of disputes; most large portfolios are reconciled daily, and dummy reconciliations before trading help.
2. Dual valuation agents each call on their own numbers, which creates disputes and delays.
3. Splitting only suits small differences and is reactive, not proactive.
4. A larger entity typically insists on being valuation agent; none of the other options aligns valuations before a call is made.

Why B is correct: Frequent reconciliation aligns the parties' trade populations, valuations and margin requirements before a call is made, preventing disputes rather than settling them.

Why the others are wrong

- **A:** Two valuation agents are used only when counterparties are similar in size, and it can lead to disputes and processing delays.
- **C:** Reactive and valid only for small differences; larger ones stay uncollateralized until resolved.
- **D:** Not a dispute-prevention tool; a larger entity usually insists on the role, and a one-sided agent does not align valuations.

LO 35.g

Explain the features of a collateralization agreement.

Q815

BT answers P2.T6.325.2 · BT notes P2.T6.325.2

In regard to collateralization, each of the following is true except which is FALSE?

A. The valuation agent is the party calling for delivery or return of collateral and thus must handle all calculations, i.e., calculation of credit exposure, market value of posted collateral, uncollateralized exposure, and the delivery or return amount

B. A counterparty who posts a dividend-paying asset as collateral (i.e., the "giver" of collateral) forfeits all dividends as the receiver immediately becomes the economic owner of the collateral

C. The threshold represents an amount of uncollateralized exposure: if an exposure is above the threshold, only the incremental exposure will be collateralized

D. An independent amount is effectively a negative threshold and is typically held as a cushion against "gap risk", i.e., the risk that a transaction's market value may gap substantially and quickly

Answer: B

Step by step

1. Test each statement against the collateral framework.
2. A: the valuation agent calls for collateral and performs the exposure, collateral value, uncollateralized exposure and delivery/return calculations - true.
3. B: the receiver of collateral is not the permanent economic owner (it becomes so only if the giver defaults) and must pass on coupons and dividends to the giver - so the giver does not forfeit them; the statement is false.
4. C: exposure below the threshold is uncollateralized; only the excess above it is collateralized - true.
5. D: the independent amount (initial margin) works in the opposite direction to the threshold and cushions gap risk - true.
6. The FALSE statement is B.

Why B is correct: Collateral is posted as security; the receiver is only a temporary holder and must pass through dividends and coupons to the giver, so the giver keeps the economic benefits.

Why the others are wrong

- **A:** True: this is the standard list of valuation agent duties, so it is not the false statement.

- **C:** True: the threshold is the level of uncollateralized exposure; picking this reflects misreading it as the level at which everything is collateralized.
- **D:** True: the independent amount acts like a negative threshold and protects against gap risk; picking it confuses it with a positive threshold.

Q816

AnalystPrep Q.1899

A financial institution trades OTC derivatives with a counterparty under a bilateral collateral agreement. From the institution's perspective, which statement about how collateral is exchanged is correct?

A. If its MtM is positive it posts collateral, and if its MtM is negative it calls for collateral.
B. If its MtM is positive it calls for collateral from the counterparty, and if its MtM is negative the counterparty calls for collateral from it.
C. If its MtM is negative it calls for collateral, and if its MtM is positive the counterparty must post collateral to it only when the counterparty has a lower rating.
D. Collateral is a constant requirement whenever the institution has a lower credit rating than the counterparty.

Answer: B

Step by step

1. Positive MtM means the institution is in the money, so the counterparty owes it value and the institution bears the credit risk.
2. The institution therefore calls for collateral from the counterparty.
3. Negative MtM means the counterparty is in the money, so the counterparty calls and the institution posts.
4. In a bilateral agreement the direction of flow follows the sign of MtM, not relative ratings.

Why B is correct: Collateral flows from the party with negative MtM to the party with positive MtM, which is the party exposed to default.

Why the others are wrong

- **A:** Reverses the direction of flow: the party that is in the money does not post.
- **C:** Reverses the direction for negative MtM and adds a rating condition; a negative-MtM party posts rather than calls.
- **D:** Confuses rating-linked terms with a constant requirement; a lower rating may tighten thresholds, but posting still depends on MtM.

Q817

AnalystPrep Q.1909

REBUILT — Option B was arguably defensible and D was weak; reworded B and D so only A is correct.

Although derivative markets increasingly use daily margining, some smaller institutions may prefer a longer margin call frequency because:

A. a longer frequency gives them more time to meet operational and funding requirements.
B. a longer frequency removes the counterparty's uncollateralized exposure between calls.
C. daily margin calls are appropriate only for markets that are highly volatile.
D. daily margin calls are prohibited under the ISDA Master Agreement for smaller institutions.

Answer: A

Step by step

1. Daily margining requires daily valuation, calls and funding of transfers.
2. Smaller institutions may struggle with those operational and funding demands.
3. A longer call frequency reduces the workload, at the cost of more uncollateralized exposure between calls.
4. Longer frequencies are practical mainly in less volatile markets, but daily has become the market norm.

Why A is correct: A longer margin call frequency mainly reduces operational and funding burden for institutions with limited capacity.

Why the others are wrong

- **B:** Backwards: a longer frequency increases uncollateralized exposure between calls.
- **C:** Reverses the logic: longer frequencies suit less volatile markets, and daily calls are the general norm.
- **D:** Invented restriction; nothing in the ISDA documentation prohibits daily calls.

Q818 AnalystPrep Q.5500

Global Finance Bank is revising its collateralization agreement for its OTC derivatives portfolio, aiming to balance operational workload against risk mitigation. Which set of features best matches the typical features of such an agreement?

A. A high threshold, a low minimum transfer amount, haircuts from the latest VaR, and interest on cash collateral at a fixed rate below the market rate.

B. A zero threshold for all counterparties regardless of rating, a high minimum transfer amount, and uniform haircuts for all collateral including cash and government bonds.

C. Thresholds and initial margins linked to counterparty credit rating, tiered haircuts reflecting collateral riskiness, and interest on cash collateral at the overnight market rate.

D. No initial margin, a high threshold linked to market volatility instead of credit quality, and daily margin calls with no minimum transfer amount.

Answer: C

Step by step

1. Thresholds and initial margins are usually tiered by credit rating: lower ratings get lower thresholds and larger initial margins.
2. Haircuts rise with collateral riskiness (cash 0%, then government bonds, corporates, structured products, equities).
3. Interest on cash collateral is paid at the overnight market rate.
4. Only C is consistent with all three conventions.

Why C is correct: C matches standard practice: rating-linked terms, risk-based haircuts and overnight-rate interest on cash.

Why the others are wrong

- **A:** A high threshold leaves large uncollateralized exposure, a low MTA raises workload, and below-market fixed interest is not standard.
- **B:** Uniform haircuts ignore collateral risk (cash carries no haircut), and a rating-blind zero threshold ignores credit-linked terms.
- **D:** Ignores rating linkage, and a no-MTA daily regime maximises operational workload.

Q819 AnalystPrep Q.5501

A bank is revising the haircuts in its collateralization agreement. Which approach is most consistent with sound practice?

A. A uniform 5% haircut across cash, government bonds and equities to simplify calculation.

B. Higher haircuts for riskier assets such as equities and commodities, and low or zero haircuts for cash and high-quality government bonds.

C. No haircuts at all, to encourage counterparties to post collateral.

D. Extremely high haircuts on all collateral regardless of liquidity or volatility.

Answer: B

Step by step

1. A haircut of x% credits only (1 - x) of the collateral's value, covering the risk that it falls in value.
2. Riskier, more volatile or less liquid assets need larger haircuts.
3. Cash has a 0% haircut, followed by government bonds, AAA corporates, structured products, then equities and commodities.

Why B is correct: Haircuts should scale with collateral volatility, liquidity and default risk.

Why the others are wrong

- **A:** A uniform haircut penalises cash needlessly and understates the risk on equities.
- **C:** No haircuts leaves the bank exposed to declines in collateral value.
- **D:** Blanket high haircuts ignore asset differences and make collateral needlessly costly.

Q820 AnalystPrep Q.5503

REBUILT — Source's correct option (tiering by transaction volume) is not a standard feature; rebuilt so the MTA is tiered by credit rating.

A bank is setting the minimum transfer amount (MTA) in its collateral agreements. Which approach is most consistent with standard practice?

A. Tier the MTA by credit rating, with higher MTAs for lower-rated counterparties.
B. Use one high MTA for all counterparties to minimise the number of transfers, whatever their credit quality.
C. Tier the MTA by credit rating, with higher MTAs for higher-rated counterparties and lower MTAs for weaker ones.
D. Eliminate the MTA so that any change in exposure is collateralized immediately.

Answer: C
Step by step
1. The MTA is the smallest amount of collateral that can be transferred; it cuts the workload of small transfers.
2. A larger MTA means more uncollateralized exposure, which is tolerable for strong credits.
3. Standard practice links MTAs to rating: higher ratings get higher amounts.
4. Weaker counterparties get lower MTAs, tightening mitigation where risk is greatest.

Why C is correct: The MTA is typically tiered by rating with higher amounts for stronger credits, balancing workload against mitigation.
Why the others are wrong
- **A:** Reverses the tiering and loosens terms for weaker credits.
- **B:** Ignores credit quality and leaves weak counterparties under-collateralized.
- **D:** No MTA maximises mitigation but also the workload of tiny transfers.

Q821

BT Mock Q72 · BT answers P2.T6.905.1

REBUILT — Sources conflicted (Mock key B, BT key A) and option B relied on reset features outside the Schweser LO; B replaced with a threshold/MTA additivity statement.

Two counterparties have a set of OTC derivative transactions and a bilateral collateral agreement. Each of the following is true EXCEPT which is false?

A. Following best practice, collateral posting should be continuous for both parties.
B. Variation margin is collateral that tracks the change in the MtM of the position over time.
C. In a bilateral agreement, a party with positive MtM calls for collateral and a party with negative MtM must post.
D. The threshold and the minimum transfer amount are additive: exposure must exceed their sum before a call can be made.

Answer: A
Key correction: BT Mock Q72 key said B and BT answers P2.T6.905.1 key said A; the two conflicted. Correct is A: continuous posting is not best practice (daily is the norm).
Step by step
1. Margin calls should be made at least daily; daily margining is the market norm, with intraday calls mainly for repos and centrally cleared products.
2. Continuous posting is not best practice because of its operational cost, so A is false.
3. Variation margin tracks MtM changes, so B is true.
4. Bilateral flows follow the sign of MtM, so C is true.
5. Threshold and MTA add, so D is true.

Why A is correct: Margining is periodic (daily at least), not continuous, to keep operational workload under control.
Why the others are wrong
- **B:** True statement: variation margin follows MtM.
- **C:** True statement: calls run both ways according to the sign of MtM.
- **D:** True statement: a call requires exposure above threshold plus MTA.

Q822

BT answers P2.T6.410.2 · BT notes P2.T6.410.2

In detailing the terms of a collateralization agreement, Barbara wants to mitigate her firm's exposure to a counterparty default as much as possible, accepting a heavier operational workload. Each of these adjustments will tend to increase mitigation EXCEPT which one?

A. Increase the independent amount
B. Increase the threshold
C. Decrease the minimum transfer amount
D. Increase the margin call frequency (e.g., from weekly to daily)

Answer: B

Step by step

1. The threshold is the exposure level below which no collateral is called, i.e. the uncollateralized exposure.
2. Raising it increases uncollateralized exposure and reduces mitigation.
3. A larger independent amount (initial margin) overcollateralizes, a lower MTA allows smaller calls, and more frequent calls shrink the gap between calls; all increase mitigation.

Why B is correct: A higher threshold leaves more exposure uncollateralized, so it reduces mitigation; a zero threshold means full collateralization.

Why the others are wrong

- **A:** Initial margin adds overcollateralization, so it improves mitigation.
- **C:** A smaller MTA lets smaller exposures be collateralized.
- **D:** Calling daily instead of weekly reduces unprotected exposure between calls.

Q823 BT answers P2.T6.906.2 · BT notes P2.T6.906.2

An institution is re-positioning a large portfolio into significant positions in the over-the-counter (OTC) derivatives market. Its primary concern with respect to the portfolio is gap risk: the risk that the portfolio value may gap substantially in a short period of time. Which of the following is the most appropriate protection against gap risk?

A. Higher haircut
B. Higher thresholds
C. Higher initial margins
D. Higher minimum transfer amounts (MTA)

Answer: C

Step by step

1. Gap risk is the risk that the exposure jumps by a large amount between the last margin call and close-out after a counterparty default.
2. Initial margin is posted upfront, independent of variation margin, and leaves the margin receiver overcollateralized; it effectively converts counterparty risk into gap risk that is covered by the cushion.
3. Therefore a larger initial margin directly absorbs a large sudden move in the portfolio value.
4. Haircuts protect against a fall in the value of the collateral asset, thresholds and MTAs leave exposure uncollateralized.

Why C is correct: Initial margin is an extra cushion that stays in place even when variation margin lags, so it is sized to cover large value movements (gaps) of the trades if the counterparty defaults.

Why the others are wrong

- **A:** A higher haircut covers price volatility and illiquidity of the posted collateral asset, not a jump in the portfolio's exposure; this is the plausible but second-best choice.
- **B:** A higher threshold increases the uncollateralized exposure below which no margin can be called, so it worsens rather than mitigates gap risk.
- **D:** A higher MTA delays small transfers (reducing operational workload) and adds to residual uncollateralized exposure, so it weakens protection.

Q824 BT notes P2.T6.906.1

Collateral agreements are increasingly about funding (FVA) and capital (KVA) costs in addition to pure counterparty risk (CVA) issues. Given the explicit recognition of these costs and the shift toward central clearing and bilateral margin requirements, what is the general impact on thresholds?

A. More zero thresholds
B. More infinite thresholds
C. No impact on thresholds
D. Thresholds often set equal to initial margin

Answer: A

Step by step

1. Regulatory (bilateral margin) requirements and central clearing require full variation margin with a zero threshold.
2. Thresholds can now be non-zero only for parties exempt from the bilateral margin requirements, so thresholds

are increasingly zero.
3. Hence the trend is toward more zero thresholds.

Why A is correct: Thresholds are increasingly zero and can be non-zero only for parties exempt from bilateral margin requirements; central clearing also requires full margining.

Why the others are wrong

- **B:** An infinite threshold means a party never posts (as in a one-way CSA); one-way CSAs have become more problematic given funding and capital costs, so this is the opposite of the trend.
- **C:** Wrong because regulation and clearing directly change threshold terms.
- **D:** Thresholds and initial margin work in opposite directions (one leaves exposure uncollateralized, the other overcollateralizes); they are not set equal, and where initial margin is present the threshold is typically zero.

Q825 AnalystPrep Q.5165

Which of the following is the most common margin call frequency in OTC derivatives markets?

A. Intraday
B. Monthly
C. Weekly
D. Daily

Answer: D

Step by step

1. Margin call frequency is the contractual period at which margin may be called and returned.
2. Daily calls have become standard in most OTC derivatives markets.
3. Intraday calls are mainly for vanilla products such as repos and for CCP-cleared derivatives; longer periods suit only smaller institutions or less volatile markets.

Why D is correct: Daily margin calls are the standard in most OTC derivative markets.

Why the others are wrong

- **A:** Intraday calls are common for repos and CCP-cleared products, not for typical bilateral OTC derivatives.
- **B:** A monthly frequency would leave exposure uncollateralized for too long; it is not standard.
- **C:** Weekly calls can be agreed with smaller or low-volatility counterparties but are not the norm.

LO 35.h

Differentiate between a two-way and one-way CSA agreement and describe how collateral parameters can be linked to credit quality.

Q826 AnalystPrep Q.6050

In the context of a derivative contract, a financial risk manager is considering a downgrade trigger as part of the credit risk mitigation strategy. This trigger requires a counterparty to post additional collateral if their credit rating falls below a predetermined level. Assuming that the counterparty's credit rating is downgraded, triggering the clause, and the counterparty has outstanding transactions, which of the following would be a likely course of action?

A. The counterparty would be granted a grace period to improve their credit rating before any action is taken.
B. The outstanding transactions would be netted to reflect a single net claim without any change in the collateral requirements.
C. The counterparty would be required to post additional collateral or potentially terminate outstanding transactions at market values.
D. Netting will be suspended, and the full gross value of all outstanding transactions becomes due immediately.

Answer: C

Step by step

1. A downgrade trigger links collateral terms to credit quality: once the rating falls below the agreed level, the clause is activated.
2. Activation tightens the collateral parameters (for example a lower threshold or added initial margin), so more collateral must be posted.
3. Alternatively the contract may allow the transactions to be terminated and settled at their market values (a

break clause).
4. Therefore C is the likely outcome; the other options delay the trigger, ignore it, or misstate how netting works.

Why C is correct: A downgrade trigger exists to give the protected party additional collateral, or the right to close out at market value, as soon as credit quality deteriorates.

Why the others are wrong

- **A:** A trigger is effective immediately; a grace period defeats its purpose of protecting against rapid deterioration.
- **B:** The stem says the clause is triggered, so collateral requirements must change; this option ignores the trigger.
- **D:** Netting is a legal feature that remains in force; triggers do not suspend it or make gross values due. Close-out, if any, is at net market value.

Q827

AnalystPrep Q.1894

REBUILT — Original option A was awkwardly worded and the stem mixed in unrelated claims; rewrote stem and options so exactly one option correctly describes a two-way agreement.

A collateral (margin) agreement between two counterparties is described as two-way. Which of the following best describes what this implies?

A. Whichever party has a negative mark-to-market value on the portfolio posts collateral to the other, so the direction of posting can change as exposure changes.

B. Both parties must post collateral at all times, regardless of the sign of the mark-to-market value.

C. Only the party with the weaker credit rating ever posts collateral, and the other party never does.

D. Either party may unilaterally change the collateral terms at any time if its counterparty's credit quality deteriorates.

Answer: A

Step by step

1. In a two-way agreement both parties are contractually obliged to post collateral, but only when they are out-of-the-money (negative MtM) so that the other party is exposed.
2. If the portfolio value flips sign, the collateral flows reverse: the previous receiver becomes the poster.
3. Hence A. One-way agreements (only one party posts) and unilateral changes to terms are not features of a two-way CSA.

Why A is correct: Two-way means either party may be the collateral provider, depending on which side of the net exposure it is on.

Why the others are wrong

- **B:** Collateral covers only the net exposure of the receiving party; a party that is in-the-money is not required to post.
- **C:** This describes a one-way agreement, in which only one party ever posts.
- **D:** Terms can be linked to credit quality only through pre-agreed provisions (e.g. rating triggers), not by unilateral amendment.

Q828

AnalystPrep Q.5502

Global Finance Bank (GFB) is reassessing the threshold levels in its collateralization agreements for OTC derivatives transactions. The reassessment comes amid changing market conditions and a diverse range of counterparties with varying credit qualities. Which of the following approaches to setting threshold levels in the collateralization agreement is most appropriate for GFB to balance risk management with operational efficiency?

A. GFB sets a universal threshold level for all counterparties, irrespective of their credit ratings, to standardize the collateral process and reduce complexity.

B. The bank decides to adopt a tiered threshold approach, where threshold levels are linked to the credit ratings of counterparties, with lower-rated counterparties having lower thresholds.

C. GFB eliminates threshold levels altogether, requiring all counterparties to fully collateralize their exposures from the first dollar, maximizing risk mitigation.

D. The bank sets extremely high threshold levels for all transactions to minimize the frequency of collateral calls, thereby reducing operational workload.

Answer: B

Step by step

1. The threshold is the amount of exposure left uncollateralised, so it should reflect counterparty credit risk.
2. Linking the threshold to rating (e.g. infinite for AAA, falling to zero for weaker ratings) reduces operational

workload for strong names while requiring collateral early from weak names.
3. So a rating-tiered schedule with lower thresholds for lower ratings balances risk control and efficiency (B).

Why B is correct: Rating-linked thresholds tie uncollateralised exposure to credit quality, minimising workload for strong counterparties and protecting against weak ones.

Why the others are wrong

- **A:** One threshold for all ignores credit differences, leaving weak names under-collateralised or burdening strong names with unnecessary calls.
- **C:** Zero thresholds for everyone maximise protection but ignore operational cost; they are unnecessary for high-quality counterparties, so there is no balance.
- **D:** Very high thresholds for all leave large uncollateralised exposure to weak counterparties, sacrificing risk mitigation for convenience.

Q829 BT answers P2.T6.905.2 · BT notes P2.T6.905.2

Two counterparties are negotiating the terms of a new Master Agreement in reference to a series of derivative transactions. They are also negotiating the parameters of a credit support annex (CSA) which might be appended to the Master Agreement. There are three approaches: two-way CSA, one-way CSA, or no CSA. About the possible CSA, which of the following statements is TRUE?

A. As CSAs are rare, the most likely approach is "No CSA"
B. If they agree to a one-way CSA, then by definition, the CSA will specify a threshold of zero
C. If they agree to a two-way CSA, then either party can unilaterally edit the CSA subsequent to signing the CSA
D. If they agree to a one-way CSA, the non-collateral posting party is likely to pay charges for the counterparty risk (CVA), funding (FVA), collateral terms (ColVA), and/or capital requirements (KVA) they impose on the other party

Answer: D

Step by step

1. A one-way CSA means only one party posts; the other receives collateral, and the poster bears additional risk and costs.
2. The non-posting party is therefore likely to be charged for the xVA it imposes: CVA, FVA, ColVA and KVA.
3. Check the others: CSAs are common among banks and financial institutions (A false); a one-way CSA implies an infinite threshold for the non-posting party (B false); a signed CSA cannot be edited unilaterally (C false).

Why D is correct: The party that does not post imposes counterparty, funding, collateral and capital costs on the poster, and is expected to compensate for them.

Why the others are wrong

- **A:** CSAs are not rare: banks almost always trade under CSAs and financial institutions often do.
- **B:** A one-way CSA gives the non-posting party an infinite threshold (it never posts), not zero; zero thresholds are typical of two-way interbank CSAs.
- **C:** A signed agreement is a contract and cannot be changed unilaterally; amendments need both parties' agreement.

Q830 Schweser Module Quiz 35.2 Q2

When dealing with a hedge fund, a bank would most likely negotiate a:

A. one-way agreement in the bank's favor given the bank's stronger credit rating.
B. one-way agreement in the bank's favor agreeing to post collateral to the hedge fund.
C. two-way agreement given the relatively small difference in credit quality between the two entities.
D. two-way agreement where both parties agree to post collateral.

Answer: A

Step by step

1. One-way CSAs are used when counterparties differ significantly in credit quality, size or risk.
2. A bank is typically of much higher credit quality than a hedge fund, so the bank can demand that only the hedge fund posts collateral.
3. Thus a one-way agreement in the bank's favor (A).

Why A is correct: A large credit-quality gap between a bank and a hedge fund supports a one-way agreement under which only the hedge fund posts.

Why the others are wrong

- **B:** Internally contradictory: a one-way agreement in the bank's favor means the hedge fund posts, not the bank.
- **C:** Two-way agreements suit similar counterparties, but the bank and hedge fund have a large, not small, credit gap.
- **D:** A two-way agreement is typical between similar parties such as banks, not for a bank facing a weaker hedge fund.

LO 35.i

Explain aspects of collateral including funding, rehypothecation, and segregation.

Q831

AnalystPrep Q.1911

A counterparty that has posted securities as collateral asks for them to be returned. Under a standard collateral agreement, this request is satisfied only if the counterparty:

A. posts a valid replacement of equal value to the collateral withdrawn, but must post the haircut amount in cash.
B. posts a replacement in exactly the same asset class as the collateral previously posted.
C. posts a replacement in a different asset class from the collateral previously posted.
D. posts an alternative amount of eligible collateral with equivalent value, after applying the relevant haircut.

Answer: D

Step by step

1. The request to recover posted collateral is a substitution request.
2. A substitution is valid when the replacement is eligible collateral under the agreement and has equivalent value.
3. Equivalent value is measured after the relevant haircut of the new asset, so more of a haircut asset must be posted.
4. Asset class and currency of the replacement need not match the original; eligibility is what matters.
5. Therefore D.

Why D is correct: Substitution requires eligible replacement collateral whose haircut-adjusted value equals that of the collateral released; the receiver cannot refuse if this is met.

Why the others are wrong

- **A:** Invents a requirement: the haircut is built into the amount of replacement collateral and need not be paid in cash.
- **B:** Substitution does not require the same asset class; any eligible collateral of equivalent value works.
- **C:** Different asset class is permitted but is not the condition; the replacement must still be eligible and equal in haircut-adjusted value.

Q832

AnalystPrep Q.4867

REBUILT — Original stem was ambiguous about whose risk is mitigated and the key mixed funding and counterparty risk; rebuilt from the receiving bank's viewpoint so exactly one option is correct.

Bank B receives collateral from Bank A under a derivatives contract. The collateral (i) does not need to be segregated, (ii) can be rehypothecated by Bank B, and (iii) has a value that is positively related to the credit quality of Bank A. Which of the following best describes the effect of this collateral for Bank B?

A. Full protection against counterparty risk and no funding benefit, because the collateral must be held until Bank A defaults.
B. A funding benefit only; it gives no counterparty risk mitigation because rehypothecable collateral is never available at default.
C. A funding benefit from reusing the collateral, but only limited counterparty risk mitigation because its value falls as Bank A's credit quality deteriorates.
D. Full counterparty risk mitigation and a funding benefit, because posted collateral always offsets the exposure completely.

Answer: C

Step by step

1. Rehypothecation lets Bank B reuse the collateral to secure its own obligations, which reduces its funding cost and its demand for high-quality collateral.

2. Collateral still reduces Bank B's exposure to Bank A, so mitigation is not zero.
3. Because the collateral's value is positively related to Bank A's credit quality, it loses value exactly when Bank A is likely to default, so protection at default is only partial.
4. Funding benefit plus partial protection gives C.

Why C is correct: Reuse (rehypothecation) provides a funding benefit, while collateral whose value falls with the poster's credit quality gives only limited protection at default.

Why the others are wrong

- **A:** Ignores the right of rehypothecation (the funding benefit) and overstates protection by ignoring the credit-quality link.
- **B:** Overstates the downside: collateral still reduces exposure; rehypothecation reduces, not eliminates, its availability.
- **D:** Overlooks that the collateral's value moves with the poster's credit quality, so mitigation is not full.

Q833

AnalystPrep Q.5491

During a risk committee meeting at GlobalBank, concerns are raised about the firm's collateral management practices for its derivatives portfolio. The Chief Risk Officer wants to differentiate between variation margin and initial margin policies. Which of the following practices would pose the highest risk from a counterparty risk perspective?

A. Rehypothecation of variation margin.
B. Segregation of initial margin.
C. Rehypothecation of initial margin.
D. Third-party custody of variation margin.

Answer: C

Step by step

1. Initial margin is extra (over-)collateral covering potential future losses; the poster expects it back if it is not needed, so it should remain protected.
2. Rehypothecating it means the receiver may be unable to return it if the receiver or the party it was reused with defaults, leaving the poster with an unsecured claim.
3. Variation margin reflects an amount currently owed, so on default it can be set off against the liability; rehypothecation adds only minimal counterparty risk.
4. Segregation and third-party custody reduce counterparty risk. The highest-risk practice is C.

Why C is correct: Initial margin is meant to be protected and recoverable, so reusing it exposes the poster to loss of collateral on default, whereas variation margin can be set off.

Why the others are wrong

- **A:** Rehypothecation of variation margin is natural and has minimal counterparty risk impact because it can be set off against the amount owed.
- **B:** Segregation is the protective practice; it lowers counterparty risk (at the cost of higher funding cost).
- **D:** Third-party custody is a form of segregation and reduces, not raises, counterparty risk.

Q834

BT Mock Q77 · BT answers P2.T6.907.2 · BT notes P2.T6.907.2

REBUILT — Merged three duplicates and replaced the title-transfer/security-interest option, which is outside the Schweser LO, with a segregation/funding statement.

Two counterparties are negotiating the credit support annex (CSA) governing collateral posting under an ISDA Master Agreement. Each of the following statements is true EXCEPT which is inaccurate?

A. If the goal is to minimize funding costs, both parties prefer to avoid rehypothecation rights.
B. With respect to variation margin, both parties probably prefer rehypothecation rights and probably will not require segregation.
C. With respect to initial margin, both parties are likely to insist on non-rehypothecation and will probably require segregation.
D. Segregation protects the collateral poster against loss of the collateral on the receiver's default, but may raise funding costs because the collateral cannot be reused.

Answer: A

Step by step

1. Rehypothecation lets the receiver reuse collateral, reducing demand for high-quality collateral and so lowering

funding cost; cost-minimizers therefore want rehypothecation rights, so A is false.
2. Variation margin reflects a current amount owed and can be set off on default, so reuse is natural and segregation is not usually required (B true).
3. Initial margin is an extra buffer that must be protected: segregated and not rehypothecated (C true).
4. Segregation works contrary to rehypothecation, so it protects the poster but forgoes reuse and raises funding cost (D true).

Why A is correct: Rehypothecation reduces funding costs, so a party minimizing funding costs would seek it rather than avoid it.

Why the others are wrong

- **B:** True: variation margin is quasi-settlement and set-off limits the counterparty risk of reuse.
- **C:** True: initial margin is an over-collateralization buffer that needs segregation and no reuse.
- **D:** True: segregation lowers counterparty risk but forgoes reuse, raising funding cost.

Q835

BT answers P2.T6.410.1 · BT notes P2.T6.410.1

REBUILT — Merged duplicates; source relied on an unreproducible exhibit and on title-transfer ownership and systemic-risk claims beyond the Schweser LO, so rebuilt around substitution, rehypothecation and segregation.

A financial institution trades a derivative with Counterparty A and hedges it with Counterparty B, who posts collateral to the institution under a CSA. The institution has the right to rehypothecate and passes the collateral on to Counterparty A. Each of the following is true EXCEPT which is not?

A. If the substituted collateral meets all eligibility criteria, a substitution request by Counterparty B cannot be refused by the institution.

B. If Counterparty A defaults and cannot return the rehypothecated collateral, the institution suffers a loss and still owes the collateral back to Counterparty B.

C. Segregation works contrary to rehypothecation and may create funding problems when replacing collateral that could have been rehypothecated.

D. If Counterparty A defaults and cannot return the rehypothecated collateral, the institution's obligation to return collateral to Counterparty B is extinguished.

Answer: D

Step by step

1. Substitution: posting equivalent value of other eligible collateral cannot be refused (A true).
2. Rehypothecation: the institution loses the collateral at Counterparty A and still has a liability to Counterparty B for its return (B true, D false).
3. Segregation legally protects posted collateral and is contrary to rehypothecation, at the cost of funding problems (C true).
4. The inaccurate statement is D.

Why D is correct: Rehypothecating collateral does not discharge the duty to return it to the poster, so a default downstream creates a double loss.

Why the others are wrong

- **A:** True: eligible substitution requests cannot be refused.
- **B:** True: this is the two-fold risk of rehypothecation (lost collateral plus liability to the poster).
- **C:** True: segregation is the opposite of rehypothecation and can cause funding issues.

LO 35.j

Explain how market risk, operational risk, and liquidity risk (including funding liquidity risk) can arise through collateralization.

Q836

AnalystPrep Q.6158

How does the process of mitigating counterparty risk through collateralization in derivatives transactions potentially affect other types of risk?

A. It reduces operational risk by simplifying trade entry and settlement processes.

B. It increases funding liquidity risk due to the need for timely payments, often within tight timescales.

C. It eliminates market risk by ensuring an objective valuation of the underlying transactions.

D. It diminishes legal risk by standardizing contractual terms across different jurisdictions.

Answer: B

Step by step

1. Collateralization converts counterparty risk into other risks rather than removing risk.
2. Variation margin must be delivered promptly (often next day or sooner) when MtM moves against the firm.
3. A firm without the liquidity or operational capacity to meet frequent calls faces funding liquidity risk, which rises when markets are illiquid.

Why B is correct: Collateral calls require liquid assets within short timescales, so counterparty risk is partly transformed into funding liquidity risk.

Why the others are wrong

- **A:** Collateral management adds missed calls, failed deliveries and human error, so operational risk increases, not decreases.
- **C:** Residual market risk remains over the margin period of risk and from thresholds/minimum transfer amounts; it is not eliminated.
- **D:** Collateral does not standardize terms across jurisdictions, and it can add legal risk (e.g., enforceability); this overstates the effect of a CSA.

Q837 BT answers P2.T6.900.2

Acme Financial Incorporated has an outstanding derivatives contract with its counterparty. Currently neither posts collateral, but they agree to modify the contract to include a bilateral collateral arrangement. Which risk has Acme reduced?

A. Acme reduced counterparty risk
B. Acme reduced operational risk
C. Acme reduced market/liquidity risk
D. All of the above (reduction in counterparty, operational, and market/liquidity risk)

Answer: A

Step by step

1. Bilateral collateral lowers exposure to the counterparty's default, so counterparty risk falls.
2. The same arrangement adds margin calls, valuation, delivery and reconciliation processes, raising operational risk.
3. Residual market risk (margin period of risk, thresholds) remains and liquidity/funding risk rises because Acme must post liquid collateral.
4. Only counterparty risk is reduced.

Why A is correct: Collateral transforms counterparty risk into operational, market and liquidity risks rather than reducing them.

Why the others are wrong

- **B:** Operational risk rises, since collateral handling creates more processes and chances for error.
- **C:** Market risk persists and funding/liquidity risk increases when collateral must be posted.
- **D:** Combines the correct item with two risks that actually increase.

Q838 AnalystPrep Q.5497

Global Finance Bank (GFB) is reviewing the operational risks associated with the collateralization of its OTC derivatives portfolio. During a period of market volatility, GFB has encountered several challenges in its collateral management process. Which scenario accurately demonstrates how operational risk can arise in the context of collateralization?

A. Operational risk emerges due to inaccurate valuation of collateral, leading to inadequate collateral postings and potential disputes with counterparties.
B. Operational risk is mitigated through the use of diversified collateral, reducing the bank's exposure to any single asset's volatility.
C. Operational risk is unaffected by collateralization practices, as it is primarily concerned with market fluctuations and credit risk management.
D. Operational risk increases as GFB engages in more OTC derivatives transactions, irrespective of the collateralization process.

Answer: A

Step by step

1. Operational risk from collateral comes from handling: valuation errors, missed calls, failed deliveries, computer or human error, fraud.
2. Inaccurate valuation causes wrong call amounts and disputes, a direct operational failure of the collateral process.
3. The other options either misattribute diversification (a market/credit risk tool), deny the link, or ignore collateralization altogether.

Why A is correct: Errors in valuing and processing collateral are classic operational failures that collateralization creates.

Why the others are wrong

- **B:** Diversification addresses market/concentration risk of the collateral, not process failures.
- **C:** Collateralization adds operational risk (missed calls, failed deliveries, errors); operational risk is not about market or credit risk.
- **D:** The question concerns the collateralization process; this option ignores it and does not describe how collateral creates operational risk.

Q839

AnalystPrep Q.5498

Global Finance Bank (GFB) is evaluating the liquidity risks associated with collateralization of its OTC derivatives. Its risk team is concerned about funding liquidity risk, the ability to meet obligations as they come due without unacceptable losses. Which scenario best illustrates how funding liquidity risk arises in the context of collateralization?

A. GFB faces funding liquidity risk when it has to post additional high-quality liquid assets as collateral during market downturns, potentially leading to a liquidity shortfall.

B. GFB experiences reduced funding liquidity risk as it receives more collateral, increasing its liquid asset holdings.

C. The use of long-term, illiquid assets as collateral by GFB minimizes funding liquidity risk by securing its obligations for an extended period.

D. Funding liquidity risk for GFB is mitigated when it diversifies its collateral portfolio, spreading the risk across different asset classes.

Answer: A

Step by step

1. Funding liquidity risk is the inability to meet margin calls when due.
2. In a downturn, MtM moves against GFB and calls rise just when liquid assets are scarce and funding costs high.
3. Only A describes a call that can exceed available liquid assets.

Why A is correct: Margin calls rise in stress, when liquid assets are hardest to obtain, so a shortfall can arise.

Why the others are wrong

- **B:** Received collateral is often owed back or restricted from reuse, and does not remove GFB's own calls.
- **C:** Posting illiquid assets does not reduce risk; eligible collateral is typically cash or high-quality securities and illiquid assets worsen liquidity strain.
- **D:** Diversification addresses concentration/market risk but not the timing of cash needs.

Q840

AnalystPrep Q.5499

REBUILT — Original keyed answer described additional collateral postings (a funding liquidity effect) as market risk; rebuilt around the margin-period-of-risk source of market risk in the LO.

Global Finance Bank (GFB) is a party to a collateralized OTC derivatives agreement that includes a threshold, a minimum transfer amount and a delay between a collateral call and receipt of the collateral. Which statement best describes how market risk arises through the collateralization?

A. Market moves between the last collateral posting and the receipt of collateral after default can leave exposure only partly covered, so some residual risk remains.

B. Posting collateral eliminates all market risk because exposure is reset to zero at every valuation date.

C. Market risk arises only when the collateral is cash held in a different currency from the trade.

D. Market risk is created only by operational failures such as missed collateral calls.

Answer: A

Step by step

1. Collateral is called and delivered only periodically, subject to threshold and minimum transfer amount.
2. Between the last posting and the time collateral is received (the margin period of risk), market moves can increase exposure.
3. Hence a residual, hard-to-hedge market risk remains even though collateral is held.

Why A is correct: Market risk is the exposure movement that occurs over the margin period of risk and within thresholds/minimum transfer amounts.

Why the others are wrong

- **B:** Ignores the delay (margin period of risk), threshold and minimum transfer amount; exposure is not reset to zero continuously.
- **C:** Currency mismatch is one source of FX risk, but it is not the source of market risk the LO describes.
- **D:** Missed calls are operational risk; they can aggravate exposure but do not define market risk.

Q841

AnalystPrep Q.6141

A bank's hedging strategy involves a mix of transactions with varying collateral requirements. The bank notices an imbalance in its collateral flows, which could affect its liquidity. Which is the most likely cause of the imbalance?

A. Inconsistent valuation methods for collateral across different transactions.

B. Discrepancy between uncollateralized client transactions and bilaterally collateralized hedges.

C. Over-collateralization of certain transactions leading to liquidity constraints.

D. Fluctuating market values affect the quality and quantity of posted collateral.

Answer: B

Step by step

1. Banks run mostly flat books: a client trade is often uncollateralized and is hedged with a trade under a CSA.
2. When the client trade moves in the client's favor (bank's loss), the bank posts margin on the hedge but receives none on the client trade.
3. This one-sided margin flow is a structural collateral imbalance and a funding liquidity cost.

Why B is correct: Hedging uncollateralized client trades with collateralized trades makes the bank fund margin without receiving offsetting margin.

Why the others are wrong

- **A:** Valuation inconsistency causes disputes (operational risk), not a systematic directional imbalance.
- **C:** Over-collateralization is a possible side effect, but not the structural cause of imbalance across the hedge book.
- **D:** Market moves change collateral values but affect both sides; they do not create a structural imbalance.

Q842

BT answers P2.T6.24.25.3

REBUILT — Original asked for the most likely risk with several defensible options (market, operational); rebuilt around a specific scenario where funding liquidity risk clearly arises from daily variation margin.

Global Manufacturing Corp. pays fixed and receives floating on a $100 million, 5-year interest rate swap with Acme Bank to hedge floating-rate debt. The swap is collateralized with daily exchange of variation margin. Interest rates fall sharply. Which risk does the collateral arrangement most directly create for Global?

A. Funding liquidity risk, because the swap has a negative MtM and Global must post liquid variation margin immediately, while the offsetting benefit (lower interest on its debt) arrives only gradually.

B. Increased credit risk to Acme, because Global's swap now has a large positive MtM to be collected from Acme.

C. No new risk, because the swap hedges the debt and daily margining removes all risks of the position.

D. Reduced operational risk, because daily margin automatically eliminates valuation disputes and failed deliveries.

Answer: A

Step by step

1. Global pays fixed and receives floating; when rates fall the swap's MtM turns negative for Global.
2. Under the CSA Global must deliver variation margin daily, in cash or eligible securities, in a short timescale.
3. The hedged debt's savings accrue slowly as coupons reset, so cash is needed now while the gain comes later: a funding liquidity strain.
4. Hence the collateral transforms credit risk into funding liquidity risk.

Why A is correct: Daily variation margin creates immediate cash needs on a losing hedge while the hedged-debt

benefit arrives over time.

Why the others are wrong

- **B:** Sign error: Global has a negative MtM and owes margin, so Acme (not Global) has the exposure.
- **C:** Collateral transforms rather than removes risk; margin calls, operations and residual market risk remain.
- **D:** Daily margining increases operational burden (calls, delivery, valuation, reconciliation), not reduces it.

Q843

BT answers P2.T6.325.3 · BT notes P2.T6.325.3

REBUILT — Source 'except' item keyed A, but a collateral issuer's default is a recognised collateral risk and the haircut/credit wording was debatable; rebuilt with one clearly correct statement matched to the LO's market, operational and liquidity risks.

Collateral management mitigates counterparty risk but converts it into other forms of risk. Which of the following correctly matches a source of risk to the risk type created by collateralization?

A. Minimum transfer amounts, thresholds and the margin period of risk leave exposure that can move before collateral is received, creating market risk.

B. Missed collateral calls, failed deliveries and human error create market risk.

C. Price impact and transaction costs when liquidating a large collateral position create operational risk.

D. Difficulty in meeting frequent, large margin calls creates operational risk.

Answer: A

Key correction: Source key (BT answers/BT notes) said A was the exception option; the haircut/credit-risk statement is debatable since collateral issuer default is a recognised collateral risk, so the item was rebuilt.

Step by step

1. Thresholds, MTAs and the delay between call and receipt (margin period of risk) mean exposure may move before collateral arrives: market risk.
2. Missed calls, failed deliveries, computer/human error and fraud are operational risk.
3. Liquidating collateral in size at a bid-offer or with price impact is liquidity/liquidation risk.
4. Meeting margin calls when they fall due is funding liquidity risk.

Why A is correct: Each source belongs to a specific risk category; only A pairs the source with its correct category.

Why the others are wrong

- **B:** Process failures are operational, not market, risk.
- **C:** Liquidation costs and price impact are liquidity (liquidation) risk.
- **D:** Inability to fund calls is funding liquidity risk; operational risk concerns process errors.

Q844

Schweser Module Quiz 35.2 Q1

REBUILT — Options B and C were weak or arguably true; reworded so only A is correct while keeping the liquidity/liquidation concept.

Collateral agreements could create multiple risks, including liquidity and liquidation risks. Which statement is most accurate?

A. Liquidation risk occurs when the amount of a security sold is large relative to its typical trading volume, which may drive down its price.

B. Liquidity risk is eliminated entirely once an appropriate haircut is applied to the collateral.

C. Liquidation risk arises only if the issuer of the collateral security defaults, not from the size of the position sold.

D. Liquidity risk occurs when there are potential pitfalls in the handling of collateral, including human error.

Answer: A

Step by step

1. Liquidating collateral incurs bid-ask and selling costs.
2. Selling an amount large relative to normal volume can move the price and cause a substantial loss.
3. Selling slowly instead exposes the holder to market volatility during liquidation.
4. Only A states this correctly.

Why A is correct: Large sales relative to normal trading volume depress price, which is the core of liquidation risk.

Why the others are wrong

- **B:** A haircut gives a buffer but does not remove price-impact losses beyond the haircut.
- **C:** Issuer default is a separate (default) risk; liquidation risk is driven mainly by size and market depth.
- **D:** Handling errors are operational risk, not liquidity risk.

LO 35.k

Describe the various regulatory capital requirements.

Q845

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A compliance analyst at a bank is preparing a briefing on the uncleared margin requirements for non-centrally-cleared derivatives between two covered entities. Which of the following statements about variation margin under these requirements is correct?

A. Variation margin may be subject to a threshold of up to EUR 50 million, provided the minimum transfer amount is zero.

B. Variation margin must be exchanged regularly (for example, daily) with a zero threshold and a minimum transfer amount not exceeding EUR 500,000.

C. Variation margin requirements are phased in over six years according to aggregate notional, and the minimum transfer amount is unrestricted.

D. Variation margin must be segregated with a third-party custodian and cannot be rehypothecated or netted.

Answer: B

Step by step

1. Variation margin: exchanged on a regular (e.g. daily) basis, full margin (zero threshold), MTA not above EUR 500,000, posted in full from the start of the rules, can be rehypothecated and netted.

Why B is correct: Option B states the variation margin rules.

Why the others are wrong

- **A:** The EUR 50 million threshold relates to initial margin, not variation margin.
- **C:** The multi-year phase-in applies to initial margin; variation margin applied in full from the start, and the MTA is capped.
- **D:** Segregation, no rehypothecation and no netting describe initial margin; variation margin may be reused and netted.

Q846

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A risk manager at a bank that is a covered entity is designing the process for determining the initial margin it will exchange on non-centrally-cleared derivatives under the uncleared margin requirements. Which of the following statements about how initial margin is calculated and implemented is correct?

A. Initial margin must be calculated using the regulatory schedule only, because internal models are not permitted for initial margin.

B. Initial margin should be calibrated to data from recent calm periods, because this reduces the procyclicality of margin over time.

C. Initial margin may be based on validated internal models or regulatory tables, using a 99%, 10-day move calibrated to a stress period.

D. Initial margin applied in full to all new and existing trades of all covered entities from the start, exactly as variation margin did.

Answer: C

Step by step

1. Initial margin covers an extreme but plausible move in portfolio value at a 99% confidence level over a 10-day horizon, on top of daily variation margin.
2. The scenario must use historical data that includes a period of significant financial stress, which keeps margin large enough and reduces procyclicality.
3. It can be calculated using validated internal models or regulatory tables.
4. Initial margin was phased in by declining aggregate notional thresholds and applies only to new transactions; variation margin applied in full from the start.

Why C is correct: Statement C matches the requirements: internal (validated) models or regulatory tables, 99% confidence, 10-day horizon, and calibration to a period of significant financial stress.

Why the others are wrong

- **A:** Wrong: the rules allow either regulatory tables or validated internal models; the mistake is treating the sched-

ule as mandatory.
- **B:** Wrong: the data must include a period of significant financial stress, and it is the stress calibration that reduces procyclicality; calm-period data would understate margin and increase procyclicality.
- **D:** Wrong: this confuses initial margin with variation margin. Initial margin was phased in by declining notional thresholds and need not be applied to existing contracts.

Q847 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

Bank A and Bank B, both covered entities, exchange initial margin on a non-centrally-cleared portfolio under the uncleared margin requirements. Which of the following statements regarding the treatment of the initial margin collected by Bank B is correct?

A. Bank B should size the margin on a 95% confidence level over a five-day horizon using historical data from calm periods.

B. Bank B may net the margin it collects against the margin it posts to Bank A, and may reuse it as it would variation margin.

C. Bank B may hold the margin in its own accounts and rehypothecate it, since initial margin is simply additional margin.

D. Bank B must segregate the margin with a third-party custodian or under other legally binding arrangements, and must not rehypothecate, repledge or reuse it.

Answer: D

Step by step

1. IM is exchanged by both parties on a gross basis (no netting).
2. The collecting party must segregate it so it is protected on its default or insolvency and must not rehypothecate, repledge or reuse it.
3. IM is based on an extreme but plausible move at 99% over 10 days, calibrated to a period of financial stress.

Why D is correct: Segregation and no reuse are required so posting IM does not raise counterparty risk.

Why the others are wrong

- **A:** The standard is 99% confidence, 10-day horizon, with stress-period data.
- **B:** IM is exchanged with no netting of amounts; only variation margin may be reused.
- **C:** Holding it in own accounts and rehypothecating contradicts segregation and no-reuse requirements.

Reading 36. Central Clearing

Jon Gregory, The xVA Challenge: Counterparty Credit Risk, Funding, Collateral, and Capital, 4th Edition (West Sussex, UK: John Wiley & Sons, 2020). Chapter 8.

LO	Learning objective	Qs
36.a	define a central counterparty (CCP) and describe the mechanics of central clearing.	3
36.b	explain the concept of novation under central clearing.	3
36.c	define netting, multilateral offset, and compression and provide examples of each.	5
36.d	describe the application and estimation of margin and default funds under central clearing.	4
36.e	discuss the risks faced by a CCP and the ways it manages its exposures.	4
36.f	provide examples of a loss waterfall.	7
36.g	explain the different methods of absorbing losses and managing the default of one or more members of a CCP.	5
36.h	compare bilateral and central clearing.	7
36.i	compare initial margin and default fund requirements for clearing members in relation to loss coverage, cost of clearing, and moral hazard.	4
36.j	describe the advantages and disadvantages of central clearing.	8

LO 36.a

Define a central counterparty (CCP) and describe the mechanics of central clearing.

Q848 AnalystPrep Q.6161

In the central clearing of over-the-counter (OTC) derivatives, which of the following correctly describes the fundamental role of a central counterparty (CCP)?

A. A CCP acts primarily as a regulatory body, setting legal and operational standards for the OTC derivatives market.

B. A CCP serves as an intermediary, stepping in between the original trading parties to assume their rights.

C. A CCP functions as a financial advisor, providing guidance on risk management and investment strategies to parties involved in OTC derivatives.

D. A CCP operates as a market maker, ensuring liquidity and setting prices for OTC derivatives.

Answer: B

Step by step

1. Central clearing replaces a bilateral trade with two trades, each facing the CCP.
2. The CCP becomes the buyer to every seller and the seller to every buyer, so the original counterparty is no longer a direct credit exposure.
3. Therefore the CCP's fundamental role is to interpose itself between the original parties (option B).

Why B is correct: Through novation the CCP stands in the middle of the trade and becomes the new counterparty to both sides, so the original counterparties no longer bear direct default risk to each other.

Why the others are wrong

- **A:** Confuses a CCP with a regulator; a CCP is a clearing entity subject to regulation, not the rule-setter for the market.
- **C:** A CCP provides no investment or risk advice; it manages its own risk through margin and default resources.
- **D:** A CCP does not make markets or set prices; trading and price formation occur between market participants, and the CCP only clears the resulting trades.

Q849 Schweser Module Quiz 36.1 Q1

There are numerous challenges when clearing over-the-counter (OTC) derivative products through a central counterparty (CCP). Which of the following lists best summarizes the key challenges for central clearing of OTC derivative products?

A. Illiquid products, jurisdictional fragmentation, presence of wrong-way risk, and legal concerns.

B. Lack of standardization, increased counterparty risk, increased dependency risk, and less transparency.

C. Jurisdictional fragmentation, increased counterparty risk, less transparency, and standardization.

D. Product complexity, illiquid products, presence of wrong-way risk, and lack of standardization.

Answer: D

Step by step

1. Recall the product characteristics that determine whether a product can be centrally cleared: standardization, complexity, liquidity, wrong-way risk and market volume.
2. Many OTC products are non-standard, complex, illiquid or subject to wrong-way risk, which makes them hard to value and margin.
3. Option D lists exactly these product-level challenges.

Why D is correct: CCPs need standardized, non-exotic, liquid products without wrong-way risk to value positions, set margin and close out defaulted positions; OTC products often fail these tests, so D is the product-challenge list.

Why the others are wrong

- **A:** Jurisdictional fragmentation and 'legal concerns' are not among the core product characteristics, and it omits complexity and lack of standardization.
- **B:** Central clearing reduces rather than increases counterparty risk and improves transparency; these are not challenges to clearing.
- **C:** Lists 'standardization' as a challenge when it is the lack of standardization that is the problem, and counterparty risk and transparency are benefits of clearing.

Q850 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A risk analyst at a derivatives dealer is reviewing the exposure created when the dealer's OTC swap portfolio is moved to a central counterparty (CCP). Which of the following statements best describes the risk position of the CCP once the trades have been cleared?

A. The CCP has a matched book and bears no net market risk, but it takes on the counterparty risk of its clearing members, which is centralized in the CCP.

B. The CCP eliminates counterparty risk from the market, because each trade is guaranteed and no party can default on the CCP.

C. The CCP bears the net market risk of all cleared trades, because it is the buyer to the seller and the seller to the buyer.

D. The CCP has a matched book and bears no risk of any kind, even if a clearing member defaults, because its positions always offset.

Answer: A

Step by step

1. The CCP is the buyer to every seller and the seller to every buyer, so its long and short positions offset: it has a matched book and no net market risk.
2. It does, however, guarantee performance, so counterparty risk is not removed but centralized in the CCP.
3. If a clearing member defaults the CCP no longer has a matched book, so it has only conditional market risk and must return to a matched book (for example by auction) and use margin and default fund resources.

Why A is correct: A CCP stands between buyers and sellers with a matched book, so it has no net market risk; it takes on and centralizes counterparty risk, with market risk arising only conditionally on a member default.

Why the others are wrong

- **B:** Counterparty risk is not eliminated; it is centralized in the CCP, which can still suffer losses if members default. Confuses transfer of risk with removal of risk.
- **C:** Because the CCP takes both sides of every trade, positions offset and it bears no net market risk. Confuses the CCP's gross positions with its net position.
- **D:** The CCP has only conditional market risk: on a member default it no longer has a matched book and may face losses. Ignores the conditional market risk and the need for margin and default fund resources.

LO 36.b

Explain the concept of novation under central clearing.

Q851 AnalystPrep Q.6091

Under central clearing, novation is a key legal step. Which of the following best describes the outcome of novation?

A. The original parties renegotiate the terms of their contract, with the CCP facilitating the process.

B. The original bilateral contract is replaced by two new contracts in which the CCP becomes the buyer to the seller and the seller to the buyer, so the original parties no longer have obligations to each other.

C. The original parties keep their mutual obligations and additionally agree to the CCP's terms.

D. The CCP guarantees payment to the original parties but does not become a party to the contracts.

Answer: B

Step by step

1. Novation replaces the original bilateral contract with new contracts.
2. The CCP steps in as counterparty to both sides: seller to the buyer and buyer to the seller.
3. The original parties' obligations to each other are extinguished; each now faces only the CCP, which bears the counterparty risk.

Why B is correct: Novation closes out the bilateral contract and substitutes the CCP as the legal counterparty to each side, so the original parties no longer owe each other anything.

Why the others are wrong

- **A:** Novation is replacement, not renegotiation; terms are not reopened.
- **C:** Confuses novation with an overlay: the original mutual obligations are discharged, not retained.
- **D:** Describes a mere guarantee; under novation the CCP is itself the contractual counterparty.

Q852 BT answers P2.T6.24.29.1 · BT notes P2.T6.24.29.1

REBUILT — Source key D paired a $10 million total with each-side $10 million and had the CCP pay fixed to the fixed payer; rewrote options so exactly one is consistent with novation.

Vintage Eagle Banking enters into a 5-year interest rate swap with Vinyl Corporation, notional $10 million. Vintage Eagle pays a fixed rate of 2.5% and receives 3-month LIBOR. The trade is then novated to a central counterparty (CCP). Which of the following is TRUE after novation?

A. The CCP has one $10 million swap with Vintage Eagle and one $10 million swap with Vinyl; it receives fixed from Vintage Eagle and pays fixed to Vinyl, and pays LIBOR to Vintage Eagle and receives LIBOR from Vinyl, leaving it with no net market exposure.

B. The original swap remains in place between Vintage Eagle and Vinyl, and the CCP guarantees its performance.

C. The CCP has two $5 million swaps, one with each party, splitting the original notional; it pays fixed to Vintage Eagle and receives floating from Vinyl.

D. The CCP has one $10 million swap with each party; it pays fixed to Vintage Eagle and receives LIBOR from Vinyl, and receives fixed from Vinyl.

Answer: A

Key correction: Source key (BT) said D; D has the CCP paying fixed to Vintage Eagle, but Vintage Eagle is the fixed payer, so the CCP must receive fixed from it. The source explanation also cited $20 million total notional against option D's $10 million.

Step by step

1. Novation replaces the single $10m swap with two $10m swaps, one between Vintage Eagle and the CCP, one between Vinyl and the CCP.
2. Vintage Eagle pays fixed 2.5% to the CCP and receives LIBOR from it.
3. Vinyl is the opposite side: it pays LIBOR to the CCP and receives fixed 2.5% from it.
4. CCP cash flows: in fixed 2.5% from Vintage, out fixed 2.5% to Vinyl; in LIBOR from Vinyl, out LIBOR to Vintage. Both net to zero, so the CCP is market neutral (gross notional $20m, net exposure zero).

Why A is correct: Novation creates two mirror-image swaps with the CCP in the middle, so each party faces only the CCP and the CCP's matched book has zero net exposure.

Why the others are wrong

- **B:** The original contract is extinguished by novation; a guarantee describes the wrong legal mechanism.
- **C:** Novation does not split the notional; each leg keeps the full $10m, and the payment directions given are also wrong.
- **D:** Reverses the CCP's fixed leg against Vintage Eagle, which is the fixed payer, and has the CCP receiving fixed from Vinyl, which is actually the fixed receiver, so the CCP would not be matched.

Q853 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

Three clearing members have the following bilateral OTC payment obligations: A owes B 90, B owes C 60, and C owes A 30 (total gross liabilities of 180). All three contracts are then novated to a CCP, which nets each member's obligations. After novation and netting, what is the net amount that Member A owes to the CCP, and what is the status of the original bilateral contracts?

A. 90; the original bilateral contracts remain in force alongside the new contracts with the CCP.
B. 60; the original bilateral contracts cease to exist and the members have no counterparty risk to each other.
C. 60; the original bilateral contracts remain in force and the CCP only guarantees them.
D. 120; the original bilateral contracts cease to exist and the members have no counterparty risk to each other.

Answer: B

Step by step

1. Novation replaces each bilateral contract with contracts between each member and the CCP; the original parties are no longer obligated to each other.
2. Member A owes 90 (to B) and is owed 30 (by C).
3. Net owed by A to the CCP = 90 - 30 = 60.
4. A is owed 90 and owes 60, so is a net receiver of 30; C is owed 60 and owes 30, so is a net receiver of 30; total net payments fall from 180 to 60.

Why B is correct: Novation extinguishes the bilateral contracts, leaving each member with a position only against the CCP, and A's net obligation is 90 - 30 = 60.

Why the others are wrong

- **A:** Gives A's gross payment of 90 and keeps the old contracts; the netting offset is ignored and novation is mis-described as an addition rather than a replacement.
- **C:** The net figure is right, but under novation the original contracts cease to exist; the CCP is the counterparty, not a mere guarantor of the old contracts.
- **D:** Adds the 30 receivable to the 90 payable (90 + 30) instead of netting it. The legal description is correct but the amount is wrong.

LO 36.c

Define netting, multilateral offset, and compression and provide examples of each.

Q854 AnalystPrep Q.6095

Compression is a key mechanism utilized in central clearing. How does compression benefit financial institutions that hold multiple derivative contracts?

A. It allows them to expand their investment portfolios while maintaining net economic exposure.
B. It consolidates derivative contracts, reducing the number of contracts needed to maintain an equivalent position.
C. It increases market liquidity by expanding the available derivatives that can be traded.
D. It separates and categorizes derivatives based on their risk profiles for better risk management.

Answer: B

Step by step

1. Compression cancels multiple offsetting contracts and replaces them with fewer contracts.
2. Gross exposure and notional fall while the net exposure is unchanged.
3. Fewer and smaller cleared notionals lower initial margin, regulatory capital and administrative work.
4. Only option B describes this.

Why B is correct: Compression cancels redundant, offsetting contracts and substitutes fewer ones, cutting gross exposure and contract count while keeping net exposure.

Why the others are wrong

- **A:** Compression shrinks the portfolio's gross size; it does not expand it. Confuses 'net exposure unchanged' with growth.
- **C:** Compression reduces the number of outstanding contracts; it does not create additional tradable derivatives or liquidity.

- **D:** Categorizing contracts by risk profile is portfolio analytics, not compression, which cancels contracts.

Q855 AnalystPrep Q.6093

Central clearing involves mechanisms such as netting, multilateral offset, and compression to manage and mitigate counterparty exposure. With respect to multilateral offset, how does it primarily serve a Central Counterparty (CCP)?

A. It allows a CCP to increase the total notional value of trades without impacting the risk exposure.
B. It provides a CCP with the leverage to negotiate better terms for each trade that it clears.
C. It lets a CCP reduce counterparty exposure by centralizing trades which can offset each other.
D. It allows a CCP to increase transparency and, thus, the predictability of market movements.

Answer: C

Step by step

1. Under novation the CCP sits between all members, so each member's positions across counterparties are combined.
2. Offsetting obligations (e.g., A owes the CCP 60 and is owed 90, net 30) cancel without cancelling any trade.
3. This lowers total liabilities and exposures, which is the main purpose of multilateral offset.
4. Option C states this.

Why C is correct: Because the CCP centralizes trades through novation, offsetting claims across members net against each other, shrinking counterparty exposure.

Why the others are wrong

- **A:** Offset reduces exposure; it is not a way to add notional without risk. Mixes up offset with growth in cleared volume.
- **B:** Offset is a risk-reduction effect, not bargaining power over trade terms.
- **D:** Transparency is a separate CCP benefit and does not make market movements predictable.

Q856 AnalystPrep Q.6094

As a key risk management practice within central clearing, netting plays a significant role in reducing the credit risk associated with multiple financial claims. What does netting accomplish for participating financial institutions?

A. It enables them to increase the number of outstanding trades without affecting risk exposure.
B. It consolidates several transactions into a singular net position, decreasing the number of settlements needed.
C. It increases the granularity of financial claims, providing more detailed information for risk assessment.
D. It allows institutions to avoid the CCP's centralized risk management system for more flexibility.

Answer: B

Step by step

1. Netting offsets multiple claims between parties into one net amount.
2. This reduces redundant transactions and the number of settlements, lowering counterparty risk.
3. Option B matches; the others describe more trades, more granularity or avoiding the CCP, none of which netting does.

Why B is correct: Netting combines offsetting claims into a single net position, cutting redundant transactions and settlements and hence credit exposure.

Why the others are wrong

- **A:** Netting reduces, not increases, the number of transactions that carry counterparty risk.
- **C:** Netting aggregates claims into fewer, less granular positions.
- **D:** Netting is part of the CCP's risk framework, not an escape from it.

Q857 AnalystPrep Q.6128

A derivatives analyst notes growth in centrally cleared OTC derivatives, but certain characteristics of OTC derivatives make central clearing difficult. For derivatives that cannot be centrally cleared, which approach brings some of the functionality of central clearing into bilateral OTC markets?

A. Implementation of standardized terms for all OTC derivative contracts to facilitate CCP clearing.
B. Use of multilateral compression to reduce notional exposure and simplify the portfolio without central clearing.
C. Application of a Central Counterparty (CCP) for all OTC transactions to centralize trade and risk management.

D. Mandatory conversion of all long-dated OTC derivatives into short-term instruments for central clearing eligibility.

Answer: B

Step by step

1. The question concerns contracts that stay bilateral, so routes that move them to a CCP (A, C, D) are excluded.
2. Portfolio compression cancels offsetting real transactions and replaces them with fewer ones, even without a CCP.
3. It reduces gross notional and operational burden while preserving net exposure, mimicking part of the CCP's offset benefit.
4. Answer B.

Why B is correct: Compression can be done in the bilateral OTC market without a CCP and reproduces some netting-type benefits of central clearing.

Why the others are wrong

- **A:** Standardizing terms is a route to central clearing, not a bilateral substitute, and non-standard trades cannot simply be standardized.
- **C:** Putting all trades through a CCP is exactly what is not feasible for these contracts.
- **D:** Converting long-dated trades into short-term ones would change the economics and is not a recognized technique.

Q858

BT answers P2.T7.20.6.1

REBUILT — The exhibit was missing and options/numbers were OCR-damaged (options A and C identical); data were reconstructed consistently and the options rebuilt.

Three counterparties (A, B, C) have the following exposures. Standard transactions can be cleared centrally; nonstandard transactions cannot. Values are the mark-to-market value to the party named (the other party has the opposite value). Ignore margin and collateral; exposure is the positive net value of each netting set.

Pair	Standard trades	Nonstandard trades
A-B	+40 to A	+120 to B
A-C	+50 to A	+80 to C
B-C	+10 to B	+30 to B

Without central clearing, bilateral netting across standard and nonstandard trades gives exposures A = 0, B = 120, C = 30 (average 50.0). If the standard transactions are now centrally cleared (nonstandard remain bilateral), what is the average exposure per counterparty including and excluding exposure to the CCP?

A. 106.7 including exposure to the CCP; 76.7 excluding it

B. 76.7 including exposure to the CCP; 106.7 excluding it

C. 136.7 including exposure to the CCP; 76.7 excluding it

D. 110.0 including exposure to the CCP; 76.7 excluding it

Answer: A

Key correction: Source key (BT answers) said C (100.0 incl, 70.0 excl); the original exhibit is unavailable and its options A and C were identical, so the question was rebuilt with reconstructed data.

Step by step

1. Pre-clearing check: A-B = 40-120 = -80 to A; A-C = 50-80 = -30 to A; B-C = 10+30 = +40 to B. So A = 0, B = 80+40 = 120, C = 30+0 = 30; average 50.0.
2. CCP nets each member's standard trades across counterparties: A = +40+50 = +90; B = -40+10 = -30; C = -50-10 = -60. Only A is owed by the CCP, so exposure to the CCP is A = 90, B = 0, C = 0.
3. Remaining bilateral nonstandard sets: A is owed nothing (-120, -80) so 0; B = 120+30 = 150; C = 80 (owed by A) with 0 on B-C.
4. Excluding the CCP: (0 + 150 + 80)/3 = 76.7.
5. Including the CCP: A = 0+90 = 90; B = 150; C = 80; total 320, average 106.7.

Why A is correct: Multilateral offset at the CCP nets A's two standard positions into one claim of 90 and turns B's and C's net standard positions into payables, while the non-cleared trades keep their bilateral exposures.

Why the others are wrong

- **B:** Swaps the two figures; central clearing here adds a CCP exposure (A's 90), so the including figure must be the larger.
- **C:** Counts B's and C's net payables to the CCP (30 and 60) as exposures instead of zero, ignoring that exposure

is only the positive net value.
- **D:** Treats cleared trades pair by pair without multilateral netting at the CCP, so B's +10 on B-C is counted as an exposure to the CCP.

LO 36.d

Describe the application and estimation of margin and default funds under central clearing.

Q859 AnalystPrep Q.6096 · AnalystPrep Q.6169

Which of the following best describes the primary reason a central counterparty (CCP) requires initial margin from its clearing members?

A. To pay clearing members interest on the excess cash they deposit.
B. To cover the CCP's operating costs, including technology and infrastructure.
C. To pre-fund potential default losses, sized to cover adverse price moves over an assumed close-out period at a high confidence level.
D. To reduce the notional value of trades cleared at the CCP without changing members' risk exposure.

Answer: C

Step by step

1. Initial margin is the up-front deposit required of every member.
2. It is sized so that, to a high confidence level (usually 99% or more), the CCP can cover the loss on a defaulting member's positions over the time needed to close out or hedge them.
3. Therefore its purpose is to absorb potential default losses before other members' resources are touched.

Why C is correct: Initial margin is a buffer against future potential default losses, calibrated to market-price moves over the close-out period and a confidence level.

Why the others are wrong

- **A:** Interest on excess collateral is a by-product of holding cash, not the reason margin is required.
- **B:** Confuses margin with fees; operating costs are funded by clearing fees, not risk collateral.
- **D:** Describes compression, a separate service that reduces gross notional; margin does not reduce notional.

Q860 AnalystPrep Q.6097 · AnalystPrep Q.6168

What is the primary purpose of the variation margin collected by a CCP from its clearing members?

A. To fund the CCP's technology and operating expenses.
B. To reflect daily mark-to-market profits and losses of open positions in cash flows between the CCP and its members.
C. To pre-fund losses beyond a defaulting member's initial margin by mutualizing them across members.
D. To guarantee a fixed final settlement price at contract maturity regardless of market movements.

Answer: B

Step by step

1. Positions are marked to market at least daily.
2. Members whose positions lost value pay variation margin; members whose positions gained receive it.
3. This keeps current exposure close to zero each day (with intraday calls during large moves), so only a short period of price change remains at risk on a default.

Why B is correct: Variation margin settles daily price changes in cash, preventing exposure from accumulating.

Why the others are wrong

- **A:** Operating costs are not a margin function.
- **C:** Describes the default fund (loss mutualization), not variation margin.
- **D:** Variation margin does not lock in a settlement price; it passes on price changes as they occur.

Q861 AnalystPrep Q.6098

What role does the default fund play in a CCP's loss-absorption structure?

A. It provides daily liquidity adjustments based on the marked-to-market value of members' portfolios.

B. It finances the CCP's investment in technology and infrastructure.

C. It is a mutualized pool of member contributions used to absorb default losses that exceed the defaulting member's initial margin.

D. It pays members interest on excess cash posted as initial margin.

Answer: C

Step by step

1. First line of defense: the defaulting member's initial margin (and variation margin).
2. If losses exceed it, the CCP draws on the default fund, to which all members contribute (loss mutualization).
3. Contributions are sized to cover potential losses with high confidence, and non-defaulting members may bear a share even with no direct dealings with the defaulter.

Why C is correct: The default fund mutualizes losses beyond the defaulter's own margin across members.

Why the others are wrong

- **A:** Daily liquidity adjustments are variation margin.
- **B:** The default fund is risk capital, not a development budget.
- **D:** Interest on excess collateral is unrelated to the default fund.

Q862 BT answers P2.T6.24.29.3 (BTA 563)

REBUILT — Original asked for the 'most appropriate' action with several defensible options (hourly variation margin, more stress testing); rebuilt as a single-answer question on the drivers of initial margin.

A CCP sets each member's initial margin to cover adverse price moves over an assumed close-out period at a given confidence level. Which of the following changes would be expected to increase the initial margin required on a member's portfolio?

A. Shortening the assumed close-out period.

B. Lowering the required confidence level from 99% to 95%.

C. Lengthening the assumed close-out period because the cleared contracts have become less liquid.

D. An upgrade of the clearing member's credit rating.

Answer: C

Step by step

1. Initial margin depends on the time the CCP is exposed to the position and on the confidence level.
2. A longer close-out period allows larger price moves, so more margin is needed; less liquid contracts take longer to close out.
3. Initial margin depends primarily on market risk, not on the member's credit quality.

Why C is correct: A longer exposure period (from lower liquidity) widens the potential loss distribution, raising initial margin.

Why the others are wrong

- **A:** A shorter period reduces potential price moves and lowers margin.
- **B:** A lower confidence level covers fewer scenarios and lowers margin.
- **D:** Initial margin is driven by market risk rather than credit quality, so a rating change does not raise it.

LO 36.e

Discuss the risks faced by a CCP and the ways it manages its exposures.

Q863 AnalystPrep Q.5484

REBUILT — Original was a vague 'what should the CCP do' item with a giveaway distractor (A) and a long correct option; rewritten around a concrete CDS wrong-way risk scenario with one defensible answer.

A central counterparty (CCP) clears single-name credit default swaps. A clearing member sells protection on a reference entity that is closely tied to the member's own financial health (for example, its parent bank). Which of the following best describes why this creates wrong-way risk (WWR) for the CCP, and how the CCP should respond?

A. The member's default makes the value of its cleared position fall sharply at the time of default, so losses may exceed initial margin sized for normal conditions; the CCP should review the adequacy of initial margin and default fund contributions for such positions.
B. Central clearing removes WWR because the CCP is the buyer to every seller, so no change to margin or default fund methodology is needed.
C. WWR arises mainly because the CCP's interest income on margin falls when credit spreads widen, so the CCP should hold less margin.
D. WWR is a purely bilateral issue that is eliminated by multilateral netting of the member's positions, so the CCP only needs to monitor the member's credit rating.

Answer: A
Step by step
1. WWR for a CCP occurs when the default of a member is adversely linked to the value of its cleared contract, so the contract loses value rapidly just as the member defaults.
2. Selling protection on an entity correlated with the seller's own health is the classic case: the position becomes a large loss at the very moment the member can no longer pay variation margin.
3. Margin and default fund sizing based on normal market moves will then understate the loss during the margin period of risk.
4. The appropriate response is to reassess the adequacy of initial margin and default fund contributions for such products (CDS is among the hardest products to clear for this reason).

Why A is correct: WWR means exposure is largest when the member is most likely to default, so standard margin calibrations may be insufficient and the CCP must reconsider margin and default fund adequacy.
Why the others are wrong
- **B:** Wrongly assumes a CCP is immune to risk; a CCP still bears member default risk and WWR, and complacency leaves margin undersized.
- **C:** Confuses WWR with investment income risk; WWR is about the link between member default and position value, and holding less margin would worsen it.
- **D:** Netting reduces exposure within a portfolio but does not remove the adverse correlation between default and position value; credit rating monitoring alone does not size margin for it.

Q864 AnalystPrep Q.6099

A central counterparty (CCP) encounters a diversity of risks by standing in the middle of financial transactions and acting as the counterparty to both the buyer and the seller. One of the primary risks faced by a CCP is member default. Which of the following is a method used by CCPs to manage this risk?

A. Increasing transaction fees to augment the default fund after a member has defaulted.
B. Macro-hedging to minimize exposure to major market risk factors and reduce the market risk faced by the CCP.
C. Mandating additional capital contributions from all members regardless of their exposure or default probability.
D. Reducing the number of transactions cleared to limit potential defaults and associated market risks.

Answer: B
Step by step
1. After a member default the CCP holds an unmatched book and must keep paying variation margin to non-defaulters.
2. In the margin period of risk the default management group first macro-hedges, using liquid transactions to neutralize major sensitivities (e.g., interest rate risk).
3. The CCP then auctions the portfolio (with its hedges) by sub-portfolio to surviving members.
4. Of the options, only macro-hedging is a recognized default management tool.

Why B is correct: Macro-hedging quickly removes most market risk from the defaulter's portfolio, limiting losses during the liquidation period and making the later auction more attractive.
Why the others are wrong
- **A:** Raising fees after the default is not a recognized loss-management tool; the waterfall uses margin, default fund and CCP capital, not ex post fee increases.
- **C:** Uncapped additional capital from all members regardless of exposure is not how CCPs manage default; contributions are set in advance and assessments come later in the waterfall.
- **D:** Clearing fewer trades defeats the CCP's purpose and is not a default management technique.

Q865

BT notes P2.T6.24.29.3

REBUILT — Original asked for 'most appropriate' enhancement with several defensible options (B and D both plausible, A and C arguable); replaced with a determinate question on the order of the CCP's default management steps.

A clearing member of ClearX, a CCP, has defaulted. ClearX decides not to port the member's trades to a surviving member. Which of the following best describes the sequence of risk-reducing actions in the margin period of risk (MPoR) and the reason for that sequence?

A. Pre-default period, then auction of the full portfolio, then macro-hedging of residual risk, because auction proceeds determine the hedge needed.

B. Pre-default period, then macro-hedging with liquid transactions, then auctions by sub-portfolio, because hedged portfolios carry little market risk and attract higher bids.

C. Pre-default period, then immediate use of non-defaulting members' default fund contributions, then macro-hedging, because the loss must be mutualized before it can be hedged.

D. Auction first, with macro-hedging used only if the auction fails, because auctions remove all risk immediately.

Answer: B

Step by step

1. The MPoR runs from the defaulter's last margin posting until positions are closed, and has three stages: pre-default, macro-hedging, auctions.
2. Once default is declared, the default management group macro-hedges using liquid instruments, reducing major risk factors quickly.
3. The hedged sub-portfolios (defaulter's trades plus hedges) are then auctioned to surviving members; low market risk helps ensure sufficiently high bids.
4. Loss absorption by the waterfall (defaulter's margin and default fund first) covers hedging and auction costs; it does not precede them.

Why B is correct: Hedging first neutralizes most market risk quickly, so the auction deals with low-risk portfolios and risk falls to zero as each auction closes.

Why the others are wrong

- **A:** Reverses the order; auctioning an unhedged book exposes the CCP and bidders to large market risk and would lower bids.
- **C:** Mutualized resources are used only after the defaulter's own margin and default fund contributions are exhausted, and they cover costs of hedging and auctions rather than replacing them.
- **D:** Auctions are the last stage; risk only reaches zero as each auction wraps up, and macro-hedging is the standard first step, not a fallback.

Q866

Schweser Module Quiz 36.2 Q1

REBUILT — Option C was arguably defensible (CCPs do earn some income on held assets); reworded so that only B is correct.

Which of the following statements regarding risk management by central counterparties (CCPs) is most accurate?

A. Netting of variation margin across different currencies is possible.

B. Only cash is accepted as variation margin.

C. The primary source of CCP income is interest earned on assets held, rather than fees per trade.

D. During the margin period of risk (MPoR), most risk is eliminated by auctions in which the defaulting member's portfolio is sold off.

Answer: B

Step by step

1. Variation margin is treated as settlement, so CCPs normally accept only cash for it (initial margin may also be highly liquid government securities).
2. Netting is possible within a single currency but not across currencies, ruling out A.
3. CCPs recover costs mainly through fees per trade; interest on assets is a secondary, indirect source, ruling out C.
4. Most risk is removed early through liquid macro-hedges; auctions finish the job later, ruling out D.

Why B is correct: Variation margin is effectively settlement of gains and losses, so it must be in cash.

Why the others are wrong

- **A:** Netting works in a single currency only; mixing currencies prevents netting for variation margin purposes.
- **C:** Reverses the order of importance: fees per trade are the direct source, interest on held assets is indirect and secondary.

- **D:** Confuses the stages; macro-hedging, not auctions, removes most risk quickly, with auctions bringing the residual to zero.

LO 36.f

Provide examples of a loss waterfall.

Q867

AnalystPrep Q.6102

REBUILT — Distractors were giveaways; reworded so each wrong option reflects a realistic misconception about the first layer of the waterfall.

In a CCP loss waterfall, the first resource applied to losses from a clearing member's default is the defaulting member's initial margin. Which statement best describes its role?

A. It is the defaulter's own collateral, sized to cover potential losses on its portfolio over the close-out period, and is consumed before any resources of the CCP or surviving members.
B. It is a pooled buffer contributed by all members, consumed pro rata with the defaulter's share first.
C. It is consumed only after the CCP has contributed its own equity (skin in the game).
D. It is drawn from non-defaulting members' margin in proportion to their positions, to spread losses evenly.

Answer: A

Step by step

1. The waterfall starts with defaulter-pays resources: the defaulter's initial margin, then its default fund contribution.
2. Initial margin is calibrated to cover losses on the defaulter's portfolio during the close-out (margin period of risk).
3. Only after these are exhausted do CCP equity, surviving members' default fund contributions and assessments come into play.

Why A is correct: Initial margin is the defaulter's own collateral and the first line of defense, so losses are borne by the defaulter before anyone else.

Why the others are wrong

- **B:** Describes the default fund (mutualised), not initial margin, which is member-specific.
- **C:** Reverses the order; skin in the game comes after the defaulter's margin and default fund contribution.
- **D:** Non-defaulting members' initial margin is not used in the waterfall; mutualisation applies only to default fund contributions.

Q868

AnalystPrep Q.6103

REBUILT — Distractors were giveaways; restated as a clear ordering question with realistic misconceptions.

After the defaulting member's initial margin is exhausted, which resource is applied next in the CCP loss waterfall?

A. The CCP's own equity (skin in the game).
B. The defaulting member's own default fund contribution.
C. The default fund contributions of surviving members.
D. Rights of assessment on surviving members.

Answer: B

Step by step

1. Defaulter-pays resources come first: initial margin, then the defaulter's default fund contribution.
2. Only then does the CCP contribute its own capital, followed by surviving members' default fund contributions and rights of assessment.

Why B is correct: The defaulter's own default fund contribution is part of defaulter-pays resources and is used right after its margin.

Why the others are wrong

- **A:** Skin in the game comes after all defaulter-pays resources, so this skips the defaulter's default fund contribution.
- **C:** Surviving members' contributions are mutualised losses and come after the CCP's own equity.
- **D:** Rights of assessment are the last mutualised layer, used only after the default fund is depleted.

Q869 AnalystPrep Q.6104

REBUILT — Distractors were giveaways; reworded to test the definition and position of skin in the game.

In the CCP loss waterfall, the CCP's 'skin in the game' refers to:

A. A contribution of the CCP's own equity, applied after the defaulter's margin and default fund contribution but before surviving members' default fund contributions.

B. A contribution of the CCP's own equity, applied before the defaulter's initial margin to give members confidence.

C. The additional contributions called from surviving members after the default fund is depleted.

D. A compensation payment made by the CCP to non-defaulting members whose positions gained value.

Answer: A

Step by step

1. Defaulter-pays resources (initial margin and default fund contribution) are used first.
2. The CCP then taps its own equity (enough that it can still function normally).
3. Only if loss remains do surviving members' default fund contributions, then assessments, absorb it.

Why A is correct: Skin in the game is the CCP's own capital placed between defaulter-pays resources and mutualised resources, giving it an incentive to set margins prudently.

Why the others are wrong

- **B:** Wrong order; the CCP's equity is not used before the defaulter's own resources.
- **C:** This describes rights of assessment.
- **D:** Not a feature of the waterfall; gains of other members are, if anything, subject to variation margin gains haircutting.

Q870 BT answers P2.T6.24.30.2

A central counterparty (CCP) has the following loss waterfall structure in place:

Resource	Amount
Defaulter's margin	\$80 million
Defaulter's default fund contribution	\$30 million
CCP's own capital (skin in the game)	\$25 million
Surviving members' default fund contributions	\$200 million
Assessment calls	Up to \$700 million

Suppose a clearing member defaults, resulting in a loss of \$275 million. How much of the surviving members' default fund contributions will be used to cover the loss?

A. \$0 million

B. \$40 million

C. \$90 million

D. \$140 million

Answer: D

Step by step

1. Loss = 275.
2. After defaulter's margin: 275 - 80 = 195.
3. After defaulter's default fund contribution: 195 - 30 = 165.
4. After CCP's own capital: 165 - 25 = 140.
5. Surviving members' default fund (200) covers the remaining 140; no assessment calls are needed.

Why D is correct: Defaulter-pays resources and CCP equity absorb 135, and the remaining 140 falls to the surviving members' default fund, which is large enough.

Why the others are wrong

- **A:** Would require the first three layers to cover the whole loss, but they total only 135.
- **B:** Not produced by the waterfall; an arbitrary mis-subtraction of the layers.
- **C:** Not produced by the waterfall; results from mis-subtracting layers (the correct residual is 140).

Q871 BT notes P2.T6.24.29.2

REBUILT — Original scenarios were inconsistent and the question vague; replaced with a self-contained waterfall calculation that reaches assessment calls.

A CCP has the following loss waterfall resources:

Resource	Amount
Defaulter's initial margin	$80 million
Defaulter's default fund contribution	$30 million
CCP's own capital (skin in the game)	$25 million
Surviving members' default fund contributions	$200 million
Rights of assessment	Up to $700 million

A clearing member defaults, producing a total loss of $420 million. How much must be raised through rights of assessment?

A. $85 million
B. $140 million
C. $285 million
D. $0 million

Answer: A

Step by step

1. Loss = 420.
2. Defaulter-pays: 80 + 30 = 110, leaving 420 - 110 = 310.
3. CCP equity: 25, leaving 285.
4. Surviving members' default fund: 200 (fully exhausted), leaving 285 - 200 = 85.
5. Rights of assessment cover 85, within the 700 cap.

Why A is correct: Resources are consumed in order; after the default fund is exhausted, the remaining $85 million is called through assessments.

Why the others are wrong

- **B:** Is the residual for a $275 million loss, not for this $420 million loss.
- **C:** Stops after CCP equity and omits the $200 million surviving members' default fund.
- **D:** Assumes the default fund absorbs the whole residual, forgetting it is capped at $200 million.

Q872 Schweser Module Quiz 36.2 Q2

Given the following three events, what is the proper order of the CCP loss waterfall?
I. Nondefaulting members' default fund contributions are exhausted.
II. Defaulting member's collateral and default fund contributions are exhausted.
III. The CCP taps an amount of its equity that enables it to function normally.

A. I, II, III.
B. II, I, III.
C. II, III, I.
D. III, II, I.

Answer: C

Step by step

1. Defaulter-pays resources (II) come first.
2. Then the CCP's own equity, skin in the game (III).
3. Then mutualisation: surviving members' default fund contributions (I), followed by rights of assessment.

Why C is correct: The waterfall runs defaulter's resources, CCP equity, then survivors' default fund.

Why the others are wrong

- **A:** Puts mutualised funds before the defaulter's own resources.
- **B:** Uses survivors' default fund before the CCP's equity.
- **D:** Uses CCP equity before the defaulter's resources.

Q873 BT answers P2.T6.24.29.2

REBUILT — Original stem gave unrelated scenario numbers and a vague 'key benefit' answer; rebuilt as a numeric loss-waterfall question with one calculable answer (fits LO 36.f).

A clearing member defaults and the CCP incurs a total realized loss of $48 million from macro-hedging and auc-

tioning its portfolio. The defaulter's initial margin is $20 million and its default fund contribution is $8 million. The CCP contributes $5 million of its own capital (skin in the game) before using the default fund contributions of non-defaulting members. What amount of the loss is absorbed by the non-defaulting members' default fund contributions?

A. $28 million
B. $20 million
C. $15 million
D. $13 million

Answer: C

Step by step

1. Defaulter's initial margin: 48 - 20 = 28 remaining.
2. Defaulter's default fund contribution: 28 - 8 = 20 remaining.
3. CCP skin in the game: 20 - 5 = 15 remaining.
4. The remaining 15 is mutualised across non-defaulting members' default fund contributions.

Why C is correct: The waterfall exhausts the defaulter's margin and default fund contribution, then the CCP's own capital, before survivors' default fund contributions are used.

Why the others are wrong

- **A:** Subtracted only the initial margin, ignoring the defaulter's default fund contribution and the CCP's capital.
- **B:** Skipped the CCP's own capital contribution (skin in the game): 48 - 20 - 8 = 20.
- **D:** Not the result of a consistent waterfall: subtracting the layers in order gives 48 - 20 - 8 - 5 = 15, and $13 million would arise only from an arithmetic slip such as deducting an extra $2 million.

LO 36.g

Explain the different methods of absorbing losses and managing the default of one or more members of a CCP.

Q874

AnalystPrep Q.6100

REBUILT — Original distractors were giveaways (marketing, profit redistribution); rewrote options as realistic confusions with other default-management tools.

A clearing member of a central counterparty (CCP) has defaulted. The CCP has already macro-hedged the defaulter's portfolio against its major risk factors. Which statement best describes the role of the auction in the CCP's default management process?

A. The CCP sells the entire unhedged portfolio to the single member offering the lowest fee, before any hedging takes place.
B. The CCP splits the hedged portfolio into sub-portfolios, and surviving clearing members submit two-way prices, with the best bidder winning each sub-portfolio, so that the defaulter's exposure is neutralised.
C. The CCP allocates the defaulter's unpaid losses across surviving members in proportion to their default fund contributions.
D. The CCP transfers the defaulter's clients' positions and margin to pre-arranged backup members at pre-agreed prices.

Answer: B

Step by step

1. After a default the CCP first macro-hedges the portfolio to remove major directional risk (e.g. interest rate sensitivity).
2. The hedged portfolio is then split into sub-portfolios (e.g. by currency) and auctioned among surviving clearing members.
3. Members submit two-way prices; the best price wins each sub-portfolio, which closes out the CCP's exposure.
4. Members are motivated to bid to avoid being hit later through default fund losses; auction costs are covered first by the defaulter's margin and default fund contribution.

Why B is correct: The auction is the step that actually transfers (neutralises) the defaulter's hedged sub-portfolios to surviving members through competitive bidding.

Why the others are wrong

- **A:** Reverses the order and structure: hedging comes first and the portfolio is auctioned in sub-portfolios, not as one unhedged block to a single member.
- **C:** Describes loss mutualisation (the survivors-pay default fund stage), not the auction, which is a trade-out

mechanism.
- **D:** Describes porting of client positions, which is a separate action from the auction of the defaulter's own portfolio.

Q875

AnalystPrep Q.6101

REBUILT — Original distractors were obviously false; rewrote them as confusions with auction, tear-up and forced allocation.

A clearing member of a CCP defaults. Which of the following correctly describes the porting of the defaulted member's client trades?

A. Surviving clearing members are compelled by the CCP to accept the client positions regardless of their own preferences.

B. The CCP cancels the trades that are opposite to the defaulter's positions in order to restore a matched book.

C. The client positions, together with the associated margin, are transferred to a surviving clearing member that accepts them, preserving the clients' positions without liquidating them.

D. The client positions are split into sub-portfolios and sold to the surviving member submitting the best price.

Answer: C

Step by step

1. Porting moves a defaulted member's client positions (and associated client margin) to a surviving clearing member, often a pre-agreed backup member.
2. It requires the receiving member to accept both the portfolio and its margin, so it is voluntary, not compelled.
3. If porting is not possible, client trades are handled in the default process together with the defaulter's own portfolio.

Why C is correct: Porting preserves clients' positions by transferring them with their margin to a willing surviving member.

Why the others are wrong

- **A:** Describes forced allocation; porting requires the surviving member to accept the portfolio and margin.
- **B:** Describes tear-up, which cancels unmatched contracts rather than transferring positions.
- **D:** Describes the auction of sub-portfolios, which is used for the defaulter's own portfolio, not the transfer of client positions.

Q876

AnalystPrep Q.6170

REBUILT — Original options B and D were both defensible and neither matched the GARP/Schweser definition (gains of members with gains are haircut; payers pay in full); rewrote with one precise correct option.

Which of the following best describes variation margin gains haircutting (VMGH) as a loss allocation method of a CCP?

A. The CCP reduces the initial margin of non-defaulting members to cover the shortfall left by the defaulter.

B. The CCP applies a haircut to the value of securities posted as margin to reflect their price volatility.

C. The CCP reduces the variation margin it pays to members whose positions have gained in value, while members who owe variation margin must still pay in full.

D. The CCP reduces the variation margin owed by all members, both those with gains and those with losses, by the same percentage.

Answer: C

Step by step

1. The defaulter's losses are matched by gains on the matched book of other members.
2. VMGH discounts the variation margin paid to members with gains, so they receive only part of the gain.
3. Amounts owed to the CCP are satisfied in full, so the CCP stays matched and funded.
4. Only variation margin gains are haircut; non-defaulting members' initial margin is not touched.

Why C is correct: VMGH allocates the shortfall to the members who made gains by paying them less than their full variation margin gain, while payers pay in full.

Why the others are wrong

- **A:** Describes initial margin haircutting, which is not an accepted way to allocate losses to non-defaulting members; VMGH applies only to variation margin gains.
- **B:** Confuses VMGH with a collateral valuation haircut on securities posted as margin, which is a risk control for collateral, not loss allocation.
- **D:** Haircuts apply only to payouts to members with gains; members owing variation margin must pay in full,

otherwise the CCP would be unfunded.

Q877 BT answers P2.T6.24.30.1 · BT notes P2.T6.24.29.3

REBUILT — Original options mixed different topics and option B was defensible; rewrote to ask for the correct sequence of actions that neutralise the defaulter's market risk.

RiskShield CCP clears interest rate swaps, credit default swaps and futures. One of its clearing members, BetalBank, defaults. Which sequence best describes the CCP's actions to neutralise the market risk of BetalBank's portfolio, including the fallback if the first approach fails?

A. Auction the entire portfolio as one block; if no bids are received, haircut the initial margin of non-defaulting members.

B. Macro-hedge the portfolio against major risk factors, auction the sub-portfolios to surviving members, and if the auction does not work, tear up unmatched contracts.

C. Tear up all contracts first, then macro-hedge, then auction the remaining positions.

D. Suspend clearing for the whole market, then call on rights of assessment before any hedging or auction.

Answer: B

Step by step

1. Step 1: macro-hedge to remove exposure to major risk factors with liquid transactions, reducing the CCP's market risk.
2. Step 2: split the hedged portfolio into sub-portfolios and auction them to surviving members who submit two-way prices.
3. Step 3: if the auction fails to neutralise the exposure, tear up unmatched contracts to restore a matched book.
4. Losses are met from the defaulter's margin and default fund first; haircutting initial margin of non-defaulting members is not an accepted approach.

Why B is correct: The standard sequence is macro-hedge, then auction sub-portfolios, with tear-up as the fallback if the auction does not work.

Why the others are wrong

- **A:** An unhedged single-block auction is inefficient, and haircutting non-defaulting members' initial margin is not an accepted fallback.
- **C:** Tear-up is a fallback after a failed auction, not a first step; it leaves members with losses and residual market risk.
- **D:** Suspending all clearing is not a standard default tool, and rights of assessment come late in the waterfall after the default fund is depleted.

Q878 BT answers P2.T6.24.30.3 · BT notes P2.T6.24.30.3

A CCP has four clearing members: W, X, Y, and Z. The current positions and unrealized gains/losses are as follows:
- Member W: Long position, unrealized gain of $80 million
- Member X: Short position, unrealized loss of $60 million
- Member Y: Long position, unrealized gain of $20 million
- Member Z: Short position, unrealized loss of $40 million

Member X defaults, and the CCP closes out their position at a premium, resulting in a total loss of $90 million. The CCP's loss waterfall is as follows:
- Member X's initial margin: $30 million
- Member X's default fund contribution: $15 million
- CCP's own capital (skin-in-the-game): $10 million

If the CCP uses VMGH to allocate the remaining loss proportionally based on the members' unrealized gains, how much of Member W's variation margin gain will be haircut?

A. $10 million

B. $20 million

C. $28 million

D. $35 million

Answer: C

Step by step

1. Remaining loss after the defaulter's resources and the CCP's capital: $90 - 30 - 15 - 10 = 35$ million.
2. Members with gains: W 80 and Y 20, total 100 million.

3. W's share of gains: $80/100 = 80\%$.
4. W's haircut: $0.80 \times 35 = 28$ million (Y's is 7); W receives $80 - 28 = 52$ million.

Why C is correct: VMGH spreads the remaining loss across members with gains in proportion to their gains, so W bears 80% of $35 million.

Why the others are wrong

- **A:** Used the CCP's own capital ($10 million), a different waterfall layer, not W's haircut.
- **B:** Used Y's gain of $20 million rather than W's proportional share of the remaining loss.
- **D:** Charged the whole remaining loss of $35 million to W instead of sharing it with Y in proportion to gains.

LO 36.h

Compare bilateral and central clearing.

Q879

MK AP Mock 3 Q24

REBUILT — Original options were vague and the keyed option did not test the GARP cost comparison; rewritten around where loss-absorbency cost falls in each market.

Which of the following statements best compares the main source of loss-absorbency cost in bilaterally cleared and centrally cleared OTC derivative markets?

A. Bilateral: mainly capital (increasingly some initial margin); central clearing: mainly initial margin and default fund contributions.

B. Bilateral: mainly initial margin and default fund contributions; central clearing: mainly capital.

C. Both markets: mainly capital, because margin only covers current exposure.

D. Bilateral: no loss-absorbency cost, since each party negotiates its own terms; central clearing: capital only.

Answer: A

Step by step

1. In a bilateral market, a dealer absorbs counterparty losses mostly with regulatory capital, although bilateral initial margin is being introduced.
2. In a CCP, members post full collateral including initial margin enforced by the CCP, plus default fund contributions.
3. Capital requirements for members' cleared exposures are generally low, so margin and default fund carry the cost.
4. So A matches the bilateral versus central comparison.

Why A is correct: Loss absorbency is capital-driven in bilateral markets but margin- and default-fund-driven in centrally cleared markets.

Why the others are wrong

- **B:** Reverses the two markets.
- **C:** Ignores that CCPs fully collateralise with initial margin and default funds.
- **D:** Negotiating terms does not remove loss-absorbency cost, and CCP cost is not capital only.

Q880

AnalystPrep Q.6105

In bilateral clearing, parties negotiate terms directly and bear the risk of counterparty default themselves. Central clearing introduces a central counterparty (CCP) that assumes the risk of both sides of each trade. How does this difference affect the liquidity of the underlying markets?

A. Bilateral clearing enhances market liquidity by allowing counterparties to negotiate credit terms tailored to their needs.

B. Central clearing standardizes and mutualizes risk, which can help increase liquidity of the underlying markets through greater predictability and confidence.

C. Bilateral clearing inherently provides greater liquidity because there are no standard margin requirements tying up capital.

D. Central clearing restricts liquidity because the rigid CCP structure cannot adapt to market changes.

Answer: B

Step by step

1. A CCP replaces many bilateral credit relationships with one standardized counterparty and common margin rules.
2. Participants can trade with any member without assessing each one's credit, and positions are easier to close out or offset.
3. This predictability can improve liquidity of the underlying market, as GARP notes.
4. B states this.

Why B is correct: Standardization, multilateral netting and mutualized risk make counterparties interchangeable, which supports liquidity.

Why the others are wrong

- **A:** Bespoke credit terms fragment the market and hinder offsetting; they do not enhance liquidity.
- **C:** Focuses only on margin cost; bilateral markets are not inherently more liquid, and tailored terms reduce fungibility.
- **D:** Reverses the effect; a CCP can improve liquidity, not restrict it.

Q881

AnalystPrep Q.6106

Opacity of risk exposures is a key point when contrasting bilateral and central clearing. How does bilateral clearing affect transparency compared with central clearing?

A. Bilateral clearing promotes transparency because individual agreements let participants directly assess counterparty risk.

B. A CCP gives a consolidated view of market exposures and can see aggregate risk concentrations, whereas bilateral clearing obscures them because it is decentralized.

C. Bilateral clearing is more opaque because it relies on credit rating agencies to disclose counterparty risk to the market.

D. Bilateral trades are customizable, so market risk assessments are clearer than the homogenized representations in central clearing.

Answer: B

Step by step

1. In a bilateral market each pair of counterparties sees only its own trades, so nobody sees total exposures or concentrations.
2. A CCP sits in the middle of all cleared trades and sees aggregate positions and risk concentrations.
3. Hence central clearing is more transparent; B states this.

Why B is correct: Centralized record-keeping at the CCP gives a system-wide view that decentralized bilateral markets lack.

Why the others are wrong

- **A:** Bilateral parties see only their own relationships, so the market as a whole is opaque.
- **C:** Opacity arises from decentralization, not from reliance on rating agencies.
- **D:** Customization makes bilateral exposures harder, not easier, to aggregate.

Q882

AnalystPrep Q.6107

REBUILT — Keyed answer implied CCPs clear complex bespoke trades, contradicting GARP that only standardised, non-exotic, liquid products can be cleared; rewritten to test product and participant eligibility.

Compared with bilateral markets, which statement about the products and participants in centrally cleared OTC derivative markets is most accurate?

A. Only standardized, plain-vanilla, liquid products can be cleared, and participants are usually large dealers (clearing members) with others clearing through them.

B. Any product the two parties agree on can be cleared, and any market participant can become a clearing member.

C. Only standardized products can be cleared, but any participant can join the CCP directly.

D. Complex, bespoke trades are cleared preferentially because CCPs are best able to manage non-standard risk.

Answer: A

Step by step

1. Central clearing requires standardized, vanilla, liquid products so the CCP can value and close out positions.
2. Clearing members are usually large dealers; other entities clear through those members.
3. Bilateral markets allow all products and all participants.

4. A is the only option consistent with both points.

Why A is correct: Product and participant restrictions in central clearing contrast with the unrestricted bilateral market.

Why the others are wrong

- **B:** Describes the bilateral market, not central clearing.
- **C:** Right on products but wrong on participants; access is through clearing members.
- **D:** Bespoke trades typically stay bilateral because they are not standard or liquid.

Q883

BT answers P2.T6.24.31.1 · BT notes P2.T6.24.30.3

In the context of OTC derivative markets, which of the following statements BEST describes the key difference between bilaterally and centrally cleared transactions?

A. Bilaterally cleared transactions are subject to stricter regulatory oversight than centrally cleared transactions.

B. Centrally cleared transactions typically involve higher counterparty risk than bilaterally cleared transactions.

C. Centrally cleared transactions offer greater transparency and operational efficiency than bilaterally cleared transactions.

D. Bilaterally cleared transactions have standardized margin requirements, while centrally cleared transactions have negotiated margin requirements.

Answer: C

Step by step

1. A CCP holds a central record, nets multilaterally and runs a coordinated default process, raising transparency and efficiency.
2. Check the others: oversight, counterparty risk and margin are each reversed.

Why C is correct: Centralization gives visibility of aggregate exposures and streamlined netting and settlement.

Why the others are wrong

- **A:** Reversed; CCPs face heavier oversight.
- **B:** Reversed; CCP margining and loss mutualization generally lower counterparty risk.
- **D:** Reversed; bilateral margin is bespoke and CCP margin is standardized and enforced.

Q884

MK GARP PE1 Q66

A junior risk analyst compares bilateral netting and central clearing of derivative trades. The bilateral exposures of three firms (each row is the first firm's exposure to the second firm) are:

Exposure	AUD million
Firm 1 to Firm 2	90
Firm 2 to Firm 1	60
Firm 1 to Firm 3	12
Firm 3 to Firm 1	70
Firm 2 to Firm 3	57
Firm 3 to Firm 2	0

Which statement is correct?

A. Under bilateral netting, Firm 1's net exposure is AUD 28 million.

B. Under bilateral netting, Firm 2's net exposure is AUD 27 million.

C. Under central clearing, Firm 3's net exposure is AUD 0 million.

D. Under central clearing, the CCP's net exposure is AUD 28 million.

Answer: D

Step by step

1. Bilateral netting (pairwise, floored at 0): Firm 1 = max(90-60,0) + max(12-70,0) = 30 + 0 = 30; Firm 2 = max(60-90,0) + max(57-0,0) = 57; Firm 3 = max(70-12,0) + max(0-57,0) = 58.
2. Central clearing nets each firm across all counterparties: Firm 1 = (90+12)-(60+70) = -28, so exposure 0; Firm 2 = (60+57)-(90+0) = 27; Firm 3 = (70+0)-(12+57) = 1.
3. CCP exposure = sum of firms' positive net exposures = 0 + 27 + 1 = 28.
4. D is correct.

Why D is correct: The CCP's exposure to the firms equals the sum of their positive net positions, 27 + 1 = 28.

Why the others are wrong

- **A:** 28 is Firm 1's net position under central clearing (a net payable), not its bilateral exposure, which is 30.
- **B:** 27 is Firm 2's exposure under central clearing; bilaterally it is 57.
- **C:** Firm 3 nets to +1 under central clearing; 0 ignores its small net receivable.

Q885 Schweser Module Quiz 36.3 Q1

Which of the following is NOT an improvement that centrally cleared markets offer relative to bilateral markets? Centrally cleared markets:

A. remain market neutral by netting trades.
B. formalize the default workout process by using a loss waterfall structure.
C. offer more flexibility in contract selection because of their collateral collecting process.
D. improve the counterparty risk picture by replacing the original counterparty with a series of counterparties.

Answer: C

Step by step

1. Central clearing allows only standardized, liquid products and fixed, nonnegotiable collateral rules.
2. So flexibility in contract choice is lower than in bilateral markets, making C not an improvement.
3. A, B and D are genuine features: multilateral netting, a formal loss waterfall, and the CCP and its members replacing the original counterparty.

Why C is correct: Central clearing trades flexibility for standardization; it does not add flexibility.

Why the others are wrong

- **A:** Netting is a real benefit of central clearing.
- **B:** The loss waterfall is a formal default process, unlike ad hoc bilateral workout.
- **D:** Replacing the original counterparty with the CCP and its members changes the counterparty risk picture and is cited as a feature.

LO 36.i

Compare initial margin and default fund requirements for clearing members in relation to loss coverage, cost of clearing, and moral hazard.

Q886 AnalystPrep Q.6109

REBUILT — Original had vague 'predictable cost structure' wording and three giveaway distractors; rebuilt to test the cost and moral hazard effects of shifting from initial margin to default fund.

A CCP is considering lowering the initial margin required of its clearing members and correspondingly raising the size of the default fund. Which statement best describes the likely effect of this change?

A. The cost of clearing rises, because mutualized default fund contributions must cover every member's default.
B. The cost of clearing falls, because under the larger-initial-margin design members hold less collateral overall.
C. The cost of clearing falls, but moral hazard rises because a larger share of default losses is borne by other members rather than the defaulter.
D. Both the cost of clearing and moral hazard fall, because losses are shared more widely among members.

Answer: C

Step by step

1. Initial margin is posted by each member against its own portfolio and must be funded in full, so it is expensive.
2. The default fund is mutualized: each member's contribution is relatively small, yet the pool gives far more loss absorbency, so substituting default fund for margin lowers the cost of clearing.
3. With less initial margin, a defaulter's losses more often exceed its own margin and fall on the default fund, weakening the defaulter-pays principle, so moral hazard rises.
4. Lower cost but higher moral hazard: C.

Why C is correct: Smaller initial margins and larger default funds make clearing cheaper through loss mutualization, but weaken the link between a member's behaviour and who pays for its default, raising moral hazard.

Why the others are wrong

- **A:** Reverses the cost effect: mutualization lowers, not raises, the cost of clearing compared with funding losses through initial margin.

- **B:** Reverses the design comparison: the larger-initial-margin design requires more collateral from each member and is the costlier one.
- **D:** Gets the cost right but moral hazard wrong: wider loss sharing weakens, not strengthens, the defaulter-pays incentive.

Q887

AnalystPrep Q.6110

REBUILT — Original option A was also defensible (two correct answers); rebuilt so exactly one option correctly describes loss coverage by initial margin versus default fund.

A CCP sets each clearing member's initial margin so that it covers losses from that member's default with a high level of confidence (for example, 99%). Which statement best describes how loss coverage is divided between initial margin and the default fund?

A. Initial margin covers the defaulter's own losses in all but rare, extreme scenarios; the mutualized default fund absorbs the extreme losses that exceed it.

B. Initial margin is pooled across all members, while the default fund is posted by the defaulter alone against its own portfolio.

C. The default fund absorbs the majority of ordinary losses, and initial margin is used only for small estimation errors.

D. A 99% initial margin covers every possible loss from a default, so a default fund is unnecessary.

Answer: A

Step by step

1. Initial margin is member-specific and sized to cover the close-out loss of that member with about 99% confidence.
2. A residual chance of a margin breach remains, and loss distributions are heavy-tailed, so losses in that case can be very large.
3. The default fund is a mutualized pool with far more loss absorbency than margin, and is used for losses beyond the defaulter's margin.
4. This is statement A.

Why A is correct: Initial margin handles the defaulter's losses with high probability, and the mutualized default fund covers the rare, extreme excess losses.

Why the others are wrong

- **B:** Reverses the roles: initial margin is the defaulter's own, and the default fund is the pooled resource.
- **C:** Reverses the layering: margin takes the first, most likely losses and the fund takes only the extreme tail.
- **D:** Ignores the tail: margin breaches are possible and losses then may be extreme, which is exactly why a default fund exists.

Q888

AnalystPrep Q.6111

REBUILT — Original was vague with giveaway distractors; rebuilt as a calculation comparing how much loss survivors bear and the moral hazard under two margin/default fund splits.

Two CCPs have identical members. A member defaults, causing a close-out loss of $70 million. At CCP X the defaulter had posted initial margin of $60 million; at CCP Y it had posted $20 million, and CCP Y has a correspondingly larger default fund. Ignore the defaulter's own default fund contribution and any CCP capital. Which statement is correct?

A. CCP X: survivors bear $10 million (about 14%) through the default fund, so CCP X has the higher moral hazard.

B. CCP Y: survivors bear $20 million (about 29%) through the default fund, and CCP Y has the higher moral hazard.

C. CCP Y: survivors bear $50 million (about 71%) through the default fund, and CCP Y has the lower moral hazard because losses are widely shared.

D. CCP Y: survivors bear $50 million (about 71%) through the default fund, and CCP Y has the higher moral hazard.

Answer: D

Step by step

1. CCP X: margin covers $60m of the $70m loss, leaving 70 - 60 = $10m for the default fund (14.3% of the loss).
2. CCP Y: margin covers $20m, leaving 70 - 20 = $50m for the default fund (71.4% of the loss).
3. The more of the loss that falls on the mutualized default fund, the less the defaulter pays for the losses it causes.
4. So CCP Y has the higher moral hazard, with survivors bearing $50m (about 71%): D.

Why D is correct: Smaller initial margins push more of a default loss onto the mutualized fund, weakening the

defaulter-pays principle and raising moral hazard.

Why the others are wrong

- **A:** Computes CCP X correctly but assigns it the higher moral hazard; X has the larger margin and therefore the lower moral hazard.
- **B:** Uses the initial margin ($20m) as the survivors' loss instead of the excess over margin (70 - 20 = 50).
- **C:** Gets the $50m correctly but claims wide sharing lowers moral hazard; sharing losses weakens the defaulter-pays incentive and raises it.

Q889

BT answers P2.T6.24.31.2

REBUILT — Original had inconsistent data (own capital $700m vs $100m), an unsupported claim that higher margin does not raise clearing cost, and a garbled option D; rebuilt as a clean comparison of cost and moral hazard.

ClearCo, a CCP, is reviewing its risk framework. Currently members post total initial margin of $500 million and contribute $200 million to the default fund. Proposal A raises initial margin requirements by 20% and leaves the default fund unchanged. Proposal B leaves initial margin unchanged and raises default fund contributions by 30%. Which statement is correct?

A. Proposal A is cheaper for members, because initial margin is returned to non-defaulting members and so has no funding cost.

B. Proposal A adds $100 million of defaulter-specific margin, keeping losses closer to the defaulter and lowering moral hazard but at a higher clearing cost; Proposal B adds $60 million of mutualized resources, which is cheaper but raises moral hazard.

C. Proposal B lowers moral hazard, because larger default fund contributions give members more skin in the game.

D. Proposal B has the higher clearing cost, because a 30% increase exceeds the 20% increase in Proposal A.

Answer: B

Key correction: Source key (BT) said A, arguing that higher initial margin does not raise clearing cost; per the reading, larger initial margins and smaller default funds carry a higher cost of clearing, so that key is not supported.

Step by step

1. Proposal A: 20% x $500m = $100m more initial margin (to $600m).
2. Proposal B: 30% x $200m = $60m more default fund (to $260m).
3. Larger initial margin means higher clearing cost but is more consistent with the defaulter paying, so moral hazard is lower.
4. A larger default fund adds mutualized loss absorbency at lower cost, but makes it more likely that others pay for a default, so moral hazard is higher.
5. Statement B matches these results.

Why B is correct: Initial margin is costly but defaulter-pays (lower moral hazard); default fund is cheaper through mutualization but raises moral hazard.

Why the others are wrong

- **A:** Ignores that posting collateral is costly even if later returned; larger margins increase the cost of clearing.
- **C:** Reverses moral hazard: mutualized contributions mean others pay for a default, which weakens incentives rather than strengthening them.
- **D:** Compares percentages rather than dollars and ignores that mutualization makes default fund resources cheaper per unit of loss absorbency.

LO 36.j

Describe the advantages and disadvantages of central clearing.

Q890

AP Mock 3 Q39

In a session on the implications of central counterparties in the OTC derivatives market, a debate arises about unintended consequences. Which of the following statements best captures the moral hazard problem that can arise in the context of CCPs?

A. CCPs, by providing a guarantee on transactions, may inadvertently encourage market participants to engage in riskier trading practices, underestimating the systemic risk.

B. CCPs, through their stringent margin requirements, deter participants from entering into derivative transac-

tions, thereby excessively limiting market liquidity.

C. CCPs, by offering advisory services on risk management and investment strategies, may lead to an over-reliance on CCP judgments, diminishing participants' responsibility.

D. CCPs, by simplifying the legal framework for derivatives, may inadvertently increase legal risk due to participants' overconfidence in CCPs' ability to manage complex arrangements.

Answer: A

Step by step

1. Moral hazard means taking less care (or more risk) because someone else bears the consequences.
2. In central clearing, netting, collateralization and loss mutualization pass most counterparty risk to the CCP.
3. Members therefore have little incentive to monitor each other's credit quality or manage counterparty risk prudently.
4. Only A describes this weakened risk discipline.

Why A is correct: Because the CCP absorbs and mutualizes counterparty losses, members have less incentive to monitor credit quality and manage risk carefully, which is moral hazard.

Why the others are wrong

- **B:** Describes a liquidity cost of margining, not moral hazard; CCPs are in fact credited with improving liquidity.
- **C:** Advisory services are not a CCP function in the reading; this is an invented over-reliance story.
- **D:** Clearing is rules-based and reduces legal costs; legal risk from overconfidence is not a listed disadvantage.

Q891

AnalystPrep Q.6112

REBUILT — Option B referred to reduced 'capital'; GARP/Schweser describe lower margin costs from multilateral offset, so B was reworded and the clearly wrong options tightened.

Central counterparties (CCPs) enhance the clearing process and facilitate trading in financial markets. Which of the following is one of the primary advantages of central clearing through a CCP?

A. Because a CCP centralizes trades, risk becomes harder to assess, which makes the trading environment more intricate but more secure.

B. Centralization allows multilateral offset of opposing positions across members, which lowers margin costs for members.

C. Centralization means each member must collect and monitor detailed private credit information about every other member.

D. Central clearing lowers the cost and complexity of transactions because the CCP is an extra layer that must be paid for.

Answer: B

Step by step

1. The CCP sees many of its members' trades, so opposing positions that were bilateral can be offset.
2. Multilateral offset removes the need for members to monitor other members' creditworthiness.
3. It also lowers margin costs for members.
4. Only B states this advantage.

Why B is correct: A CCP's transparency over members' trades enables multilateral offset, which lowers margin costs and reduces the need to monitor other members.

Why the others are wrong

- **A:** Reverses the advantage: a CCP sees aggregate risk concentrations, so risk becomes more transparent, not harder to assess.
- **C:** Offset eliminates the need to monitor other members; the claim reverses this.
- **D:** Legal and operational efficiency means lower, not higher, cost and complexity; this describes a drawback the reading does not list.

Q892

AnalystPrep Q.6113

While central clearing through CCPs offers benefits such as risk management and improved market liquidity, which of the following is a potential disadvantage of central clearing?

A. Central clearing prevents bilateral risk assessment, leading to over-reliance on CCPs and potential neglect of due diligence by market participants.

B. It simplifies the legal documentation process, leading to a reduction in clarity and an increase in market ambiguity.

C. The aggregated risk exposure makes it difficult for CCPs to manage the risk effectively, often resulting in higher losses during defaults.
D. CCPs reduce market participation by introducing stringent standards that exclude smaller market players from participating.

Answer: A
Step by step
1. Listed disadvantages: moral hazard, adverse selection, bifurcations, procyclicality.
2. Moral hazard: members have little incentive to vet the creditworthiness of others because netting, collateral and loss mutualization protect them.
3. A describes exactly this neglect of due diligence.

Why A is correct: Netting, collateralization and loss mutualization weaken members' incentive to vet each other's credit quality, which is moral hazard, a listed disadvantage.
Why the others are wrong
- **B:** Rules-based clearing minimizes legal costs and improves clarity; ambiguity is not a listed disadvantage.
- **C:** Aggregation gives the CCP transparency and the loss waterfall formalizes default handling; this is not a listed disadvantage.
- **D:** Exclusion of smaller players is not a listed disadvantage; the market-structure issue is bifurcation of cleared vs non-cleared trades.

Q893

AnalystPrep Q.6163

REBUILT — The original key (B, loss mutualisation concentrating systemic risk) conflicts with the reading, which lists loss mutualization as an advantage; rebuilt around procyclicality, a listed disadvantage.

In light of the evolving dynamics in OTC derivatives markets, the advantages and disadvantages of central counterparties (CCPs) are being scrutinized. Which of the following best describes a disadvantage of central clearing of OTC derivatives?

A. Members must monitor each other's creditworthiness more closely because of the CCP's multilateral offset.
B. When markets and the economy become more volatile, CCPs generally raise collateral requirements, which can worsen a potential default scenario.
C. Loss mutualization through the loss waterfall increases the market impact of a member default.
D. A CCP has less information on aggregate member exposures than counterparties have in bilateral markets.

Answer: B
Key correction: Source key (B, mutualising losses concentrates systemic risk) is not a disadvantage listed in the reading, where loss mutualization is an advantage; the rebuilt item has a different correct option (procyclicality).
Step by step
1. Disadvantages in the reading: moral hazard, adverse selection, bifurcations, procyclicality.
2. Procyclicality: CCPs raise collateral requirements as volatility rises, which can exacerbate a default scenario.
3. B states this; A, C, D each contradict an advantage of CCPs.

Why B is correct: Raising margin and collateral in stressed markets is procyclicality, a listed disadvantage that can worsen default scenarios.
Why the others are wrong
- **A:** Multilateral offset removes the need to monitor other members; the opposite problem (too little monitoring) is moral hazard.
- **C:** Loss mutualization reduces the market impact of a default and is a listed advantage.
- **D:** A CCP sees many members' trades and so has more aggregate information; transparency is a listed advantage.

Q894

BT answers P2.T6.24.31.3 · BT notes P2.T6.24.31.3

REBUILT — Original 'main advantage and disadvantage' was subjective and used figures irrelevant to the LO; rebuilt as a clear classification of CCP features into the listed advantages and disadvantages.

Total Clear, a central counterparty (CCP), is expanding its clearing services to a new class of derivatives. A risk analyst lists four observations about the expanded service. Which of the following correctly matches an observation to the advantage or disadvantage of central clearing that it illustrates?

A. Clearing members stop reviewing one another's creditworthiness because losses are netted, collateralized and mutualized: moral hazard (a disadvantage).
B. Total Clear sees a member's extreme aggregate exposure and limits its trading: adverse selection (a disadvan-

tage).

C. Only standardized contracts are cleared, leaving customized trades in the bilateral market: multilateral offset (an advantage).

D. Total Clear raises margin requirements as market volatility rises: loss mutualization (an advantage).

Answer: A

Key correction: Source key (A: reduced counterparty risk vs higher member cost) rested on a subjective 'main' judgement and higher member cost is not a listed disadvantage; the item was rebuilt with a different key.

Step by step

1. Moral hazard: little incentive for members to vet others' credit because netting, collateral and loss mutualization protect them. A is correct.
2. B is transparency (an advantage), not adverse selection.
3. C is bifurcation (a disadvantage), which reduces multilateral offset.
4. D is procyclicality (a disadvantage).

Why A is correct: Reduced member monitoring due to netting, collateral and loss mutualization is the textbook moral hazard disadvantage.

Why the others are wrong

- **B:** Mislabels transparency (CCP spotting concentrations) as adverse selection; adverse selection is members knowing more than the CCP.
- **C:** Mislabels bifurcation as an advantage; splitting cleared and non-cleared trades reduces offset benefits.
- **D:** Confuses procyclicality with loss mutualization; raising collateral in volatile markets is a disadvantage.

Q895

Mock 1 Q29

A risk manager is examining capital requirements for central counterparties (CCPs). After his assessment, he discusses the advantages and disadvantages of CCPs with a colleague. Regarding this discussion, why would adverse selection be a disadvantage in the CCP's centrally cleared structure?

A. In performing extensive member due diligence, the CCP may omit a member firm in error.

B. Only member firms can transact with a CCP, which limits the pool of available transactions.

C. A CCP has a transparent viewpoint from which to see all transactions in its network, which puts an information gap advantage squarely on the side of the CCP.

D. The trading desks at major financial institutions know derivatives products inside and out, which may create a one-sided information gap that puts the CCP at a disadvantage.

Answer: D

Step by step

1. Adverse selection arises from asymmetric information.
2. Member firms specialize in derivatives and may know pricing and risk better than the CCP.
3. They may route riskier or mispriced trades to CCPs that underprice them.
4. D describes this one-sided gap against the CCP.

Why D is correct: Members' superior knowledge of pricing and risk lets them selectively trade with CCPs that underprice risk, which is adverse selection.

Why the others are wrong

- **A:** An operational error in member screening; unrelated to information asymmetry.
- **B:** A membership design feature, not an information problem.
- **C:** Transparency is a CCP advantage and gives the CCP, not the members, the information edge; direction reversed.

Q896

Mock 2 Q1

REBUILT — Option D credited liquidity to margin calculations, whereas the reading attributes liquidity to multilateral offset and transparency; D reworded so the correct option matches the reading.

A risk analyst is preparing a briefing on the benefits that central counterparties (CCPs) bring to the over-the-counter derivatives market. Which of the following is most accurately described as a benefit of central clearing?

A. Counterparty risk is eliminated under a CCP structure.

B. Loss sharing through the loss waterfall protects members in the event that a CCP fails.

C. There is more incentive for end users to monitor the creditworthiness of clearing members.

D. Market liquidity is enhanced because multilateral offset and greater transparency make it easier to enter and

terminate trades.

Answer: D

Step by step

1. The reading lists liquidity as an advantage: easier trading and multilateral offset improve transparency and liquidity.
2. A: CCPs reduce but do not eliminate counterparty risk.
3. B: the waterfall handles member defaults, not CCP failure.
4. C: clearing lowers the incentive to monitor (moral hazard).

Why D is correct: Multilateral offset and transparency make it easier for members to trade, improving market liquidity.

Why the others are wrong

- **A:** Overstates the benefit; counterparty risk is reduced and concentrated, not eliminated.
- **B:** Misstates the loss waterfall, which allocates a member's default losses; it does not protect members if the CCP itself fails.
- **C:** Reversed; netting and mutualization reduce the incentive to monitor, which is moral hazard.

Q897 Schweser Module Quiz 36.3 Q2

Following the financial crisis of 2007-2009, the role of central counterparties (CCPs) was increased to reduce systemic risk through a centralized clearing process. Which of the following actions is not an advantage of the CCP in the centralized clearing process?

A. Increase transparency.
B. Manage loss mutualization.
C. Eliminate counterparty risk.
D. Improve operational efficiency.

Answer: C

Step by step

1. Advantages: transparency, multilateral offset, loss mutualization, legal and operational efficiency, liquidity, default management.
2. A, B and D are on the list.
3. Counterparty risk is greatly reduced but not eliminated, so C is not an advantage.

Why C is correct: CCPs reduce counterparty risk but cannot eliminate it, so 'eliminate' is not a valid advantage.

Why the others are wrong

- **A:** Transparency is a listed advantage.
- **B:** Loss mutualization is a listed advantage.
- **D:** Legal and operational efficiency is a listed advantage.

Reading 37. Future Value and Exposure

Jon Gregory, The xVA Challenge: Counterparty Credit Risk, Funding, Collateral, and Capital, 4th Edition (West Sussex, UK: John Wiley & Sons, 2020). Chapter 11.

LO	Learning objective	Qs
37.a	describe and calculate the following metrics for credit exposure: expected mark-to-market, expected exposure, potential future exposure, expected positive exposure and negative exposure, effective expected positive exposure, and maximum exposure.	33
37.b	compare the characterization of credit exposure to VaR methods and describe additional considerations used in the determination of credit exposure.	3
37.c	identify factors that affect the calculation of the credit exposure profile and summarize the impact of collateral on exposure.	3
37.d	identify typical credit exposure profiles for various derivative contracts and combination profiles.	15
37.e	explain how payment frequencies and exercise dates affect the exposure profile of various securities.	3
37.f	explain the general impact of aggregation on exposure, and the impact of aggregation on exposure when there is correlation between transaction values.	10
37.g	describe the differences between funding exposure and credit exposure.	4
37.h	explain the impact of collateralization on exposure and assess the risk associated with the remargining period, threshold, and minimum transfer amount.	15
37.i	assess the impact of collateral on counterparty risk and funding, with and without segregation or rehypothecation.	3

LO 37.a

Describe and calculate the following metrics for credit exposure: expected mark-to-market, expected exposure, potential future exposure, expected positive exposure and negative exposure, effective expected positive exposure, and maximum exposure.

Q898

BT answers P2.T6.901.1 · BT notes P2.T6.901.1 · BT Mock Q52

Freecone Financial entered into an interest rate swap in which it pays semi-annual fixed payments at 3.0% per annum and receives floating payments on a notional of $50.0 million. The floating rate is six-month SOFR, which is also the discount rate. The swap has a remaining life of 1.25 years. The floating rate for the exchange in three months was set three months ago at 2.40%. As with a LIBOR-based swap, the floating rate is observed at the start of each six-month period and paid at its end. The SOFR term structure is currently flat at 2.60% per annum, continuously compounded. Which of the following is nearest to Freecone's current exposure on this swap?

A. Zero
B. -$335,900
C. +$335,900
D. +$1,335,900

Answer: A

Step by step

1. Remaining payment dates: t = 0.25, 0.75, 1.25. Fixed payment = 50m x 3.0%/2 = $750,000 each.
2. Floating payments: first = 50m x 2.40%/2 = $600,000 (already set); the other two = 50m x (e^{0.026 x 0.5} - 1) = $654,243 each (forward rate from the flat curve).
3. Net (floating - fixed) PVs: (600,000-750,000)e^{-0.026x0.25} = -149,028; (654,243-750,000)e^{-0.026x0.75} = -93,907; (654,243-750,000)e^{-0.026x1.25} = -92,695.
4. Swap MtM to Freecone = -149,028 - 93,907 - 92,695 = about -$335,600, i.e. nearest to -$335,900.
5. Current exposure = max(MtM, 0) = max(-335,600, 0) = 0.

Why A is correct: Exposure is the loss if the counterparty defaults now, so only positive MtM counts; Freecone's swap is worth about -$336k, so its exposure is zero.

Why the others are wrong

- **B:** Reports the negative MtM as the exposure; exposure is floored at zero because a negative MtM is not a credit loss to Freecone on counterparty default.
- **C:** Flips the sign of the MtM (treats Freecone as the fixed receiver, or mis-signs fixed minus floating); the 2.40% setting below the 3.0% fixed rate makes Freecone a net payer.
- **D:** Not supported by the cash-flow valuation: the swap's PV is only a few hundred thousand dollars in magnitude, so this overstates by about $1m (a gross arithmetic/scale error).

Q899

BT answers P2.T6.321.1 · BT notes P2.T6.321.1

REBUILT — Original key relied on 'approximately zero' (true only at inception) and option B equated expected MtM with replacement cost (really EE); rewrote the options so exactly one statement is false.

Which of the following statements about expected mark-to-market (expected future value) and expected exposure (EE) for a derivatives position at a future date is FALSE?

A. Expected MtM can be negative.
B. EE always equals expected MtM.
C. EE is greater than or equal to expected MtM.
D. Expected MtM can differ significantly from the current MtM.

Answer: B

Key correction: Source key (D, 'expected MtM approximately zero' is false) is defensible only loosely, and source option B (expected MtM approximately equals expected replacement cost) is also questionable; the item was rebuilt.

Step by step

1. Expected MtM averages all outcomes, positive and negative; EE averages max(V,0), i.e. negatives set to zero.
2. Since max(V,0) >= V in every scenario, EE >= expected MtM, with equality only if no outcome is negative.
3. Hence B (always equal) is false; A, C, D are true (drift, cash flows and long horizons move expected MtM away from today's value).

Why B is correct: EE floors negative values at zero while expected MtM does not, so EE is generally larger.

Why the others are wrong

- **A:** True: if the position is expected to lose value, expected MtM is negative; picking it confuses MtM with exposure, which is never negative.
- **C:** True: because negative outcomes are set to zero in EE, EE cannot be below expected MtM.
- **D:** True: long horizons and the specific cash-flow pattern cause expected MtM to drift from the current MtM; 'about zero' holds only at inception.

Q900

AnalystPrep Q.1936

REBUILT — Original option A said 'confidence interval' and was loosely worded; rewrote around the GARP definition (exceeded with probability no more than 1 minus the confidence level).

Which of the following correctly defines the 99% potential future exposure (PFE) of a portfolio at a given future date?

A. The average of the positive exposures across all scenarios at that date.
B. The average of the portfolio values across all scenarios at that date, including negative values.
C. The exposure level that would be exceeded at that date with a probability of no more than 1%.
D. The average, over time, of the expected exposure out to that date.

Answer: C

Step by step

1. PFE is the worst exposure at a given date at a high confidence level, analogous to VaR.
2. At 99% confidence, the PFE is the 99th percentile of exposure: exceeded with probability at most 100% - 99% = 1%.
3. Statement C matches; A, B, D define EE, expected future value and EPE respectively.

Why C is correct: PFE is a high-percentile (VaR-like) exposure at a single date.

Why the others are wrong

- **A:** Describes expected exposure (EE) at the date, an average, not a tail quantile.
- **B:** Describes expected future value (expected MtM), which includes negative values.
- **D:** Describes expected positive exposure (EPE), a time average of EE, not a single-date quantile.

Q901

AnalystPrep Q.1938

REBUILT — Original relied on a missing graph (points X, Y, Z); replaced with a data table of 95% PFE by month.

The 95% potential future exposure (PFE) profile of a swap is shown below.

Month	0	12	24	36	48	60	72
95% PFE ($ millions)	0	1.8	4.2	4.6	3.1	1.2	0

What is the maximum PFE over the life of the swap, and when does it occur?

A. $4.2 million at month 24

B. $2.1 million (average of the profile)
C. $3.1 million at month 48
D. $4.6 million at month 36

Answer: D
Step by step
1. Maximum PFE is the highest PFE value across the horizon.
2. Scan values: 0, 1.8, 4.2, 4.6, 3.1, 1.2, 0.
3. The highest is $4.6 million at month 36.

Why D is correct: Maximum PFE is the peak of the PFE profile, not an average or an earlier local high.
Why the others are wrong
- **A:** Stops at the third point, mistaking an earlier value for the peak.
- **B:** Averages the profile ((0+1.8+4.2+4.6+3.1+1.2+0)/7 = 2.13); an average is not the maximum PFE.
- **C:** Picks a value after the peak, where the profile is declining as payments amortise.

Q902

AnalystPrep Q.1944

REBUILT — Original option B ('considering only positive values') was ambiguous; reworded options to separate EE (negatives set to zero within the average) from the conditional mean of positives.

Bank Y pays fixed on a 5-year $50 million interest rate swap. What does Bank Y's 12-month expected exposure (EE) represent?

A. The average, across all scenarios 12 months ahead, of the swap's value to Y with negative values replaced by zero.
B. The average, across all scenarios 12 months ahead, of the swap's value to Y including negative values.
C. The average of only those scenarios in which the swap's value to Y is strictly positive.
D. The average of the expected exposures from inception to month 12.

Answer: A
Step by step
1. EE at a date is the mean of max(V,0) over all scenarios at that date.
2. Zero and negative outcomes still enter the average, contributing zero with their probability.
3. That is option A.

Why A is correct: EE counts every scenario, setting losses to zero rather than excluding them.
Why the others are wrong
- **B:** Is expected MtM (expected future value); it lets negatives offset positives.
- **C:** Conditional mean of positive outcomes; it ignores the probability of zero exposure and overstates EE.
- **D:** Is expected positive exposure (EPE), a time average of EE.

Q903

AnalystPrep Q.1945

Banks X and Y enter into a 5-year $50 million interest rate swap in which X pays floating and Y pays fixed. Bank X has a 95% 12-month potential future exposure (PFE) of $3 million. This is equivalent to saying that:

A. 12 months ahead, we are 95% confident that X's gain on the swap will be $3 million or more.
B. 12 months ahead, we are 95% confident that X's gain on the swap will be $3 million or less.
C. 12 months ahead, X's loss on the swap will not exceed $3 million.
D. 12 months ahead, X's gain on the swap will be exactly $3 million with probability 95%.

Answer: B
Step by step
1. PFE is the exposure exceeded with probability no more than 1 - confidence = 5%.
2. X's exposure is its gain (positive MtM) on the swap.
3. So with 95% confidence, X's 12-month gain is at most $3 million.

Why B is correct: A 95% PFE is the 95th percentile of the gain distribution, an upper bound at that confidence.
Why the others are wrong
- **A:** Reverses the inequality; it would describe a 5% tail bound.
- **C:** Confuses exposure (gain, credit risk) with X's own loss; PFE concerns positive MtM.
- **D:** Treats the PFE as a point estimate with a probability attached, rather than a percentile bound.

Q904

AnalystPrep Q.1946

REBUILT — Original claimed PFE 'must' exceed EE, which is not generally true (e.g. a deeply out-of-the-money portfolio); replaced with a numeric normal-distribution case.

The future value of a portfolio at a given date is normally distributed with mean $0 and standard deviation $10 million. Which row correctly gives expected future value, expected exposure (EE), and 95% PFE (in $ millions)?

A. 0, 16.45, 3.99
B. 0, 10.00, 16.45
C. 0, 3.99, 16.45
D. 0, 3.99, 23.26

Answer: C

Step by step

1. Expected future value = mean = 0.
2. EE = E[max(V,0)] = $\sigma\,\phi(0) = 10 \times 0.3989 = 3.99$.
3. 95% PFE = $\mu + 1.645\sigma = 16.45$.
4. Ordering: expected value 0 < EE 3.99 < PFE 16.45.

Why C is correct: For a zero-mean normal, EE is about 0.4 sigma while a 95% PFE is 1.645 sigma.

Why the others are wrong

- **A:** Swaps EE and PFE; EE is an average and must be smaller than a tail quantile here.
- **B:** Uses EE = sigma, ignoring the factor phi(0) = 0.399.
- **D:** Uses the 99% multiplier 2.326 instead of 1.645 for the 95% PFE.

Q905

AnalystPrep Q.2902

REBUILT — Original options were badly worded (option A ungrammatical, C vague about what 'gain' means); rewrote options so exactly one correctly states the PFE meaning.

Banks X and Y enter into a vanilla 5-year interest rate swap with a notional of $100 million and semiannual payments. Bank X pays six-month Libor (floating) and Bank Y pays the fixed swap rate of 5%. Bank X calculates a 12-month potential future exposure (PFE) on the swap of $7.5 million at the 99% confidence level. Which statement is the correct interpretation?

A. There is a 1% probability that Bank X's exposure to Y in 12 months will be $7.5 million or less.
B. Bank X is 99% confident that its loss on the swap in 12 months will be no more than $7.5 million because of market moves alone.
C. There is at most a 1% probability that Bank X's positive mark-to-market gain on the swap (its exposure to Bank Y) will exceed $7.5 million in 12 months.
D. Bank X is 99% confident that its exposure to Bank Y in 12 months will be at least $7.5 million.

Answer: C

Step by step

1. PFE is the worst exposure at a future date at a given confidence level, analogous to VaR.
2. Exposure is positive MtM (a gain for X that is lost if Y defaults): max(MtM, 0).
3. A 99% PFE of $7.5m means the exposure exceeds $7.5m with probability no more than $100\% - 99\% = 1\%$; equivalently X is 99% confident the exposure will not exceed $7.5m.
4. Only C states this; the fixed/floating roles and notional are not needed.

Why C is correct: PFE is a high-percentile (upper-tail) quantile of positive exposure, so only 1% of outcomes exceed it.

Why the others are wrong

- **A:** Puts the 1% probability on the wrong side: the 99th percentile is exceeded 1% of the time, not 'at or below' 1% of the time.
- **B:** Describes a loss; credit exposure and PFE measure a gain (positive MtM) that is at risk if the counterparty defaults.
- **D:** Reverses the inequality: PFE is a ceiling the exposure stays under with 99% confidence, not a floor.

Q906 BT answers P2.T6.25.1.2

Over a period of 5 months, the bank has had counterparty exposures as follows:

Period	EE (Expected Exposure) $m	PFE (95.0% Potential Future Exposure) $m	EEE (Effective non-decreasing EE) $m
1	5	9.8	5
2	10	19.6	10
3	9	17.64	10
4	5	9.8	10
5	4	7.84	10

Which row correctly gives the Expected Positive Exposure (EPE), the Effective Expected Positive Exposure (EEPE) and the maximum potential future exposure?

A. EPE = $6.6m, EEPE = $9m and Maximum PFE = $19.6m
B. EPE = $9m, EEPE = $6.6m and Maximum PFE = $7.84m
C. EPE = $6.6m, EEPE = $10m and Maximum PFE = $19.6m
D. EPE = $6.6m, EEPE = $12.9m and Maximum PFE = $19.6m

Answer: A

Step by step

1. EPE is the average of the EE profile: $(5 + 10 + 9 + 5 + 4)/5 = 6.6$.
2. EEPE is the average of the effective (non-decreasing) EE: $(5 + 10 + 10 + 10 + 10)/5 = 9.0$.
3. Maximum PFE is the highest PFE over the period: $\max(9.8, 19.6, 17.64, 9.8, 7.84) = 19.6$.
4. Row A matches all three.

Why A is correct: EPE averages EE, EEPE averages the non-decreasing EE (so it is at least EPE), and maximum PFE is the largest PFE value, not an average.

Why the others are wrong

- **B:** Swaps EPE and EEPE (EEPE cannot be below EPE) and picks the smallest PFE (7.84) instead of the largest.
- **C:** Uses the peak effective EE (10) as EEPE instead of averaging the five effective EE values.
- **D:** Computes 12.9 as the average of the PFE values $(9.8 + 19.6 + 17.64 + 9.8 + 7.84)/5 = 12.94$, a different metric from the effective EE average.

Q907 BT answers P2.T6.318.2 · BT notes P2.T6.318.2

Credit exposure is asymmetric and, according to Gregory, can be likened to an option position. Which of the following best describes this?

A. Credit exposure = min(MtM, 0), a long option position
B. Credit exposure = min(-MtM, 0), a long option position
C. Credit exposure = max(MtM, 0), a short option position
D. Credit exposure = max(-MtM, 0), a long protective put

Answer: C

Step by step

1. Positive exposure is $\max(\text{MtM}, 0)$: the party loses if the value is positive (in its favour, counterparty defaults) and does not gain if it is negative.
2. The payoff has an option shape (a call-like payoff on MtM struck at zero), so options-pricing ideas such as volatility apply.
3. From the surviving party's perspective it is a short option position, because the exposure constitutes a loss: the defaulting counterparty holds the option not to pay.
4. Only C has the correct payoff and the short position.

Why C is correct: Exposure is $\max(\text{MtM}, 0)$, and because the surviving party bears the loss the counterparty effectively holds the option, making it a short position.

Why the others are wrong

- **A:** $\min(\text{MtM}, 0)$ is the negative exposure (what the party owes), not credit exposure, and it is not a long option.
- **B:** $\min(-\text{MtM}, 0) = -\max(\text{MtM}, 0)$ is the negative of the exposure; it carries the wrong sign and is not a long option payoff.
- **D:** $\max(-\text{MtM}, 0)$ is the counterparty's exposure (the party's negative exposure in absolute value), and 'protective put' is not how credit exposure is characterised.

Q908 BT answers P2.T6.318.3 · BT notes P2.T6.318.3

REBUILT — Original option D ('PFE must be at least zero') is debatable because PFE can be defined at low confidence levels on the negative side; replaced with an unambiguous true statement about maximum PFE.

Each of the following is true about potential future exposure (PFE) EXCEPT which is false?

A. PFE varies with the confidence level, like value at risk (VaR).
B. PFE will approximately equal the notional at contract inception.
C. PFE is an estimate of credit exposure on a future date.
D. Maximum PFE is the highest PFE value over a stated time frame.

Answer: B

Step by step

1. PFE is the worst exposure at a future date at a given confidence level, equivalent to VaR for exposure (A and C true).
2. Maximum PFE is the peak of the PFE profile over the time frame (D true).
3. At inception a typical derivative such as a swap or forward has zero MtM and no time for values to move, so PFE is near zero, not the notional.
4. B is therefore the false statement.

Why B is correct: PFE depends on volatility, maturity and confidence level, not directly on notional, and it starts near zero for a new at-market contract.

Why the others are wrong

- **A:** True: a 99% PFE is larger than a 95% PFE for the same distribution, just as VaR rises with confidence.
- **C:** True: PFE is measured at a specific future date (the horizon).
- **D:** True: maximum PFE is by definition the highest PFE across the dates in the profile.

Q909 BT answers P2.T6.321.2 · BT notes P2.T6.321.2

Suppose a mark-to-market (MtM) is normally distributed with mean $\mu = 2.0\%$ and standard deviation $\sigma = 5.0\%$. Each of the following is true about the expected exposure (EE) EXCEPT which is FALSE?

A. The EE is greater than 2.0%.
B. The EE is greater than the 95% potential future exposure (PFE).
C. An increase in the mean assumption will increase the EE.
D. An increase in the standard deviation assumption will increase the EE.

Answer: B

Step by step

1. EE is the expected positive part of MtM: $EE = \mu\Phi(\mu/\sigma) + \sigma\phi(\mu/\sigma) = 3.15\%$.
2. 95% PFE $= \mu + 1.645\sigma = 2 + 8.22 = 10.22\%$.
3. EE (3.15%) is below the 95% PFE (10.22%), so B is false.
4. A: 3.15% > 2.0%, true (setting negative values to zero lifts the average above the mean). C and D: EE rises with μ (3.84% at $\mu = 3\%$) and with σ (3.53% at $\sigma = 6\%$).

Why B is correct: PFE is a far-tail percentile of the distribution while EE is an average of the positive part, so EE sits well below a 95% PFE.

Why the others are wrong

- **A:** True: EE averages max(MtM,0), which is greater than the unconstrained mean of 2.0%.
- **C:** True: shifting the distribution right raises every positive outcome, so EE increases.
- **D:** True: greater dispersion fattens the positive tail and, with losses truncated at zero, raises EE (option-like convexity).

Q910 BT answers P2.T6.321.3 · BT notes P2.T6.321.3

Suppose a mark-to-market (MtM) is normally distributed with mean $\mu = 1.0\%$ and standard deviation $\sigma = 3.0\%$. Each of the following is true about the potential future exposure (PFE) EXCEPT which is FALSE?

A. PFE is similar to value at risk (VaR), except it is defined further into the future and is associated with a future gain rather than a loss.
B. The 99% PFE is greater than the 95% PFE, which is greater than the expected exposure (EE).
C. If the mean is reduced by 1.0% to $\mu = 0$, the PFE will increase by 1.0%.
D. The PFE is defined only at a specific future point in time; it does not incorporate or average the entire life of

the position.

Answer: C

Step by step

1. 99% PFE = 1 + 2.326(3) = 7.98%; 95% PFE = 1 + 1.645(3) = 5.93%; EE = 1(0.631) + 3(0.377) = 1.76%. So 7.98 > 5.93 > 1.76 and B is true.
2. With $\mu = 0$: 99% PFE = 2.326(3) = 6.98%, which is 1.0% lower, not higher. C is false.
3. A and D are the standard definitions: PFE is a point-in-time upper quantile (gain), unlike EPE which averages across time.

Why C is correct: PFE = $\mu + z\sigma$, so changing the mean shifts PFE one-for-one in the same direction; a lower mean lowers PFE.

Why the others are wrong

- **A:** True: PFE is the VaR-like quantile of the gain side at a longer horizon.
- **B:** True: 7.98% > 5.93% > 1.76% as computed.
- **D:** True: PFE is a single-date measure, unlike EPE which averages EE through time.

Q911

BT answers P2.T6.322.1 · BT notes P2.T6.322.1

REBUILT — Original depended on a missing chart; replaced with a data table of EE and asked for the effective EE series.

A one-year trade has the following expected exposure (EE) profile (in USD millions):

Time (years)	0.2	0.4	0.6	0.8	1.0
Expected exposure (EE)	0.20	0.35	0.30	0.40	0.25

Which series is the effective expected exposure (effective EE) at the five dates?

A. 0.20, 0.35, 0.30, 0.40, 0.25
B. 0.20, 0.35, 0.35, 0.40, 0.40
C. 0.34, 0.34, 0.34, 0.34, 0.34
D. 0.20, 0.35, 0.35, 0.40, 0.25

Answer: B

Step by step

1. Effective EE is the non-decreasing version of EE: $\text{EffEE}_t = \max(\text{EffEE}_{t-1}, \text{EE}_t)$.
2. t=0.2: 0.20. t=0.4: max(0.20, 0.35) = 0.35.
3. t=0.6: EE falls to 0.30, so effective EE stays at 0.35.
4. t=0.8: max(0.35, 0.40) = 0.40. t=1.0: EE falls to 0.25, so effective EE stays at 0.40.
5. Series: 0.20, 0.35, 0.35, 0.40, 0.40.

Why B is correct: Effective EE can never decline over time; it carries forward the highest EE so far, which captures rollover risk on short-dated trades.

Why the others are wrong

- **A:** This is the raw EE profile; it ignores the non-decreasing requirement.
- **C:** A flat line at 0.34 is the effective EPE (average of effective EE), not the time series of effective EE.
- **D:** Applies the non-decreasing rule only up to t=0.8 and then lets the series fall back to 0.25 at t=1.0, which effective EE does not allow.

Q912

BT answers P2.T6.322.2 · BT notes P2.T6.322.2

REBUILT — Original depended on a missing chart and was a giveaway definition; converted to a calculation distinguishing EPE from effective EPE.

A one-year trade has the following expected exposure (EE) profile (in USD millions), observed at five equally spaced dates:

Time (years)	0.2	0.4	0.6	0.8	1.0
Expected exposure (EE)	0.20	0.35	0.30	0.40	0.25

Which is the effective expected positive exposure (effective EPE)?

A. 0.30
B. 0.34
C. 0.35
D. 0.40

Answer: B

Step by step

1. Build effective EE as the running maximum of EE: 0.20, 0.35, 0.35, 0.40, 0.40.
2. Effective EPE is the average of effective EE: $(0.20 + 0.35 + 0.35 + 0.40 + 0.40)/5 = 1.70/5 = 0.34$.
3. For comparison, EPE $= (0.20 + 0.35 + 0.30 + 0.40 + 0.25)/5 = 0.30$.

Why B is correct: Effective EPE is the time average of the non-decreasing effective EE, so it is at least as large as EPE.

Why the others are wrong

- **A:** This is the ordinary EPE, the average of raw EE; it skips the non-decreasing adjustment.
- **C:** Takes the effective EE at t=0.6 (or the peak of the first hump) rather than averaging.
- **D:** Takes the maximum effective EE (0.40), a maximum-type measure rather than an average.

Q913

BT answers P2.T6.322.3 · BT notes P2.T6.322.3

REBUILT — Removed dependence on the missing chart; statements stand on their own.

Each of the following is true about the standard credit exposure metrics, except which is FALSE?

A. At any point in time, effective EE cannot be less than EE

B. (Effective) EPE is the average (effective) EE over time

C. Effective EPE cannot be less than EPE

D. For each point in time, there is a different maximum PFE, so maximum PFE does not represent a single value

Answer: D

Step by step

1. Effective EE is the running maximum of EE, so it is always >= EE (A true).
2. EPE averages EE and effective EPE averages effective EE (B true).
3. Since effective EE >= EE at every date, its average is also >= EPE (C true).
4. Maximum PFE is the highest PFE over the whole horizon, so it is one number; PFE (not maximum PFE) varies by date. D is false.

Why D is correct: Maximum PFE is the single highest value of the PFE profile over the stated horizon, while PFE itself is date-specific.

Why the others are wrong

- **A:** True: effective EE is non-decreasing and starts from EE, so it is never below EE.
- **B:** True: EPE and effective EPE are time averages of EE and effective EE.
- **C:** True: averaging a series that is pointwise >= EE gives an average >= EPE.

Q914

BT answers P2.T6.329.1 · BT notes P2.T6.329.1

Gregory gives the following approximation for expected exposure when the mark-to-market is normally distributed with zero mean: $EE(t) \approx 0.4\,\sigma\sqrt{t}$. An exposure is normally distributed with zero mean and volatility of 9.0% per annum. Using 250 trading days per year, which is NEAREST to the expected exposure at the end of 10 days?

A. 0.72%

B. 1.45%

C. 3.60%

D. 7.18%

Answer: A

Step by step

1. $t = 10/250 = 0.04$ years, so $\sqrt{t} = 0.20$.
2. $EE \approx 0.4 \times 9.0\% \times 0.20 = 0.72\%$.

Why A is correct: For zero-mean normal MtM, $EE = \sigma\sqrt{t}/\sqrt{2\pi} \approx 0.4\sigma\sqrt{t}$; the square-root-of-time scaling shrinks annual volatility to the 10-day horizon.

Why the others are wrong

- **B:** About double the right answer (as if the coefficient were 0.8 instead of 0.4).
- **C:** $0.4 \times 9\% = 3.60\%$: omits the $\sqrt{t}$ time scaling.
- **D:** About ten times the right answer, from mis-scaling the horizon (using the number of days rather than the square root of the year fraction).

Q915 BT answers P2.T6.411.2 · BT notes P2.T6.411.2

Which of the following statements is TRUE about expected exposure (EE)?

A. An increase in the confidence level (e.g., from 95% to 99%) will increase the expected exposure
B. If the drift of the position is negative (i.e., if future expected value is negative), the expected exposure is zero
C. Expected exposure is an increasing function of the variance of the future value's distribution
D. Expected exposure is less than potential future exposure at all (any) confidence levels

Answer: C

Step by step

1. EE is the expected value of $\max(V, 0)$: no confidence level is involved, so A is false.
2. With negative drift but positive variance, some outcomes are still positive, so EE > 0; B is false.
3. For normal $V \sim N(\mu, \sigma^2)$, $EE = \mu\Phi(\mu/\sigma) + \sigma\varphi(\mu/\sigma)$, which rises with σ; C is true.
4. PFE depends on the confidence level; at a low level (e.g., 50%) PFE can fall below EE, so D is false.

Why C is correct: EE is the mean of the positive part of the MtM, and a wider distribution puts more weight on large positive outcomes, raising EE.

Why the others are wrong

- **A:** Confuses EE with PFE; only PFE depends on a confidence level.
- **B:** Negative expected MtM lowers but does not eliminate EE, since positive outcomes still occur; EE is zero only if MtM is never positive.
- **D:** Holds only for sufficiently high confidence levels; at low confidence PFE can be below EE.

Q916 BT answers P2.T6.411.3 · BT notes P2.T6.411.3

A position has a future mark-to-market value that is normally distributed. Acme Bank has the following exposure metrics at the 99% confidence level (values in millions):

Parameter	Value
Mu (mean)	2.0
Standard deviation (sigma)	4.0
Confidence level	99%
Expected exposure (EE)	2.79
Potential future exposure (PFE)	11.31

Each of the following statements about Acme Bank's PFE is true, EXCEPT which is not true?

A. There is a 1.0% probability that the future (mark-to-market) value will be greater than +$11.31 million
B. If the mean (mu) is zero, there is no impact on PFE, as PFE is conditional on gains
C. At a 95% confidence level, the PFE is less than $11.31 million
D. At a 50% confidence level, the PFE is 2.0

Answer: B

Step by step

1. PFE at the 99% level is the 99th percentile of future value: $2.0 + 2.326 \times 4.0 = 11.31$.
2. A: by definition only 1% of outcomes lie above the 99th percentile, so A is true.
3. C: at 95%, $2.0 + 1.645 \times 4.0 = 8.58 < 11.31$, so C is true (a lower confidence level gives a lower PFE).
4. D: at 50%, the quantile is the mean, $2.0 + 0 \times 4.0 = 2.0$, so D is true.
5. B: with $\mu = 0$, PFE(99%) = $0 + 2.326 \times 4.0 = 9.31$, a fall of 2.0. PFE depends on the mean, so B is false.

Why B is correct: PFE is a quantile of the whole future-value distribution (like VaR), so it shifts one-for-one with the mean. It is not conditional on gains; that description fits EE, not PFE.

Why the others are wrong

- **A:** True: the 99% PFE is the point above which only 1% of outcomes lie. Rejecting it confuses the tail probability with the confidence level.
- **C:** True: 8.58 is below 11.31 because a lower confidence level gives a smaller quantile. Rejecting it ignores that PFE rises with confidence.
- **D:** True: the 50th percentile of a normal distribution equals its mean, 2.0. Rejecting it forgets that z = 0 at the median.

Q917 BT answers P2.T6.412.1 · BT notes P2.T6.411.3/2 (original 412.1)

REBUILT — Original relied on a missing exposure chart (lines A, B, C) and the notes source gave no answer; rebuilt with an explicit quarterly EE profile and numeric statements testing the same EE, EPE, effective EE, EEPE and maximum PFE concepts.

A one-year derivative has the following expected exposure (EE) profile, measured at equally spaced quarterly dates (in millions):

Time (years)	0.25	0.50	0.75	1.00
EE	0.20	0.50	0.30	0.10

Each of the following statements is correct, EXCEPT which is not correct?

A. The effective EE at 0.75 years is 0.50, because effective EE is the non-decreasing version of EE

B. Because the EE profile has a single peak of 0.50, the maximum PFE over the year is also 0.50

C. The expected positive exposure (EPE), the "loan equivalent", is 0.275

D. The effective EPE is 0.425, which is greater than the EPE

Answer: B

Step by step

1. Effective EE is the running maximum of EE: 0.20, 0.50, 0.50, 0.50. At 0.75 years it is 0.50, so A is correct.
2. EPE is the average of EE: $(0.20 + 0.50 + 0.30 + 0.10)/4 = 0.275$, so C is correct.
3. Effective EPE is the average of effective EE: $(0.20 + 0.50 + 0.50 + 0.50)/4 = 0.425 > 0.275$, so D is correct.
4. Maximum PFE is the highest PFE (a high-quantile exposure) over the horizon, not the highest EE. PFE lies above EE, so the maximum PFE exceeds 0.50. B is not correct.

Why B is correct: EE is an average exposure, while PFE is a high-confidence worst-case exposure; the peak of the EE profile therefore understates the maximum PFE regardless of the profile's shape.

Why the others are wrong

- **A:** Correct statement: the non-decreasing constraint carries the 0.50 peak forward. Rejecting it ignores that effective EE never falls.
- **C:** Correct statement: 0.275 is the simple average of the four EE points. Rejecting it usually comes from averaging effective EE (0.425) or taking only the peak.
- **D:** Correct statement: effective EPE averages effective EE (0.425), which exceeds EPE because EE declines after the peak. Rejecting it assumes the two measures are equal.

Q918 BT answers P2.T6.412.2 · BT notes P2.T6.412.2

In order to determine the counterparty credit risk capital charge, the Basel III framework requires an effective expected positive exposure (effective EPE or EEPE) calculated with stressed parameters. Each of the following statements about the EEPE is true, EXCEPT which is FALSE?

A. If expected exposure (EE) increases over a few months to a peak, then declines toward zero by the end of the year, EEPE must be greater than EPE

B. If expected exposure (EE) is monotonically increasing, EEPE is identical to EPE

C. The intent of the non-decreasing constraint (in effective EE and EEPE) is to capture roll-off or rollover risk, for transactions close to maturity that are likely to be replaced in practice

D. Because EEPE is a weighted average over time, it is a single number, but it depends on a confidence level, typically 99%, which can be set at more conservative levels

Answer: D

Key correction: Sources key D is consistent: D is the false statement the question asks for (the BT answers text 'D is CORRECT' means D is the correct choice for EXCEPT).

Step by step

1. Effective EE is the non-decreasing EE; EEPE is the average of effective EE, and EPE is the average of EE.
2. A: after the peak, effective EE stays at the peak while EE falls, so effective EE exceeds EE at those dates and EEPE > EPE. True.
3. B: if EE never falls, effective EE = EE at every date, so EEPE = EPE. True.
4. C: holding exposure non-decreasing captures rollover of short-term trades that would be replaced. True.
5. D: EEPE is an average of expected exposures; it has no confidence level (that belongs to PFE). D is false.

Why D is correct: EEPE averages expected (mean) exposures, so it involves no quantile or confidence level; only PFE-type measures depend on a confidence level.

Why the others are wrong

- **A:** True: the non-decreasing floor lifts effective EE above EE after the peak. Rejecting it forgets that the floor

only matters when EE falls.
- **B:** True: with a rising EE the non-decreasing constraint is never binding. Rejecting it assumes effective EE always differs from EE.
- **C:** True: this is the stated purpose of effective EE (rollover risk for short-dated trades). Rejecting it confuses the constraint with a conservatism add-on.

Q919 BT answers P2.T6.908.1 · BT notes P2.T6.908.1

The probability distribution of the future value (in millions) of a derivative position is shown below:

Future Value	Probability
$10.0	5.0%
$6.0	10.0%
$3.0	15.0%
$1.0	20.0%
($1.0)	20.0%
($3.0)	15.0%
($6.0)	10.0%
($10.0)	5.0%

What is the expected exposure (EE) of this position?

A. Zero
B. 1.75
C. 3.50
D. 5.25

Answer: B

Step by step

1. Exposure is $\max(V, 0)$, so negative values contribute zero but keep their probability weight.
2. $EE = 0.05(10) + 0.10(6) + 0.15(3) + 0.20(1) = 0.50 + 0.60 + 0.45 + 0.20 = 1.75$.

Why B is correct: EE is the probability-weighted average of positive future values with negative outcomes set to zero (not removed).

Why the others are wrong

- **A:** Zero is the expected future value (expected MtM), since the distribution is symmetric; it includes the negative values instead of flooring them at zero.
- **C:** 3.50 is the average of positive values conditional on being positive (1.75 / 0.50), or the mean absolute value; it drops the 50% probability of zero exposure.
- **D:** 5.25 matches no valid exposure metric; it comes from scaling the correct figure by an incorrect factor (1.5 x 3.50).

Q920 BT answers P2.T6.908.3 · BT notes P2.T6.908.3

REBUILT — Merged the full-table and truncated copies; two options had OCR-uncertain values (-79/+73.25), so the distractors were rebuilt as realistic mistakes (wrong tail, normal approximation, maximum).

The probability distribution of the future value (FV) of a derivative position is shown below. The mean is $3.0 and the standard deviation is $12.56; the distribution is symmetric but not exactly normal.

Future Value	Probability
$33.0	2.5%
$25.0	5.0%
$18.0	7.5%
$12.0	15.0%
$7.0	20.0%
($1.0)	20.0%
($6.0)	15.0%
($12.0)	7.5%
($19.0)	5.0%
($27.0)	2.5%

What is the 95% potential future exposure (PFE) of this position?

A. $18.00
B. $23.66

C. $25.00
D. $33.00

Answer: C
Step by step
1. PFE at 95% is the exposure level that is exceeded with probability no more than $1 - 95\% = 5\%$.
2. Accumulate probability from the top: P(FV > 33) = 0%; P(FV > 25) = 2.5%; P(FV > 18) = 2.5% + 5.0% = 7.5%.
3. P(FV > 25) = 2.5% <= 5%, but P(FV > 18) = 7.5% > 5%, so 18 is not high enough; the smallest value meeting the 5% limit is 25.
4. Cross-check from the bottom: cumulative probability up to 18 is 92.5% (<95%) and up to 25 is 97.5% (>=95%), so the 95th percentile is $25.00.
5. PFE (95%) = $25.00.

Why C is correct: PFE is a high percentile of the future-value distribution; the 95th percentile is the first point whose cumulative probability reaches 95%, which is $25.00 in this discrete distribution.

Why the others are wrong
- **A:** $18.00 is the 90% PFE (7.5% of outcomes exceed it, which is more than the 5% allowed); it comes from stopping the tail accumulation too early.
- **B:** $3 + 1.645 \times 12.56 = 23.66$ is the normal-approximation PFE; the stem gives the actual distribution (not exactly normal), so the percentile should be read from the table.
- **D:** $33.00 is the maximum outcome (roughly the 100% level), not the 95% level; it ignores that 5% of probability is allowed above the PFE.

Q921

BT answers P2.T6.909.2 · BT notes P2.T6.909.2

REBUILT — Original EXCEPT item rested on EAD conventions outside LO 37.a and had a debatable option D; rewritten to test the LO 37.a metrics (EFV, EE, PFE, EPE, effective EPE, maximum PFE) with one clearly false statement.

Which of the following statements about credit exposure metrics is INCORRECT?

A. Expected positive exposure (EPE) is the average of the expected exposure (EE) profile across time horizons; with unequal spacing it is the time-weighted average.
B. Effective EE is the non-decreasing version of EE, so effective EPE can never be lower than EPE for the same exposure profile.
C. Expected future value (expected mark-to-market) can be negative, whereas EE, which counts only positive values (negative ones set to zero), cannot.
D. Maximum PFE is the average of the PFE values across all time horizons over the life of the transaction.

Answer: D
Key correction: Source key (A) said the inaccurate statement was the EAD = EE option; that item was rebuilt around the LO 37.a metrics, where the incorrect statement is D.
Step by step
1. Check each statement against the definitions.
2. A: EPE is the average (time-weighted if intervals differ) of EE across time - true.
3. B: Effective EE at each date is the running maximum of EE, so it is >= EE at every date; its average (effective EPE) is therefore >= EPE - true.
4. C: Expected MtM averages both gains and losses so it can be negative; EE sets negative values to zero and so is non-negative - true.
5. D: Maximum PFE is the highest PFE value over the stated time frame (the peak of the PFE profile), not an average - false.

Why D is correct: Maximum PFE is the peak of the PFE profile; averaging exposures over time is what EPE and effective EPE do, not maximum PFE.

Why the others are wrong
- **A:** A is a correct statement of EPE; picking it would mean confusing EPE with a peak measure.
- **B:** B is correct: taking the running maximum can only raise EE, so effective EPE >= EPE; rejecting it ignores that effective EE is non-decreasing.
- **C:** C is correct: expected MtM includes negative values while EE floors them at zero; rejecting it confuses expected MtM with EE.

Q922 BT answers P2.T6.909.3 · BT notes P2.T6.909.3

REBUILT — The chart exhibit was missing and option D was garbled; rebuilt as a self-contained calculation of effective EPE from a tabulated EE profile.

The expected exposure (EE) of a short-dated position, measured at the end of each month for twelve months, is below (in $ millions).

Month	1	2	3	4	5	6	7	8	9	10	11	12
EE	2	4	6	8	10	9	8	7	6	5	4	3

What is the effective expected positive exposure (effective EPE) of this position, in $ millions?

A. 6.00
B. 6.67
C. 8.33
D. 10.00

Answer: C

Step by step

1. Effective EE is the non-decreasing EE, i.e. the running maximum: 2, 4, 6, 8, 10, 10, 10, 10, 10, 10, 10, 10.
2. Sum of effective EE = $2 + 4 + 6 + 8 + 8 \times 10 = 100$.
3. Effective EPE = average of effective EE = $100/12 = 8.33$.

Why C is correct: Effective EPE averages the non-decreasing (running-maximum) EE, which captures rollover risk for short-dated trades; after the month-5 peak of 10 the effective EE stays at 10.

Why the others are wrong

- **A:** 6.00 = 72/12 is plain EPE, the average of EE itself; it ignores the non-decreasing adjustment.
- **B:** 6.67 averages only the first six effective EE values (2, 4, 6, 8, 10, 10) instead of all twelve months.
- **D:** 10.00 is the peak EE (and the final effective EE), not the average of the effective EE profile.

Q923 BT notes P2.T6.407.2

Each of the following statements about credit exposure, current exposure and potential exposure is necessarily true, EXCEPT which is false?

A. Credit exposure is conditional on counterparty default
B. If the mark-to-market (MtM) of a contract is negative, current exposure is zero
C. If the mark-to-market (MtM) of a contract is negative, credit exposure is zero
D. Credit exposure encompasses (includes) both current exposure and potential exposure

Answer: C

Step by step

1. Exposure at a date is $\max(V, 0)$: a party loses only if the counterparty defaults while the contract has positive value - A is true.
2. With negative MtM today, current exposure is $\max(V, 0) = 0$ - B is true.
3. Credit exposure also contains potential future exposure: the MtM may turn positive later, so a negative MtM today does not make credit exposure zero - C is false.
4. Credit exposure = current exposure + potential (future) exposure - D is true.

Why C is correct: A negative current MtM only sets current exposure to zero; potential future exposure can still be positive, so credit exposure is not necessarily zero.

Why the others are wrong

- **A:** True: exposure matters only if the counterparty defaults; rejecting it overlooks the conditional nature of credit exposure.
- **B:** True: current exposure is max(MtM, 0), which is zero when MtM is negative.
- **D:** True: credit exposure is the sum of what is at risk now (current) and what may be at risk later (potential).

Q924 BT notes P2.T6.909.1 · BT answers P2.T6.909.1

REBUILT — Chart missing and the two copies had inconsistent, unrecoverable EE values (18 vs 78 months); merged and rebuilt as a self-contained time-weighted EPE calculation.

The expected exposure (EE) profile of a position is given below, with EE treated as constant over the period ending at each observation date.

Observation date (years)	0.5	1	2	3	5
EE ($ millions)	3	5	8	6	2

What is the expected positive exposure (EPE), computed as the time-weighted average of the EE profile, in $ millions?

A. 4.40
B. 4.80
C. 5.50
D. 8.00

Answer: A

Step by step

1. Interval lengths: 0-0.5 = 0.5, 0.5-1 = 0.5, 1-2 = 1, 2-3 = 1, 3-5 = 2; total 5 years.
2. Weighted sum = $3(0.5) + 5(0.5) + 8(1) + 6(1) + 2(2) = 1.5 + 2.5 + 8 + 6 + 4 = 22$.
3. EPE = $22/5 = 4.40$.

Why A is correct: EPE is the average of EE across time; with unequally spaced dates each EE must be weighted by the length of its interval.

Why the others are wrong

- **B:** $(3 + 5 + 8 + 6 + 2)/5 = 4.80$ is the unweighted average, ignoring the unequal interval lengths.
- **C:** $(3 + 5 + 8 + 6)/4 = 5.50$ drops the final (longest, lowest-EE) period.
- **D:** 8.00 is the peak EE, not the average.

Q925

BT notes P2.T7.3.2

REBUILT — Option labels used non-standard terms (AECE, WCE, ECE); rewritten with GARP terminology (EE at a date, PFE, EPE).

Which counterparty credit exposure metric is typically used as the basis for the loan equivalent amount (even if a multiplier may then be applied)?

A. Current exposure (the larger of current MtM and zero)
B. Expected exposure (EE) at a single future date
C. Potential future exposure (PFE) at a high confidence level, such as 95%
D. Expected positive exposure (EPE), the average of the EE profile over time

Answer: D

Step by step

1. The loan equivalent is a single number representing the average amount effectively lent to the counterparty over the life of the trade.
2. That is the time-average of expected exposure, i.e. EPE (average EPE).
3. A single-date or peak measure (current exposure, EE at one date, PFE) does not average over time, so it is not the basis.

Why D is correct: The loan equivalent is the average amount effectively lent to the counterparty, which is exactly EPE, the time-average of the EE profile.

Why the others are wrong

- **A:** Current exposure is only today's positive MtM, ignoring how exposure evolves.
- **B:** EE at one future date captures one point on the profile, not the average over the life.
- **C:** PFE is a high-percentile worst-case measure (its peak is maximum PFE), not an average amount lent.

Q926

Schweser Module Quiz 37.1 Q1

REBUILT — The exhibit with the '?' was missing; replaced it with a verbal description of the plotted line so the item is self-contained.

A risk report plots a derivative's expected exposure (EE) profile against future dates. A single horizontal line, labeled '?', is drawn at the average of the EE values across all dates through the life of the trade. Which credit exposure metric does the '?' represent?

A. Expected positive exposure (EPE).
B. Potential future exposure (PFE).
C. Effective expected exposure (effective EE).
D. Effective expected positive exposure (effective EPE).

Answer: A

Step by step

1. EE is the expected positive MtM at each future date, so the plotted curve is the EE profile.
2. EPE is defined as the average of EE through time, i.e., a single number summarizing the whole profile.
3. A horizontal line at the time-average of the EE profile is therefore EPE.

Why A is correct: EPE collapses the EE profile into one amount by averaging EE through time; it is the 'loan equivalent' single-number exposure.

Why the others are wrong

- **B:** PFE is a high-confidence percentile of the exposure distribution at each date (worst case), not an average of EE; it would plot as a curve above EE.
- **C:** Effective EE is the non-decreasing version of the EE curve; it is itself a profile (a curve), not a single average line.
- **D:** Effective EPE is the average of the effective (non-decreasing) EE, not of the raw EE profile; it is equal to or higher than EPE. Choosing it confuses the raw average with the average of the effective profile.

Q927

AnalystPrep Q.1939

The Basel Committee (2005) introduced effective expected exposure (effective EE) and effective expected positive exposure (effective EPE) to supplement EE and EPE. What was the reason for these changes?

A. EE and EPE may underestimate exposure for short-dated transactions because they do not take into account roll-over risk.

B. EE and EPE may overestimate exposure for short-dated transactions because they do not take into account roll-over risk.

C. EE and EPE may underestimate exposure for long-dated transactions because they do not take into account roll-over risk.

D. EE and EPE may underestimate exposure for short-dated transactions because they take into account only roll-over risk.

Answer: A

Step by step

1. Short-dated trades (under one year) fall in EE as they approach maturity, so EE and EPE look small.
2. In practice such trades are rolled over or replaced with the same counterparty, so exposure persists.
3. Effective EE forces EE to be non-decreasing and effective EPE is the average of effective EE, which captures roll-over risk and avoids the understatement.

Why A is correct: EE declines near maturity of short-dated trades but the exposure is effectively renewed through roll-over, so unadjusted EE/EPE understate it; the effective measures correct this.

Why the others are wrong

- **B:** Overestimation is the wrong direction: ignoring roll-over lowers, not raises, measured exposure.
- **C:** The problem concerns short-dated transactions (under one year); long-dated trades are not the target of the adjustment.
- **D:** EE and EPE ignore roll-over risk; they do not 'only' consider it, which is a reversal of the actual shortcoming.

Q928

AnalystPrep Q.1995

REBUILT — The original options B and C were nearly indistinguishable and the stem asked for a verbal definition; rebuilt as a calculation that separates EE from expected MtM, PFE and the average of positives only.

At a single future date, a netting set has five equally likely MtM outcomes (in $ millions) to the bank: 40, 20, 0, -20, -30. What is the expected exposure (EE) at that date?

A. $2 million

B. $12 million

C. $30 million

D. $40 million

Answer: B

Step by step

1. Exposure is the positive part of MtM: max(V, 0) = 40, 20, 0, 0, 0.
2. EE = probability-weighted mean of these = (40 + 20 + 0 + 0 + 0)/5.
3. EE = 60/5 = $12 million.

Why B is correct: EE is the mean of the exposure distribution at a date, with negative MtM values set to zero but still counted in the probability weighting.

Why the others are wrong

- **A:** $2 million is expected MtM: (40+20+0-20-30)/5; it lets negative values offset positives instead of flooring them at zero.
- **C:** $30 million is the average of only the positive outcomes, 60/2; zero and negative scenarios must still be counted in the denominator.
- **D:** $40 million is the largest outcome (PFE at a very high confidence level), a worst-case figure rather than an expected value.

Q929

AnalystPrep Q.2003

REBUILT — The source defined EPE inconsistently (average over a distribution vs time-weighted average of EE); rebuilt as a calculation using the Schweser definition (EPE = average of EE through time).

A bank estimates a swap's expected exposure (EE) at four equally spaced future dates as $10 million, $18 million, $14 million and $6 million. Using the definition of expected positive exposure (EPE) as the average of EE through time, what is the EPE?

A. $12 million
B. $14 million
C. $16 million
D. $18 million

Answer: A

Step by step

1. EPE is the time-average of the EE profile; dates are equally spaced, so use a simple average.
2. Sum = 10 + 18 + 14 + 6 = 48.
3. EPE = 48/4 = $12 million.

Why A is correct: EPE summarizes the EE profile as one number by averaging EE across time horizons.

Why the others are wrong

- **B:** $14 million is just the third EE point; it is not the average of all four dates.
- **C:** $16 million is the effective EPE: average of the non-decreasing EE (10, 18, 18, 18); that is a different measure from EPE.
- **D:** $18 million is the maximum EE (the peak of the profile), not its average.

Q930

BT answers P2.T6.24.32.2 · BT notes P2.T6.24.32.2

REBUILT — The source table was garbled (probabilities sum to 80% and per-period FV x probability is not a valid EE), and its key of 0.6 was the peak EE rather than an average; rebuilt with the EE profile given directly.

A risk manager estimates a derivative portfolio's expected exposure (EE, in $ millions) at four quarterly dates over the next year.

Quarter	Q1	Q2	Q3	Q4
Expected exposure	0.50	0.60	0.45	0.20

What is the effective expected positive exposure (effective EPE) over the year, and what does it represent?

A. 0.575; the average of the non-decreasing (effective) EE profile, which captures roll-over risk.
B. 0.600; the average of the effective EE profile, which captures roll-over risk.
C. 0.575; the maximum EE reached over the period, emphasizing peak risk.
D. 0.4375; the average of the raw EE profile, which captures roll-over risk.

Answer: A

Key correction: Source key (BT answers/notes) said B (0.6); correct is A because 0.6 is the maximum effective EE, whereas effective EPE is its average, 0.575.

Step by step

1. Effective EE is the non-decreasing EE: running maximum of 0.50, 0.60, 0.45, 0.20 = 0.50, 0.60, 0.60, 0.60.
2. Effective EPE = average of effective EE = (0.50 + 0.60 + 0.60 + 0.60)/4 = 2.30/4.
3. Effective EPE = 0.575; it averages the effective profile to reflect roll-over of short-dated trades.

Why A is correct: Effective EPE averages the non-decreasing EE profile, so later low EE values are replaced by the earlier peak; the average (0.575) lies below the peak (0.60).

Why the others are wrong

- **B:** 0.60 is the peak effective EE (maximum), not its average; the Q1 value of 0.50 pulls the average down to 0.575.
- **C:** The value is right but the description is wrong: 0.575 is an average of effective EE, whereas a maximum is a peak measure (maximum EE / maximum PFE-type metric).
- **D:** 0.4375 is the average of the raw EE (0.50+0.60+0.45+0.20)/4; it ignores the non-decreasing adjustment that defines effective EPE, and so is plain EPE.

LO 37.b

Compare the characterization of credit exposure to VaR methods and describe additional considerations used in the determination of credit exposure.

Q931

AnalystPrep Q.1933

REBUILT — Original referred to a diagram that was not included and had a giveaway correct option; rewrote as a self-contained conceptual question on why future exposure is characterized probabilistically.

A risk manager plots a derivative portfolio's exposure to a counterparty on a time axis split into history, today, and the future. Which statement best explains why the future part of the chart is expressed as a distribution (potential future exposure) rather than a single line, in the same way that VaR characterizes market risk?

A. Past and current exposure are known with certainty, whereas future exposure depends on uncertain future market movements and can only be described by probabilistic estimates.

B. Future exposure can be projected deterministically by extrapolating the trend of past and present exposure.

C. Both current and future exposure are uncertain, because today's mark-to-market value is itself only an estimate.

D. Past exposure is known with certainty, but current and future exposure are both subject to probabilistic estimation.

Answer: A

Step by step

1. Exposure on a given date is the positive part of the portfolio value, $\max(V, 0)$, so it is known once the value is observed.
2. History and today's mark-to-market value are observable, so past and current exposure are known.
3. Future exposure depends on future market moves, which are unknown, so it must be described by a distribution (e.g., PFE at a confidence level), as VaR does for market risk.
4. This matches option A.

Why A is correct: Only the future is unobserved; its dependence on random market movements is why exposure is characterized by probabilistic metrics like PFE, analogous to VaR.

Why the others are wrong

- **B:** Treats the future as predictable from the past; ignores that future market moves are random, which is the whole reason for a distribution.
- **C:** Wrongly says current exposure is uncertain; today's value is observed (or at least a base value), the uncertainty lies only in the future.
- **D:** Wrongly says current exposure is uncertain; only future exposure is uncertain.

Q932

AnalystPrep Q.1934

In financial risk management, value at risk (VaR) and exposure are often compared because both involve quantifying the market risk of a portfolio. What is the primary reason characterizing exposure is more complex than characterizing VaR?

A. Exposure must be defined over multiple time horizons, often far in the future, unlike VaR, which uses a short horizon.

B. Exposure can take positive and negative values, whereas VaR is always positive.

C. Future VaR is always known with certainty, but future exposure is not.

D. It is easier to model future VaR than future exposure, because VaR has no risk mitigants.

Answer: A

Step by step

1. VaR is computed over a short horizon (e.g., 10 days), so drift, cash flows, exercise decisions and ageing of trades can be ignored.
2. Exposure must be defined over many horizons, often years ahead, so drift, volatility, co-dependence, cash flows and path dependency all matter.
3. Exposure also involves risk mitigants (netting, collateral) and is used for pricing as well as risk, but the primary stated difference is the multiple, long time horizons.
4. Answer: A.

Why A is correct: Because exposure is needed at many future dates, the ageing of trades, drift and path dependency must be modelled, which VaR's short horizon lets one neglect.

Why the others are wrong

- **B:** Sign is not the source of complexity; exposure is defined as the positive part of value (floored at zero) and VaR is a loss quantile.
- **C:** No risk measure is known with certainty for the future; both VaR and exposure are probabilistic.
- **D:** This restates the conclusion rather than giving a reason, and its premise is backwards: risk mitigants such as netting and collateral are an extra complexity for exposure, not a feature of VaR; the primary driver is the multiple, long horizons.

Q933 BT answers P2.T6.412.3 · BT notes P2.T6.412.3

Credit exposure, a fundamental credit risk measure, is compared with value at risk (VaR), a fundamental market risk measure. Each of the following is a valid comparison EXCEPT which is FALSE?

A. Unlike VaR, credit exposure needs to be defined over multiple time horizons (often far in the future) to capture the impact of time and the specifics of the underlying contracts.

B. Unlike VaR, credit exposure often must model complex and/or subjective factors (including legal interpretations) because of risk mitigants such as netting and collateral.

C. Unlike VaR, which depends critically on the portfolio's expected future value (drift), credit exposure is measured with a simulation that does not depend on expected future value.

D. VaR is a risk measure only, whereas credit exposure is defined for both risk management and valuation (pricing), which may imply different calculations for each.

Answer: C

Step by step

1. A: true. Exposure needs many (long) horizons; VaR uses a short one.
2. B: true. Netting and collateral add legal and subjective modelling complexity.
3. C: false. It reverses the relationship: over long horizons the drift (expected future value) matters for credit exposure, while VaR can ignore drift because its horizon is short.
4. D: true. Exposure is used for pricing (xVA) and risk, so different calculations may be needed.
5. The false statement is C.

Why C is correct: Drift and expected future value are relevant to long-horizon exposure but negligible for short-horizon VaR, so C reverses the truth.

Why the others are wrong

- **A:** True statement (time horizon difference), so not the exception.
- **B:** True statement (risk mitigants add complexity and subjectivity), so not the exception.
- **D:** True statement (exposure serves both pricing and risk, VaR only risk), so not the exception.

LO 37.c

Identify factors that affect the calculation of the credit exposure profile and summarize the impact of collateral on exposure.

Q934 AnalystPrep Q.1904

REBUILT — Original relied on a missing exposure-profile figure and had a vague statement 4; rewrote as a self-contained statement-evaluation item with exactly one correct combination.

A risk manager compares a counterparty's exposure profile with and without a collateral agreement and notes that collateral cannot remove exposure completely. Consider the following statements about why collateral does not perfectly mitigate exposure:

I. A threshold means that exposure below the threshold is never called for and so remains uncollateralized.

II. Delays in receiving collateral (the margin period of risk) allow exposure to rise before the additional collateral is actually received.

III. Collateral value is fixed once posted, so any residual risk arises only from movements in the exposure itself.

IV. Once an exposure exceeds the threshold, the collateral agreement always calls for exactly the full exposure, so no residual risk remains above the threshold.

Which statements are correct reasons?

A. I and II only
B. II and III only
C. I and III only
D. III and IV only

Answer: A

Step by step

1. Collateral reduces but does not eliminate exposure; the reasons are the agreement terms and timing/valuation frictions.
2. I is correct: a threshold is an exposure level below which collateral is not called, so that amount stays uncollateralized.
3. II is correct: the margin period of risk (delay in receiving collateral) lets exposure move after the last margin call.
4. III is false: for non-cash collateral the value itself varies, which is another source of residual risk.
5. IV is false: minimum transfer amounts and rounding (the granularity effect) and delays mean the call is not exactly the full exposure.
6. Only I and II are correct, so A.

Why A is correct: Thresholds leave exposure uncollateralized and collateral delays let exposure change before collateral arrives; both leave residual risk even under a collateral agreement.

Why the others are wrong

- **B:** Includes III, which wrongly assumes collateral value is stable; non-cash collateral value fluctuates. It also omits the threshold effect.
- **C:** Includes III (collateral value does not vary) and drops the delay effect (II), the key timing risk.
- **D:** Both III and IV are false: collateral value can vary, and minimum transfer amounts, rounding and delays prevent perfect calling.

Q935 BT answers P2.T6.910.3

REBUILT — Option C (floating payer in an upward-sloping curve) was ambiguous about 'greater credit exposure'; replaced with a clearly false statement about swap exposure rising to maturity.

Credit exposure profiles vary according to the instrument and can be substantially altered by the specific nature of the cash flows in a transaction. With respect to credit exposure profiles, which of the following statements is TRUE?

A. As in market value-at-risk (VaR) measurement, quantification of counterparty credit exposure for xVA can safely ignore the drift of the risk factors, since drift is negligible at all horizons
B. For a single-name credit default swap, expected exposure generally decreases as the volatility of the credit spread rises
C. For an interest rate swap, exposure rises steadily until maturity, because periodic payments do not reduce future uncertainty
D. For a simple forward contract with a single cash flow on an underlying whose returns are independent and identically distributed, exposure is approximately proportional to the square root of time

Answer: D

Step by step

1. Single cash flow at maturity: uncertainty about the final exchange grows with the time horizon.
2. With i.i.d. returns, the standard deviation of the underlying scales with $\sqrt{t}$, so exposure $\propto \sqrt{t}$ $(t \leq T)$.
3. Check A: VaR uses short horizons, so drift can be ignored; xVA needs long horizons, where drift matters.
4. Check B: higher spread volatility raises the chance of large positive values, so CDS exposure increases.
5. Check C: swap payments reduce the remaining cash flows, so the profile is hump-shaped and falls toward zero at maturity.
6. Only D is true.

Why D is correct: A single-payout forward has exposure driven by the volatility of the underlying over the horizon, which grows with the square root of time under i.i.d. returns.

Why the others are wrong

- **A:** Drift can be ignored only in short-horizon VaR; for xVA exposure over long horizons the drift is relevant.
- **B:** Reverses the relationship: exposure is an increasing, not decreasing, function of spread volatility.
- **C:** Ignores the amortization effect of periodic cash flows: exposure peaks mid-life and falls toward zero at maturity (not monotonic).

Q936

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A counterparty risk analyst at a bank reviews a netting set with a hedge fund. The credit support annex provides for cash collateral only, with no threshold, a minimum transfer amount of USD 1.0 million, and daily margin calls. Since the last call, the mark-to-market exposure to the hedge fund has risen by USD 0.6 million, and the hedge fund has met every previous call in full and on time. Which of the following best explains why the bank is still not fully collateralized at the end of the day?

A. Variation in collateral value: the cash collateral held by the bank has lost value since it was posted.

B. Path dependency: the call amount is reduced because collateral was received late in earlier periods.

C. The granularity effect: the minimum transfer amount prevents the bank from calling for all of the collateral it is actually owed.

D. The margin period of risk: the hedge fund has a grace period in which it may delay delivery of an already-valid call.

Answer: C

Step by step

1. Collateral is cash, so its value cannot vary; this removes the collateral-value effect.
2. The hedge fund has always paid in full and on time, so no delay or path-dependent shortfall has arisen.
3. The USD 0.6 million increase is below the USD 1.0 million minimum transfer amount, so no valid call can be made.
4. The uncollateralized USD 0.6 million is therefore a granularity effect: key parameters prevent asking for all collateral actually required.

Why C is correct: The increase in exposure is smaller than the minimum transfer amount, so the bank cannot call for it; this is the granularity effect.

Why the others are wrong

- **A:** Cash does not change in value, so this reason applies only to non-cash collateral. Error: treating all collateral as subject to value variation.
- **B:** Path dependency arises when the amount called depends on collateral collected in the past; the hedge fund has always met calls on time, so there is no such history. Error: inventing a history of late delivery.
- **D:** The margin period of risk concerns delay between a call and receipt; here no call has been made at all because of the MTA. Error: confusing a non-call with a delayed call.

LO 37.d

Identify typical credit exposure profiles for various derivative contracts and combination profiles.

Q937

Mock 3 Q35

REBUILT — Original distractor values (9.45, 19.63) were inconsistent with the option's value; replaced with Black-Scholes value, discounted-strike bound and intrinsic value.

A derivative trading firm sells (writes) a European-style call option on stock JKJ with 9 months to expiration, a

strike of EUR 45, an underlying price of EUR 67 and implied annual volatility of 27%. The annual risk-free rate is 2.5%. The premium has been received in full at inception. What is the firm's counterparty credit exposure from this transaction?

A. EUR 0.
B. EUR 23.02.
C. EUR 22.84.
D. EUR 22.00.

Answer: A

Step by step

1. The firm wrote the option and has already received the premium, so the buyer owes it nothing further.
2. Its MtM is $V = -c \leq 0$ at all times, so credit exposure is $\max(V, 0) = 0$.
3. The Black-Scholes value is $c \approx 23.02$ (with $S = 67, K = 45, T = 0.75, r = 2.5\%, \sigma = 27\%$), but it is a liability of the firm, not an asset.
4. Only the option buyer faces counterparty exposure (long option profile).

Why A is correct: A written option with an upfront premium is always a liability to the writer, so the writer has no credit exposure; the buyer bears all of it.

Why the others are wrong

- **B:** EUR 23.02 is the Black-Scholes value of the call, i.e. the buyer's exposure; the writer's MtM is the negative of this.
- **C:** EUR 22.84 is S minus the PV of the strike, a lower bound on the call value; it again measures the buyer's exposure.
- **D:** EUR 22.00 is the intrinsic value S - K, treating the writer as if it held the option.

Q938

AnalystPrep Q.1943

Two banks, X and Y, enter into an interest rate swap in which X pays floating and Y pays fixed on a notional of $50 million. The swap has a term of 5 years and is struck at the market swap rate. Which of the following is most likely true at inception?

A. The market value would be $50 million to both X and Y.
B. The market value would be zero to both X and Y.
C. X would have a market value of zero while Y would have a market value of $50 million.
D. Y would have a market value of zero while X would have a market value of $50 million.

Answer: B

Step by step

1. A par swap is set so that the PV of the fixed payments equals the PV of the expected floating payments.
2. Thus $V_X = V_Y = 0$ at inception and the exposure profile starts at zero.
3. Notional is never exchanged in an interest rate swap, so $50 million is not a value of the contract.

Why B is correct: At the market swap rate the fixed and floating legs have equal PV, so the swap starts with zero MtM for both parties and exposure builds from zero.

Why the others are wrong

- **A:** Confuses the notional with the market value; notional is not exchanged and the parties' MtMs must sum to zero.
- **C:** Treats the notional as the value for one party; the values to the two parties are equal and opposite, and here both are zero.
- **D:** Same notional/value confusion with the parties reversed.

Q939

AnalystPrep Q.2901

REBUILT — Original asked which rate/vol change raises PFE, with questionable reasoning; rebuilt to test the FX forward exposure profile shape and its driver.

Wiki Bank has entered into a 2-year FX forward with a counterparty (GSO), agreeing to sell GBP against JPY at the market forward rate. Which of the following best describes the bank's potential future exposure (PFE) profile to GSO over the life of the contract?

A. Bell-shaped, peaking at about one year and falling toward zero at maturity because payments roll off.
B. Monotonically increasing and largest near maturity, driven mainly by FX volatility and the single final exchange of notional.
C. Flat at roughly 100% of notional throughout, as for a loan or repo.

D. Highest at inception and declining thereafter as the forward's time value decays.

Answer: B

Step by step

1. An FX forward has one payment, at maturity, and no intermediate cash flows that roll off.
2. FX volatility is high, so the dispersion of the exchange rate widens roughly with $\sqrt{t}$.
3. The contract is struck at market, so MtM is zero at inception and PFE grows continuously, reaching its maximum at or just before maturity.
4. This is the monotonically increasing profile typical of FX products.

Why B is correct: With a single final notional exchange and highly volatile FX rates, exposure uncertainty keeps growing, so the PFE rises monotonically to maturity.

Why the others are wrong

- **A:** Describes an interest rate swap, where roll-off of payments offsets growing uncertainty; the forward has no periodic payments.
- **C:** Describes bonds, loans and repos, whose PFE is about the notional; an FX forward starts with zero exposure.
- **D:** A market-struck forward has zero MtM at inception; its exposure starts at zero, not at its maximum.

Q940 AnalystPrep Q.5496

A risk analyst wants to identify which instrument is most likely to exhibit a bell-shaped (peaked) potential future exposure profile over time. Which instrument best fits?

A. Equity-linked note
B. Interest rate cap
C. Plain vanilla interest rate swap
D. Credit default swap on a corporate bond

Answer: C

Step by step

1. Swap PFE balances two forces: growing uncertainty about rates as time passes, and shrinking remaining cash flows (roll-off) as payments are made.
2. Early on, uncertainty dominates and PFE rises; later, roll-off dominates and PFE falls to zero at maturity.
3. The result is a peaked profile. A cap is a long option (increasing), a note is about its notional, and a CDS shows a jump at default.

Why C is correct: Growing rate uncertainty offset by amortizing remaining payments produces the peak-shaped exposure of a vanilla swap.

Why the others are wrong

- **A:** An equity-linked note is a bond-like claim with exposure about its notional, not peaked.
- **B:** A cap is a long option with an upfront premium; its PFE increases with time rather than falling back.
- **D:** A CDS profile is driven by spread widening and a jump to (1 - recovery) at default; roll-off does not shape it.

Q941 BT answers P2.T6.327.1 · BT notes P2.T6.327.1

REBUILT — The PFE chart exhibit was missing; replaced it with a textual description of the profile.

The 99% PFE profile of a single-name credit default swap is described as follows: it rises gradually over the first years and then jumps abruptly to 60% of notional once the reference entity's cumulative default probability exceeds 1%. (The 95% PFE shows only the gradual rise over the same horizon.) Which position is most likely represented?

A. Long CDS (buy protection) with recovery = 40%
B. Long CDS (buy protection) with recovery = 60%
C. Short CDS (sell protection) with recovery = 40%
D. Short CDS (sell protection) with recovery = 60%

Answer: A

Step by step

1. A jump in PFE on default of the reference entity means the counterparty's default-contingent payoff is owed to us, which is the protection buyer's position.
2. The exposure after a credit event is the payout $1 - R$ of notional.
3. $1 - R = 60\% \Rightarrow R = 40\%$.

4. The jump appears at 99% when default probability exceeds $1 - 99\% = 1\%$; the early gradual rise is spread widening.

Why A is correct: Only the protection buyer receives a large contingent payoff on default, equal to $1 - R$ of notional, so a 60% jump implies a long CDS with 40% recovery.

Why the others are wrong

- **B:** With R = 60% the post-default exposure would be 40% of notional, not 60%.
- **C:** A protection seller's exposure is limited to the PV of premiums owed to it and has no jump to (1 - R); the seller pays out on default.
- **D:** Wrong side (seller) and wrong recovery (the jump would be 40% if R were 60%).

Q942

BT answers P2.T6.413.3 · BT notes P2.T6.413.3

REBUILT — The CDS exposure chart was missing; replaced it with a textual description and restated the options so exactly one is false.

A bank buys protection on a single-name CDS (assumed recovery 30%). The 95% PFE rises gradually over time, whereas the 98% PFE rises gradually and then jumps abruptly to a much higher level. Each of the following is true (or at least plausible) about these profiles EXCEPT:

A. The maximum exposure of the position is 70% of notional, reached on default of the reference entity.

B. For a given confidence level, PFE is less than or equal to expected shortfall at that level, since ES is the average exposure beyond the PFE quantile.

C. The gradual early increase in exposure corresponds to scenarios in which the CDS spread widens.

D. The abrupt jump in the 98% PFE is caused by the wrong-way risk correlation parameter; setting it to zero would remove the jump.

Answer: D

Step by step

1. A: payout on default is $1 - R = 70\%$ of notional, the maximum exposure. True.
2. B: ES is the expected exposure conditional on exceeding the PFE quantile, so ES $\geq$ PFE. True.
3. C: spread widening raises the replacement value of protection bought earlier. True.
4. D: the jump arises because default is a discrete event; once the reference entity's default probability exceeds $1 - 98\% = 2\%$, the quantile falls in the default scenarios and PFE jumps to $1 - R$. Zero correlation would not remove it. False.

Why D is correct: The jump comes from the discrete default payoff when default probability exceeds one minus the confidence level, not from a correlation parameter, so D is the false statement.

Why the others are wrong

- **A:** A true statement (1 - 0.30 = 70%), so not the exception.
- **B:** A true statement; ES is by construction at least as large as the PFE quantile at the same confidence.
- **C:** A true statement; this is the spread-widening diffusion before default.

Q943

BT answers P2.T6.407.1

REBUILT — Clarified the stem (underlying price unbounded above) and corrected the source explanation, which referenced a bull spread not among the options.

A bank trades with a counterparty on an underlying whose price can rise without limit. Which of the following positions offers the greatest potential future credit exposure to the counterparty?

A. Long call

B. Long put

C. Short forward

D. Short credit default swap (selling protection)

Answer: A

Step by step

1. Credit exposure arises only when the position has positive MtM.
2. Long call: value rises with the underlying and has no upper bound, so potential exposure is unbounded.
3. Long put: maximum value is the strike (underlying falls to zero). Short forward: maximum gain is the forward price (underlying falls to zero).
4. Short CDS (protection seller): exposure is only the PV of remaining premiums if spreads tighten, a small amount.

Why A is correct: Only the long call has an unbounded upside, so only it has unbounded potential exposure.

Why the others are wrong

- **B:** A put's gain is capped at the strike because the price cannot fall below zero.
- **C:** A short forward gains only if the underlying falls, capped at the forward price.
- **D:** The protection seller's asset is the premium stream, small relative to notional; it mostly owes on default.

Q944

BT answers P2.T6.910.1 · BT notes P2.T6.910.1

REBUILT — The original relied on an unreadable percent-of-notional figure (3.0%) and a garbled PFE bonus; rewritten as a self-contained qualitative question on exposure just before maturity.

Consider the credit exposure profiles of six positions, each with five years to maturity:
I. Fixed-rate payer in a vanilla interest rate swap (IRS)
II. Cross-currency swap (CCS) with a final exchange of notional
III. Long call option (premium paid up front)
IV. Short (written) call option (premium received up front)
V. Long CDS (protection buyer)
VI. Short CDS (protection seller)
Just before the five-year maturity, for which position(s) is the expected exposure (EE) likely to remain clearly above zero?

A. II (CCS) and III (long call) only
B. IV (short call) and VI (short CDS) only
C. I (fixed-rate payer IRS), II (CCS), IV (short call) and V (long CDS)
D. All six positions

Answer: A

Step by step

1. Swap-like profiles (IRS, and the CDS, whose EE has a typical swap-like shape) roll off as payments are made, so EE converges to about zero at maturity (I, V, VI).
2. A CCS has a final exchange of notional driven by high FX volatility, so its exposure rises monotonically and is still large at maturity (II).
3. A long option with an up-front premium has exposure that grows over time because it can end deep in the money; EE at expiry equals expected intrinsic value, which is positive (III).
4. A written option has no exposure to the buyer's default risk, since the premium was received up front (IV).
5. Positions with clearly positive EE at maturity: II and III only.

Why A is correct: Exposure stays high at maturity only when there is a single large uncertain payoff at the end (final FX notional exchange, option intrinsic value); periodic-payment swaps roll off and the option writer has nothing owed to it.

Why the others are wrong

- **B:** Reverses the logic: the written call owes, rather than is owed, and the protection seller's swap-like exposure amortizes to zero.
- **C:** Treats the IRS and long CDS as having terminal exposure and the written call as exposed; it misses the long call and mixes up which side of the option has exposure.
- **D:** Ignores roll-off of swap-like profiles and the zero exposure of the option writer after receiving the premium.

Q945

BT answers P2.T6.910.2 · BT notes P2.T6.910.2

REBUILT — Option D (higher correlation) can also reduce exposure per GARP (low correlation increases CCS exposure), giving two defensible answers; replaced with a clearly exposure-increasing option.

For a cross-currency swap, which of the following tends to DECREASE both the expected exposure (EE) and the potential future exposure (PFE)?

A. A longer maturity
B. Higher foreign exchange (FX) volatility
C. Paying in the currency with the lower interest rate (receiving the higher-rate currency)
D. Paying in the currency with the higher interest rate (receiving the lower-rate currency)

Answer: C

Step by step

1. A CCS combines an IRS and an FX forward; exposure is driven mainly by FX volatility and the final notional exchange.

2. The payer of the high-rate currency pays large coupons expected to be offset by the expected gain on the final notional exchange, producing a large positive drift and exposure (D).
3. The payer of the low-rate currency is increasingly likely to have a negative future value, a negative drift, so exposure is much lower (C).
4. Longer maturity and higher FX volatility add exposure (A, B).

Why C is correct: Paying the lower-interest-rate currency creates a negative drift in the swap's value, which pushes the distribution of future value down and reduces EE and PFE.

Why the others are wrong

- **A:** Longer maturity adds more payments and time for FX to move, so exposure rises, not falls.
- **B:** Higher FX volatility widens the distribution of the final notional exchange and raises exposure.
- **D:** This is the opposite side: paying the high-rate currency gives a positive drift and significant exposure.

Q946

BT answers P2.T6.911.1 · BT notes P2.T6.911.1

REBUILT — The exhibit was missing; the profile is now described in words (99% PFE jumps to 60% from year 3 with 40% recovery).

A long-protection single-name CDS has a five-year life. Its EE profile looks like that of an ordinary swap (rising early as the spread widens, then amortizing). Its 99% PFE profile, however, shows an abrupt jump to 60% of notional beginning around year 3, assuming a 40% recovery rate. Which statement is TRUE?

A. The key parameter that determines the shape of the EE profile is the recovery rate

B. The EE profile is incorrectly drawn; it should also show an abrupt jump like the PFE

C. From year 3 on, the default probability of the reference entity exceeds 1 minus the PFE confidence level (PD > 1%), so the 99th percentile scenario includes default

D. To force the PFE curve to overlap the EE curve, keep lowering the interest rate parameter until PFE falls to EE

Answer: C

Step by step

1. PFE at 99% is the 99th percentile of exposure; the jump to 60% = 1 − recovery $= 1 - 40\% = 60\%$ is the payoff on a credit event.
2. The jump appears once the probability of default exceeds 1 - 99% = 1%, so the default scenario falls inside the tail (C).
3. EE is an average over scenarios, so the rare default jump is diluted and EE stays swap-like; it is driven by spread movements, not recovery (A, B).
4. To bring PFE down toward EE you would lower the confidence level, not change interest rates (D).

Why C is correct: PFE is a percentile, so it jumps to the default payoff $(1 - R)$ once default probability exceeds 1 - C; EE is a mean and does not show the discrete jump.

Why the others are wrong

- **A:** Recovery sets the height of the PFE jump $(1 - R)$, not the shape of EE, which follows spread movements.
- **B:** EE is an expectation, so the low-probability default is averaged away and EE stays swap-like; it should not jump.
- **D:** Interest rates do not drive the jump; only lowering the confidence level (or using a measure like expected shortfall) would reduce the gap.

Q947

BT notes P2.T6.326.1 · BT answers P2.T6.326.1

REBUILT — The colour-coded chart was missing; replaced by a description of the profile shape so the instrument can be identified without the exhibit.

A risk analyst reviews four PFE-type profiles for 10-year trades. One profile rises monotonically to its maximum at final maturity. Most of its exposure comes from uncertainty about a large final notional exchange driven by FX volatility, with a smaller added contribution from interest-rate payments. Which instrument is it?

A. PFE of an interest rate swap

B. PFE of an FX forward contract

C. PFE of a cross-currency swap

D. Average PFE of an interest rate swap

Answer: C

Step by step

1. An IRS PFE is peaked: it rises with diffusion $(\sqrt{t})$ and falls as payments roll off $(T - t)$, ending at zero (A).
2. An FX forward has a single final payment, so exposure grows with $\sqrt{t}$, but it has no interest-rate component (B).

3. A CCS is an IRS plus an FX forward: the FX forward dominates, with a smaller IRS contribution on top, so the profile is monotonically increasing and above that of a plain FX forward (C).
4. The average PFE of an IRS is a single flat number across the life (D).

Why C is correct: A cross-currency swap is a combination profile: the monotonic FX-forward component plus a smaller IRS component, giving a rising profile above that of a plain FX forward.

Why the others are wrong

- **A:** An IRS profile peaks at about one-third of maturity and falls to zero; it is not monotonic.
- **B:** An FX forward is monotonic but has no interest-rate payments, so it lacks the extra contribution described.
- **D:** The average PFE is a flat constant, not a rising profile.

Q948

BT notes P2.T6.910.3

Credit exposure profiles vary by instrument and can be substantially altered by the specific nature of the cash flows. Which of the following statements is TRUE?

A. As with market VaR, in most uses of counterparty exposure quantification (xVA) the drift of the risk factor can safely be assumed to be zero

B. For a single-name CDS, expected exposure generally decreases as spread volatility rises

C. For an interest rate swap with an upward-sloping yield curve, the floating-rate payer has a positive expected future value and the greater credit exposure

D. For a simple forward contract with i.i.d. returns on the underlying market variable, the exposure profile is approximately proportional to the square root of time

Answer: D

Step by step

1. A single-cash-flow forward with i.i.d. returns has exposure proportional to $\sqrt{t}$, for $t \leq T$ (D).
2. VaR can ignore drift because its horizon is short; for long exposure horizons drift is relevant, so A is false.
3. Higher spread volatility widens the distribution of CDS value and raises EE, so B is false.
4. With an upward-sloping curve the fixed-rate payer (floating receiver) has the higher EPE; the receiver of fixed has the lower EPE, so C is false.

Why D is correct: With i.i.d. returns the standard deviation of a forward's value grows with $\sqrt{t}$, so exposure follows a square-root-of-time rule.

Why the others are wrong

- **A:** Drift can be ignored in short-horizon VaR but matters over long exposure horizons.
- **B:** Higher volatility increases, not decreases, EE.
- **C:** Reverses the sides: the fixed-rate payer, not the floating payer, has the greater exposure.

Q949

BT answers P2.T6.413.1

REBUILT — Removed the reference to an unavailable chart; the stem is now self-contained as a description of a 10-year IRS exposure profile.

The expected exposure profile of a 10-year interest rate swap is hump-shaped. Let T be the 10-year maturity and t the horizon between today and maturity. Each of the following is true EXCEPT:

A. Because the swap's maturity is ten years, the exposure peaks at about $\frac{1}{3} \times 10 = 3.33$ years

B. With a typical upward-sloping yield curve, the receiver swap (counterparty who pays floating) has greater exposure than the payer swap (counterparty who pays fixed)

C. The profile initially increases because of the $\sqrt{t}$ term, called the diffusion effect

D. The profile later decreases because of the $(T - t)$ term, called the amortization effect

Answer: B

Step by step

1. IRS exposure is approximately proportional to $(T-t)\sqrt{t}$; the maximum occurs at $t = T/3 = 10/3 = 3.33$ years (A true).
2. Early on the $\sqrt{t}$ term dominates (diffusion), so exposure rises (C true).
3. Later the shrinking remaining maturity $(T - t)$ dominates (amortization), so exposure falls (D true).
4. With an upward-sloping curve the receiver of fixed (pays floating) expects net cash inflows early and outflows later, so its EPE is lower; the payer swap has the greater exposure. B is therefore false.

Why B is correct: The receiver of fixed is expected to receive positive net cash flows early, giving it a negative expected future value and lower EPE than the payer, so B inverts the relationship.

Why the others are wrong

- **A:** True statement (maximum at one-third of maturity), so not the exception.
- **C:** True statement (diffusion effect), so not the exception.
- **D:** True statement (amortization effect), so not the exception.

Q950 BT answers P2.T6.413.2 · BT notes P2.T6.413.2

REBUILT — The PFE exhibit was missing from both sources; the stem now describes the profile in words (rising, largest near the 10-year maturity, never returning toward zero) and the duplicate records were merged.

A bank reviews the 99% potential future exposure (PFE) profile of a 10-year cross-currency swap with final exchange of notional. The PFE rises steadily through time, is largest close to the 10-year maturity, and does not decline toward zero as maturity approaches. The risk manager decomposes the profile into an interest rate swap component and an FX forward component. Which of the following statements about this profile is LEAST plausible?

A. The profile can be represented as a combination of an interest rate swap profile, an FX forward profile, and an assumption about the correlation between the interest rates and the FX rate.

B. The FX forward component contributes more to the total exposure than the interest rate swap component.

C. Because the PFE does not fall back toward zero at maturity, the swap must have had a significantly positive value at inception.

D. An increase in the assumed FX volatility would increase the exposure.

Answer: C

Step by step

1. A cross-currency swap is essentially an interest rate swap combined with an FX forward (final exchange of notionals), with the correlation between interest rates and the FX rate also affecting the combined exposure, so A is plausible.
2. The interest rate swap component has the usual peak-and-decay shape (roll-off of payments), while the FX forward component rises monotonically because FX volatility is high and the large notional is exchanged at maturity; the FX component therefore dominates, so B is plausible.
3. The exposure stays high at the end because the final notional exchange is still outstanding and FX uncertainty keeps growing with time. This arises even for a par swap with zero initial value, so the shape does not imply a positive initial value; C is the implausible statement.
4. Higher FX volatility widens the distribution of the FX rate and so raises the 99% PFE, so D is plausible.

Why C is correct: The non-vanishing exposure at maturity is caused by the final exchange of notional and growing FX uncertainty, not by the swap starting in the money; a par cross-currency swap has this same shape.

Why the others are wrong

- **A:** This is true: a cross-currency swap combines an interest rate swap and an FX forward, with the correlation as a driver. Choosing it means rejecting a standard decomposition.
- **B:** This is true: the FX component dominates because of high FX volatility and the final notional exchange. Choosing it confuses the cross-currency swap with an ordinary interest rate swap.
- **D:** This is true: greater FX volatility widens the exposure distribution and raises PFE. Choosing it misreads the sign of the volatility effect.

Q951 Schweser Module Quiz 37.2 Q2

Which of the following security types will most likely result in a peaked shape for the exposure profile represented by potential future exposure (PFE)?

A. Long option position.

B. Foreign exchange product.

C. 10-year loan with a floating rate payment.

D. Swap.

Answer: D

Step by step

1. A swap involves periodic exchange of payments, so the shape depends on two opposing forces.
2. Growing uncertainty: the longer the horizon, the more the swap's value can diverge, which raises exposure (roughly proportional to the square root of time).
3. Roll-off: as payments are exchanged, the remaining life and number of remaining payments fall, which shrinks

exposure (roughly proportional to remaining maturity).
4. The product of the rising and falling effects gives exposure that rises, peaks part-way through the life, and then declines to zero at maturity: a peaked profile.

Why D is correct: Periodic payment exchange makes the swap's remaining exposure roll off over time while uncertainty keeps growing, so the profile rises and then falls.

Why the others are wrong

- **A:** A long option's potential exposure keeps increasing with time (no periodic cash flows to reverse it); the error is treating the option like a swap.
- **B:** An FX product (e.g. forward) has a single payment at maturity, so uncertainty grows monotonically with no roll-off before maturity; it is not peaked.
- **C:** A loan's exposure is approximately its notional (about 100%) throughout, since the lender is owed principal; floating payments do not create a peak. Confusing it with a swap overlooks that principal is exchanged only at the end.

LO 37.e

Explain how payment frequencies and exercise dates affect the exposure profile of various securities.

Q952

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A risk analyst compares two 5-year interest rate swaps with the same notional, both with the same fixed rate and term. In Swap 1, fixed and floating payments are both exchanged semiannually. In Swap 2, the bank pays fixed semiannually but receives floating quarterly. The analyst then considers Swap 3, in which the bank pays floating quarterly but receives fixed only semiannually. Relative to the equal-frequency profile of Swap 1, which of the following is correct about the bank's potential future exposure (PFE)?

A. Swap 2 shows higher exposure and Swap 3 shows lower exposure.

B. Swap 2 and Swap 3 both show the same exposure as Swap 1, because payment frequency affects only cash flow timing and not exposure.

C. Swap 2 and Swap 3 both show lower exposure, because more frequent payments in either direction reduce uncertainty.

D. Swap 2 shows lower exposure and Swap 3 shows higher exposure.

Answer: D

Step by step

1. When payments are received more frequently than they are made (Swap 2), the amounts owed to the bank are settled sooner, so less value accumulates unpaid and exposure falls below the equal-payment case.
2. When payments are made more frequently than they are received (Swap 3), the bank pays out sooner while its receipts lag, so exposure is greater than in the equal-payment case.
3. The effect therefore reverses with the direction of the frequency mismatch.

Why D is correct: Receiving more frequently than paying reduces exposure (Swap 2); paying more frequently than receiving increases it (Swap 3).

Why the others are wrong

- **A:** This reverses the direction of the effect. Error: assuming that frequent receipts increase exposure.
- **B:** The reading shows unequal payment frequencies change the exposure profile. Error: ignoring settlement timing as a driver of exposure.
- **C:** Frequency mismatch is directional; paying more often than receiving raises exposure. Error: treating frequency as always beneficial.

Q953

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A bank buys a European swaption from a corporate client that is swap-settled at an exercise date one year from now, and also enters into a forward swap with an identical underlying swap with another client. Which of the following correctly compares the potential future exposure of the swaption with that of the forward swap?

A. Before the exercise date the swaption has greater exposure, and after the exercise date the forward swap has greater exposure.

B. The swaption has greater exposure both before and after the exercise date, because the bank has paid a premium.
C. The two have identical exposure at all times, because the swaption is exercised into the same underlying swap.
D. Before the exercise date the forward swap has greater exposure, and after the exercise date the swaption has greater exposure.

Answer: A
Step by step
1. Before exercise, the swaption holder can only gain: the option value is never negative, so the swaption exposure is at least that of the forward swap, which can be negative in some scenarios.
2. After exercise, the forward swap still exists in all scenarios, including those where its value later turns positive but the swaption was not exercised.
3. Hence after the exercise date the forward swap has the greater exposure; the relationship reverses.

Why A is correct: The swaption dominates before exercise; the forward swap dominates afterwards, as it is live even in scenarios in which the swaption was not exercised.
Why the others are wrong
- **B:** The relationship reverses after exercise; the forward swap then has the greater exposure because it remains live in scenarios where the swaption is not exercised. Error: ignoring the exercise decision.
- **C:** The swaps are identical only in scenarios where the option is exercised; the swaption is not exercised in others. Error: ignoring the exercise decision.
- **D:** This reverses both periods. Error: ignoring that the option cannot be worth less than zero before exercise.

Q954 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.
A credit analyst reviews the exposure of two European swaptions with a one-year exercise date on the same underlying interest rate swap. Swaption X is cash-settled and Swaption Y is physically settled (swap-settled). Which of the following statements about their exposure profiles is correct?
A. Swaption X has greater exposure than Swaption Y before the exercise date, because the cash settlement amount is certain.
B. The two have identical exposure until the exercise date; afterwards Swaption X has zero exposure while Swaption Y retains exposure on the underlying swap.
C. After the exercise date both swaptions have zero exposure, because the exercise decision has been made.
D. Swaption Y has zero exposure after the exercise date, while Swaption X retains exposure until the underlying swap matures.

Answer: B
Step by step
1. Before the exercise date both swaptions are options on the same underlying swap, so their exposure profiles are identical.
2. Cash settlement ends the contract at exercise with a single payment, so exposure drops to zero afterwards.
3. Physical settlement leaves the underlying swap in place when exercised, so exposure continues, with the usual swap profile, until its maturity.

Why B is correct: Cash settlement terminates exposure at the exercise date; physical settlement converts the swaption into a live swap with continuing exposure.
Why the others are wrong
- **A:** Until exercise the two profiles are identical. Error: believing settlement type changes pre-exercise exposure.
- **C:** Physical settlement leaves a live swap, so exposure persists. Error: assuming exercise ends exposure in both cases.
- **D:** This reverses the two settlement types. Error: confusing which one leaves a swap outstanding.

LO 37.f

Explain the general impact of aggregation on exposure, and the impact of aggregation on exposure when there is correlation between transaction values.

Q955

AP Mock 3 Q67

REBUILT — Original relied on qualitative dominance without stating the model; rebuilt to apply the netting factor formula explicitly and compare quantitatively (retagged to 37.f, which covers netting and correlation).

A trading desk holds all of its equity positions under one netting agreement. Assume exposures are jointly normal with identical volatilities and a common pairwise correlation, so the netting factor is $\sqrt{n + n(n-1)\rho}\,/\,n$. The desk currently has 9 positions with average correlation 0.35. The chief trader proposes four alternative combinations:

Combination	Number of positions	Average correlation
K	4	0.25
Y	7	-0.08
E	10	-0.11
W	5	0.55

Which combination would increase the desk's expected netting benefit the most relative to the current position?

A. Combination K.
B. Combination Y.
C. Combination E.
D. Combination W.

Answer: C

Step by step

1. Current: $\sqrt{9 + 9 \cdot 8 \cdot 0.35}/9 = \sqrt{34.2}/9 = 65.0\%$.
2. K: $\sqrt{4 + 12 \cdot 0.25}/4 = 66.1\%$ (worse than current).
3. Y: $\sqrt{7 + 42 \cdot (-0.08)}/7 = 27.3\%$.
4. E: $\sqrt{10 + 90 \cdot (-0.11)}/10 = \sqrt{0.1}/10 = 3.2\%$; W: $\sqrt{5 + 20 \cdot 0.55}/5 = 80.0\%$ (worse).
5. The lowest netting factor, E at 3.2%, gives the largest benefit; all correlations satisfy $\rho \geq -1/(n-1)$.

Why C is correct: A lower netting factor means greater netting benefit, achieved with more positions and lower (negative) correlation; E has the lowest factor.

Why the others are wrong

- **A:** K has fewer positions and still positive correlation; its factor (66.1%) is actually slightly worse than today's 65.0%.
- **B:** Y improves the factor to 27.3%, but with fewer positions and weaker negative correlation than E it is less effective.
- **D:** W has fewer positions and a much higher correlation (0.55); its factor of 80.0% is worse than the current position.

Q956

AnalystPrep Q.2899

REBUILT — Original distractors were arbitrary numbers; replaced them with results of realistic formula errors, answer unchanged.

Assume future values follow a multivariate normal distribution with 21 exposures in a netting set and an average pairwise correlation of 0.3. What is the netting factor?

A. 0.33
B. 0.44
C. 0.58
D. 0.70

Answer: C

Step by step

1. Netting factor $= \sqrt{n + n(n-1)\bar{\rho}}\,/\,n$ with $n = 21,\ \bar{\rho} = 0.3$.
2. $n(n-1)\bar{\rho} = 21 \times 20 \times 0.3 = 126.0$
3. $n + n(n-1)\bar{\rho} = 147.0$, so the square root is 12.1244.
4. Netting factor $= 12.1244/21 = 0.5774 \approx 0.58$.

Why C is correct: Under multivariate normality, the aggregate exposure scales with the portfolio standard deviation, so the netting factor is $\sqrt{n + n(n-1)\bar{\rho}}\,/\,n$; it equals 100% only when correlation is 1 and falls with more exposures and lower correlation.

Why the others are wrong

- **A:** Linear approximation $(1 + (n-1)\bar{\rho})/n = 0.33$: forgot the square root in the portfolio volatility.
- **B:** Counted distinct pairs $n(n-1)/2$ instead of ordered pairs $n(n-1)$, giving $\sqrt{21 + 210 \times 0.3}/21 = 0.44$.
- **D:** Used $1 - \bar{\rho} = 0.70$, ignoring the number of exposures.

Q957

AnalystPrep Q.2905

REBUILT — Original distractors were arbitrary numbers; replaced them with results of realistic formula errors, answer unchanged.

Assume future values follow a multivariate normal distribution with 17 exposures in a netting set and an average pairwise correlation of 0.7. What is the netting factor?

A. 0.62
B. 0.85
C. 0.72
D. 0.30

Answer: B

Step by step

1. Netting factor $= \sqrt{n + n(n-1)\bar{\rho}}\,/\,n$ with $n = 17,\ \bar{\rho} = 0.7$.
2. $n(n-1)\bar{\rho} = 17 \times 16 \times 0.7 = 190.4$
3. $n + n(n-1)\bar{\rho} = 207.4$, so the square root is 14.4014.
4. Netting factor $= 14.4014/17 = 0.8471 \approx 0.85$.

Why B is correct: Under multivariate normality, the aggregate exposure scales with the portfolio standard deviation, so the netting factor is $\sqrt{n + n(n-1)\bar{\rho}}\,/\,n$; it equals 100% only when correlation is 1 and falls with more exposures and lower correlation.

Why the others are wrong

- **A:** Counted distinct pairs $n(n-1)/2$ instead of ordered pairs, giving $\sqrt{17 + 136 \times 0.7}/17 = 0.62$.
- **C:** Linear approximation $(1 + (n-1)\bar{\rho})/n = 0.72$: forgot the square root.
- **D:** Used $1 - \bar{\rho} = 0.30$, ignoring the number of exposures.

Q958

BT answers P2.T6.911.2 · BT notes P2.T6.911.2

Gregory defines the netting factor as the ratio of net to gross exposure, so a netting factor of 100% means no netting benefit and 0% means maximum netting benefit. Assume eight independent transactions with equal volatility, i.e., $n = 8$, average correlation $\rho = 0$, and the same σ_i for each. What is the implied netting factor?

A. 12.5%
B. 35.4%
C. 75.0%
D. 87.5%

Answer: B

Step by step

1. $\text{NF} = \dfrac{\sqrt{n + n(n-1)\rho}}{n}$
2. With $\rho = 0$: $\text{NF} = \sqrt{n}/n = 1/\sqrt{n}$.
3. $1/\sqrt{8} = 1/2.8284 = 0.3536$, i.e., 35.4%.

Why B is correct: With zero correlation the exposures partly diversify: net exposure grows with the square root of n while gross exposure grows with n, so the factor is 1/sqrt(n).

Why the others are wrong

- **A:** 12.5% is 1/n = 1/8, which omits the square root (it would imply perfectly offsetting exposures).
- **C:** 75.0% has no basis in the formula; it is an arbitrary round number (e.g., 6/8).
- **D:** 87.5% is 1 - 1/n = 1 - 1/8, mixing up the netting factor with a benefit-style 1/n adjustment.

Q959

BT answers P2.T6.911.3 · BT notes P2.T6.911.3

REBUILT — Source tables had conflicting OCR values and blank netting columns; used the consistent data set and made the question self-contained with distractors from realistic errors.

Consider two trades under five equally likely scenarios at a single future date:

Scenario	Trade 1 FV	Trade 2 FV
Scenario 1	20.0	-15.0
Scenario 2	14.0	-5.0
Scenario 3	3.0	3.0
Scenario 4	-5.0	-10.0
Scenario 5	-15.0	-30.0

Exposure is the positive part of value, and expected exposure (EE) is the average across scenarios. What is the netting factor (EE with netting divided by EE without netting)?

A. 25.0%
B. 50.0%
C. 75.0%
D. 200.0%

Answer: B

Step by step

1. Without netting, exposure = max(T1,0) + max(T2,0) = 20, 14, 6, 0, 0; EE = 40/5 = 8.0.
2. With netting, exposure = max(T1+T2,0) = 5, 9, 6, 0, 0; EE = 20/5 = 4.0.
3. Netting factor = 4.0 / 8.0 = 50.0%.

Why B is correct: Netting helps only in scenarios where the trades have opposite signs (Scenarios 1 and 2), cutting total exposure there from 34 to 14 and halving EE overall.

Why the others are wrong

- **A:** 25.0% comes from dividing the net EE by an inflated denominator (e.g., 4.0/16.0); the correct ratio is 4.0/8.0.
- **C:** 75.0% is 1 minus 25%, or the result of crediting netting in only one of the two opposite-sign scenarios; both Scenarios 1 and 2 benefit.
- **D:** 200.0% inverts the ratio (8.0/4.0); netting cannot raise exposure, so the factor cannot exceed 100%.

Q960

BT notes P2.T6.328.1

REBUILT — Original options included arbitrary non-diagnostic values (9.92%, 37.50%, 60.88%) and the exposure columns were blank; kept the data and rebuilt options from realistic calculation errors.

Two trades are evaluated under five equally likely scenarios (a variation on Gregory's aggregation example). Exposure is the positive part of value, and the netting factor is EE with netting divided by EE without netting.

Scenario	Trade 1 MtM	Trade 2 MtM
Scenario 1	25	15
Scenario 2	15	-10
Scenario 3	10	-5
Scenario 4	-5	5
Scenario 5	-15	-3

What is the netting factor?

A. 14.29%
B. 28.57%
C. 71.43%
D. 140.00%

Answer: C

Step by step

1. Without netting: max(T1,0)+max(T2,0) = 40, 15, 10, 5, 0; sum 70, EE = 14.0.
2. With netting: max(T1+T2,0) = 40, 5, 5, 0, 0 (Scenario 4 nets to 0 and Scenario 5 to -18, both floored at 0); sum 50, EE = 10.0.
3. Netting factor = 10.0 / 14.0 = 71.43%.

Why C is correct: Only Scenarios 2, 3 and 4 have opposite-signed trades, so netting removes just 20 of 70 units of gross exposure, leaving a high factor.

Why the others are wrong

- **A:** 14.29% = 10/70: net EE (already divided by 5) compared with the gross sum that was not divided by 5.
- **B:** 28.57% is the netting benefit (1 - 71.43%), not the netting factor.
- **D:** 140.00% inverts the ratio (14/10); netting cannot raise exposure so the factor cannot exceed 100%.

Q961

BT notes P2.T6.328.2

REBUILT — Original relied on a marginal-EE table outside 37.f; rebuilt around standalone EEs, netted EE and the effect of correlation using the same normal-MtM data.

Two trades have jointly normal future MtM values. Trade 1 has mean 6% and standard deviation 10%; Trade 2 has mean -10% and standard deviation 30%; the correlation is 10%. Their standalone expected exposures (EE) are

7.69% and 7.63%. Which of the following statements about the two-trade portfolio is FALSE?

A. EE without netting is about 15.31%.
B. EE with netting is about 11.09%.
C. If the correlation increases, the EE with netting increases.
D. If the correlation increases, the EE without netting increases.

Answer: D

Step by step

1. Without netting, EE is the sum of standalone EEs: 7.69% + 7.63% ≈ 15.31% (unrounded 15.314%) (A true).
2. With netting, portfolio MtM is normal with mean $6\%-10\% = -4\%$ and variance $10^2+30^2+2(0.1)(10)(30) = 1060$, so $\sigma = 32.56\%$.
3. $\text{EE} = \mu\,\Phi(\mu/\sigma) + \sigma\,\phi(\mu/\sigma) = -4(0.4511) + 32.56(0.3959) \approx 11.09\%$ (B true).
4. Higher correlation raises σ of the netted portfolio, and EE of a normal variable increases with σ, so netted EE rises (C true).
5. Without netting each trade is measured separately, so correlation has no effect on EE (D false).

Why D is correct: Netting makes exposure depend on the distribution of the summed MtM, so correlation matters; without netting exposures simply add and correlation is irrelevant.

Why the others are wrong

- **A:** True (7.69% + 7.63% is about 15.31% using unrounded values), so it is not the false statement.
- **B:** True (netted EE with mean -4% and sigma 32.56% is 11.09%), so it is not the false statement.
- **C:** True: higher correlation reduces offsetting, pushing netted EE toward the no-netting figure; picking it assumes netting benefit grows with correlation.

Q962

Mock 2 Q28

REBUILT — Cleaned the stem and replaced the arbitrary 53% distractor with 52% (the result of omitting the square root).

Counterparty A has five exposures to counterparty B covered by a single netting agreement. The exposures have equal volatility and an average correlation of 40%. What is the netting factor?

A. 28%
B. 40%
C. 52%
D. 72%

Answer: D

Step by step

1. $\text{NF} = \dfrac{\sqrt{n + n(n-1)\rho}}{n}$
2. With $n = 5$, $\rho = 0.40$: $n + n(n-1)\rho = 5 + 5(4)(0.4) = 13$.
3. $\text{NF} = \sqrt{13}/5 = 3.6056/5 = 0.7211 \approx 72\%$.

Why D is correct: The netting factor is the volatility of the netted sum relative to the gross sum; moderate positive correlation leaves most exposure un-netted.

Why the others are wrong

- **A:** 28% is the netting benefit (1 - 72%), not the factor.
- **B:** 40% simply repeats the correlation input.
- **C:** 52% = 13/25: forgot to take the square root.

Q963

Schweser Module Quiz 37.2 Q1

Miven Corp. has two trades outstanding with one of its counterparties. Which of the following scenarios would result in the greatest netting advantage for Miven?

A. The two trades have a strong positive correlation.
B. The two trades have a weak positive correlation.
C. The two trades are uncorrelated with each other.
D. The two trades have a strong negative correlation.

Answer: D

Step by step

1. Netting helps only when one trade's value is positive while the other's is negative.
2. Lower (more negative) correlation makes opposite signs more likely, so the netting factor falls.
3. Strong negative correlation therefore gives the largest offset.

Why D is correct: Strong negative correlation makes the trades' values most likely to have opposite signs, so positive exposures are offset by negative values.

Why the others are wrong

- **A:** Strong positive correlation gives same-signed values, hence the least netting benefit.
- **B:** Weak positive correlation still gives more same-sign outcomes than zero or negative correlation.
- **C:** Zero correlation gives some offset (factor of about 71% for two trades) but less than negative correlation.

Q964 Schweser Module Quiz 37.2 Q3

Which of the following statements best describes the benefit of netting risk exposures? The benefits of netting are realized when:

A. marked-to-market (MtM) values have high structural correlations for two trades.
B. marked-to-market (MtM) values have opposite signs for two trades.
C. expected exposure (EE) values are minimal.
D. expected future exposure (EFE) values have zero correlation.

Answer: B

Step by step

1. Close-out netting offsets positive and negative values within a netting set.
2. A benefit arises only in scenarios where trade MtMs have opposite signs.
3. So B is correct.

Why B is correct: Close-out netting reduces exposure only where positive and negative MtMs offset each other.

Why the others are wrong

- **A:** High positive correlation produces same-signed values and little or no netting benefit.
- **C:** Low EE is a result, not the condition that creates a netting benefit.
- **D:** Zero correlation gives only a partial benefit; the full benefit requires opposite-signed values.

LO 37.g

Describe the differences between funding exposure and credit exposure.

Q965 AnalystPrep Q.2904

REBUILT — Original did not say the initial margin is exchanged both ways and had unexplainable distractor values; clarified the stem and rebuilt options from realistic formula errors.

A portfolio has a current mark-to-market value of +15 (in the firm's favor). The firm holds 10 of variation margin from the counterparty. In addition, initial margin of 6 is exchanged bilaterally: the firm has posted 6 and received 6, and all initial margin is segregated (no reuse or rehypothecation). Ignoring close-out costs, what are the counterparty credit risk exposure and the funding exposure, respectively?

A. Counterparty exposure 0; funding exposure 11
B. Counterparty exposure 5; funding exposure 11
C. Counterparty exposure 0; funding exposure 0
D. Counterparty exposure 11; funding exposure 0

Answer: A

Step by step

1. Credit (default) exposure: max(value − margin received + non-segregated margin posted, 0). Received margin of any type counts: VM 10 and IM 6. Posted IM is segregated, so it is not added.
2. CCR exposure = $\max(15 - 10 - 6, 0) = \max(-1, 0) = 0$.
3. Funding exposure: value − reusable margin received + all margin posted. Received VM (10) is reusable; received segregated IM (6) is not; posted IM (6) counts in full.
4. Funding = $15 - 10 + 6 = 11$.

Why A is correct: Segregated initial margin received protects against default (reducing credit exposure) but cannot

be reused, and initial margin posted is a funding cost even though it is segregated; so credit exposure can be zero while funding exposure remains positive.

Why the others are wrong

- **B:** Ignores the 6 of received initial margin in the credit exposure (15 - 10 = 5), although the funding figure of 11 is right.
- **C:** Floors funding at zero by netting received IM against value (15 - 10 - 6 = -1), forgetting that segregated IM received cannot be reused and posted IM must be funded.
- **D:** Reverses the two measures: 11 is the funding figure, while segregated posted IM does not add to default exposure.

Q966

BT notes P2.T6.912.1

REBUILT — Clarified that the segregated initial margin is exchanged bilaterally (posted and received) so the credit and funding formulas are unambiguous.

The mark-to-market value of a portfolio is +$30.0 million and $27.0 million of variation margin is held. Initial margin of $8.0 million is exchanged bilaterally (posted and received) and is segregated. What are, respectively, the current counterparty credit risk (CCR) exposure and the funding exposure?

A. CCR exposure = zero; funding exposure = $11.0 million
B. CCR exposure = $3.0 million; funding exposure = zero
C. CCR exposure = $3.0 million; funding exposure = $3.0 million
D. CCR exposure = $11.0 million; funding exposure = $19.0 million

Answer: A

Step by step

1. CCR $= \max(30 - 27 - 8, 0) = \max(-5, 0) = 0$ (received VM and IM both reduce default exposure).
2. Funding $= 30 - 27 + 8 = 11$: received VM is reusable, posted IM must be funded; received segregated IM is not reusable.

Why A is correct: Margin held covers the whole MTM in default, so credit exposure is zero, but posted segregated IM is a funded asset that cannot be reused, leaving a funding exposure of 11.

Why the others are wrong

- **B:** Nets only variation margin (30 - 27 = 3) and ignores initial margin in both measures.
- **C:** Applies 30 - 27 = 3 to both measures, ignoring IM entirely.
- **D:** Adds IM to the credit exposure (30 - 27 + 8 = 11, wrong sign) and builds funding as 27 - 8 = 19, mixing up the margin terms.

Q967

BT notes P2.T6.912.2

REBUILT — Source omitted units and left the direction of margin and the initial margin exchange ambiguous; stated them explicitly.

The mark-to-market value of a portfolio is -$36.0 million (negative for the firm). The firm has posted $40.0 million of variation margin to the counterparty (not segregated). Initial margin of $3.0 million is exchanged bilaterally (posted and received) and is segregated. What are, respectively, the current counterparty credit risk (CCR) exposure and the funding exposure?

A. CCR exposure = zero; funding exposure = zero
B. CCR exposure = zero; funding exposure = $3.0 million
C. CCR exposure = $1.0 million; funding exposure = $7.0 million
D. CCR exposure = $3.0 million; funding exposure = $33.0 million

Answer: C

Step by step

1. Over-posted VM: the firm has posted 40 against a liability of 36, so 4 is owed back by the counterparty and is at risk in default.
2. CCR $= \max(-36 - 3 + 40, 0) = \max(1, 0) = 1$ (received IM of 3 offsets; posted IM is segregated, so not added).
3. Funding $= -36 - 0 + 40 + 3 = 7$: posted VM (40) and posted IM (3) are funded; received IM is not reusable.

Why C is correct: Excess posted variation margin is a credit exposure net of received IM, while funding captures all margin posted against the negative value, giving 1 and 7.

Why the others are wrong

- **A:** Floors both measures at zero because MTM is negative, ignoring the over-posted margin.
- **B:** Treats only the IM posted as the funding amount and ignores the unsecured excess VM.

- **D:** Takes CCR as the IM (3) and funding as 36 - 3 = 33, ignoring the posted VM.

Q968 BT answers P2.T6.24.32.1 · BT notes P2.T6.24.32.1

REBUILT — Original distractors were vague giveaways and the stem wordy; tightened the stem and rebuilt distractors as realistic reversals of GARP's differences.

A risk manager reviewing a book of collateralized and uncollateralized swaps must distinguish funding exposure from credit exposure. Which statement correctly describes how they differ?

A. Credit exposure arises only in default scenarios, is measured at the netting set level and depends on subjective close-out assumptions; funding exposure arises in non-default scenarios, is objective and is considered at the overall portfolio level.

B. Funding exposure depends on close-out assumptions and a default-contingent margin period of risk, whereas credit exposure is an objective value measured at the overall portfolio level.

C. Both use the same margin period of risk, so the funding value adjustment is zero whenever the credit value adjustment is zero and vice versa.

D. Wrong-way risk is a key consideration for funding exposure but not for credit exposure, while segregation affects only credit exposure.

Answer: A

Step by step

1. Credit exposure is defined in default scenarios: value rests on subjective close-out assumptions, netting applies at the netting set level, MPoR assumes counterparty default and wrong-way risk matters.
2. Funding exposure arises in non-default scenarios: value is objective, the whole portfolio may be considered (margin can be reused across counterparties), the funding delay is not default-contingent and wrong-way risk is of little importance.
3. Segregation affects both, since it prevents reuse of margin (funding) while still helping in default (credit).
4. Only A states these correctly.

Why A is correct: Funding is a going-concern measure at portfolio level with objective values, whereas credit exposure is a default-scenario, netting-set measure depending on close-out valuation.

Why the others are wrong

- **B:** Swaps the roles: close-out subjectivity and default-contingent MPoR belong to credit exposure, and netting set level applies to credit, not funding.
- **C:** Funding MPoR is not tied to default, so a collateralized derivative can have zero FVA while CVA is non-zero.
- **D:** Reverses wrong-way risk (a credit exposure concept, not key for funding) and wrongly says segregation does not affect funding; it blocks margin reuse.

LO 37.h

Explain the impact of collateralization on exposure and assess the risk associated with the remargining period, threshold, and minimum transfer amount.

Q969 AnalystPrep Q.6216

REBUILT — Original did not say which party posted collateral or what the cure period implied; rewrote so the collateral posted by the bank (based on the value one margin period of risk earlier) is explicit.

A bank and a counterparty have a two-way collateral agreement with zero threshold and a margin period of risk (cure period) of 20 days. On one simulation trial, the value of the outstanding transactions to the bank is -$50 million at time T (the counterparty defaults) and was -$45 million 20 days earlier. Collateral at time T is based on the value 20 days earlier, so the bank had posted $45 million, none of which is returned on default. What is the bank's credit exposure to the counterparty at time T?

A. $5 million

B. $0

C. $50 million

D. $45 million

Answer: B

Step by step

1. Value to the bank at T: $V = -50$, so the bank owes the counterparty \$50 million.
2. The bank posted $C = 45$ based on the value 20 days earlier (-45); the counterparty posted nothing to the bank.
3. The bank's claim is its positive net position: $\max(V + C_{\text{posted}}, 0) = \max(-50 + 45, 0) = \max(-5, 0) = 0$.
4. The bank still owes a net \$5 million after its posted collateral is applied, so it has no credit exposure.

Why B is correct: Exposure arises only when the bank is owed more than it holds. Collateral already posted by the bank is applied against its own obligation and the remaining \$5 million shortfall is owed by the bank, not to it, so exposure is zero.

Why the others are wrong

- **A:** The \$5 million is the gap between -50 and -45, but that is an amount the bank still owes, not an amount owed to it.
- **C:** Takes the gross value of the transactions and ignores both the sign (bank owes) and the collateral.
- **D:** Reports the collateral the bank posted as if it were exposure; posted collateral is exposure only if it exceeds what the bank owes.

Q970

AP Mock 2 Q43

REBUILT — Source option D (USD 683,200) was a rounded value of the exact result and the other options were weak; recomputed exactly and rebuilt distractors from realistic mistakes.

A coffee trader has purchased 10,000 bags of coffee for delivery in one year at USD 500 per bag, so the contract is worth zero today. The annual volatility of the coffee price return is 25% (normally distributed, mean zero). A margin call is made whenever credit exposure exceeds a threshold of USD 500,000, and collateral is received within a two-day remargining period. Assume 252 trading days per year and that the margin agreement is fully enforced. What is the approximate 95% credit value at risk (maximum credit exposure) over the one-year life?

A. USD 683,185
B. USD 183,185
C. USD 516,319
D. USD 2,056,250

Answer: A

Step by step

1. Notional $N = 10{,}000 \times 500 = 5{,}000{,}000$.
2. Two-day volatility: $\sigma_{2d} = 0.25\sqrt{2/252} = 0.02227$.
3. 95% one-sided move over the remargining period: $1.645 \times 0.02227 \times 5{,}000{,}000 = 183{,}185$.
4. With a fully enforced margin agreement, exposure can exceed the threshold only by the value change during the remargining period: $500{,}000 + 183{,}185 = 683{,}185$.

Why A is correct: Collateral caps exposure at the threshold plus whatever the position can gain during the two-day remargining period, which is the 95% two-day move.

Why the others are wrong

- **B:** Omits the 500,000 threshold; the uncollateralized threshold amount is part of the exposure.
- **C:** Scales volatility by $2/252$ instead of the square root of $2/252$, understating the two-day move ($500{,}000 + 16{,}319$).
- **D:** Ignores the margin agreement and uses the full one-year 95% move $1.645 \times 0.25 \times 5{,}000{,}000$.

Q971

AnalystPrep Q.5493

Quantum Investments trades derivatives actively in a volatile market and currently has a three-day remargining period. The risk department is considering shortening it. How would a shorter remargining period affect the firm's credit exposure?

A. It lowers operational costs by reducing the frequency of collateral adjustments.
B. It increases exposure to market volatility and counterparty risk.
C. It decreases exposure to market volatility because collateral is adjusted more frequently, shrinking the period during which exposure can rise uncollateralized.
D. It has no significant impact on exposure.

Answer: C

Step by step

1. The remargining period (margin period of risk) is the time from a collateral call to actual delivery; exposure that increases during it is uncollateralized.

2. A shorter period gives the exposure less time to move before collateral arrives, so the uncollateralized increase falls.
3. The effect is largest when market volatility is high, since larger moves fit into any given period.
4. Hence a shorter period lowers exposure (at the cost of more operational effort).

Why C is correct: The margin period of risk creates exposure equal to the change in value that can occur before collateral is received; shortening it reduces that change.

Why the others are wrong

- **A:** More frequent collateral adjustment raises, not lowers, operational cost.
- **B:** Reverses the relationship; a longer, not shorter, period increases exposure.
- **D:** The length of the margin period is a primary driver of residual exposure, especially in volatile markets.

Q972

AnalystPrep Q.5494

Brightstar Trading is reassessing the threshold in its collateral agreements for a portfolio of complex derivatives. How does the threshold affect its credit risk?

A. Increasing the threshold reduces credit risk because more collateral is posted.
B. Lowering the threshold reduces credit risk because collateral is called for smaller exposures.
C. The threshold has no significant impact on credit risk; it only affects operational efficiency.
D. A higher threshold reduces credit risk because it lowers the frequency of collateral calls.

Answer: B

Step by step

1. The threshold is the exposure level below which collateral is not called, so it is a permanently uncollateralized amount.
2. A lower threshold means collateral is requested sooner and more of the exposure is covered, reducing credit risk.
3. A higher threshold leaves more exposure uncollateralized (it approximately caps the collateral benefit) and so raises credit risk, though it reduces call frequency.

Why B is correct: The threshold is exposure left uncollateralized by design; the lower it is, the less exposure remains.

Why the others are wrong

- **A:** A higher threshold means less collateral is called, not more.
- **C:** The threshold directly determines uncollateralized exposure.
- **D:** A higher threshold does reduce call frequency and administrative cost, but it increases, not decreases, credit risk.

Q973

AnalystPrep Q.6211

REBUILT — Original was qualitative with an ambiguous perspective (client vs bank exposure); rebuilt as a numeric case with clear collateral timing.

A bank has a swap with a corporate client under a netting agreement with a two-way zero-threshold collateral agreement and a margin period of risk of 15 days. Collateral at any time reflects the swap value as of 15 days earlier. Fifteen days before the client defaults, the swap was worth +$8 million to the bank and the client posted $8 million of collateral. At default (time T) the swap is worth +$14 million to the bank. What is the bank's credit exposure at default?

A. $14 million
B. $8 million
C. $22 million
D. $6 million

Answer: D

Step by step

1. Value to the bank at default: $V(T) = 14$.
2. Collateral held reflects the value one margin period earlier: $C = 8$.
3. Exposure $= \max(V(T) - C, 0) = 14 - 8 = 6$ million.

Why D is correct: Collateral lags the value by the margin period of risk, so only the 15-day value increase of $6 million is uncovered.

Why the others are wrong

- **A:** Ignores the collateral held.

- **B:** Reports the collateral held (the 15-day-old value) rather than the uncovered amount.
- **C:** Adds collateral to the current value instead of subtracting it.

Q974 AnalystPrep Q.6215

A bank has a collateral agreement with zero threshold and a margin period of risk of 20 days, with both parties posting collateral equal to their exposure. On a simulation trial, the value of the outstanding transactions to the bank is -$50 million at time T and was -$55 million 20 days earlier. Collateral at T reflects the value 20 days earlier, and in the event of the counterparty's default at T none of the collateral previously posted by the bank is returned. What is the bank's exposure at T?

A. $0
B. $5 million
C. $50 million
D. $55 million

Answer: B

Step by step

1. The bank posted $C = 55$ (value 20 days earlier was -55).
2. At T the bank owes 50, so $V = -50$.
3. Posted collateral is not returned, so the bank's claim is $C + V = 55 - 50 = 5$.
4. Exposure $= \max(5, 0) = \$5$ million (over-posted collateral: posting margin can itself create positive exposure).

Why B is correct: The value moved 5 in the bank's favor during the margin period, so the bank has over-posted by $5 million, which is lost as an unsecured claim.

Why the others are wrong

- **A:** Applies max(V,0) and ignores that the posted collateral exceeds the current obligation.
- **C:** Uses the gross value of the transactions, but the bank owes that amount.
- **D:** Takes the full posted collateral as exposure, forgetting that $50 million of it covers the bank's own obligation.

Q975 BT answers P2.T6.329.3 · BT notes P2.T6.329.3

Regarding the modeling of the credit exposure of collateralized positions, each of the following is true EXCEPT which is false?

A. If a position is strongly collateralized, the primary concerns are the length of the remargin period and the volatilities of the exposure and the collateral.
B. For risk assessment, the remargin period will equal the legal margin call frequency, i.e. typically one day.
C. Imperfect collateralization at a given date can arise from the terms of the collateral agreement (threshold, minimum transfer amount, rounding), which prevent calling the full amount.
D. Exposure may increase between margin calls, and that portion cannot be collateralized.

Answer: B

Step by step

1. Strongly collateralized (zero threshold, no initial margin): residual risk comes from the margin period of risk and from volatility of exposure and collateral, so A is true.
2. The remargin period (MPoR) runs from the call to actual receipt and includes valuation, receipt, settlement, grace and close-out steps, so it is longer than the contractual call frequency; B is false.
3. Threshold, minimum transfer amount and rounding cause imperfect collateralization (C true).
4. Exposure that rises between calls is not covered by collateral (D true).

Why B is correct: The margin period of risk includes operational and legal delays on top of the call frequency, so it is not equal to it.

Why the others are wrong

- **A:** True statement, so not the exception.
- **C:** True statement, so not the exception.
- **D:** True statement, so not the exception.

Q976 BT notes P2.T6.907.3 · BT answers P2.T6.907.3

REBUILT — Merged two near-identical versions; stated the approximation inside the stem, removed the worked example text and replaced distractors with results of realistic errors.

The margin period of risk (MPoR) is the time between a counterparty ceasing to post collateral and the close-out or replacement of the transactions, and it is the period in which any increase in exposure remains uncollateralized. Assuming only variation margin, an approximation for the best-case reduction in counterparty risk is that uncollateralized EPE is about $0.5\sqrt{T/\text{MPoR}}$ times collateralized EPE, so the best-case reduction is $1 - 1/\text{ratio}$. Assuming 250 trading days per year, what is the approximate best-case reduction for a three-year OTC derivatives portfolio with an MPoR of 23 days?

A. 35.0%
B. 65.0%
C. 82.5%
D. 93.9%

Answer: B

Step by step

1. Maturity in days: $T = 3 \times 250 = 750$; MPoR $= 23$.
2. Ratio $= 0.5\sqrt{750/23} = 0.5 \times 5.710 = 2.855$.
3. Best-case reduction $= 1 - 1/2.855 = 1 - 0.3502 = 0.6498 \approx 65.0\%$.

Why B is correct: A longer MPoR leaves more exposure uncollateralized, so the reduction depends on the maturity-to-MPoR ratio through its square root.

Why the others are wrong

- **A:** Reports 1/ratio (0.350) instead of 1 - 1/ratio.
- **C:** Omits the factor 0.5: $1 - 1/\sqrt{750/23} = 82.5\%$.
- **D:** Omits the square root: $1 - 1/(0.5 \times 750/23) = 93.9\%$.

Q977 GARP PE1 Q78 · Mock 3 Q3

A financial institution has a two-way credit support annex (CSA) with a counterparty covering a portfolio valued at JPY 400 million. The margining terms of the collateralized portfolio include a threshold of JPY 180 million, a minimum transfer amount of JPY 30 million, and a margin period of risk of 10 days. Which of the following is correct?

A. A lower threshold implies that a larger portion of exposure is protected by collateral.
B. A shorter margin period of risk implies that a smaller portion of exposure is protected by collateral.
C. A lower independent amount (initial margin) implies that a larger portion of exposure is protected by collateral.
D. The protection from collateral specified in the CSA is uniform throughout the life of the exposure profile.

Answer: A

Step by step

1. The threshold is the exposure level below which collateral is not called, so it is uncollateralized exposure by construction.
2. Lowering the threshold means collateral is called on a larger share of the exposure, so more of the exposure is protected (A true).
3. A shorter MPoR means collateral arrives sooner after a call, so less exposure builds up uncollateralized (B reverses this).
4. Initial margin / independent amount is collateral held regardless of MtM; a lower amount gives less protection (C reverses this).
5. The effect of the threshold, minimum transfer amount and MPoR depends on how large the exposure is at each point in time, so protection varies over the profile (D false).

Why A is correct: The threshold is an amount of exposure that is never collateralized, so reducing it increases the share of exposure that is covered by collateral.

Why the others are wrong

- **B:** Reverses the MPoR effect: a shorter MPoR means less time for exposure to grow before collateral arrives, hence more protection.
- **C:** Reverses the effect of initial margin: lower initial margin means a smaller collateral cushion, not a larger one.
- **D:** Collateral is path-dependent and the threshold/MTA bind differently as exposure changes, so protection is not uniform over time.

Q978 AnalystPrep Q.5492

ACME Capital, a hedge fund, collateralizes its derivative trades with several counterparties. Its collateral agreements specify a remargining period, a threshold for collateral calls, and a minimum transfer amount. Which of the following statements about the impact of these terms is correct?

A. A longer remargining period reduces both operational burden and exposure to market value changes of the derivatives.

B. A higher threshold reduces the frequency of collateral exchanges and administrative cost but increases uncollateralized credit risk.

C. A lower minimum transfer amount increases the frequency of collateral movements, which reduces both exposure and transaction costs.

D. Shortening the remargining period and lowering the threshold and minimum transfer amount have no significant impact on exposure.

Answer: B

Step by step

1. Threshold: collateral is only called for exposure above it, so a higher threshold means fewer calls but a larger permanently uncollateralized amount (B true).
2. Remargining period: a longer period gives more time for exposure to rise before collateral is received, so exposure increases (A wrong on exposure).
3. Minimum transfer amount: a lower MTA means smaller calls are honoured, so collateral moves more often at higher operational cost, with less residual uncollateralized exposure (C wrongly says transaction costs fall).
4. All three terms create exposure that remains uncollateralized, so changing them has a material effect (D false).

Why B is correct: A higher threshold leaves more exposure uncollateralized (more credit risk) while reducing the number of collateral calls and associated administration.

Why the others are wrong

- **A:** A longer remargining period lengthens the MPoR and therefore increases, not reduces, exposure; the operational saving does not offset this.
- **C:** Right that calls become more frequent, but this raises operational cost; exposure falls only at the price of higher transaction costs.
- **D:** Treats the terms as immaterial; threshold, MTA and MPoR are the main sources of residual exposure under collateralization.

Q979 BT answers P2.T6.329.2 · BT notes P2.T6.329.2

REBUILT — Source distractors (8.77%, 23.64%) did not correspond to identifiable errors and the formula was garbled by OCR; restated the formula and rebuilt distractors from realistic mistakes.

For a normal distribution with zero mean, the potential future exposure of a collateralized position over the margin period of risk is PFE $= z_\alpha \sigma\sqrt{T_M}$, where σ is the annual volatility of the collateralized exposure and T_M is the remargin period in years. An exposure has an annual volatility of 15.0% and is perfectly collateralized by cash, with a remargin period (including worst-case delays) of 30 days. Assume 250 days per year and $z_{99\%} = 2.33$. Which is NEAREST to the 99% PFE at the end of the 30 days?

A. 4.19%

B. 10.02%

C. 12.11%

D. 34.95%

Answer: C

Step by step

1. $T_M = 30/250 = 0.12$ years.
2. $\sqrt{0.12} = 0.3464$.
3. PFE $= 2.33 \times 15.0\% \times 0.3464 = 12.11\%$.

Why C is correct: Volatility scales with the square root of time, so the 99% exposure over the 30-day MPoR is 2.33 times the 30-day volatility of 5.20%.

Why the others are wrong

- **A:** Scaled volatility linearly in time (30/250) instead of by the square root: 2.33 x 15% x 0.12 = 4.19%.
- **B:** Used 365 calendar days instead of the stated 250 days: 2.33 x 15% x sqrt(30/365) = 10.02%.
- **D:** Ignored the time scaling entirely: 2.33 x 15% = 34.95%, which is a one-year PFE.

Q980

BT answers P2.T6.330.1 · BT notes P2.T6.330.1

REBUILT — Original relied on a chart that was not available; replaced it with a data table of the same kind of profile and rewrote the options so exactly one is inconsistent.

The table below shows one simulated path for a collateralized interest rate swap from Counterparty A's point of view. MtM is the value to A (positive means B owes A). Collateral is positive if held by A and negative if posted by A. Exposure is max(MtM - Collateral, 0). All figures are percent of notional.

Year	MtM	Collateral	Exposure
1.0	1.10	0.00	1.10
1.5	1.60	0.60	1.00
2.0	2.00	1.00	1.00
3.0	0.50	0.00	0.50
4.0	-1.80	-0.80	0.00

Each party has a threshold and there is a single minimum transfer amount. Each of the following is consistent with the table EXCEPT which?

A. Counterparty A's threshold is 1.0%.
B. The minimum transfer amount is zero, and collateral is delivered with no delay.
C. Counterparty B's threshold is 1.0%.
D. Collateral is delivered with no delay in this path.

Answer: B

Step by step

1. B's threshold: at years 1.5 and 2.0 collateral held equals MtM minus 1.0 (0.60 and 1.00), so B's threshold is 1.0% (C consistent).
2. A's threshold: at year 4, MtM is -1.80 and A has posted 0.80, i.e. the amount beyond 1.0 (A consistent).
3. MTA: at year 1.0 MtM exceeds B's 1.0% threshold by 0.10 yet no collateral is held. Collateral at the other dates equals MtM less the threshold exactly (no lag), so the missing 0.10 cannot be a delay; with a zero MTA and no delay a call of 0.10 would have been made. Option B is therefore inconsistent.
4. Delay: collateral equals MtM less threshold at each date shown, so no lag is evident (D consistent).

Why B is correct: With zero MTA and no delay, the 0.10% excess over the threshold at year 1 would have been collateralized; its absence shows a positive minimum transfer amount (above 0.10).

Why the others are wrong

- **A:** Consistent: A posts 0.80 when MtM is -1.80, which equals the excess over a 1.0% threshold.
- **C:** Consistent: collateral held at years 1.5 and 2.0 equals MtM minus 1.0.
- **D:** Consistent: collateral equals MtM less threshold at every date, so there is no visible lag.

Q981

BT answers P2.T6.330.2 · BT notes P2.T6.330.2

REBUILT — Stem wording ('EXCEPT for which would NOT') was confusing; restated as a single negative question and called the independent amount initial margin.

Assume a collateralized exposure with positive and increasing mark-to-market. Which of the following would NOT tend to increase the risk of the expected exposure (EE) profile?

A. A longer remargin period
B. A higher threshold
C. A larger independent amount (initial margin) posted by the counterparty
D. A larger minimum transfer amount

Answer: C

Step by step

1. A longer remargin period lets more exposure accumulate before collateral arrives, so exposure rises.
2. A higher threshold leaves a bigger amount permanently uncollateralized, so exposure rises.
3. A larger minimum transfer amount allows a larger shortfall to go uncalled, so exposure rises.
4. An independent amount / initial margin held against the exposure is extra collateral that offsets exposure, so it reduces EE. It is the exception (C).

Why C is correct: Initial margin held is a cushion against exposure, whereas the other three terms each create exposure that stays uncollateralized.

Why the others are wrong

- **A:** Increases risk: longer MPoR means exposure can grow for longer before collateral is received.
- **B:** Increases risk: the threshold is exposure that is never collateralized.

- **D:** Increases risk: exposure below the MTA is not called, so a larger MTA means a larger uncollateralized gap.

Q982

Mock 1 Q13

REBUILT — Original options were printed as negative percentages and the stem was ambiguous about what was measured; restated as expected exposure as a percentage of the position over the remargin period and rebuilt distractors.

A risk analyst models the change in value of a collateralized position over a 10-day remargining period as a zero-mean normal distribution. The annual volatility of the position is 10% and there are 252 trading days per year. The analyst also notes a 99% PFE confidence level. Which is NEAREST to the expected exposure (EE) over the remargining period, as a percentage of the position?

A. 0.66%
B. 0.80%
C. 1.99%
D. 4.63%

Answer: B

Key correction: Source options were printed as negative percentages (-0.8% keyed); exposure is non-negative, so the correct answer is +0.80%.

Step by step

1. 10-day volatility: $\sigma_T = 10\% \times \sqrt{10/252} = 1.992\%$.
2. For a zero-mean normal, $\text{EE} = E[\max(X, 0)] = \sigma_T/\sqrt{2\pi}$.
3. $\text{EE} = 1.992\% \times 0.3989 = 0.795\% \approx 0.80\%$.
4. The 99% confidence level is only needed for PFE and is irrelevant to EE.

Why B is correct: EE is the mean of the positive part of a zero-mean normal, $\sigma_T/\sqrt{2\pi} \approx 0.4\,\sigma_T$.

Why the others are wrong

- **A:** Used 365 days: 10% x sqrt(10/365) / sqrt(2 pi) = 0.66%.
- **C:** Reported the 10-day volatility itself and forgot the 1/sqrt(2 pi) factor.
- **D:** Computed the 99% PFE (2.326 x 1.992% = 4.63%) instead of the EE.

Q983

Schweser Module Quiz 37.3 Q1

REBUILT — Original asked for the 'except' item but its keyed answer (posting collateral) is not a distinct listed step and overlaps with 'receiving collateral'; rebuilt so exactly one option is clearly not a component of the margin period of risk.

The margin period of risk (MPoR) is the time from a collateral call to the point at which the position has been closed out and re-hedged following the default of the party that must post collateral. Which of the following is NOT one of the time steps that enter into the calculation of the number of days in the MPoR?

A. Valuation of the exposure and collateral, and the margin call
B. Grace period allowed to the counterparty before it is deemed to have failed to pay
C. Settlement of the collateral that is delivered
D. Annual review and renegotiation of the collateral threshold in the credit support annex

Answer: D

Key correction: Source key (Schweser) said B (posting collateral); the underlying reading lists receiving collateral as an MPoR step, so that option was ambiguous. The item was rebuilt.

Step by step

1. The MPoR consists of five time steps: (1) valuation and margin call, (2) receiving collateral, (3) settlement of the collateral, (4) grace period, (5) liquidation/close-out and re-hedge.
2. Option A is step 1, option B is step 4, option C is step 3.
3. Option D is a periodic contractual/administrative review of the CSA terms; it occurs outside the default-and-close-out window, so it adds no days to the MPoR.

Why D is correct: The MPoR measures only the operational and legal delays between the margin call and the completed close-out and re-hedge; a periodic review of the threshold is not a step in that chain.

Why the others are wrong

- **A:** This is step 1 of the MPoR (time to calculate exposure and collateral value and make a valid call); it is included, not excluded.
- **B:** The grace period before a failure to pay becomes a default is step 4 of the MPoR; excluding it understates the MPoR.

- **C:** Settlement time (cash intraday, government bonds about one day, corporate bonds about three days) is step 3 of the MPoR.

LO 37.i

Assess the impact of collateral on counterparty risk and funding, with and without segregation or rehypothecation.

Q984

BT answers P2.T6.912.1

REBUILT — Source treated posted segregated initial margin as reducing credit exposure while also adding it to funding (inconsistent); stem clarified (IM posted by the bank, segregated, VM received) and options rebuilt.

A bank's derivatives portfolio with a counterparty has a current mark-to-market (MTM) value of +$30.0 million to the bank. The bank holds $27.0 million of variation margin received (reusable, not segregated). The bank has also posted $8.0 million of initial margin to the counterparty under bilateral rules, and this initial margin is segregated. What are, respectively, the counterparty credit exposure and the funding exposure?

A. Credit exposure = zero; funding exposure = $11.0 million
B. Credit exposure = $3.0 million; funding exposure = zero
C. Credit exposure = $3.0 million; funding exposure = $11.0 million
D. Credit exposure = $11.0 million; funding exposure = $3.0 million

Answer: C

Key correction: Source key (A) said credit exposure is zero by subtracting the IM as if it were received; the IM was posted and segregated, so it does not affect credit exposure. Correct is C (3.0 and 11.0).

Step by step

1. Credit exposure = max(MTM − margin received + non-segregated margin posted, 0).
2. Margin received (VM) = 27.0 reduces exposure; the posted IM is segregated, so it is bankruptcy-remote and adds nothing.
3. Credit exposure = $\max(30 - 27 + 0, 0) = 3.0$ million.
4. Funding exposure = MTM - reusable margin received + all margin posted (irrespective of segregation).
5. Funding = $30 - 27 + 8 = 11.0$ million.

Why C is correct: Segregated posted margin is retrievable on default so it creates no counterparty risk, but the bank must still fund it, so it increases funding exposure.

Why the others are wrong

- **A:** Subtracts the posted IM from exposure (30-27-8 < 0), treating it as margin received; posted segregated IM does not reduce credit exposure.
- **B:** Gets credit exposure right but ignores that the bank must fund the net MTM asset and the posted IM; funding is 30-27+8, not zero.
- **D:** Adds the segregated IM to credit exposure (30-27+8) and omits it from funding; segregation excludes it from credit exposure but it is included in funding.

Q985

BT answers P2.T6.912.2

REBUILT — Source subtracted posted segregated IM from credit exposure and ignored that over-posted non-segregated VM adds exposure; stem clarified and options rebuilt.

A bank's portfolio with a counterparty has an MTM value of -$36.0 million to the bank (a liability). The bank has posted $40.0 million of variation margin, which is not segregated (transferred by title). The bank has also posted $3.0 million of initial margin under bilateral rules, which is segregated. What are, respectively, the bank's counterparty credit exposure and funding exposure?

A. Credit exposure = $4.0 million; funding exposure = $7.0 million
B. Credit exposure = $1.0 million; funding exposure = $7.0 million
C. Credit exposure = zero; funding exposure = $7.0 million
D. Credit exposure = $4.0 million; funding exposure = $4.0 million

Answer: A

Key correction: Source key said credit exposure $1.0m and funding $7.0m, subtracting the posted IM from credit exposure; segregated posted IM does not reduce exposure, so credit exposure is $4.0m (option A here).

Step by step

1. Credit exposure = max(MTM − margin received + non-segregated margin posted, 0).
2. Nothing received; posted VM of 40.0 is not segregated so it is at risk; posted IM of 3.0 is segregated so it is excluded.
3. Credit exposure = max(−36 − 0 + 40, 0) = 4.0 million (the over-posted VM).
4. Funding exposure = MTM - reusable margin received + all margin posted = −36 − 0 + (40 + 3) = 7.0 million.

Why A is correct: Only non-segregated posted margin (the 4.0 of VM above the liability) is exposed to the counterparty's default, while all posted margin, including segregated IM, must be funded.

Why the others are wrong

- **B:** Subtracts the segregated IM (-36+40-3 = 1); posted IM is not margin received and does not reduce exposure.
- **C:** Assumes a liability position means no credit exposure, ignoring that the 4.0 of non-segregated VM posted above the liability is unsecured exposure.
- **D:** Gets credit exposure right but leaves the segregated IM out of funding (-36+40); all posted margin counts toward funding.

Q986

BT answers P2.T6.912.3 · BT notes P2.T6.912.3

In a certain credit portfolio, Acme Bank holds four large over-the-counter (OTC) derivative positions, each with a different counterparty. The mark-to-market value of each position is significantly positive (above the threshold), so Acme Bank holds collateral for each position. The collateral types are:

I. Cash that does not need to be segregated
II. Securities that can be rehypothecated
III. Cash/securities that must be segregated (cannot be rehypothecated)
IV. The counterparty is posting its own bonds, and they can be rehypothecated

Which of the following statements about counterparty risk reduction and funding benefits is TRUE?

A. All four provide counterparty risk reduction and funding benefits
B. I. and II. provide both counterparty risk reduction and funding benefits; III. provides only counterparty risk reduction benefits; and IV. provides only funding benefit
C. I. provides both counterparty risk reduction and funding benefits; II. and III. provide only funding benefit; and IV. provides only counterparty risk reduction benefit
D. None except I. provide either counterparty risk reduction or funding benefit

Answer: B

Step by step

1. Margin reduces counterparty risk if it can be monetised on default; it gives a funding benefit only if it can be reused (not segregated, rehypothecable) in non-default scenarios.
2. I: non-segregated cash is usable in default and reusable, so both benefits. II: rehypothecable securities (with adequate haircuts) likewise give both.
3. III: segregated margin can be monetised on default (risk reduction) but cannot be reused, so no funding benefit.
4. IV: the counterparty's own bonds will be nearly worthless when it defaults (wrong-way risk), so risk reduction is questionable, but being rehypothecable they provide a funding benefit.

Why B is correct: Counterparty risk reduction depends on value retained at default; funding benefit depends on reusability, so each collateral type is classified on those two tests.

Why the others are wrong

- **A:** Ignores segregation (III has no funding benefit) and wrong-way risk (IV gives little risk reduction).
- **C:** Reverses the logic: rehypothecable securities (II) give both benefits, segregated collateral (III) gives risk reduction rather than funding, and own bonds (IV) give funding rather than risk reduction.
- **D:** Too restrictive: rehypothecable securities (II) also give both benefits, and III and IV each give one.

Reading 38. CVA

Jon Gregory, The xVA Challenge: Counterparty Credit Risk, Funding, Collateral, and Capital, 4th Edition (West Sussex, UK: John Wiley & Sons, 2020). Chapter 17.

LO	Learning objective	Qs
38.a	explain the motivation for and the challenges of pricing counterparty risk.	3
38.b	describe credit value adjustment (CVA).	4
38.c	calculate CVA and CVA as a spread with no wrong-way risk, netting, or collateralization.	11
38.d	evaluate the impact of changes in the credit spread and recovery rate assumptions on CVA.	4
38.e	describe debt value adjustment (DVA) and bilateral CVA (BCVA).	7
38.f	explain the differences between unilateral CVA (UCVA) and BCVA, and between unilateral DVA (UDVA) and BCVA.	3
38.g	calculate DVA, BCVA, and BCVA as a spread.	10
38.h	explain how netting can be incorporated into the CVA calculation.	3
38.i	define and calculate incremental CVA and marginal CVA and explain how to convert CVA into a running spread.	13
38.j	explain the impact of incorporating collateralization into the CVA calculation, including the impact of margin period of risk, thresholds, and initial margins.	3
38.k	describe wrong-way risk and contrast it with right-way risk.	3
38.l	identify examples of wrong-way risk and examples of right-way risk.	15
38.m	discuss the impact of collateral on wrong-way risk.	3
38.n	identify examples of wrong-way collateral.	3
38.o	discuss the impact of wrong-way risk on central counterparties (CCPs).	3
38.p	describe the various wrong-way modeling methods including hazard rate approaches, structural approaches, parametric approaches, and jump approaches.	4
38.q	explain the implications of central clearing on wrong-way risk.	3

LO 38.a

Explain the motivation for and the challenges of pricing counterparty risk.

Q987

BT answers P2.T6.331.2

REBUILT — Original EXCEPT item relied on a 'centralized desk' claim and an additivity claim not supported by the GARP chapter and had a contestable false statement; rebuilt as a single-answer item on why swap counterparty risk is harder to price than bond credit risk.

A bank's risk manager explains why quantifying counterparty risk on an interest rate swap is significantly more difficult than pricing the credit risk of a bond issued by the same counterparty. Which statement best captures the reason?

A. A swap has bilateral cash flows, so a given cash flow is not fully at risk in default because it can be partly offset by opposing cash flows, and the exposure must therefore be modeled rather than read off the contract

B. A swap has a notional that is exchanged at maturity, so the amount at risk is always the full notional

C. A bond has no recovery value in default, whereas a swap always has a positive recovery rate

D. A swap's counterparty default probability cannot be estimated, whereas a bond issuer's can

Answer: A

Step by step

1. For a bond, payments are one-way: the investor is owed the cash flows, so credit risk is handled by accounting for default when discounting and adding any recovery payments.
2. For a swap, fixed, floating or contingent payments flow in both directions, so the net value can be positive or negative and varies with market variables.
3. A cash flow owed to the bank is therefore not fully at risk, since it may be offset by current and future opposing payments; the exposure is max(V,0) at the default date and has to be modeled (simulated).
4. This bilateral nature is what makes counterparty risk quantification significantly more difficult for derivatives, which is the challenge in LO 38.a.

Why A is correct: Bilateral cash flows make derivative exposure uncertain and partly offsetting, so it cannot be priced like the one-way cash flows of a bond.

Why the others are wrong

- **B:** Interest rate swaps do not exchange notional; the exposure is the (positive) replacement value, far below notional. This confuses a swap with a bond principal repayment.
- **C:** Both instruments have an LGD/recovery concept; recovery is not the source of the difficulty. This is a misstatement of what differs.

- **D:** Default probability is estimated the same way (e.g., market CDS or proxy spreads) for both; the difficulty lies in exposure, not default probability.

Q988 BT answers P2.T6.320.2 · BT notes P2.T6.320.2

The price of a financial instrument can generally be defined in one of two ways:
I. Actuarial price: the price represents an expected value of future cashflows, incorporating some adjustment for the risk that is being taken (the risk premium).
II. Risk-neutral price: the price is the cost of an associated hedging strategy.
Into which camp does the credit valuation adjustment (CVA) fall?

A. I only
B. II only
C. Both, because the actual price must equal the risk-neutral price
D. Both, although the prices are likely to differ, perhaps dramatically

Answer: D

Step by step

1. Historically CVA was treated as an actuarial reserve/provision computed with historical (real-world) parameters, i.e., an expected-loss price (camp I).
2. Under current accounting (exit price concept) CVA is computed with market-implied (risk-neutral) parameters, reflecting the cost of hedging (camp II).
3. Market-implied default probabilities are significantly higher than real-world ones and default usually cannot be hedged with liquid single-name CDS, so the two prices can differ greatly.
4. So CVA can be seen in both camps, but the two prices need not agree.

Why D is correct: CVA can be viewed both as an actuarial expected loss and as a risk-neutral hedging cost, and because market-implied default probabilities exceed real-world ones the two values can differ dramatically.

Why the others are wrong

- **A:** Ignores the risk-neutral/exit-price view now used for pricing and accounting.
- **B:** Ignores the actuarial reserve view that banks historically used and some still use.
- **C:** Wrongly asserts the two prices coincide; real-world and risk-neutral default probabilities differ, so the prices generally do not.

Q989 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

The finance director of a regional bank asks the counterparty risk manager why the bank charges a separate adjustment for counterparty risk on a newly executed 7-year interest rate swap with a corporate client, rather than simply relying on the swap's quoted price. Which of the following is the most accurate response?

A. The swap price already embeds the client's default risk, so the adjustment is only a presentation item that does not generate any reserve.
B. The adjustment should be computed on the swap in isolation, ignoring netting and collateral, so that the reserve is as conservative as possible.
C. The adjustment is a price of counterparty risk on all outstanding positions with the client, charged in addition to the price of the swap itself, and it builds reserves to absorb potential default losses.
D. The adjustment is needed mainly because the swap's full notional is exchanged at maturity and is therefore wholly at risk in default.

Answer: C

Step by step

1. Counterparty risk pricing is a function of the credit exposure to the counterparty and its default probability.
2. The resulting price covers all outstanding positions with the counterparty and exists in addition to the price of the instrument itself (e.g., the swap).
3. Charging it generates reserves that absorb potential losses from the counterparty's default, and it must reflect risk mitigants such as netting and collateral.

Why C is correct: The price of counterparty risk is a separate charge on top of the instrument's own price, reflects all positions with the counterparty, and funds reserves for default losses while recognizing risk mitigants.

Why the others are wrong

- **A:** The instrument price is the risk-free price; the counterparty-risk adjustment is separate and does generate reserves. Treating it as embedded and cosmetic is the error.
- **B:** Pricing must account for risk mitigants such as netting and collateral; ignoring them overstates the charge and is not best practice.
- **D:** Swap notionals are not exchanged, and the amount at risk is the (netted, collateral-adjusted) exposure, not the notional; confusing a swap with a bond is the mistake.

LO 38.b

Describe credit value adjustment (CVA).

Q990

AnalystPrep Q.6208

REBUILT — Original answer (PFE) conflicts with the reading, where CVA uses expected (positive) exposure; stem was vague about 'accuracy', so it was rebuilt as a clear question on the inputs of the CVA formula.

A bank is computing the unilateral credit value adjustment (CVA) on an uncollateralized portfolio of swaps with one counterparty, assuming no wrong-way risk. Which of the following correctly lists the inputs that the standard discrete CVA formula, $\text{CVA} = -\text{LGD}\sum_{i=1}^{m}\text{EPE}(t_i)\,\text{PD}(t_{i-1},t_i)$, combines?

A. Loss given default, discounted expected positive exposure at each future date, and the marginal default probability in each interval.

B. Loss given default, the 97.5th-percentile potential future exposure at each future date, and the marginal default probability in each interval.

C. Loss given default, the current mark-to-market value of the portfolio only, and the cumulative default probability to maturity.

D. Loss given default, discounted expected negative exposure at each future date, and the marginal default probability of the bank itself.

Answer: A

Key correction: Source key (AnalystPrep) said C (incorporate PFE); the reading builds CVA from expected positive exposure (EPE), not a high-percentile PFE, so PFE is not the correct input.

Step by step

1. CVA is the price of counterparty credit risk: $\text{CVA} = -\text{LGD}\sum\text{EPE}(t_i)\,\text{PD}(t_{i-1},t_i)$.
2. Three drivers: (1) LGD, (2) the exposure profile, measured as discounted expected positive exposure (risk-free discounting is usually built into the EPE), (3) the marginal default probability in each interval.
3. Exposure matters only when positive (counterparty owes the bank) and is averaged (expected), not taken at a high percentile.
4. Hence A matches the formula.

Why A is correct: CVA is the expected loss from counterparty default: LGD times the sum over intervals of discounted EPE times the marginal PD for that interval.

Why the others are wrong

- **B:** PFE is a high-percentile (worst-case) exposure used for limits, not the average exposure that prices expected loss; using it overstates CVA.
- **C:** A single current MtM ignores how exposure evolves over the life of the trade, and the cumulative PD is not matched to each exposure date; the formula needs the whole profile and interval PDs.
- **D:** Expected negative exposure and the bank's own default probability drive DVA, not the counterparty CVA.

Q991

AnalystPrep Q.5474

REBUILT — Option D was self-contradictory (it credited the direct formula with a path-wise feature); rewrote it as a clear distractor so only C is correct.

Eastpointe Bank has entered into a 10-year receiver interest rate swap with Counterparty A (it receives fixed and pays floating). The risk team is computing the credit value adjustment (CVA) for this uncollateralized swap by Monte Carlo, choosing between a direct CVA calculation and a path-wise CVA calculation. The bank wants to minimize computational time without sacrificing accuracy. Which approach is more effective?

A. A path-wise CVA formula, because it is more intuitive as a sum over the EPE profile and therefore must be cheaper to compute.

B. A path-wise CVA formula, because valuations at successive simulation points have low correlation, so it converges quickly.

C. A direct CVA formula, because assuming independence of default time, exposure at default and LGD lets one simulate default-time exposure directly with a single average LGD, giving lower variance for the same number of valuations.

D. A direct CVA formula, because it requires multiple valuations of the portfolio at every time interval of each simulation, which improves accuracy.

Answer: C

Step by step

1. The path-wise formula CVA $= -\text{LGD}\sum \text{EPE}(t_i)\text{PD}(t_{i-1}, t_i)$ is intuitive, but each simulation needs valuations at all m time intervals.
2. Valuations at successive points are highly correlated, so the extra valuations add little information and convergence is slow.
3. The direct approach assumes default time, exposure at default and LGD are independent (no wrong-way risk), so a single average LGD is used and only the exposure at the default time needs to be simulated.
4. This gives a much smaller standard deviation for the same number of valuations (about six times smaller in the reading's example, i.e. far fewer valuations for equal accuracy). So C.

Why C is correct: Independence (no WWR) lets the direct method value the portfolio only at the simulated default time with an average LGD, avoiding the highly correlated multi-date valuations of the path-wise method.

Why the others are wrong

- **A:** Intuitiveness is a real advantage of the path-wise formula but says nothing about computational efficiency; it is actually the slower method.
- **B:** Wrong premise: successive path-wise points are highly (not lowly) correlated, which is why path-wise converges slowly.
- **D:** Requiring multiple valuations at every interval is a feature of the path-wise method and is its computational burden, not an advantage of the direct method.

Q992

BT answers P2.T6.417.1

REBUILT — Original depended on an unavailable chart; rebuilt with explicit constant hazard rate and numeric marginal PDs, testing the same point (the exposure input is EE/EPE, marginal PD falls over time).

The CVA of a five-year interest rate swap is built from four components: LGD, discount factors, the expected (positive) exposure profile, and the marginal default probability in each annual interval. Assume a constant hazard rate (default intensity) of 5% per year, so the marginal (unconditional) default probability for year i is $e^{-0.05(i-1)} - e^{-0.05i}$. Which of the following statements is correct?

A. The marginal default probability is 5.00% in every year, because the hazard rate is constant.

B. The marginal default probability falls from 4.88% in year 1 to 3.99% in year 5, because survival to each interval declines.

C. The marginal default probability rises from 4.88% in year 1 to 22.12% in year 5, because default risk accumulates.

D. The exposure input to CVA is the potential future exposure (PFE) profile, which is why it is plotted alongside the marginal default probability.

Answer: B

Step by step

1. Year 1: $1 - e^{-0.05} = 4.88\%$.
2. Year 5: $e^{-0.20} - e^{-0.25} = 3.99\%$; each year's marginal PD equals the previous one times $e^{-0.05}$, so it declines geometrically.
3. The cumulative PD to year 5 is $1 - e^{-0.25} = 22.12\%$, which is a different quantity from the marginal PD.
4. The hazard rate is a conditional (given survival) default rate and is constant; the unconditional probability of defaulting in a given interval falls as survival probability falls.
5. CVA uses expected exposure (EE/EPE), not PFE, so D is false.

Why B is correct: With constant hazard rate the probability of surviving to each interval shrinks, so the unconditional probability of defaulting in a given interval declines over time.

Why the others are wrong

- **A:** Treats the conditional hazard rate as the unconditional marginal PD; it ignores that the firm must survive to the interval first.
- **C:** Confuses marginal PD with the cumulative PD (22.12% at year 5, not a one-year figure); the cumulative rises, the marginal falls.

- **D:** CVA uses the expected (positive) exposure profile; PFE is a high-percentile measure and is not a CVA input.

Q993 BT Mock Q42 · BT answers P2.T6.914.1

REBUILT — The source 'principles' (risk-neutrality either way, inseparability) are not supported by the GARP chapter and the two sources keyed different answers; rebuilt as a single question on what CVA is and how it is computed, using GARP's statements.

Which of the following statements about credit value adjustment (CVA) is correct?

A. CVA is generally computed with market-implied (risk-neutral) parameters where practical, which supports the exit-price concept required by accounting standards.

B. Accounting standards require CVA to be computed with historical default probabilities and volatilities, so that it acts as an actuarial reserve.

C. CVA depends only on the counterparty's probability of default and is independent of the exposure profile.

D. CVA is proportional to the counterparty's exposure and likelihood of default, but not to the loss given default.

Answer: A

Key correction: Source keys disagreed (BT Mock Q42: Risk-neutrality misstated; BT P2.T6.914.1: Inseparability incorrect); the GARP chapter supports neither framing, so the question was rebuilt on its own statements.

Step by step

1. CVA is the price of counterparty risk: the difference between the value of the derivative with and without counterparty default.
2. Historically CVA was an actuarial 'reserve' using historical parameters; accounting rules have since made it an exit price computed with market-implied (risk-neutral) parameters wherever practical (historical use remains only in rare cases).
3. CVA is driven by PD, exposure and LGD: CVA $= -\text{LGD}\sum \text{EPE}\,\text{PD}$, so it depends on all three.
4. Only A is correct.

Why A is correct: CVA is an exit-price concept and so is computed with market-implied (risk-neutral) parameters where practical, giving a price that can be hedged.

Why the others are wrong

- **B:** Reverses the history: historical parameters and the reserve view were the old approach; accounting now points to market-implied exit prices.
- **C:** Wrong: CVA is the product of exposure (EPE), default probability and LGD, so it depends heavily on the exposure profile; default probability alone is not enough.
- **D:** CVA is directly proportional to LGD as well as PD and exposure; omitting LGD is wrong.

LO 38.c

Calculate CVA and CVA as a spread with no wrong-way risk, netting, or collateralization.

Q994 BT answers P2.T6.331.3

REBUILT — Stem referred to an unseen 'swap' context and listed EE at maturity (700 bps) above the maximum PFE (600 bps), an inconsistent/OCR-damaged value; made the trade explicit, set EE at maturity to 100 bps and rebuilt the distractors as realistic wrong-metric errors.

A trader needs a very quick estimate of the CVA on an interest rate swap and has no time for complex calculations. The exposure management group reports: expected exposure (EE) at maturity = 100 bps; potential future exposure (PFE) at maturity = 300 bps; average expected positive exposure (EPE) over the life of the swap = 250 bps; and maximum PFE = 600 bps (all as a percentage of notional). The counterparty's credit spread is 200 bps per annum. Which is nearest to an approximation of the CVA as a running spread?

A. 2.0 bps

B. 5.0 bps

C. 6.0 bps

D. 12.0 bps

Answer: B

Step by step

1. The running-spread approximation is $\text{CVA}_{spread} \approx -\text{EPE}_{avg} \times s$, assuming no wrong-way risk and a roughly constant EPE profile.

2. The correct exposure measure is the average EPE = 250 bps = 2.5%.
3. 2.5% × 200 bps = 0.025 × 200 = 5.0 bps.
4. The CVA is a charge, so it is -5.0 bps (a 5.0 bps running spread).

Why B is correct: CVA as a spread is the product of the average expected positive exposure and the counterparty credit spread; EE at maturity, PFE and maximum PFE are not the exposure measure used.

Why the others are wrong

- **A:** Used EE at maturity: 1% × 200 = 2.0 bps. A single-date EE does not represent the average exposure over the life.
- **C:** Used PFE at maturity: 3% × 200 = 6.0 bps. PFE is a high-percentile exposure, not the expected exposure.
- **D:** Used the maximum PFE: 6% × 200 = 12.0 bps. The peak tail exposure overstates the average exposure.

Q995

AnalystPrep Q.1973

Given a credit spread of 500 basis points and 1.6% EPE, the credit value adjustment is approximately equal to:

A. -312.5 bps
B. -80 bps
C. -31.25 bps
D. -8 bps

Answer: D

Step by step

1. Approximation: $\text{CVA}_{spread} \approx -\text{EPE} \times s$.
2. EPE = 1.6% = 0.016; s = 500 bps.
3. 0.016 × 500 = 8 bps, so CVA = -8 bps.

Why D is correct: With a constant EPE, CVA expressed as a running spread equals minus the EPE multiplied by the credit spread.

Why the others are wrong

- **A:** Divided the spread by EPE: 500/1.6 = 312.5. The two quantities must be multiplied.
- **B:** Treated EPE as 16% (decimal slip): 0.16 × 500 = 80.
- **C:** Divided the spread by 16 (decimal slip on EPE plus division): 500/16 = 31.25.

Q996

AnalystPrep Q.1977

REBUILT — Original distractors (-30, -8, -5) did not correspond to clear errors; replaced with options generated by realistic mistakes.

Given a credit spread of 250 basis points and 1.2% EPE, the credit value adjustment is approximately equal to:

A. -300 bps
B. -208 bps
C. -30 bps
D. -3 bps

Answer: D

Step by step

1. $\text{CVA}_{spread} \approx -\text{EPE} \times s$.
2. 0.012 × 250 = 3 bps, so CVA = -3 bps.

Why D is correct: CVA as a running spread is minus the product of EPE (1.2%) and the credit spread (250 bps).

Why the others are wrong

- **A:** Treated 1.2% as 1.2 (i.e. 120%): 1.2 × 250 = 300.
- **B:** Divided the spread by EPE: 250/1.2 ≈ 208. The two must be multiplied.
- **C:** Treated EPE as 12% (decimal slip): 0.12 × 250 = 30.

Q997

AnalystPrep Q.2699

REBUILT — Original distractors (-800, -167, -200) were not tied to clear errors; replaced with options generated by realistic mistakes.

In a swap transaction, the counterparty's expected positive exposure (EPE) is 5% and its credit spread is 300 basis points. Calculate the CVA as a running spread.

A. -1,500 bps
B. -167 bps

C. -60 bps
D. -15 bps

Answer: D
Step by step
1. $\text{CVA}_{spread} \approx -\text{EPE} \times s$.
2. $0.05 \times 300 = 15$ bps, so CVA = -15 bps.

Why D is correct: The running-spread approximation multiplies the average EPE (5%) by the credit spread (300 bps).
Why the others are wrong
- **A:** Used EPE = 5 instead of 5%: $5 \times 300 = 1{,}500$.
- **B:** Divided EPE by the spread in mismatched units: $5/3 \approx 1.67 \rightarrow 167$.
- **C:** Divided the spread by the EPE in percent: $300/5 = 60$.

Q998
AnalystPrep Q.3085

Jon Boyle is a trader at a big German bank and needs a quick approximation of the CVA spread on a swap. The risk management group at the bank comes up with an expected positive exposure of 13%. The counterparty's credit spread is around 225 basis points per year. Based on this information, the CVA, as a running spread in percentage terms, is closest to:
A. -29.25%
B. -5.77%
C. -0.2925%
D. -0.577%

Answer: C
Step by step
1. $\text{CVA}_{spread} \approx -\text{EPE} \times s$.
2. 0.13×225 bps $= 29.25$ bps.
3. Convert: 29.25 bps = 0.2925%, so CVA = -0.2925%.

Why C is correct: Multiplying EPE (13%) by the spread (225 bps) gives 29.25 bps, which is 0.2925% once expressed in percentage terms.
Why the others are wrong
- **A:** Right magnitude but read 29.25 bps as 29.25% (forgot to convert basis points to percent).
- **B:** Divided EPE by the spread in percent: $13/2.25 \approx 5.77$.
- **D:** Divided with a further decimal slip: $1.3/2.25 \approx 0.577$.

Q999
BT Mock Q27 · BT answers P2.T6.915.1

REBUILT — Merged two identical questions; stem now states the single-period approximation (CVA = LGD x PD(0,T) x current value) explicitly and distractors were rebuilt from realistic errors.

Strongcore Financial has purchased a 3-year European at-the-money call option from a risky counterparty. S = K = USD 25.0 million, volatility = 20.0%, risk-free rate = 3.0% (continuous), T = 3.0 years, N(d1) = 0.66750, N(d2) = 0.53450. The counterparty has a constant hazard rate of 6.0% and may default at any time during the three years; recovery on default is 30.0% (LGD = 70.0%). Using the single-period approximation CVA $\approx$ LGD $\times$ PD$(0,T) \times V$, where V is the current value of the option (treated as the expected exposure) and PD(0,T) is the cumulative default probability, which is nearest to the CVA charge?
A. USD 188,000
B. USD 221,000
C. USD 516,000
D. USD 737,000

Answer: C
Step by step
1. Option value: $V = 25\text{m}(0.66750) - 25\text{m}\,e^{-0.03\times3}(0.53450) = 16.6875\text{m} - 12.2124\text{m} \approx 4.475\text{m}$.
2. Cumulative PD: $\text{PD}(0,3) = 1 - e^{-0.06\times3} = 1 - e^{-0.18} = 16.473\%$.
3. $\text{LGD} = 1 - 30\% = 70\%$.

4. CVA ≈ 0.70 × 0.16473 × 4.475m ≈ USD 516,000.

Why C is correct: CVA is the exposure (option value) multiplied by the probability the counterparty defaults before maturity and by the fraction lost on default (LGD).

Why the others are wrong

- **A:** Used the annual hazard rate 6% instead of the 3-year cumulative PD: 4.475m × 0.06 × 0.70 ≈ 188,000.
- **B:** Used the recovery rate (30%) instead of LGD (70%): 4.475m × 0.16473 × 0.30 ≈ 221,000.
- **D:** Omitted the LGD factor: 4.475m × 0.16473 ≈ 737,000, which ignores the recovered amount.

Q1000

BT answers P2.T6.417.3

REBUILT — Clarified that PD is over the option life and rebuilt distractors from realistic errors; corrected the garbled source explanation (put value verified at about 4.00).

CityBank holds a long position in a 5-year OTC European put option on an asset with current price 70.00 and strike 50.00. Asset volatility is 30.0% and the continuously compounded risk-free rate is 4.0%. N(d1) = 0.870 and N(d2) = 0.680. CityBank assumes the present-valued expected exposure to the counterparty equals the option's present value. The counterparty's probability of default over the option's life is 8.0% and LGD is 75.0%. Using the approximation CVA ≈ LGD × PD × V, which is nearest to the credit risk-adjusted value of the long put (risk-free value less CVA)?

A. 3.68
B. 3.76
C. 3.92
D. 4.24

Answer: B

Step by step

1. Put value: $p = Ke^{-rT}N(-d_2) - S\,N(-d_1)$ with $N(-d_1) = 0.130$ and $N(-d_2) = 0.320$.
2. $50e^{-0.2}(0.320) - 70(0.130) = 50(0.81873)(0.320) - 9.10 = 13.10 - 9.10 \approx 4.00$.
3. CVA ≈ 0.75 × 0.08 × 4.00 = 0.24.
4. Risky value = 4.00 - 0.24 = 3.76.

Why B is correct: Risky value equals the risk-free put value minus the CVA, which is LGD times PD times the option value.

Why the others are wrong

- **A:** Omitted the LGD: CVA = 0.08 × 4.00 = 0.32, giving 4.00 − 0.32 = 3.68.
- **C:** Used the recovery rate (25%) instead of LGD (75%): CVA = 0.25 × 0.08 × 4.00 = 0.08, giving 3.92.
- **D:** Added the CVA to the risk-free value (4.00 + 0.24) instead of subtracting; counterparty risk reduces value.

Q1001

BT answers P2.T6.914.3

REBUILT — Original included a CSA/collateral (outside LO 38.c, and the collateral figure was OCR-damaged); removed collateral so the question tests the plain CVA sum, stated the credit-triangle hazard assumption, and rebuilt distractors.

The CRO of Global Financial wants to price the counterparty risk of a 3-year derivative exposure to a major counterparty. There is no netting, collateral or wrong-way risk. The counterparty's default follows a constant hazard rate process, with the hazard rate implied by each year's data as $\lambda = s/(1-R)$. Forecast data:

Year 1	Year 2	Year 3	
Expected exposure (USD millions)	32.0	32.0	32.0
CDS spread (bps)	100	200	300
Recovery rate (%)	90%	80%	70%

The risk-free rate is 4.0% (continuous, flat term structure) and exposures are discounted at the end of each year. Which is nearest to the CVA for this position?

A. USD 2.30 million
B. USD 1.60 million
C. USD 1.54 million
D. USD 1.46 million

Answer: D

Step by step

1. Hazard rate each year: 1%/10% = 2%/20% = 3%/30% = 10% (constant).

2. Survival $e^{-0.1t}$: 0.9048, 0.8187, 0.7408, so marginal PDs are 9.516%, 8.611%, 7.791%.
3. LGD = 10%, 20%, 30%; discount factors $e^{-0.04t}$ = 0.9608, 0.9231, 0.8869.
4. CVA $= \sum \text{LGD}_t \cdot DF_t \cdot EE \cdot PD_t$: Year 1 = 0.10(0.9608)(32)(0.09516) = 0.2926m; Year 2 = 0.20(0.9231)(32)(0.08611) = 0.5087m; Year 3 = 0.30(0.8869)(32)(0.07791) = 0.6633m.
5. Total ≈ 0.2926 + 0.5087 + 0.6633 = 1.465 million.

Why D is correct: Each year's loss is LGD times discounted exposure times the marginal (interval) default probability; summing the three terms gives about USD 1.46 million.

Why the others are wrong

- **A:** Applied the Year-3 LGD (30%) to every year: total ≈ 2.30m. Each year has its own LGD.
- **B:** Did not discount the exposures ($DF = 1$): total ≈ 1.60m.
- **C:** Used one average LGD of 20% for all years instead of year-specific LGDs: total ≈ 1.54m.

Q1002

BT answers P2.T6.331.1

REBUILT — Formula image was missing from the stem; reproduced the standard CVA formula and numbered components explicitly.

The simplified credit value adjustment (CVA) with no wrong-way risk is $\text{CVA} = -\underbrace{\text{LGD}}_{(1)} \sum_{j=1}^{m} \underbrace{DF(t_j)}_{(2)} \underbrace{\text{EE}(t_j)}_{(3)} \underbrace{\text{PD}(t_{j-1}, t_j)}_{(4)}$. Each of the following statements about the four components is accurate, except which is inaccurate?

A. The first component is loss given default, i.e. one minus the expected recovery rate.
B. The second component is the risk-free discount factor for date t(j).
C. The third component is the expected exposure at the relevant future dates.
D. The fourth component is the conditional default probability, i.e. the probability of default in the interval given survival to t(j-1).

Answer: D

Step by step

1. Component (1) is LGD = 1 - recovery rate: A is accurate.
2. Component (2) is the risk-free discount factor at t_j: B is accurate.
3. Component (3) is the expected exposure at each date: C is accurate.
4. Component (4) is $\text{PD}(t_{j-1}, t_j)$, the marginal (unconditional) probability of default in the interval as seen today, not conditional on survival to t_{j-1}. D is inaccurate.

Why D is correct: The default probability in the formula is the marginal probability that default occurs within the interval, measured from today; the conditional probability given survival would be larger and would double-count survival.

Why the others are wrong

- **A:** Accurate: component (1) is LGD = 1 - R, so this is not the exception.
- **B:** Accurate: the second component is the risk-free discount factor, so this is not the exception.
- **C:** Accurate: the third component is the expected exposure profile, so this is not the exception.

Q1003

BT answers P2.T6.914.2

REBUILT — Stated hazard rate (6.7%) was inconsistent with a 400 bp spread and 50% recovery (should be 8%), and option D was a giveaway; corrected the data and rebuilt the distractors from realistic formula mistakes.

A risk manager needs a quick approximation of the CVA charge on a derivative, expressed as a running spread. The assumptions are: expected exposure is constant, so the expected positive exposure (EPE) is 5.0% of notional; the counterparty's credit spread is 400 bps per annum; the recovery rate is 50%; the implied constant hazard rate is 8.0%; and the maturity is about 5 years. There is no wrong-way risk, netting, or collateral. Which of the following is closest to the CVA as a running spread?

A. 10 bps per annum
B. 20 bps per annum
C. 40 bps per annum
D. 100 bps per annum

Answer: B

Key correction: Source key (A, 20 bps) was right, but the stem's 6.7% hazard rate was inconsistent with the spread and recovery inputs; it has been corrected to 8.0%.

Step by step

1. With constant EPE and constant default probability, CVA spread ≈ EPE × credit spread.
2. The credit spread already embeds loss given default (spread ≈ $\lambda \times$ LGD = 8% × 50% = 4%).
3. CVA spread = 5.0% × 400 bps = 0.05 × 400 = 20 bps per annum.

Why B is correct: For a running-spread approximation, the CVA charge is the average EPE multiplied by the counterparty's credit spread, which already reflects both default probability and LGD.

Why the others are wrong

- **A:** 10 bps multiplies by LGD (50%) again, double-counting recovery because the spread already includes LGD.
- **C:** 40 bps multiplies EPE by the hazard rate (5% x 8%) instead of the credit spread, ignoring LGD.
- **D:** 100 bps multiplies EPE x spread by the 5-year maturity (5% x 4% x 5 = 1.0% of notional). That is a cumulative, not per-annum, charge.

Q1004

GARP PE2 Q48

A risk analyst at a regional bank is estimating the unilateral credit valuation adjustment (UCVA) for an interest rate swap position with a counterparty bank. The analyst gathers the following information:
1-year risk-neutral default probability of the counterparty: 6.5%
Annual continuously compounded risk-free rate: 2.6%
Annual yield on the counterparty's credit instrument underlying the swap: 5.8%
Loss given default of the swap position: 75.0%
Average expected positive exposure (EPE) for the swap position: 5.0% (constant over time)
What is the best estimate of the position's UCVA as a spread?

A. -33 bps
B. -29 bps
C. -16 bps
D. -12 bps

Answer: C

Step by step

1. Credit spread = 5.8% − 2.6% = 3.2% (yield minus risk-free rate).
2. UCVA ≈ −average EPE × spread.
3. = −0.05 × 0.032 = −0.0016 = −16 bps.
4. The default probability and LGD are already embedded in the spread, so they are not used again.

Why C is correct: The running-spread approximation multiplies average EPE by the credit spread, measured as the credit yield less the risk-free rate.

Why the others are wrong

- **A:** -33 bps uses the 1-year default probability: 5.0% x 6.5% = 32.5 bps (about 33). A probability is not a spread.
- **B:** -29 bps uses the full 5.8% yield instead of the spread: 5.0% x 5.8% = 29 bps; the risk-free component was not removed.
- **D:** -12 bps multiplies by LGD again: 5.0% x 3.2% x 75% = 12 bps. The spread already reflects LGD, so this double-counts it.

LO 38.d

Evaluate the impact of changes in the credit spread and recovery rate assumptions on CVA.

Q1005

AP Mock 1 Q25

REBUILT — Original did not say whose CVA was meant and had giveaway options C/D; clarified that Panamba (the dealer) charges a unilateral CVA on Baroda and rewrote distractors as realistic mistakes.

Bank of Baroda often enters into swap contracts with Bank of Panamba, a major swap dealer. Panamba charges Baroda a unilateral credit value adjustment (CVA) that reflects Panamba's expected loss from a Baroda default on swaps where Baroda owes Panamba. Recently Panamba was downgraded from A to A-, and its credit spread rose from 38 bps to 128 bps. Baroda was downgraded from BB+ to BB, and its credit spread rose from 118 bps to 228 bps. Assuming exposures and recovery rates are unchanged, which of the following is the most likely response with respect to the CVA charge?

A. Panamba requests an increase in the CVA charge, because Baroda's wider credit spread implies a higher default

probability.

B. Baroda requests a reduction in the CVA charge, because Panamba's own spread has widened.

C. Panamba requests a reduction in the CVA charge, because the spread widening of both banks offsets in the CVA calculation.

D. No change, because both banks were downgraded by only one notch and the rating moves do not alter CVA.

Answer: A

Step by step

1. Unilateral CVA = LGD x sum of discounted EE x marginal default probability of the counterparty (Baroda) only.
2. Baroda's spread widened from 118 to 228 bps, an increase of 110 bps, so its implied default probability rises.
3. With exposure and recovery unchanged, a higher default probability raises CVA (LO 38.d: CVA generally increases with credit spread).
4. Panamba's own spread (38 to 128 bps) is not an input to a unilateral CVA, so it does not offset anything.
5. Therefore Panamba will ask for a larger CVA charge.

Why A is correct: Unilateral CVA depends on the counterparty's default probability; Baroda's spread rose by 110 bps, so the CVA charge on Baroda should rise.

Why the others are wrong

- **B:** Uses the wrong party's credit: the dealer's own spread does not enter the unilateral CVA the client pays, and Baroda's own deterioration pushes the charge up, not down.
- **C:** Offsets the two spread moves against each other; unilateral CVA ignores the dealer's own credit, and Baroda's move is the relevant one and is upward.
- **D:** Treats the downgrades as immaterial; a 110 bps spread widening on a BB-rated name is a large rise in implied default probability and CVA is directly spread-sensitive.

Q1006

AnalystPrep Q.1969

REBUILT — Original options 'reasonably small' vs 'insignificant' were not clearly distinguishable and gave no reasoning; reframed around the actual/market LGD cancellation with mechanism-based options.

A bank calibrates a counterparty's default probability from its CDS spread and then computes CVA. It assumes the actual LGD on the derivatives claim equals the market LGD used to imply default probability from the spread (same seniority), and it raises both from 50% to 60% (spread held constant). Which statement best describes the effect on CVA?

A. CVA rises sharply, because a higher LGD directly scales up the loss given default.

B. CVA changes only modestly, because a higher LGD lowers the market-implied default probability while raising the loss given default, and the two effects largely offset.

C. CVA is exactly unchanged, because the LGD terms cancel in the CVA formula to all orders.

D. CVA rises sharply, because a higher LGD raises both the implied default probability and the loss given default.

Answer: B

Step by step

1. CVA $\approx \text{LGD}_{actual} \sum \text{DF} \cdot \text{EE} \cdot \Delta PD$, with $PD \approx 1 - e^{-st/\text{LGD}_{mkt}}$.
2. Raising LGD_mkt lowers the implied default probability (the same spread is explained by a larger loss per default).
3. Raising LGD_actual increases the loss if default occurs.
4. With LGD_actual = LGD_mkt, the two effects cancel to first order (LGD times hazard rate is about the spread s).
5. The net effect is only second order: GARP's example shows changing LGD from 60% to 50% changes CVA by less than 2%. Modest, not zero.

Why B is correct: Higher LGD lowers market-implied default probability but raises loss in default; with equal actual and market LGD these largely cancel, leaving only a small second-order change.

Why the others are wrong

- **A:** Counts only the loss-severity effect and forgets that the implied default probability falls when LGD rises.
- **C:** Overstates the cancellation: it holds only to first order, so CVA still moves slightly (less than 2% for a 10-point LGD change in the GARP example).
- **D:** Gets the sign of the default-probability effect wrong: a higher recovery (lower LGD) raises the implied default probability, a higher LGD lowers it.

Q1007 AnalystPrep Q.5488

Quantum Financial Consulting is reassessing the risk factors that affect their Credit Value Adjustment (CVA) calculations across their derivative portfolio. They especially want to refine the sensitivity of their CVA to dynamic market conditions. How should Quantum approach the impact of varying credit spread and recovery rate assumptions on their CVA calculations?

A. Quantum should anticipate that CVA will likely decrease with increasing credit spread and increasing recovery rate, resulting from the offsetting effects.

B. Quantum should primarily focus on changes in recovery rate assumptions, as credit spread has minimal impact on CVA calculations according to sensitivity.

C. Quantum should expect that an increase in credit spread will generally lead to an increase in CVA, reflecting heightened credit risk and potential losses.

D. Quantum should adjust their models with the presumption that CVA does not change meaningfully with fluctuations in credit spread and recovery rate under normal market conditions.

Answer: C

Step by step

1. Wider credit spread means a higher implied default probability, so CVA generally rises (the relation is non-linear and reverses only for names very close to default).
2. Changing the recovery rate has a comparatively modest effect on CVA, because the implied default probability moves in the opposite direction to the loss given default.
3. Option C states the spread effect correctly. The other options misstate the spread or the combined sensitivity.

Why C is correct: CVA is increasing in credit spread in normal conditions because the counterparty's default probability rises, so heightened credit risk means a larger CVA.

Why the others are wrong

- **A:** Claims CVA falls when the spread rises; the spread effect has the opposite sign (CVA generally increases with spread), and the recovery effect is small rather than offsetting the spread effect.
- **B:** Treats the credit spread as unimportant; CVA generally increases with credit spread, whereas the LGD/recovery assumption has only a modest net effect.
- **D:** Assumes CVA is insensitive to both inputs; CVA is clearly sensitive to the credit spread, and only modestly to LGD.

Q1008 BT answers P2.T6.915.2

REBUILT — Formula exhibit was missing and option B was garbled; stated the CVA formula in the stem and rewrote option B as a plausible misconception.

Gregory's CVA approximation for a plain vanilla interest rate swap is $\text{CVA} \approx \text{LGD}_{actual} \sum_i \text{DF}(t_i)\,\text{EE}(t_i)\,[\,e^{-s_{i-1}t_{i-1}/\text{LGD}_{mkt}} - e^{-s_i t_i/\text{LGD}_{mkt}}\,]$, where s_i is the credit spread to t_i. Which of the following statements is TRUE?

A. CVA has a linear relationship with the credit spread.

B. CVA increases with the credit spread at every spread level, so the most distressed counterparty always has the largest CVA.

C. For a given cumulative default probability, CVA does not depend on the shape of the credit curve.

D. If settled (market) LGD is assumed equal to actual LGD, changing this common LGD has only a small effect on CVA.

Answer: D

Step by step

1. A and B false: CVA depends on the spread through the exponential default-probability terms, so it is not linear in the spread; CVA generally increases with spread but, at very large spreads, default is nearly certain early on and CVA converges toward the current exposure (zero for a par swap), so it falls. The most distressed name need not have the largest CVA.
2. C false: curves with the same 5-year cumulative default probability but different shapes (upward-sloping, flat, inverted) have different marginal default probabilities, so CVA differs; GARP shows the upward-sloping curve gives the largest CVA and the inverted curve the smallest.
3. D true: LGD_actual rises but the market-implied default probability falls, so the effects largely cancel; GARP notes changing LGD from 60% to 50% changes CVA by less than 2%.

Why D is correct: With equal actual and market LGD the loss-severity and implied-default-probability effects largely offset, leaving only a small second-order effect on CVA.

Why the others are wrong

- **A:** Assumes proportionality; default probability is a bounded exponential function of the spread, so CVA is non-linear in the spread.
- **B:** Extends the usual positive spread effect to all levels; for a counterparty very close to default CVA actually falls (and is zero once default has occurred).
- **C:** Ignores timing: with the same cumulative default probability, default probability concentrated early or late interacts differently with the EE profile, so curve shape changes CVA.

LO 38.e

Describe debt value adjustment (DVA) and bilateral CVA (BCVA).

Q1009

AnalystPrep Q.1930

Which of the following best describes the concept of bilateral exposure in any contract between an institution and the counterparty?

A. This is the exposure owed to the counterparty.

B. When the counterparty defaults, the institution is only paid a recovery fraction of their exposure which is a loss for the institution. On the other hand, when the institution defaults, the counterparty receives a recovery fraction of the negative exposure which is a gain for the institution.

C. This is the worst exposure an institution could have at a certain time in the future measured at a specified level of confidence.

D. This is the amount expected to be lost if the counterparty defaults when the mark-to-market is positive.

Answer: B

Step by step

1. Bilateral counterparty risk assumes that BOTH parties can default.
2. If the counterparty defaults while the institution's MtM is positive, the institution recovers only a fraction of what it is owed: a loss (the CVA side).
3. If the institution defaults while its MtM is negative, it pays only a recovery fraction of what it owes: a gain to the institution (the DVA side).
4. Only option B captures both sides.

Why B is correct: Bilateral exposure recognizes that each party faces the other's default: losses on positive exposure when the counterparty defaults and gains on negative exposure when the institution itself defaults.

Why the others are wrong

- **A:** Describes only the negative-exposure (liability) side; it ignores the recovery mechanics and the counterparty-default side, so it is not the bilateral concept.
- **C:** This is potential future exposure (PFE), a high-percentile exposure measure, not bilateral exposure.
- **D:** This is a one-sided (unilateral) expected loss / CVA-type definition that considers only the counterparty's default.

Q1010

BT answers P2.T6.333.2

REBUILT — Original option B tied the sign of BCVA to risky vs risk-free value, which flips with sign convention (Schweser vs GARP) and made B arguably false too; reworded B convention-free and tightened D.

Each of the following statements about the bilateral credit value adjustment (BCVA) is true EXCEPT which is not necessarily true?

A. The second BCVA term is the debt value adjustment (DVA); it corresponds to the institution defaulting first, when it books a gain if its mark-to-market is negative.

B. Unlike a unilateral CVA, which can only reduce the value of a position, BCVA can increase it when the DVA term outweighs the CVA term, so the risky value can exceed the risk-free value.

C. Netting does not always benefit the institution under BCVA; it can be a disadvantage when the DVA term dominates, which normally corresponds to the institution being riskier than its counterparty.

D. If all market participants agree on the BCVA approach and parameters, the total counterparty risk in the market (the sum of all BCVAs) equals the sum of all expected exposures.

Answer: D

Step by step

1. Under bilateral risk, BCVA = CVA + DVA; the DVA term mirrors CVA using the institution's own default probability and the expected negative exposure.
2. A: DVA is a gain when the institution defaults first with negative MtM: true.
3. B: because DVA offsets CVA, the net adjustment can favor the institution (DVA > CVA): true.
4. C: netting can hurt when DVA dominates (institution riskier): true.
5. D: with agreed parameters, one party's BCVA is +X and the other's is -X, so the sum of all BCVAs is zero, not the sum of expected exposures: false.

Why D is correct: BCVA is symmetric: each party's adjustment is the exact opposite of its counterparty's, so market-wide BCVAs net to zero in theory.

Why the others are wrong

- **A:** True: it is the standard description of DVA. Rejecting it would confuse DVA with a loss.
- **B:** True: DVA can outweigh CVA, so BCVA can raise the risky value above the risk-free value; unilateral CVA never can.
- **C:** True: with netting, the DVA benefit of discarding negative-MtM trades is lost when DVA dominates, i.e. the institution is the riskier party.

Q1011

AnalystPrep Q.5489

Acme Hedging Solutions is optimizing its valuation methods for derivative instruments, keen on incorporating both the firm's own credit risk and that of its counterparties. In light of this, how should Acme proceed with respect to understanding the implications of Debt Value Adjustment (DVA) and Bilateral Credit Value Adjustment (BCVA)?

A. Focus only on CVA because DVA is no longer a recognized method for adjusting valuations due to changes in the firm's own credit risk.

B. Implement DVA adjustments to reflect changes in the market's perception of Acme's credit risk, applying it in conjunction with CVA to determine BCVA for comprehensive valuation.

C. Use DVA exclusively for valuation, considering that it has now superseded CVA as the industry standard for incorporating credit risk in valuations.

D. Calculate BCVA as an extension of CVA but without including DVA, as including the firm's own credit risk may lead to double-counting in the valuation adjustments.

Answer: B

Step by step

1. Bilateral risk means both the counterparty and Acme can default.
2. CVA prices the counterparty's default on Acme's positive exposure; DVA prices Acme's own default on its negative exposure.
3. BCVA = CVA + DVA, so both are needed.
4. Only B combines the two.

Why B is correct: BCVA is by definition CVA plus DVA, capturing both the counterparty's and the institution's own credit risk.

Why the others are wrong

- **A:** DVA is recognized (accounting rules since FAS 157 / IFRS 13 require own credit risk); ignoring it reverts to unilateral CVA.
- **C:** DVA complements rather than replaces CVA; using it alone ignores counterparty default.
- **D:** A BCVA without DVA is just a unilateral CVA. DVA is the mirror-image term for the institution's own default, not double counting.

Q1012

BT answers P2.T6.333.1

REBUILT — Original formula exhibit was missing; the stem now states the first BCVA term explicitly so the question is self-contained, and distractors made more realistic.

The first term of Gregory's bilateral CVA (BCVA) formula, from the viewpoint of an Institution (I) with a Counterparty (C), is $-\text{LGD}_C \sum_j \text{EE}(t_j)\, S_I(t_{j-1})\, \text{PD}_C(t_{j-1}, t_j)$. It generalizes the unilateral CVA by adding the multiplicative factor $S_I(t_{j-1})$. What does $S_I(t_{j-1})$ represent?

A. The probability that the Institution has survived (not defaulted) up to time t_{j-1}

B. The probability that the Counterparty has survived up to time t_{j-1}

C. The probability that the Institution has defaulted before time t_{j-1}

D. The recovery rate on the Institution's collateral at time t_{j-1}

Answer: A

Step by step

1. Notation: S is a survival probability and the subscript I identifies the Institution.
2. The CVA term is a loss on the counterparty's default between t_{j-1} and t_j, which is captured by PD_C.
3. That loss only occurs if the Institution has not itself defaulted earlier, so the term is weighted by the Institution's survival probability $S_I(t_{j-1})$.
4. So the factor is the Institution's survival probability, and the term shrinks relative to unilateral CVA.

Why A is correct: Counterparty default only hurts the Institution if the Institution is still alive at that time, so the CVA term is conditioned on the Institution's survival.

Why the others are wrong

- **B:** The counterparty's default probability is already in PD_C; its survival probability appears in the second (DVA) term instead.
- **C:** This is $1 - S_I$; using it would increase rather than reduce the CVA term, reversing the first-to-default logic.
- **D:** S denotes a survival probability, not a recovery rate; recovery enters through LGD $= 1 -$ recovery.

Q1013

BT answers P2.T6.916.3

REBUILT — Option D (funding-benefit view of DVA) is not in the Schweser LO coverage; replaced it with a true statement on DVA as the mirror image of CVA.

Each of the following statements about the debt value adjustment (DVA) is true EXCEPT which is false?

A. Bilateral CVA is the sum of the credit value adjustment (CVA) and DVA.

B. Compared with the risky value that includes only the unilateral CVA term, adding DVA increases the value of the risky position.

C. DVA is the formalization of a term that incorporates specific wrong-way risk (i.e., dependence between exposure and counterparty default) in the CVA framework.

D. DVA is the mirror image of CVA: it is based on the expected negative exposure, the institution's own default probability and its loss given default.

Answer: C

Step by step

1. BCVA = CVA + DVA: A is true.
2. CVA is a cost (reduces value) and DVA is a gain, so adding DVA offsets part of the CVA and raises the risky value: B is true.
3. DVA mirrors CVA using ENE and the institution's own default probability: D is true.
4. DVA reflects the institution's own default, not any dependence between exposure and default (wrong-way risk), which is a separate concept: C is false.

Why C is correct: DVA is the bilateral complement to CVA for the institution's own default; wrong-way risk is a separate dependence effect that is not what DVA formalizes.

Why the others are wrong

- **A:** True by definition of BCVA; rejecting it confuses BCVA with a unilateral measure.
- **B:** True: DVA is a gain that offsets CVA, so risky value is higher than under unilateral CVA alone.
- **D:** True: this is exactly the mirror-image structure of the CVA formula, with ENE and the institution's own default probability.

Q1014

AP Mock 1 Q4

REBUILT — Original asked about 'DVA and CVA on both sides' loosely; stem now fixes that exposure profiles are unchanged so only the spread effect is tested, and the options are worded in terms of magnitude.

Northern Star Bank (NSB) and Horizon Financial Group (HFG) entered into a 4-year interest rate swap in which NSB pays a fixed rate of 4.5% and receives 6-month SOFR plus a spread. Since inception, both entities' credit ratings have improved: NSB's credit spread fell from 90 bps to 40 bps and HFG's from 130 bps to 100 bps. Assume the SOFR curve is unchanged, so the exposure profiles (EPE and ENE) of an identical 4-year swap initiated today are the same as at inception. Which of the following statements is most likely correct for the new swap?

A. Because NSB's spread fell by more than HFG's, NSB's DVA will decrease and HFG's DVA will increase.

B. Because NSB's spread fell by more than HFG's, NSB's CVA will decrease and HFG's CVA will increase.

C. Because both spreads fell, the CVA and DVA recorded by each party will be lower in magnitude.

D. Because both spreads fell, the CVA and DVA recorded by each party will be higher in magnitude.

Answer: C

Step by step

1. Each party's CVA is driven by the counterparty's spread: CVA $\approx -\text{spread}_{cp} \times \text{EPE}$.
2. Each party's DVA is driven by its own spread: DVA $\approx \text{spread}_{own} \times \text{ENE}$.
3. NSB's CVA depends on HFG's spread (130 -> 100 bps, lower); NSB's DVA depends on NSB's own spread (90 -> 40 bps, lower).
4. HFG's CVA depends on NSB's spread (lower); HFG's DVA depends on HFG's own spread (lower).
5. With exposures unchanged, all four adjustments shrink in magnitude. How much each spread fell relative to the other has no bearing on the direction.

Why C is correct: CVA and DVA are each proportional to a credit spread times an exposure measure; with both spreads lower and exposures unchanged, every adjustment falls in magnitude for both parties.

Why the others are wrong

- **A:** A party's DVA depends only on its own spread, which fell for both NSB and HFG, so HFG's DVA cannot rise; this compares the size of the two spread changes, which is irrelevant.
- **B:** HFG's CVA is driven by NSB's spread, which fell, so it decreases; the error is netting the two spread changes against each other instead of applying each spread to its own term.
- **D:** Lower spreads mean lower default probabilities, so adjustments shrink; higher adjustments would require spreads to widen.

Q1015

BT notes P2.T5.507.3

REBUILT — Original relied on FVA and collateral rate adjustment (not in Schweser LOs); rebuilt as a CVA/DVA/BCVA statement question with one false statement.

Which of the following statements about credit value adjustment (CVA), debt value adjustment (DVA) and bilateral CVA (BCVA) is FALSE?

A. CVA is the cost to a party arising from the possibility that its counterparty defaults.

B. DVA is the benefit to a party arising from the possibility that it defaults itself, and (with both parties using the same inputs) equals the counterparty's CVA on the same position.

C. BCVA equals CVA plus DVA, so it is smaller in absolute terms than unilateral CVA when DVA is positive.

D. All else equal, a widening of a party's own credit spread reduces its DVA.

Answer: D

Step by step

1. Check each statement against the definitions.
2. A: CVA is the expected loss from the counterparty's default, a cost to the party. True.
3. B: DVA is the mirror image of the counterparty's CVA: the benefit of not paying in full if the party itself defaults. True.
4. C: BCVA $=$ CVA $+$ DVA with CVA negative and DVA positive, so the net charge is smaller than unilateral CVA. True.
5. D: a wider own spread means a higher own default probability, so the benefit from defaulting is larger and DVA increases. The statement says it falls, so it is false.

Why D is correct: DVA rises when the party's own default probability rises (wider own spread), so a statement that it falls is false.

Why the others are wrong

- **A:** True definition of CVA; selecting it shows confusion about which party bears the cost.
- **B:** True; DVA is the counterparty's CVA seen from the other side. Rejecting it confuses DVA with a cost.
- **C:** True; DVA offsets CVA in BCVA. Rejecting it forgets DVA enters with the opposite sign.

LO 38.f

Explain the differences between unilateral CVA (UCVA) and BCVA, and between unilateral DVA (UDVA) and BCVA.

Q1016

BT answers P2.T6.24.34.3

REBUILT — Original options were easily eliminated (third-party default, role reversals); rebuilt to test the real UCVA/UDVA/BCVA distinction, including that BCVA is not simply UCVA plus UDVA because of the first-to-default (survival) adjustment.

A bank values a 5-year, $50 million interest rate swap with a corporate client. The risk team compares the unilateral CVA (UCVA), unilateral DVA (UDVA) and bilateral CVA (BCVA) of the trade. Which of the following statements correctly distinguishes them?

A. UCVA reflects default of both parties; UDVA reflects only the client's default; BCVA is exactly UCVA plus UDVA.

B. UCVA reflects only the client's default and UDVA only the bank's own default; BCVA is exactly UCVA plus UDVA because the two defaults are treated independently.

C. UCVA reflects only the client's default and UDVA only the bank's own default; BCVA reflects both, with each default conditioned on the other party not having defaulted first, so its CVA and DVA components are smaller in magnitude than UCVA and UDVA.

D. UCVA reflects only the bank's own default and UDVA only the client's default; BCVA reflects both defaults with first-to-default survival adjustments.

Answer: C

Step by step

1. UCVA assumes the bank itself is default-free and charges only for the client's default: UCVA $= -\text{LGD}_C \sum \text{PD}_C$ EPE, with no survival term for the bank.
2. UDVA is the mirror image: it assumes the client is default-free and credits the bank for its own default on negative exposure (ENE).
3. BCVA = CVA + DVA includes both parties. Each term is multiplied by the probability that the other party has survived (first-to-default effect), so each component is smaller in magnitude than its unilateral counterpart.
4. Therefore BCVA does not equal UCVA + UDVA (GARP's example: UCVA + UDVA = -21.8 bps vs BCVA = -21.5 bps).

Why C is correct: The unilateral measures each ignore the other party's default, whereas BCVA conditions each default on the other party surviving, so it is not a simple sum of the two unilateral numbers.

Why the others are wrong

- **A:** UCVA only captures the client's default and UDVA only the bank's own; saying UCVA covers both parties and that BCVA is a plain sum mixes up the definitions.
- **B:** The definitions of UCVA and UDVA are right, but treating the defaults as independent and simply adding ignores the survival (first-to-default) conditioning in BCVA, so the sum overstates the components.
- **D:** This reverses the roles: UCVA is the charge for the counterparty's default, and UDVA is the benefit from the bank's own default.

Q1017

Mock 3 Q43 · GARP PE1 Q37 (same question per source note)

REBUILT — Wording of 'change in the charge' was ambiguous and one distractor (141,600) had no derivation; stem now specifies survival-adjusted BCVA and the decrease in the charge, and distractors map to real errors.

The head of a CVA desk finds that a junior analyst may have reported the 1-year unilateral CVA (UCVA) instead of the bilateral CVA (BCVA) on a derivatives portfolio. The discounted expected positive exposure of the financial institution to the counterparty is AUD 48 million, equal to the discounted expected positive exposure of the counterparty to the institution (so ENE = 48 million in magnitude). Other inputs for the coming year:

Parameter	Financial institution	Counterparty
Annual probability of default	3.3%	1.5%
Expected recovery rate	75%	90%

BCVA is computed with first-to-default (survival) adjustments: each party's term is multiplied by the other party's survival probability. From the institution's perspective, by approximately how much does the CVA charge decrease when BCVA is used instead of UCVA?

A. AUD 320,400

B. AUD 390,100

C. AUD 392,400

D. AUD 462,100

Answer: C

Step by step

1. UCVA charge $= \text{LGD}_C \times \text{PD}_C \times \text{EPE} = 0.10 \times 0.015 \times 48{,}000{,}000 = 72{,}000$.
2. CVA leg in BCVA $= 72{,}000 \times (1 - 0.033) = 69{,}624$ (institution must survive).

3. DVA leg $= \text{LGD}_{FI} \times \text{PD}_{FI} \times (1 - \text{PD}_C) \times \text{ENE} = 0.25 \times 0.033 \times 0.985 \times 48{,}000{,}000 = 390{,}060$.
4. BCVA charge $= 69{,}624 - 390{,}060 = -320{,}436$ (a net benefit).
5. Change $= -320{,}436 - 72{,}000 = -392{,}436$, i.e. the charge falls by about AUD 392,400.

Why C is correct: Moving from UCVA to BCVA slightly shrinks the CVA leg and subtracts a much larger DVA leg (the institution is riskier and has a lower recovery), so the charge falls by about 392,400.

Why the others are wrong

- **A:** 320,400 is the level of the BCVA (a 320,436 benefit), not the change; the UCVA charge of 72,000 was not subtracted.
- **B:** 390,100 is only the DVA leg (390,060); it ignores that the CVA leg also changes (72,000 -> 69,624) and that the comparison base is UCVA.
- **D:** 462,100 adds the DVA leg to the UCVA (390,060 + 72,000) instead of subtracting the CVA, a sign error on the CVA component.

Q1018

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A bank's own credit spread is much wider than that of a highly rated corporate client. The bank's valuation team compares the unilateral CVA (UCVA) and the bilateral CVA (BCVA) of a swap portfolio it holds with the client. Which of the following statements is correct?

A. BCVA is always more negative than UCVA, because it adds the cost of the bank's own default to the cost of the client's default.

B. UCVA can turn positive when the bank's own credit risk is large, whereas BCVA is always a cost to the bank.

C. UCVA and BCVA will be equal, because the bank's own default cannot affect the value of a trade in which the client is the only party at risk.

D. BCVA can be positive because the debt value adjustment (DVA) may exceed the CVA, whereas UCVA can only be a cost to the bank.

Answer: D

Step by step

1. UCVA reflects only the client's default, so it is a cost (negative adjustment) to the bank.
2. BCVA adds the DVA term, which is a gain to the bank because on its own default it would pay only a recovery amount on what it owes.
3. When the bank is much riskier than the client, DVA can dominate CVA, so BCVA is positive and the risky value exceeds the risk-free value.

Why D is correct: BCVA contains both CVA (a cost) and DVA (a gain); with a much riskier bank, DVA can dominate and BCVA turns positive, which cannot happen with UCVA alone.

Why the others are wrong

- **A:** DVA is a gain, not an additional cost, so BCVA is generally less negative than UCVA and may even be positive; adding the two as costs is the error.
- **B:** Reversed: the unilateral CVA can only be a cost, while the bilateral measure can be positive.
- **C:** In a bilateral setting both parties can default; the bank's own default gives rise to DVA, so BCVA differs from UCVA.

LO 38.g

Calculate DVA, BCVA, and BCVA as a spread.

Q1019

AnalystPrep Q.3084

REBUILT — Original treated BCVA spread as counterparty spread minus own spread with ambiguous party pairing and no exposure inputs; rebuilt as a clear BCVA-spread change question using the EPE/ENE formula.

Bank X has a swap with Bank Y. From Bank X's perspective the expected positive exposure (EPE) is 8% and the magnitude of the expected negative exposure (ENE) is 5%. Bank Y's credit spread is 200 bps and Bank X's own credit spread is initially 120 bps. Bank X is then downgraded and its own credit spread widens to 180 bps; Bank Y's spread and both exposures are unchanged. Using the approximation $\text{BCVA} = -\text{EPE} \times s_Y - \text{ENE} \times s_X$ with ENE entered as a negative number, by how much does Bank X's BCVA (as a running spread) change?

A. +3 bps (BCVA becomes less negative)

B. -3 bps (BCVA becomes more negative)
C. +4.8 bps
D. -7 bps

Answer: A
Step by step
1. Before: BCVA $= -0.08 \times 200 - (-0.05) \times 120 = -16 + 6 = -10$ bps.
2. After: BCVA $= -0.08 \times 200 + 0.05 \times 180 = -16 + 9 = -7$ bps.
3. Change $= -7 - (-10) = +3$ bps.
4. Only the DVA term moves: $0.05 \times (180 - 120) = +3$ bps. Bank X's own deterioration raises its DVA benefit.

Why A is correct: BCVA = CVA + DVA; DVA is proportional to the firm's own spread times ENE, so a wider own spread raises DVA and BCVA rises (less negative) while CVA is unchanged.

Why the others are wrong
- **B:** Sign error: assumes a downgrade of one's own credit must make BCVA worse, but the DVA leg is a gain that grows when own spread widens.
- **C:** Applied EPE (8%) instead of ENE (5%) to the change in own spread: $0.08 \times 60 = 4.8$.
- **D:** This is the new BCVA level (-7 bps), not the change from the original -10 bps.

Q1020

Mock 4 Q49

REBUILT — Source treated BCVA as a negative that was keyed by magnitude; made sign and survival convention explicit and replaced unexplained distractor.

A senior risk manager at Bank Gamma is presenting to newly hired risk analysts on calculating bilateral CVA (BCVA). Bank Gamma and Bank Phi are the only counterparties to each other. The discounted expected positive exposure of Gamma to Phi is CNY 60 million and the discounted expected negative exposure (magnitude) is CNY 45 million. Use the convention BCVA = CVA + DVA, where CVA is a negative number (a cost) and DVA is positive for the firm, and apply the first-to-default adjustment (each leg is multiplied by the survival probability of the other party), assuming independent defaults.

Parameter	Bank Gamma	Bank Phi
Annual probability of default	2.5%	1.8%
Recovery rate	82%	92%

What is the BCVA from Bank Gamma's perspective?

A. CNY 84,240
B. CNY 114,615
C. CNY 198,855
D. CNY 283,095

Answer: B
Key correction: Source labelled the BCVA -114,615 (CVA - DVA) but keyed the magnitude. Under BCVA = CVA + DVA with CVA negative, it is +114,615 (a net benefit to Gamma).
Step by step
1. CVA $= -60{,}000{,}000 \times (1 - 0.92) \times 0.018 \times (1 - 0.025) = -84{,}240$.
2. DVA $= +45{,}000{,}000 \times (1 - 0.82) \times 0.025 \times (1 - 0.018) = +198{,}855$.
3. BCVA $= -84{,}240 + 198{,}855 = +114{,}615$, a net benefit to Gamma.

Why B is correct: Gamma is the weaker credit with a large ENE, so its DVA gain exceeds its CVA cost and the BCVA is a positive CNY 114,615.

Why the others are wrong
- **A:** CVA leg only (cost); ignored DVA.
- **C:** DVA leg only (benefit); ignored the CVA cost.
- **D:** Added the two legs instead of netting: 84,240 + 198,855.

Q1021

AP Mock 2 Q77

REBUILT — Added explicit survival-adjustment convention to the stem and replaced an unexplained distractor with the DVA-as-cost error.

ABC Bank and Prime Bank are counterparties. Prime Bank's discounted expected positive exposure to ABC is USD 2,500,000 and its discounted expected negative exposure (magnitude) is USD 1,800,000. Use the convention BCVA = CVA + DVA, where CVA is a negative number (a cost) and DVA is positive for the firm, and apply the first-to-default adjustment (each leg is multiplied by the survival probability of the other party), assuming independent defaults.

Parameter	Prime Bank	ABC Bank
Annual probability of default	3.1%	2.3%
Recovery rate	87%	91%

What is Prime Bank's bilateral CVA?

A. USD 5,015 (cost to Prime)
B. USD 7,087 (benefit to Prime)
C. USD 2,073 (net benefit to Prime)
D. USD 12,102 (net cost to Prime)

Answer: C

Key correction: Source key (C) is retained; the source labelled the result as -2,072 while the option is positive. Under BCVA = CVA + DVA with CVA negative, the result is +2,073, a net benefit to Prime.

Step by step

1. CVA (Prime's loss from ABC defaulting) $= -\text{EPE} \times \text{LGD}_{ABC} \times \text{PD}_{ABC} \times (1 - \text{PD}_{Prime}) = -2{,}500{,}000 \times 0.09 \times 0.023 \times 0.969 = -5{,}014.6$.
2. DVA (Prime's gain from its own default) $= +1{,}800{,}000 \times \text{LGD}_{Prime} \times \text{PD}_{Prime} \times (1 - \text{PD}_{ABC}) = 1{,}800{,}000 \times 0.13 \times 0.031 \times 0.977 = +7{,}087.2$.
3. BCVA $= -5{,}014.6 + 7{,}087.2 = +2{,}072.6 \approx$ USD 2,073, a net benefit to Prime.

Why C is correct: BCVA nets the CVA cost against the DVA gain; each leg uses the other party's exposure with own LGD/PD of the party that defaults, and DVA exceeds CVA here, so BCVA is positive.

Why the others are wrong

- **A:** CVA leg alone; ignores the DVA benefit.
- **B:** DVA leg alone; ignores the CVA cost.
- **D:** Added the two legs (treated DVA as a further cost) instead of netting CVA against DVA: 5,015 + 7,087.

Q1022

BT answers P2.T6.901.3

Herman, Ferry, and Quitzon LLC is a market maker with a position in an over-the-counter (OTC) derivatives contract. For this position, they have estimated the following values:
- The riskfree value of their position is +$32.0 million
- The credit valuation adjustment (CVA) is $5.0 million
- The debt valuation adjustment (DVA) is $8.0 million

Based only on this information, which of the following is the best estimate of the RISKY VALUE of the position from Herman's perspective?

A. Risky value = $27.0 million
B. Risky value = $29.0 million
C. Risky value = $35.0 million
D. Without estimates for the counterparty's default probability (PD) and loss given default (LGD), we cannot estimate a risky value

Answer: C

Step by step

1. Risky value = riskfree value +BCVA, where BCVA $= -\text{CVA} + \text{DVA}$ (CVA entered as a cost, DVA as a benefit).
2. BCVA $= -5.0 + 8.0 = +3.0$ million.
3. Risky value $= 32.0 + 3.0 = 35.0$ million.

Why C is correct: The firm's own default option (DVA) is worth more than the cost of counterparty default (CVA), so the risky value is above the riskfree value.

Why the others are wrong

- **A:** Deducted only the CVA (32 - 5) and ignored DVA.
- **B:** Reversed the signs: 32 - 8 + 5, treating DVA as a cost and CVA as a gain.
- **D:** CVA and DVA are already given as dollar amounts, so no further PD/LGD inputs are needed.

Q1023

AP Mock 3 Q66

REBUILT — Two distractors were unexplained; ENE entered as -0.09 beside a percentage EPE; rebuilt with ENE stated as -9.0% and distractors tied to specific mistakes.

A trading firm needs a quick estimate of a swap's bilateral credit value adjustment. The expected positive exposure for a trade of this type is 13.6% and the expected negative exposure is -9.0%. The counterparty's credit spread is 267 bps per annum and the firm's own credit spread is 191 bps per annum. Using BCVA $= -\text{EPE} \times s_C - \text{ENE} \times s_P$, which is nearest to the BCVA?

A. -53.502 bps
B. -19.122 bps
C. -1.946 bps
D. -36.312 bps

Answer: B

Step by step

1. CVA term: $-0.136 \times 267 = -36.312$ bps.
2. DVA term: $-(-0.09) \times 191 = +17.190$ bps.
3. BCVA $= -36.312 + 17.190 = -19.122$ bps.

Why B is correct: BCVA as a spread nets the counterparty-default cost (EPE x counterparty spread) against the own-default benefit (ENE x own spread).

Why the others are wrong

- **A:** Treated the DVA term as a further cost: -36.312 - 17.190.
- **C:** Swapped the spreads (EPE x own spread, ENE x counterparty spread): -0.136x191 + 0.09x267.
- **D:** CVA leg only; omitted the DVA benefit.

Q1024

BT answers P2.T6.24.34.2

REBUILT — Source arithmetic and options did not reconcile (inputs give -21.7 bps, no option matched); fixed options and units.

A bank has a portfolio of interest rate swaps with the following characteristics: the average expected positive exposure (EPE) is 9.8% and the average expected negative exposure (ENE) magnitude is 7.4%. The credit spread of the counterparties is 750 bps and the bank's own credit spread is 700 bps. Using BCVA $= -\text{EPE} \times s_C - \text{ENE} \times s_P$ (ENE entered as a negative number), calculate the BCVA for the portfolio.

A. -73.5 bps
B. -21.7 bps
C. +21.7 bps
D. -125.3 bps

Answer: B

Key correction: Source key (B, -7.3) does not follow from the stated inputs, which give -21.7 bps; options were rebuilt around the correct figure.

Step by step

1. CVA term: $-0.098 \times 750 = -73.5$ bps.
2. DVA term: $-(-0.074) \times 700 = +51.8$ bps.
3. BCVA $= -73.5 + 51.8 = -21.7$ bps.

Why B is correct: The own-default benefit (ENE x own spread) partially offsets the counterparty-default cost, leaving a net cost of 21.7 bps.

Why the others are wrong

- **A:** CVA leg only; ignored DVA.
- **C:** Sign error: reported the net adjustment as a gain.
- **D:** Treated DVA as a cost: -73.5 - 51.8.

Q1025

BT answers P2.T6.333.3

REBUILT — OCR-damaged options (73/77 bps) and ambiguous sign convention; rebuilt options with signs and error-based distractors.

A trader needs a very quick estimate of the BCVA on a swap. The EPE is 7.0% and the ENE magnitude is 4.0%. The counterparty's credit spread is 300 bps per annum and the credit spread of the trader's own institution is 200 bps per annum. Using BCVA $= -\text{EPE} \times s_C - \text{ENE} \times s_P$ with ENE entered as a negative number, which is nearest to the BCVA?

A. -29 bps
B. -13 bps
C. +13 bps
D. -2 bps

Answer: B
Key correction: Source key (C, 13 bps) is the magnitude of the charge; under the BCVA = CVA + DVA sign convention the value is -13 bps.
Step by step
1. CVA term: $-0.07 \times 300 = -21$ bps.
2. DVA term: $-(-0.04) \times 200 = +8$ bps.
3. BCVA $= -21 + 8 = -13$ bps (a net cost to the trader).

Why B is correct: The DVA benefit of 8 bps reduces the 21 bps CVA cost to a net 13 bps cost.
Why the others are wrong
- **A:** Treated DVA as a cost: -21 - 8.
- **C:** Sign error: magnitude right, but the net adjustment is a cost, not a gain.
- **D:** Swapped the spreads: -0.07x200 + 0.04x300.

Q1026

BT answers P2.T6.417.2

REBUILT — Options were positive magnitudes with no sign convention and 3 unexplained values; rebuilt with signed options and mistake-based distractors (PFE kept as a red herring).

CityBank is going to enter into a swap with FinCorp. From CityBank's perspective, the EPE is 6.00% and the ENE magnitude is 4.20%. CityBank's credit spread is 110 bps per annum, while FinCorp's credit spread is 250 bps. The maximum PFE is 18.0%. Using BCVA $= -\text{EPE} \times s_C - \text{ENE} \times s_P$ with ENE entered as a negative number, which is nearest to CityBank's BCVA?

A. -19.62 bps
B. -10.38 bps
C. +10.38 bps
D. +3.90 bps

Answer: B
Key correction: Source key (B, 10.38 bps) is the magnitude of the charge; under the BCVA = CVA + DVA sign convention the value is -10.38 bps.
Step by step
1. CVA term (FinCorp default): $-0.06 \times 250 = -15.00$ bps.
2. DVA term (CityBank default): $-(-0.042) \times 110 = +4.62$ bps.
3. BCVA $= -15.00 + 4.62 = -10.38$ bps. Maximum PFE is not used.

Why B is correct: BCVA nets the 15 bps CVA cost against the 4.62 bps DVA gain; PFE is a tail measure and does not enter the expected-exposure formula.
Why the others are wrong
- **A:** Treated DVA as a cost: -15.00 - 4.62.
- **C:** Sign error on the net result.
- **D:** Swapped the spreads: -0.06x110 + 0.042x250.

Q1027

GARP PE1 Q37

REBUILT — Added explicit survival-adjustment convention and replaced unexplained distractor with an error-based one.

The manager of the CVA desk of a financial institution is reviewing a 1-year bilateral CVA (BCVA) on derivative positions with a counterparty. The manager suspects the analyst computed the unilateral CVA (UCVA) instead of the BCVA. The discounted expected positive exposure of the financial institution to the counterparty is AUD 48 million for the coming year, and equals the discounted expected positive exposure of the counterparty to the financial institution. Use the convention BCVA = CVA + DVA, where CVA is a negative number (a cost) and DVA is positive for the firm, and apply the first-to-default adjustment (each leg is multiplied by the survival probability of the other party), assuming independent defaults.

Parameter	Financial Institution	Counterparty
Annual probability of default	3.3%	1.5%
Expected recovery rate	75%	90%

From the institution's perspective, approximately what is the change in the charge if the analyst derives the BCVA instead of the UCVA?

A. AUD 248,400
B. AUD 320,400
C. AUD 390,100
D. AUD 392,400

Answer: D

Step by step

1. UCVA $= -48{,}000{,}000 \times 0.10 \times 0.015 = -72{,}000$ (no survival adjustment in the unilateral measure).
2. CVA leg in BCVA $= -72{,}000 \times (1 - 0.033) = -69{,}624$.
3. DVA leg $= 48{,}000{,}000 \times 0.25 \times 0.033 \times (1 - 0.015) = +390{,}060$.
4. BCVA $= -69{,}624 + 390{,}060 = +320{,}436$.
5. Change $= 320{,}436 - (-72{,}000) = 392{,}436 \approx$ AUD 392,400.

Why D is correct: Moving from UCVA to BCVA adds the DVA benefit and slightly shrinks the CVA cost, so the position improves by the difference between the two measures, not just the BCVA level.

Why the others are wrong

- **A:** Subtracted instead of adding the UCVA cost: 320,436 - 72,000.
- **B:** The BCVA level (320,436), not the change from the UCVA.
- **C:** The DVA leg alone (390,060); ignored the survival effect on the CVA leg and the removal of the UCVA cost.

Q1028

Mock 1 Q26

REBUILT — Options C and D were both defensible (DVA is positive by convention); replaced D with a false statement so exactly one option is correct.

A firm has entered into a portfolio of derivative transactions with a single counterparty. The expected positive exposure is 7% of notional and the expected negative exposure magnitude is 5% of notional. The counterparty's credit spread is 330 bps and the firm's own credit spread is 250 bps. Using BCVA $= -\text{EPE} \times s_C - \text{ENE} \times s_P$ (ENE entered as negative), which of the following is correct regarding the firm's BCVA?

A. The CVA component will be positive for the firm.
B. The BCVA will be positive for the firm.
C. The BCVA will be negative for the firm.
D. The DVA component will exceed the CVA component in magnitude.

Answer: C

Step by step

1. CVA term: $-0.07 \times 330 = -23.1$ bps.
2. DVA term: $+0.05 \times 250 = +12.5$ bps.
3. BCVA $= -23.1 + 12.5 = -10.6$ bps, which is negative.

Why C is correct: The CVA cost (23.1 bps) outweighs the DVA benefit (12.5 bps) because the counterparty is weaker and the positive exposure is larger.

Why the others are wrong

- **A:** Standalone CVA can only be negative (a cost); it is never a gain.
- **B:** BCVA is positive only if DVA exceeds CVA; here 12.5 < 23.1.
- **D:** DVA of 12.5 bps is smaller than CVA of 23.1 bps.

LO 38.h

Explain how netting can be incorporated into the CVA calculation.

Q1029 AnalystPrep Q.1884

REBUILT — Original stem was poorly worded with near-synonymous options (additive/non-additive/multiplicative); rewritten as a clear conceptual question on netting-set CVA with exactly one correct option.

A bank has several OTC derivative trades with one counterparty, all covered by a single legally enforceable netting agreement. Which of the following statements about incorporating netting into the bank's CVA calculation is most accurate?

A. CVA must be calculated at the netting-set level from the netted exposure profile, so the netting-set CVA is not the sum of the trades' standalone CVAs and is never higher than that sum.

B. CVA can be calculated trade by trade and summed, because expected exposure is additive across the trades in a netting set.

C. Netting increases CVA, because offsetting positions concentrate the remaining exposure in fewer trades.

D. Netting changes CVA only through the counterparty's default probability, since exposure at default is unaffected by offsetting positions.

Answer: A

Step by step

1. Under a netting agreement, on default the positive and negative trade values are offset, so exposure is $\max(\sum_k V_k, 0)$ rather than $\sum_k \max(V_k, 0)$.
2. Because $\max(\sum_k V_k, 0) \leq \sum_k \max(V_k, 0)$, netted expected exposure (EE) is at most the sum of standalone EEs.
3. EE (and therefore CVA, which is a weighted sum of EE) is non-additive across trades in a netting set, so CVA must be computed on the netting-set exposure profile.
4. Hence netting-set CVA $\leq \sum$ standalone CVAs; netting reduces CVA.

Why A is correct: Netting lets gains on some trades offset losses on others, so exposure and CVA are computed for the netting set as a whole and are non-additive, with netting-set CVA never above the sum of standalone CVAs.

Why the others are wrong

- **B:** Wrong: EE is non-additive within a netting set; summing trade-level CVAs ignores offsetting and overstates CVA.
- **C:** Wrong: netting can only reduce (never increase) exposure, so it cannot raise CVA; this confuses concentration with offsetting.
- **D:** Wrong: netting acts on the exposure side (EE falls); default probability and recovery are unchanged by netting.

Q1030 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A bank already holds a payer swap with a client under an enforceable netting agreement. The standalone CVA charge on that swap is EUR 410,000. The bank is considering a second trade whose standalone CVA charge is EUR 360,000. The CVA charge on the netting set containing both trades is EUR 280,000. Which of the following is the incremental CVA charge of the second trade?

A. A decrease of EUR 130,000

B. A decrease of EUR 490,000

C. An increase of EUR 280,000

D. An increase of EUR 360,000

Answer: A

Step by step

1. Incremental CVA = CVA of the netting set after the trade minus CVA of the netting set before the trade.
2. Incremental charge = 280,000 - 410,000 = -130,000.
3. The second trade therefore reduces the netting-set CVA charge by EUR 130,000 even though its standalone charge is EUR 360,000.

Why A is correct: Incremental CVA is measured at the netting-set level as the change in the total, which here is a decrease of EUR 130,000.

Why the others are wrong

- **B:** 490,000 (410,000 + 360,000 - 280,000) is the total netting benefit across both trades, not the incremental effect of the second trade.
- **C:** 280,000 is the post-trade netting-set charge, not the change from the pre-trade level.
- **D:** 360,000 is the standalone charge of the new trade, which ignores netting against the existing swap.

Q1031 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A bank has an existing receive-fixed swap with a client under an enforceable netting agreement and is pricing a new pay-fixed swap that largely offsets it. Which of the following statements regarding the incremental CVA of the new trade is correct?

A. It can never be below zero, because adding any trade to a netting set must raise the netting set's expected positive exposure.

B. It can be a benefit rather than a cost, because the new trade can reduce the netting set's expected positive exposure, which allows the bank to offer more favorable terms than to a client with no existing trades.

C. It equals the new trade's standalone CVA, because the netting agreement affects only the counterparty's default probability and not the exposure.

D. It becomes larger in its netting benefit as the new trade's notional is increased, because greater size produces more offsetting cash flows.

Answer: B

Step by step

1. Incremental CVA is the change in netting-set CVA caused by the new trade, computed from the netting-set exposure with and without it.
2. A trade that offsets existing positions can make incremental EPE negative, so incremental CVA is a benefit and not a cost.
3. Because of this, a bank with existing trades can quote more competitively on offsetting trades.

Why B is correct: Netting works through exposure: offsetting trades can lower netting-set EPE, so incremental CVA can be a benefit and is never worse than standalone CVA.

Why the others are wrong

- **A:** Incremental EPE can be negative because of beneficial netting, so incremental CVA can be a benefit; the premise that EPE must rise is false.
- **C:** Netting changes exposure, not default probability or LGD; incremental CVA is generally lower in cost than standalone CVA.
- **D:** As the new trade's size increases, the netting benefit is lost (the trade eventually dominates), so incremental CVA moves toward the standalone value.

LO 38.i

Define and calculate incremental CVA and marginal CVA and explain how to convert CVA into a running spread.

Q1032 AnalystPrep Q.5486

REBUILT — Original option A was incoherent (subtracting netting-set CVA from each standalone CVA) and C/D were obvious giveaways; options rewritten as realistic mistakes while keeping the correct answer (incremental CVA with/without the trade).

MetroQuant Advisors has a portfolio of OTC transactions with a counterparty under a netting agreement. The portfolio contains both directional and offsetting positions. A new trade is being priced with this counterparty. How should MetroQuant incorporate netting into its CVA calculation for the new trade?

A. Compute the new trade's standalone CVA and add it to the existing netting-set CVA, since CVA is additive across trades.

B. Compute the CVA of the netting set with and without the new trade, and charge the difference (incremental CVA), which reflects the netting benefit.

C. Compute the CVA of each trade using only its positive exposure, then sum these, so that netting is ignored conservatively.

D. Compute the CVA of the existing netting set only, since the new trade's effect is captured by the counterparty's credit spread.

Answer: B

Step by step

1. With netting, a new trade changes the exposure of the whole netting set, not just its own exposure.
2. Calculate CVA of the netting set before the trade: CVA_{NS}.
3. Calculate CVA of the netting set after adding the trade: CVA_{NS+1}.

4. Incremental CVA $= \text{CVA}_{NS+1} - \text{CVA}_{NS}$ (equivalently, CVA using the change in EE, ΔEE); the trade should earn at least this to be worthwhile.
5. It is no greater than the standalone CVA, and can be negative if the trade offsets existing exposure.

Why B is correct: Netting makes CVA non-additive, so the new trade's cost is measured by the change in netting-set CVA with versus without the trade, which captures the offset against existing positions.

Why the others are wrong

- **A:** Wrong: standalone CVA ignores netting and overstates the charge; it equals incremental CVA only when the trade is so large that the netting benefit vanishes.
- **C:** Wrong: summing positive-exposure CVAs ignores netting entirely (overstates risk) rather than incorporating it.
- **D:** Wrong: ignores the new trade's effect on netting-set exposure; credit spread reflects counterparty default risk, not trade-level exposure change.

Q1033

AP Mock 1 Q8

REBUILT — Original had no option that actually describes incremental CVA (A, B and D describe marginal CVA per GARP); rewrote the stem and options so exactly one option describes incremental CVA.

Alliance Derivatives Group (ADG) has an existing netting set with a counterparty and is deciding which CVA measure to use in different situations. In which situation would it be most appropriate for ADG to use incremental CVA?

A. Splitting the netting-set CVA at a given date into trade-level contributions that sum to the total, for accounting purposes.
B. Pricing several trades executed simultaneously with the counterparty as part of the same deal.
C. Charging the desk that has just executed a new trade with the change in netting-set CVA caused by that trade.
D. Analysing each trade's CVA contribution in a way that does not depend on the order in which trades were executed.

Answer: C

Step by step

1. Incremental CVA = CVA of the netting set including the trade minus CVA of the netting set without it (GARP Eq. 20.11).
2. It captures a trade's contribution to total CVA at the time it is executed, so it depends on execution order but does not change when later trades are added.
3. This is why it fits charging CVA to individual traders, businesses and clients, and pricing a new trade (also unwinds, restructurings and changed margin terms).
4. Allocating CVA to trades at a given date, pricing simultaneous trades and order-independent contributions are all marginal CVA uses.

Why C is correct: Incremental CVA measures the change in netting-set CVA caused by one specific change (new trade, unwind, restructuring), which is exactly the amount to charge the originating desk or client.

Why the others are wrong

- **A:** Describes marginal CVA: contributions that sum to total CVA at a point in time (accounting, restructuring decisions). Incremental CVAs are tied to execution sequence.
- **B:** Pricing simultaneous trades is the marginal CVA use case; incremental CVA would depend arbitrarily on which of the simultaneous trades is treated as 'first'.
- **D:** Order-independence is the defining feature of marginal CVA; incremental CVA is explicitly order-dependent.

Q1034

BT answers P2.T6.418.3

Your colleague Peter is evaluating whether to use marginal or incremental CVA. He makes the following two statements:

I. Incremental CVA depends on the order in which trades are executed but does not change due to subsequent trades. It makes the most sense when CVA needs to be charged to individual traders and businesses.

II. Marginal CVA does change when new trades are executed. While it probably should not be used to price new transactions, it is good for pricing simultaneous transactions and for decomposing a total CVA into its trade-level contributions.

Which of the above statements are (is) TRUE?

A. Neither
B. I. only
C. Both

D. II. only

Answer: C

Step by step

1. Statement I: incremental CVA reflects the order of execution and is fixed once the trade is done; it suits charging individual traders, businesses and clients. True.
2. Statement II: marginal CVA of existing trades is revised when new trades arrive, so it is unsuitable for pricing a new trade, but it suits simultaneous trades and decomposing total CVA. True.
3. Both are true.

Why C is correct: Both statements restate GARP's description of incremental CVA (order-dependent, stable, used for charging) and marginal CVA (changes with new trades, additive decomposition).

Why the others are wrong

- **A:** Wrongly rejects both; each statement matches GARP.
- **B:** Accepts I but wrongly doubts II; marginal CVA does change with new trades.
- **D:** Accepts II but wrongly doubts I; incremental CVA is stable after execution and is used for charging desks.

Q1035 AnalystPrep Q.1970

In which of the following cases is the use of the marginal CVA most appropriate?

A. Pricing new transactions.
B. Pricing trades transacted at different times.
C. Pricing long-dated positions.
D. Calculating the trade-level CVA contributions at a given time.

Answer: D

Step by step

1. Marginal CVA breaks the netting-set CVA into transaction-level contributions that sum to total CVA.
2. Because it does not depend on execution order, it is the appropriate way to allocate CVA to trades at a given time (e.g. accounting) and to price simultaneous trades.
3. It is not used to price new trades because the marginal CVA of existing trades changes whenever a new trade is executed.
4. Hence D.

Why D is correct: Marginal CVA allocates the total CVA across trades at a given time so that contributions add up to the total.

Why the others are wrong

- **A:** New trades are priced with incremental CVA; marginal CVA of existing trades changes when a trade is added.
- **B:** Trades done at different times are naturally handled sequentially by incremental CVA, which reflects the timing of each trade.
- **C:** Trade maturity is not what distinguishes the two measures; this is a plausible-sounding but irrelevant criterion.

Q1036 AnalystPrep Q.1971

REBUILT — Source said only 'duration'; changed to 'risky duration' to match GARP's risky-annuity conversion.

An interest rate swap is expected to have a risky duration of 4 years. The upfront CVA is 1% of the notional. The CVA as a running spread is approximately equal to:

A. 400 bps
B. 4 bps
C. 25 bps
D. 2.5 bps

Answer: C

Step by step

1. CVA as a running spread = upfront CVA / risky duration (risky annuity).
2. Upfront CVA = 1% of notional; risky duration = 4 years.
3. Running spread = $1\%/4 = 0.25\% = 25$ bps.
4. Convert to basis points (1% = 100 bps).

Why C is correct: Dividing the upfront CVA by the risky annuity spreads the one-off cost into an equivalent per-annum running charge.

Why the others are wrong

- **A:** Multiplied by the duration (1% x 4 = 4% = 400 bps) instead of dividing.
- **B:** Multiplied (1% x 4 = 4%) and then misread 4% as 4 bps.
- **D:** Divided correctly (0.25%) but converted to bps wrongly (0.025% = 2.5 bps).

Q1037

AnalystPrep Q.1976

REBUILT — Source said only 'duration'; changed to 'risky duration' to match GARP's risky-annuity conversion.

An interest rate swap that references a $10 million notional is expected to have a risky duration of 3 years. The upfront CVA is 0.75% of the notional. The CVA as a running spread is approximately equal to:

A. 250 bps
B. 75 bps
C. 12.5 bps
D. 25 bps

Answer: D

Step by step

1. CVA as a running spread = upfront CVA / risky duration (risky annuity).
2. Upfront CVA = 0.75% of notional; risky duration = 3 years.
3. Running spread = 0.75%/3 = 0.25% = 25 bps (the $10 million notional cancels: upfront CVA = $75,000, annuity = 3 years).
4. Convert to basis points (1% = 100 bps).

Why D is correct: Dividing the upfront CVA by the risky annuity spreads the one-off cost into an equivalent per-annum running charge.

Why the others are wrong

- **A:** Decimal slip: 0.25% read as 2.50% = 250 bps.
- **B:** Ignored the duration and quoted the upfront CVA (75 bps) as the running spread.
- **C:** Divided the correct answer by 2 (or used a duration of 6); not supported by the data.

Q1038

BT Mock Q7

REBUILT — Original distractors (-1.100, -2.220) had no traceable error; replaced with options reflecting realistic mistakes.

Consider a netting set with two derivative positions: an interest rate swap (IRS) with a marginal CVA of −1.450 and a cross-currency swap (CCS) with a marginal CVA of −1.550. If the first transaction executed is the cross-currency swap and its incremental CVA is −2.100, then what is the incremental CVA of the IRS?

A. −0.550
B. −0.900
C. −1.450
D. −5.100

Answer: B

Step by step

1. Marginal CVAs sum to total CVA: $-1.450 + (-1.550) = -3.000$.
2. Incremental CVAs, in order of execution, also sum to total CVA: $-2.100 + x = -3.000$.
3. $x = -3.000 - (-2.100) = -0.900$.
4. The first-executed trade's incremental CVA equals its stand-alone CVA, so the IRS (executed second) gets only the remainder.

Why B is correct: Both marginal and sequential incremental allocations decompose the same total CVA, so the second trade's incremental CVA is total minus the first trade's incremental CVA.

Why the others are wrong

- **A:** Subtracted the CCS marginal CVA from its incremental CVA (-2.100 + 1.550), mixing the two allocation methods.
- **C:** Reported the IRS marginal CVA, ignoring that incremental CVA is order-dependent.
- **D:** Added instead of subtracted (-3.000 + -2.100).

Q1039

BT answers P2.T6.332.1

REBUILT — Incremental CVA formula exhibit was missing and the sign convention was unstated; defined incremental CVA in the stem and stated that CVA is quoted as a positive cost.

Incremental CVA of a new trade is the CVA of the netting set including the new trade minus the CVA of the existing netting set (CVA expressed as a positive cost). Each of the following statements about incremental CVA is true EXCEPT which is FALSE?

A. It depends on the order in which trades are executed.
B. It is less than or equal to the stand-alone CVA of the new trade.
C. It must be greater than or equal to zero; it cannot be negative.
D. It is useful for the impact of adding (or removing) a trade to (from) the existing netting set.

Answer: C

Step by step

1. Netting can only reduce CVA, so the incremental CVA of a trade is at most its stand-alone CVA (B true).
2. Incremental CVA is tied to the sequence of trades; each is measured given the trades already in the set (A true).
3. It is designed for new trades, unwinds and restructurings (D true).
4. If the new trade offsets existing exposure strongly enough, netting-set EPE falls and CVA falls: incremental CVA is negative, i.e. a benefit. So C is false.

Why C is correct: Incremental EPE, and hence incremental CVA, can be negative when the new trade sufficiently offsets the existing netting set, so there is no non-negativity floor.

Why the others are wrong

- **A:** True: order of execution changes the impact of each trade.
- **B:** True: netting never makes incremental CVA exceed the stand-alone CVA.
- **D:** True: this is the primary purpose of incremental CVA.

Q1040

BT answers P2.T6.332.2

Each of the following statements about marginal CVA is true EXCEPT which is FALSE?

A. By definition, marginal CVA must be less than or equal to incremental CVA.
B. Unlike incremental CVA, marginal CVA is additive: the sum of marginal CVAs is the total CVA.
C. Marginal CVA is more relevant (than incremental CVA) for apportioning CVA contributions fairly across existing trades or assessing the CVA of more than one new trade.
D. Whereas it is difficult to price simultaneous trades with incremental CVA, marginal CVA is the appropriate way to calculate the trade-level CVA contributions of several trades at the same time.

Answer: A

Step by step

1. Marginal CVA allocates total CVA to trades so that contributions sum to total CVA (B true).
2. Because it is order-independent, it is the fairer allocation for existing trades and for simultaneous trades (C and D true).
3. There is no ordering relationship between the two: in GARP Table 20.5 the IRS marginal CVA is -17.7 versus incremental -26.3 (IRS first) and -13.7 (XCCY first), so marginal lies between the incremental values.
4. Hence the claim that marginal is always less than or equal to incremental is false.

Why A is correct: Marginal CVA is an order-independent allocation, so it can be above or below a given incremental CVA depending on execution order; no inequality holds by definition.

Why the others are wrong

- **B:** True: additivity (sum equals total CVA) is the defining feature of marginal CVA.
- **C:** True: order-independence makes it the 'fair' allocation.
- **D:** True: simultaneous trades are a stated use of marginal CVA.

Q1041

BT answers P2.T6.418.1

REBUILT — Table OCR was garbled and 'strictly lower' claims were not guaranteed; reconstructed table layout and stated the assumption that netting benefits are strictly positive.

The table shows stand-alone CVA for four transactions (three interest rate swaps and one cross-currency swap). Some incremental and marginal CVAs are shown as variables. Assume the trades partly offset each other, so any trade executed after another gains a strictly positive netting benefit. Two execution orders are shown: 1-2-3-4 and 4-1-2-3.

Trade	Stand-alone	Incremental (1-2-3-4)	Incremental (4-1-2-3)	Marginal (1-2-3-4)	Marginal (4-1-2-3)
(1) Payer IRS GBP 5Y	92,593	S	T	V	W
(2) Payer IRS GBP 6Y	124,816	-	-	-	-
(3) Payer IRS EUR 5Y	76,006	-	-	-	-
(4) CCS GBPUSD 5Y	99,558	-	-	-	-
Total	392,973	X	X	Y	Y

Each of the following is true EXCEPT which is false?

A. S = 92,593; the incremental CVA of the trade executed first equals its stand-alone CVA.
B. T < S; trade 1 has a lower incremental CVA if executed second than if executed first.
C. W < V; trade 1 has a lower marginal CVA if executed second than if executed first.
D. X < 392,973 and Y < 392,973; total CVA is lower than the sum of stand-alone CVAs.

Answer: C

Step by step

1. The first trade in an order has nothing to net against, so its incremental CVA is its stand-alone CVA: S = 92,593 (A true).
2. In order 4-1-2-3 trade 1 is netted against trade 4, so T < S under the stated assumption (B true).
3. Marginal CVA does not depend on execution order, so the two marginal columns are identical: W = V, not W < V (C false).
4. Total CVA (X = Y) is less than the 392,973 sum of stand-alone CVAs because of netting benefits (D true).

Why C is correct: Marginal CVA is order-independent, hence W equals V; only incremental CVA changes with the order.

Why the others are wrong

- **A:** True: first-executed trade has incremental = stand-alone CVA.
- **B:** True: executing second lets trade 1 benefit from netting with trade 4.
- **D:** True: both allocation methods sum to the same netted total, which is below the stand-alone sum.

Q1042

BT answers P2.T6.418.2

REBUILT — Replaced distractors that had no traceable error with realistic mistakes (decimal slip, no duration, multiplying).

An interest rate swap that references a $100.0 million notional has a risky duration of 3.50 years. The upfront CVA is 0.90% of the notional. Which is nearest to the approximate CVA as a running spread?

A. 2.6 bps
B. 25.7 bps
C. 90.0 bps
D. 315.0 bps

Answer: B

Step by step

1. CVA as a running spread = upfront CVA / risky duration (risky annuity).
2. Upfront CVA = 0.9% of notional; risky duration = 3.5 years.
3. Running spread = 0.90%/3.50 = 0.2571% = 25.7 bps (upfront CVA = $0.9 million).
4. Convert to basis points (1% = 100 bps).

Why B is correct: Dividing the upfront CVA by the risky annuity spreads the one-off cost into an equivalent per-annum running charge.

Why the others are wrong

- **A:** Decimal slip: 0.2571% read as 0.0257% = 2.6 bps.
- **C:** Ignored the duration and quoted the upfront CVA (90 bps) as the running spread.
- **D:** Multiplied by the duration (0.90% x 3.5 = 3.15% = 315 bps) instead of dividing.

Q1043

Schweser Module Quiz 38.2 Q1

With respect to the CVA calculation, which of the following statements is correct when a risk manager wishes to understand which trades have the greatest impact on a counterparty's CVA? The manager would use:

A. incremental CVA because it accounts for the change in CVA once the new trade is priced, accounting for netting.
B. marginal CVA because he could break down netted trades into trade-level contributions.
C. incremental CVA because he could break down netted trades into trade-level contributions.

D. marginal CVA because it accounts for the change in CVA once the new trade is priced, accounting for netting.

Answer: B

Step by step

1. The goal is attribution: which existing trades within a netted portfolio drive the counterparty's CVA.
2. Marginal CVA uses each trade's marginal expected exposure, so the netting-set CVA is split into trade-level contributions that sum to the total.
3. Incremental CVA instead measures the change in CVA when a new trade is added to the existing netting set, which is used for pricing that new trade.
4. Hence the tool is marginal CVA, and the correct justification is its trade-level breakdown: B.

Why B is correct: Marginal CVA replaces EE with marginal EE, allocating the netted CVA to individual trades, which is exactly what is needed to see which trades contribute most.

Why the others are wrong

- **A:** Wrong tool: incremental CVA is the right measure for pricing a new trade, but it does not attribute CVA across existing trades; the reason given is a correct description of incremental CVA, not an answer to this question.
- **C:** Right-sounding reason attached to the wrong measure: the trade-level breakdown is a property of marginal CVA, not incremental CVA.
- **D:** Correct measure but wrong justification: the change in CVA from pricing a new trade describes incremental CVA, not marginal CVA.

Q1044

BT answers P2.T6.916.1

REBUILT — Distractors -1.100 and -2.220 were arbitrary giveaways; replaced with the marginal-CVA and sum-of-incrementals mistakes.

A netting set contains two derivatives with a counterparty: an interest rate swap (IRS) with a marginal CVA of -1.450 and a cross-currency swap (CCS) with a marginal CVA of -1.550. If the CCS is the first transaction executed and its incremental CVA is -2.100, what is the incremental CVA of the IRS (executed second)?

A. -1.550
B. -0.900
C. -1.450
D. -5.100

Answer: B

Step by step

1. Marginal CVAs sum to the total netted CVA: $-1.450 + (-1.550) = -3.000$.
2. Incremental CVAs, taken in order of execution, also sum to the total: $\text{CVA}_{CCS,\,first} + \text{CVA}_{IRS,\,second} = -3.000$.
3. The first trade's incremental CVA equals its standalone CVA: -2.100.
4. $\text{CVA}_{IRS,\,incr} = -3.000 - (-2.100) = -0.900$.

Why B is correct: Both marginal and sequential incremental CVAs allocate the same total netted CVA; once the first trade's incremental CVA is fixed, the second trade is the remainder, -0.900 (a smaller charge because the IRS nets against the CCS).

Why the others are wrong

- **A:** -1.550 is the CCS marginal CVA, i.e., the wrong trade's figure.
- **C:** -1.450 is the IRS marginal CVA; marginal and incremental CVA differ, because incremental depends on the execution order.
- **D:** -5.100 adds the CCS incremental CVA to the total ($-3.000 + (-2.100)$) instead of subtracting it.

LO 38.j

Explain the impact of incorporating collateralization into the CVA calculation, including the impact of margin period of risk, thresholds, and initial margins.

Q1045

BT answers P2.T6.24.34.1

A bank is considering a derivative transaction with a counterparty. The uncollateralized expected positive exposure (EPE) for the transaction is $10 million, and the counterparty's default probability (PD) is 2%. The loss given default (LGD) is estimated at 60%. The bank and the counterparty agree to post variation margin with a zero threshold and an initial margin to reduce the exposure. The margin period of risk (MPR) is set at 10 days. If the collateralized EPE,

considering the MPR, is reduced to $2 million, what is the approximate reduction in CVA due to collateralization?

A. $96,000
B. $120,000
C. $160,000
D. $200,000

Answer: A

Step by step

1. Use CVA = EPE × PD × LGD; collateral changes only the exposure term, not PD or LGD.
2. Uncollateralized CVA = 10,000,000 × 0.02 × 0.60 = 120,000.
3. Collateralized CVA = 2,000,000 × 0.02 × 0.60 = 24,000.
4. Reduction = 120,000 − 24,000 = 96,000.

Why A is correct: Collateralization reduces CVA only by shrinking expected exposure (here by $8 million), so the CVA falls by the exposure reduction times PD times LGD.

Why the others are wrong

- **B:** This is the uncollateralized CVA itself (10m x 2% x 60%); it ignores the remaining $24,000 of collateralized CVA.
- **C:** Uses the exposure reduction of $8m times PD (0.02) but forgets to multiply by LGD (60%).
- **D:** Uses the full uncollateralized EPE times PD ($10m x 2%) and omits both LGD and the remaining collateralized exposure.

Q1046

BT answers P2.T6.332.3

REBUILT — The CVA formula exhibit was missing, so the stem now states the four components explicitly and asks which one is changed by collateralization.

A bank computes unilateral CVA with no wrong-way risk as $\text{CVA} = -\text{LGD}\sum_i \text{DF}(t_i)\,\text{EE}(t_i)\,\text{PD}(t_{i-1}, t_i)$. The four components are (1) the loss given default, LGD, (2) the discount factor, DF, (3) the expected exposure, EE, and (4) the marginal default probability, PD. When the bank begins to collateralize the trades with this counterparty, which component(s) are changed by the collateral?

A. (1) only
B. (3) only
C. (4) only
D. All except (2)

Answer: B

Step by step

1. Collateral is a claim on assets that can be seized at default, so it reduces the amount the bank stands to lose, that is, the exposure.
2. The counterparty's default probability depends on its creditworthiness and is unaffected by whether the bank holds collateral, so component (4) is unchanged.
3. The discount factor depends on interest rates only, so (2) is unchanged.
4. LGD in the formula is the loss on the counterparty's unsecured claim (1 - recovery) and is unchanged; the collateral effect is captured by lowering EE, so only (3) changes.

Why B is correct: Collateralization reduces CVA by changing only the counterparty's expected exposure, not its default probability, recovery or the discounting.

Why the others are wrong

- **A:** Treats collateral as lowering the loss rate; in the standard formula collateral is reflected in the exposure term and LGD stays tied to the counterparty's recovery.
- **C:** Confuses collateral with credit protection (CDS), which reflects the counterparty's default likelihood; collateral does not change PD.
- **D:** Assumes collateral alters LGD, EE and PD together; only EE is affected, and PD in particular is independent of collateral.

Q1047

Schweser Module Quiz 38.1 Q4

When incorporating netting and collateralization into the CVA calculation, which of the following statements is incorrect?

I. Netting increases the CVA price because it reduces exposure when trades are settled.
II. Collateralization does not change the CVA because it only changes the counterparty's expected exposure.

A. I only.
B. II only.
C. Both I and II.
D. Neither I nor II.

Answer: C
Step by step
1. Statement I: netting reduces exposure, so it reduces (not increases) the CVA. Statement I is incorrect.
2. Statement II: CVA is driven by expected exposure, so a change in EE changes the CVA; collateralization lowers EE and therefore lowers CVA. Statement II is incorrect.
3. Both statements are incorrect, so the answer is Both I and II.

Why C is correct: Netting and collateralization both reduce expected exposure and hence CVA; collateral leaves the default probability unchanged but still lowers CVA through EE.
Why the others are wrong
- **A:** Correctly spots I as wrong but accepts II; the error is thinking that changing only EE leaves CVA unchanged, when CVA depends directly on EE.
- **B:** Correctly spots II as wrong but accepts I; the error is believing netting raises CVA, when lower netted exposure lowers it.
- **D:** Treats both statements as true, missing that netting lowers CVA and that collateral lowers CVA through EE.

LO 38.k

Describe wrong-way risk and contrast it with right-way risk.

Q1048

AP Mock 3 Q2

REBUILT — Stem clarified to ask about the value of the protection to the investor (CDS seller B is the counterparty) and options reworded so exactly one is defensible.

An investor holds a USD 2 million bond issued by Issuer A and buys credit protection on Issuer A through a CDS from Issuer B (the protection seller). The investor will be paid only if Issuer A defaults and Issuer B is able to honor its obligation. If the correlation between the default of Issuer A and the default of Issuer B increases, what is the impact on the value of this CDS protection to the investor (i.e., the price the investor should be willing to pay)?

A. It decreases, because a default of A becomes more likely to coincide with a default of B (wrong-way risk), lowering the expected recovery of the protection payment.
B. It increases, because a higher chance of joint default means the protection is more likely to be needed.
C. It is unchanged, because the CDS value depends only on the default probability of the reference entity A.
D. It may rise or fall depending on the economic conditions of the country, so the direction cannot be determined.

Answer: A
Step by step
1. The CDS pays only if reference entity A defaults, and the investor collects only if seller B has not also defaulted.
2. Higher A-B correlation raises P(B defaults | A defaults), so the chance that protection fails exactly when it is needed rises.
3. This is wrong-way risk: exposure to B (the CDS payoff) is high exactly when B's default is more likely, which raises the CVA on the position.
4. The counterparty-risk-adjusted value of the protection = risk-free value minus CVA, so it falls; in the limit of perfect correlation the protection is worthless.

Why A is correct: Higher correlation between the reference entity and the protection seller increases wrong-way risk, so the expected payoff net of counterparty default (and thus the value of protection) falls.
Why the others are wrong
- **B:** Confuses the probability of A defaulting (unchanged) with the reliability of payment; joint default means the protection fails, it does not add value.
- **C:** Ignores counterparty risk on B; the stem is about the dependence between A and B, which does change the adjusted value.
- **D:** Direction is determinate: positive dependence between exposure and counterparty default is always wrong-way risk and reduces value.

Q1049 AnalystPrep Q.2688

The risk that the exposure to a counterparty is adversely correlated with the credit quality of that counterparty is known as:

A. Concentration risk
B. Settlement risk
C. Wrong-way risk
D. Credit migration risk

Answer: C

Step by step

1. Identify the definition: dependence between exposure and counterparty creditworthiness that raises overall counterparty risk (higher CVA).
2. Exposure rising when the counterparty's default probability also rises is an unfavorable dependence.
3. That is the definition of wrong-way risk.

Why C is correct: Wrong-way risk is any unfavorable dependence between exposure and counterparty credit quality, increasing CVA.

Why the others are wrong

- **A:** Concentration risk is exposure to a single name, sector or region, not a dependence between exposure and credit quality.
- **B:** Settlement risk is the risk that one leg of a transaction is delivered without the other being received.
- **D:** Credit migration risk is the risk of a rating/credit-quality change, with no link to the size of exposure.

Q1050 AnalystPrep Q.1984

One of the challenges experienced when attempting to quantify wrong-way risk has a lot to do with:

A. Difficult and time-consuming calculations
B. The need for high-level computer software that can sometimes be expensive
C. A lack of well-developed models
D. Uninformative historical data

Answer: D

Step by step

1. GARP lists the problems in quantifying WWR: direction unclear, misspecified relationship, and uninformative historical data.
2. WWR may be subtle and not revealed by empirical analysis such as historical correlations (e.g., zero correlation does not imply independence).
3. So historical data is the key limitation; computation, software and model availability are not the stated challenges.

Why D is correct: Wrong-way risk may not show up in historical time series or correlations, so history gives an incomplete or misleading picture.

Why the others are wrong

- **A:** Computation can be heavy, but it is not the fundamental challenge identified.
- **B:** Software cost is a practical matter, not a conceptual obstacle.
- **C:** Models exist (hazard rate, structural, parametric, jump); the difficulty is calibrating them without informative data.

LO 38.l

Identify examples of wrong-way risk and examples of right-way risk.

Q1051 AnalystPrep Q.1982

Cartogenes Investments is contemplating the purchase of a unique over-the-counter put option on GigaBank's stock from Alamine Bank, another financial institution. This decision could be unsuitable because of wrong-way risk. What is the most accurate reason?

A. The put option will only be valuable if the stock of GigaBank goes up, in which case the counterparty's credit

quality will likely be deteriorating.

B. The put option will only be valuable if the stock of GigaBank goes down, in which case the counterparty's credit quality will likely be improving.

C. The performance of GigaBank and Alamine Bank is likely to be correlated, and an increase in the value of the put will most likely coincide with a deterioration in the credit quality of Alamine Bank.

D. Such a purchase makes the buyer prone to specific risks affecting banks.

Answer: C

Step by step

1. Wrong-way risk (WWR) exists when exposure to the counterparty rises together with the counterparty's default probability.
2. The put pays off when GigaBank's stock falls, so Cartogenes' exposure to Alamine is highest in that state.
3. Bank stocks and bank credit quality share macroeconomic drivers, so a fall in GigaBank's stock is likely to coincide with deteriorating credit quality of Alamine Bank.
4. Exposure up and counterparty credit quality down is exactly WWR; a put on a bank's stock bought from another bank is the standard example.

Why C is correct: Positive dependence between the put's value and the counterparty's default probability increases overall counterparty risk, which is WWR.

Why the others are wrong

- **A:** Reverses the payoff: a put gains value when the stock falls, not when it rises.
- **B:** Gets the put payoff right but pairs it with improving credit quality; that would be right-way risk, not WWR.
- **D:** Too vague: it names a generic sector concentration rather than the link between exposure and counterparty credit quality that defines WWR.

Q1052

AnalystPrep Q.1987

Which of the following situations would lead to an increase in wrong-way risk?
I. Players in a given industry buying cover from a particular monoline insurer.
II. A borrower enlisting a guarantor who is also the borrower's business partner.

A. II only

B. I only

C. Both I and II

D. Neither I nor II

Answer: C

Step by step

1. Situation I: the monoline's obligations are concentrated in one industry. When the industry suffers, claims (exposure) surge and the monoline's credit quality deteriorates at the same time, so protection buyers face WWR.
2. Situation II: a guarantor sharing ownership or business with the borrower is hit by the same distress that causes the borrower to default, so the guarantee is least reliable when needed (WWR).
3. Both situations link exposure and counterparty credit quality unfavorably, so both increase WWR.

Why C is correct: In each case the same adverse event raises the exposure and weakens the credit quality of the party providing protection.

Why the others are wrong

- **A:** Treats only the guarantor case as WWR and misses the concentration of monoline claims in the industry being insured.
- **B:** Treats only the monoline case as WWR and misses that a guarantor with shared ownership or business defaults together with the borrower.
- **D:** Assumes insurance and guarantees remove credit risk; in fact both fail when most needed if the provider is correlated with the risk insured.

Q1053

AnalystPrep Q.5477

REBUILT — Source key B (selling CDS protection on own sovereign) is not WWR for the seller (GARP calls it a right-way trade); rewrote options so exactly one is a WWR example.

Oracle Financial Group has a large book of OTC derivatives. Its risk committee wants to prioritize contracts for wrong-way risk (WWR) review. Which of the following contracts should be prioritized?

A. Buying CDS protection on mortgage-backed securities from a monoline insurer whose guarantees are concentrated in the mortgage market.
B. Selling CDS protection to a counterparty on Oracle's own sovereign, because the strong link between Oracle and the sovereign creates extreme WWR for the seller.
C. Derivatives with a commodity producer that is hedging its output price, because the producer owes money mainly when prices are high.
D. Equity call options bought from a bank on that same bank's stock.

Answer: A

Key correction: Source key (AnalystPrep) said B; correct is A because selling protection on a strongly linked entity is a right-way trade for the seller, while buying protection from a concentrated monoline is WWR.

Step by step

1. WWR arises when the exposure and the counterparty's default probability rise together.
2. A: if mortgage securities fall, the CDS gains value (exposure up) and the concentrated monoline faces a flood of claims (default probability up). This is the classic 2007-2009 WWR (e.g. AMBAC, MBIA).
3. B: as seller, Oracle owes the payout when the sovereign weakens; with a strong link this is a right-way trade for the seller, not WWR.
4. C: a hedging producer owes only when prices (and profits) are high, which is right-way risk.
5. D: a call bought on the bank's own stock gains when the stock rises, when the bank is stronger; equity calls are right-way products.

Why A is correct: Protection bought from a counterparty concentrated in the very risk it insures delivers its gains exactly when the seller is least able to pay.

Why the others are wrong

- **B:** Confuses the buyer's and seller's positions: strong linkage is WWR for the protection buyer, but selling protection on a linked entity is right-way risk for the seller.
- **C:** Hedging by commodity producers creates right-way risk because the producer owes when its business is most profitable.
- **D:** Call payoffs rise with the stock, when the counterparty's credit quality is likely improving; this is right-way, the opposite of a put.

Q1054

AnalystPrep Q.5478

REBUILT — Original option C (selling protection on own sovereign) was itself defensibly right-way and option D was a giveaway; rewrote distractors so only the mining-company hedge is right-way.

Vertex Trading is reviewing its OTC trades for wrong-way and right-way risk. Which of the following trades would Vertex typically expect to exhibit right-way risk?

A. A derivative with a mining company that hedges its copper price risk, so the company owes money mainly when copper prices are high.
B. Buying CDS protection from a seller whose credit quality is strongly linked to that of the reference entity.
C. Buying an out-of-the-money put on one bank's stock from another bank.
D. An FX forward with a sovereign counterparty in which Vertex pays the sovereign's local currency and receives dollars.

Answer: A

Step by step

1. Right-way risk: exposure and counterparty default probability move in opposite directions, so overall counterparty risk falls.
2. A: the hedging producer owes only when copper is high, when its business should be more profitable and default is less likely. Right-way risk arises from hedging, not speculation.
3. B: exposure rises when the reference entity's spread widens, which tends to coincide with the seller weakening: extreme WWR.
4. C: the put gains when bank stocks fall, when the counterparty bank is weakening: WWR.
5. D: Vertex gains as the local currency depreciates, which coincides with sovereign credit stress: WWR.

Why A is correct: A hedging commodity producer owes money only in states where it is most profitable, so exposure is high when default risk is low.

Why the others are wrong

- **B:** Strong credit linkage between reference entity and seller produces WWR, not right-way risk.
- **C:** Put on a bank's stock from another bank is the textbook WWR; both banks are hit by the same shocks.

- **D:** Paying a sovereign's weak local currency leaves Vertex with gains exactly when the sovereign's credit deteriorates, which is WWR.

Q1055

AnalystPrep Q.5479

At a quarterly strategy meeting, the CRO of Concord Wealth Management asks which of the firm's positions typically poses the most concern for wrong-way risk (WWR). Which position should be the focus of further analysis?

A. Credit default swaps where Concord buys protection, which carry extreme WWR if the reference entity's credit quality is strongly related to that of the protection seller.

B. Mortgage derivatives during an economic boom, because a boom increases the likelihood of borrower defaults.

C. Hedging positions in commodity derivatives, which show right-way risk when commodity prices and business profitability are high.

D. Credit default swaps where Concord sells protection on other financial institutions, because financial institutions have countercyclical default probabilities, reducing WWR.

Answer: A

Step by step

1. Buying CDS protection: exposure grows as the reference entity's spread widens, and the seller's default probability is tied to the same credit environment.
2. If reference entity and seller are strongly related, both rise together: extreme WWR.
3. B, C and D each misstate the economics: defaults fall in a boom, hedges are right-way, and financial institutions' defaults are procyclical.

Why A is correct: Gains on protection bought from a seller that weakens when the reference entity does will not be collected when needed.

Why the others are wrong

- **B:** Mortgage-related WWR arises in recessions (falling property prices and higher defaults), not in booms.
- **C:** Describes right-way risk, which is not a concern for WWR review.
- **D:** Financial institutions' defaults are clustered in downturns (procyclical), not countercyclical; and as protection seller Concord's exposure is limited.

Q1056

AnalystPrep Q.5482

REBUILT — Source answer A recommended puts on other banks' stocks, which GARP identifies as WWR when bought from a bank; rebuilt as a single-best-answer question on choosing a hedge with low linkage to the counterparty.

Apex Portfolio Management wants to buy OTC put protection on its equity holdings from a dealer bank and wants to limit wrong-way risk (WWR). Which choice best limits WWR?

A. Buy a put on the stock of a company in an unrelated industry, whose fortunes have little link to the dealer bank's credit quality.

B. Buy a put on the stock of another large bank from the dealer bank.

C. Buy a put on a financial-sector index from the dealer bank, since the index diversifies across many banks.

D. Buy a put on the dealer bank's own stock, struck in the money rather than out of the money.

Answer: A

Key correction: Source key (AnalystPrep) said A (puts on other banks' stocks); that is wrong because buying a put on one bank's stock from another bank is GARP's standard WWR example.

Step by step

1. WWR on an equity put arises when the underlying's fortunes are strongly correlated with the counterparty's credit quality: the put pays off when the stock falls, when the counterparty is likely to be weak.
2. B: another bank's stock moves with the same macro factors as the dealer bank; GARP cites 'buying a put on one bank's stock from another bank' as an obvious WWR case.
3. C: a financial-sector index is still highly correlated with the dealer bank's credit quality.
4. D: the dealer's own stock has the strongest linkage; moneyness only modestly changes this.
5. A: choosing an underlying with little link to the counterparty reduces the dependence between exposure and default probability.

Why A is correct: WWR is reduced by choosing underlyings whose value is not driven by the same factors as the counterparty's creditworthiness.

Why the others are wrong

- **B:** Another bank shares the same macro shocks as the counterparty; this is the standard WWR example.
- **C:** Diversification across banks does not remove the common financial-sector factor; the index still falls when the dealer weakens.
- **D:** The stock of the counterparty itself gives the strongest (specific) WWR; being in the money does not remove the link.

Q1057

BT Mock Q2 · BT Mock Q32

Among its financial positions, Zapex Financial Corporation has written a call, entered into an interest rate swap, and written a credit default swap. Among the following four positions, which is the best example of wrong-way risk?

A. Zapex has written (sold) a call option to a counterparty in the same industry.

B. Zapex is the fixed-rate payer in an interest rate swap during an economic recovery.

C. Zapex has written (sold) a credit default swap to a counterparty whose credit quality is positively correlated with the reference entity.

D. Zapex has a position in a forward currency contract where its mark-to-market gain is positively correlated with the counterparty's default probability.

Answer: D

Step by step

1. Wrong-way risk requires Zapex to have positive exposure to the counterparty that rises as the counterparty's default probability rises.
2. D states this directly: gains (exposure) positively correlated with counterparty default probability.
3. A: having written the call, Zapex owes the option payoff; it holds no claim on the buyer beyond the premium already received.
4. C: as protection seller Zapex owes the payout; the link is between the counterparty (buyer) and the reference entity, not a rising claim of Zapex on the counterparty.
5. B: paying fixed during a recovery (rates rising) is typically right-way risk.

Why D is correct: Only D describes a claim on the counterparty that grows as the counterparty becomes more likely to default.

Why the others are wrong

- **A:** A seller of an option has an obligation, not an asset, so there is no meaningful credit exposure to the buyer.
- **B:** In a recovery rates rise, the fixed payer gains while counterparty credit quality tends to improve, a right-way situation.
- **C:** Looks like WWR because of the positive correlation, but it is Zapex who owes under the CDS, so it has no growing exposure to the buyer.

Q1058

BT answers P2.T6.419.1

REBUILT — Original stem used a confusing 'each ... EXCEPT which ... is likely to benefit' double structure; reworded to ask directly for the right-way position.

From the perspective of Acme Financial Institution, which of the following positions is most likely to exhibit right-way risk rather than wrong-way risk?

A. Acme buys an out-of-the-money put option on a bank's stock from another bank.

B. Acme is the floating-rate payer (fixed-rate receiver) in a vanilla interest rate swap with a bank, and Acme's economist has shown that low rates signal a recession.

C. Acme is long an oil commodity swap with an oil producer: Acme pays a fixed oil price and receives the spot price of oil.

D. Acme buys CDS protection where the credit quality of the reference entity is strongly positively related to that of the protection seller.

Answer: C

Step by step

1. C: Acme gains when the oil price rises. The producer is hedging, so it owes money mainly when oil is high, i.e. when its business is most profitable and default is least likely. Exposure up with credit risk down: right-way risk.
2. A: put gains when bank stocks fall, when the counterparty bank weakens: WWR.
3. B: fixed-rate receiver gains when rates fall; falling rates signal recession and weaker counterparty: WWR.
4. D: protection gains as the reference entity weakens, when the seller is also weakening: extreme WWR.

Why C is correct: A hedging producer owes only when prices are high and its business is strong, so exposure and default probability move oppositely.

Why the others are wrong

- **A:** A put on a bank's stock bought from another bank is the textbook WWR example.
- **B:** A receiver swap gains in a downturn when low rates coincide with higher counterparty default risk.
- **D:** Strong linkage between reference entity and seller makes the protection least reliable when it is most valuable.

Q1059

Schweser Module Quiz 38.3 Q1

REBUILT — Statement 4 was vague ('may or may not' lower risk); reworded it as an 'always' claim so it is unambiguously false, and numbered the statements.

How many of the following statements regarding wrong-way risk (WWR) and right-way risk (RWR) are correct?

1. Co-movement in exposure and counterparty default probability that produces a decline in overall counterparty risk is an example of WWR.
2. Co-movement in exposure and counterparty default probability that produces an increase in overall counterparty risk is an example of RWR.
3. Co-movement in exposure and counterparty default probability that produces neither a decline nor an increase in overall counterparty risk is an example of WWR.
4. A decline in exposure combined with an increase in counterparty default probability always produces RWR.

A. None.
B. All.
C. Two.
D. Three.

Answer: A

Step by step

1. WWR/RWR is classified by the net effect on overall counterparty risk: an overall increase = WWR (CVA up); an overall decrease = RWR (CVA down).
2. Statement 1: a decline in overall risk is RWR, not WWR. False.
3. Statement 2: an increase in overall risk is WWR, not RWR. False.
4. Statement 3: no net change means neither WWR nor RWR (the independence case). False.
5. Statement 4: falling exposure and rising default probability push in opposite directions, so the net effect depends on their relative size; RWR is not guaranteed. False.
6. Number correct = 0.

Why A is correct: The label depends only on whether the interaction of exposure and default probability raises (WWR) or lowers (RWR) overall counterparty risk; all four statements violate this, so none is correct.

Why the others are wrong

- **B:** All: assumes the statements are right by reversing the WWR/RWR definitions (treating a decline as WWR and an increase as RWR).
- **C:** Two: accepts the swapped definitions in statements 1 and 2 as correct.
- **D:** Three: accepts statement 3 (no net change) as WWR and statement 4 as automatically RWR, plus one of the swapped definitions; none of these follows from the definition.

Q1060

Schweser Module Quiz 38.3 Q2

Which of the following events would likely lead to an increase in wrong-way risk (WWR)?

I. The borrower and the guarantor are business partners (share ownership or the same industry).

II. A monoline insurer has sold protection concentrated in a single business or industry.

A. I only.
B. II only.
C. Both I and II.
D. Neither I nor II.

Answer: C

Step by step

1. Test each event: does it make exposure and the counterparty's default probability rise together?
2. I: if borrower and guarantor are linked, the same shock can cause the borrower to default (guarantee exposure rises) and the guarantor to default too, so the guarantee may be worthless. WWR increases.
3. II: a concentrated monoline faces a flood of claims in the same shock that raises the value of the protection

bought from it. Exposure and its default probability rise together; WWR increases.
4. Both events raise WWR.

Why C is correct: Both situations create a positive link between the exposure and the counterparty's default probability (a guarantor tied to the borrower; a protection seller concentrated in the risk it insures), which is the definition of WWR.

Why the others are wrong

- **A:** I only: overlooks that sector concentration of a protection seller (as with monolines in 2007-2009) also raises WWR.
- **B:** II only: overlooks that a guarantor correlated with the borrower provides little protection exactly when needed.
- **D:** Neither: treats the two parties as independent; this is the no-WWR case, not applicable here.

Q1061

Schweser Module Quiz 38.3 Q3

Which of the following statements regarding WWR and RWR is correct?

A. A long put option is subject to WWR if both risk exposure and counterparty default probability decrease.

B. A long call option experiences RWR if the interaction between risk exposure and counterparty default probability produces an overall decline in counterparty risk.

C. A declining local currency can decrease the position gain in a foreign currency transaction while increasing the risk exposure of the counterparty.

D. The 2007-2009 credit crisis provides an example of WWR from the perspective of a long who had sold credit default swaps (CDSs) as protection against bond issuers' default.

Answer: B

Step by step

1. Classify by net effect on counterparty risk: increase = WWR, decrease = RWR.
2. A: for a long put, WWR arises when exposure and default probability both INCREASE (underlying falls, put gains, short is distressed). A states decrease. False.
3. B: for a long call, an overall decline in counterparty risk (e.g., underlying rises while the short's credit improves) is RWR. True.
4. C: a falling local currency INCREASES the gain of the developed-market party, raising exposure. C states decrease. False.
5. D: in the crisis, the WWR was borne by those who BOUGHT CDS protection (monoline sellers were overwhelmed), not by sellers. False.

Why B is correct: RWR is defined by an overall decline in counterparty risk, and only B states this correctly for a long call.

Why the others are wrong

- **A:** Reverses the direction: WWR on a long put requires exposure and default probability to rise together, not fall.
- **C:** Reverses the sign of the FX effect: depreciation of the counterparty's currency increases the gain and the exposure.
- **D:** Confuses buyer and seller: the buyer of protection held the exposure to the failing monolines; the seller owed the payments.

Q1062

Schweser Module Quiz 38.3 Q4

REBUILT — Statement 2 (historical attention to WWR vs RWR) is not a GARP-based fact; replaced with a testable WWR/RWR-CVA relationship and numbered the statements.

How many of the following statements regarding counterparty risk are correct?

1. Speculation in normally functioning derivatives markets automatically produces RWR.
2. Relative to the no-dependence case, RWR increases CVA while WWR reduces CVA.
3. The counterparty default probability does not enter the estimate of overall counterparty risk.
4. Unlike exposure to OTC derivatives, which is normally assumed to be a fixed amount over a specified period, exposure to bank loans fluctuates with market conditions.

A. None.

B. All.

C. Two.

D. Three.

Answer: A

Step by step

1. Statement 1: it is hedging (e.g., a commodity producer hedging price risk), not speculation, that automatically produces RWR. False.
2. Statement 2: WWR raises CVA and RWR lowers CVA relative to independence; the statement reverses this. False.
3. Statement 3: counterparty risk is roughly exposure x default probability x loss given default, so default probability is a key input. False.
4. Statement 4: reversed. Loan exposure is roughly fixed, whereas OTC derivative exposure fluctuates with market conditions. False.
5. Number correct = 0.

Why A is correct: Each statement misstates a basic fact: hedging (not speculation) gives RWR, WWR (not RWR) raises CVA, default probability is an input, and it is derivatives exposure that fluctuates.

Why the others are wrong

- **B:** All: every statement is a reversal or misstatement of the correct fact.
- **C:** Two: accepts a pair of the reversed statements (for example the CVA direction and the loan/derivative exposure swap).
- **D:** Three: accepts all but one of the false statements; none of them is correct.

Q1063

Schweser Module Quiz 38.3 Q5

Which of the following statements is correct?

I. After the default of Lehman Brothers, depreciation of the yen against the dollar gave Japanese banks a substantial gain on their currency swap positions used to obtain dollar funding.

II. A fixed-rate receiver in an interest rate swap gains value to the extent that the swap rate increases.

A. I only.

B. II only.

C. Both I and II.

D. Neither I nor II.

Answer: D

Step by step

1. I: Japanese banks pledged yen to receive dollars. After Lehman the yen APPRECIATED, so the pledged yen bought more dollars and U.S. counterparties owed more, a gain with rising exposure to weakened U.S. banks (WWR). Depreciation would have produced a loss. False.
2. II: a receiver of fixed locks in a coupon; when the swap rate falls the fixed leg is worth more, so the receiver gains when rates DECLINE. False.
3. Neither statement is correct.

Why D is correct: Yen appreciation (not depreciation) created the gain, and receiver swaps gain when rates fall (the recession scenario that also hurts counterparty credit).

Why the others are wrong

- **A:** I only: gets the direction of the yen move wrong; depreciation would have reduced the value of the pledged yen.
- **B:** II only: confuses receiver and payer; a payer of fixed gains when the swap rate rises.
- **C:** Both: makes both direction errors.

Q1064

GARP PE2 Q58

REBUILT — Original was a composite item mixing LCR/NSFR (outside CR) with a CDS point and an erratum in the key; rebuilt as a focused, self-contained WWR identification question.

VLT Bank (VLTB), a Singapore bank, has bought a 5-year credit default swap (CDS) on the EUR-denominated bonds of ConSol Corp, a Singapore manufacturer that buys locally produced inputs in SGD and earns most of its revenue in EUR. The SGD then appreciates sharply against all major currencies, and ConSol's credit spread widens substantially.

VLTB is comparing two possible protection sellers:

Seller X: a German bank with a diversified global book and no meaningful exposure to Singapore manufacturers.

Seller Y: a Singapore bank whose loan book is concentrated in Singapore exporters that earn revenue in foreign currencies.

Which statement is most accurate?

A. Both CDS trades show WWR, because any purchased protection gains value when the reference spread widens.
B. The CDS with Seller Y shows WWR, whereas the CDS with Seller X does not.
C. The CDS with Seller X shows WWR, because a weaker euro damages a German bank, whereas the CDS with Seller Y does not.
D. Neither trade shows WWR, because CDS exposure depends only on ConSol and is independent of the seller's credit quality.

Answer: B

Key correction: Source key (GARP PE2 Q58) said B, but its own explanation supports A (the long CDS gains value when ConSol's spread widens); NSFR actually rises, so B was wrong. The rebuilt item tests the WWR aspect only.

Step by step

1. The SGD appreciation hurts ConSol (EUR revenue buys fewer SGD, costs fixed in SGD), so ConSol's spread widens and VLTB's bought CDS gains value: exposure to the seller rises.
2. WWR exists only if the seller's default probability rises in the same scenario.
3. Seller Y's loans are concentrated in exporters with the same weakness as ConSol, so the same shock raises Y's default probability: exposure and PD rise together = WWR.
4. Seller X is diversified with no stated link to the shock: exposure rises but its PD is not tied to it, so no meaningful WWR.
5. Answer: B.

Why B is correct: WWR needs a positive link between the exposure and the seller's default probability; only Seller Y, concentrated in the sector hurt by the SGD move, has it.

Why the others are wrong

- **A:** Treats a rising CDS value as WWR by itself; this ignores the need for a link to the seller's credit quality (the same error would label every hedge as WWR).
- **C:** Wrongly attributes WWR to the diversified German bank by assuming a euro decline hurts it, which the stem excludes, and misses Seller Y's concentration.
- **D:** Assumes independence between exposure and seller credit quality; that fails for Y, whose creditworthiness depends on the same exporter stress that raises the CDS value.

Q1065

Mock 2 Q61

An investor has recently purchased several over-the-counter (OTC) puts on the S&P 500 exchange-traded fund (ETF). He uses options to hedge risk and to overweight or underweight asset classes in an efficient and risk conscious fashion. Which of the following statements best describes the use of long OTC puts in the context of wrong-way risk (WWR) and right-way risk (RWR)?

A. Out-of-the-money put options have less WWR than in-the-money put options.
B. Out-of-the-money put options have more WWR than in-the-money put options.
C. In-the-money put options do not have any more or less WWR or RWR than out-of-the-money put options.
D. A long put option is subject to WWR if both risk exposure and counterparty default probability decrease.

Answer: B

Step by step

1. A long put pays only if the underlying falls; the same macro stress that drops the underlying raises the writer's default probability.
2. An out-of-the-money put has little or no exposure today and only gains value in a large fall, exactly the stressed state in which the writer is likely to default, so exposure and default probability are strongly positively linked.
3. An in-the-money put already has exposure, so a smaller share of its exposure is conditional on the stress scenario; WWR is less pronounced.
4. WWR means exposure and counterparty default probability move in the same direction upward (positive dependence), not both falling. Hence B.

Why B is correct: OTM puts gain value mainly in severe market falls, when the writer is most likely to default, so their exposure is more strongly tied to default than that of ITM puts.

Why the others are wrong

- **A:** Reverses the relationship; ITM puts already have exposure, so they are less dependent on the stress scenario.
- **C:** Moneyness matters: the dependence between exposure and default differs, so ITM and OTM puts do not carry equal WWR.
- **D:** WWR requires exposure to rise when default probability rises; both falling would describe a benign positive co-movement and is not the WWR definition.

LO 38.m

Discuss the impact of collateral on wrong-way risk.

Q1066

AnalystPrep Q.1985

As a risk manager at Prestige Global Bank, you have been analyzing various derivative agreements to assess the potential impact of wrong-way risk (WWR). One of the agreements under review is an interest rate swap with SovereignX, a small country with an emerging economy. In the swap, Prestige Global Bank receives a fixed rate and pays a variable rate to SovereignX. The collateral backing the swap is sovereign bonds issued by SovereignX itself. Amidst your analysis, news breaks out that there's increasing political instability in SovereignX, which might lead to drastic economic decisions. Given the nature of your bank's agreement and the potential scenarios that could unfold, you are concerned about the role of collateral in managing the WWR. Which of the following best describes the impact of collateral on wrong-way risk, particularly in the context of potential rapid changes in SovereignX's economic conditions?

A. Collateral will effectively counterbalance any WWR, regardless of the speed of exposure increase.
B. The potential for a sudden increase in exposure means collateral might not provide significant mitigation against WWR.
C. Collateral's effectiveness in reducing WWR is directly proportional to the rate of exposure increase.
D. A rapid increase in exposure will enhance the role of collateral in mitigating WWR.

Answer: B

Step by step

1. Collateral reduces exposure, but only if it can actually be called and received before default.
2. If exposure rises gradually, extra collateral is easy to request and receive, so collateral mitigates WWR.
3. If exposure jumps suddenly (e.g., sovereign devaluation or default), there is no time to collect more collateral, so the benefit is very limited.
4. Here SovereignX's political instability could trigger a sudden jump, and the collateral is SovereignX's own bonds, which would lose value at the same time (wrong-way collateral). So collateral gives little mitigation: B.

Why B is correct: Collateral only mitigates WWR when exposure builds gradually enough for margin to be received; a sudden exposure jump leaves little benefit, especially with wrong-way collateral.

Why the others are wrong

- **A:** Wrongly assumes collateral always offsets WWR; ignores the speed of the exposure increase and the wrong-way nature of the sovereign-bond collateral.
- **C:** Reverses the relationship: the faster the exposure rises, the less collateral can be collected, so effectiveness is not proportional to the rate of increase.
- **D:** Backwards: a rapid exposure increase reduces, not enhances, the role of collateral because it cannot be called in time.

Q1067

AnalystPrep Q.5480

Fidelity Finance Corp is a trading firm that actively partakes in the OTC derivatives market, applying margin to manage exposures. The firm is appraising its margin policy effectiveness in response to potential wrong-way risk (WWR). As part of this strategic evaluation, they need an understanding of the impact of margin on WWR. Given current market dynamics and the nature of WWR, how effective is the margin likely to be in mitigating WWR for Fidelity Finance Corp?

A. Margin can effectively mitigate WWR regardless of market conditions, as it serves as a buffer that absorbs potential increases in exposure, irrespective of the timing.
B. Margin's effectiveness is timing-dependent, and only if the exposure increases gradually prior to default can margin be received and offer a risk mitigation benefit for WWR.
C. Margin has a universal mitigating effect on WWR due to the regular and standardized posting intervals which prevent significant exposure increases at any point in time.
D. Margin is most effective during periods of rapid market decline, as it provides instantaneous liquidity which can cover any exposure increases due to WWR.

Answer: B

Step by step

1. Margin reduces exposure only to the extent it is actually received before the counterparty defaults.

2. With gradual exposure growth, additional margin can be requested and received, so margin offers a mitigation benefit against WWR.
3. With a jump in exposure near default, margin cannot be collected in time and the benefit is very limited.
4. Hence margin's effectiveness is timing-dependent: B.

Why B is correct: Margin helps against WWR only if exposure builds gradually before default so that margin can be called and received; a sudden jump defeats it.

Why the others are wrong

- **A:** Treats margin as a universal buffer; ignores that timing matters and that margin cannot be received after a sudden exposure jump.
- **C:** Assumes regular posting intervals prevent large exposure moves, but exposure can jump between margin calls and margin takes time to receive.
- **D:** Claims margin is most effective in rapid declines and instantaneous; in fact margin is not instantaneous, and rapid moves are when it works worst.

Q1068 BT answers P2.T6.907.1

Acme Bank's position in a derivative contract has a current exposure of $100.0 million; i.e., the bank has a mark-to-market, unrealized gain of $100.0 million. The position's price volatility is 20.0%. Consider the following four scenarios:
- Scenario A. Acme leaves the exposure unhedged
- Scenario B. Acme hedges with $80.0 in collateral that itself has price volatility of 10.0% and is uncorrelated to the exposure
- Scenario C. Acme hedges with $80.0 in collateral that itself has price volatility of 10.0% and has a negative correlation of -0.290 to the exposure
- Scenario D. Acme hedges with $70.0 in collateral that itself has price volatility of 10.0% and has a positive correlation of +0.420 to the exposure

In which of the above scenarios is the DOLLAR volatility of Acme Bank's net position the highest?

A. Scenario A; unhedged
B. Scenario B; $80.0 million collateral with ρ = zero
C. Scenario C; $80.0 million collateral with ρ = -0.290
D. Scenario D; $70.0 million collateral with ρ = +0.420

Answer: C

Step by step

1. Net position = exposure E minus collateral C, so $\sigma_{net} = \sqrt{E^2\sigma_E^2 + C^2\sigma_C^2 - 2EC\sigma_E\sigma_C\rho}$.
2. A: unhedged, $\sigma = 100 \times 0.20 = 20.0$.
3. B: $\sqrt{20^2 + 8^2 - 0} = \sqrt{464} = 21.54$.
4. C: $\sqrt{400 + 64 - 2(100)(80)(0.20)(0.10)(-0.29)} = \sqrt{400 + 64 + 92.8} = 23.60$.
5. D: $\sqrt{400 + 49 - 2(100)(70)(0.20)(0.10)(0.42)} = \sqrt{449 - 117.6} = 18.20$.
6. Highest is C (23.60): negative correlation between collateral and exposure is wrong-way collateral, which adds volatility instead of hedging it.

Why C is correct: When collateral is negatively correlated with the exposure (wrong-way collateral), its value falls as exposure rises, so net dollar volatility exceeds even the unhedged position.

Why the others are wrong

- **A:** Unhedged volatility is 20.0, below C (23.60) because the collateral in C is wrong-way and adds variance; assuming 'no hedge = riskiest' ignores the negative-correlation effect.
- **B:** 21.54 assumes zero correlation; it is higher than unhedged but below C because the -0.29 correlation adds a further 92.8 to variance.
- **D:** 18.20 is the lowest: positive correlation of +0.42 makes collateral move with the exposure, reducing net volatility. Flipping the correlation sign by mistake would wrongly make this the highest.

LO 38.n

Identify examples of wrong-way collateral.

Q1069 BT answers P2.T6.24.33.1

REBUILT — Original had Moe's Bank as protection buyer yet spoke of 'losses on the CDS' and had a trivially small $20 million collateral; rewrote so the protection seller posts collateral correlated with the reference entity and the options differ clearly in how much protection remains.

Moe's Bank buys credit protection through a credit default swap (CDS) with a notional of $700 million on the bonds of Sully's Corporation. The protection seller, Larry's Securities, posts as collateral $300 million of bonds issued by DEF Inc., a wholly owned subsidiary of Sully's Corporation. If Sully's Corporation defaults, which of the following outcomes is most likely?

A. The DEF Inc. bonds will rise in value as investors move into related debt, so the collateral will exceed the CDS exposure.

B. The DEF Inc. bonds will hold their value, so the collateral will continue to cover the CDS exposure.

C. The DEF Inc. bonds will fall moderately, but the collateral will still cover most of the CDS exposure.

D. The DEF Inc. bonds will fall sharply just as the CDS exposure jumps, so the collateral will cover only a small part of the exposure.

Answer: D

Step by step

1. Identify the exposure: Moe's Bank is owed a payout by Larry's Securities if Sully's defaults, so the CDS exposure jumps from near zero to roughly notional × (1 − recovery) at the default.
2. Identify the collateral: DEF Inc. is a subsidiary of the reference entity, so its credit quality is strongly linked to Sully's own.
3. A default of the parent signals distress across the group, so DEF bonds are likely to fall sharply in price (and may default or be downgraded) at the same moment the exposure rises.
4. Collateral value is therefore lowest exactly when exposure is highest, which is the definition of wrong-way collateral; because the exposure jumps rather than builds gradually, extra collateral cannot be called in time.
5. Answer D.

Why D is correct: Collateral issued by an entity closely tied to the reference entity (or counterparty) loses value when the exposure it secures rises, so it is wrong-way collateral and gives little protection.

Why the others are wrong

- **A:** Assumes a flight to related debt; a group member's bonds fall, not rise, when the parent defaults, so this reverses the direction of the dependence.
- **B:** Treats the collateral as independent of the reference entity (ordinary right-way/neutral collateral); ignores the credit link between parent and subsidiary.
- **C:** Recognises a fall but understates it; the strong dependence means the decline is severe, and with a jump in exposure the collateral cannot be topped up in time, so it covers only a small part.

Q1070 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A dealer enters a 10-year pay-fixed interest rate swap with a hedge fund, and the hedge fund posts high-quality government bonds as collateral against the dealer's exposure. Which of the following statements about this collateral arrangement is correct?

A. It is right-way collateral, because government bonds are high quality and so retain their value in a rate shock.

B. It would be wrong-way collateral if the dealer were instead receiving fixed, because a fall in rates would then raise the exposure and reduce the bonds' value.

C. It is wrong-way collateral, because a rise in interest rates increases the swap's value to the dealer while reducing the value of the government bonds held.

D. It is neutral, because the collateral and the swap are both rate-driven and therefore always change in value by the same amount.

Answer: C

Step by step

1. A pay-fixed swap gains value when interest rates rise, so the dealer's exposure to the hedge fund increases with rates.
2. Government bond prices fall when rates rise, so the collateral value declines at the same time.
3. Exposure up and collateral down together is general wrong-way collateral.

Why C is correct: With a pay-fixed swap collateralized by government bonds, a rate rise raises exposure and reduces collateral value simultaneously.

Why the others are wrong

- **A:** Credit quality does not matter here; the issue is that bond values and swap exposure both move with interest rates, in opposite directions for the dealer.
- **B:** Reversed: for a receive-fixed swap a rate fall raises the swap value but also raises bond prices, which is a right-way (beneficial) arrangement.
- **D:** The two values move in opposite directions for the dealer, which is why protection is eroded rather than preserved.

Q1071 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A risk manager reviews a derivative with a corporate counterparty that posts its own senior bonds as collateral. Which of the following best describes the wrong-way risk associated with this collateral?

A. There is none, because the counterparty's own bonds are liquid and are valued daily at market prices.

B. The risk is severe because the bonds lose all value as soon as the derivative's exposure increases, regardless of the counterparty's credit quality.

C. The risk arises only if the derivative exposure itself rises in an interest rate shock, since the bonds are unrelated to the counterparty's own default.

D. The value of the bonds is likely to fall when the counterparty's credit quality deteriorates, so the collateral weakens exactly as the default probability rises.

Answer: D

Step by step

1. Collateral consisting of the counterparty's own bonds has a direct link between the collateral's value and the poster's credit quality.
2. If the counterparty's credit worsens, bond prices fall and default is more likely, so protection is lowest when most needed.
3. This is a direct but comparatively weak form of wrong-way collateral.

Why D is correct: An entity posting its own bonds is a specific wrong-way collateral example because the collateral value is tied to the poster's own credit quality.

Why the others are wrong

- **A:** Daily valuation does not remove the link; the bonds still lose value as the issuer's credit deteriorates.
- **B:** The bonds do not lose all value merely because exposure rises; the mechanism is the poster's credit deterioration.
- **C:** The wrong-way link is to the poster's credit quality, not to interest rate exposure, and the bonds are clearly related to its default.

LO 38.o

Discuss the impact of wrong-way risk on central counterparties (CCPs).

Q1072 AnalystPrep Q.5475

REBUILT — Original stem was vague and its keyed option B (CVA/DVA trending to zero) was overstated; rebuilt so exactly one option reflects that initial margin reduces but does not eliminate CCP wrong-way risk.

A central counterparty (CCP) raises the initial margin it collects from clearing members. A risk manager at a clearing member notes that the Nasdaq default showed clearing members can lose money through default fund contributions even when the CCP itself is not insolvent. Which of the following best describes the effect of the higher initial margin on the wrong-way risk (WWR) faced by the CCP and its members?

A. It reduces exposure and the CVA to the defaulter, but does not eliminate WWR, because extreme market moves and difficulty auctioning a defaulted portfolio can still produce losses exceeding the margin.

B. It eliminates WWR entirely, because continuous full collateralization removes any possibility of loss to the CCP or its clearing members.

C. It increases the CCP's exposure, because margin posted by members has an opportunity cost and therefore raises member credit risk.

D. It has no effect on exposure, because initial margin only matters for uncollateralized counterparties and clearing members are already fully collateralized.

Answer: A

Key correction: Source key (AnalystPrep) said B (CVA and DVA trend toward zero, lessening losses); the direction is right but the option is overstated and does not capture the residual WWR, so the item was rebuilt with the correct statement as A.

Step by step

1. Initial margin is collateral posted up front against potential future exposure, so it reduces the CCP's exposure (and the CVA to the defaulter) towards zero.
2. Towards zero is not zero: margins are set at a high but finite coverage level, so large, correlated market moves at the time of a default can exceed them.
3. Losses can also arise because it is hard to auction a defaulted portfolio (as in the Nasdaq case); such losses fall on the defaulter-pays resources and then on default funds mutualized across non-defaulting members.
4. Hence higher initial margin mitigates but does not eliminate CCP WWR, which is option A.

Why A is correct: Initial margin lowers exposure and CVA but only to a finite coverage level, so WWR (large market moves coinciding with member default, and auction difficulty) can still exhaust it and hit default funds.

Why the others are wrong

- **B:** Overstates the benefit: 'eliminates' ignores the finite coverage level of margin and the Nasdaq-type losses to default funds.
- **C:** Reverses the effect: opportunity cost of collateral is a cost to the poster, not additional exposure to the CCP; margin reduces exposure.
- **D:** Wrong on both counts: margin reduces exposure for any counterparty, and members are not assumed fully collateralized (default fund losses show residual risk).

Q1073

BT answers P2.T6.24.33.3

REBUILT — Original had two giveaway distractors and no stated remedy; rebuilt around collateral WWR at a CCP (members posting their own-issuer bonds) and the haircut mitigant from the reading.

A central counterparty (CCP) clearing interest rate swaps allows clearing members to post corporate bonds as initial margin. Several members have begun posting bonds issued by themselves or their affiliates, which are illiquid. Which of the following is the most appropriate way for the CCP to address the wrong-way risk (WWR) this creates?

A. Impose higher haircuts on such assets, since their value is likely to fall exactly when the posting member is more likely to default.

B. Apply lower haircuts to the bonds, since a larger accepted collateral value reduces the exposure to the member.

C. Take no action, since corporate bond prices are driven by interest rates and are unrelated to the member's credit quality.

D. Reduce the members' default fund contributions, since the bonds already cover the losses that would arise from their default.

Answer: A

Step by step

1. A CCP relies on the defaulting member's posted margin and default fund contribution to absorb losses, so collateral value at default matters.
2. Bonds issued by the member or its affiliates are positively dependent on its credit quality: if the member defaults, the bonds are likely to lose value or become illiquid (wrong-way collateral).
3. The reading's mitigant is for the CCP to impose higher haircuts on specific assets accepted as collateral that carry WWR.
4. So higher haircuts (A) is correct.

Why A is correct: Own-issuer bonds lose value precisely when the member defaults, so the CCP should discount them more heavily (higher haircuts) to keep the loss waterfall adequate.

Why the others are wrong

- **B:** Lower haircuts credit the bonds with more value and worsen the shortfall when they fall at default.
- **C:** Ignores that issuer credit quality drives the bond's value; the correlation with the poster is exactly the source of WWR.
- **D:** Cutting default fund contributions weakens the loss waterfall further; the bonds' value is unreliable at default.

Q1074

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A risk manager at a central counterparty (CCP) reviews how initial margins and default fund contributions are set. Margins are driven primarily by the market risk of each member's portfolio, and members must meet a minimum

credit standard to be admitted. A colleague argues that wrong-way risk (WWR) is therefore adequately captured. Which of the following is the most appropriate response?

A. She is incorrect, because WWR tends to increase with credit quality, so the CCP implicitly ignores it and should arguably require higher margin and default fund contributions from stronger, more systemic members.

B. She is correct, because margin set from portfolio market risk already captures the jump in exposure at the default of a systemic member.

C. She is incorrect, because WWR falls as the credit quality of a member rises, so CCPs should require less margin from stronger members.

D. She is correct, because the minimum credit standard ensures that members' defaults are independent of their exposures.

Answer: A

Step by step

1. CCPs tend to disassociate credit quality from exposure, charging margin mainly from portfolio market risk.
2. WWR increases with credit quality (a large, strong dealer implies more WWR), so this approach implicitly ignores it.
3. It could be argued that CCPs should demand higher margin and default fund contributions from members with better credit quality and greater systemic importance.

Why A is correct: WWR rises with credit quality, so market-risk-based margins implicitly ignore it; higher margin and default fund contributions from stronger members are warranted.

Why the others are wrong

- **B:** Market-risk margin does not account for jumps and stressed volatility around a systemic member's default, so it understates WWR.
- **C:** The relationship is the opposite: WWR increases with credit quality, so stronger members warrant higher, not lower, margin.
- **D:** A minimum credit standard does not make default independent of exposure; WWR persists, especially for large systemic members.

LO 38.p

Describe the various wrong-way modeling methods including hazard rate approaches, structural approaches, parametric approaches, and jump approaches.

Q1075

BT answers P2.T6.419.2

REBUILT — Original relied on a missing CVA-equation exhibit; equation added to stem and options tightened.

The standard CVA expression with no wrong-way risk is $\text{CVA} = (1 - \text{Rec}) \sum_i \text{DF}(t_i)\, \text{EE}(t_i)\, \text{PD}(t_{i-1}, t_i)$, where EE is the unconditional expected exposure. When wrong-way risk is incorporated, which adjustment to this expression is appropriate?

A. Remove the (1-Rec) term to reflect the most conservative recovery assumption.

B. Increase the discount factor DF(t) to reflect the correlation.

C. Replace EE(t) with the expected exposure conditional on default of the counterparty.

D. No adjustment is possible; the expression cannot be used when wrong-way risk is present.

Answer: C

Step by step

1. The standard formula multiplies unconditional EE by PD, which implicitly assumes exposure and default are independent.
2. With wrong-way risk, exposure at the time of default differs from average exposure.
3. The fix is to use EE(t) conditional on the counterparty defaulting at t, which captures the dependence while keeping the formula's structure.
4. If WWR is present, conditional EE exceeds unconditional EE, so CVA rises.

Why C is correct: Wrong-way risk is built into the CVA formula by using exposure conditional on counterparty default instead of unconditional expected exposure.

Why the others are wrong

- **A:** Recovery is not the dependence being modeled; removing (1-Rec) would set LGD to 100% for all cases and not address exposure-default linkage.

- **B:** The discount factor is a risk-free rate term and has no role in capturing exposure-default correlation.
- **D:** The formula remains usable; it only requires conditional exposure (e.g., via the hazard rate approach) rather than being discarded.

Q1076 AnalystPrep Q.5476

LuxeBank is reassessing its wrong-way risk (WWR) models and is debating whether to enhance its hazard rate, structural, parametric, or jump approach. Given the findings of Chung and Gregory (2019), which enhancement should LuxeBank prioritize if its goal is to better capture scenarios in which exposure can increase significantly and abruptly?

A. Hazard rate approaches, as they offer transparent correlation parameters that are easy to calibrate, making them ideal for precise WWR estimation.

B. Structural approaches, because they provide a strong correlation effect and suit situations where gradual changes in exposure are expected.

C. Parametric approaches, due to their simplicity and use of historical data for calibration, suitably addressing moderate changes in exposure over time.

D. Jump approaches, since they account for abrupt changes in exposure, which is critical for capturing the risk of sudden market shifts.

Answer: D

Step by step

1. Goal: capture exposure that rises significantly and abruptly, which is the specific WWR case (e.g., an FX rate jumping at counterparty default).
2. The jump approach assumes the relevant market variable jumps at the counterparty default time, so it directly models a sudden exposure increase.
3. Chung and Gregory (2019) found material implied jumps (financial institutions about 13.5%, non-financial corporates about 8%, sovereigns about 38.4%), supporting jump modeling.
4. Hazard rate, structural and parametric approaches are continuous/gradual in nature and do not build in a discrete jump at default, so D is the best choice.

Why D is correct: Jump approaches embed a discrete move in the exposure driver at default, so they capture abrupt exposure increases that continuous approaches understate.

Why the others are wrong

- **A:** The hazard (intensity) approach is easy to implement but generally implies only weak dependence between default and exposure, so it tends to underestimate WWR; the parameter-transparency claim is also not its defining feature.
- **B:** The structural approach's correlation parameter is opaque and hard to calibrate, and it describes gradual rather than abrupt exposure changes.
- **C:** The parametric approach links historical exposure to credit spreads; it is simple but depends on history reflecting current conditions and does not capture sudden jumps.

Q1077 AnalystPrep Q.5483

REBUILT — Option A conceded a true statement about continuous models; reworded so the wrong options are clearly wrong for Hudson's observation.

Hudson Capital has observed that collateral becomes much less effective during market stress, particularly on its FX portfolio, where exposure tends to gap upward when a counterparty defaults. Which WWR modeling approach would best reflect this observation, and why?

A. Continuous approaches such as the intensity and structural approaches, because they show margin to be an effective mitigant against WWR.

B. Jump approaches, because they tend to show collateralization as nearly useless when exposure jumps at default.

C. Parametric approaches, because they give a clear-cut measure of collateral effectiveness without needing to consider timing of exposure changes.

D. Hudson's current approach is adequate, because WWR is not materially affected by how quickly exposure increases relative to margin calls.

Answer: B

Step by step

1. Observation: collateral fails when exposure jumps during stress on an FX portfolio.

2. Margin only helps if exposure rises gradually enough for collateral to be received before default; a jump in exposure makes margin useless.
3. Jump approaches (Chung and Gregory 2019 on FX portfolios) show collateralization as near useless against WWR, matching the observation.
4. Continuous (intensity, structural) approaches suggest margin is effective, which contradicts what Hudson sees. Answer B.

Why B is correct: The jump approach models exposure gapping at default, leaving no time for margin to be collected, which matches the observed collateral ineffectiveness.

Why the others are wrong

- **A:** Continuous approaches do suggest margin is effective, which is exactly why they fail to reproduce Hudson's experience; they smooth exposure growth so collateral can keep up.
- **C:** Parametric approaches rely on the historical exposure/spread relationship and have no mechanism for the timing of a jump; 'clear-cut' collateral measurement is not their feature.
- **D:** Timing matters: collateral effectiveness is very timing dependent, so ignoring the speed of exposure growth ignores the key driver of WWR with margin.

Q1078

BT answers P2.T6.24.33.2

REBUILT — Original had giveaway distractors and a vague 'jump intensities and sizes' answer; rebuilt around GARP/Schweser characteristics of the jump approach versus the other WWR methods.

A bank holds a cross-currency swap with a sovereign counterparty and models wrong-way risk with the jump approach. Which of the following statements best describes this approach?

A. It assumes the relevant FX rate jumps by a factor related to a residual value at the counterparty's default time; the jump is typically larger for highly rated, systemically important sovereigns.

B. It simulates a stochastic process for credit spreads and computes conditional exposure on paths where default occurs; it is the easiest method but tends to overstate WWR.

C. It maps the default-time and exposure distributions onto a bivariate distribution, so that an early default coupled with a high exposure signals WWR.

D. It infers WWR from the historical relationship between portfolio value and credit spreads, and is reliable only when history reflects current scenarios.

Answer: A

Step by step

1. Recall the four WWR methods: hazard rate (stochastic spreads), structural (bivariate mapping of default time and exposure), parametric (historical link of exposure and spreads), jump (discrete move at default).
2. The jump approach assumes the FX rate jumps at default; the FX rate depreciates by (1 - RV), where RV is the residual value factor.
3. Evidence (Levy and Levin; Chung and Gregory) shows larger jumps for large firms and highly rated sovereigns because the shock is more significant. This matches A.
4. B describes the hazard rate approach, C the structural approach, D the parametric approach.

Why A is correct: The jump approach is defined by a discrete move in the exposure driver (e.g., FX) at default, with a larger jump for higher-quality, systemically important names.

Why the others are wrong

- **B:** This is the hazard rate (intensity) approach; it is easy to implement but generally shows weak default-exposure dependence, so it tends to understate (not overstate) WWR.
- **C:** This is the structural approach, which maps default and exposure distributions onto a bivariate distribution.
- **D:** This is the parametric approach, which examines the historical link between portfolio exposure and credit spreads.

LO 38.q

Explain the implications of central clearing on wrong-way risk.

Q1079 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A clearing member's risk analyst explains why a central counterparty (CCP) may be particularly prone to wrong-way risk (WWR). Which of the following is the best explanation?

A. Novation to the CCP removes the original counterparty entirely, so any CDS exposure that remains is a residual risk that is not related to the defaulting member.

B. The CCP relies mainly on the defaulting member's margin and default fund contribution, and in a WWR scenario these defaulter-pays resources may prove insufficient, which would push losses onto surviving members and could threaten the CCP's own solvency.

C. Multilateral netting by the CCP eliminates the exposure on CDS positions, so WWR arises only from the CCP's equity capital.

D. The CCP sets initial margin primarily from members' credit ratings, which makes margins unstable when ratings change.

Answer: B

Step by step

1. CCPs depend on posted margin (collateral) and default fund contributions, and a key aim is that losses stay within the defaulting member's resources (defaulter pays).
2. When exposure rises as the member's credit quality deteriorates (WWR), the defaulter's resources may be insufficient to cover the losses.
3. The CCP would then use its own equity and the default fund contributions of non-defaulting members, imposing losses on surviving members, and could itself become insolvent.

Why B is correct: Heavy reliance on collateral and default fund contributions, combined with the possibility that defaulter-pays resources are insufficient under WWR, creates WWR that can reach surviving members and the CCP.

Why the others are wrong

- **A:** Novation replaces the original counterparty with the CCP; it does not remove the defaulting member's risk to the CCP, and the exposure to that member's default remains.
- **C:** Netting reduces but does not eliminate exposure, and WWR arises from the margin and default resources of the defaulting member, not merely the CCP's equity.
- **D:** CCP margins are driven primarily by portfolio market risk and not by external credit ratings; that is why WWR may be ignored.

Q1080 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A CCP sets each clearing member's initial margin and default fund contribution primarily from the market risk of that member's portfolio. A member of the CCP's risk committee warns that, because these requirements do not incorporate wrong-way risk (WWR), the CCP's loss waterfall may be insufficient if a member defaults. Which of the following changes is most consistent with addressing this concern?

A. Use the default fund contributions of non-defaulting members before the defaulting member's margin, so that the defaulter's collateral is preserved for later losses.

B. Reduce initial margin for the highest-credit-quality members, because they are less likely to default and their own resources will therefore always cover their losses.

C. Require higher initial margin and default fund contributions from members with higher credit quality, because WWR tends to increase with increasing credit quality.

D. Set initial margins solely from members' external credit ratings, because ratings fully capture the WWR in each member's portfolio.

Answer: C

Step by step

1. The CCP first relies on the defaulting member's posted margin and default fund contribution; if these do not cover losses, it uses its own equity and the default funds of non-defaulting members.
2. If initial margins and default fund contributions are driven only by portfolio market risk, WWR is implicitly ignored and the waterfall may be insufficient.
3. Since WWR tends to increase with increasing credit quality (e.g. a large, systemically important dealer), it can be argued that the CCP should demand higher margin and default fund contributions from higher-credit-quality members.

Why C is correct: Incorporating WWR means charging more to the members whose default would create the most WWR, and because WWR increases with credit quality, those are the better-credit-quality members.

Why the others are wrong

- **A:** This reverses the loss waterfall: the defaulter-pays resources (the defaulting member's margin and default fund contribution) are used first, and the non-defaulting members' default funds only after these are exhausted.
- **B:** This confuses default probability with WWR: better-credit-quality members are less likely to default, but WWR rises with credit quality, so the impact if they default is likely to be more severe; lowering their margin worsens the problem.
- **D:** CCPs set membership credit standards with their own criteria rather than external ratings, and ratings do not capture the jump in exposure associated with WWR; this would not make the waterfall sufficient.

Q1081 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A CCP is under pressure from clearing members to widen the range of securities it accepts as initial margin. A risk manager warns that this would increase the CCP's wrong-way risk (WWR). Which of the following is the most likely reason?

A. A larger list of eligible assets means that haircuts no longer matter, because every accepted asset is then assumed to be risk-free.

B. Accepting more securities lowers the CCP's liquidity risk, which makes margin calls less frequent and so increases the CCP's exposure to members.

C. Wider eligibility reduces the diversity of collateral, so any WWR is concentrated in a single asset class.

D. Members would tend to post the assets with the greatest risk relative to their haircut, which may also present the greatest WWR to the CCP, and underestimated haircuts would add to the risk.

Answer: D

Step by step

1. CCPs admitting a wide range of securities become exposed to adverse selection.
2. Members choose to post margin with the greatest risk relative to its haircut, which can also be the asset with the greatest WWR (e.g., a European bank posting European sovereign debt).
3. Riskier, illiquid assets also put more emphasis on haircut calculations, which add risk if underestimated.

Why D is correct: Broader eligibility invites adverse selection: members post the assets whose risk is highest relative to the haircut, which may carry the greatest WWR.

Why the others are wrong

- **A:** Haircuts matter more when risky and illiquid assets are accepted, and errors in them can increase risk.
- **B:** Accepting riskier and less liquid assets raises, not lowers, collateral risk; liquidity is not improved by widening eligibility.
- **C:** Wider eligibility increases the range of assets, and the concern is adverse selection toward risky assets, not concentration in one asset class.

Reading 39. The Evolution of Stress Testing Counterparty Exposures

Akhtar Siddique and Iftekhar Hasan (eds.), Stress Testing: Approaches, Methods, and Applications (London, UK: Risk Books, 2013). Chapter 4.

LO	Learning objective	Qs
39.a	differentiate among current exposure, peak exposure, expected exposure, and expected positive exposure.	4
39.b	explain the treatment of counterparty credit risk (CCR) both as a credit risk and as a market risk and describe its implications for trading activities and risk management for a financial institution.	3
39.c	describe a stress test that can be performed on a loan portfolio, and on a derivative portfolio.	7
39.d	differentiate between stressed expected loss and stress loss of a credit portfolio, and calculate the stress loss on a loan portfolio and the stress loss on a derivative portfolio.	7
39.e	describe a stress test that can be performed on CVA.	4
39.f	calculate the stressed CVA and the stress loss on CVA.	3
39.g	calculate the DVA and explain how stressing DVA enters into aggregating stress tests of CCR.	8
39.h	describe the common pitfalls in stress testing CCR.	3

LO 39.a

Differentiate among current exposure, peak exposure, expected exposure, and expected positive exposure.

Q1082

AnalystPrep Q.2700

Which of the following statements correctly defines peak exposure?

A. The larger of zero and the market value of a transaction (or portfolio of transactions) that would be lost if the counterparty defaulted with no recovery.

B. A high percentile (typically 95% or 99%) of the distribution of exposures at a given future date before the maturity of the longest transaction in the netting set.

C. The mean of the distribution of exposures at a given future date before the maturity of the longest transaction in the netting set.

D. The weighted average over time of the expected exposures, with weights equal to the proportion of the time interval each represents.

Answer: B

Step by step

1. Peak exposure is a tail measure: it looks at a high percentile of the exposure distribution, like a VaR-type quantile.
2. It is computed at a particular future date, and typically for many future dates up to the longest maturity in the netting set.
3. Match each option to a named measure: A = current exposure, B = peak exposure, C = expected exposure, D = expected positive exposure (EPE).
4. Only B describes a 95% or 99% percentile, so B is correct.

Why B is correct: Peak exposure is the high-percentile (95% or 99%) point of the exposure distribution at a future date, so it captures the tail rather than the average.

Why the others are wrong

- **A:** This is current exposure (replacement cost), max(0, MtM) today; it is a present value, not a future percentile.
- **C:** This is expected exposure, the mean of the distribution at a future date; the error is using the average instead of a tail percentile.
- **D:** This is expected positive exposure, a time-weighted average of expected exposures; it is an average over time, not a percentile at a date.

Q1083

Schweser Module Quiz 39.1 Q1

Which of the following exposure measures reflects the average (mean) of the distribution of exposures at a specific future date prior to the maturity of the longest-maturity transaction within a netting set?

A. Peak exposure.

B. Current exposure.

C. Expected exposure.

D. Expected positive exposure.

Answer: C

Step by step

1. The key features are: mean of the distribution, at a specific future date.
2. Expected exposure (EE) is the mean of the exposure distribution at a given future date, generated for many dates up to the longest maturity.
3. Compare with the alternatives: peak exposure uses a high percentile, current exposure is today's max(0, MtM), and EPE averages EE across dates.
4. The answer is expected exposure.

Why C is correct: Expected exposure is, by definition, the mean of the exposure distribution at a given future date; the other measures are a percentile, a current value, or a time average.

Why the others are wrong

- **A:** Peak exposure is a 95% or 99% percentile of the distribution at a future date, not the mean.
- **B:** Current exposure is the larger of zero and today's market value; it has no future-date distribution.
- **D:** EPE is a time-weighted average of the expected exposures across dates, not the mean at one specific date; mixing up averaging across scenarios (EE) with averaging across time (EPE) gives this.

Q1084

BT answers P2.T6.25.1.1

REBUILT — Source used non-standard EE/EPE definitions and inconsistent options (-$70m); rebuilt as a self-contained exposure profile with standard definitions (current exposure floored at zero, peak = max percentile, EPE = time-weighted average of EE).

A bank has a single netting set with a counterparty. Its current net market value is -$3m (the bank owes the counterparty). Simulation of the netting set over the next year gives the following exposure profile:

Period end	Interval length (years)	Expected exposure ($m)	95th percentile exposure ($m)
0.25 years	0.25	4	11
0.50 years	0.25	10	19
1.00 year	0.50	6	15

Which row correctly gives the current exposure, the expected positive exposure (EPE) over the one-year horizon, and the maximum peak exposure?

A. Current exposure $0m; EPE $6.5m; peak exposure $19m

B. Current exposure $3m; EPE $6.5m; peak exposure $19m

C. Current exposure $0m; EPE $6.67m; peak exposure $19m

D. Current exposure $0m; EPE $6.5m; peak exposure $10m

Answer: A

Key correction: Source key (A) is retained, but its EE/EPE figures ($5.38m and $8.6m, and a -$70m current exposure) rested on non-standard definitions and did not match the printed table; EPE is the time-weighted average of expected exposures.

Step by step

1. Current exposure = $\max(0, \text{MtM}) = \max(0, -3) = 0$, because the bank would lose nothing if the counterparty defaulted.
2. EPE is the time-weighted average of EE: $0.25(4) + 0.25(10) + 0.50(6) = 1.0 + 2.5 + 3.0 = 6.5$.
3. Peak exposure is the 95th percentile exposure; the maximum across dates is 19 (at 0.5 years).
4. Combining: current exposure 0, EPE 6.5, peak 19, which is option A.

Why A is correct: Current exposure is floored at zero, EPE weights each expected exposure by the share of the year it represents, and peak exposure is the highest 95th-percentile value across future dates.

Why the others are wrong

- **B:** Takes the absolute value of the negative MtM (3) instead of flooring at zero; a negative value means no loss on default.
- **C:** Uses the simple average of the three EEs, $(4 + 10 + 6)/3 = 6.67$, ignoring that the last interval is twice as long.
- **D:** Reports the maximum expected exposure (10) as the peak exposure; peak exposure uses the 95th percentile, whose maximum is 19.

Q1085 BT answers P2.T6.707.1

REBUILT — Original depended on a missing chart and on effective EPE/PFE terminology not in the LO; rebuilt as a conceptual question on the four exposure measures with definitions per the reading.

Which of the following statements about counterparty credit exposure measures is correct?

A. Current exposure is the mean of the distribution of exposures at a given future date.

B. Peak exposure is a high percentile (typically 95% or 99%) of the exposure distribution at a future date, and is usually generated for many future dates.

C. Expected positive exposure is the maximum of the expected exposures across all future dates.

D. Expected exposure is the larger of zero and the current market value of the netting set.

Answer: B

Step by step

1. Recall the four definitions: current exposure = max(0, MtM today); peak exposure = 95% or 99% percentile at a future date; expected exposure = mean at a future date; EPE = time-weighted average of EE.
2. A describes expected exposure, not current exposure.
3. C describes a maximum, but EPE is a weighted average.
4. D describes current exposure, not expected exposure. Only B matches its definition.

Why B is correct: Peak exposure is the high-percentile measure of the future exposure distribution and is computed for many dates, which is exactly what B states.

Why the others are wrong

- **A:** Swaps current exposure with expected exposure; current exposure is today's max(0, MtM), with no distribution.
- **C:** Confuses EPE with a peak (maximum) measure; EPE is the time-weighted average of expected exposures, with weights proportional to interval length.
- **D:** Swaps expected exposure with current exposure; the max(0, MtM) definition is current exposure (replacement cost).

LO 39.b

Explain the treatment of counterparty credit risk (CCR) both as a credit risk and as a market risk and describe its implications for trading activities and risk management for a financial institution.

Q1086 BT answers P2.T6.25.1.3

REBUILT — Original had OCR-damaged options and ambiguous risk labels; rebuilt as a clean two-number calculation of credit-view expected loss and market-view CVA after a spread widening.

A bank has a 5-year interest rate swap with a counterparty. Assume a credit exposure of $60 million, a loss given default of 50%, and a one-period default probability of 1.50%. Treating counterparty credit risk (CCR) as a credit risk, the bank measures its expected loss. Treating CCR as a market risk, the bank values the CVA after credit spreads widen so that the default probability rises to 1.70%, using the same exposure and LGD. Which option gives the expected loss (credit view) and the stressed CVA (market view)?

A. Expected loss $450,000; stressed CVA $510,000

B. Expected loss $900,000; stressed CVA $1,020,000

C. Expected loss $450,000; stressed CVA $60,000

D. Expected loss $510,000; stressed CVA $450,000

Answer: A

Step by step

1. Credit view: expected loss = EAD × LGD × PD.
2. 60,000,000 × 0.50 × 0.015 = 450,000.
3. Market view: CVA reprices with the credit spread, so use the widened PD of 1.70%.
4. 60,000,000 × 0.50 × 0.017 = 510,000.
5. The CVA rises by $60,000 because of the spread widening, an effect the credit-only view misses.

Why A is correct: Under the credit view, loss is measured with the current PD; under the market view, CVA is the market value of CCR and moves with credit spreads, so the higher PD raises it.

Why the others are wrong

- **B:** Omits the 50% LGD (60m x PD), giving 900,000 and 1,020,000.
- **C:** Correct EL, but reports only the change in CVA (60m x 0.5 x 0.2%) rather than the level of stressed CVA.
- **D:** Swaps the two measures: assigns the stressed PD to the credit view and the base PD to the market view.

Q1087 BT answers P2.T6.707.2 · BT notes P2.T6.707.2

Rick is the bank's CCR manager preparing a stress test report for a credit portfolio with positions with five major counterparties. In addition to default risk and credit deterioration, each position exposes the bank to fluctuations in its credit valuation adjustment (CVA). Rick wants to summarize all relevant values without showing more than necessary. Which of the following is the MOST LIKELY number of stress test results he will need to present?

A. One value for the aggregate portfolio
B. Six values: one for the credit risk of each counterparty plus one for credit risk of the portfolio
C. Ten values: for each counterparty, both market and credit risk (but overall portfolio values will be superfluous)
D. At least eleven values: both credit risk and market risk of each counterparty plus the portfolio

Answer: D

Step by step

1. Treating CCR as both a credit risk and a market risk gives two stress views (credit and CVA/market).
2. Stresses are run for each of the 5 counterparties under each view, plus the aggregate portfolio.
3. GARP: the number of stress results is at least twice the number of counterparties, plus one.
4. $2 \times 5 + 1 = 11$, so at least eleven values are needed.
5. The count would at least double again if stressed risk measures were shown in addition to instantaneous shocks, hence 'at least'.

Why D is correct: CCR viewed as both credit and market risk requires at least twice the number of counterparties plus one stress results (here at least 11), including the portfolio.

Why the others are wrong

- **A:** Ignores counterparty-level results and the dual credit/market view.
- **B:** Counts only the credit view (6), omitting the CVA/market risk stresses.
- **C:** Counts both views for the 5 counterparties (10) but wrongly dismisses the portfolio-level result, which is needed for aggregation.

Q1088 Schweser Module Quiz 39.1 Q2

Is the following statement on the treatment of counterparty credit risk (CCR) correct? "Treating CCR as a market risk does not allow an institution to hedge market risk losses, and it exposes the institution to declines in counterparty creditworthiness and default."

A. The statement is correct with regard to both hedging market risk losses and counterparty creditworthiness and default.
B. The statement is incorrect with regard to both hedging market risk losses and counterparty creditworthiness and default.
C. The statement is correct with regard to hedging market risk losses only.
D. The statement is correct with regard to counterparty creditworthiness and default only.

Answer: D

Step by step

1. Part 1: treating CCR as market risk does allow hedging of market risk losses (via CVA hedging), so the claim that it does not is incorrect.
2. Part 2: it leaves the institution exposed to declines in counterparty creditworthiness and default, so this part is correct.
3. Only the second part is correct.

Why D is correct: The market-risk view permits hedging of CVA market risk but leaves default and credit deterioration exposure unaddressed.

Why the others are wrong

- **A:** Wrongly accepts the first part; the market-risk view does permit hedging market risk losses.
- **B:** Wrongly rejects the second part; default and creditworthiness exposure remains.
- **C:** Reverses the correct and incorrect halves of the statement.

LO 39.c

Describe a stress test that can be performed on a loan portfolio, and on a derivative portfolio.

Q1089

BT answers P2.T6.25.2.1

REBUILT — OCR-damaged table (missing rating, garbled option figures $272/$748m) was reconstructed and the options rewritten so exactly one is correct.

A bank has stress tested its largest counterparty balances. Current exposure is the greater of MtM minus collateral and zero; the stressed current exposure is measured after the stress scenario is applied.

Counterparty	Rating	MtM ($m)	Collateral ($m)	Stressed Current Exposure ($m)
G	BBB	-20	0	100
H	BB	-100	0	48
I	A	70	30	34
J	BB	48	36	22

Based on the above, which of the following is the most accurate statement?

A. The portfolio's current exposure is $52m, of which about 77% pertains to Counterparty I.

B. The portfolio's current exposure is -$68m, so no stress test is required.

C. The stressed current exposure total exceeds the portfolio's current exposure by $204m.

D. The largest contributor to the increase from current to stressed exposure is Counterparty H, at $48m.

Answer: A

Step by step

1. Current exposure per counterparty = max(MtM − collateral, 0): G = max(-20,0) = 0; H = max(-100,0) = 0; I = 70 - 30 = 40; J = 48 - 36 = 12.
2. Portfolio current exposure = 0 + 0 + 40 + 12 = $52m; Counterparty I share = 40/52 = 76.9%, about 77%.
3. Stressed current exposure total = 100 + 48 + 34 + 22 = $204m; increase over current = 204 - 52 = $152m.
4. Increase by counterparty: G = 100 - 0 = 100; H = 48 - 0 = 48; I = 34 - 40 = -6; J = 22 - 12 = 10, so G is the largest contributor.

Why A is correct: Exposure is floored at zero for each counterparty because a negative MtM is a liability, not a claim, so only I and J contribute to the $52m.

Why the others are wrong

- **B:** Nets negative MtMs against positive exposures (-20 - 100 + 40 + 12 = -68); exposure cannot be negative per counterparty, and stress testing is still needed.
- **C:** Reports the stressed total ($204m) instead of the difference; the increase is 204 - 52 = $152m.
- **D:** Uses H's stressed exposure of $48m, but G's increase is $100m (100 - 0), which is larger.

Q1090

AnalystPrep Q.1999

Which of the following is a shortcoming of stress tests on current exposure?

A. A lack of relevant financial data.

B. A lack of models tailored to simulate specific events.

C. They provide no information on wrong-way risk.

D. The costs involved are unjustifiably high.

Answer: C

Step by step

1. Stress tests of current exposure re-price the portfolio under a scenario of risk-factor changes, counterparty by counterparty.
2. The two recognised shortcomings are that aggregating results is problematic and that credit quality of the counterparty is not reflected.
3. Because credit quality is omitted, the test cannot show how exposure correlates with the counterparty's creditworthiness, i.e. it gives no information on wrong-way risk.

Why C is correct: Stressed current exposure reflects only trade values, so it says nothing about the correlation between exposure and counterparty credit quality (wrong-way risk).

Why the others are wrong

- **A:** Data are the same market and trade data used for ordinary exposure measurement; scarcity is not a named shortcoming.
- **B:** Re-pricing under a risk-factor scenario uses existing pricing models; model availability is not the stated limi-

tation.
- **D:** Cost is not identified as a problem; the stress tests have been run by institutions for a long time.

Q1091 AnalystPrep Q.2000

Financial institutions often stress test current exposure. What is done to create a stressed current exposure?

A. Assume a scenario of changes in underlying risk factors and re-price the portfolio under that scenario.
B. Select the major counterparties and perform detailed credit analysis of their recent financial information.
C. Estimate the worst-case loss at a given confidence level.
D. Randomly select a few counterparties and re-price their exposures for a given change in one market risk factor.

Answer: A

Step by step

1. A scenario of risk-factor changes (e.g., equity market down 25%) is specified.
2. The portfolio with each counterparty is re-priced under the scenario to give a stressed MtM and stressed current exposure.
3. The procedure is applied to every counterparty, not a sample.

Why A is correct: Stressed current exposure is obtained by scenario-based repricing of the portfolio, applied to each counterparty.

Why the others are wrong

- **B:** Describes credit review of counterparties, which does not involve a risk-factor scenario or repricing.
- **C:** Describes a statistical measure such as VaR or peak exposure, not a scenario-based stress.
- **D:** Stresses are applied to every counterparty, not a random sample.

Q1092 Schweser Module Quiz 39.2 Q1

An analyst notes that stress testing current exposure is problematic because aggregating results is typically not meaningful, although it is easy to account for the credit quality of the counterparty. Are the analyst's statements correct?

A. Correct with regard to both aggregating results and credit quality.
B. Correct with regard to aggregating results only.
C. Correct with regard to credit quality only.
D. Incorrect with regard to both aggregating results and credit quality.

Answer: B

Step by step

1. Aggregation: summing stressed exposures equals the loss if all counterparties defaulted simultaneously, an unlikely scenario, so aggregation is not meaningful. This statement is correct.
2. Credit quality: stressed current exposures reflect only trade values, not the counterparty's ability or willingness to pay, so they do not account for credit quality. The statement that it is easy to account for is incorrect.
3. So only the first statement is correct.

Why B is correct: Aggregation is a genuine shortcoming, but credit quality is not accounted for by stressed current exposure, so the 'easy' statement is wrong.

Why the others are wrong

- **A:** Accepts that credit quality is easy to incorporate; stressed current exposure does not account for it.
- **C:** Reverses the two statements.
- **D:** Rejects the aggregation problem, which is one of the two main shortcomings.

Q1093 AnalystPrep Q.1997

A bank applies a 30% equity market crash stress test to each of its counterparties. How best would this stress test benefit the bank?

A. It allows the bank to set aside sufficient balance sheet provisions.
B. It ranks all counterparties from least to most risky.
C. It estimates when the next equity crash will occur.
D. It identifies the counterparties of concern in such a stress event and shows how much each would owe the bank.

Answer: D

Step by step

1. The portfolio with each counterparty is re-priced under the equity crash to give stressed current exposure.
2. Ranking the stressed exposures shows which counterparties are of concern in that scenario and the amount each would owe.
3. The test does not predict timing, set provisions, or give an overall risk ranking including credit quality.

Why D is correct: Per-counterparty stressed exposure shows which counterparties matter in the scenario and how much they would owe the bank.

Why the others are wrong

- **A:** Provisions relate to expected loss; stressed exposures are not aggregated into a loss estimate and ignore credit quality.
- **B:** The ranking is only for this scenario and ignores credit quality, so it is not an overall risk ranking.
- **C:** Stress tests assess the impact of a scenario; they do not forecast when it occurs.

Q1094

AnalystPrep Q.2001

Stress tests of current exposure are useful for individual counterparties but create problems when aggregated. If a bank sums the stressed current exposures to obtain an aggregate stress exposure, the result represents the loss if every counterparty defaulted in the stress scenario. Is such an aggregate result a meaningful measure of stress loss?

A. No, the aggregate would clearly be an exaggeration of likely losses.

B. Yes, it provides a bird's-eye view of the maximum exposure and is therefore a sound stress loss.

C. Yes, it effectively simulates a global crisis like that of 2007/2008.

D. No, because summing exposures ignores the individual counterparties' risks, which a portfolio model would capture exactly.

Answer: A

Step by step

1. Summing stressed current exposures assumes all counterparties default simultaneously in the scenario.
2. Unless the scenario is extreme, this clearly exaggerates likely losses; default probabilities and correlations are ignored.
3. So the aggregate is not a meaningful stress loss measure; a proper aggregate needs expected exposures plus credit quality information.

Why A is correct: The sum assumes simultaneous default of all counterparties, ignoring default likelihood and correlation, so it overstates likely stress loss.

Why the others are wrong

- **B:** Treats the worst-case sum as a valid stress loss; it is an overstatement, not a loss estimate.
- **C:** Confuses a simultaneous-default sum with a crisis simulation; even in a crisis not every counterparty defaults.
- **D:** Reaches 'No' for the wrong reason: the issue is the simultaneous default assumption, and a simple sum does not use a portfolio model.

Q1095

Mock 2 Q62

REBUILT — Original key (collateral omitted) is not a pitfall in the reading, which lists missing counterparty credit quality and wrong-way information; options rewritten so exactly one is correct.

A risk manager stress tests the bank's current counterparty exposures under a scenario where global equity markets fall 40%. The counterparties are ranked by stressed current exposure and the table is presented to the credit committee. Which of the following is the most significant shortcoming of this approach?

A. The 40% equity decline is implausible as a scenario.

B. It ranks counterparties on stressed current exposure alone, ignoring counterparty credit quality and giving no information on wrong-way risk.

C. It considers only a single risk factor rather than several.

D. It can be run only at the aggregate portfolio level and not for individual counterparties.

Answer: B

Key correction: Source key (Mock 2) said D (collateral not incorporated); per GARP the stressed current exposure table already shows collateral, and the identified weaknesses are aggregation, no credit quality and no wrong-way information, so B is correct.

Step by step
1. Stress tests of current exposure are meaningful per counterparty but have recognised weaknesses.
2. They reflect only the value of trades, not the counterparty's probability of default, so exposure is not weighed by credit quality.
3. They also give no information on wrong-way risk, and aggregation of the results is problematic.
4. Option B states these shortcomings.

Why B is correct: Current exposure stress tests ignore counterparty credit quality and wrong-way risk, so a ranking by exposure alone can mislead on where losses are likely.

Why the others are wrong
- **A:** Scenario severity is a design choice, not a structural defect of the method.
- **C:** Single-factor scenarios are standard stress tests; this is not the main pitfall.
- **D:** Backwards: individual counterparty results are meaningful; aggregation is the problem.

LO 39.d

Differentiate between stressed expected loss and stress loss of a credit portfolio, and calculate the stress loss on a loan portfolio and the stress loss on a derivative portfolio.

Q1096

AnalystPrep Q.3088

REBUILT — Stem mentioned an irrelevant credit VaR definition and two options ($51,094.6, $362,755.4) were implausible; reworded to ask for stress loss and replaced distractors with realistic mistakes.

A portfolio consists of two bonds. Taylor Pharma's bond has a value of $7 million, a default probability of 6% and a recovery rate of 80%. Simon Chemicals' bond has a value of $9 million, a default probability of 5% and a recovery rate of 90%. Under a stress scenario the default probabilities of Taylor and Simon rise to 12% and 18%, respectively; recovery rates are unchanged. What is the stress loss on the portfolio?

A. $129,000
B. $330,000
C. $459,000
D. $201,000

Answer: D

Step by step
1. LGD: Taylor = 1 - 0.80 = 20%; Simon = 1 - 0.90 = 10%.
2. Expected loss: $7{,}000{,}000 \times 0.06 \times 0.20 + 9{,}000{,}000 \times 0.05 \times 0.10 = 84{,}000 + 45{,}000 = 129{,}000$.
3. Stressed expected loss: $7{,}000{,}000 \times 0.12 \times 0.20 + 9{,}000{,}000 \times 0.18 \times 0.10 = 168{,}000 + 162{,}000 = 330{,}000$.
4. Stress loss = $EL^S - EL = 330{,}000 - 129{,}000 = 201{,}000$.

Why D is correct: Stress loss is the increase in expected loss caused by the stress, i.e. stressed expected loss minus (unstressed) expected loss.

Why the others are wrong
- **A:** This is the unstressed expected loss; it ignores the stress entirely.
- **B:** This is the stressed expected loss EL^S; the question asks for the stress loss, so the base EL must be subtracted.
- **C:** Adds the stressed EL and the base EL (330,000 + 129,000) instead of subtracting.

Q1097

BT answers P2.T6.25.2.2

A bank estimates expected loss (EL) and stressed expected loss for its top counterparties under normal and stressed conditions. LGD is assumed to be unchanged under stress and EPE is unchanged (alpha = 1).

Counterparty	PD (%)	EPE (US$m)	LGD	Stressed PD (%)
AA	0.05	190	55%	0.5
BB	0.03	202.5	60%	0.3
CC	0.45	75	70%	0.62
DD	0.9	30	65%	1.2

Which counterparty contributes the most to the stress loss?

A. AA
B. BB
C. CC

D. DD

Answer: A

Step by step

1. Stress loss = EPE × LGD × $(PD^S - PD)$.
2. AA: $190 \times 0.55 \times (0.005 - 0.0005) = 0.470$ m.
3. BB: $202.5 \times 0.60 \times (0.003 - 0.0003) = 0.328$ m.
4. CC: $75 \times 0.70 \times (0.0062 - 0.0045) = 0.089$ m; DD: $30 \times 0.65 \times (0.012 - 0.009) = 0.059$ m.
5. AA has the largest stress loss.

Why A is correct: Stress loss depends on the PD increase scaled by exposure and LGD; AA's tenfold PD jump on a large EPE dominates.

Why the others are wrong

- **B:** BB has the largest EPE, but its PD increase is only 0.27 percentage points, giving 0.328m, below AA.
- **C:** CC has a high base PD but a small PD increase (0.17 pts) on a small EPE; this confuses PD level with PD change.
- **D:** DD has the highest stressed PD, but the increase is small and EPE lowest; picking it uses the stressed PD level instead of the stress loss.

Q1098

BT answers P2.T6.25.2.3

REBUILT — Options were 'closest' with two duplicate $750 values and a $750,000,000 giveaway; rebuilt options around realistic errors with the exact stress loss of about $331,000, and removed the collateral/MtM clutter.

A bank stresses its OTC derivatives portfolio with an equity market shock. For each counterparty, expected loss is EPE x LGD x PD (alpha = 1), and the stress affects PD, LGD and EPE.

Item	Counterparty A	Counterparty B
PD	3%	18%
Stressed PD	4%	23%
LGD	5%	20%
Stressed LGD	10%	40%
EPE ($m)	4.37	3.99
Stressed EPE ($m)	5.46	4.99

What is the stress loss on the portfolio?

A. $481,000

B. $331,000

C. $150,000

D. $42,000

Answer: B

Step by step

1. EL(A) = $4.37 \times 0.05 \times 0.03 = 0.006555$ m; EL(B) = $3.99 \times 0.20 \times 0.18 = 0.14364$ m; total EL = 0.150195 m.
2. Stressed EL(A) = $5.46 \times 0.10 \times 0.04 = 0.02184$ m; Stressed EL(B) = $4.99 \times 0.40 \times 0.23 = 0.45908$ m; total = 0.48092 m.
3. Stress loss = 0.48092 - 0.150195 = 0.330725 m, about $331,000.

Why B is correct: Stress loss = total stressed EL minus total base EL, with PD, LGD and EPE all stressed.

Why the others are wrong

- **A:** This is the stressed expected loss ($481k) before subtracting the base EL.
- **C:** This is the unstressed expected loss ($150k).
- **D:** Stresses only PD while keeping base EPE and LGD (stressed EL 0.1923m less base EL 0.1502m = 0.042m); it ignores the stress to EPE and LGD.

Q1099

BT answers P2.T6.707.3 · BT notes P2.T6.707.2

REBUILT — Merged two identical items; source gave no key for one. Replaced odd distractors with realistic mistakes.

A bank re-examines its stress test of counterparty exposure within an expected-loss framework (PD stress, dotcom crash).

Counterparty	PD	EPE ($mm)	LGD	EL ($mm)	Stressed PD	Stressed EL ($mm)	Stress Loss ($mm)
AA	0.05%	213.00	70.0%	0.0746	0.50%	0.7455	0.6710
BB	0.03%	202.50	60.0%	0.0365	0.30%	0.3645	0.3281
CC	0.45%	75.00	70.0%	0.2363	0.62%	0.3255	0.0893
DD	1.00%	300.00	65.0%	1.9500	1.60%	?	?

What is the stress loss for counterparty DD?

A. $1,950,000
B. $3,120,000
C. $1,170,000
D. $1,800,000

Answer: C

Step by step

1. EL = 0.010 × 300 × 0.65 = 1.95 m.
2. Stressed EL = 0.016 × 300 × 0.65 = 3.12 m.
3. Stress loss = 3.12 - 1.95 = 1.17 m = $1,170,000.

Why C is correct: Stress loss is stressed EL minus EL; only PD changes, so it equals EPE x LGD x (1.6% - 1.0%).

Why the others are wrong

- **A:** This is the base EL, not the stress loss.
- **B:** This is the stressed EL; the base EL was not subtracted.
- **D:** Omits LGD: 300 x (1.6% - 1.0%) = 1.8m, treating the whole exposure as lost.

Q1100

BT answers P2.T6.708.1 · BT notes P2.T6.708.1

A CCR manager analyzes a portfolio of funded loans plus a sub-portfolio of unfunded, collateralized credit derivative positions. For each loan, expected loss is $p_i \times EAD_i \times LGD_i$, summed over the loans. The stress loss is $SL = EL^S - EL$, and expected and stressed expected losses are summed across the two sub-portfolios. Starting from the loan EL formula, he modifies it to produce a CCR-type stressed EL for the derivatives sub-portfolio. Each of the following modifications is likely EXCEPT which one, which is probably NOT appropriate?

A. For both sub-portfolios, retain default probability p_i and stress it to a higher stressed probability p_i^S.
B. For the derivatives sub-portfolio, replace EAD_i with alpha multiplied by expected positive exposure, $\alpha \times EPE_i$.
C. For the derivatives sub-portfolio, stress exposure at default EAD_i to a higher stressed exposure at default EAD_i^S.
D. For the derivatives sub-portfolio, stress expected positive exposure EPE_i to a higher stressed EPE_i^S.

Answer: C

Step by step

1. For derivatives, EAD is stochastic and driven by market factors, so it is replaced by $\alpha \times EPE_i$ (B is appropriate).
2. Stressing is then applied through the market variables that drive EPE, giving $\alpha \times EPE_i^S$, with alpha held constant (D appropriate).
3. PD is stressed in both sub-portfolios (A appropriate), LGD is retained.
4. A separate stressed EAD no longer exists in the derivatives formula, so C is the inappropriate modification.

Why C is correct: In the derivatives EL, EAD is replaced by $\alpha \times EPE$, so the stress is applied to EPE (and PD), not to a stand-alone EAD.

Why the others are wrong

- **A:** Appropriate: stressing PD is the standard stress for both loans and derivatives.
- **B:** Appropriate: this is how CCR exposure enters a portfolio credit model.
- **D:** Appropriate: market-variable shocks flow through the stressed EPE.

Q1101

Mock 1 Q37

Bank Q's regulator requires all banks to calculate the stress loss on loan portfolios. The CFO compiles the following data.

Parameter	Value
Loan portfolio	USD 250 million
Recovery rate	60%
PD, normal conditions	2.5%
PD, stressed conditions	7.0%
Correlation	0.25

What is the stress loss for Bank Q?

A. USD 2.5 million.
B. USD 4.5 million.
C. USD 7.0 million.
D. USD 9.5 million.

Answer: B

Step by step

1. LGD = 1 - 0.60 = 40%.
2. EL = 250 × 0.025 × 0.40 = 2.5 m.
3. Stressed EL = 250 × 0.07 × 0.40 = 7.0 m.
4. Stress loss = 7.0 - 2.5 = 4.5 m. Correlation is not used.

Why B is correct: Stress loss is the stressed EL less the EL; the correlation input is irrelevant in this expected-loss framework.

Why the others are wrong

- **A:** This is the base EL.
- **C:** This is the stressed EL without subtracting the base EL.
- **D:** Adds the two ELs (2.5 + 7.0) instead of subtracting.

Q1102

Schweser Module Quiz 39.2 Q2

REBUILT — Option B was a near-giveaway contradiction of the premise; rewrote distractors to be plausible while keeping the tested point.

Which of the following best explains why a financial institution that stresses EPE does not need to separately consider aggregating those stresses with its loan portfolio?

A. Loans are insensitive to market variables, so their exposures do not change when market variables are shocked.
B. EPE stresses are applied only to current exposure, which has no loan equivalent.
C. EPE and the loan portfolio are negatively correlated, so the stresses offset.
D. EPE and the loan portfolio are positively correlated, so the stresses are already captured in loan losses.

Answer: A

Step by step

1. EPE stresses shock market variables (swap rates, equity prices) that drive derivative exposure.
2. Loan exposures do not depend on these variables, so the shocks produce no change in loan exposure.
3. Hence no separate aggregation of the EPE stresses with the loan portfolio is needed.

Why A is correct: Loan exposures are not sensitive to market variables, so market-variable shocks leave them unchanged.

Why the others are wrong

- **B:** The shocks are indeed to current exposure, but that is not the stated reason; the reason is loans' insensitivity to market variables.
- **C:** Correlation is not the reason; offsetting would still require aggregation.
- **D:** Positive correlation would, if anything, argue for aggregation; it is not the stated reason.

LO 39.e

Describe a stress test that can be performed on CVA.

Q1103 AnalystPrep Q.2007

A bank stress tests its unilateral CVA using the simplified formula (which omits wrong-way risk): $\text{CVA}_n = \text{LGD}_n^* \sum_{j=1}^{T} \text{EE}_n^*(t_j)\, q_n^*(t_{j-1}, t_j)$. In this equation, what does $\text{EE}_n^*(t_j)$ represent?

A. The expected exposure at time j, calculated under a real-world measure for counterparty n.
B. The potential exposure at time j, calculated under a risk-neutral measure for counterparty n.
C. The discounted expected exposure during the jth time period, calculated under a real-world measure for counterparty n.
D. The discounted expected exposure during the jth time period, calculated under a risk-neutral measure for counterparty n.

Answer: D

Step by step

1. CVA is a market-based (fair value) adjustment, so all inputs (LGD, marginal default probability q, exposure) are risk-neutral, not physical.
2. The exposure term is the discounted expected exposure (EE) over the jth time interval, not peak/potential exposure.
3. Hence $\text{EE}_n^*(t_j)$ is the discounted expected exposure in period j under the risk-neutral measure for counterparty n.

Why D is correct: CVA is the risk-neutral, discounted expected loss over the life of the trades, so the exposure input is the discounted expected exposure under the risk-neutral measure.

Why the others are wrong

- **A:** Uses a real-world measure and omits discounting; CVA uses risk-neutral values (physical values are for credit-risk expected loss).
- **B:** Confuses expected exposure with potential (peak) exposure; CVA uses the average, not a high-percentile, exposure.
- **C:** Discounting is right but the measure is wrong; a real-world measure is used for credit-risk EL, not CVA.

Q1104 AnalystPrep Q.2309

REBUILT — Original stem was vague and its correct option only loosely supported; rebuilt to test how a CVA stress test is actually performed (instantaneous shock to market variables).

A bank stress tests the CVA on its derivatives portfolio. Which description of the stress test is most accurate?

A. Apply an instantaneous shock to market variables (credit spreads, market spreads, derivative values), recompute CVA, and take the stress loss as the stressed CVA minus the current CVA.
B. Re-estimate each counterparty's physical probability of default over a one-year horizon from historical default data and recompute expected loss.
C. Hold all market variables constant and increase only the bank's own default probability, then measure the change in CVA.
D. Apply a gradual shock to market variables over the life of each trade and take the stress loss as the current CVA minus the stressed CVA.

Answer: A

Step by step

1. CVA depends on market variables: marginal default probabilities come from credit spreads, LGD from convention or market spreads, and EE from derivative values.
2. A CVA stress applies an instantaneous shock to these variables, which changes the discounted EE and/or the risk-neutral marginal default probability.
3. Stressed CVA is recomputed with the shocked inputs; stress loss = stressed CVA minus current CVA.

Why A is correct: CVA is a market-based measure, so the stress is an instantaneous market shock, and the stress loss is the increase in CVA.

Why the others are wrong

- **B:** Describes a credit-risk expected-loss view (physical PD over a fixed horizon); CVA uses risk-neutral values over the life of the trades.
- **C:** The bank's own default probability enters only via DVA/bilateral CVA; unilateral CVA stress shocks counterparty spreads and exposures.
- **D:** The shock is instantaneous, and the stress loss is stressed CVA minus current CVA (a rise in CVA is a loss), so the sign is reversed.

Q1105

BT answers P2.T6.708.2 · BT notes P2.T6.708.2

REBUILT — Source had a missing/garbled formula; reconstructed the simplified CVA formula and noted it omits wrong-way risk.

The simplified (unilateral) CVA for a counterparty is $\text{CVA}_n = \text{LGD}_n^* \sum_{j=1}^{T} \text{EE}_n^*(t_j)\, q_n^*(t_{j-1}, t_j)$. It is applied to a single derivative with a five-year life, with an interval of three months between dates. Each of the following statements is true EXCEPT which is false?

A. A stressed CVA is likely to shock EE to a stressed expected exposure.

B. To capture wrong-way risk, the formula should add a debt value adjustment (DVA) term.

C. A stressed CVA is likely to shock the marginal default probability q to a stressed risk-neutral marginal default probability.

D. The expected exposure and default probability will be estimated at twenty points in time.

Answer: B

Step by step

1. Number of dates: 5 years / 0.25 years = 20, so D is true.
2. Stress shocks market variables, which move the discounted EE and the risk-neutral marginal PD (from credit spreads); A and C are true.
3. The formula omits wrong-way risk (dependence between exposure and default). Capturing it requires a conditional EE (exposure-PD correlation), not a DVA term.
4. DVA turns unilateral CVA into bilateral CVA (own default); it is unrelated to wrong-way risk, so B is false.

Why B is correct: Wrong-way risk is captured by modelling exposure conditional on default (exposure-PD correlation), whereas DVA reflects the bank's own default and creates bilateral CVA.

Why the others are wrong

- **A:** True: the stress applies shocks to the exposure input.
- **C:** True: credit spread shocks stress the risk-neutral marginal default probability.
- **D:** True: 5 x 4 = 20 quarterly intervals; the mistake would be counting 5 or 60.

Q1106

BT notes P2.T6.708.3

REBUILT — Formula was garbled and option B oddly worded; restated BCVA in words and reworded B to the GARP point about current vs expected exposure.

Acme Bank measures counterparty credit exposure using bilateral CVA (BCVA) and will stress test it. BCVA = unilateral CVA (counterparty default, weighted by Acme's survival probability) minus a DVA term (Acme's default, using negative expected exposure and weighted by the counterparty's survival probability). Each of the following statements is true EXCEPT which is false?

A. If Acme's own credit quality improves, Acme can experience a loss on the BCVA.

B. For stress testing BCVA, expected exposures should be replaced by current exposures.

C. Compared with unilateral CVA, BCVA adds survival probabilities of Acme and of its counterparty.

D. BCVA allows CCR to be treated as a market risk, so CCR stress losses can be added to market risk stress losses if the same shocks are applied.

Answer: B

Step by step

1. Improved own credit quality lowers the value of Acme's option to default (DVA falls), which can produce a loss: A is true.
2. BCVA includes Acme's survival probability in the CVA part and the counterparty's in the DVA part: C is true.
3. BCVA is a market-value measure, so with the same shocks applied to trading and BCVA, results aggregate by simple addition: D is true.
4. Using current exposure instead of expected exposure (or EPE) can cause substantial errors, particularly at the money; EE/EPE should be used. B is false.

Why B is correct: Current exposure is not an appropriate substitute for expected exposure in CCR stress testing; EE or EPE should be used.

Why the others are wrong

- **A:** True: a counterintuitive result of DVA - own credit improvement reduces DVA gains.
- **C:** True: both survival probabilities enter the bilateral formula.
- **D:** True: coherent, additive aggregation with market-risk stress tests is a key advantage of BCVA.

LO 39.f

Calculate the stressed CVA and the stress loss on CVA.

Q1107

AnalystPrep Q.3045

REBUILT — Original distractors had no traceable error; replaced them with options that each follow from a realistic mistake and clarified the wording.

A risk analyst is calculating the stress loss on a bond portfolio's credit value adjustment (CVA). Under normal market conditions the portfolio CVA is $4.3 million. After an instantaneous 30% fall in markets, the stressed CVA is $76.0 million. What is the stress loss on the portfolio CVA?

A. $4.3 million
B. $71.7 million
C. $76.0 million
D. $80.3 million

Answer: B

Step by step

1. Stress loss = stressed CVA - current CVA, i.e. $\text{CVA}^S - \text{CVA}$.
2. $76.0 - 4.3 = 71.7$ million.

Why B is correct: The stress loss is the increase in CVA caused by the shock, i.e. the difference between the stressed and the unstressed CVA.

Why the others are wrong

- **A:** This is the unstressed CVA; it ignores the shock entirely.
- **C:** This is the stressed CVA itself; the current CVA was not subtracted.
- **D:** Added the two CVAs (76.0 + 4.3) instead of subtracting.

Q1108

BT answers P2.T6.25.3.1

A bank runs a CVA stress test. Counterparty 1 has an expected exposure (with netting) of $5 million, LGD of 60%, and a marginal default probability q^* of 1.5%. After a market shock that widens credit spreads, exposure is unchanged at $5 million but q^* rises to 3%. What is the stress loss on the CVA, and by what percentage does the stressed CVA exceed the initial CVA?

A. $90,000; 50%
B. $45,000; 50%
C. $45,000; 100%
D. $90,000; 100%

Answer: C

Step by step

1. Initial CVA $= 0.60 \times 5{,}000{,}000 \times 0.015 = 45{,}000$.
2. Stressed CVA $= 0.60 \times 5{,}000{,}000 \times 0.03 = 90{,}000$.
3. Stress loss $= 90{,}000 - 45{,}000 = 45{,}000$.
4. Percentage increase $= 45{,}000/45{,}000 = 100\%$.

Why C is correct: With exposure and LGD unchanged, CVA is proportional to q*, so doubling q* doubles CVA: the loss equals the initial CVA (100% increase).

Why the others are wrong

- **A:** Reports the stressed CVA ($90,000) as the loss and measures the 50% against the stressed level (45/90).
- **B:** Loss amount correct, but 50% divides the loss by the stressed CVA instead of the initial CVA.
- **D:** Reports the stressed CVA ($90,000) as the stress loss rather than subtracting the initial CVA.

Q1109

BT answers P2.T6.25.3.2

REBUILT — Original exhibit was missing; reconstructed a three-counterparty data table and recomputed the options.

A bank stresses the CVA on its OTC derivative receivables, with and without netting agreements. LGD is 60% for all counterparties, and CVA for each counterparty = LGD x EE x q*. The shock widens credit spreads (exposures unchanged) and raises q* as shown.

Counterparty	EE no netting ($m)	EE with netting ($m)	Initial q*	Stressed q*
1	20	5	1.5%	3.0%
2	10	4	2.0%	4.0%
3	8	2	1.0%	2.5%

Which statement gives the total stressed CVA across the three counterparties?

A. With netting $0.216m; without netting $0.720m
B. With netting $0.105m; without netting $0.348m
C. With netting $0.111m; without netting $0.372m
D. With netting $0.720m; without netting $0.216m

Answer: A

Step by step

1. Stressed CVA without netting = $0.6\,(20 \times 0.03 + 10 \times 0.04 + 8 \times 0.025) = 0.6 \times 1.2 = 0.720$ m.
2. Stressed CVA with netting = $0.6\,(5 \times 0.03 + 4 \times 0.04 + 2 \times 0.025) = 0.6 \times 0.36 = 0.216$ m.
3. Netting reduces exposure, so the netted figure is the smaller one.

Why A is correct: Stressed CVA uses the stressed q* with each counterparty's exposure; netting lowers EE so the netted total is lower ($0.216m vs $0.720m).

Why the others are wrong

- **B:** Used the initial q* (unstressed CVA: 0.105 and 0.348) rather than the stressed q*.
- **C:** Computed the stress loss (stressed minus initial CVA: 0.111 and 0.372), not the stressed CVA.
- **D:** Reversed the netting logic; netting cannot increase exposure.

LO 39.g

Calculate the DVA and explain how stressing DVA enters into aggregating stress tests of CCR.

Q1110

AnalystPrep Q.2005

When stress testing counterparty credit risk, a financial institution also considers scenarios in which the institution itself, as well as its counterparty, can default on its obligations. This approach effectively constitutes:

A. Unilateral CVA
B. Bilateral CVA
C. Negative exposure from the institution's point of view
D. General wrong-way risk

Answer: B

Step by step

1. Unilateral CVA reflects only the possibility that the counterparty defaults on the institution.
2. Allowing the institution to default on its counterparty as well adds the institution's option to default (the DVA component).
3. CVA plus this DVA-type term is the bilateral CVA (BCVA), so the answer is B.

Why B is correct: BCVA captures both the loss from the counterparty's default (CVA) and the value of the institution's own option to default (DVA), so it is the measure that matches a two-sided default scenario.

Why the others are wrong

- **A:** Unilateral CVA considers only the counterparty's default; it ignores the institution's own default.
- **C:** Negative exposure (NEE) is only one input to the DVA leg, not the overall approach of letting both parties default.
- **D:** Wrong-way risk concerns the correlation between exposure and the counterparty's default probability, not two-sided default.

Q1111

BT answers P2.T6.25.3.3

REBUILT — Original exhibit was missing; rebuilt with a self-contained single-period data table and recalculated every statement.

A bank (LGD of 60%, marginal default probability of 2%, survival probability $S_I = 0.97$) stress tests three counterparties over one period using unilateral CVA = $\text{LGD}_n \times EE \times q_n$ and bilateral CVA = $\text{LGD}_n \times EE \times q_n \times S_I - \text{LGD}_I \times NEE \times q_I \times S_n$. Amounts are in $m.

Counterparty	EE	NEE	LGD_n	q_n	S_n
1	40	10	60%	2%	0.95
2	100	30	60%	4%	0.90
3	50	0	50%	1%	0.98

Which of the following statements about the CVA analysis is incorrect?

A. Counterparty 2 requires the largest charge under both the unilateral CVA and the BCVA.
B. In aggregate, the unilateral CVA (USD 3.13m) exceeds the BCVA (USD 2.60m).
C. The BCVA of Counterparty 3 equals its unilateral CVA (USD 0.25m) because its NEE is zero.
D. The DVA component for Counterparty 1 is USD 0.114m.

Answer: C

Key correction: Source key (A) referred to an exhibit that is not available; the item was rebuilt and the incorrect statement is now C.

Step by step

1. Unilateral CVA: CP1 $0.6 \times 40 \times 0.02 = 0.480$; CP2 $0.6 \times 100 \times 0.04 = 2.400$; CP3 $0.5 \times 50 \times 0.01 = 0.250$; total 3.130.
2. BCVA credit leg adds $S_I = 0.97$: CP1 0.4656; CP2 2.3280; CP3 0.2425.
3. DVA leg $0.6 \times NEE \times 0.02 \times S_n$: CP1 $0.6 \times 10 \times 0.02 \times 0.95 = 0.114$; CP2 $0.6 \times 30 \times 0.02 \times 0.90 = 0.324$; CP3 0.
4. BCVA: CP1 0.3516; CP2 2.0040; CP3 0.2425; total 2.598, about 2.60.
5. Even with zero NEE, CP3's BCVA (0.2425) is below its unilateral CVA (0.25) because the CVA leg is multiplied by the institution's survival probability 0.97, so C is false.

Why C is correct: In the BCVA formula the loss from the counterparty's default is conditional on the institution surviving first, so even with no DVA the charge is scaled by $S_I < 1$.

Why the others are wrong

- **A:** True: CP2 has the largest charge under both measures (2.400 unilateral; 2.004 bilateral).
- **B:** True: 3.130 vs 2.598; the DVA offsets and the survival factor reduce the bilateral figure.
- **D:** True: 0.6 x 10 x 0.02 x 0.95 = 0.114.

Q1112

BT answers P2.T6.25.4.1

REBUILT — Original exhibit (exposures, spreads, CVA/DVA/BCVA summary) was missing; rebuilt with explicit inputs so BCVA must be computed and ranked.

Xavier Bank has exposure to four counterparties, which also hold credit balances with the bank (negative exposure). For the bank, LGD is 50%, marginal default probability is 1% and survival probability is $S_I = 0.98$. Single-period data, in $m:

Counterparty	EE	NEE	LGD_n	q_n	S_n
Alpha Pie	60	25	60%	2.0%	0.97
Bravo Ltd	90	70	60%	3.0%	0.96
Charlie Ltd	120	150	50%	3.0%	0.98
Delta Ltd	40	40	50%	5.0%	0.90

Using BCVA $= \text{LGD}_n\, EE\, q_n S_I - \text{LGD}_I\, NEE\, q_I S_n$, which counterparty has the highest BCVA adjustment?

A. Alpha Pie
B. Delta Ltd
C. Bravo Ltd
D. Charlie Ltd

Answer: C

Step by step

1. CVA leg $\text{LGD}_n\, EE\, q_n\, S_I$: Alpha 0.7056; Bravo 1.5876; Charlie 1.7640; Delta 0.9800.
2. DVA leg $0.5 \times NEE \times 0.01 \times S_n$: Alpha 0.1212; Bravo 0.3360; Charlie 0.7350; Delta 0.1800.
3. BCVA = CVA leg - DVA leg: Alpha 0.5844; Bravo 1.2516; Charlie 1.0290; Delta 0.8000.
4. Bravo Ltd has the highest BCVA.

Why C is correct: Charlie has the largest CVA leg, but its large NEE produces a large DVA offset; net of that offset Bravo has the highest bilateral adjustment.

Why the others are wrong

- **A:** Alpha's BCVA (0.584) is the lowest of the four; it is chosen only by mis-ranking.
- **B:** Delta has the highest counterparty default probability (5%), but its small EE keeps BCVA at 0.800.
- **D:** Charlie has the highest unilateral CVA (1.80) and CVA leg (1.764); ignoring the 0.735 DVA offset gives this

answer.

Q1113 BT answers P2.T6.25.4.2

REBUILT — Original exhibit and garbled stem were unusable; rebuilt as a single-counterparty BCVA stress-loss calculation with explicit inputs.

A bank has a single-period exposure to a counterparty with EE = \$150m, NEE = \$100m, $\text{LGD}_n = 60\%$, $q_n = 2\%$ and $S_n = 0.97$. The bank's own LGD is 50%. Currently the bank's marginal default probability is $q_I = 1\%$ and its survival probability is $S_I = 0.98$. In a stress scenario the bank's credit spread widens, so $q_I = 5\%$ and $S_I = 0.94$; all counterparty inputs and exposures are unchanged. Using BCVA $= \text{LGD}_n\, EE\, q_n S_I - \text{LGD}_I\, NEE\, q_I S_n$ and stress loss = stressed BCVA - current BCVA, what is the result of the stress?

A. Stress loss of \$2.01m
B. Stress gain of \$0.73m
C. Stress gain of \$1.94m
D. Stress gain of \$2.01m

Answer: D

Step by step

1. Current BCVA: $0.6 \times 150 \times 0.02 \times 0.98 - 0.5 \times 100 \times 0.01 \times 0.97 = 1.7640 - 0.4850 = 1.2790$.
2. Stressed BCVA: $0.6 \times 150 \times 0.02 \times 0.94 - 0.5 \times 100 \times 0.05 \times 0.97 = 1.6920 - 2.4250 = -0.7330$.
3. Stress loss $= -0.733 - 1.279 = -2.012$. BCVA is a charge, so a fall of 2.01 is a gain of about \$2.01m.

Why D is correct: The wider own spread raises the DVA (value of the option to default) far more than it lowers the CVA leg, so BCVA falls and the stress produces a counterintuitive gain; the reverse (own credit improving) would produce a loss.

Why the others are wrong

- **A:** Sign error: treats the fall in the BCVA charge as a loss; since BCVA is a charge, a decrease is a gain.
- **B:** Reports the stressed BCVA (-0.73) itself rather than subtracting the current BCVA.
- **C:** Leaves the CVA leg unchanged (keeps S_I = 0.98), giving -0.661 - 1.279 = -1.94; it omits the effect of the lower S_I on the CVA leg.

Q1114 BT answers P2.T6.708.3

The bilateral credit value adjustment (BCVA) formula is: $\text{BCVA} = \sum_{n=1}^{N} \text{LGD}_n \sum_{j=1}^{T} EE_n(t_j)\, q_n(t_{j-1}, t_j)\, S_I(t_{j-1}) - \sum_{n=1}^{N} \text{LGD}_I \sum_{j=1}^{T} NEE_n(t_j)\, q_I(t_{j-1}, t_j)\, S_n(t_{j-1})$.

Acme Bank applies this formula to measure its counterparty credit exposure and will stress test the BCVA. Each of the following statements is true EXCEPT which one?

A. If Acme's own credit quality improves, Acme can experience a loss on the BCVA.
B. For stress testing the BCVA, expected exposures should be replaced by observable current exposures.
C. Compared with unilateral CVA, BCVA adds the survival probabilities of Acme, S(I), and of its counterparty, S(n).
D. BCVA allows CCR to be treated as a market risk, so CCR can be incorporated into market risk stress testing coherently and additively.

Answer: B

Step by step

1. Check each statement against the BCVA framework.
2. A: losses depend on the firm's own credit spread, so improved own credit (lower DVA) can cause a loss: true.
3. C: S_I scales the CVA leg and S_n scales the DVA leg: true. D: applying the same shocks to market variables lets BCVA stress results be added to market risk stress losses: true.
4. B: using current exposure can lead to significant errors (especially at-the-money); expected exposure or EPE should be used, so B is false.

Why B is correct: Stress tests of CCR should be built on expected exposure or EPE, not current exposure, so replacing expected exposures with current exposures is the false statement.

Why the others are wrong

- **A:** True: the counterintuitive result that improved own credit quality can cause a BCVA loss is a known feature.
- **C:** True: BCVA brings S_I into the CVA leg and S_n into the DVA leg.
- **D:** True: BCVA treats CCR as market risk, so its stress gains and losses can be added to market risk stress results.

Q1115 GARP PE2 Q75

REBUILT — Original was a multi-subject stem (operational, liquidity, cyber items) outside LO 39.g; rebuilt to test only BCVA stress-testing practice.

A bank is designing the stress testing of its counterparty credit risk using the bilateral CVA (BCVA). Which of the following approaches is consistent with best practice?

A. Stress only the counterparty's credit spread, since the bank's own spread affects only the DVA, which is not relevant to stress losses.

B. Calculate the stress loss as the current BCVA minus the stressed BCVA, and keep BCVA stresses separate from market risk stress tests.

C. Assume that an improvement in the bank's own credit quality can never produce a loss on the BCVA.

D. Stress the bank's own credit spread together with counterparty spreads, consider their correlation, and apply the same market shocks as for the trading portfolio so the BCVA stress loss can be added to market risk stress results.

Answer: D

Step by step

1. BCVA depends on both the counterparty's and the bank's own spreads (survival probabilities and DVA), so both must be stressed.
2. The bank must also consider how its own credit spread is correlated with its counterparties' spreads.
3. Stress loss = stressed BCVA - current BCVA.
4. Because BCVA treats CCR as a market risk, applying the same shocks to the trading portfolio and the BCVA lets the BCVA stress loss be added to market risk stress results.

Why D is correct: BCVA depends on both parties' credit, and applying common shocks allows coherent aggregation with market risk stress tests.

Why the others are wrong

- **A:** Ignores the DVA and the S_I terms; the bilateral approach requires the bank's own spread to be stressed.
- **B:** Reverses the definition (stress loss = stressed - current) and forgoes the key benefit that BCVA stress results can be aggregated with market risk.
- **C:** Wrong: losses depend on the firm's own credit spread and can arise when its own credit quality improves (lower DVA).

Q1116 Schweser Module Quiz 39.2 Q3

Is the following statement on bilateral credit value adjustment (BCVA) correct? "The formula for BCVA is similar to the formula for CVA, except that the BCVA formula uses expected positive exposure (EPE) and it incorporates the probability of the counterparty's survival."

A. The statement is correct with regard to both EPE and probability of survival.

B. The statement is correct with regard to EPE only.

C. The statement is correct with regard to probability of survival only.

D. The statement is incorrect with regard to both EPE and probability of survival.

Answer: C

Step by step

1. BCVA differs from CVA in two ways.
2. First, it adds expected negative exposure (ENE/NEE), measured from the counterparty's perspective, not EPE: the EPE part of the statement is wrong.
3. Second, the option to default depends on the counterparty surviving first, so the counterparty's survival probability enters: that part is correct.

Why C is correct: The new exposure term is ENE (the DVA leg) and the survival probability is added, so only the survival part is correct.

Why the others are wrong

- **A:** EPE is already used in the CVA formula; the new term in BCVA is ENE.
- **B:** Reverses the facts: the EPE part is wrong, the survival part is right.
- **D:** Survival probability is indeed incorporated, so the statement is not wrong on both.

Q1117 BT answers P2.T6.25.4.3 701

REBUILT — Original relied on a missing exhibit with inconsistent figures; rebuilt with self-contained CVA/DVA data testing the stressed BCVA loss and its counter-intuitive sign.

A bank's current CVA on a counterparty is 4.0m and its DVA is 3.0m. Under a stress scenario in which the bank's own credit spread widens sharply, the CVA rises to 6.5m and the DVA rises to 9.0m. Using BCVA = CVA - DVA, which of the following best describes the stress result?

A. A stress loss of 2.5m, being the increase in CVA.
B. A stress gain of 3.5m, counterintuitive because it arises from the deterioration of the bank's own credit quality.
C. A stress loss of 3.5m, because the stressed BCVA is below the current BCVA.
D. A stress loss of 8.5m, from stressed BCVA of 15.5m less current BCVA of 7.0m.

Answer: B

Key correction: Source key (BT answers) said C for the original exhibit-based item; the item was rebuilt so that key is not carried over.

Step by step

1. Current BCVA = 4.0 - 3.0 = 1.0m.
2. Stressed BCVA = 6.5 - 9.0 = -2.5m.
3. Stress loss = stressed BCVA - current BCVA = -2.5 - 1.0 = -3.5m, i.e. a gain of 3.5m (the net adjustment falls).
4. The gain arises because the bank's own spread widened (DVA up), a counterintuitive result that must be flagged.

Why B is correct: Stress loss on BCVA is stressed minus current BCVA; a negative value is a gain driven by the bank's own credit deterioration (higher DVA).

Why the others are wrong

- **A:** Looked only at the CVA increase (6.5 - 4.0) and ignored DVA.
- **C:** Right magnitude but wrong sign: a fall in BCVA is a gain, not a loss.
- **D:** Added DVA to CVA instead of subtracting (15.5 - 7.0).

LO 39.h

Describe the common pitfalls in stress testing CCR.

Q1118 Mock 4 Q76

REBUILT — Original items on margin period of risk and collateral exclusion are outside the Reading 39 pitfalls and the 'sound' item was only loosely tied to the LO; rewritten around the stated pitfalls in stress testing CCR.

An internal auditor at Bank Alpha is reviewing how the bank stress tests counterparty credit risk (CCR) and aggregates the results with its loan portfolio stress tests. Which of the following practices is consistent with sound practice?

A. Running stressed current exposures through a portfolio credit risk model to aggregate CCR with the loan portfolio.
B. Revaluing an at-the-money derivative portfolio under a large market shock using delta sensitivities only.
C. Summing the stressed current exposures across counterparties to obtain the aggregate stress loss.
D. Using expected exposure (or expected positive exposure), not current exposure, when combining CCR stress results with loan portfolio stress results.

Answer: D

Key correction: Source key (Mock 4) said A for the original item set; the original items were replaced, so that key no longer applies.

Step by step

1. Recall the pitfalls: stressed current exposures cannot be meaningfully aggregated, and current exposure ignores how exposure can change before default.
2. Portfolio credit risk models should be fed expected exposures (EE/EPE), not current exposures.
3. Summing stressed current exposures equals the loss if every counterparty defaulted in the scenario, a clear overstatement.
4. Delta approximations are poor for CCR because exposure is highly nonlinear (floored at zero), especially for at-the-money portfolios and large shocks.
5. Only D uses the correct exposure measure for aggregation.

Why D is correct: Expected exposure reflects the possible future exposure at default and is the appropriate input when aggregating CCR with loan or trading stress results; current exposure is not.

Why the others are wrong

- **A:** Pitfall: feeding current exposures into a portfolio credit risk model is incorrect; expected exposures should be used.
- **B:** Pitfall: delta linearisation is highly misleading for nonlinear CCR exposures, especially at-the-money with large shocks.
- **C:** Pitfall: the sum equals the loss if all counterparties default simultaneously, which overstates loss; default likelihood and correlation are ignored.

Q1119

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A risk analyst at a large bank is preparing the bank's counterparty credit risk (CCR) stress test for a portfolio of at-the-money interest rate swaps with a single counterparty. The current mark-to-market value of the portfolio is zero. Under the stress scenario, the mark-to-market value of the portfolio rises instantly by USD 40 million. The analyst plans to use the resulting change in current exposure to measure the increase in exposure that will be combined with the bank's loan portfolio stress losses. Using the expected exposure to the same shock instead would show an increase of approximately USD 20 million. Which of the following best describes the effect of using current exposure rather than expected exposure?

A. It overstates the increase in exposure by approximately USD 20 million, because for at-the-money exposures the difference between current exposure and expected exposure is almost half the size of the shock.

B. It has no material effect on the result, because for at-the-money exposures current exposure and expected exposure respond equally to a market shock.

C. It overstates the increase in exposure by approximately USD 40 million, because current exposure recognizes the full shock while expected exposure recognizes none of it.

D. It understates the increase in exposure by approximately USD 20 million, because current exposure ignores the potential for the portfolio value to rise further before maturity.

Answer: A

Step by step

1. Current exposure is max(MtM, 0). The portfolio starts at zero, so the USD 40 million shock raises current exposure by the full USD 40 million.
2. Expected exposure for an at-the-money position responds only partly: the increase is approximately USD 20 million.
3. Difference = 40 - 20 = USD 20 million, which is almost half the shock, consistent with the GARP chapter's result for at-the-money exposures.
4. Current exposure therefore overstates the increase in exposure by approximately USD 20 million.

Why A is correct: At-the-money, current exposure jumps by the whole shock while expected exposure moves by only about half of it. The gap is approximately USD 20 million, so stressing current exposure overstates the exposure to be aggregated with loan losses.

Why the others are wrong

- **B:** Wrong. The GARP chapter shows the two measures diverge most for at-the-money exposures (the mistake is treating current exposure as a good proxy for expected exposure).
- **C:** Wrong. Expected exposure does recognize part of the shock (about USD 20 million), so the overstatement is the difference of 20, not the whole 40 (the mistake is ignoring the change in expected exposure).
- **D:** Wrong direction. Current exposure rises by the full USD 40 million while expected exposure rises by about USD 20 million, so current exposure overstates rather than understates the increase (the mistake is reversing the direction of the error).

Q1120

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A market risk manager at a bank wants to reduce the computational cost of stressing the bank's derivatives portfolio for counterparty credit risk by using delta sensitivities to approximate the change in exposure under large market moves. The portfolio is currently close to at-the-money with respect to its main risk factors. Which of the following statements best describes a pitfall of this approach?

A. Delta sensitivities are accurate for CCR because exposure is a linear function of the underlying risk factors, so

the approximation is reliable even for large price moves.

B. Exposure is highly nonlinear in the risk factors, so linearizing with delta can introduce errors that are not obvious and can be highly misleading, especially for at-the-money portfolios subject to large price moves.

C. Delta approximations are unreliable only for deep out-of-the-money portfolios, where exposure is close to zero, and are acceptable for at-the-money portfolios.

D. The errors from delta approximations are large but easy to detect, because they always cause the stressed exposure to be understated relative to full revaluation.

Answer: B

Step by step

1. Counterparty exposure is the positive part of portfolio value, so it behaves like an option and is highly nonlinear in the risk factors.
2. A delta (first-order) approximation ignores this curvature; the error grows with the size of the move.
3. The GARP chapter notes the errors are not obvious and linearization can be highly misleading, with at-the-money portfolios and large moves especially prone to error.
4. Only B states this correctly.

Why B is correct: Because exposure is highly nonlinear, delta linearization can be highly misleading, and at-the-money portfolios under large price moves are especially prone to error. The errors are not obvious.

Why the others are wrong

- **A:** Wrong. Treats exposure as linear; the chapter says delta use is especially problematic for CCR because it is highly nonlinear.
- **C:** Wrong. Reverses the risk; at-the-money portfolios, not only out-of-the-money ones, are especially prone to error under large moves.
- **D:** Wrong. The chapter says the errors are not obvious, not easy to detect, and it does not say they always understate exposure.

Reading 40. Structured Credit Risk

Allan Malz, Financial Risk Management: Models, History, and Institutions (Hoboken, NJ: John Wiley & Sons, 2011). Chapter 9.

LO	Learning objective	Qs
40.a	describe common types of structured products.	6
40.b	describe tranching and the distribution of credit losses in a securitization.	7
40.c	describe a waterfall structure in a securitization.	3
40.d	identify the key participants in the securitization process and describe conflicts of interest that can arise in the process.	8
40.e	calculate and evaluate one or two iterations of interim cash flows in a three-tiered securitization structure.	6
40.f	describe the treatment of excess spread in a securitization structure and estimate the value of the overcollateralization account at the end of each year.	9
40.g	explain the tests on the excess spread that a custodian must go through at the end of each year to determine the cash flow to the overcollateralization account and to the equity noteholders.	3
40.h	describe a simulation approach to calculating credit losses for different tranches in a securitization.	3
40.i	explain how the default probabilities and default correlations affect the credit risk in a securitization.	11
40.j	explain how default sensitivities for tranches are measured.	3
40.k	describe risk factors that impact structured products.	3
40.l	define implied correlation and describe how it can be measured.	3
40.m	identify the motivations for using structured credit products.	3

LO 40.a

Describe common types of structured products.

Q1121 AnalystPrep Q.1849

Some financial instruments are created by securitizing tranches of other collateralized debt obligations (CDOs). Such instruments are commonly known as:

A. Collateralized loan obligations (CLOs)
B. Synthetic CDOs
C. CDO squared
D. Mortgage-backed securities (MBS)

Answer: C

Step by step

1. Identify the collateral: the pool consists of tranches (liabilities) of other CDOs.
2. A securitization whose collateral pool consists of CDO liabilities adds another level of securitization and is called a CDO-squared (CDO of CDOs).

Why C is correct: A CDO-squared repackages tranches of other CDOs, adding a second layer of securitization.

Why the others are wrong

- **A:** A CLO is backed by a pool of corporate loans, not CDO tranches; confusing the collateral type.
- **B:** A synthetic CDO gets its credit exposure through credit derivatives such as CDS, not by holding CDO tranches.
- **D:** An MBS is backed by mortgage loans; it is a single-level securitization.

Q1122 AnalystPrep Q.4395

Troy Dean, FRM, has invested in a mortgage-backed security that entitles him to a pro rata share of all principal and interest payments made on the underlying pool of mortgage loans. Which of the following types of structured products has Dean invested in?

A. Mortgage pass-through
B. Collateralized mortgage obligation
C. Covered bond
D. Collateralized debt obligation

Answer: A

Step by step

1. A pro rata share of all pool cash flows with no tranching means investors receive the pool's cash flows less servicing fees.
2. This is the definition of a mortgage pass-through security.

Why A is correct: Pass-throughs pass pool cash flows pro rata to investors; no cash-flow tranching is involved.

Why the others are wrong

- **B:** A CMO divides pool cash flows into tranches by predetermined rules (e.g., sequential pay), so investors do not get a pro rata share.
- **C:** A covered bond stays on the originator's balance sheet and is paid and guaranteed by the originator, not passed through from the pool.
- **D:** A CDO tranches credit risk of a pool of debt instruments; holders do not receive pro rata pool cash flows.

Q1123

AnalystPrep Q.5526

REBUILT — Original was a giveaway (CDO versus corporate bonds, T-bills and loans); rebuilt to distinguish CDOs from covered bonds, pass-throughs and CMOs.

Which of the following structured products is a true off-balance-sheet securitization that tranches a pool of credit-risky debt so that junior tranches absorb losses first and senior tranches have the lowest probability of write-down?

A. Covered bond
B. Mortgage pass-through security
C. Collateralized debt obligation (CDO)
D. Sequential-pay collateralized mortgage obligation (CMO) in which tranches differ in prepayment risk

Answer: C

Step by step

1. Covered bonds stay on the originator's balance sheet and are not true securitizations: rule out A.
2. Pass-throughs are true securitizations but pass cash flows pro rata without credit tranching: rule out B.
3. A sequential-pay CMO tranches by timing of principal and prepayment risk, not by credit-loss priority: rule out D.
4. A CDO pools risky debt and creates tranches with different credit risk, with equity first-loss and senior last: C.

Why C is correct: Only a CDO combines a true (bankruptcy-remote) securitization with tranches ranked by credit-loss priority.

Why the others are wrong

- **A:** On-balance-sheet, backed by the originator's guarantee; not a true securitization and not tranched by credit risk.
- **B:** True securitization but no tranching, so all holders share losses pro rata.
- **D:** Tranches differ in maturity and prepayment risk rather than in credit-loss priority.

Q1124

AnalystPrep Q.5531

Global Finance Bank (GFB) is exploring a complex structured product. It would create a pool from conventional debt obligations such as mortgages or loans, as well as other securitized assets such as asset-backed securities (ABS), mortgage-backed securities (MBS), and collateralized loan obligations (CLOs), and then issue new securities backed by this pool. What type of structured product is GFB considering?

A. Mortgage-backed security (MBS)
B. Asset-backed commercial paper (ABCP)
C. Structured investment vehicle (SIV)
D. Collateralized debt obligation (CDO)

Answer: D

Step by step

1. The collateral is a mix of debt obligations and other securitizations (ABS, MBS, CLOs).
2. Securitizations that repackage such collateral and issue tranched bonds against it are CDOs.

Why D is correct: A CDO issues securities against a pool of debt obligations and/or other securitized products.

Why the others are wrong

- **A:** An MBS is backed only by mortgages, not by a mix including ABS and CLOs.
- **B:** ABCP is short-term paper backed by assets; it does not describe repackaging securitization tranches into new tranched securities.
- **C:** An SIV funds long-term assets with short-term borrowing; it is not defined by tranching a pool of securitized assets.

Q1125 BT notes P2.T6.312.1

REBUILT — Reworded the double-negative stem into a single 'NOT a valid difference' and aligned option wording with the reading.

Which of the following is NOT a valid difference between a covered bond and a true securitization?

A. In a covered bond the cover pool remains on the originator's balance sheet, whereas in a true securitization the assets are removed from it

B. Covered bond principal and interest are paid and guaranteed by the originator, whereas in a securitization they depend on the performance of the collateral pool

C. A covered bond involves no true sale of assets to a bankruptcy-remote special purpose vehicle, whereas a true securitization does

D. A covered bond does not issue securities or raise funds in the capital markets, whereas a true securitization does

Answer: D

Step by step

1. A: covered bonds are on-balance-sheet securitizations; true securitizations are off-balance sheet. Valid difference.
2. B: covered bond payments are guaranteed by the originator and do not depend on the pool's performance. Valid difference.
3. C: covered bond assets are not in a bankruptcy-remote structure and investors have recourse to the originator. Valid difference.
4. D: covered bonds are bonds, so they do issue securities and raise funding. Not a valid difference, so D is the exception.

Why D is correct: Both covered bonds and securitizations issue securities to raise funds; the real differences are balance-sheet treatment, recourse and payment source.

Why the others are wrong

- **A:** True statement, so it is a valid difference and not the exception.
- **B:** True statement; the originator's guarantee is what separates covered bonds from true securitizations.
- **C:** True statement; absence of a true sale to a bankruptcy-remote SPV is why covered bonds are not true securitizations.

Q1126 BT answers L2.T5.56.1

What is the essential difference between a pass-through mortgage backed security (MBS) and a collateralized mortgage obligation (CMO)?

A. The nature and type of collateral in a CMO is more varied

B. The pass-through MBS can shield some investors from prepayment risk

C. CMO issues classes with different risk/return/cash-flow properties

D. An MBS is backed by an agency (Fannie Mae, Freddie Mac, Ginnie Mae)

Answer: C

Step by step

1. A pass-through passes the pool net cash flows pro rata to all holders (single class).
2. A CMO divides (tranches) the mortgage pool cash flows into separate classes with different risk, return, and cash-flow characteristics.
3. This tranching is the essential difference.

Why C is correct: A CMO tranches mortgage cash flows into classes with different risk and payment priority, whereas pass-through holders share cash flows pro rata.

Why the others are wrong

- **A:** Both are backed by mortgage pools; collateral type is not the defining difference.
- **B:** Reversed: pro rata pass-through holders bear prepayment risk equally, whereas CMO tranching can shield some classes.
- **D:** Most pass-throughs are agency MBS, but agency backing is not what separates them from CMOs.

LO 40.b

Describe tranching and the distribution of credit losses in a securitization.

Q1127 AnalystPrep Q.1850

In a securitization structure, which segment is characterized by not receiving a fixed coupon payment but bearing the full brunt of default risks?

A. Equity
B. Junior debt
C. Senior debt
D. Capital stack

Answer: A

Step by step

1. The equity tranche has the lowest priority and absorbs the first losses up to a prespecified level.
2. It is paid only from residual cash flows after all other claims are satisfied, so it carries no fixed coupon.
3. Equity is therefore the segment that bears the first and full brunt of defaults.

Why A is correct: The equity tranche is the first-loss piece and receives only residual cash flows, so its return is variable rather than a fixed coupon.

Why the others are wrong

- **B:** Junior (mezzanine) debt absorbs losses only after equity is fully written down and pays a relatively high fixed coupon or spread.
- **C:** Senior debt has the highest priority, is protected by subordination, and receives the lowest coupon, which is still a stated coupon.
- **D:** The capital stack is the whole set of tranches (the structure), not a segment of it.

Q1128 AnalystPrep Q.1854

REBUILT — Options B and D were identical in the source and the stem was ambiguous about counting; rewrote with four distinct options stated in words.

In a securitization with equity, mezzanine (junior debt), and senior tranches, which of the following correctly states which attachment and detachment points each tranche has?

A. Equity: attachment point only; Mezzanine: both; Senior: detachment point only
B. Equity: both an attachment and a detachment point; Mezzanine: both; Senior: attachment point only
C. Equity: detachment point only; Mezzanine: both an attachment and a detachment point; Senior: attachment point only
D. Equity: detachment point only; Mezzanine: attachment point only; Senior: attachment point only

Answer: C

Step by step

1. The boundary between two tranches is the attachment point of the more senior tranche and the detachment point of the more junior tranche.
2. Equity is the bottom tranche: it has no tranche below it, so no attachment point, and it ends at its boundary with the mezzanine (detachment point).
3. Mezzanine sits between two boundaries, so it has an attachment point (with equity) and a detachment point (with senior).
4. Senior is the top tranche: it has an attachment point (with mezzanine) but nothing above it, so no detachment point.

Why C is correct: Only the interior tranche has a boundary on both sides; the bottom tranche has just a detachment point and the top tranche just an attachment point.

Why the others are wrong

- **A:** Reverses the logic: equity has no tranche below it, so it cannot have an attachment point, and senior has no tranche above it.
- **B:** Treats the 0% start of equity as an attachment point; by convention the equity tranche has only a detachment point.
- **D:** Forgets that the mezzanine detaches from the senior tranche above; it has both points, not only an attachment point.

Q1129 AnalystPrep Q.2707

A structured solution worth $2 billion has the following capital structure:

Tranche	Size
Equity tranche	$500 million
Junior debt	$700 million
Senior debt	$800 million

Which of the following statements correctly describes how an $800 million loss will be absorbed by the different tranches?

A. All of the losses are absorbed by the senior tranche.
B. $700 million is absorbed by the junior tranche and $100 million is absorbed by the equity tranche.
C. $500 million is absorbed by the equity tranche and $300 million is absorbed by the senior tranche.
D. $500 million is absorbed by the equity tranche and $300 million is absorbed by the junior tranche.

Answer: D

Step by step

1. Losses are absorbed from the bottom of the capital structure upward: equity first.
2. Equity absorbs $\min(\$800, \$500) = \$500$ million, leaving $\$800 - \$500 = \$300$ million.
3. Junior debt absorbs $\min(\$300, \$700) = \$300$ million, leaving zero for the senior tranche.

Why D is correct: Losses hit the equity tranche until it is wiped out, then the junior tranche; the senior tranche is untouched because the loss does not exceed $1.2 billion (equity plus junior).

Why the others are wrong

- **A:** Puts the loss on the senior tranche first, reversing the loss priority.
- **B:** Loads the junior tranche first and gives equity the remainder; equity is always first-loss.
- **C:** Skips the junior tranche and charges the remainder to senior, ignoring that junior has $700m of capacity.

Q1130 AnalystPrep Q.5519

REBUILT — Stem referred to an exhibit not provided and had a giveaway option; made self-contained with a numeric loss example and plausible distractors.

A $400 million collateral pool funds three tranches: senior (75%, $300 million), mezzanine (20%, $80 million), and equity (5%, $20 million). The pool suffers credit losses of $70 million. How are the losses allocated?

A. Equity $3.5 million; Mezzanine $14 million; Senior $52.5 million
B. Equity $0; Mezzanine $70 million; Senior $0
C. Equity $20 million; Mezzanine $50 million; Senior $0
D. Equity $0; Mezzanine $0; Senior $70 million

Answer: C

Step by step

1. Equity ($20m) is first-loss: it absorbs $\min(\$70, \$20) = \$20$ million, leaving $50 million.
2. Mezzanine ($80m) absorbs $\min(\$50, \$80) = \$50$ million, leaving zero.
3. Senior absorbs $0 because the loss is below the $100m (equity plus mezzanine) of subordination.

Why C is correct: Losses are allocated strictly by seniority from the bottom up, so equity is exhausted first, then mezzanine takes the remainder, and senior is protected.

Why the others are wrong

- **A:** Allocates the loss pro rata to tranche size (5%/20%/75%); losses follow priority, not size.
- **B:** Hits the mezzanine before equity; equity is the first-loss tranche.
- **D:** Hits the senior tranche first, the reverse of the loss waterfall.

Q1131 AnalystPrep Q.5527

Which of the following statements correctly describes tranching?

A. The process of pooling together different loans or debt securities to create a diversified portfolio.
B. Dividing the cash flows generated from the collateral pool into different securities with varying levels of credit risk.
C. Transferring the default risk of the collateral pool to a third-party guarantor.
D. The process of converting illiquid assets into marketable securities.

Answer: B

Step by step

1. Securitization first pools the credit-sensitive assets.
2. Tranching then splits the liabilities/cash flows of the pool into a capital structure of securities with different priority and so different credit risk.
3. Option B describes this split.

Why B is correct: Tranching slices the pool cash flows into securities (equity, mezzanine, senior) with different loss priority and therefore different credit risk.

Why the others are wrong

- **A:** Describes pooling (the collateral side), not the splitting of liabilities into tranches.
- **C:** Describes credit protection/guarantee, which is not what tranching is.
- **D:** Describes securitization/liquidity transformation in general, not the tranche structure.

Q1132

BT answers L2.T6.106.2

REBUILT — Source relied on a case-study exhibit (with an inconsistent width of 77.55%) that is not available; rebuilt as a self-contained attachment/detachment width question.

A securitization has the following tranches:

Tranche	Attachment point	Detachment point
Equity	0%	6%
Mezzanine	6%	14%
Senior	14%	100%

Which tranche is the widest (thickest), and what is its width?

A. Equity, 6%

B. Mezzanine, 8%

C. Senior, 14%

D. Senior, 86%

Answer: D

Step by step

1. Width = detachment point minus attachment point.
2. Equity: $6\% - 0\% = 6\%$; Mezzanine: $14\% - 6\% = 8\%$; Senior: $100\% - 14\% = 86\%$.
3. The senior tranche is widest at 86%; widths sum to 100%.

Why D is correct: A tranche's width is its detachment point minus its attachment point; senior tranches typically occupy most of the capital structure.

Why the others are wrong

- **A:** Equity is the thinnest slice, not the widest.
- **B:** Mezzanine is purposefully thin (8%), far narrower than senior.
- **C:** Uses the senior attachment point (14%) as its width instead of $100\% - 14\% = 86\%$.

Q1133

Schweser Module Quiz 40.1 Q1

How many of the following statements concerning the capital structure in a securitization are most likely correct?

I. The mezzanine tranche is typically the smallest tranche size.

II. The mezzanine and equity tranches typically offer fixed coupons.

III. The senior tranche typically receives the lowest coupon.

A. No statements are correct.

B. One statement is correct.

C. Two statements are correct.

D. Three statements are correct.

Answer: B

Step by step

1. I is false: the equity tranche is typically the smallest part of the capital structure (mezzanine is thin, but equity is thinnest).
2. II is false: equity receives residual cash flows and typically carries no fixed coupon.
3. III is true: senior is safest and receives the lowest coupon.

4. Exactly one statement is correct.

Why B is correct: Only III is correct: priority and safety rise with seniority while coupon falls, and the equity tranche is the thinnest, residual-claimant slice.

Why the others are wrong

- **A:** Misses that III is correct (lowest-risk tranche, lowest coupon).
- **C:** Wrongly counts I (mezzanine smallest) or II (fixed equity coupon) as true.
- **D:** Counts all three, treating equity as a fixed-coupon, or mezzanine as the smallest, tranche.

LO 40.c

Describe a waterfall structure in a securitization.

Q1134 AnalystPrep Q.2886

What best describes a waterfall structure in a securitization?

A. The excess spread when there is no default
B. A financial intermediary that aggregates underlying loans and designs the securitization structure
C. Rules on how cash flows from the collateral are distributed to the various securities in the capital structure
D. The process of creating a securitized credit product

Answer: C

Step by step

1. The waterfall is the set of rules that allocates collateral cash flows (after fees and costs) among the securities in the capital structure.
2. Payments are made in sequence, top down: senior coupons first, then mezzanine, with the residual going to equity (and, above the overcollateralization trigger, to the trust account).
3. This matches option C.

Why C is correct: The waterfall is the payment priority rule set: cash from the collateral is distributed to the tranches in order of seniority, top down.

Why the others are wrong

- **A:** Excess spread is the residual cash flow left after senior and mezzanine claims are paid; it is an output of the waterfall, not the waterfall itself.
- **B:** This describes the arranger/underwriter (or special purpose vehicle) that pools loans and structures the deal, not the payment rules.
- **D:** This is securitization itself (pooling credit-sensitive assets and issuing new securities); the waterfall is only one feature of the structure.

Q1135 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A structurer at a bank is modeling the first-period cash flows of a securitization. The collateral is 400 identical loans of $1 million each, paying the market reference rate (currently 4%) plus 300 bps. The senior, mezzanine, and equity tranches are 70%, 25%, and 5% of the pool, with the senior and mezzanine tranches paying the market reference rate plus 1.5% and plus 5%, respectively. The equity tranche may receive at most $2.0 million per period; any residual above that amount is diverted to an excess trust account. The structurer assumes that 5% of the loans default and that the defaulted loans produce no interest. How much cash flows into the excess trust account in the first period?

A. $2.20 million
B. $1.60 million
C. $0.20 million
D. $4.20 million

Answer: C

Step by step

1. Interest collected = 400 x $1 million x 7% x (1 - 0.05) = $26.6 million.
2. Senior coupon = $280 million x 5.5% = $15.4 million; mezzanine coupon = $100 million x 9.0% = $9.0 million; total bond coupons = $24.4 million.

3. Residual = 26.6 - 24.4 = $2.2 million. Equity receives the capped $2.0 million, so $0.2 million is diverted to the trust account.

Why C is correct: After paying the senior and mezzanine coupons in order, the residual of $2.2 million exceeds the $2.0 million equity cap, so the $0.2 million excess goes to the trust account.

Why the others are wrong

- **A:** Treats the entire residual of $2.2 million as excess and forgets that equity first receives up to its $2.0 million cap.
- **B:** Ignores the 5% default haircut: 28.0 - 24.4 = 3.6 residual, less the $2.0 million equity cap, gives 1.6. Defaulted loans pay no interest, so collections must be reduced.
- **D:** Forgets to add the market reference rate to the mezzanine spread (coupon of 5% instead of 9%), understating bond coupons at $20.4 million and overstating the residual (6.2 - 2.0 = 4.2).

Q1136 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A securitization has a waterfall that pays the senior coupon first, then the mezzanine coupon, and then the equity tranche, with no reserves yet in the excess trust account. In a given period the senior coupon due is $15.4 million and the mezzanine coupon due is $9.0 million. Because of a higher-than-assumed default rate, the loan interest collected is only $23.8 million. Which of the following correctly describes the distribution of this period's cash flows?

A. The senior and mezzanine tranches share the $0.6 million shortfall pro rata to their coupons, so each is paid slightly less than its promised coupon.

B. The senior and mezzanine tranches are paid in full, with the $0.6 million shortfall funded by the equity holders.

C. The senior tranche receives $14.8 million and the mezzanine receives $9.0 million, because the most senior claim absorbs the first shortfall.

D. The senior tranche receives $15.4 million, the mezzanine receives $8.4 million, and neither the equity tranche nor the trust account receives anything.

Answer: D

Step by step

1. Senior is paid first: 15.4 of the 23.8 available, leaving 8.4.
2. Mezzanine is owed 9.0 but only 8.4 remains, so it bears the $0.6 million shortfall.
3. Nothing remains for equity or for the trust account.

Why D is correct: The waterfall pays top down: the senior tranche is paid in full before the mezzanine receives anything, so the shortfall falls on the mezzanine and the equity gets nothing.

Why the others are wrong

- **A:** Assumes pari passu sharing (senior would get about $15.02 million and mezzanine about $8.78 million). A waterfall is sequential, not pro rata.
- **B:** Equity holders are residual claimants and have no obligation to fund shortfalls; they simply receive nothing when cash is insufficient.
- **C:** Reverses the priority: it makes the senior tranche bear the first shortfall, when shortfalls reach the junior claims first.

LO 40.d

Identify the key participants in the securitization process and describe conflicts of interest that can arise in the process.

Q1137 AnalystPrep Q.5524

Global Finance Bank (GFB) is auditing its securitization activities for conflicts of interest among participants. Which of the following scenarios correctly describes a conflict of interest in the securitization process?

A. The rating agency, paid by the issuer, gives a high rating to the securitized assets to protect its business relationship with GFB.

B. Investors demand higher yields for buying the senior tranches of the securitized assets.

C. GFB, as originator, funds a reserve account to enhance the credit quality of the lower tranches, which may reduce the yield of the pool.

D. The underwriter performs due diligence to check the quality of the mortgage loans in the pool.

Answer: A

Step by step

1. Rating agencies are typically paid by the issuer and also influence the structure (attachment points, required enhancement).
2. This creates a conflict between winning and keeping rating business (profit, growth) and providing an objective assessment; the agency has an incentive to require less enhancement.
3. Scenario A matches this conflict exactly; B, C and D describe ordinary market pricing, credit enhancement and diligence, none involving a clash of interests.

Why A is correct: An issuer-paid rating agency that also shapes the structure is tempted to be lenient to keep business, which is the textbook conflict of interest in securitization.

Why the others are wrong

- **B:** Investors wanting compensation for risk is normal pricing, not a conflict between parties' duties and incentives.
- **C:** Credit enhancement by the originator is a structural feature (and retained risk actually aligns interests); lower yield is a trade-off, not a conflict.
- **D:** Due diligence is the underwriter doing its job; it works against, not for, a conflict.

Q1138

AnalystPrep Q.5535

REBUILT — Original described fiduciary/covenant-enforcement duties not covered by the GARP chapter; restated around the administrative role GARP assigns to trustees and custodians.

In a securitization, which participants are the economically minor players tasked with keeping records, verifying documentation, moving cash flows among deal accounts and paying noteholders?

A. Trustee and custodian
B. Rating agency and underwriter
C. Servicer and manager
D. Originator and sponsor

Answer: A

Step by step

1. Trustees and custodians play an administrative role: record keeping, verifying documents, transferring funds between accounts, and paying noteholders.
2. They take no credit or structuring decisions and have little economic exposure, so they are the minor players.
3. The other pairs have active economic roles: structuring/rating, collecting and managing collateral, or creating the loans.

Why A is correct: Trustees and custodians are the administrative back-office participants who verify documents and move cash according to the deal rules.

Why the others are wrong

- **B:** The rating agency assesses credit quality and shapes subordination; the underwriter structures, markets and warehouses - both are active and exposed to conflicts.
- **C:** The servicer collects/distributes cash and resolves distress; the manager selects and monitors collateral - both make economically important decisions.
- **D:** The originator creates the loans and the sponsor supplies most of the collateral; they are the source of the assets, not administrators.

Q1139

BT answers P2.T6.312.3 · BT notes P2.T6.312.3

In a securitization, which key participant is most likely to incur warehousing risk?

A. Originator
B. Underwriter
C. Rating agency
D. Servicer

Answer: B

Step by step

1. During the aggregation phase the underwriter (arranger) accumulates the loans on its balance sheet before the securities are sold.
2. Warehousing risk is the risk that the deal will not be completed or that the value of the accumulated collateral falls.
3. So the underwriter bears it; the others do not hold the unsold collateral awaiting issuance.

Why B is correct: The underwriter warehouses the collateral and holds finished liabilities until investors buy them, so it bears the risk of an unmarketed deal or falling collateral value.

Why the others are wrong

- **A:** The originator creates and funds the loans, but it is the underwriter that holds the pool during aggregation for issuance.
- **C:** The rating agency only assesses and assigns ratings; it holds no collateral.
- **D:** The servicer collects and distributes cash flows and manages distressed loans; it does not warehouse the pool.

Q1140 Schweser Module Quiz 40.2 Q1

Which of the following pairs of participants in the securitization process is least likely to face a conflict of interest?

A. Credit rating agency and servicer
B. Servicer and underwriter
C. Custodian and trustee
D. Trustee and manager

Answer: C

Step by step

1. Rating agency: paid by issuers and influences structure, so it is conflicted.
2. Servicer: may delay foreclosure to earn fees and must choose between bondholder classes, so it is conflicted.
3. Manager: tempted to minimize monitoring effort, hence first-loss retention; also conflicted.
4. Custodian and trustee play only an administrative role, so this is the only pair with no such conflict. Pairs A, B and D each contain at least one conflicted party.

Why C is correct: Custodians and trustees only verify documents and move funds, so they have little discretion to favour their own interests.

Why the others are wrong

- **A:** Both are conflicted: the CRA wants issuer business and the servicer wants fees from delayed foreclosure.
- **B:** The servicer faces conflicts with bondholders, and the underwriter has structuring and warehousing incentives.
- **D:** The manager has an incentive to minimize credit-monitoring effort, which is why first-loss retention is used; this makes the pair conflicted.

Q1141 BT answers L2.T6.103.6

REBUILT — Original asked for the Ashcraft friction label (asset manager vs investor = principal-agent), which is outside the 2026 Schweser/GARP text; rebuilt to test the in-scope manager-investor conflict of interest and its remedy.

A securitization pool is actively managed. Investors delegate the ongoing monitoring of the pool's credit quality to the manager, who would naturally prefer to minimize monitoring effort. Which mechanism is commonly used to align the manager's incentives with those of investors?

A. Requiring the manager to hold a portion of the most senior tranche
B. Requiring the manager to hold a first-loss (equity) portion of the deal
C. Paying the manager a fixed fee independent of pool performance
D. Having the rating agency monitor the pool on the manager's behalf

Answer: B

Step by step

1. Identify the conflict: investors delegate monitoring to the manager, who bears the cost of effort but not the losses (principal-agent problem).
2. Remedy: tie the manager's payoff to pool performance.
3. The manager holding the first-loss portion means the manager absorbs the earliest credit losses, so it has a direct incentive to monitor credit quality.
4. This retention is also enshrined in Dodd-Frank risk retention rules.

Why B is correct: First-loss retention gives the manager skin in the game, so poor monitoring directly reduces the manager's own payoff.

Why the others are wrong

- **A:** A senior piece is last to absorb losses and is nearly insulated from pool performance, so it creates little incentive to monitor.
- **C:** A fixed fee unrelated to performance gives no incentive; it worsens the effort problem.
- **D:** Rating agencies assess ratings and structure, they do not act as the investors' monitor, and they have their own conflict (issuer-paid).

Q1142

BT answers L2.T6.103.7

REBUILT — Original asked for the Ashcraft friction label (investor vs CRA = model error), which is outside the Schweser/GARP text; rebuilt to test the in-scope CRA conflict of interest.

Credit rating agencies in a securitization are typically paid by the issuer and also help determine the deal structure. Which conflict of interest does this create?

A. The agency has an incentive to require more credit enhancement so that fewer tranches are rated investment grade

B. The agency has an incentive to require less credit enhancement, allowing the issuer to create a larger set of investment-grade tranches

C. The agency has an incentive to delay assigning ratings to increase its fees

D. The agency has an incentive to hold the first-loss piece of the deal

Answer: B

Step by step

1. Issuers pay the agency, so the agency wants to win rating assignments and grow business.
2. The agency also tells the issuer how much enhancement (subordination) is needed for the top tranches to be investment grade.
3. Requiring less enhancement lets the issuer create a larger senior, investment-grade portion with lower spreads and a wider investor base.
4. So the conflict is a bias toward lenient (lower) enhancement; investors cope by demanding wider spreads or doing their own credit review.

Why B is correct: Issuer-pay compensation combined with a role in setting attachment points rewards the agency for requiring less enhancement.

Why the others are wrong

- **A:** Reverses the incentive: more enhancement would shrink the investment-grade portion and displease issuers.
- **C:** Delay does not serve the agency's business-winning incentive and is not the conflict described.
- **D:** Rating agencies do not hold tranches; first-loss retention is a remedy applied to managers, not to rating agencies.

Q1143

BT answers L2.T6.103.2

REBUILT — Original asked which friction is susceptible to fraud, with no parseable key and an inconsistent explanation; rebuilt to test the in-scope servicer conflict of interest.

A servicer of a CMBS deal must decide how to resolve one distressed loan: foreclose or extend the loan. Which statement best describes the resulting conflict of interest?

A. Foreclosure helps the junior bond avoid an immediate writedown, so juniors favor it

B. Extension may increase the risk to the senior bond because a larger property loss erodes credit enhancement, while it gives the junior bond some chance of recovery

C. Foreclosure causes an immediate writedown of the senior bond, so seniors favor extension

D. The servicer faces no conflict because it must maximize the loan's present value

Answer: B

Step by step

1. Foreclosure: the senior bond is repaid early (probably no writedown), but the junior bond may take an immediate writedown if the property is sold at a loss.
2. Extension: the junior bond avoids the immediate loss and keeps some chance of recovery.
3. But the senior bond then faces the risk of a greater eventual property loss that erodes credit enhancement.
4. Whatever the servicer does, it aligns better with some bond classes than others, so a conflict remains.

Why B is correct: Either resolution favors some tranches over others: foreclosure favors seniors, extension favors juniors.

Why the others are wrong

- **A:** Reversed: foreclosure can trigger an immediate junior writedown, so juniors prefer extension.
- **C:** Reversed: foreclosure usually brings seniors early principal repayment rather than a writedown.
- **D:** Maximizing present value is the servicer's obligation, but the action chosen still favors some classes over others, so the conflict persists.

Q1144

BT answers L2.T6.104.1

REBUILT — Original asked about Ashcraft-Schuermann's seven frictions (not in the Schweser/GARP reading, ambiguous 'least responsible'); rebuilt to test the servicer conflict of interest from LO 40.d.

A commercial mortgage-backed securitization holds a distressed loan. The servicer must decide between foreclosing on the property (expected to be sold at a loss) and extending the loan. Which of the following statements best describes the conflict of interest the servicer faces?

A. Foreclosure favors the junior bondholders, because they avoid an immediate writedown while the senior bonds absorb the loss.

B. Extension favors the junior bondholders, who avoid an immediate writedown and keep a chance of recovery, while the senior bondholders risk a larger property loss that erodes their credit enhancement.

C. Because the servicer is obliged to maximize the present value of the loan, its decision affects all bondholder classes identically and no conflict arises.

D. Extension favors the senior bondholders, because they receive an earlier-than-anticipated repayment of principal, while the junior bondholders bear the property loss.

Answer: B

Key correction: Source key (A) answered a different, Ashcraft-based question (servicer frictions not blamed for the subprime crisis); that framework is not part of the 2026 reading, so the item was replaced.

Step by step

1. Foreclosure: the senior bonds are unlikely to be written down and instead receive principal back early, even if the property sells at a loss; the junior bond may suffer an immediate credit writedown.
2. Extension: the junior bond avoids the immediate loss and keeps a small positive probability of recovering value; the senior bond faces the risk that the eventual property loss is larger, eroding the credit enhancement beneath it.
3. The servicer must maximize the total present value of the loan, but whichever action it takes is better aligned with some bond classes than others.
4. Hence the conflict is between bondholder classes (and the servicer), and extension is the choice that favors the juniors: B.

Why B is correct: Resolving a distressed loan redistributes losses across the capital structure: foreclosure benefits seniors (early par repayment) while extension benefits juniors (no immediate writedown), so any servicer action favors some classes over others.

Why the others are wrong

- **A:** Reverses the effect: it is foreclosure that triggers the junior writedown, so it favors seniors, not juniors.
- **C:** Maximizing present value is the servicer's duty, but the loss is allocated unequally by seniority, so the action still helps some classes and hurts others.
- **D:** Attributes early repayment to extension; early principal repayment results from foreclosure, and extension actually puts the senior's enhancement at risk.

LO 40.e

Calculate and evaluate one or two iterations of interim cash flows in a three-tiered securitization structure.

Q1145

Schweser Module Quiz 40.1 Q2

REBUILT — Source omitted the answer options (open-ended); converted to a 4-option MCQ with the same numbers and a clarified stem (junior = mezzanine, spreads over the reference rate).

A securitization pool has 100 identical loans of $500,000 each. Each loan pays the market reference rate plus a 300 bp spread, reset annually; the market reference rate is 4%. The senior, mezzanine and equity tranches are 75%,

20% and 5% of the pool. The senior and mezzanine tranches pay the reference rate plus spreads of 2% and 6%, respectively. Equity receives the residual, but any residual cash flow above $1,000,000 is diverted to the excess trust account. The default rate is 2% and defaulted loans pay no interest. What are the first-period cash flows to the mezzanine tranche and the excess trust account?

A. Mezzanine $600,000; trust account $0
B. Mezzanine $1,000,000; trust account $180,000
C. Mezzanine $1,000,000; trust account $0
D. Mezzanine $180,000; trust account $0

Answer: C

Step by step

1. Pool = 100 × $500,000 = $50,000,000; loan rate = 4% + 3% = 7%.
2. Collateral cash flow = 50,000,000 × 7% × (1 − 0.02) = $3,430,000.
3. Senior coupon = 4% + 2% = 6%: 50,000,000 × 0.75 × 6% = $2,250,000.
4. Mezzanine coupon = 4% + 6% = 10%: 50,000,000 × 0.20 × 10% = $1,000,000 (fully covered).
5. Residual = 3,430,000 - 2,250,000 - 1,000,000 = $180,000.
6. $180,000 is below the $1,000,000 trigger, so all of it goes to equity and nothing is diverted: trust account = $0.

Why C is correct: The mezzanine coupon is fully paid from the collateral cash flow, and the residual ($180,000) is below the $1,000,000 overcollateralization trigger, so nothing is diverted to the trust.

Why the others are wrong

- **A:** Used only the 6% spread for the mezzanine coupon and omitted the 4% reference rate (10,000,000 × 6% = $600,000).
- **B:** Correct mezzanine payment, but treated the $180,000 residual as diverted to the trust; the trust only receives residual cash above the $1,000,000 trigger.
- **D:** Confused the mezzanine payment with the residual (equity) cash flow of $180,000 and so understated the mezzanine coupon.

Q1146 AnalystPrep Q.1855

A swap curve is flat at 5%. The collateral, mezzanine and senior tranches of a securitization have the following (principal, spread over the swap rate): Collateral: ($100m, 350 bps); Mezzanine: ($10m, 500 bps); Senior: ($85m, 50 bps). How do the annual interest amounts compare?

A. Mezzanine > Collateral > Senior
B. Collateral > Mezzanine > Senior
C. Collateral > Senior > Mezzanine
D. Senior > Collateral > Mezzanine

Answer: C

Step by step

1. Collateral coupon = 5% + 3.5% = 8.5%, so interest = 0.085 × 100 = $8.500m.
2. Senior coupon = 5% + 0.5% = 5.5%, so interest = 0.055 × 85 = $4.675m.
3. Mezzanine coupon = 5% + 5% = 10%, so interest = 0.10 × 10 = $1.000m.
4. Ranking: Collateral (8.500) > Senior (4.675) > Mezzanine (1.000).

Why C is correct: Annual interest is principal times (swap rate + spread). The large senior tranche earns more dollars than the small mezzanine tranche despite its low spread, and the collateral earns more than the combined liabilities (5.675), leaving excess spread.

Why the others are wrong

- **A:** Ranks by spread (500 > 350 > 50) and ignores principal sizes.
- **B:** Puts mezzanine second because of its high coupon rate; its $10m principal makes its interest only $1.0m, below the senior $4.675m.
- **D:** Puts senior above collateral; senior interest ($4.675m) is below collateral interest ($8.5m), which has both a higher coupon and larger principal.

Q1147 AnalystPrep Q.5536

A three-tier securitization has a senior tranche ($100 million, 4% coupon), a mezzanine tranche ($40 million, 6% coupon) and an equity tranche ($10 million). Interest is paid semi-annually. The pool has performed well, no defaults have occurred, and collateral interest is sufficient to pay all coupons. What total interest is paid to the mezzanine tranche over the first year?

A. $4 million
B. $1.2 million
C. $2.4 million
D. $6 million

Answer: C

Step by step

1. Annual mezzanine coupon = 6% × 40 = $2.4m.
2. Semi-annual payments: 2 × 1.2 = $2.4m over one year.
3. With no defaults and sufficient collateral interest, both mezzanine coupons are paid in full.

Why C is correct: With no losses the waterfall pays each coupon in full: the mezzanine receives its coupon rate times its notional, which over two semi-annual periods is $2.4m.

Why the others are wrong

- **A:** This is the senior tranche annual interest (4% × 100), not the mezzanine.
- **B:** Only one semi-annual coupon (2.4/2); the question asks for a full year.
- **D:** Applied the mezzanine 6% to the senior notional of $100m.

Q1148 BT answers P2.T6.313.1 · BT notes P2.T6.313.1

REBUILT — Exhibit was OCR-garbled (loan counts, tranche sizes, cap); reconstructed a clean table and re-verified every figure by recomputation.

Consider the three-tier securitization of Malz section 9.2: 100 identical leveraged loans of $1.0m par paying a fixed 8.5% (LIBOR 5.0% + 350 bps); loans and liabilities mature in 5 years; senior bonds $85.0m pay LIBOR + 50 bps; mezzanine bonds $10.0m pay LIBOR + 500 bps; equity contributes $5.0m. The default rate is 2.0% per year (2 loans), recovery is 40%, and the money market rate is 5.0%. Results ($ millions except defaults):

t	Defaults d(t)	Survivors	Loan interest L(t)	Excess spread	OC(t)	Recovery R(t)	OC+R	Equity flow	OC a/c balance
0	| 100	|	|	| (5.000)					
1	2	98	8.330	2.655	1.750	0.800	2.550	0.905	2.550
2	2	96	8.160	2.485	1.750	0.800	2.550	0.735	5.228
3	2	94	7.990	2.315	1.750	0.800	2.550	0.565	8.039
4	2	92	7.820	2.145	1.750	0.800	2.550	0.395	10.991
5	2	90	7.650	|	0.800	| 9.315	11.540		

Total terminal available funds = 109.990; owed to bond tranches = 100.675; equity terminal cash flow = 9.315.

Each of the following statements about this structure is true EXCEPT which?

A. The bondholders are paid interest of $5.675 million in each of the five years, plus principal in year 5.
B. The amount diverted to the overcollateralization account from excess spread is capped at $1.750 million per year.
C. The recovery assumption is 40%.
D. Recovered funds are paid out to the equity holders each year as they are received.

Answer: D

Step by step

1. Bond coupons: senior 0.055 × 85 = 4.675; mezzanine 0.10 × 10 = 1.000; total $B = 5.675$. A is true.
2. Year 1 excess spread = 8.330 − 5.675 = 2.655 > K, so OC(t) = K = 1.750 and equity gets 2.655 − 1.750 = 0.905. B is true.
3. Recovery R(t) = 0.8 on 2 defaults = 0.4 × 2. C is true.
4. Year-1 OC account = $OC + R = 1.750 + 0.800 = 2.550$; year 2 = 2.550 × 1.05 + 2.550 = 5.228. Recoveries go into the OC account; equity receives only $L - B - OC$. D is false.
5. Terminal check: 7.650 + 90 + 0.8 + 11.540 = 109.990; less 100.675 owed gives equity 9.315.

Why D is correct: In the interim years recoveries R(t) are deposited into the overcollateralization account together with the diverted excess spread, where they earn interest; equity receives only the excess spread left after the OC diversion.

Why the others are wrong

- **A:** True: coupons are 4.675 + 1.000 = 5.675 each year with principal repaid at maturity, so it is not the exception.
- **B:** True: OC(t) = 1.750 in every year in which excess spread exceeds 1.750, so it is not the exception.
- **C:** True: R(t)/d(t) = 0.8/2 = 40%, so it is not the exception.

Q1149

BT answers P2.T6.313.2 · BT notes P2.T6.313.2

REBUILT — OCR-garbled exhibit and a misprinted default rate repaired; table re-derived by simulation and the source explanation inconsistent numbers replaced with recomputed shortfalls.

Same structure as the Malz section 9.2 example (100 loans of $1.0m paying 8.5%; senior $85.0m at LIBOR + 50 bps; mezzanine $10.0m at LIBOR + 500 bps; equity $5.0m; LIBOR 5.0%; recovery 40%; OC cap $1.750m), but defaults are much higher: 10, 9, 8, 7 and 7 loans in years 1 to 5. Results ($ millions except defaults):

t	Defaults d(t)	Survivors	Loan interest L(t)	Excess spread	OC(t)	Recovery R(t)	OC+R	Equity flow	OC a/c balance
0	| 100	|	|	| (5.000)					
1	10	90	7.650	1.975	1.750	4.000	5.750	0.225	5.750
2	9	81	6.885	1.210	1.210	3.600	4.810	0	10.848
3	8	73	6.205	0.530	0.530	3.200	3.730	0	15.120
4	7	66	5.610	(0.065)	(0.065)	2.800	2.735	0	18.611
5	7	59	5.015	|	2.800	| 0	19.541		

Total terminal available funds = 86.356; owed to bond tranches = 100.675.

Which statement is TRUE?

A. There is never a year in which either the mezzanine or the senior bonds are paid their full interest.

B. Both bondholder classes receive all of their interest in the first four years, but neither is repaid its full obligation in year 5.

C. Mezzanine bondholders suffer interest and principal shortfalls, but senior bondholders receive all their interest and principal.

D. Both bondholder classes receive all interest and get their full principal back.

Answer: B

Step by step

1. Coupon requirement $B = 5.675$. Years 1-3: $L(t) = 7.650, 6.885, 6.205 > 5.675$, so coupons are fully paid from interest.
2. Year 4: $L = 5.610 < 5.675$, a shortfall of 0.065, covered from the OC account ($15.120 \times 1.05 + 2.800 - 0.065 = 18.611$). Coupons are therefore paid in full in years 1-4.
3. Year 5 terminal funds = 5.015 + 59 + 2.800 + 19.541 = 86.356 (19.541 is the interest-grown OC balance).
4. Owed: senior 85 + 4.675 = 89.675 plus mezzanine 10 + 1.000 = 11.000, total 100.675. Shortfall = 100.675 − 86.356 = 14.319.
5. Senior is paid first: it receives all 86.356 and is short 89.675 − 86.356 = 3.319; mezzanine receives nothing (short 11.000); equity gets 0. Hence B.

Why B is correct: The OC account built up from early excess spread and recoveries covers the small year-4 coupon gap, so interest is met through year 4; but in year 5 total available funds of 86.356 are below the 100.675 owed, so even the senior tranche is not repaid in full.

Why the others are wrong

- **A:** Coupons are paid in full in years 1-4 (positive excess spread in years 1-3; the OC account covers the 0.065 gap in year 4).
- **C:** The mezzanine does lose everything in year 5, but the senior tranche is also short (86.356 available vs 89.675 owed, about 3.319), so senior does not escape.
- **D:** Terminal funds (86.356) are well below the 100.675 owed, so a shortfall must occur.

Q1150

BT answers P2.T6.313.3 · BT notes P2.T6.313.3

REBUILT — OCR-garbled exhibit and tranche sizes repaired and re-derived; distractors replaced because the originals (2.09%, -33.3%, 56.25%) corresponded to no identifiable error.

Same Malz section 9.2 structure (senior $85.0m, mezzanine $10.0m, equity contribution $5.0m, collateral coupon 8.5%, LIBOR 5.0%, recovery 40%), but the annual default rate is 3.0% (3 loans per year). Results ($ millions except defaults):

t	Defaults d(t)	Survivors	Loan interest L(t)	Excess spread	OC(t)	Recovery R(t)	OC+R	Equity flow	OC a/c balance
0	| 100	|	|	| (5.000)					
1	3	97	8.245	2.570	1.750	1.200	2.950	0.820	2.950
2	3	94	7.990	2.315	1.750	1.200	2.950	0.565	6.048
3	3	91	7.735	2.060	1.750	1.200	2.950	0.310	9.300
4	3	88	7.480	1.805	1.750	1.200	2.950	0.055	12.715
5	3	85	7.225	|	1.200	| 6.101	13.351		

Total terminal available funds = 106.776; owed to bond tranches = 100.675; equity terminal cash flow = 6.101. Which is nearest to the equity holders internal rate of return?

A. 4.06%
B. 11.44%
C. 31.40%
D. 57.02%

Answer: B

Step by step

1. Equity cash flows (t = 0 to 5): $-5.000, 0.820, 0.565, 0.310, 0.055, 6.101$, where the year-5 flow is $106.776 - 100.675 = 6.101$.
2. Solve $-5 + \frac{0.820}{1+y} + \frac{0.565}{(1+y)^2} + \frac{0.310}{(1+y)^3} + \frac{0.055}{(1+y)^4} + \frac{6.101}{(1+y)^5} = 0$.
3. Numerically (root finder) $y = 11.44\%$; at this rate the discounted inflows sum to 5.000.

Why B is correct: The IRR is the discount rate at which the present value of the equity inflows equals the \$5.0m outlay; the dated flows give 11.44%.

Why the others are wrong

- **A:** Ignores the interim flows and annualizes only the terminal flow: $(6.101/5)^{1/5} - 1 = 4.06\%$.
- **C:** Average annual inflow over the outlay: $(7.851/5)/5 = 31.4\%$; ignores timing and compounding.
- **D:** Total undiscounted profit over the outlay, $(7.851 - 5)/5 = 57.0\%$; not annualized or time-weighted.

LO 40.f

Describe the treatment of excess spread in a securitization structure and estimate the value of the overcollateralization account at the end of each year.

Q1151

AnalystPrep Q.4398

REBUILT — Source key diverted the 2.05m excess to the trust as 2.0m->equity/50k->trust, reversing the cap rule; restated the cap as the maximum diverted amount K and reset the options so exactly one is correct.

A collateralized loan obligation holds 100 identical leveraged loans with par value \$1,000,000 each, priced at par. The loans pay a fixed spread of 4% over the one-year swap rate (r). The senior, mezzanine and equity tranches are 85%, 10% and 5% of the pool. The senior tranche pays r + 200 bps and the mezzanine tranche pays r + 500 bps. Each year, up to K = \$2,000,000 of excess spread is diverted to the overcollateralization (OC) account, and any excess spread above K is paid to the equity tranche. The swap curve is flat at 5%, there are no fees, no defaults, and all payments occur at year-end. What are the first-year cash flow to the mezzanine tranche and the amount diverted to the OC account?

A. Mezzanine \$1,000,000; OC account \$50,000
B. Mezzanine \$2,050,000; OC account \$0
C. Mezzanine \$1,000,000; OC account \$2,000,000
D. Mezzanine \$5,950,000; OC account \$2,000,000

Answer: C

Key correction: Source key (A) said mezzanine \$1,000,000 and trust \$50,000, treating the \$2,000,000 as what goes to equity; under the standard rule the cap K is the most that can be diverted to the OC account and the amount above K goes to equity, so the correct pair is \$1,000,000 and \$2,000,000.

Step by step

1. Collateral interest: $100 \times \$1{,}000{,}000 \times (5\% + 4\%) = \$9{,}000{,}000$
2. Senior coupon: $\$85\text{m} \times (5\% + 2\%) = \$5{,}950{,}000$
3. Mezzanine coupon: $\$10\text{m} \times (5\% + 5\%) = \$1{,}000{,}000$

4. Excess spread: $L_1 - B = 9{,}000{,}000 - 6{,}950{,}000 = \$2{,}050{,}000$
5. Since $L_1 - B \geq K = 2{,}000{,}000$, the OC account receives $K = \$2{,}000{,}000$ and the remaining \$50,000 goes to equity.

Why C is correct: Positive excess spread is first diverted to the OC account up to the cap K; only the amount above K is released to equity. Here the excess of 2.05m exceeds K, so OC gets 2.0m and equity 0.05m.

Why the others are wrong

- **A:** Reverses the rule: gives the cap to equity and only the 50k excess above the cap to the OC account.
- **B:** Adds the whole excess spread to the mezzanine coupon and sends nothing to the OC account; excess spread is not paid to the mezzanine.
- **D:** Uses the senior coupon (5.95m) as the mezzanine cash flow; the OC figure is right but the mezzanine amount is wrong.

Q1152

AnalystPrep Q.5522

REBUILT — Stem said the OC account itself reaches a cap; in the standard framework the cap K limits the annual amount diverted from excess spread, so the stem was reworded.

In a securitization, excess spread is diverted each year into an overcollateralization (OC) account, up to a maximum annual amount K. In a year when the excess spread $L_t - B$ exceeds K, what happens to the amount above K?

A. It is paid to the equity noteholders.
B. It is also deposited in the OC account to further enhance credit quality.
C. It is shared pro rata among all tranches.
D. It is used to repay the principal of the senior bonds.

Answer: A

Step by step

1. If $L_t - B \geq K$, the OC account receives $OC_t = K$.
2. The remainder, $L_t - B - K$, is the cash flow to equity: it is released as residual cash flow.
3. Hence the amount above the cap goes to the equity noteholders.

Why A is correct: The cap limits how much soft credit enhancement is built each year; surplus beyond it is the equity tranche's residual.

Why the others are wrong

- **B:** Ignores the cap; if all excess were retained there would be no limit K.
- **C:** Excess spread is not allocated across tranches; bondholders only receive their contractual coupons.
- **D:** The structure does not use excess spread to amortize senior principal during the interim years; principal is repaid at maturity.

Q1153

AnalystPrep Q.5525

REBUILT — Options A and C were both defensible and the correct description lacked the cap/equity mechanics; rewrote options around the diversion-up-to-K rule.

Which of the following best describes the treatment of positive excess spread in a three-tiered securitization with an overcollateralization (OC) mechanism?

A. All excess spread is paid to the equity tranche each year.
B. Excess spread is paid to the senior bonds to retire principal early.
C. Excess spread is shared equally among senior, mezzanine and equity tranches.
D. Up to a maximum annual amount K is diverted to the OC account to absorb future shortfalls, and any remainder goes to equity.

Answer: D

Step by step

1. Test whether $L_t - B \geq 0$ (interest covers bond coupons).
2. If $L_t - B \geq K$, K goes to the OC account and $L_t - B - K$ to equity; if $L_t - B < K$, the whole excess goes to the OC account and equity gets nothing.
3. The OC account is then available to pay bond coupons in later years when excess spread turns negative.

Why D is correct: Excess spread is a soft form of credit enhancement: it is first used to build a reserve (up to K per year) that protects bondholders, with equity receiving the residual.

Why the others are wrong

- **A:** Ignores the diversion to the OC account that protects the bonds, which has first priority up to K.

- **B:** Interim excess spread funds the OC account, not early principal repayment of the senior bonds.
- **C:** Tranches receive only their contractual coupons; excess spread is not distributed equally.

Q1154

BT answers L2.T6.106.4

REBUILT — Original referred to an unseen case study and had OCR-damaged numbers; made self-contained using the corrected 51 bp and 13 bp figures.

A securitization trust holds collateral earning a weighted average interest rate of 8.30%. The tranches pay a weighted average coupon of LIBOR + 23 bps, with LIBOR at 5.32%. The trust also pays servicing fees of 51 bps and a payment to its swap counterparty of 13 bps (all as annual percentages of the collateral balance). What is the initial excess spread?

A. 2.11%
B. 2.24%
C. 2.62%
D. 2.75%

Answer: A

Step by step

1. Weighted average tranche coupon: 5.32% + 0.23% = 5.55%
2. Collateral yield less coupons: 8.30% - 5.55% = 2.75%
3. Less servicing fee: 2.75% - 0.51% = 2.24%
4. Less swap payment: 2.24% - 0.13% = 2.11%

Why A is correct: Excess spread is the collateral income left after all payments out of the trust (tranche coupons, servicing fees and swap payments).

Why the others are wrong

- **B:** Stops after subtracting the servicing fee and omits the 13 bp swap payment.
- **C:** 2.75% - 0.13%: subtracts the swap payment but omits the 51 bp servicing fee.
- **D:** Compares only collateral yield with the tranche coupon and ignores both the servicing fee and swap payment.

Q1155

BT answers P2.T6.24.19.1

REBUILT — Cash-flow exhibit was garbled and year-3 defaults were not stated; rebuilt with explicit parameters and defaults for years 1-3 (8, 7, 7), consistent with the original's years 1-2 and answer.

A CLO holds 100 identical loans of $1,000,000 each, priced at par. The senior tranche is 85% of the pool and the junior tranche 10%. The market reference rate (MRR) is 3.5%; loans pay 9.5% in total; the senior tranche pays MRR + 3.5% and the junior tranche pays MRR + 6.0%, so total bond coupons due each year are $6.90 million. Loans that default in a year pay no interest that year; interest is paid at year-end by surviving loans only. The numbers of defaults are 8 in year 1, 7 in year 2 and 7 in year 3 (all interim years). What is the excess spread at the end of year 3?

A. $1.46 million
B. $1.175 million
C. $0.51 million
D. $0.415 million

Answer: C

Step by step

1. Cumulative defaults after year 3: 8 + 7 + 7 = 22, so 78 loans survive
2. Loan interest: $L_3 = 78 \times \$1\text{m} \times 9.5\% = \$7.41\text{m}$
3. Bond coupons: $B = 85 \times (3.5\% + 3.5\%) + 10 \times (3.5\% + 6.0\%) = 5.95 + 0.95 = \$6.90\text{m}$
4. Excess spread: $L_3 - B = 7.41 - 6.90 = \$0.51\text{m}$

Why C is correct: Interest comes only from loans that have survived all prior defaults, so the excess spread shrinks each year as the pool erodes.

Why the others are wrong

- **A:** Subtracts only the senior coupon (5.95m) from 7.41m, forgetting the junior coupon.
- **B:** Uses the year-2 survivors (85 loans, interest 8.075m); ignores year-3 defaults.
- **D:** Assumes 8 defaults in year 3 (77 loans, 7.315m) instead of 7.

Q1156 BT answers P2.T6.312.2 · BT notes P2.T6.312.2

REBUILT — Merged duplicates; replaced the giveaway 'negative' option and unexplained distractors with options from realistic errors.

A CLO holds 100 identical floating-rate loans of $1.0 million each, priced at par, paying LIBOR + 300 bps. The capital structure is: senior debt $75.0 million paying LIBOR + 100 bps; junior (mezzanine) debt $15.0 million paying LIBOR + 500 bps; and an equity tranche. Coupons are annual, there are no fees, no defaults, and the LIBOR swap curve is flat at 2.0%. What is the annual excess spread, i.e. collateral interest minus tranche coupon payments?

A. $1.50 million
B. $1.70 million
C. $2.75 million
D. $3.95 million

Answer: B

Step by step

1. Collateral interest: $100 \times (2\% + 3\%) = \$5.000\text{m}$
2. Senior coupon: $75 \times (2\% + 1\%) = \$2.250\text{m}$
3. Junior coupon: $15 \times (2\% + 5\%) = \$1.050\text{m}$
4. Excess spread: $5.000 - 2.250 - 1.050 = \$1.700\text{m}$

Why B is correct: With no defaults, excess spread is full collateral interest less the senior and junior coupons; the 10m equity tranche earns the LIBOR component of the pool without paying a coupon.

Why the others are wrong

- **A:** Compares spreads only (3.0m - 0.75m - 0.75m) and ignores the 2% LIBOR earned on the $10m not funded by debt (0.2m).
- **C:** Subtracts only the senior coupon (5.0 - 2.25) and forgets the junior coupon.
- **D:** Subtracts only the junior coupon (5.0 - 1.05) and forgets the senior coupon.

Q1157 GARP PE1 Q72 · Mock 3 Q9

A manager of collateralized loan obligations (CLOs) is reviewing a CLO with a pool of 50 identical loans, each priced at its par value of GBP 1 million. The loans are floating-rate obligations that pay a fixed spread of 150 bps over LIBOR. Coupons and interest payments on the liabilities are made annually at year-end:

Liabilities	Amount (GBP)	Coupon
Senior debt	37,500,000	LIBOR + 45 bps
Mezzanine debt	10,000,000	LIBOR + 300 bps
Equity	2,500,000	-

The CLO initially has no overcollateralization, and the annual excess spread flowing into the overcollateralization account is limited to GBP 250,000. Suppose LIBOR remains flat at 4% in the first year, there are no defaults, and there are no management or transaction fees. What amounts are posted to the overcollateralization account and to the equity tranche after the first year?

A. OC account GBP 0 and equity GBP 0
B. OC account GBP 0 and equity GBP 381,250
C. OC account GBP 250,000 and equity GBP 131,250
D. OC account GBP 381,250 and equity GBP 0

Answer: C

Step by step

1. Collateral interest: $50\text{m} \times (4\% + 1.5\%) = \text{GBP } 2{,}750{,}000$
2. Senior coupon: $37.5\text{m} \times 4.45\% = 1{,}668{,}750$; mezzanine coupon: $10\text{m} \times 7.0\% = 700{,}000$; $B = 2{,}368{,}750$
3. Excess spread: $2{,}750{,}000 - 2{,}368{,}750 = 381{,}250$
4. Since $381{,}250 > K = 250{,}000$: OC account = 250,000; equity = $381{,}250 - 250{,}000 = 131{,}250$

Why C is correct: Excess spread is positive and above the cap K, so K goes to the OC account and the remainder to equity.

Why the others are wrong

- **A:** Treats the excess spread as zero, ignoring the positive 381,250.
- **B:** Skips the OC test and pays all excess spread to equity.
- **D:** Ignores the 250,000 cap and diverts the entire excess spread to the OC account.

Q1158 GARP PE2 Q45

REBUILT — Original stated year-1 interest (5.7m) below coupons (6.3m) yet an OC balance of zero, an inconsistency; changed coupons to EUR 5.5m so year-1 OC is 200,000 and options now test interest on the balance, recoveries and the shortfall.

A manager of a structured credit desk illustrates cash flows in a securitization using the following information (all liabilities and the collateral have a maturity of 5 years):
Initial number of loans in the collateral pool: 100
Principal amount of each loan: EUR 1,000,000
Total coupon interest paid annually on all junior and senior bonds: EUR 5,500,000
Maximum annual amount flowing from excess spread into the overcollateralization (OC) account: EUR 1,500,000
Swap rate for all maturities: 3.5%
Recovery rate on defaulted loans: 45%
The loans pay a fixed spread of 2.2% over the swap curve. Defaulted loans pay no interest in the year of default. There were no defaults in year 1, and recoveries are deposited in the OC account, which earns the swap rate. What is the value of the OC account at the end of year 2 if there are 8 defaults in year 2?

A. EUR 3,544,000
B. EUR 3,551,000
C. EUR 3,807,000
D. EUR 4,063,000

Answer: B

Key correction: Source key (C, EUR 2,544,000) relied on data that were internally inconsistent (year-1 interest below coupons yet OC of zero); with coupons restated the result is EUR 3,551,000, computed as above.

Step by step

1. Loan rate = 3.5% + 2.2% = 5.7%. Year 1: $L_1 = 100 \times 5.7\% \times 1\text{m} = 5{,}700{,}000$; excess spread = 5.7m - 5.5m = 200,000 (below K), so $OC_1 = 200{,}000$
2. Year 2: $L_2 = 92 \times 5.7\% \times 1\text{m} = 5{,}244{,}000$; excess spread $L_2 - B = -256{,}000$ (shortfall paid from the OC account)
3. Recovery: $R_2 = 8 \times 0.45 \times 1{,}000{,}000 = 3{,}600{,}000$
4. Prior balance with interest: $200{,}000 \times 1.035 = 207{,}000$
5. OC account at end of year 2: $207{,}000 + 3{,}600{,}000 - 256{,}000 =$ EUR 3,551,000

Why B is correct: The OC account grows by interest on the prior balance and by recoveries, and is drawn down to cover the bond-coupon shortfall when excess spread is negative.

Why the others are wrong

- **A:** Omits one year of interest on the opening balance (adds 200,000 instead of 207,000).
- **C:** Ignores the 256,000 shortfall: 207,000 + 3,600,000.
- **D:** Adds the negative excess spread instead of subtracting it: 207,000 + 3,600,000 + 256,000.

Q1159 Mock 4 Q46

A collateralized loan obligation (CLO) holds a pool of 100 loans, each priced at its par value of EUR 1 million. The coupon interest paid annually on the CLO's junior and senior debt tranches totals EUR 6.3 million. The maximum annual amount of excess spread that may flow into the overcollateralization (OC) account is EUR 1.5 million. The swap rate is 3.5% and the loans pay a spread of 2.2% over the swap rate. The recovery rate on defaulted loans is 45%.
There were no defaults in Year 1 and the balance of the OC account at the end of Year 1 was zero. In Year 2, 8 loans in the pool default. What is the balance of the OC account at the end of Year 2?

A. EUR 600,000.
B. EUR 1,056,000.
C. EUR 2,544,000.
D. EUR 3,600,000.

Answer: C

Step by step

1. Loan interest from the 92 surviving loans (defaulted loans pay nothing): $L_2 = (3.5\%+2.2\%)\times 92\times 1{,}000{,}000 = 5{,}244{,}000$.
2. Excess spread: $L_2 - B = 5{,}244{,}000 - 6{,}300{,}000 = -1{,}056{,}000$, so it is negative; nothing is diverted into the OC account ($OC_2 = 0$) and the cap of EUR 1.5 million is irrelevant.
3. Recoveries are deposited into the OC account: $R_2 = 0.45 \times 8 \times 1{,}000{,}000 = 3{,}600{,}000$.

4. Custodian test: prior OC balance (0, so no interest earned) plus recoveries = 3,600,000 ≥ shortfall of 1,056,000, so the full shortfall is paid out of the OC account.
5. OC balance at end of Year 2 = 0 + 3,600,000 - 1,056,000 = EUR 2,544,000.

Why C is correct: When excess spread is negative, the bond-coupon shortfall is met from the OC account, which is first credited with the recoveries on the year's defaults; what remains is the closing balance.

Why the others are wrong

- **A:** EUR 600,000 is the Year 1 shortfall (100 x 5.7% = 5.7m vs 6.3m coupons); it ignores the Year 2 defaults and recoveries altogether.
- **B:** EUR 1,056,000 is the Year 2 shortfall itself; it is the amount paid out of the account, not the balance left after recoveries are added.
- **D:** EUR 3,600,000 is the recovery deposit alone; it forgets to pay the EUR 1,056,000 coupon shortfall out of the OC account.

LO 40.g

Explain the tests on the excess spread that a custodian must go through at the end of each year to determine the cash flow to the overcollateralization account and to the equity noteholders.

Q1160

AnalystPrep Q.4399

REBUILT — Stem's diversion rule was reversed relative to the GARP/Schweser custodian test (up to K goes to the trust account, the remainder to equity) so the source key of $0 was wrong; rule restated and options rebuilt.

A collateralized loan obligation holds 100 identical leveraged loans with a par value of $1,000,000 each, priced at par. The loans pay a fixed spread of 400 bps over the one-year reference rate. The capital structure is equity 5%, junior (mezzanine) 10% and senior 85% of the pool. The senior and junior tranche coupons are the reference rate plus 200 bps and 500 bps, respectively; equity receives only the residual. At the end of each year the custodian diverts excess spread (loan interest minus senior and junior coupons) into the overcollateralization (trust) account, up to a maximum of $2,000,000 per year; any excess spread above that maximum is paid to the equity noteholders. Assumptions: the reference rate is flat at 5%; no upfront, management or trustee fees; all interest and coupons are annual and paid at year-end; 5% of the original loans default during the first year and pay no interest for that year; recoveries are zero. What amount is deposited into the overcollateralization account at the end of the first year?

A. $2,000,000
B. $1,600,000
C. $1,172,500
D. $0

Answer: B

Key correction: Source key (D, $0) came from reading the rule as 'only spread above $2,000,000 goes to the trust'; under the standard custodian test, any excess spread up to the cap goes to the trust, so correct is B ($1,600,000).

Step by step

1. Loan rate = 5% + 4% = 9%. Surviving loans = 100 × (1 − 0.05) = 95, so $L_1 = 95 \times \$1{,}000{,}000 \times 9\% = \$8{,}550{,}000$.
2. Senior coupon = $85m × (5% + 2%) = $5,950,000; junior coupon = $10m × (5% + 5%) = $1,000,000; $B = \$6{,}950{,}000$.
3. Test 1: $L_1 - B = \$1{,}600{,}000 \geq 0$, so coupons are covered.
4. Test 2: is $L_1 - B \geq K = \$2{,}000{,}000$? No (1.6m < 2.0m), so the entire excess spread flows to the overcollateralization account: $OC_1 = L_1 - B = \$1{,}600{,}000$; nothing is left for equity.

Why B is correct: Excess spread is positive but below the cap K, so the custodian diverts all of it ($OC_t = \min(L_t - B, K)$) to the overcollateralization account.

Why the others are wrong

- **A:** Ignores the default haircut (interest 9.0m, excess 2.05m) and then applies the cap, or simply assumes the cap is always diverted; the cap only binds when excess spread exceeds K.
- **C:** Applies the 5% default haircut twice (collateral interest 100m × 9% × 0.95^2 = 8.1225m), understating the excess spread.
- **D:** Reverses the rule, treating only spread above 2,000,000 as diverted to the trust; the cap K limits what is diverted, it is not a threshold that must be exceeded.

Q1161 BT answers P2.T6.24.19.2

REBUILT — Original had an equity coupon, an incoherent trust/equity-account rule and a key that did not reconcile; rebuilt as a standard GARP-style custodian test with a single cap K and a clear default convention.

A CLO holds 200 identical loans of $1,000,000 each, priced at par, paying a fixed spread of 5.50% over a reference rate that is flat at 3.50%. The senior tranche (75% of the pool) pays the reference rate plus 300 bps, and the junior tranche (20% of the pool) pays the reference rate plus 600 bps. The remaining 5% is the equity tranche, which receives only residual cash flows. Each year the custodian first pays the senior and junior coupons, then diverts excess spread to the overcollateralization account up to a maximum of $1,000,000 per year; any remainder is paid to equity. Each year 4% of the original loans (8 loans) default; a loan that defaults during a year pays no interest for that year, and recoveries are zero. All payments occur at year-end. What is the cash flow to the equity noteholders at the end of year 2?

A. $1,000,000
B. $2,730,000
C. $3,010,000
D. $2,010,000

Answer: D

Key correction: Source key (D, $2,365,500) did not reconcile with its own working (inconsistent principal balance and account treatment); the rebuilt item uses a consistent structure.

Step by step

1. Loan rate = 3.5% + 5.5% = 9%. After 2 years of 8 defaults each, surviving loans = 200 − 16 = 184, so $L_2 = 184 \times \$1{,}000{,}000 \times 9\% = \$16{,}560{,}000$.
2. Senior coupon = $\$150\text{m} \times 6.5\% = \$9{,}750{,}000$; junior coupon = $\$40\text{m} \times 9.5\% = \$3{,}800{,}000$; $B = \$13{,}550{,}000$.
3. Test 1: $L_2 - B = \$3{,}010{,}000 > 0$.
4. Test 2: $L_2 - B \geq K = \$1{,}000{,}000$, so $OC_2 = K = \$1{,}000{,}000$ is diverted.
5. Equity cash flow = $L_2 - B - K = \$3{,}010{,}000 - \$1{,}000{,}000 = \$2{,}010{,}000$.

Why D is correct: Excess spread of 3.01m exceeds the cap, so exactly K goes to the overcollateralization account and the remainder goes to equity.

Why the others are wrong

- **A:** This is the amount diverted to the overcollateralization account (K), not the amount paid to equity.
- **B:** Uses 192 surviving loans (only one year of defaults), giving excess spread 3.73m and equity 2.73m; two years of defaults (16 loans) have occurred by year 2.
- **C:** Reports the whole excess spread without deducting the 1,000,000 diverted to the overcollateralization account.

Q1162 BT answers P2.T6.24.19.3

REBUILT — Source key ($200,000) contradicted its own working and omitted recoveries; rebuilt with explicit assumption (ignore account interest) and options containing the correct $800,000.

A securitization has aggregate loan interest of $7,000,000 for the current year, while bond coupon interest due to the senior and mezzanine bondholders is $7,200,000. The overcollateralization account balance at the end of the previous year was $300,000, and recoveries on defaults during the year of $700,000 are deposited into that account. Ignoring interest earned on the account, how much remains in the overcollateralization account after the custodian performs the tests on the excess spread at year-end?

A. $100,000
B. $1,000,000
C. $800,000
D. $200,000

Answer: C

Key correction: Source key (D, $200,000) was inconsistent with its own working; the account holds 300,000 + 700,000 = 1,000,000, and the 200,000 shortfall is paid out of it, leaving $800,000.

Step by step

1. Excess spread $L_t - B = \$7{,}000{,}000 - \$7{,}200{,}000 = -\$200{,}000 < 0$: interest does not cover the coupons.
2. Test whether the account can cover the shortfall: funds available = prior balance + recoveries = $\$300{,}000 + \$700{,}000 = \$1{,}000{,}000$.
3. Since $\$1{,}000{,}000 \geq \$200{,}000$, bondholders are paid in full and no write-down occurs; no amount is diverted from excess spread ($OC_t = 0$).

4. Remaining balance = $1,000,000 − $200,000 = $800,000.

Why C is correct: With negative excess spread, the custodian draws the shortfall from the account's prior balance plus current recoveries; the balance is sufficient, so the remainder stays in the account.

Why the others are wrong

- **A:** Draws the shortfall from the prior balance only (300,000 − 200,000) and ignores the 700,000 of recoveries deposited.
- **B:** Adds recoveries to the balance but forgets to pay out the 200,000 shortfall to bondholders.
- **D:** Confuses the size of the shortfall with the remaining balance (the shortfall itself is 200,000).

LO 40.h

Describe a simulation approach to calculating credit losses for different tranches in a securitization.

Q1163

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A risk analyst at an asset manager is documenting how credit losses for the tranches of a CLO are calculated using a simulation approach. Which of the following correctly describes the sequence of the procedure?

A. Estimate the default probabilities and the correlation parameter, generate correlated default times for each loan in the collateral pool, and then run the resulting default counts through the waterfall to compute losses for each tranche in every simulation thread.

B. Estimate the default probabilities, run the waterfall once at the expected default rate to compute tranche losses, and then adjust each tranche loss upward to reflect the correlation parameter.

C. Estimate the correlation parameter, generate default times for the equity tranche only, and then infer the losses of the mezzanine and senior tranches from the equity losses.

D. Generate default times for each loan, estimate the default probabilities and correlation from the simulated default times, and then run the average number of defaults through the waterfall to compute tranche losses.

Answer: A

Step by step

1. Step 1: estimate parameters (default intensity and correlation).
2. Step 2: use those parameters in a copula to simulate default times for every loan.
3. Step 3: count defaults per period in each thread, pass the cash flows through the waterfall, and obtain tranche losses (and thus loss distributions and Credit VaR).

Why A is correct: The parameters must be estimated first, they drive the simulated default times, and the waterfall is applied to each simulated thread to give a loss for every tranche.

Why the others are wrong

- **B:** Describes a single scenario analysis with a correlation add-on; it ignores correlation inside the simulation and yields no distribution of tranche losses.
- **C:** Default times are simulated for every loan in the pool, not for a tranche; tranche losses come from running the collateral cash flows through the waterfall.
- **D:** Reverses steps 1 and 2 (parameters cannot be estimated from simulated data they generate) and averages over threads, which discards the loss distribution.

Q1164

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A risk manager simulates 1,000 default threads for a CLO and passes each through the waterfall. For the mezzanine bond, the mean loss across the threads is $0.62 million, the 10th highest simulated loss is $3.40 million, and the 50th highest simulated loss is $1.90 million. Using the approach of measuring Credit VaR as the difference between the expected loss and the relevant quantile of the simulated losses, what is the 99% Credit VaR of the mezzanine bond?

A. $1.28 million

B. $2.78 million

C. $1.90 million

D. $3.40 million

Answer: B

Step by step

1. With 1,000 threads, the 99% quantile is the 10th highest loss (1% of threads): $3.40 million.
2. Credit VaR = quantile loss - expected loss = 3.40 - 0.62 = $2.78 million.

Why B is correct: The 99% Credit VaR is the 10th highest loss less the mean loss, which is $2.78 million.

Why the others are wrong

- **A:** Uses the 50th highest loss (the 95% quantile) and subtracts the mean: 1.90 - 0.62 = 1.28. That is the 95% Credit VaR.
- **C:** Uses the 95% quantile loss and does not subtract the expected loss.
- **D:** Uses the correct 99% quantile loss but fails to subtract the expected loss; that figure is the total tail loss, not Credit VaR as defined.

Q1165 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A structured credit analyst must simulate losses on the tranches of a CLO. Default probabilities for the loans can be inferred from market CDS spreads, but there are almost no market data from which to estimate the pairwise default correlation. Which of the following is the most appropriate way to proceed?

A. Use the observed equity return correlation as the copula correlation, because the copula correlation has a direct economic interpretation as an equity relationship.

B. Infer the correlation from the tranche ratings and estimate the default intensities from a single assumed correlation value.

C. Infer the default intensities from spread data and run the simulation across a range of default probability and correlation pairs to examine how tranche results change.

D. Set the correlation to zero, since correlation cannot be observed, and rely on the diversification of the pool to produce the tranche loss distribution.

Answer: C

Step by step

1. Step 1 needs default intensities (hazard rates) and correlations.
2. Intensities can be estimated from spreads; correlation is hard to estimate because the copula correlation is not intuitive and there is little data.
3. Therefore run sensitivity analysis over a grid of default probability and correlation pairs.

Why C is correct: Because the copula correlation cannot be estimated reliably, the analyst performs sensitivity analysis over many (default probability, correlation) pairs.

Why the others are wrong

- **A:** The copula correlation has no direct economic interpretation; equity correlation is at best a proxy for asset correlation, not the copula parameter.
- **B:** Reverses the logic: spread data inform the default intensities, not the correlation, and a single assumed correlation avoids the needed sensitivity analysis.
- **D:** Zero correlation understates clustering of defaults and the risk to the senior tranche; losses would be mismeasured.

LO 40.i

Explain how the default probabilities and default correlations affect the credit risk in a securitization.

Q1166 AP Mock 1 Q29 · GARP PE1 Q46 · Mock 3 Q34

A financial firm has sold default protection on the most senior tranche of a CDO. If the default correlation between the assets held in the CDO decreases sharply from the correlation used in pricing the tranches, assuming everything else is unchanged, how will the position of the financial firm be affected?

A. It will either increase or decrease, depending on the pricing model used and market conditions.

B. It will gain significant value, since the probability of the protection being exercised falls.

C. It will lose significant value, since the protection it sold will gain value.

D. It will neither gain nor lose value, since only expected default losses matter and correlation does not affect expected default losses.

Answer: B

Step by step

1. The protection seller is long the senior tranche: it pays out only if pool losses exceed the subordination below the senior tranche.
2. Such losses require many defaults at the same time, which is a clustering (high-correlation) event.
3. Lower correlation makes defaults more independent, so the number of defaults stays near its expectation and rarely breaches the subordination.
4. The probability that the protection is called falls, so the liability of the seller shrinks and the position gains value.

Why B is correct: Senior tranches lose value as correlation rises and gain value as it falls; a protection seller is long the tranche and therefore benefits from lower correlation.

Why the others are wrong

- **A:** The direction is determined, not model-dependent: senior tranche value falls as correlation rises, and the reverse holds.
- **C:** Reverses the seller's side: the protection gains value only when it is more likely to be called, which happens when correlation rises, not falls.
- **D:** Correlation does not change the pool's expected loss, but it redistributes that loss across tranches; the senior tranche's expected loss does change.

Q1167

AP Mock 1 Q30

An investment firm has sold default protection on the most senior tranche of a CDO. Suppose the default correlation between the assets held in the CDO increases sharply, with everything else unchanged. How will the firm's position be affected?

A. It will neither lose nor gain value, because correlation does not affect default losses.
B. It will lose significant value, since the probability of the protection being exercised increases.
C. It will either increase or decrease, depending on market conditions.
D. It will gain significant value, because the probability of the protection being exercised falls.

Answer: B

Step by step

1. The senior tranche is hit only when pool losses exceed the subordination beneath it.
2. Higher correlation makes defaults cluster, so extreme-loss scenarios become more likely.
3. The probability that the senior tranche is impaired, and the protection called, rises.
4. The protection seller is long the tranche and so loses value.

Why B is correct: Rising correlation fattens the tail of the pool loss distribution, raising the chance that subordination is breached, which hurts the senior tranche and its protection seller.

Why the others are wrong

- **A:** Confuses expected pool loss (unchanged by correlation) with the allocation of loss to tranches, which does change.
- **C:** The effect on a senior tranche is unambiguous: its value falls as correlation rises.
- **D:** Gets the sign backwards; it would apply if correlation fell. Higher correlation raises, not lowers, the chance of exercise.

Q1168

AP Mock 1 Q34

REBUILT — Original answer (buy senior, sell equity) ignored that higher default probability also hurts the senior tranche; rewritten to ask which tranche is unambiguously hurt.

A CDO analyst expects the default probability of the underlying assets to rise significantly while the default correlation falls, with all else equal. Which tranche is unambiguously reduced in value by BOTH changes?

A. The senior tranche
B. The equity tranche
C. The mezzanine tranche
D. None of the tranches, because the two effects always offset

Answer: B

Step by step

1. Higher default probability lowers the value of every tranche for a given correlation.

2. Lower correlation reduces the chance of the few-default scenarios that benefit equity, so the equity tranche is also hurt by the fall in correlation: both effects are negative.
3. For the senior tranche, lower correlation helps (breach of subordination is less likely) and partly offsets the higher default probability, so the net effect is mixed.
4. For the mezzanine tranche the correlation effect depends on the default-rate level, so the net effect is also mixed.
5. Only the equity tranche is hurt by both changes.

Why B is correct: Equity gains from correlation because it benefits from clustering in the low-default scenario; lower correlation and higher default probability both reduce its value.

Why the others are wrong

- **A:** Senior is hurt by higher default probability but helped by lower correlation, so it is not hurt by both.
- **C:** Mezzanine's correlation effect switches sign with the default rate, so it is not unambiguously hurt by both.
- **D:** The effects offset for senior and mezzanine, but for equity they reinforce each other.

Q1169

AnalystPrep Q.5520

REBUILT — Source claimed both higher default probability and correlation raise risk in a securitization, ignoring tranche dependence; rebuilt to test the tranche-specific effects.

Which of the following statements about the effects of default probability and default correlation on tranche values in a securitization is CORRECT?

A. A higher default probability lowers the value of all tranches, and a higher correlation lowers the value of the senior tranche but raises the value of the equity tranche.

B. A higher default probability lowers the value of all tranches, and a higher correlation lowers the value of every tranche.

C. A higher default probability lowers the value of the equity tranche but raises the value of the senior tranche, and a higher correlation lowers the value of the senior tranche.

D. A higher default probability lowers the value of all tranches, and a higher correlation raises the value of the senior tranche but lowers the value of the equity tranche.

Answer: A

Step by step

1. Default probability: for a given correlation, more defaults reduce cash flows, so all tranche values fall.
2. Correlation, senior tranche: higher correlation makes defaults cluster, so subordination is more likely to be breached and senior value falls.
3. Correlation, equity tranche: higher correlation makes the zero- or low-default outcome more likely, so mean equity value rises.
4. Statement A matches both results.

Why A is correct: Default probability hurts every tranche, while correlation shifts loss risk toward the tail: it hurts the senior tranche and helps the equity tranche (mezzanine is mixed).

Why the others are wrong

- **B:** Treats correlation as uniformly harmful; equity actually benefits from the higher chance of few defaults.
- **C:** Wrongly claims higher default probability raises the senior value; every tranche is hurt.
- **D:** Reverses the correlation effects: senior falls and equity rises with correlation.

Q1170

AnalystPrep Q.5530

REBUILT — Tightened the stem and reworded option B, which was arguably true generally, so exactly one option is correct.

In a CDO structure, how does systematic risk, expressed as high default correlation among the underlying loans, affect the senior bonds?

A. It raises liquidity risk, making senior bonds hard to trade.

B. It lowers the expected loss on senior bonds as correlation rises.

C. It raises the likelihood of large losses on senior bonds, shown by a higher credit VaR.

D. It lowers the credit risk of senior bonds, because the diversification of the collateral is greater.

Answer: C

Step by step

1. High correlation makes defaults cluster, so extreme pool-loss scenarios become more likely.
2. Senior bonds are impaired only in these extreme scenarios, so their probability of loss and expected loss rise.

3. The tail loss, measured by credit VaR, rises consistently with correlation.
4. So senior-bond credit risk increases.

Why C is correct: Correlation fattens the tail of the pool loss distribution, increasing the chance that subordination is exhausted and raising senior credit VaR.

Why the others are wrong

- **A:** Liquidity is not the mechanism; correlation affects the default-loss distribution, not trading liquidity.
- **B:** Gets the direction wrong: senior expected loss increases with correlation as breaches of subordination become likelier.
- **D:** Higher correlation means less effective diversification, not more.

Q1171

BT answers P2.T6.314.1 · BT notes P2.T6.314.1

REBUILT — Original relied on a missing histogram (Malz Figure 9.2); restated as a self-contained question on equity tranche value and credit VaR sensitivities.

In a simulation of a securitization, the credit VaR of the equity tranche is measured as the shortfall of a low-percentile simulated value below the mean value. Regarding the equity tranche, each of the following statements is true EXCEPT which is false?

A. For a given correlation, an increase in the default rate lowers the expected equity tranche value.

B. For a given default rate, an increase in correlation raises the expected equity tranche value.

C. For a given default rate, an increase in correlation decreases the equity credit VaR; for a given correlation, a higher default rate increases the equity credit VaR.

D. At low correlations, the equity value is substantially positively convex in default rates.

Answer: C

Step by step

1. A: more defaults reduce equity cash flows, so value falls. True.
2. B: higher correlation raises the chance of very few defaults, so mean equity value rises. True.
3. C: higher correlation widens the distribution (large losses or very small losses), so equity credit VaR increases with correlation; a higher default rate squeezes the distribution toward zero value, so equity VaR falls. Both halves are reversed. False.
4. D: at low correlation, equity loses value rapidly at low default rates and more slowly later (positive convexity). True.

Why C is correct: Equity credit VaR rises with correlation (more variable outcomes) and falls with default probability (expected loss is already high), the opposite of statement C.

Why the others are wrong

- **A:** True statement: default rate hurts equity for every correlation.
- **B:** True statement: equity benefits from correlation as the low-default scenario becomes more probable.
- **D:** True statement: the decline in value is steep at first and then flattens (positive convexity).

Q1172

BT answers P2.T6.314.2 · BT notes P2.T6.314.2

REBUILT — Original relied on a missing figure and a default probability (0.0150) not in the reference grid; rebuilt with a self-contained exhibit using the GARP/Malz senior-bond VaR grid.

A simulated three-tranche securitization produces the following 99% Credit VaR for the senior bond tranche ($ million; VaR = expected loss minus the 10th highest simulated loss, so a larger number means more risk). Rows are the default probability π and columns the default correlation ρ.

π	$\rho = 0.00$	$\rho = 0.30$	$\rho = 0.90$
0.0225	0.00	17.77	59.99
0.0525	0.00	35.74	56.23
0.0975	8.33	42.60	47.79

With respect to the senior bond tranche, each of the following is true EXCEPT which is false?

A. For a given default rate, an increase in default correlation increases senior bond losses.

B. For a given positive default correlation, an increase in the default rate generally increases expected senior bond losses.

C. The low-default, high-correlation case ($\pi = 0.0225, \rho = 0.90$) has a lower senior 99% Credit VaR than the high-default, zero-correlation case ($\pi = 0.0975, \rho = 0$).

D. At $\rho = 0.90$ the senior 99% Credit VaR is very large (roughly $48 to $60 million in the exhibit) and not highly sensitive to the default rate.

Answer: C

Key correction: Source keys said D in their option ordering; that is the same statement as option C here (the 'low-default, high-correlation gives lower VaR' claim), so the keys agree.

Step by step

1. Read the exhibit: the $\rho = 0.90$ column is 59.99, 56.23, 47.79 for $\pi = 0.0225, 0.0525, 0.0975$.
2. Option C compares (0.0225, 0.90) = 59.99 with (0.0975, 0) = 8.33. Since 59.99 > 8.33, the claim that the first is lower is false.
3. Option A: along each row VaR rises with correlation (e.g. 0.00, 17.77, 59.99), so it is true.
4. Option B: higher default probability raises senior losses (every tranche is hurt by more defaults; the senior default01 is positive), so it is true.
5. Option D: at $\rho = 0.90$ the VaR stays in the 48 to 60 band whatever π, so it is true. C is the false statement.

Why C is correct: Correlation is a more important driver of senior tail risk than default probability: clustering of defaults breaches subordination, while independent defaults (rho = 0) stay near their expected level and rarely reach the senior tranche.

Why the others are wrong

- **A:** True statement, so not the answer: higher correlation clusters defaults and raises senior losses; choosing it means misreading the direction of the correlation effect.
- **B:** True statement, so not the answer: more defaults hurt every tranche; choosing it confuses the senior bond with the equity tranche, whose VaR can fall as default rates rise.
- **D:** True statement, so not the answer: at high correlation VaRs are similar whatever the default rate (48 to 60 in the exhibit); choosing it reflects thinking senior VaR must keep rising with default probability.

Q1173 BT answers P2.T6.314.3 · BT notes P2.T6.314.3

In Malz's simulation, the mezzanine tranche has a width of only 10%. It is typical for the mezzanine tranche to be thinner than the senior bond. In regard to the mezzanine tranche, each of the following is true EXCEPT which is false?

A. At low default rates, an increase in correlation increases losses on the mezzanine bond (behaves like a senior bond).

B. At high default rates, an increase in correlation decreases losses on the mezzanine bond (behaves like equity).

C. The mezzanine tranche has negative convexity in default rates for low default rates (behaves like a senior bond) but is positively convex for high default rates (behaves like equity).

D. At low correlations, for example, $\rho = 0$, the mezzanine tranche is essentially insensitive to changes in the default rate (behaves like a senior bond).

Answer: D

Step by step

1. Recall the mezzanine pattern: senior-like at low default rates, equity-like at high default rates.
2. A and B are the two halves of the correlation effect: at low default rates more correlation means more mezzanine losses; at high default rates more correlation means fewer expected losses (few-default outcomes become likelier).
3. C is the convexity pattern: negative at low default rates, positive at high default rates.
4. D claims insensitivity at $\rho = 0$. Because the tranche is thin, at zero correlation a small rise in the default rate moves the pool loss from below the mezzanine attachment point to through it: mezzanine default01 is largest at $\rho = 0$ (about 0.26 percent of par per bp at default probability 0.0675). D is false.

Why D is correct: A thin tranche is hit hard once pool losses reach its attachment point, so with independent defaults ($\rho = 0$) a modest rise in default rate moves it from no loss to heavy loss; it is highly sensitive, not insensitive.

Why the others are wrong

- **A:** True, so not the answer: it mirrors the senior-bond correlation effect; picking it means thinking correlation always helps junior tranches.
- **B:** True, so not the answer: at high default rates higher correlation makes few-default scenarios likelier, as for equity; picking it ignores the equity-like regime.
- **C:** True, so not the answer: convexity switches from senior-like (negative) to equity-like (positive); picking it confuses the sign of convexity.

Q1174 BT answers P2.T6.315.2 · BT notes P2.T6.315.2

In the three-tiered (senior bond, mezzanine, and equity tranche) securitization structure, Malz tends to measure the risk of a tranche by its 95% and/or 99% Credit VaR. In regard to tranche risks, each of the following is true EXCEPT which is false?

A. Credit VaR is the worst-case loss at the 0.05 or 0.01 quantile minus the expected loss (EL).

B. High systemic risk can be expressed as high default correlation, which tends to increase the risk (Credit VaR) of senior bonds.

C. Because the mezzanine tranche is thick, at high correlations and high default rates, the mezzanine tranche behaves like the senior bond with a bell-curve-shaped loss distribution.

D. An increase in granularity can significantly diminish securitization risk.

Answer: C

Step by step

1. A: Credit VaR is the quantile loss minus expected loss; true.
2. B: higher correlation clusters defaults and raises senior VaR; true.
3. D: low granularity (few large loans) raises the probability of tail events, so a more granular pool is less risky; true.
4. C: the mezzanine tranche is thin, not thick, and at high correlation and default rates it behaves like equity with a U-shaped (all-or-nothing) loss distribution, not a bell curve. C is false.

Why C is correct: Mezzanine is a thin tranche; at high correlation and default rates losses are either small or wipe it out, giving an equity-like U-shaped distribution.

Why the others are wrong

- **A:** True, so not the answer: it is the definition of Credit VaR; picking it confuses VaR with the raw quantile loss.
- **B:** True, so not the answer: systematic risk expressed as high correlation raises the likelihood of senior write-downs; picking it assumes senior bonds are immune to correlation.
- **D:** True, so not the answer: more loans of smaller size reduce tail events; picking it ignores granularity as a risk factor.

Q1175 Mock 1 Q63

Investor A has sold protection on the most junior tranche of a collateralized debt obligation (CDO). Investor B has sold default protection on the most senior tranche of the same CDO. If default correlations decrease, which of the following statements is incorrect?

I. Investor A's position will lose value.

II. Investor B's position will gain value.

A. I only.

B. II only.

C. Both I and II.

D. Neither I nor II.

Answer: D

Step by step

1. Selling protection on a tranche is economically the same as holding that tranche's credit risk (long the tranche).
2. Equity/junior tranche value rises with correlation (more chance of very few defaults), so when correlation falls the junior position loses value: I is correct.
3. Senior tranche value falls with correlation (clustered defaults breach subordination), so when correlation falls the senior position gains value: II is correct.
4. Both statements are correct, so neither is incorrect: D.

Why D is correct: Junior tranches are long correlation and senior tranches are short correlation, so falling correlation hurts the junior protection seller and helps the senior one.

Why the others are wrong

- **A:** Treats I as incorrect, e.g. assuming lower correlation helps equity; in fact equity loses its chance of a near-zero-loss outcome.
- **B:** Treats II as incorrect, e.g. assuming the senior protection seller loses whenever correlation moves; it actually gains as tail clustering fades.
- **C:** Treats both as incorrect, reversing the correlation exposure of both tranches.

Q1176 Schweser Module Quiz 40.3 Q1

Which of the following statements about portfolio losses and default correlation is most likely correct?
I. Increasing default correlation decreases senior tranche values but increases equity tranche values.
II. At high default rates, increasing default correlation decreases mezzanine bond prices.

A. I only.
B. II only.
C. Both I and II.
D. Neither I nor II.

Answer: A

Step by step

1. I: higher correlation clusters defaults, so senior tranches are breached more often (value falls); equity gains because the few-default outcome becomes likelier. True.
2. II: mezzanine is senior-like at low default rates (correlation lowers value) but equity-like at high default rates, where more correlation raises its value. So the statement is false.
3. Only I is correct: A.

Why A is correct: Correlation hurts senior tranches, helps equity, and for mezzanine the sign depends on the default rate: negative at low rates, positive at high rates.

Why the others are wrong

- **B:** Selects only II, which has the mezzanine effect backwards at high default rates (it behaves like equity and gains).
- **C:** Accepts II as well, applying the low-default-rate (senior-like) mezzanine effect to high default rates.
- **D:** Rejects I, wrongly believing higher correlation lowers all tranche values; equity actually benefits from the higher chance of few defaults.

LO 40.j

Explain how default sensitivities for tranches are measured.

Q1177 BT answers P2.T6.315.1 · BT notes P2.T6.315.1

According to Malz, "The default01 measures the impact of an increase of one (1) basis point in the default probability. It is analogous to the DV01 and the spread01 and is calculated numerically in a similar way. To compute the default01, we increase and decrease default probability 10 bps and revalue each tranche at these new [default probability] values ... Each default01 is expressed as a positive number and expresses the decline in value or increase in loss resulting from a 1-basis point rise in default probability."
Each of the following is true about default01 EXCEPT, which is false?

A. For all tranches, in all cases, default01 is positive regardless of the initial value of default probability or correlation since equity and bond values decrease monotonically as the default probability rises
B. Because default01 "models correlation as constant according to an implicit ceteris paribus," default01 is invariant (insensitive) to different correlation levels
C. The default01 sensitivity converges to zero for all the tranches for very high default rates
D. The peak default01 is at a default probability of zero for the equity tranche (its attachment point) and occurs at a lower default rate for the mezzanine than for the senior tranche because it has a lower attachment point

Answer: B

Step by step

1. Default01 is the change in a tranche's value (or increase in loss) for a 1 bp rise in default probability π, estimated by shocking π up and down 10 bps and revaluing each tranche.
2. A: all tranches lose value as default probability rises, so default01 is positive for every combination of default probability and correlation - true.
3. B: the default01 plots are drawn for different constant pairwise correlations and differ materially across them (for example, the equity default01 varies most when correlation is low). 'Constant' correlation only means correlation is held fixed while π is shocked; the measure is not insensitive to the correlation level - FALSE.
4. C: once losses are already extremely high, an extra default adds little, so default01 tends to zero for very high default rates - true.

5. D: the peak default01 occurs where pool losses approach a tranche's attachment point: at zero default probability for equity (attachment 0), earlier for mezzanine than for senior - true.

Why B is correct: Default01 is computed holding correlation fixed, but its level and shape depend strongly on the correlation assumed, so claiming it is invariant to correlation is the false statement.

Why the others are wrong

- **A:** True, so not the EXCEPT answer: equity and bond values fall monotonically as default probability rises, so default01 is always positive. Choosing it confuses the sign convention (default01 is quoted as a positive number for a decline in value).
- **C:** True: at very high default rates losses are already large and marginal defaults matter little. Choosing it reflects thinking sensitivity keeps growing with default probability.
- **D:** True: the largest sensitivity starts where pool losses reach the tranche's attachment point, so equity peaks at zero and mezzanine peaks at a lower default rate than senior. Choosing it misreads the attachment-point logic.

Q1178 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A portfolio manager measures the default sensitivity of a mezzanine bond in a CLO. Revaluing the bond with default probabilities shocked up and down by 10 bps gives mean losses of 7.9% of par at the higher default probability and 5.5% of par at the lower default probability. Which of the following is the correct default01 of the bond?

A. 2.40% of par per basis point
B. 0.24% of par per basis point
C. 1.20% of par per basis point
D. 0.12% of par per basis point

Answer: D

Step by step

1. Default01 = [loss(pi + 10 bps) - loss(pi - 10 bps)] / 20.
2. (7.9% - 5.5%) / 20 = 0.12% of par per bp.

Why D is correct: The 2.4% change in mean loss occurs over a 20 bp total shock, so the default01 is 0.12% of par per bp.

Why the others are wrong

- **A:** Does not divide at all, so the result is the loss change over 20 bps rather than per 1 bp.
- **B:** Divides by 10 (the size of one shock) instead of 20, the full width between the up and down shocks.
- **C:** Divides by 2, confusing the central difference with an average of the two losses.

Q1179 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

An analyst computes the default01 of each CLO tranche using Monte Carlo simulation. Which of the following procedures is consistent with how default sensitivities are measured?

A. Shock the default probability up and down by 10 bps, reuse the same correlated normal draws, recompute the default counts and subsequent cash flows, and divide the difference in mean values or losses by 20.
B. Shock the default probability by 1 bp, redraw all random numbers for the revaluation, and report the result as a negative number for bonds because their values fall.
C. Shock the correlation parameter up and down by 10 bps, redraw the normal variables for each shock, and divide the difference in mean losses by 20.
D. Shock the default probability up and down by 10 bps, keep the original default counts fixed, and revalue each tranche from the existing simulated cash flows.

Answer: A

Step by step

1. Default01 is the analog of DV01 for default probability.
2. Both shocks revalue each tranche from the point where default times are generated, so default counts and cash flows must be recomputed.
3. Reusing the same random numbers avoids adding simulation noise; the result is expressed as a positive number.

Why A is correct: The procedure shocks the default probability, keeps the same random draws, recomputes defaults and cash flows, and scales the difference by 20.

Why the others are wrong

- **B:** A fresh set of draws introduces simulation noise that could swamp a 1 bp effect, and default01 is reported as a positive number (decline in value or increase in loss).
- **C:** Shocks correlation rather than default probability; default01 holds correlation constant.
- **D:** Fixed default counts would not change with the shock, giving a zero sensitivity; the default counts must be recomputed.

LO 40.k

Describe risk factors that impact structured products.

Q1180

AnalystPrep Q.5528

What is the primary concern associated with tranche thinness in structured credit products like MBS and CDOs?

A. The potential for market liquidity risk to increase due to the small size of the tranches.
B. The likelihood of large loss severity within thin tranches once a loss event breaches these levels.
C. The difficulty in accurately pricing the tranches due to their small size relative to the total pool.
D. The challenge in finding enough investors willing to invest in smaller-sized tranches.

Answer: B

Step by step

1. A thin tranche has a narrow gap between its attachment and detachment points.
2. Once collateral losses reach the attachment point, they pass through the narrow band quickly, so the tranche is largely wiped out.
3. Hence, conditional on the tranche suffering any loss, the loss is likely to be large; this shows up as a small gap between the 95% and 99% credit VaR of the mezzanine tranche.

Why B is correct: Because the loss band is narrow, any breach of the attachment point quickly consumes most of the tranche, so loss severity given a loss is very large.

Why the others are wrong

- **A:** Liquidity risk is not the risk the reading attributes to thinness; it confuses thinness with market-structure issues.
- **C:** Pricing difficulty is not the stated concern; thinness is about loss severity, not valuation mechanics.
- **D:** Investor demand is a marketing consideration, not the credit-risk consequence of thinness.

Q1181

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

Two CLO pools each have $1 billion of par and identical average default probability, recovery rate, and pairwise default correlation. Pool X holds 1,000 loans of $1 million each and Pool Y holds 50 loans of $20 million each. Which of the following statements is correct?

A. Pool Y has lower granularity, but because the pool parameters are identical the loss distributions and Credit VaR of the tranches are the same.
B. Pool Y has lower granularity and a greater probability of extreme loss outcomes, so its tranches have greater tail risk than those of Pool X.
C. Pool X has higher granularity, so its tranches face greater tail risk because defaults are spread across many loans.
D. Granularity affects only the expected loss of the tranches and has no effect on the tails of the loss distribution.

Answer: B

Step by step

1. Granularity concerns loan-level diversification.
2. With few large loans, a single default removes a large share of the pool, so losses are lumpy.
3. Lumpy pools have a higher probability of tail events than equal-sized pools of many smaller loans.

Why B is correct: Pool Y is lumpy, so the probability of extreme outcomes (and tail risk for the tranches) is higher than for Pool X.

Why the others are wrong

- **A:** Equal average parameters do not equalize the loss distributions; the size of individual loans changes the tail.
- **C:** Reverses the effect: more granular pools are better diversified and have thinner tails.
- **D:** Granularity mainly changes the tails, not the expected loss, so the claim is backwards.

Q1182 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A pension fund investment officer states that the senior bond of a CLO is essentially risk free because the underlying pool of 100 loans is diversified across lenders, geographies, and industries. A credit analyst responds that the senior bond remains exposed to systematic risk. Which of the following best supports the analyst's view?

A. Diversification across lenders, geographies, and industries increases the pool's default correlation and therefore the probability of senior bond writedowns.

B. The senior bond bears the same loss as the average loan in the pool, so diversification of the collateral does not change its risk.

C. If pairwise default correlation is high, there is a material probability of a state with an unusually large number of defaults, so the senior bond can suffer a large loss even though the pool is diversified.

D. Systematic risk affects only the equity and mezzanine tranches, so the senior bond is exposed only if its tranche is thin.

Answer: C

Step by step

1. Diversification cannot eliminate systematic risk in a credit portfolio.
2. High pairwise correlations mean defaults cluster in bad states of the world.
3. In those states losses can pass through the subordination and reach the senior bond.

Why C is correct: High correlation (systematic risk) raises the likelihood of senior writedowns despite a well-diversified pool.

Why the others are wrong

- **A:** Diversification generally lowers, not raises, correlation; the point is that it cannot remove systematic risk.
- **B:** The senior bond's expected loss is typically lower than the pool's because of subordination, though its tail loss can still be large.
- **D:** Systematic risk affects all tranches; the senior bond is exposed when correlation is high.

LO 40.l

Define implied correlation and describe how it can be measured.

Q1183 BT answers P2.T7.507.2

REBUILT — Original asked about the 2005 equity/mezzanine hedge-fund trade (outside the LO text and with a debatable key); rebuilt to test the LO 40.l inference of implied correlation from a tranche price move.

An analyst backs out the implied correlation of the CDX equity tranche using a Gaussian copula, constant pairwise correlation, and risk-neutral default probabilities extracted from the constituent CDS spreads. The equity tranche market price rises from \$3.0 million to \$3.2 million while the constituent CDS spreads (and hence the risk-neutral default probabilities) are unchanged. Which of the following is the most appropriate conclusion?

A. The implied correlation has increased, because equity holders benefit from higher correlation.

B. The implied correlation has decreased, because lower correlation reduces the probability of large clustered losses.

C. The implied correlation is unchanged, because the default probabilities are unchanged.

D. The risk-neutral default probabilities have fallen, so the model price no longer needs a correlation adjustment.

Answer: A

Step by step

1. Implied correlation is the single correlation input that makes the model equity tranche price equal the observed market price, given the CDS-implied risk-neutral default probabilities.
2. The default probabilities are held fixed (CDS spreads unchanged), so the only free input that can explain a price change is correlation.

3. Higher correlation fattens the tails of the loss distribution: more scenarios with no or few defaults, which helps the first-loss (equity) holder, whose expected loss falls.
4. Equity price up from 3.0 to 3.2 therefore implies the market-implied correlation rose.

Why A is correct: With default probabilities fixed, a higher equity tranche price can only be reconciled by a higher implied correlation, because equity value increases with correlation.

Why the others are wrong

- **B:** Reverses the sign of the relationship; lower correlation raises equity expected loss and lowers its price.
- **C:** Ignores that the default probabilities are only one of two model inputs; correlation is the unknown that is solved for, so it must move when the price moves.
- **D:** Contradicts the stem: CDS spreads are unchanged, so the risk-neutral default probabilities have not changed.

Q1184

Schweser Module Quiz 40.3 Q2

REBUILT — Option D ('risk-adjusted' default probabilities) was ambiguous against the correct risk-neutral inputs and option A/B wording was loose; reworded options so only C is correct.

Which of the following statements best describes the calculation of implied correlation for a securitized tranche?

A. The implied correlation is computed separately for each tranche using a constant pairwise correlation, so the resulting values are identical across all tranches.

B. Observed CDS prices are inserted directly into the pricing function as the tranche values, and no correlation is solved for.

C. The tranche pricing function is calibrated so that the model price matches the observed market price of the tranche, with the correlation as the unknown.

D. Historical (real-world) default probabilities are fed into the copula, and the correlation is chosen to match the market tranche price.

Answer: C

Step by step

1. Implied correlation works like Black-Scholes implied volatility: start from an observed market price and a pricing function.
2. Extract risk-neutral default probabilities (and possibly recovery rates) from the CDS term structure.
3. Assume constant pairwise correlation and feed the default estimates into a copula.
4. Solve for the unique correlation at which the model tranche price equals the market price; the result differs by tranche (correlation skew).

Why C is correct: Implied correlation is the input that calibrates the model tranche price to the observed market price, analogous to implied volatility.

Why the others are wrong

- **A:** The pairwise correlation is assumed constant within a tranche's calculation, but the implied values differ across tranches (correlation skew); they are not identical.
- **B:** CDS prices are used to infer default probabilities, not the tranche values; tranche prices are observed separately and the correlation is solved for.
- **D:** Implied correlation uses risk-neutral default probabilities from CDS spreads, not historical probabilities.

Q1185

New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

An analyst backs out implied correlations for the CDX equity (0-3%), junior mezzanine (3-7%), and senior mezzanine (7-10%) tranches using a Gaussian copula with constant pairwise correlation and the same risk-neutral default probabilities extracted from the constituent CDS spreads. The calibrated values are different for each tranche. Which of the following is the most appropriate interpretation?

A. The differing values should be averaged, since the average is the true implied correlation of the index portfolio.

B. The differing values show a calibration error, because a constant pairwise correlation implies the same value must match every tranche.

C. The differing values arise because the analyst used historical default probabilities for some tranches and risk-neutral probabilities for others.

D. The differing values reflect a correlation skew; each implied (base) correlation is the value that makes the model price match the market price of that tranche.

Answer: D

Step by step

1. Implied correlation is solved separately for each tranche so that the model price equals the observed tranche price.
2. The solutions generally differ across tranches (base correlation), a phenomenon called correlation skew.

Why D is correct: Tranche-specific calibration normally yields different implied correlations, which is the correlation skew.

Why the others are wrong

- **A:** There is no meaningful index-level average; each tranche's value is the unique number that matches that tranche's price.
- **B:** Constant pairwise correlation is a modeling assumption inside the copula, not a guarantee that different tranches will give the same value; the differences reflect the market.
- **C:** All tranches use the same risk-neutral probabilities from CDS; the differences do not come from mixing probability measures.

LO 40.m

Identify the motivations for using structured credit products.

Q1186

AnalystPrep Q.5532

Which of the following is a common motivation among investors for using structured credit products?

A. Diversification of credit risk.
B. Increase in interest rate risk exposure.
C. Reduction in credit enhancement.
D. Simplification of the investment process.

Answer: A

Step by step

1. Investor motivations (Malz): access to diversified loan pools (mortgages, credit card, auto loans) that capital-markets investors could not otherwise hold, and tranching to select a desired risk-return profile.
2. Option A (credit diversification) matches the first motivation.
3. Options B, C and D describe no investor motive in the reading: B is unrelated to credit, C is an issuer-side feature, D contradicts the due-diligence burden structured products impose.

Why A is correct: Securitization lets investors hold a diversified pool of credits (and pick a tranche risk-return profile) that would otherwise be available only to banks.

Why the others are wrong

- **B:** Structured credit is about credit exposure, not seeking more interest rate risk; this confuses a side effect with a motivation.
- **C:** Credit enhancement (subordination) protects senior investors, and lowering it is something issuers may achieve with good pools, not an investor motive.
- **D:** Structured products are complex and require extra due diligence to understand the return distribution, so they do not simplify investing.

Q1187

BT answers P2.T6.315.3 · BT notes P2.T6.315.3

In regard to the motivations of issuers or investors in a securitization, according to Malz, each of the following is true EXCEPT, which is not?

A. Securitizations undertaken primarily to capture the spread between the underlying loan interest and the coupon rates of the liabilities are sometimes called arbitrage CDOs, while securitizations motivated largely for balance sheet relief are termed balance-sheet CDOs
B. An important motive of the issuer for securitization is that it provides technology for maturity matching; i.e., for providing term funding for the underlying loans
C. Tranching technology is primarily motivated by issuers; investors do not require it
D. Mezzanine tranches have less of a natural investor base than other securitized credit products because they may behave like a senior tranche (i.e., low probability of loss but high systematic risk), or they may behave more like an equity tranche (i.e., high probability of impairment)

Answer: C

Step by step

1. Test each statement against Malz: issuers' needs are met by pooling and securitization; investors, not issuers, motivate tranching, because tranching tailors return distributions to desired risk profiles.
2. A: arbitrage CDOs capture the spread; balance-sheet CDOs seek balance sheet relief - true.
3. B: term funding / maturity matching is an important issuer motive - true.
4. C: reverses who motivates tranching - false, so it is the EXCEPT answer.
5. D: mezzanine tranches can act like senior or equity tranches, so have a weak natural investor base - true.

Why C is correct: Tranching is demanded by investors who want tailored risk profiles; issuers are satisfied by pooling and securitization alone.

Why the others are wrong

- **A:** True statement; choosing it means missing that the question asks for the false statement.
- **B:** True statement; maturity matching/term funding is a key issuer motive.
- **D:** True statement; mezzanine behavior is ambiguous (senior-like or equity-like), so it lacks a natural investor base.

Q1188 AnalystPrep Q.5521

Global Finance Bank (GFB) is considering expanding its investment portfolio by incorporating structured credit products. The investment committee is discussing the potential motivations for including these products in the bank's asset mix. Which of the following options best represents the motivations for GFB to use structured credit products in its investment strategy?

A. To gain exposure to high-frequency trading opportunities and capitalize on short-term market volatilities.

B. To achieve portfolio diversification, enhance yields, and obtain specific risk-return profiles that match the bank's investment objectives.

C. To primarily hedge against currency and interest rate risks inherent in the bank's existing investment portfolio.

D. To invest in highly liquid assets that can be quickly converted to cash for short-term liquidity needs.

Answer: B

Step by step

1. Investor motivations (Malz): diversified pool access, tranching to choose a risk-return profile (equity tranches high, senior low), and yield pick-up for institutions needing return targets.
2. B combines diversification, yield and tailored risk-return - all three appear.
3. A, C and D concern trading, market-risk hedging and liquidity, none of which is a structured-credit investor motive.

Why B is correct: Structured credit gives investors diversified credit exposure and, via tranching, a risk-return profile and yield tailored to their objectives.

Why the others are wrong

- **A:** Structured credit positions are typically held for credit return; short-term volatility trading is not the motivation.
- **C:** The products carry credit risk; they are not instruments for hedging currency or interest rate risk.
- **D:** Structured credit tranches, especially mezzanine and equity, are often illiquid and risky, so they are not a liquidity-management tool.

Reading 41. An Introduction to Securitization

Moorad Choudhry, Structured Credit Products: Credit Derivatives & Synthetic Securitization, 2nd Edition (New York, NY: John Wiley & Sons, 2010). Chapter 12.

LO	Learning objective	Qs
41.a	define securitization, describe the securitization process, and explain the roles of participants in the process.	10
41.b	explain the terms over-collateralization, first-loss piece, equity piece, and cash waterfall within the securitization process.	9
41.c	analyze the differences in the mechanics of issuing securitized products using a trust versus a special purpose vehicle (SPV) and distinguish between the three main SPV structures: amortizing, revolving, and master trust.	5
41.d	explain the reasons for and the benefits of undertaking securitization.	3
41.e	describe and assess the various types of credit enhancements.	4
41.f	explain the various performance analysis tools for securitized structures and identify the asset classes they are most applicable to.	4
41.g	define and calculate the delinquency ratio, default ratio, monthly payment rate (MPR), debt service coverage ratio (DSCR), the weighted average coupon (WAC), the weighted average maturity (WAM), and the weighted average life (WAL) for relevant securitized structures.	11
41.h	explain the prepayment forecasting methodologies and calculate the constant prepayment rate (CPR) and the Public Securities Association (PSA) rate.	13

LO 41.a

Define securitization, describe the securitization process, and explain the roles of participants in the process.

Q1189 AnalystPrep Q.5533

At a seminar on the securitization process, a case study of a mortgage-backed security (MBS) issuance highlights one participant whose role is not immediately obvious. This participant examines the underlying assets, assesses their quality, and provides an independent evaluation of the risk involved. Its assessment drives the perceived creditworthiness of the securitization; in the case study it led to a higher rating for the MBS, which attracted a broader range of investors because of the perceived lower risk. Which participant is being described?

A. Originator
B. Servicer
C. Underwriter
D. Rating agency

Answer: D

Step by step

1. The role described is an independent analysis of asset quality and risk that produces a formal credit rating for the notes.
2. Providing formal credit ratings for each securitization (and analyzing the legal structure and cash flows of the SPV) is the function of credit rating agencies such as Moody's, S&P and Fitch.
3. A higher rating lowers perceived risk and widens the investor base, matching the description.

Why D is correct: Rating agencies provide independent, formal credit ratings of the securitized notes, and the rating drives pricing and investor demand.

Why the others are wrong

- **A:** The originator creates the assets and seeks to convert them to cash; it is not independent and does not rate them.
- **B:** The servicer collects and remits payments on the pool; it does not evaluate credit quality for investors.
- **C:** The underwriter/structuring side designs, prices and sells the notes; it is not an independent assessor of risk.

Q1190 AnalystPrep Q.5534

During a training session on securitization, attention turns to a key player in the issuance of a new asset-backed security (ABS). This participant structures the deal, determines pricing, and manages the sale of the securities to investors. In a recent issuance it worked closely with the originator to structure the tranches, balancing risk and return to appeal to a diverse investor base, and used its knowledge of market conditions and investor appetite to distribute the securities. Which participant is being described?

A. Servicer
B. Trustee
C. Underwriter
D. Legal advisor

Answer: C
Step by step
1. Structuring the tranches, setting the price and selling the securities to investors are the functions of the underwriter (structuring/arranging agent).
2. The servicer instead collects payments on the pool; the trustee monitors the asset pool and protects investors' interests; legal advisors deal with legal structure and compliance.
3. Only the underwriter combines structuring, pricing and distribution.

Why C is correct: The underwriter/structuring agent designs the note structure with the originator, prices the notes and places them with investors.

Why the others are wrong
- **A:** The servicer administers the underlying loans (collections, remittance); it has no role in structuring or selling the notes.
- **B:** The trustee has a fiduciary duty to investors and monitors pool conditions such as minimum credit quality and delinquency ratios; it does not price or sell notes.
- **D:** Legal advisors handle the legality and documentation of the transfer and SPV, not pricing and distribution.

Q1191

BT answers L2.T6.102.2

REBUILT — Original stem relied on the GSAMP assignment and left the SPV (which also ends up holding the loans) as a defensible answer; stem reworded so only the arranger fits.

In a subprime mortgage securitization, one party buys the loans from the originator, performs due diligence on the originator, and arranges and underwrites the sale of the resulting securities to investors, before the loans are placed in the issuing trust. Which party is being described?

A. Credit rating agency
B. Servicer
C. Arranger (structuring agent)
D. Special purpose vehicle/entity (SPV/SPE)

Answer: C
Step by step
1. Identify the function: acquiring the pool, vetting the originator, and structuring/underwriting the securities for sale.
2. That is the arranger (structuring agent), the de facto advisor responsible for security design and placement; in the GSAMP Trust example it was Goldman Sachs.
3. The SPV only holds the assets and issues the notes; the rating agency rates; the servicer collects.

Why C is correct: The arranger/structuring agent sources the loans, conducts due diligence and underwrites the securities distributed to investors.

Why the others are wrong
- **A:** A rating agency assigns ratings to the notes and does not buy loans or underwrite securities.
- **B:** The servicer collects and remits loan payments and handles delinquencies; it does not underwrite securities.
- **D:** The SPV is the bankruptcy-remote issuer that holds the assets; it does not perform due diligence or market the securities. It is the tempting choice because it ultimately owns the loans.

Q1192

BT answers L2.T6.102.3

REBUILT — Added securitization context to the stem (original referred to an unnamed case study).

In a mortgage securitization, which party is responsible for collecting and remitting loan payments, holding escrow funds for taxes and insurance, contacting delinquent borrowers, and supervising foreclosures and property dispositions?

A. Servicer
B. Arranger
C. Special purpose vehicle/entity (SPV/SPE)

D. Originator

Answer: A

Step by step

1. List the duties: collecting and remitting payments, escrow management, delinquency follow-up, foreclosure supervision.
2. These are the day-to-day loan administration functions of the servicer.

Why A is correct: Servicing is the ongoing administration of the loan pool, performed by the servicer (the originator may act as initial servicer).

Why the others are wrong

- **B:** The arranger structures and sells the securities; it does not administer loans.
- **C:** The SPV is a legal shell that owns the pool and issues notes; it has no operating loan-administration role.
- **D:** The originator creates the loans; although it is often the initial servicer, the duties listed define the servicer role, and after the true sale the originator no longer owns the pool.

Q1193 BT answers L2.T6.102.4

Which party, by purchasing the credit-sensitive assets, is essential to the credit risk transfer because it keeps the assets bankruptcy remote, i.e., protected from claims against the originator if the originator enters bankruptcy?

A. Servicer
B. Arranger
C. Special purpose vehicle/entity (SPV/SPE)
D. Originator

Answer: C

Step by step

1. The SPV is a separate legal trust or company set up to buy the assets in a true sale.
2. This separates the asset pool from the originator's other assets, so the pool is unaffected by originator insolvency.
3. Hence investors look to the quality of the pool, not the originator.

Why C is correct: A true sale of assets to a separate SPV makes the pool bankruptcy remote, which is the basis of the credit risk transfer.

Why the others are wrong

- **A:** The servicer administers loans; it does not hold title to the assets.
- **B:** The arranger structures and places the deal; it does not hold the pool.
- **D:** The originator is the party whose bankruptcy the pool is being isolated from; it is the seller of the assets, not the protector.

Q1194 BT answers L2.T6.102.5

REBUILT — Original wording 'underwrites' was ambiguous between loan underwriting and securities underwriting; reworded to describe origination unambiguously.

In a mortgage securitization, which party assesses the borrowers' credit, makes the loans and initially funds them, and later sells the loans to convert them into cash?

A. Servicer
B. Arranger
C. Special purpose vehicle/entity (SPV/SPE)
D. Originator

Answer: D

Step by step

1. Making and funding the loans is origination.
2. The originator is the entity that wants to convert its credit-sensitive assets into cash and transfer the credit risk by selling them to the issuer/SPV.

Why D is correct: The originator creates and initially funds the loans, then sells them to the SPV to obtain cash.

Why the others are wrong

- **A:** The servicer administers loans after origination; it does not fund them.

- **B:** The arranger underwrites the securities issue, not the loans; the word 'underwrite' invites this confusion.
- **C:** The SPV purchases the loans after they have been originated; it does not originate or initially fund them.

Q1195 BT answers P2.T5.102.1

Each of the following is true about the securitization of residential mortgage-backed securities (RMBS) EXCEPT:

A. The originator pools mortgage loans and sells them to an issuer or special purpose vehicle (SPV), which issues securities to investors

B. Key benefits of securitization, including RMBS, include transfer of credit risk to investors and diversification via pooling of credit-sensitive assets

C. By selling mortgage loans to issuers in a securitization, originators raise cash to fund new loans

D. The mortgage servicer ensures that the SPV delivers on its contractual obligations

Answer: D

Step by step

1. A is the basic structure: originator sells assets to the SPV, which issues notes.
2. B and C are standard benefits: credit risk transfer, diversification, and cash to originate new loans.
3. D is false: the trustee, charged with a fiduciary duty to investors, monitors the pool and acts for investors if the SPV defaults. The servicer collects payments and handles delinquencies and foreclosures.

Why D is correct: Protecting investors and monitoring compliance with pool conditions is the trustee's job, not the servicer's.

Why the others are wrong

- **A:** True description of the securitization structure.
- **B:** True; credit risk transfer and diversification are recognized benefits.
- **C:** True; sale of loans produces cash that can fund new lending.

Q1196 BT answers P2.T6.610.1 · BT notes P2.T6.610.1

REBUILT — Source exhibit (linkage diagram A-D) was missing; the question was rewritten as a self-contained, text-only EXCEPT question on the same flow of assets, notes and proceeds.

Which of the following statements about the securitization process is INCORRECT?

A. The originator sells its pool of assets to the SPV in a true sale, which removes the assets from the originator's balance sheet without recourse

B. Because the SPV is bankruptcy remote, investors in the notes remain primarily exposed to the creditworthiness of the originator

C. The SPV funds its purchase of the assets by issuing notes, and the proceeds of the note sales are passed to the originator as payment for the assets

D. The notes are divided into tranches to meet the credit quality needs of different investors and are then placed with investors in the capital markets

Answer: B

Step by step

1. A is correct: a true sale to a distinct legal entity transfers the assets off the originator's balance sheet with no recourse.
2. C and D are correct: the SPV issues tranched notes to investors and uses the proceeds to pay for the assets.
3. B is incorrect: the SPV separates the pool from the originator, so investors are concerned with the credit quality of the securitized issue, not the originator's financial strength.

Why B is correct: Bankruptcy remoteness means note investors look to the asset pool and structure, not to the originator's credit.

Why the others are wrong

- **A:** Correct statement of a true sale; not the answer to an EXCEPT question.
- **C:** Correct; note proceeds fund the purchase price paid to the originator.
- **D:** Correct; tranching customizes the notes and they are placed with investors.

Q1197 Mock 2 Q29

A bank is establishing a special purpose vehicle (SPV) to securitize a portfolio of receivables. The bank's structuring team is explaining to the treasury department why the SPV is necessary and what characteristics it must possess. Which of the following is the most important feature the SPV must have for the securitization to achieve its intended purpose?

A. The SPV must be domiciled in the same jurisdiction as the originating bank so that a single legal regime governs the transaction.

B. The SPV must be consolidated onto the originating bank's balance sheet so that investors have recourse to the bank's capital.

C. The SPV must be bankruptcy remote, so that the insolvency of the originating bank does not give its creditors a claim on the transferred assets.

D. The SPV must retain the originating bank as servicer so that the credit quality of the pool is actively managed.

Answer: C

Step by step

1. The purpose of securitization is to make investors' exposure depend on the asset pool, not on the originator's solvency.
2. This requires a true sale of the assets to a separate legal entity (the SPV) with no recourse to the originator.
3. The SPV is therefore bankruptcy remote: if the originator fails, its creditors cannot claim the transferred assets.
4. Check each option against this requirement: only C states it.

Why C is correct: Bankruptcy remoteness (via a true sale) separates the asset pool from the originator's credit risk, which is the core reason for using an SPV.

Why the others are wrong

- **A:** Same-jurisdiction domicile is not required; SPVs are often set up offshore (e.g., Cayman Islands) for tax and legal reasons.
- **B:** Consolidation with recourse to the bank would defeat the true sale and the off-balance-sheet treatment, tying investors back to the originator's credit.
- **D:** The servicer role is an operational arrangement and does not provide the legal isolation; keeping the originator involved is not the defining requirement.

Q1198 Schweser Module Quiz 41.1 Q1

Which of the following statements most accurately describes the effect of selling a loan without recourse?

A. The bank that sells the loan retains a contingent liability.

B. The bank that sells the loan bears a specified percentage of the credit risk.

C. The loan is removed from the balance sheet of the bank that sells the loan.

D. The purchaser has the right to sell the loan back to the bank that originated the loan.

Answer: C

Step by step

1. Selling without recourse means the buyer has no claim on the seller if the borrower defaults.
2. The seller therefore keeps no credit risk or contingent liability on the loan.
3. Hence the sale is a true sale and the loan leaves the seller's balance sheet (off-balance-sheet treatment).

Why C is correct: Without recourse, all credit risk transfers to the purchaser, so the loan is derecognized from the seller's balance sheet.

Why the others are wrong

- **A:** A contingent liability is what arises with recourse, where the seller may have to compensate the buyer for losses.
- **B:** Retaining a specified share of credit risk is a partial-recourse (or retained-interest) arrangement, not a sale without recourse.
- **D:** A put-back right to the originator is a recourse feature; without recourse the purchaser has no such right.

LO 41.b

Explain the terms over-collateralization, first-loss piece, equity piece, and cash waterfall within the securitization process.

Q1199 BT answers P2.T6.609.1

REBUILT — Item IV (post-crisis risk retention rule) is not covered by the Schweser LOs; replaced with the first-loss piece (LO 41.b) and re-keyed the options.

Consider the following four elements of a classic securitization:

I. The slicing of the notes issued to investors into different seniorities and risk profiles

II. The priority scheme that determines how cash flows from the asset pool are distributed to the different note holders

III. The bankruptcy-remote legal entity that purchases the credit-sensitive assets and issues the notes

IV. The most junior class, which absorbs the first losses and is often held by the originator

Which sequence correctly identifies the elements?

A. I. Cash waterfall; II. First-loss piece; III. SPV; IV. Tranches

B. I. Cash waterfall; II. SPV; III. First-loss piece; IV. Tranches

C. I. Tranches; II. Cash waterfall; III. SPV; IV. First-loss piece

D. I. First-loss piece; II. Tranches; III. Cash waterfall; IV. SPV

Answer: C

Step by step

1. I: dividing the notes into classes of differing seniority is tranching.
2. II: the order in which pool cash flows are paid, senior before junior, is the cash waterfall.
3. III: the separate legal entity that buys the assets and issues notes is the SPV.
4. IV: the lowest class that absorbs losses first is the first-loss (equity) piece.
5. The only sequence matching is Tranches, Cash waterfall, SPV, First-loss piece.

Why C is correct: Tranches, cash waterfall, SPV and first-loss piece map to the four descriptions respectively.

Why the others are wrong

- **A:** Gets III right but labels the slicing as the waterfall, the payment priority as the first-loss piece, and the junior class as the tranches.
- **B:** Labels the slicing as the waterfall, the payment priority as the SPV, and the legal entity as the first-loss piece.
- **D:** Labels the slicing as the first-loss piece, the payment priority as tranches, the legal entity as the waterfall, and the junior class as the SPV.

Q1200 AnalystPrep Q.1851

REBUILT — Source key (whole portfolio) was conceptually wrong and the stem vague; reworded so overcollateralization is tied to the lower-rated notes, as in the Schweser text.

In a securitization, overcollateralization (the principal value of notes issued is lower than the principal value of the assets in the pool) is primarily used to enhance the credit quality of which of the following?

A. The senior tranche only, because it is paid first

B. The lower-rated (junior) notes, which gain a cushion that absorbs losses

C. The reference portfolio, whose assets are guaranteed by the excess collateral

D. The originator's other unsecuritized assets

Answer: B

Key correction: Source key (AnalystPrep) said D, the whole portfolio; the readings identify the lower-rated/junior notes as the ones enhanced, so the correct answer is B.

Step by step

1. Overcollateralization means notes issued have a principal value below the pool's assets (e.g., 101 mortgages backing notes worth 100 mortgages).
2. The excess collateral is a cushion that must be depleted by losses before note holders suffer economic loss.
3. Schweser presents overcollateralization as enhancing the credit quality of the lowest (lower-rated) notes, so it is the junior notes whose credit quality is enhanced.

Why B is correct: The excess collateral sits beneath the notes and absorbs initial losses, improving the credit quality of the lower-rated (junior) notes.

Why the others are wrong

- **A:** The senior tranche is protected mainly by subordination and is not the class OC is aimed at; being paid first is a waterfall feature, not OC.
- **C:** The reference portfolio is the asset pool itself; it is not a tranche and is not guaranteed by OC.
- **D:** Overcollateralization concerns only assets inside the SPV; unsecuritized assets are unaffected.

Q1201 AnalystPrep Q.1852

Which tranche of a securitization portfolio will be paid after all other obligations of the SPE are satisfied?

A. Capital stack
B. Junior debt
C. Senior debt
D. Equity

Answer: D

Step by step

1. The cash waterfall pays senior tranches first, then the junior tranches.
2. The equity (first-loss) piece is the most junior claim.
3. It is therefore paid only after all other obligations are met, and it absorbs losses first.

Why D is correct: The equity piece is the residual claim at the bottom of the waterfall: last to be paid and first to absorb losses.

Why the others are wrong

- **A:** The capital stack is the whole structure of financing, not a single tranche.
- **B:** Junior debt is subordinated to senior debt but still ranks ahead of equity.
- **C:** Senior debt is paid first in the waterfall, the opposite of last.

Q1202 AnalystPrep Q.2047

In securitization, the notes issued are structured to reflect the risk areas of the asset pool. The most junior note is hit by losses first and is therefore called the first-loss piece. What other name is given to the first-loss piece?

A. Equity piece
B. At-risk note
C. First-risk note
D. Junior piece

Answer: A

Step by step

1. The most junior note absorbs initial losses and is often non-rated.
2. Because it is usually retained by the originator, it is also called the equity piece (equity tranche/note), even though it is a bond.

Why A is correct: The first-loss piece is the equity piece/tranche, held by the originator and exposed to losses first.

Why the others are wrong

- **B:** 'At-risk note' is not a standard term.
- **C:** 'First-risk note' is an invented variation of 'first-loss'.
- **D:** 'Junior piece' describes seniority generally, but the recognized alternative name for the first-loss piece is the equity piece.

Q1203 AnalystPrep Q.2709

An ABS based on home equity loans is structured as follows:

Tranche	Principal
Subordinated tranche 1	$50 million
Subordinated tranche 2	$50 million
Senior tranche	$100 million
Collateral	$220 million

What portion of a loss of $120 million on the collateral will be absorbed by the senior tranche?

A. $120 million
B. $20 million
C. $0
D. $70 million

Answer: C

Step by step

1. Notes total 50 + 50 + 100 = $200m; collateral is $220m, so overcollateralization = 220 - 200 = $20m.
2. Loss of $120m: the first $20m is absorbed by overcollateralization.
3. The remaining $100m is absorbed by the two subordinated tranches (50 + 50).
4. Loss left for the senior tranche = 120 - 20 - 100 = $0.

Why C is correct: Overcollateralization plus the two subordinated tranches provide $120m of loss absorption, exactly the loss, so the senior tranche is untouched.

Why the others are wrong

- **A:** Assigns the entire loss to the senior tranche, ignoring OC and subordination.
- **B:** Reports the OC amount as the senior loss; OC is a buffer that absorbs loss, not a loss borne by seniors.
- **D:** Computes 120 - 50: counts only one subordinated tranche and ignores the OC and the second subordinated tranche.

Q1204

AnalystPrep Q.3047

Consider the following asset-backed security (ABS) structure:

Tranche	Principal
Senior tranche	$150 million
Subordinated tranche A	$60 million
Subordinated tranche B	$20 million

The asset pool is worth $250,000,000. What is the amount of overcollateralization, and at what amount of losses will senior tranche investors begin to lose money?

A. Overcollateralization: $20,000,000; senior losses begin above $100,000,000
B. Overcollateralization: $40,000,000; senior losses begin above $80,000,000
C. Overcollateralization: $20,000,000; senior losses begin above $80,000,000
D. Overcollateralization: $60,000,000; senior losses begin above $40,000,000

Answer: A

Step by step

1. Notes = 150 + 60 + 20 = $230m.
2. Overcollateralization = 250 - 230 = $20m.
3. Loss buffer below the senior tranche = OC 20 + tranche B 20 + tranche A 60 = $100m.
4. Senior investors begin to lose money only when pool losses exceed $100m.

Why A is correct: Losses must exhaust the overcollateralization and then both subordinated tranches before reaching the senior tranche.

Why the others are wrong

- **B:** OC of 40 comes from mis-adding the notes (e.g., 250 - 210, dropping tranche B), and the 80 threshold is inconsistent with the structure.
- **C:** OC is right, but the threshold omits the OC (20 + 60 = 80) or tranche B, forgetting that OC absorbs losses first.
- **D:** Treats tranche A's $60m as the OC and gives a threshold that ignores the subordinated tranches.

Q1205

BT answers L2.T6.106.1

REBUILT — Original relied on an unsourced case study (GSAMP Trust 2006-NC2 class X rating details); rewritten generically on the equity class created by over-collateralization, same concept.

A securitization's most junior (equity) class is created through overcollateralization, so that asset value exceeds the value of the notes senior to it. Each of the following is true about this equity class EXCEPT:

A. It is typically retained by the originator
B. It is typically non-rated
C. It is rated AAA because it is backed by the largest share of collateral
D. It absorbs the first losses on the asset pool

Answer: C

Step by step

1. The equity class is the first-loss piece: the most junior, lowest credit quality class, often non-rated.
2. It is usually retained by the originator and absorbs initial losses.

3. A AAA rating belongs to the senior tranche; the equity class cannot be AAA, so C is the false statement.

Why C is correct: The equity/first-loss class is non-rated or lowest-rated and absorbs losses first; AAA describes the senior class.

Why the others are wrong

- **A:** True: the originator typically keeps the first-loss piece.
- **B:** True: the first-loss piece is often non-rated.
- **D:** True: first-loss means it is hit by losses before any other class.

Q1206

BT answers P2.T6.610.2 · BT notes P2.T6.610.2

If the nominal value of the credit-sensitive assets in the collateral pool is exactly equal to the nominal value of issued securities, which of the following is equal to zero?

A. First-loss piece
B. Excess spread
C. External credit enhancement
D. Over-collateralization

Answer: D

Step by step

1. Overcollateralization = nominal value of pool assets - nominal value of issued securities.
2. Assets equal securities, so the difference is 0.

Why D is correct: Overcollateralization is the excess of asset value over issued note value; with equality the excess is zero.

Why the others are wrong

- **A:** The first-loss piece still exists as the most junior tranche regardless of OC.
- **B:** Excess spread depends on coupon income versus costs, not on nominal values.
- **C:** External credit enhancement (guarantees, insurance) is unrelated to nominal asset/liability equality.

Q1207

Schweser Module Quiz 41.2 Q1

A major benefit of securitization for a financial institution is the ability to remove assets from the balance sheet, which lowers risk and the required regulatory capital. While a large portion of the risk is removed, the originator often keeps a portion of it. Which of the following terms best identifies the risk that is maintained by the originator?

A. Correlation
B. Excess spread
C. First-loss piece
D. Guarantor of collateral value

Answer: C

Step by step

1. Originators typically retain ownership of the most junior class.
2. That class is the first-loss piece, which absorbs initial losses.
3. So the retained risk is the first-loss piece.

Why C is correct: The originator holds the first-loss (equity) piece, the most junior, lowest-quality class that bears losses first.

Why the others are wrong

- **A:** Correlation is a modeling input for losses, not a retained piece of the structure.
- **B:** Excess spread is a cash-flow credit enhancement, not the risk retained by the originator.
- **D:** A guarantor of collateral value is a third party, not the originator.

LO 41.c

Analyze the differences in the mechanics of issuing securitized products using a trust versus a special purpose vehicle (SPV) and distinguish between the three main SPV structures: amortizing, revolving, and master trust.

Q1208 AnalystPrep Q.2929

REBUILT — Original 'all of the above' item listed securitization itself as an extra use despite 'besides'; rewrote as a single-best-answer item on the non-securitization uses of an SPV (credit-linked notes).

Besides cash flow securitization, a special purpose vehicle (SPV) can also be used to issue credit-linked notes (CLNs). Which of the following correctly describes a CLN issued by an SPV?

A. The note's performance depends on the credit of the sponsoring entity that set up the SPV.
B. The note's performance depends on the performance of assets that have been sold to the SPV, not on the sponsoring entity.
C. The note is issued directly by the originator, which keeps the reference assets on its own balance sheet.
D. The note can only be issued if the SPV is not legally separate from the sponsoring entity.

Answer: B

Key correction: Source key (AnalystPrep) said D (all of the above); the original stem said 'besides being used to securitize' yet option A restated securitization itself, so the item was rebuilt.

Step by step

1. Recall that an SPV is a separate legal entity that holds assets sold to it by the sponsor.
2. A CLN issued by the SPV is linked to those assets, so investors take the credit risk of the asset pool.
3. Therefore the CLN has no credit linkage to the sponsoring entity (unlike a CLN issued by the originator directly).

Why B is correct: Because the SPV is legally separate and owns the assets, a CLN it issues is linked to the performance of those assets, not to the sponsor's credit.

Why the others are wrong

- **A:** This describes a CLN issued by the originator directly; SPV-issued CLNs have no credit linkage to the sponsor.
- **C:** Confuses the SPV with the originator; an SPV-issued CLN is issued by the SPV, which owns the reference assets.
- **D:** Reverses the logic: the SPV's separateness (true sale, bankruptcy remoteness) is what delinks the note from the sponsor.

Q1209 BT answers P2.T6.610.3 · BT notes P2.T6.610.3

Which of the following structures is MOST LIKELY to be used by an infrequent issuer of securities whose asset pool consists of credit card debt, i.e., short-dated assets with a relatively high prepayment speed?

A. Master trust
B. Amortizing structure
C. Revolving structure
D. None of the above

Answer: C

Step by step

1. Identify the asset: short-dated receivables with high prepayment speed and no fixed amortization schedule.
2. Such assets suit a revolving structure, where principal collections buy new receivables during the revolving period.
3. The issuer is infrequent, so the master trust (designed for frequent issuers making multiple issues) is not needed.

Why C is correct: Revolving structures fit short-dated, fast-prepaying assets such as credit card debt, and the master trust is reserved for frequent issuers.

Why the others are wrong

- **A:** A master trust is used by frequent issuers who issue multiple series from one pool; the issuer here is infrequent.
- **B:** Amortizing (pass-through) structures suit assets with scheduled amortization such as mortgages and consumer loans, not short-dated revolving receivables.
- **D:** Incorrect because the revolving structure fits the description.

Q1210 GARP PE1 Q21 · GARP PE2 Q60

REBUILT — Original option D described a master-trust feature (multiple issues) and was arguably defensible; rewrote D so that exactly one option (B) is correct.

A bank plans to securitize car loans using an SPV. The bank would sell the loans to the SPV through a "true sale" and the SPV would issue securities under a revolving securitization structure. Which of the following statements is correct?

A. The credit quality of the securitized car loan assets is enhanced if the principal value of the securities issued is higher than the principal value of the assets.
B. The SPV can give the bank access to cheaper funding if the credit quality of the securitized car loan assets is higher than the credit quality of the bank's balance sheet.
C. Under a revolving structure, prepayment assumptions are not incorporated, which typically results in principal being paid to investors coupon-by-coupon throughout the life of the security.
D. Under a revolving structure, principal collected on the loans is passed through to investors as it is received, so the notes are priced and traded on the weighted-average life of the loans.

Answer: B
Step by step
1. A true sale places the assets in a separate SPV, so investors price the credit quality of the asset pool, not the bank.
2. If the car loans are better credit than the bank's balance sheet, the notes can be funded at a lower cost than the bank's own unsecured debt: B is true.
3. Overcollateralization means notes issued are LESS than the assets, so A is backwards.
4. Coupon-by-coupon principal payment with prepayment-based WAL pricing describes an amortizing (pass-through) structure; in a revolving structure principal collections buy new receivables, so C and D are wrong.

Why B is correct: True-sale securitization lets the funding cost reflect the asset pool's credit quality, which can be better than the originator's own.
Why the others are wrong
- **A:** Reverses overcollateralization: credit is enhanced when the notes issued are less than the assets, not more.
- **C:** Describes an amortizing (pass-through) structure; in a revolving structure principal is reinvested in new receivables and later repaid through controlled amortization or a soft bullet.
- **D:** Describes the pass-through amortizing structure; in a revolving structure principal is not simply passed through but used to purchase new receivables.

Q1211 Mock 3 Q59

REBUILT — Original correct option mixed in master-trust and amortizing-style wording ('priced on weighted-average life'); rewrote the correct option to state the defining revolving feature (reinvestment of principal) cleanly.

A bank is planning to securitize a portfolio of automobile loans and is considering an amortizing or a revolving structure for the special purpose vehicle. Which of the following statements correctly describes a revolving securitization structure?

A. Principal received from the underlying loans is passed through to investors as it is collected.
B. The pool of underlying assets is fixed at closing and cannot be replenished.
C. Investors are exposed to the credit risk of the originating bank rather than of the underlying loan pool.
D. During the revolving period, principal collections are used to purchase new receivables, and investors are later repaid through controlled amortization or a single soft bullet payment.

Answer: D
Step by step
1. In a revolving structure principal collections are not passed through; they are used to buy new receivables meeting the eligibility criteria.
2. After the revolving period, principal is repaid either in equal instalments (controlled amortization) or trapped and paid as a lump sum (soft bullet).
3. A and B describe an amortizing structure; C contradicts the true sale to a bankruptcy-remote SPV.

Why D is correct: The defining feature of a revolving structure is reinvestment of principal collections in new receivables, with repayment later by controlled amortization or soft bullet.
Why the others are wrong
- **A:** Describes an amortizing (pass-through) structure.
- **B:** A fixed, non-replenished pool is the amortizing case; revolving pools are replenished.
- **C:** Contradicts the true sale: the SPV separates the assets, so investors bear the asset pool's risk, not the originator's.

Q1212 Schweser Module Quiz 41.2 Q2

Securitized products are often customized to meet the needs of the investor as well as the originator. What type of asset-backed securities (ABSs) typically uses a revolving structure?

A. Residential mortgage.
B. Credit card debt.
C. Commercial mortgage.
D. Commercial paper.

Answer: B

Step by step

1. A revolving structure suits assets repaid on a revolving basis with short horizons and high prepayment, with no pre-specified amortization schedule.
2. Credit card debt (and auto loans) fit; principal is returned in lump sums, not on an amortizing schedule.

Why B is correct: Credit card receivables are short-dated, prepay quickly and have no fixed amortization schedule, so they use a revolving structure.

Why the others are wrong

- **A:** Residential mortgages have amortization schedules and use the amortizing (pass-through) structure.
- **C:** Commercial mortgages likewise amortize on a schedule and use the amortizing structure.
- **D:** Commercial paper is a short-term funding instrument, not the typical revolving-structure collateral type tested here.

LO 41.d

Explain the reasons for and the benefits of undertaking securitization.

Q1213 BT answers P2.T6.611.1 · BT notes P2.T6.611.1

Roundcore Bank is going to securitize a portfolio of long- and medium-term commercial real estate loans that it originated during its most recent fiscal year. Which of the following is the LEAST likely to be the bank's motivation?

A. Diversify its funding mix
B. Increase return on equity (ROE)
C. Increase average duration of balance sheet assets
D. Remove certain credit risks from the balance sheet

Answer: C

Step by step

1. Recall the three main reasons to securitize: funding (diversified funding mix, lower cost, reduced maturity mismatch), balance sheet/capital management (regulatory capital relief, higher ROE), and risk management (removing credit risk and non-performing assets).
2. A: selling the loans to an SPV that issues notes gives a new funding source, so the funding mix is diversified: a valid motivation.
3. B: removing assets lowers required capital, so the equity base shrinks relative to earnings and ROE rises: a valid motivation.
4. D: credit risk on the transferred loans leaves the balance sheet (apart from any retained first-loss piece): a valid motivation.
5. C: the long- and medium-term loans leave the balance sheet and are replaced by cash, and notes issued by the SPV match the asset horizon, so securitization shortens, not lengthens, average asset duration and reduces duration mismatch. This is the least likely motivation.

Why C is correct: Securitization converts long-dated illiquid loans into cash, reducing the duration of balance sheet assets and the funding/duration mismatch, so increasing duration is not a motivation.

Why the others are wrong

- **A:** A genuine motivation: securitization adds a funding source and lowers funding cost; picking it confuses 'least likely' with 'most likely'.
- **B:** A genuine motivation: capital relief from removing assets raises ROE; it is wrongly chosen if one thinks securitization only affects funding.
- **D:** A genuine motivation: credit risk (and non-performing assets) is transferred to the SPV and note investors; choosing it ignores the risk management benefit.

Q1214 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

The treasurer of a regional bank rated BBB plans to fund a new pool of auto loans by transferring them to a special purpose vehicle (SPV) that will issue asset-backed securities (ABS). A board member objects that the secondary market for ABS is less liquid than the market for the bank's own senior bonds, and that ABS investors will therefore demand a higher yield. Which of the following is the most accurate response regarding the funding cost of the securitization?

A. The cost can still be lower because the notes are rated on the quality of the asset pool, and are typically rated higher than the bank's own bonds.

B. The cost will be higher because illiquidity always outweighs any difference in credit ratings between the ABS and the originator's bonds.

C. The cost will be lower because the SPV is not subject to any credit risk once the loans are transferred.

D. The cost will be the same because the ABS carry the originator's credit rating, which is applied to all notes issued by the SPV.

Answer: A

Step by step

1. Securitization de-links the rating of the issued notes from the rating of the originator.
2. Most SPV notes are rated higher than bonds issued directly by the originator (here BBB).
3. The ABS market is less liquid, which adds to the yield, but the benefit of the higher rating is frequently larger than the illiquidity premium.
4. Net: issuing through the SPV is frequently the cheaper funding route, which diversifies the funding mix and creates a cost saving.

Why A is correct: The notes are rated on the asset pool and structure rather than on the originator, so they typically earn higher ratings than the originator's bonds; this usually more than offsets the illiquidity premium.

Why the others are wrong

- **B:** Wrong: GARP states that the lower liquidity adds to the ABS yield but it is frequently the case that the originator's cost is still lower because of the higher rating; 'always' is false.
- **C:** Wrong: the SPV's notes still carry credit risk from the asset pool (absorbed by tranching and credit enhancement); the lower cost comes from the higher rating, not from an absence of risk.
- **D:** Wrong: this ignores the de-linking of the notes' rating from the originator's rating, which is the main source of the funding benefit.

Q1215 New question written for this bank (GARP practice-exam style)

NEW — No source question covered this LO adequately; written from the 2026 reading (content per the GARP 2026 book) in GARP practice-exam style.

A bank holds a large portfolio of loans that are each risk-weighted at 100% under Basel I, so that it must hold capital of 8% against them. The bank sells the loans to an SPV that issues notes, and the bank retains the first-loss (equity) piece of the issue. Which of the following statements best describes the effect of this securitization on the bank?

A. Regulatory capital requirements on the pool fall to zero because the SPV is not a bank, and ROE rises.

B. Regulatory capital requirements on the pool are significantly reduced but not eliminated, and ROE is likely to improve.

C. Regulatory capital requirements are unchanged because the retained first-loss piece carries the same risk as the whole pool, and ROE is unchanged.

D. Regulatory capital requirements fall, but ROE declines because the bank must now issue more preferred stock to replace the capital released.

Answer: B

Step by step

1. The SPV is not categorized as a bank, so it is not subject to the 8% capital requirement on the transferred assets.
2. The bank keeps the first-loss piece, so it retains a portion of the exposure and a capital charge on that piece.
3. Hence capital requirements fall significantly but are not eliminated.
4. Lower capital requires less preferred stock and equity issuance, which raises ROE.

Why B is correct: Retaining the first-loss piece leaves the bank with some capital charge, so relief is large but partial; needing less capital increases ROE.

Why the others are wrong

- **A:** Wrong: ignores the retained first-loss piece, which keeps part of the exposure (and capital charge) with the bank; relief is not 100%.

- **C:** Wrong: the retained piece is only a small fraction of the pool, so the charge falls substantially, and lower capital raises ROE.
- **D:** Wrong: reverses the effect; securitization reduces the need to issue preferred stock and other equity, so ROE increases, not declines.

LO 41.e

Describe and assess the various types of credit enhancements.

Q1216

AnalystPrep Q.2710

REBUILT — Original relied on internal/external enhancement labels and items (direct equity issue, holdback, letter of credit) outside the Schweser/GARP list of enhancements; rebuilt around the five standard enhancements, asking which is provided by a third party.

In structuring a securitization, which of the following credit enhancements is provided by a third party outside the SPV (originator and note structure) rather than being built into the pool or note structure itself?

A. Over-collateralization
B. Pool insurance
C. Excess spread
D. Subordination of Class B notes to Class A notes

Answer: B

Step by step

1. Over-collateralization: the assets in the pool exceed the notes issued, so it is built into the structure.
2. Excess spread: asset return minus note interest, generated by the pool itself and trapped in a reserve account.
3. Subordination: junior notes absorb losses before senior notes; this is a feature of the note structure.
4. Pool insurance: a policy from a composite insurance company covering loss of principal in the collateral pool; the insurer is an external party whose claims-paying rating matters for the issue's rating.

Why B is correct: Pool insurance is the only listed enhancement that depends on an outside insurer; the others come from the pool and note structure.

Why the others are wrong

- **A:** Over-collateralization is created by the structurer issuing fewer notes than the asset value; no outside party is involved.
- **C:** Excess spread arises from the asset yield exceeding note coupons; it is internal to the SPV cash flows.
- **D:** Subordination is a tranching feature of the notes; the structure itself provides the protection.

Q1217

AnalystPrep Q.2048

Which of the following credit enhancement techniques would be most appropriate when the nominal value of assets in the pool is greater than the nominal value of issued securities?

A. Over-collateralization
B. Pool insurance
C. Excess spread
D. Margin step-up

Answer: A

Step by step

1. The situation described is assets in the pool exceeding notes issued in nominal value.
2. That excess collateral is the definition of over-collateralization; it absorbs initial losses with no impact on investors.

Why A is correct: Over-collateralization means pool assets exceed the nominal value of the securities, giving a first cushion against losses.

Why the others are wrong

- **B:** Pool insurance is an insurance policy from a composite insurer covering principal loss; it does not depend on the asset/note size gap.
- **C:** Excess spread is the difference between asset return and note interest, a cash-flow measure, not a nominal-value gap (confuses balance with income).

- **D:** Margin step-up raises the coupon after a call date to encourage refinancing; it has nothing to do with collateral size.

Q1218 BT answers P2.T6.611.2 · BT notes P2.T6.611.2

Among the following types of credit enhancement, which is the difference between the return on the underlying assets and the interest rate payable on the issued notes (liabilities)?

A. Excess spread
B. Over-collateralization
C. Step-up margin
D. Return on equity (ROE)

Answer: A

Step by step

1. The definition given is asset return minus interest payable on the notes.
2. That is excess spread; it covers expenses and losses, and any surplus is held in a reserve account.

Why A is correct: Excess spread is by definition the asset return less note interest, available to absorb expenses and losses.

Why the others are wrong

- **B:** Over-collateralization is a balance-based cushion (assets exceed notes), not a rate difference.
- **C:** Step-up margin is a coupon increase after a call date, not an asset/liability return difference.
- **D:** ROE is a profitability ratio for equity holders and is not a credit enhancement.

Q1219 Schweser Module Quiz 41.2 Q3

Which of the following statements regarding credit enhancements in the process of structuring a securitization through a special purpose vehicle (SPV) is correct?

A. The securitization process is structured such that the asset side of the SPV has a lower cost than the liability side of the SPV.
B. Credit enhancements are typically only associated with mortgage-backed securities (MBSs) and are not used in other types of asset-backed securities (ABSs).
C. The most senior class of notes is often overcollateralized in order to reduce the risk of the asset-backed security (ABS).
D. A margin step-up is sometimes used by asset-backed securities (ABSs) where the coupon structure increases after a call date.

Answer: D

Step by step

1. A: the SPV is structured so liabilities (notes) cost less than the asset yield, creating excess spread; A reverses this.
2. B: enhancements are used for both ABS and MBS.
3. C: it is the lowest (junior) class that is overcollateralized.
4. D: a margin step-up raises the coupon after the call date, an incentive for investors and for the issuer to call; this is correct.

Why D is correct: Margin step-ups raising the coupon after a call date are a recognized ABS enhancement feature.

Why the others are wrong

- **A:** Reverses the cost relationship; the liability side is cheaper, which produces excess spread.
- **B:** Wrongly restricts enhancements to MBS; they are used across ABS and MBS.
- **C:** Confuses senior with lowest class; overcollateralization protects the lowest class of notes.

LO 41.f

Explain the various performance analysis tools for securitized structures and identify the asset classes they are most applicable to.

Q1220 Mock 4 Q61

A revolving securitization of credit card receivables includes an early amortization trigger that is activated if the monthly payment rate (MPR) falls below a minimum threshold. A risk analyst at an investor is explaining the purpose and effect of this feature to a client. Which of the following statements is correct?

A. The trigger allows the originator to extend the revolving period if the MPR deteriorates, giving the pool time to recover.

B. The trigger ends the revolving period early and begins returning principal to investors, protecting them from further deterioration in the pool.

C. The trigger requires the originator to substitute higher-quality receivables into the pool until the MPR returns above the threshold.

D. The trigger converts the transaction from a revolving to an amortizing structure while leaving the expected weighted-average life of the notes unchanged.

Answer: B

Step by step

1. The delinquency ratio, default ratio and MPR are monitored against thresholds and serve as triggers that signal early amortization of the receivables pool.
2. If the threshold is breached, reinvestment of principal collections stops and principal is passed to noteholders.
3. This protects investors from further pool deterioration and shortens the notes' life.

Why B is correct: A breached trigger ends the revolving period early and repays principal to investors, limiting their exposure to a deteriorating pool.

Why the others are wrong

- **A:** Reverses the effect: the trigger shortens, not extends, the revolving period.
- **C:** Substituting collateral is not the trigger mechanism; the response is to stop revolving and amortize.
- **D:** Amortizing early repays principal sooner, so the expected weighted-average life shortens, not stays unchanged.

Q1221 AnalystPrep Q.2053

REBUILT — Original options were loosely worded (half secured, partly secured); rebuilt so the non-recourse/DSCR link is explicit and exactly one option is correct.

A risk analyst contrasts commercial mortgage-backed securities (CMBS) with residential mortgage-backed securities (RMBS). Which statement best describes the notable difference and its implication for performance analysis?

A. CMBS loans are recourse to the borrower, so the delinquency ratio is the key performance measure.

B. CMBS loans are non-recourse because they are fully secured by the underlying property, so the debt service coverage ratio (DSCR) becomes crucial for evaluating credit risk.

C. CMBS loans are non-recourse because they are only partly secured by the property, so the monthly payment rate (MPR) becomes crucial for evaluating credit risk.

D. CMBS loans are recourse to the borrower because they are unsecured, so the loss curve is the key performance measure.

Answer: B

Step by step

1. CMBS are backed by loans on commercial properties and are non-recourse: the lender can look only to the property, not to the borrower's other assets.
2. The loan is fully secured by the property, so repayment depends on the cash the property generates.
3. The measure of that ability to repay is DSCR = net operating income / debt payments (DSCR < 1 means income does not cover debt service).
4. Hence B.

Why B is correct: Because CMBS are non-recourse and secured only by the property, credit risk rests on the property's income, which DSCR captures.

Why the others are wrong

- **A:** CMBS are non-recourse, not recourse; the delinquency ratio is a credit card ABS measure.
- **C:** Non-recourse is right but the loan is fully, not partly, secured, and MPR is a credit card measure.
- **D:** CMBS are secured by property, not unsecured, and non-recourse; the loss curve is an auto loan measure.

Q1222 BT answers P2.T6.611.3 · BT notes P2.T6.611.3

Consider the following three pairs of performance indicators:
I. Delinquency ratio and default ratio
II. Loss curves and absolute prepayment speed
III. Debt service coverage ratio and weighted average maturity (WAM)
Which sequence below correctly associates the collateral type with its primary performance indicators?

A. I. Auto loans, II. Commercial mortgages, III. Credit cards
B. I. Auto loans, II. Residential mortgages, III. Credit cards
C. I. Credit cards, II. Auto loans, III. Commercial mortgages
D. I. Commercial mortgages, II. Credit cards, III. Auto loans

Answer: C

Step by step

1. I: delinquency ratio (90+ days past due / receivables pool) and default ratio (write-offs / pool) are credit card ABS measures.
2. II: loss curves (expected cumulative loss) and absolute prepayment speed (APS) are auto loan ABS measures.
3. III: DSCR (NOI / debt payments) and WAM apply to mortgage pools, with DSCR specific to commercial mortgages.
4. Sequence: credit cards, auto loans, commercial mortgages = C.

Why C is correct: Each indicator pair is tied to the asset class it was designed for: delinquency/default to credit cards, loss curve/APS to auto loans, DSCR/WAM to commercial mortgages.

Why the others are wrong

- **A:** Puts delinquency/default on auto loans and loss curves on commercial mortgages, and DSCR on credit cards; all three are mismatched.
- **B:** Same mismatches, with loss curves/APS assigned to residential mortgages and DSCR to credit cards.
- **D:** Assigns delinquency/default to commercial mortgages, loss curves/APS to credit cards, and DSCR/WAM to auto loans; all reversed.

Q1223 Schweser Module Quiz 41.4 Q1

Which of the following sets of measures is most likely to be used for a securitized product backed by student loans?

A. Single monthly mortality (SMM), constant prepayment rate (CPR), and Public Securities Association (PSA) rate
B. Loss curves and absolute prepayment speed (APS)
C. Weighted average life (WAL), weighted average maturity (WAM), and weighted average coupon (WAC)
D. Debt service coverage ratio (DSCR) and monthly payment rate (MPR)

Answer: A

Step by step

1. CPR/SMM and the PSA benchmark are the prepayment methodologies used for pools of mortgages and student loans.
2. APS is used for auto loans, WAL/WAM/WAC are pool descriptors, DSCR is for commercial real estate and MPR for credit cards.

Why A is correct: Student-loan pools prepay like mortgage pools, so SMM, CPR and PSA are the standard prepayment tools.

Why the others are wrong

- **B:** APS is used for auto-loan ABS, not student loans; loss curves are a credit measure.
- **C:** These describe pool characteristics, not prepayment forecasting methodologies.
- **D:** DSCR is a commercial real-estate measure and MPR a credit-card measure.

LO 41.g

Define and calculate the delinquency ratio, default ratio, monthly payment rate (MPR), debt service coverage ratio (DSCR), the weighted average coupon (WAC), the weighted average maturity (WAM), and the weighted average life (WAL) for relevant securitized structures.

Q1224 AP Mock 3 Q12

REBUILT — Option A (2.93%) had no identifiable error behind it; replaced with 3.59% (90-day bucket left out of the denominator), and the stem now states the buckets are mutually exclusive.

An ABS based on credit card receivables has the following composition:

Category	Amount (USD)
Current receivables	25,200,000
Receivables over 30 days past due	8,250,000
Receivables over 60 days past due	2,750,000
Receivables over 90 days past due	1,300,000

Assuming the buckets are mutually exclusive, calculate the delinquency ratio of the ABS.

A. 3.59%
B. 3.47%
C. 5.16%
D. 7.33%

Answer: B

Step by step

1. Delinquency ratio = receivables more than 90 days past due / total receivables.
2. Total receivables = 25.20 + 8.25 + 2.75 + 1.30 = USD 37.50 million (buckets are mutually exclusive).
3. $1.30/37.50 = 3.467\% \approx 3.47\%$.

Why B is correct: The delinquency ratio uses only the receivables overdue by more than 90 days, divided by the whole receivables pool (current plus all past-due buckets).

Why the others are wrong

- **A:** Leaves the 90-day bucket out of the denominator: $1.30/36.20 = 3.59\%$.
- **C:** Divides by current receivables only: $1.30/25.20 = 5.16\%$; the denominator must be the whole pool.
- **D:** Uses the 60-day bucket in the numerator: $2.75/37.50 = 7.33\%$; only the over-90-day bucket counts.

Q1225 AnalystPrep Q.2052

Credit card ABS are monitored using the delinquency ratio, the monthly payment rate (MPR), and the default ratio. Which of the following statements correctly defines the default ratio?

A. The value of credit card receivables overdue by more than 90 days as a proportion of total credit card receivables.
B. The value of credit card receivables written off during the period as a proportion of total credit card receivables.
C. The proportion of the pool's interest and principal that is repaid in a particular period.
D. The value of credit card receivables written off as a proportion of total liabilities.

Answer: B

Step by step

1. Default ratio = receivables written off during the period / total credit card receivables.
2. It measures realized credit losses; with the delinquency ratio it indicates pool credit loss and is tied to early-amortization triggers.
3. Only B matches this definition.

Why B is correct: The default ratio captures actual charge-offs (written-off receivables) relative to the total receivables pool.

Why the others are wrong

- **A:** This is the delinquency ratio (over-90-day receivables), an early-warning measure rather than realized losses.
- **C:** This is the monthly payment rate (MPR), a payment-speed measure.
- **D:** Uses total liabilities as the denominator; the denominator is the receivables pool.

Q1226 AnalystPrep Q.3046

REBUILT — Distractor pairs rebuilt so that each wrong option reflects a specific realistic error (wrong bucket, swapped ratios, wrong denominator).

Tahoma Bank issues credit cards to US consumers. The aging analysis of its receivables is:

Age bucket	Receivables (USD million)
0-30 days	600
31-60 days	250
61-90 days	170
91-120 days	130
121 days and more	60
Total	1,210

The amount written off during the period is USD 35 million. The delinquency ratio and default ratio are closest to:

A. Delinquency ratio: 4.96%; Default ratio: 2.89%
B. Delinquency ratio: 2.89%; Default ratio: 15.7%
C. Delinquency ratio: 15.7%; Default ratio: 2.89%
D. Delinquency ratio: 15.7%; Default ratio: 18.4%

Answer: C

Step by step

1. Delinquency ratio = receivables over 90 days / total = (130 + 60)/1,210 = 190/1,210 = 15.70%.
2. Default ratio = written off / total receivables = 35/1,210 = 2.89%.
3. Both ratios use total receivables (1,210) as the denominator.

Why C is correct: The over-90-day buckets (91-120 and 121+) form the delinquency numerator, and the write-off amount forms the default numerator, each over total receivables.

Why the others are wrong

- **A:** Counts only the 121+ bucket as delinquent: 60/1,210 = 4.96%; the 91-120 bucket is also over 90 days.
- **B:** Swaps the two ratios: 2.89% is the default ratio and 15.7% the delinquency ratio.
- **D:** Delinquency is right but the default ratio divides write-offs by delinquent receivables (35/190 = 18.4%) instead of total receivables.

Q1227

AnalystPrep Q.3089

A manager of commercial mortgage-backed securities has a pool with net operating income of USD 68 million from the mortgaged properties. Total debt payments on the notes issued against these mortgages are USD 54 million. The mortgage pool is worth about USD 5 billion. The debt service coverage ratio (DSCR) is closest to:

A. 1.26
B. 0.79
C. 0.0136
D. 0.0108

Answer: A

Step by step

1. DSCR = net operating income / total debt payments.
2. 68/54 = 1.259 ≈ 1.26.
3. The USD 5 billion pool value is irrelevant to DSCR.
4. DSCR above 1 means property cash flow covers debt payments.

Why A is correct: DSCR compares the cash the properties generate (NOI) with required debt service; 1.26 means cash flow covers debt payments 1.26 times.

Why the others are wrong

- **B:** Inverts the ratio: 54/68 = 0.79.
- **C:** Divides NOI by pool value: 68/5,000 = 0.0136.
- **D:** Divides debt payments by pool value: 54/5,000 = 0.0108.

Q1228

AnalystPrep Q.3090

REBUILT — Distractors C and D had no identifiable error behind them; replaced with realistic mistakes (omitted pool; largest pool's yield).

An MBS comprises four mortgage pools:

Pool	Balance (USD million)	Yield
Apartment	24	8.4%
Villa	32	7.0%
Penthouse	16	6.8%
Farmhouse	8	5.6%

The weighted average coupon (WAC) of the MBS is closest to:

A. 6.95%
B. 7.24%
C. 7.42%
D. 7.00%

Answer: B

Step by step

1. Total balance = 24 + 32 + 16 + 8 = USD 80 million.
2. Weighted sum = $24(8.4\%)+32(7.0\%)+16(6.8\%)+8(5.6\%) = 2.016+2.240+1.088+0.448 = 5.792$ million.
3. WAC = $5.792/80 = 7.24\%$.

Why B is correct: WAC weights each pool's mortgage rate by its share of the total outstanding balance.

Why the others are wrong

- **A:** Simple average of the four yields: $(8.4 + 7.0 + 6.8 + 5.6)/4 = 6.95\%$, ignoring balances.
- **C:** Leaves out the farmhouse pool: $5.344/72 = 7.42\%$.
- **D:** Takes the yield of the largest pool (villa, 7.0%) instead of weighting all pools.

Q1229

AnalystPrep Q.3091

REBUILT — Distractors A and D had no identifiable error behind them; replaced with realistic mistakes.

A servicer's MBS portfolio comprises four pools of commercial mortgages:

Pool	Balance (USD million)	Maturity (days)
Superstore	26	82
Office space	67	169
Cinema	22	253
Restaurant	17	336

The weighted average maturity (WAM) of the pools is closest to:

A. 169 days
B. 210 days
C. 187 days
D. 215 days

Answer: C

Step by step

1. Total balance = 26 + 67 + 22 + 17 = USD 132 million.
2. Weighted sum = 26(82) + 67(169) + 22(253) + 17(336) = 2,132 + 11,323 + 5,566 + 5,712 = 24,733.
3. WAM = $24{,}733/132 = 187.4 \approx 187$ days.

Why C is correct: WAM is the balance-weighted average of remaining maturities, so larger pools (office space) pull the result toward their maturity.

Why the others are wrong

- **A:** Takes the maturity of the largest pool (169 days) rather than weighting all pools.
- **B:** Simple average of the four maturities: $(82 + 169 + 253 + 336)/4 = 210$, ignoring balances.
- **D:** Divides by the wrong total balance (USD 115 million, dropping the restaurant pool): $24{,}733/115 = 215$.

Q1230

Schweser Module Quiz 41.3 Q1

REBUILT — Source explanation lost in OCR and options A and C had no identifiable error; rewrote working and replaced A and C with realistic mistakes.

An MBS is composed of four pools of mortgages:

Balance (USD million)	Maturity (days)
2	90
3	180
5	270
10	360

The weighted average maturity (WAM) of these mortgage pools is closest to:

A. 207 days
B. 225 days
C. 270 days
D. 284 days

Answer: D

Step by step

1. Total balance = 2 + 3 + 5 + 10 = USD 20 million.
2. Weighted sum = $2(90) + 3(180) + 5(270) + 10(360) = 180 + 540 + 1{,}350 + 3{,}600 = 5{,}670$.
3. WAM $= 5{,}670/20 = 283.5 \approx 284$ days.

Why D is correct: Weighting each maturity by its pool's share of the balance gives 283.5 days; the large 360-day pool dominates.

Why the others are wrong

- **A:** Omits the largest (USD 10 million, 360-day) pool: $2{,}070/10 = 207$.
- **B:** Simple average of maturities: $(90 + 180 + 270 + 360)/4 = 225$, ignoring balances.
- **C:** Takes the 270-day maturity of the middle pool instead of computing a weighted average.

Q1231

BT answers P2.T6.612.1 · BT notes P2.T6.612.1

REBUILT — Original EXCEPT question had two false statements (the WAL does not depend on the discount yield, and duration falls when yield rises) and a garbled table; rebuilt as a self-contained WAL calculation.

A USD 1,000,000 mortgage loan fully amortizes over ten years with level annual payments of USD 135,868 (rate 6.0%). Annual principal repayments are:

Year t	Principal repaid (USD)
1	75,868
2	80,420
3	85,245
4	90,360
5	95,782
6	101,528
7	107,620
8	114,077
9	120,922
10	128,177

The weighted average life (WAL) of the loan, in years, is closest to:

A. 5.98
B. 5.50
C. 5.02
D. 4.17

Answer: A

Key correction: Source key (B) said the false statement was 'duration rises to 5.09 at 7%'; duration does fall, but statement D (WAL rises to 6.05) is also false because WAL does not depend on the discount yield, so the original had two defensible answers.

Step by step

1. WAL = $\sum t \times$ (share of principal repaid in year t), i.e. each date weighted by the fraction of principal returned then.
2. $\sum t\,\mathrm{PPMT}_t = 1(75{,}868) + 2(80{,}420) + \cdots + 10(128{,}177) = 5{,}977{,}985$.
3. WAL $= 5{,}977{,}985/1{,}000{,}000 = 5.98$ years.
4. Principal repayments sum to USD 1,000,000, so the weights are the PPMT/L fractions.

Why A is correct: WAL weights each payment date by the fraction of principal returned on that date; because repayments are back-loaded, WAL exceeds the 5.5-year midpoint.

Why the others are wrong

- **B:** Equal-weights the ten payment dates: (1+...+10)/10 = 5.5, ignoring that principal repayments grow over time.
- **C:** Weights dates by the present value of the whole payment (principal plus interest) discounted at 6%: this is the Macaulay duration (5.02), not WAL.
- **D:** Weights dates by interest paid rather than principal repaid: $\sum t\,\mathrm{IPMT}_t / \sum \mathrm{IPMT}_t = 4.17$.

Q1232

GARP PE2 Q32 · Mock 4 Q34

A financial analyst is calculating the weighted average life (WAL) of the loans in an MBS pool. The initial notional value outstanding of the MBS pool is USD 50 million. The payment schedule and the assumed pool factor for the MBS are provided in the table below:

Coupon payment	Date	Actual days	Pool factor	Principal payment (USD million)
0	November 21, 2023	66	1.00	-
1	January 26, 2024	181	0.76	12
2	July 26, 2024	184	0.54	11
3	January 26, 2025	183	0.38	8
4	July 25, 2025	182	0.26	6
5	January 24, 2026	183	0.18	4
6	July 25, 2026	-	-	9

Assuming coupon payments are made semi-annually and there are 365 days in a year, what is the WAL of the MBS pool?

A. 1.06 years
B. 1.24 years
C. 2.12 years
D. 2.68 years

Answer: B

Step by step

1. WAL $= \sum t \times PF$, with t = actual days/365 for each row and PF the pool factor in that row.
2. t values: 66/365 = 0.1808; 181/365 = 0.4959; 184/365 = 0.5041; 183/365 = 0.5014; 182/365 = 0.4986; 183/365 = 0.5014.
3. Products: 0.1808(1.00) = 0.1808; 0.4959(0.76) = 0.3769; 0.5041(0.54) = 0.2722; 0.5014(0.38) = 0.1905; 0.4986(0.26) = 0.1296; 0.5014(0.18) = 0.0903.
4. Sum = 0.1808 + 0.3769 + 0.2722 + 0.1905 + 0.1296 + 0.0903 = 1.2403 years.
5. Cross-check: weighting cumulative time (0.1808, 0.6767, 1.1808, 1.6822, 2.1808, 2.6822) by principal paid (12, 11, 8, 6, 4, 9 of USD 50 million) also gives 62.02/50 = 1.24 years.

Why B is correct: WAL is the time-weighted average of the pool factor outstanding over each period, so each period's length in years is multiplied by the pool factor in force and the products are summed.

Why the others are wrong

- **A:** 1.06 years omits the time-0 row (66 days, PF 1.00), dropping 0.1808 from the sum.
- **C:** 2.12 is the sum of the pool factors for rows 1 to 5 (0.76 + 0.54 + 0.38 + 0.26 + 0.18) with no weighting by period length, which is not a time measure.
- **D:** 2.68 years is the sum of the t values (total elapsed time) with no pool-factor weighting.

Q1233

GARP PE2 Q44 · Mock 4 Q45

A fixed-income portfolio manager is conducting a performance analysis on an agency-MBS and the underlying pool of commercial mortgages. The manager obtains the following information:
The constant prepayment rate (CPR) of the commercial mortgages loan pool: 14.38%
Debt payments per year on the commercial mortgage loans: USD 730,000
Net operating income per year of the commercial mortgage property: USD 1,825,000
The manager uses the following metrics as benchmarks for the commercial mortgage sector analysis:
Debt service coverage ratio (DSCR) of a benchmark obligor: 2.5
Single monthly mortality (SMM) rate of a benchmark security: 0.85%
Which of the following statements is correct?

A. The prepayments on the commercial mortgage measured by the CPR shows that the reduction in yield on the MBS will likely be lower compared to that on a benchmark security.
B. The prepayments on the commercial mortgage measured by the CPR shows that the reduction in yield on the

MBS will likely be higher compared to that on a benchmark security.

C. The estimated DSCR of the commercial mortgage implies that the mortgage borrower's ability to repay the mortgage loan is stronger than that of a comparable benchmark obligor.

D. The estimated DSCR of the commercial mortgage implies that the mortgage borrower's ability to repay the mortgage loan is weaker than that of a comparable benchmark obligor.

Answer: B

Key correction: Mock 4 Q45 quoted the benchmark CPR as 9.76%; the correct value is 9.74% (conclusion unchanged).

Step by step

1. DSCR = NOI / debt payments = 1,825,000 / 730,000 = 2.50, equal to the benchmark 2.5.
2. Convert the benchmark SMM to CPR: $\text{CPR} = 1 - (1 - \text{SMM})^{12} = 1 - (1 - 0.0085)^{12} = 9.74\%$.
3. Pool CPR 14.38% > benchmark 9.74%, so the pool prepays faster.
4. Prepayments at a given price reduce the MBS yield, so faster prepayment means a larger yield reduction than for the benchmark.

Why B is correct: The pool's CPR exceeds the benchmark's annualised SMM-equivalent, so prepayment is faster and the yield on the MBS falls by more; the DSCR is merely equal to the benchmark.

Why the others are wrong

- **A:** Reverses the comparison; a lower yield reduction would need a CPR below the 9.74% benchmark.
- **C:** DSCR is exactly 2.5, equal to (not above) the benchmark, so repayment ability is not stronger.
- **D:** DSCR is exactly 2.5, equal to (not below) the benchmark, so repayment ability is not weaker.

Q1234

Mock 2 Q60

REBUILT — Original options included an out-of-LO CPR option and unsupported values (delinquency 3.39%, MPR 0.81% from principal only); stated the pool-denominator convention and rebuilt distractors as realistic mistakes.

TRL Security is an asset-backed security supported by a portfolio of three- to five-year auto loans. The receivables pool is the sum of the four outstanding categories below (written-off receivables are excluded from the pool). Information on the receivables and the monthly payments is:

Receivable category	Amount
Current receivables	$35,200,000
Over 30 days past due	$4,600,000
Over 60 days past due	$2,450,000
Over 90 days past due	$905,000
Receivables written off	$560,000
Principal and interest	Amount
Monthly principal payments	$350,000
Monthly interest payments	$860,000

Which of the following statements about TRL's key ratios is most accurate?

A. The default ratio is 1.30%.

B. The monthly payment rate is 0.81%.

C. The delinquency ratio is 18.43%.

D. The monthly payment rate is 3.44%.

Answer: A

Step by step

1. Pool = 35,200,000 + 4,600,000 + 2,450,000 + 905,000 = $43,155,000.
2. Default ratio = written off / pool = 560,000 / 43,155,000 = 1.30%.
3. MPR = (principal + interest) / pool = (350,000 + 860,000) / 43,155,000 = 2.80%.
4. Delinquency ratio = receivables over 90 days past due / pool = 905,000 / 43,155,000 = 2.10%.

Why A is correct: The default ratio is write-offs divided by the receivables pool: 560,000/43,155,000 = 1.30%.

Why the others are wrong

- **B:** 0.81% = 350,000/43,155,000 uses principal payments only; MPR includes both principal and interest (correct 2.80%).
- **C:** 18.43% divides all past-due buckets (7,955,000) by the pool; the delinquency ratio uses only receivables over 90 days (2.10%).
- **D:** 3.44% = 1,210,000/35,200,000 divides by current receivables only instead of the full pool (correct MPR 2.80%).

LO 41.h

Explain the prepayment forecasting methodologies and calculate the constant prepayment rate (CPR) and the Public Securities Association (PSA) rate.

Q1235

AnalystPrep Q.2703

REBUILT — Original distractors (4.87%, 5.40%, 4.50%) were not tied to realistic errors; replaced with options that each reflect a specific mistake.

A mortgage-backed security has a single monthly mortality (SMM) rate of 0.45%. The constant prepayment rate (CPR) is closest to:

A. 5.54%
B. 5.27%
C. 5.40%
D. 0.45%

Answer: B

Step by step

1. Annualize the monthly rate: $\text{CPR} = 1 - (1 - \text{SMM})^{12}$.
2. $(1 - 0.0045)^{12} = 0.9955^{12} = 0.94732$.
3. $\text{CPR} = 1 - 0.94732 = 5.27\%$.

Why B is correct: CPR compounds the monthly survival probability over 12 months, so it is slightly less than 12 times the SMM.

Why the others are wrong

- **A:** Compounds upward, $(1 + \text{SMM})^{12} - 1 = 5.54\%$, instead of compounding the survival rate.
- **C:** Simple annualization: $12 \times 0.45\% = 5.40\%$, ignoring compounding.
- **D:** Reports the monthly SMM itself as the annual CPR.

Q1236

AnalystPrep Q.3048

REBUILT — Original wrong options did not map to specific errors; rebuilt options around CPR/12 and full-balance mistakes.

A $100,000 investment sits in a seasoned mortgage pool with a scheduled principal payment of $28.61 this month. The pool's conditional prepayment rate (CPR) is 6%. The single monthly mortality (SMM) rate and the estimated prepayment this month are closest to:

A. SMM: 0.005000; prepayment: $499.86
B. SMM: 0.005143; prepayment: $514.30
C. SMM: 0.005000; prepayment: $500.00
D. SMM: 0.005143; prepayment: $514.15

Answer: D

Step by step

1. Invert the CPR formula: $\text{SMM} = 1 - (1 - 0.06)^{1/12} = 0.005143$.
2. SMM applies to the balance after scheduled principal: $100{,}000 - 28.61 = 99{,}971.39$.
3. Prepayment $= 0.005143 \times 99{,}971.39 = \$514.15$.

Why D is correct: SMM is derived by de-compounding the CPR, and it applies to the balance remaining after the scheduled principal payment.

Why the others are wrong

- **A:** Uses CPR/12 = 0.5% as SMM, ignoring compounding (prepayment 0.005 x 99,971.39 = 499.86).
- **B:** Correct SMM, but applies it to the full $100,000, forgetting to subtract the scheduled principal.
- **C:** Uses CPR/12 and also the full balance; both errors.

Q1237

AnalystPrep Q.3049

A mortgage pool has a $24 million outstanding principal balance. Assuming a prepayment speed of 175% PSA, the single monthly mortality (SMM) rate in month 10 is closest to:

A. 0.002363
B. 0.002793
C. 0.002965
D. 0.002467

Answer: C

Step by step

1. 100% PSA CPR in month 10: $0.2\% \times 10 = 2.0\%$.
2. 175% PSA: $1.75 \times 2.0\% = 3.5\%$.
3. SMM $= 1 - (1 - 0.035)^{1/12} = 0.002965$.

Why C is correct: PSA scales the seasoning ramp (0.2% per month up to month 30) by the PSA multiple, then the annual CPR is converted to a monthly SMM.

Why the others are wrong

- **A:** An SMM of 0.002363 corresponds to a CPR of 2.80%, not the 3.5% CPR that 175% PSA implies in month 10 (wrong multiplier or ramp).
- **B:** An SMM of 0.002793 corresponds to a CPR of 3.30%, not the 3.5% CPR that 175% PSA implies in month 10 (wrong multiplier or ramp).
- **D:** An SMM of 0.002467 corresponds to a CPR of 2.92%, not the 3.5% CPR that 175% PSA implies in month 10 (wrong multiplier or ramp).

Q1238 AnalystPrep Q.3050

A risk analyst computes single monthly mortality (SMM) rates using the PSA benchmark. For month 22 at 140% PSA she obtains an SMM of 0.37%. For month 200 at 90% PSA she obtains an SMM of 0.46%. Which statement is correct?

A. She is correct for both calculations.
B. She is correct for month 22, but incorrect for month 200.
C. She is incorrect for both calculations.
D. She is incorrect for month 22, but correct for month 200.

Answer: D

Step by step

1. Month 22, 100% PSA CPR: $0.2\% \times 22 = 4.4\%$; at 140% PSA: $1.4 \times 4.4\% = 6.16\%$.
2. $\text{SMM}_{22} = 1 - (1 - 0.0616)^{1/12} = 0.528\%$, not 0.37%.
3. Month 200 is past month 30, so 100% PSA CPR is 6%; at 90% PSA: 5.4%.
4. $\text{SMM}_{200} = 1 - (1 - 0.054)^{1/12} = 0.462\% \approx 0.46\%$, correct.

Why D is correct: The month-22 figure omits the 140% multiplier (0.37% is the SMM for 100% PSA, CPR 4.4%), while the month-200 figure correctly uses the 6% plateau times 90%.

Why the others are wrong

- **A:** Wrong for month 22: the correct SMM is 0.53%, not 0.37%, so she is not correct for both.
- **B:** Reverses the result: she is wrong in month 22 (SMM 0.53%, not 0.37%) and right in month 200.
- **C:** Month 200 is actually right: 90% of the 6% plateau gives 5.4% CPR and 0.46% SMM, so she is not incorrect for both.

Q1239 BT answers P2.T5.107.1

REBUILT — Source applied the prepayment SMM to the full pool balance and used ambiguous wording; rebuilt to apply SMM to the balance net of scheduled principal (GARP convention) with clearer data and distractors.

A mortgage pool has an \$800 million balance and a weighted average coupon (WAC) of 7.2%. A sequential-pay CMO splits it into Tranche A (\$300 million), Tranche B (\$200 million), Tranche C (\$175 million) and Tranche D (\$125 million), each paying a 6.0% annual coupon. All principal (scheduled and prepaid) goes to Tranche A until it is retired. In month 1 the pool's total scheduled payment (interest plus scheduled principal) is \$5.430 million, and prepayments run at 200% PSA. The total cash flow to Tranche A in month 1 is closest to:

A. \$2.13 million
B. \$2.40 million
C. \$2.26 million
D. \$5.33 million

Answer: B

Step by step

1. Tranche A interest: $0.06/12 \times 300 = \$1.500$ million.
2. Pool interest $= 0.072/12 \times 800 = 4.800$, so scheduled principal $= 5.430 - 4.800 = \$0.630$ million.
3. Month 1 CPR at 200% PSA: $2 \times 0.2\% \times 1 = 0.4\%$; SMM $= 1 - (1 - 0.004)^{1/12} = 0.03339\%$.

4. Prepayment $= 0.0003339 \times (800 - 0.630) = \$0.267$ million.
5. Tranche A cash flow $= 1.500 + 0.630 + 0.267 = \$2.397 \approx \$2.40$ million.

Why B is correct: Under sequential pay, Tranche A receives all principal (scheduled plus prepaid) in addition to its own interest; prepayment follows from the PSA-derived SMM on the post-scheduled-principal balance.

Why the others are wrong

- **A:** Omits prepayments: 1.500 + 0.630 = 2.13.
- **C:** Uses 100% PSA (CPR 0.2%) instead of 200% PSA, giving prepayment of only \$0.133 million.
- **D:** Treats the 0.4% CPR as a monthly rate: prepayment 0.004 x 800 = 3.2, so 1.5 + 0.63 + 3.2 = 5.33.

Q1240

BT answers P2.T5.104.1

REBUILT — Original option C (6.00%) was a giveaway and 4.70% was a rounded form of 4.696%; replaced distractors with results of realistic errors and used 4.70% rounded.

A mortgage pool has a single monthly mortality (SMM) rate of 0.40%. What is the annualized constant prepayment rate (CPR)?

A. 0.033%
B. 4.70%
C. 4.80%
D. 95.30%

Answer: B

Step by step

1. $\text{CPR} = 1 - (1 - \text{SMM})^{12}$
2. $(1 - 0.004)^{12} = 0.99600^{12} = 0.9530$
3. $\text{CPR} = 1 - 0.9530 = 0.04696 \approx 4.70\%$

Why B is correct: CPR compounds the monthly survival probability (1 - SMM) over 12 months and takes the complement; it is not a simple multiple of SMM.

Why the others are wrong

- **A:** Divided SMM by 12 (0.40%/12) instead of annualizing it.
- **C:** Simple multiplication $12 \times 0.40\%$ ignores compounding of the shrinking balance.
- **D:** This is the 12-month survival factor $(1 - \text{SMM})^{12}$; forgot to subtract it from 1.

Q1241

BT answers P2.T5.104.2

A pass-through MBS assumes a 200% PSA prepayment speed. The pool has an original weighted average maturity (WAM) of 360 months and is now 60 months old. What is the model's single monthly mortality (SMM) rate for month 60?

A. 0.51%
B. 1.00%
C. 1.06%
D. 2.26%

Answer: C

Step by step

1. 100% PSA: CPR rises 0.2% per month to 6% at month 30, then stays at 6%. At month 60 the 100% PSA CPR is 6%.
2. 200% PSA: $\text{CPR} = 2 \times 6\% = 12\%$
3. $\text{SMM} = 1 - (1 - 0.12)^{1/12} = 1 - 0.88^{0.08333} = 0.010596 \approx 1.06\%$

Why C is correct: After month 30 the benchmark CPR is flat at 6%, so 200% PSA gives 12% CPR, which is converted to a monthly rate by $\text{SMM} = 1 - (1 - \text{CPR})^{1/12}$.

Why the others are wrong

- **A:** This is the SMM for 100% PSA (6% CPR); forgot to multiply by 200%.
- **B:** Divided the 12% CPR by 12 instead of using the compounding formula.
- **D:** Doubled the CPR twice (24% CPR) - applied 200% to the already-doubled 12% rate.

Q1242 BT answers P2.T5.104.3

With respect to prepayment measures, EACH of the following statements is true EXCEPT:

A. PSA assumes lower prepayment rates for young mortgage pools

B. 100% PSA assumes a CPR of 0.2% for the first month, increasing by 0.2% per month for the next 29 months when it reaches 6% per year, and remains at 6% per year thereafter

C. At month 30, the remaining principal of a mortgage pool will be lower under a 150% PSA assumption than under 100% PSA

D. The single monthly mortality rate (SMM) under 100% PSA is exactly 50% of the SMM under 200% PSA

Answer: D

Step by step

1. A and B restate the PSA benchmark: CPR starts at 0.2% and ramps by 0.2% per month to 6% at month 30, so young pools prepay slowly.
2. C: 150% PSA has higher CPR in every month, so more principal has prepaid by month 30 and the balance is lower. True.
3. D: PSA scales CPR, not SMM. 200% PSA after month 30: SMM $= 1 - 0.88^{1/12} = 1.0596\%$; 100% PSA: $1 - 0.94^{1/12} = 0.5143\%$.
4. Ratio $0.5143/1.0596 = 48.5\% \neq 50\%$, so D is false.

Why D is correct: Because SMM is a nonlinear (compounded) transformation of CPR, doubling the PSA multiple does not exactly double SMM.

Why the others are wrong

- **A:** True: the PSA ramp makes young pools prepay at lower rates.
- **B:** True: this is the exact definition of the 100% PSA ramp.
- **C:** True: a faster speed (150% PSA) means more cumulative prepayments and a lower balance at month 30.

Q1243 BT answers P2.T6.612.2 · BT notes P2.T6.612.2

REBUILT — Original options on value and duration depend on discount-rate/premium assumptions beyond the LO; restated to test the PSA/CPR/SMM mechanics and WAL direction with exactly one correct statement.

A pass-through MBS backed by a 30-year mortgage pool assumes a 200% PSA prepayment speed. Which of the following statements is TRUE?

A. Raising the assumption to 250% PSA would lengthen the security's weighted average life (WAL).

B. At month 30 under 200% PSA, the CPR is 6%.

C. At month 30 under 200% PSA, the SMM is about 1.06%, equal to $1 - (1 - 2 \times 6\%)^{1/12}$.

D. At month 30 under 200% PSA, the SMM is exactly 1.00%, equal to 12% divided by 12.

Answer: C

Step by step

1. 100% PSA CPR reaches 6% at month 30; 200% PSA therefore gives CPR = 12% at month 30.
2. SMM $= 1 - (1 - 0.12)^{1/12} = 1 - 0.88^{1/12} = 1.0596\% \approx 1.06\%$
3. Faster prepayment (250% PSA) returns principal sooner and shortens WAL, so A is false.

Why C is correct: Month-30 CPR under 200% PSA is 12% and SMM $= 1 - (1 - \text{CPR})^{1/12} \approx 1.06\%$.

Why the others are wrong

- **A:** Higher PSA means faster prepayments, which shorten (not lengthen) the WAL.
- **B:** 6% is the 100% PSA CPR; forgot to scale by 200%.
- **D:** CPR/12 ignores compounding; the correct conversion gives 1.06%.

Q1244 Mock 4 Q33

REBUILT — Replaced weak distractor 0.60% with 0.79% (result of capping/using 9% CPR) so each wrong option stems from a realistic error.

An analyst is modelling prepayments on a pool of 30-year residential mortgages. The pool is currently in month 20 of its life and prepayments follow 150% PSA. Under the standard PSA benchmark, CPR rises linearly from 0% to 6% over the first 30 months and stays at 6% thereafter. What is the single monthly mortality (SMM) rate in month 20?

A. 0.34%

B. 0.50%

C. 0.51%

D. 0.78%

Answer: C

Step by step

1. 100% PSA CPR at month 20: $0.2\% \times 20 = 4\%$
2. 150% PSA: $1.5 \times 4\% = 6\%$
3. SMM $= 1 - (1 - 0.06)^{1/12} = 1 - 0.94^{0.08333} = 0.5143\% \approx 0.51\%$

Why C is correct: Scale the month-20 benchmark CPR by 150% and convert to a monthly rate by compounding.

Why the others are wrong

- **A:** SMM at 100% PSA (4% CPR); forgot the 150% multiplier.
- **B:** Used CPR/12 = 6%/12 instead of the compounding formula.
- **D:** Applied 150% to the 6% cap (9% CPR), ignoring that the pool is only in month 20.

Q1245 AnalystPrep Q.3868

Loans may be terminated or paid off ahead of schedule. This is a particular problem with residential mortgages and other consumer loans pooled as collateral in securitized assets. This risk is known as:

A. Liquidity risk
B. Prepayment risk
C. Call risk
D. Business risk

Answer: B

Step by step

1. Early repayment of pooled loans changes expected cash flows and the WAL of the securitization.
2. That risk is called prepayment risk.

Why B is correct: Prepayment risk is the risk that borrowers repay early, altering the timing of the pool's cash flows.

Why the others are wrong

- **A:** Liquidity risk concerns the ability to trade or fund positions, not early loan repayment.
- **C:** Call risk is the issuer retiring a bond early, not borrowers repaying pooled loans.
- **D:** Business risk relates to operating performance of the firm, not loan pool cash flows.

Q1246 BT answers P2.T5.103.2

REBUILT — Original tested Veronesi mortgage-refinancing economics (outside the 41.h scope, garbled OCR, and the given key conflicted with the options); rebuilt as a self-contained question on the CPR/SMM/PSA prepayment conventions.

An analyst is forecasting prepayments on a pool of 30-year mortgages backing an MBS. Which of the following statements about prepayment forecasting is correct?

A. A 100% PSA pool has a CPR that starts at 0% and rises by 0.2% each month until it reaches 6% in month 30, and stays at 6% thereafter.
B. A 100% PSA pool has a CPR that starts at 0% and rises by 0.2% each month until it reaches 6% in month 60, and stays at 6% thereafter.
C. A monthly SMM of 0.65% implies an annual CPR of exactly 12 × 0.65% = 7.80%.
D. A 150% PSA pool has a CPR that rises by 0.2% each month and reaches a plateau of 9% in month 45.

Answer: A

Key correction: Source key (B) was based on a garbled stem and conflicted with the option text; the original item is outside the 41.h scope and was replaced.

Step by step

1. 100% PSA benchmark: CPR rises 0.2% per month for 30 months, so month 30 CPR = $0.2\% \times 30 = 6\%$; it is constant at 6% afterward.
2. Other speeds are multiples of this path: 150% PSA rises 0.3% per month, so it reaches $0.3\% \times 30 = 9\%$ in month 30.
3. CPR is the compounded annual rate: CPR $= 1-(1-\text{SMM})^{12}$. For SMM = 0.65%, CPR $= 1-(0.9935)^{12} = 7.53\%$, not 7.80%.
4. Only statement A correctly describes the 100% PSA convention.

Why A is correct: The 100% PSA benchmark ramps CPR by 0.2% per month over 30 months to a 6% plateau; other speeds scale this path proportionally, and CPR compounds SMM rather than multiplying it by 12.

Why the others are wrong

- **B:** Wrong seasoning period: the ramp lasts 30 months (0.2% x 30 = 6%), not 60; 60 months would imply only 0.1% per month for 6%.
- **C:** Multiplies SMM by 12 (simple annualization) instead of compounding; the correct CPR is $1-(1-0.0065)^{12} = 7.53\%$.
- **D:** 150% PSA scales the increment to 0.3% per month (not 0.2%) and plateaus at 9% in month 30, not 45.

Q1247

BT answers P2.T5.107.3

REBUILT — Original tested IO/PO strip valuation and duration, which is not covered by the 41.h CPR/PSA calculations; rebuilt as a PSA-speed calculation.

A mortgage pool is 20 months old and is currently prepaying at a CPR of 6.0% per year. Using the standard PSA benchmark (CPR rising 0.2% per month to 6% at month 30), what PSA speed does this pool's current prepayment experience represent?

A. 100% PSA
B. 150% PSA
C. 67% PSA
D. 300% PSA

Answer: B

Key correction: Source key (A) addressed an IO/PO duration question that was out of scope; replaced.

Step by step

1. 100% PSA CPR at month 20: $0.2\% \times 20 = 4.0\%$.
2. $\text{PSA} = \dfrac{\text{CPR}}{0.2\% \times \text{months}} \times 100 = \dfrac{6.0\%}{4.0\%} \times 100 = 150\%$.
3. Check: 150% PSA at month 20 is $0.3\% \times 20 = 6.0\%$, matching the observed CPR.

Why B is correct: PSA speed is the ratio of the observed CPR to the benchmark CPR at the pool's age: 6.0% / 4.0% = 1.5.

Why the others are wrong

- **A:** Used the 30-month plateau (6%) as the benchmark, giving 6%/6% = 100%, ignoring that the pool is only 20 months old.
- **C:** Inverted the ratio: 4.0%/6.0% = 67%.
- **D:** Used a 0.1% monthly increment (the 50% PSA path) as the base: 6%/(0.1% x 20) = 300%.